IHP

SG13G2

SG13G2 DC CV PMOS LV

Rev. 1.2 (190924)

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1 Documentation

1.1 Included Effects in Simulation Model

General	Date, Version	24.09.2019
	Processversion	SG13G2
	Simulator	ADS 2019, Spectre 18.1
	Simulation model	PSP 103.6
	Warning Boundaries	On

Statistic	Corner	x
	Process variations for Monte Carlo Simulation	e
	Mismatch	x
Transistor	DC behavior	x
	Drain diode	x
	Overlap cap.	x
	Junction cap.	x
	1/f Noise	x_1
	HF / Thermal Noise	-
	Substrate resistance (RF)	x_2
	Gate resistance (RF)	x_2
Temperature	-40°C, 25°C, 70°C, 125°C	x

- x measured and modelled
- x_1 parameters taken from SG13
- x_2 measurements are done from a different wafer run than DC model which leads to slight offsets
- e estimated and modelled
- o not included
- - not applicable

1.2 Time range measurement

The verification of the time range measurements is done by the standard IHP verification procedure.

2 Setup

2.1 Measurement Conditions

2.1.1 Measurement Conditions for Transistors

	Sweep	Start[V]	Stop[V]	No. Points	List or f(Vth)
idvg	vg	-500m	1.35	38	
	vb	0	-1.2	7	
	vd	List		3	0.05, 0.6, 1.2
	vs	0		1	
idvd	vd	0	1.35	28	
	vg	Lin f(Vth)	1.35	5	Ref. Setup: idvg Idref: 100.0n Offset: 100.0m Resolution: 1.000m
	vb	0		1	
	vs	0		1	
idvd_vbmin	vd	0	1.35	28	
	vg	Lin f(Vth)	1.35	5	Ref. Setup: idvg Idref: 100.0n Offset: 100.0m Resolution: 1.000m
	vb	-1.2		1	
	vs	0		1	
idvd_vth	vd	0	1.35	28	
	vg	List f(Vth)		3	Ref. Setup: idvg Idref: 100.0n Offset: -200.0m Resolution: 1.000m Steps: 200.0m 200.0m
	vb	0		1	
	vs	0		1	
idvd_vth_vbmin	vd	0	1.35	28	
	vg	List		3	Ref. Setup: idvg

		f(Vth)			Idref: 100.0n Offset: -200.0m Resolution: 1.000m Steps: 200.0m 200.0m
	vb	-1.2		1	
	vs	0		1	
RF idvg	vg	0	1.35	28	
	vd	List		3	0.05, 0.6, 1.2
RF idvd	vd	0	1.35	28	
	vg	0	1.35	5	
RF sdut	freq	100MEG	47.6G	31	
	vd	0	1.35	10	
	vg	0	1.35	10	

2.1.2 Measurement Conditions for Capacitance

	Sweep	Start[V]	Step[V]	Stop[V]	High Node	Low Node
Junction Bulk-Drain	1 VHI	5	-200m	-200m	B	D
Junction Bulk-Source	1 VHI	1	-20m	-300m	B	S
Oxide	1 VHI	-1.35	50m	1.35	B	G
Overlap	1 VHI	1.35	-50m	-1.35	G	D

2.2 Devices

2.2.1 Setup of DC Test Devices

Name	W[um]	L[um]	AD[um²]	AS[um²]	PD[um]	PS[um]	NRD	NRS	NF	Comment	Size Category
W10u0_L10u0_S549_5	10	10	3.4	3.4	20.7	20.7	0	0	1		Large

W0u15_L10u0_S555_3	150 m	10	101 m	101 m	1.34	1.34	0	0	1		Narrow
W10u0_L0u13_S548_2	10	130 m	3.4	3.4	20.7	20.7	0	0	1		Short
W0u15_L0u13_S554_3	150 m	130 m	101 m	101 m	1.34	1.34	0	0	1		Small
W10u0_L0u14_S548_3	10	140 m	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L0u18_S548_5	10	180 m	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L0u5_S549_1	10	500 m	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L1u2_S549_2	10	1.2	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L5u0_S549_4	10	5	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L2u0_S549_3	10	2	3.4	3.4	20.7	20.7	0	0	1		L Scale
W10u0_L0u15_S548_4	10	150 m	3.4	3.4	20.7	20.7	0	0	1		L Scale
W0u6_L10u0_S552_1	600 m	10	204 m	204 m	1.88	1.88	0	0	1		W Scale
W0u18_L10u0_S554_1	180 m	10	103 m	103 m	1.34	1.34	0	0	1		W Scale
W0u3_L10u0_S553_2	300 m	10	102 m	102 m	1.28	1.28	0	0	1		W Scale
W05u0_L10u0_S550_3	5	10	1.7	1.7	10.7	10.7	0	0	1		W Scale
W02u0_L10u0_S551_1	2	10	680 m	680 m	4.68	4.68	0	0	1		W Scale
W02u0_L2u0_S550_5	2	2	680 m	680 m	4.68	4.68	0	0	1		LW Scale
W0u18_L0u14_S553_4	180 m	140 m	103 m	103 m	1.34	1.34	0	0	1		LW Scale
W0u18_L0u15_S553_5	180 m	150 m	103 m	103 m	1.34	1.34	0	0	1		LW Scale
W0u6_L0u5_S551_4	600 m	500 m	204 m	204 m	1.88	1.88	0	0	1		LW Scale
W0u6_L1u2_S551_5	600 m	1.2	204 m	204 m	1.88	1.88	0	0	1		LW Scale
W0u3_L0u14_S552_3	300	140	102	102	1.2	1.2	0	0	1		LW

	m	m	m	m	8	8					Scale
W0u6_L0u13_S551_2	600 m	130 m	204 m	204 m	1.8 8	1.8 8	0	0	1		LW Scale
W0u6_L0u18_S551_3	600 m	180 m	204 m	204 m	1.8 8	1.8 8	0	0	1		LW Scale
W0u3_L0u15_S552_4	300 m	150 m	102 m	102 m	1.2 8	1.2 8	0	0	1		LW Scale
W02u0_L0u13_S550_4	2	130 m	680 m	680 m	4.6 8	4.6 8	0	0	1		LW Scale
W0u15_L0u14_S554_4	150 m	140 m	101 m	101 m	1.3 4	1.3 4	0	0	1		LW Scale
W05u0_L0u13_S550_1	5	130 m	1.7	1.7	10. 7	10. 7	0	0	1		LW Scale
W0u15_L0u15_S554_5	150 m	150 m	101 m	101 m	1.3 4	1.3 4	0	0	1		LW Scale
W0u3_L0u18_S552_5	300 m	180 m	102 m	102 m	1.2 8	1.2 8	0	0	1		LW Scale
W0u3_L0u13_S552_2	300 m	130 m	102 m	102 m	1.2 8	1.2 8	0	0	1		LW Scale
W05u0_L5u0_S550_2	5	5	1.7	1.7	10. 7	10. 7	0	0	1		LW Scale
W0u15_L0u5_S555_2	150 m	500 m	101 m	101 m	1.3 4	1.3 4	0	0	1		LW Scale
W0u3_L0u5_S553_1	300 m	500 m	102 m	102 m	1.2 8	1.2 8	0	0	1		LW Scale
W0u15_L0u18_S555_1	150 m	180 m	101 m	101 m	1.3 4	1.3 4	0	0	1		LW Scale
W0u18_L0u13_S553_3	180 m	130 m	103 m	103 m	1.3 4	1.3 4	0	0	1		LW Scale
W0u15_L0u12_S554_2	150 m	120 m	101 m	101 m	1.3 4	1.3 4	0	0	1		Additional
W10u0_L0u12_S548_1	10	120 m	3.4	3.4	20. 7	20. 7	0	0	1		Additional
W10u0_L0u13_S382_2	10	130 m	3.4	3.4	20. 7	20. 7	0	0	1		Short
W10u0_L0u13_T356_S382_2	10	130 m	0	0	0	0	0	0	1		Short

2.2.2 Setup of CV Test Devices

Name	W[um]	L[um]	NF	AD[um²]	AS[um²]	PD[um]	PS[um]	Comment	Size Category
Junction Capacitance Bulk - Drain									
CJBDarea_S351B_1			1	37.5 K		850		B-High, S/D-Low	BD Area
CJBDperim_S351B_2			1	28.6 K		102K		B-High, S/D-Low	BD Perim
Junction Capacitance Bulk - Source									
CJBSarea_S351B_1			1	0		0		B-High, S/D-Low	BS Area
CJBSperim_S351B_2			1	0		0		B-High, S/D-Low	BS Perim
Oxide Capacitance									
Cox_S346K_4	160	250	1					B-High, G-Low	Oxide
Overlap Capacitance									
Cgds3_S351B_13	400	10	40	79.8	79.8	436	436	G-High, S/D-Low, B+Sub-GND	Overlap 2 GDS
Cgds1_S351B_11	8K	130 m	800	1.52 K	1.52 K	8.32 K	8.32 K	G-High, S/D-Low, B+Sub-GND	Overlap GDS
Cgds2_S351B_12	6.6	250	66	1.26	1.26	6.87	6.87	G-	Overlap

	K	m	0	K	K	K	K	High, S/D-Low, B+Sub-GND	1 GDS
Cgdsb2_S351B_12	6.6K	250m	660	1.26K	1.26K	6.87K	6.87K	G-High, S/D+B-Low	Overlap 1 GDSB
Cgdsb1_S351B_11	8K	130m	800	1.52K	1.52K	8.32K	8.32K	G-High, S/D+B-Low	Overlap GDSB
Cgdsb3_S351B_13	400	10	40	79.8	79.8	436	436	G-High, S/D+B-Low	Overlap 2 GDSB

3 Corner Devices

3.1 Target Values vs. Simulated Values

3.1.1 W = 10.00u, L = 130.0n, NF = 1

	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	-520m	-524m	-0.73	-460m	-460m	0.1	-410m	-411m	-0.16
Ron*W	-	-	-	2.2	2.18	-0.76	-	-	-
Ion/W	-160	-161	-0.87	-205	-203	0.93	-260	-261	-0.48
log(Ioff/W)	-11.5	-11.6	1.24	-10.4	-10.5	1.08	-9.5	-9.51	0.22

3.1.2 W = 10.00u, L = 10.00u, NF = 1

	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	-400m	-403m	-0.67	-350m	-352m	-0.72	-300m	-303m	-0.86
Ion/W	-2.45	-2.45	0	-2.7	-2.69	0.36	-2.95	-2.96	-0.37

3.1.3 W = 150.0n, L = 130.0n, NF = 1

	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
VT	-560m	-564m	-0.73	-470m	-473m	-0.65	-380m	-378m	0.64
Ion/W	-240	-240	0.1	-315	-318	-0.78	-390	-391	-0.19

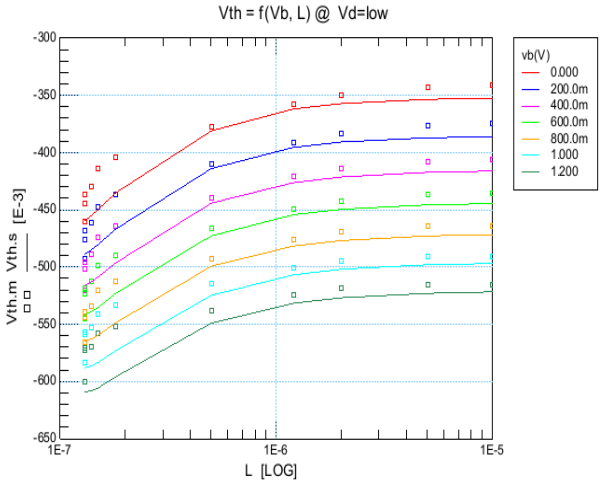
3.1.4 W = 150.0n, L = 10.00u, NF = 1

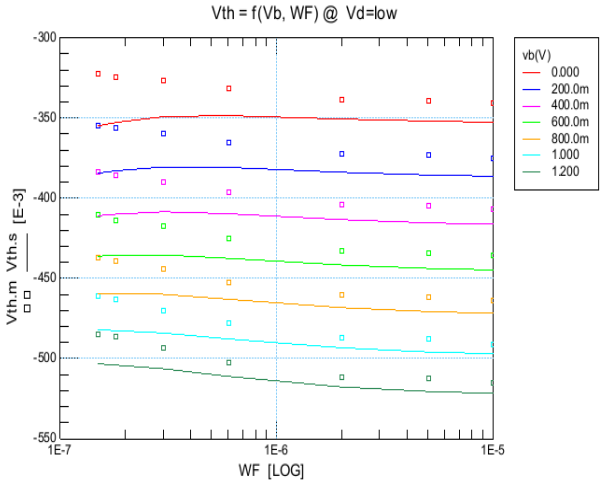
	Slow			Typic.			Fast		
	Target	Sim.	Error [%]	Target	Sim.	Error [%]	Target	Sim.	Error [%]
Ion/W	-3.25	-3.23	0.54	-3.85	-3.85	0.14	-4.45	-4.48	-0.67

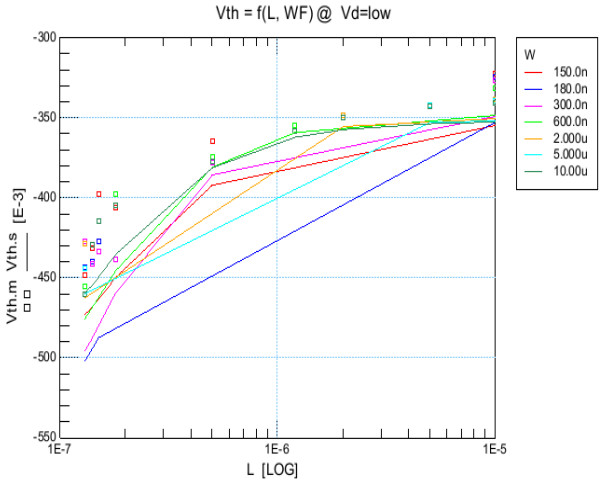
4 DC Transistors

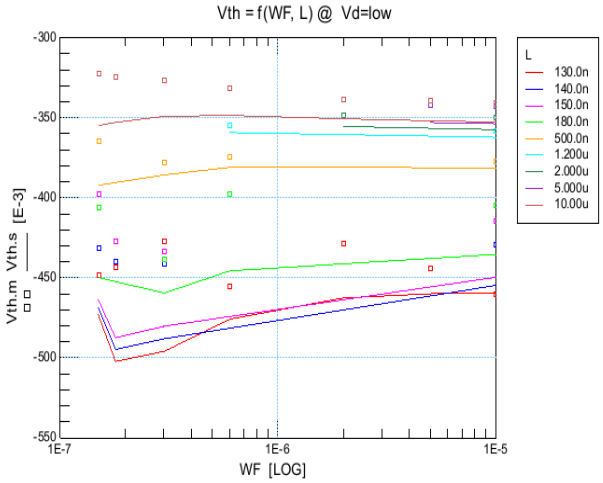
4.1 Summarized Results

4.1.1 Threshold Voltage Linear Region

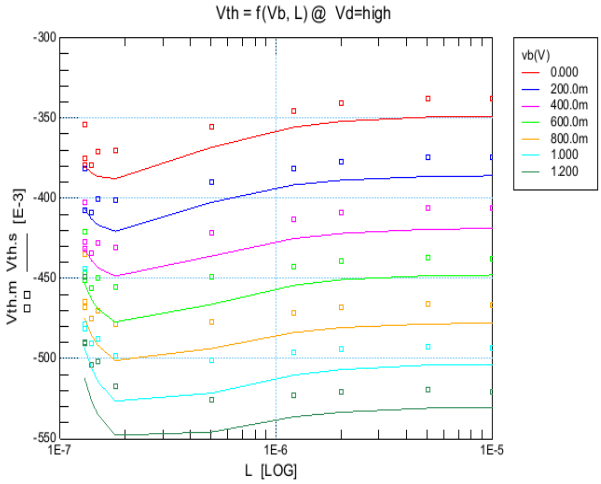


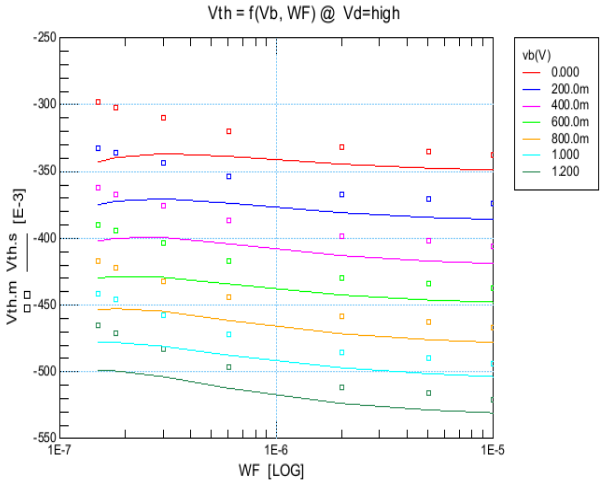


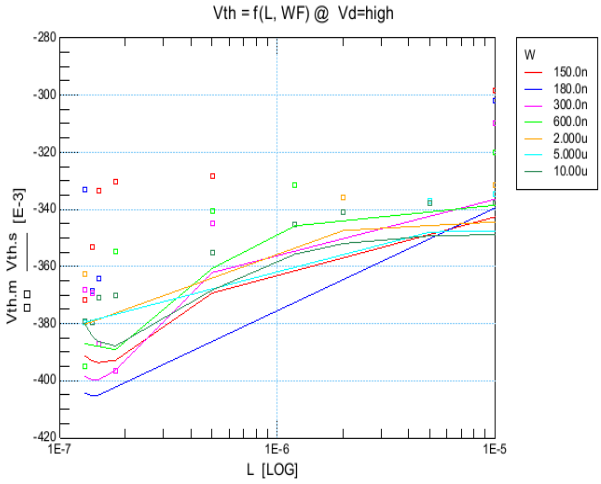


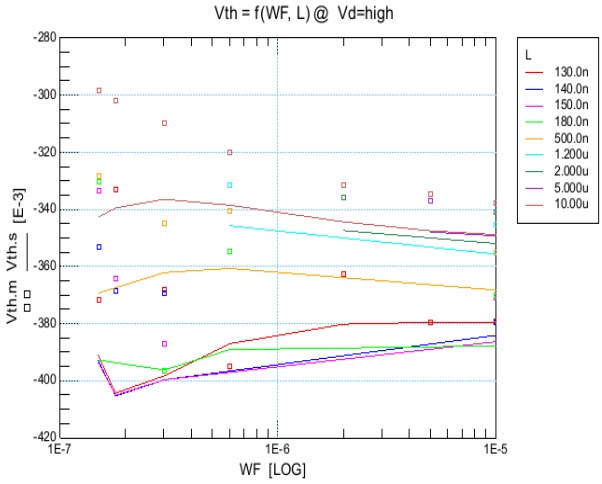


4.1.2 Threshold Voltage in Saturation

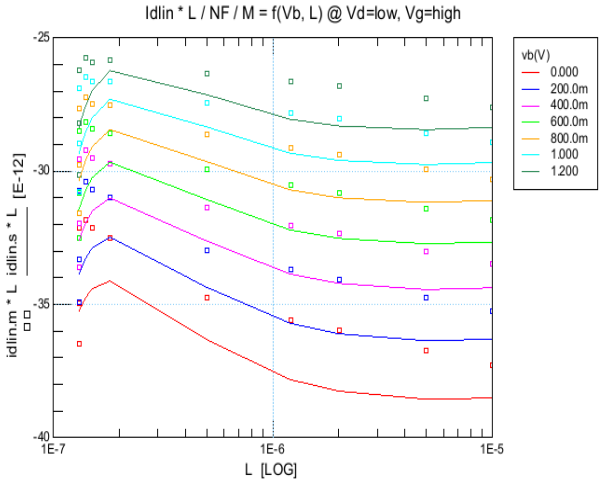


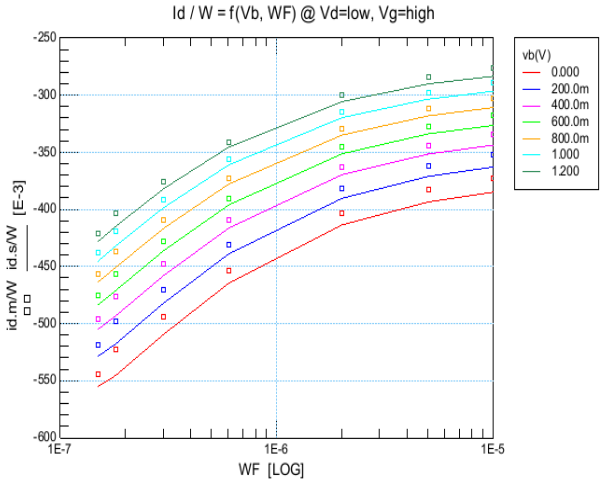


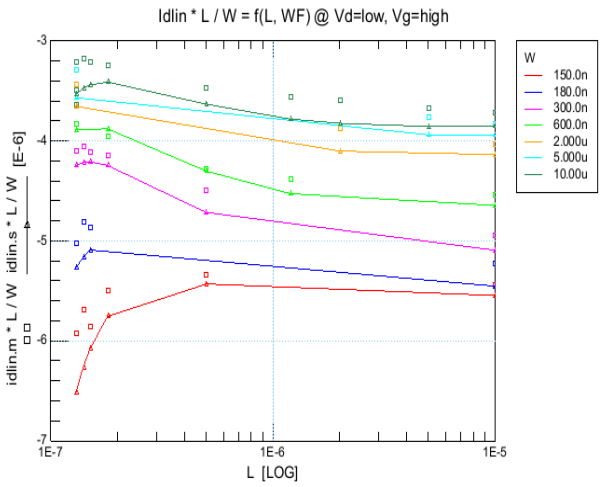


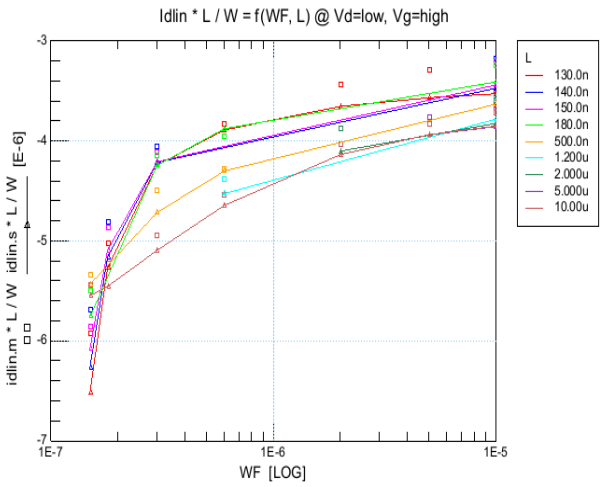


4.1.3 Maximum Drain Current in Linear Region

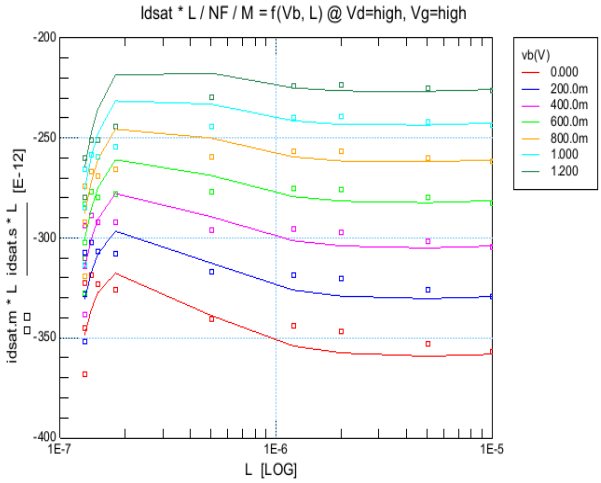


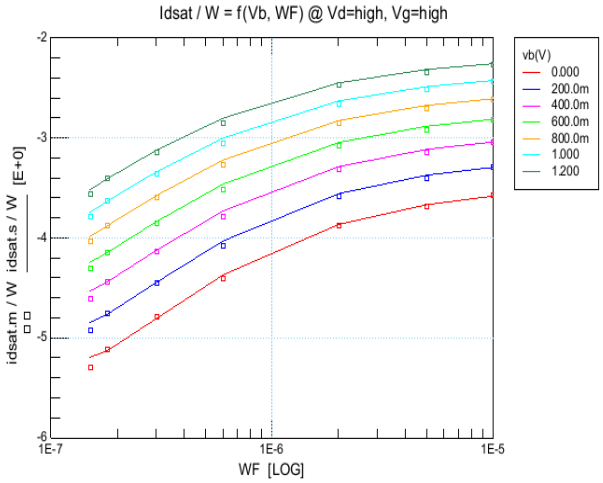


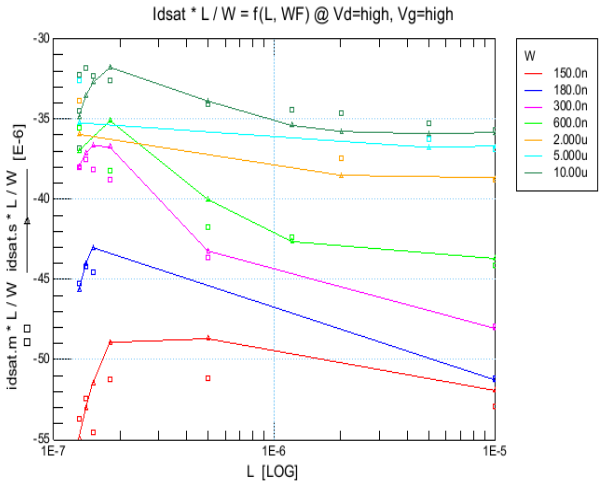


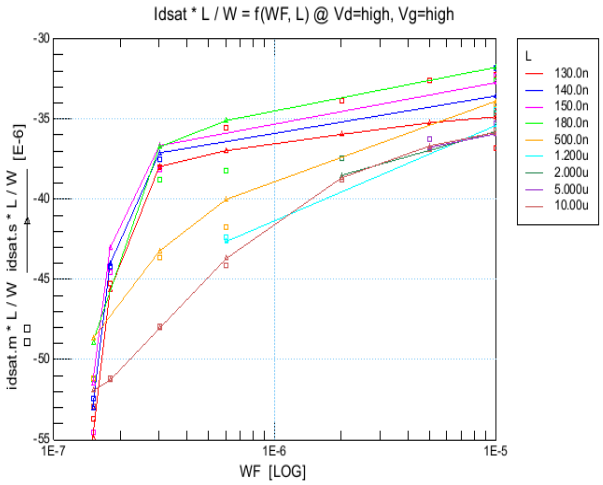


4.1.4 Maximum Drain Current in Saturation Region



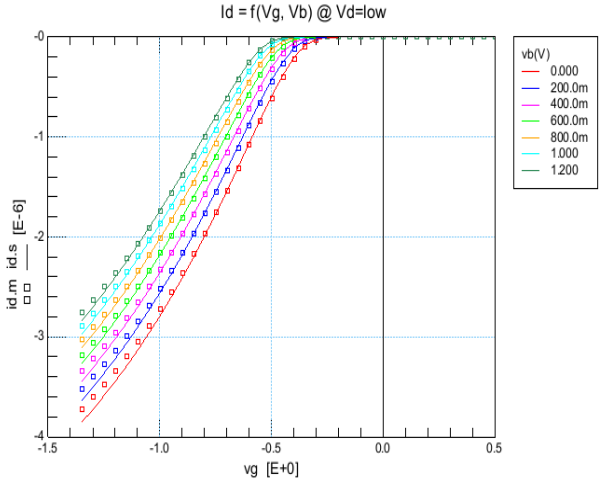


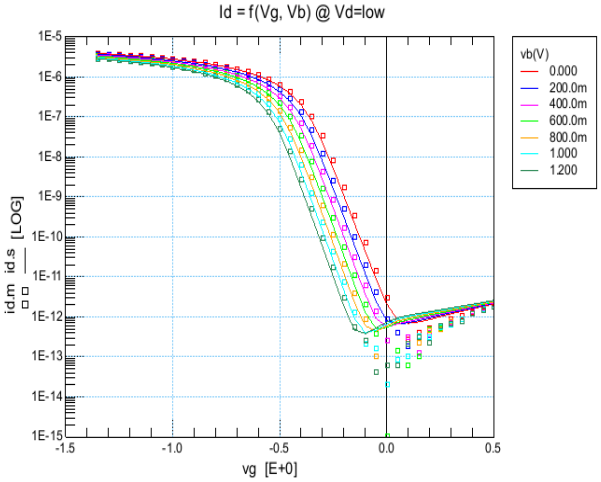


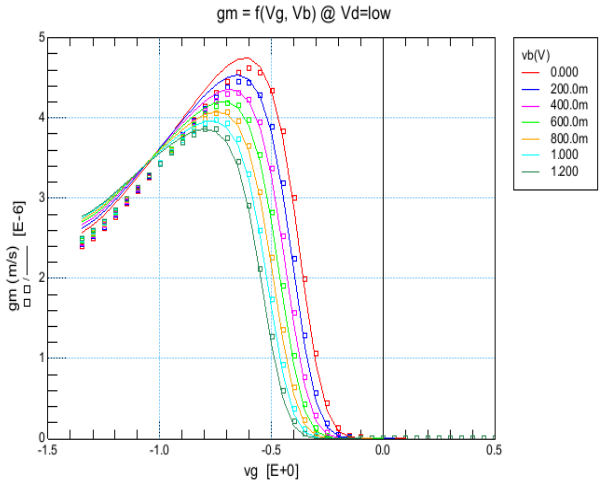


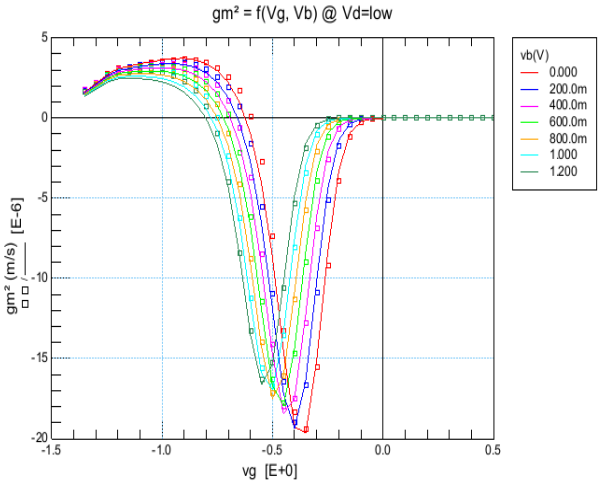
4.2 W10u0_L10u0_S549_5

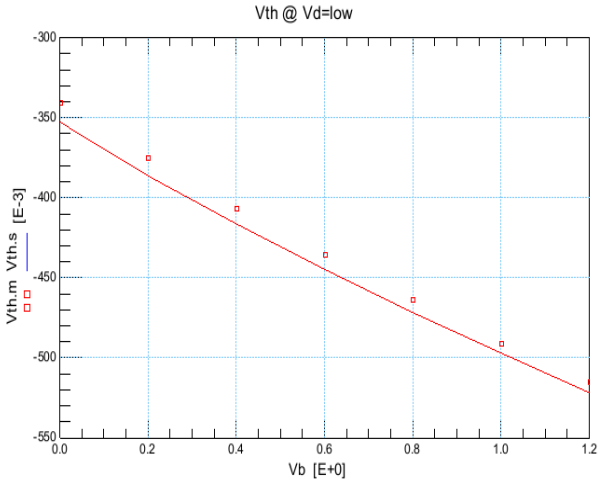
idvg, Vd = -0.05V, T = 27°C

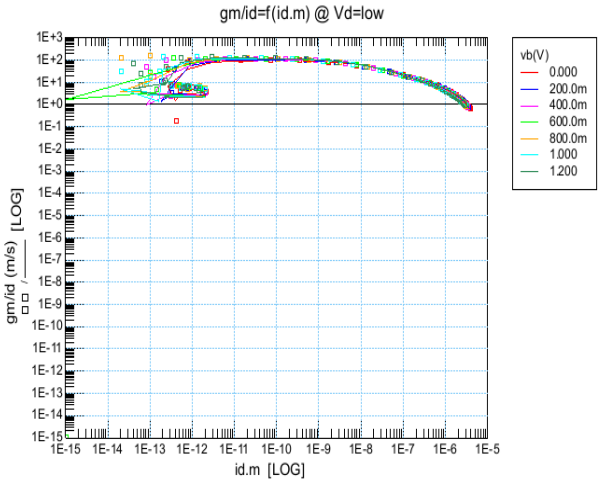


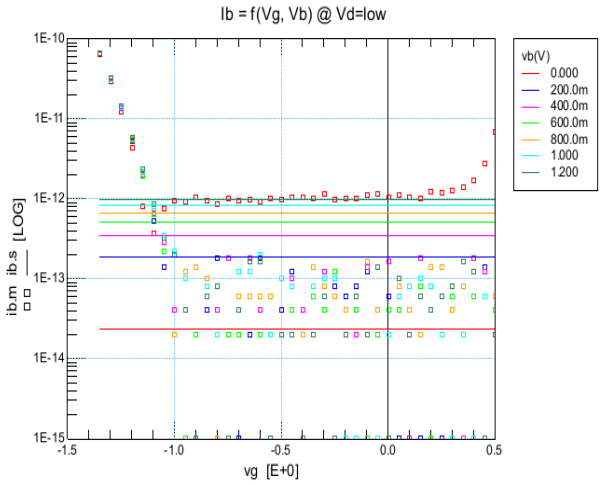


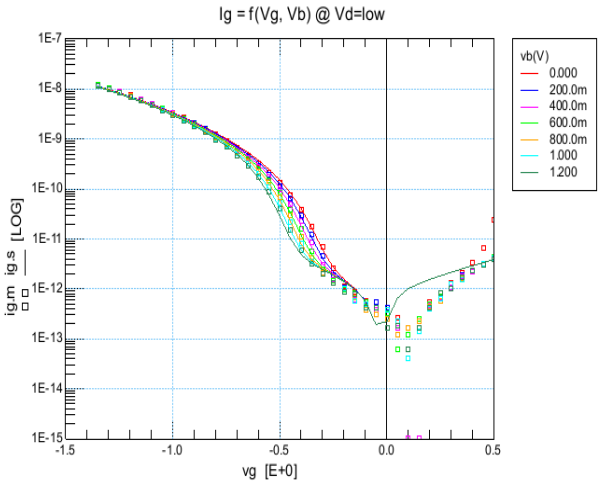




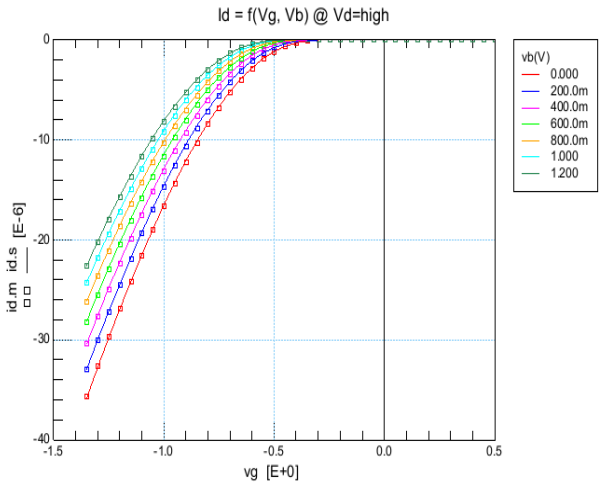


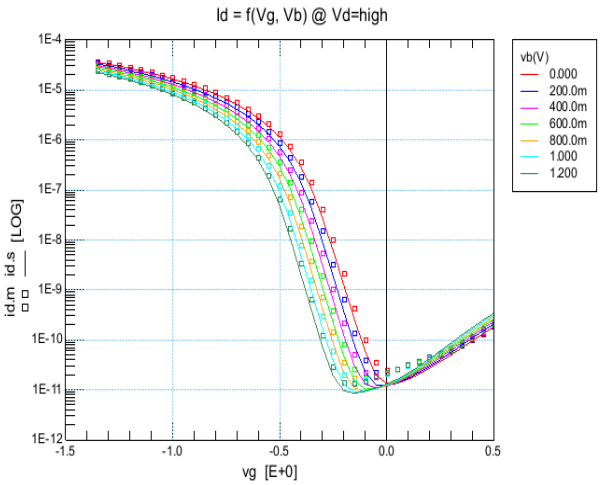


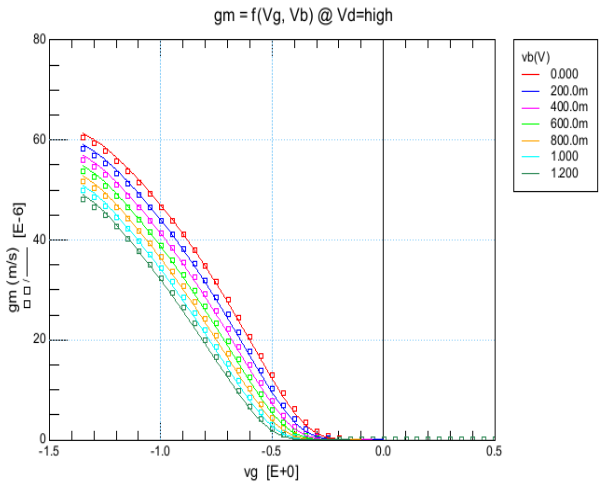


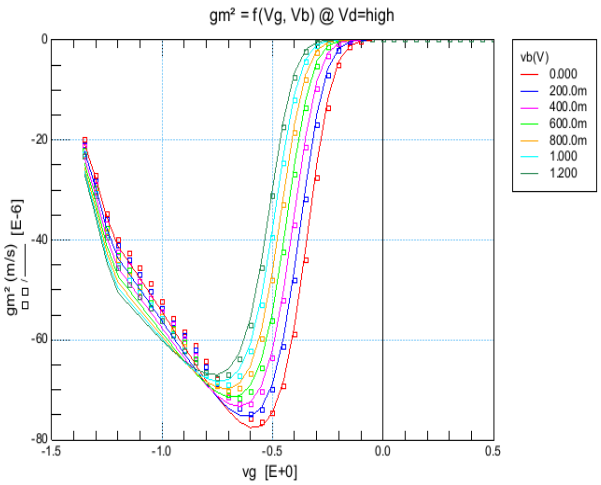


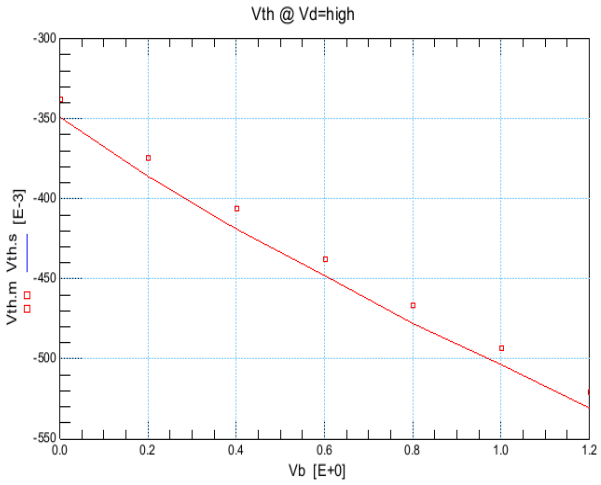
idvg, Vd = -1.2V, T = 27°C

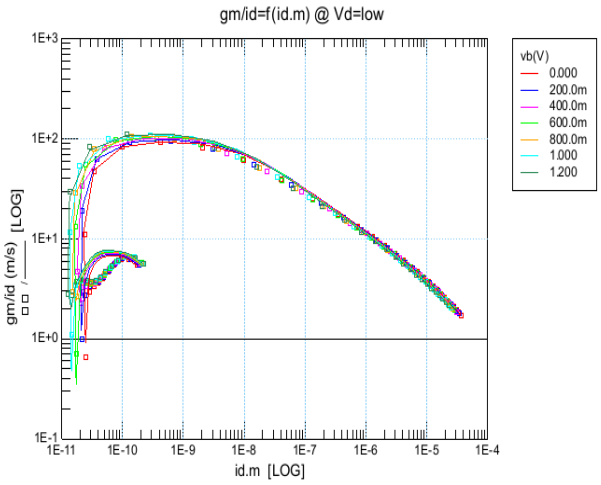


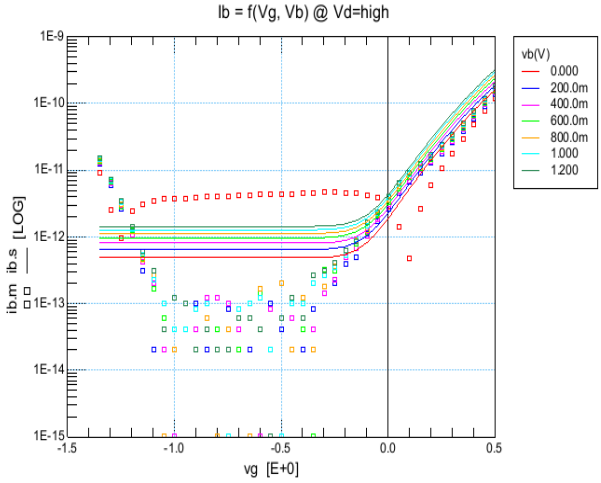


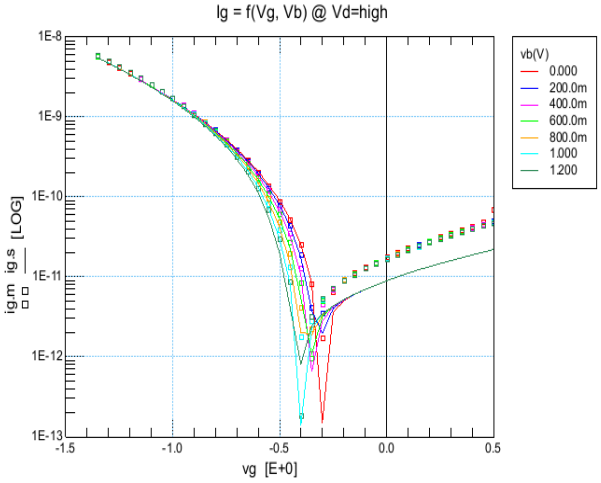




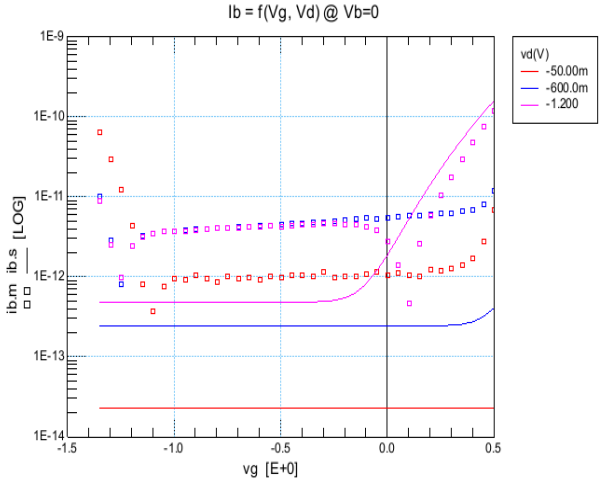


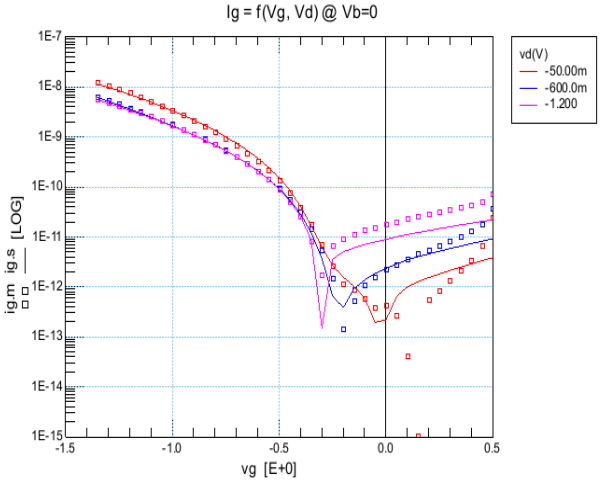




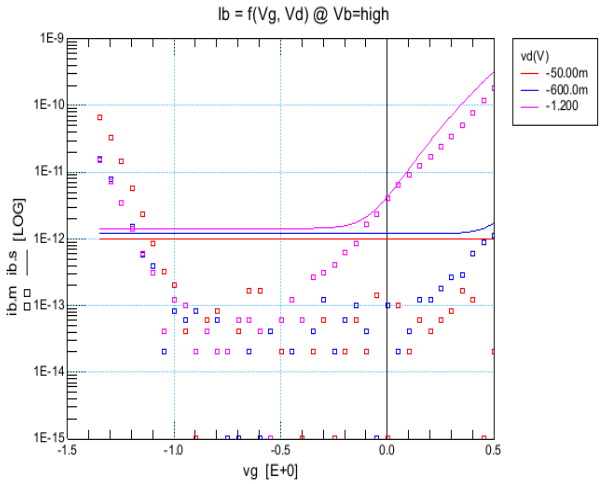


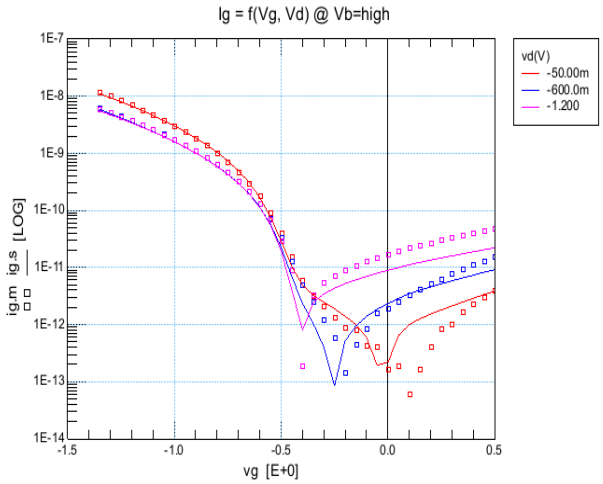
idvg, Ib,Ig, Vb = 0V, T = 27°C



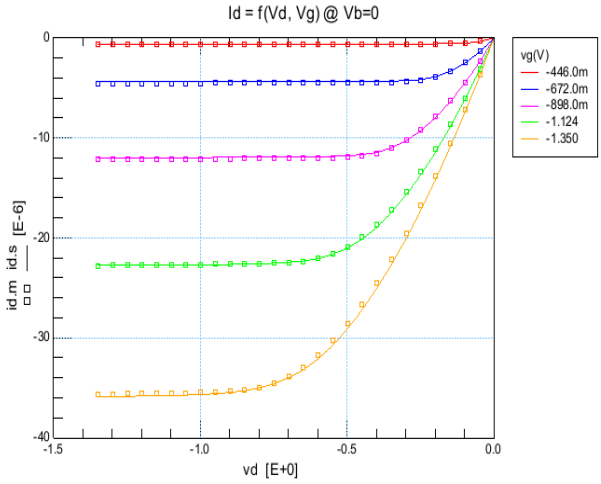


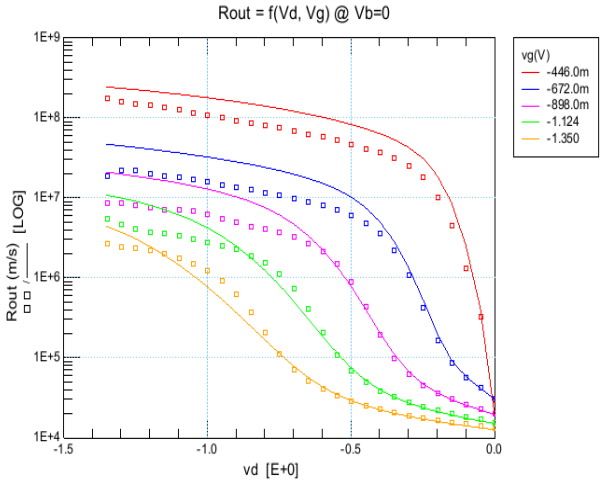
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

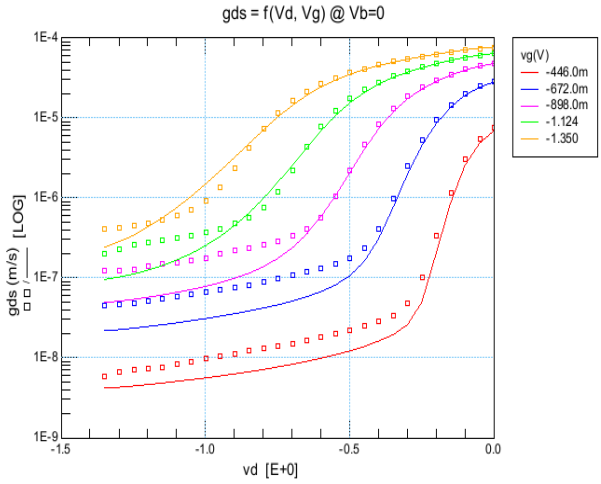




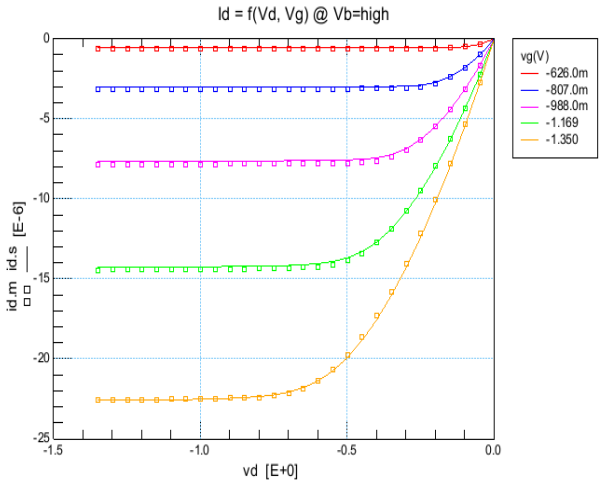
idvd, Vb = 0V, T = 27°C

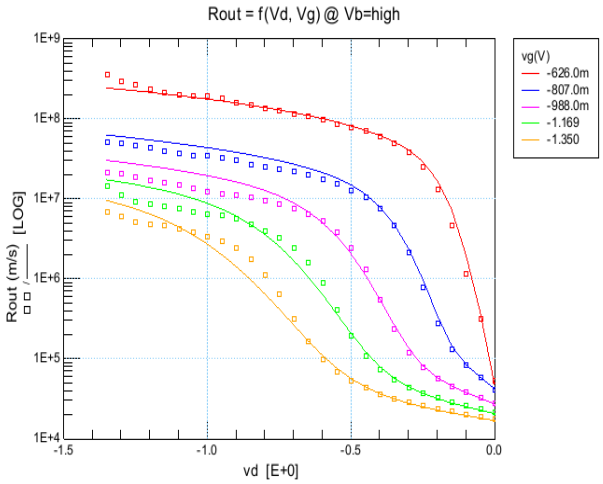


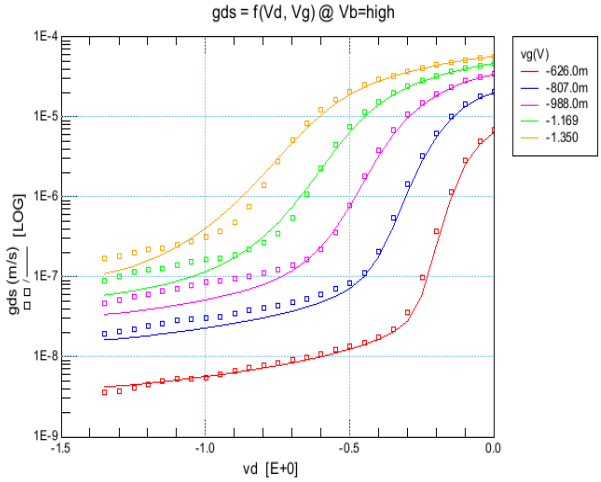




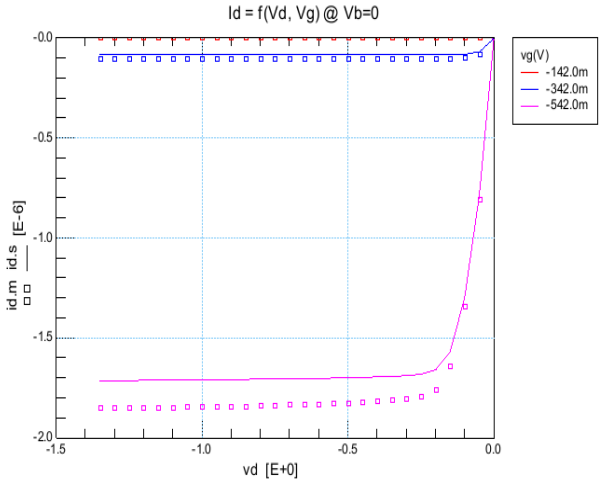
idvd_vbmin, Vb = 1.2V, T = 27°C

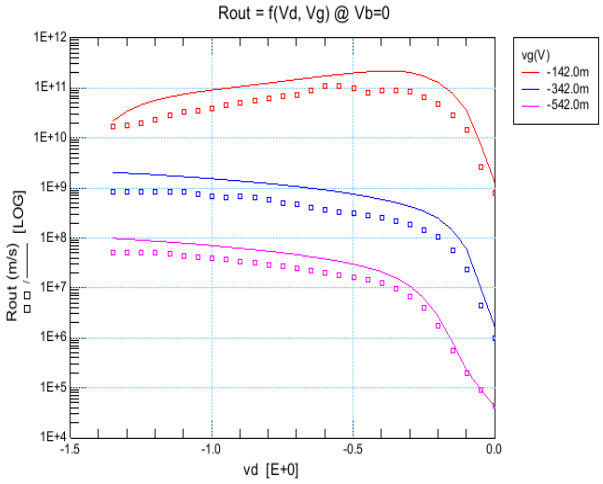


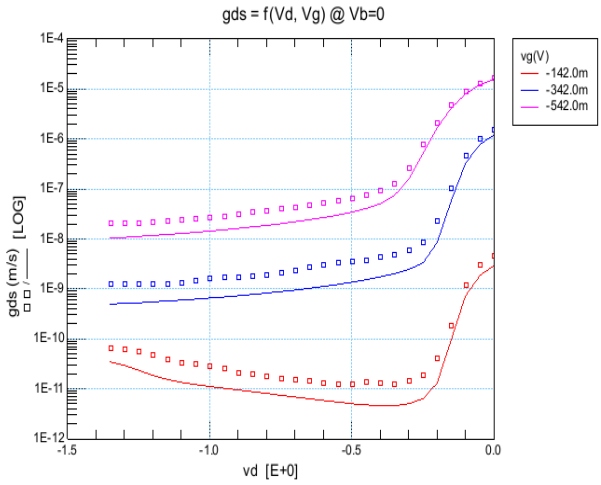




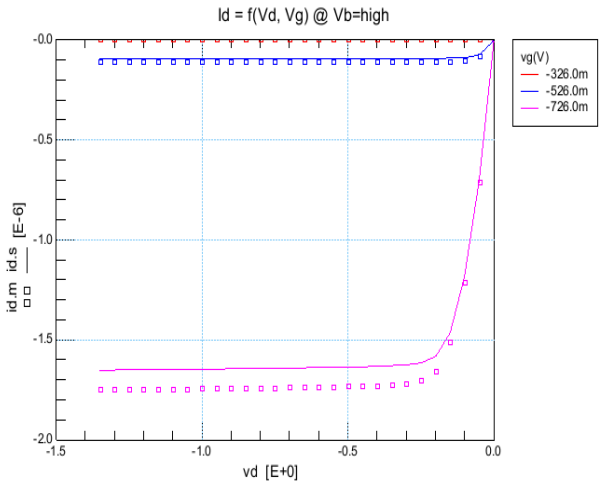
idvd_vth, Vb = 0V, T = 27°C

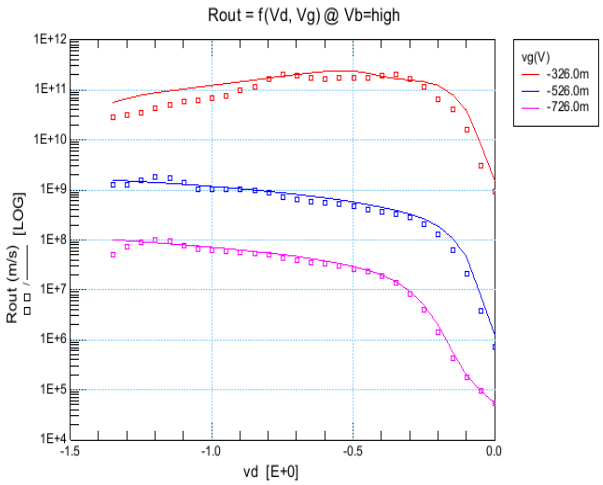


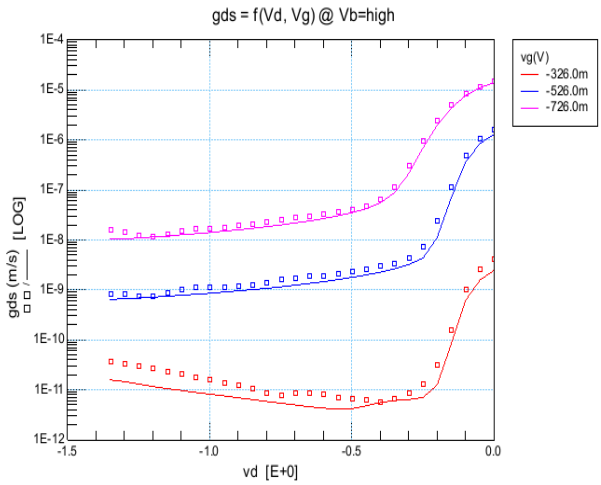




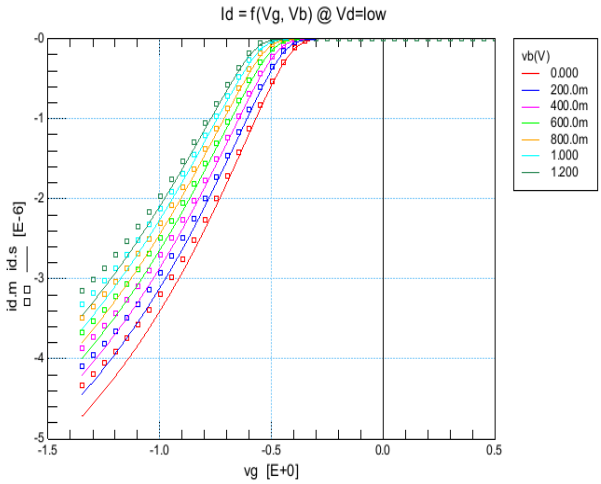
idvd_vth_vbmin, Vb = 1.2V, T = 27°C

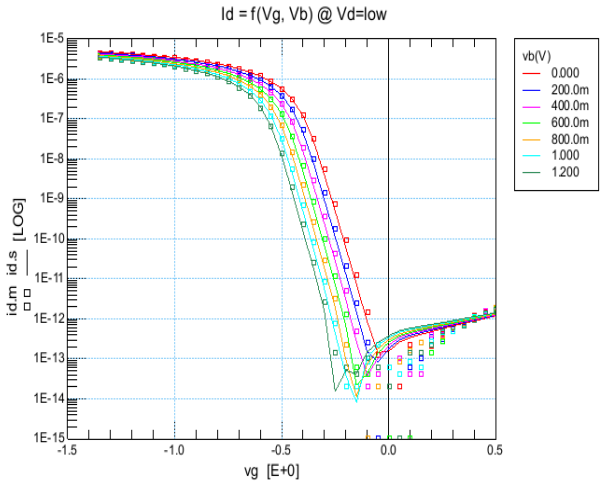


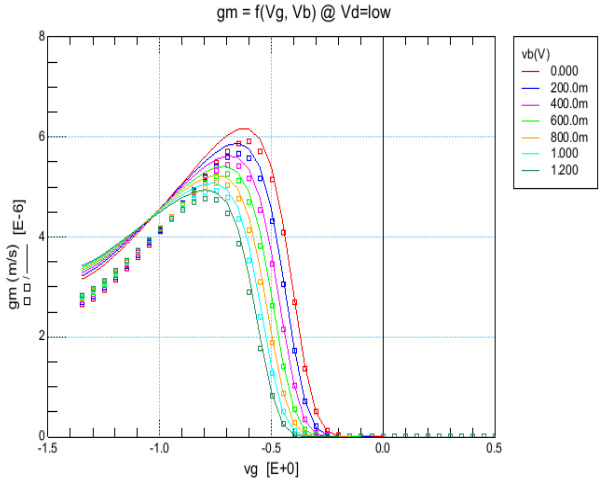


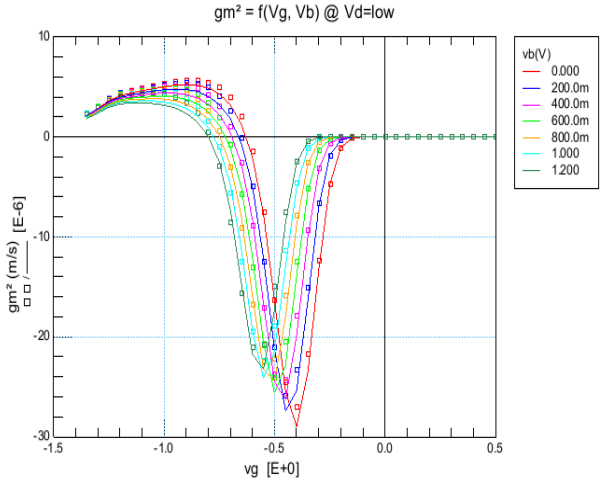


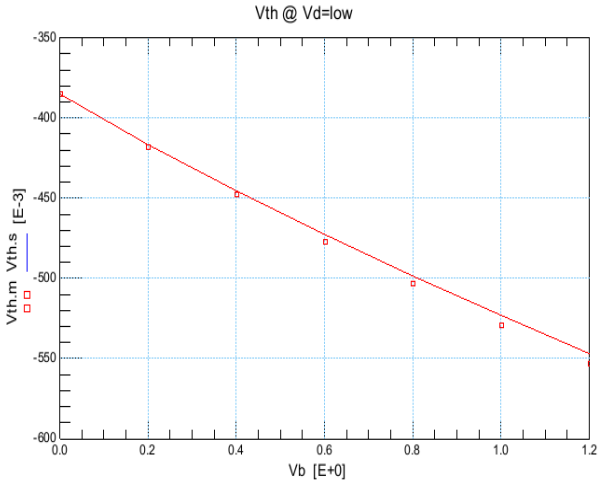
idvg, Vd = -0.05V, T = -40°C

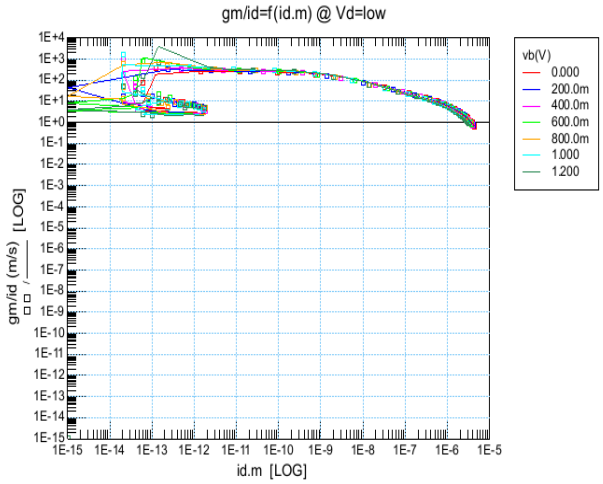


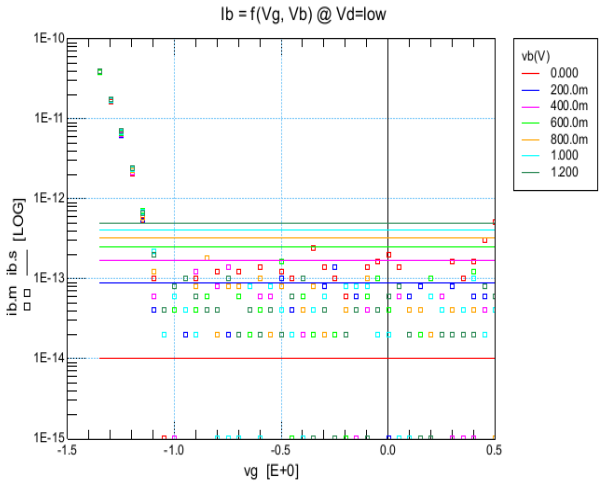


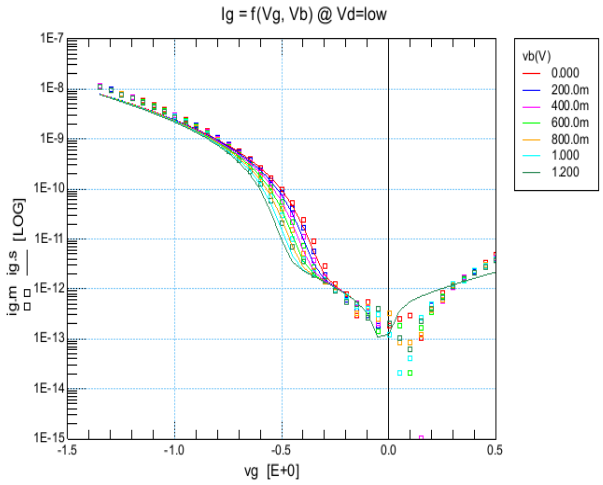




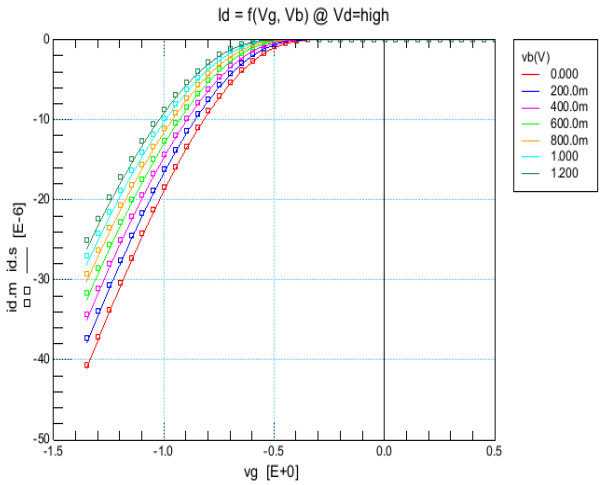


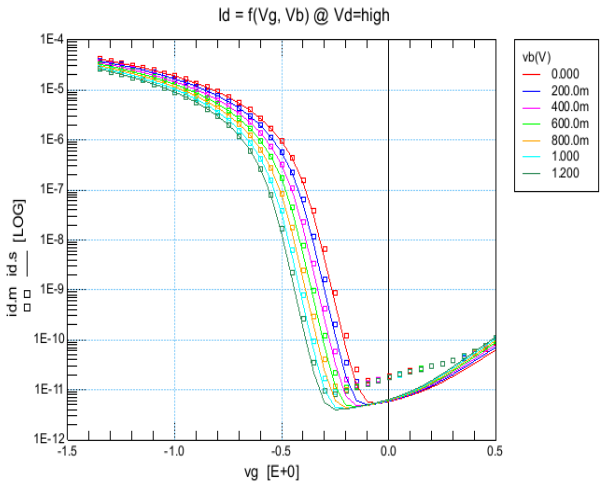


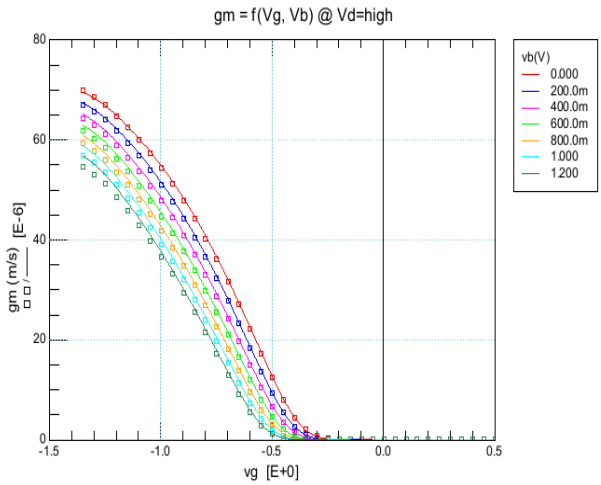


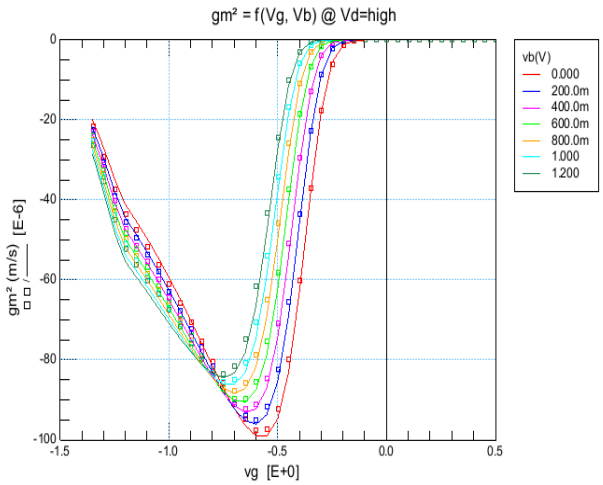


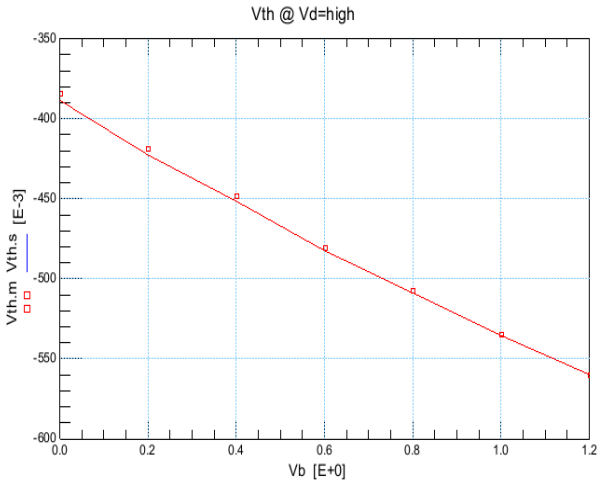
idvg, Vd = -1.2V, T = -40°C

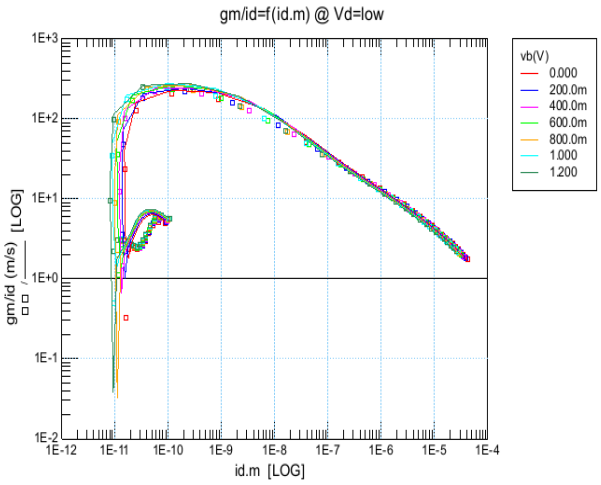


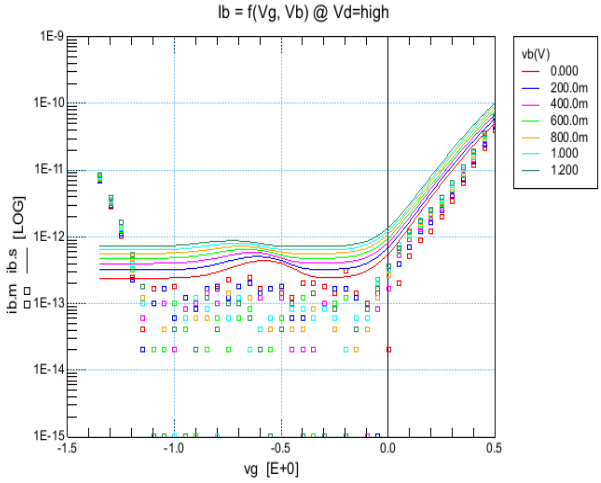


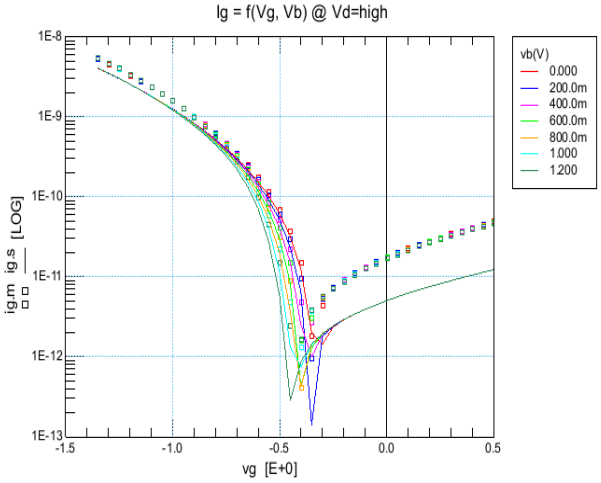




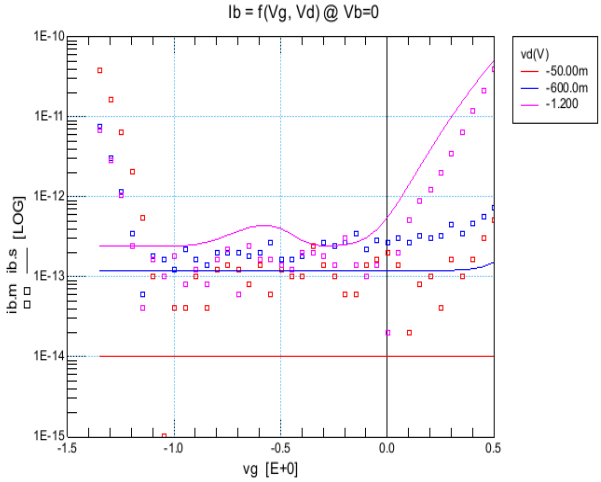


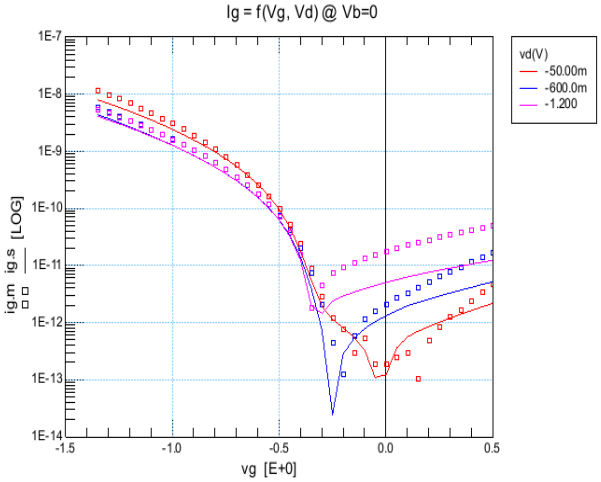




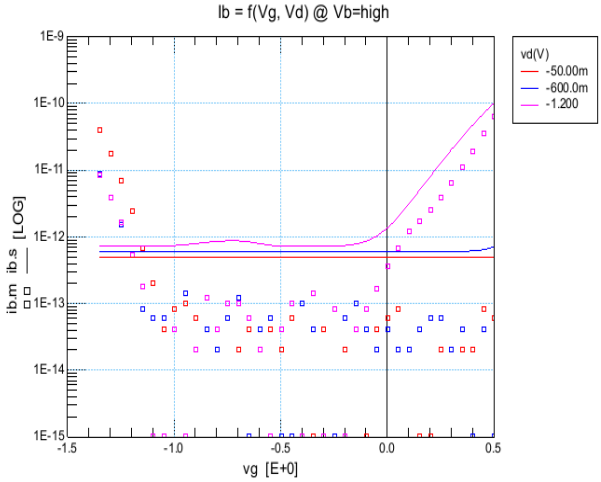


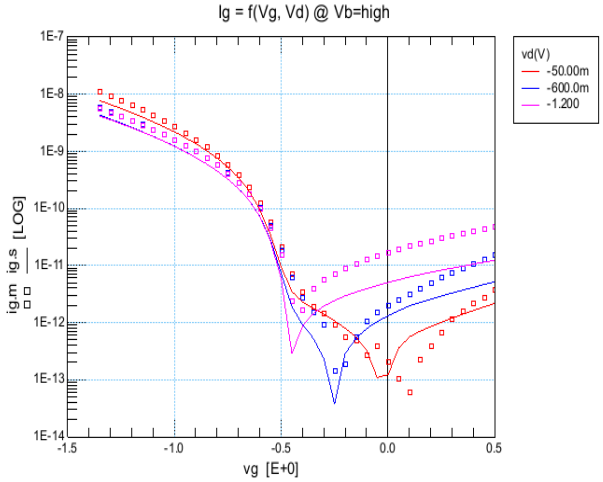
idvg, Ib,Ig, Vb = 0V, T = -40°C



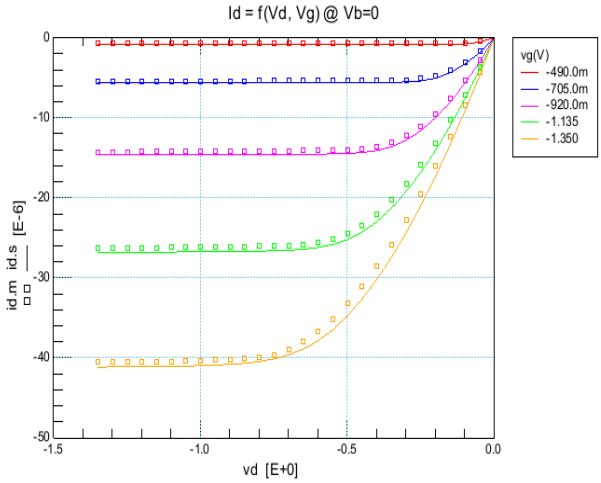


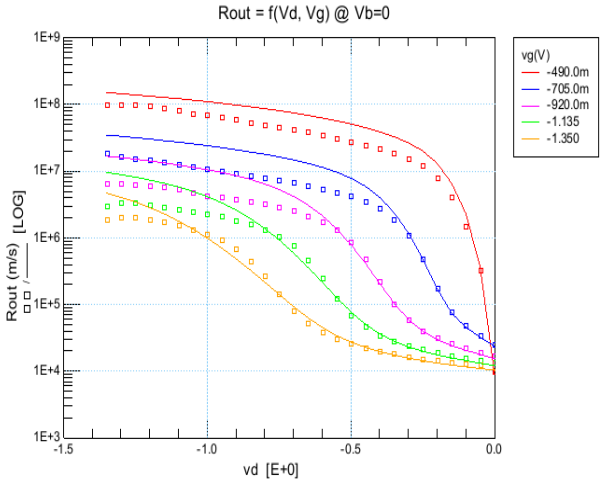
idvg, Ib,Ig, Vb = 1.2V, T = -40°C

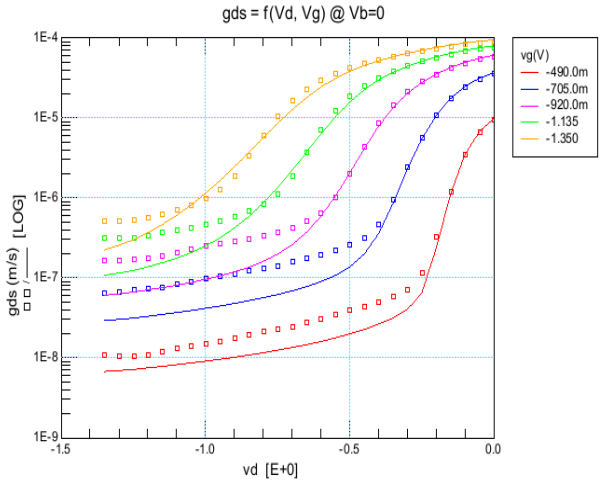




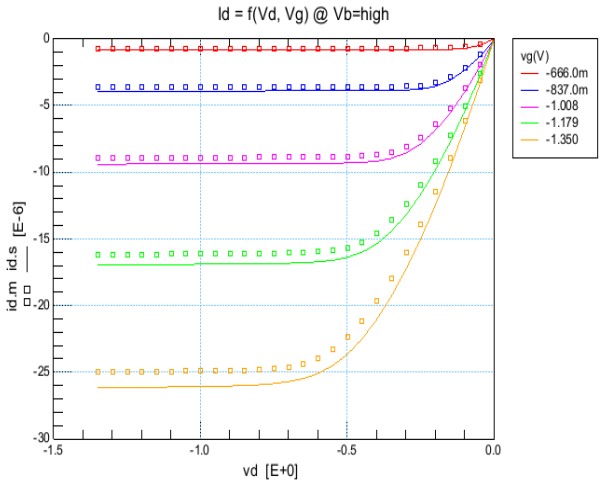
idvd, Vb = 0V, T = -40°C

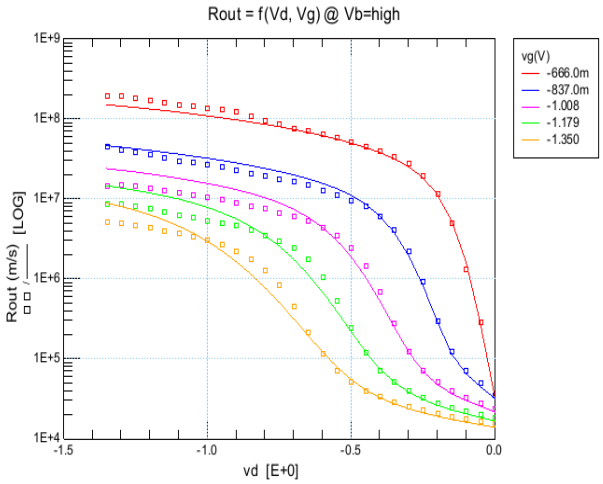


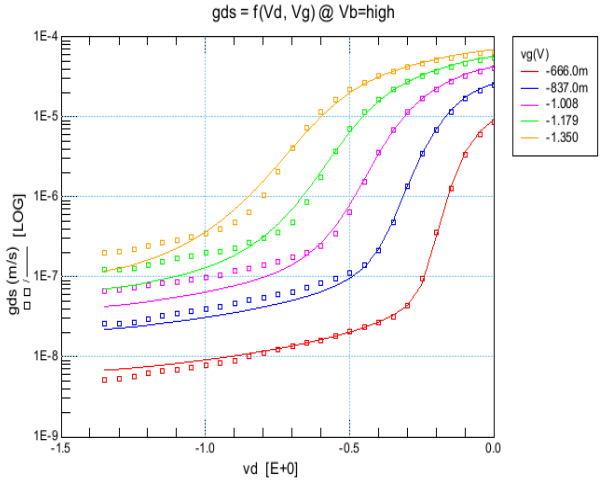




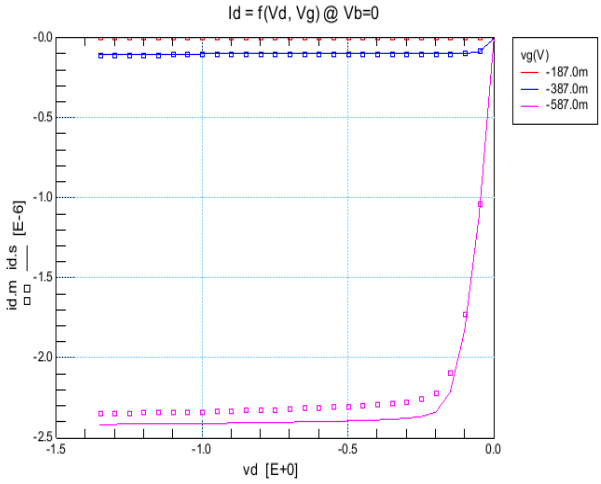
idvd_vbmin, Vb = 1.2V, T = -40°C

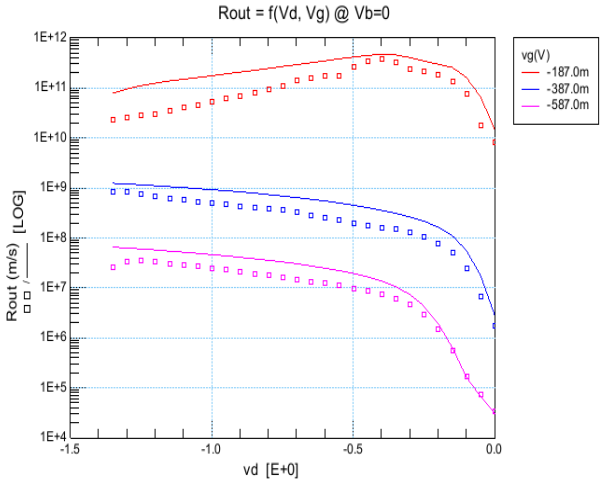


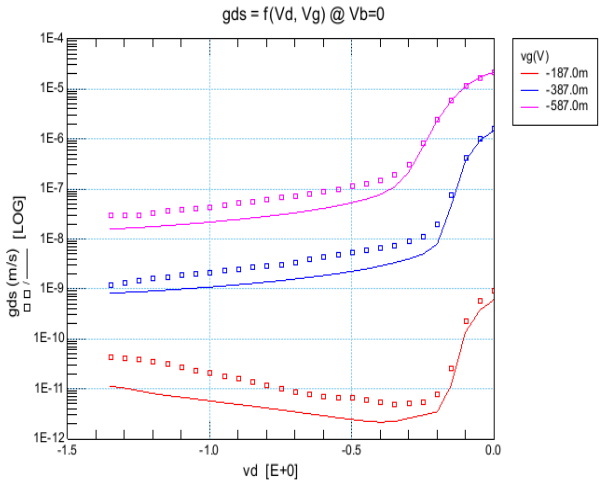




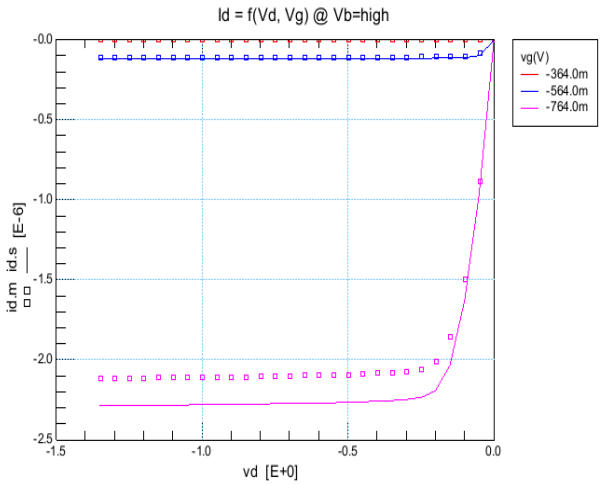
idvd_vth, Vb = 0V, T = -40°C

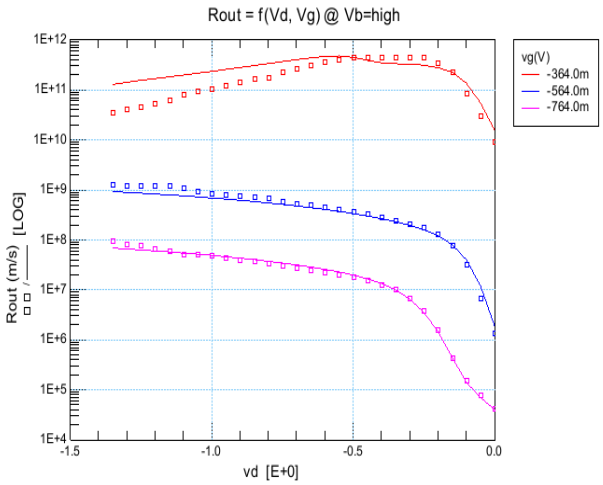


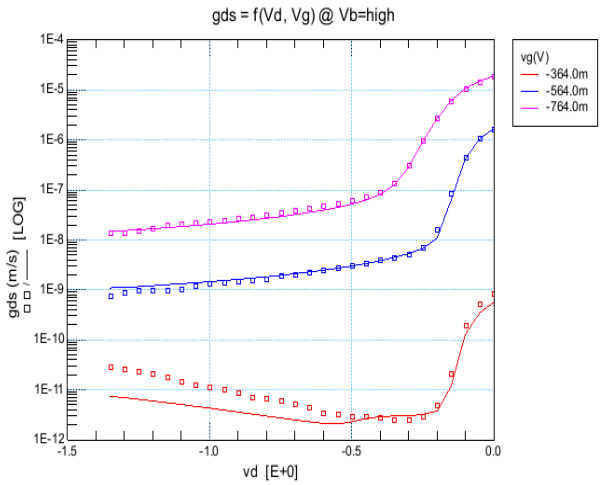




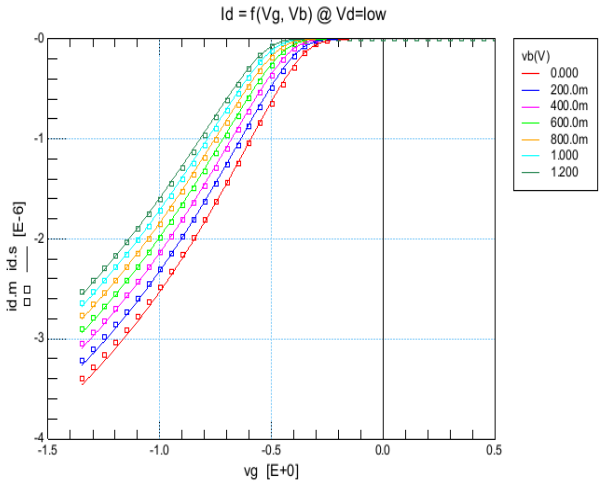
idvd_vth_vbmin, Vb = 1.2V, T = -40°C

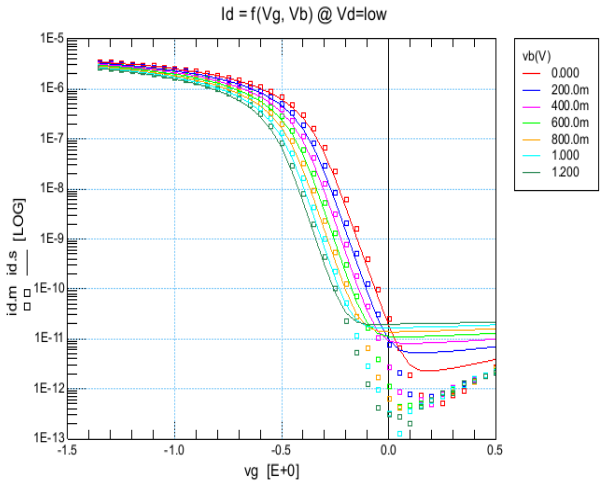


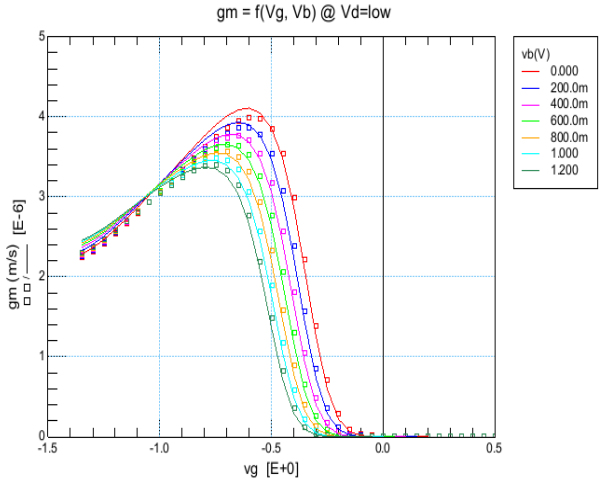


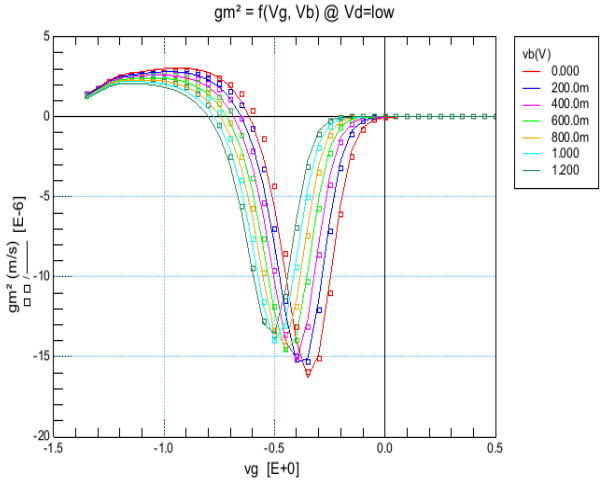


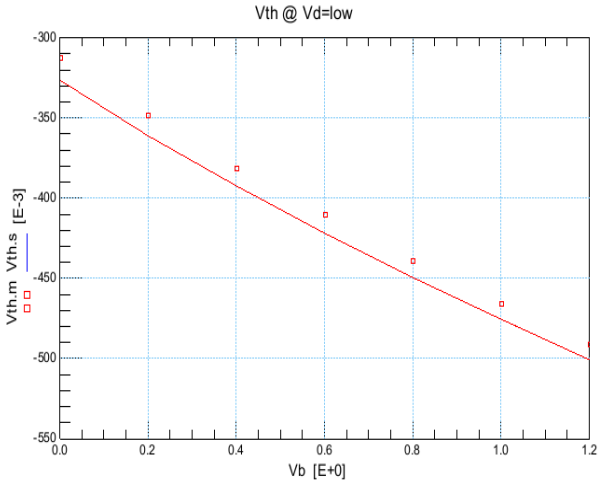
idvg, Vd = -0.05V, T = 70°C

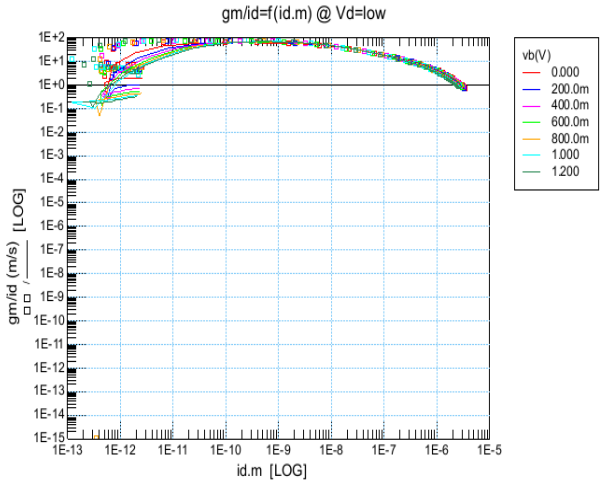


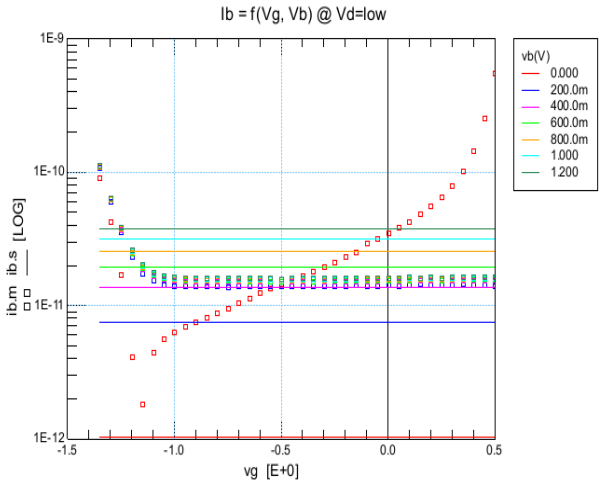


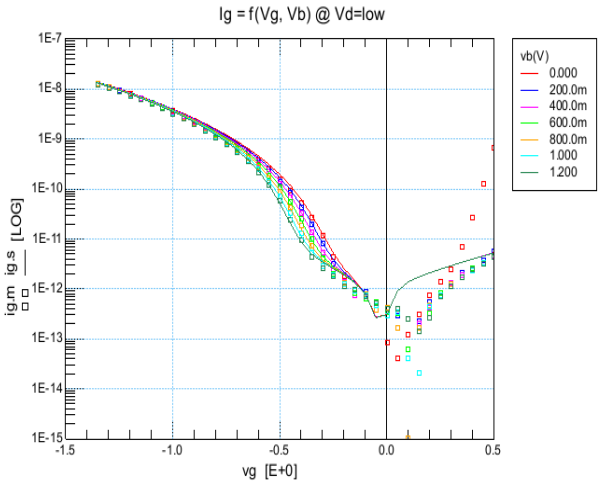




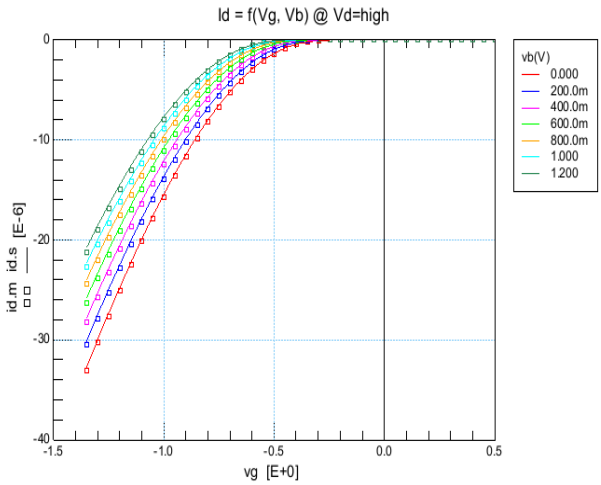


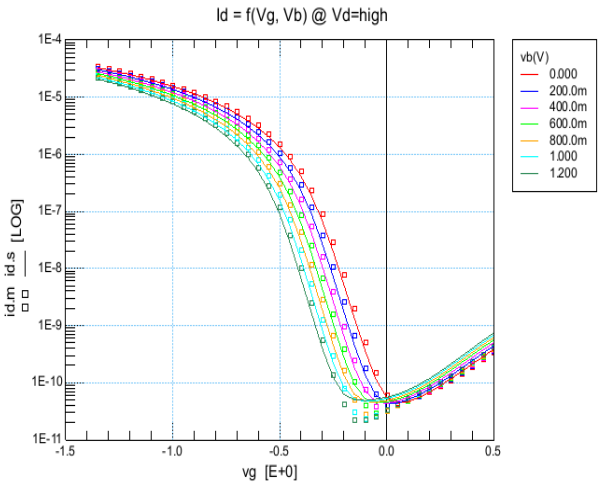


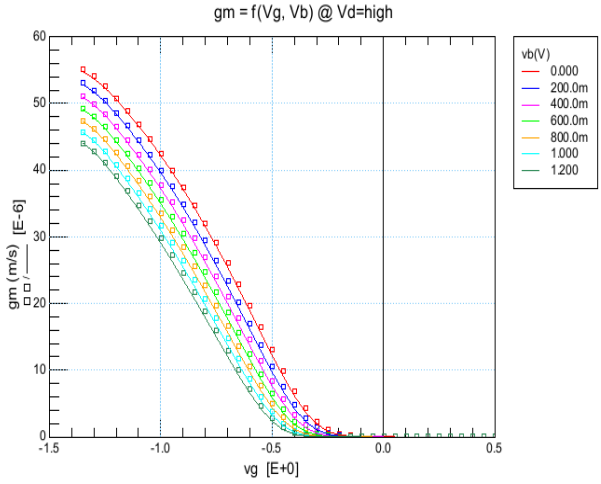


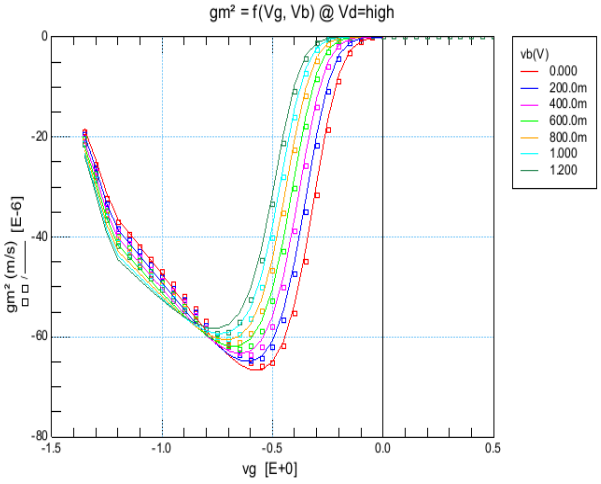


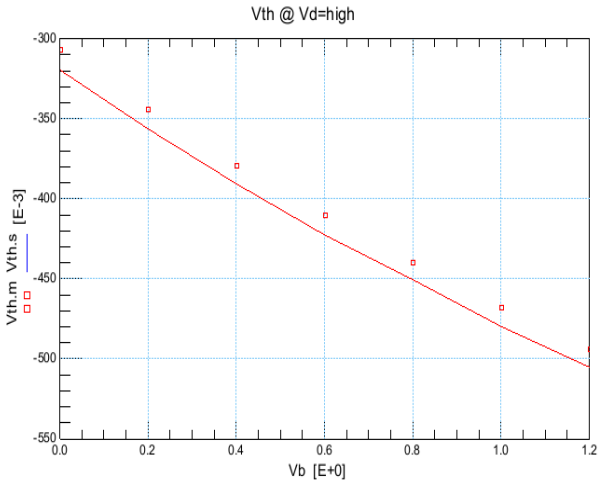
idvg, Vd = -1.2V, T = 70°C

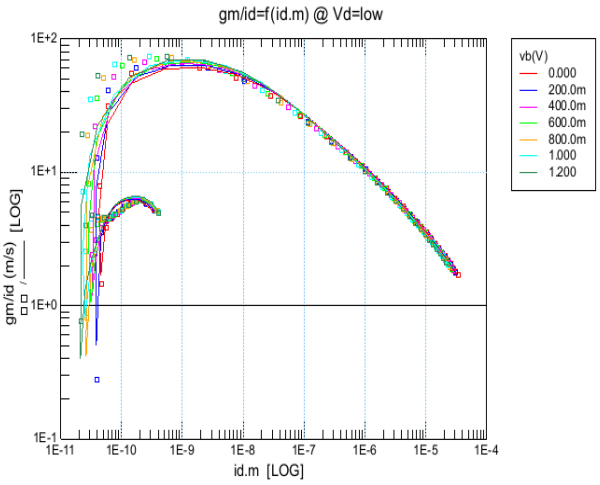


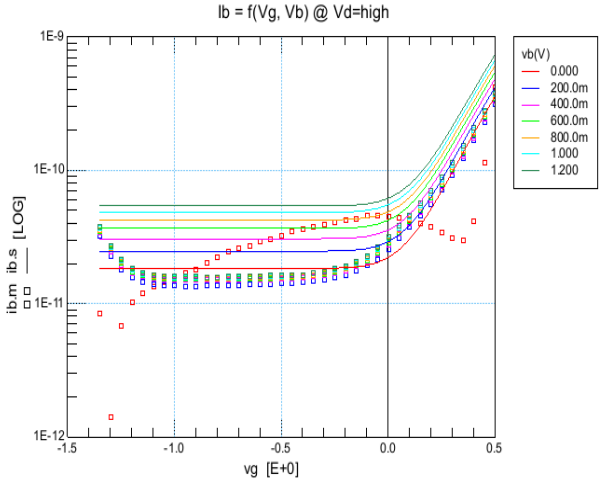


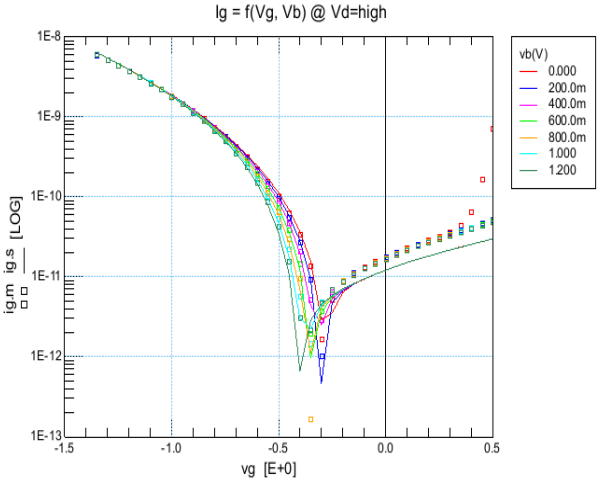




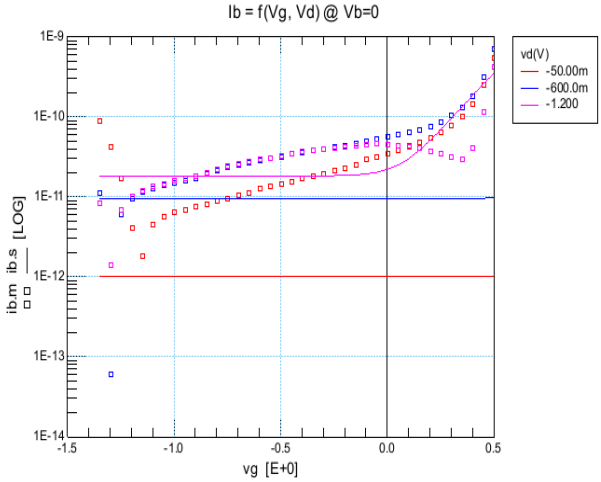


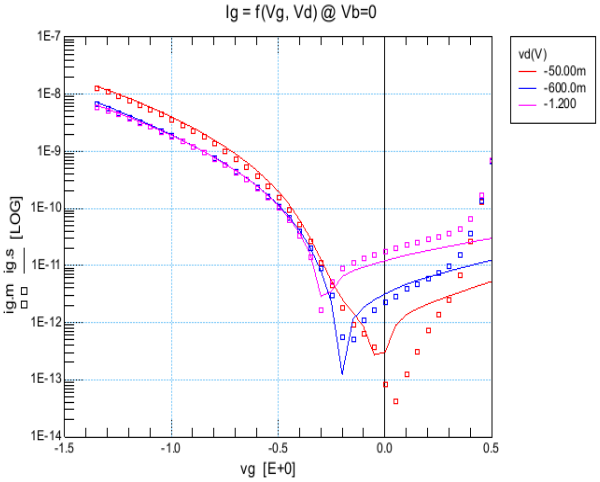




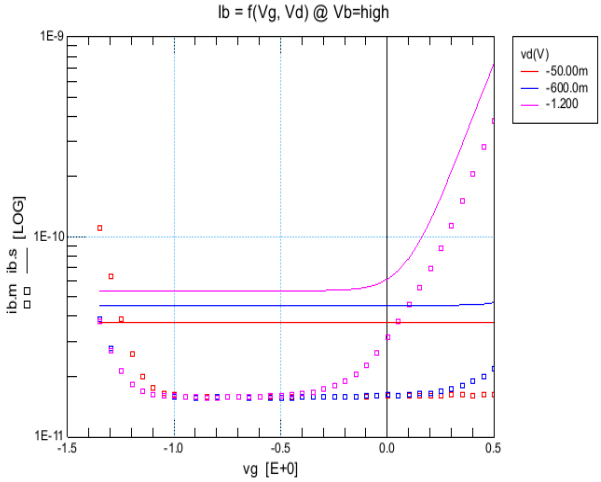


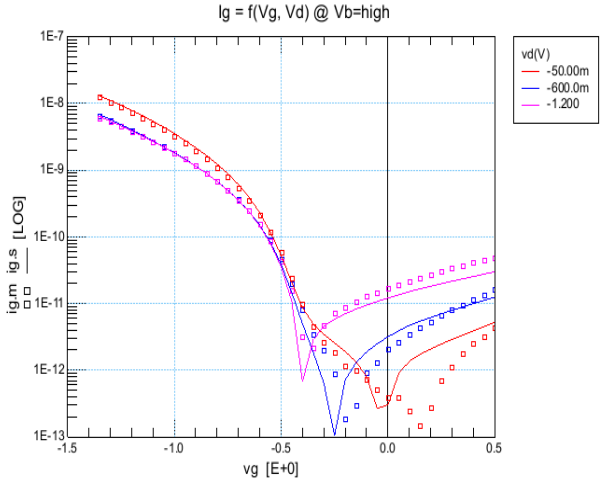
idvg, Ib,Ig, Vb = 0V, T = 70°C



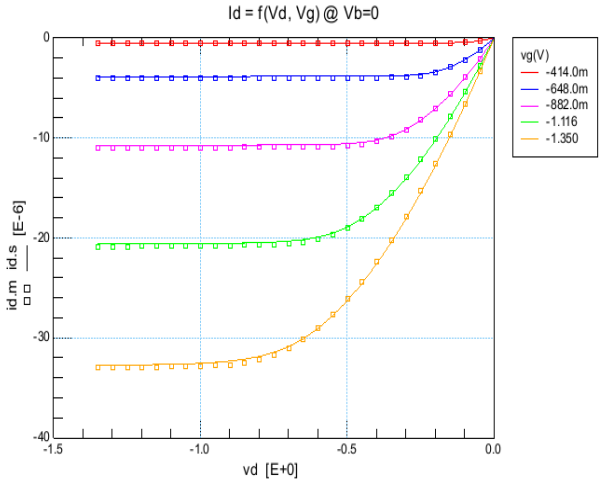


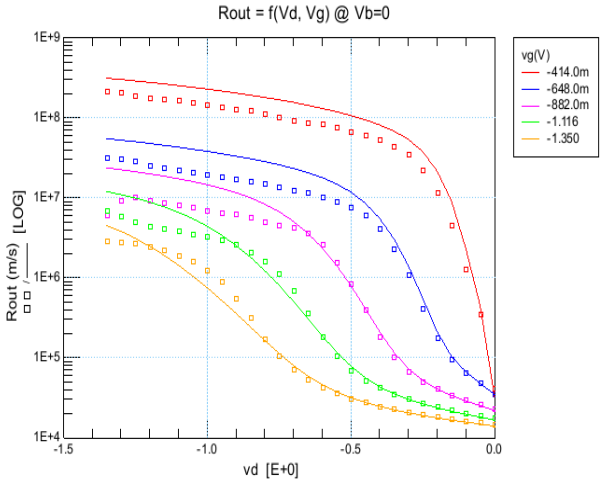
idvg, Ib,Ig, Vb = 1.2V, T = 70°C

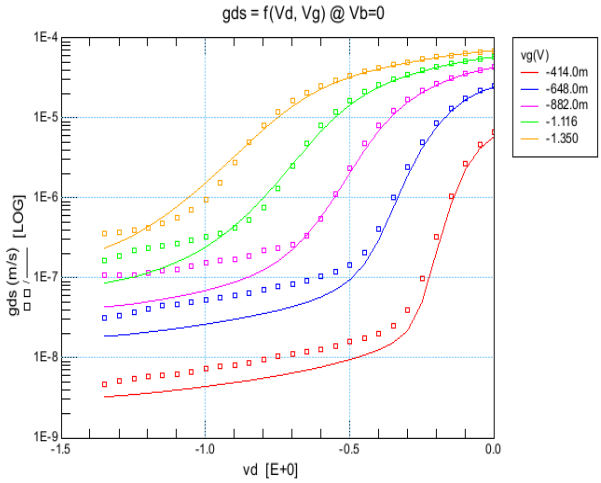




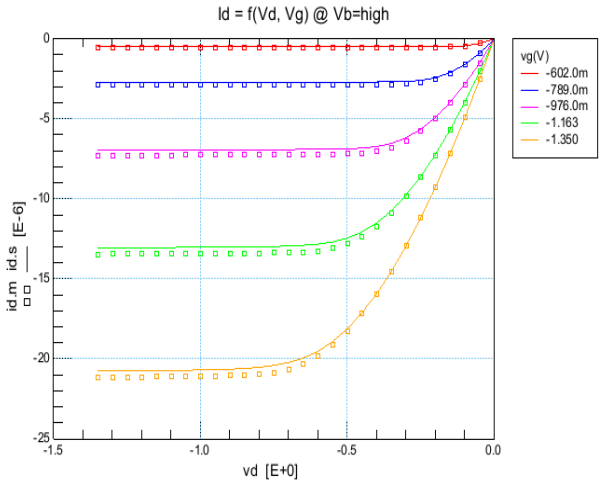
idvd, Vb = 0V, T = 70°C

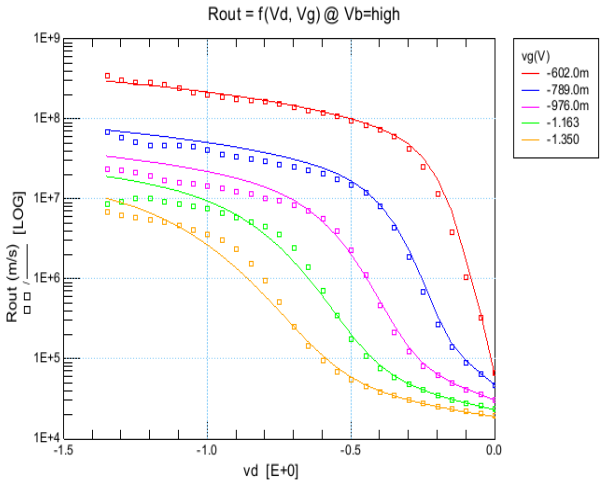


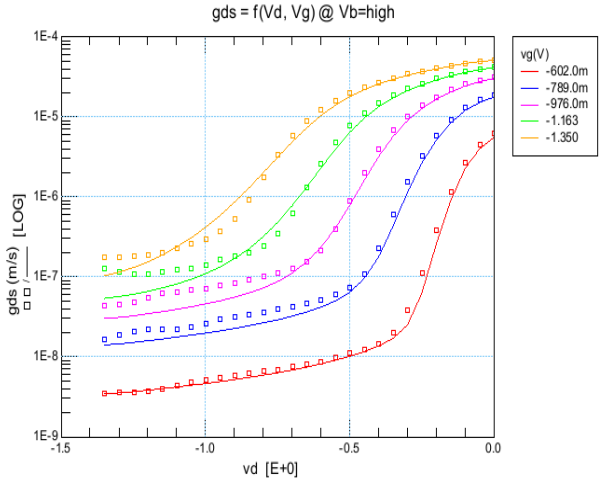




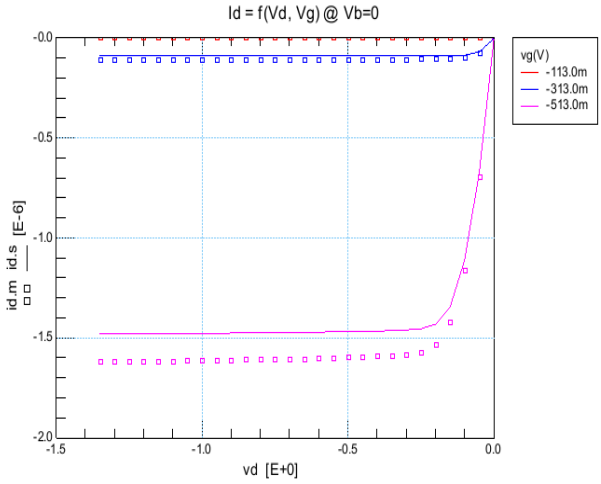
idvd_vbmin, Vb = 1.2V, T = 70°C

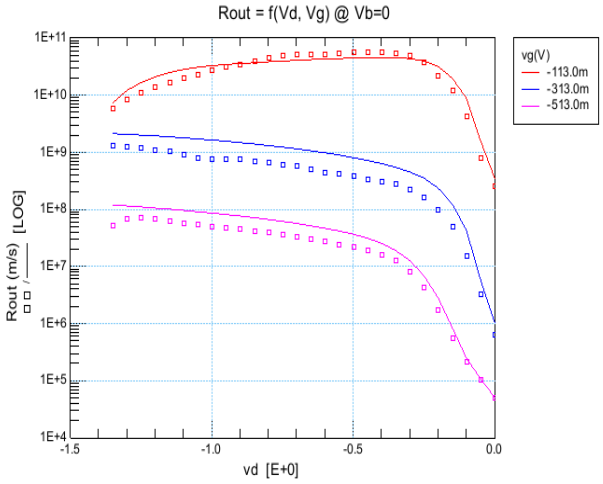


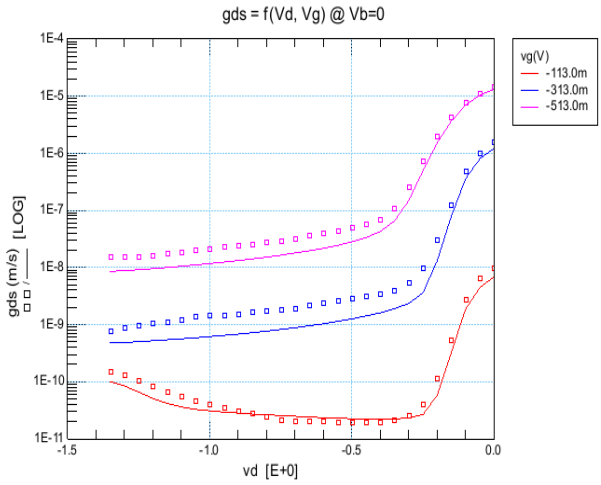




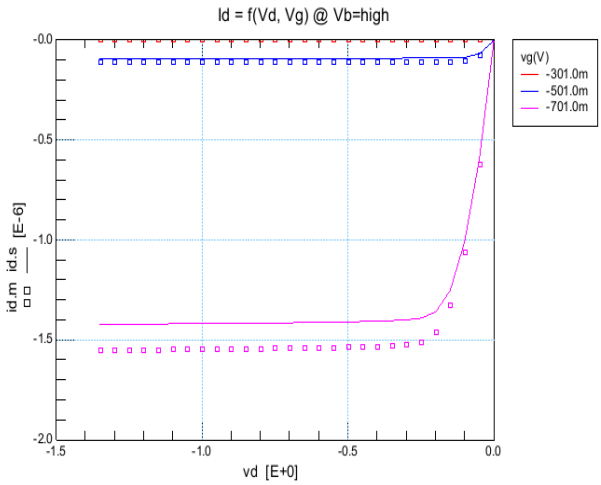
idvd_vth, Vb = 0V, T = 70°C

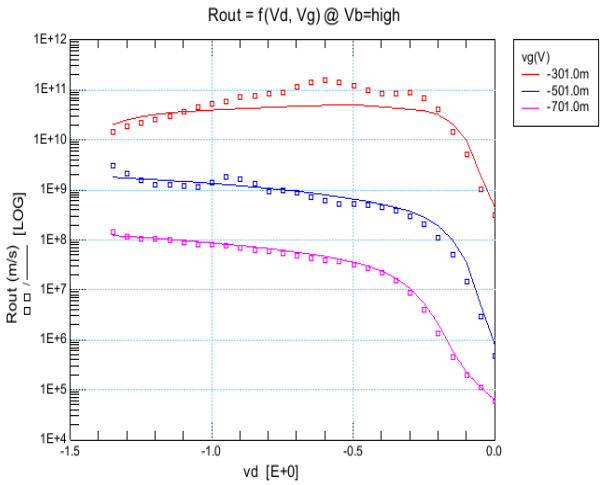


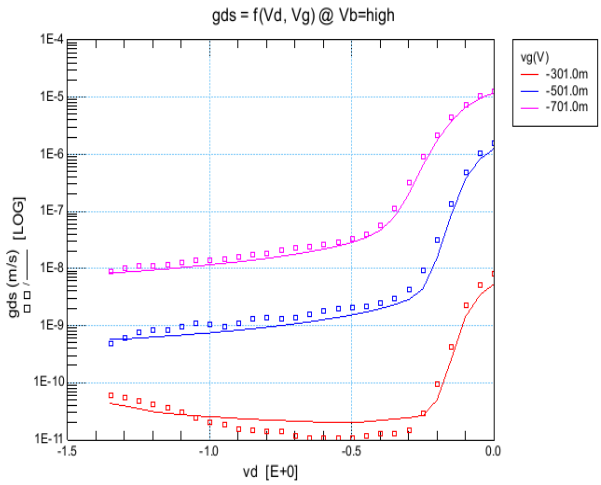




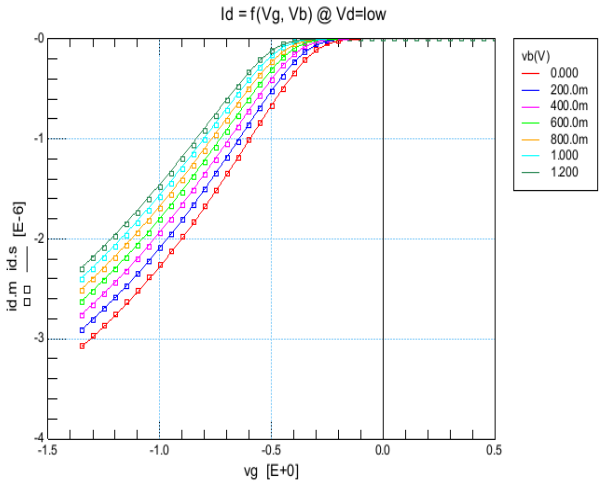
idvd_vth_vbmin, Vb = 1.2V, T = 70°C

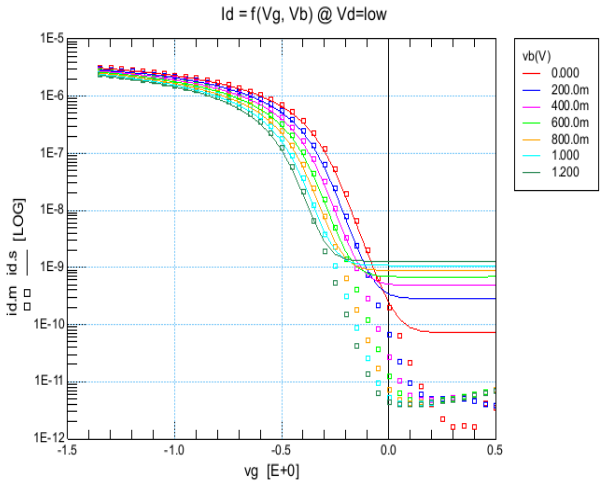


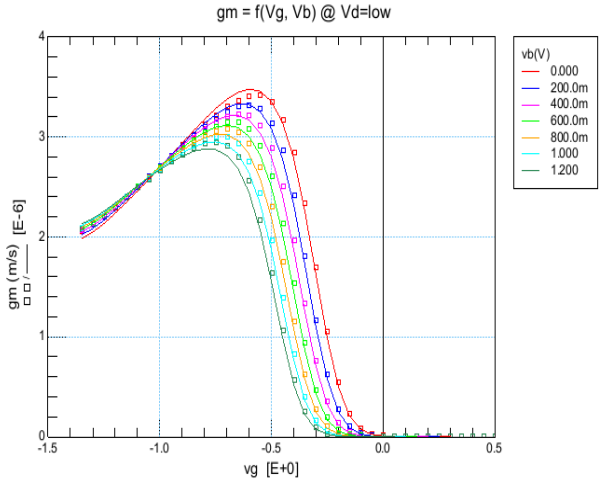


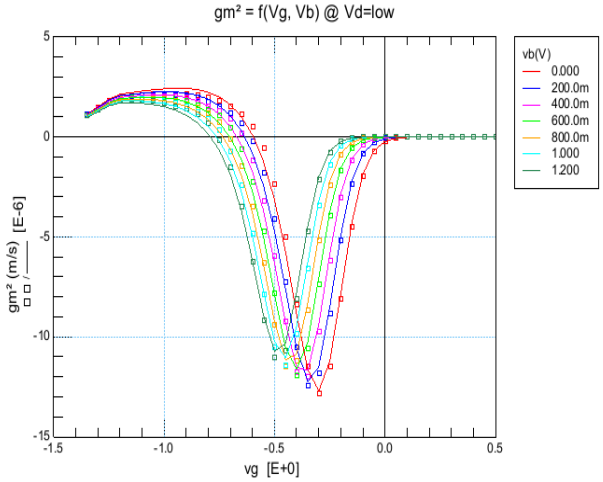


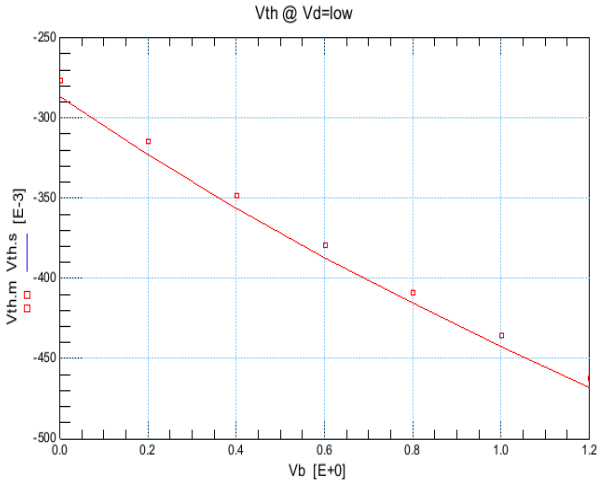
idvg, Vd = -0.05V, T = 125°C

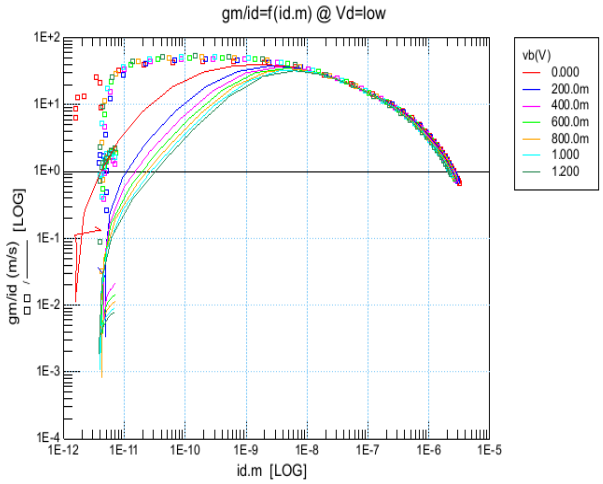


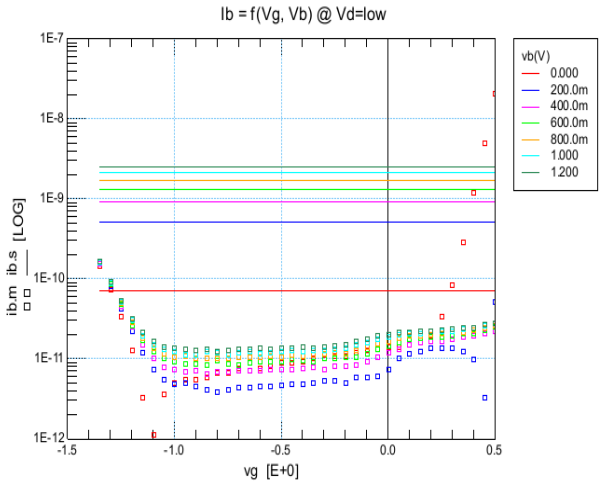


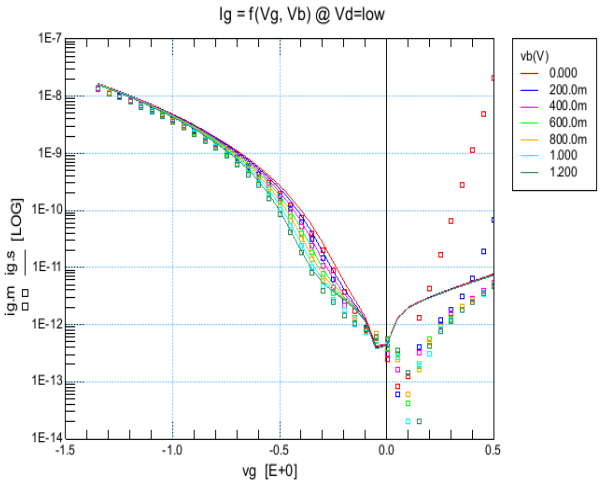




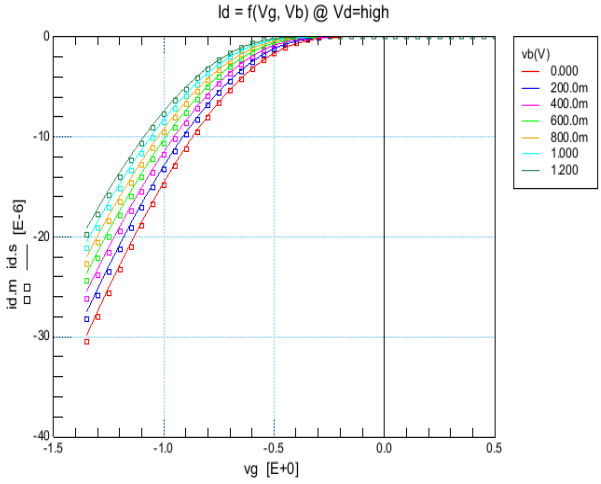


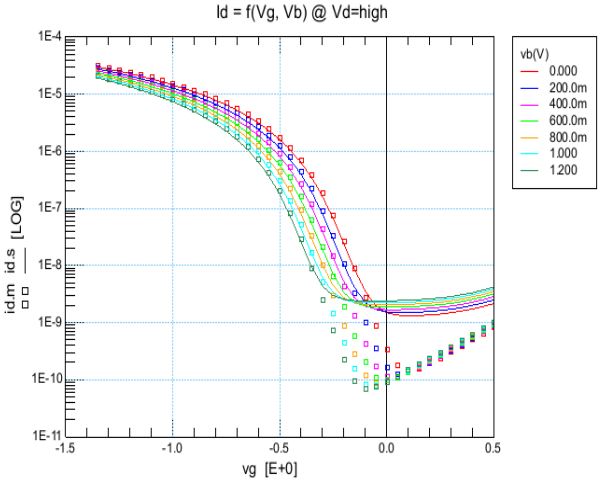


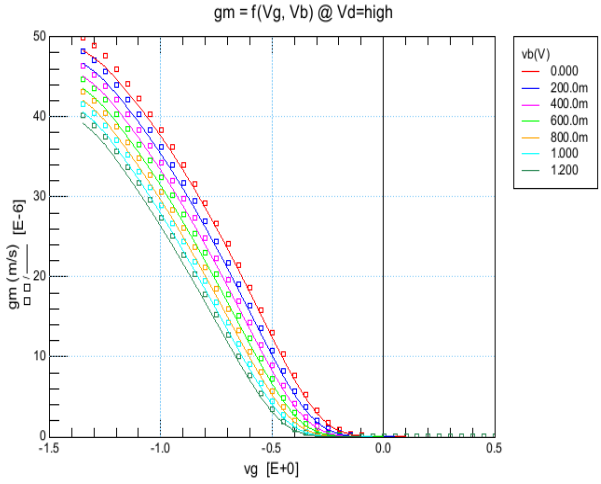


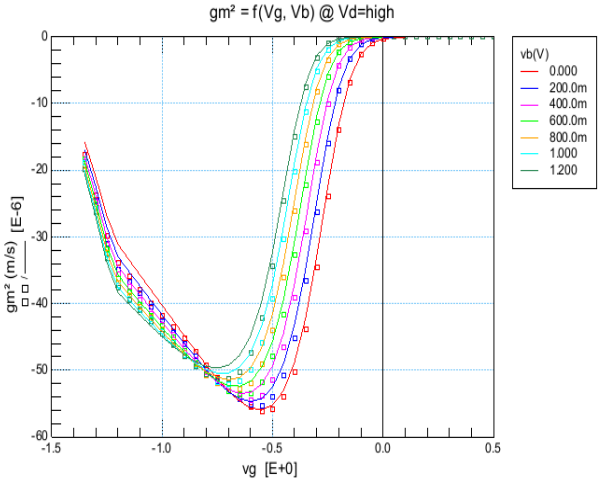


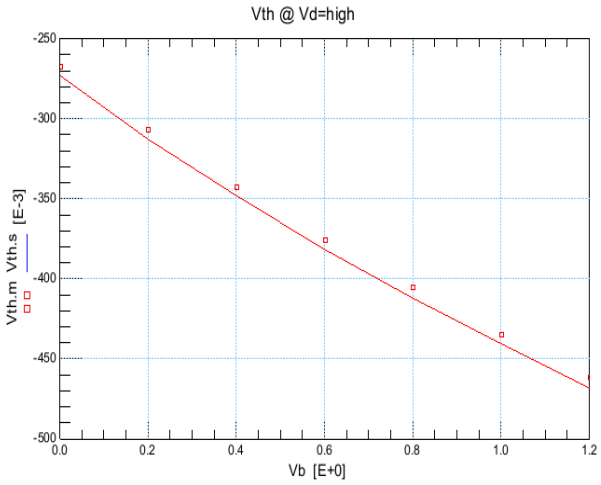
idvg, Vd = -1.2V, T = 125°C

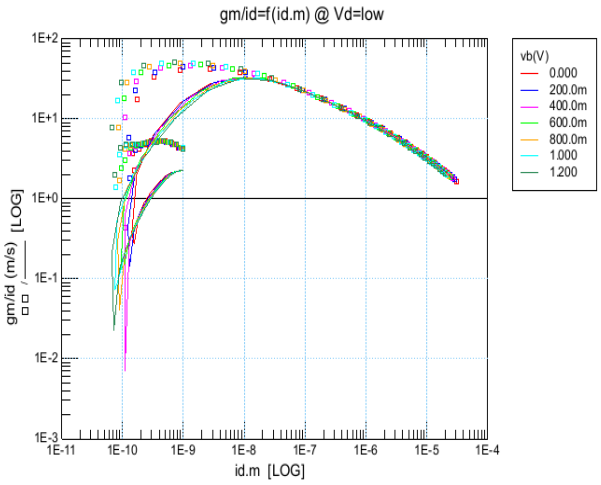


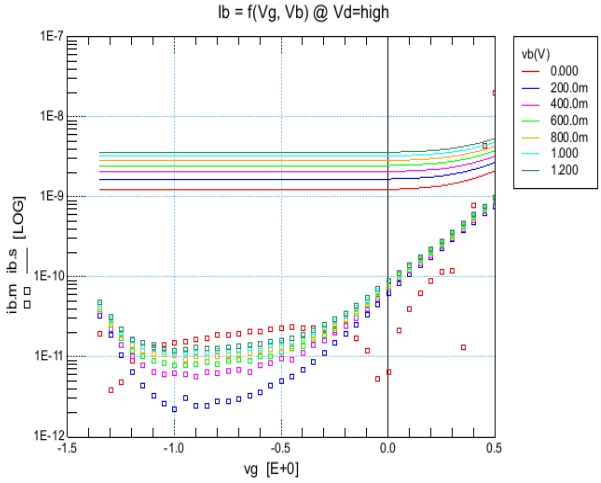


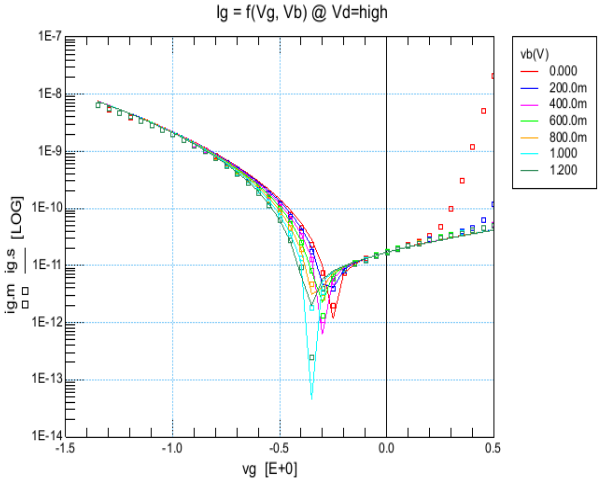




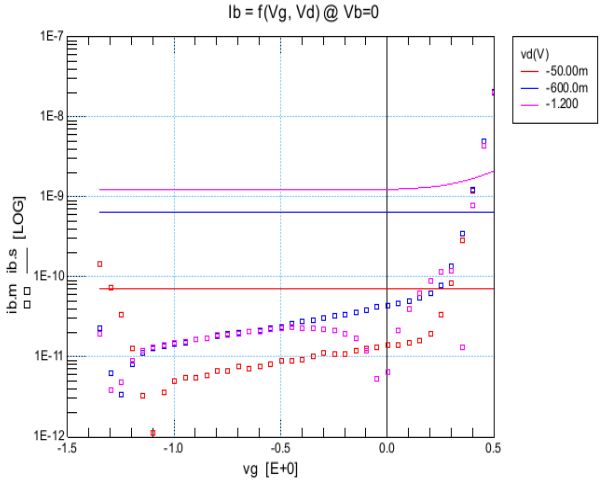


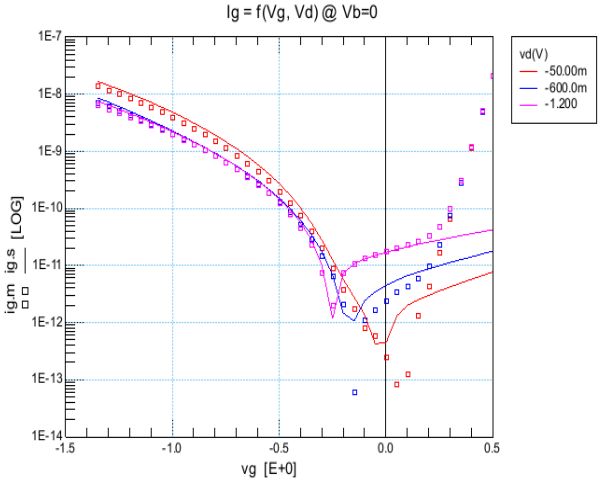




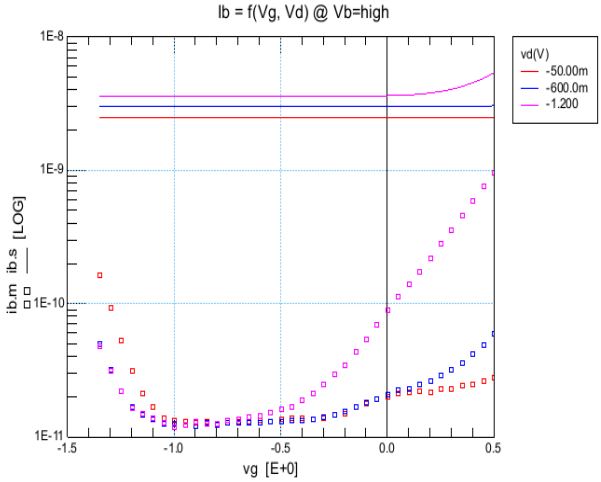


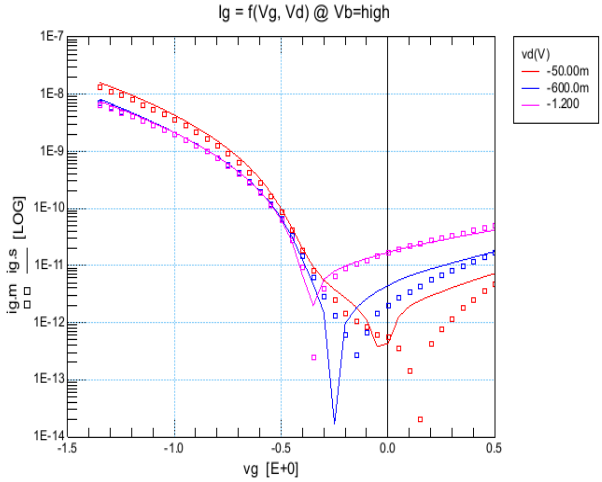
idvg, Ib,Ig, Vb = 0V, T = 125°C



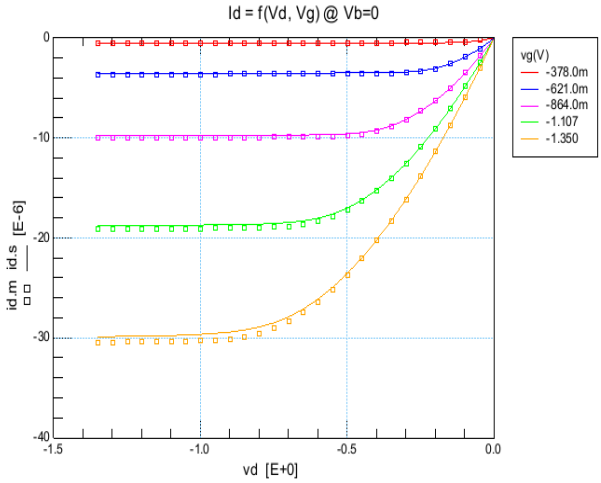


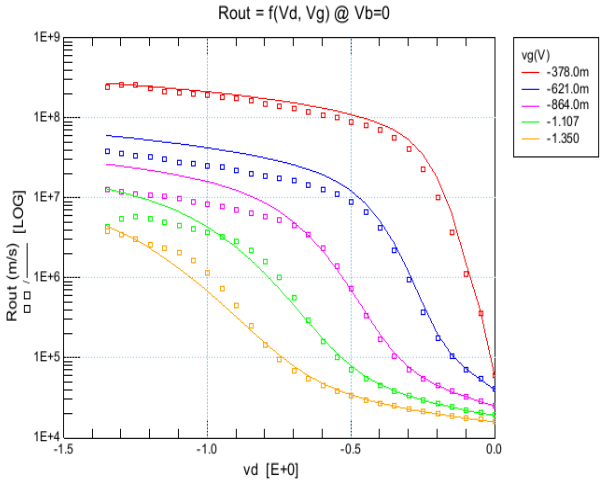
idvg, Ib,Ig, Vb = 1.2V, T = 125°C

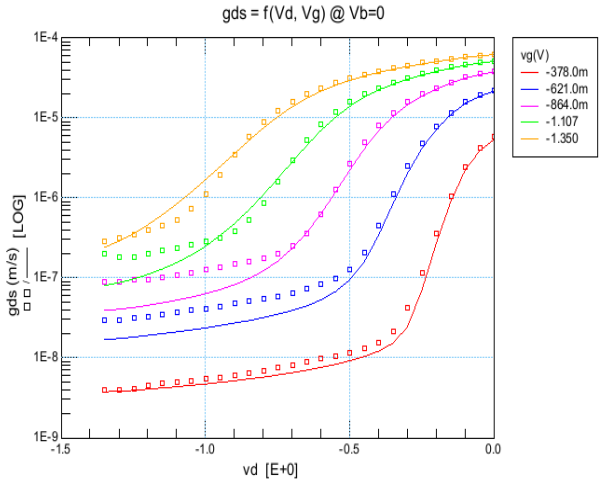




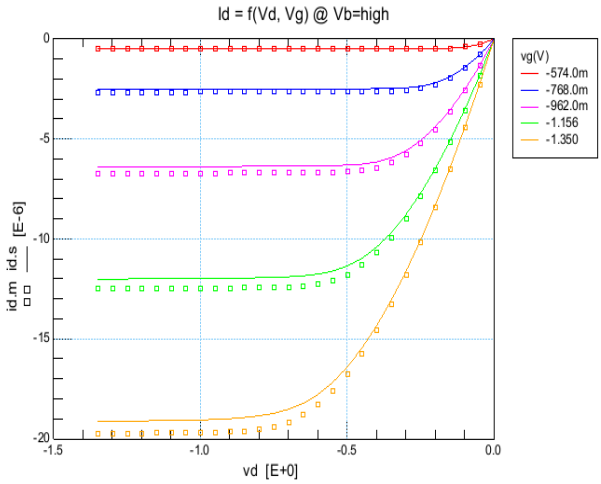
idvd, Vb = 0V, T = 125°C

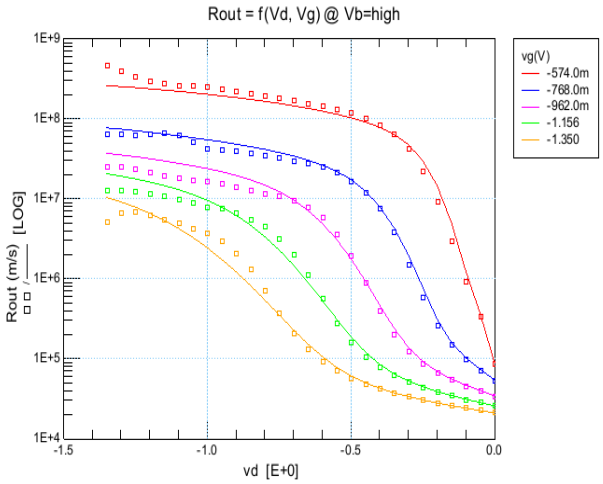


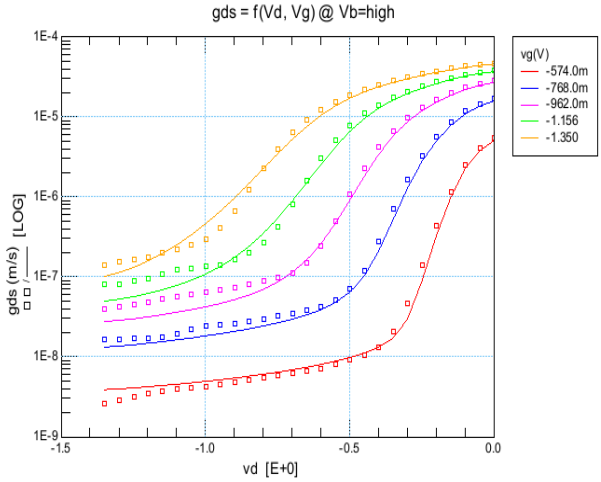




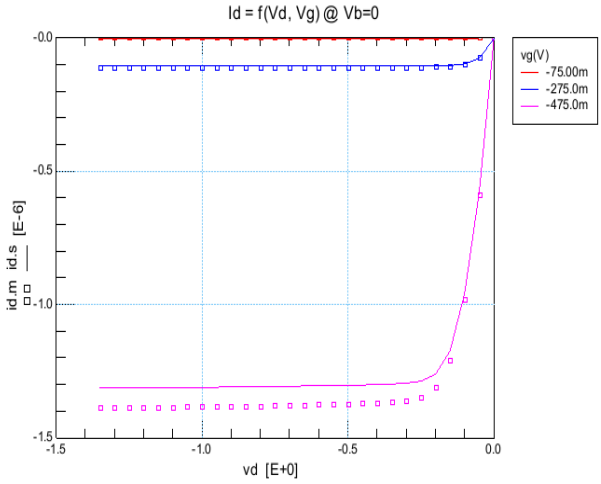
idvd_vbmin, Vb = 1.2V, T = 125°C

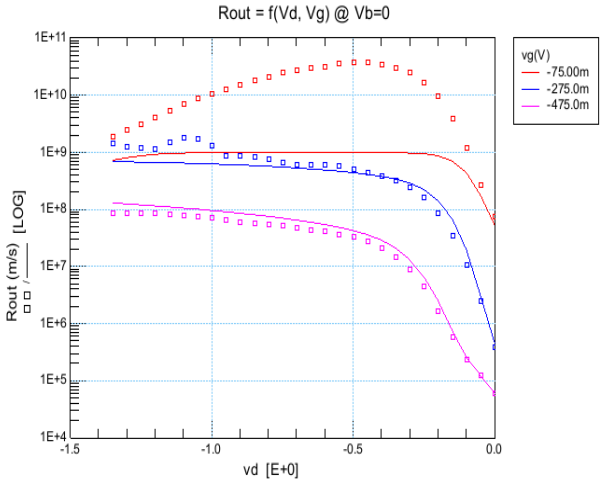


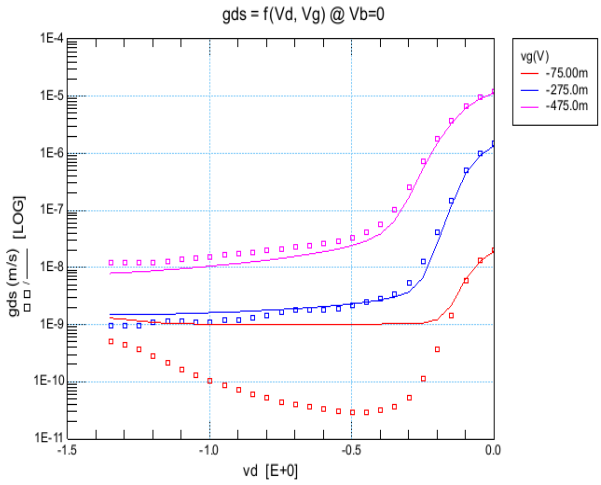




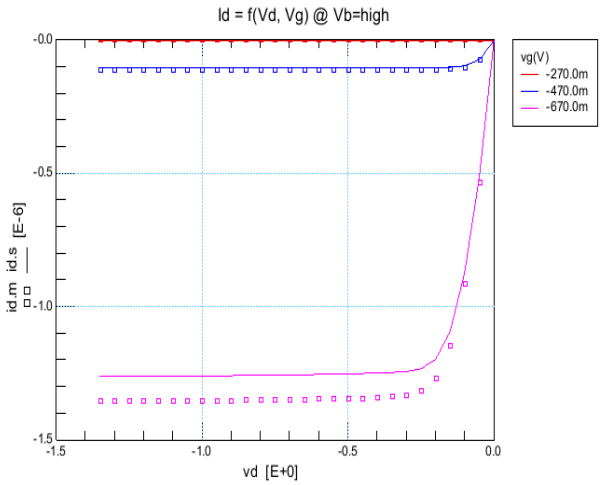
idvd_vth, Vb = 0V, T = 125°C

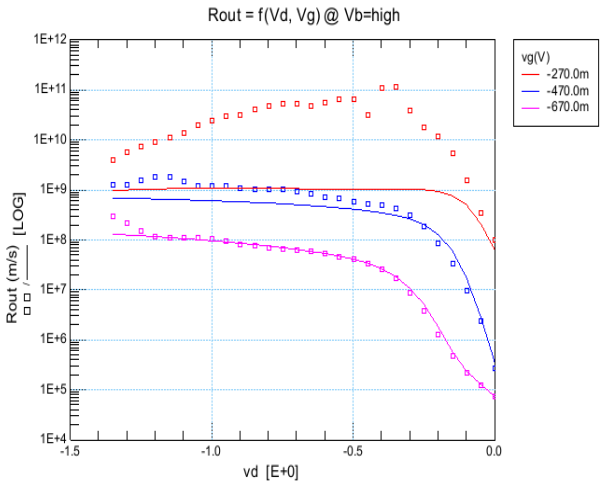


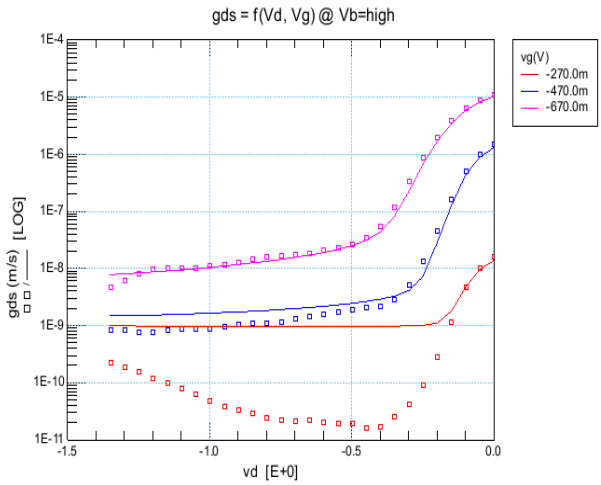




idvd_vth_vbmin, Vb = 1.2V, T = 125°C

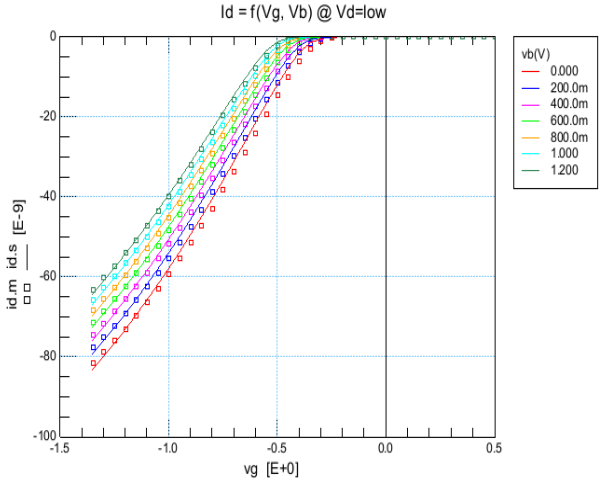


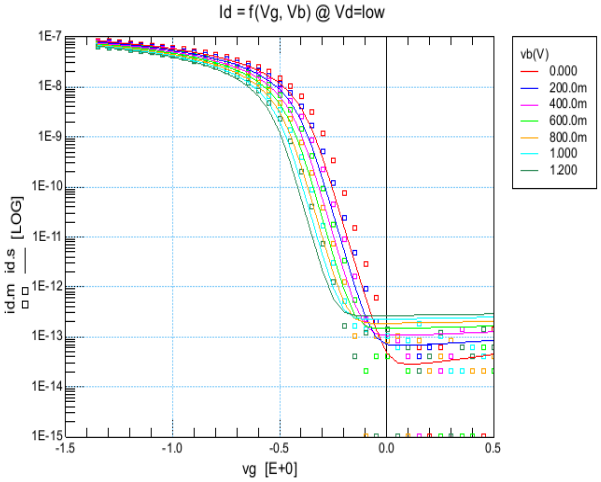


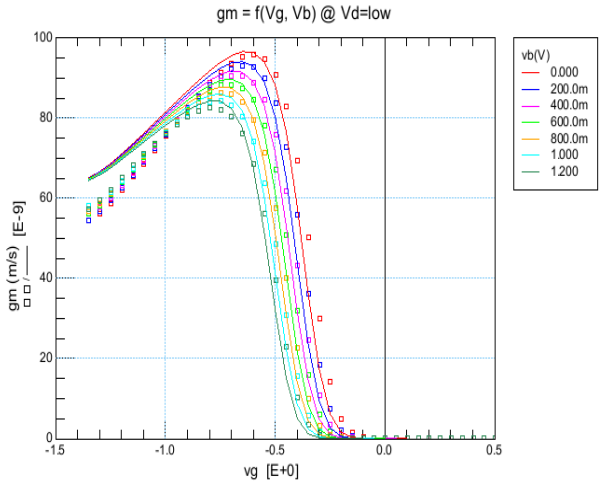


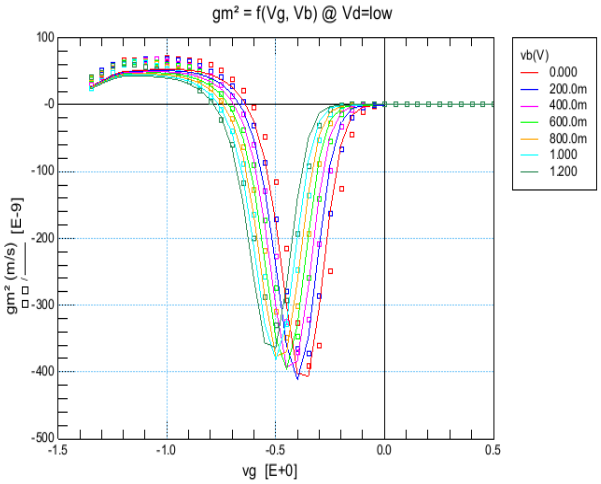
4.3 W0u15_L10u0_S555_3

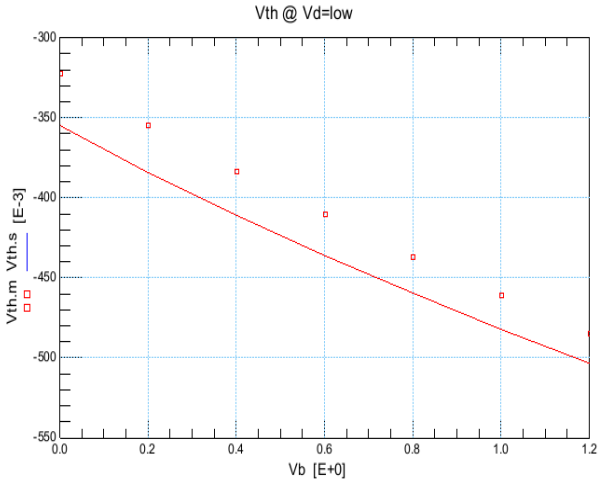
idvg, Vd = -0.05V, T = 27°C

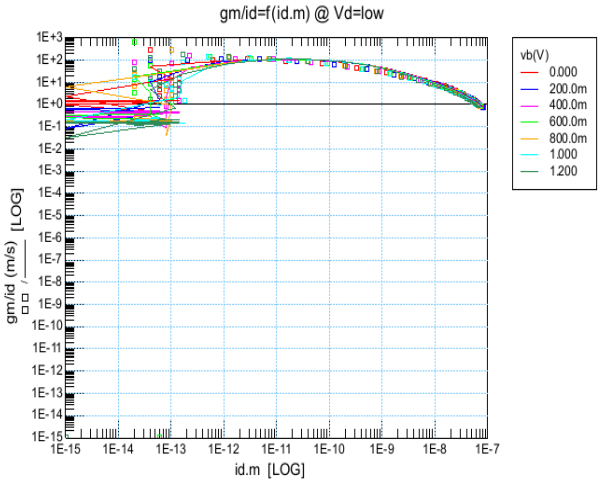


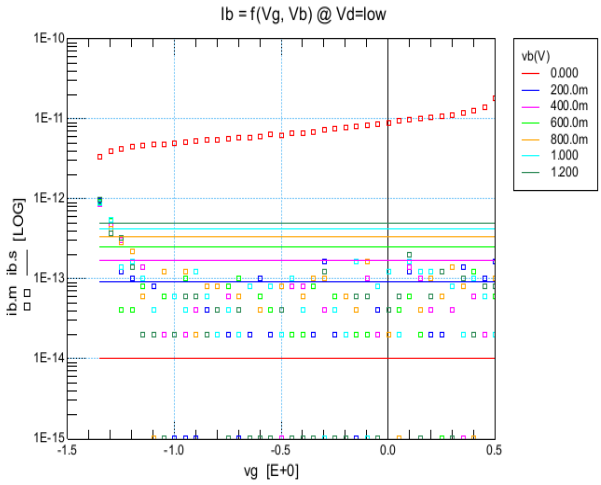


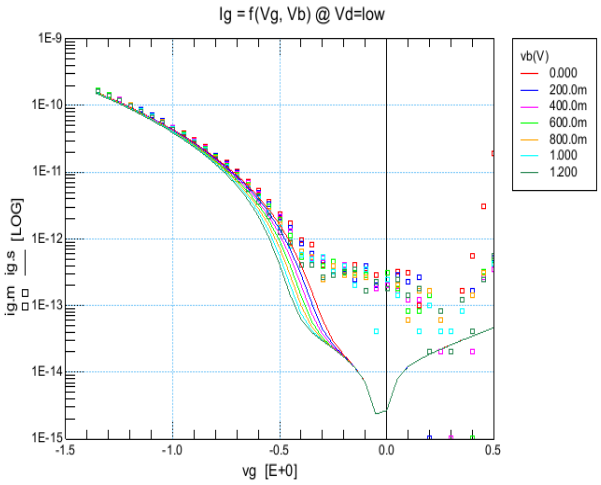




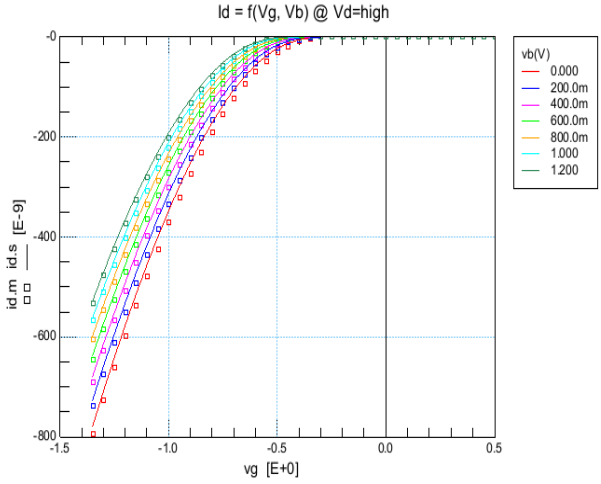


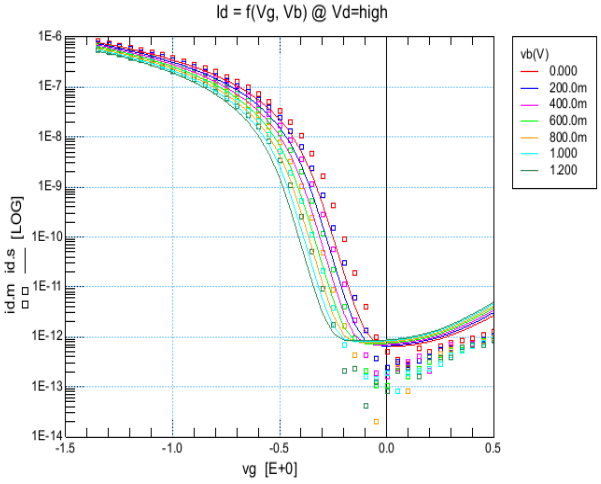


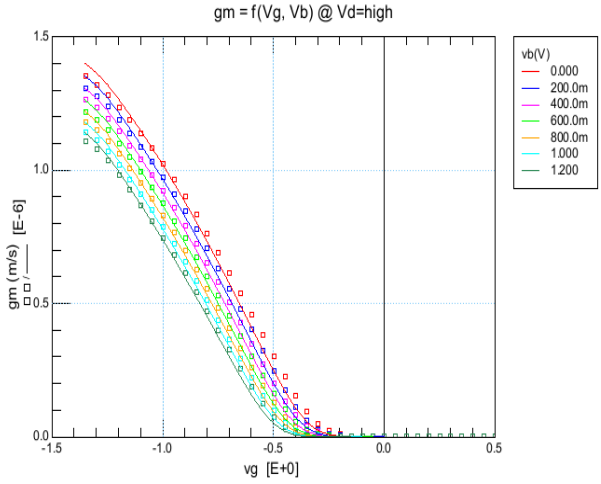


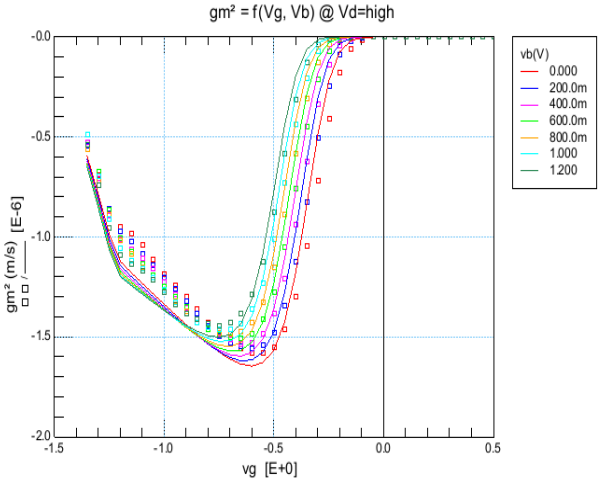


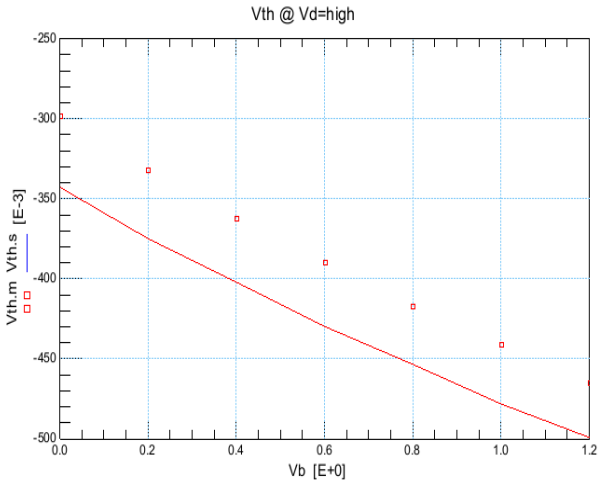
idvg, Vd = -1.2V, T = 27°C

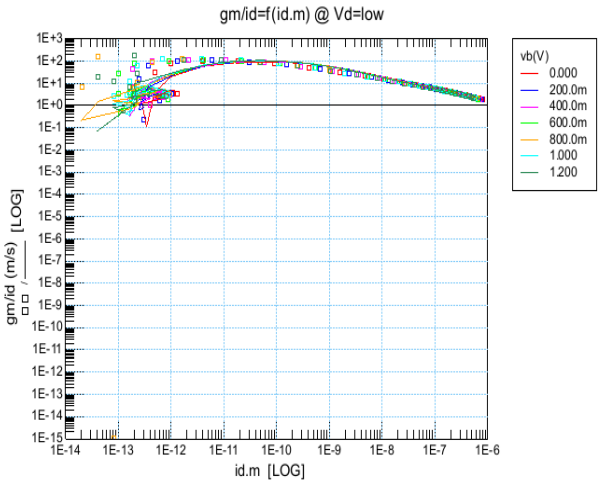


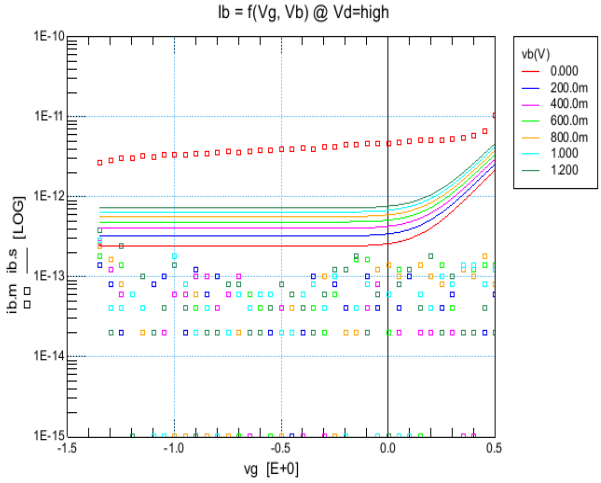


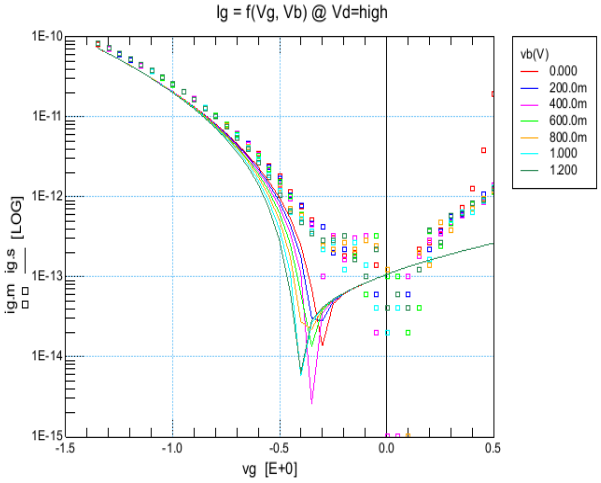




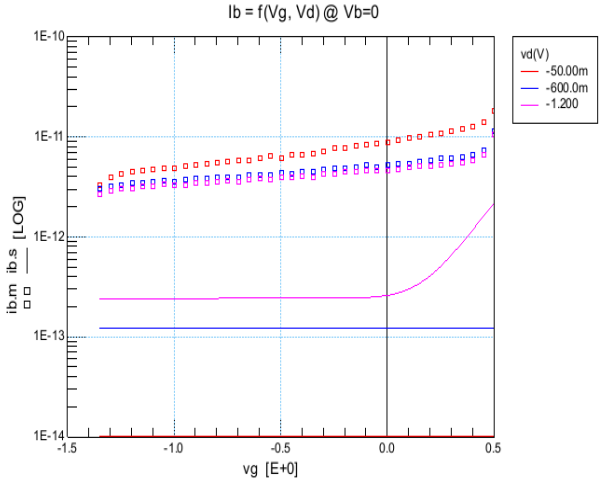


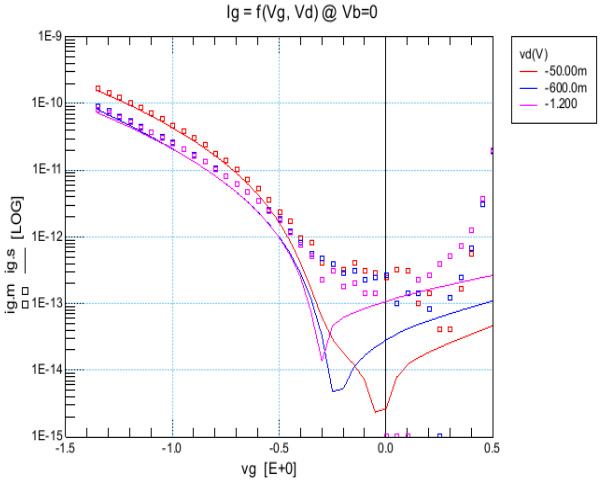




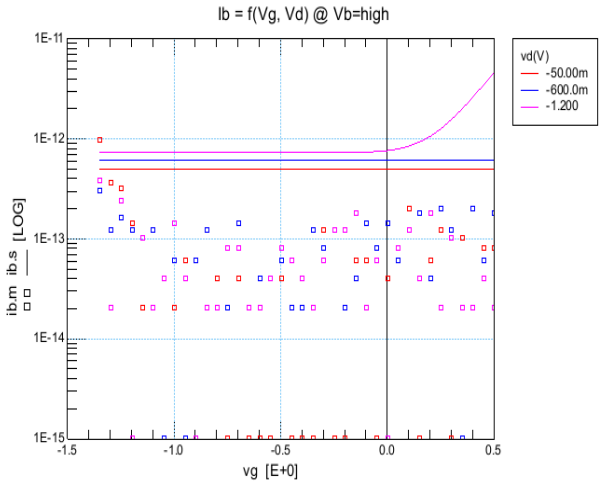


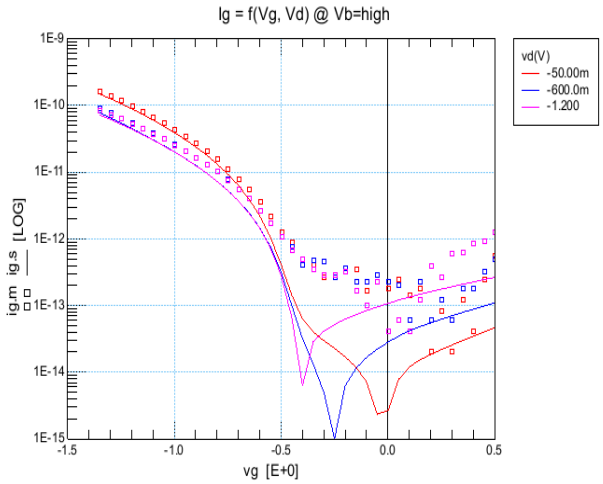
idvg, Ib,Ig, Vb = 0V, T = 27°C



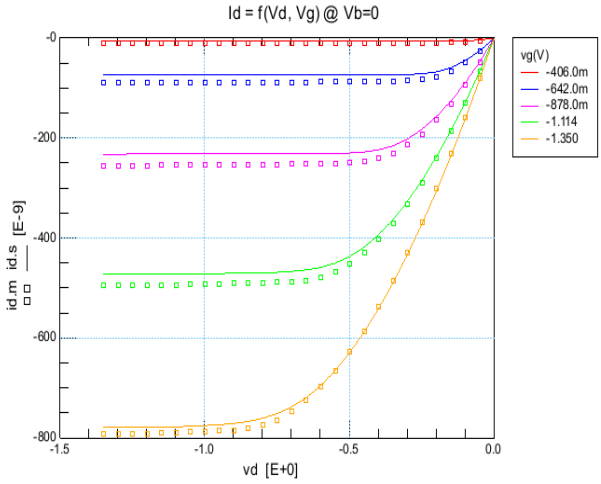


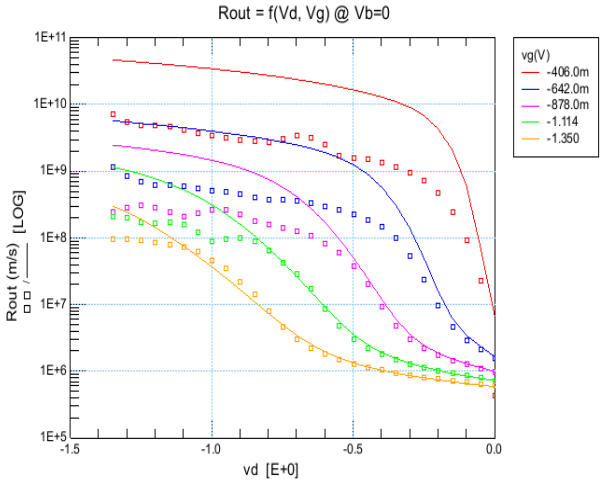
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

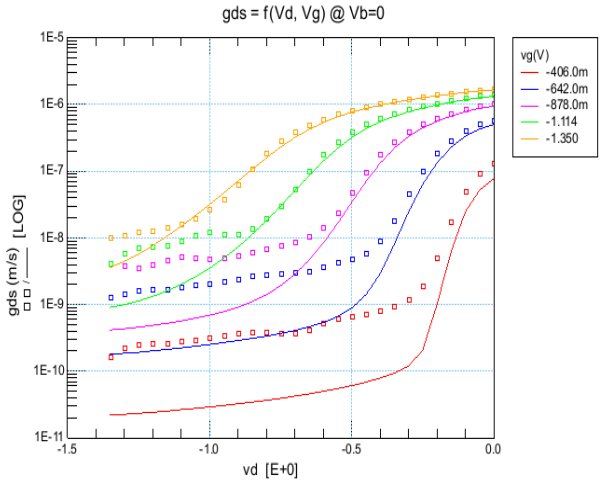




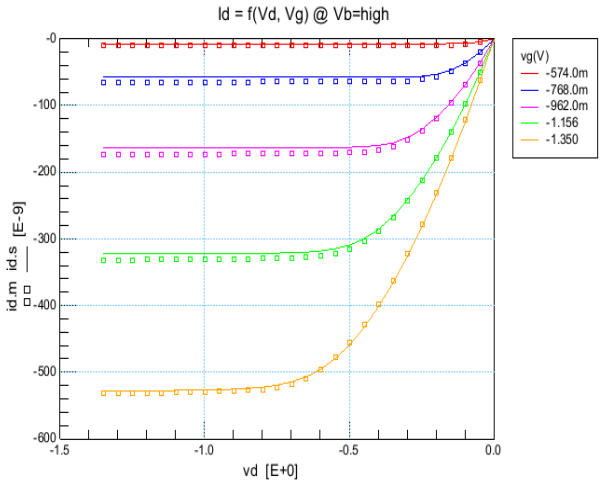
idvd, Vb = 0V, T = 27°C

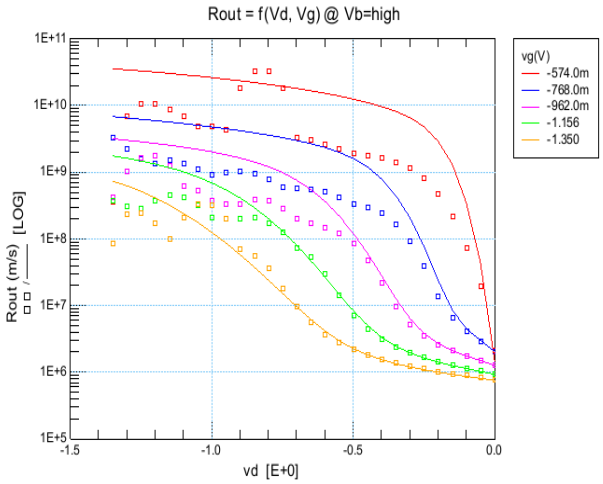


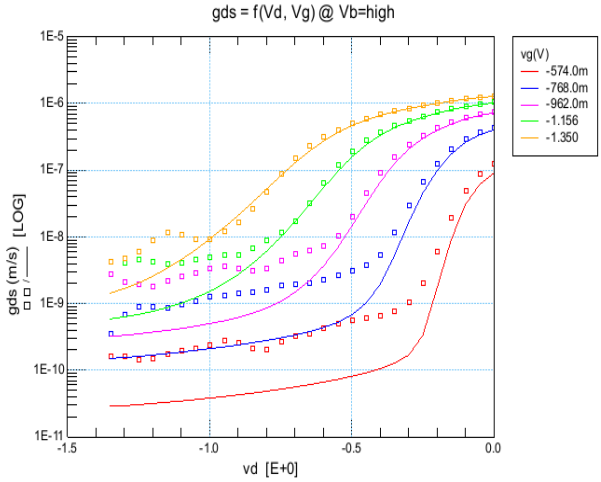




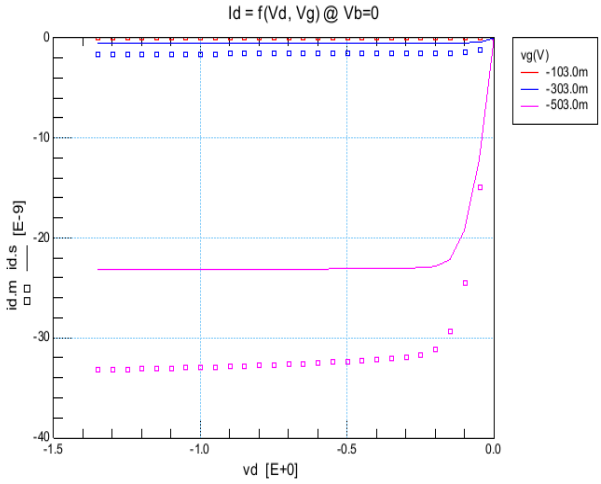
idvd_vbmin, Vb = 1.2V, T = 27°C

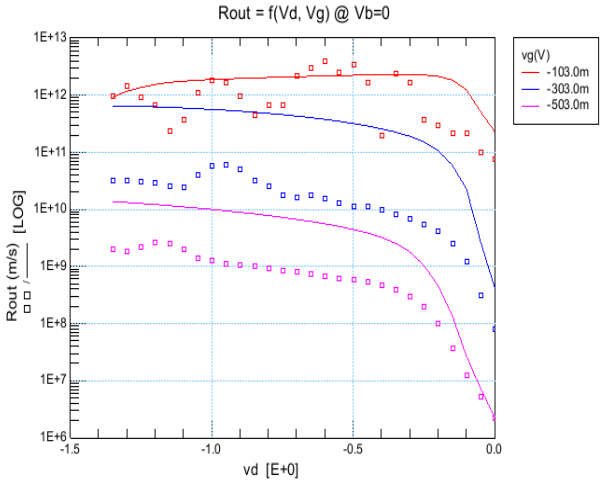


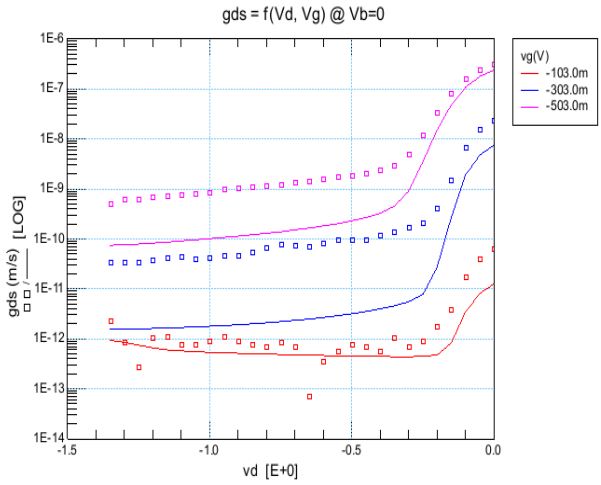




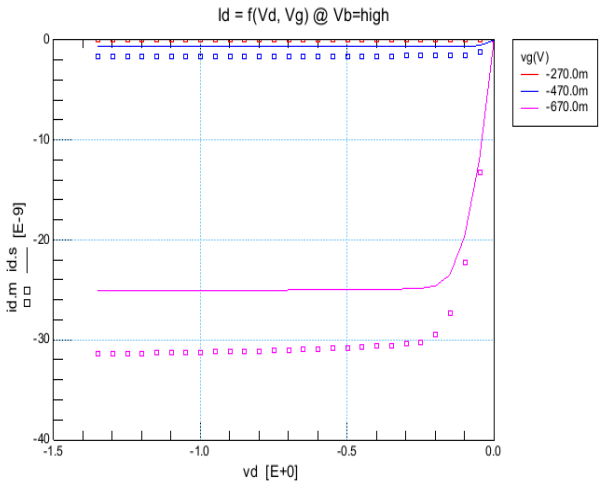
idvd_vth, Vb = 0V, T = 27°C

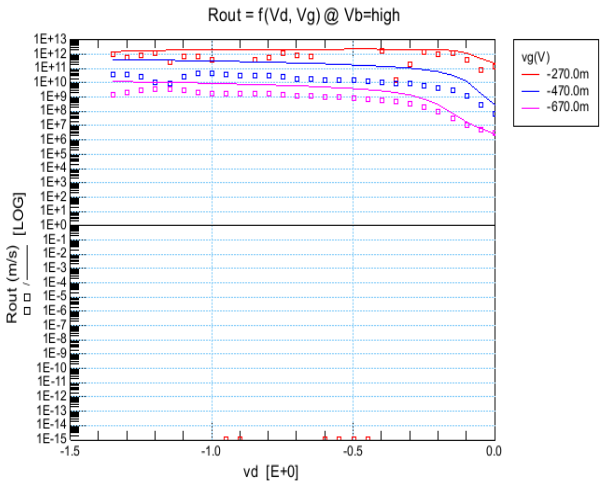


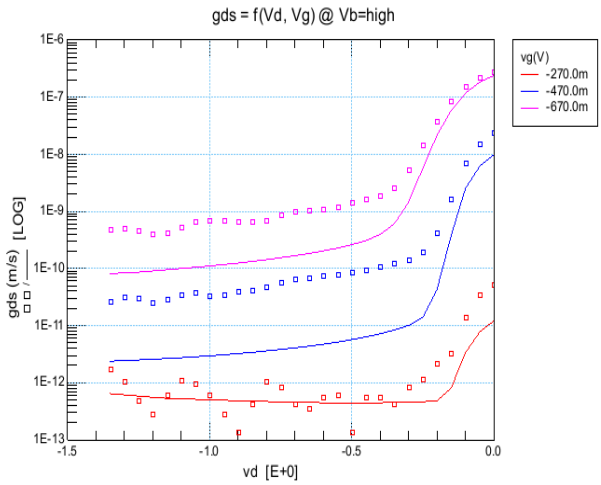




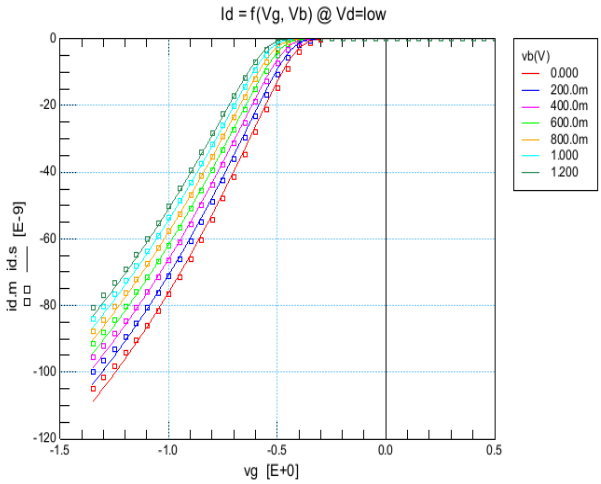
idvd_vth_vbmin, Vb = 1.2V, T = 27°C

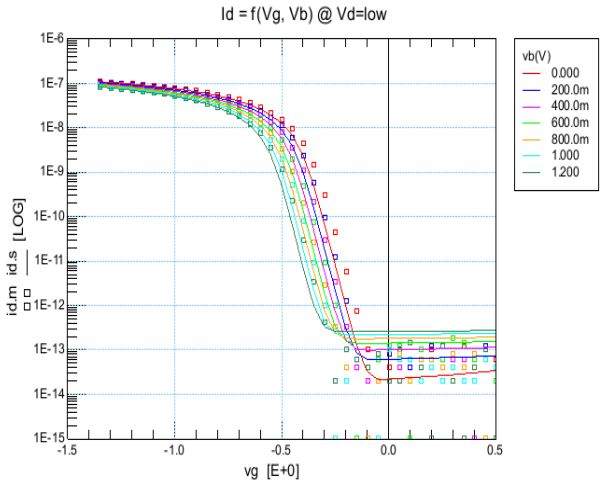


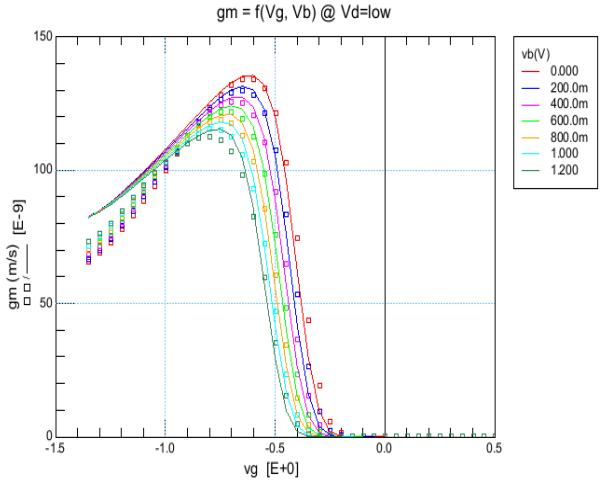


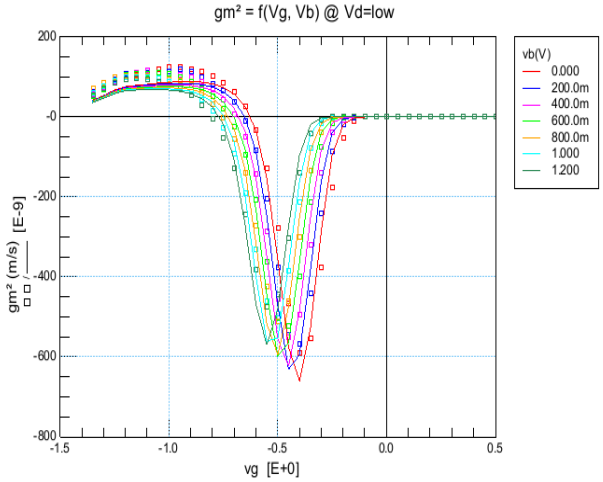


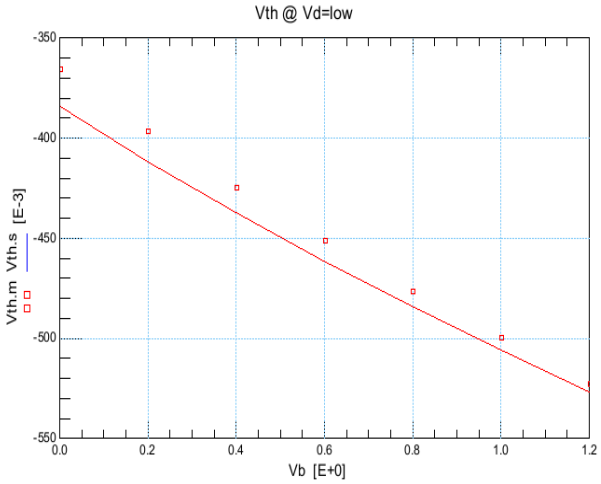
idvg, Vd = -0.05V, T = -40°C

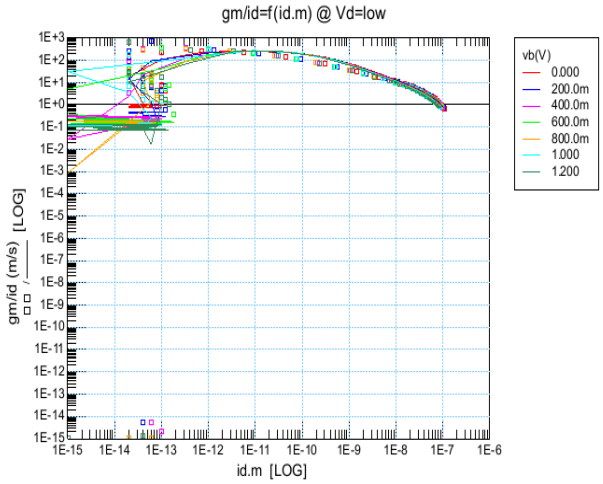


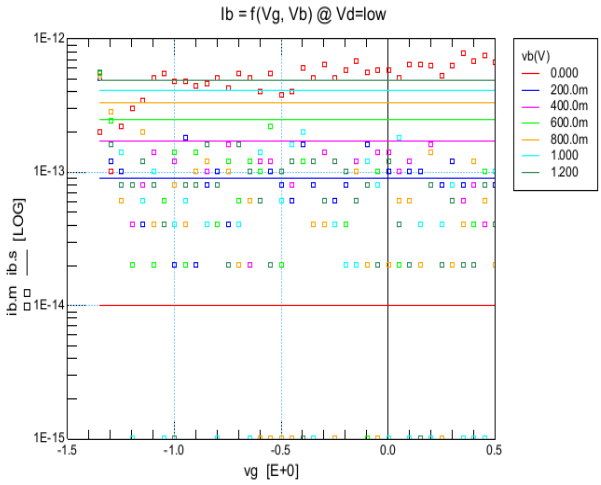


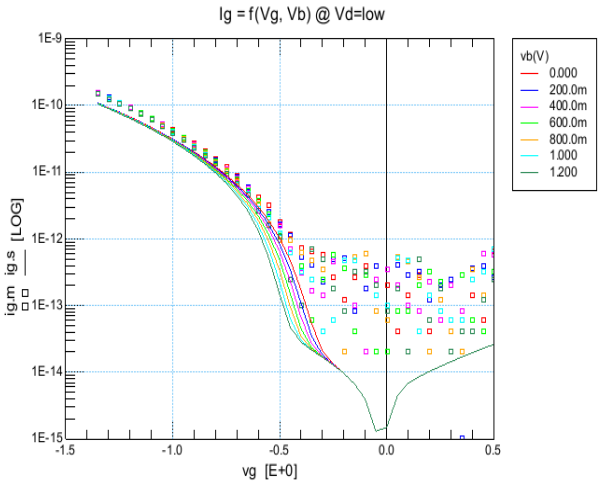




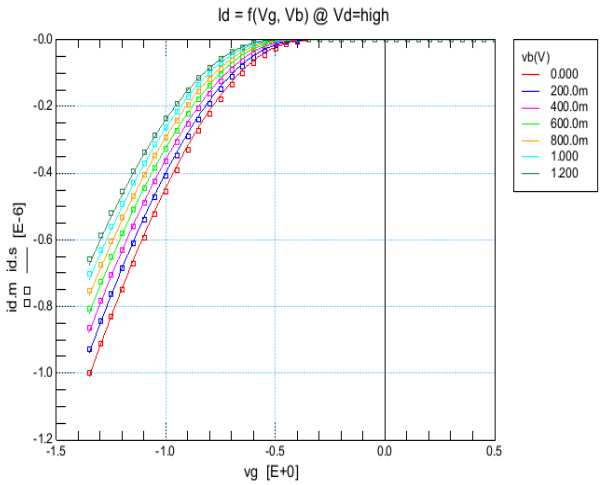


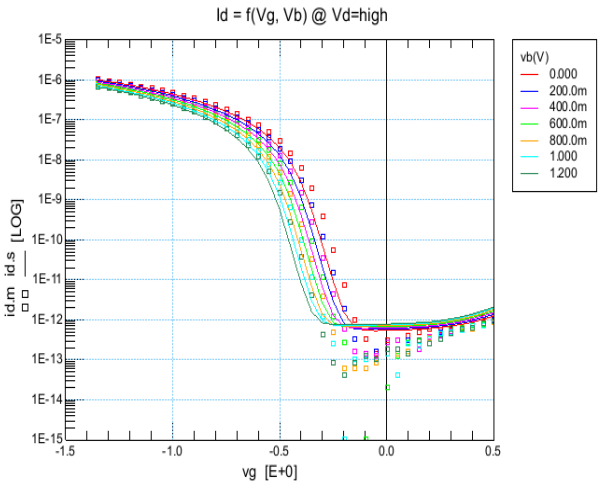


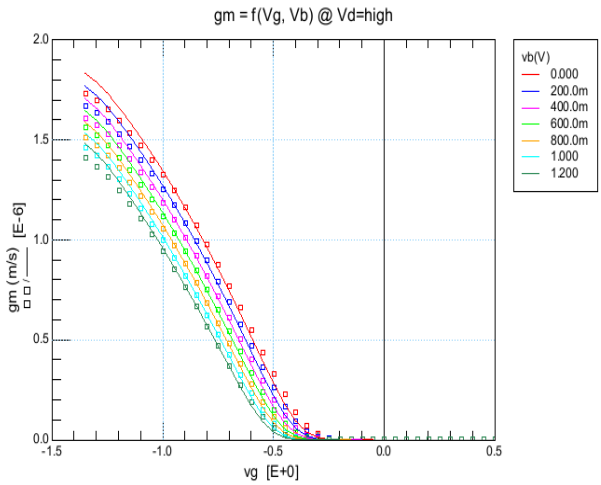


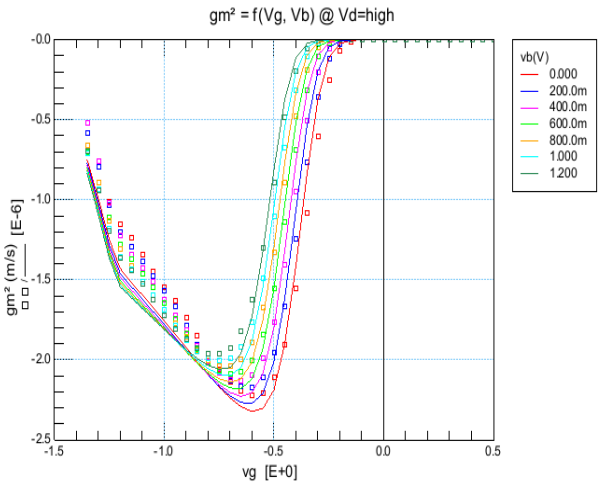


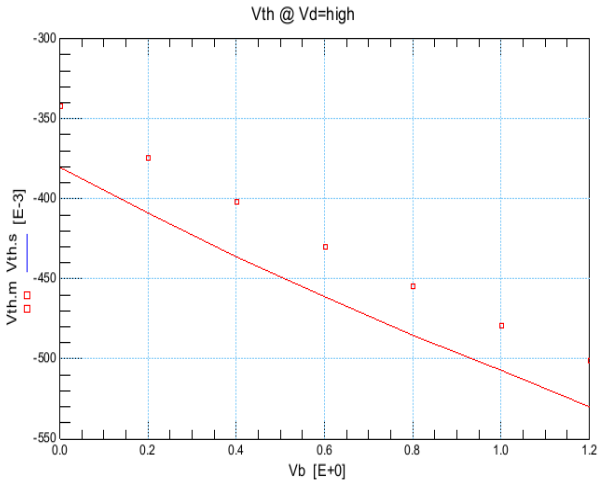
idvg, Vd = -1.2V, T = -40°C

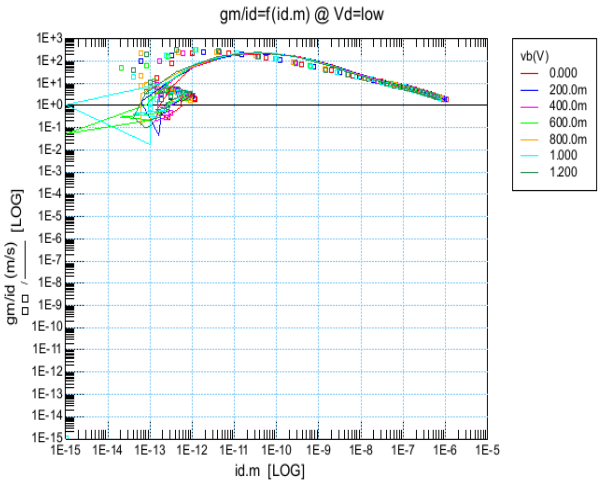


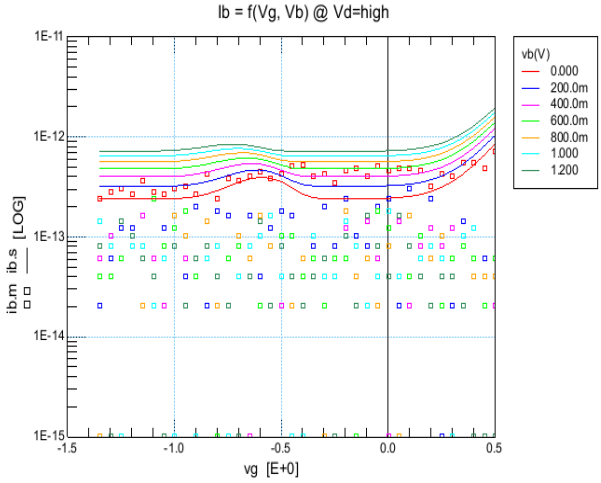


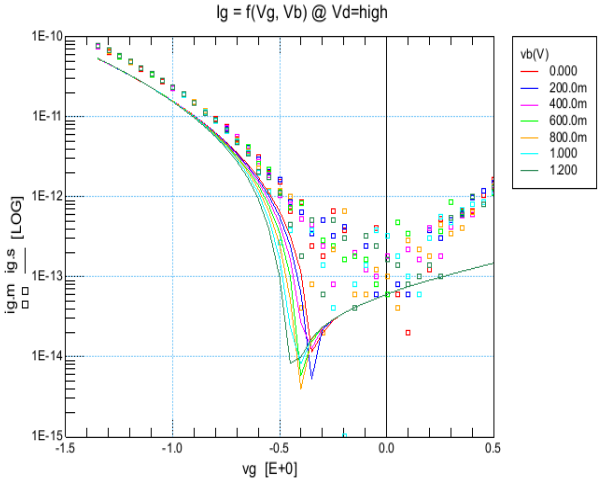




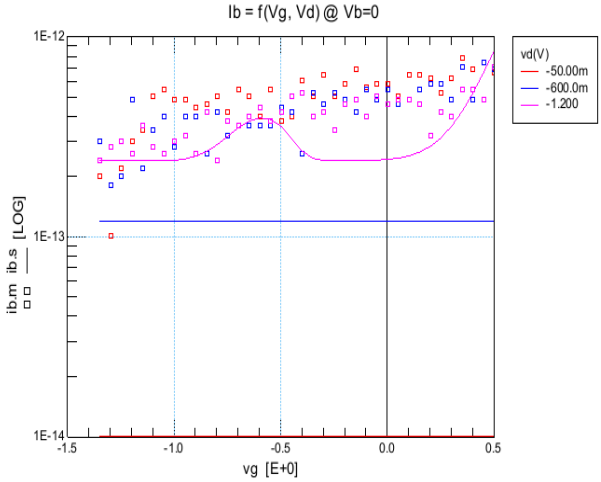


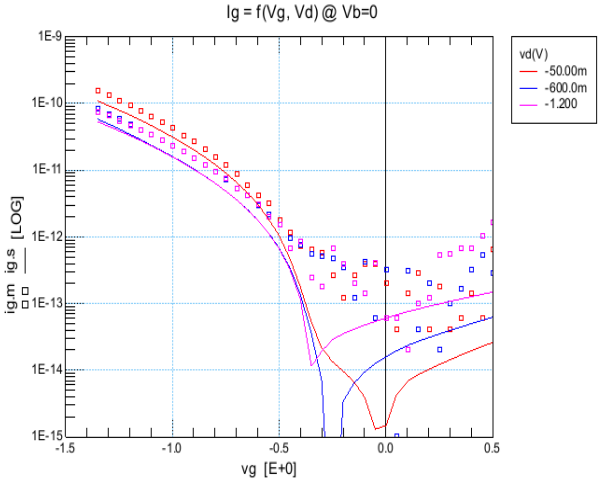




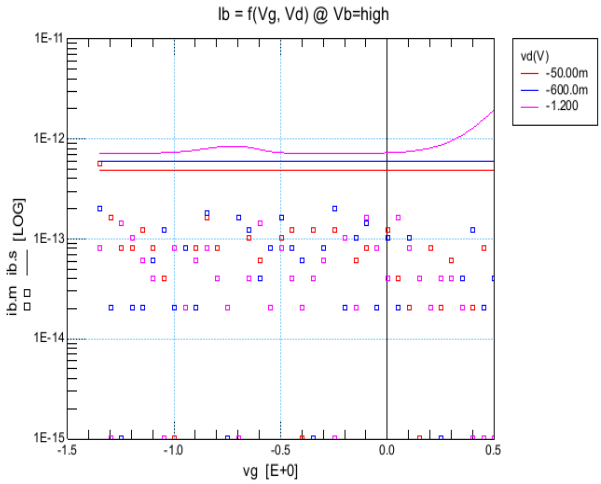


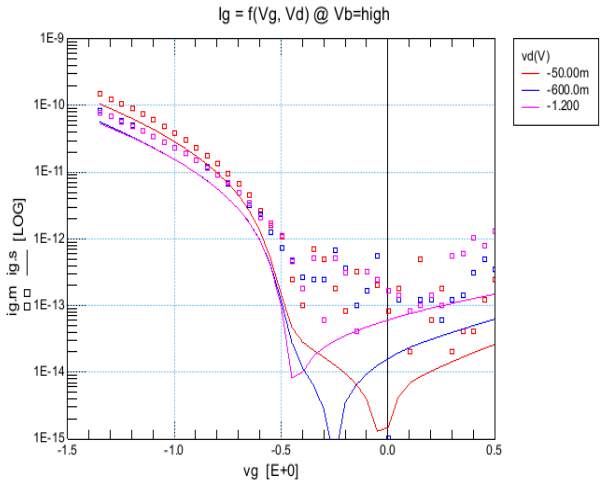
idvg, Ib,Ig, Vb = 0V, T = -40°C



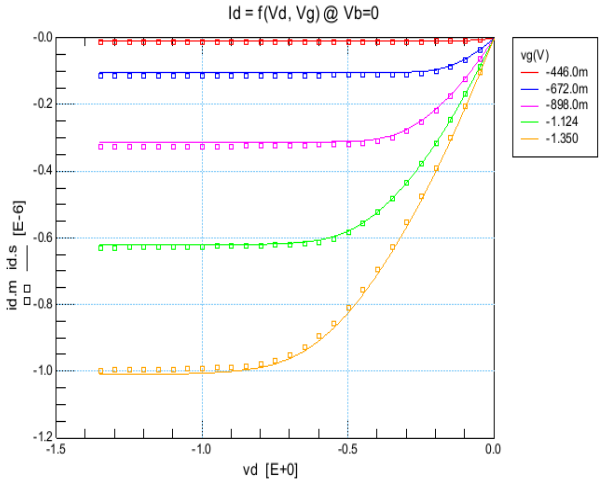


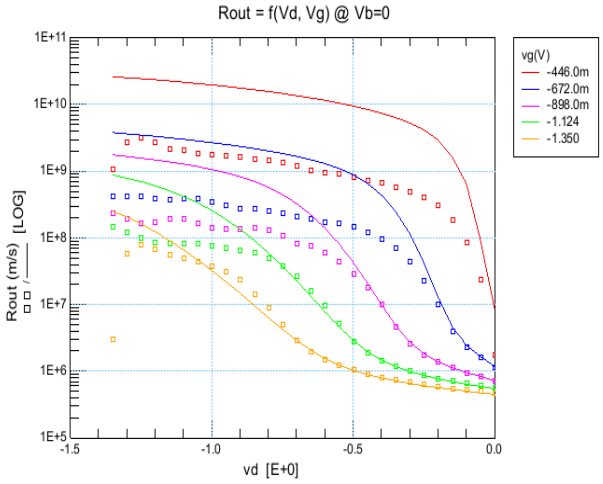
idvg, Ib,Ig, Vb = 1.2V, T = -40°C

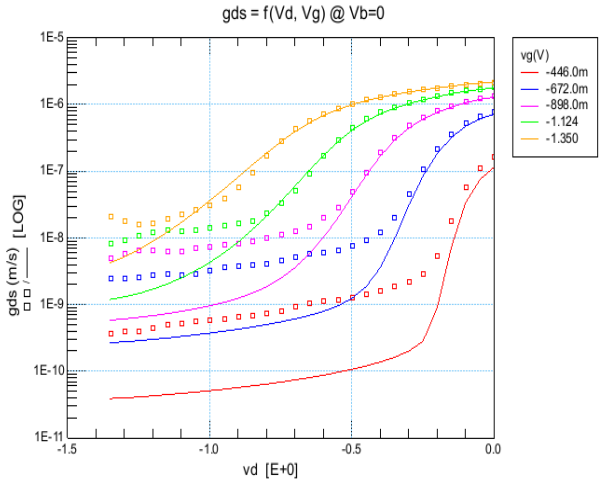




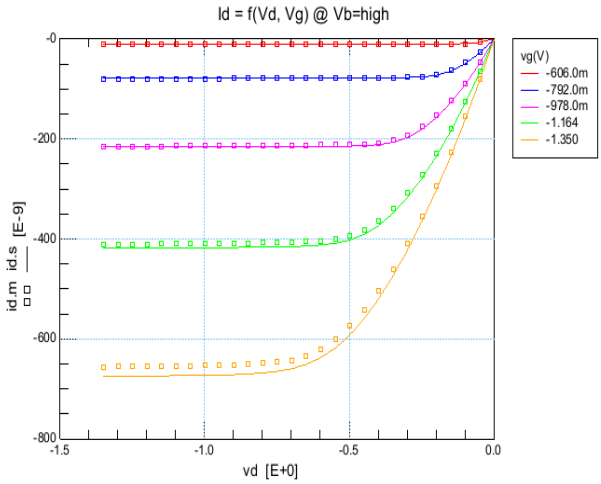
idvd, Vb = 0V, T = -40°C

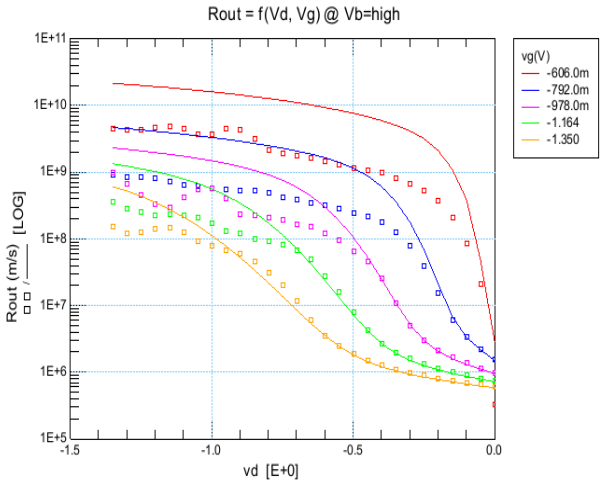


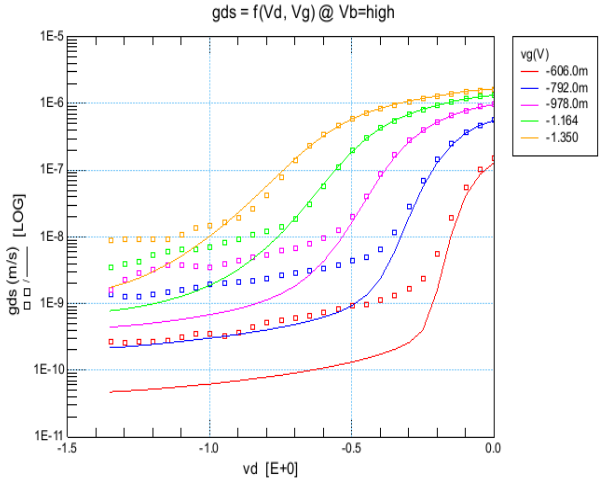




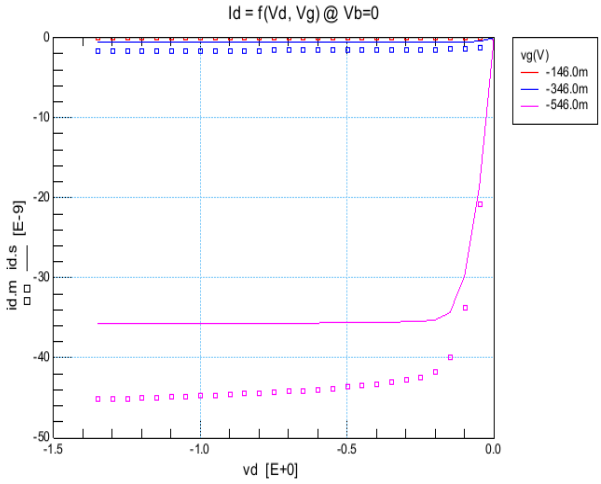
idvd_vbmin, Vb = 1.2V, T = -40°C

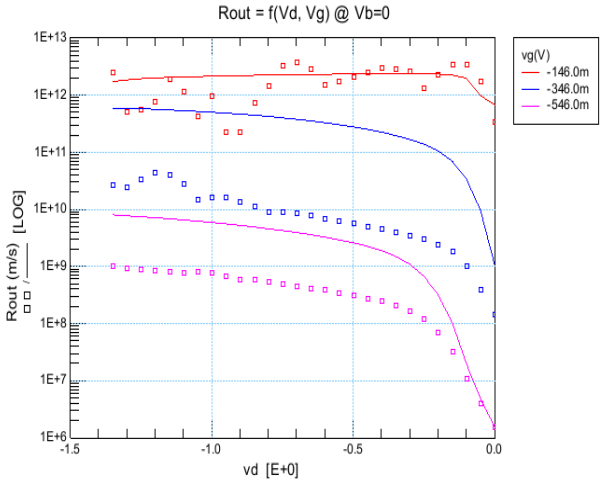


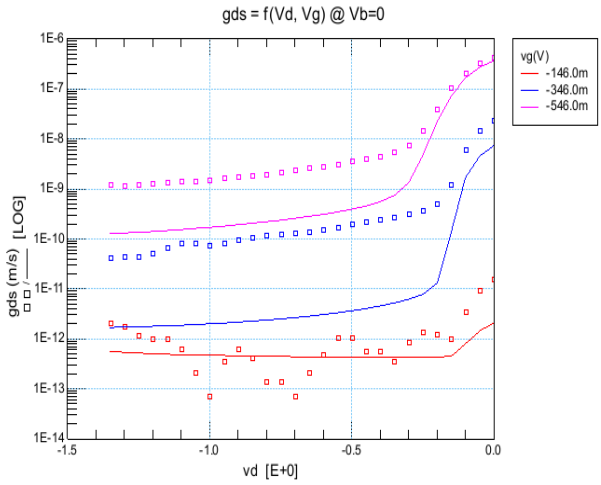




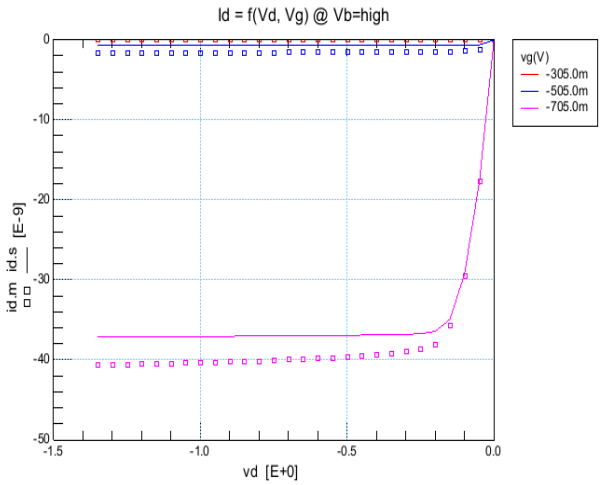
idvd_vth, Vb = 0V, T = -40°C

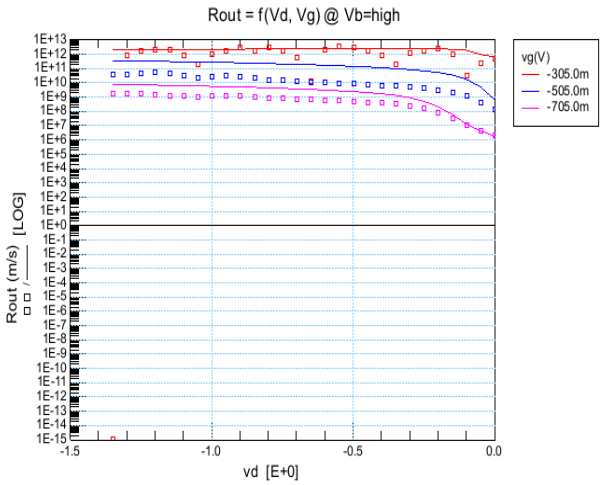


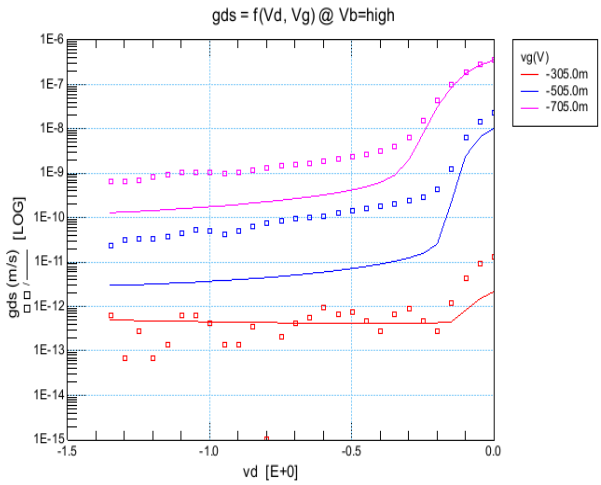




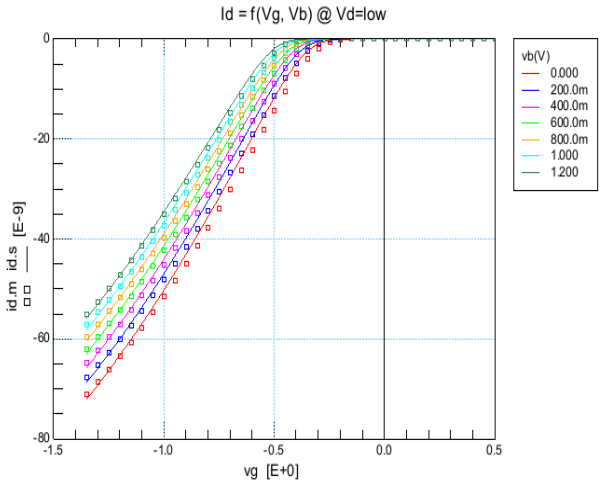
idvd_vth_vbmin, Vb = 1.2V, T = -40°C

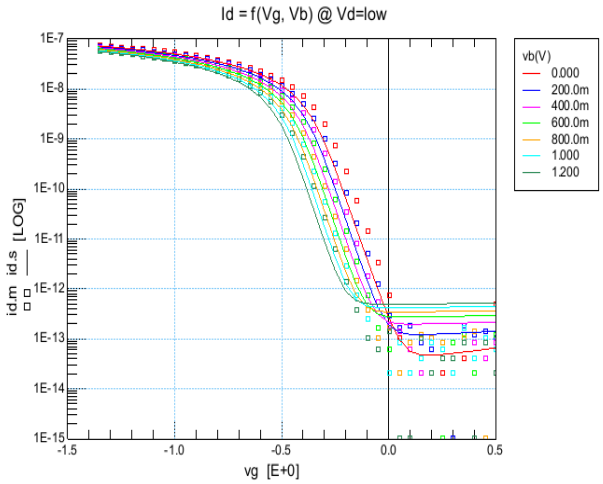


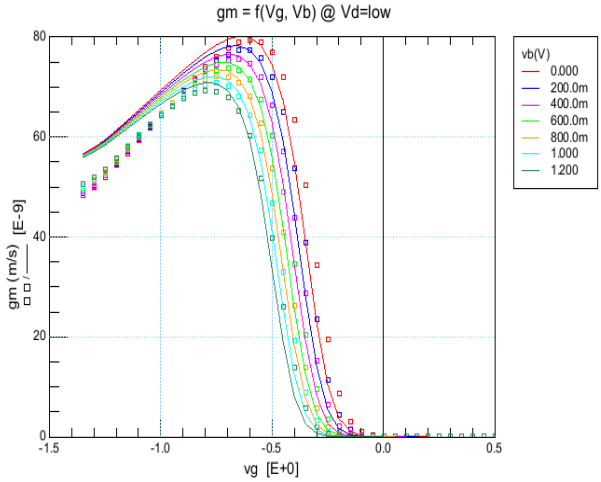


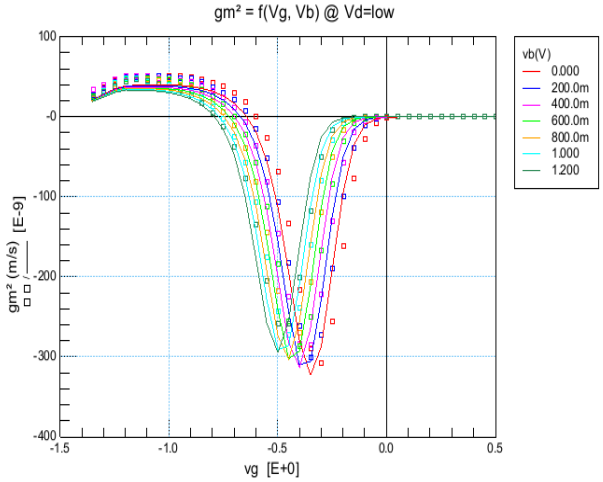


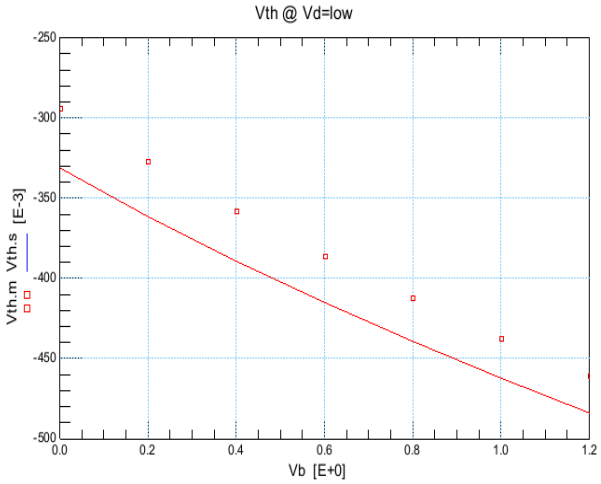
idvg, Vd = -0.05V, T = 70°C

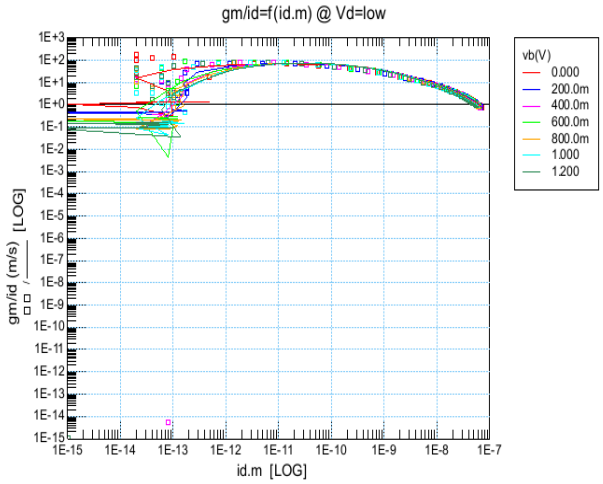


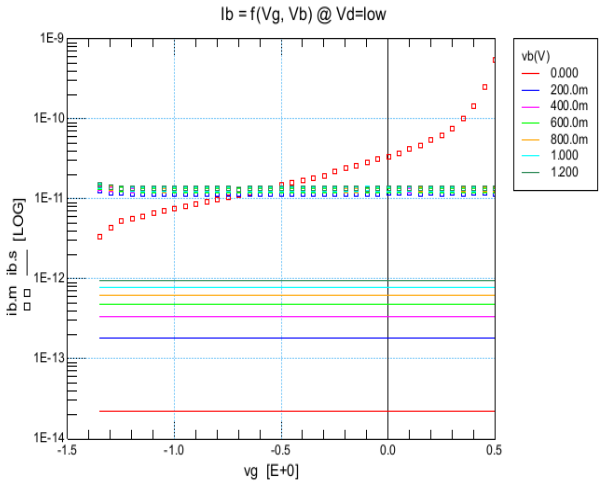


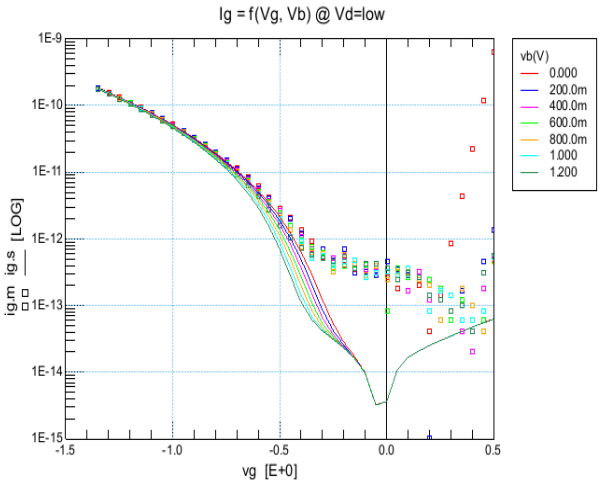




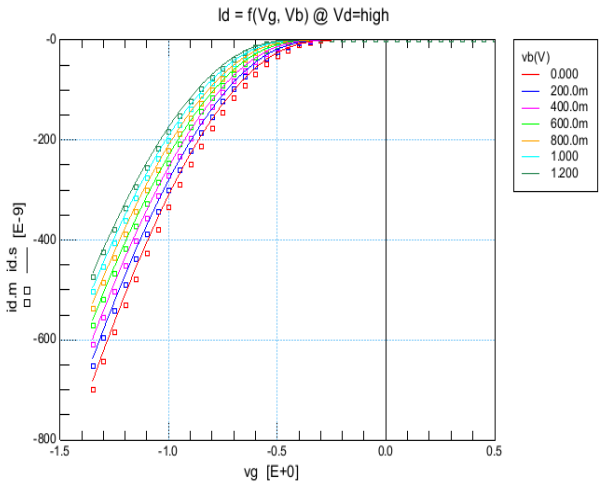


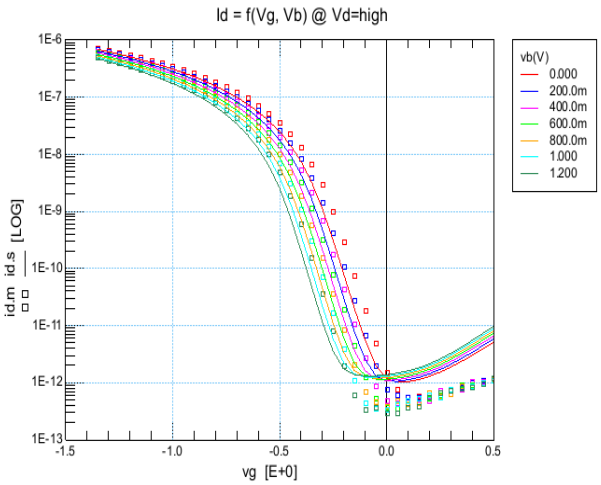


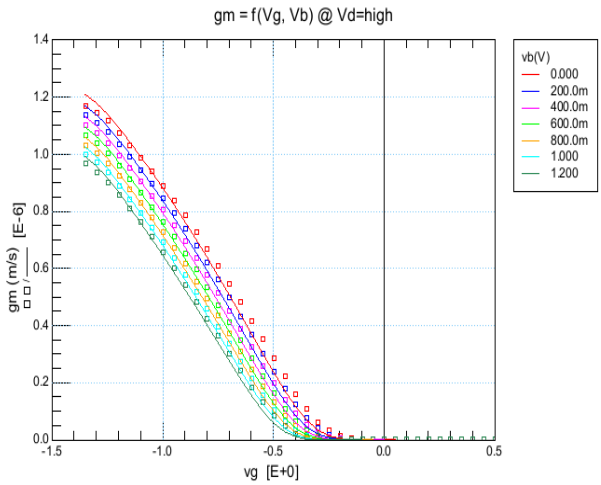


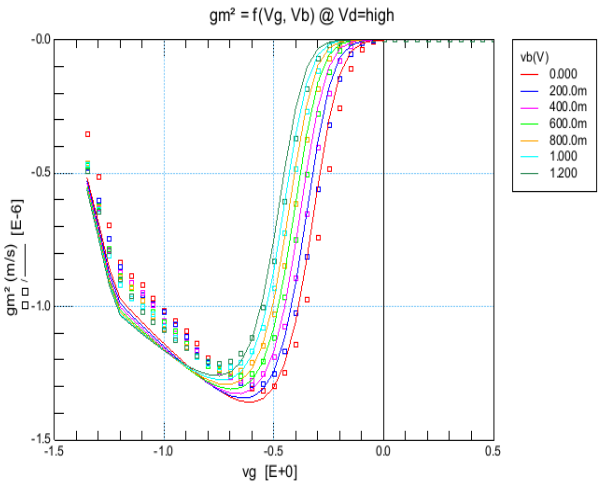


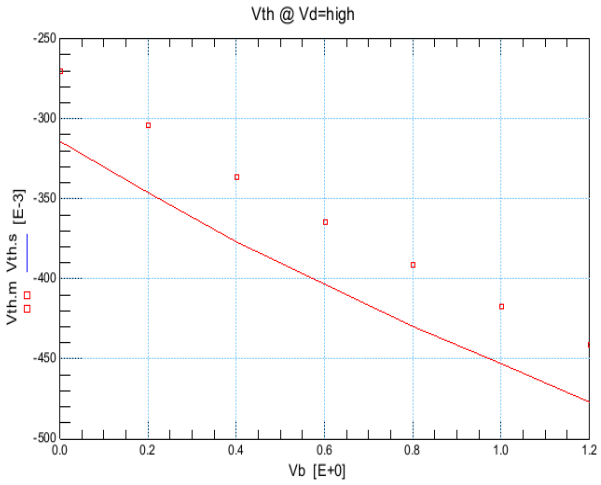
idvg, Vd = -1.2V, T = 70°C

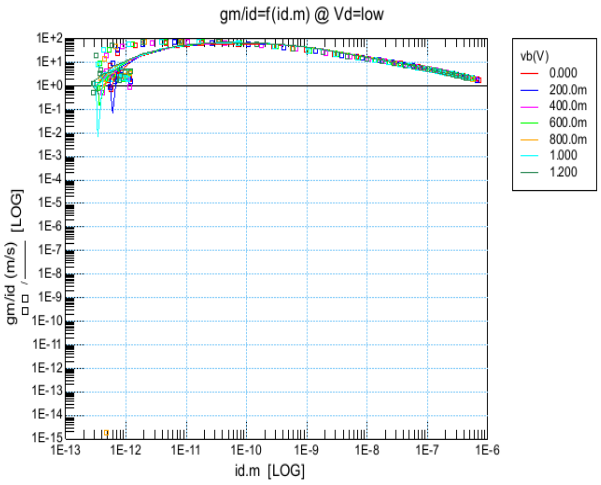


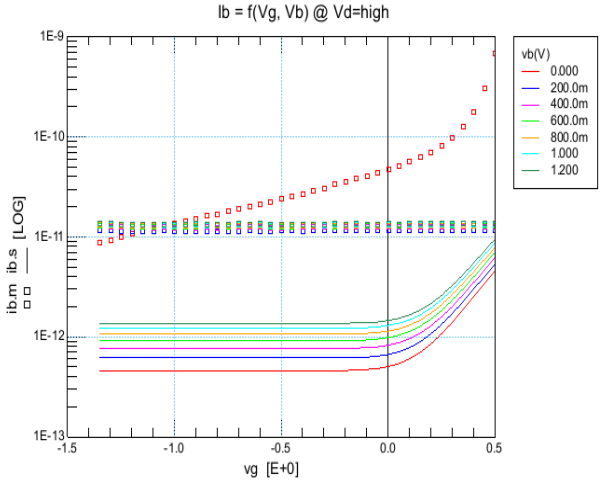


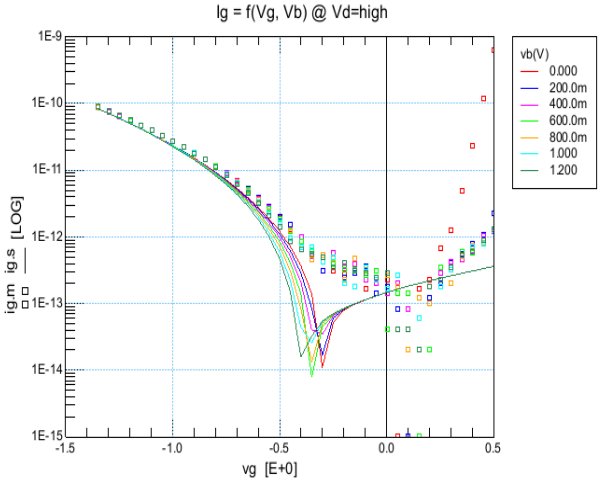




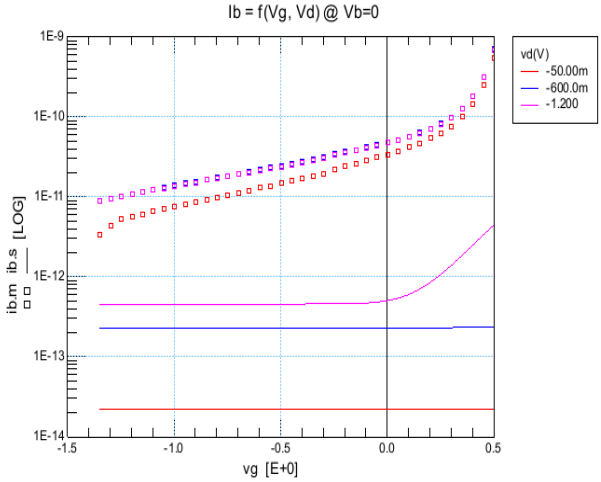


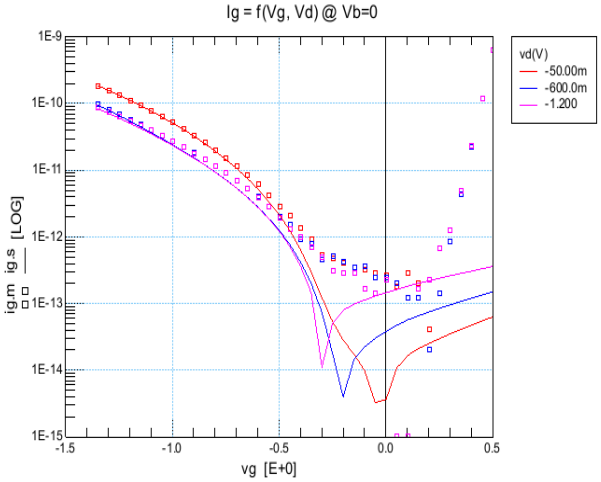




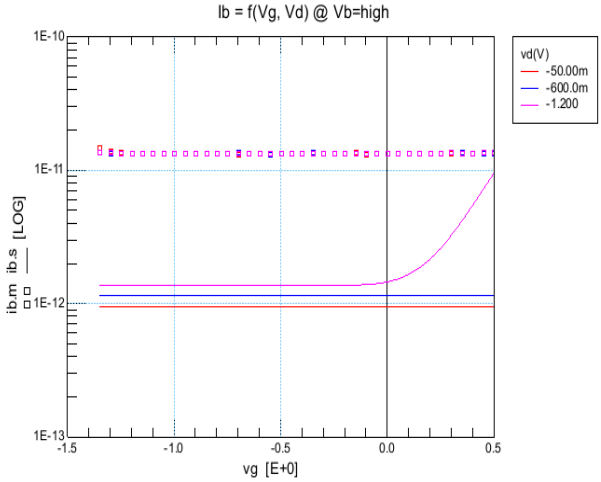


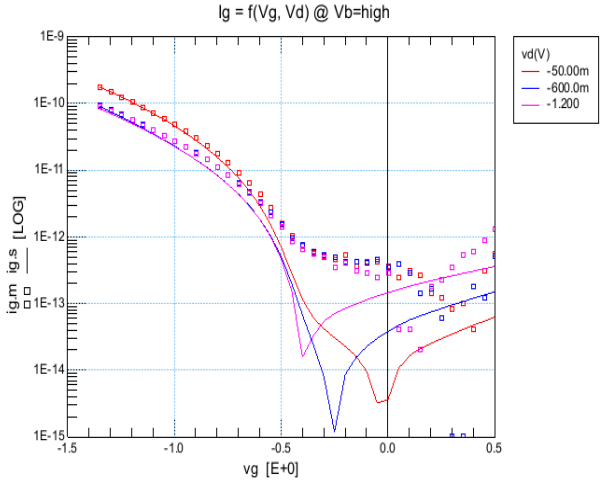
idvg, Ib,Ig, Vb = 0V, T = 70°C



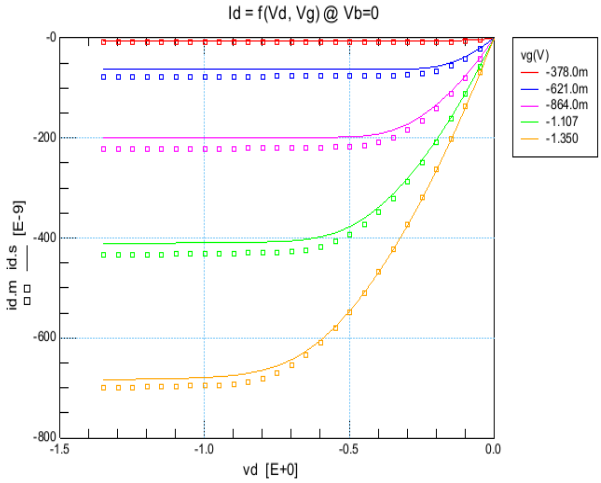


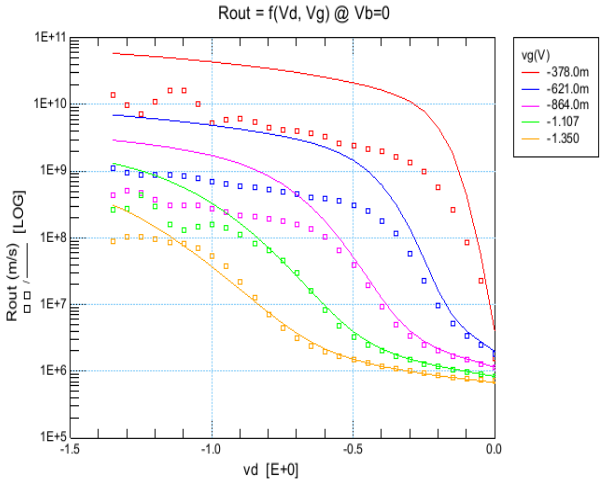
idvg, Ib,Ig, Vb = 1.2V, T = 70°C

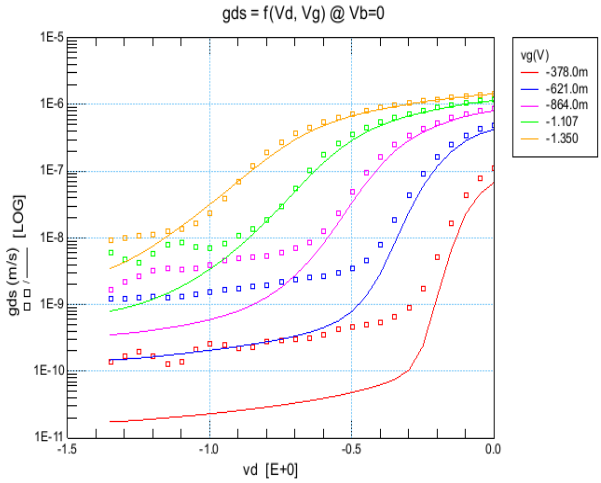




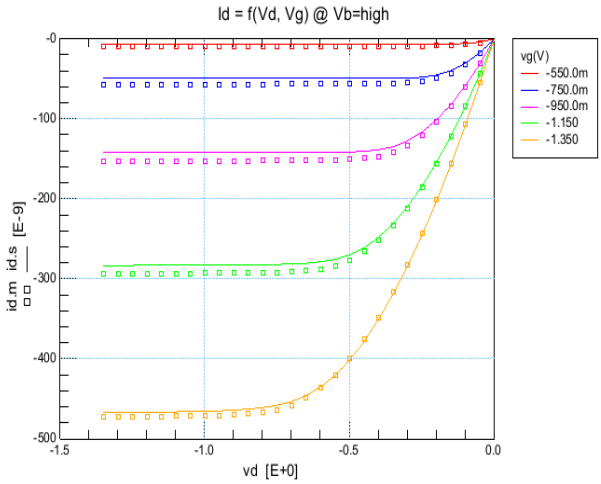
idvd, Vb = 0V, T = 70°C

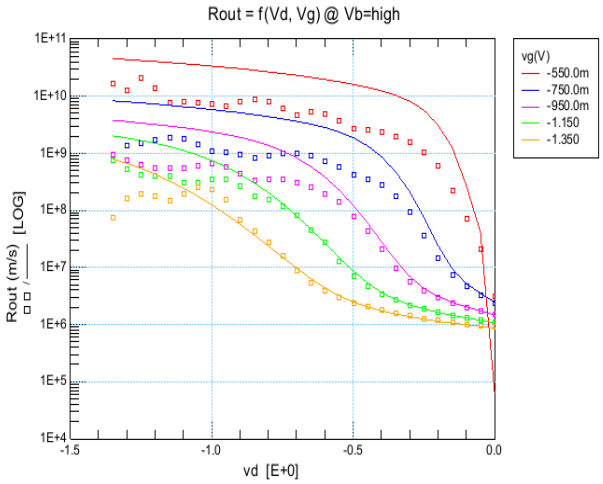


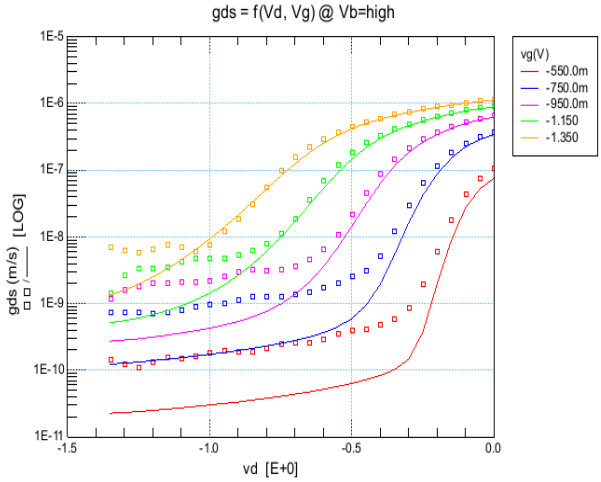




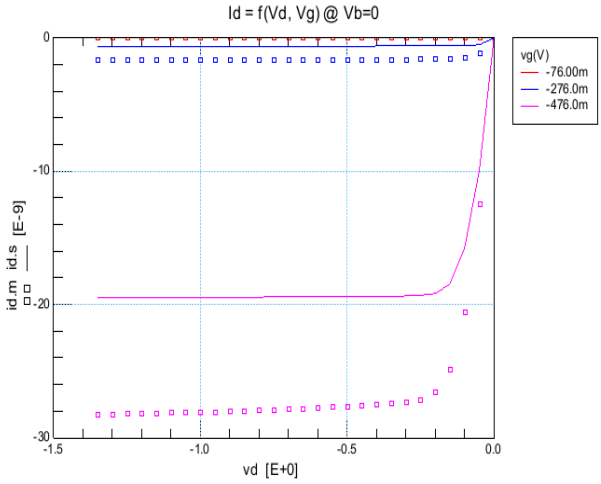
idvd_vbmin, Vb = 1.2V, T = 70°C

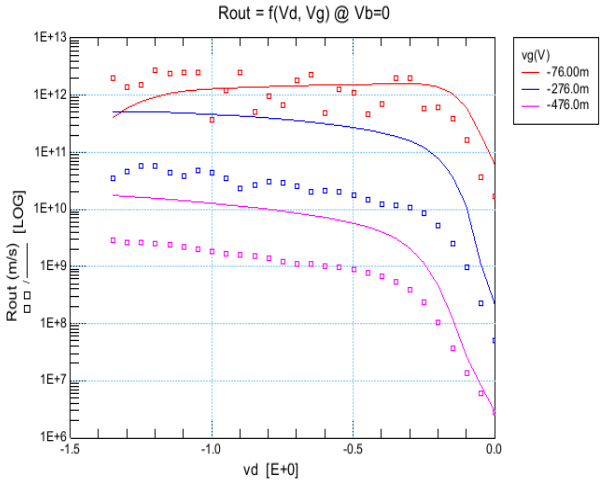


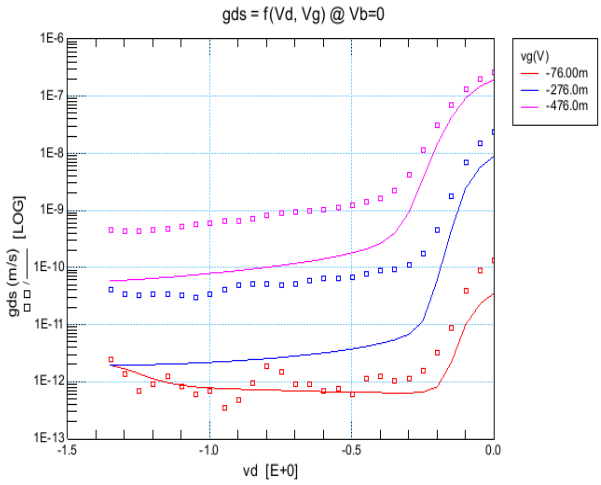




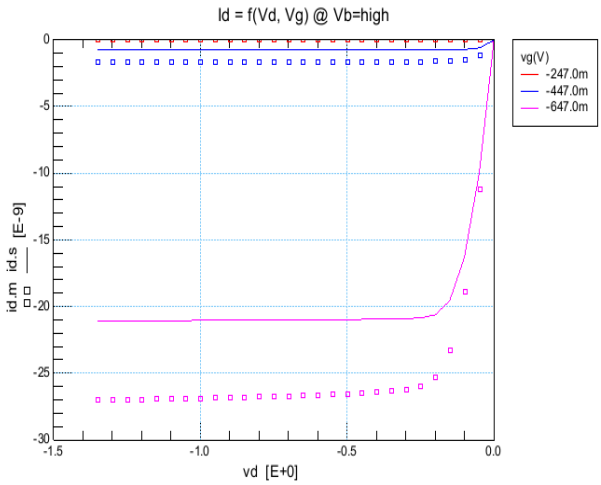
idvd_vth, Vb = 0V, T = 70°C

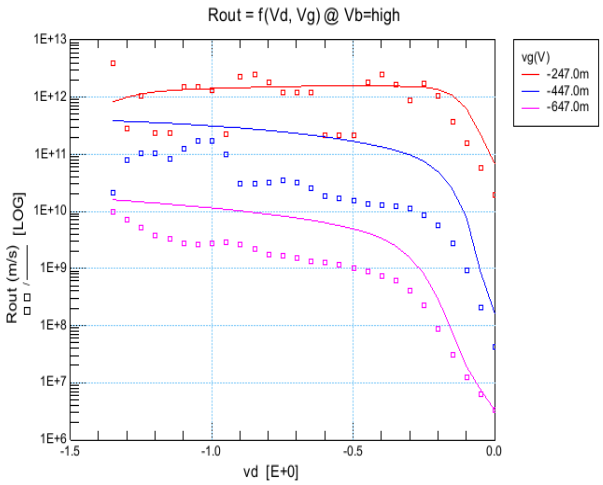


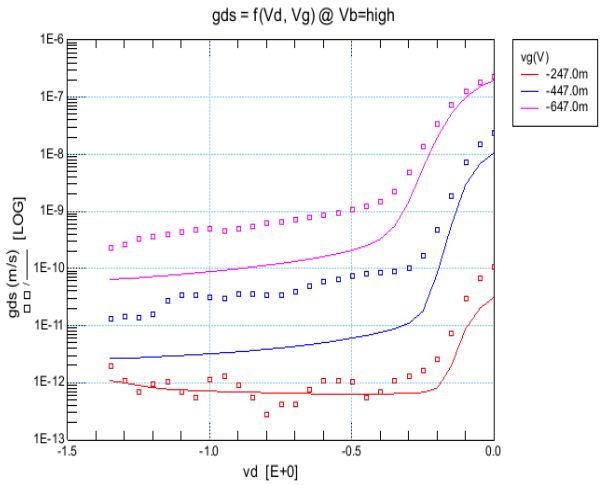




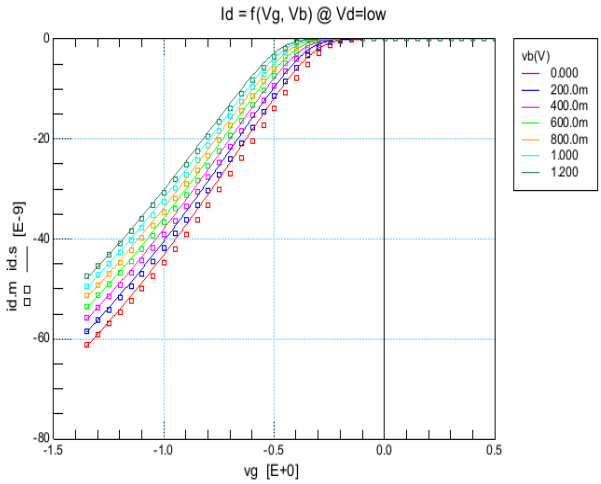
idvd_vth_vbmin, Vb = 1.2V, T = 70°C

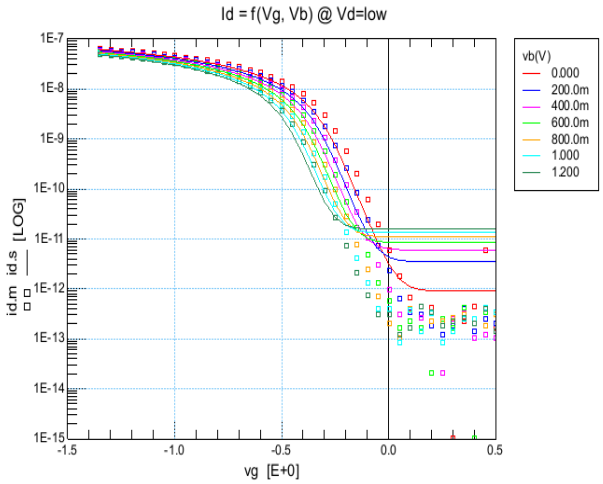


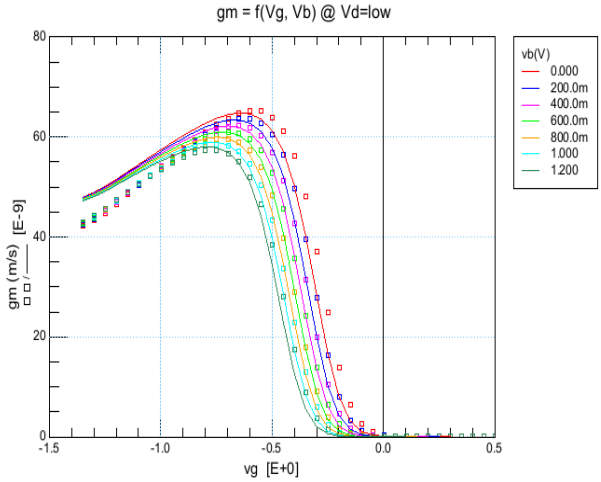


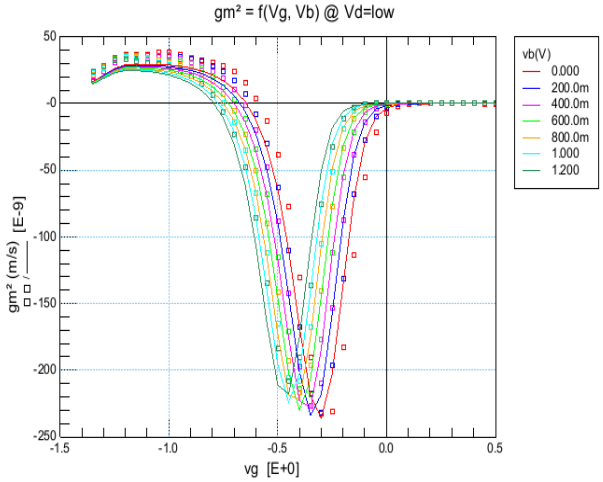


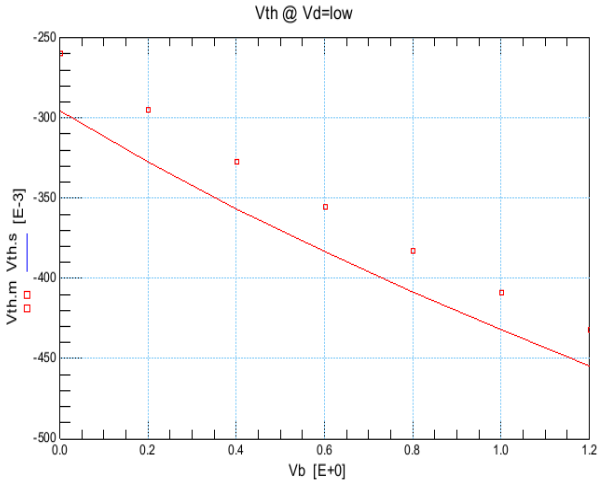
idvg, Vd = -0.05V, T = 125°C

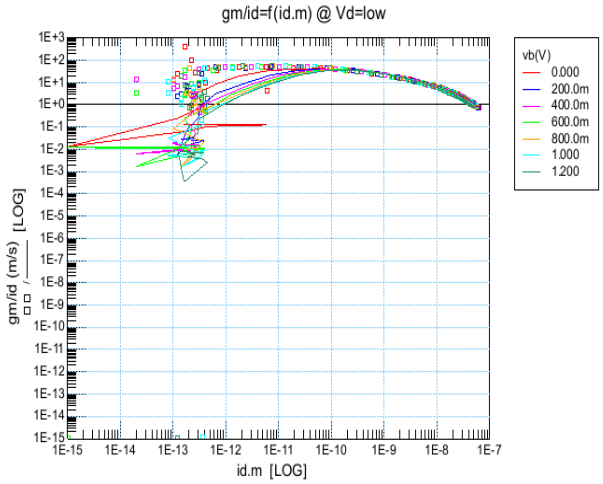


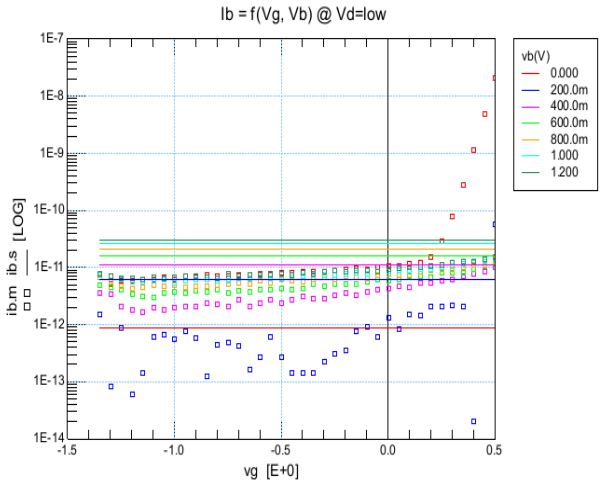


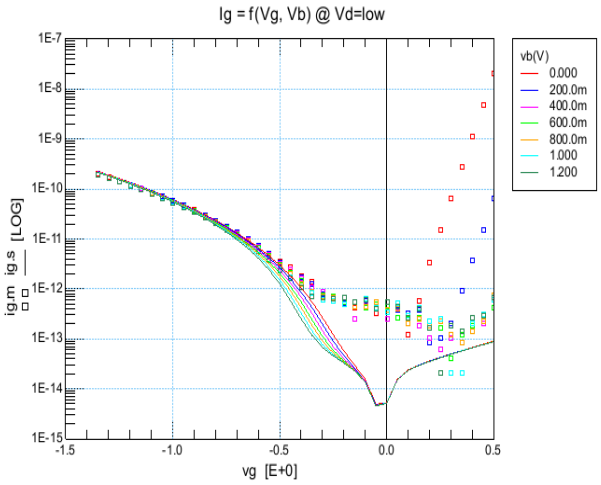




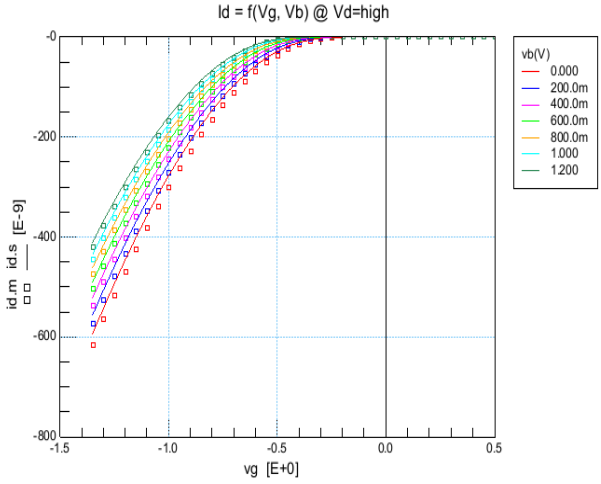


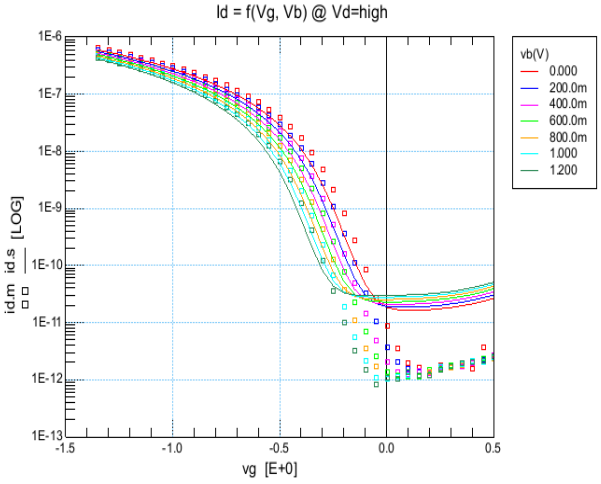


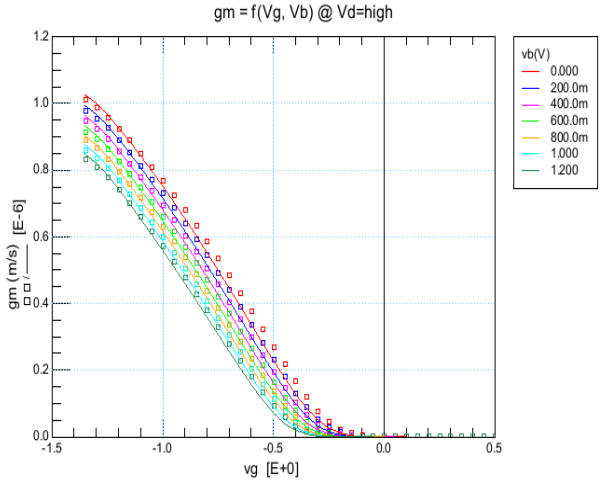


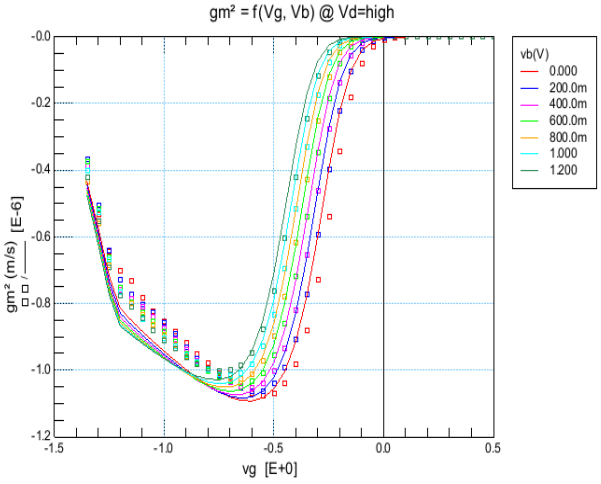


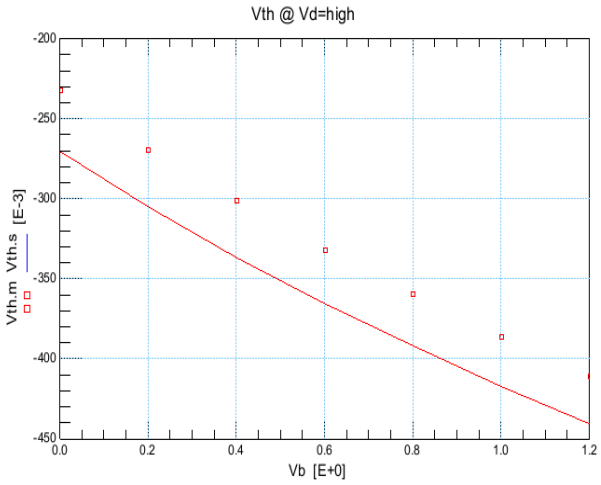
idvg, Vd = -1.2V, T = 125°C

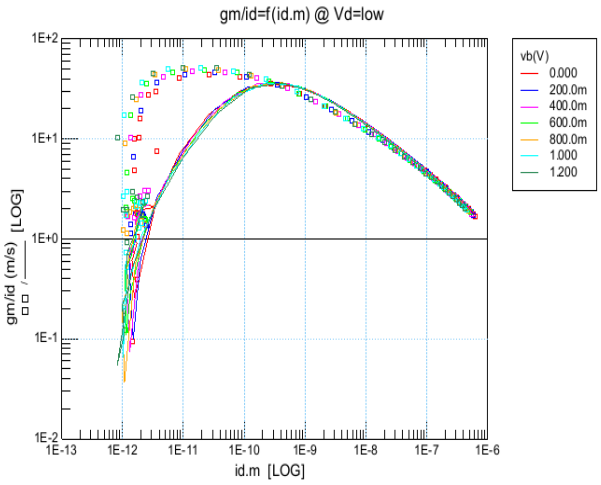


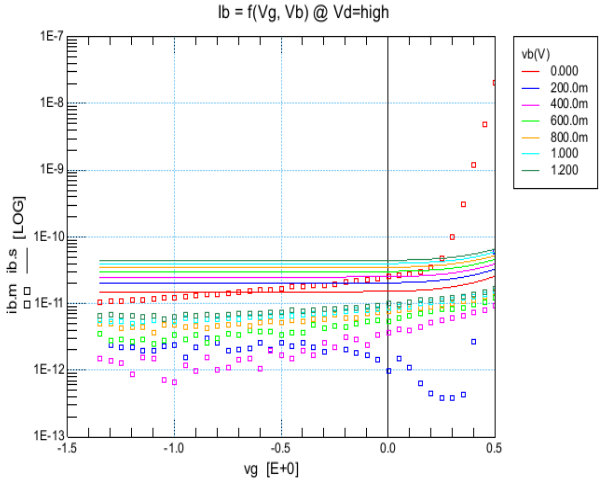


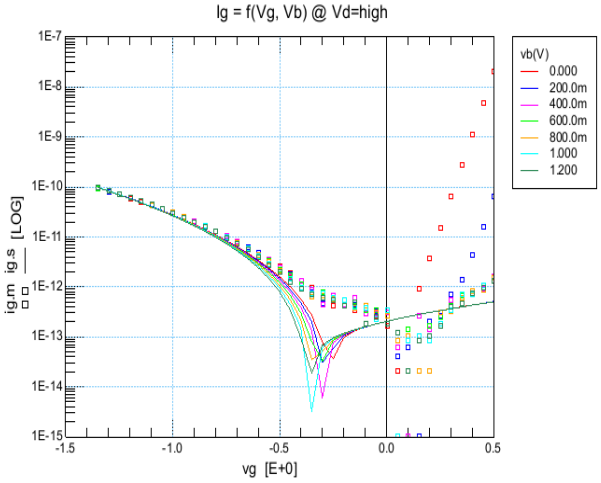




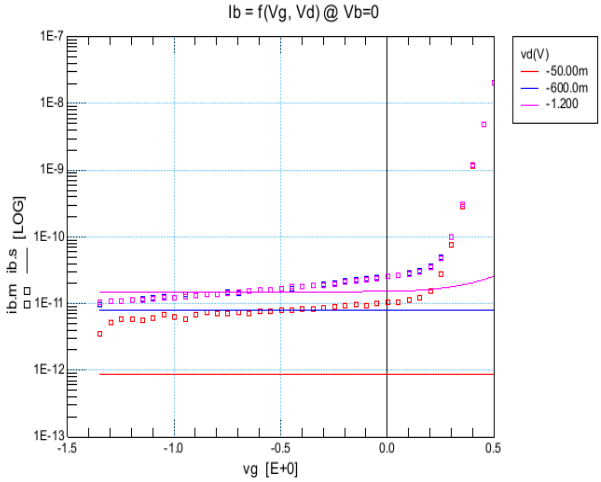


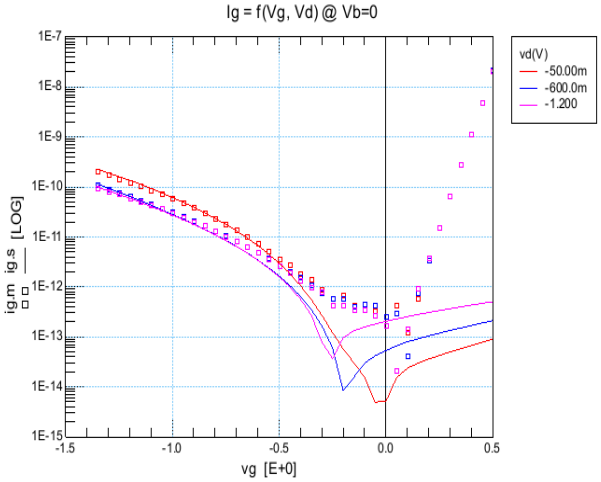




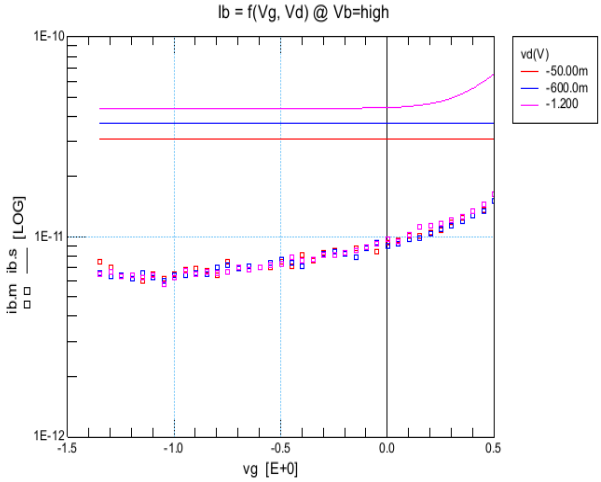


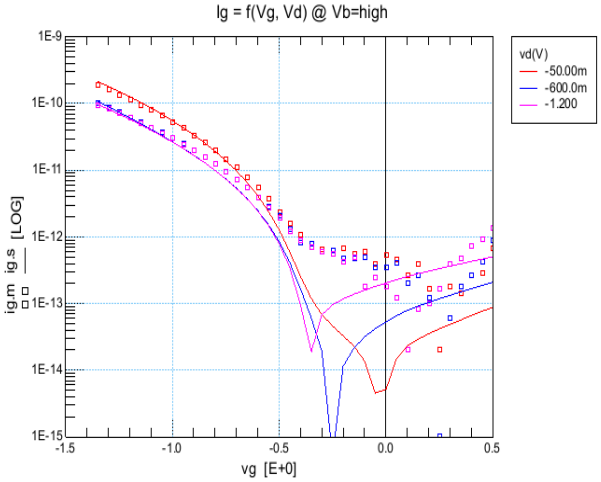
idvg, Ib,Ig, Vb = 0V, T = 125°C



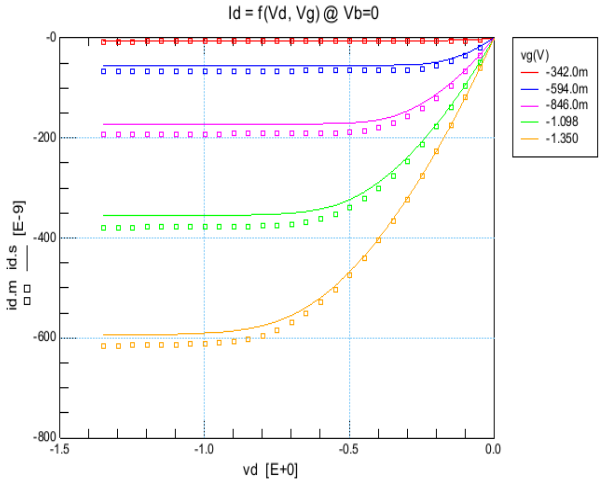


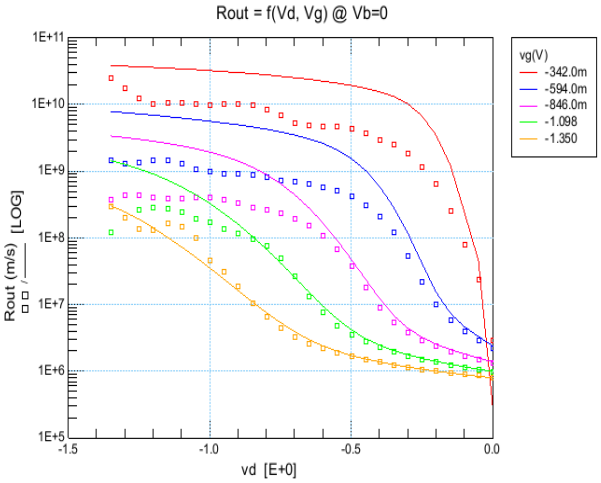
idvg, Ib,Ig, Vb = 1.2V, T = 125°C

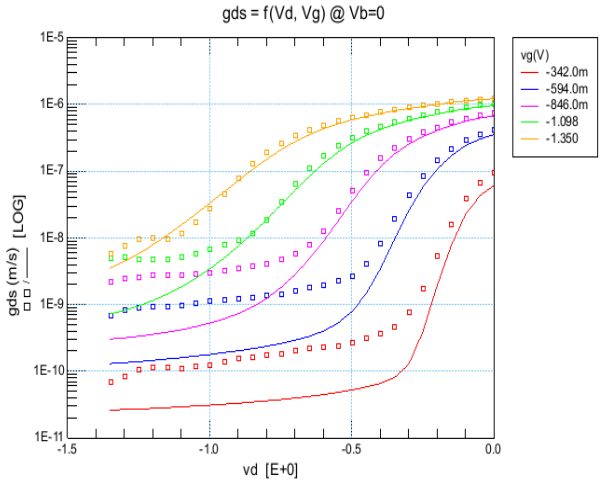




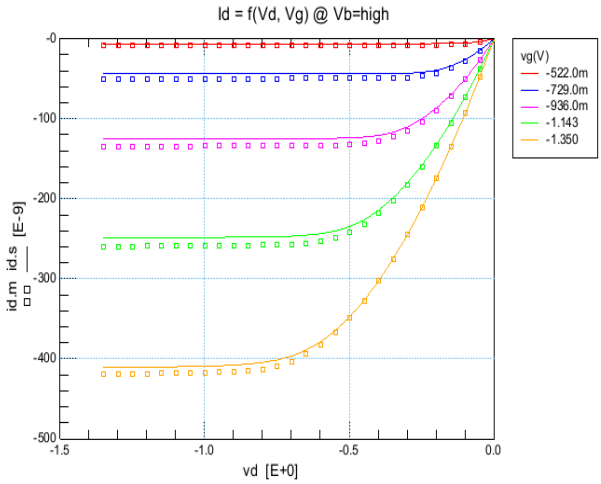
idvd, Vb = 0V, T = 125°C

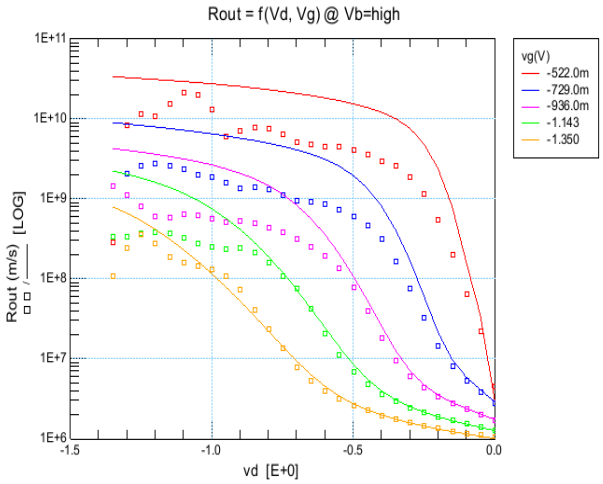


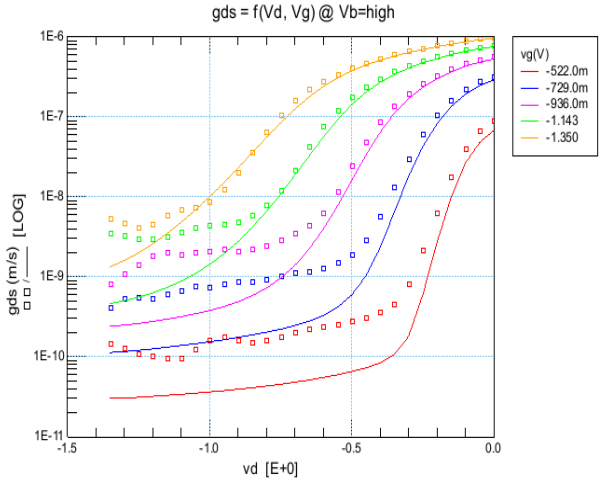




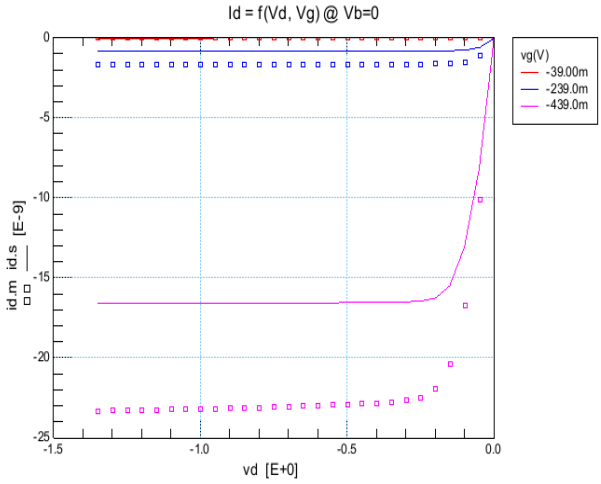
idvd_vbmin, Vb = 1.2V, T = 125°C

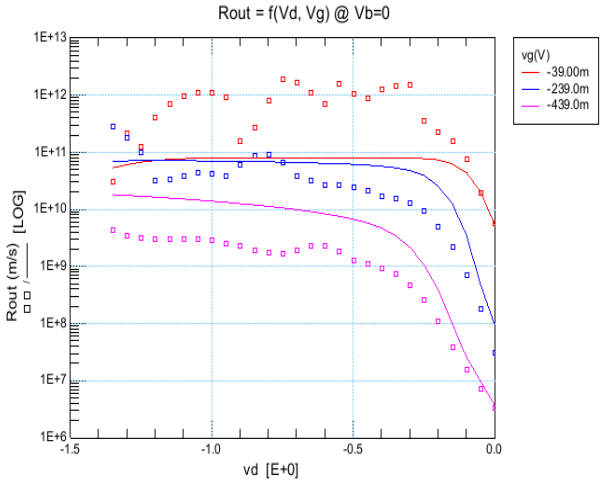


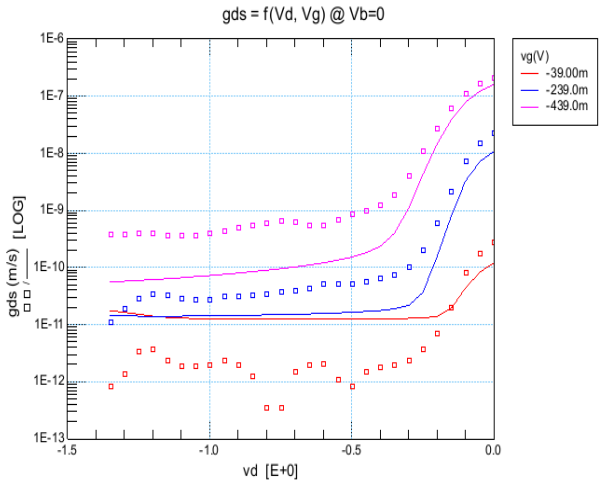




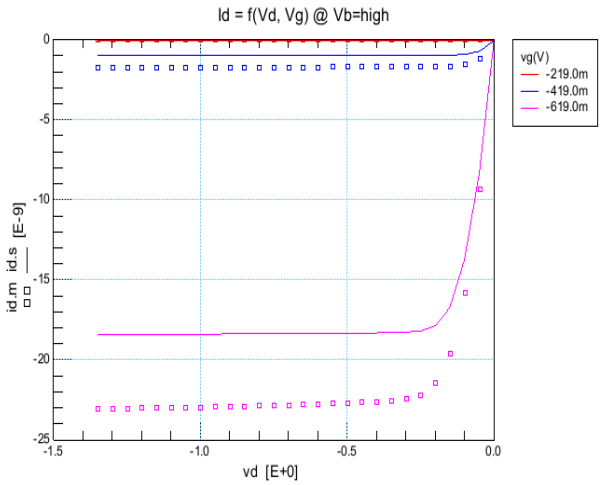
idvd_vth, Vb = 0V, T = 125°C

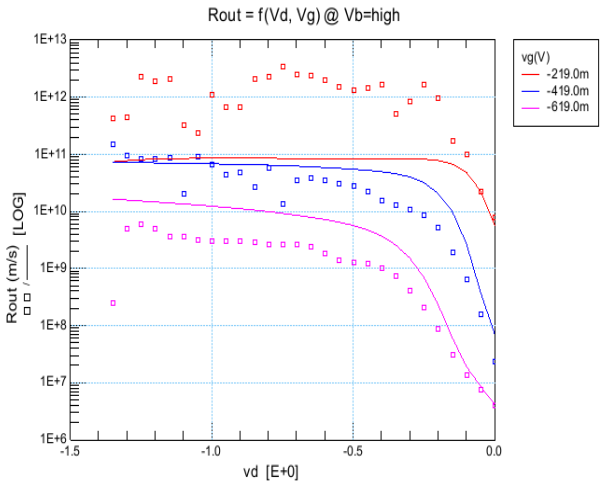


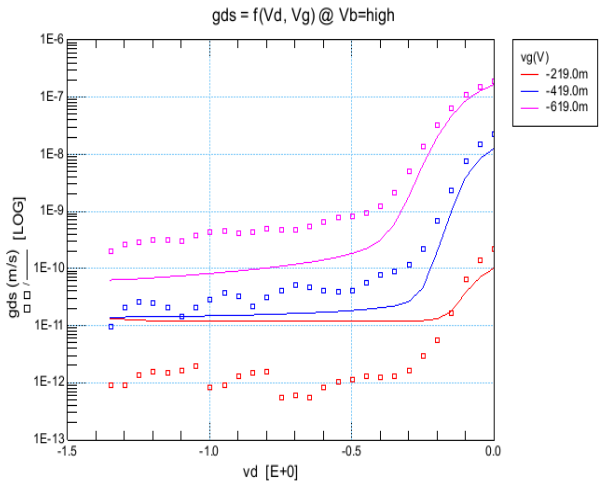




idvd_vth_vbmin, Vb = 1.2V, T = 125°C

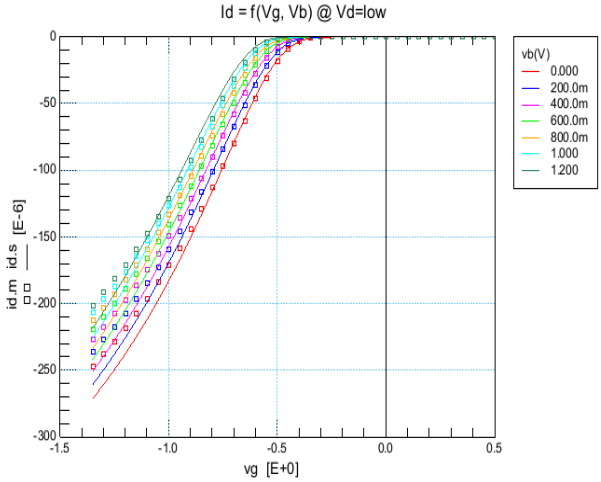


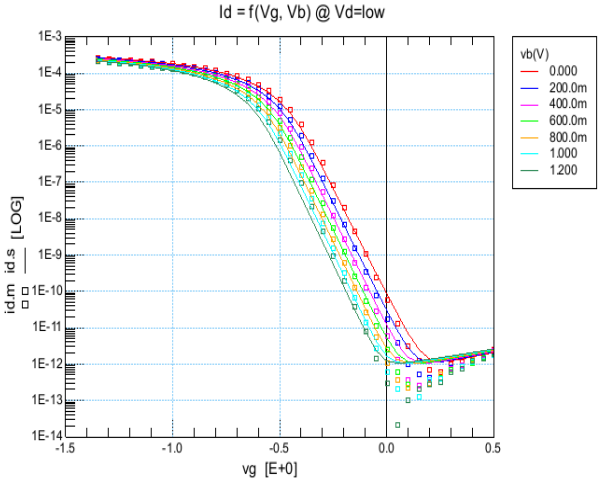


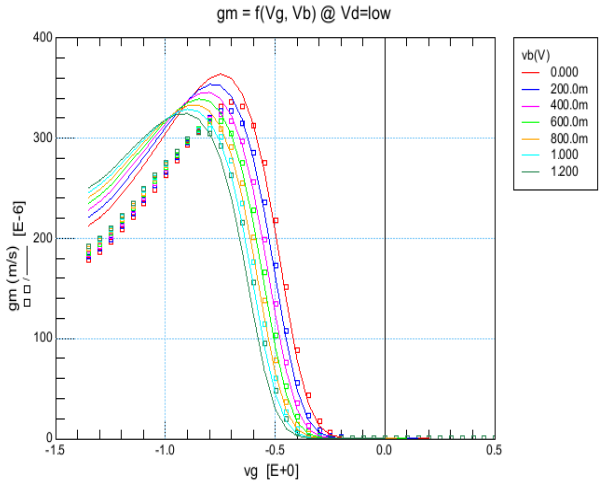


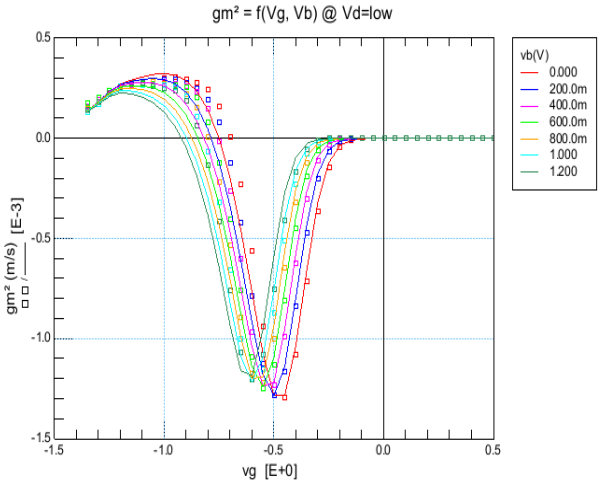
4.4 W10u0_L0u13_S548_2

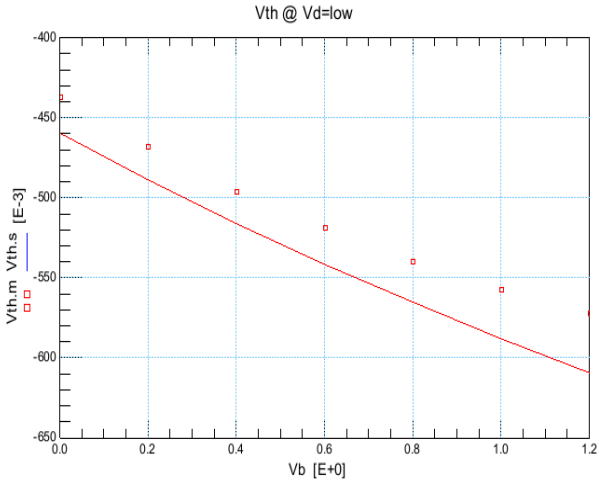
idvg, Vd = -0.05V, T = 27°C

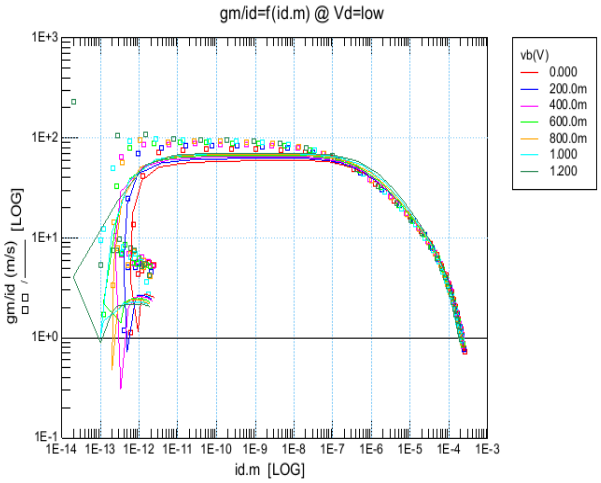


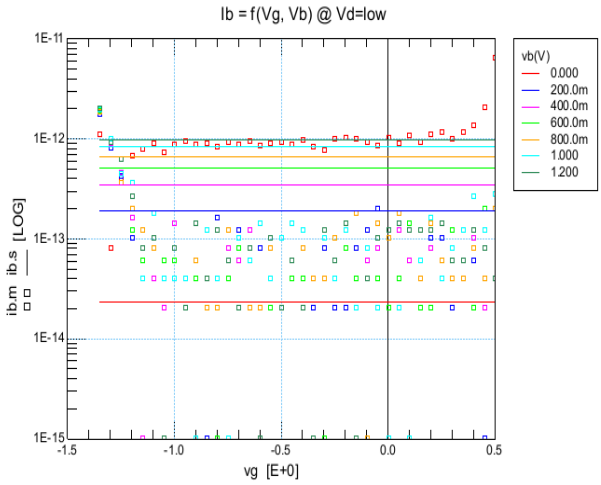


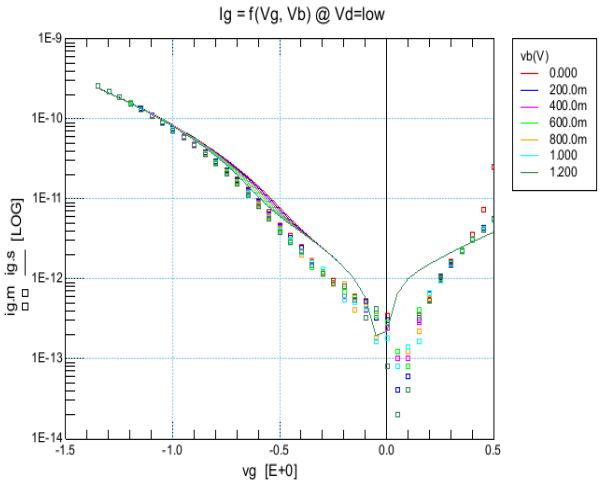




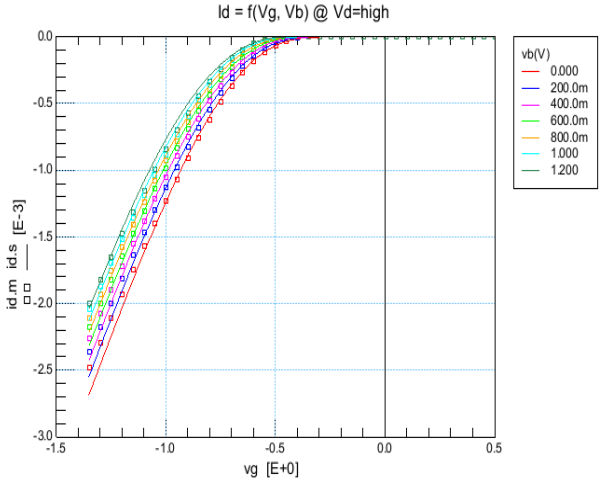


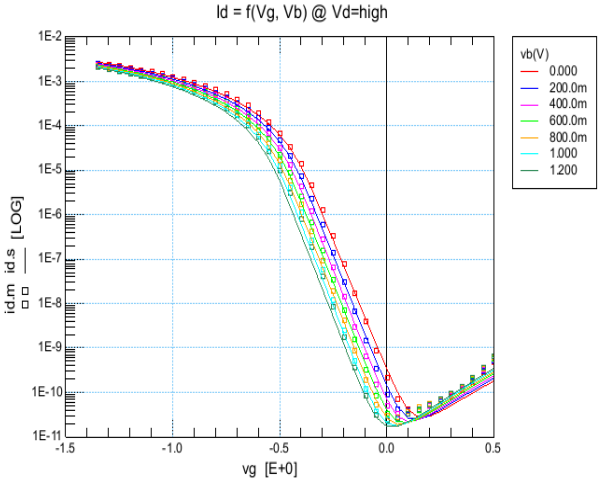


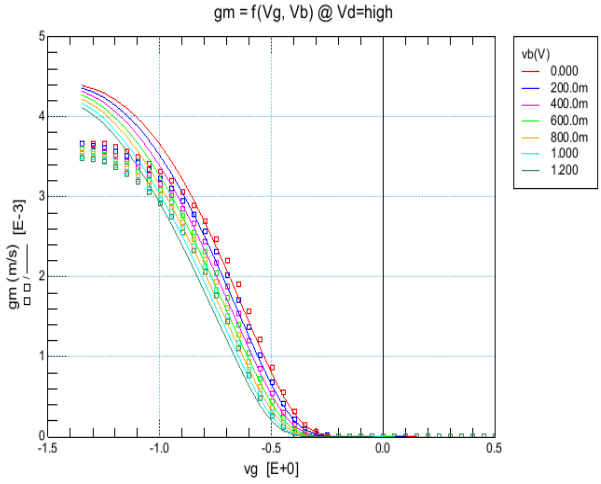


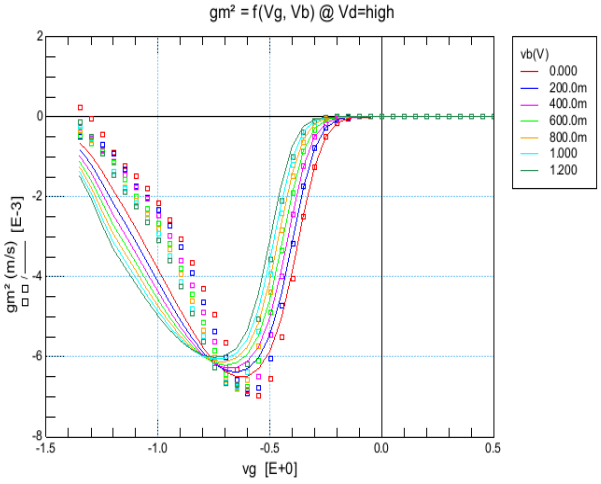


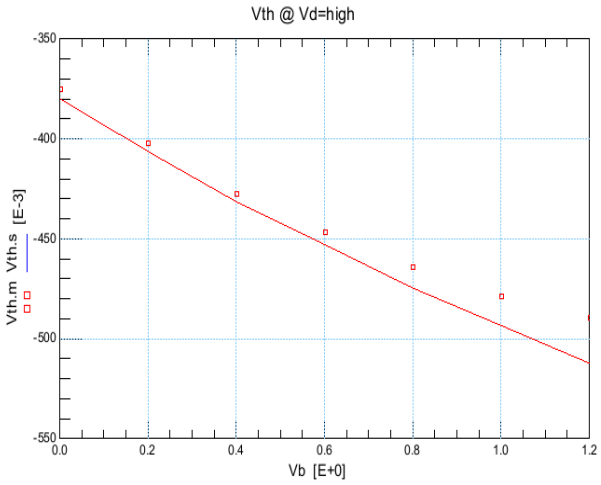
idvg, Vd = -1.2V, T = 27°C

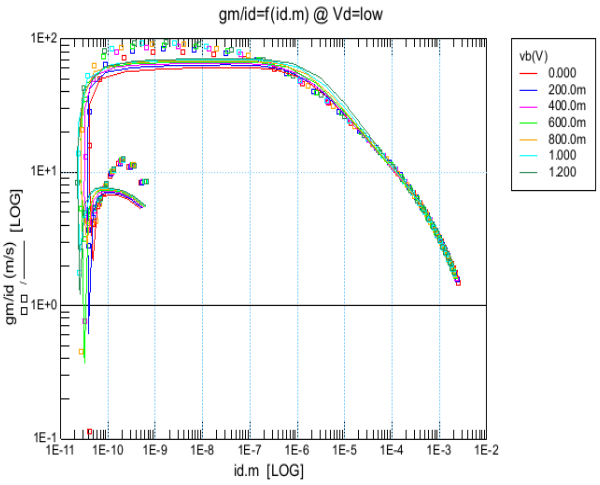


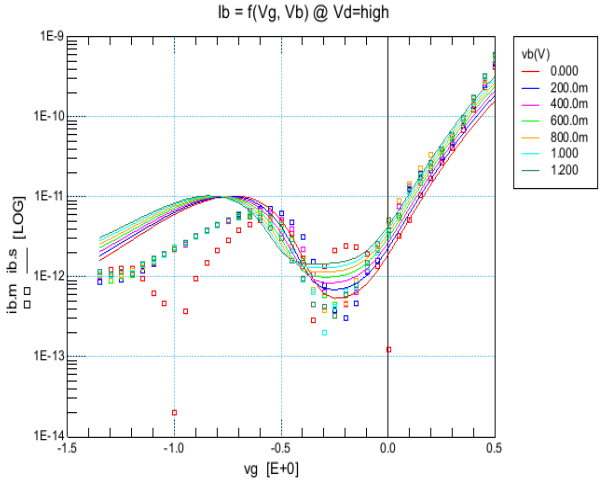


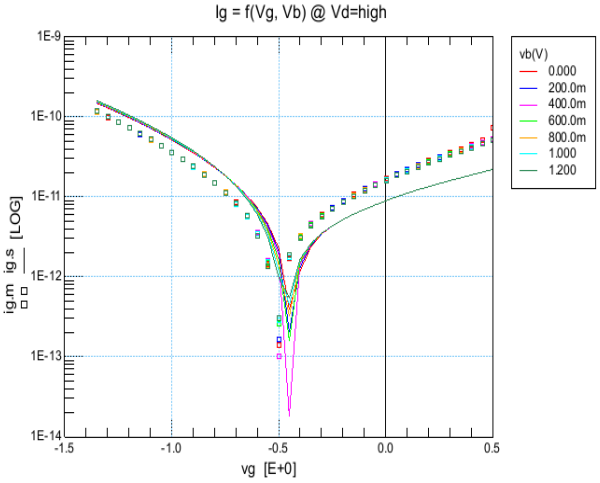




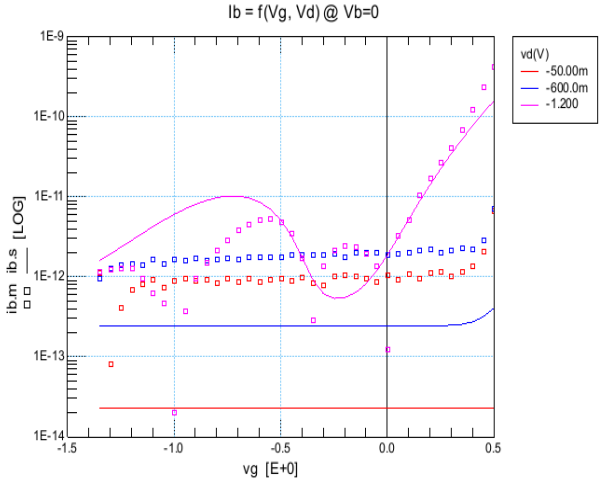


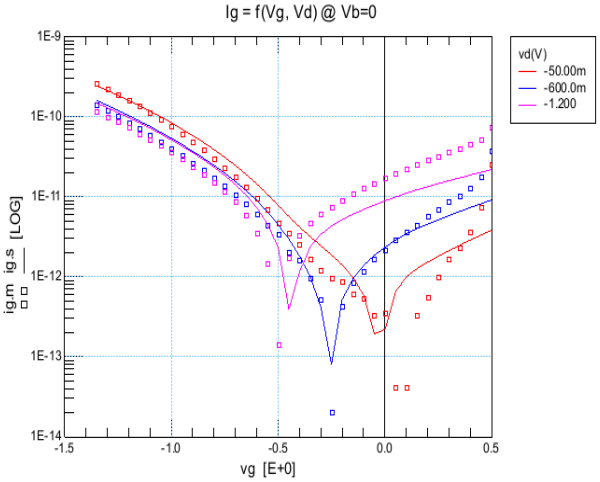




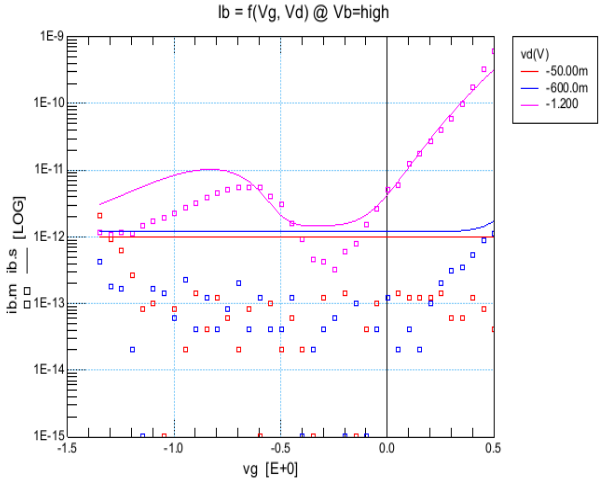


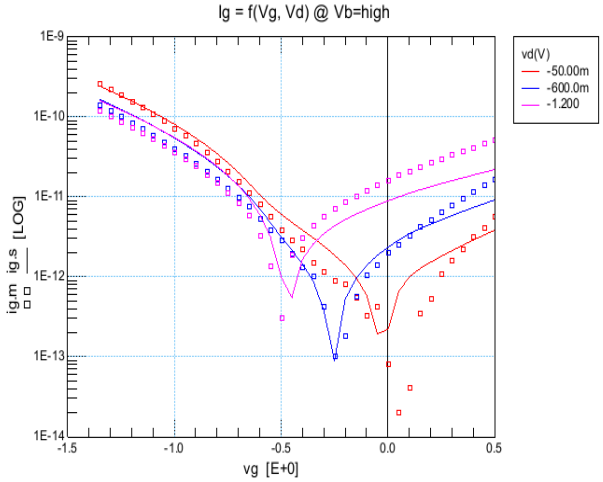
idvg, Ib,Ig, Vb = 0V, T = 27°C



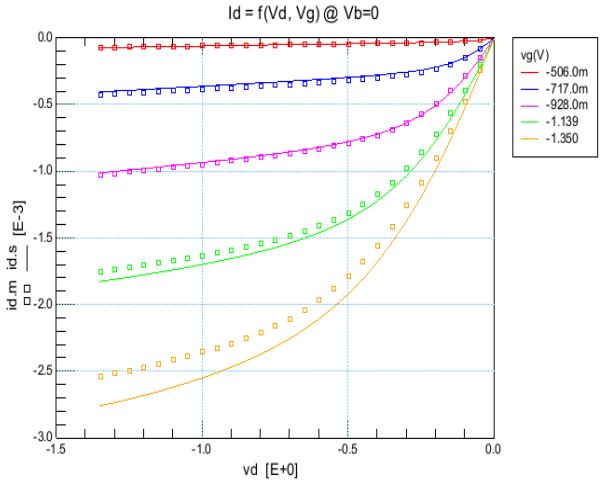


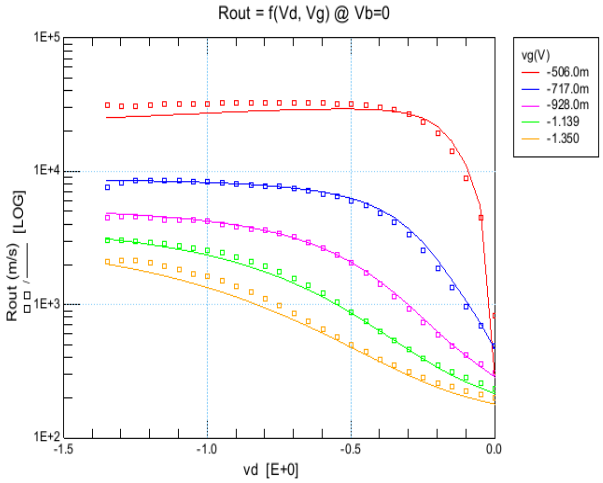
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

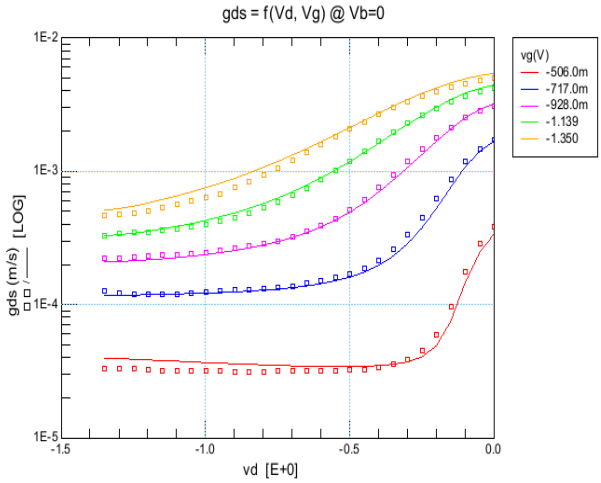




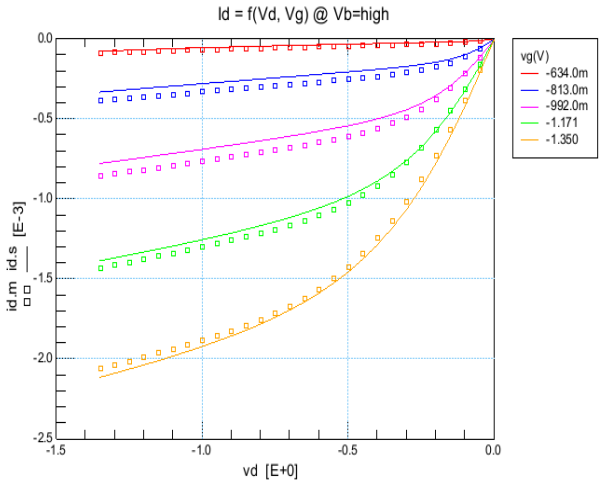
idvd, Vb = 0V, T = 27°C

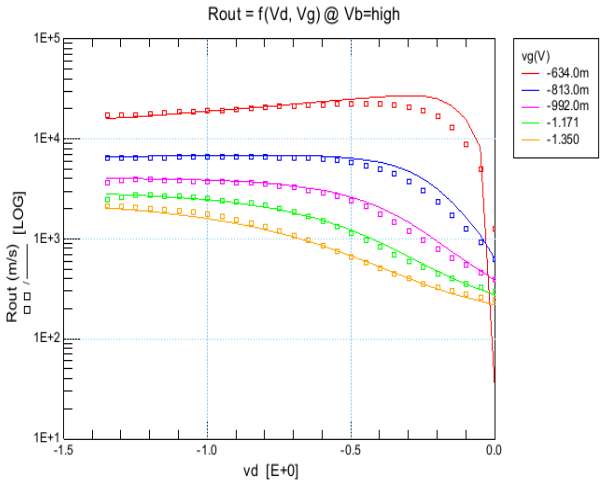


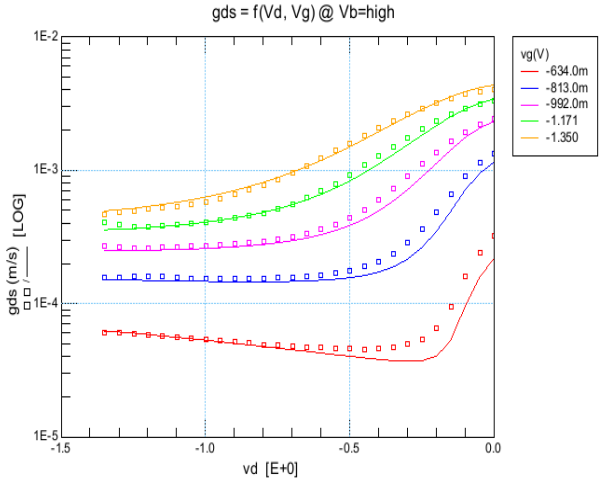




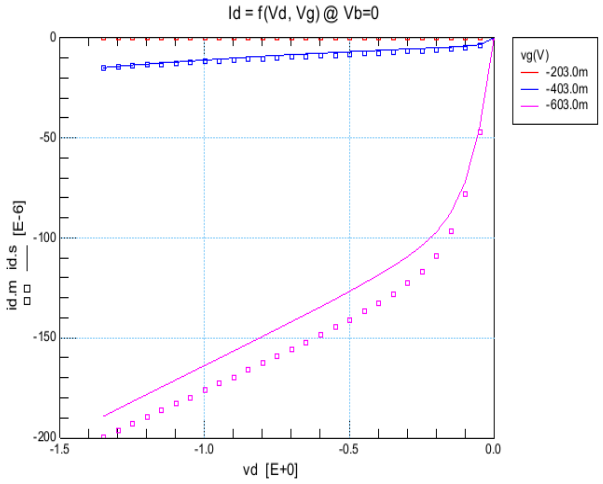
idvd_vbmin, Vb = 1.2V, T = 27°C

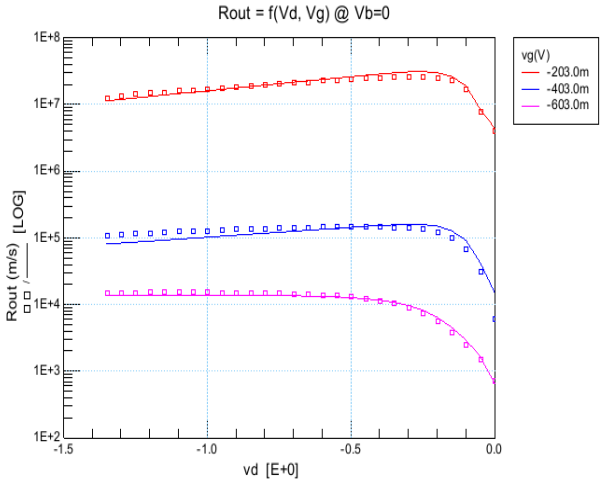


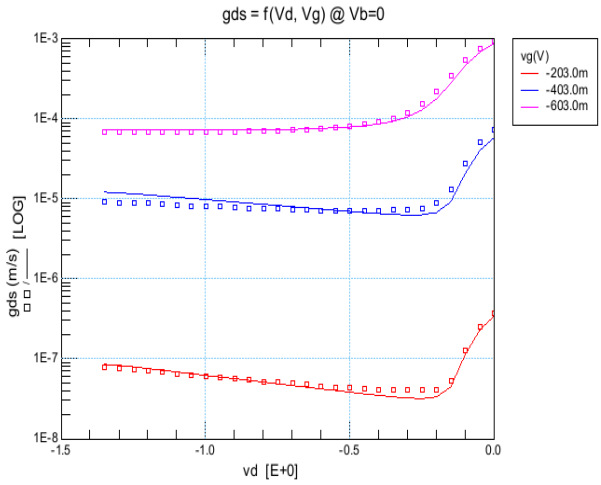




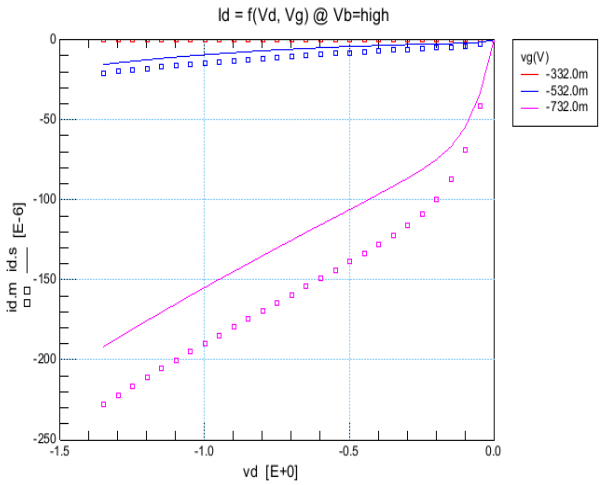
idvd_vth, Vb = 0V, T = 27°C

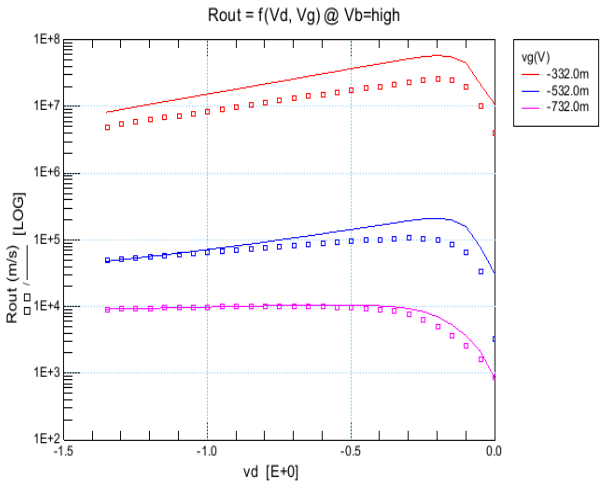


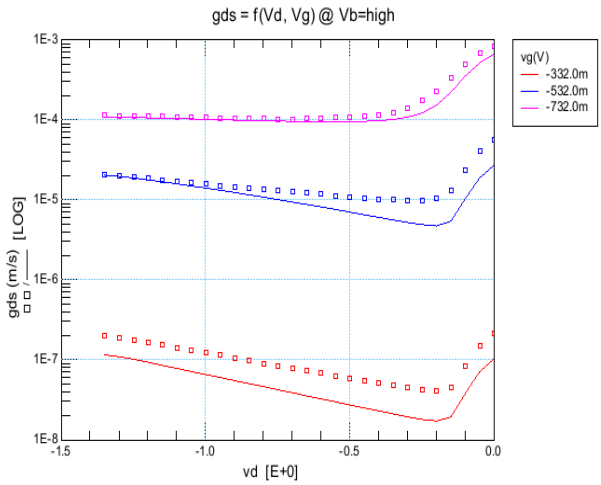




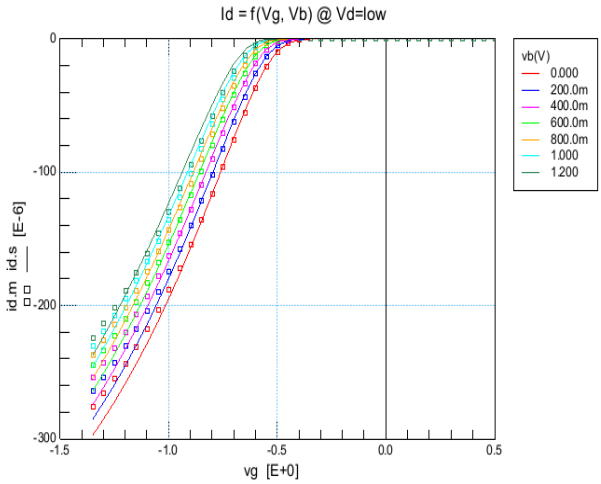
idvd_vth_vbmin, Vb = 1.2V, T = 27°C

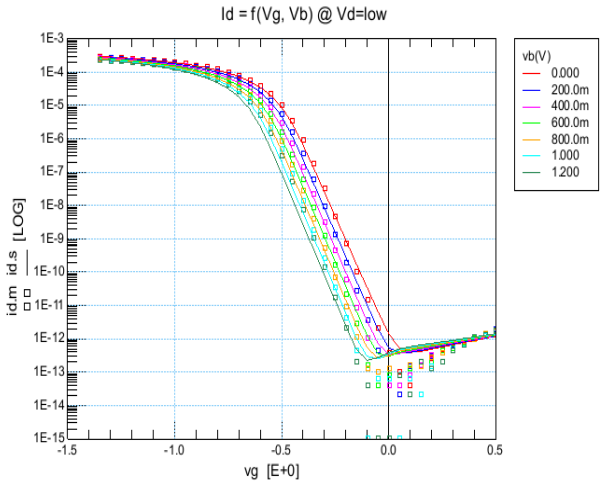


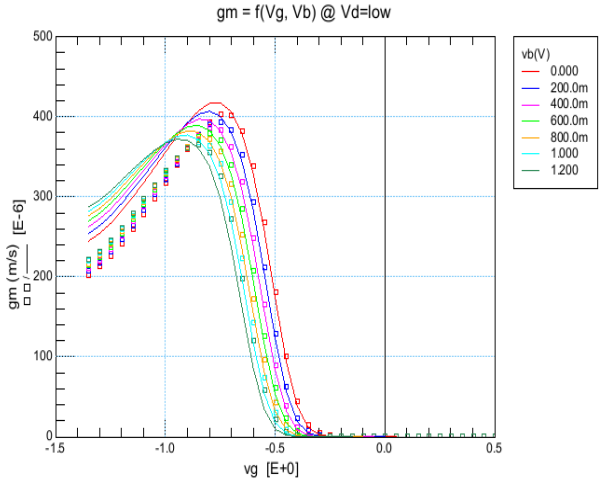


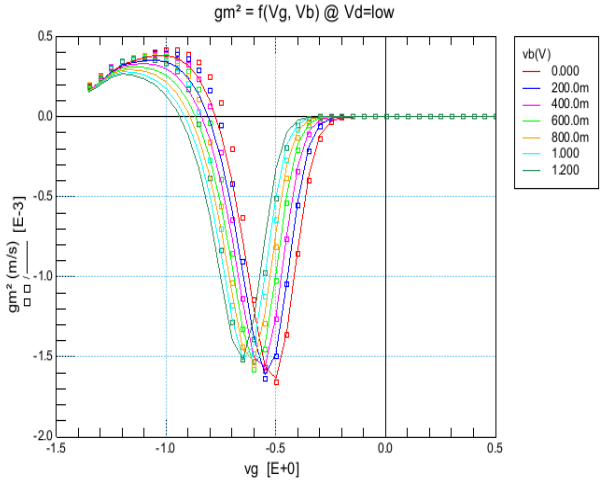


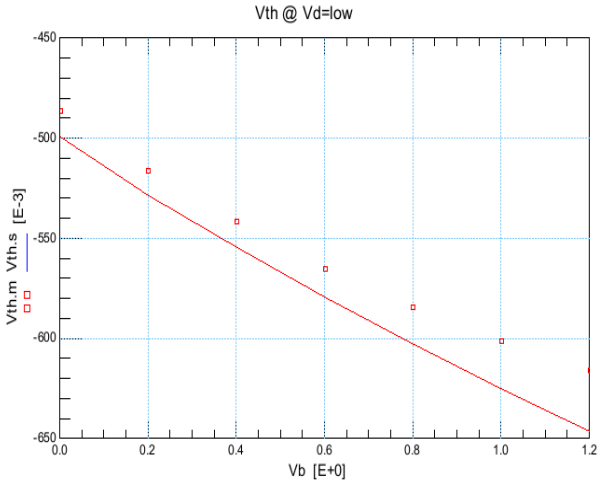
idvg, Vd = -0.05V, T = -40°C

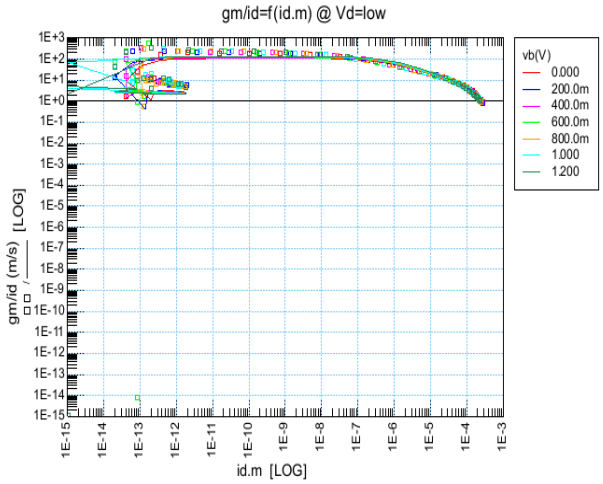


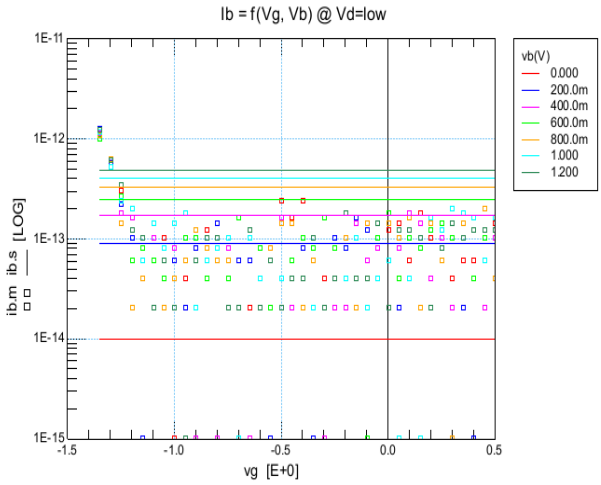


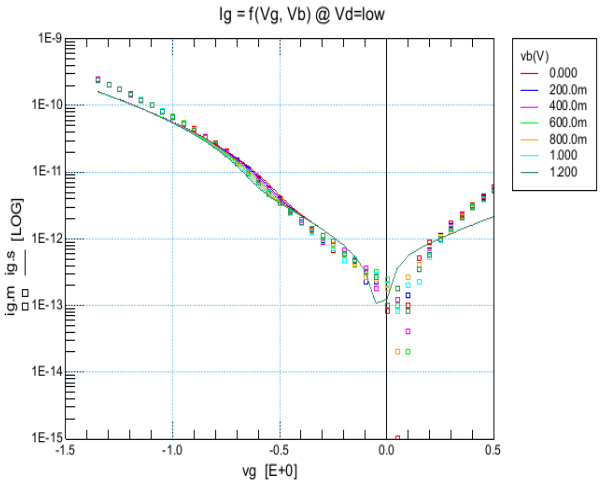




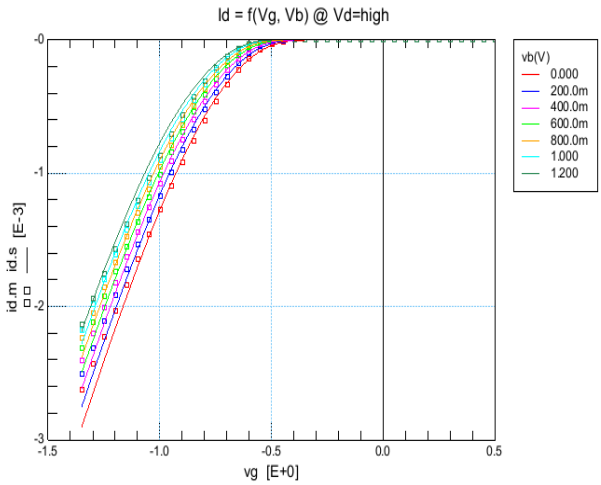


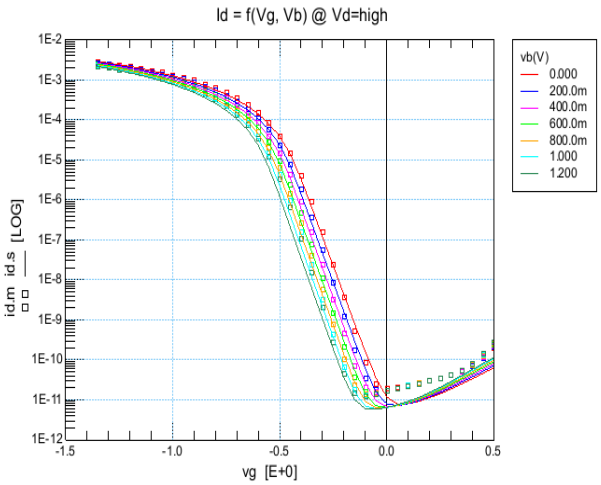


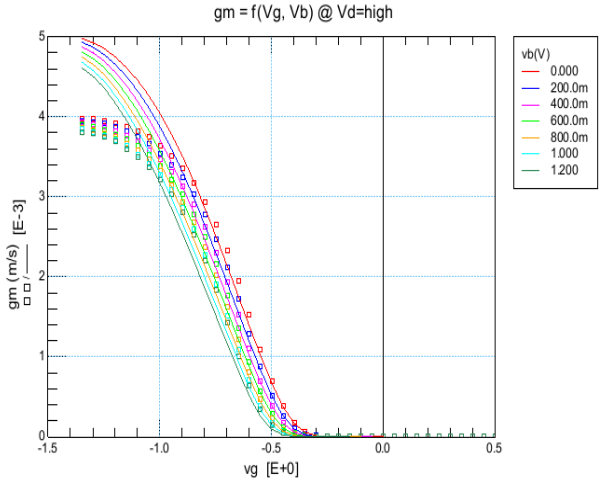


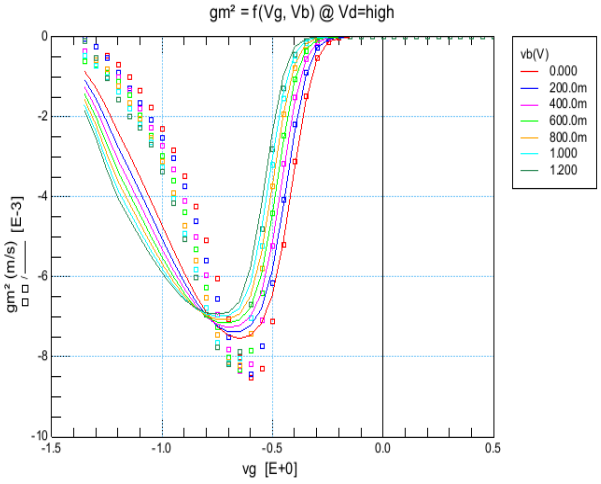


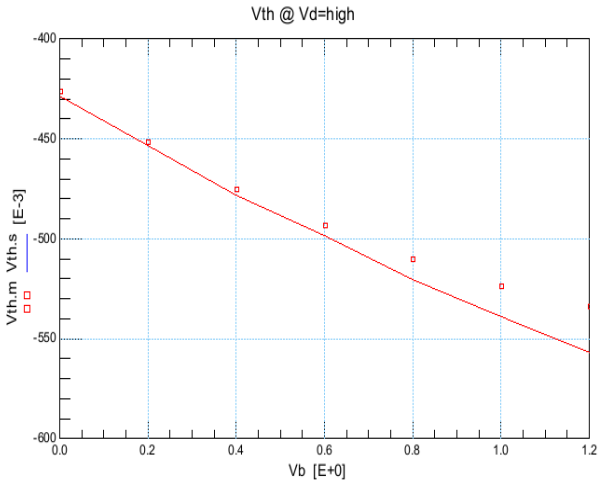
idvg, Vd = -1.2V, T = -40°C

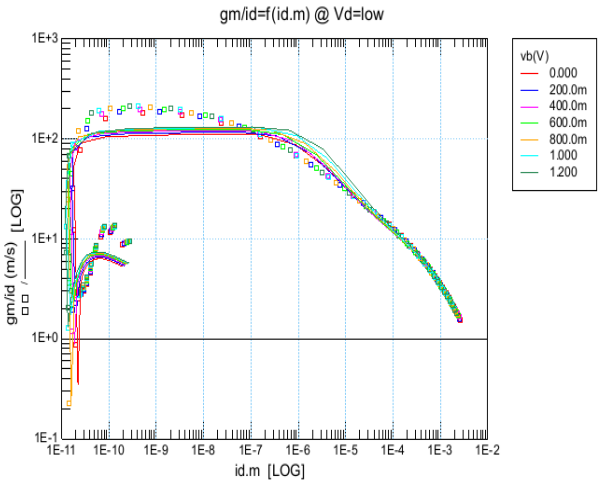


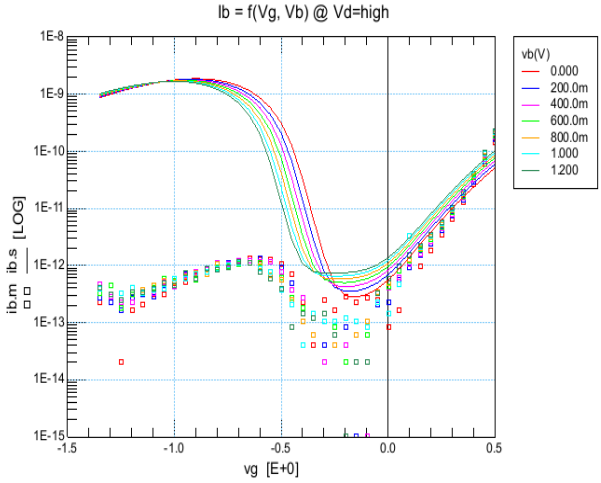


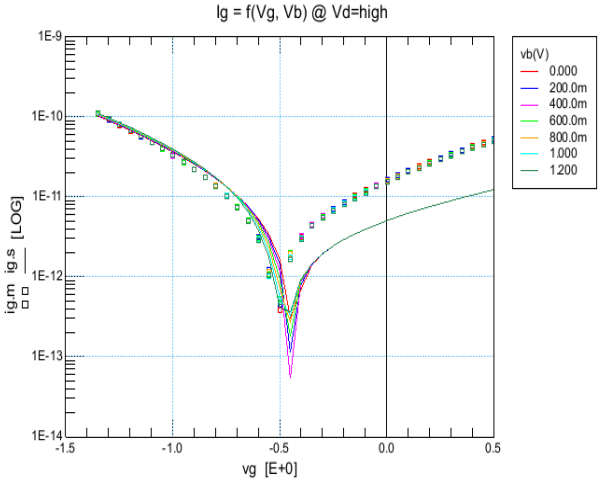




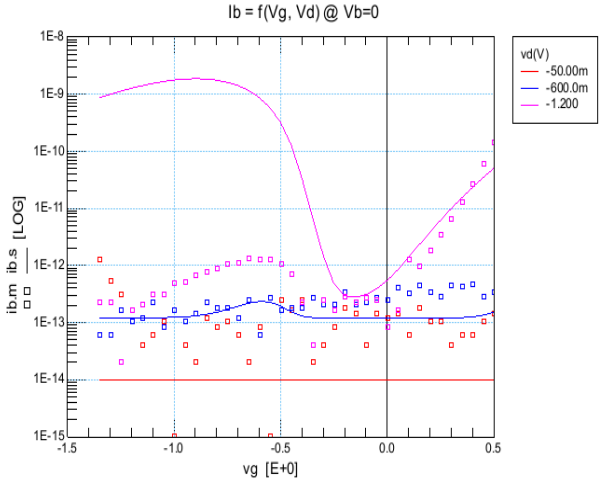


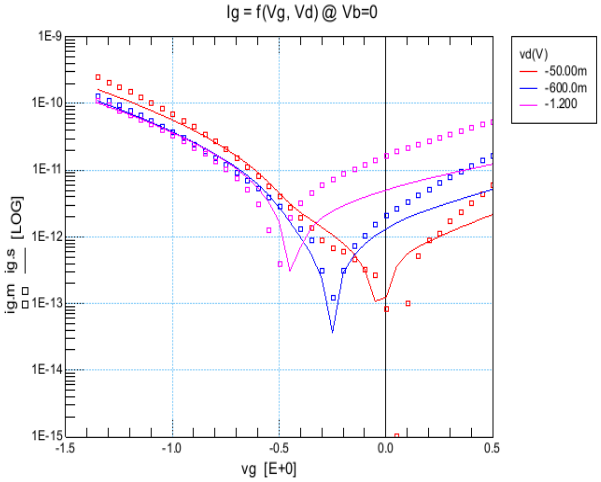




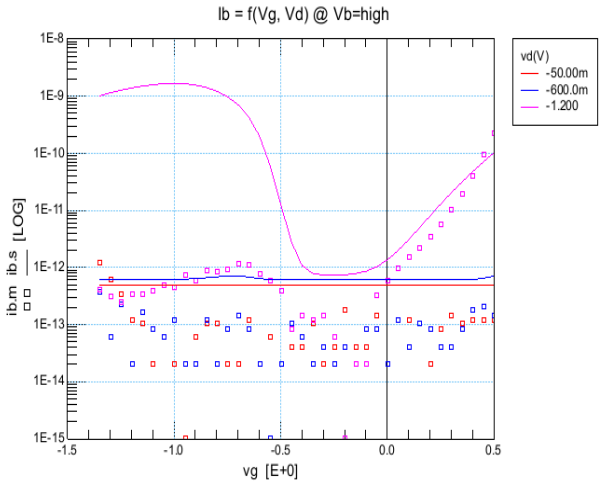


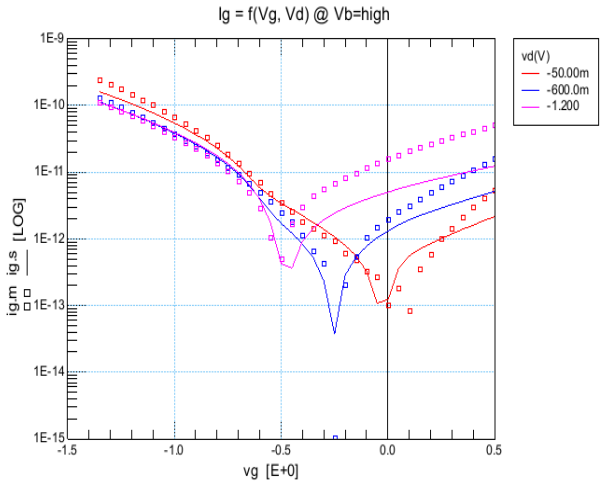
idvg, Ib,Ig, Vb = 0V, T = -40°C



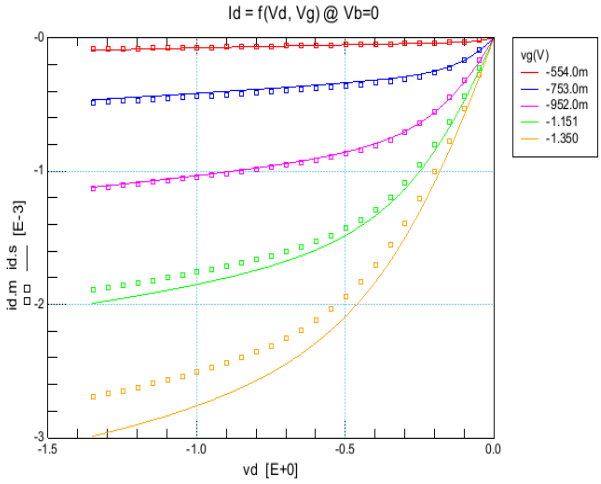


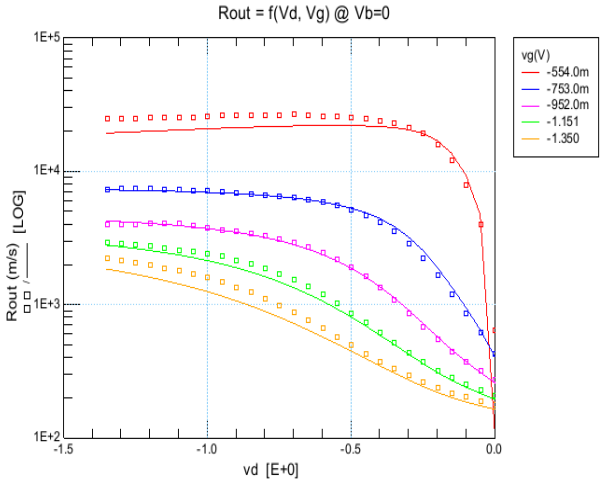
idvg, Ib,Ig, Vb = 1.2V, T = -40°C

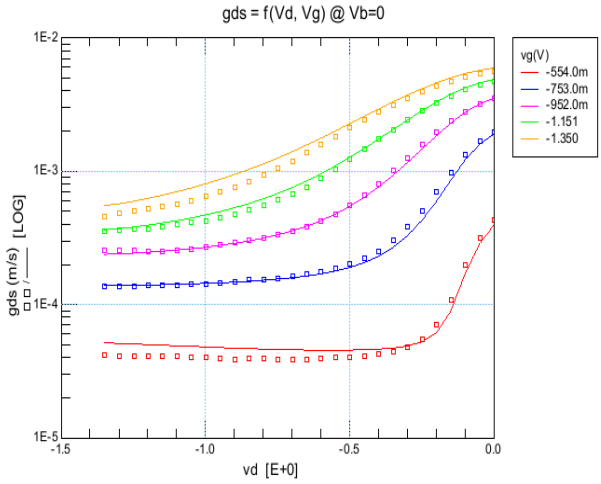




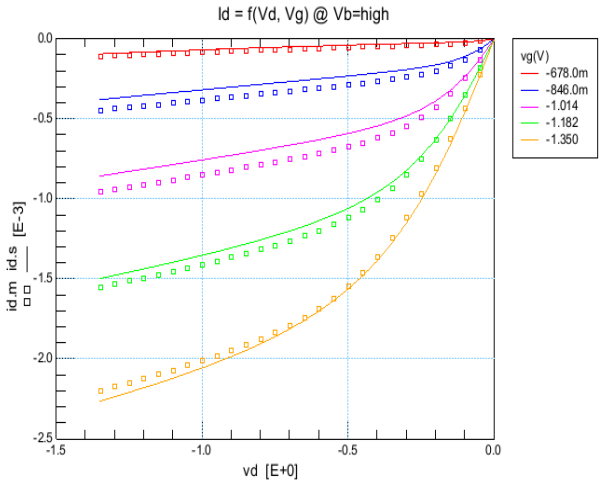
idvd, Vb = 0V, T = -40°C

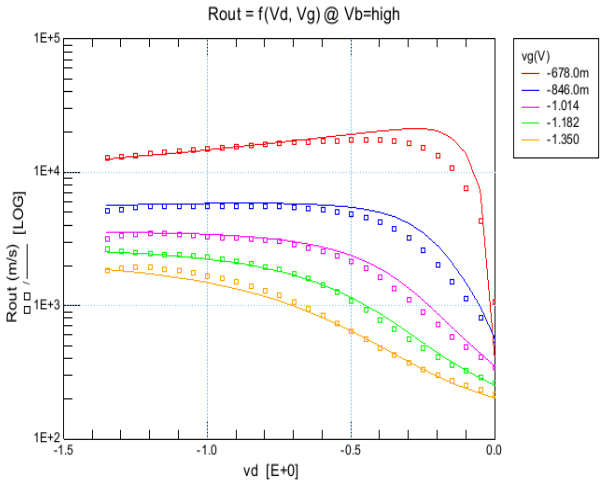


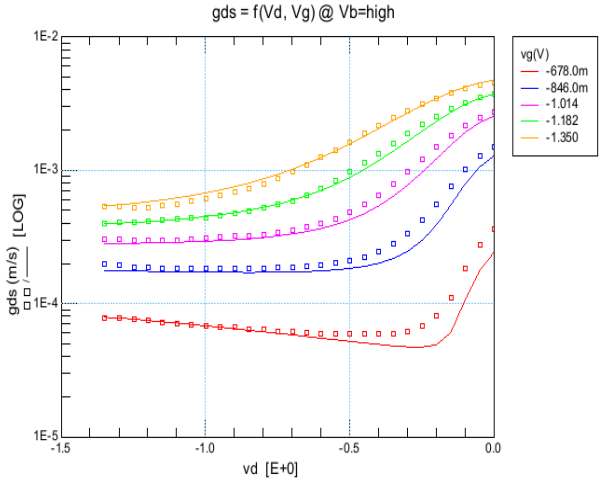




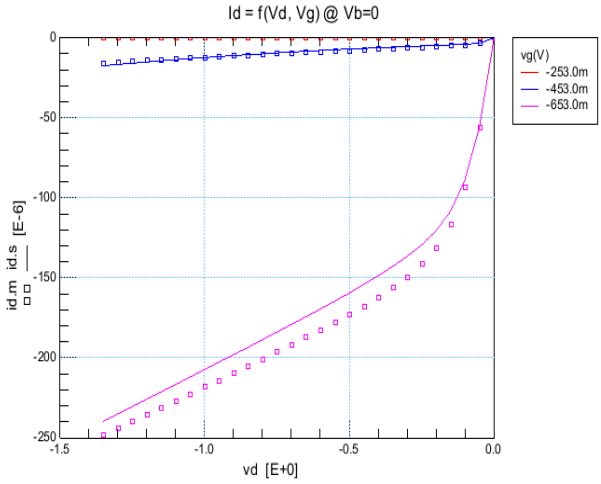
idvd_vbmin, Vb = 1.2V, T = -40°C

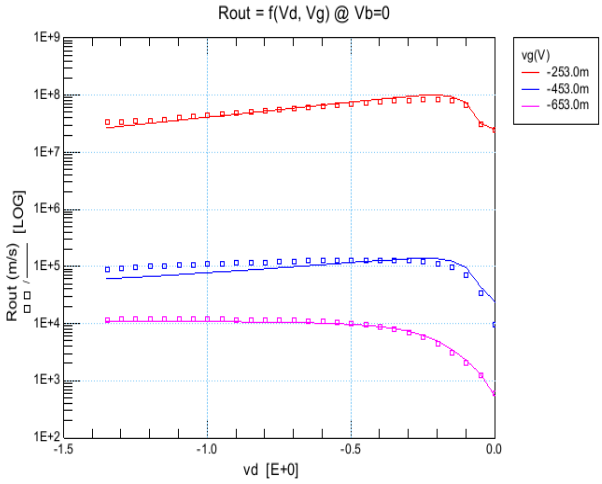


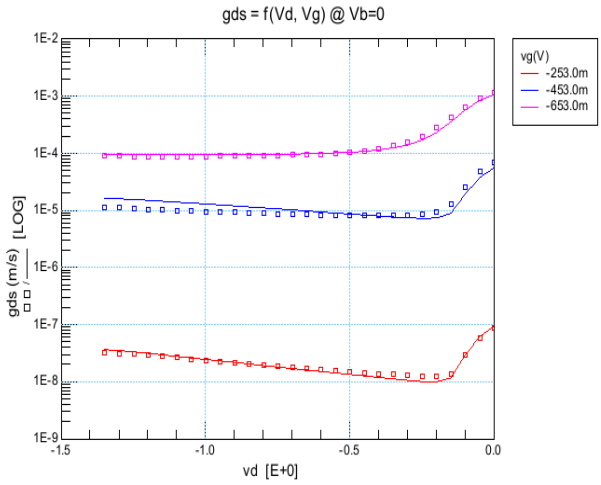




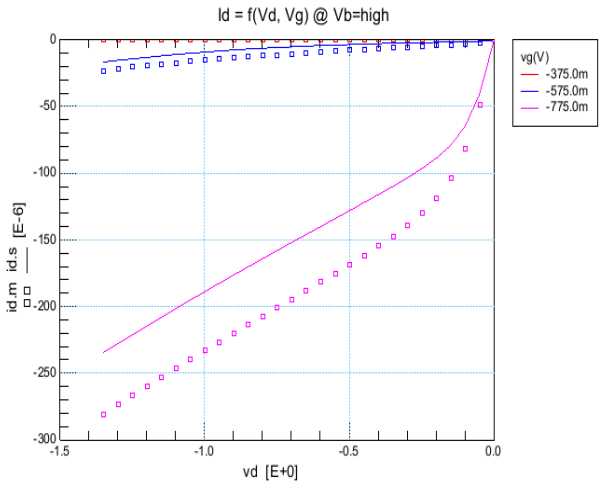
idvd_vth, Vb = 0V, T = -40°C

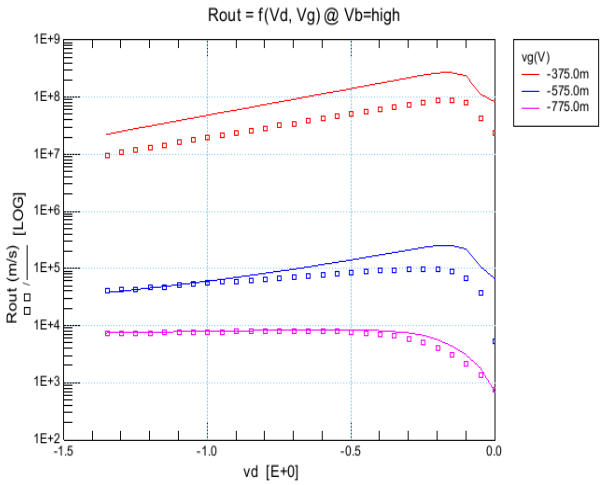


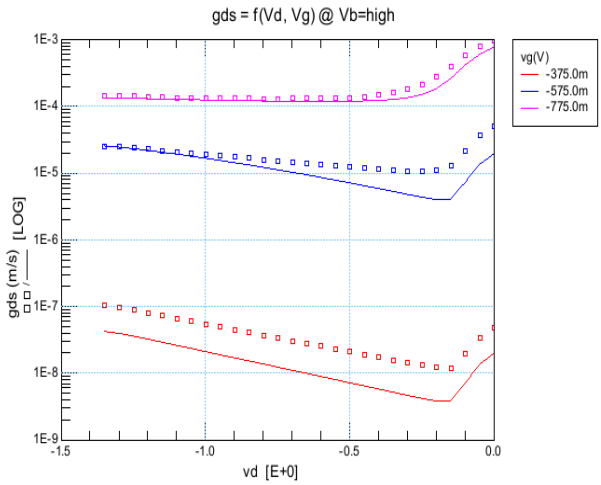




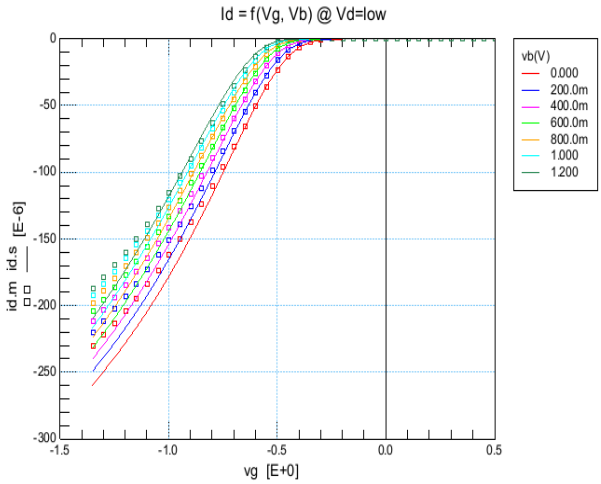
idvd_vth_vbmin, Vb = 1.2V, T = -40°C

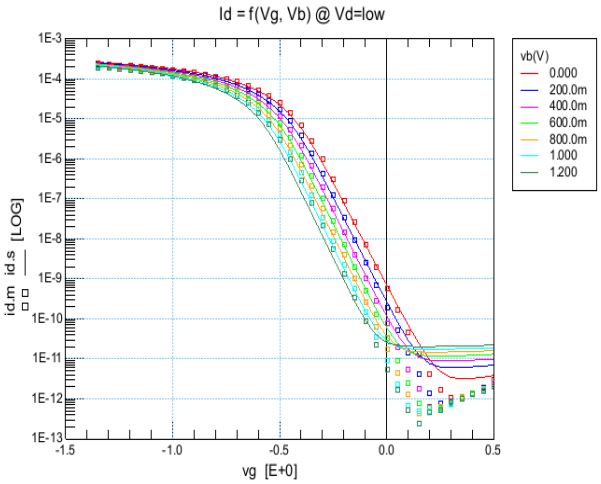


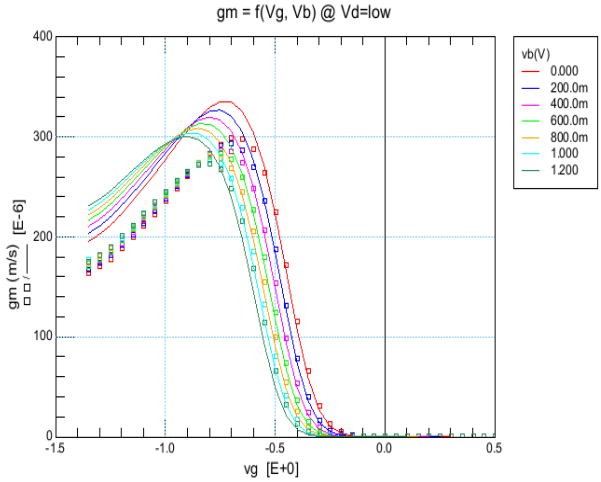


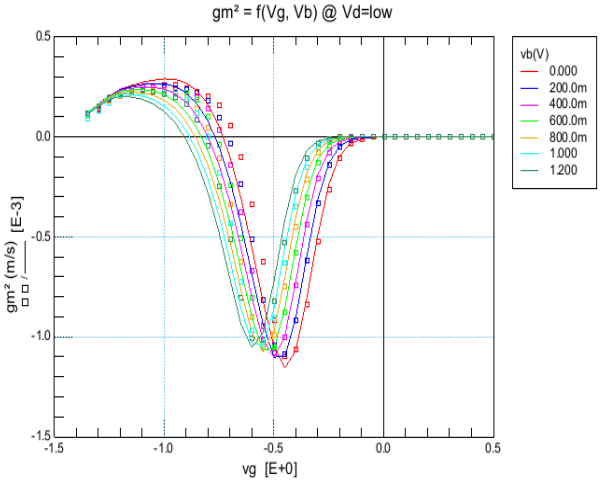


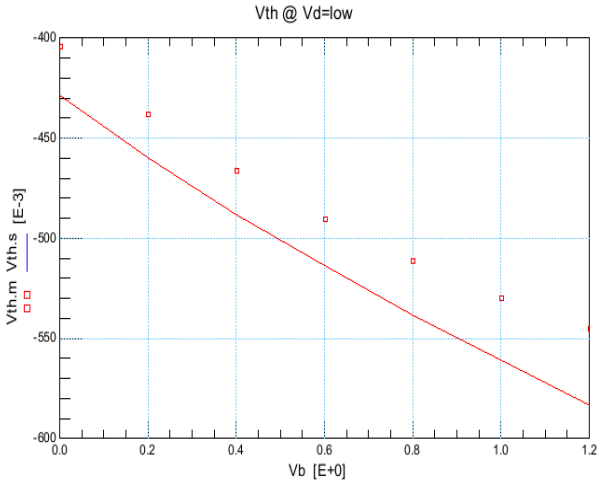
idvg, Vd = -0.05V, T = 70°C

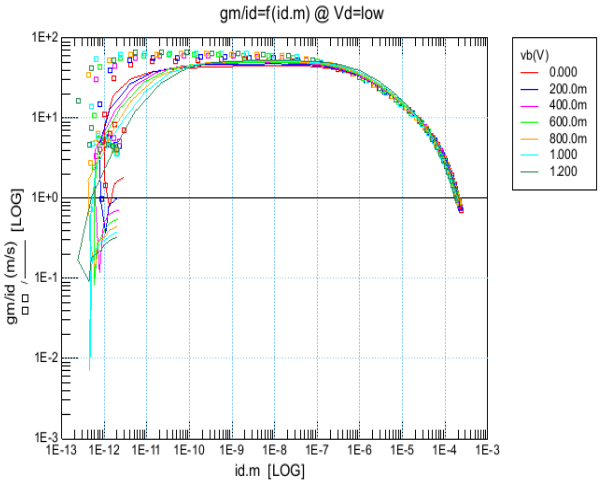


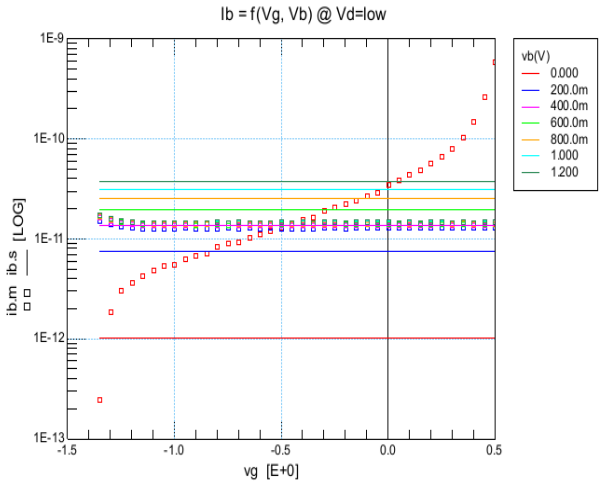


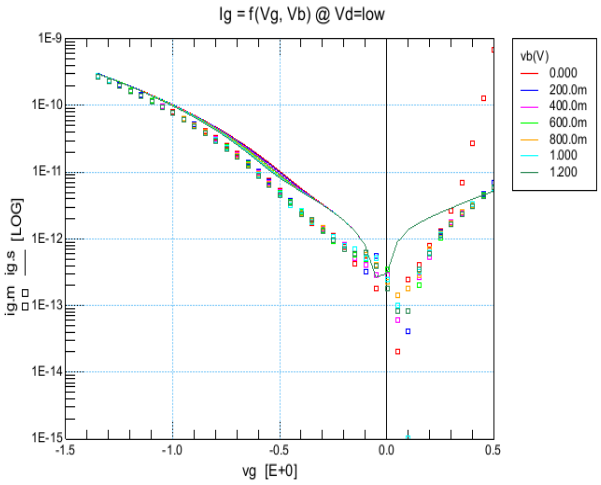




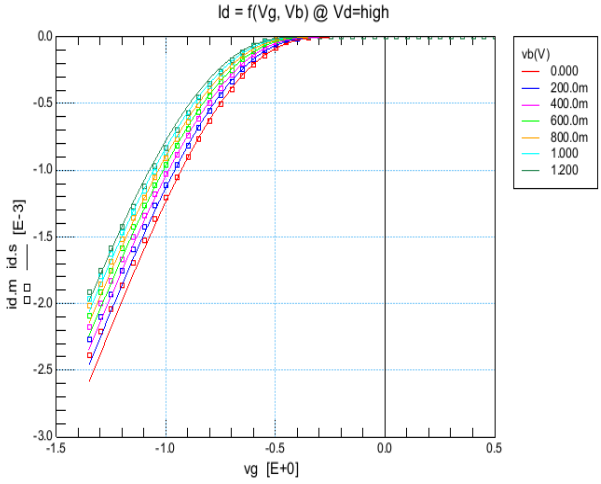


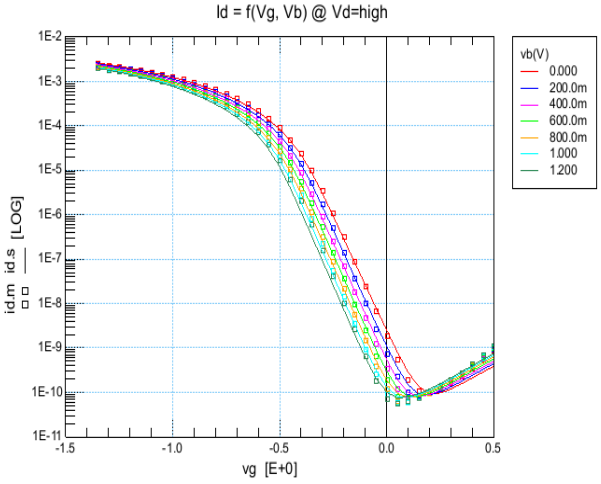


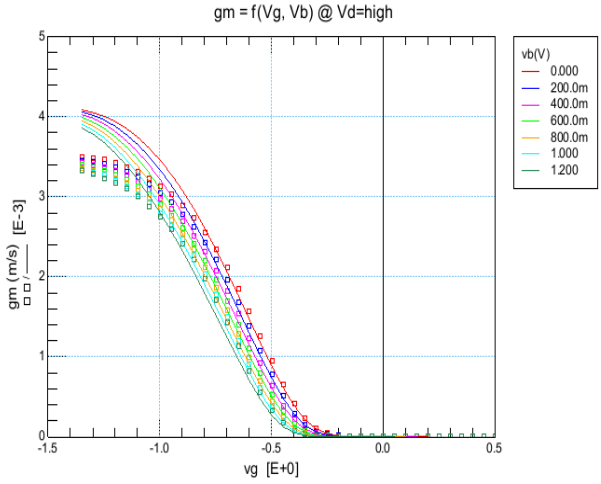


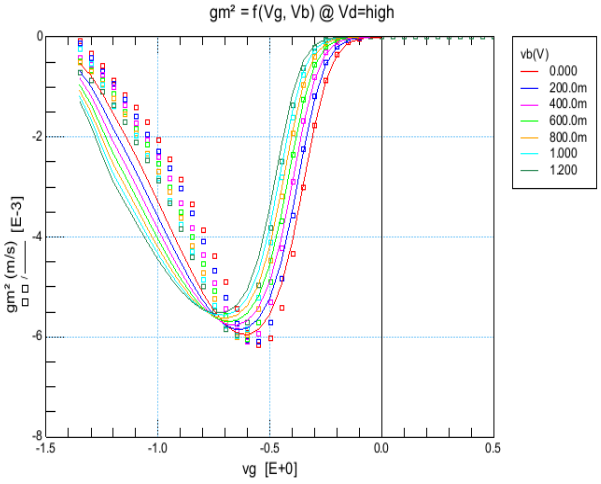


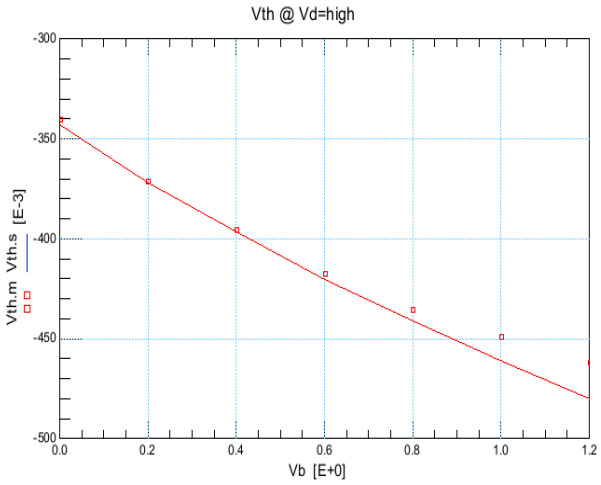
idvg, Vd = -1.2V, T = 70°C

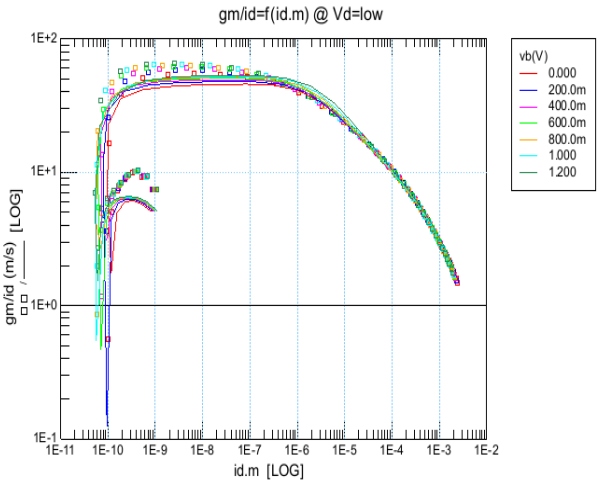


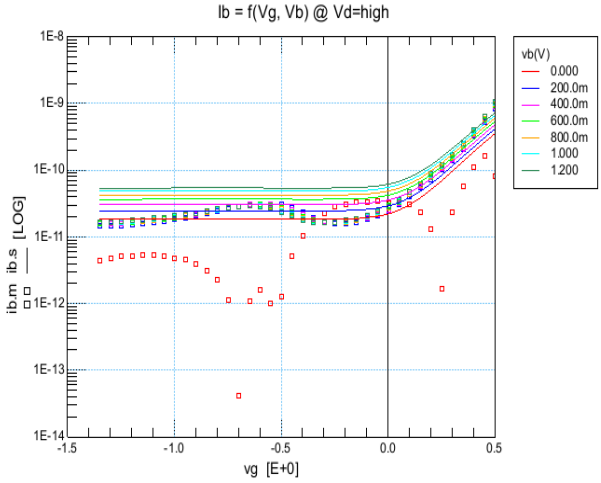


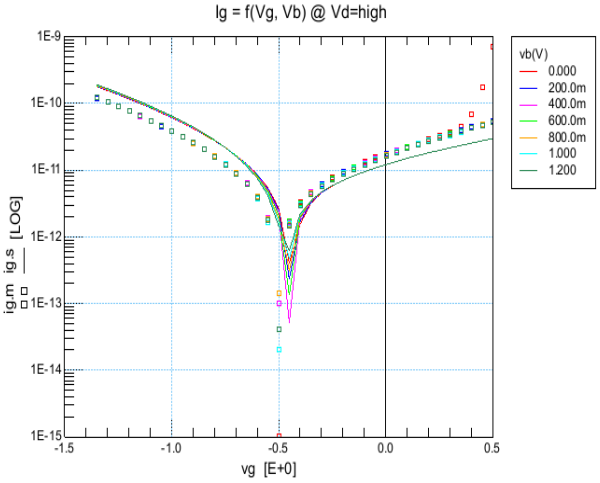




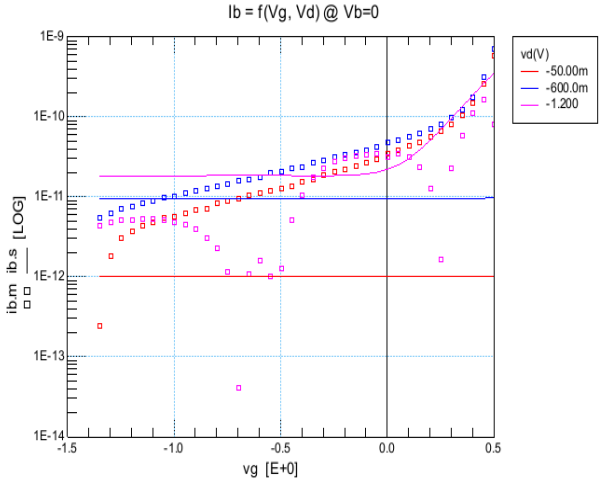


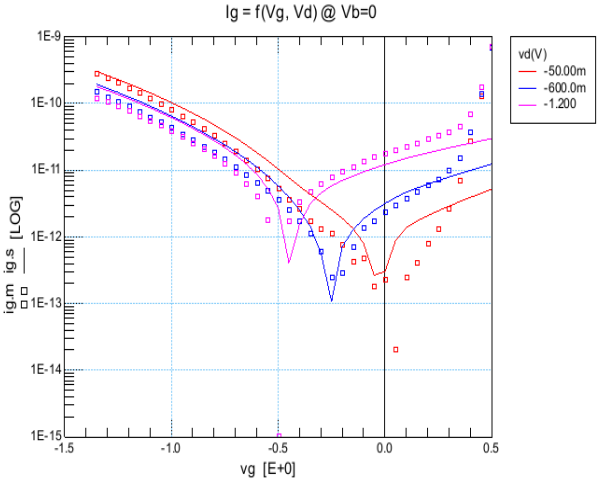




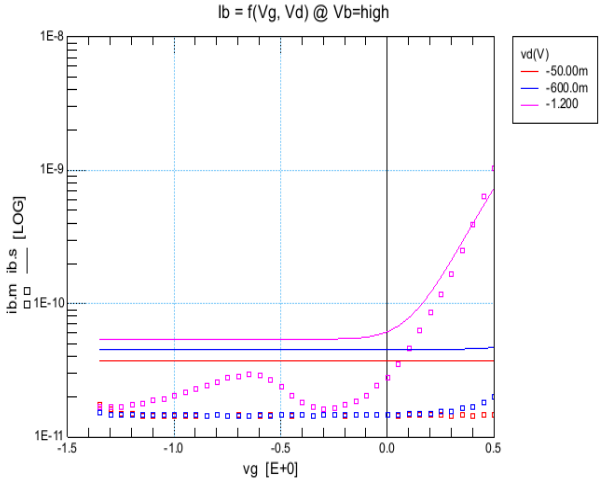


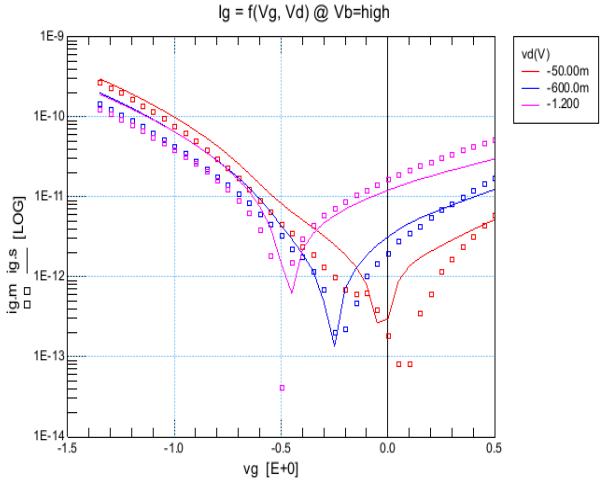
idvg, Ib,Ig, Vb = 0V, T = 70°C



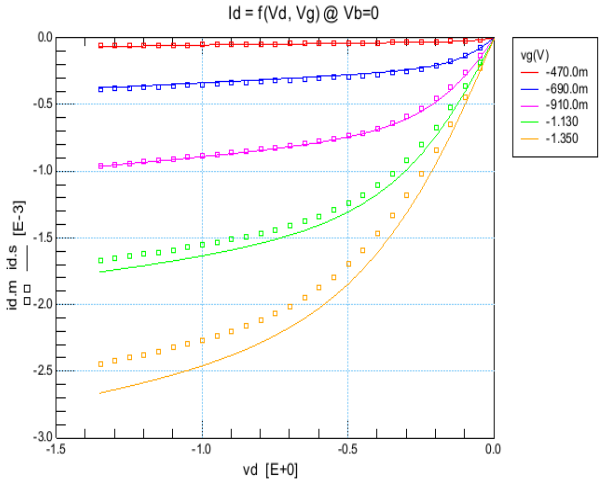


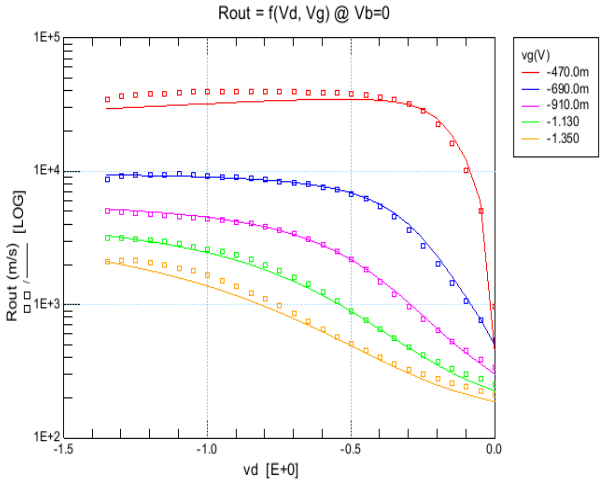
idvg, Ib,Ig, Vb = 1.2V, T = 70°C

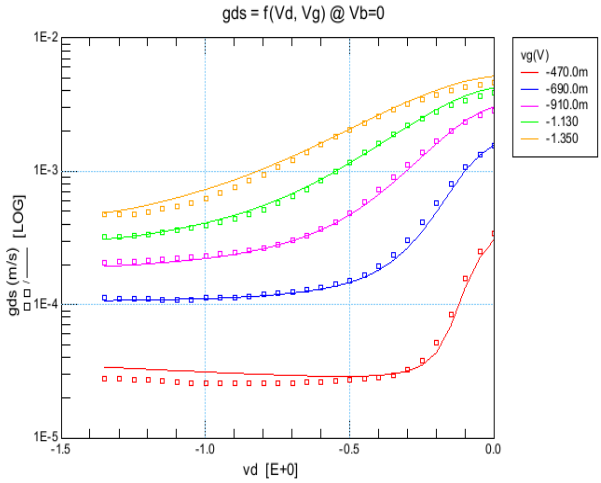




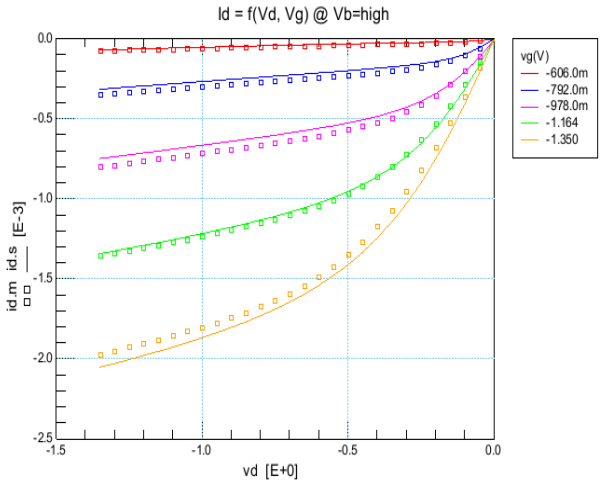
idvd, Vb = 0V, T = 70°C

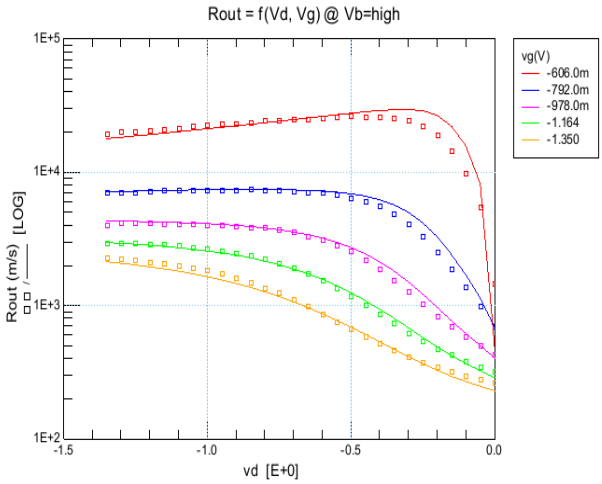


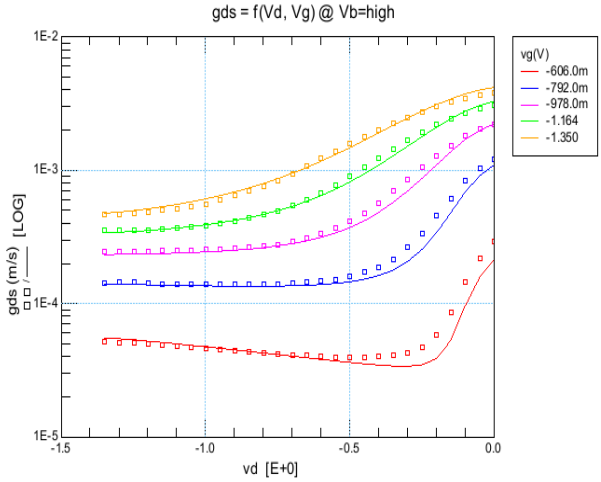




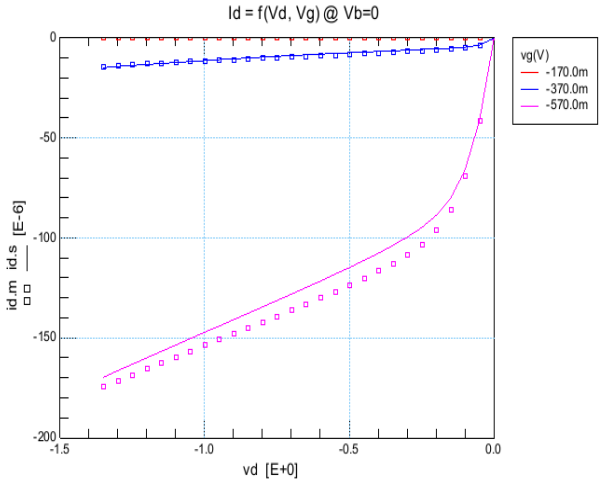
idvd_vbmin, Vb = 1.2V, T = 70°C

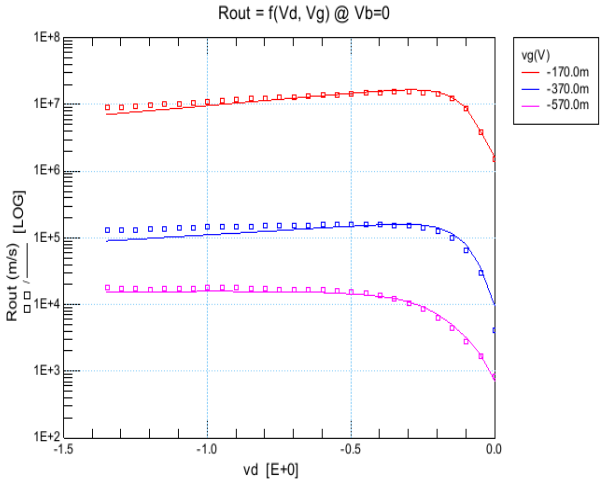


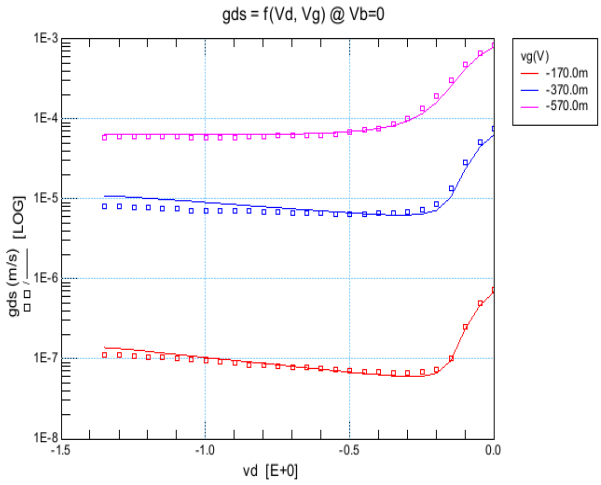




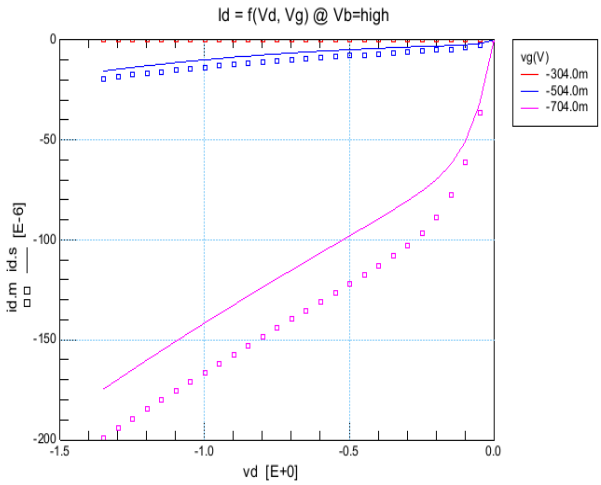
idvd_vth, Vb = 0V, T = 70°C

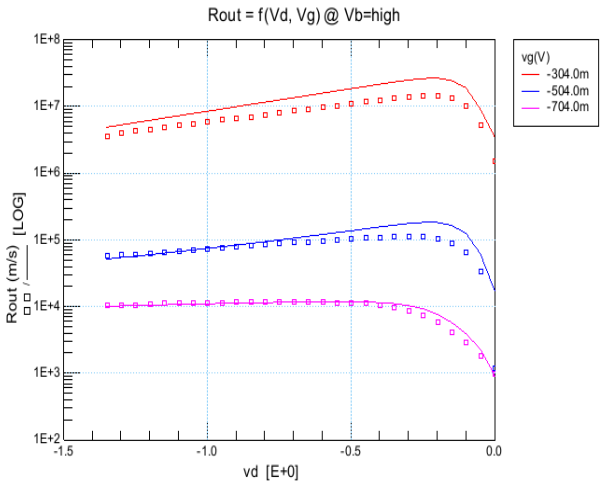


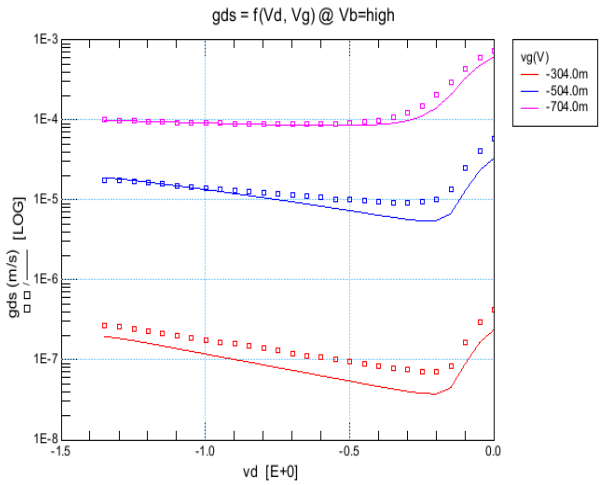




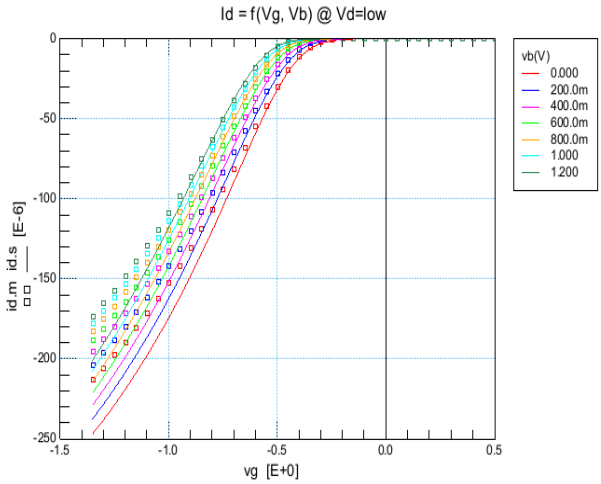
idvd_vth_vbmin, Vb = 1.2V, T = 70°C

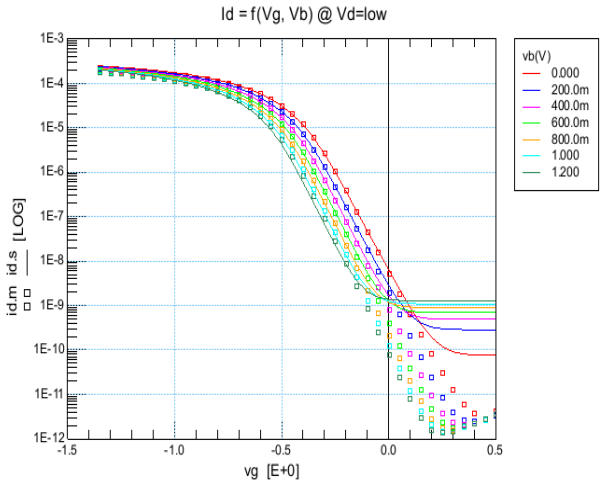


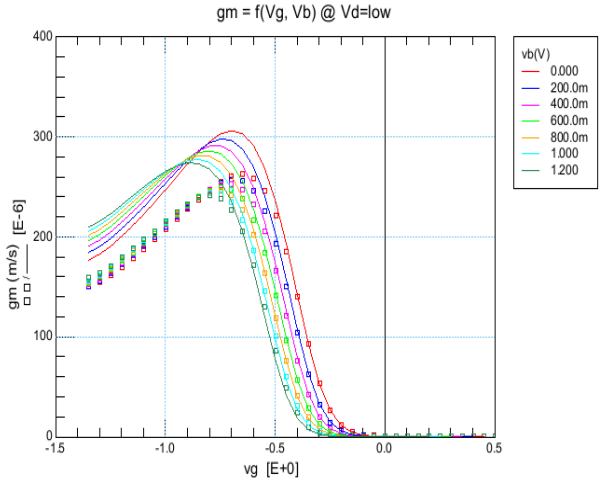


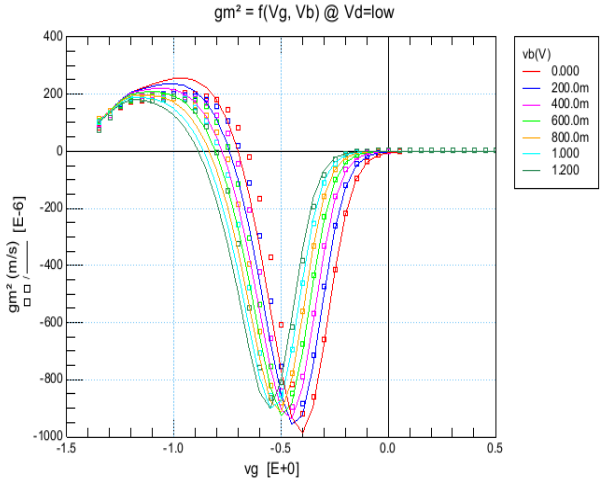


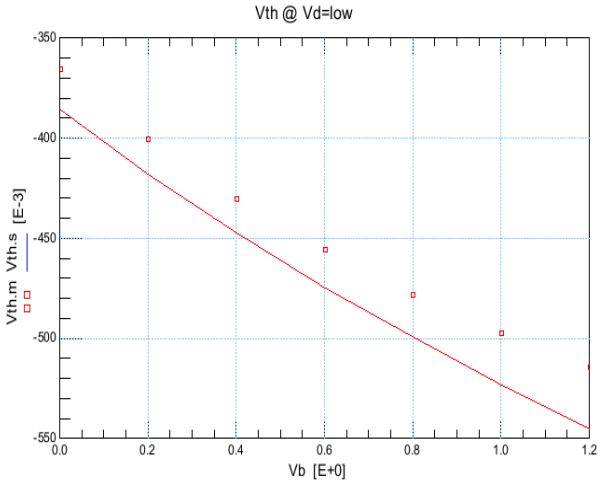
idvg, Vd = -0.05V, T = 125°C

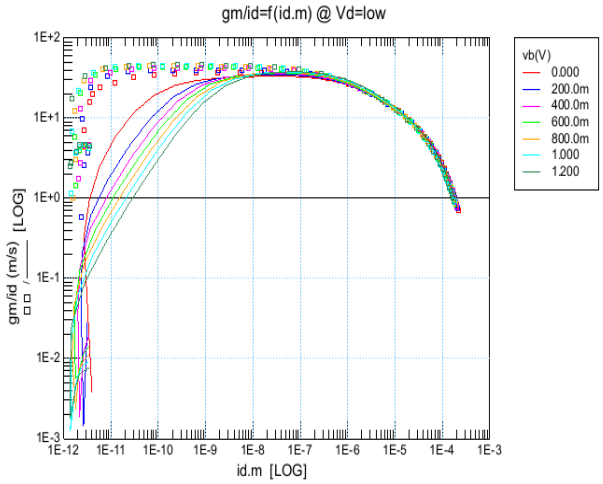


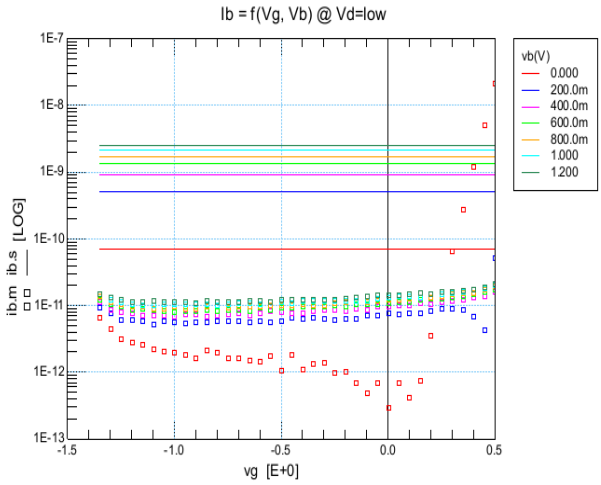


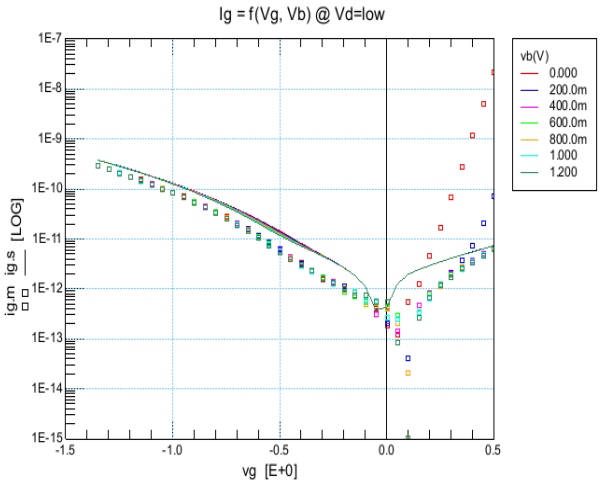




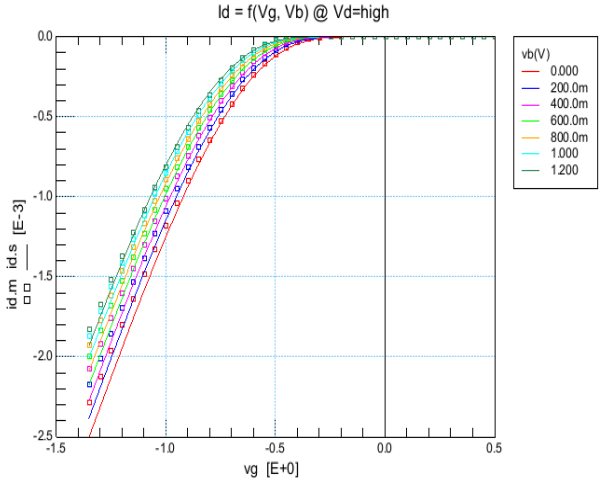


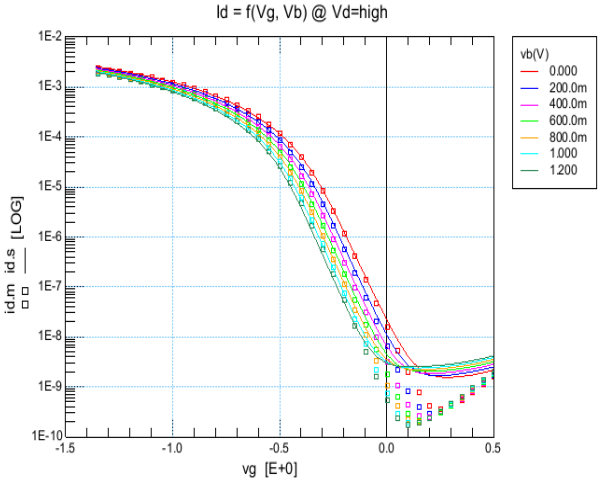


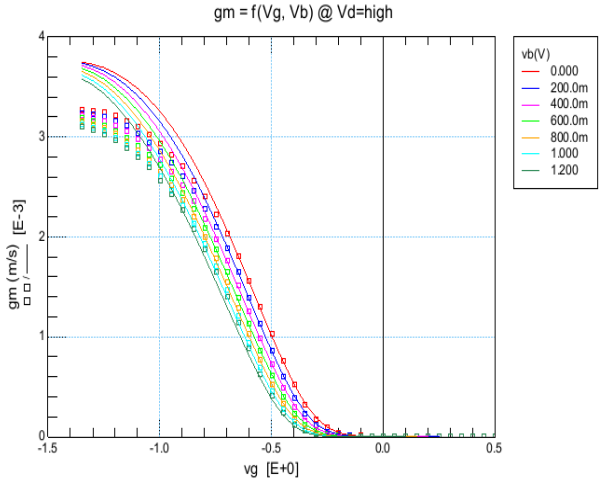


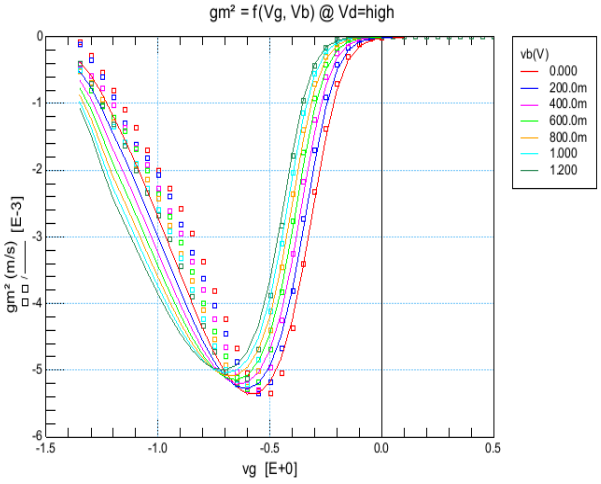


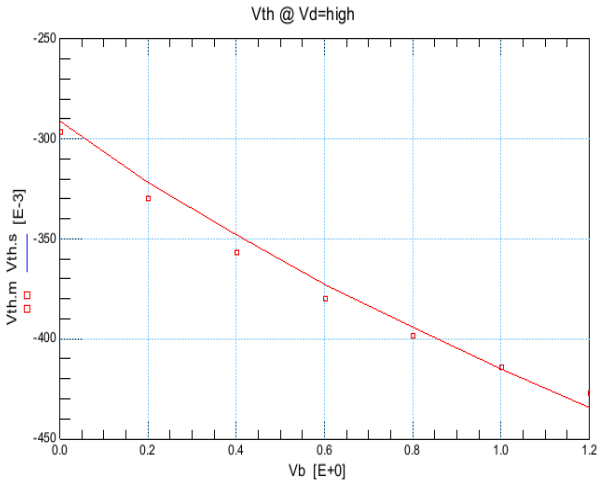
idvg, Vd = -1.2V, T = 125°C

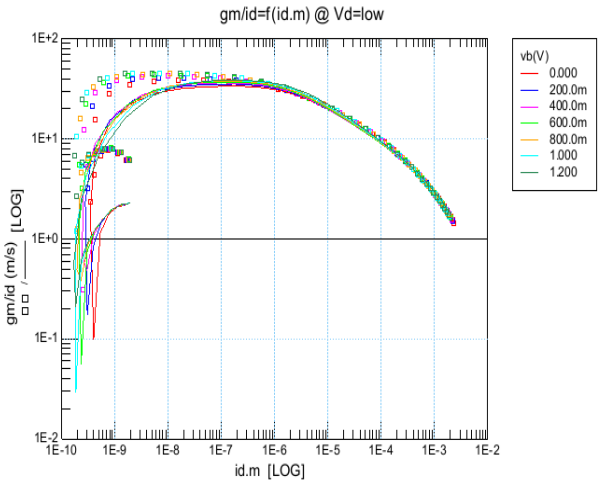


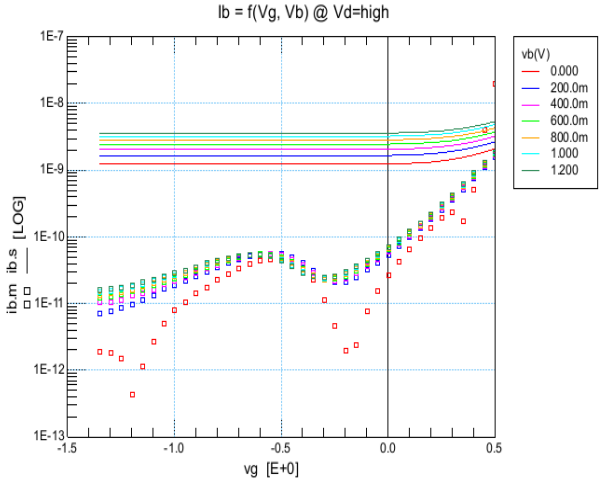


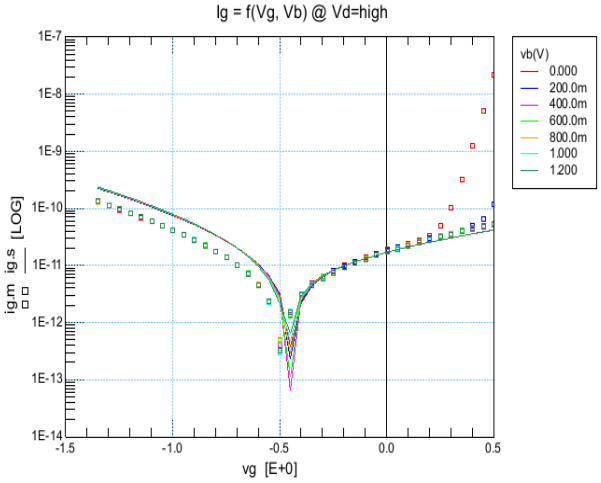




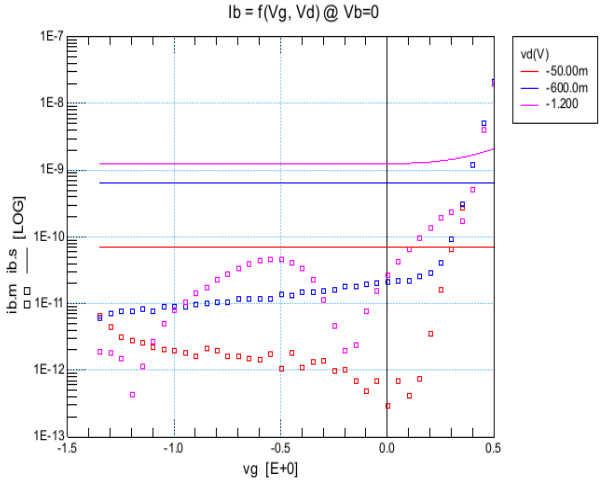


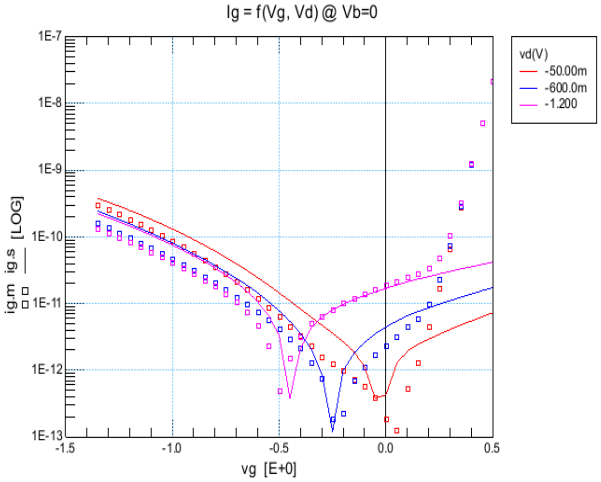




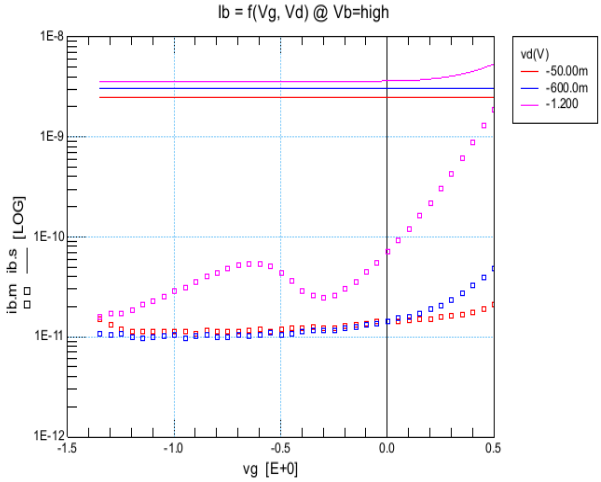


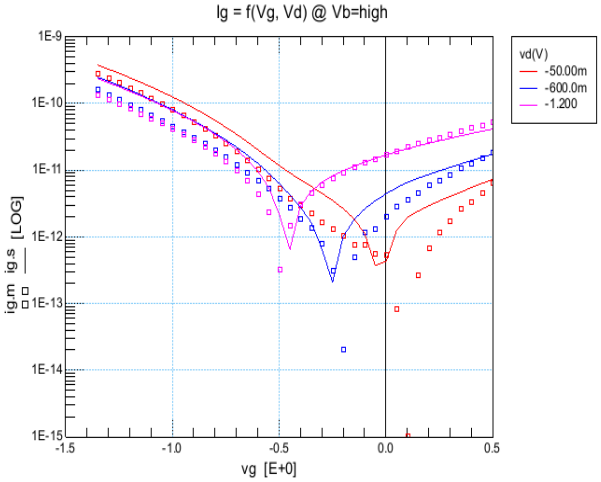
idvg, Ib,Ig, Vb = 0V, T = 125°C



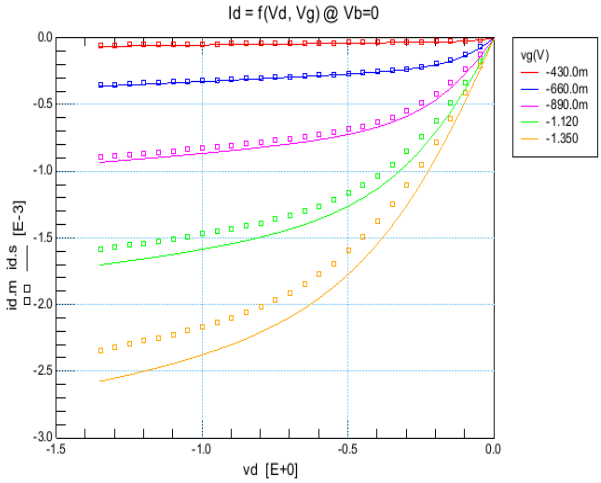


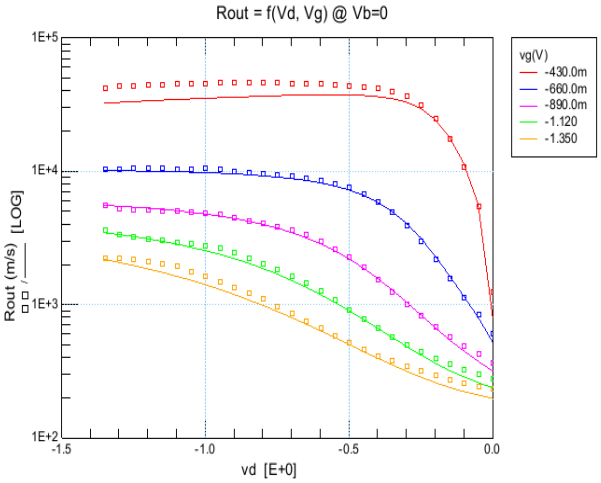
idvg, Ib,Ig, Vb = 1.2V, T = 125°C

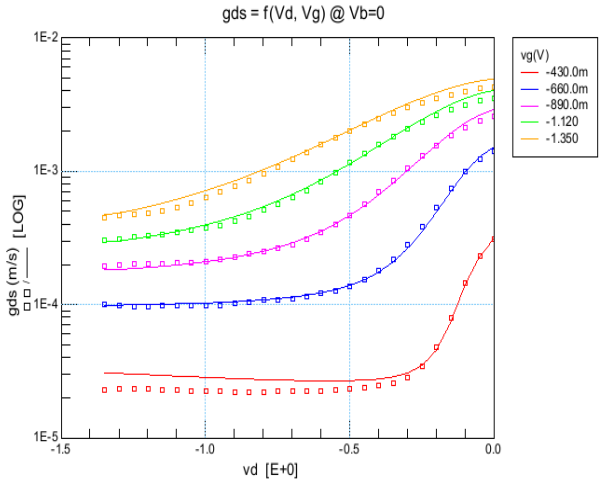




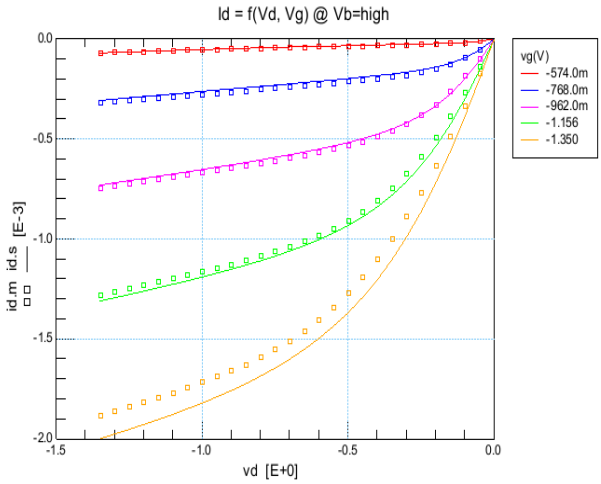
idvd, Vb = 0V, T = 125°C

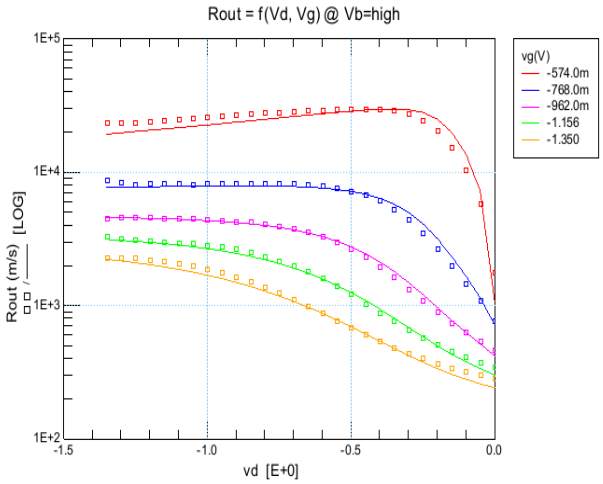


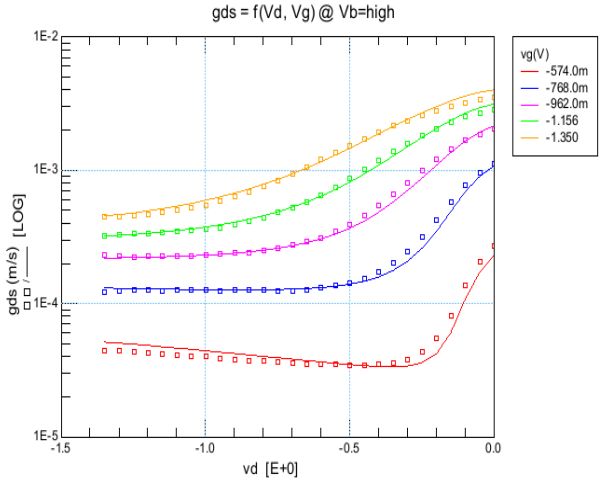




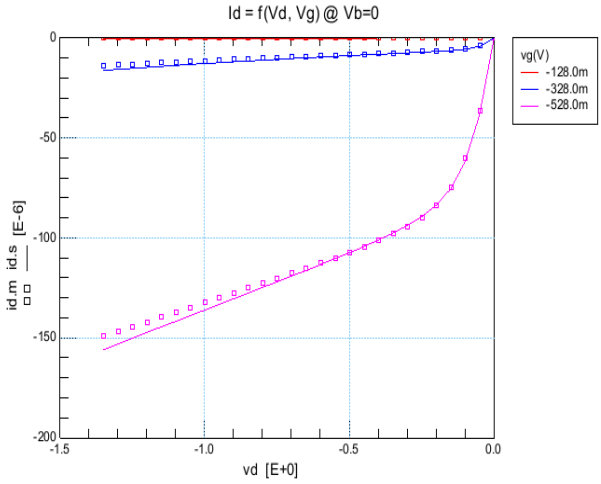
idvd_vbmin, Vb = 1.2V, T = 125°C

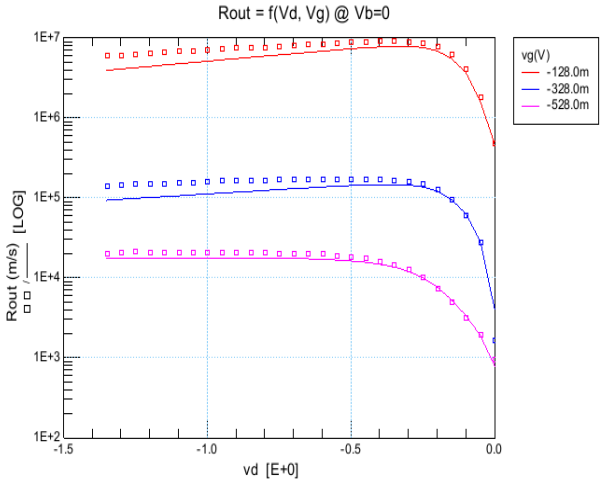


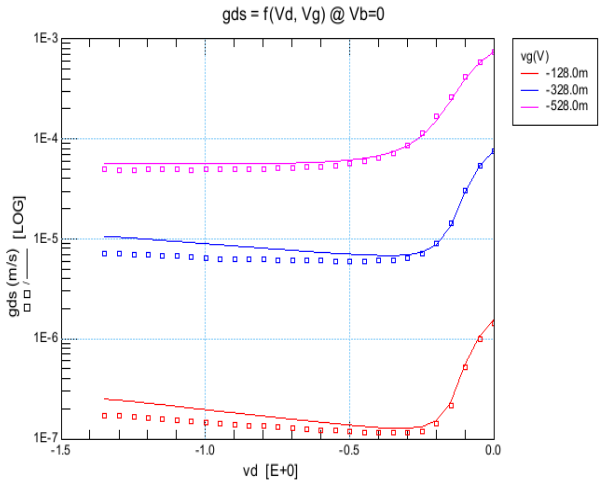




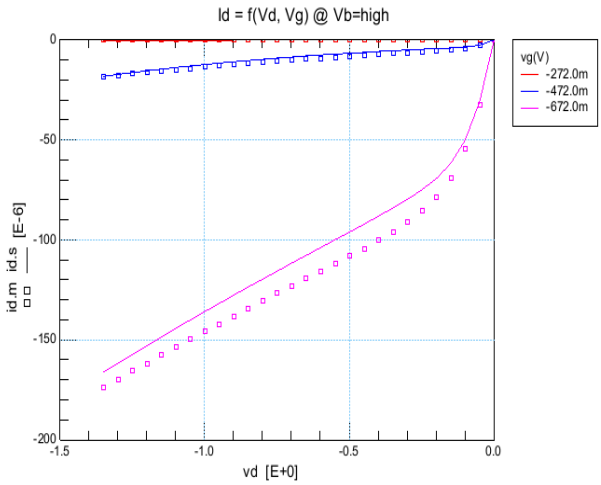
idvd_vth, Vb = 0V, T = 125°C

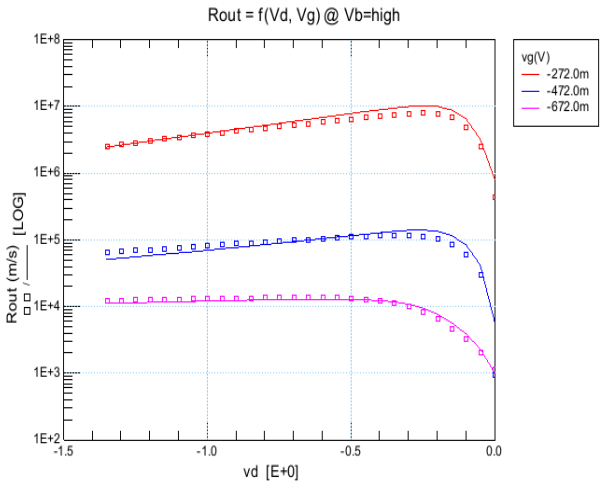


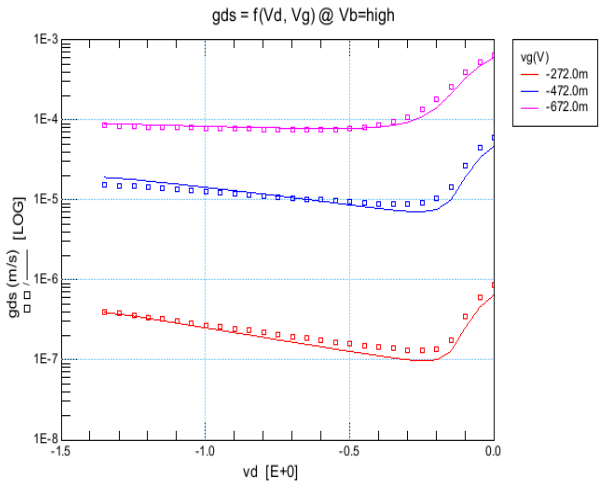




idvd_vth_vbmin, Vb = 1.2V, T = 125°C

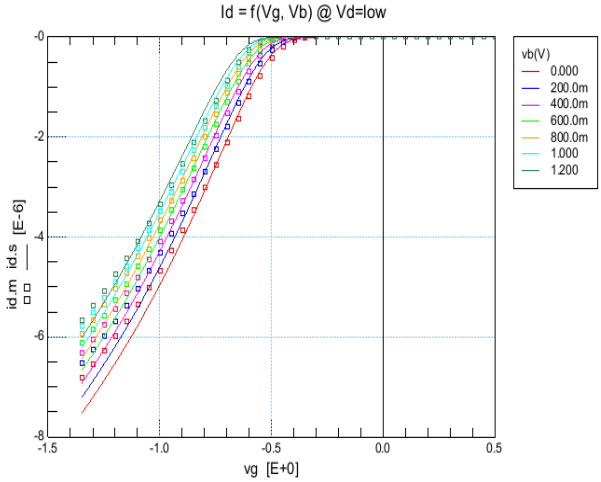


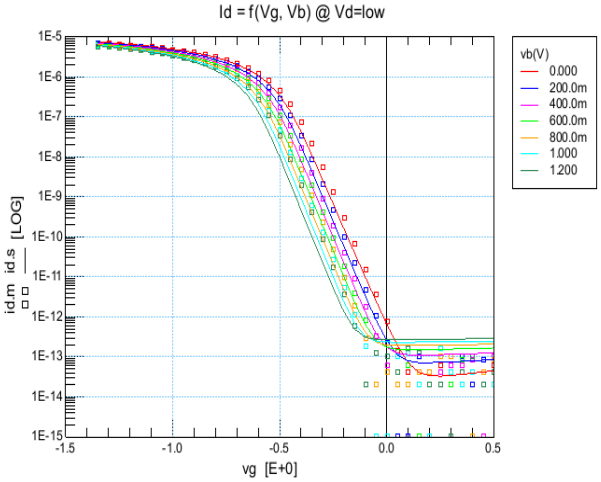


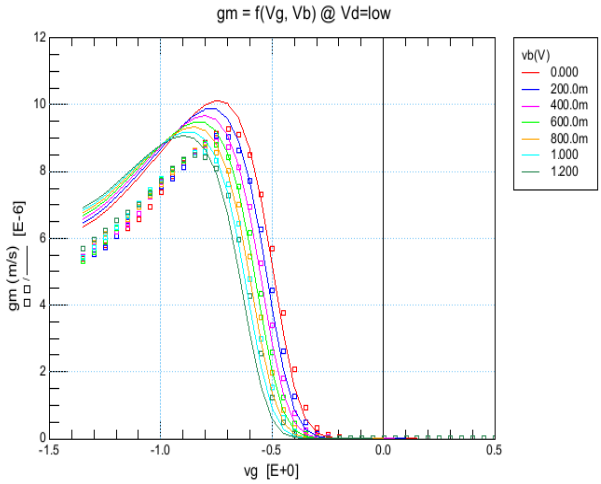


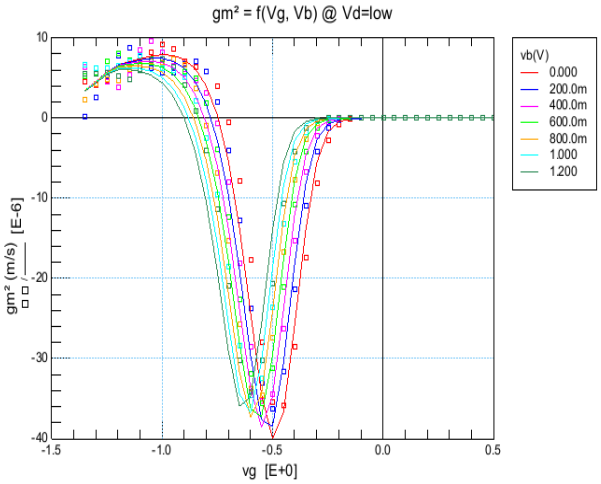
4.5 W0u15_L0u13_S554_3

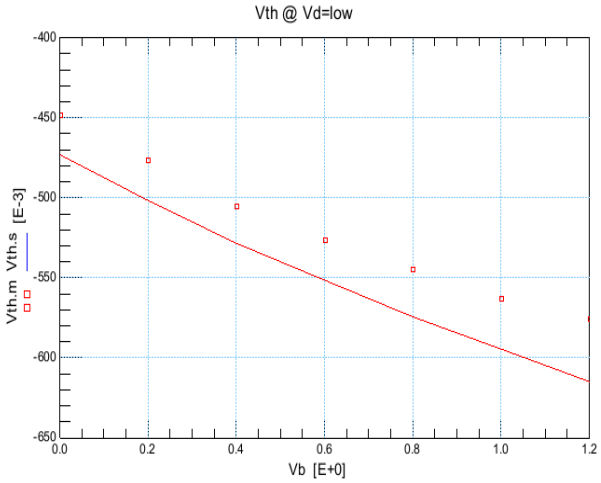
idvg, Vd = -0.05V, T = 27°C

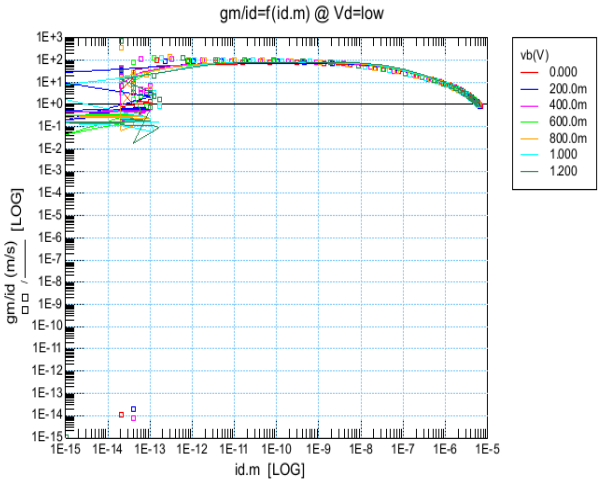


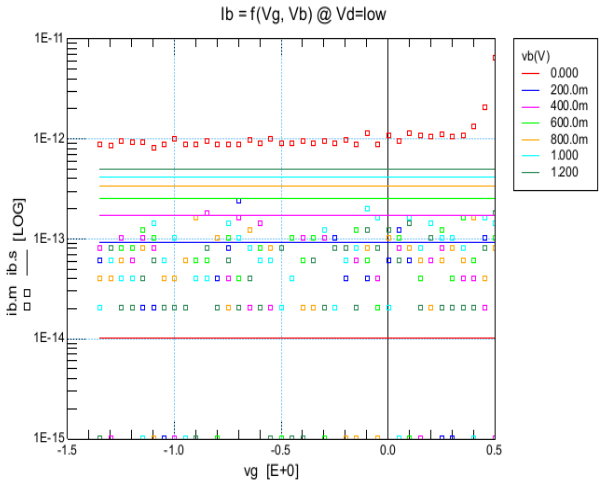


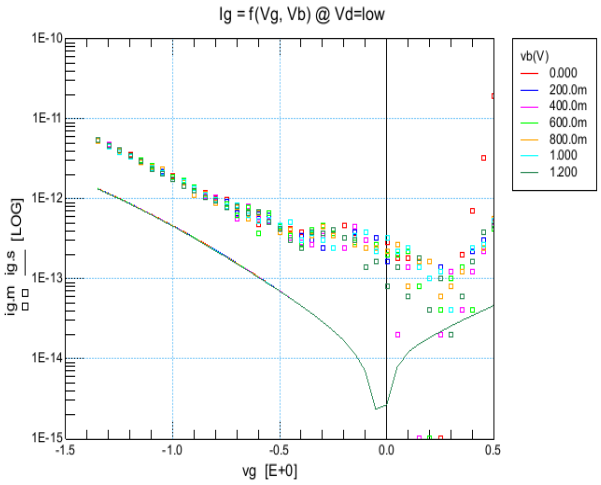




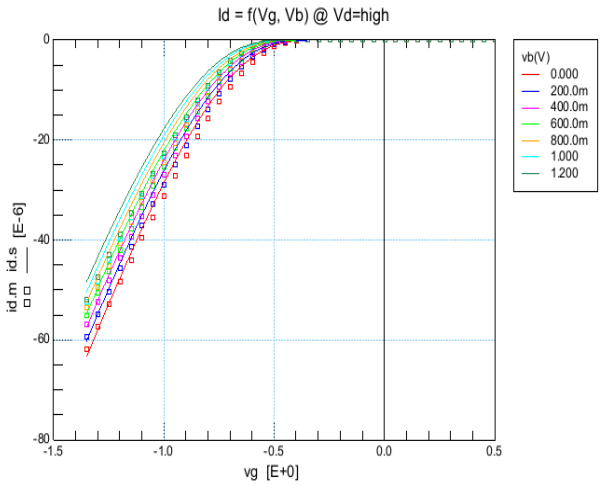


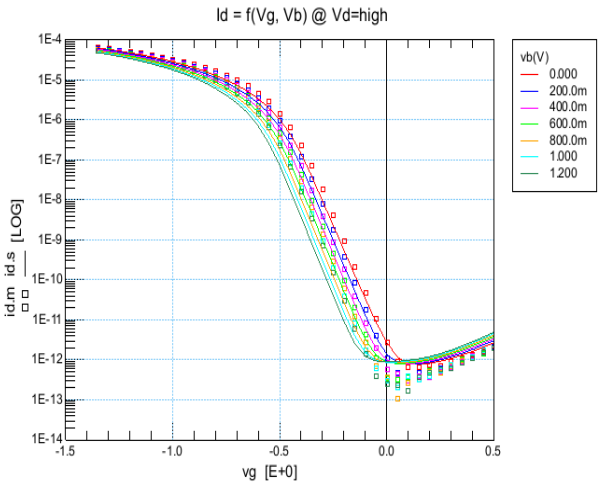


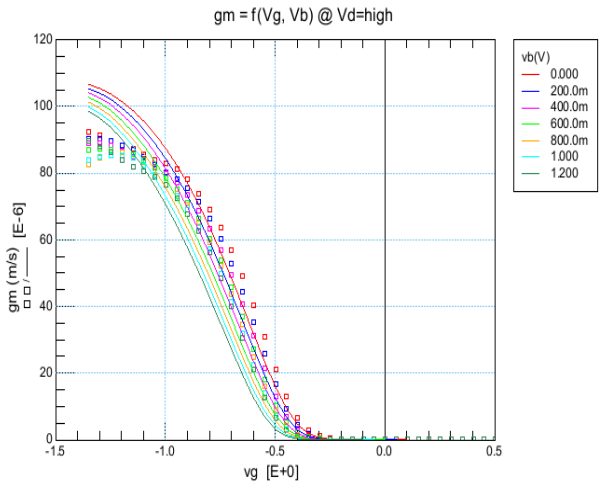


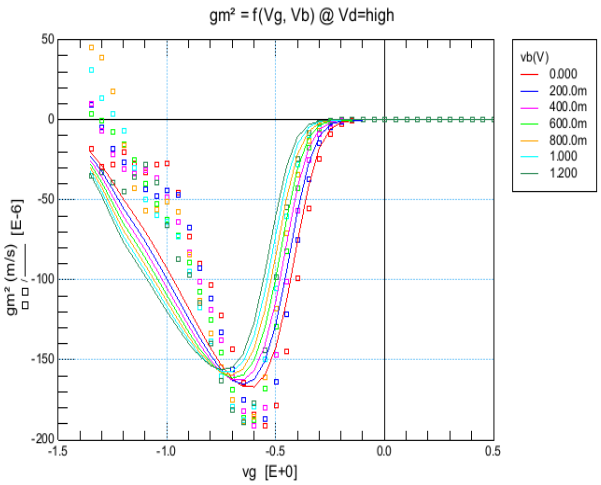


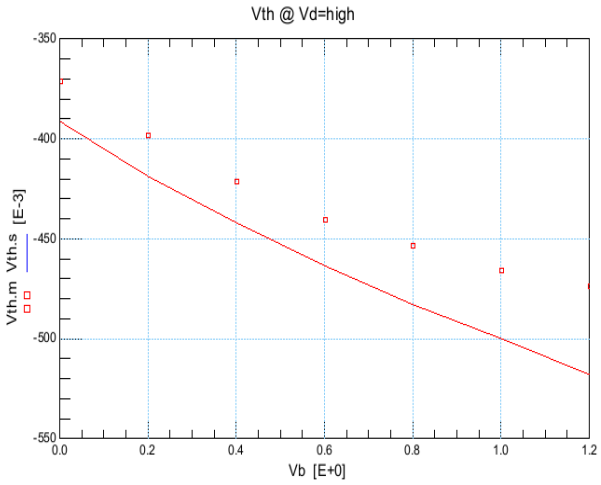
idvg, Vd = -1.2V, T = 27°C

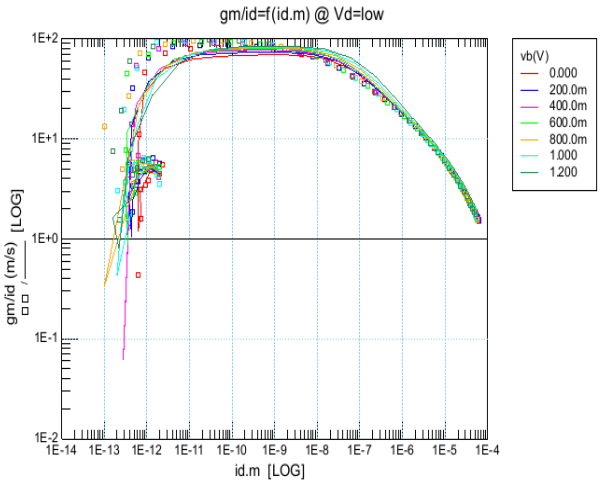


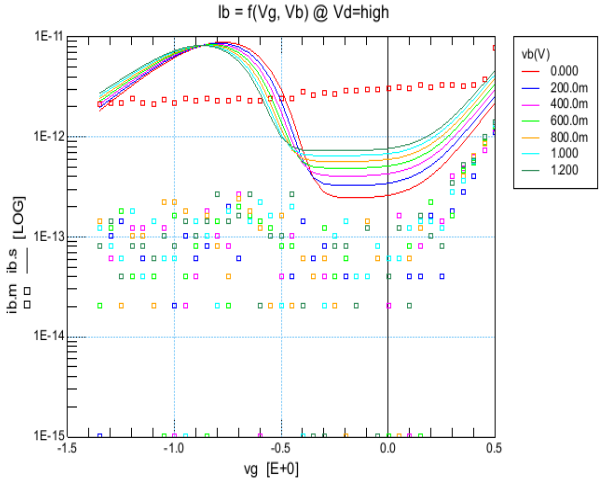


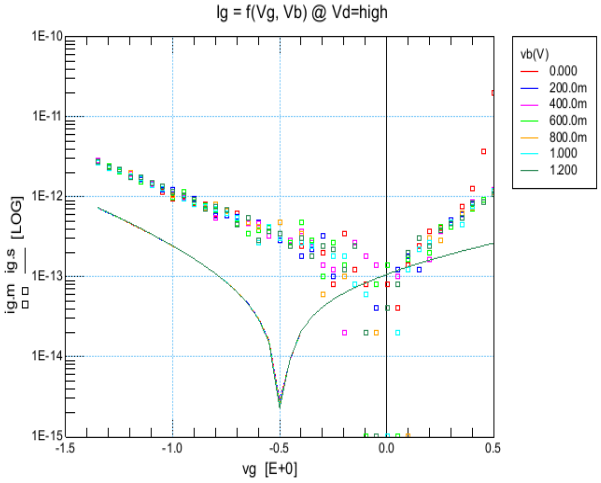




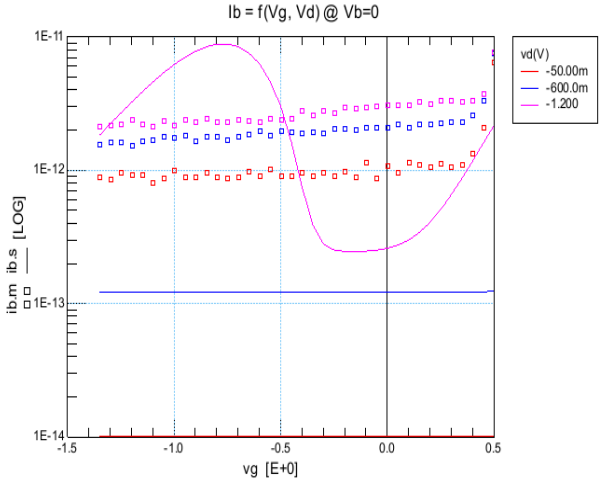


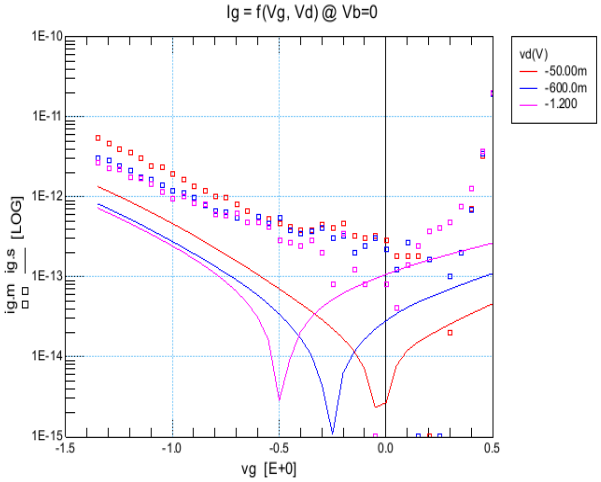




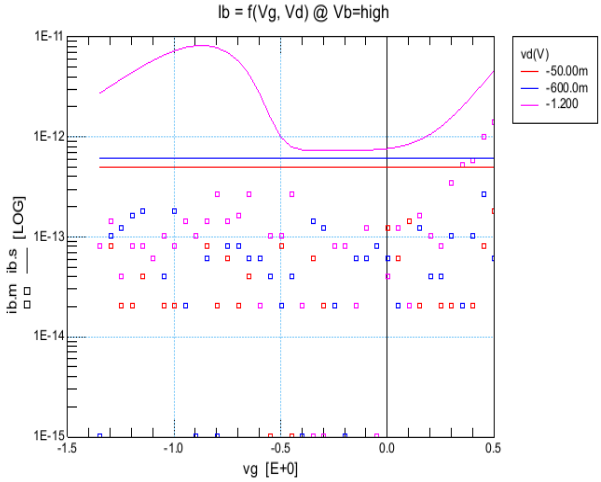


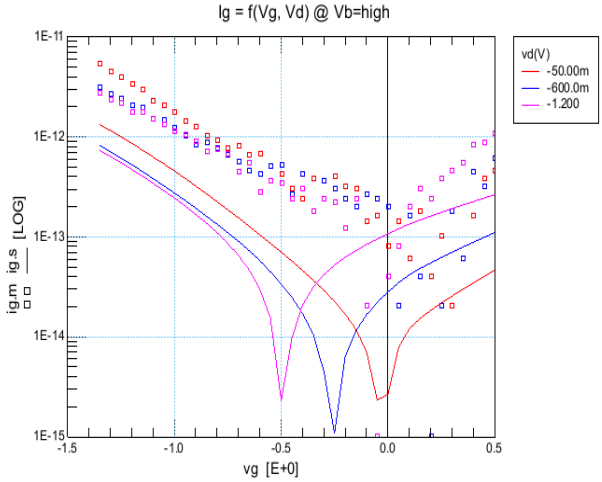
idvg, Ib,Ig, Vb = 0V, T = 27°C



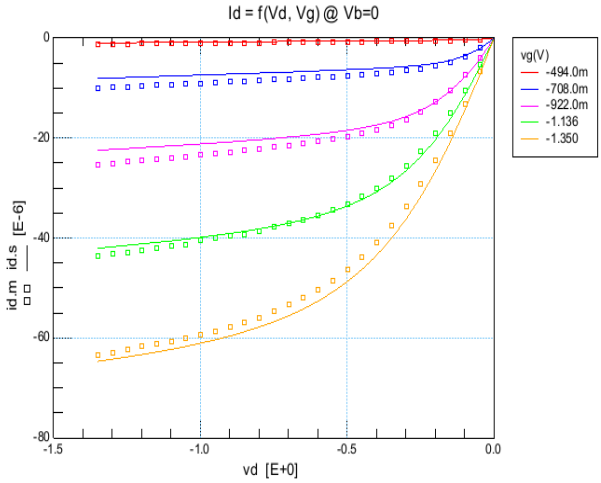


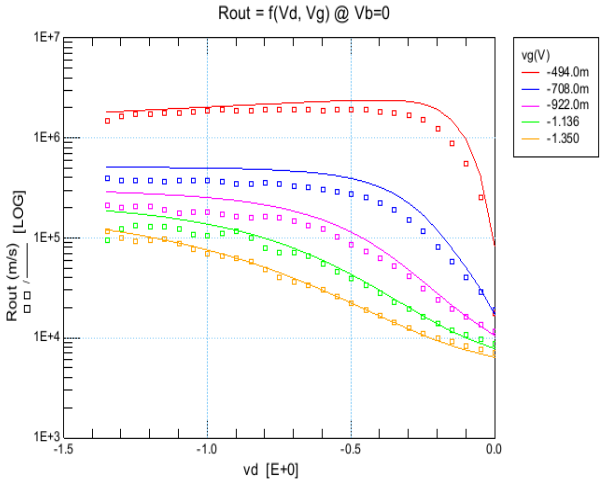
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

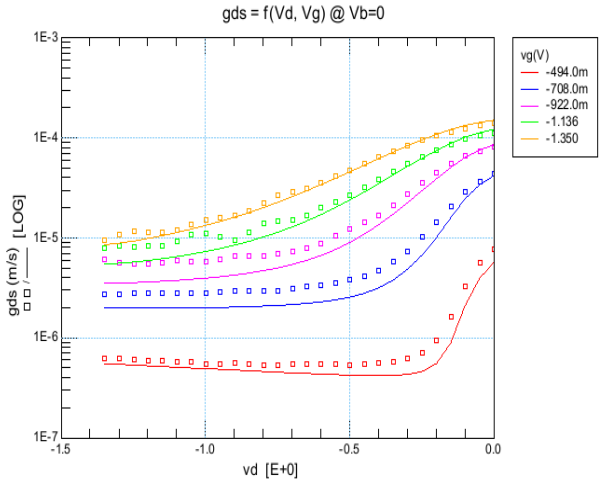




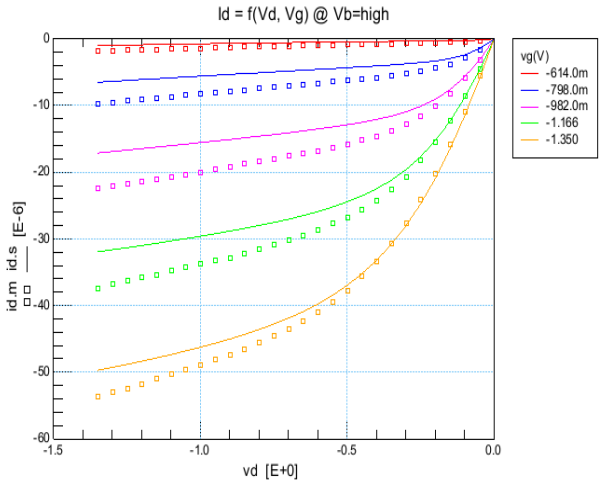
idvd, Vb = 0V, T = 27°C

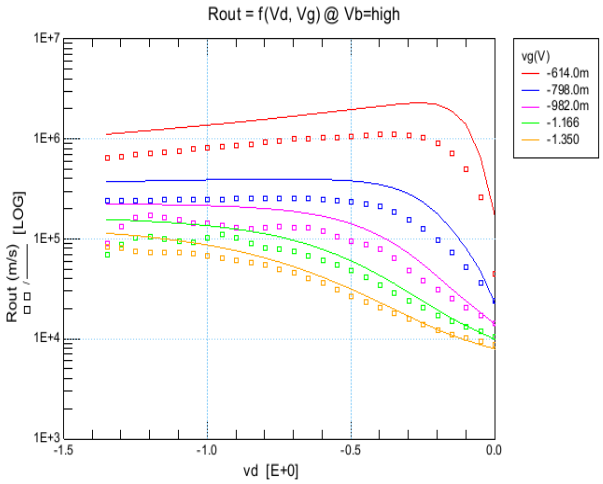


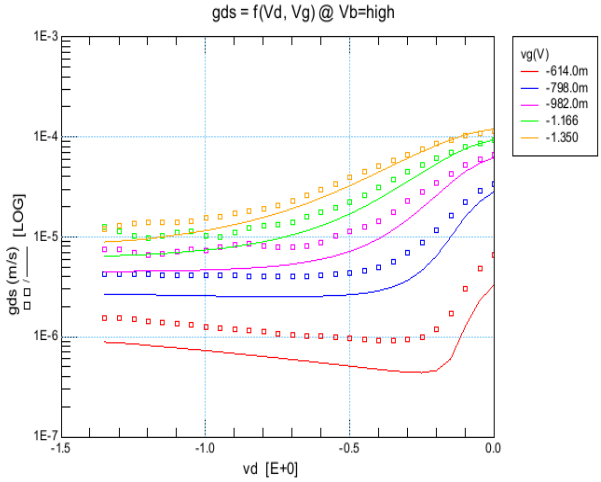




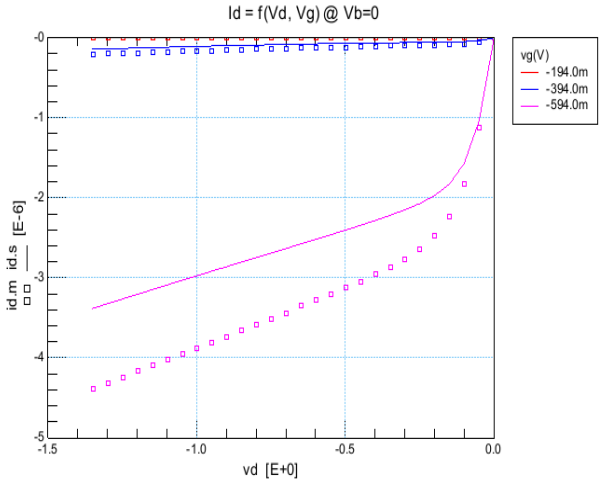
idvd_vbmin, Vb = 1.2V, T = 27°C

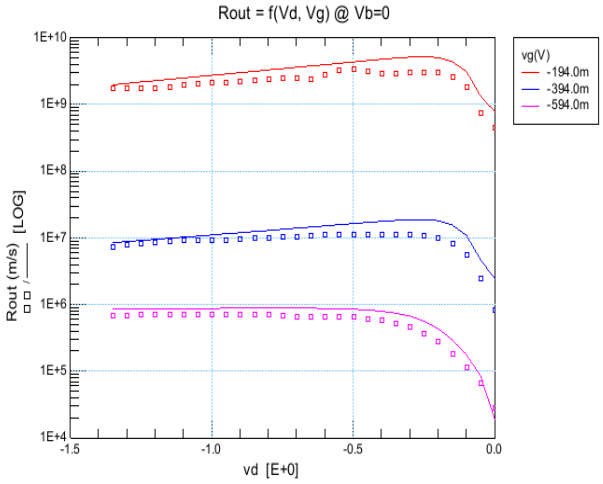


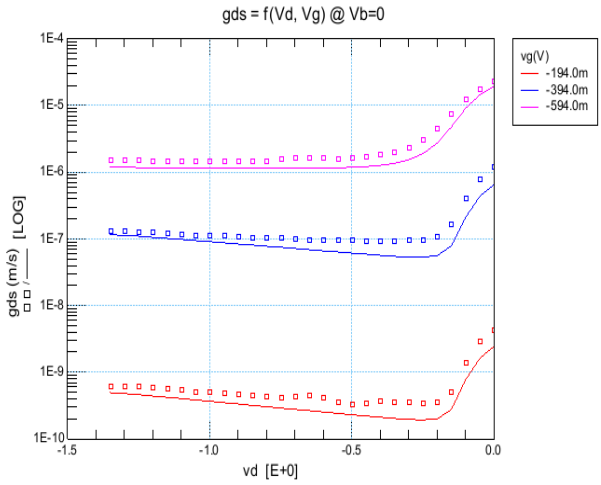




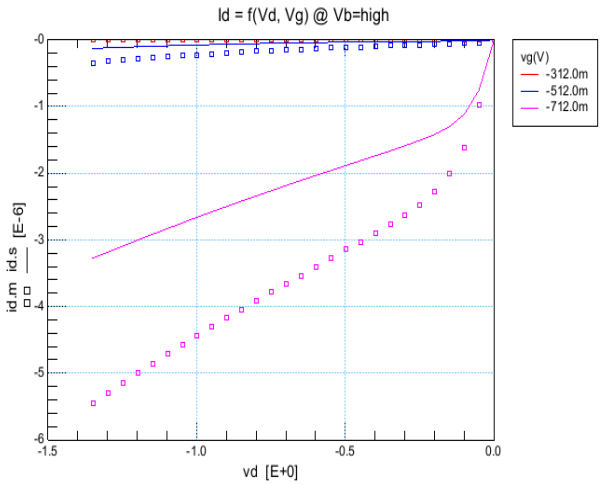
idvd_vth, Vb = 0V, T = 27°C

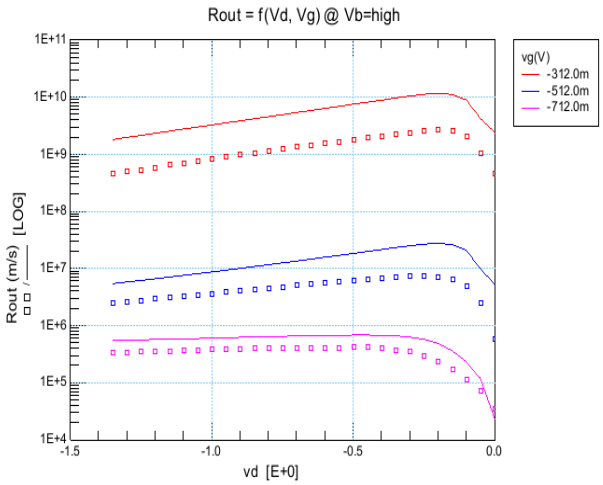


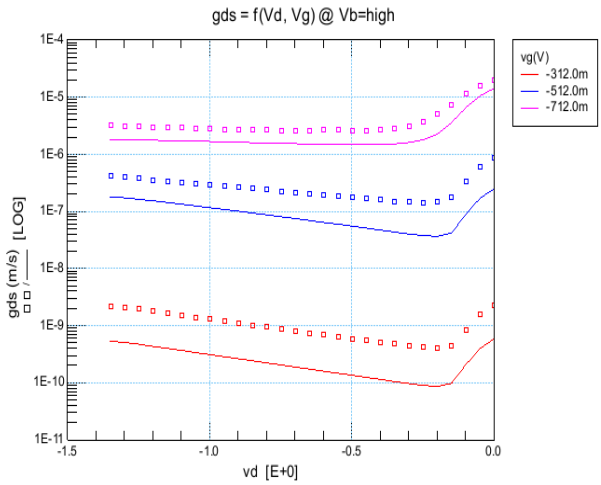




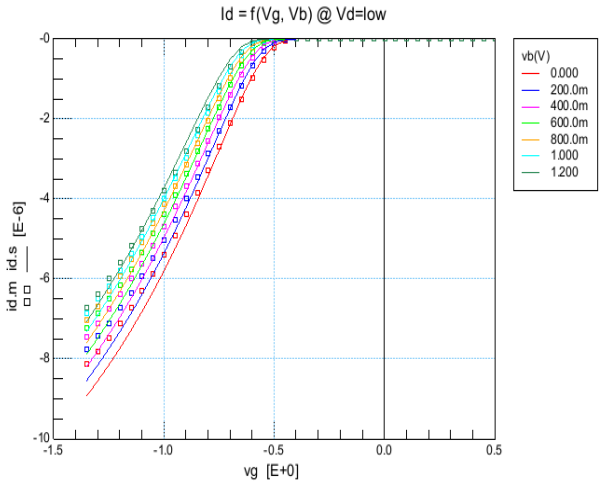
idvd_vth_vbmin, Vb = 1.2V, T = 27°C

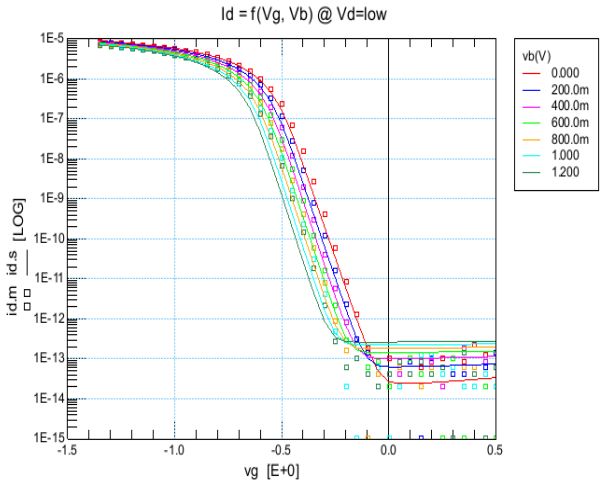


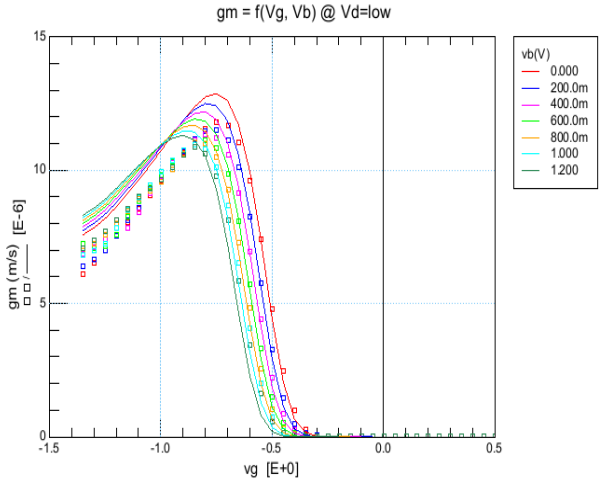


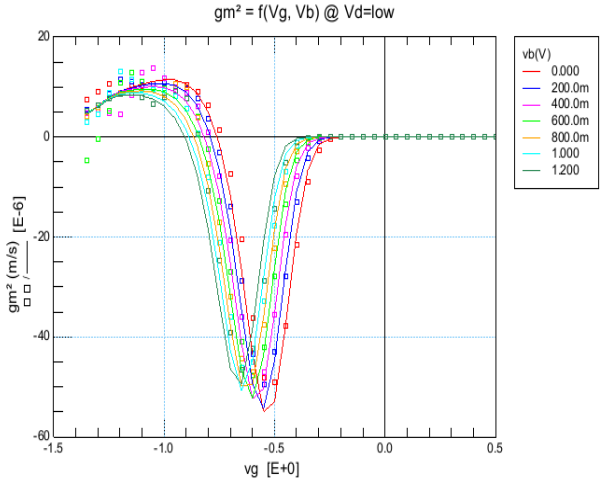


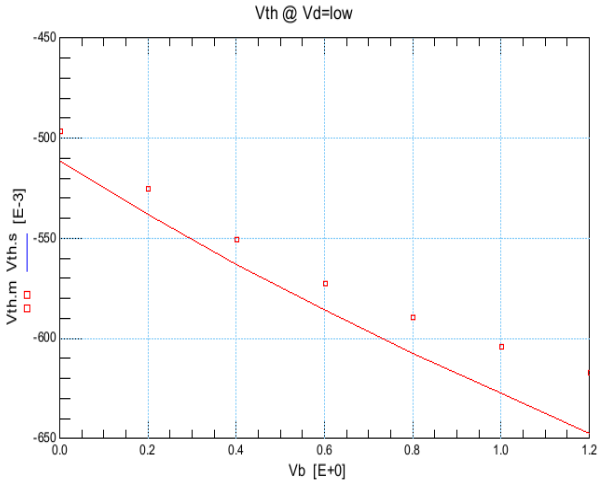
idvg, Vd = -0.05V, T = -40°C

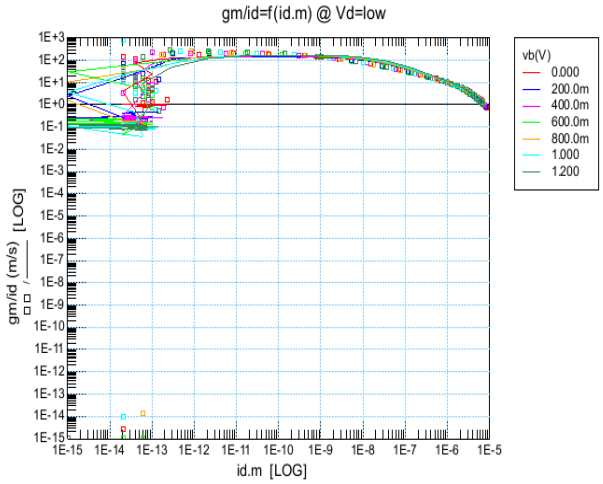


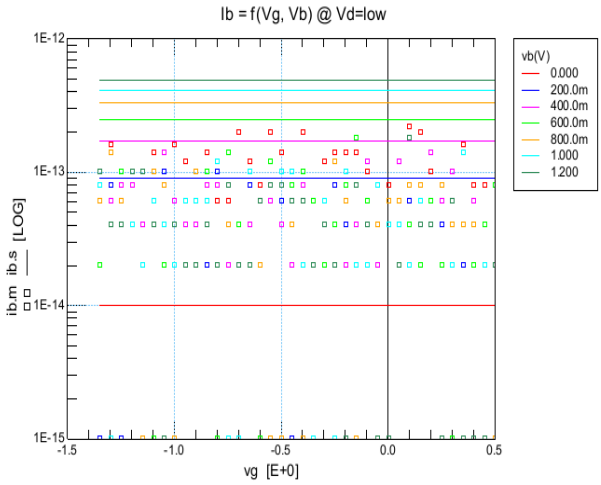


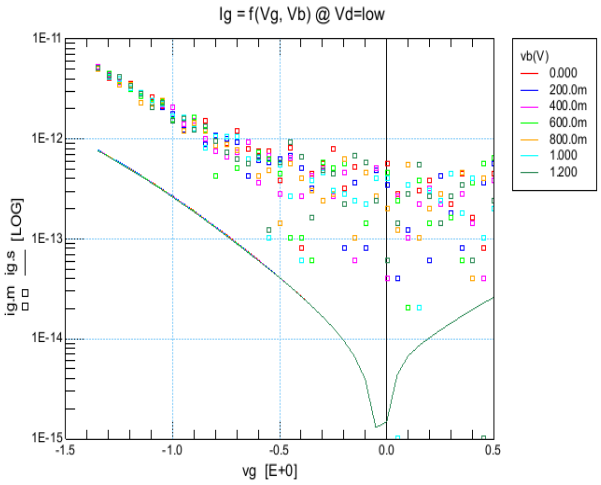




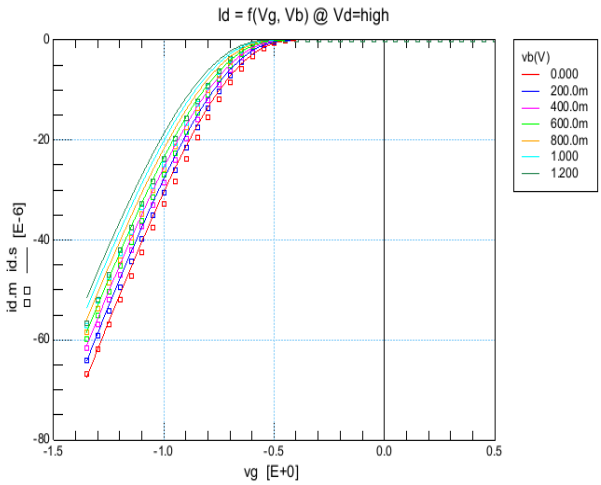


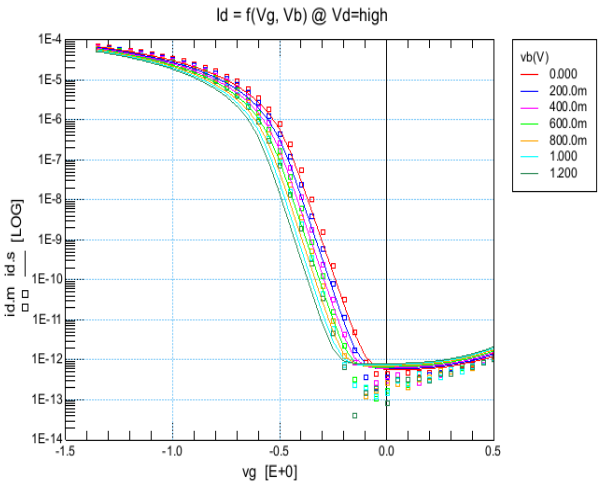


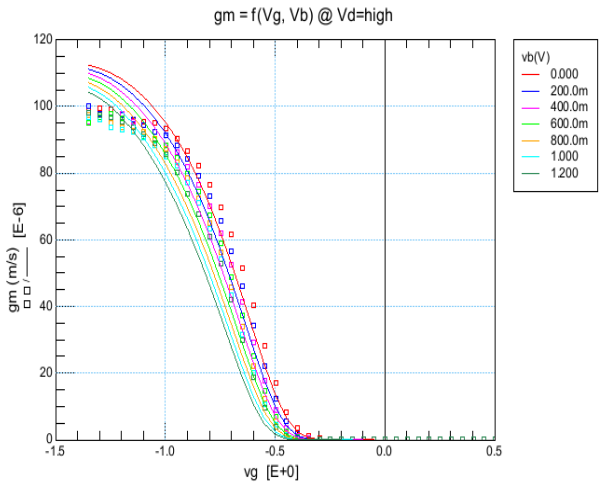


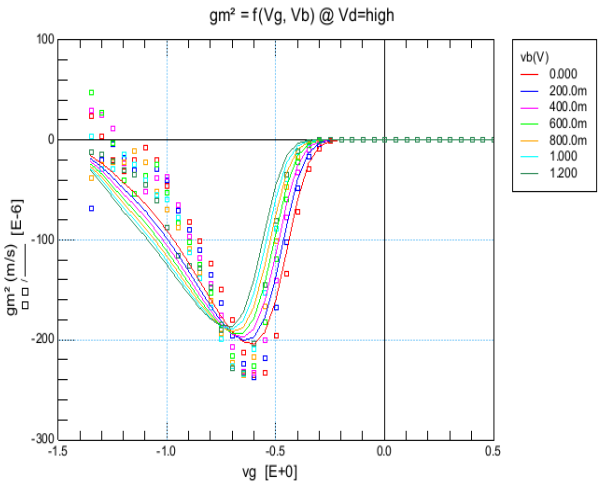


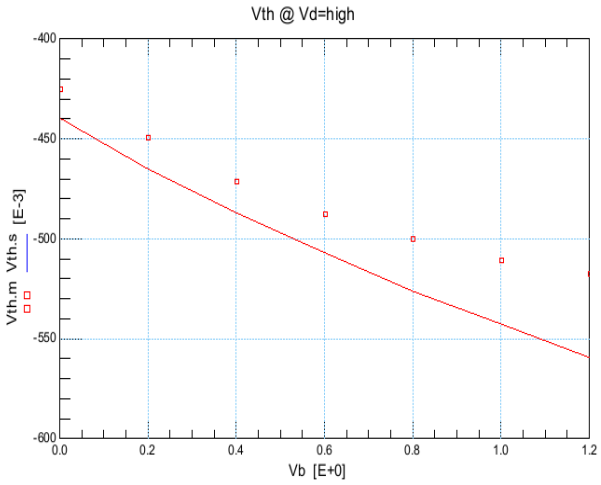
idvg, Vd = -1.2V, T = -40°C

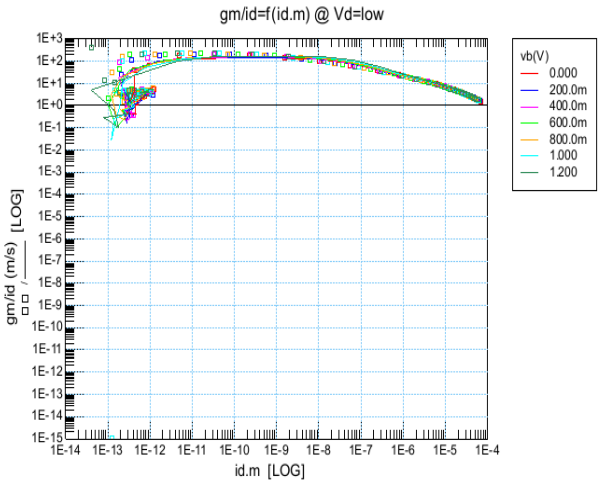


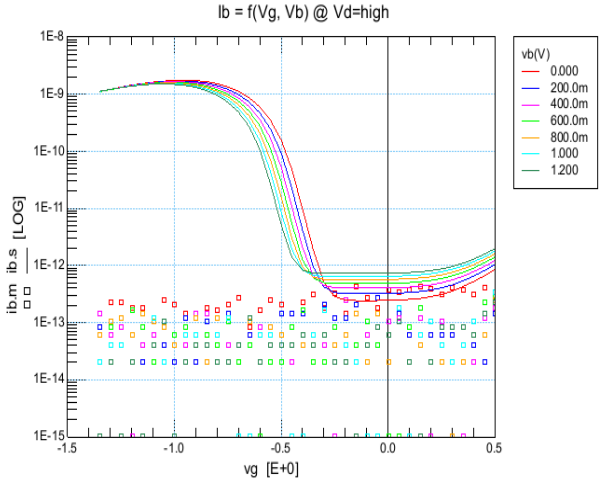


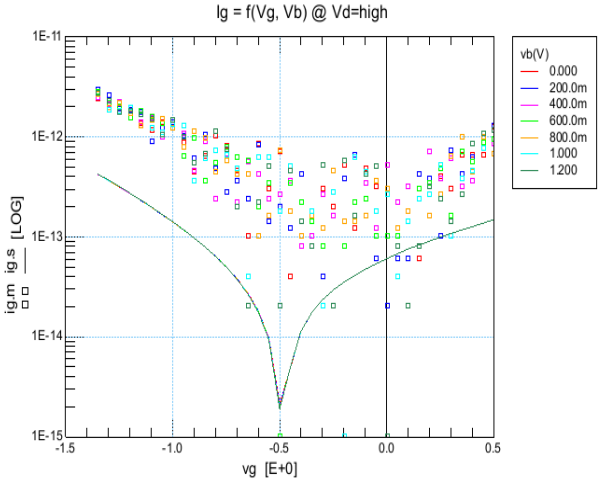




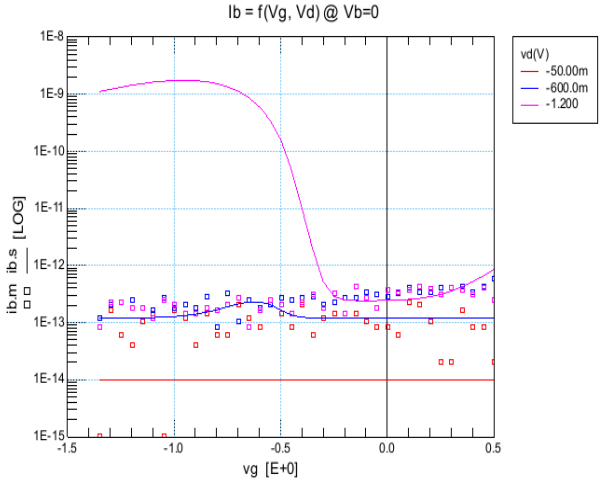


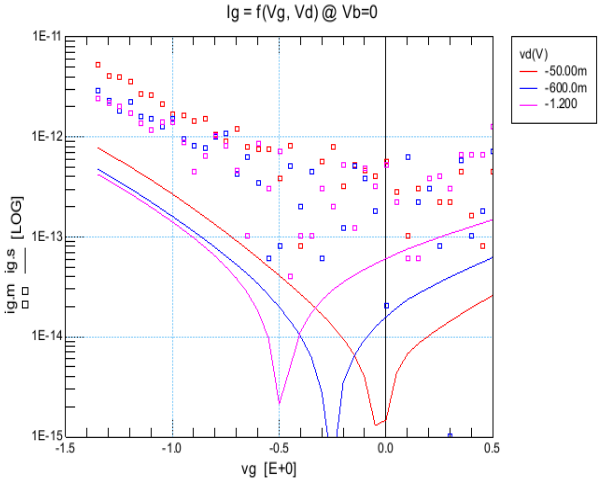




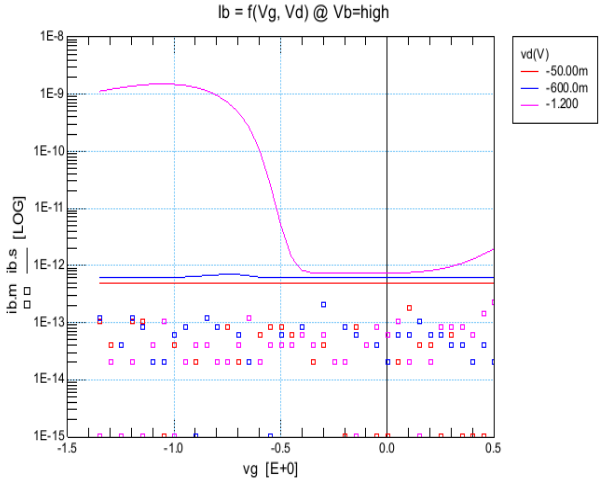


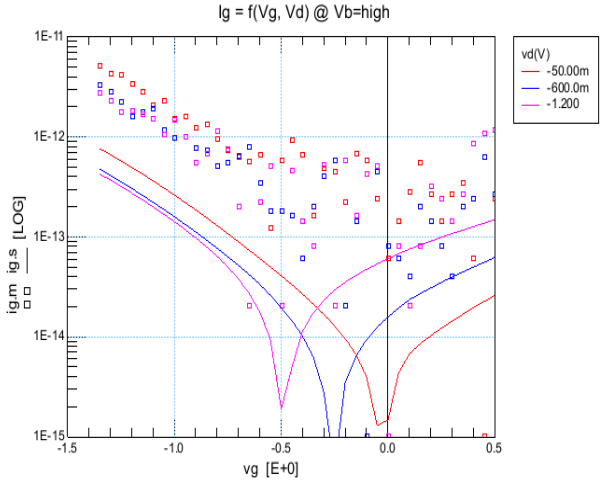
idvg, Ib,Ig, Vb = 0V, T = -40°C



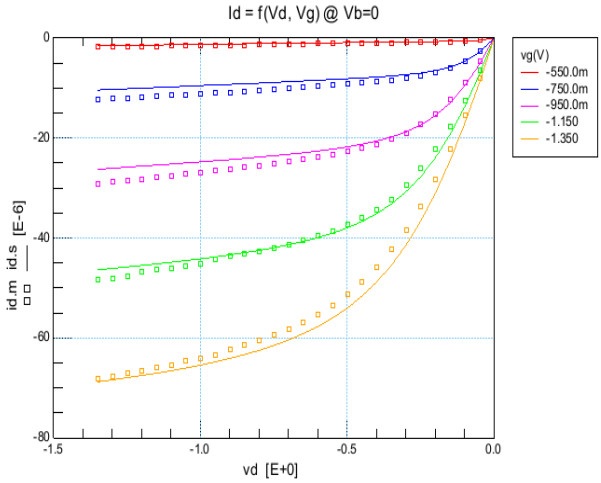


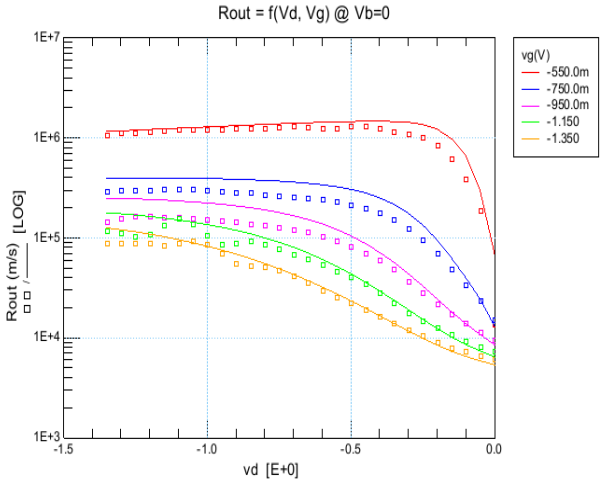
idvg, Ib,Ig, Vb = 1.2V, T = -40°C

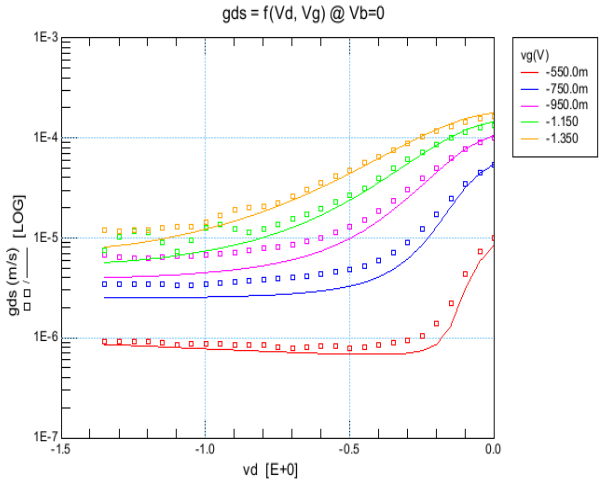




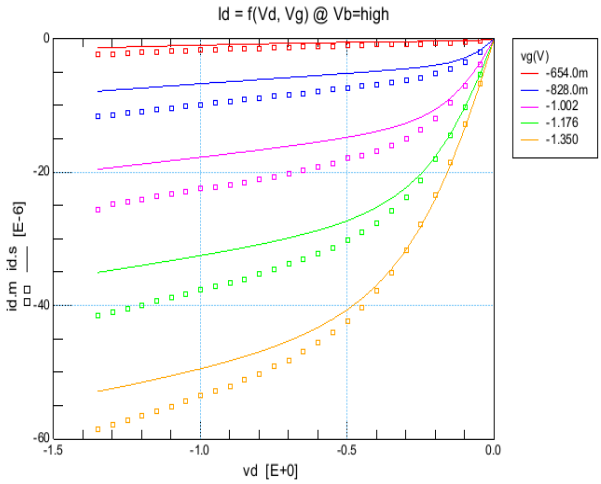
idvd, Vb = 0V, T = -40°C

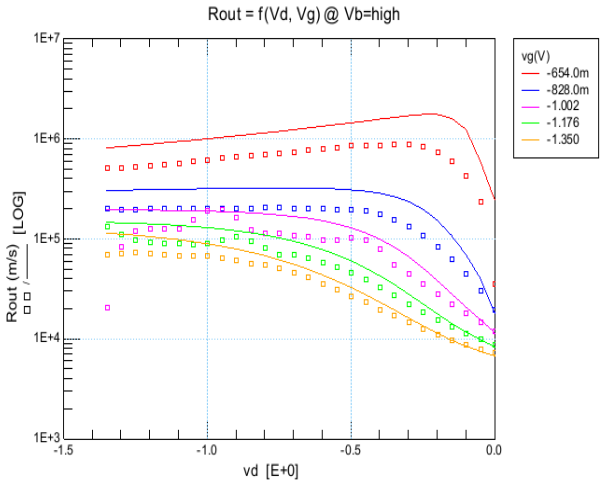


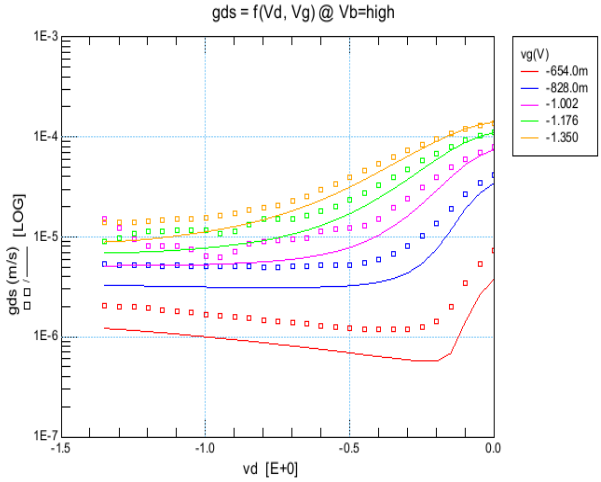




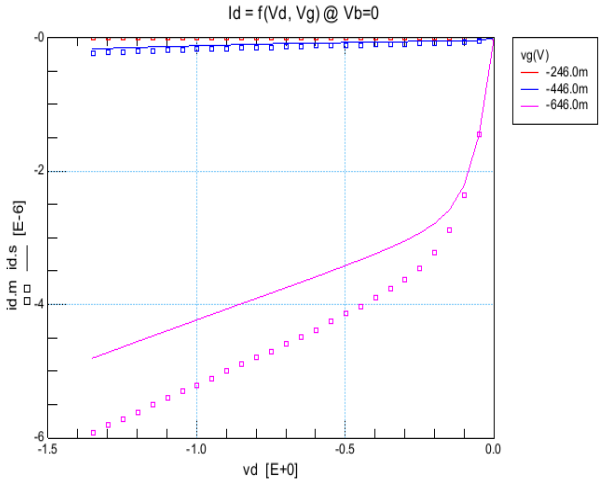
idvd_vbmin, Vb = 1.2V, T = -40°C

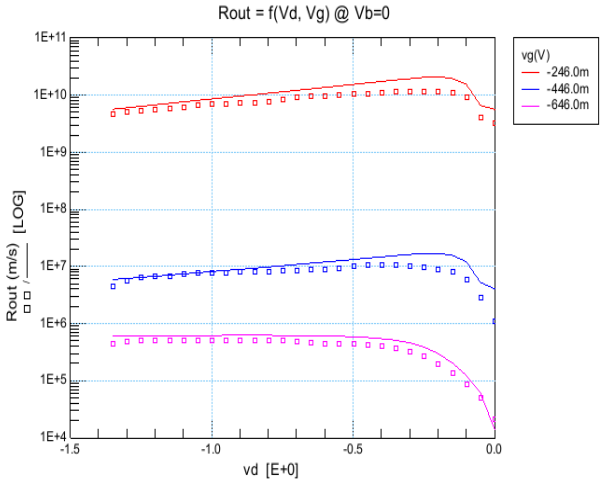


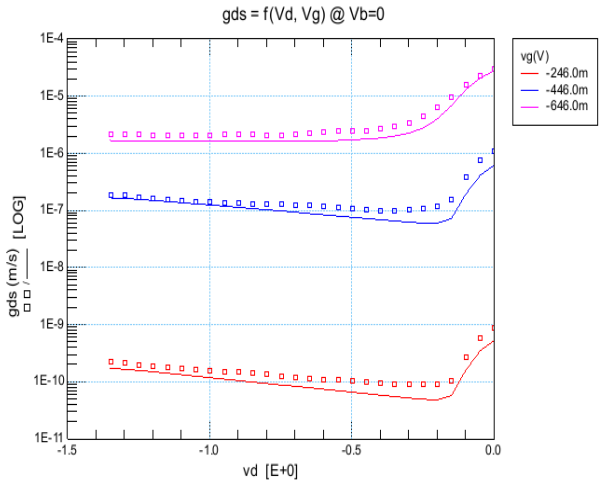




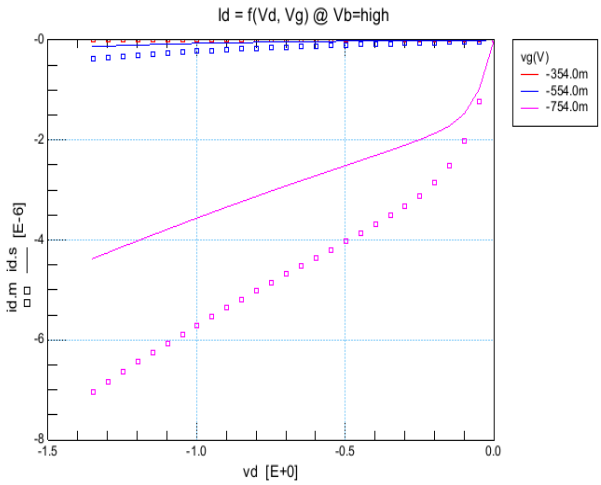
idvd_vth, Vb = 0V, T = -40°C

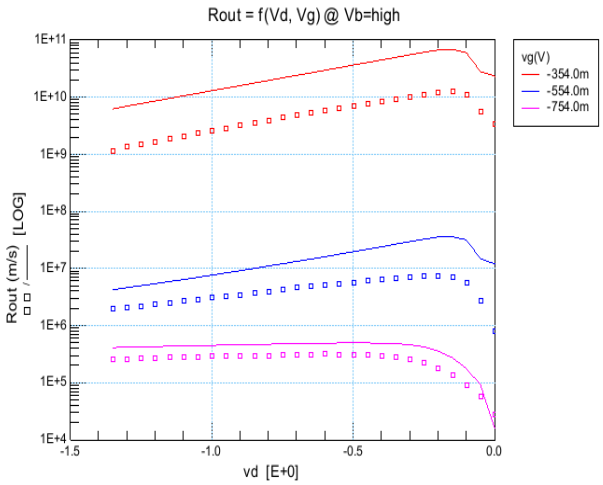


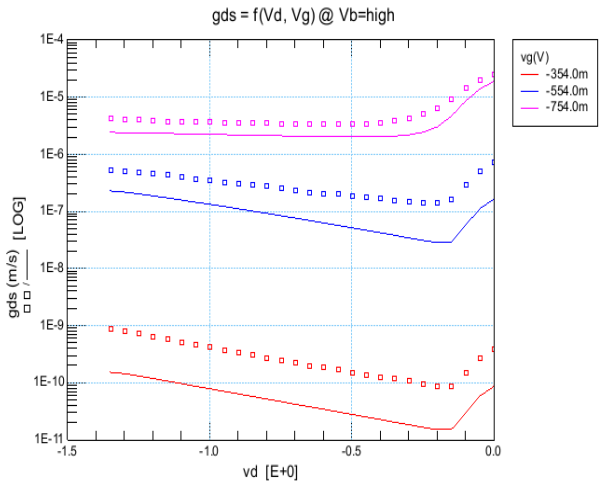




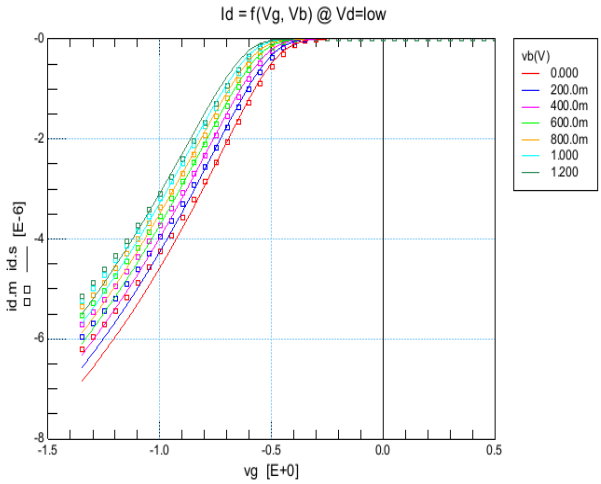
idvd_vth_vbmin, Vb = 1.2V, T = -40°C

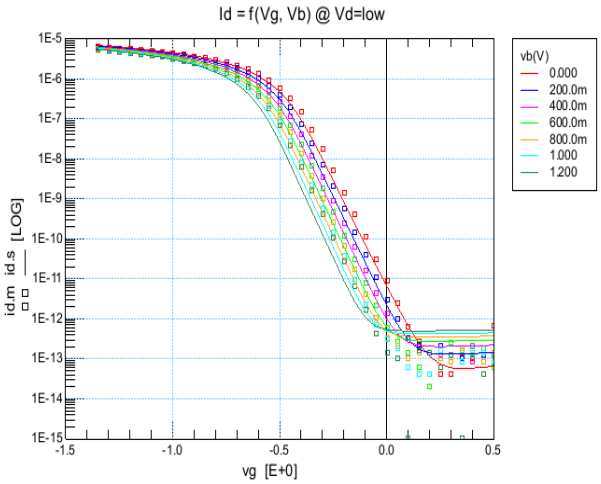


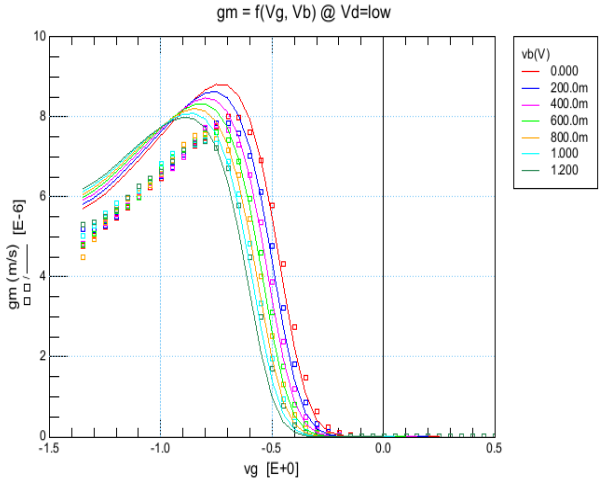


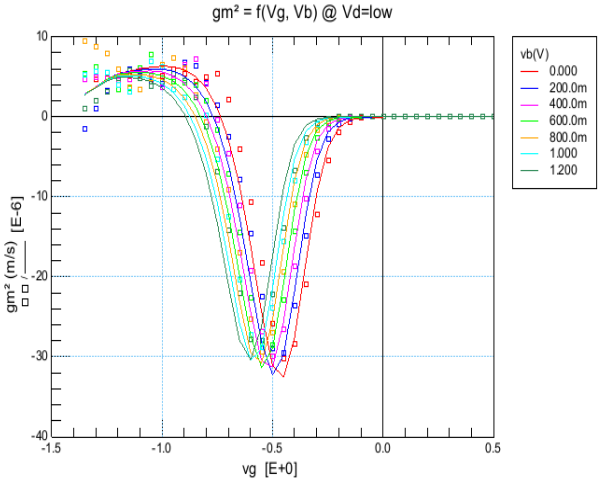


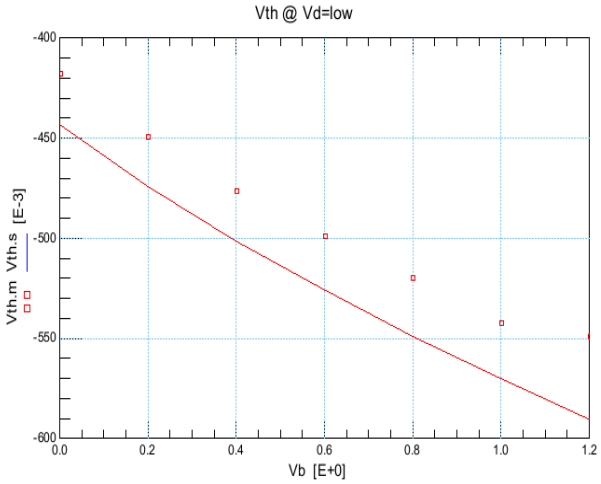
idvg, Vd = -0.05V, T = 70°C

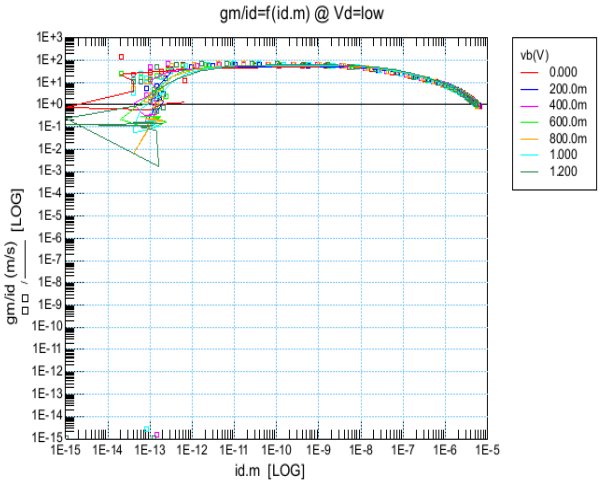


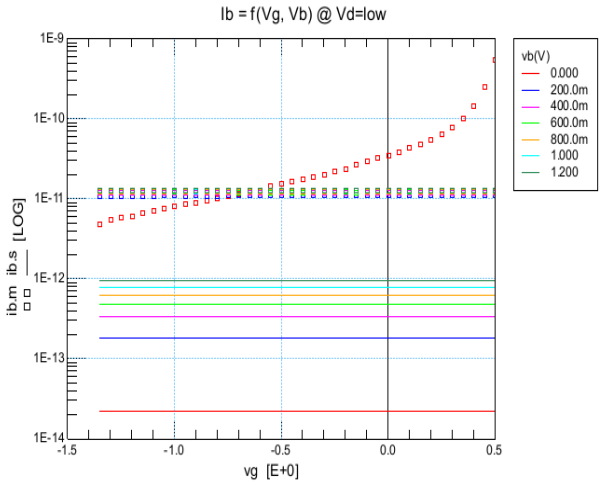


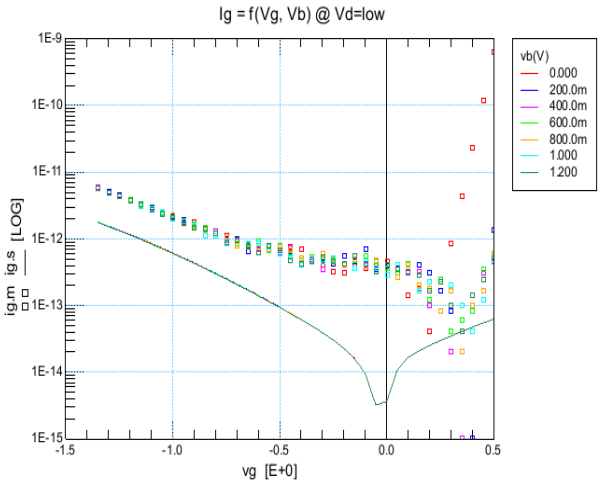




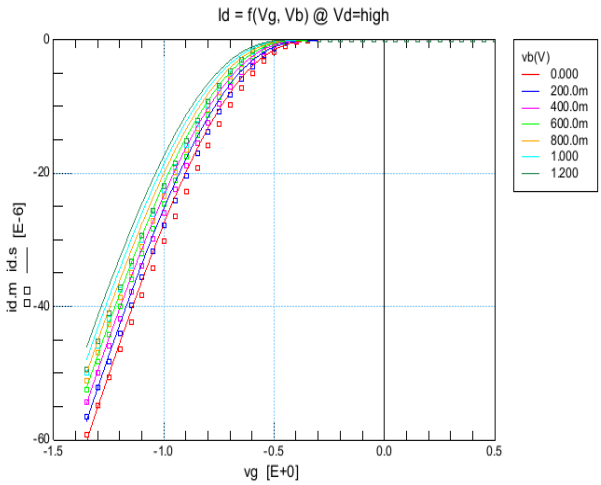


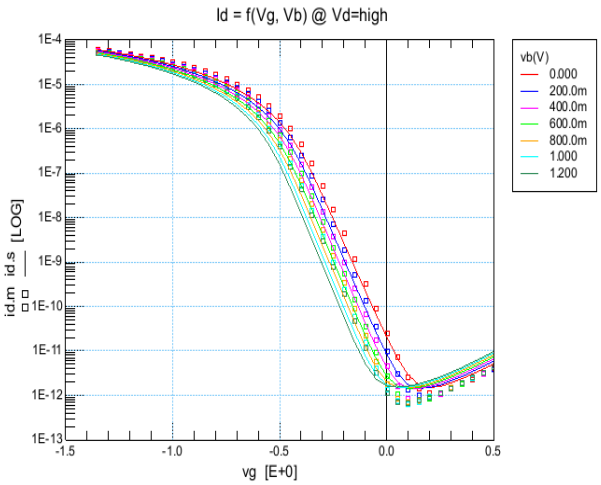


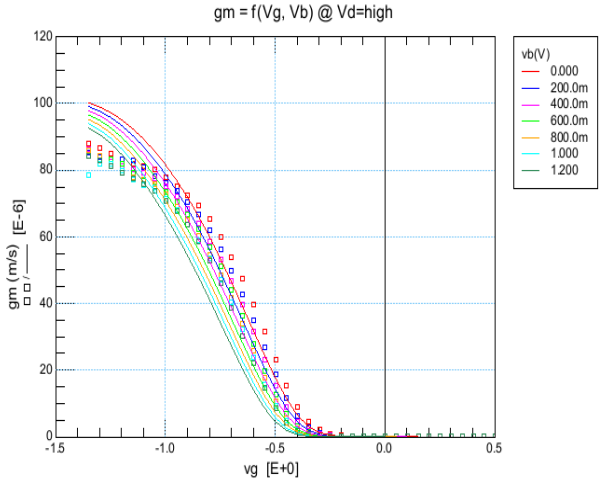


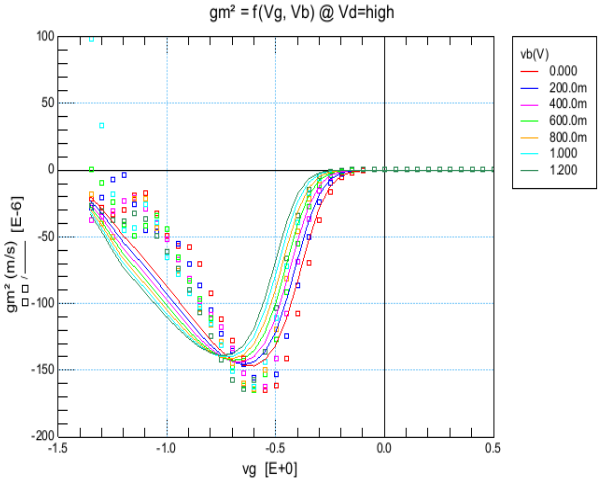


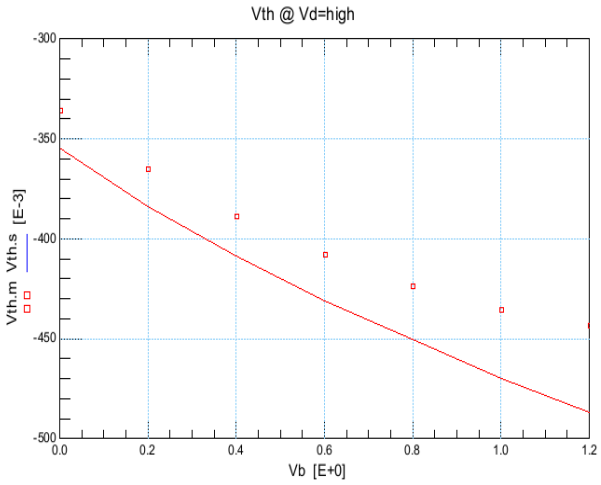
idvg, Vd = -1.2V, T = 70°C

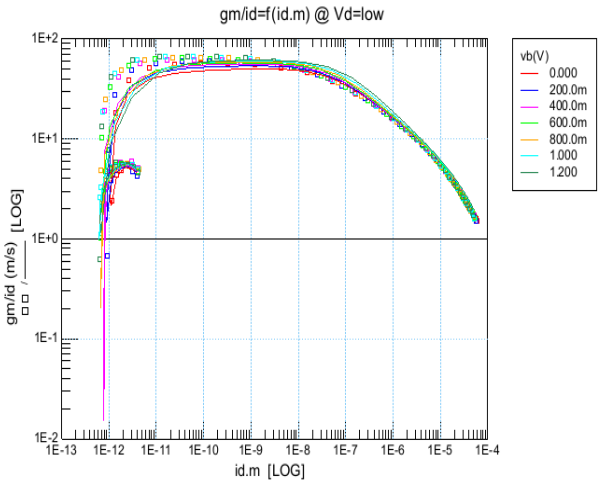


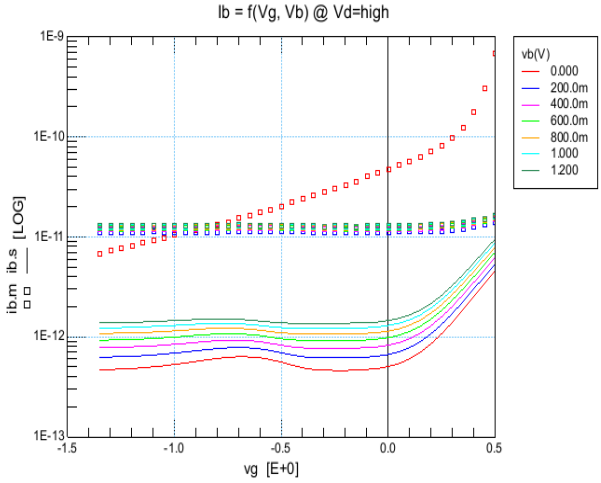


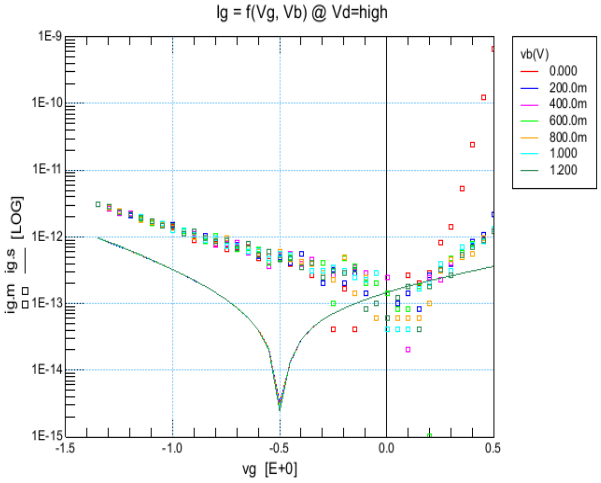




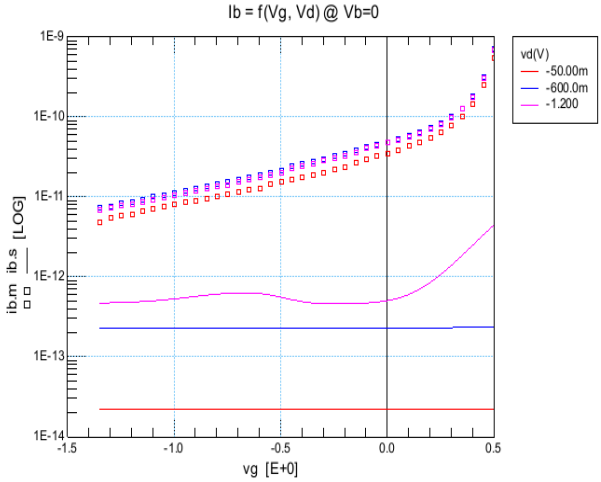


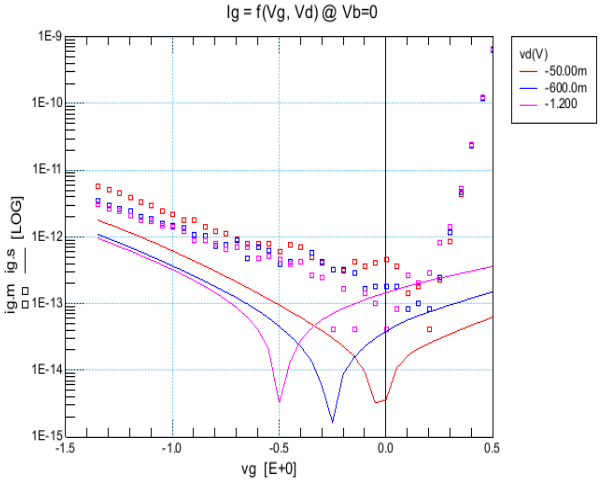




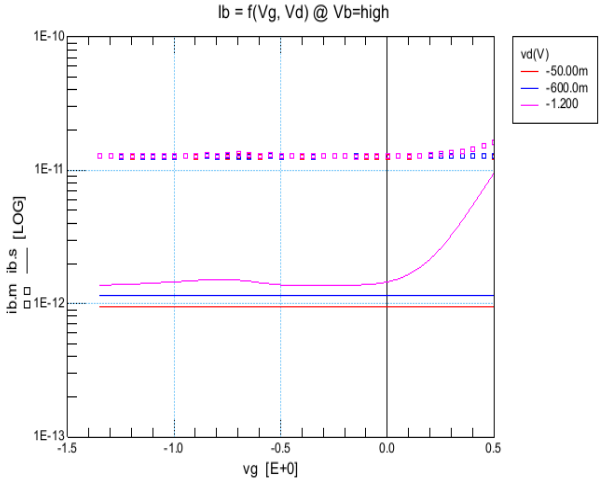


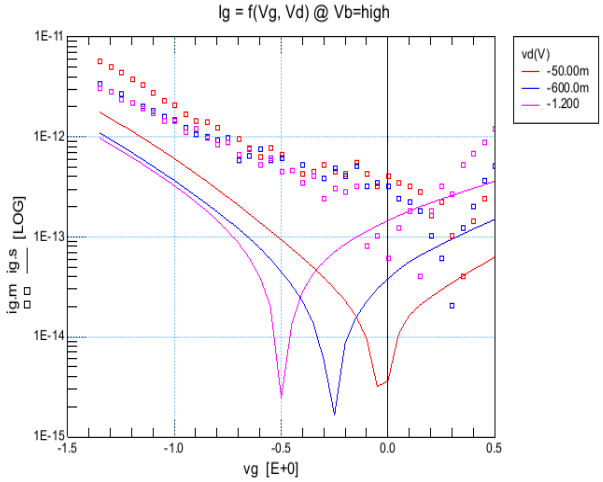
idvg, Ib,Ig, Vb = 0V, T = 70°C



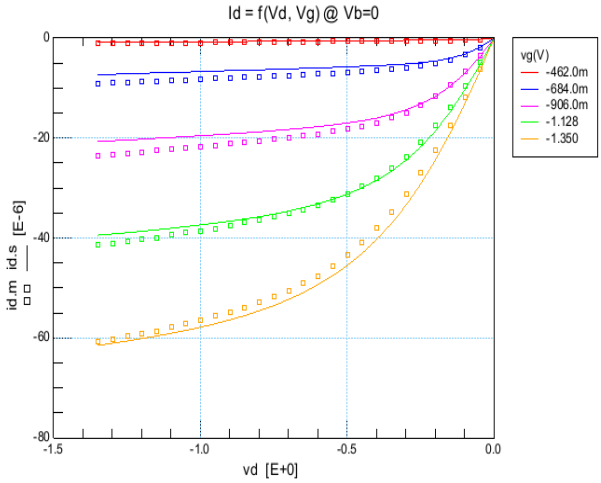


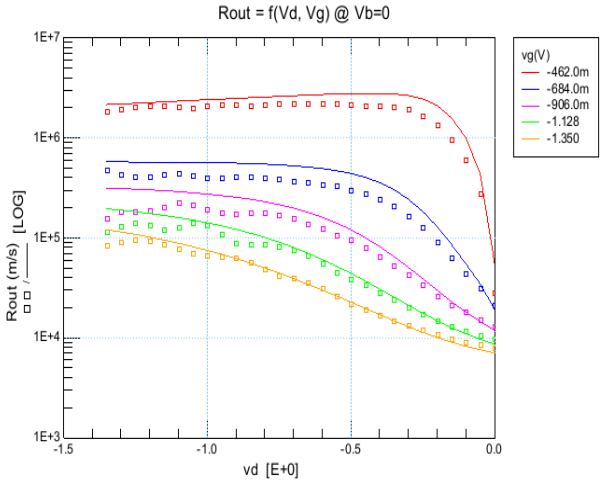
idvg, Ib,Ig, Vb = 1.2V, T = 70°C

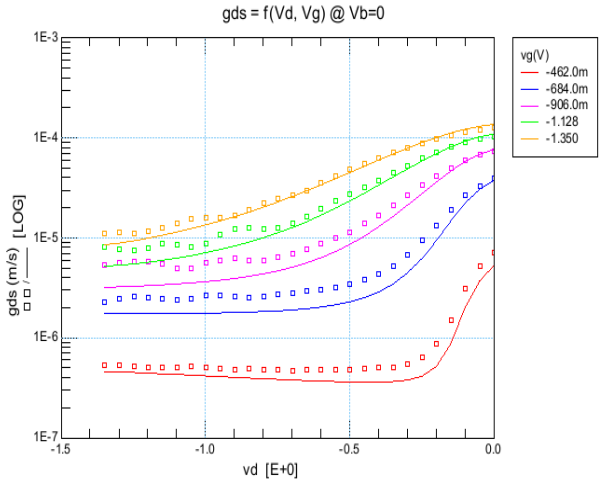




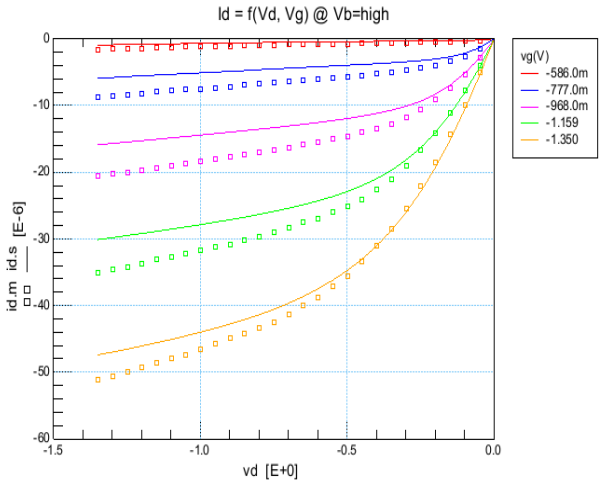
idvd, Vb = 0V, T = 70°C

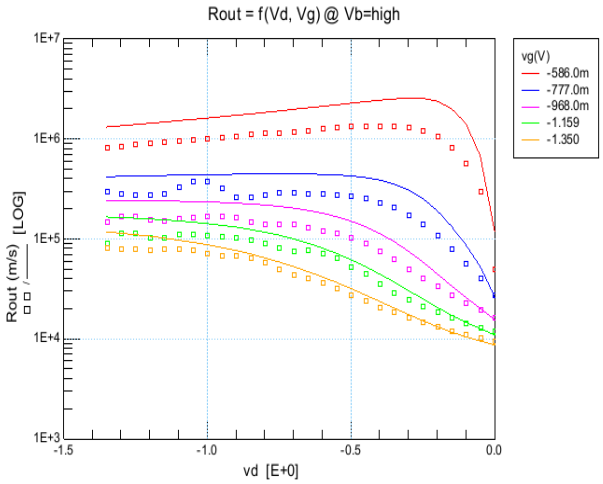


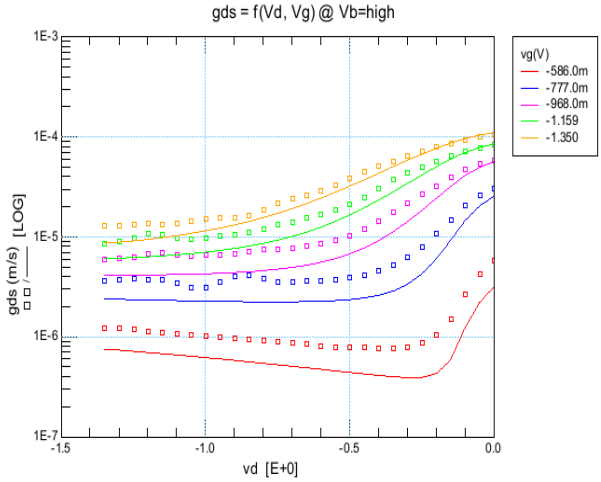




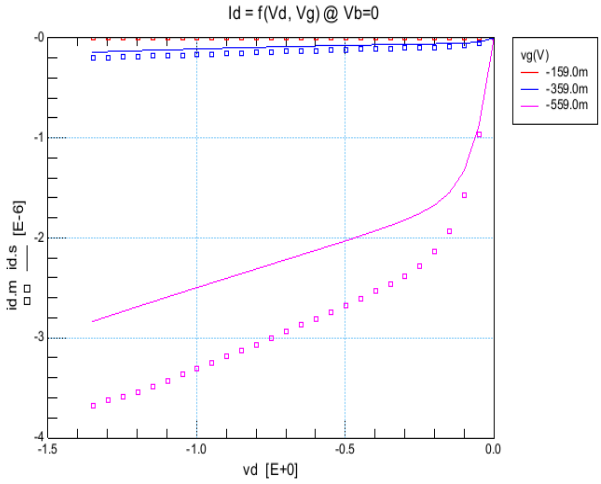
idvd_vbmin, Vb = 1.2V, T = 70°C

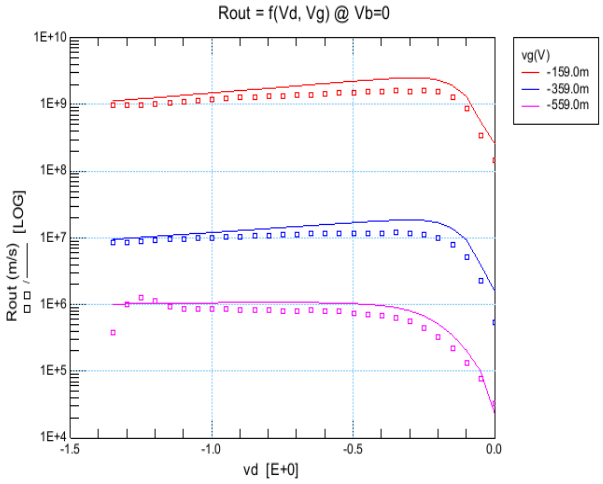


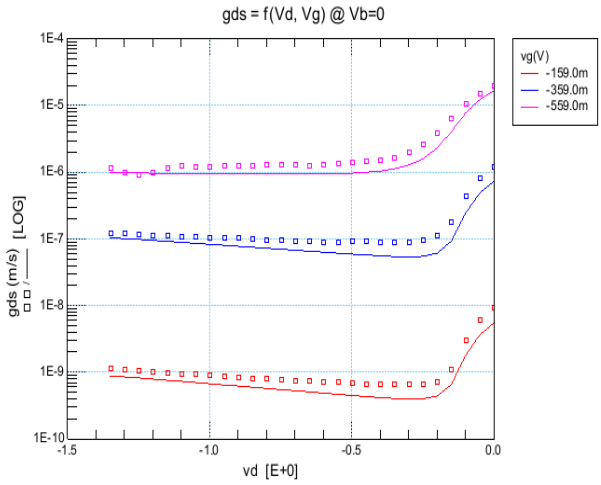




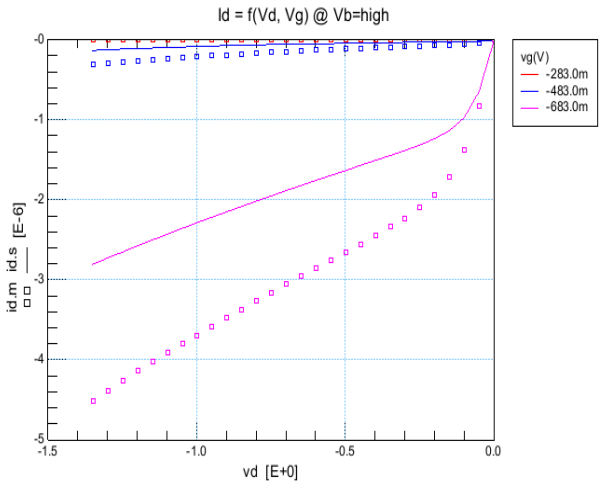
idvd_vth, Vb = 0V, T = 70°C

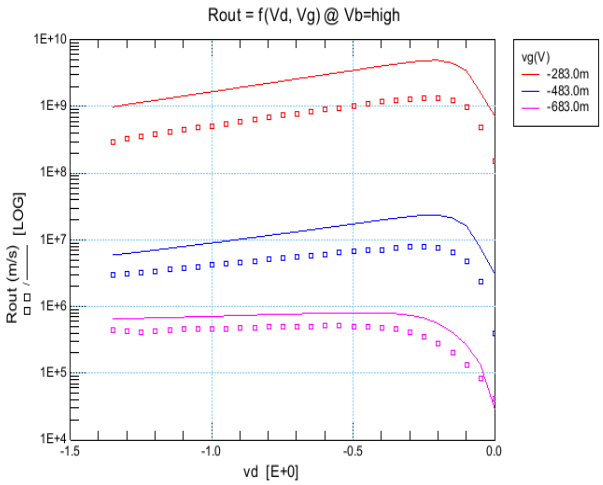


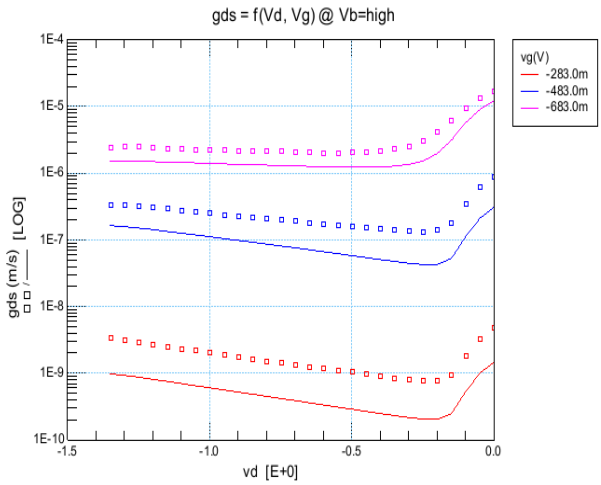




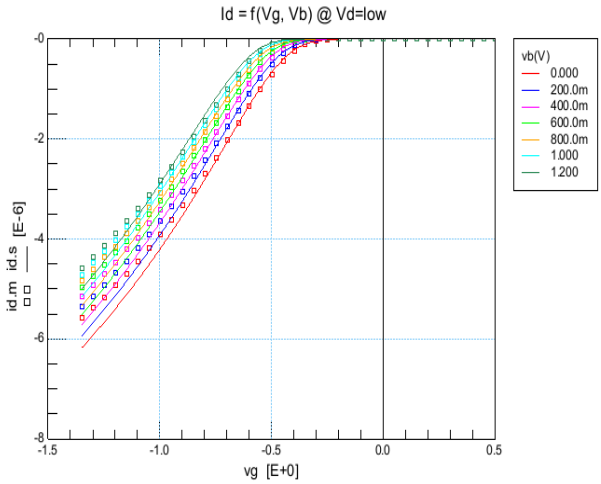
idvd_vth_vbmin, Vb = 1.2V, T = 70°C

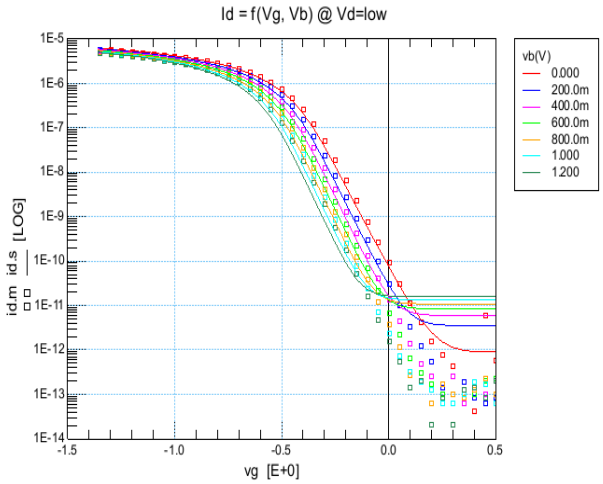


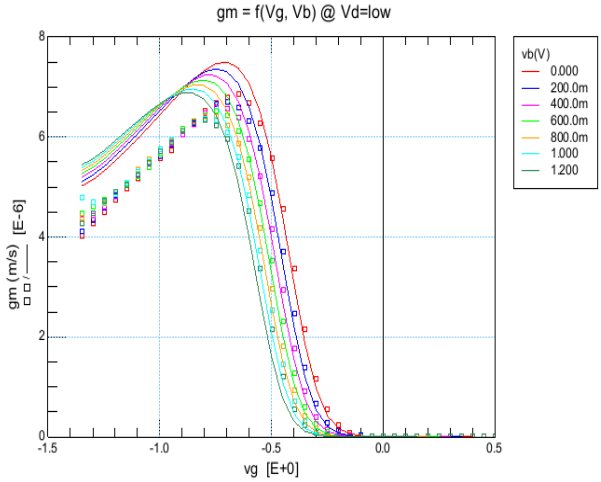


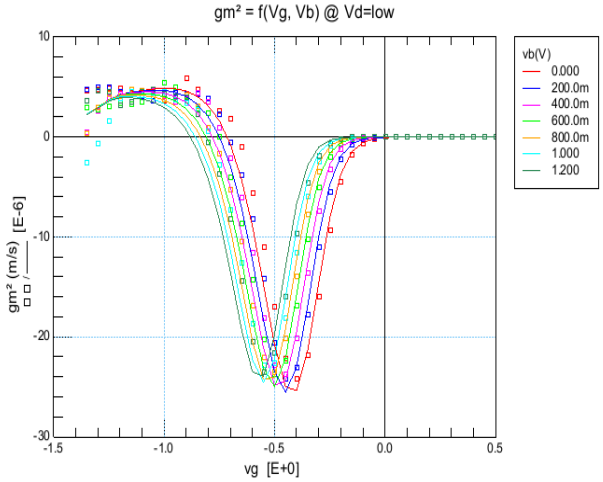


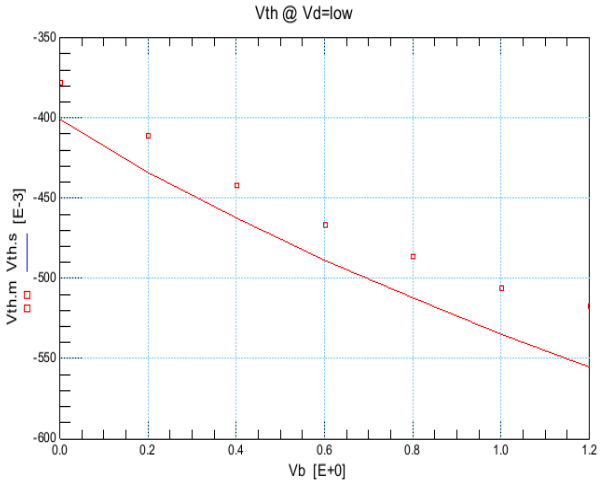
idvg, Vd = -0.05V, T = 125°C

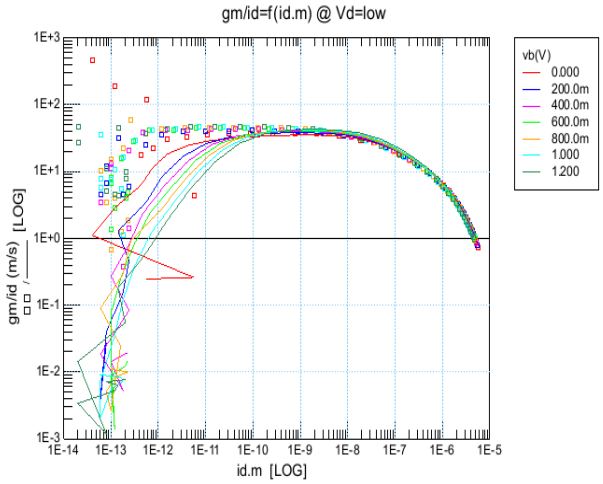


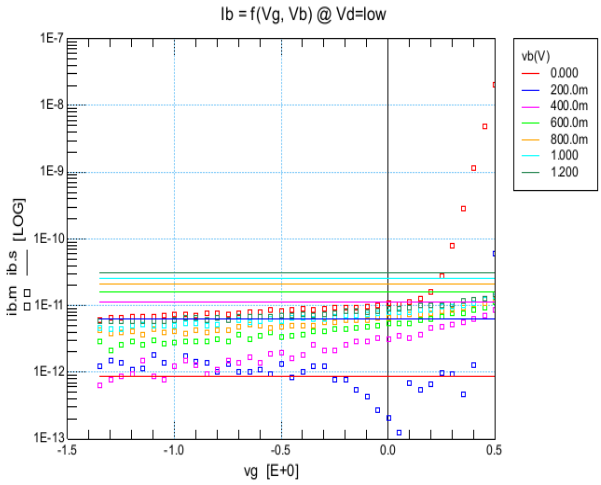


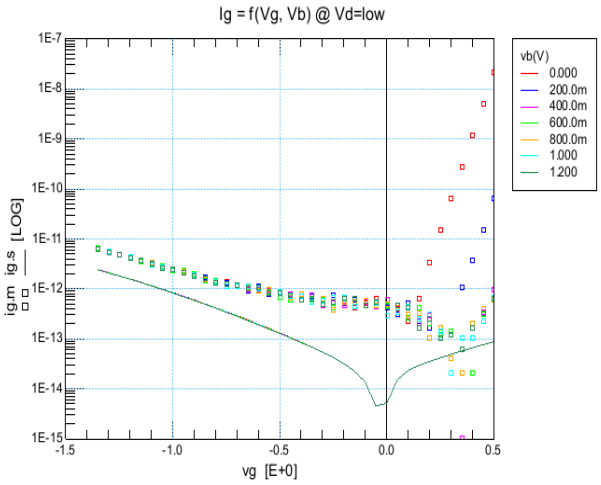




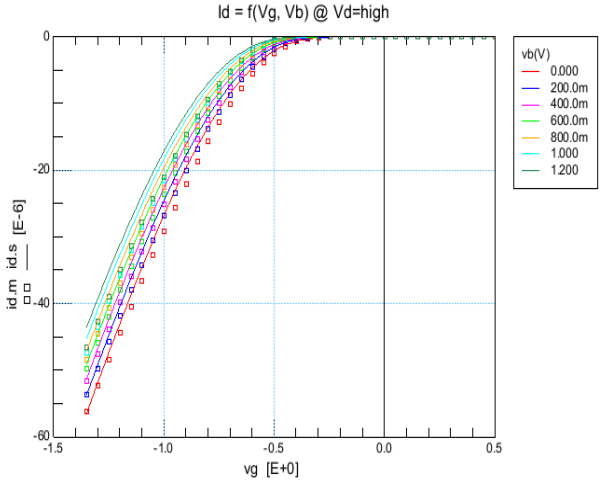


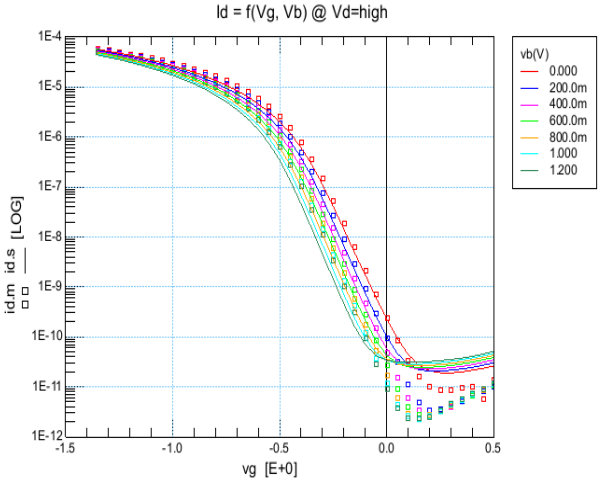


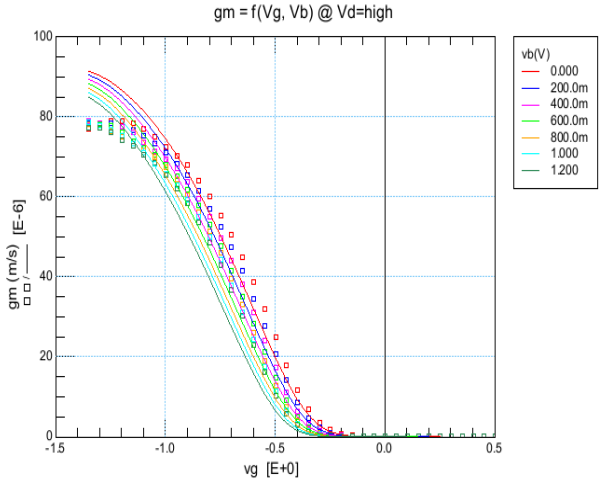


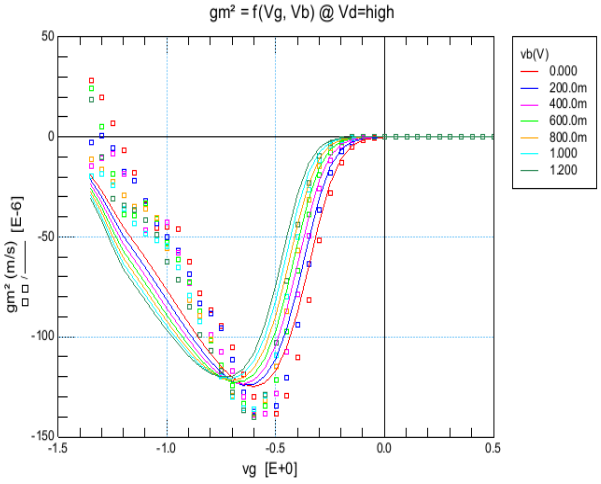


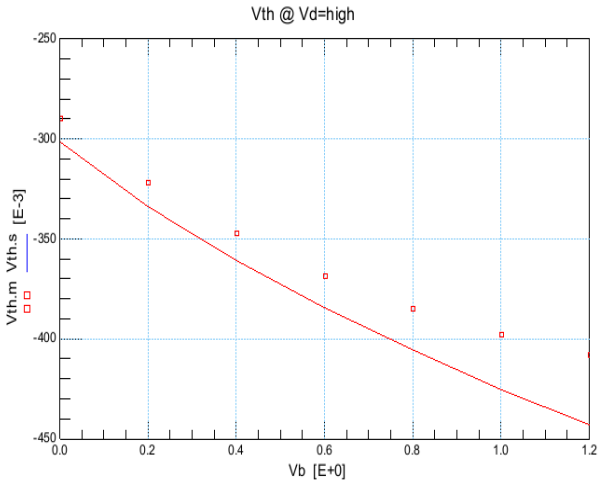
idvg, Vd = -1.2V, T = 125°C

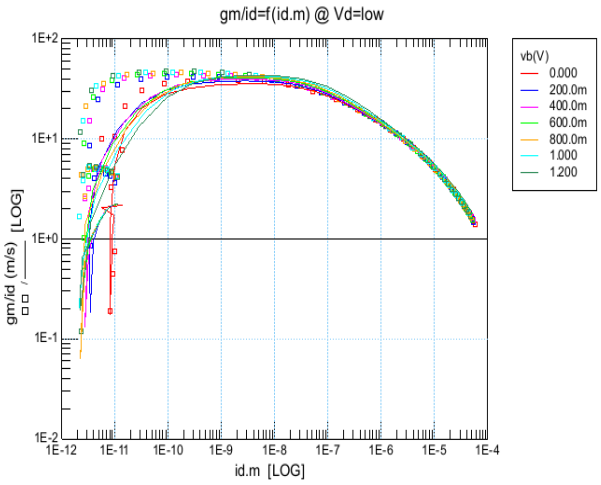


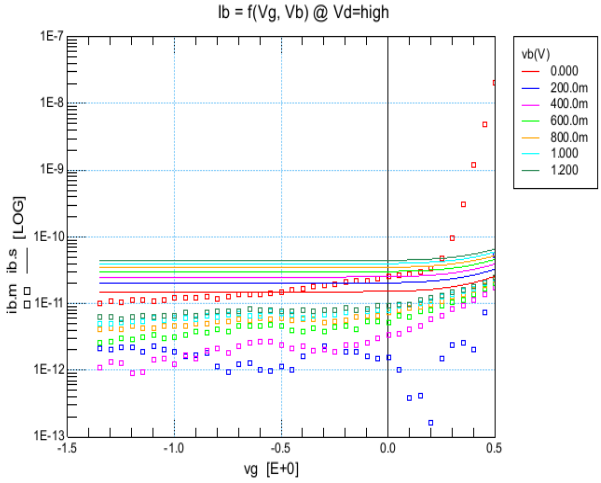


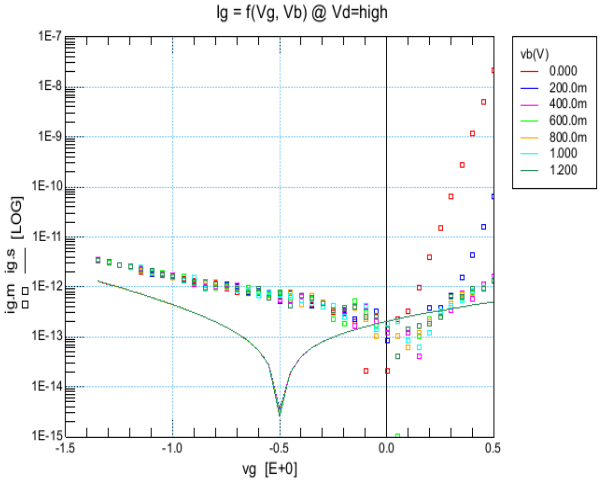




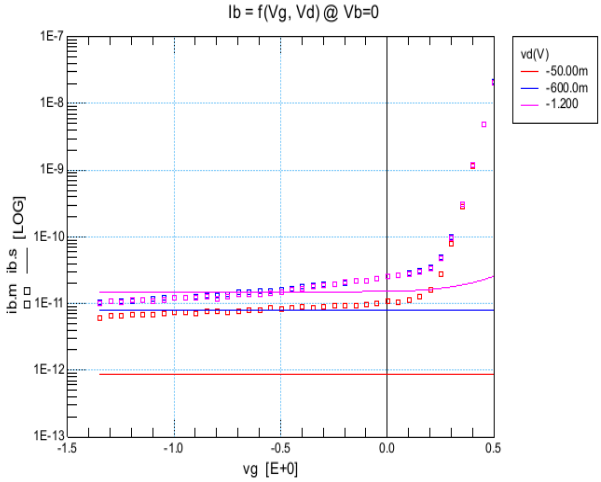


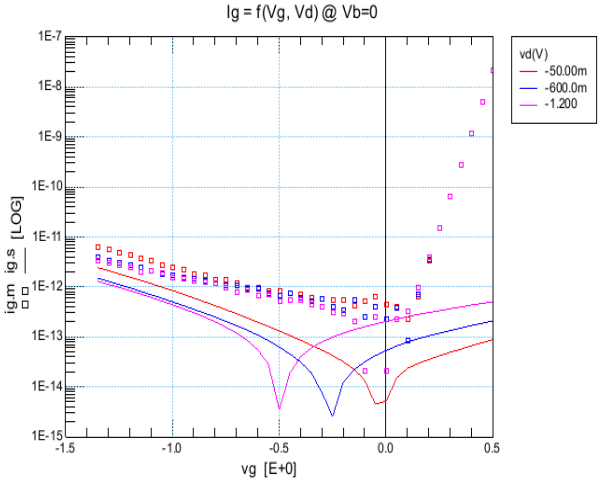




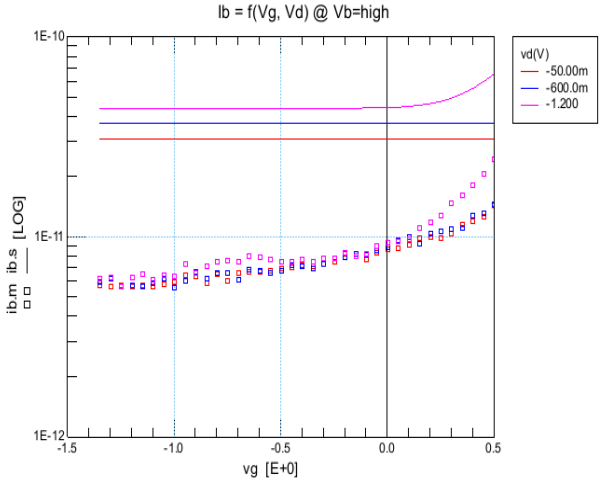


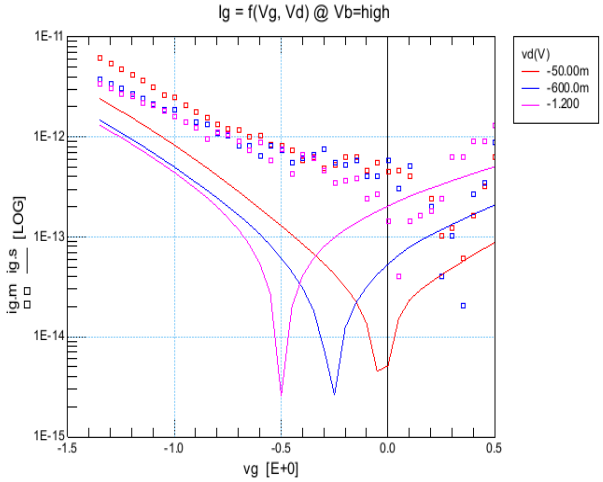
idvg, Ib,Ig, Vb = 0V, T = 125°C



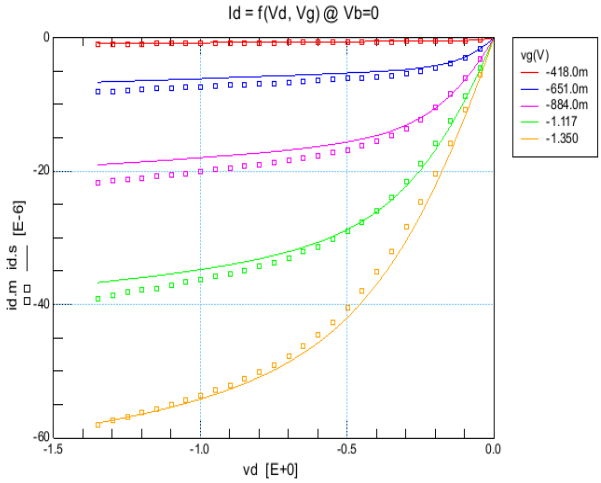


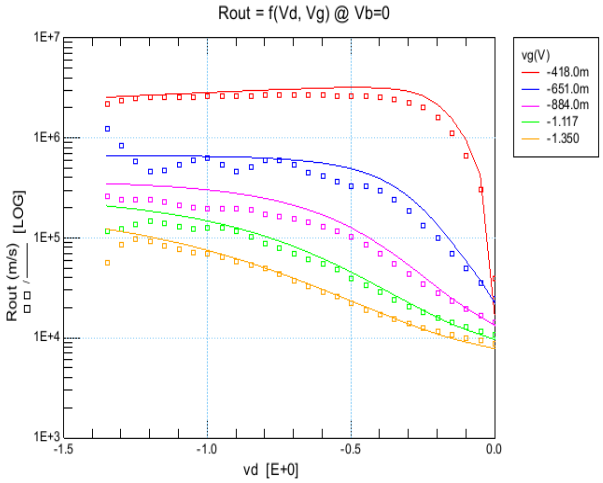
idvg, Ib,Ig, Vb = 1.2V, T = 125°C

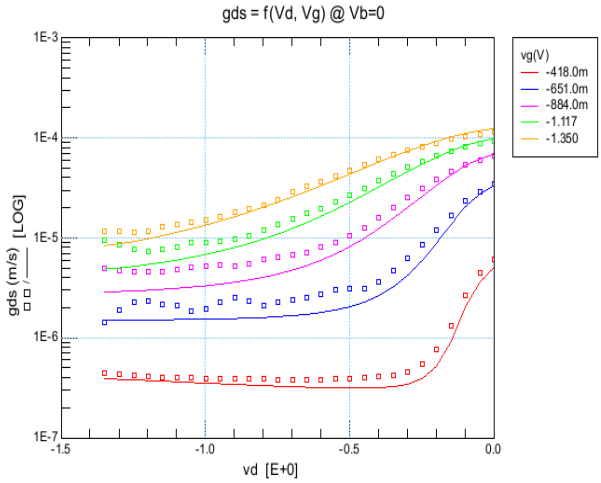




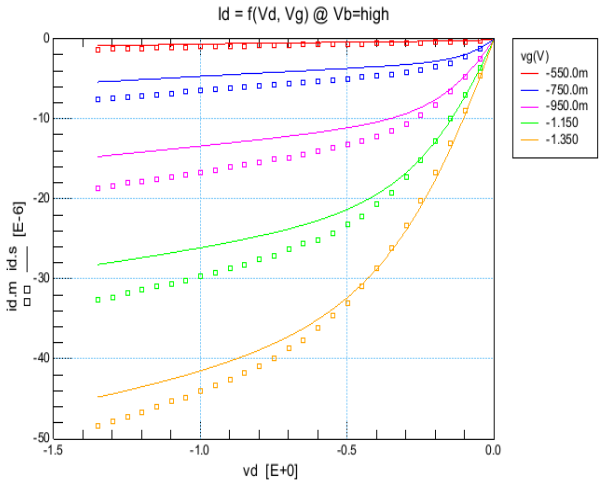
idvd, Vb = 0V, T = 125°C

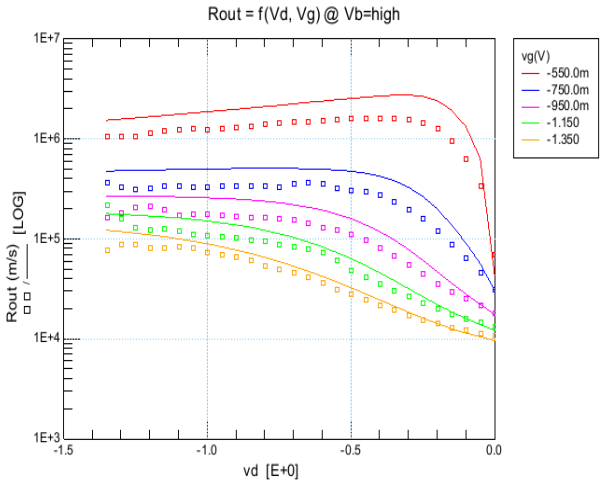


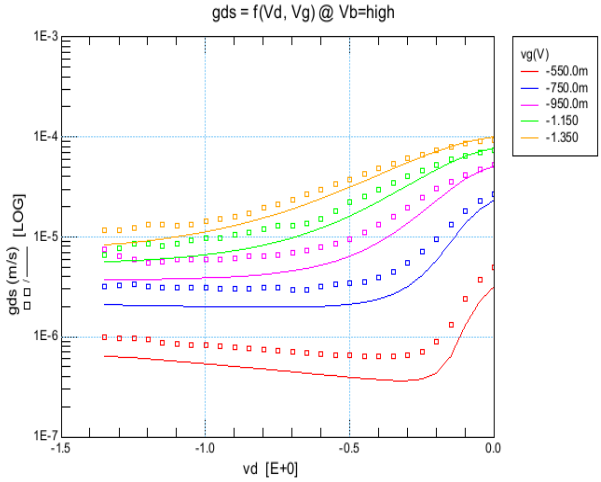




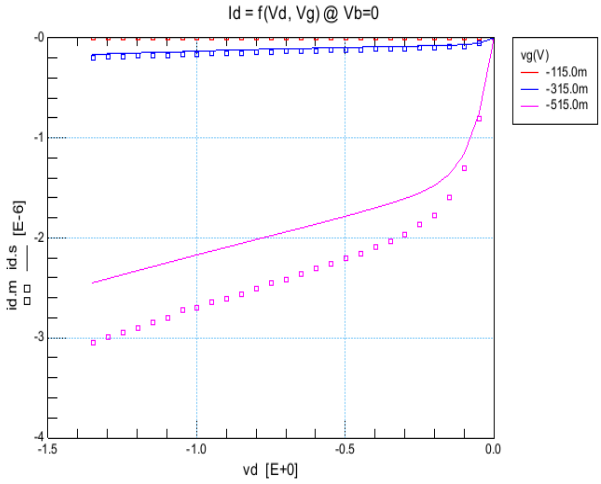
idvd_vbmin, Vb = 1.2V, T = 125°C

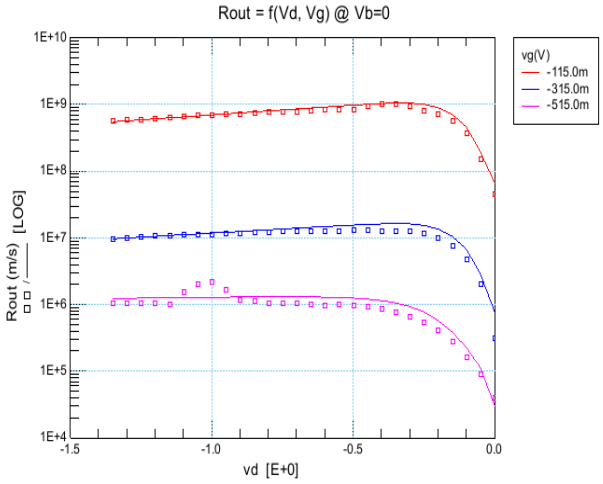


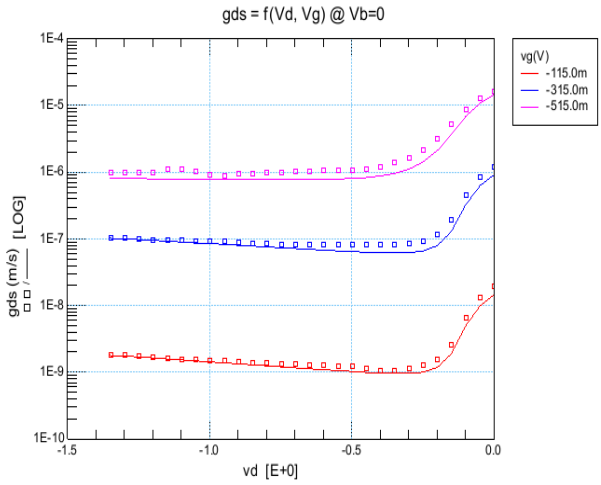




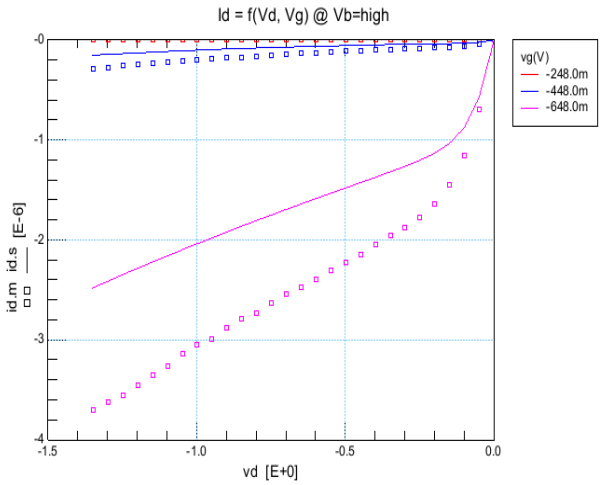
idvd_vth, Vb = 0V, T = 125°C

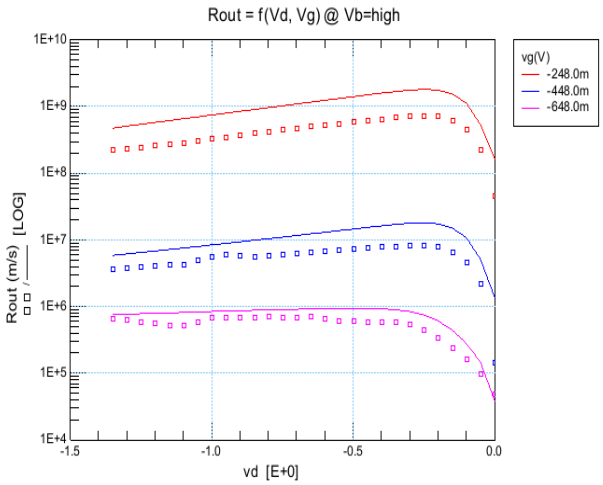


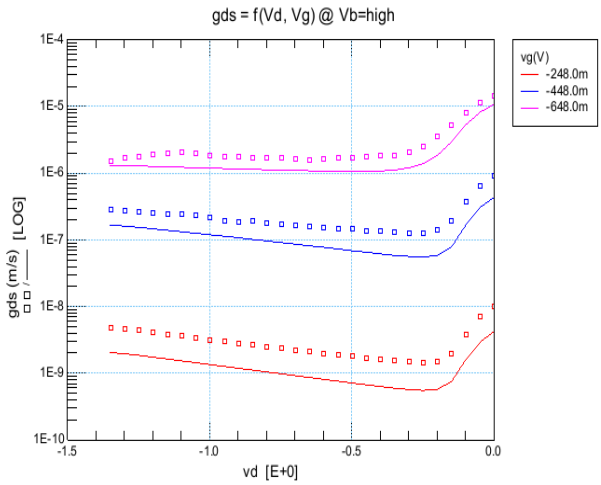




idvd_vth_vbmin, Vb = 1.2V, T = 125°C

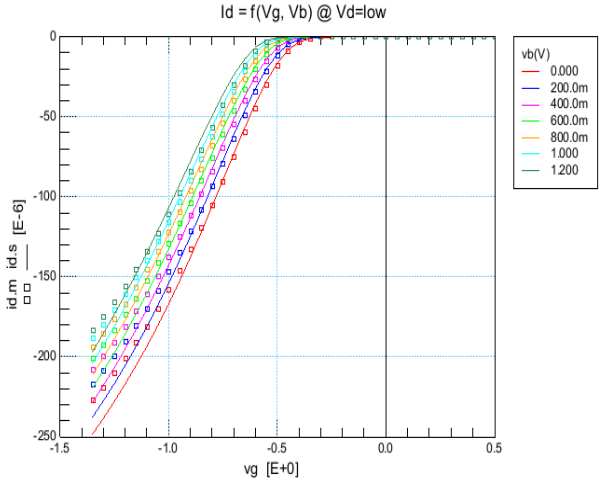


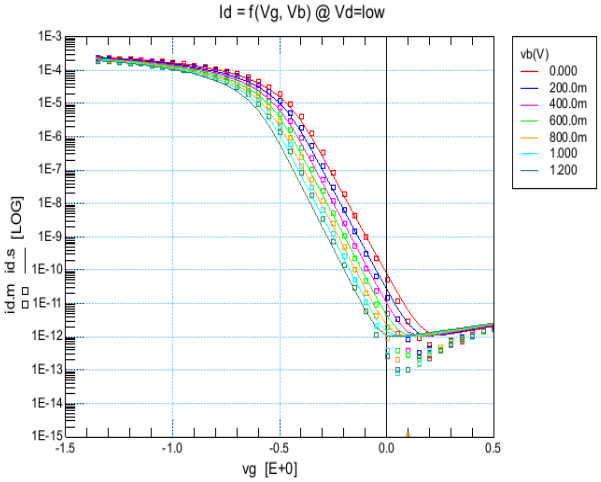


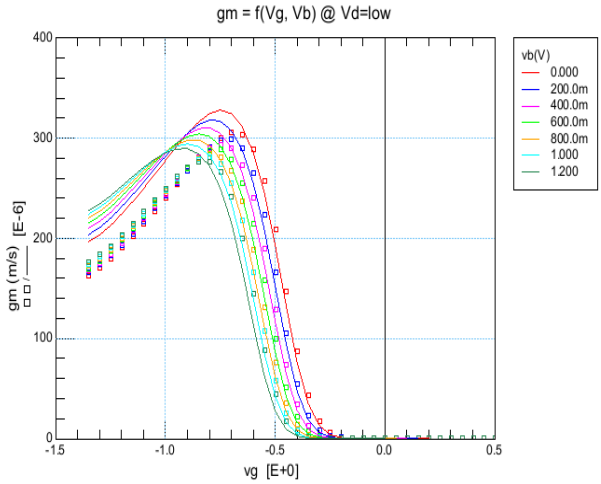


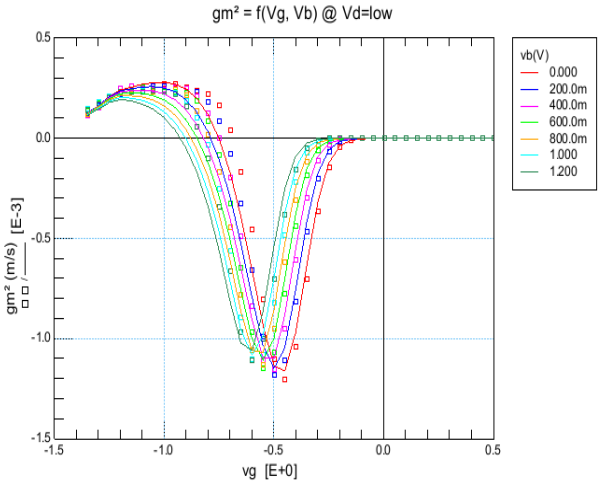
4.6 W10u0_L0u14_S548_3

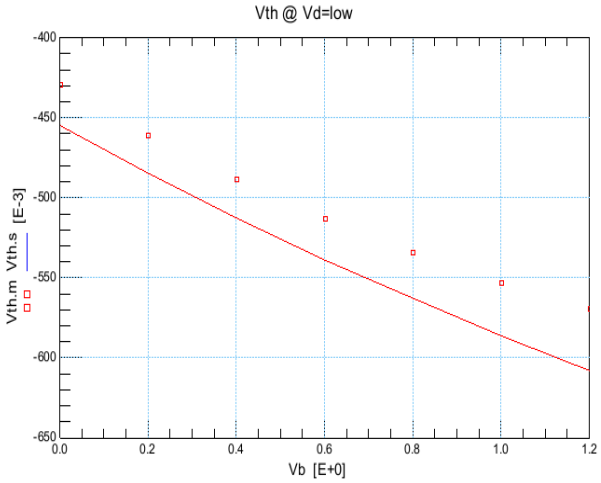
idvg, Vd = -0.05V, T = 27°C

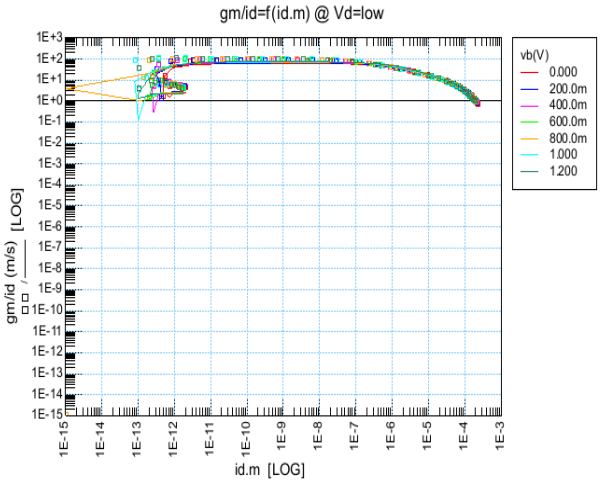


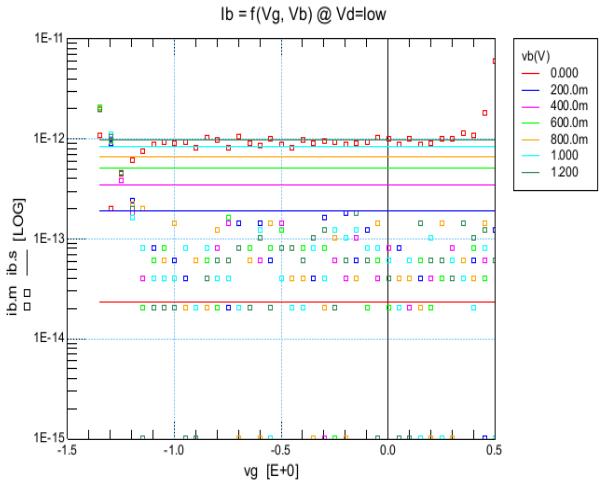


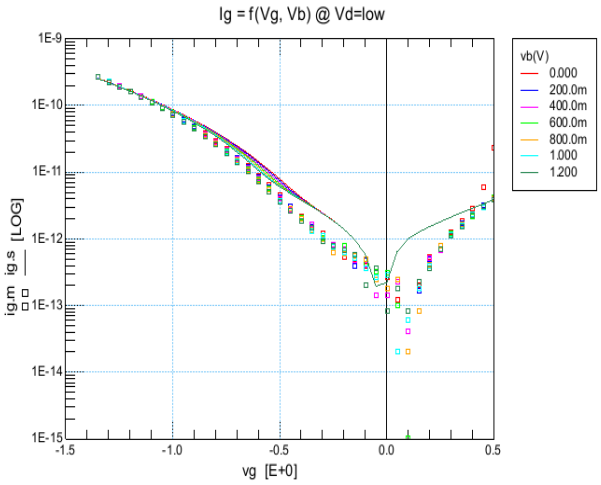




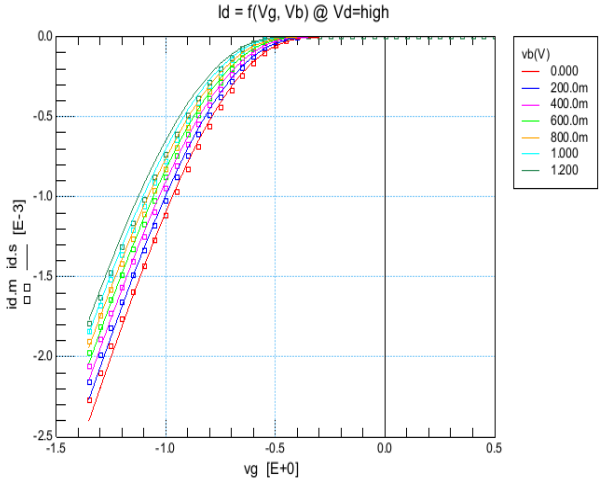


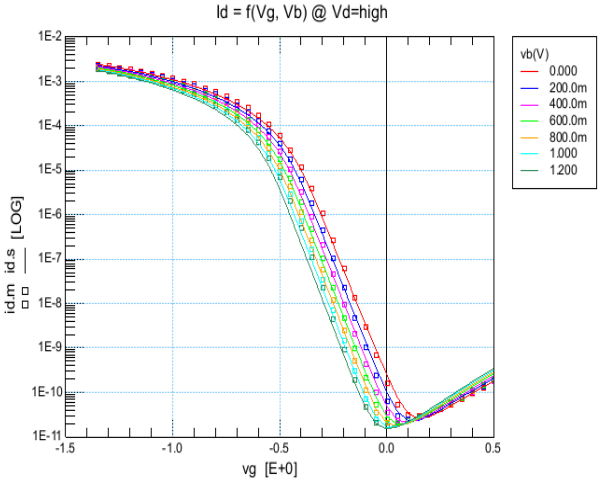


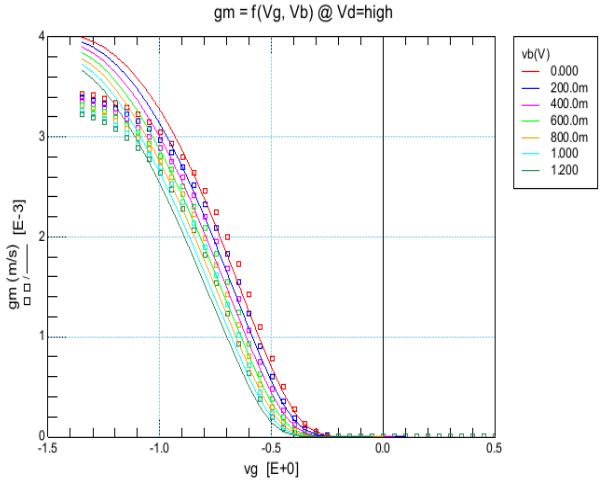


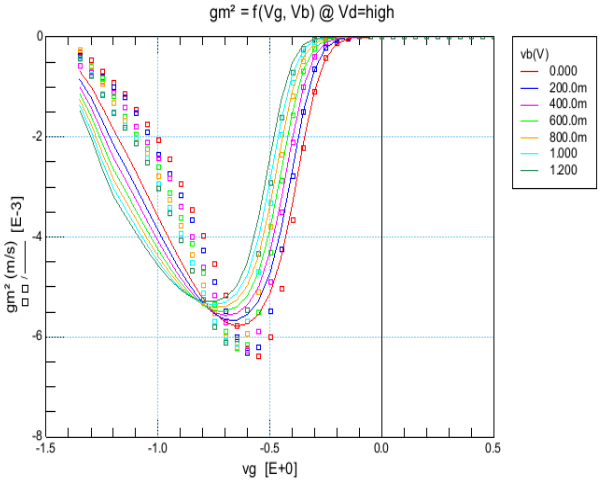


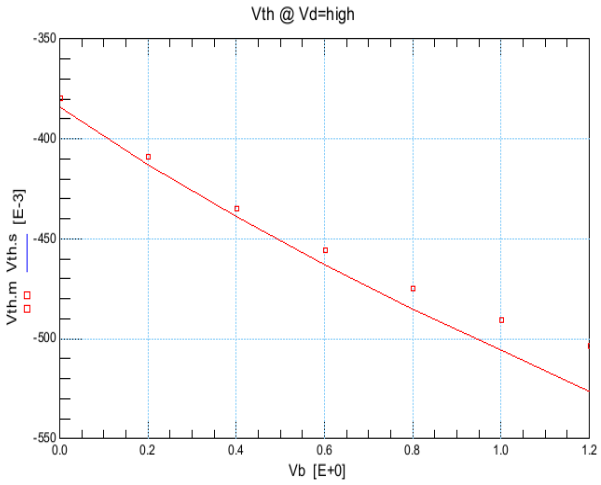
idvg, Vd = -1.2V, T = 27°C

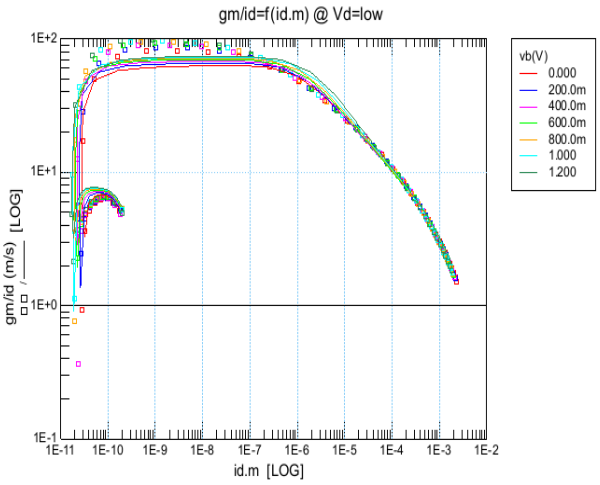


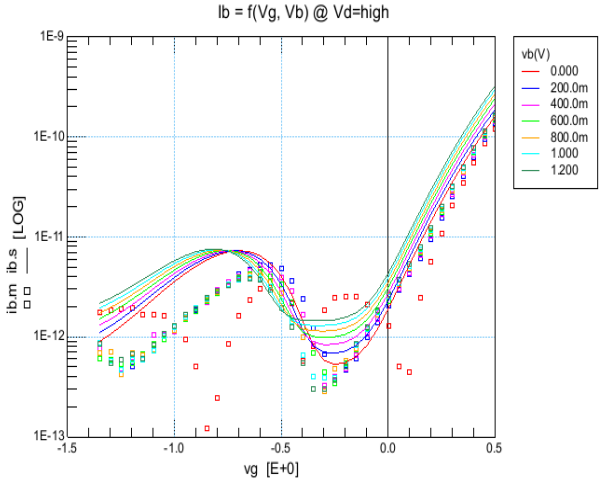


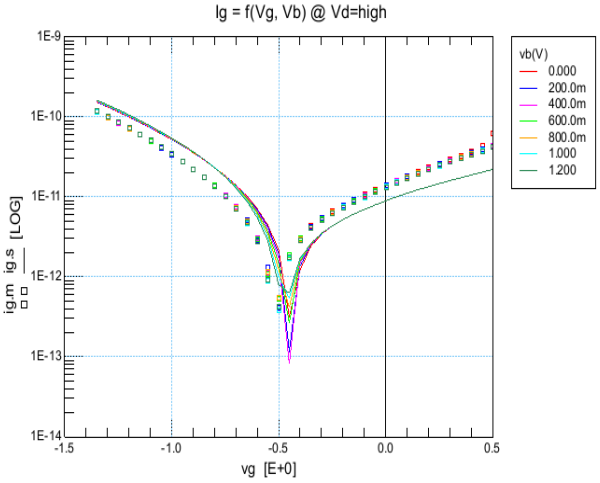




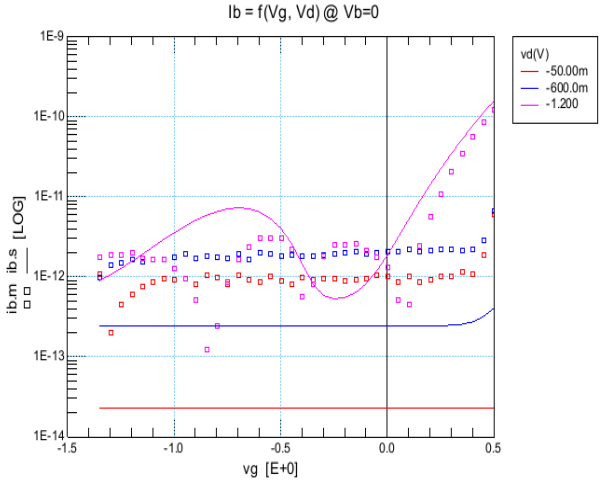


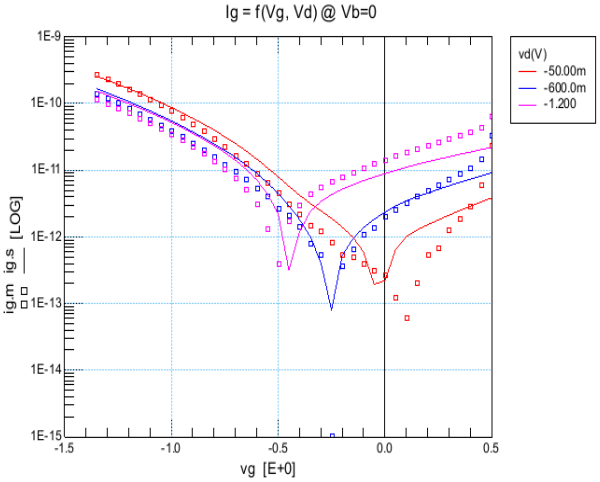




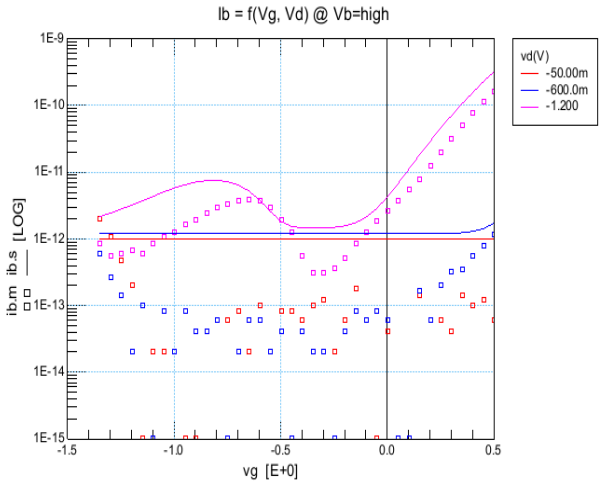


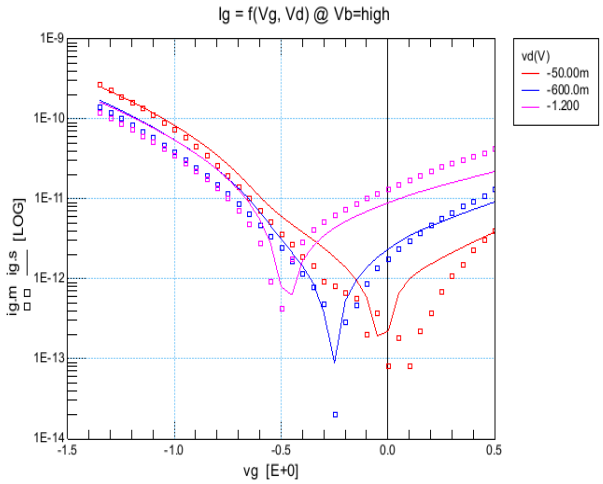
idvg, Ib,Ig, Vb = 0V, T = 27°C



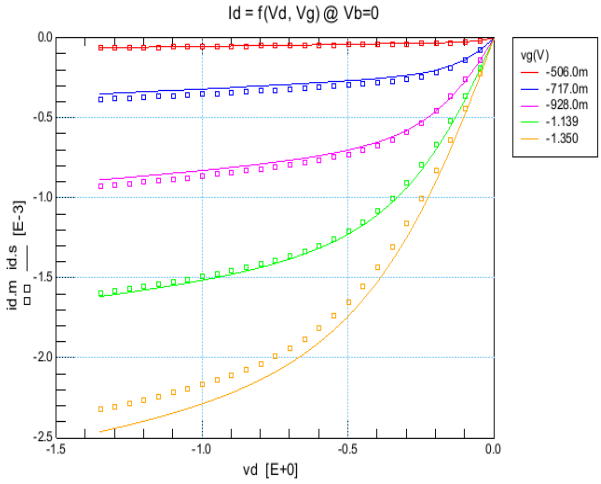


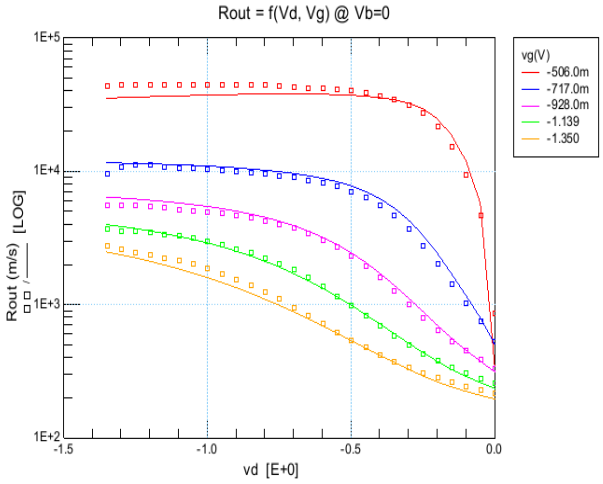
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

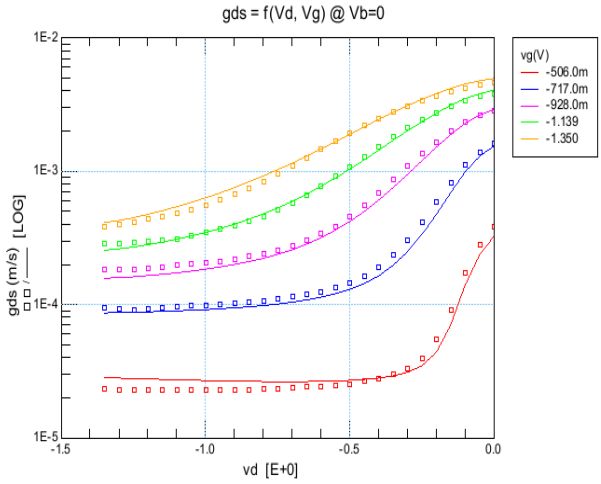




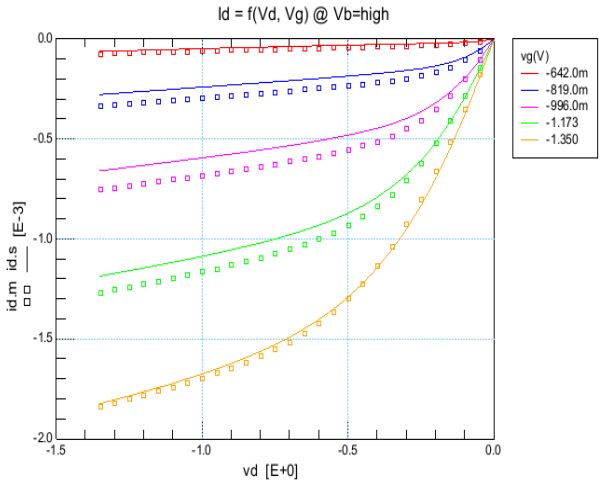
idvd, Vb = 0V, T = 27°C

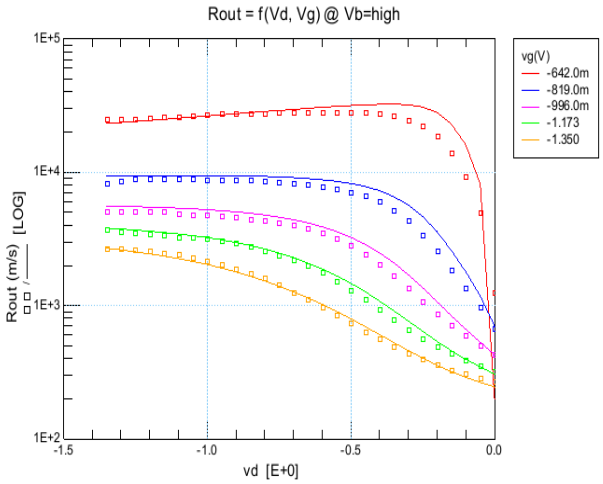


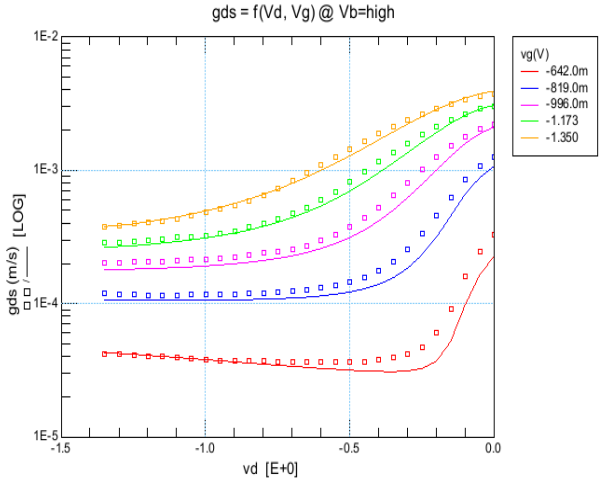




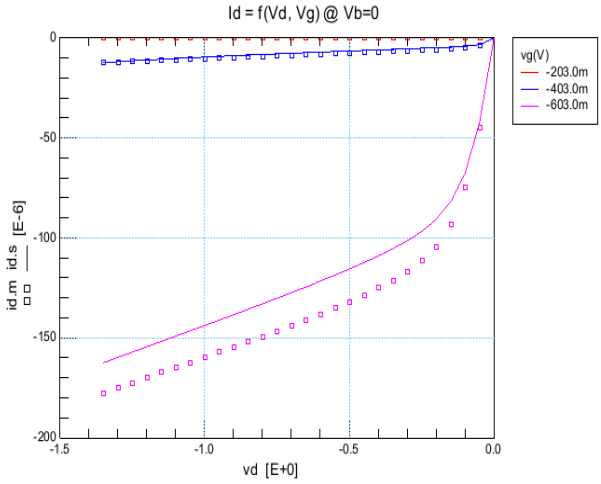
idvd_vbmin, Vb = 1.2V, T = 27°C

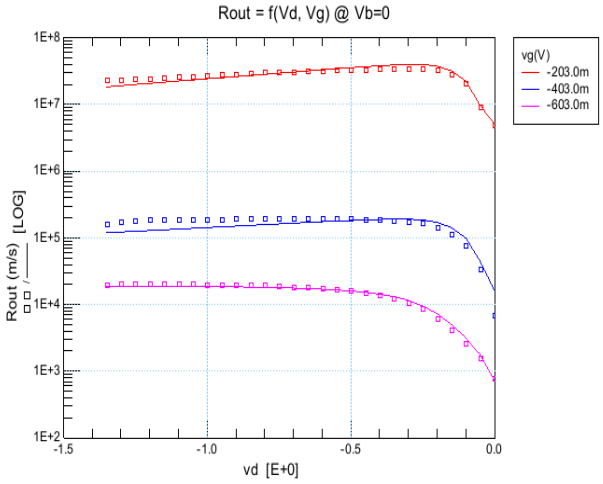


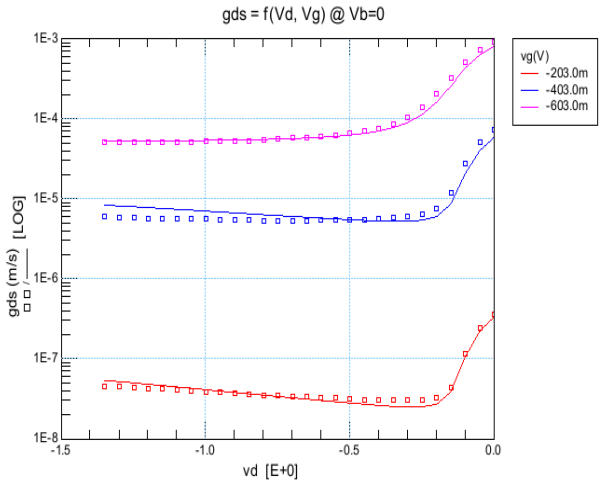




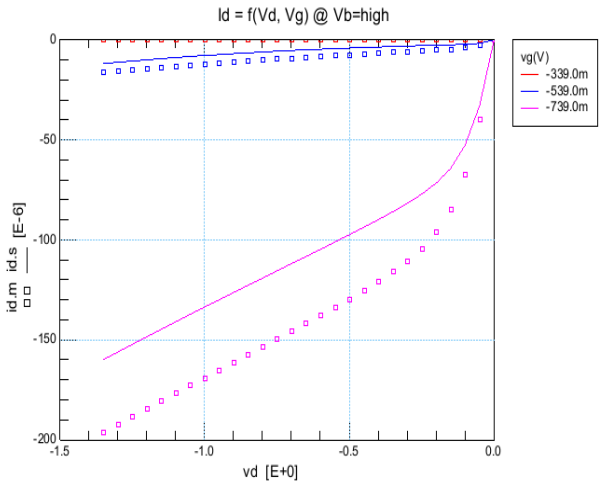
idvd_vth, Vb = 0V, T = 27°C

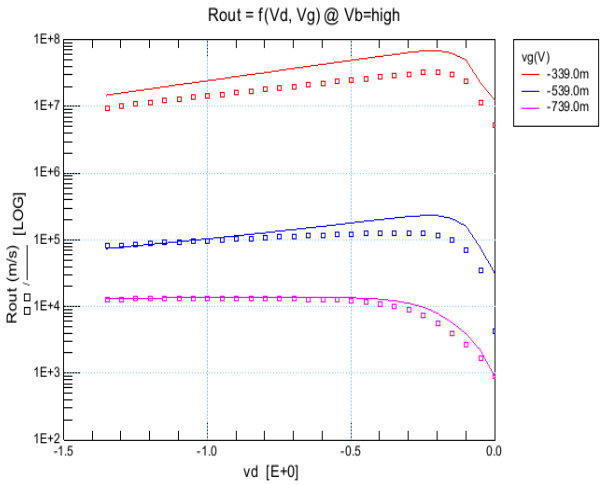


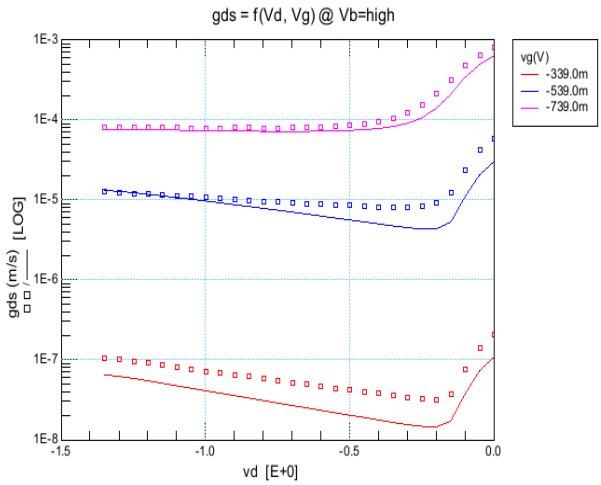




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

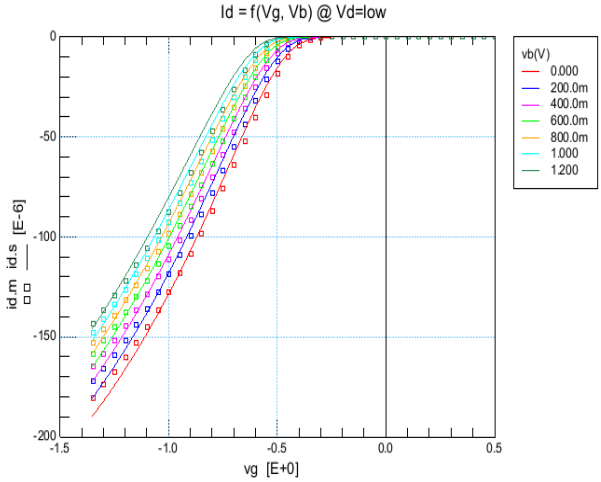


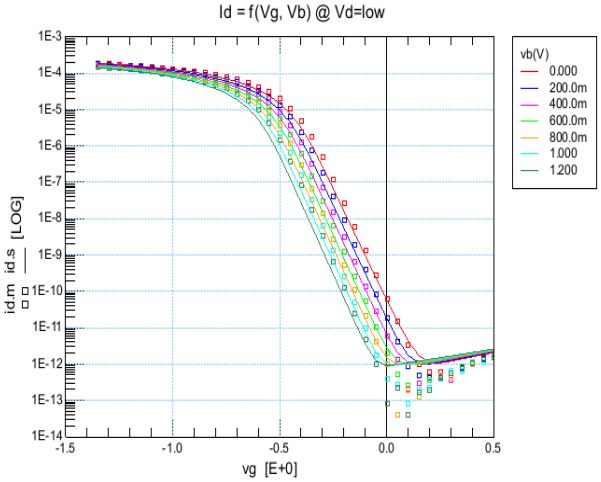


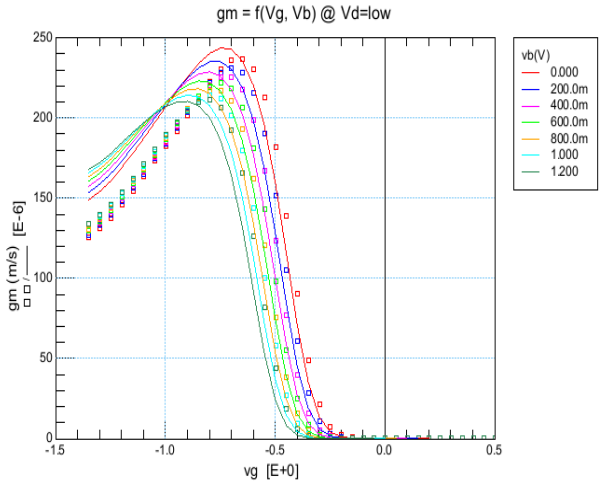


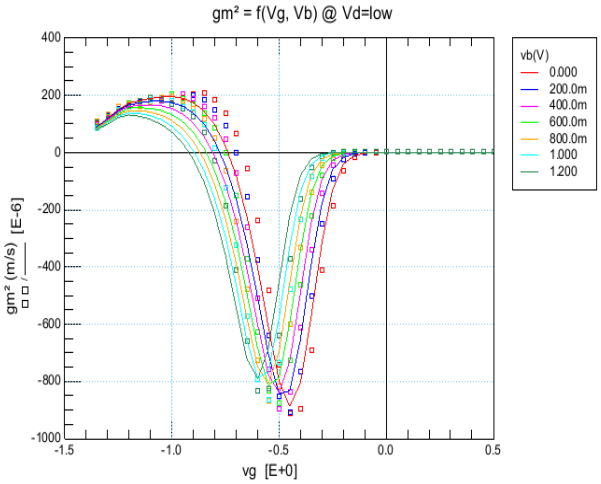
4.7 W10u0_L0u18_S548_5

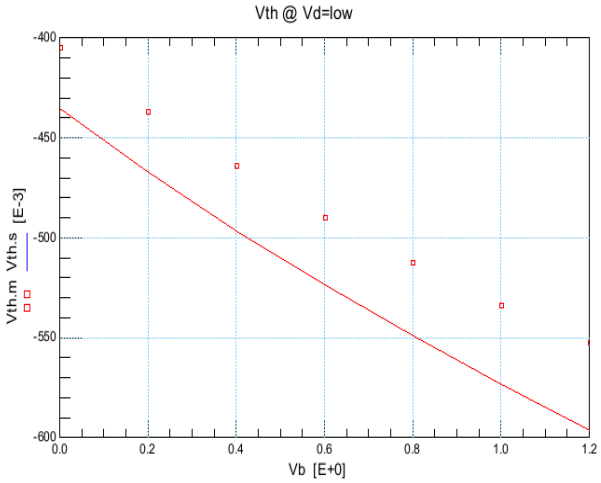
idvg, Vd = -0.05V, T = 27°C

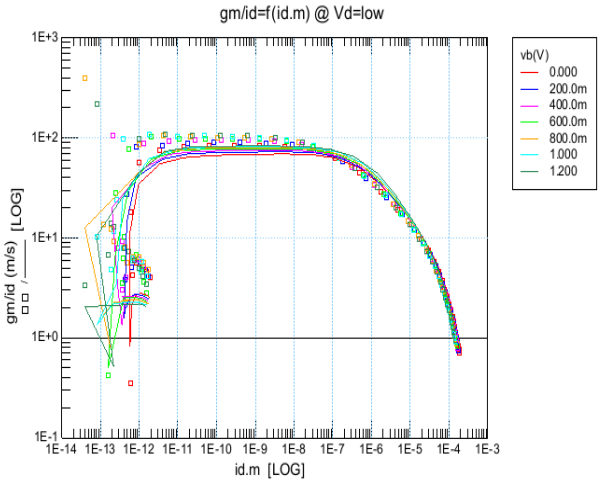


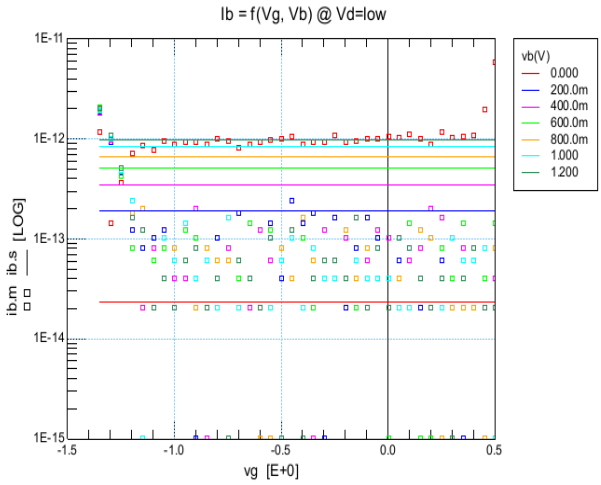


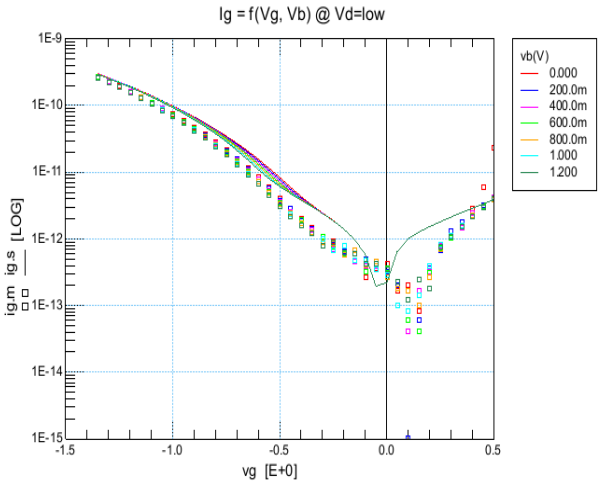




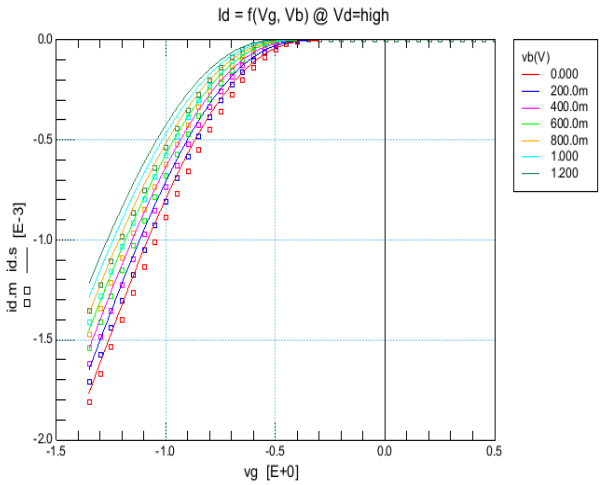


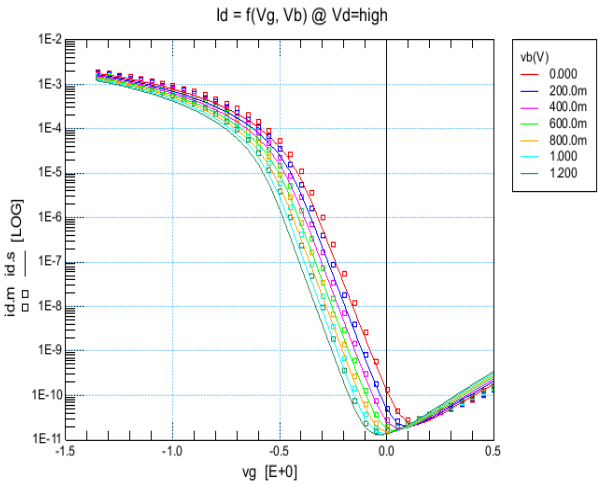


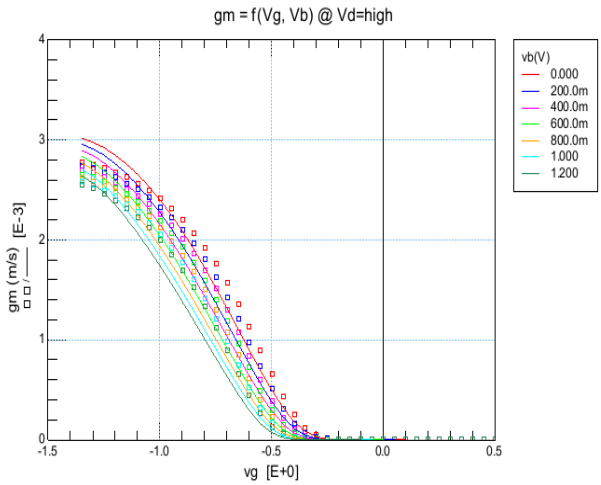


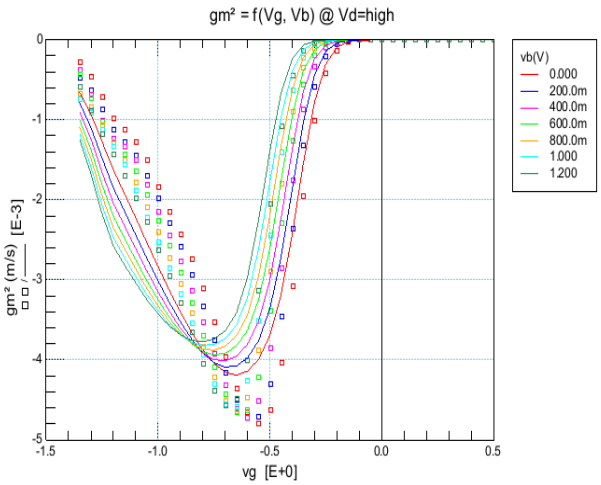


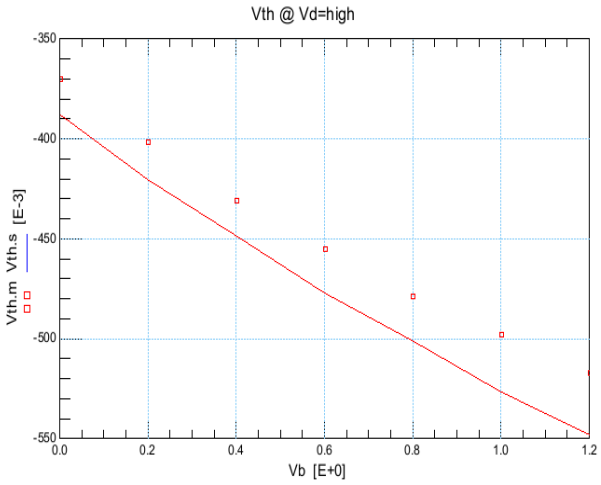
idvg, Vd = -1.2V, T = 27°C

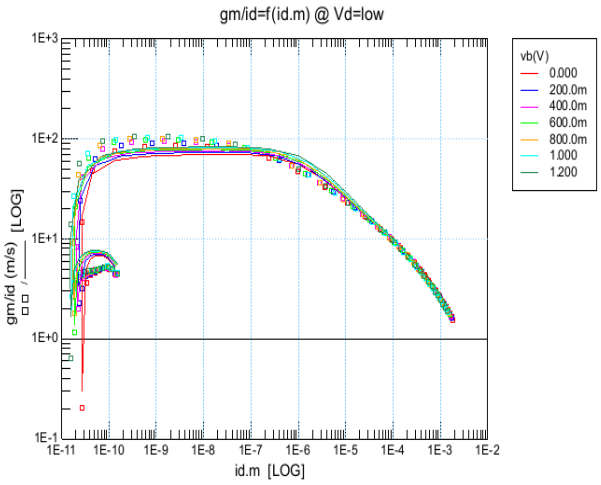


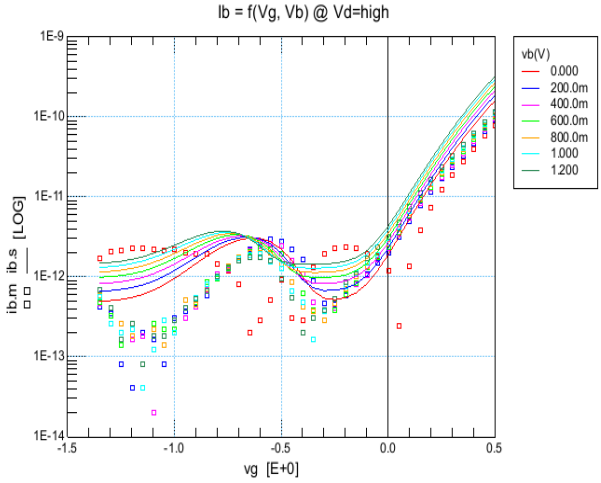


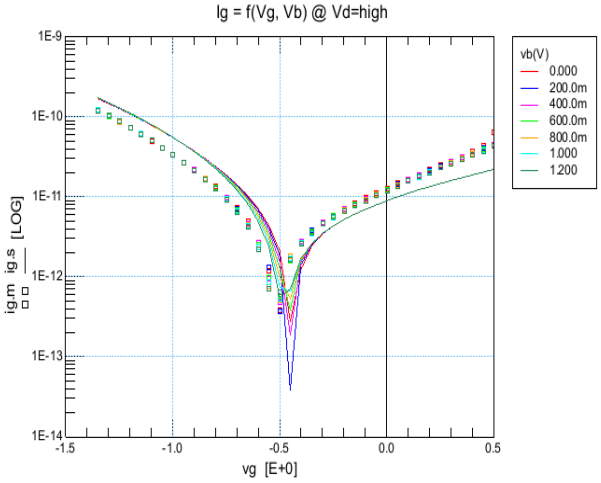




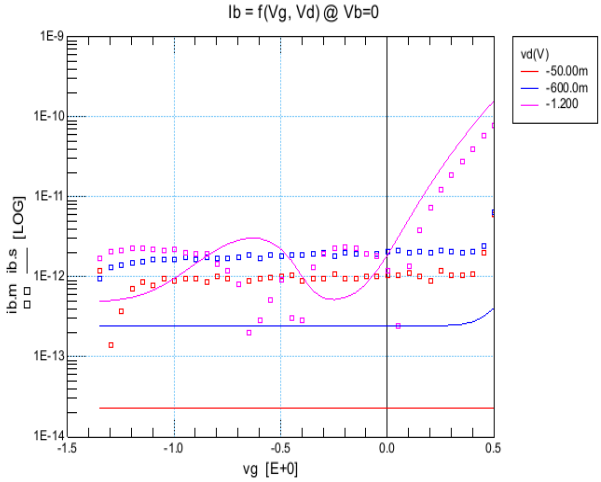


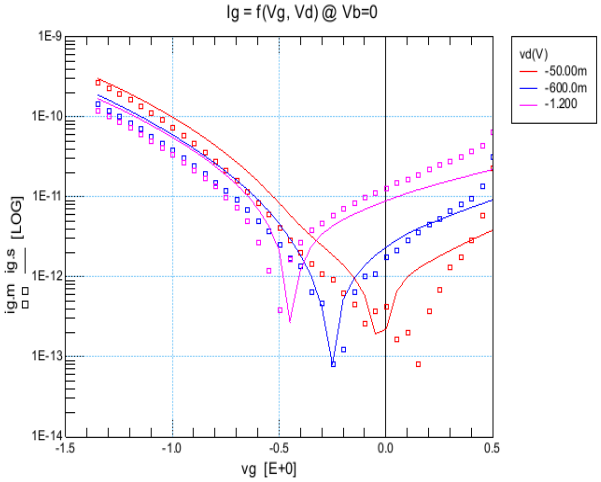




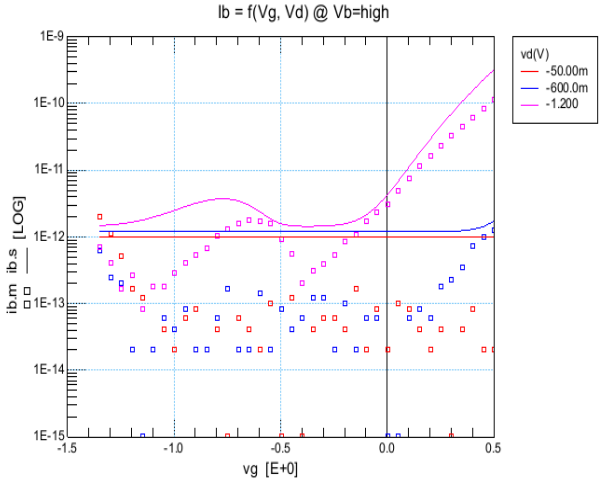


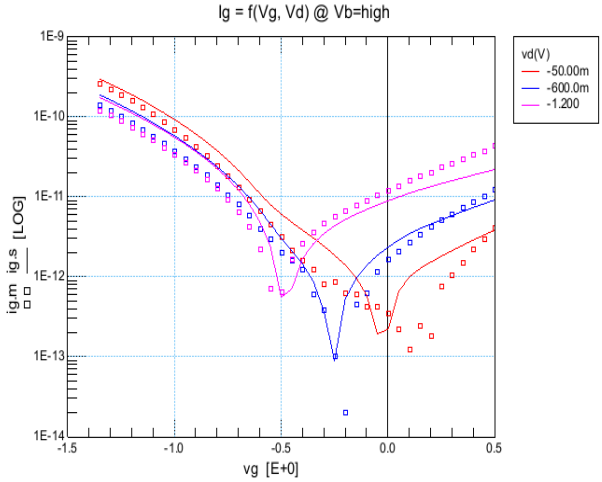
idvg, Ib,Ig, Vb = 0V, T = 27°C



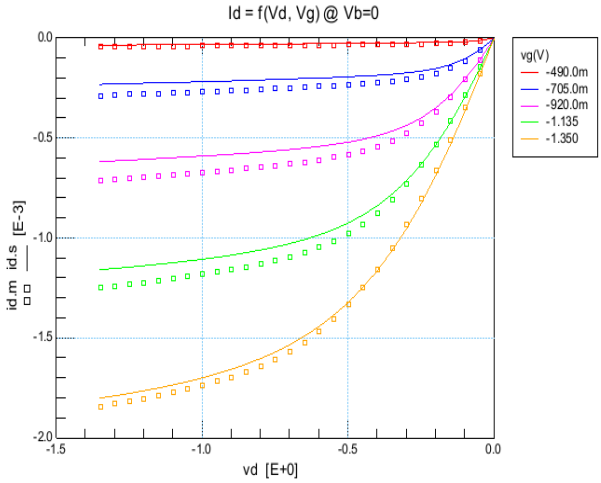


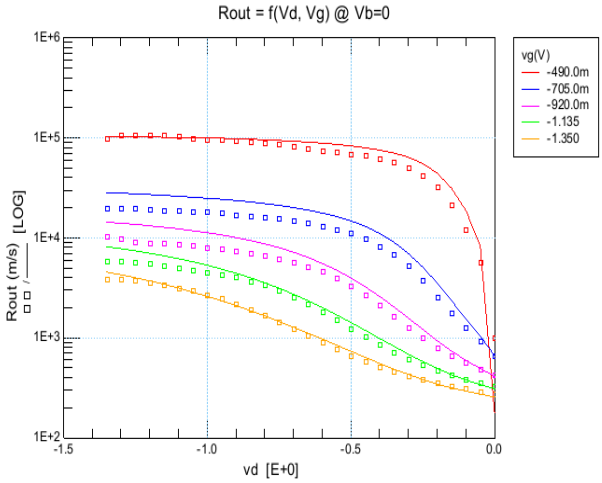
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

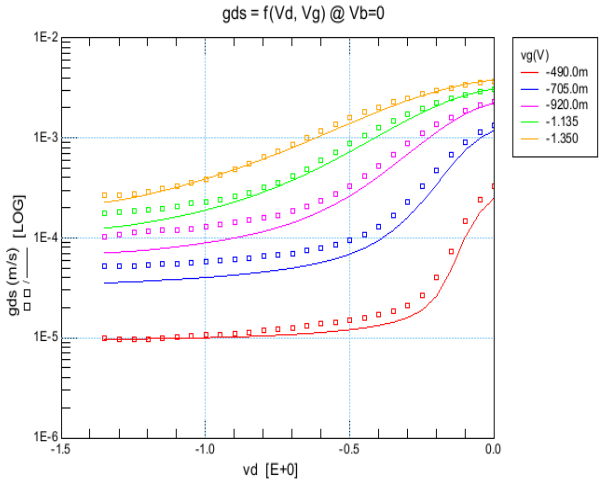




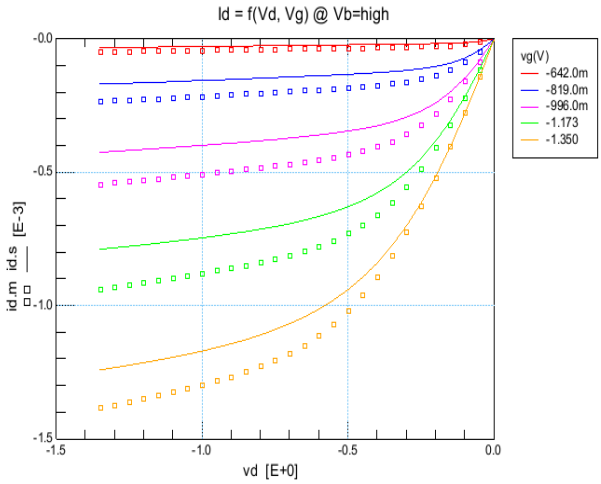
idvd, Vb = 0V, T = 27°C

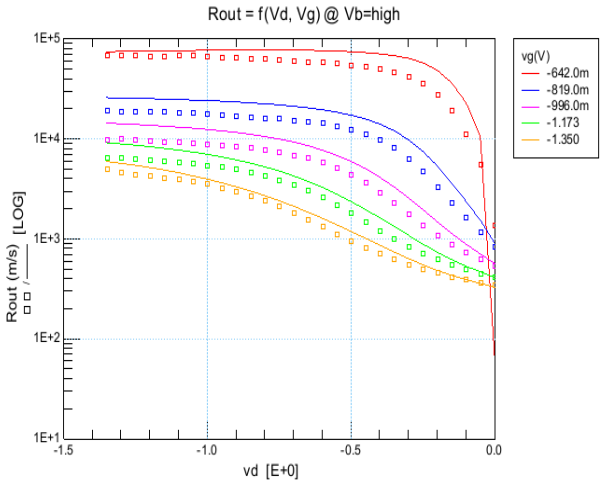


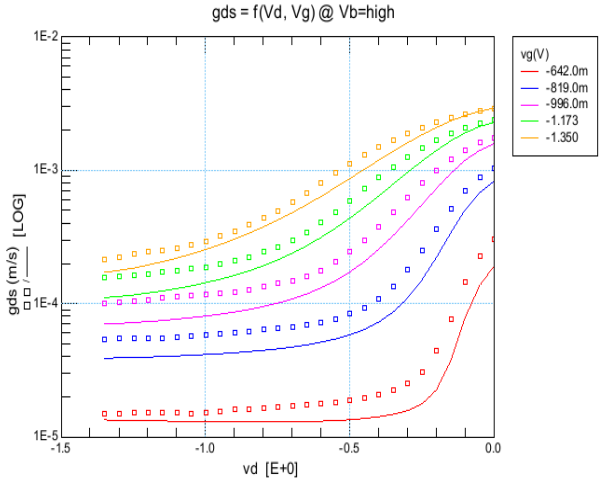




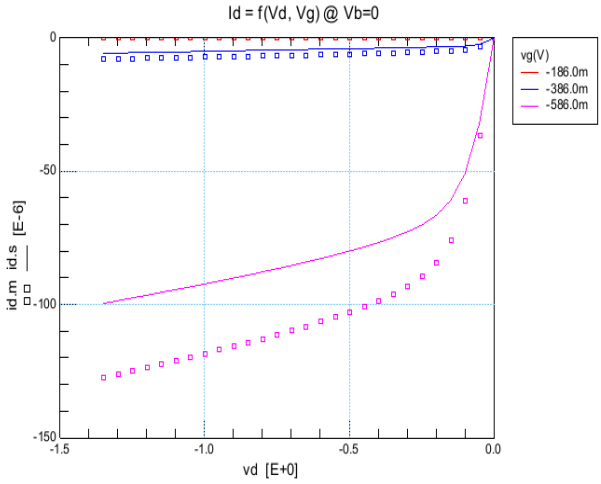
idvd_vbmin, Vb = 1.2V, T = 27°C

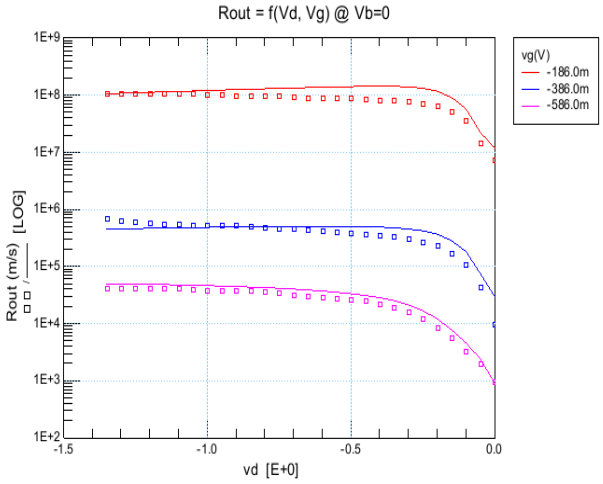


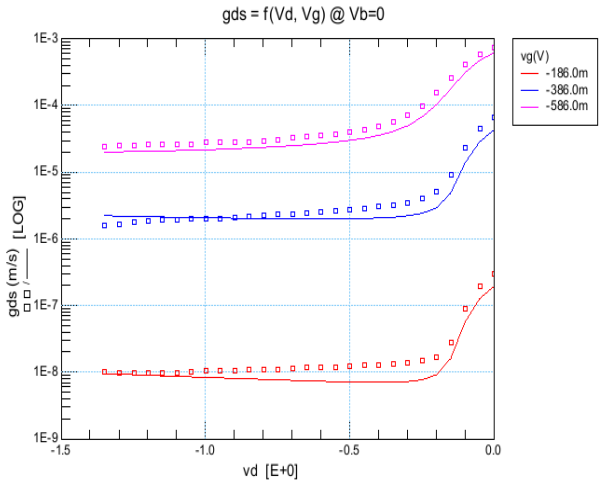




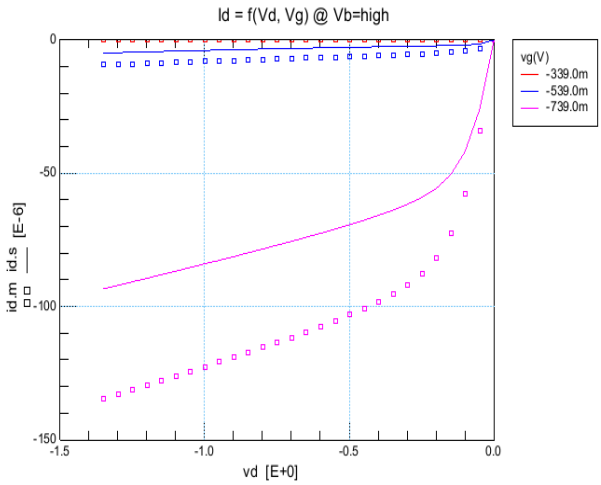
idvd_vth, Vb = 0V, T = 27°C

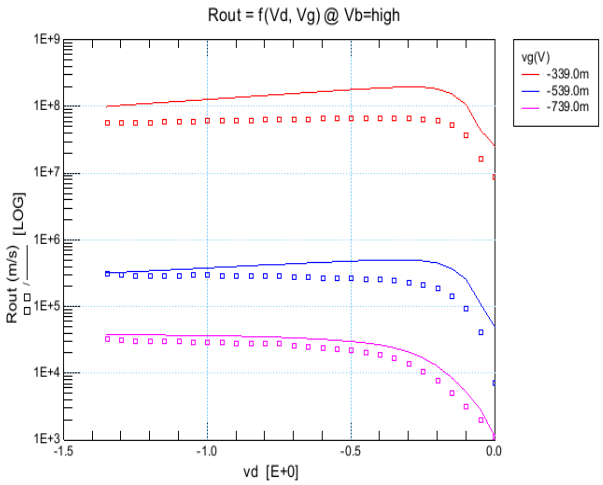


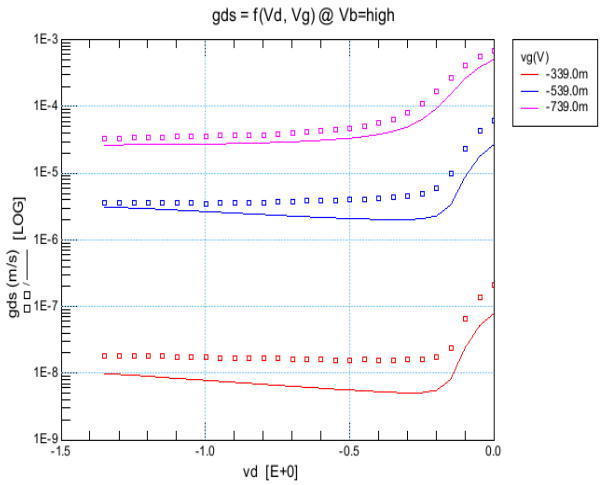




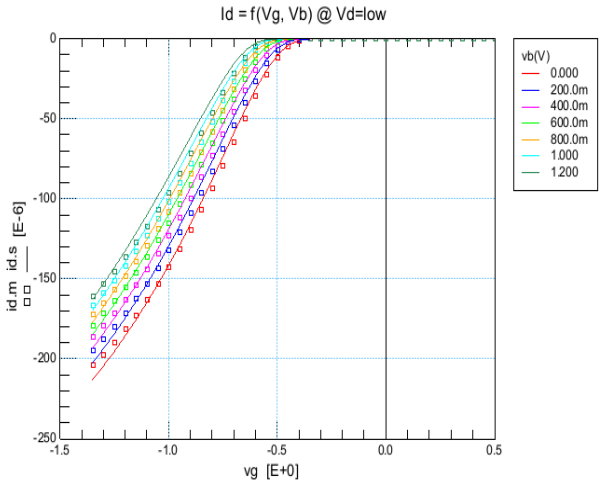
idvd_vth_vbmin, Vb = 1.2V, T = 27°C

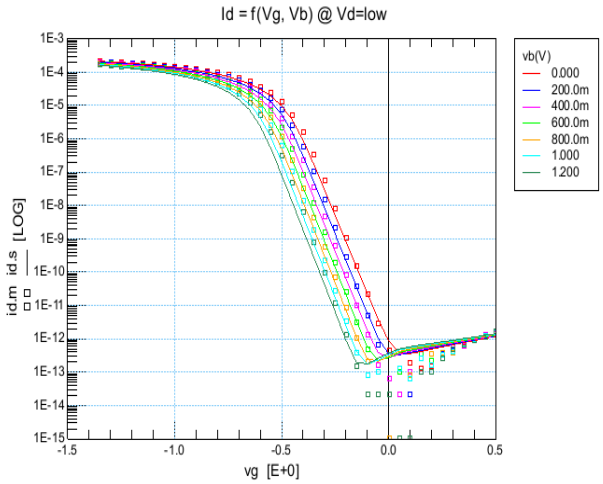


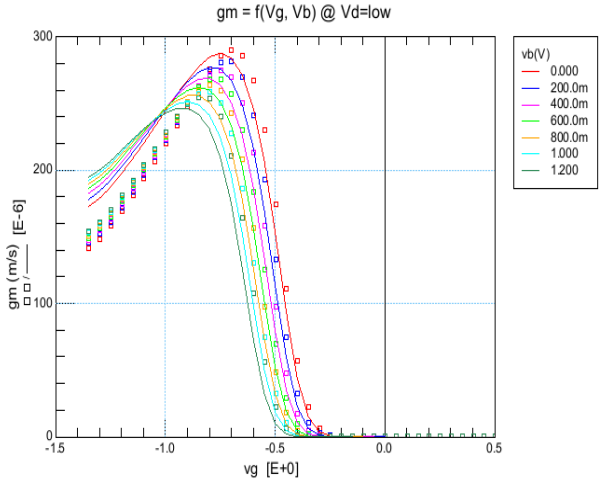


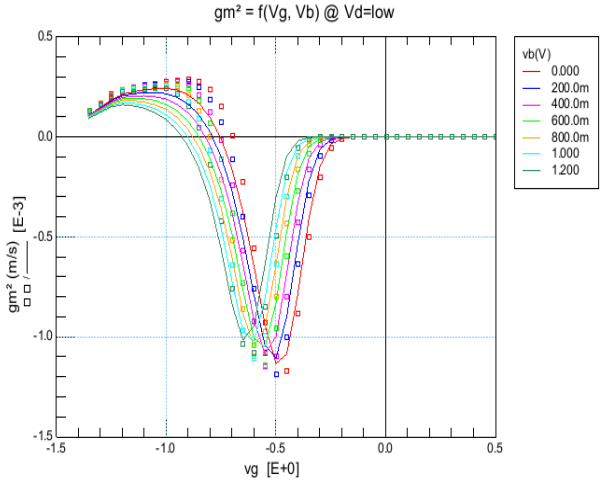


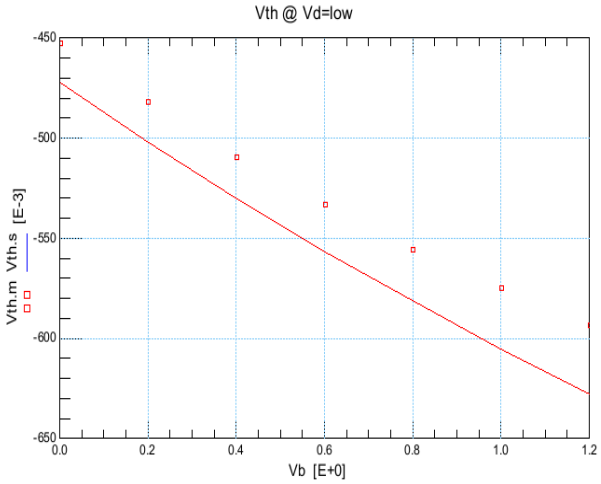
idvg, Vd = -0.05V, T = -40°C

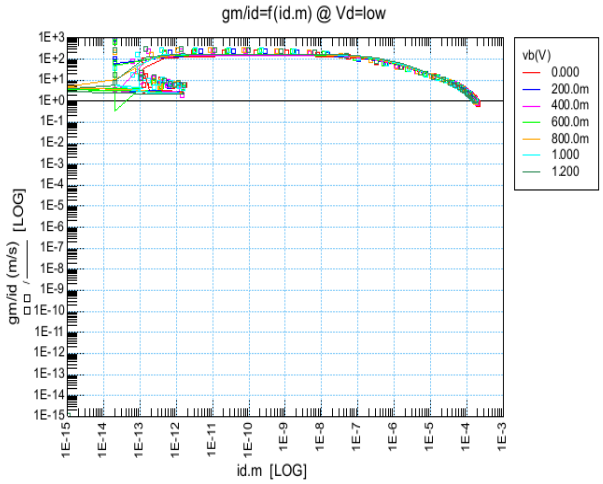


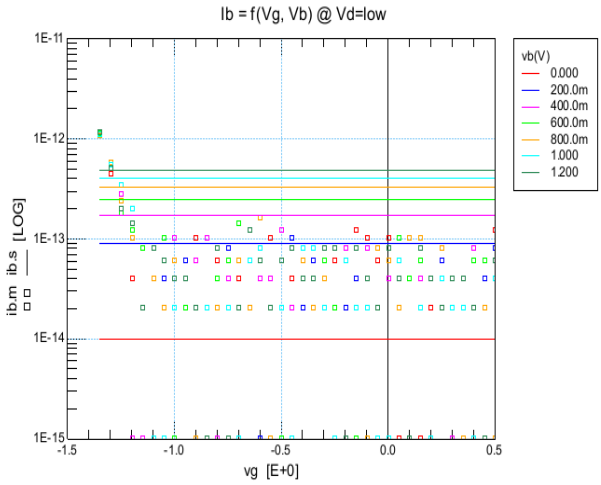


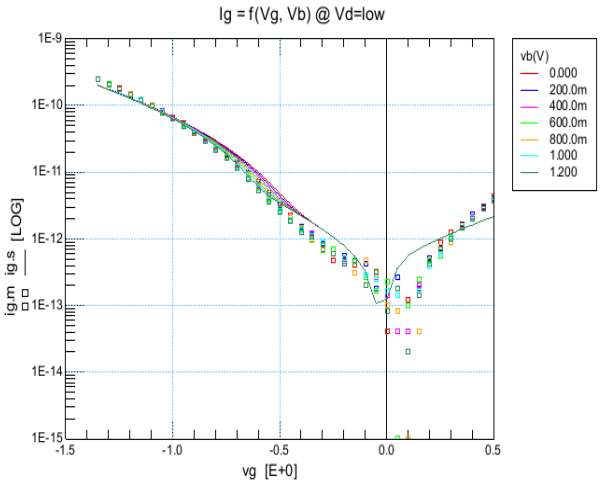




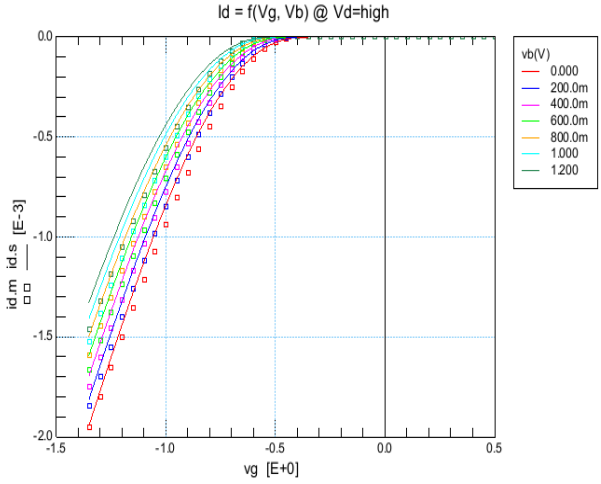


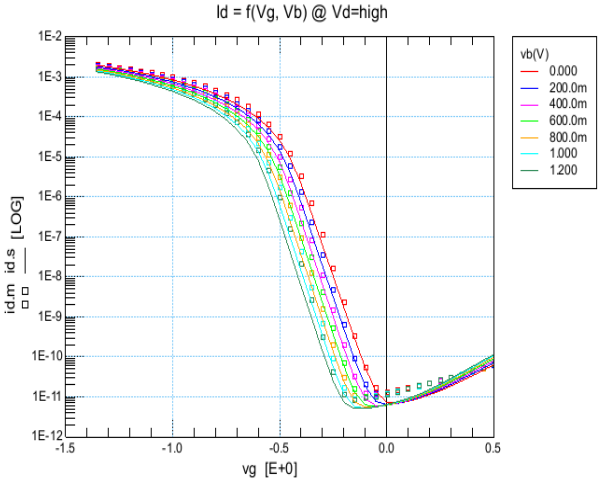


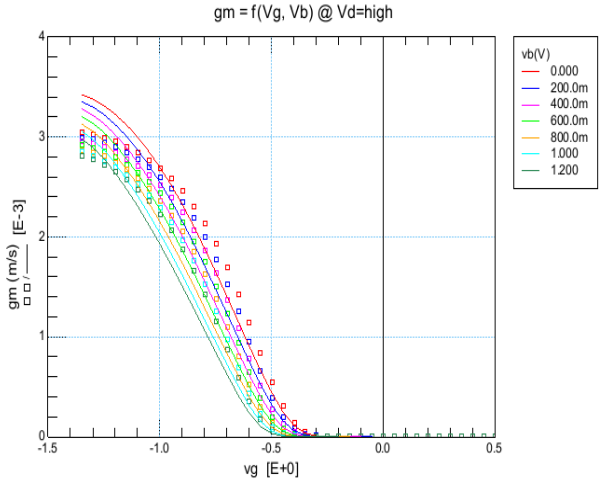


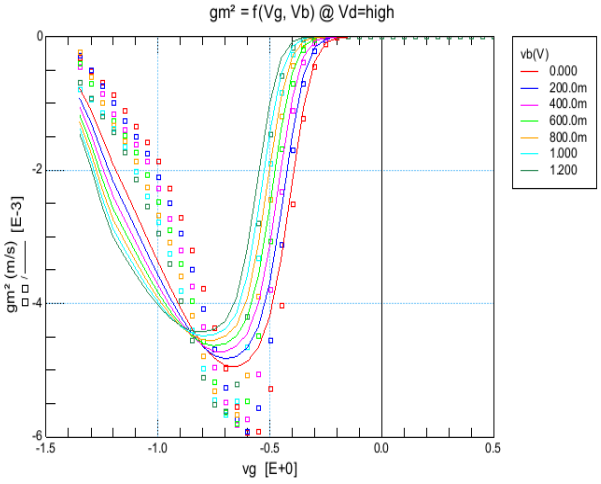


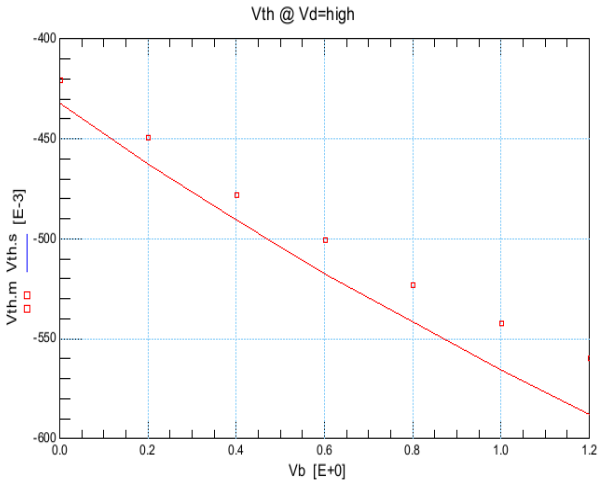
idvg, Vd = -1.2V, T = -40°C

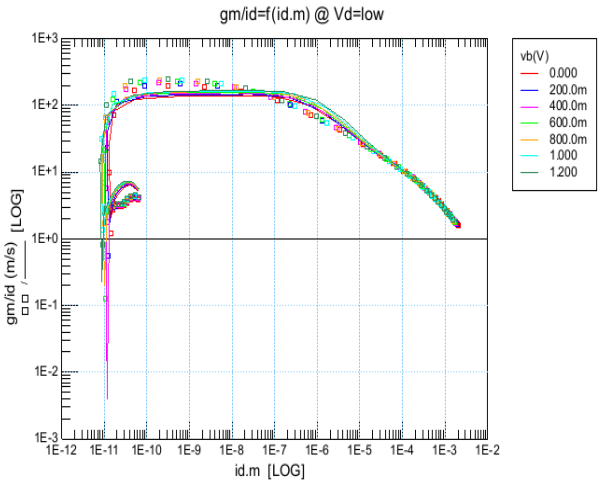


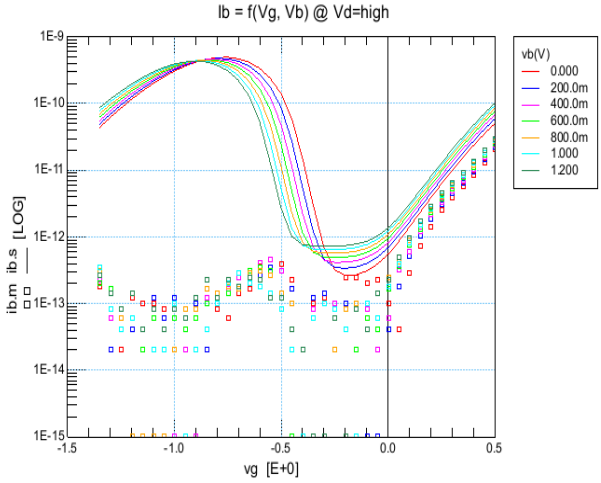


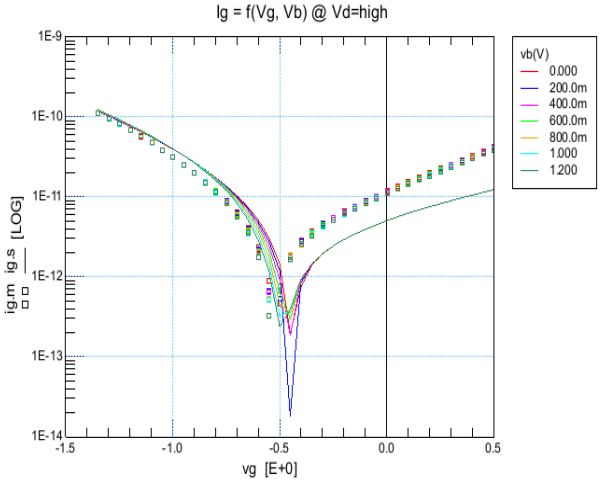




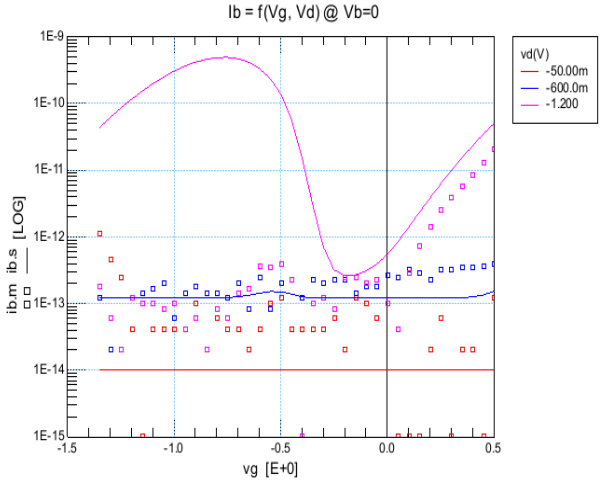


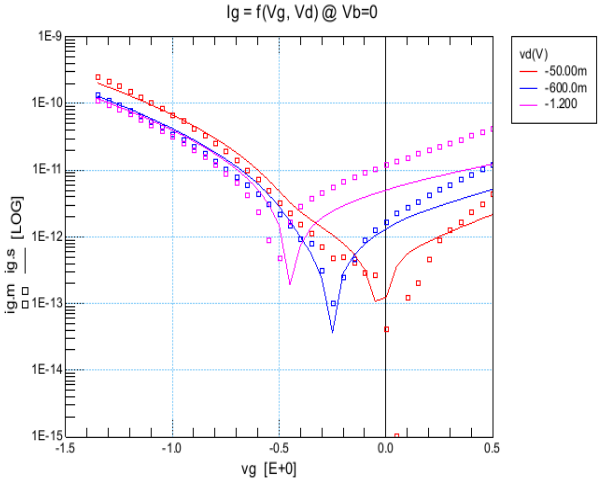




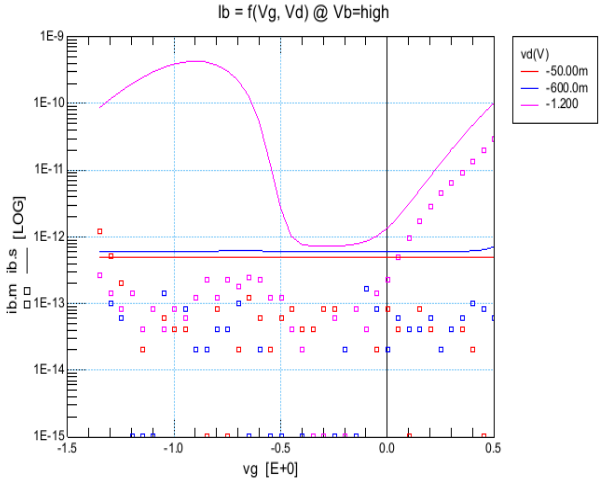


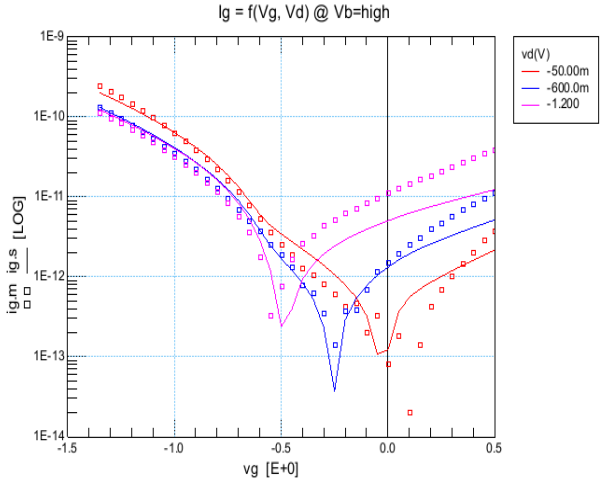
idvg, Ib,Ig, Vb = 0V, T = -40°C



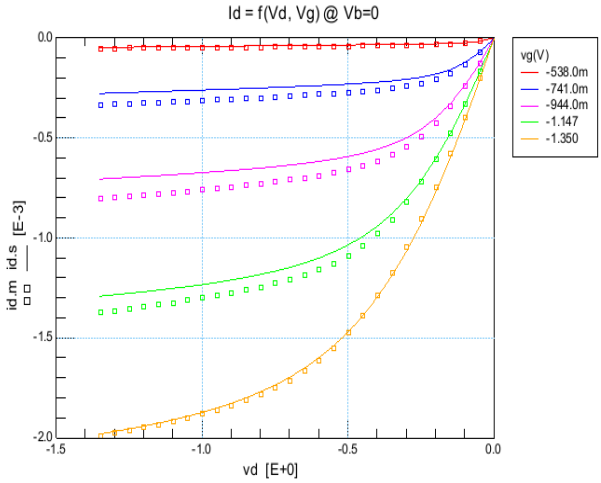


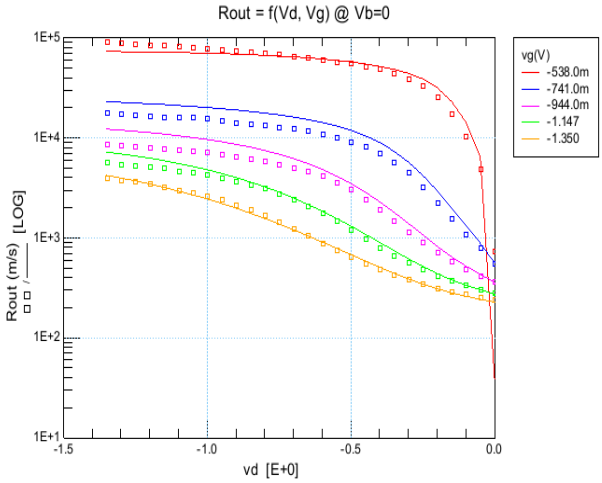
idvg, Ib,Ig, Vb = 1.2V, T = -40°C

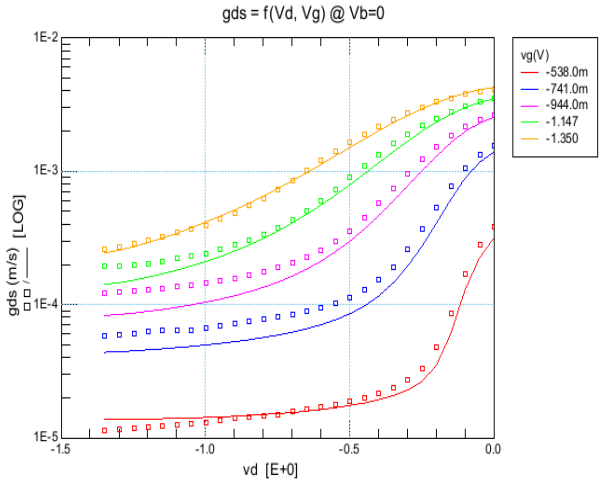




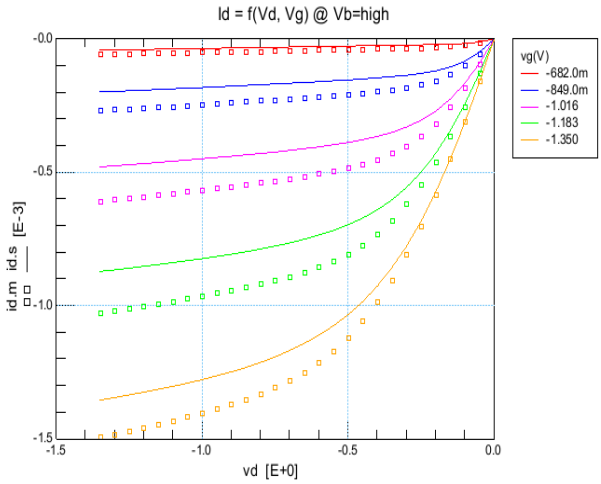
idvd, Vb = 0V, T = -40°C

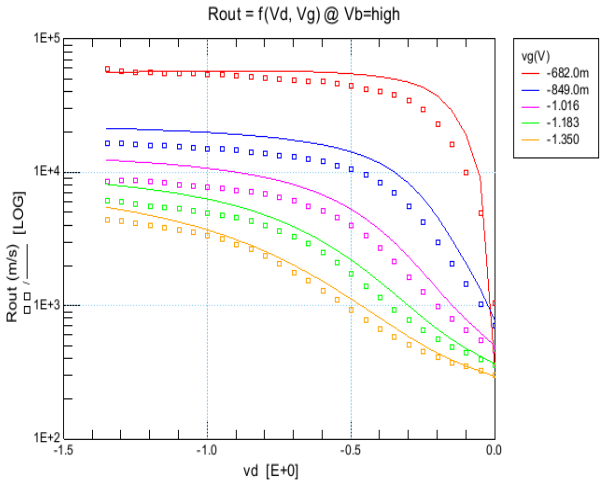


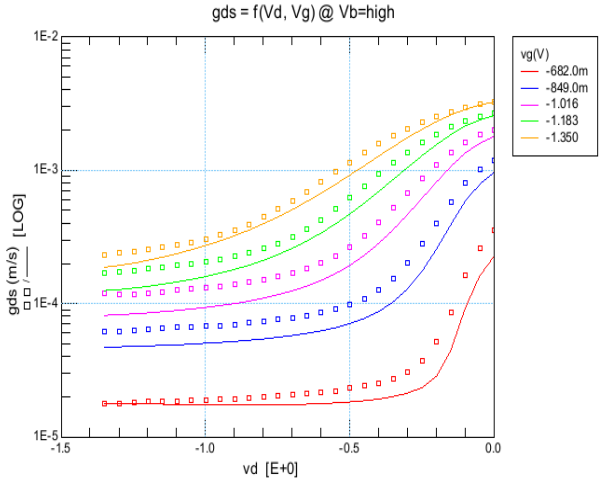




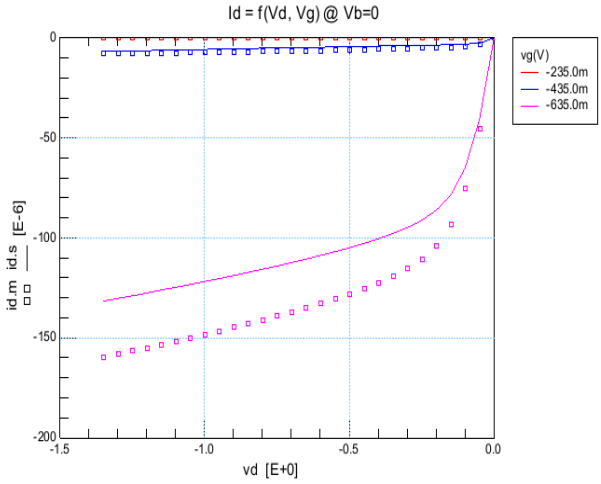
idvd_vbmin, Vb = 1.2V, T = -40°C

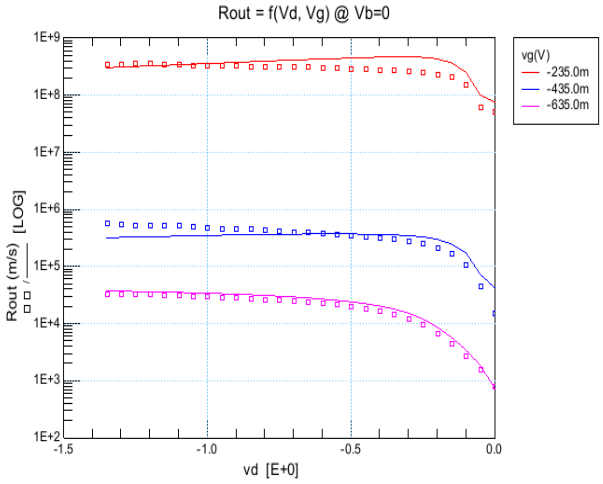


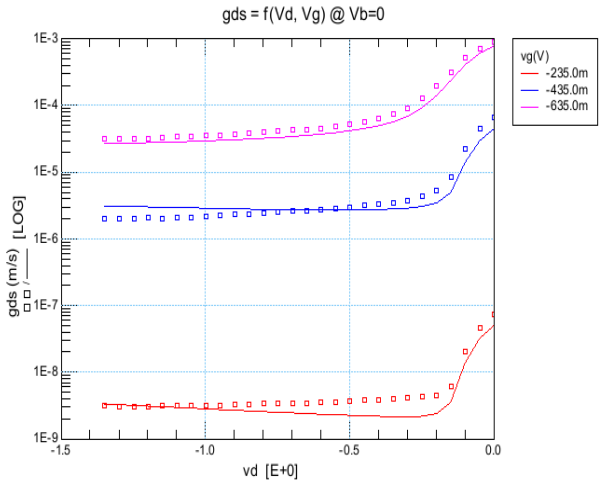




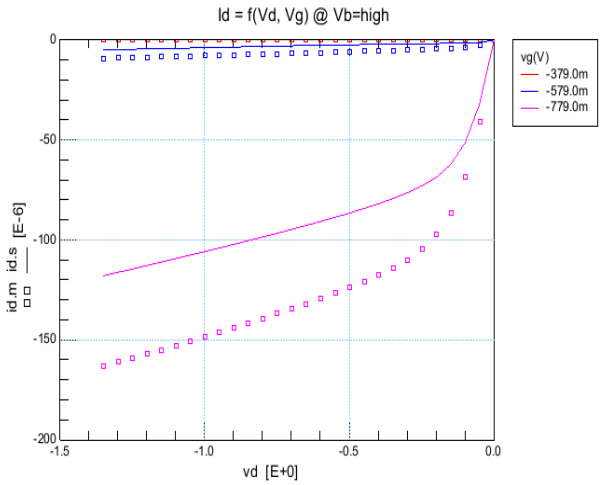
idvd_vth, Vb = 0V, T = -40°C

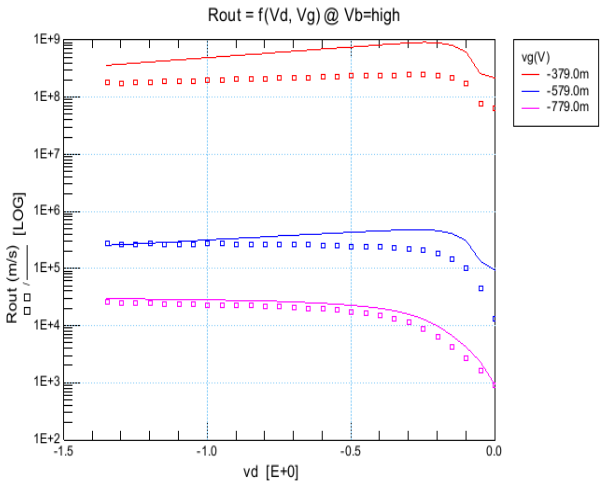


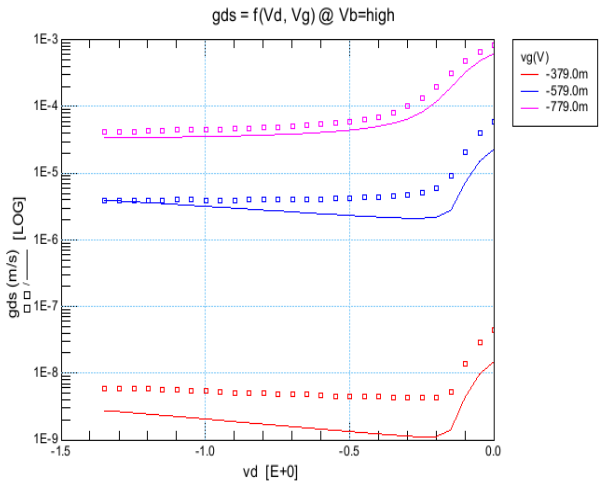




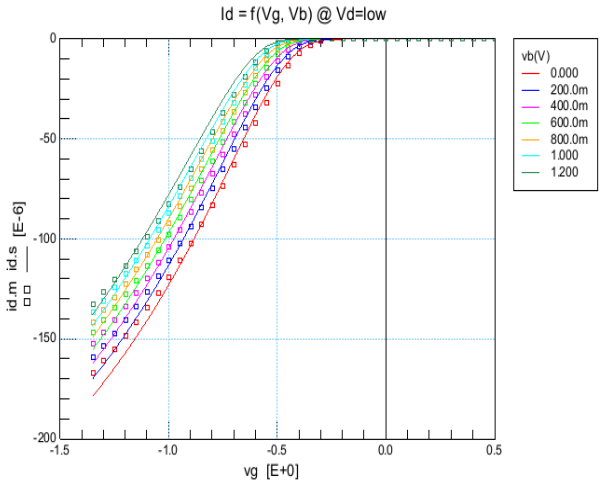
idvd_vth_vbmin, Vb = 1.2V, T = -40°C

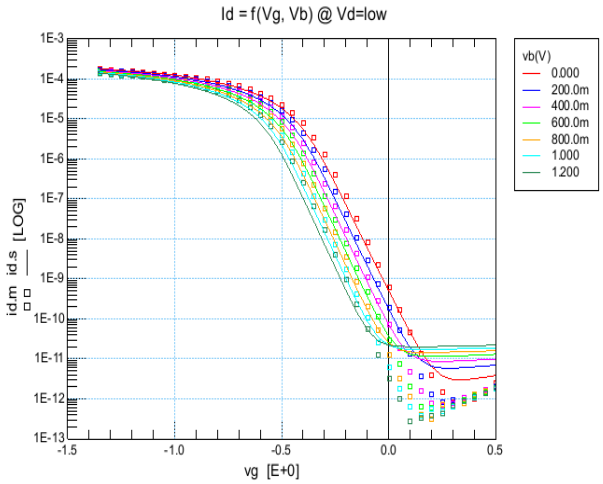


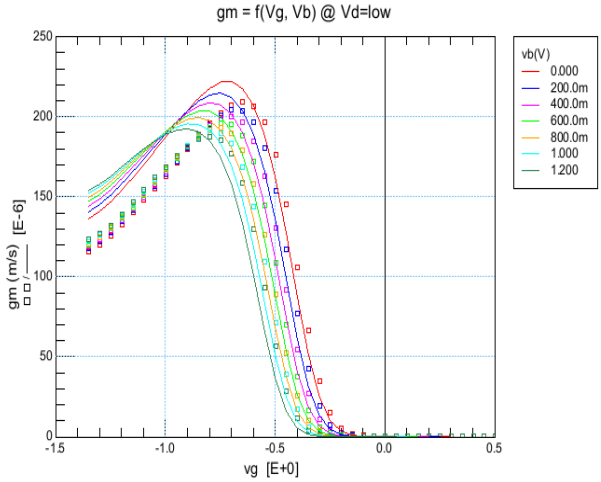


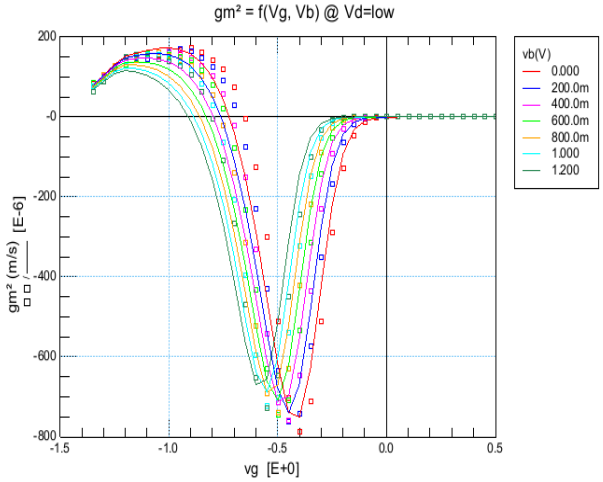


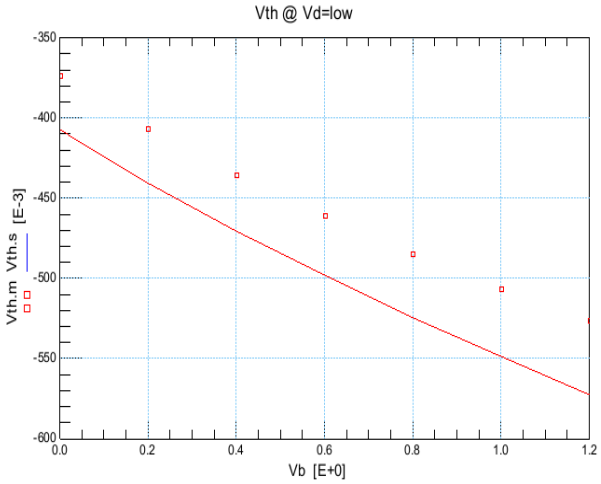
idvg, Vd = -0.05V, T = 70°C

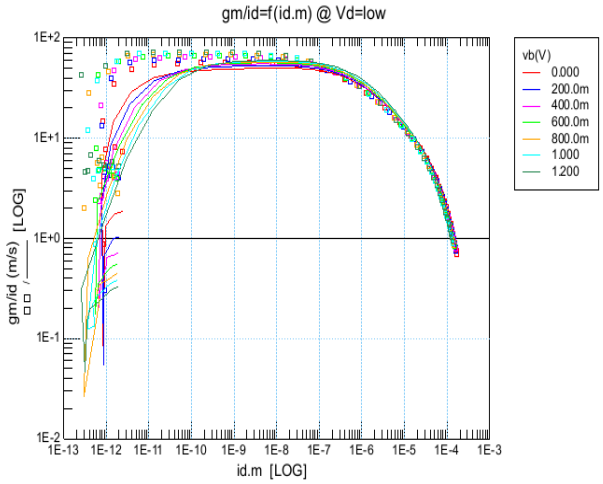


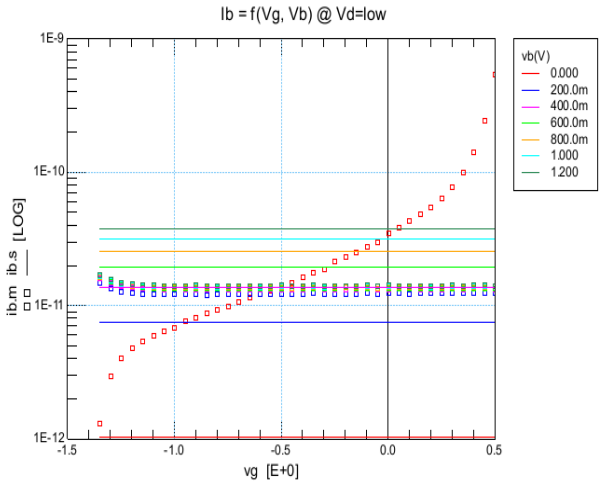


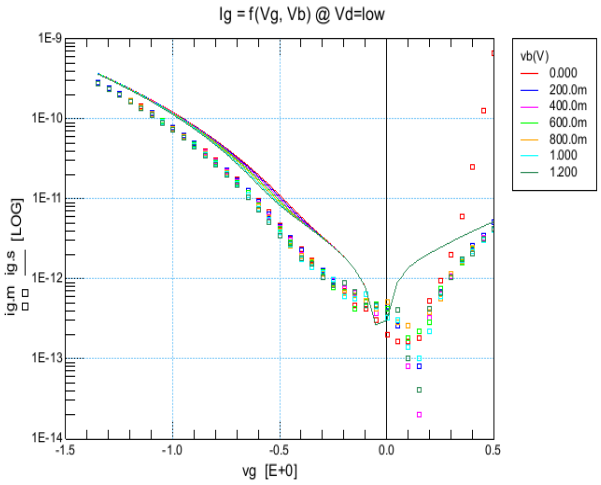




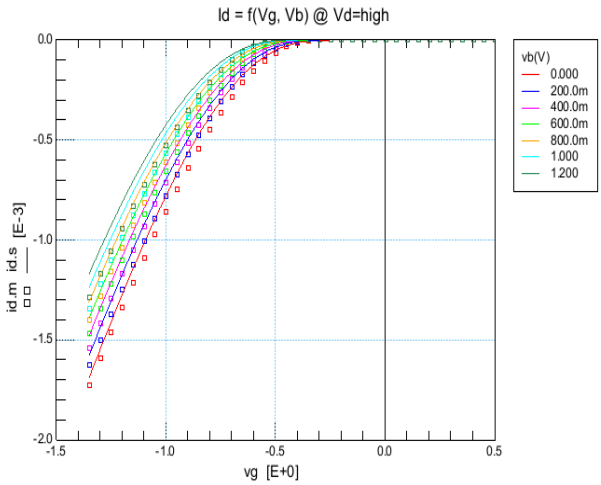


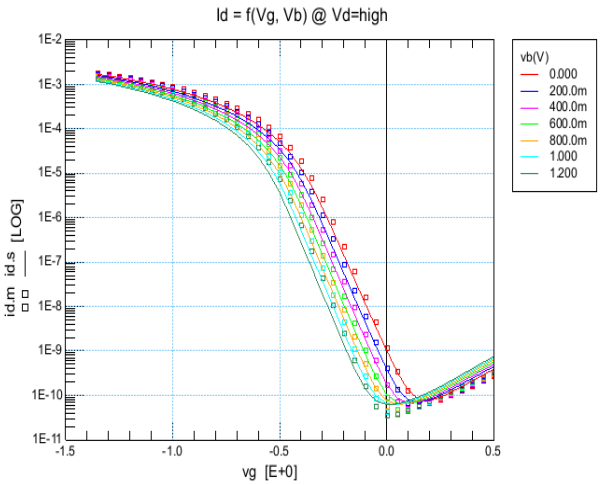


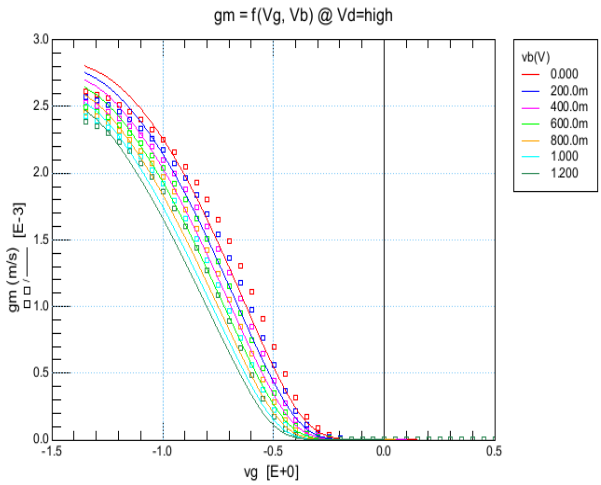


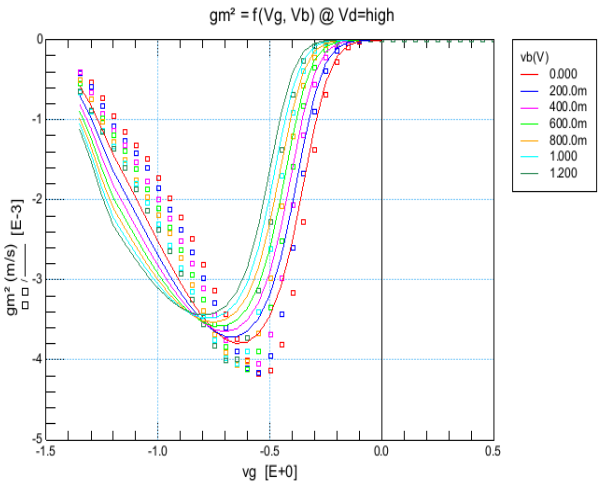


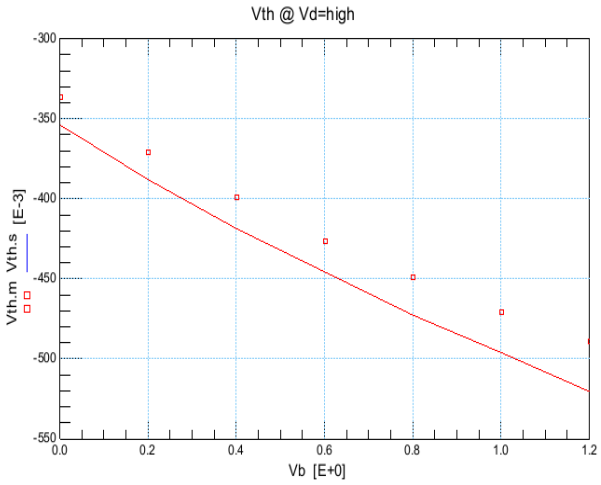
idvg, Vd = -1.2V, T = 70°C

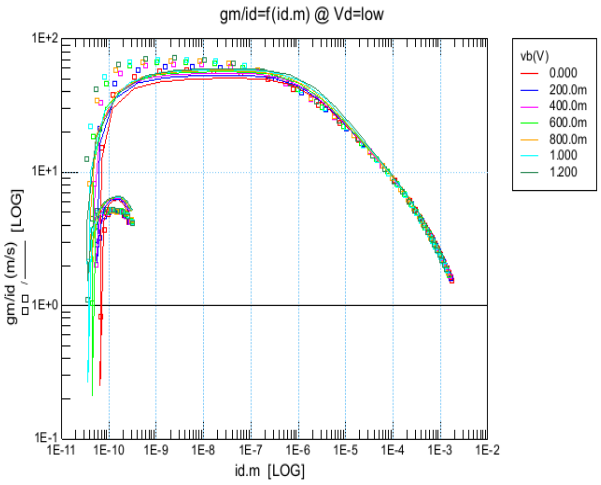


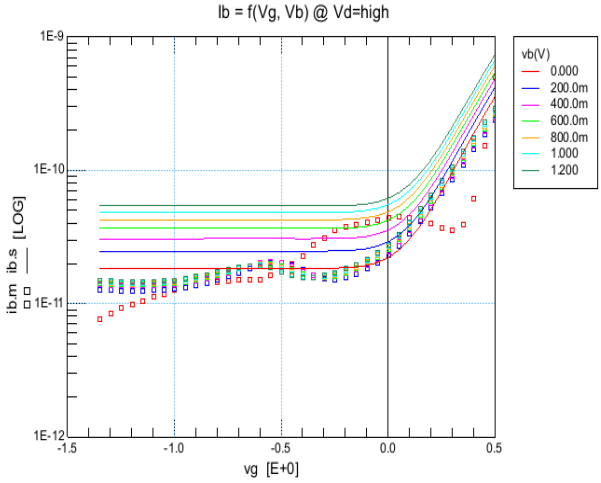


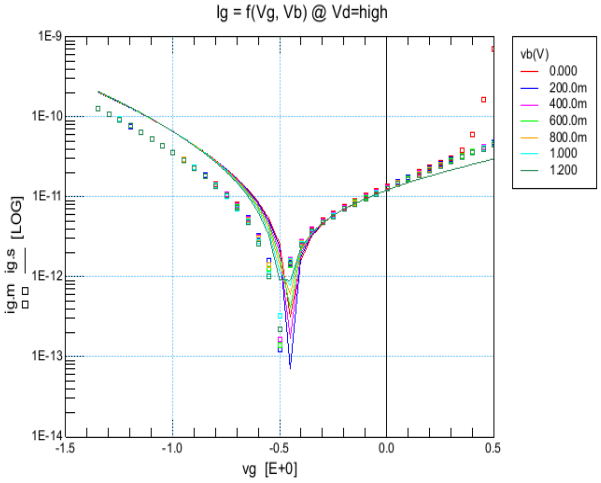




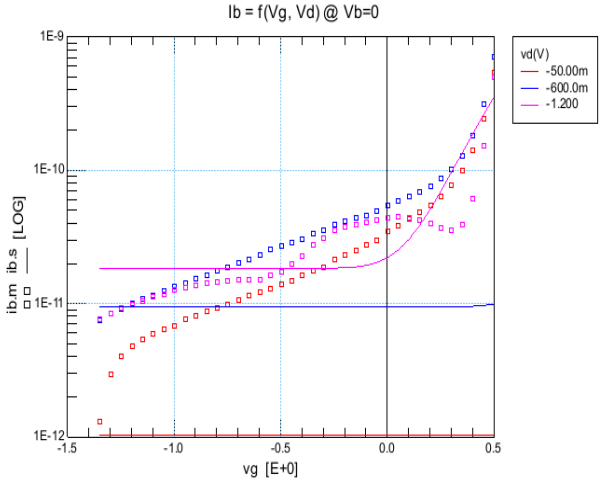


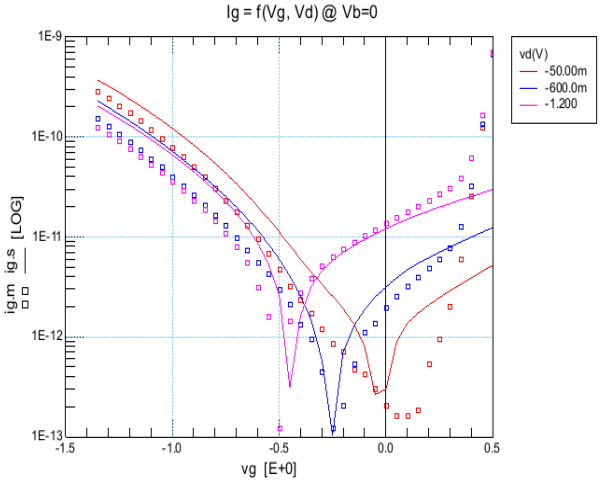




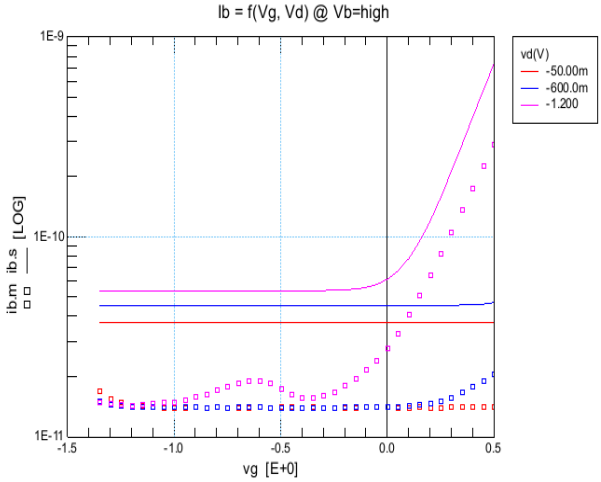


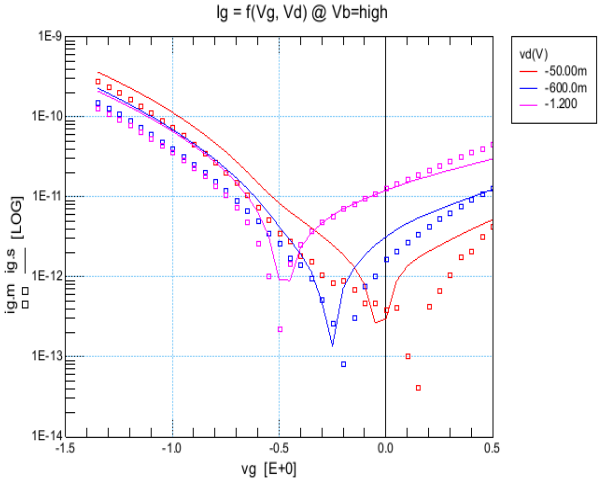
idvg, Ib,Ig, Vb = 0V, T = 70°C



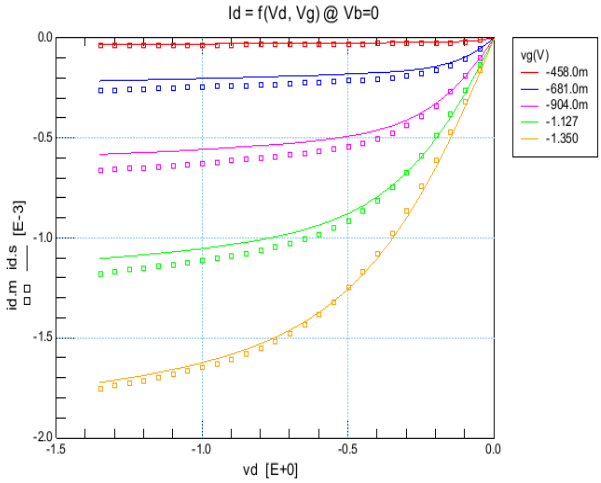


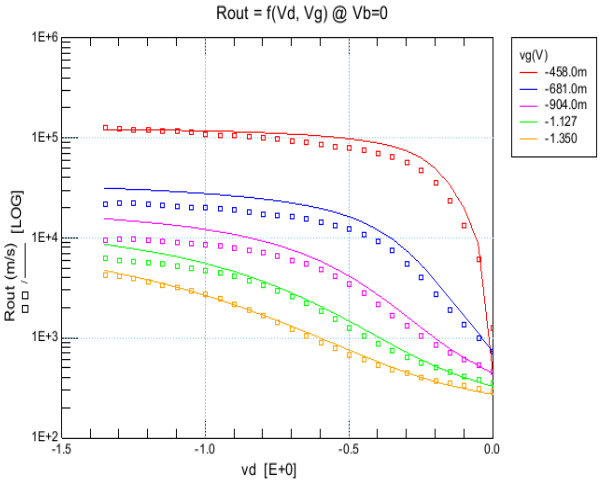
idvg, Ib,Ig, Vb = 1.2V, T = 70°C

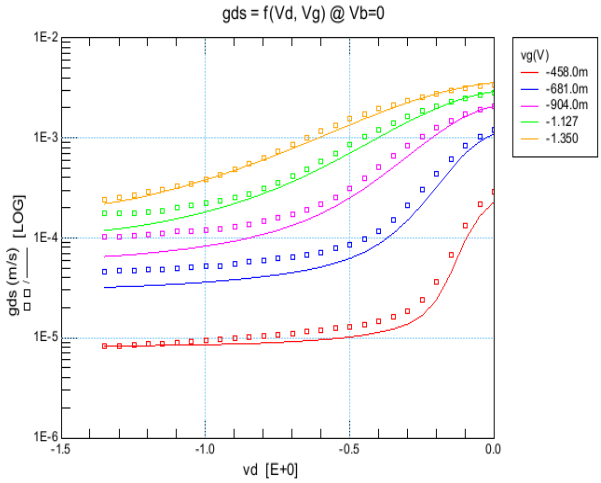




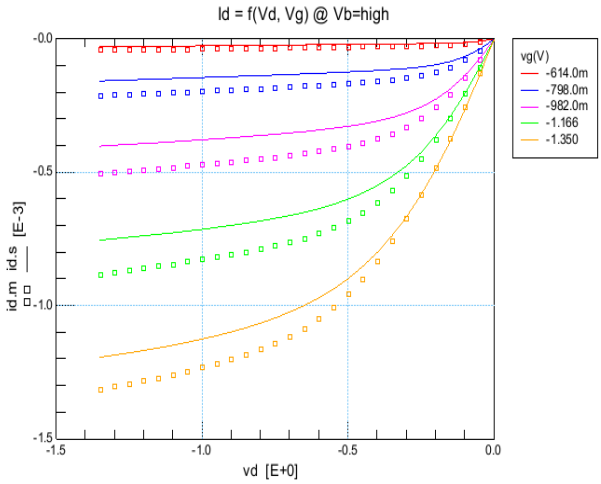
idvd, Vb = 0V, T = 70°C

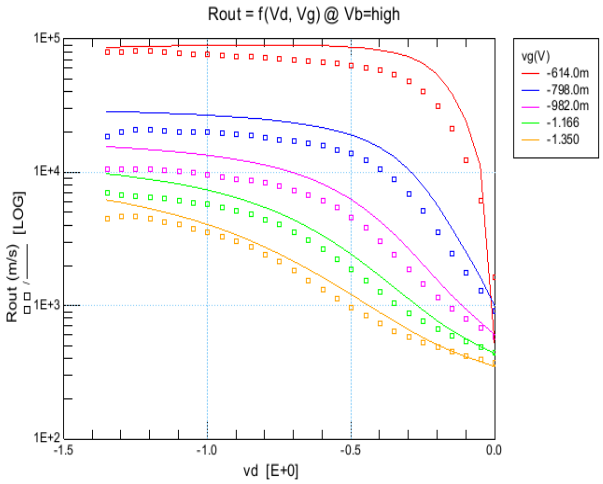


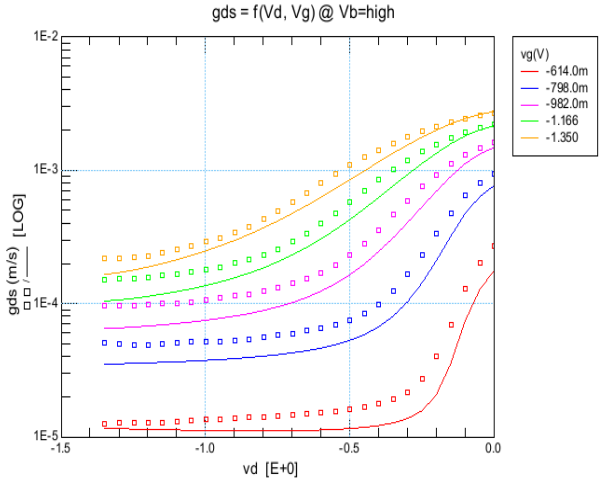




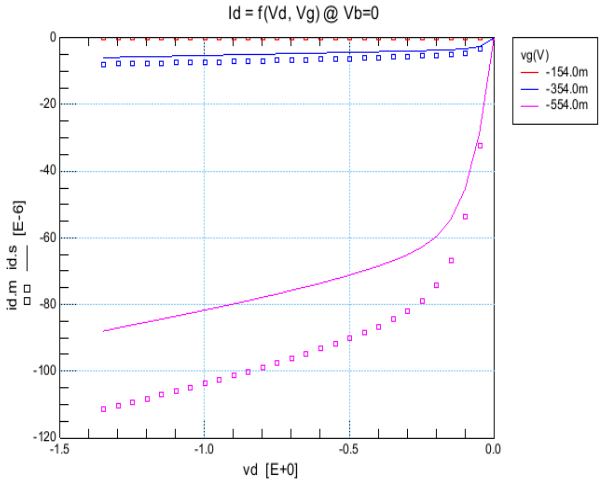
idvd_vbmin, Vb = 1.2V, T = 70°C

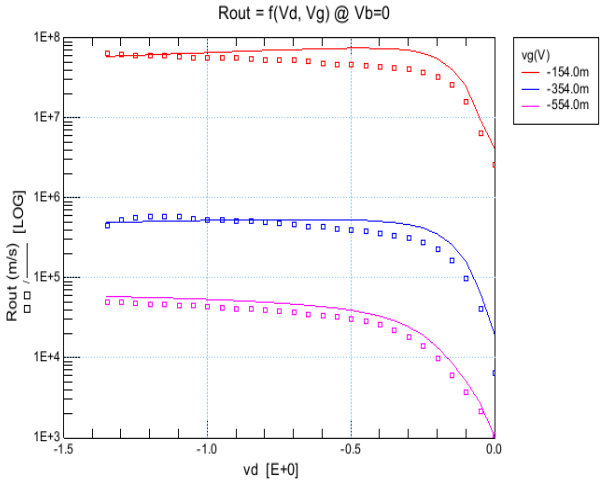


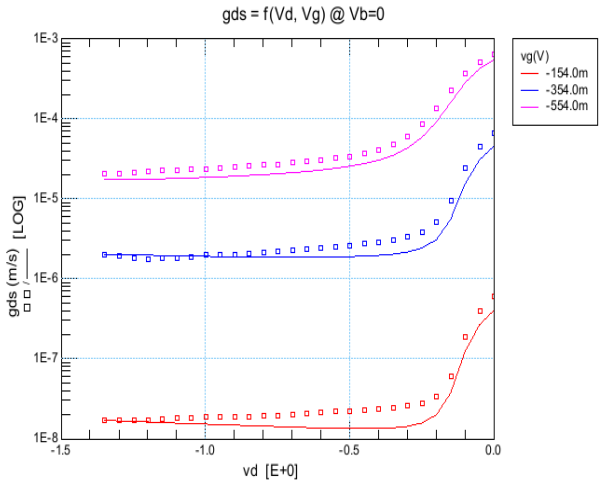




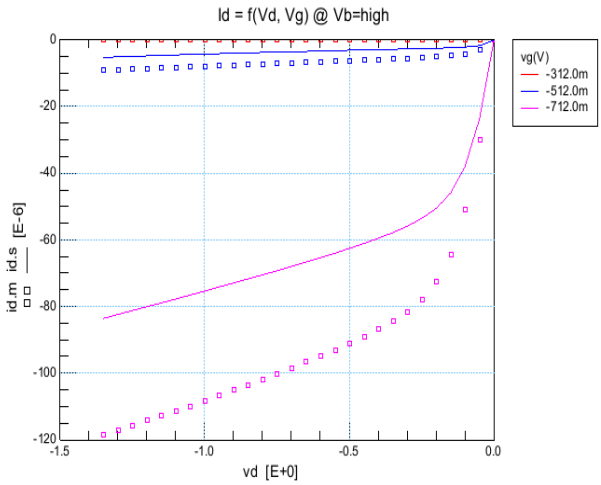
idvd_vth, Vb = 0V, T = 70°C

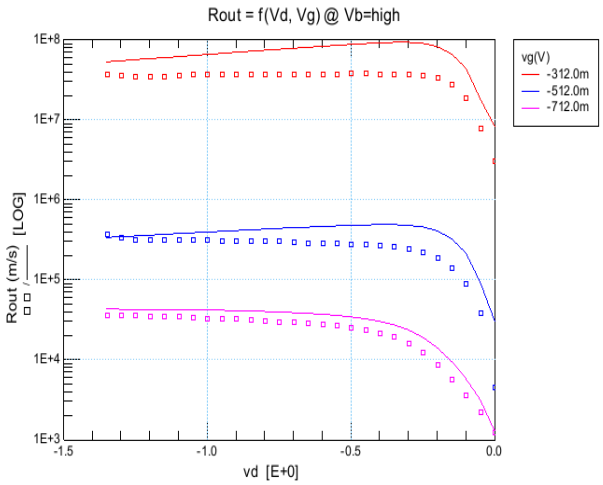


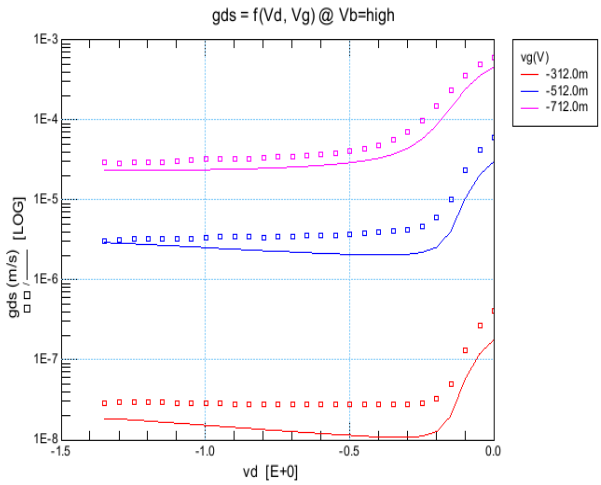




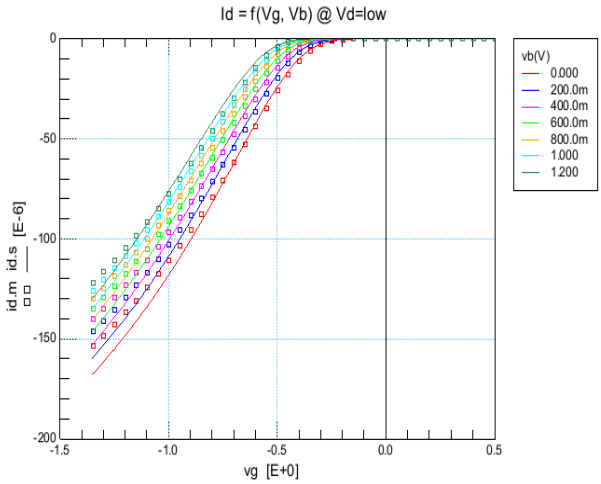
idvd_vth_vbmin, Vb = 1.2V, T = 70°C

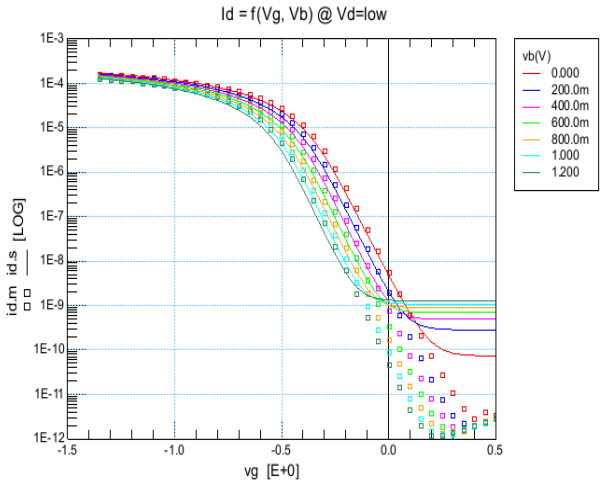


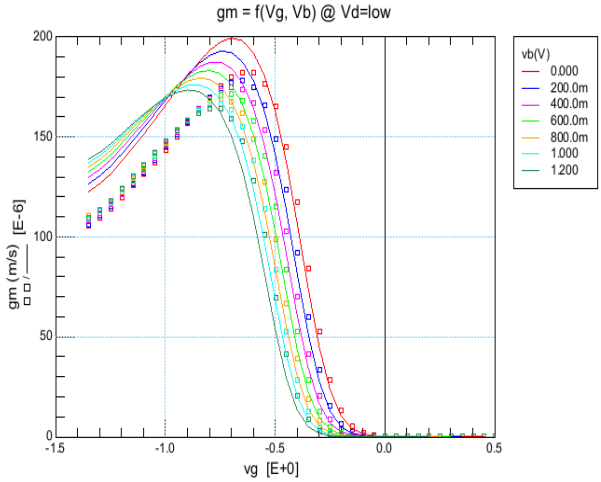


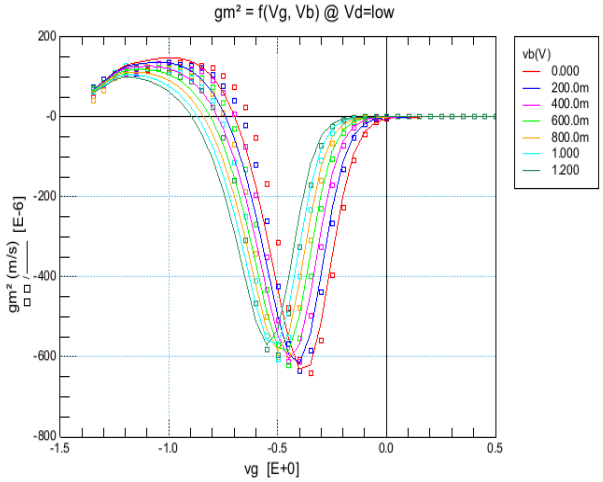


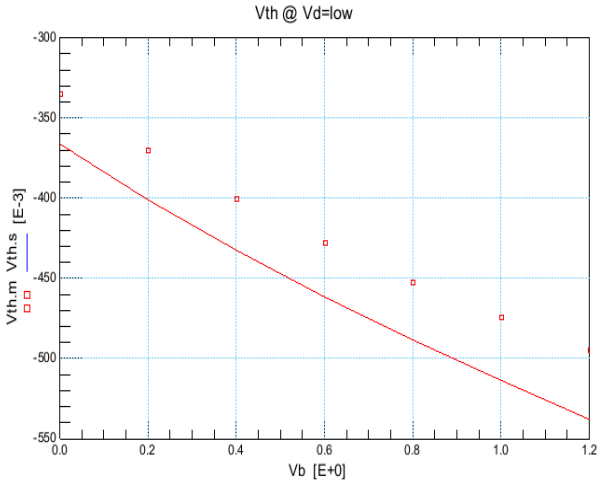
idvg, Vd = -0.05V, T = 125°C

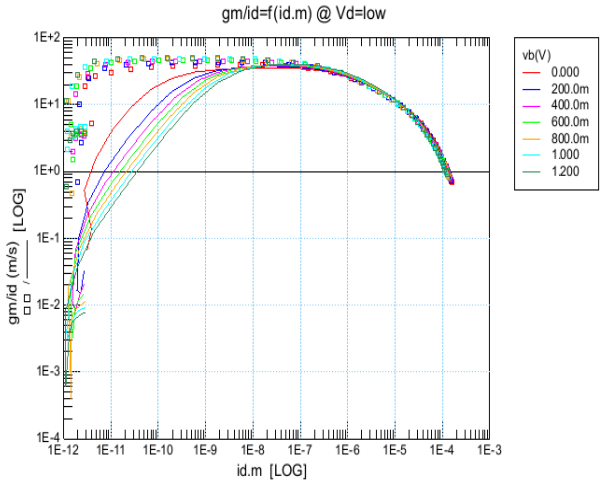


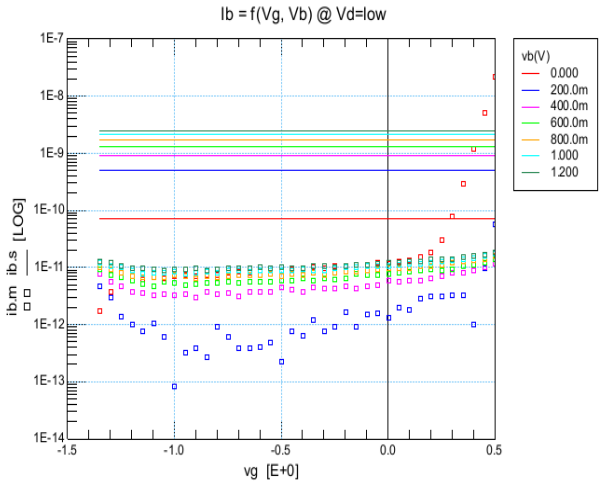


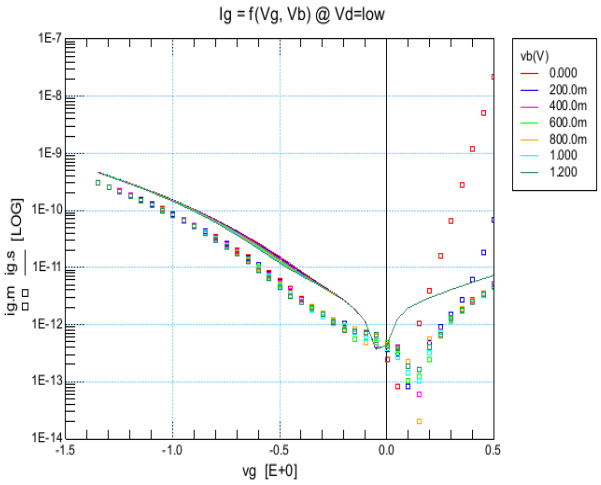




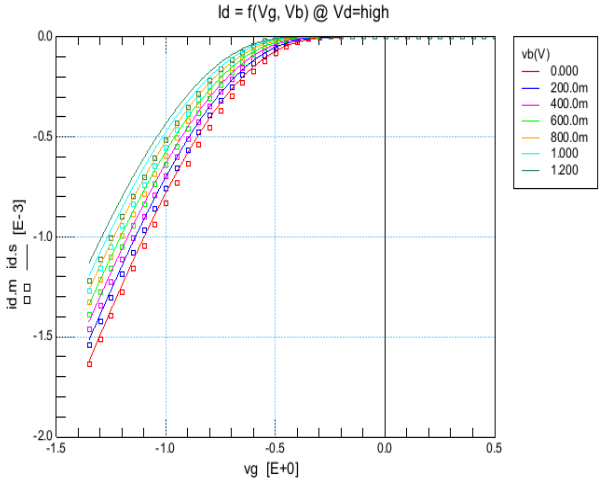


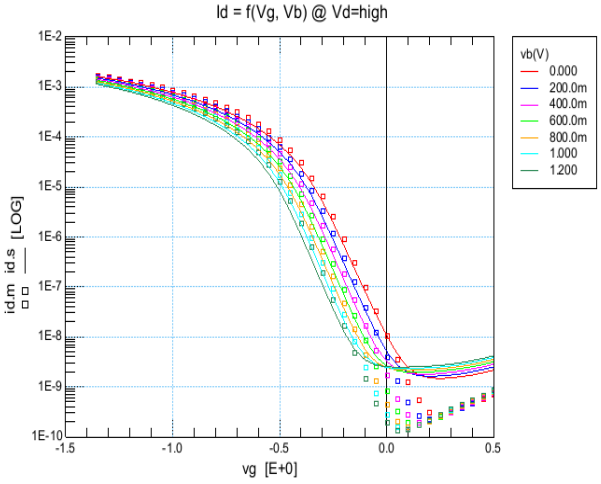


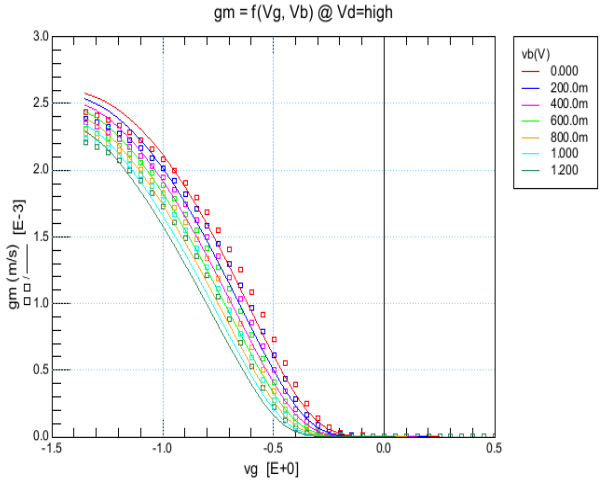


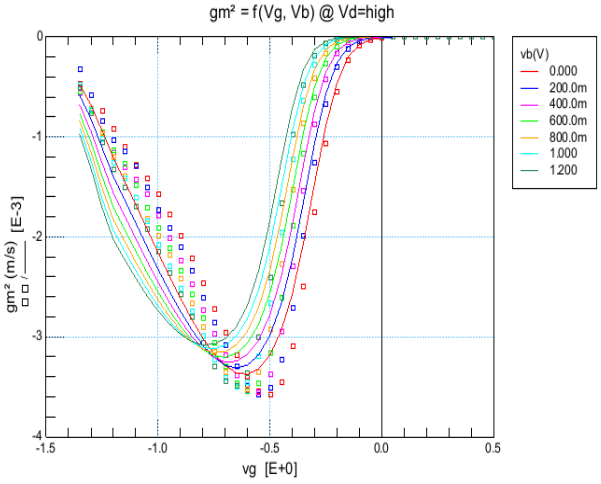


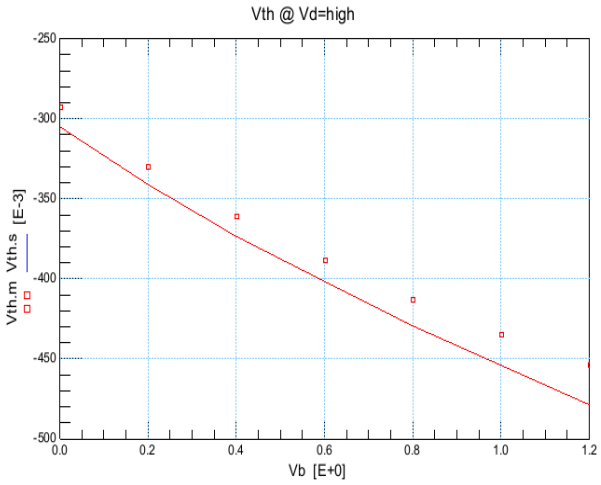
idvg, Vd = -1.2V, T = 125°C

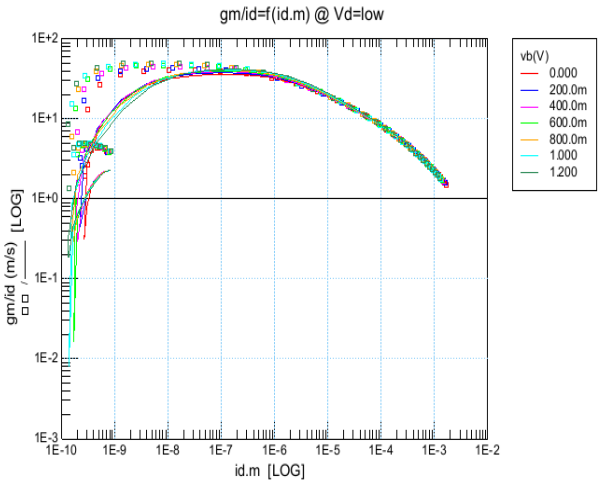


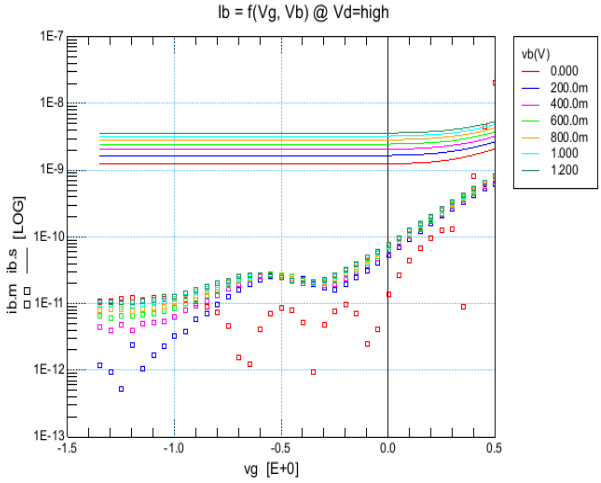


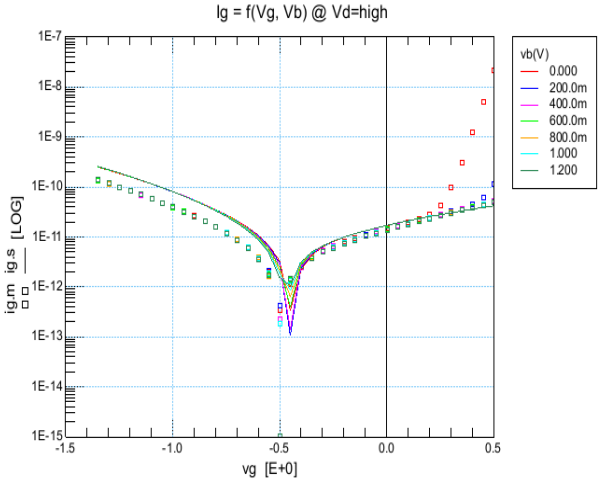




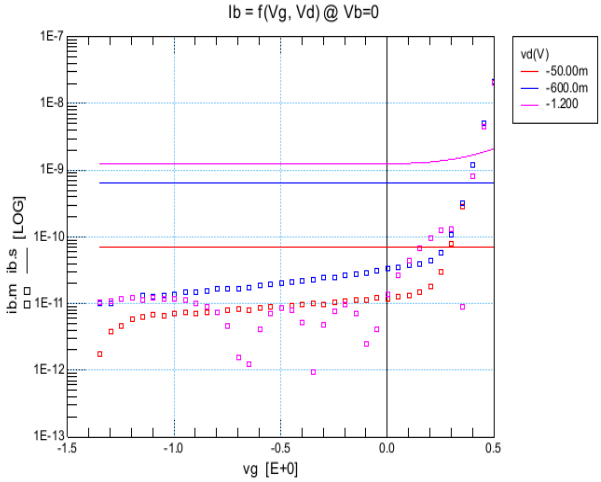


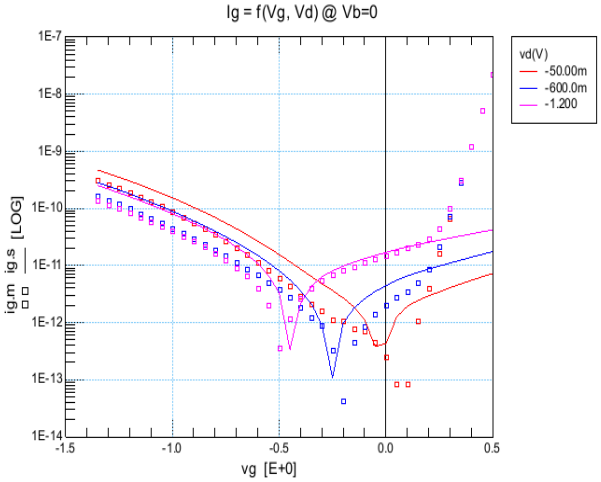




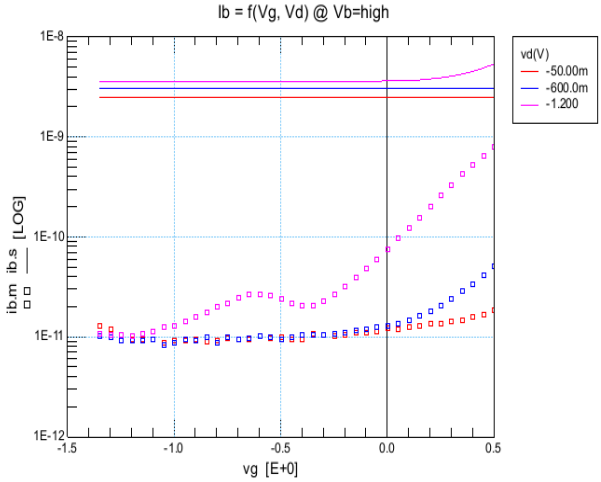


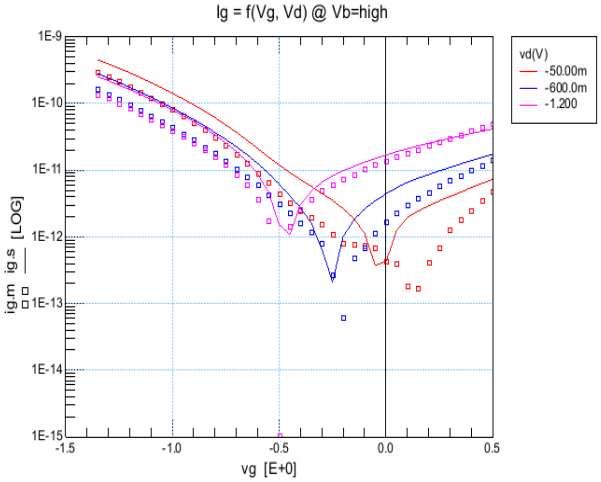
idvg, Ib,Ig, Vb = 0V, T = 125°C



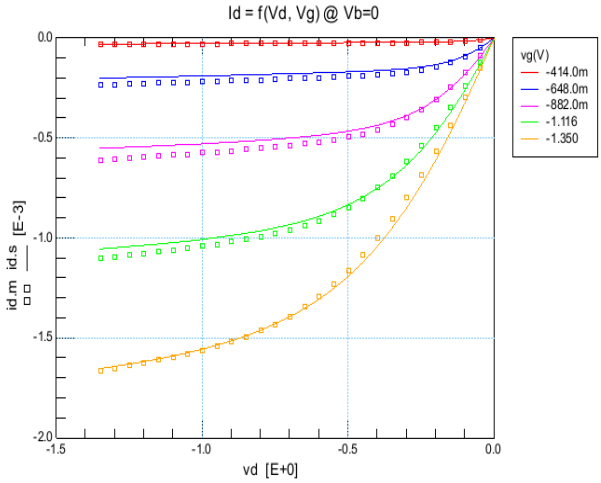


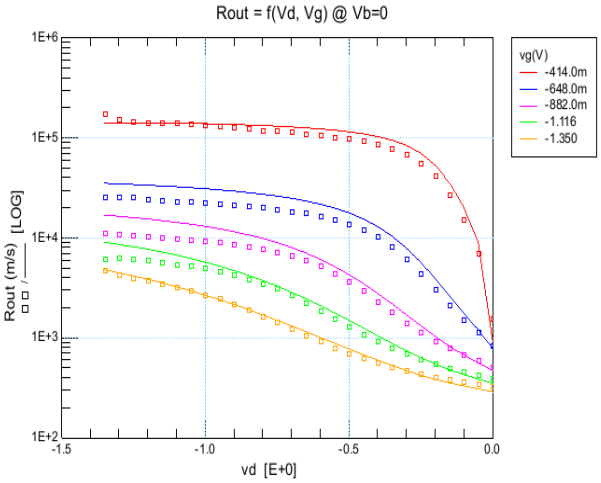
idvg, Ib,Ig, Vb = 1.2V, T = 125°C

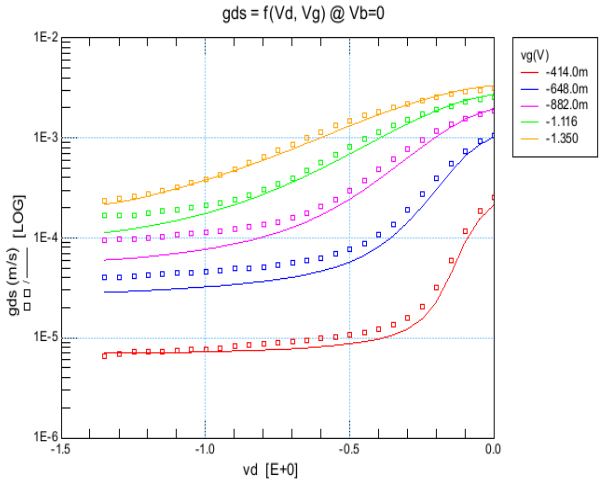




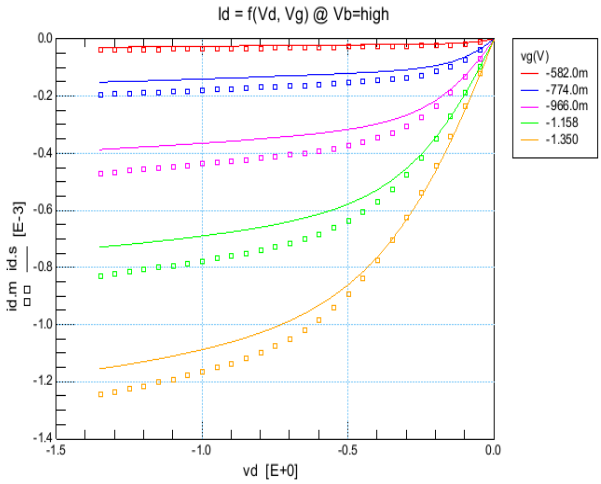
idvd, Vb = 0V, T = 125°C

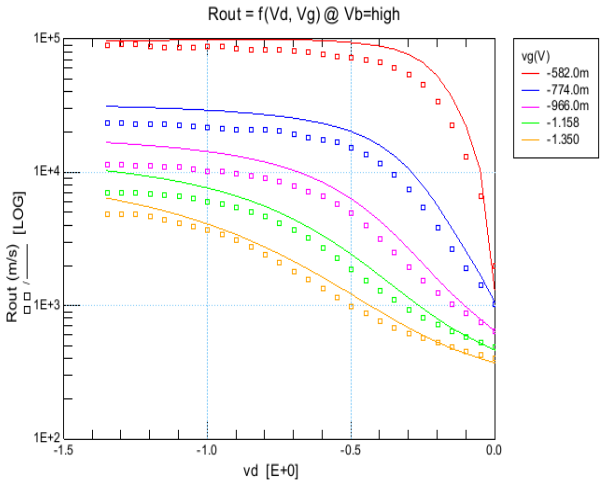


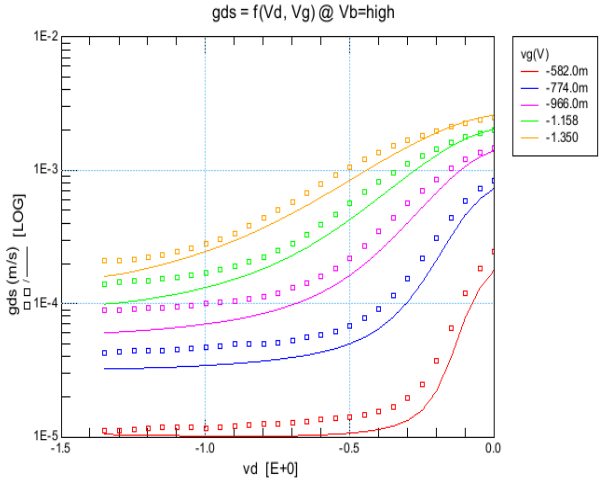




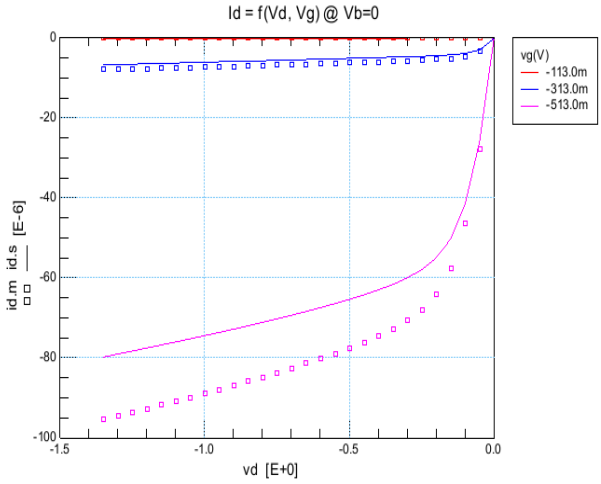
idvd_vbmin, Vb = 1.2V, T = 125°C

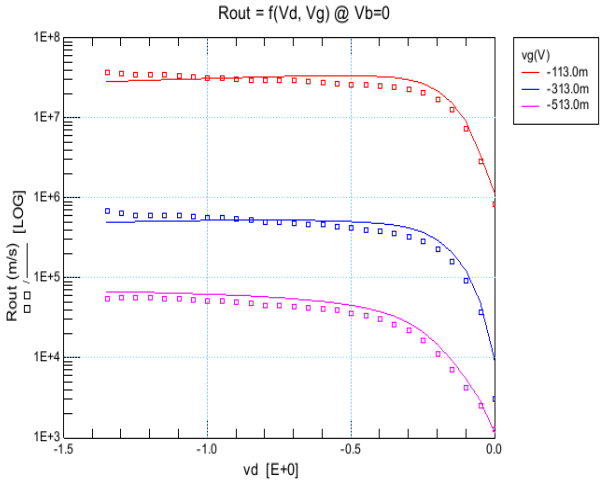


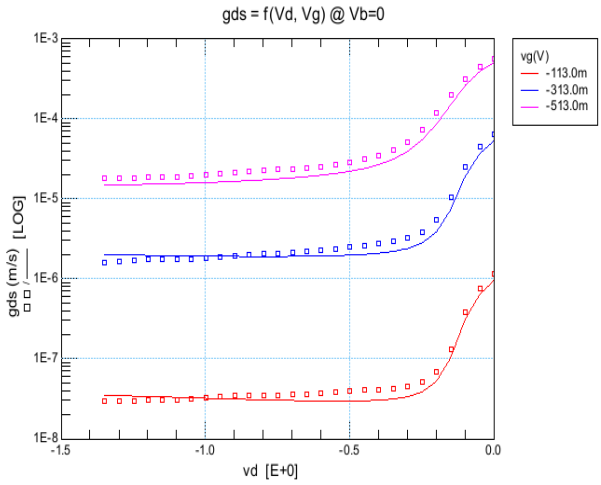




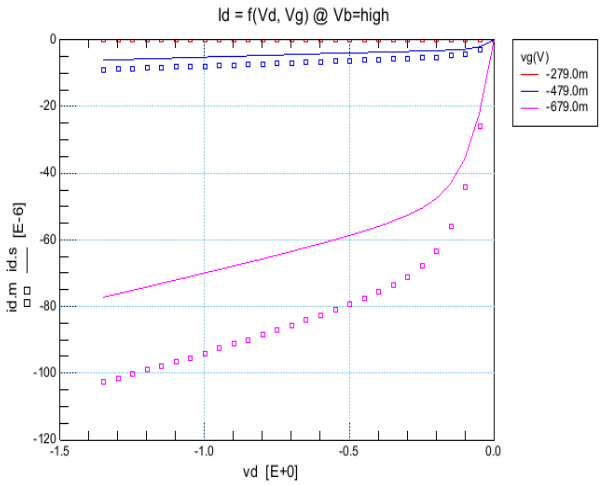
idvd_vth, Vb = 0V, T = 125°C

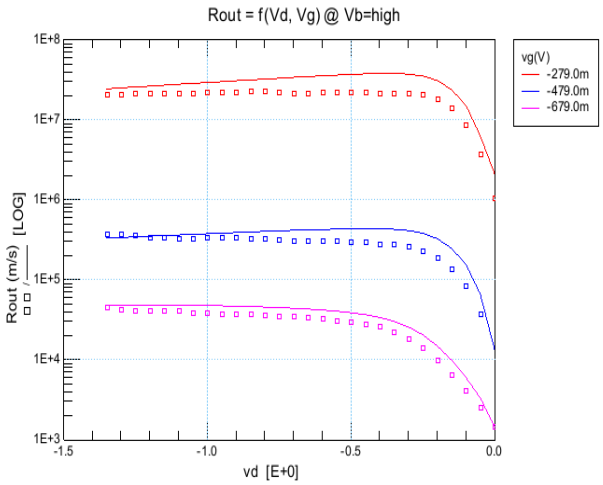


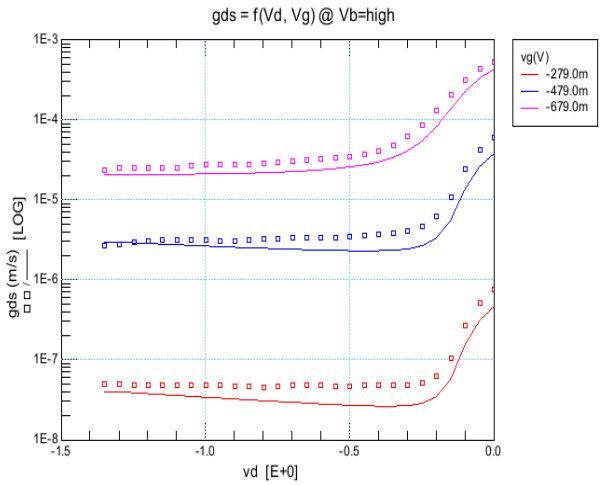




idvd_vth_vbmin, Vb = 1.2V, T = 125°C

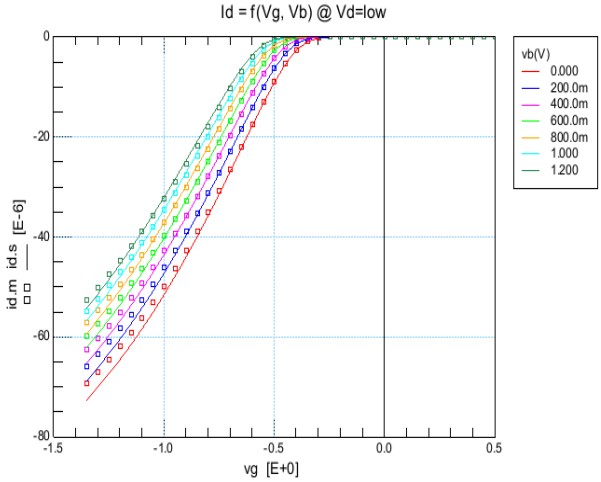


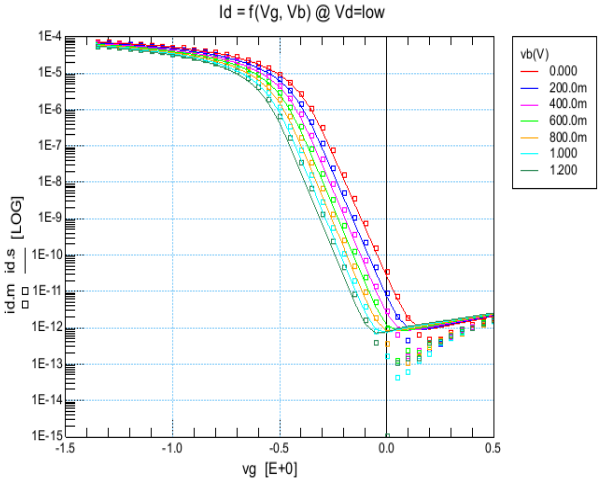


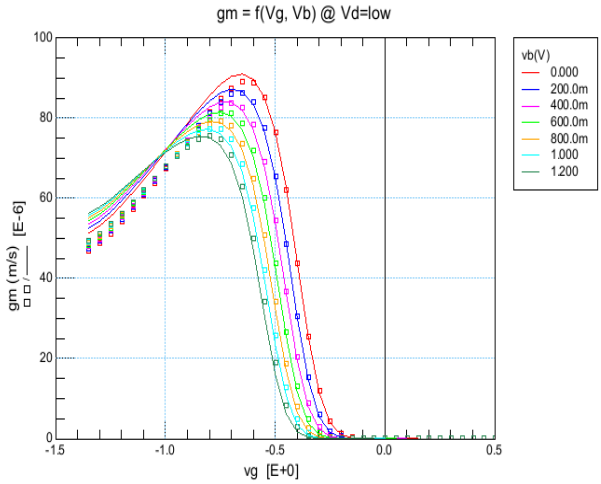


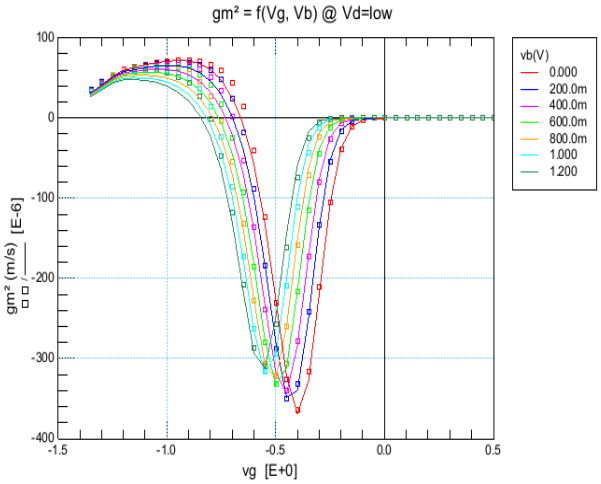
4.8 W10u0_L0u5_S549_1

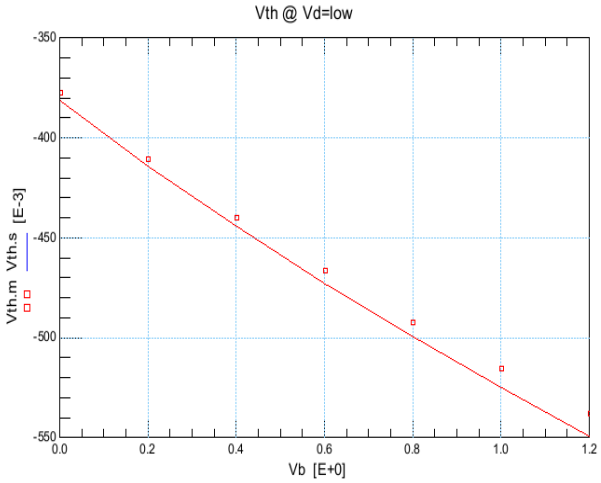
idvg, Vd = -0.05V, T = 27°C

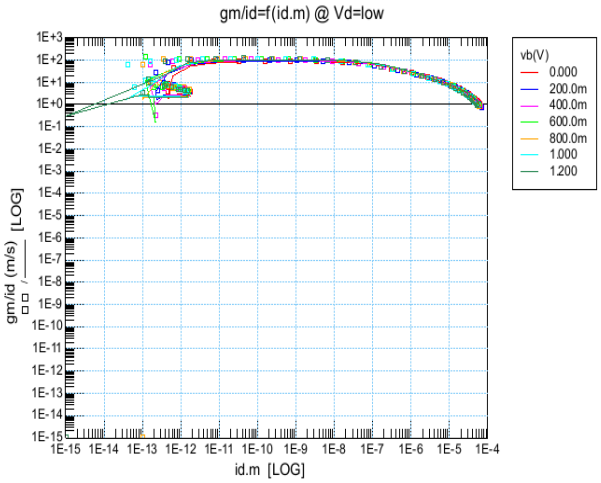


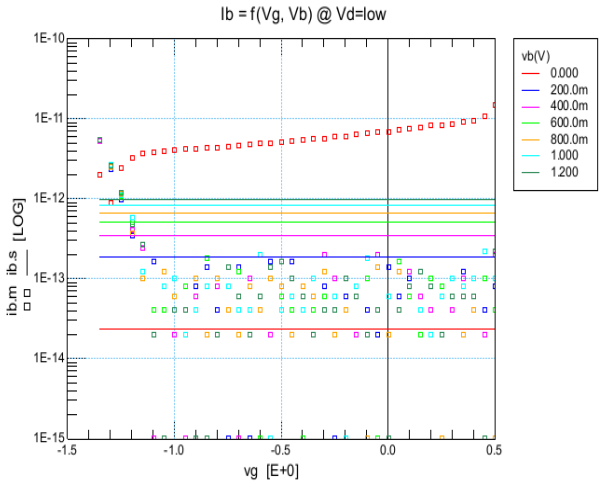


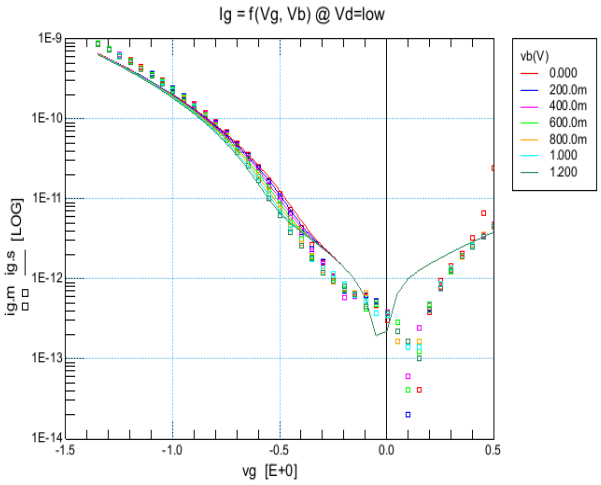




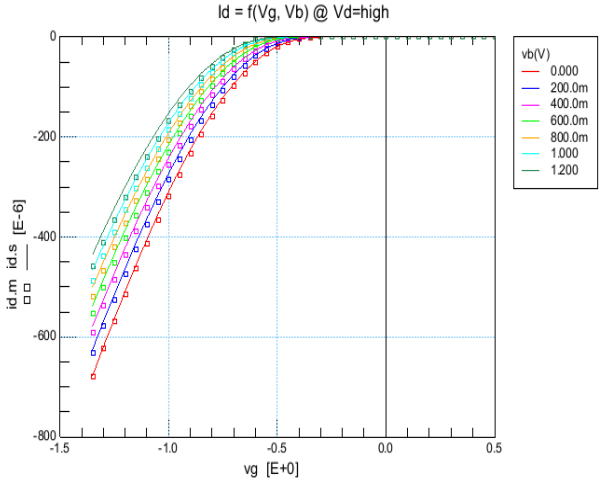


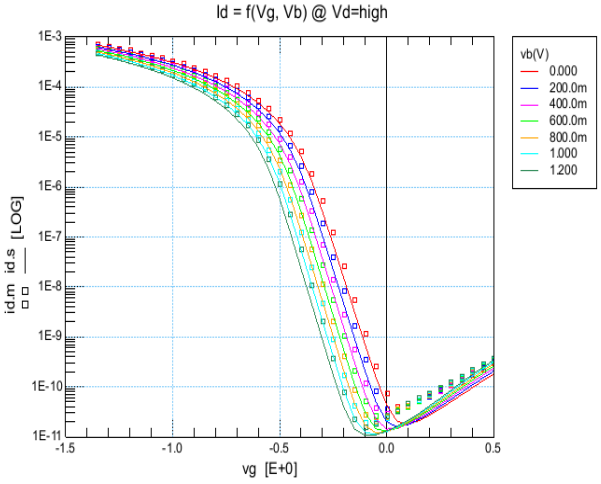


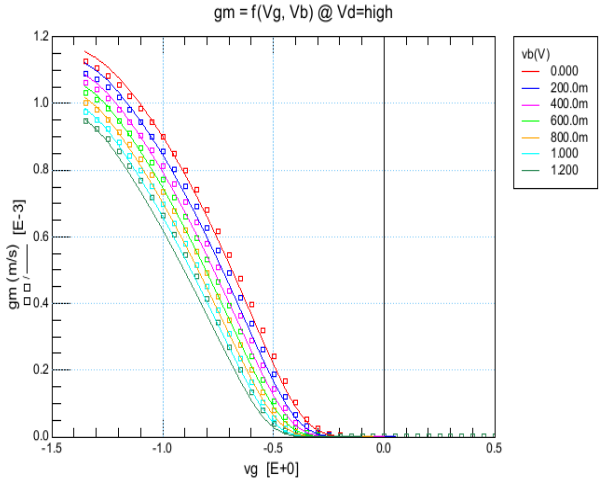


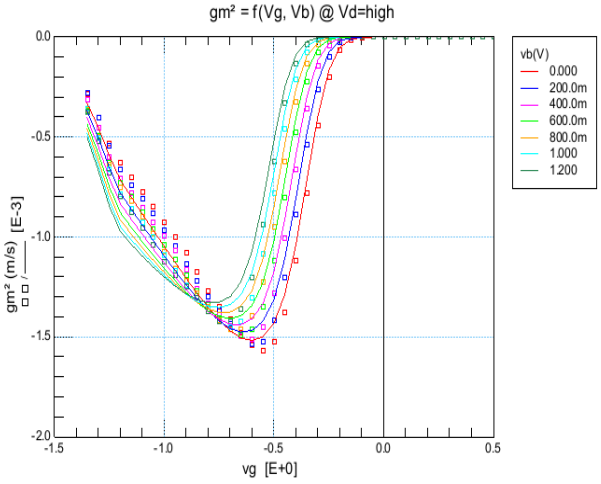


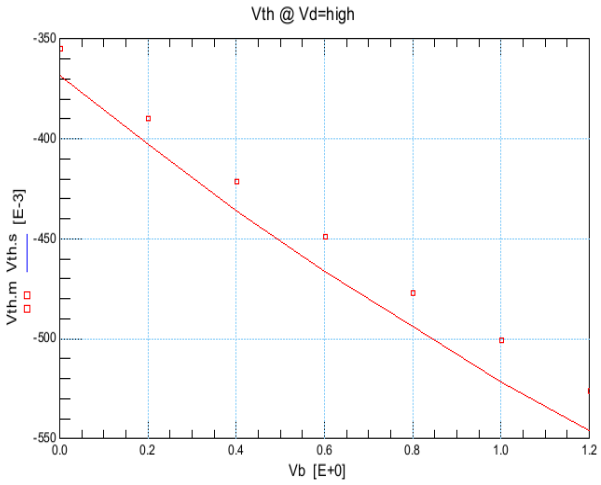
idvg, Vd = -1.2V, T = 27°C

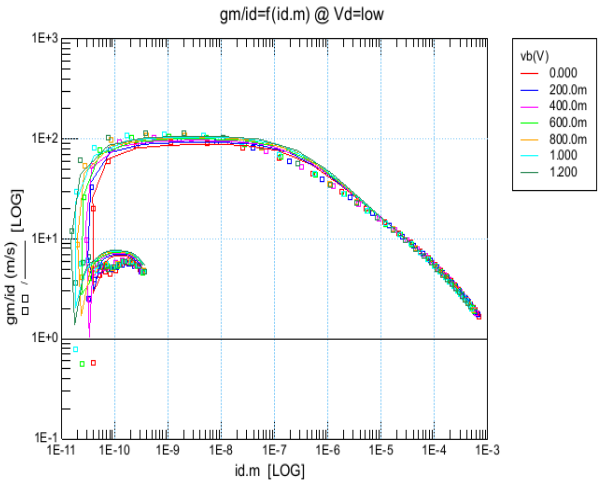


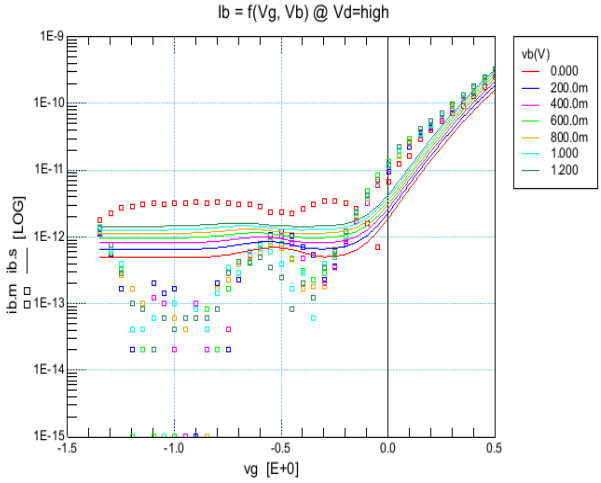


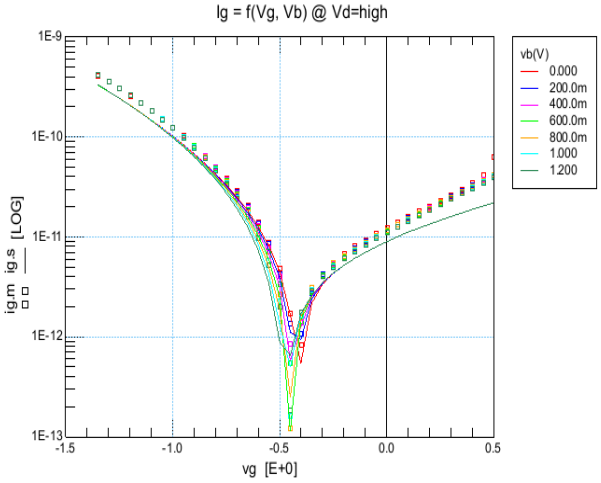




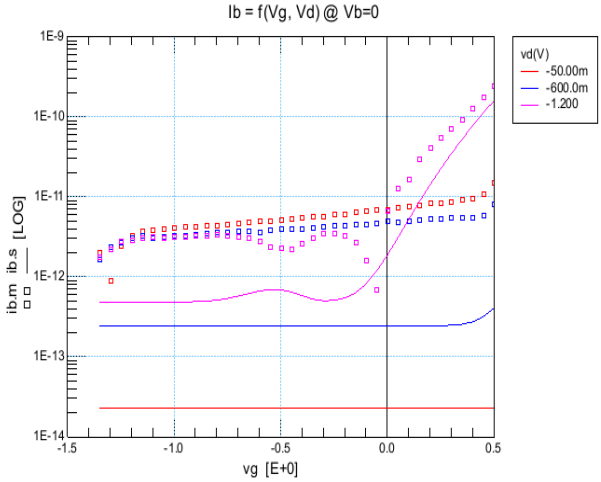


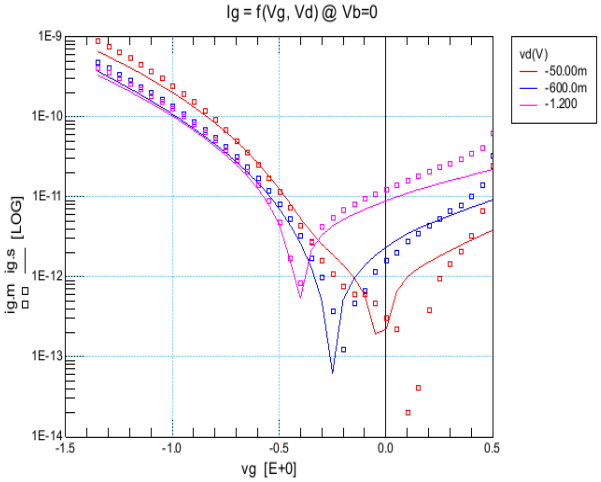




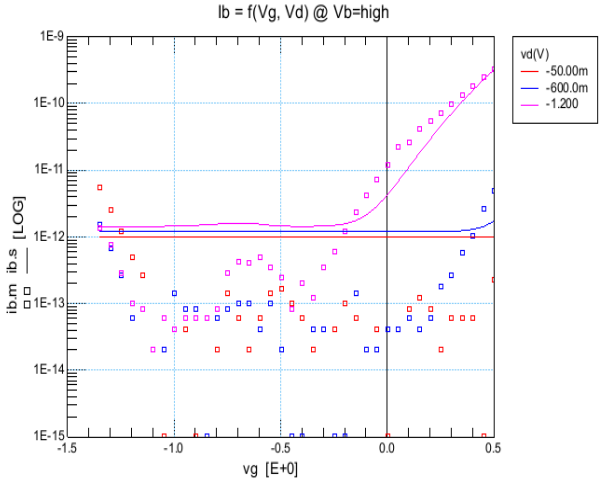


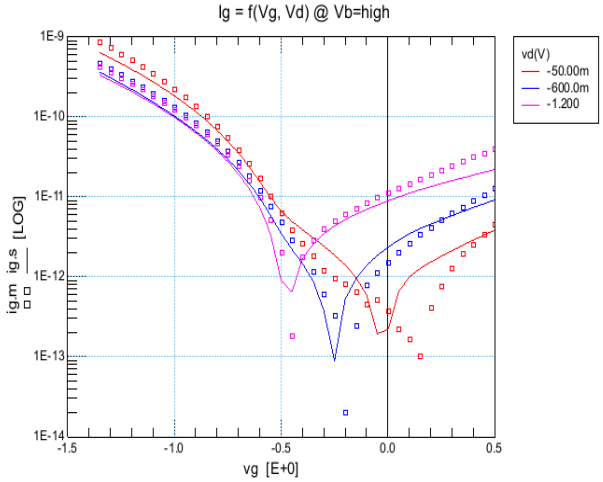
idvg, Ib,Ig, Vb = 0V, T = 27°C



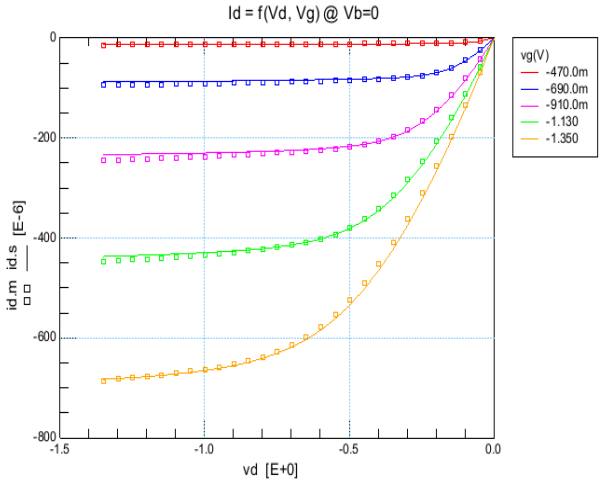


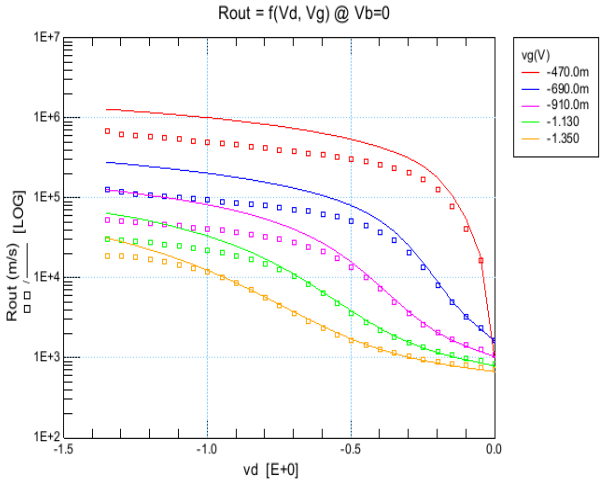
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

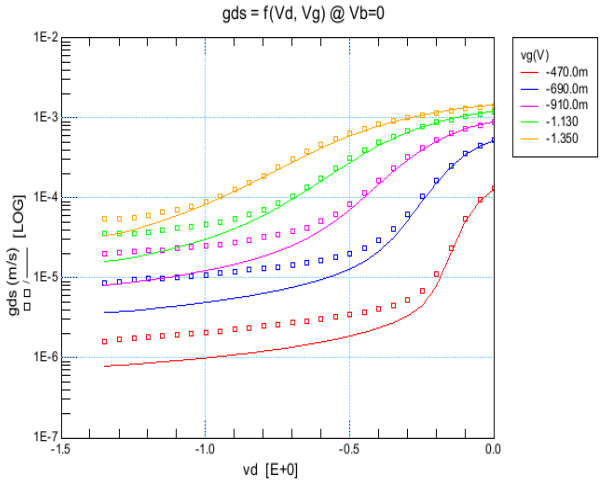




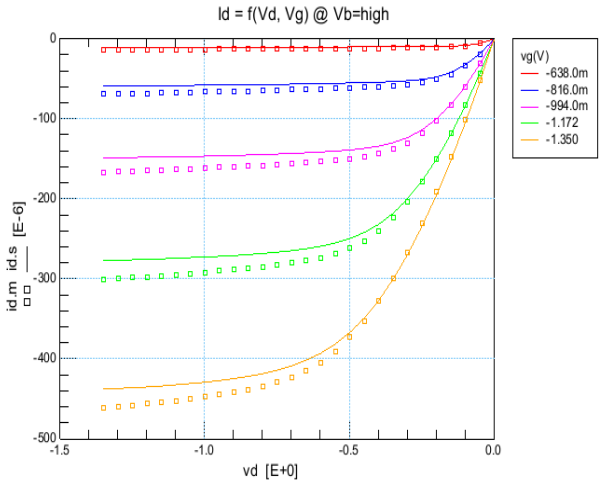
idvd, Vb = 0V, T = 27°C

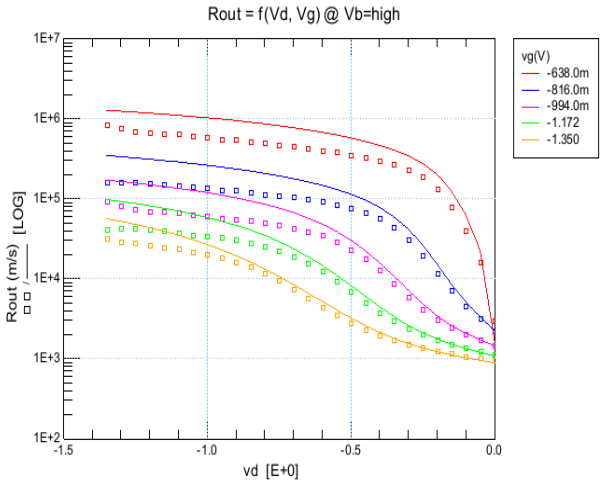


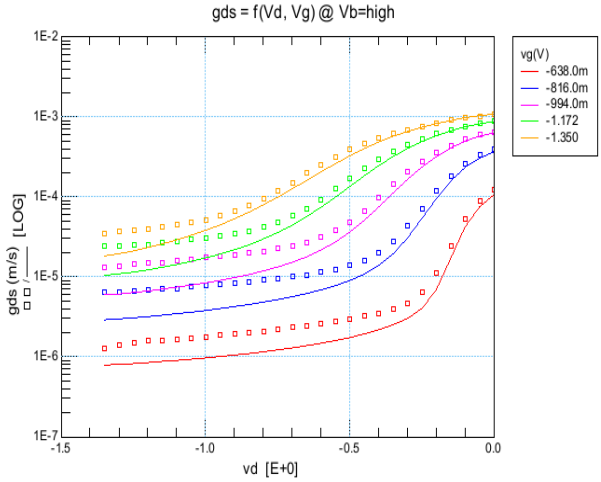




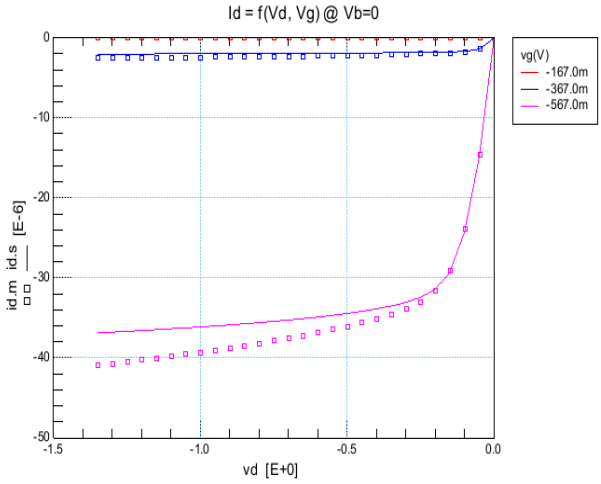
idvd_vbmin, Vb = 1.2V, T = 27°C

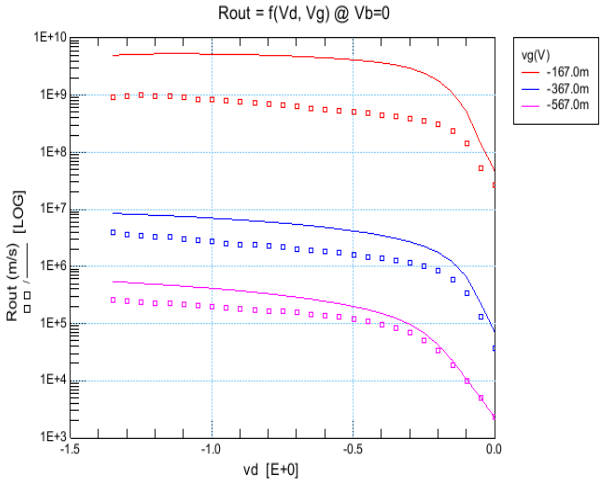


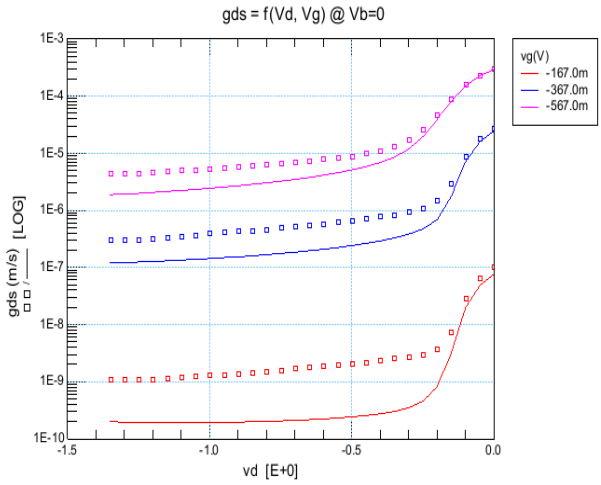




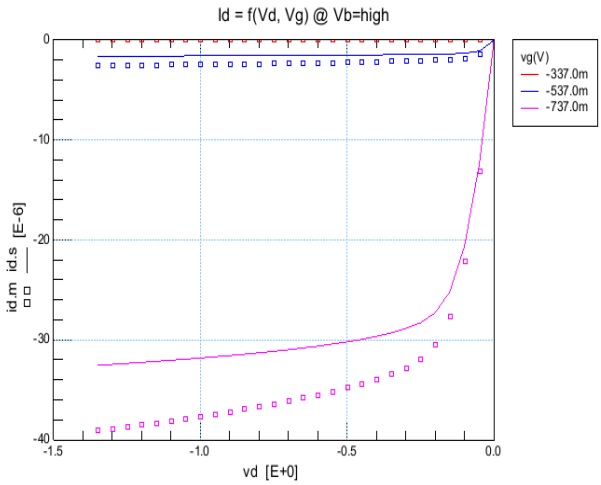
idvd_vth, Vb = 0V, T = 27°C

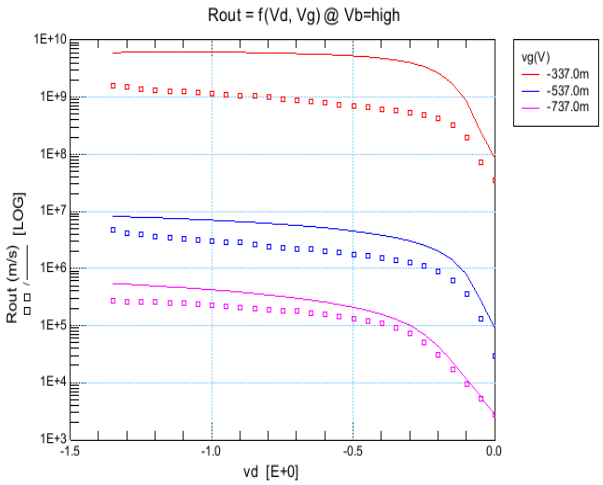


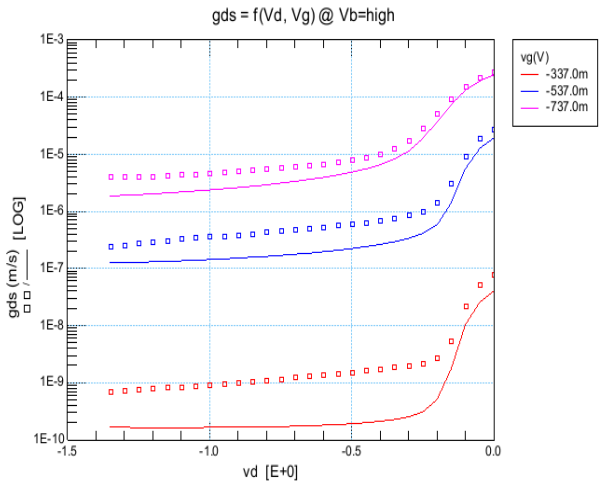




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

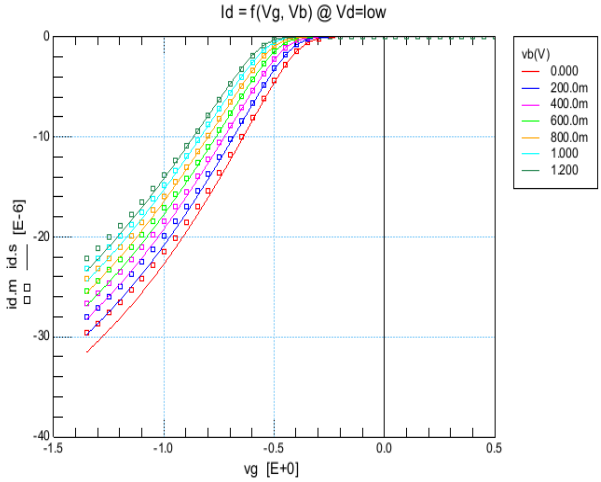


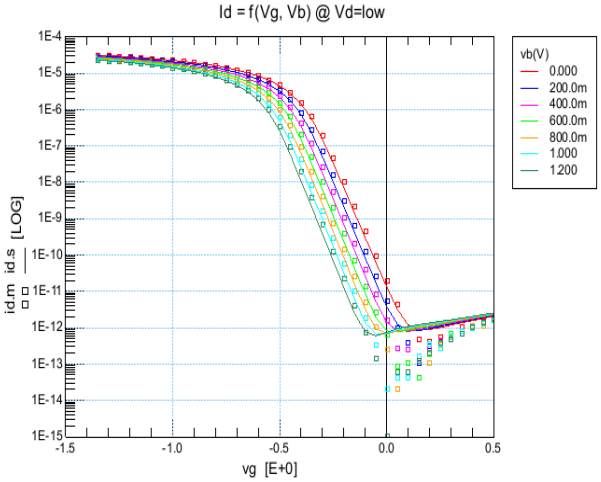


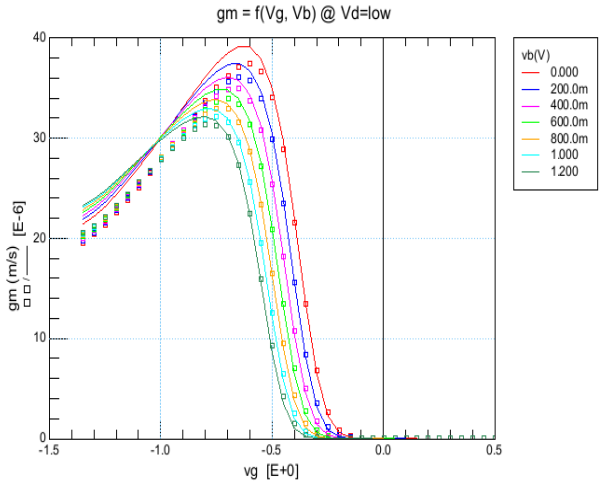


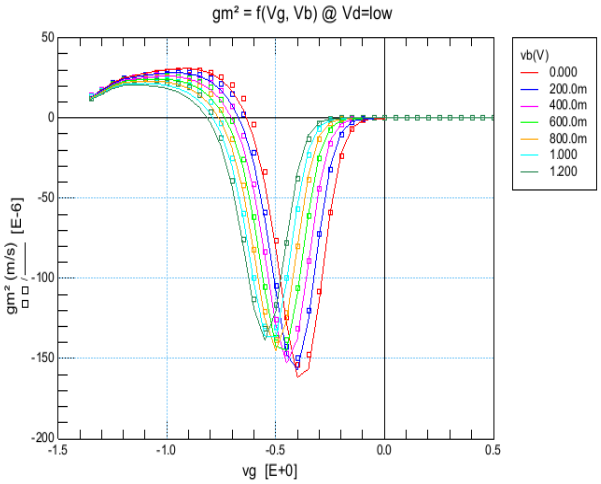
4.9 W10u0_L1u2_S549_2

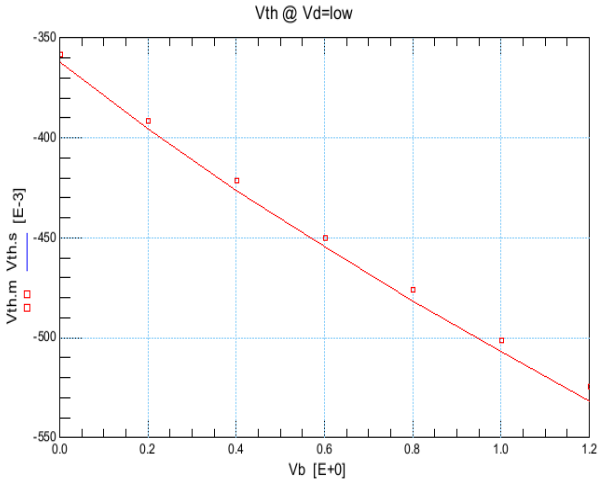
idvg, Vd = -0.05V, T = 27°C

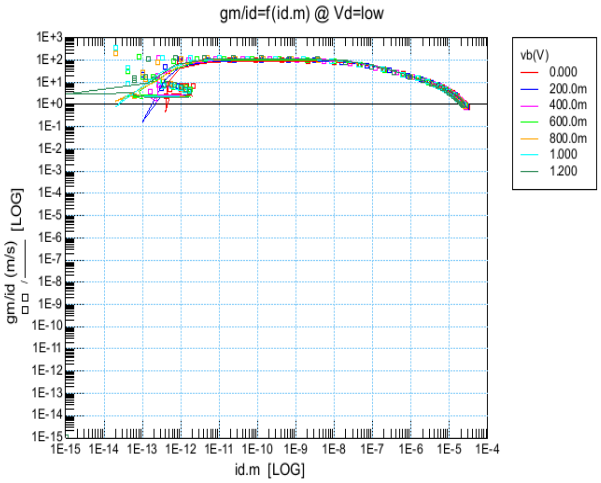


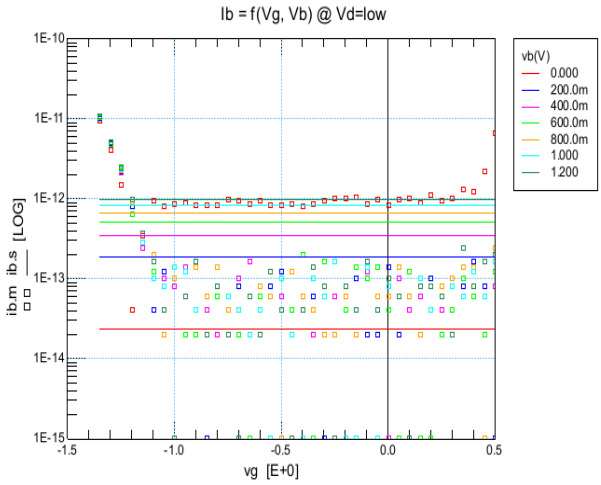


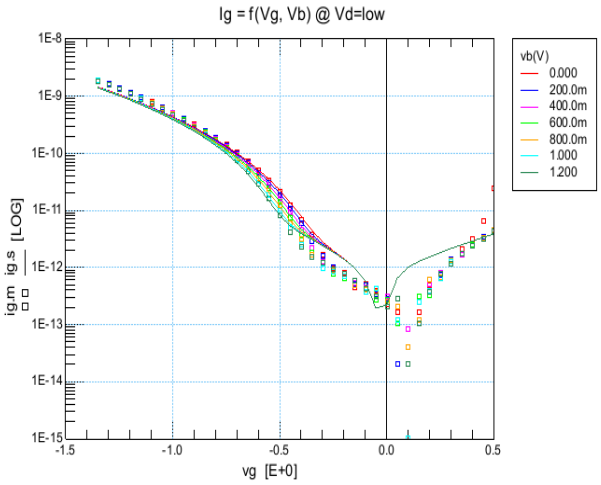




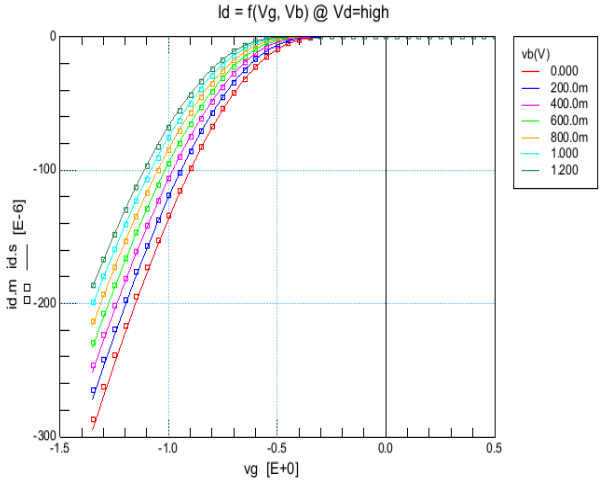


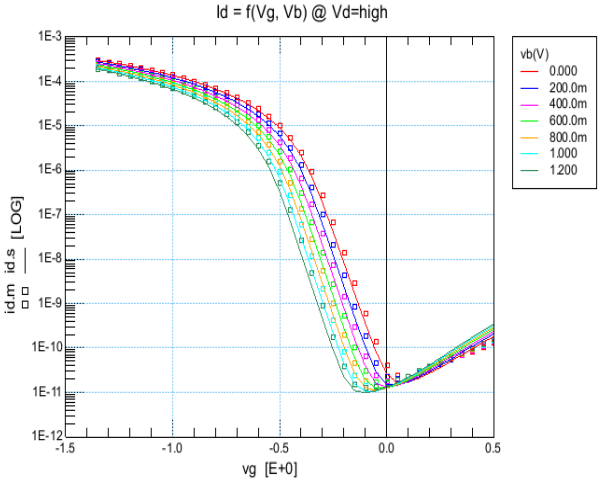


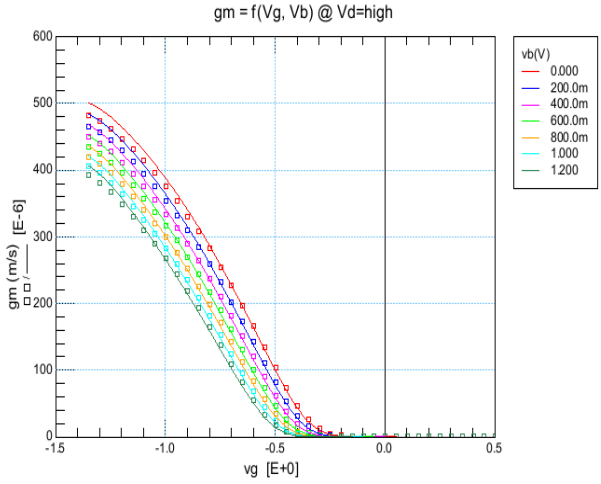


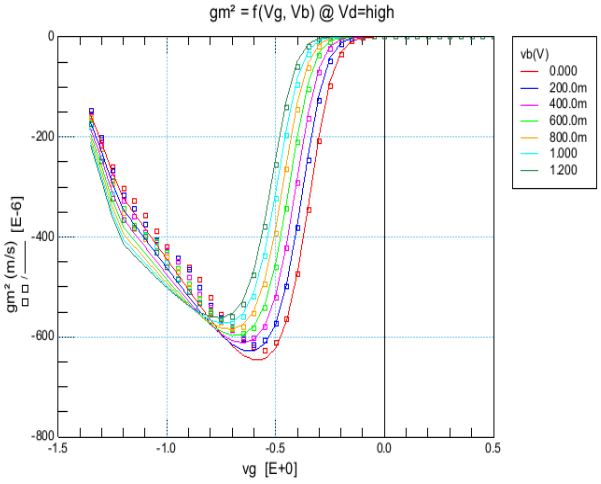


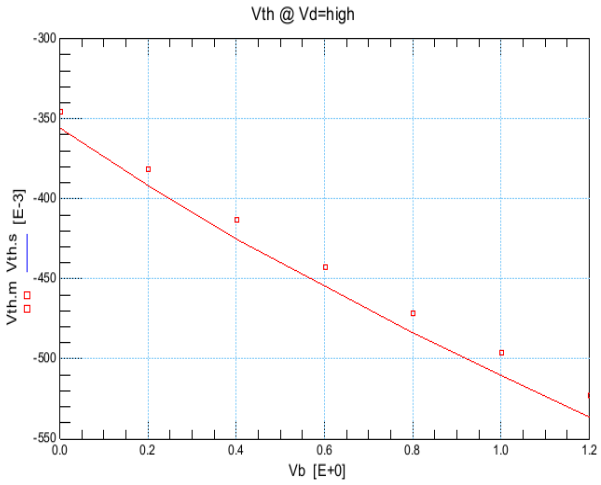
idvg, Vd = -1.2V, T = 27°C

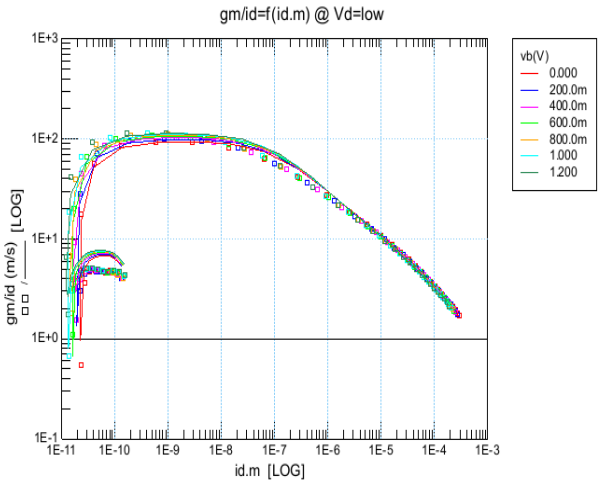


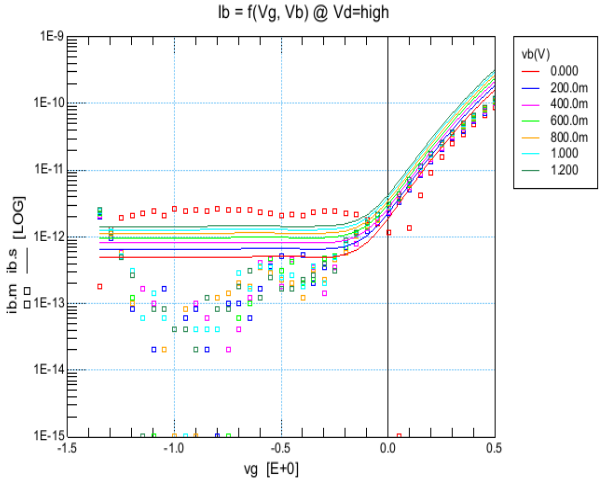


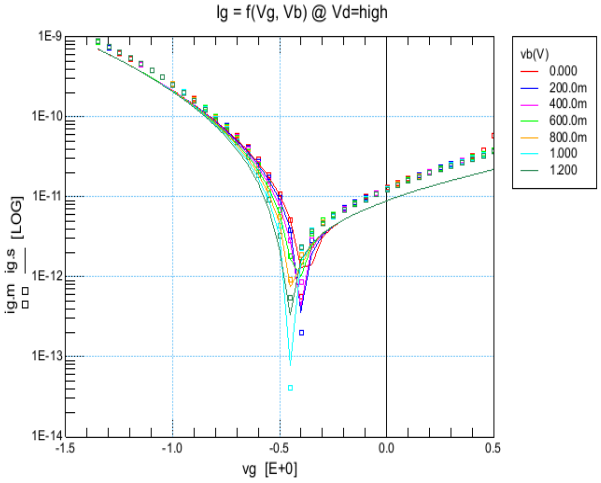




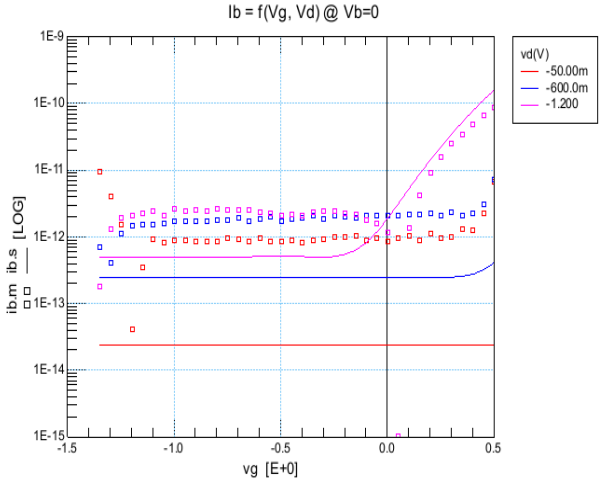


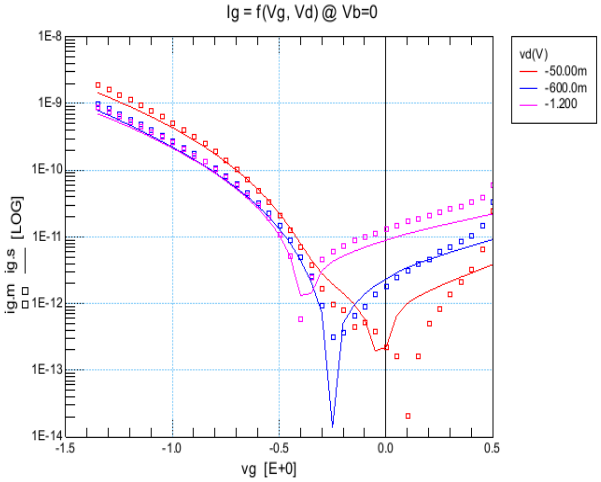




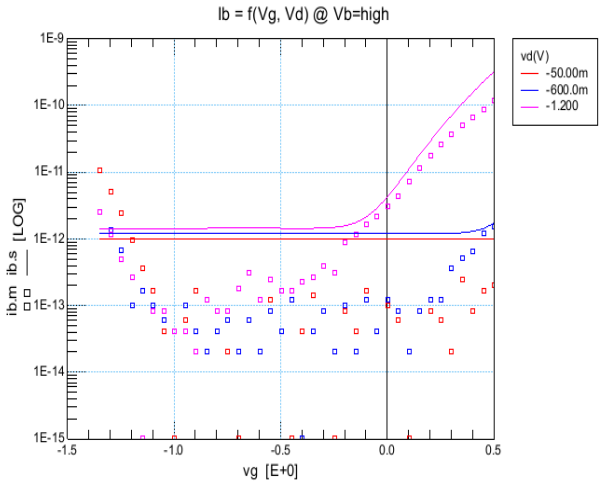


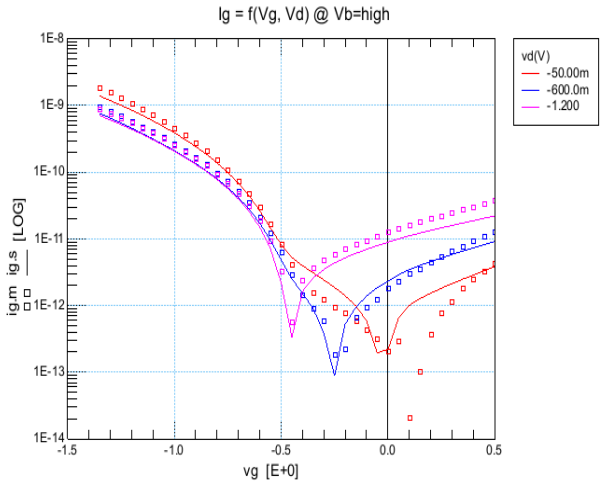
idvg, Ib,Ig, Vb = 0V, T = 27°C



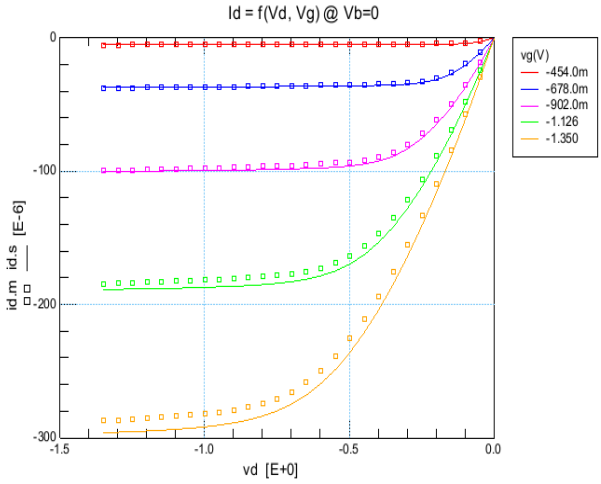


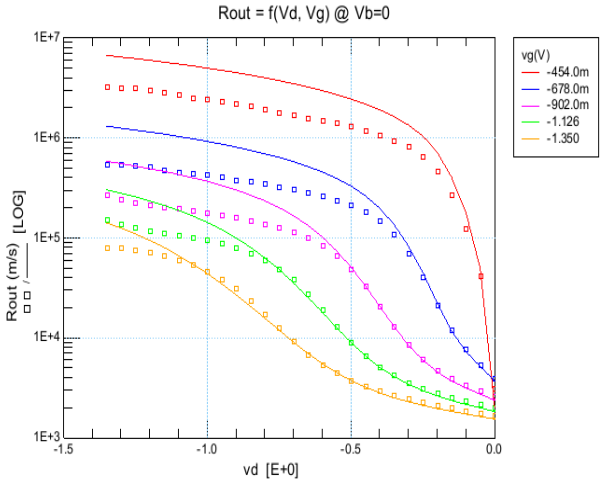
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

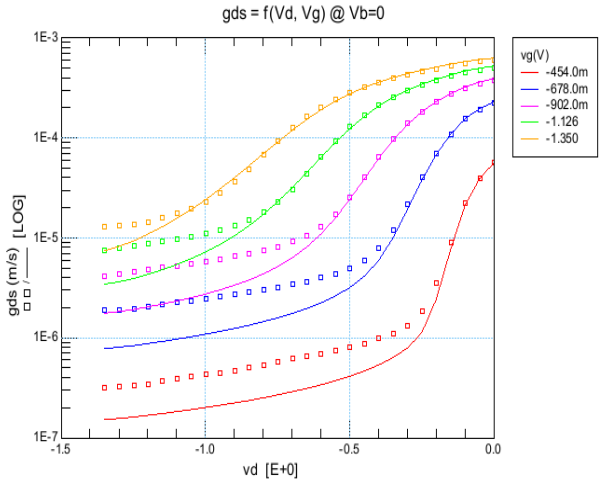




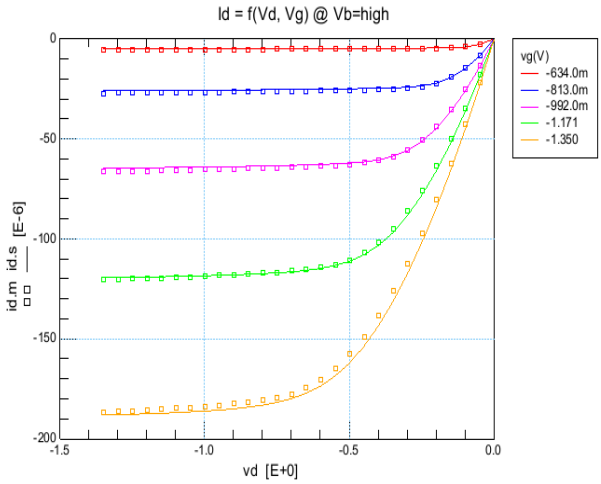
idvd, Vb = 0V, T = 27°C

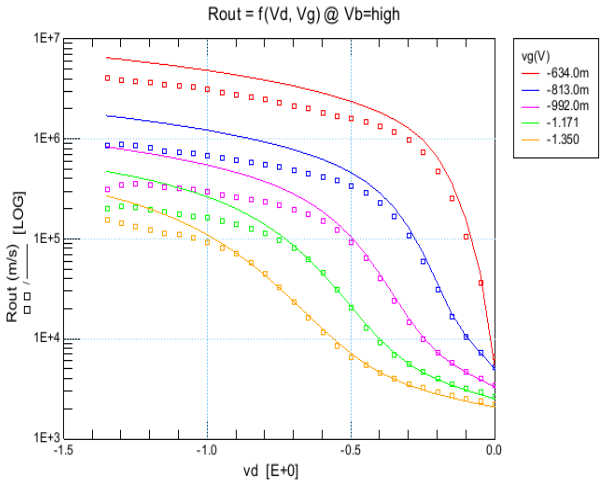


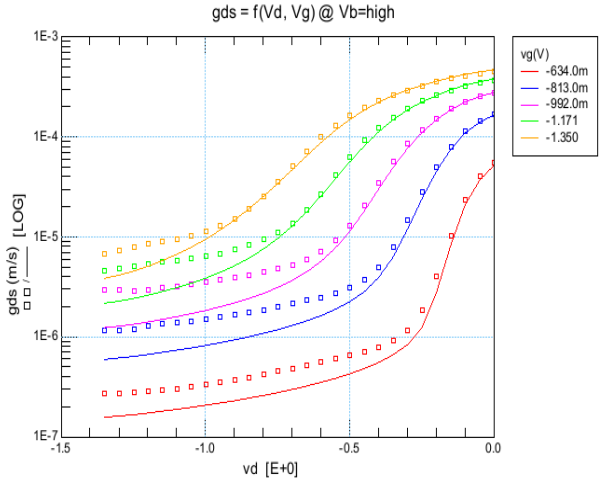




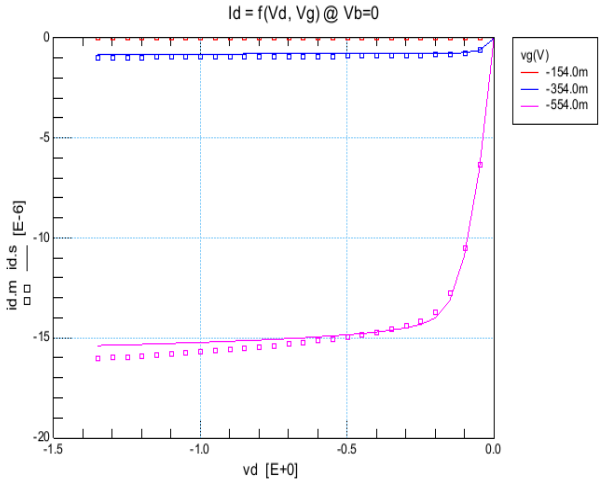
idvd_vbmin, Vb = 1.2V, T = 27°C

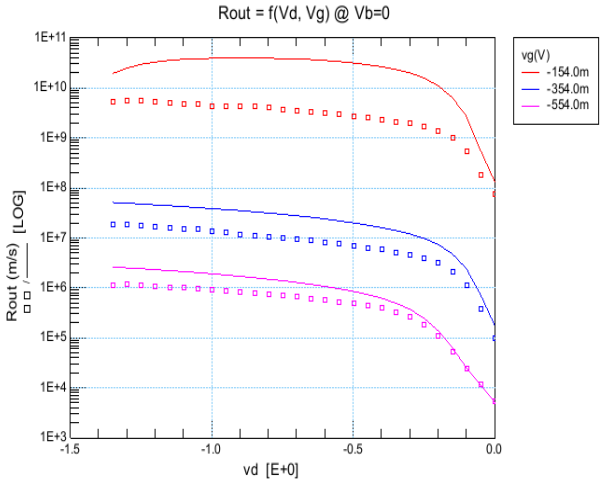


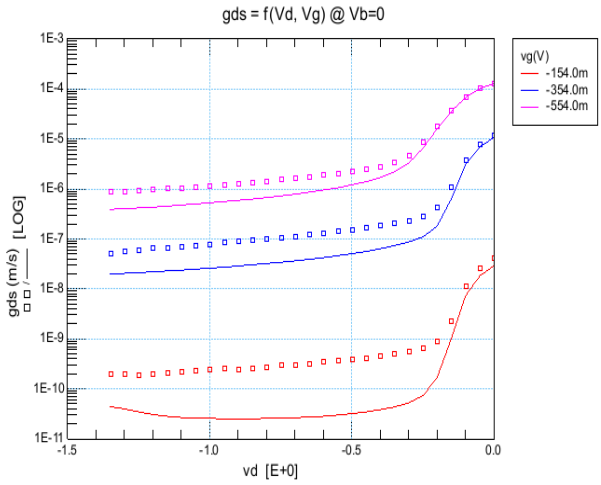




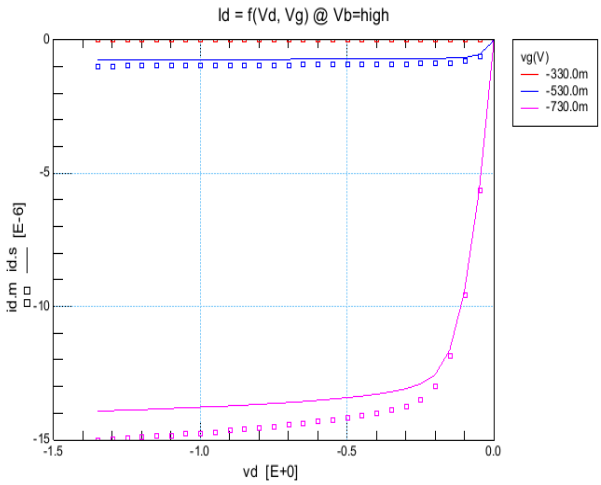
idvd_vth, Vb = 0V, T = 27°C

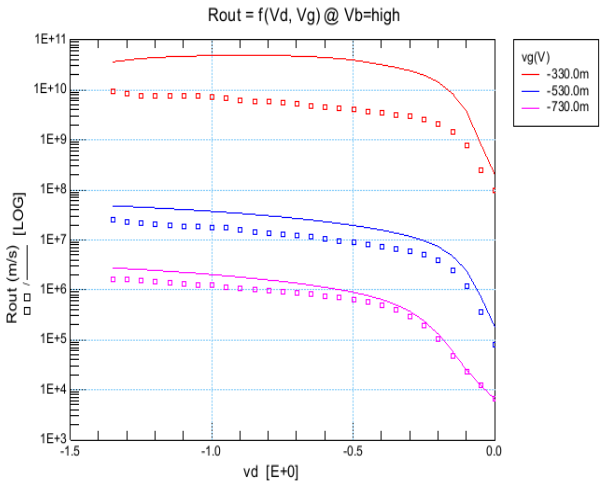


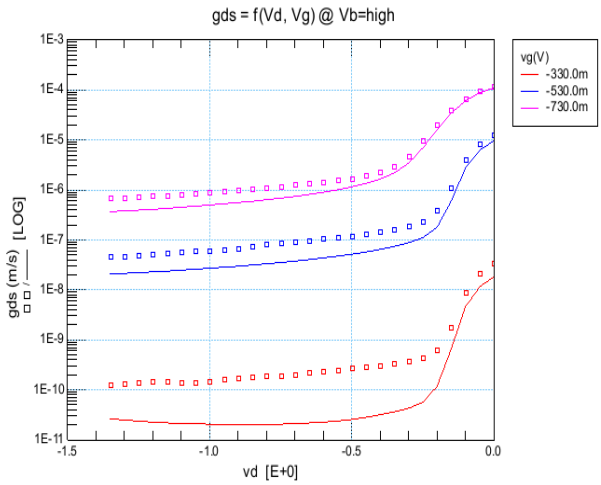




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

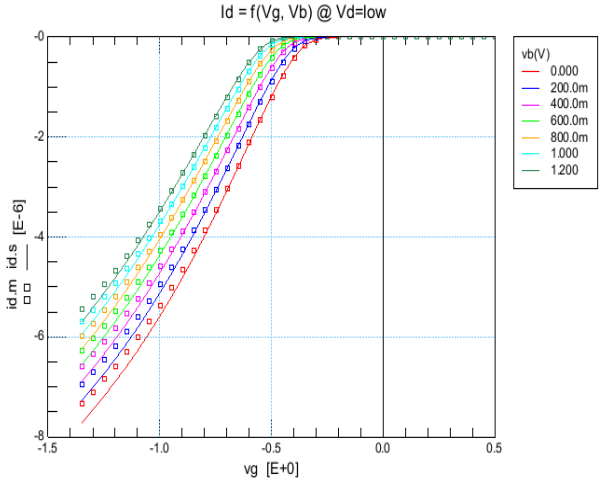


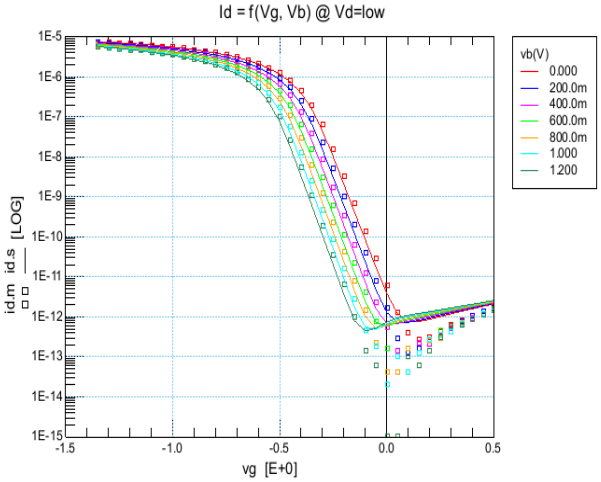


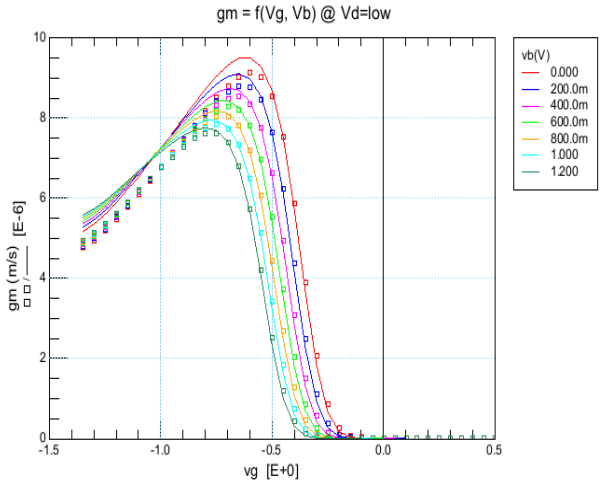


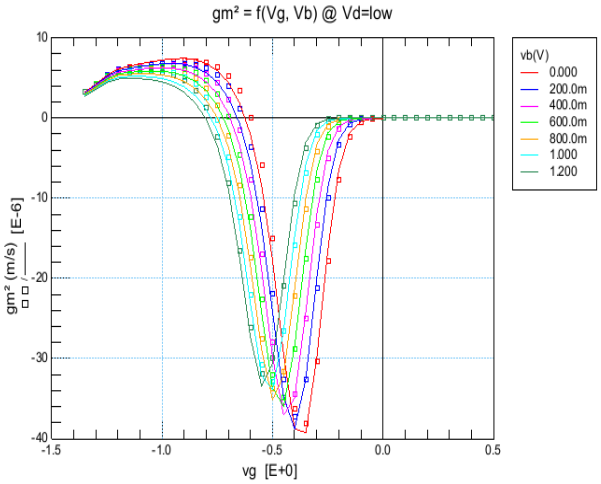
4.10W10u0_L5u0_S549_4

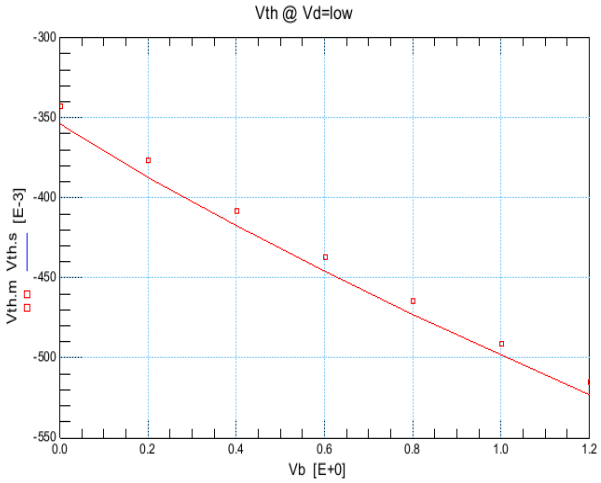
idvg, Vd = -0.05V, T = 27°C

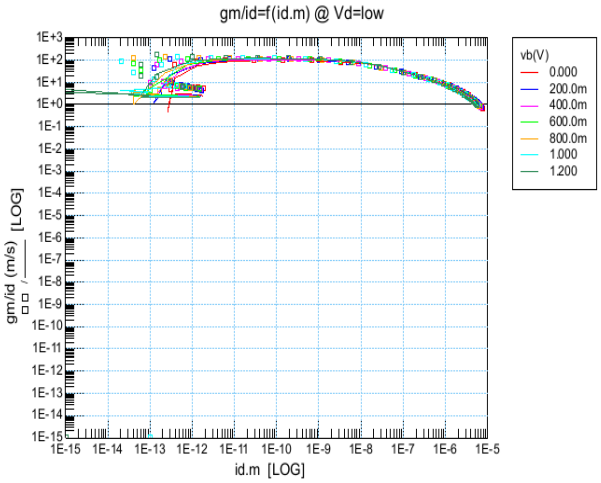


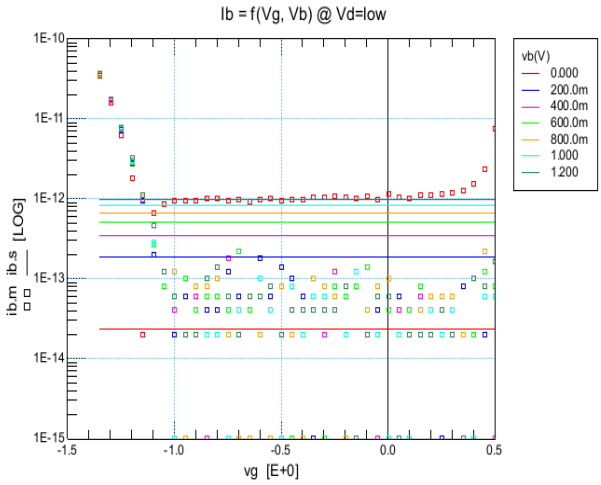


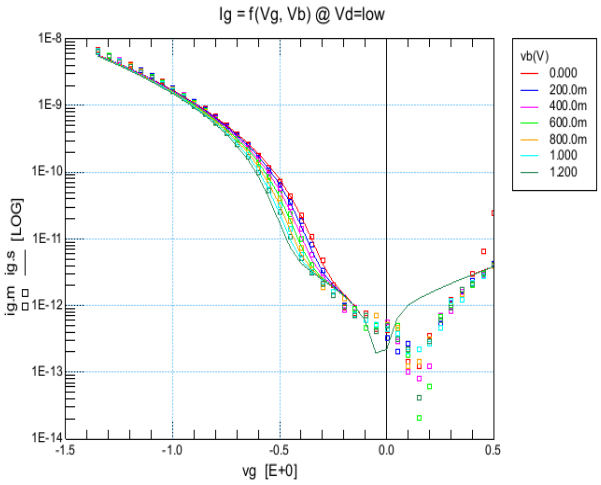




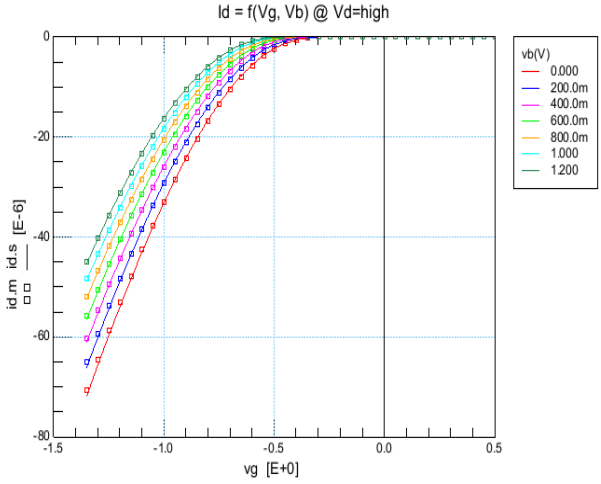


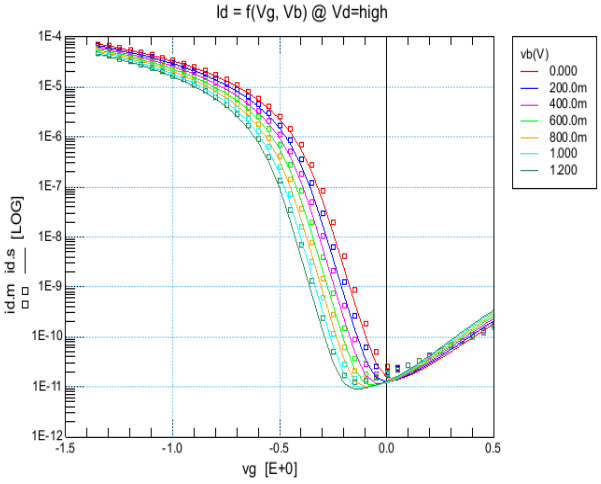


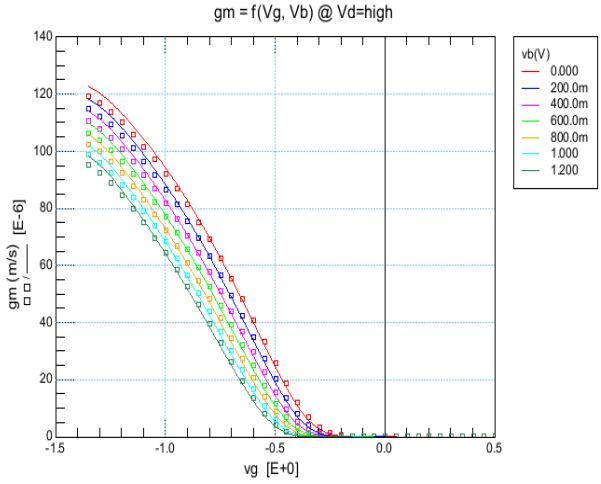


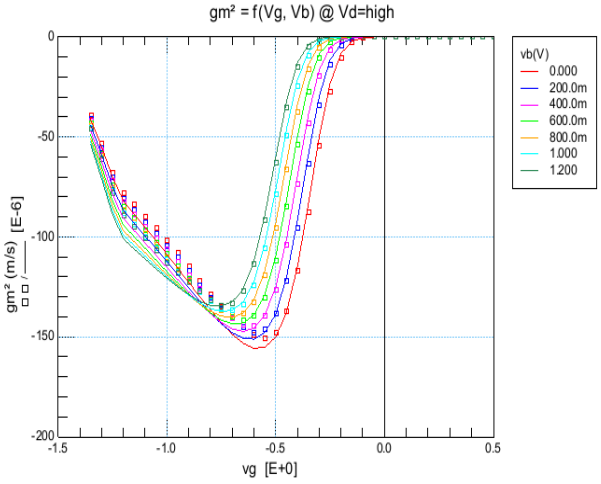


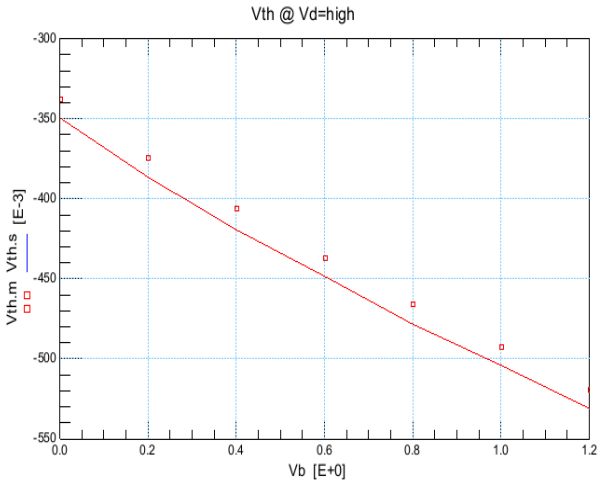
idvg, Vd = -1.2V, T = 27°C

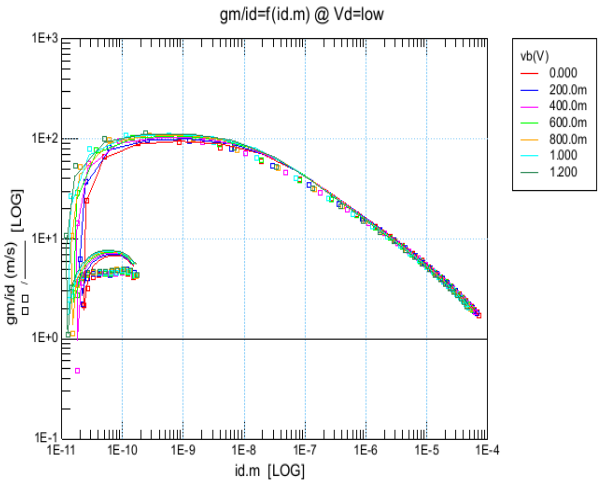


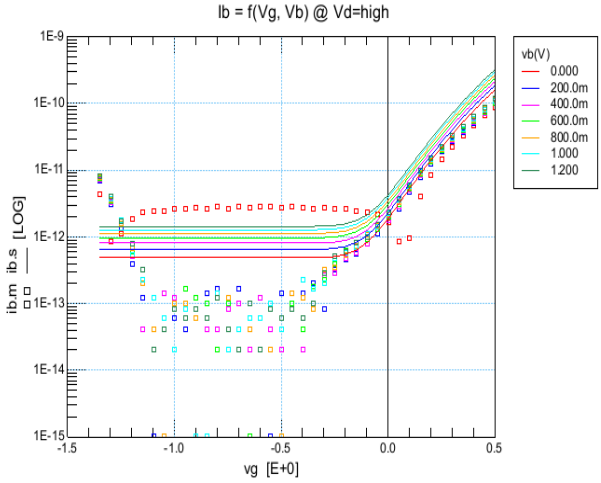


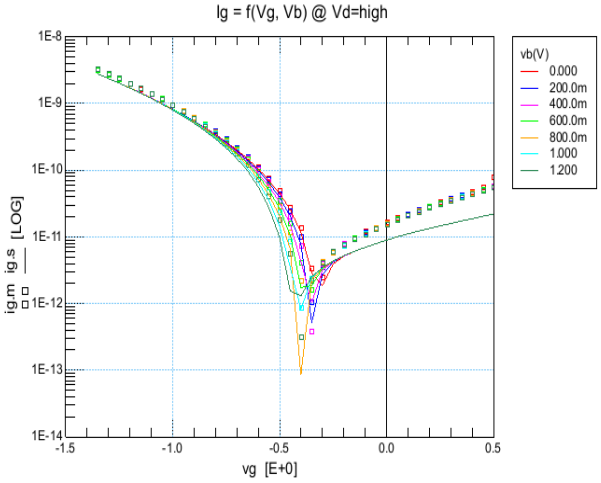




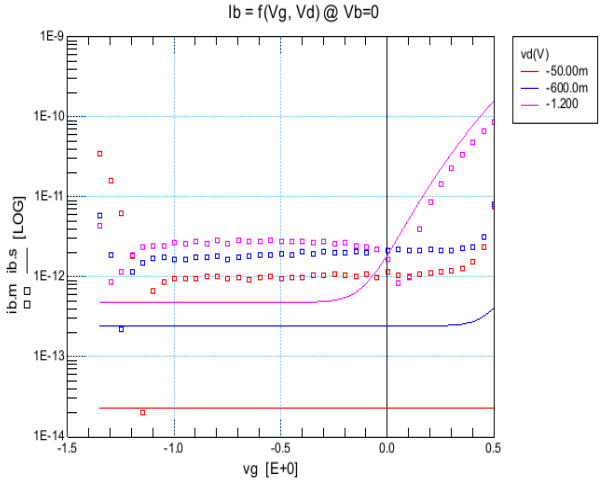


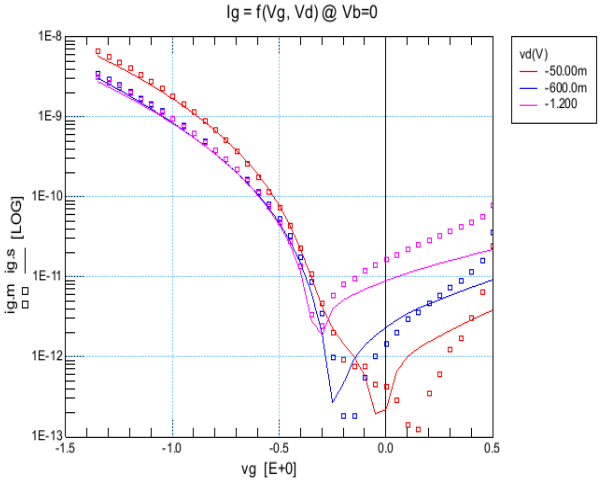




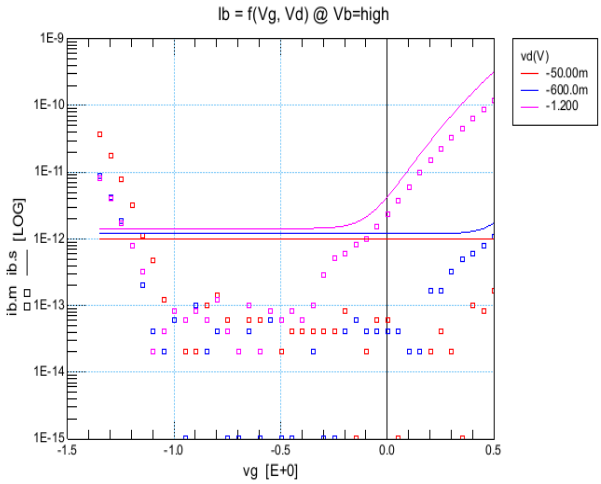


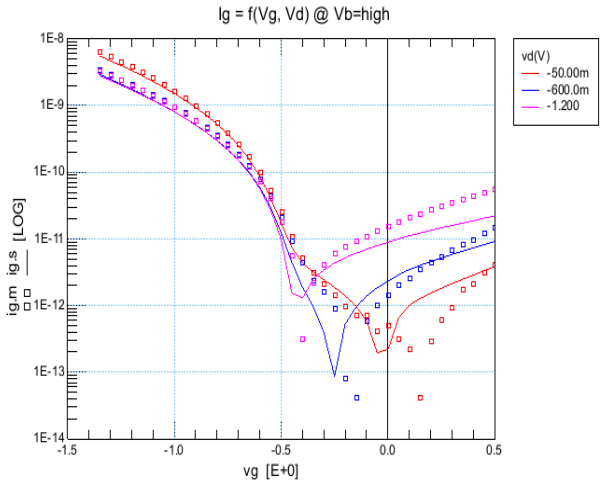
idvg, Ib,Ig, Vb = 0V, T = 27°C



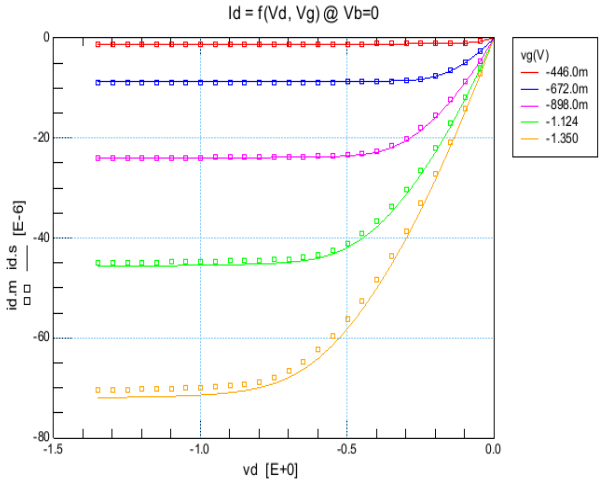


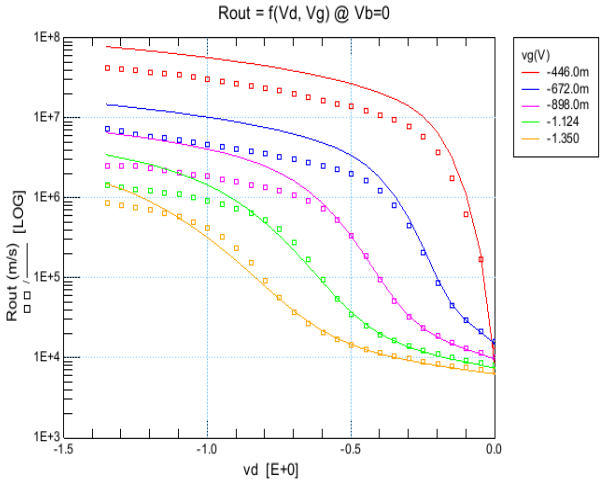
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

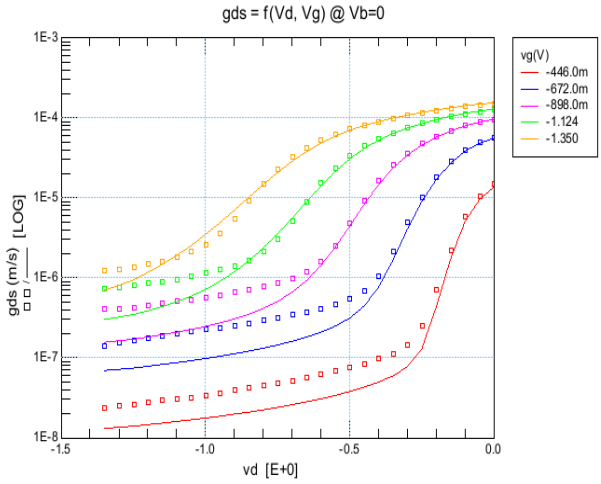




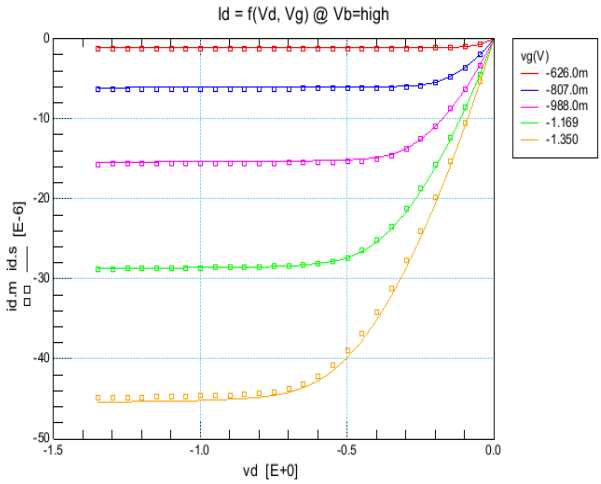
idvd, Vb = 0V, T = 27°C

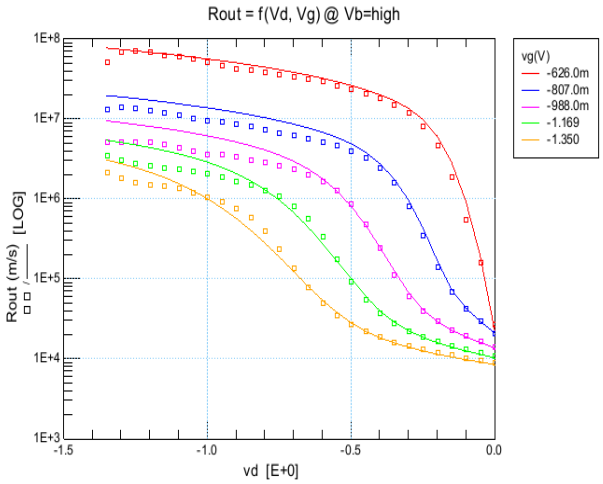


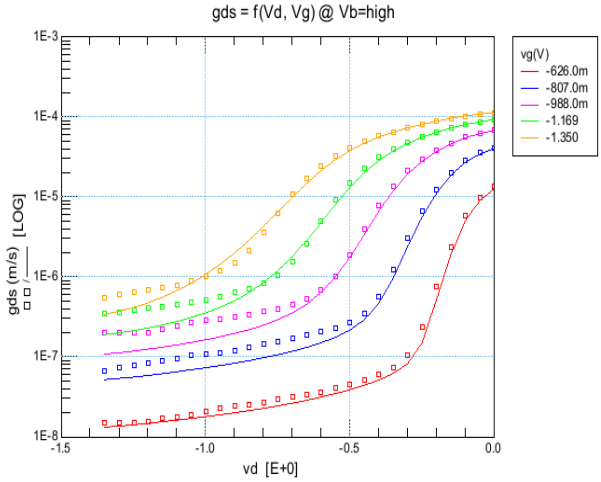




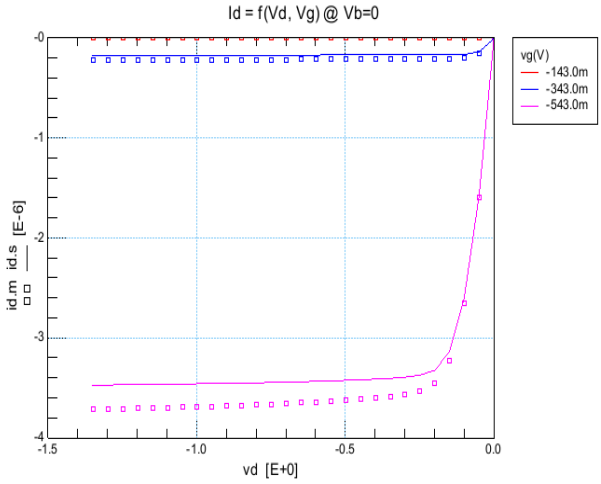
idvd_vbmin, Vb = 1.2V, T = 27°C

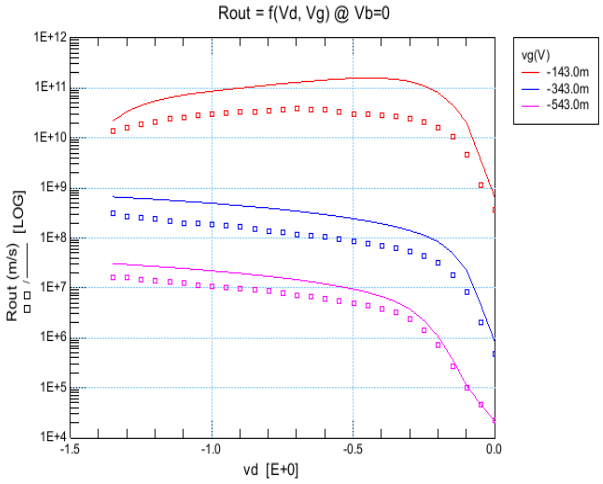


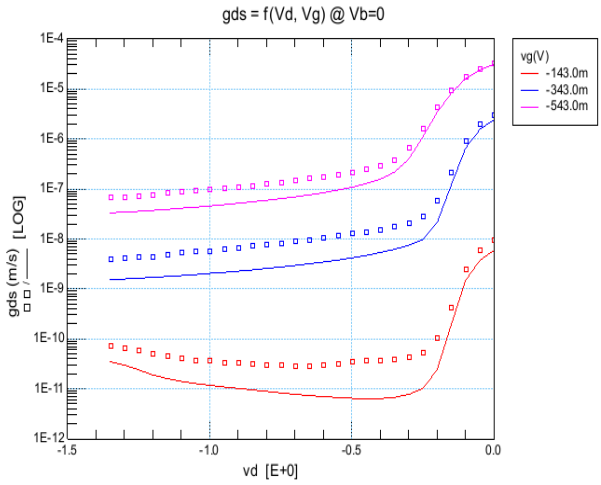




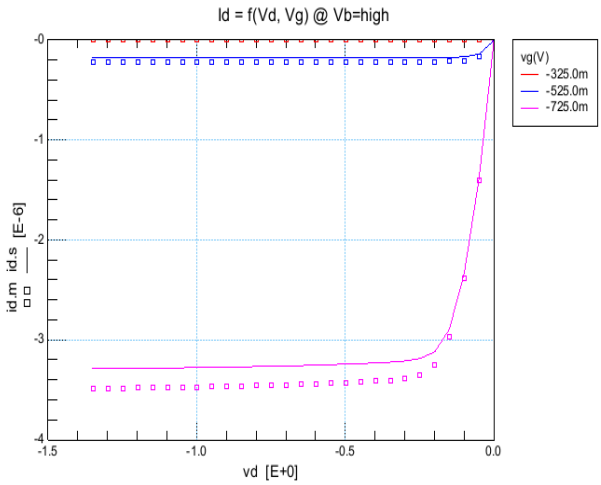
idvd_vth, Vb = 0V, T = 27°C

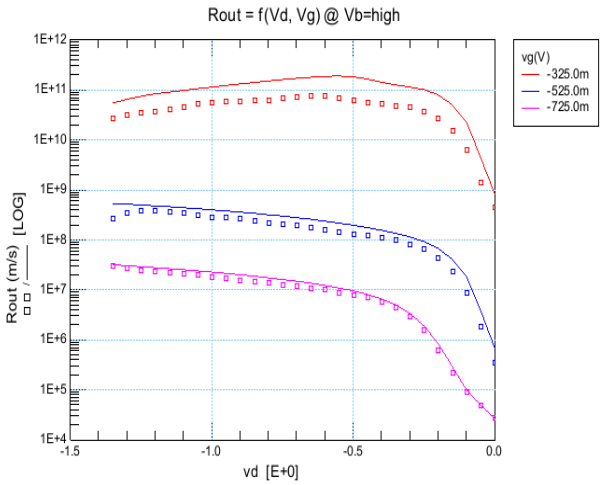


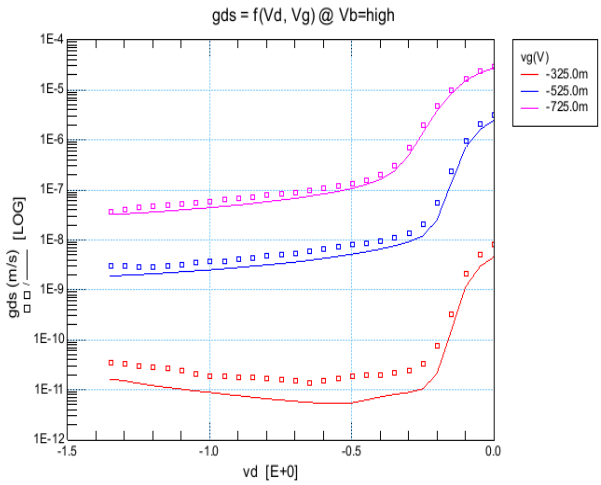




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

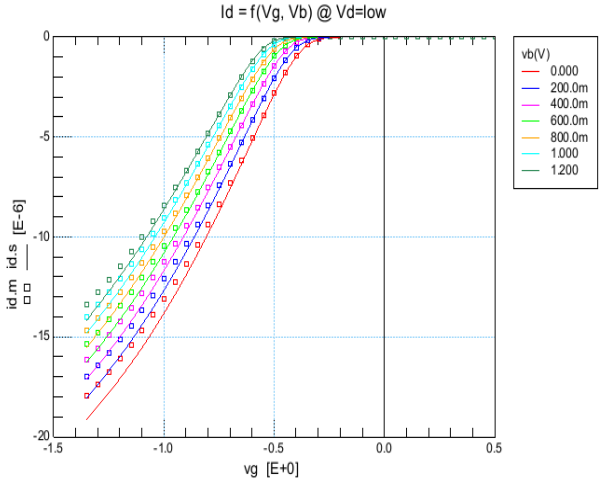


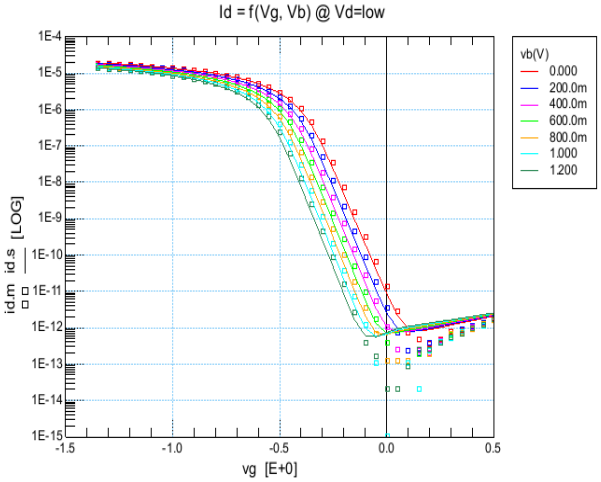


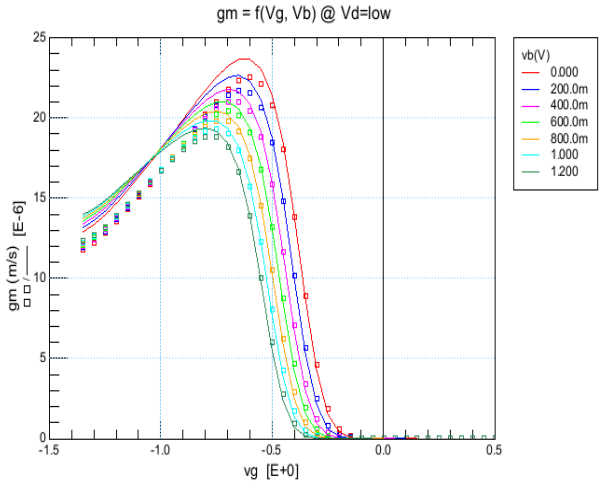


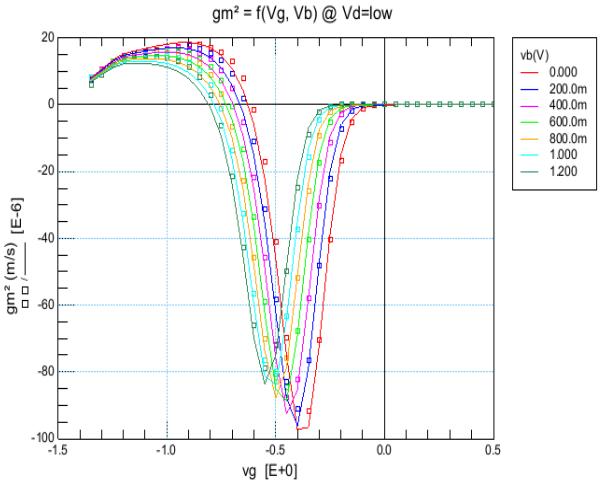
4.11 W10u0_L2u0_S549_3

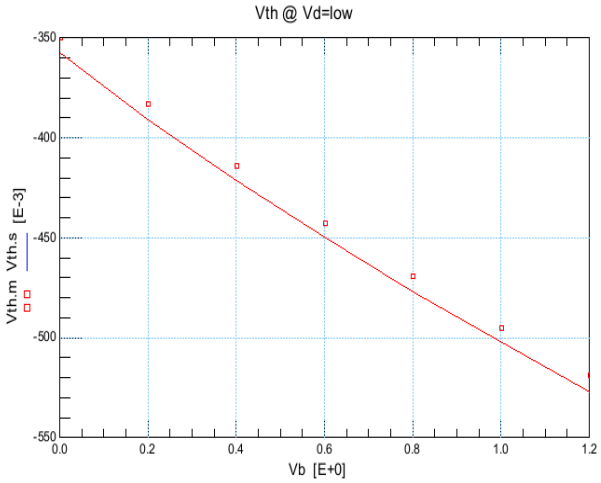
idvg, Vd = -0.05V, T = 27°C

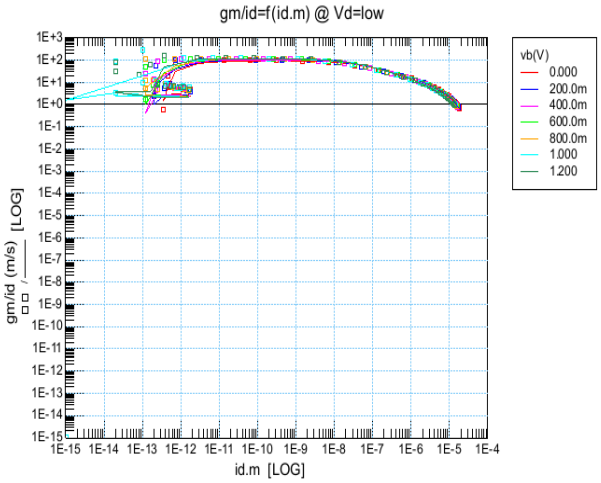


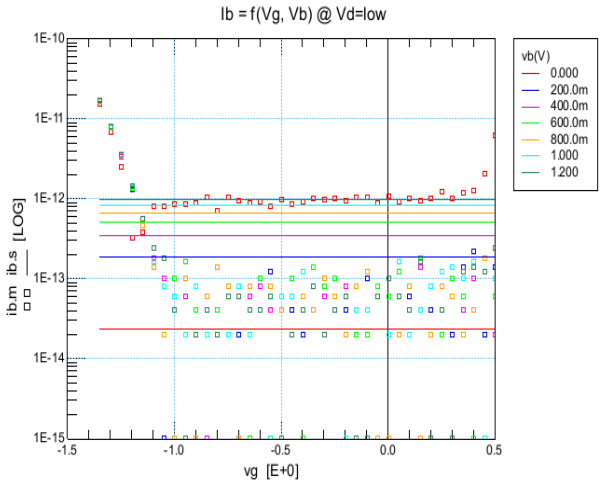


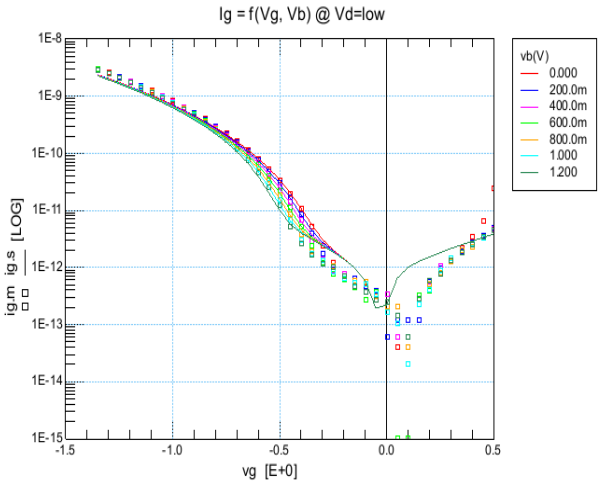




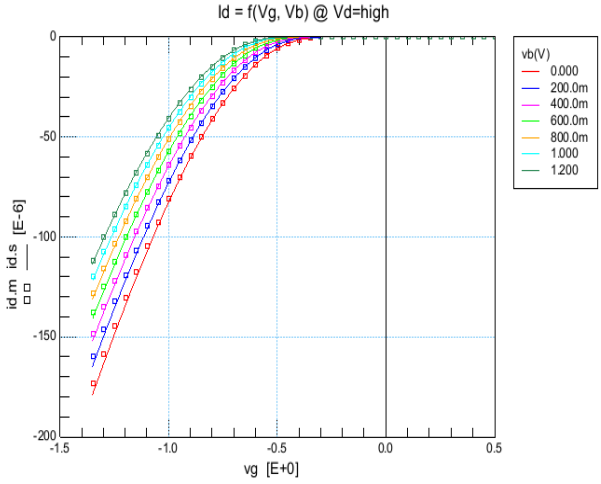


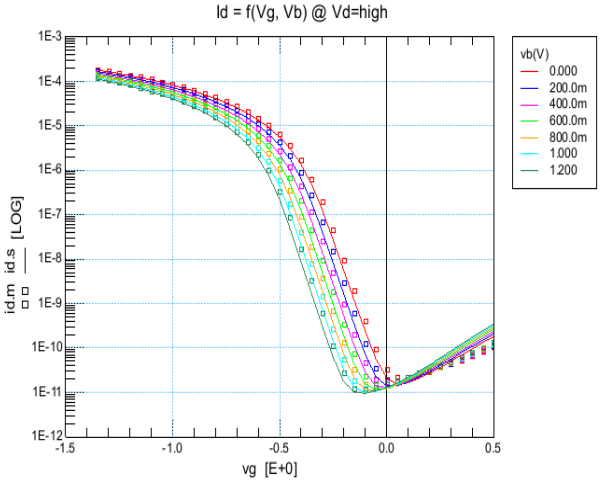


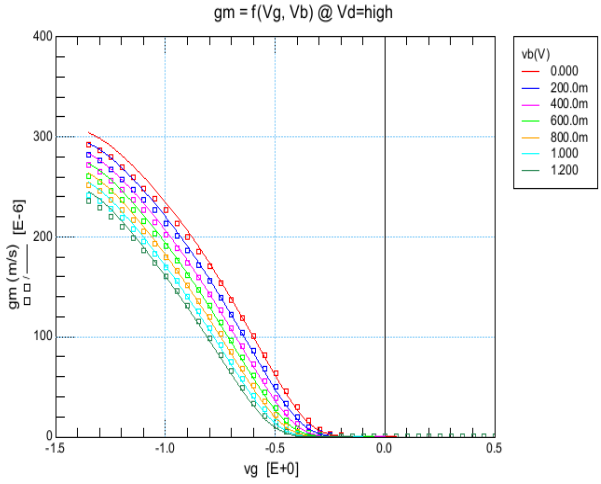


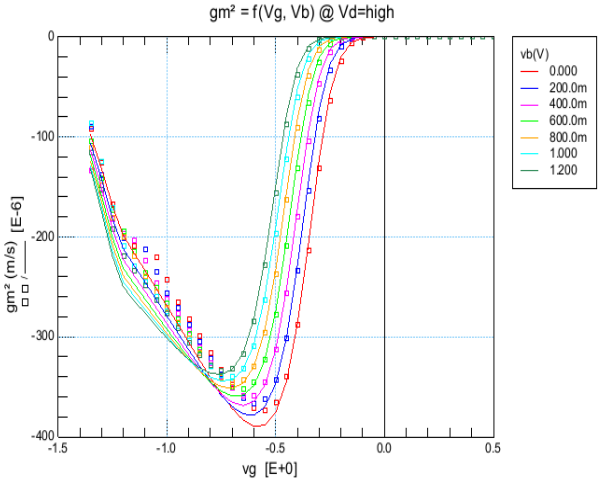


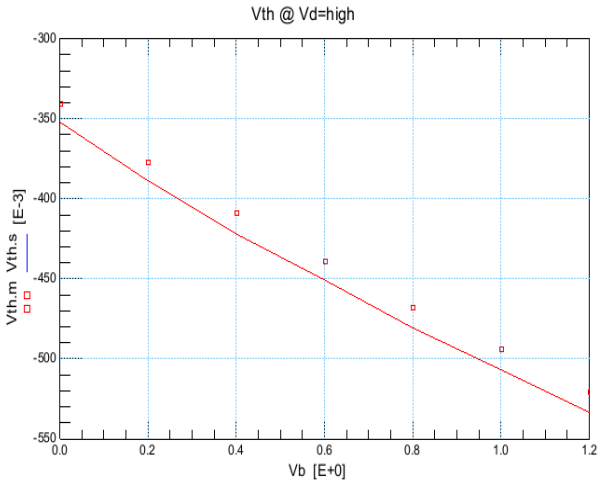
idvg, Vd = -1.2V, T = 27°C

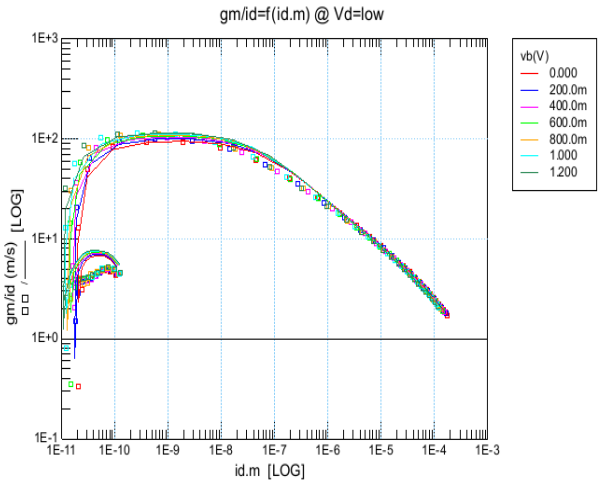


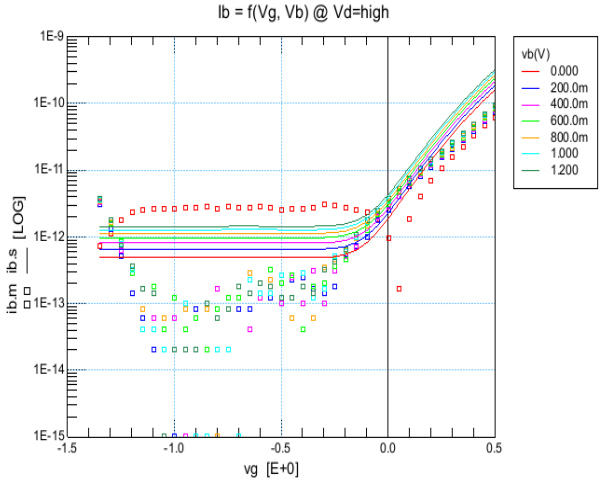


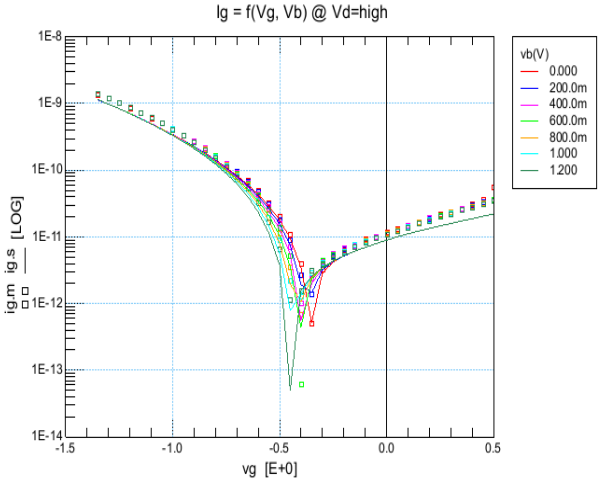




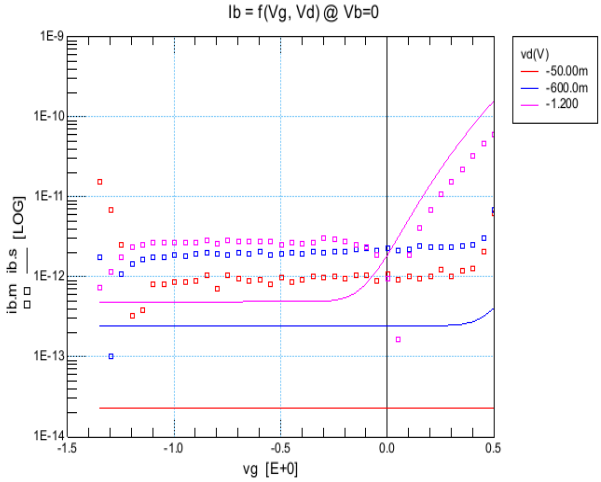


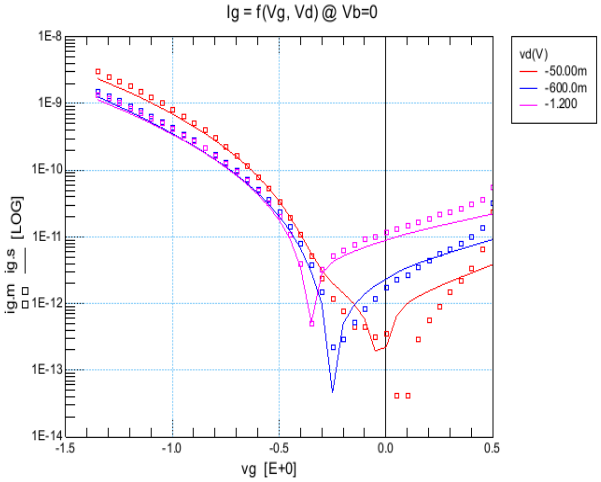




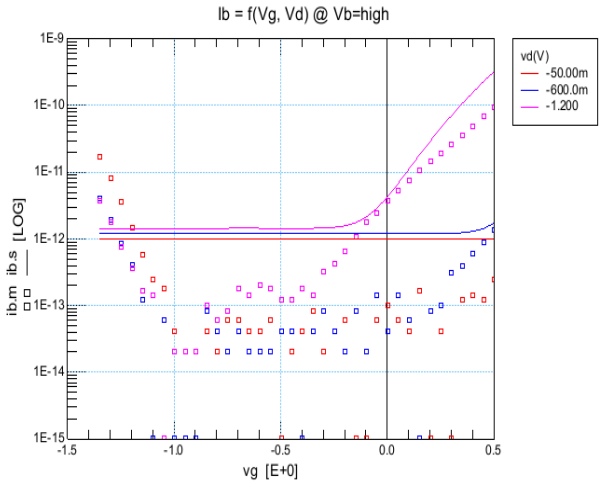


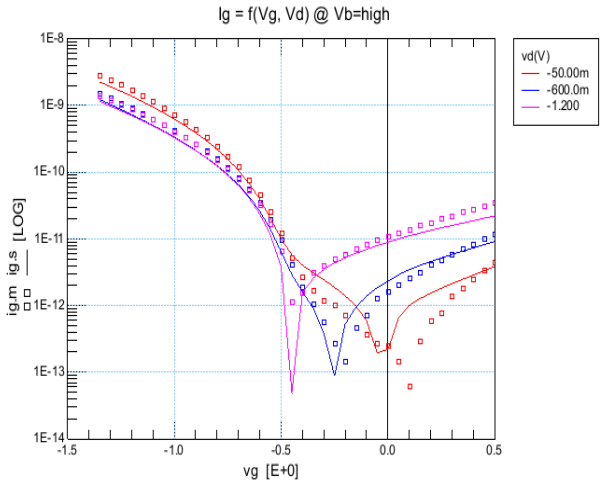
idvg, Ib,Ig, Vb = 0V, T = 27°C



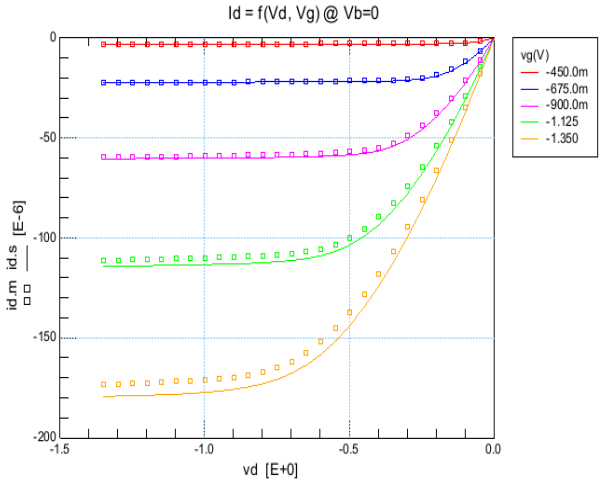


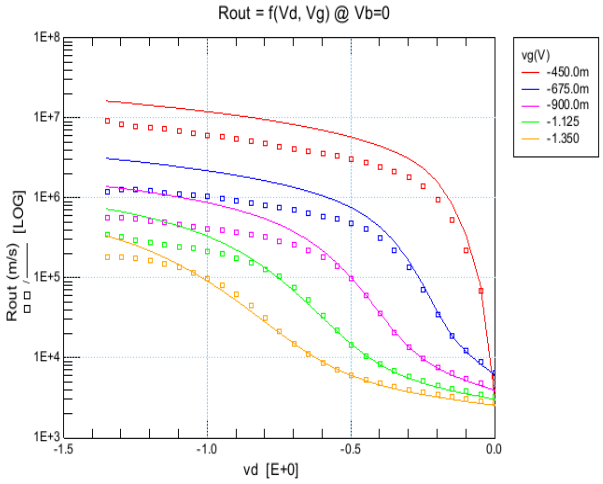
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

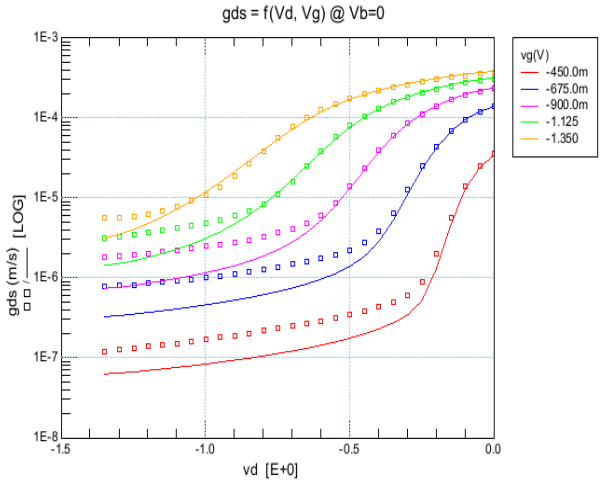




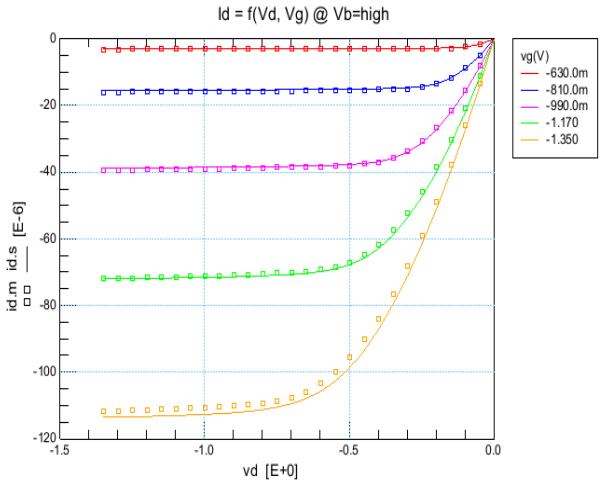
idvd, Vb = 0V, T = 27°C

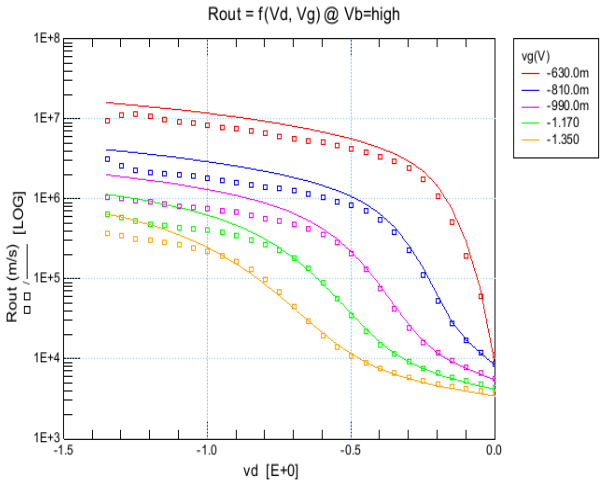


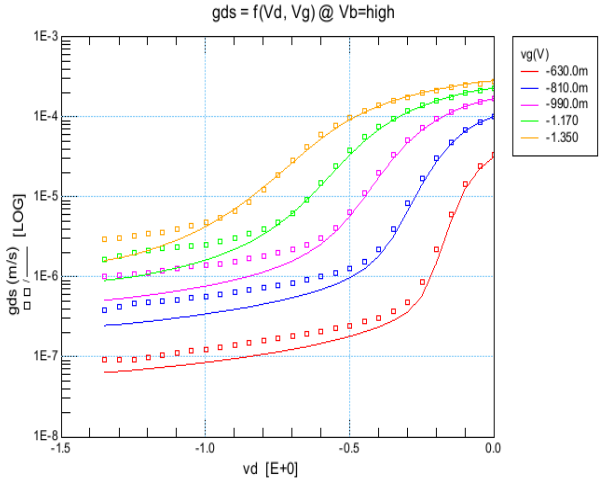




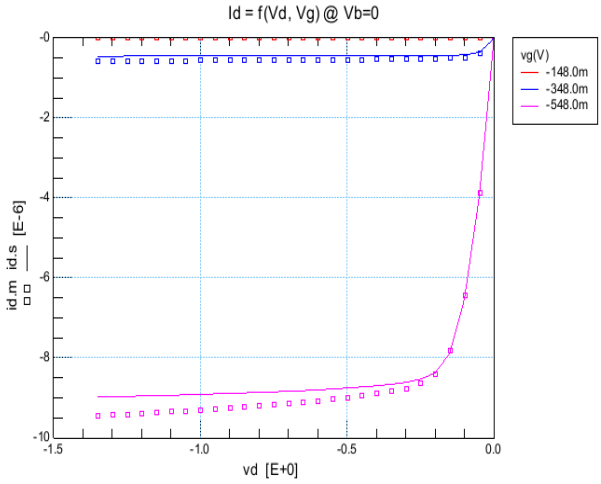
idvd_vbmin, Vb = 1.2V, T = 27°C

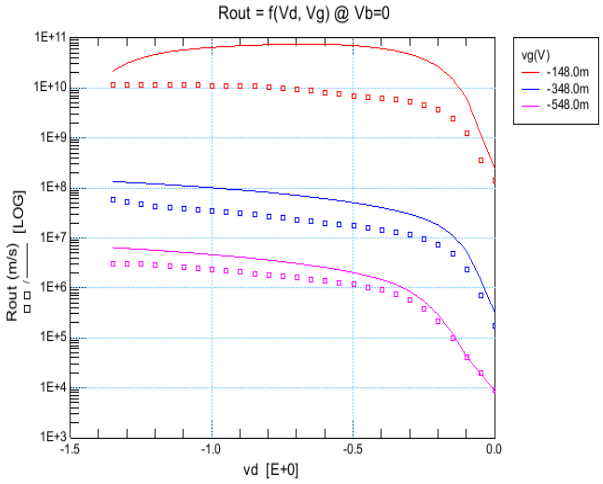


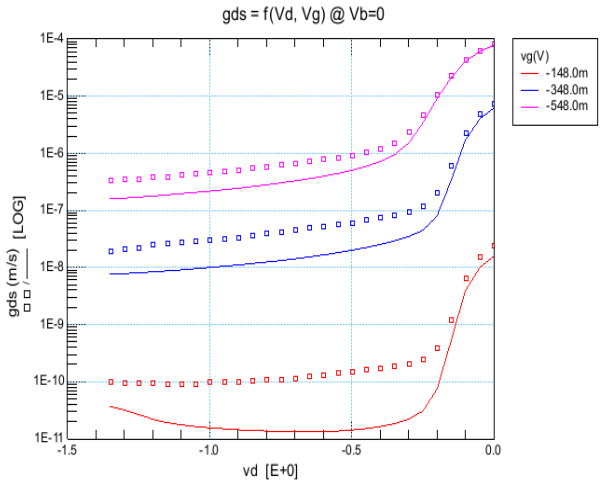




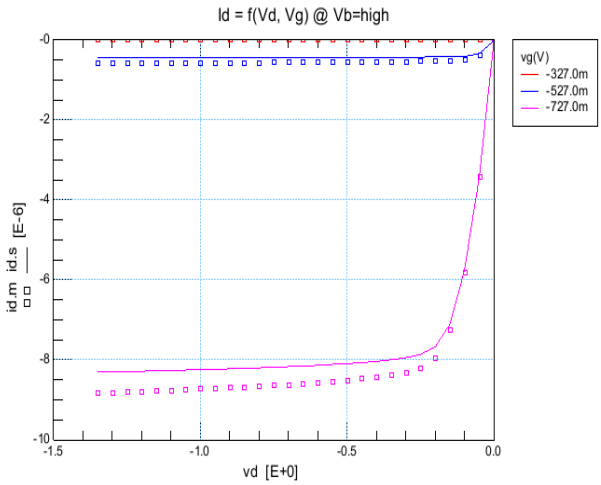
idvd_vth, Vb = 0V, T = 27°C

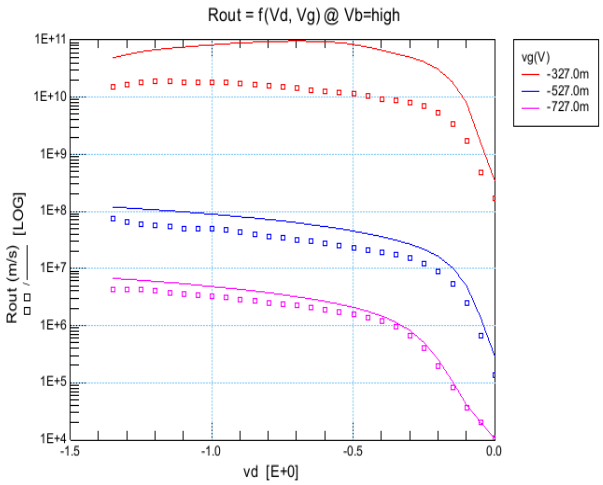


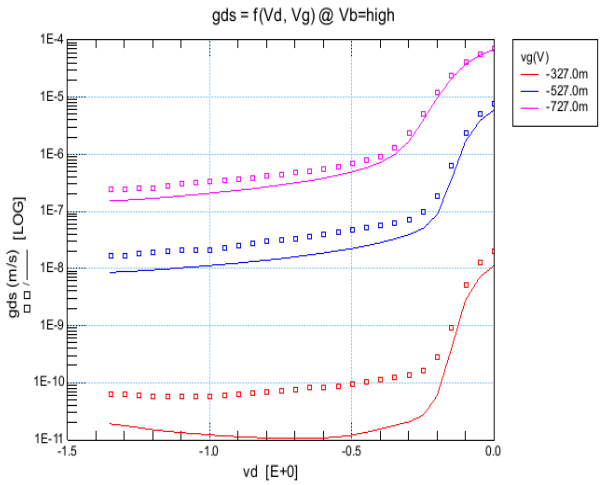




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

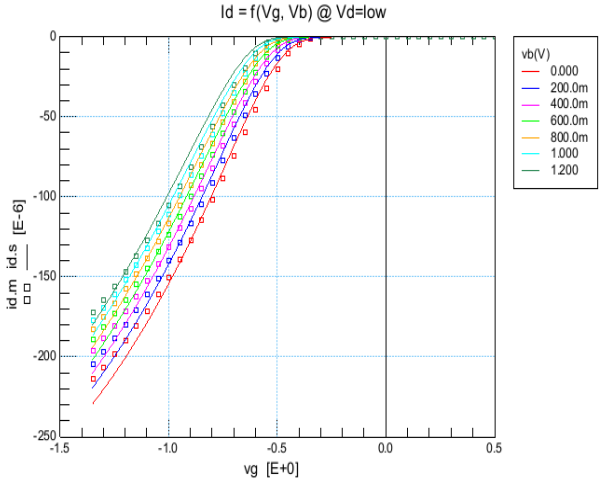


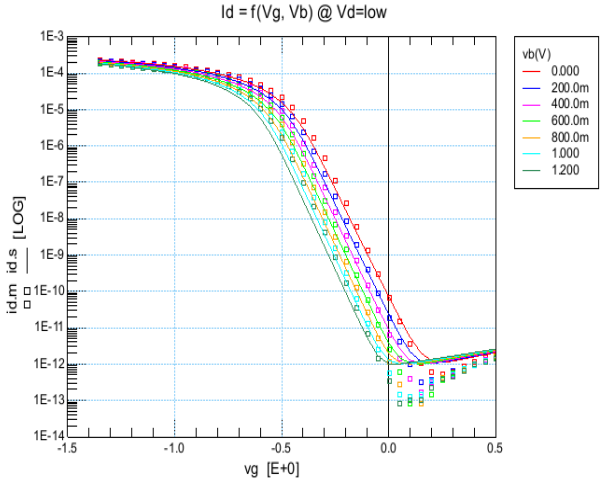


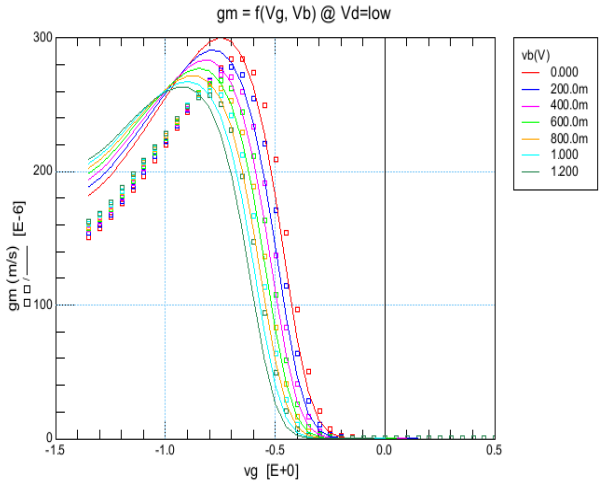


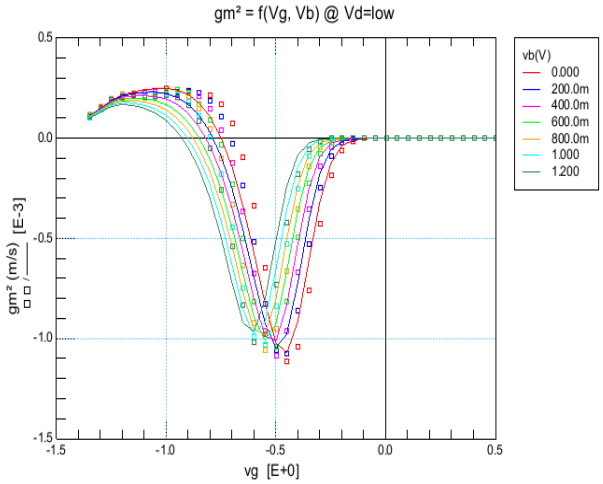
4.12 W10u0_L0u15_S548_4

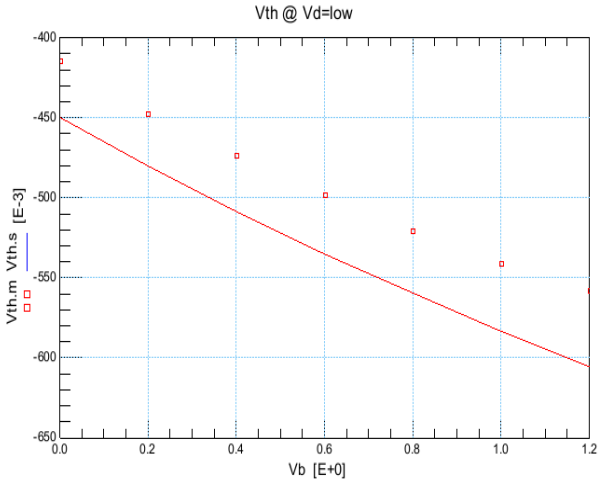
idvg, Vd = -0.05V, T = 27°C

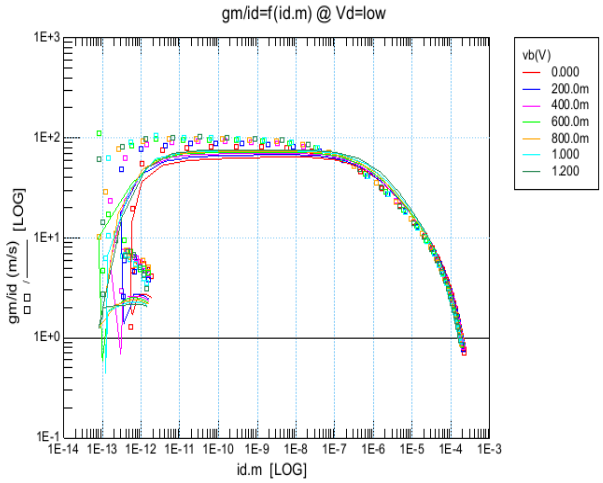


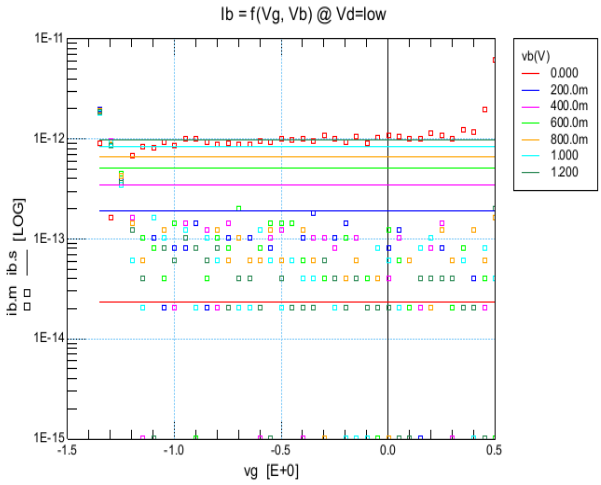


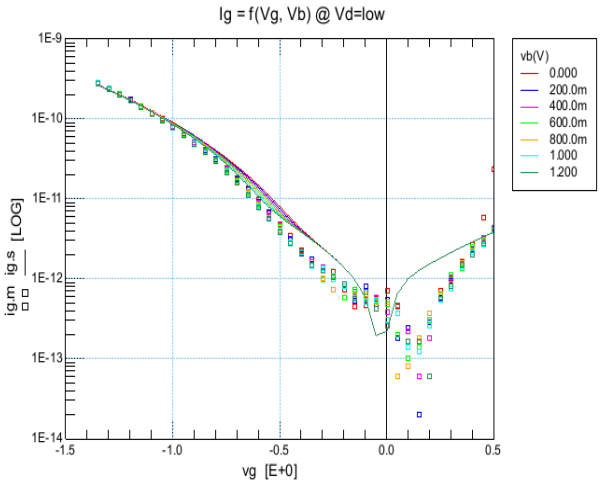




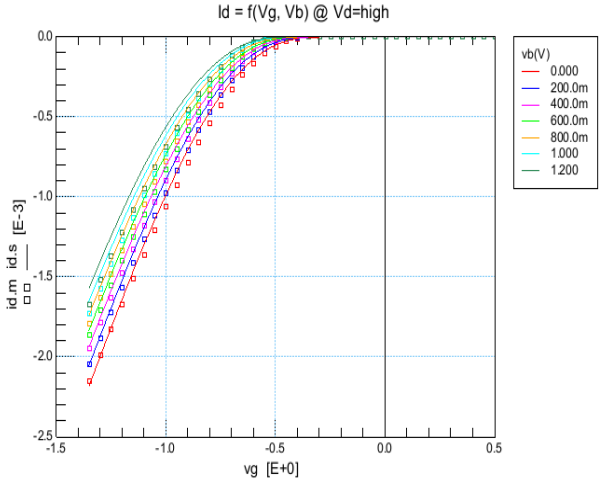


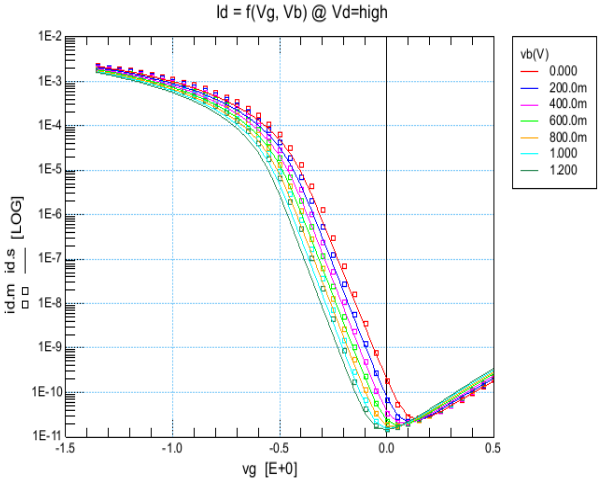


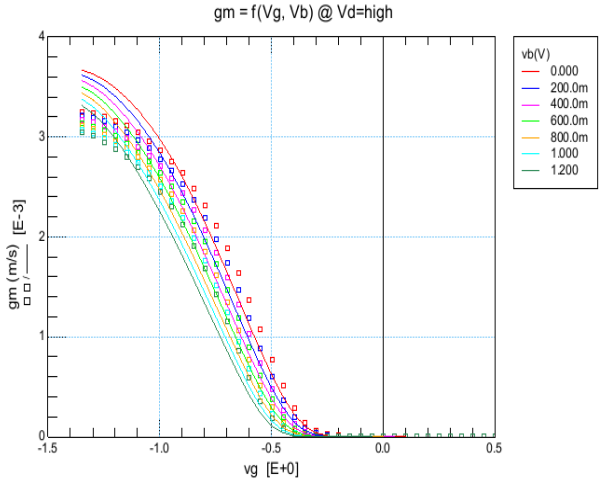


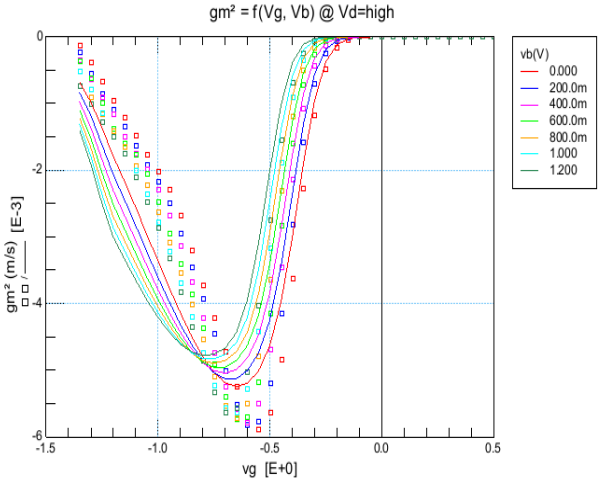


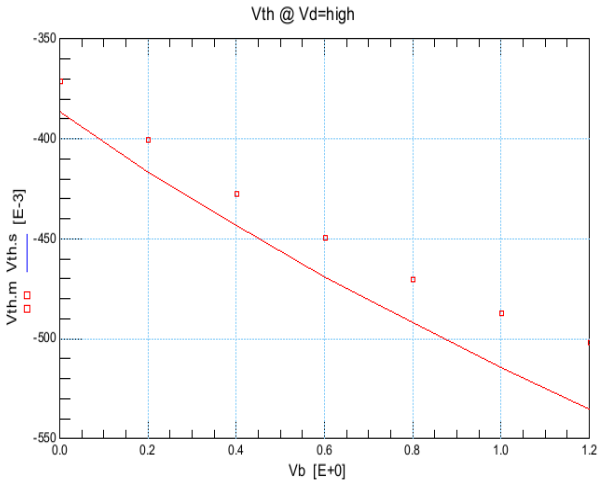
idvg, Vd = -1.2V, T = 27°C

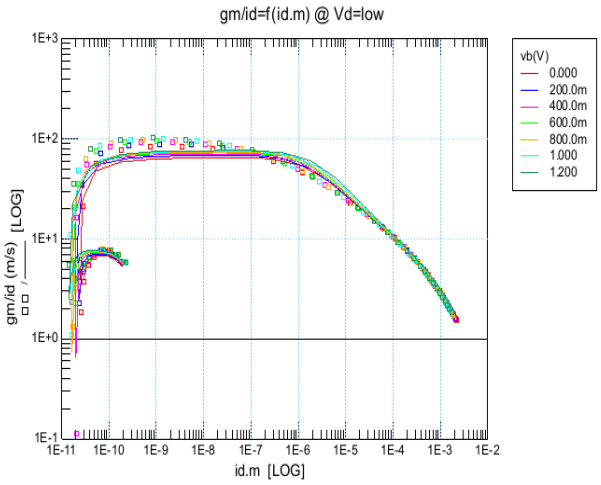


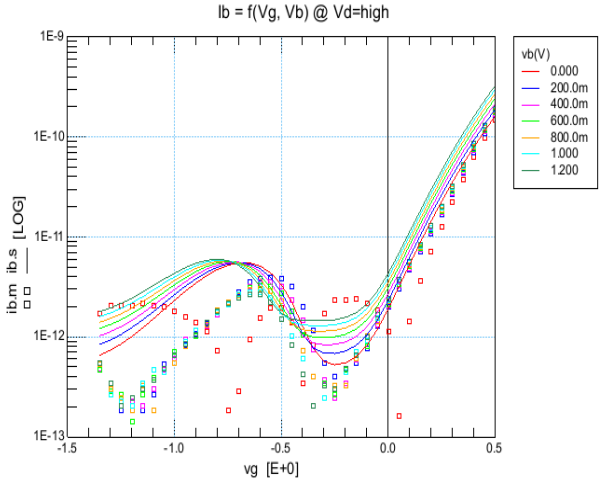


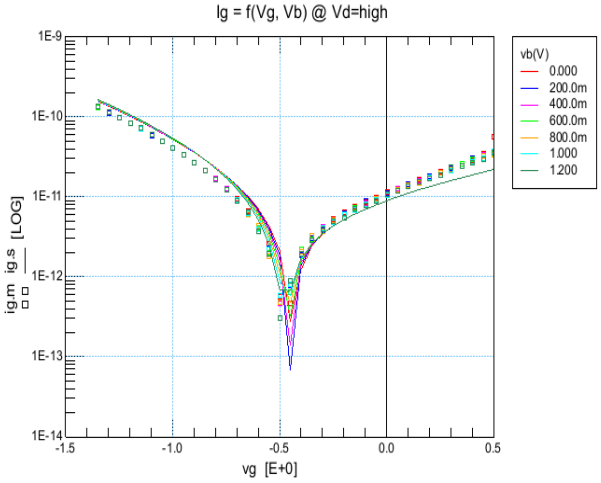




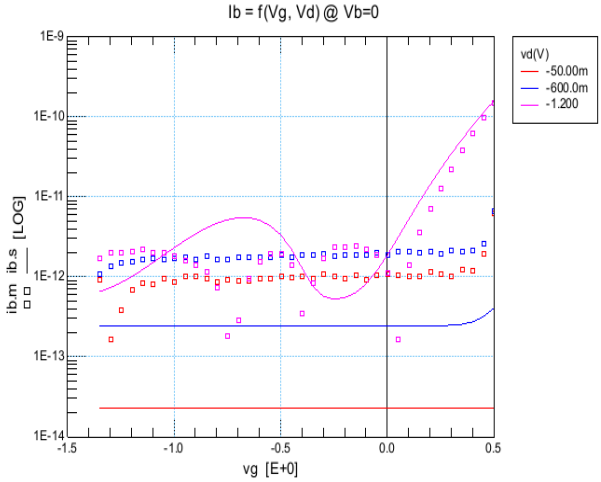


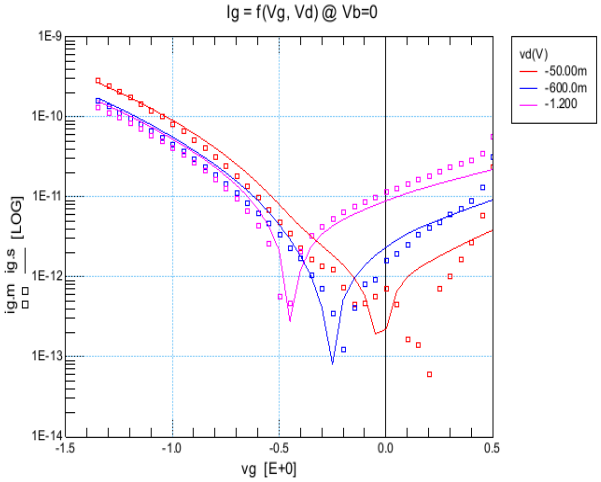




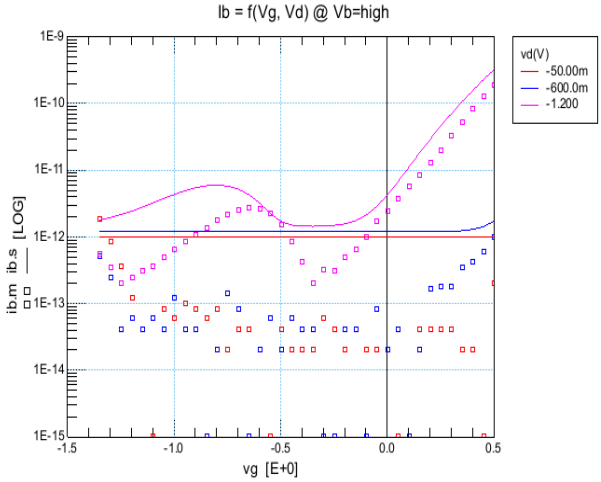


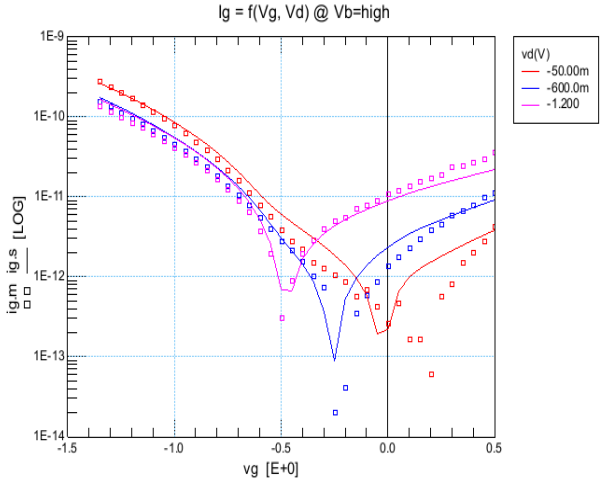
idvg, Ib,Ig, Vb = 0V, T = 27°C



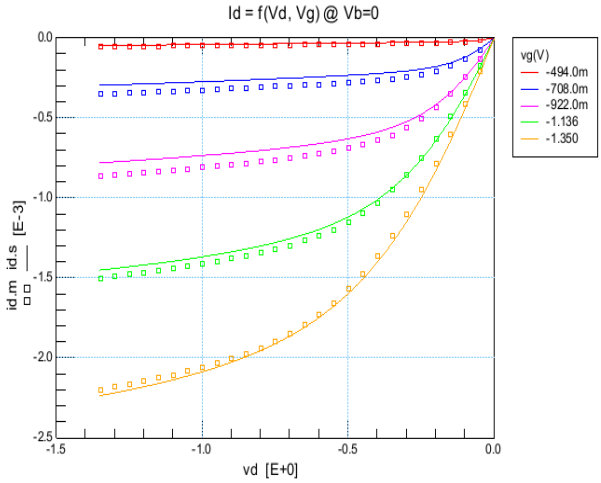


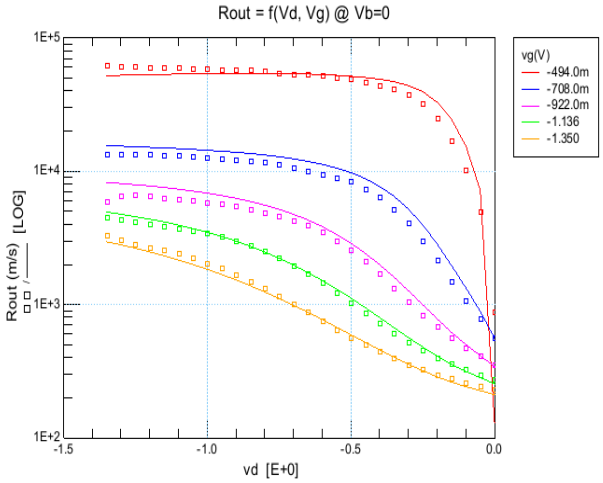
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

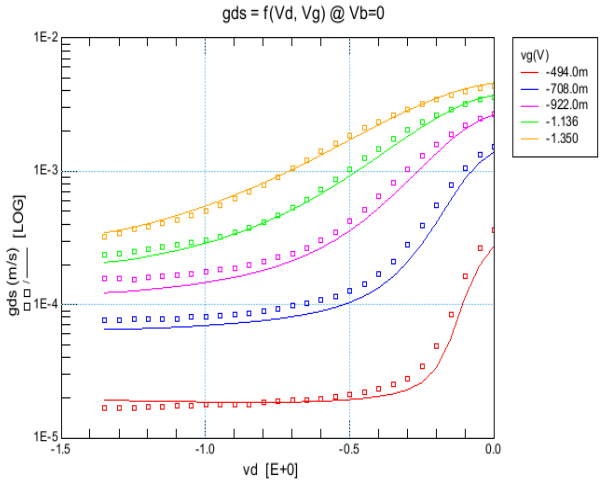




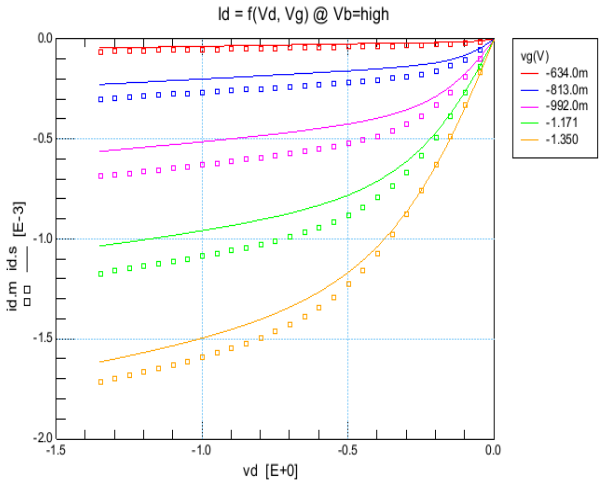
idvd, Vb = 0V, T = 27°C

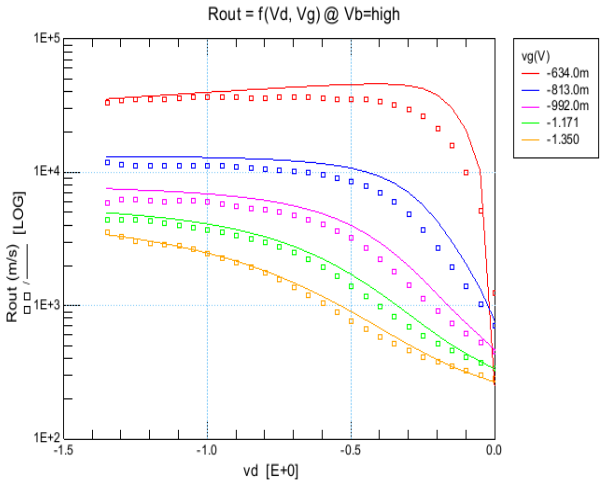


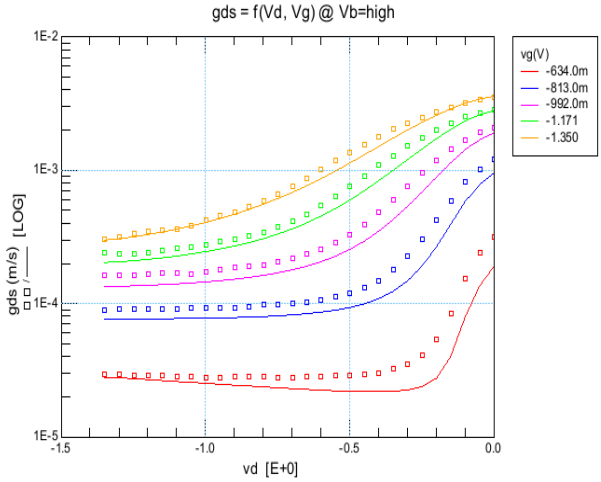




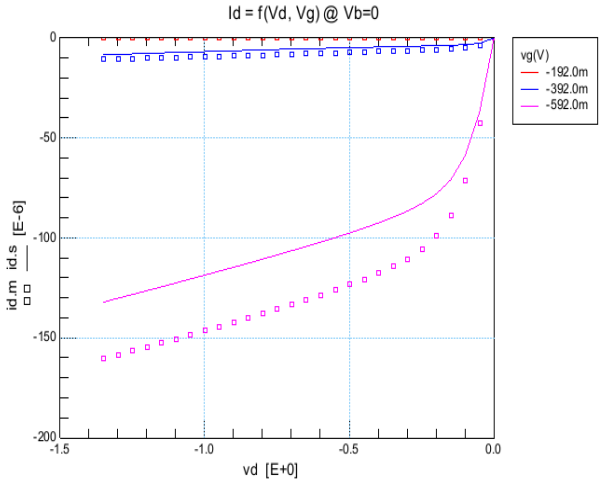
idvd_vbmin, Vb = 1.2V, T = 27°C

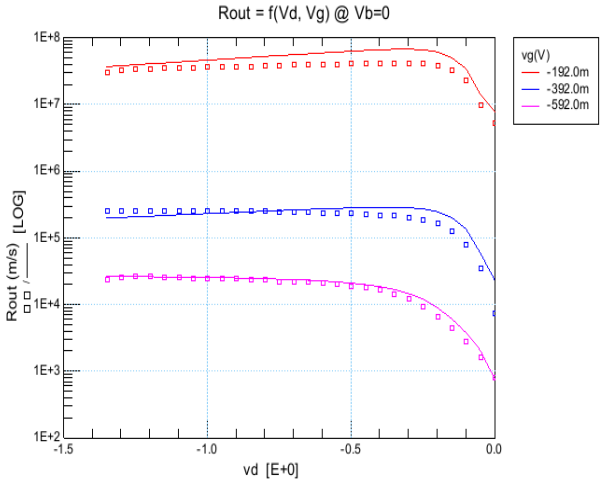


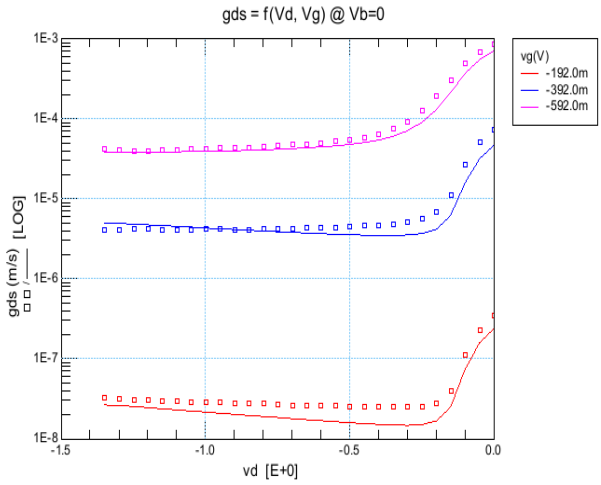




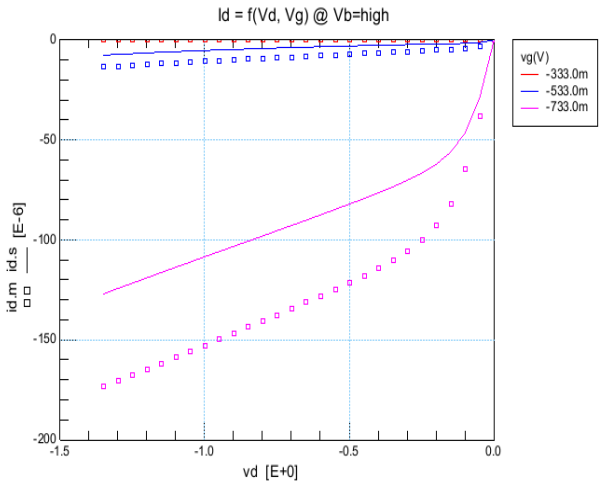
idvd_vth, Vb = 0V, T = 27°C

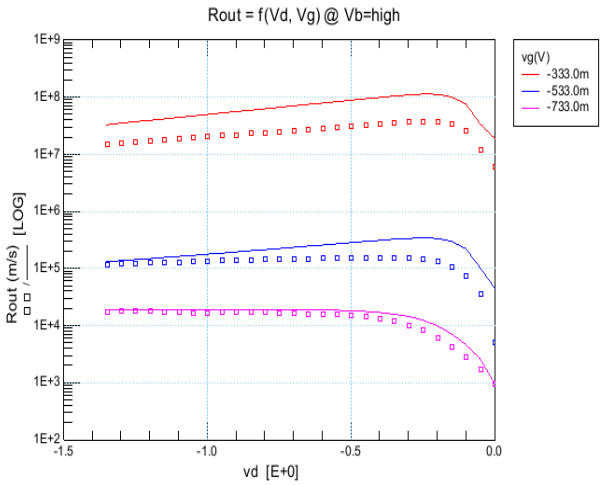


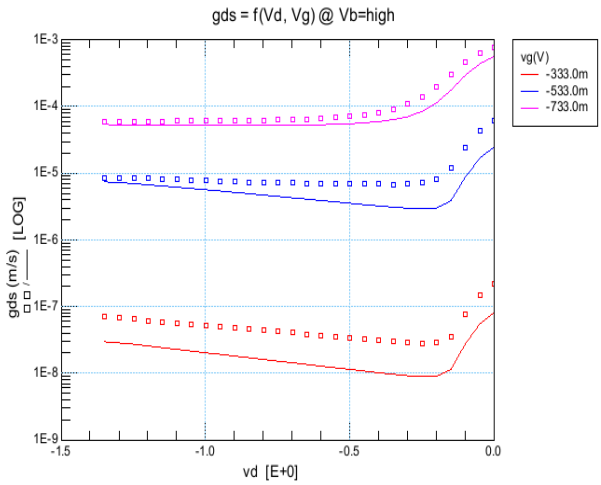




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

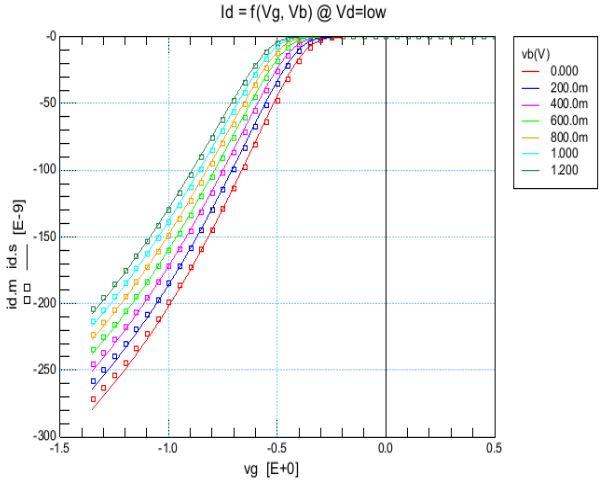


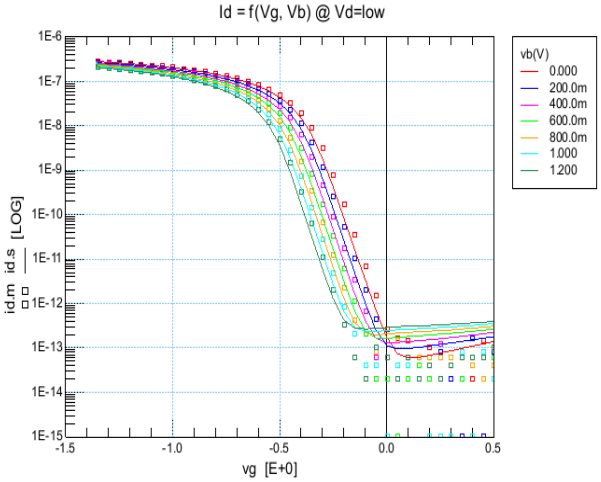


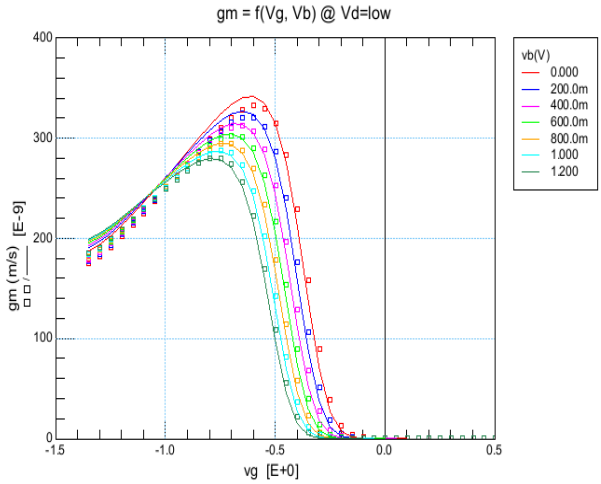


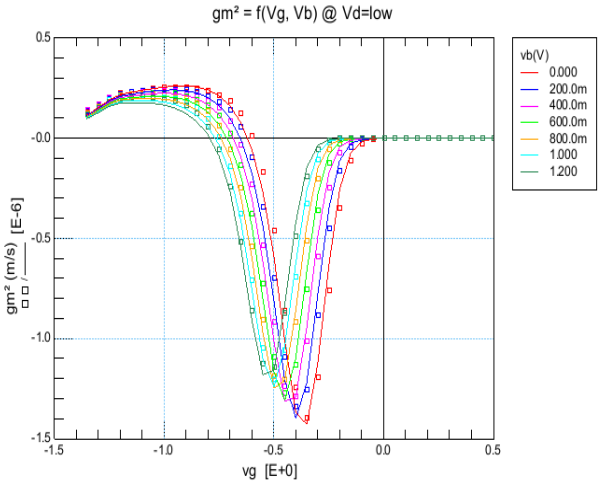
4.13 W0u6_L10u0_S552_1

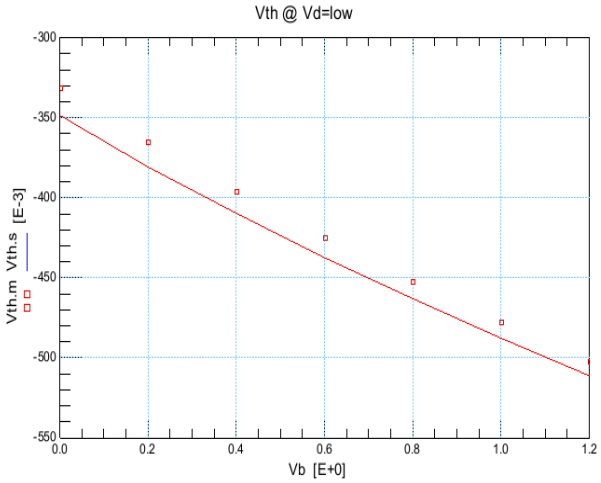
idvg, Vd = -0.05V, T = 27°C

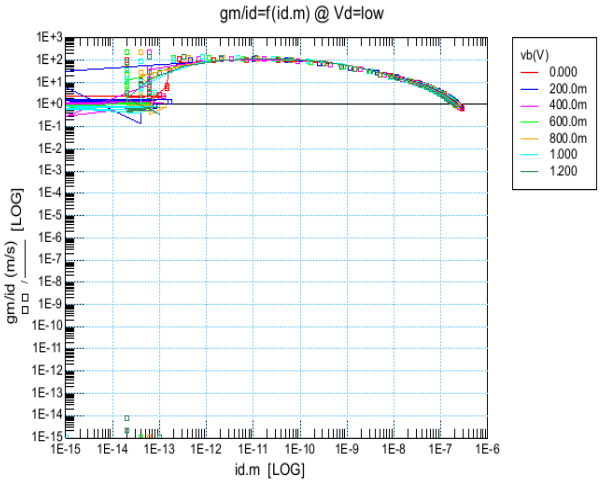


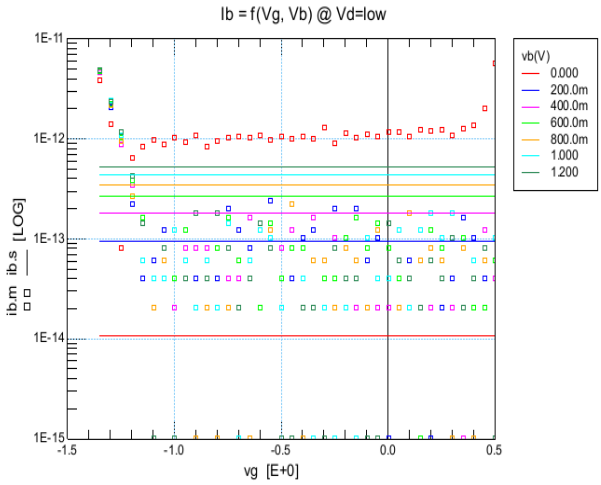


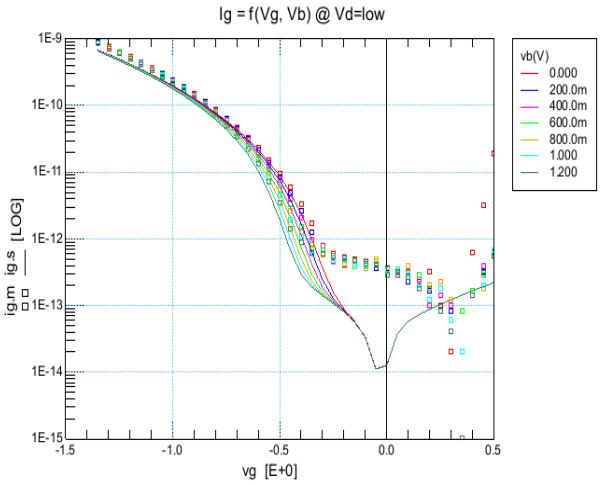




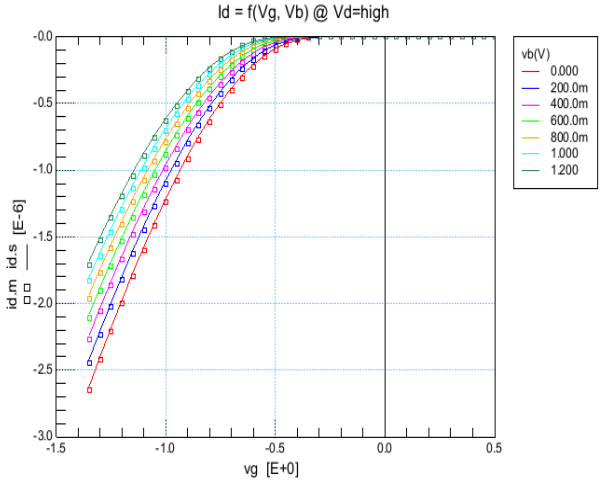


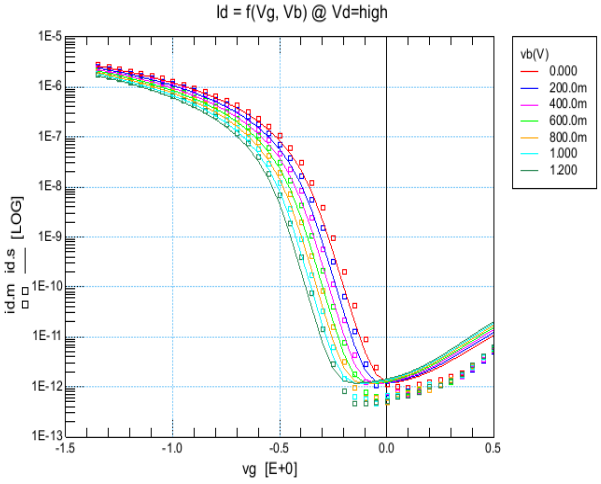


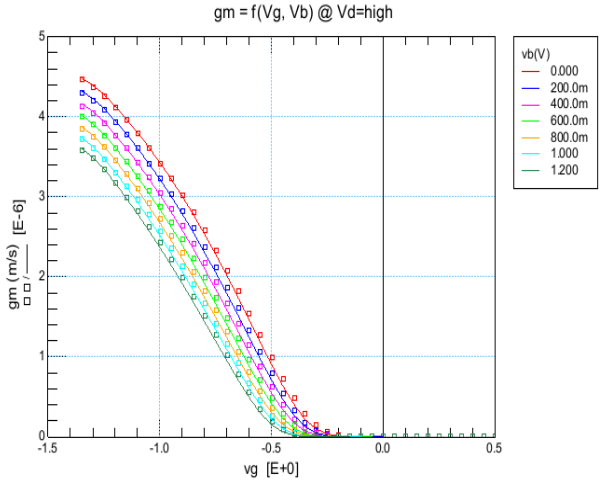


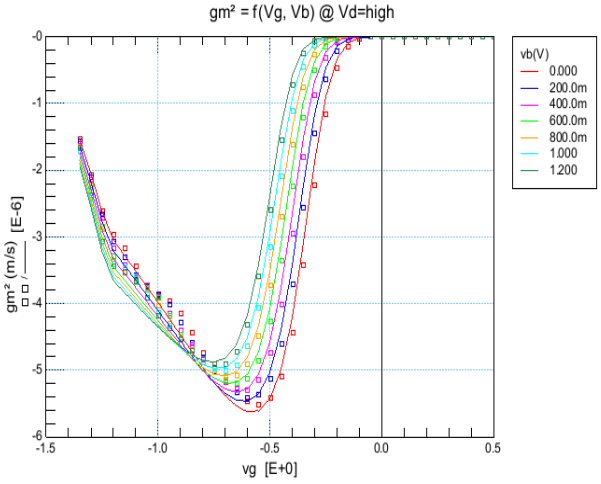


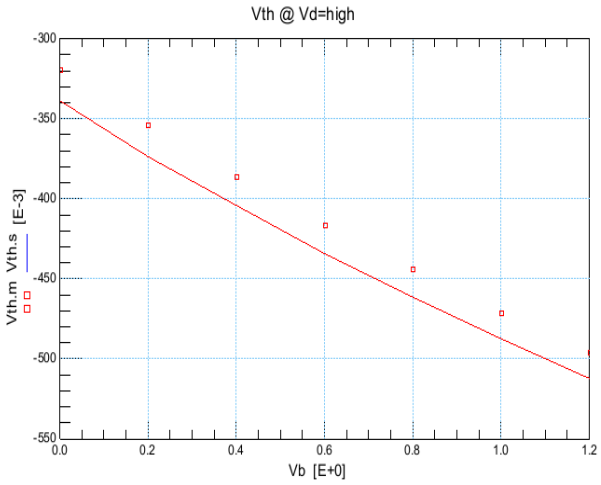
idvg, Vd = -1.2V, T = 27°C

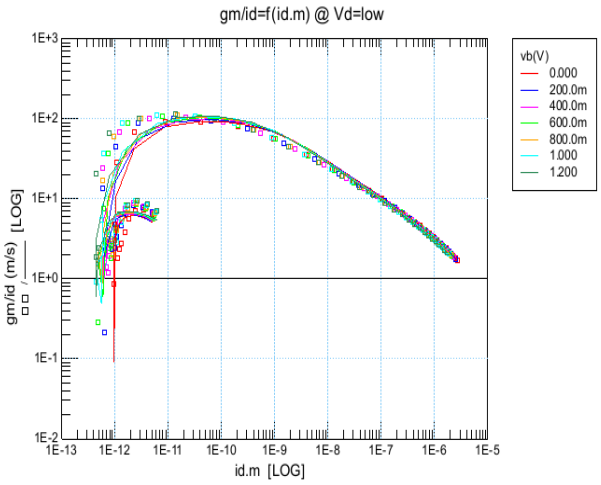


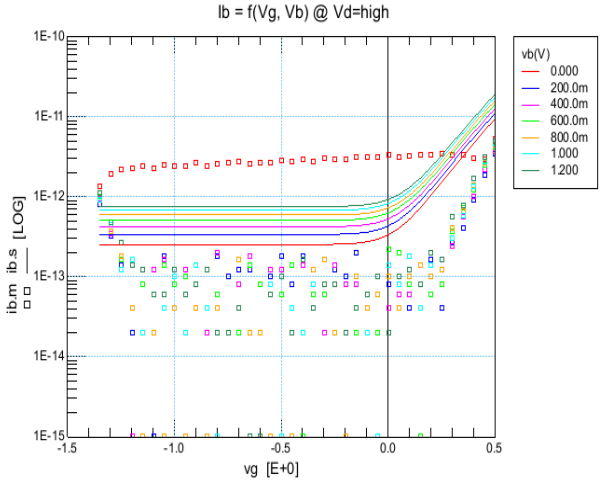


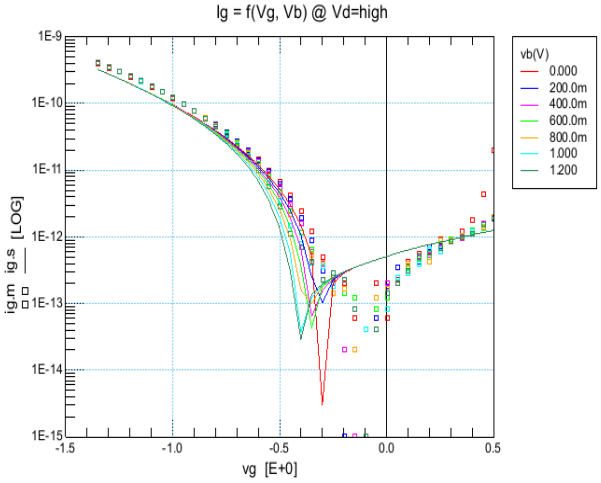




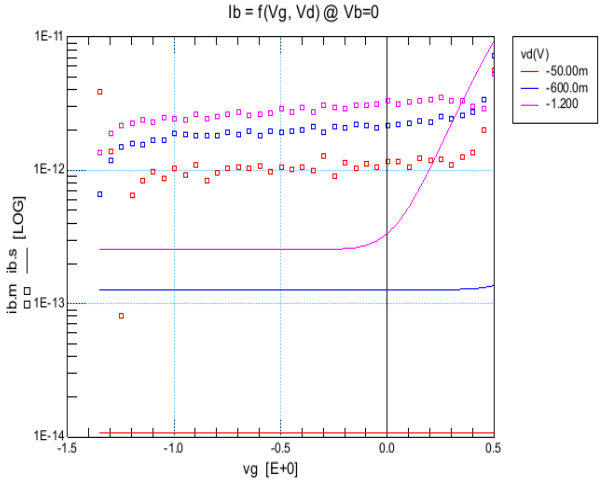


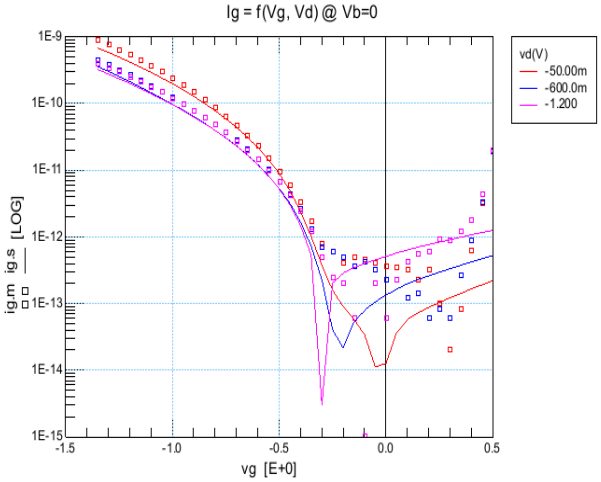




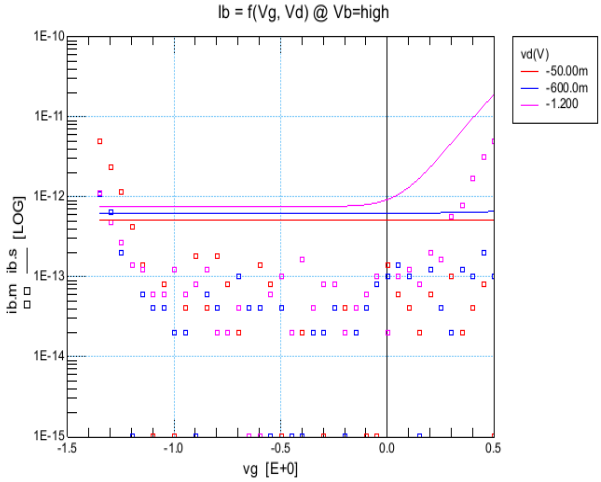


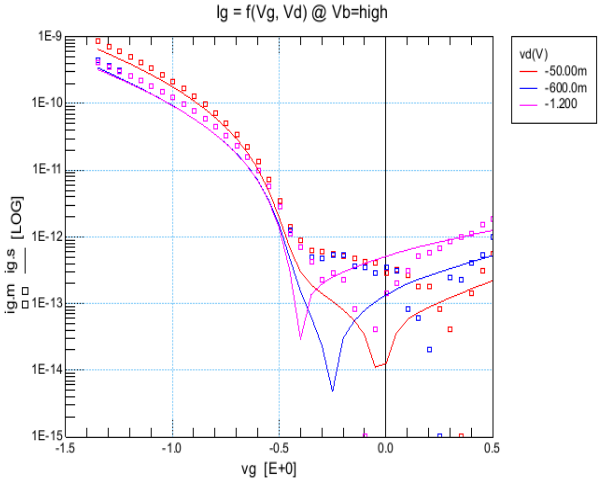
idvg, Ib,Ig, Vb = 0V, T = 27°C



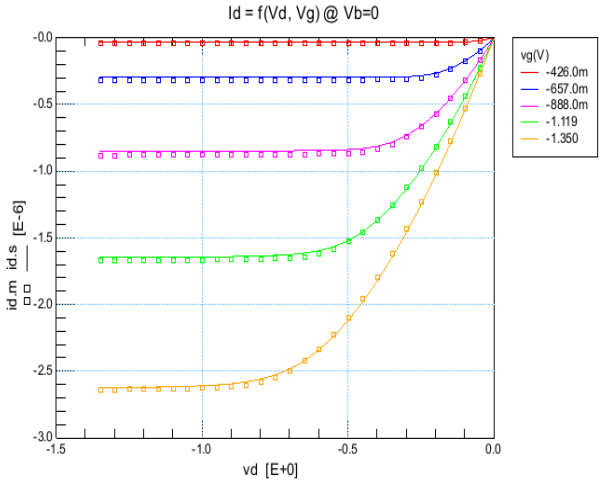


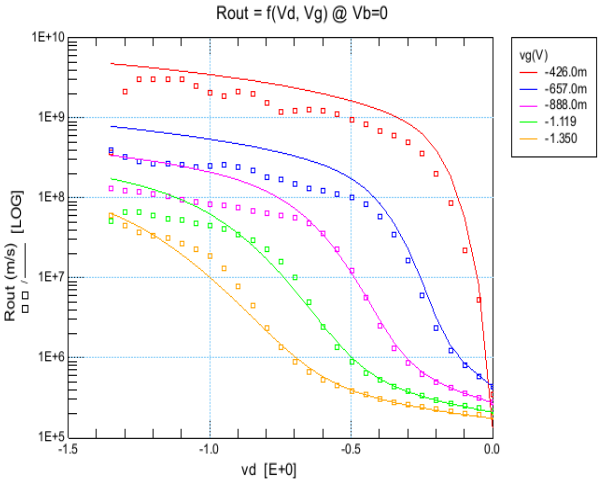
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

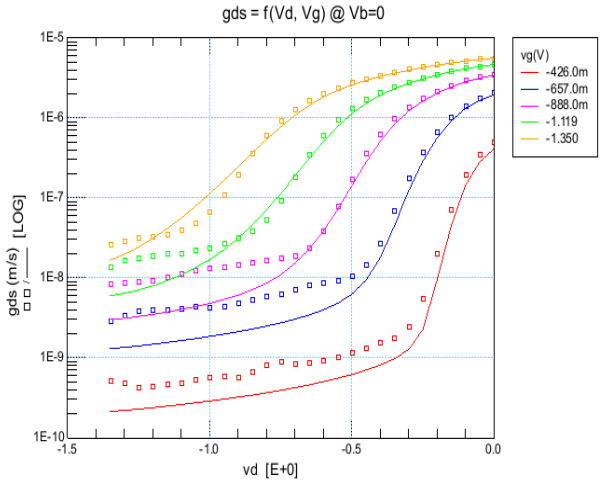




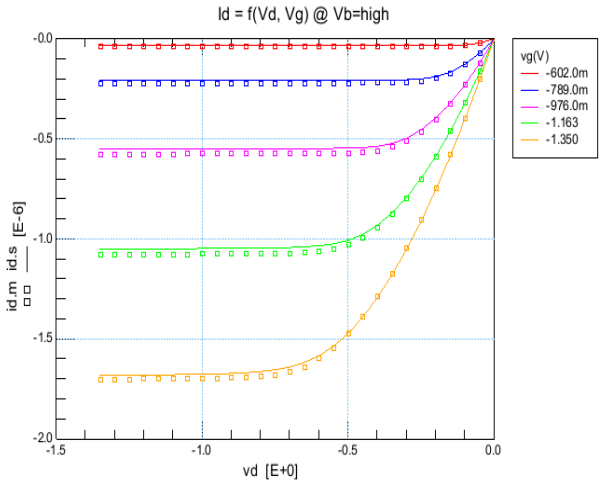
idvd, Vb = 0V, T = 27°C

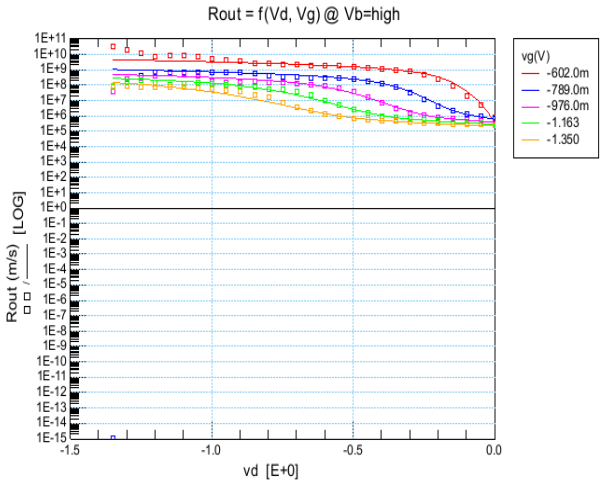


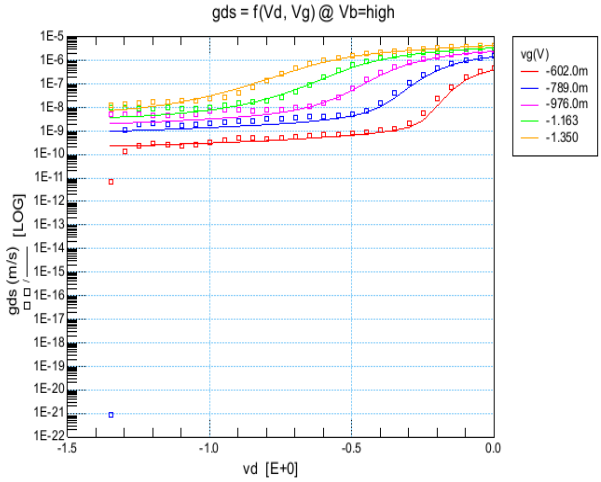




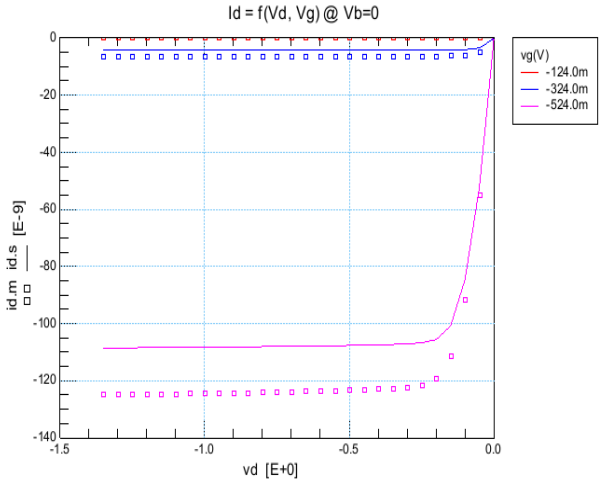
idvd_vbmin, Vb = 1.2V, T = 27°C

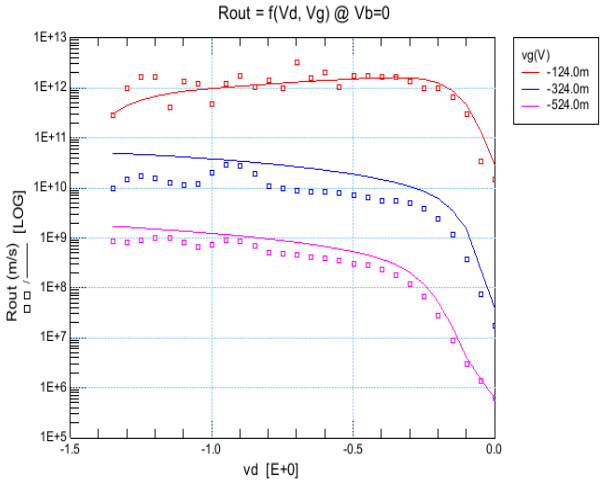


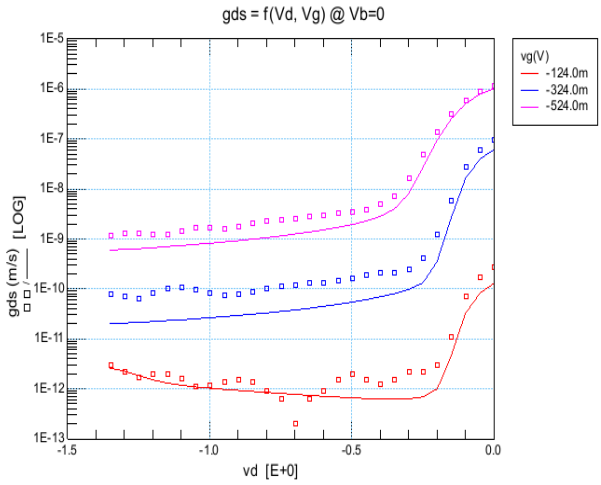




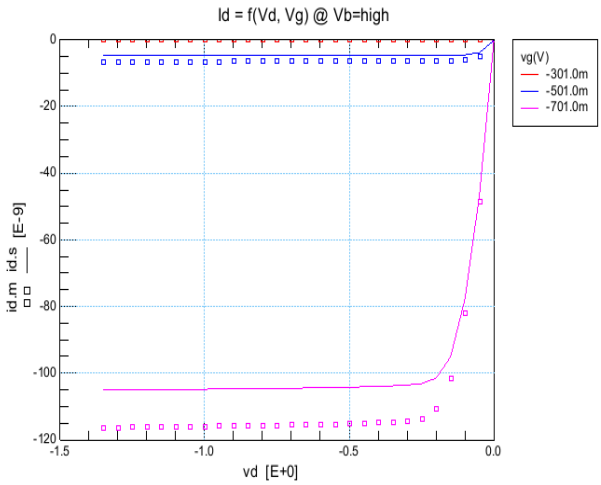
idvd_vth, Vb = 0V, T = 27°C

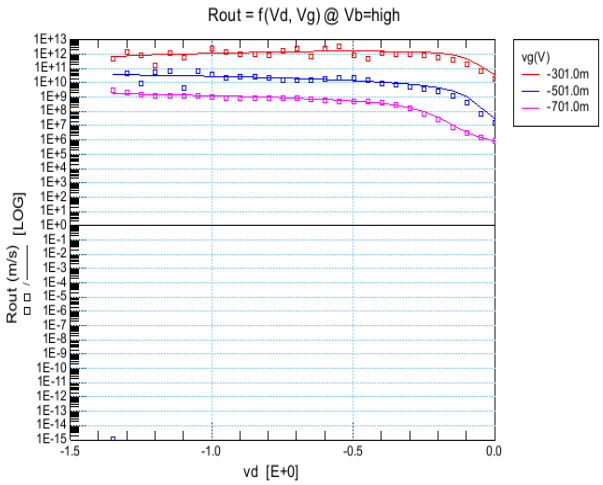


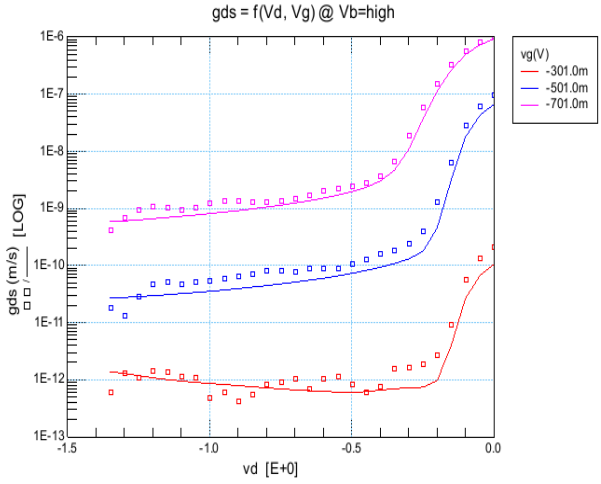




idvd_vth_vbmin, Vb = 1.2V, T = 27°C



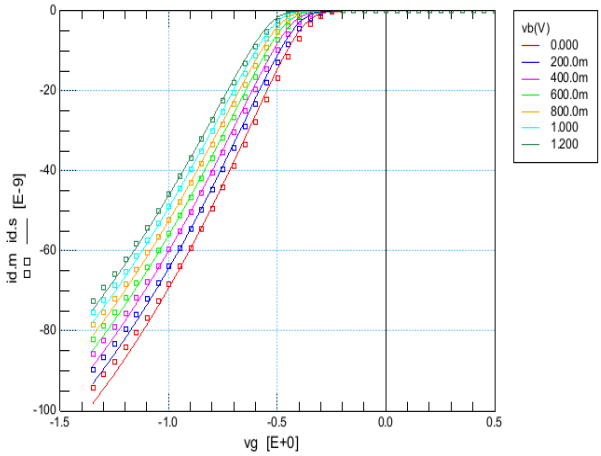




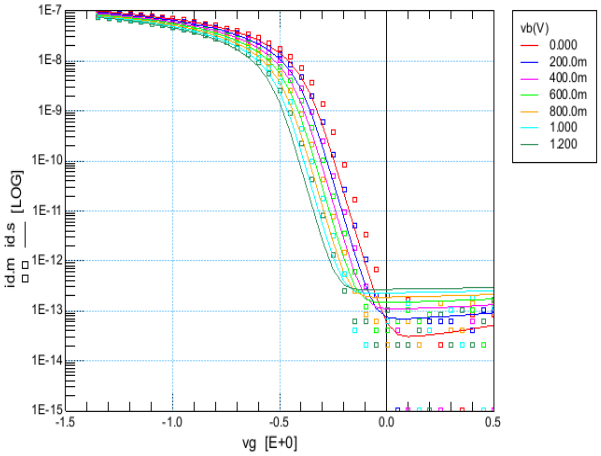
4.14 W0u18_L10u0_S554_1

idvg, Vd = -0.05V, T = 27°C

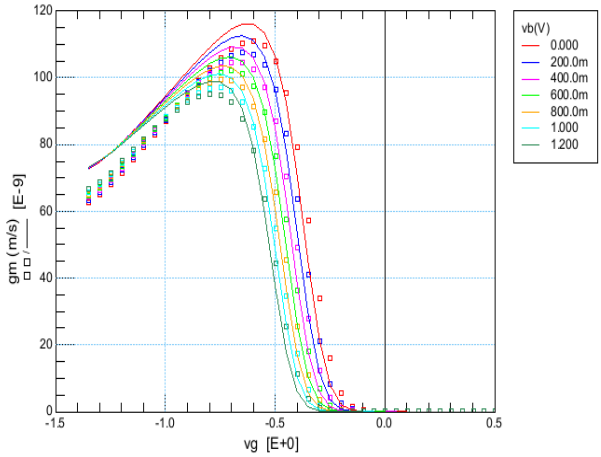
Id = f(Vg, Vb) @ Vd=low



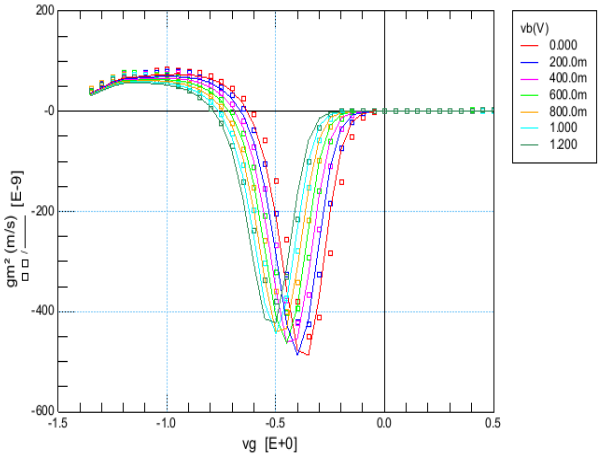
Id = f(Vg, Vb) @ Vd=low



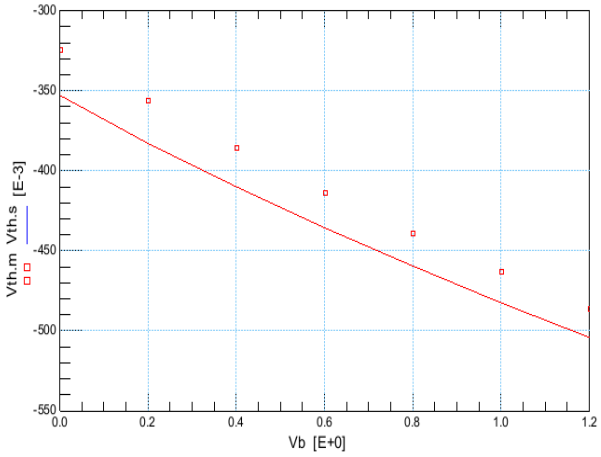
gm = f(Vg, Vb) @ Vd=low



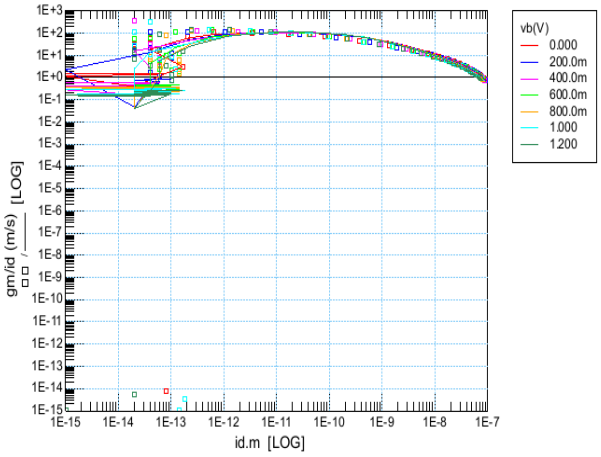
gm² = f(Vg, Vb) @ Vd=low

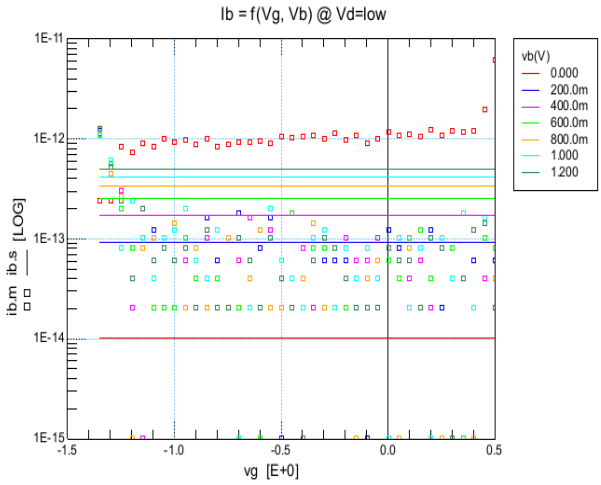


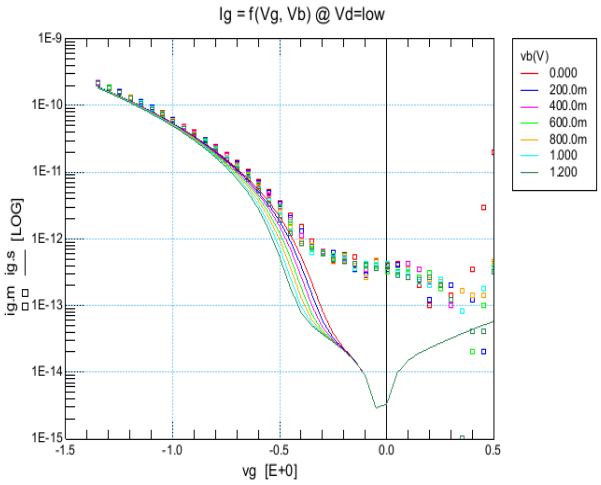
Vth @ Vd=low



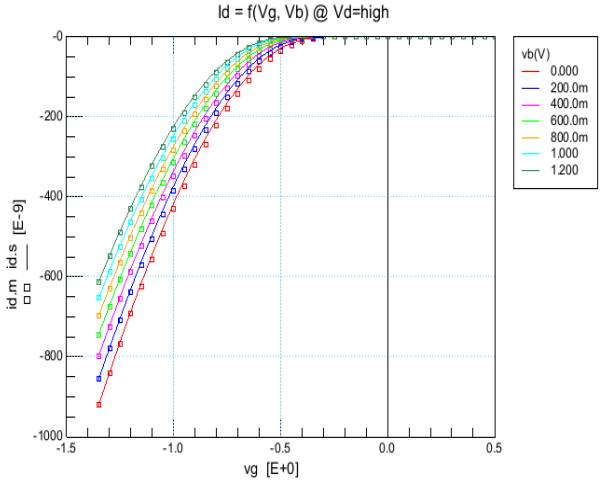
gm/id=f(id.m) @ Vd=low

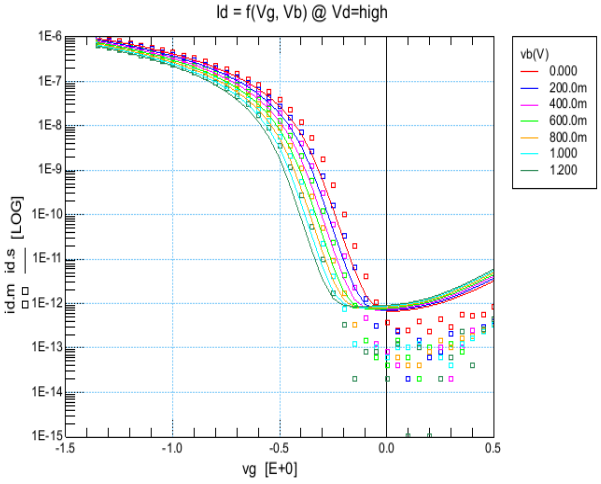


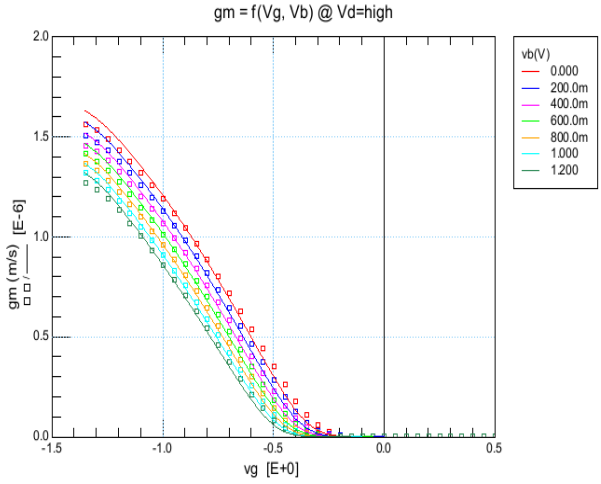


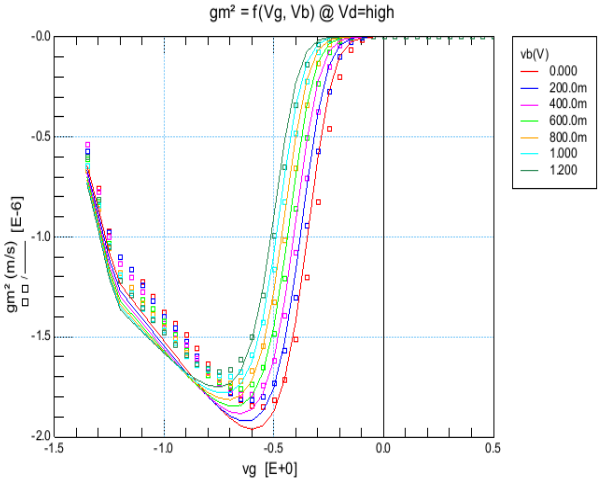


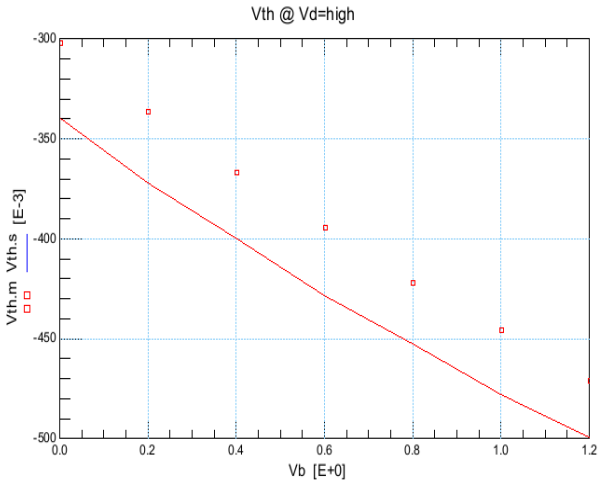
idvg, Vd = -1.2V, T = 27°C

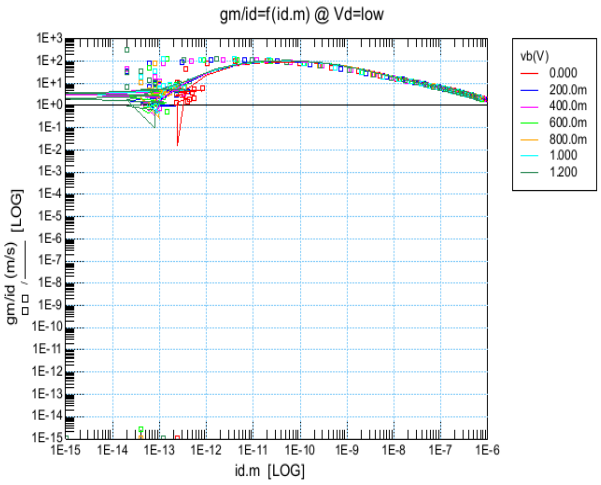


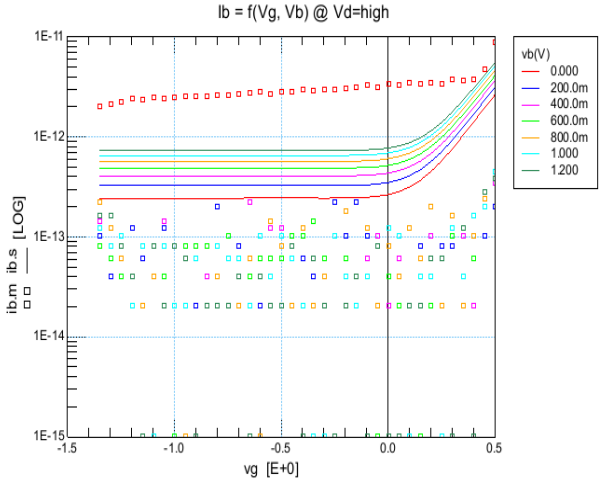


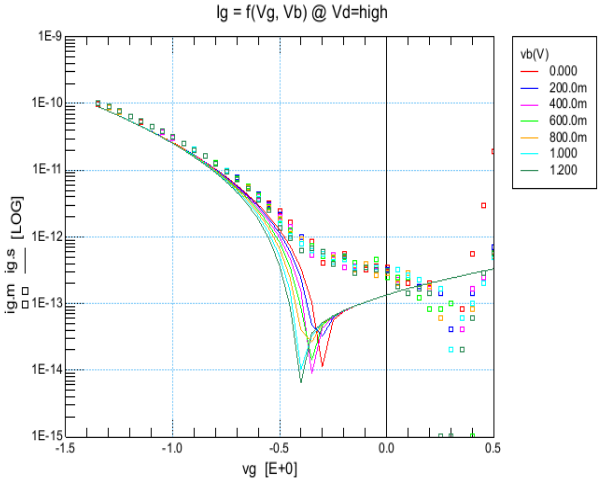




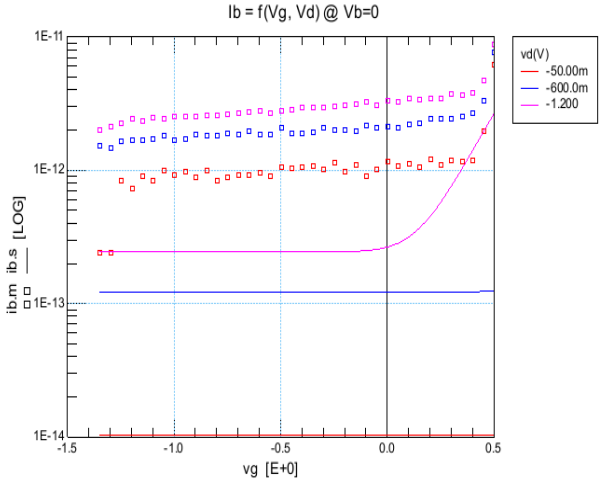


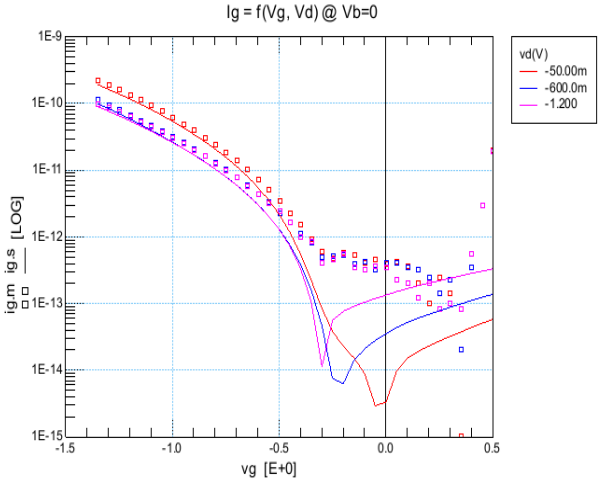




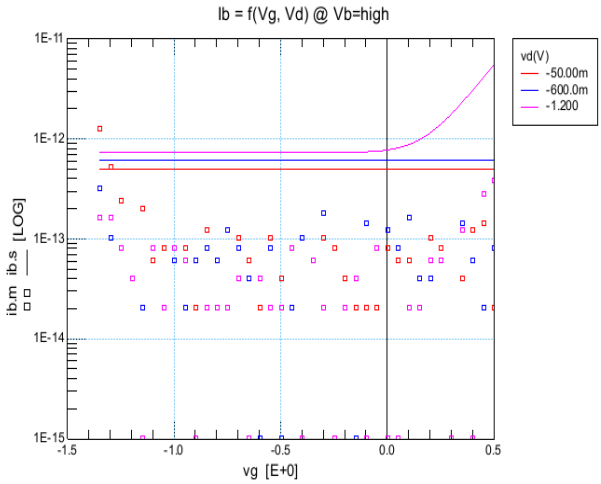


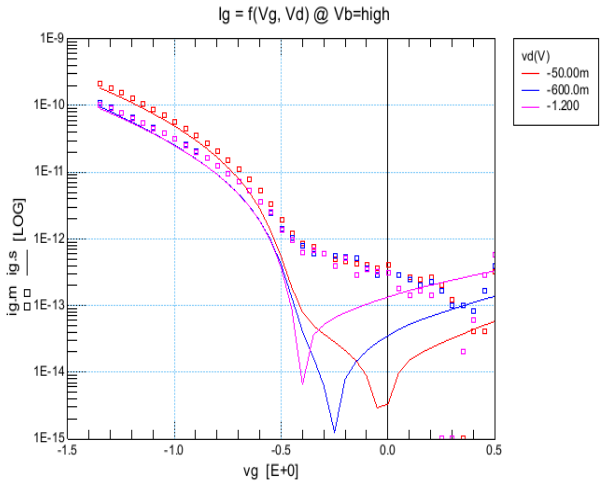
idvg, Ib,Ig, Vb = 0V, T = 27°C



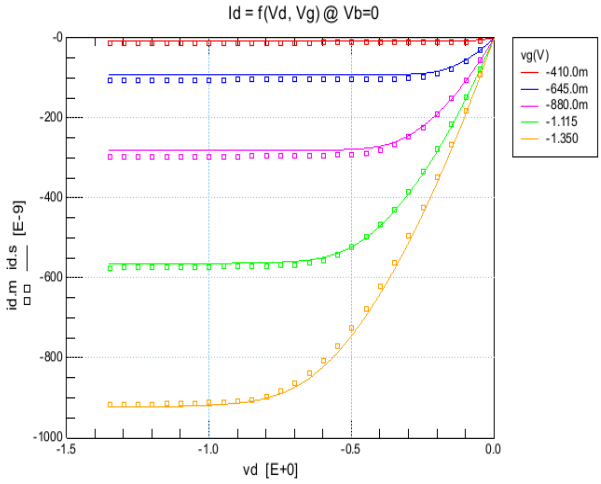


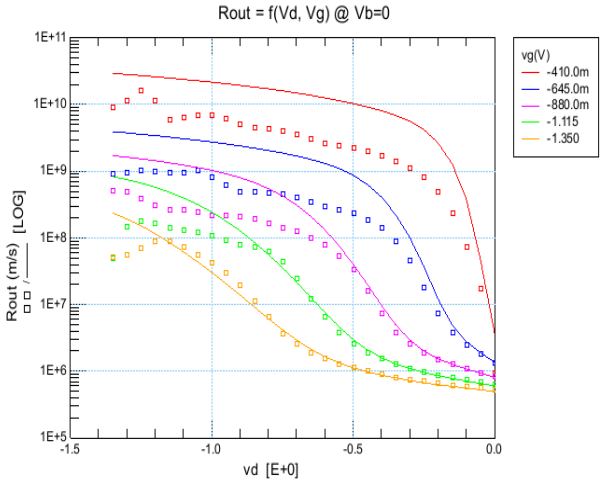
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

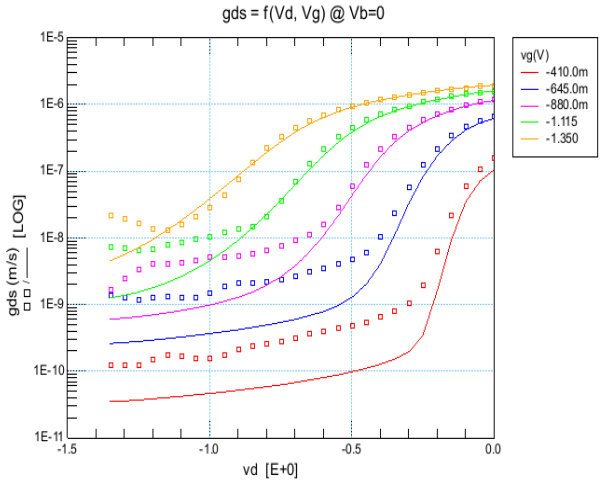




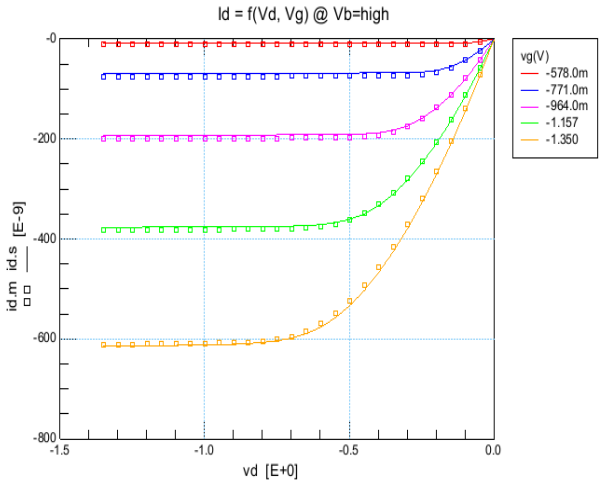
idvd, Vb = 0V, T = 27°C

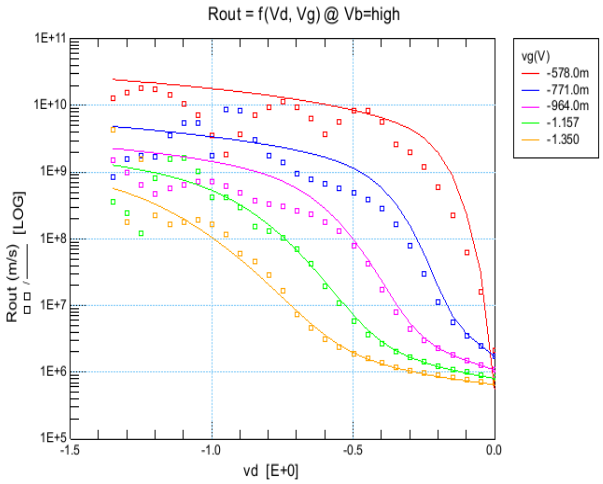


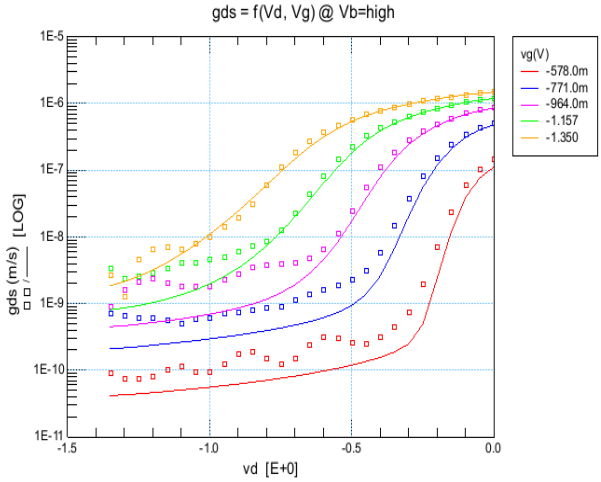




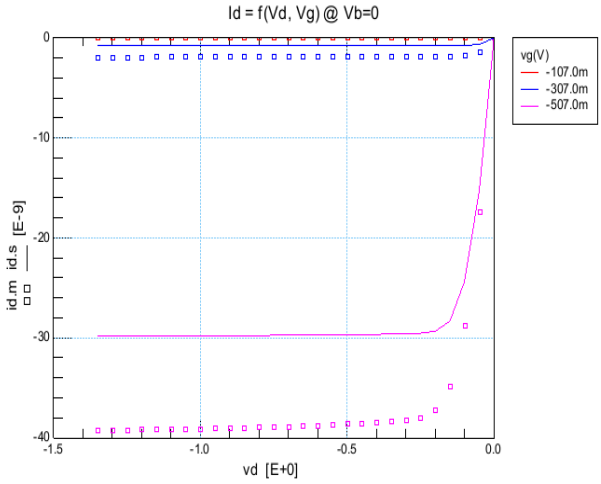
idvd_vbmin, Vb = 1.2V, T = 27°C

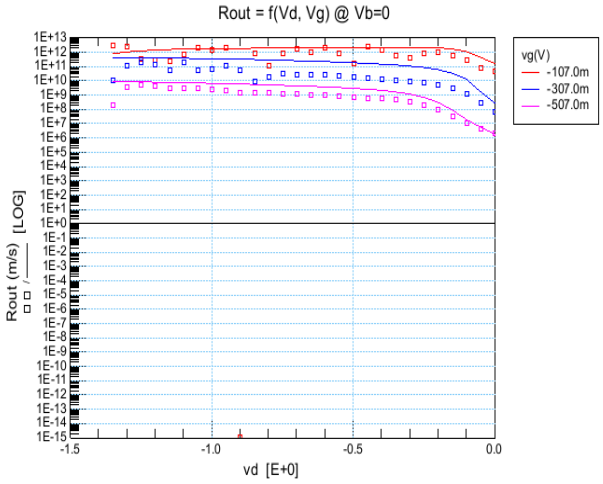


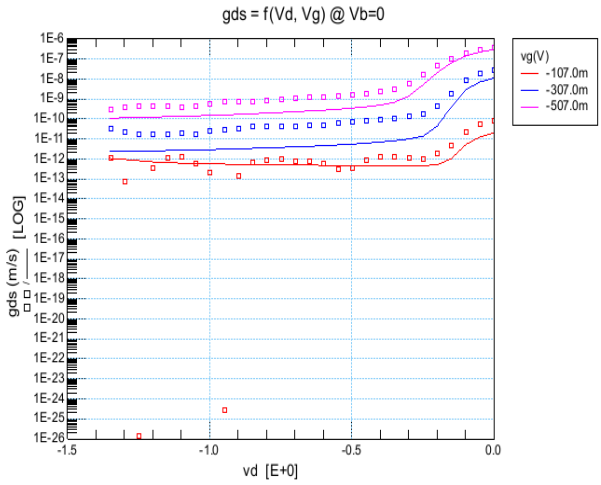




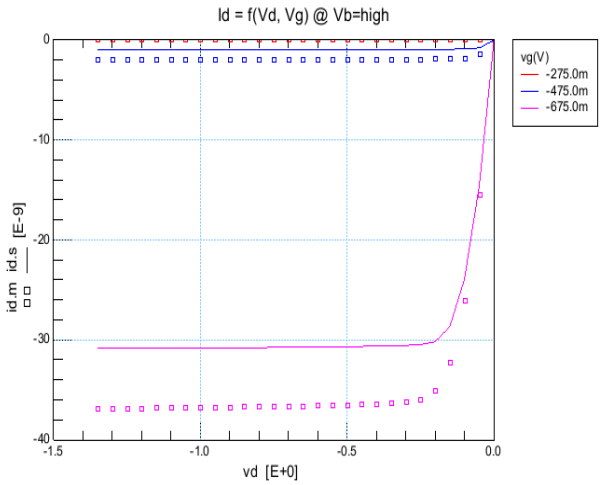
idvd_vth, Vb = 0V, T = 27°C

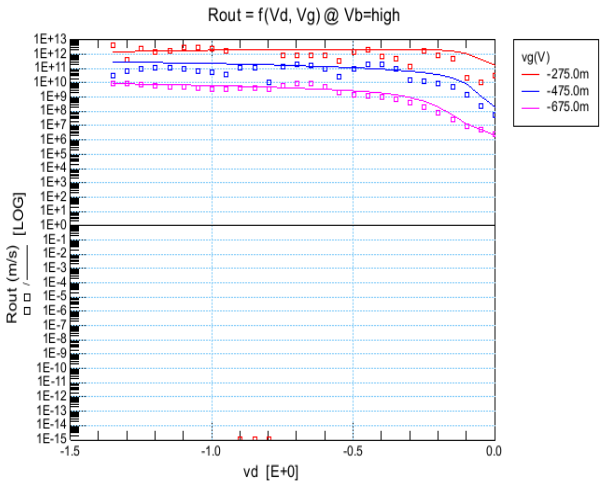


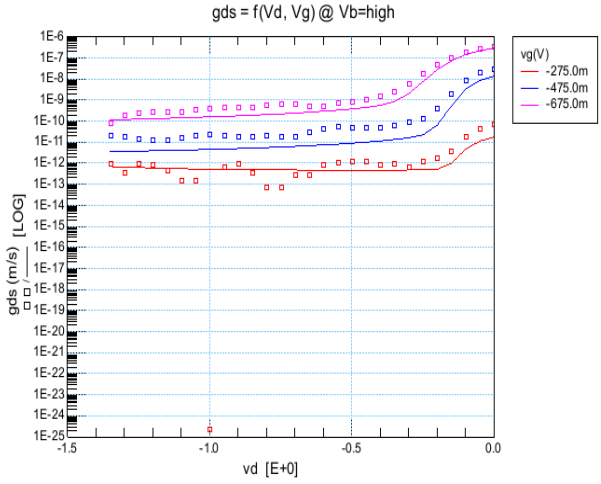




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

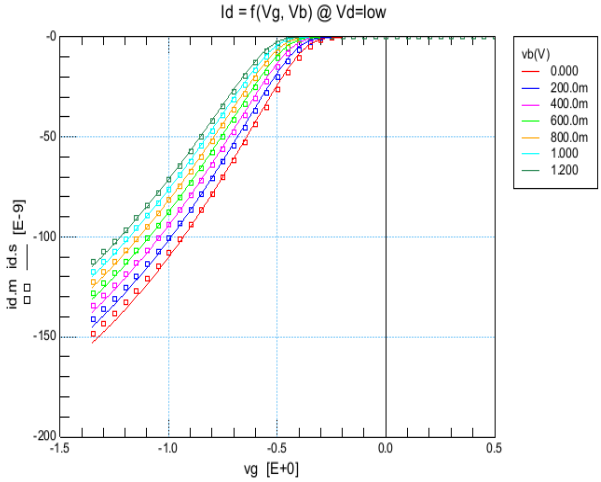


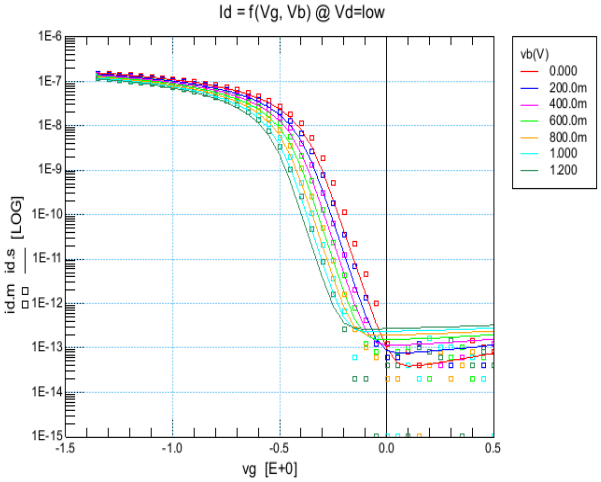


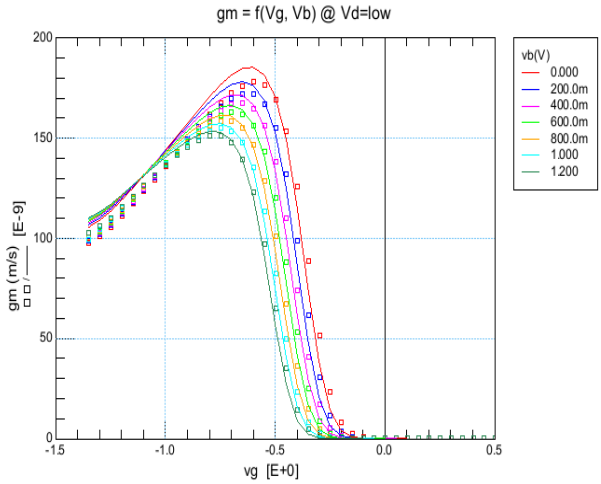


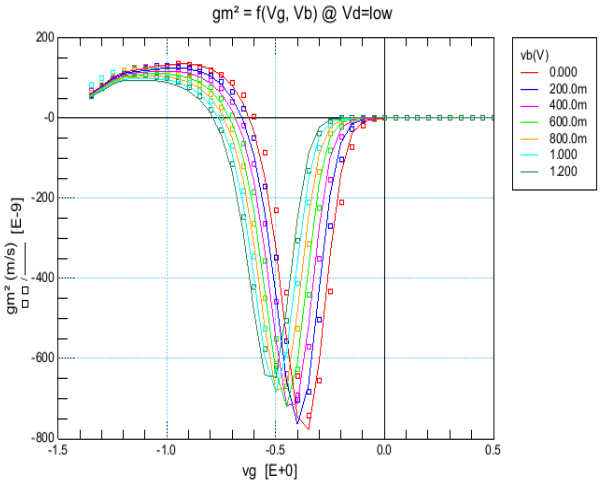
4.15W0u3_L10u0_S553_2

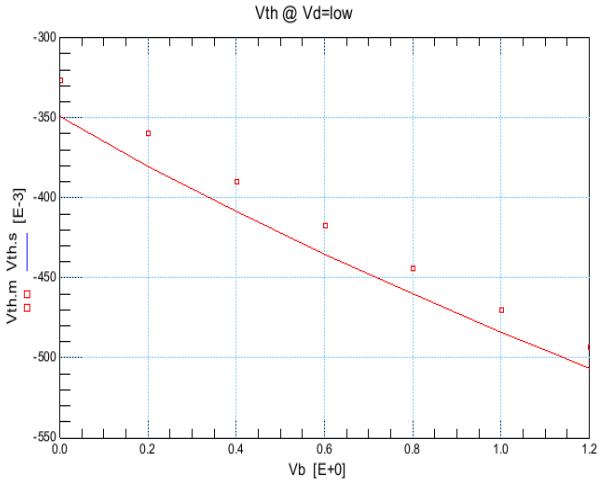
idvg, Vd = -0.05V, T = 27°C

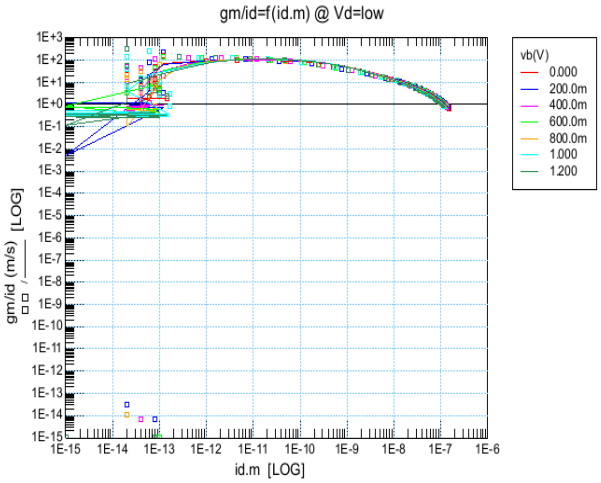


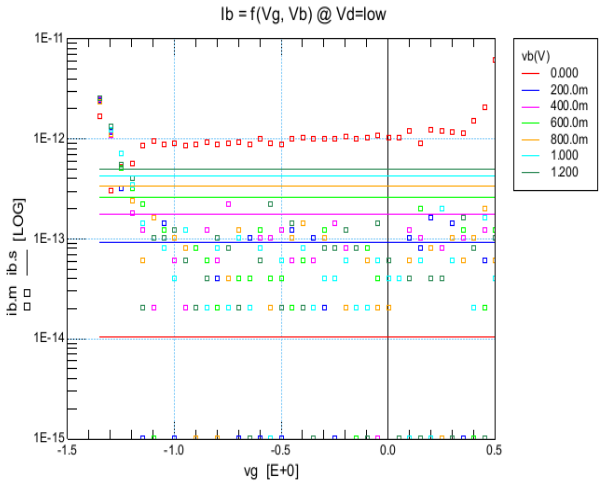


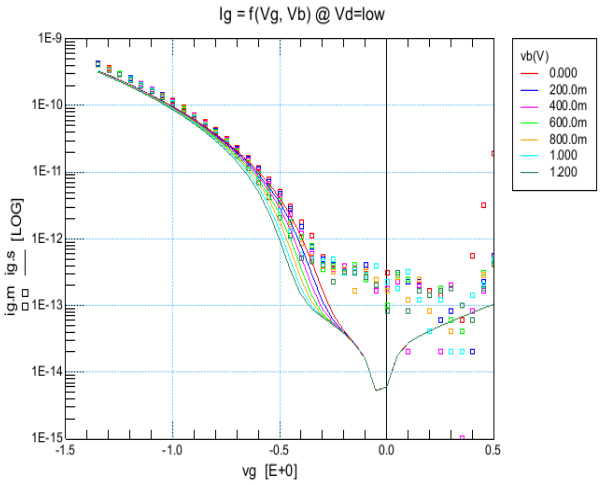




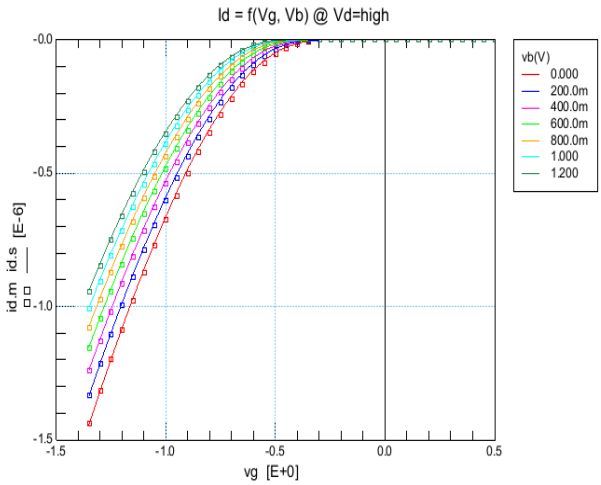


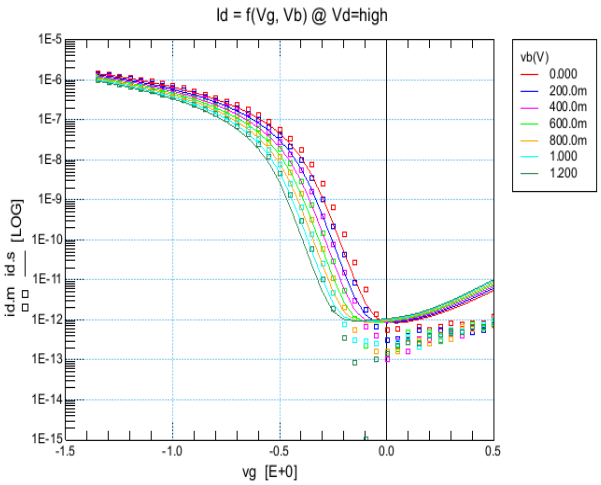


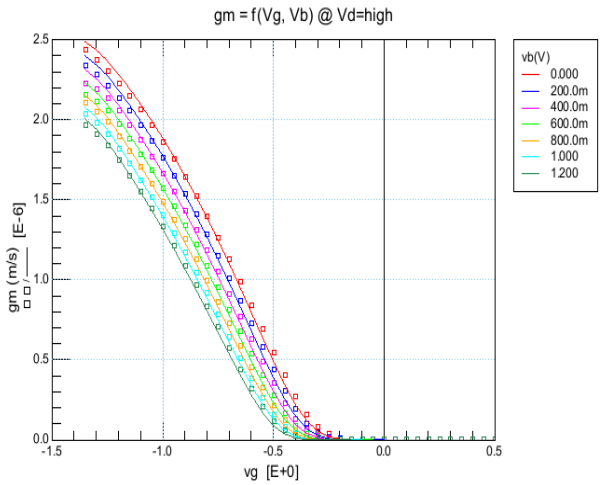


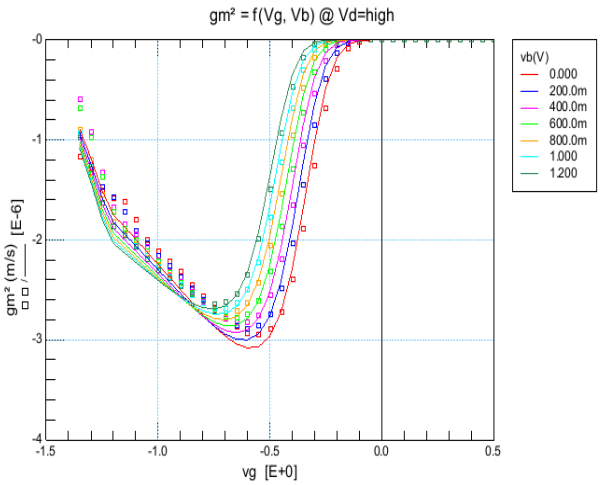


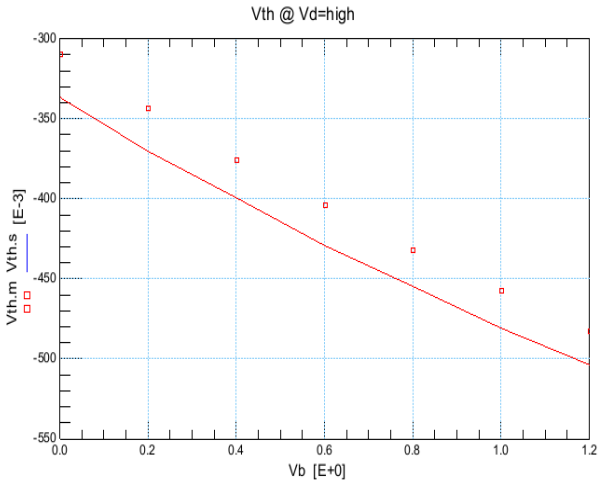
idvg, Vd = -1.2V, T = 27°C

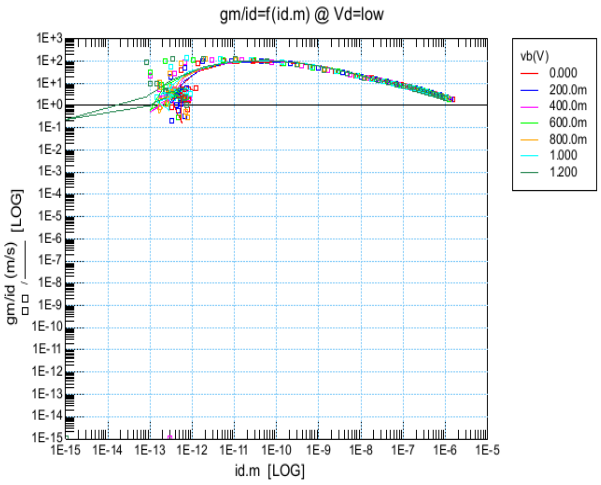


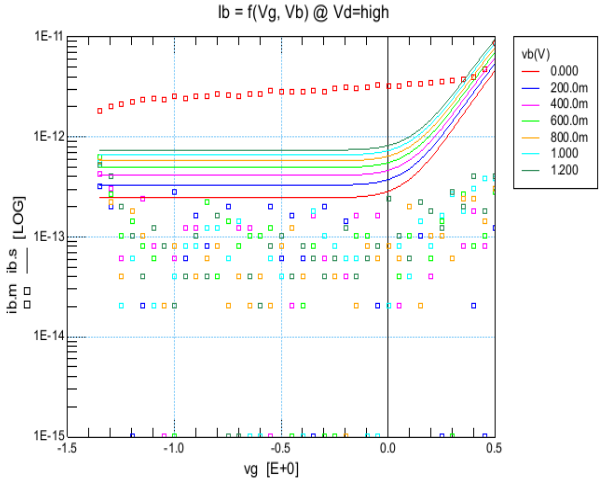


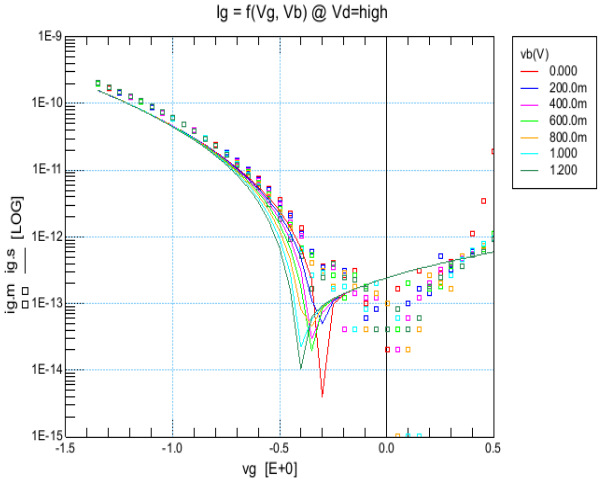




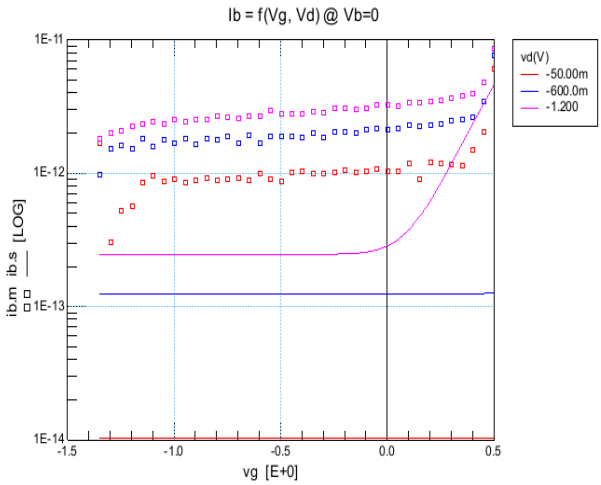


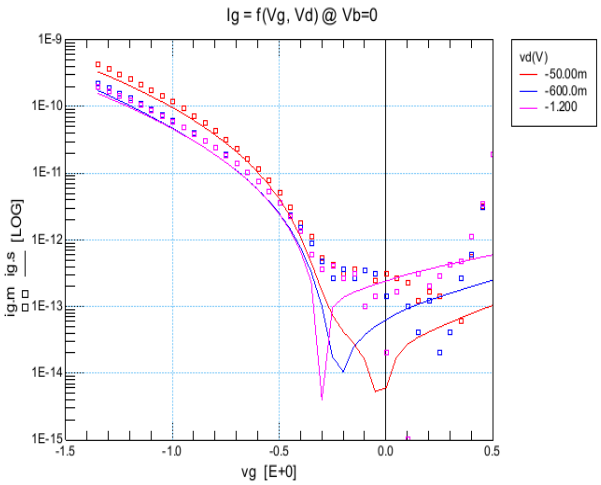




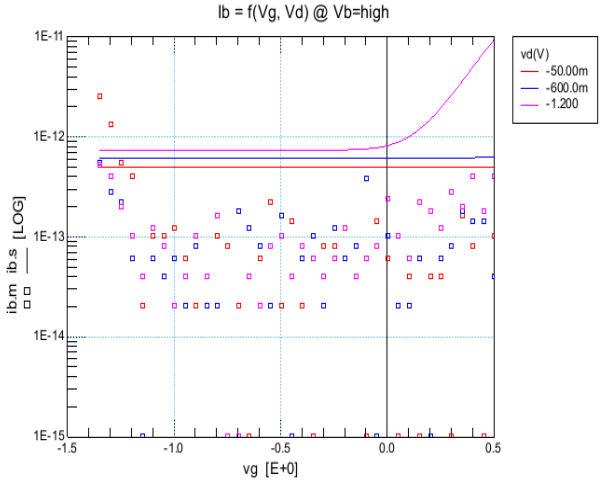


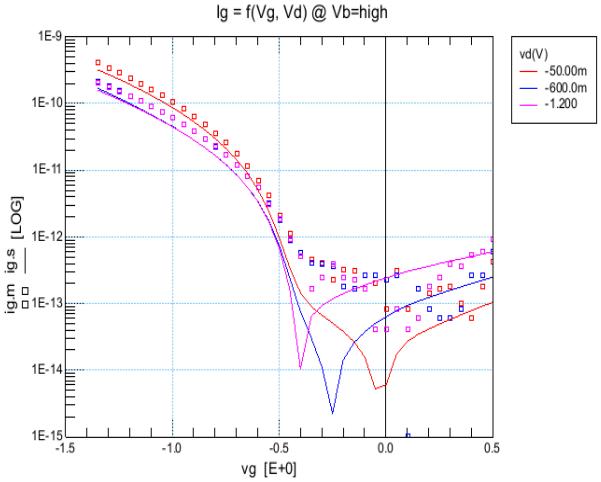
idvg, Ib,Ig, Vb = 0V, T = 27°C



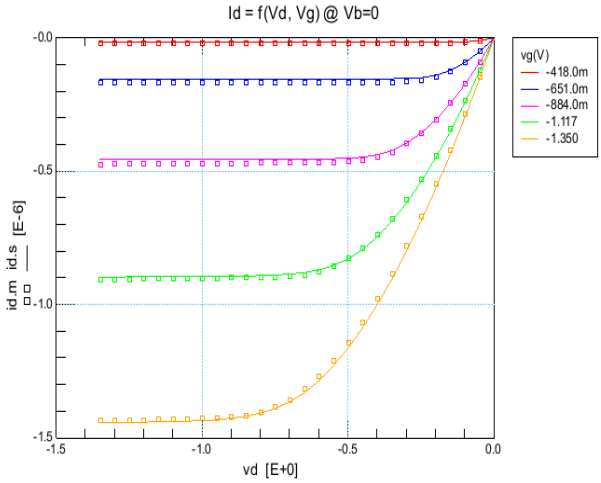


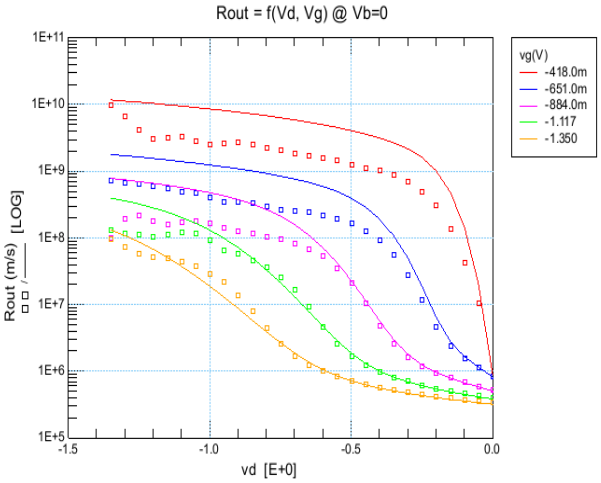
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

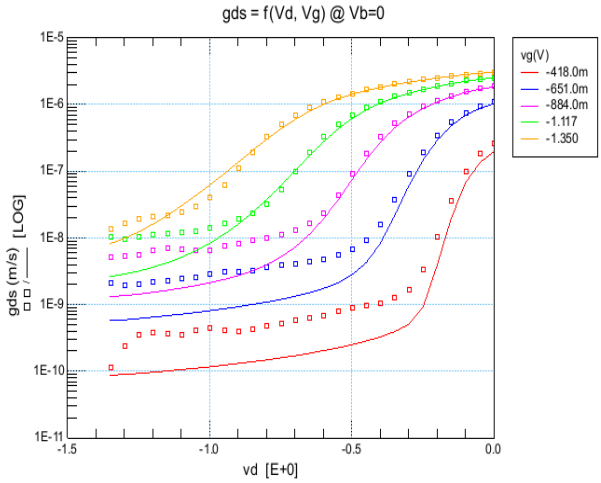




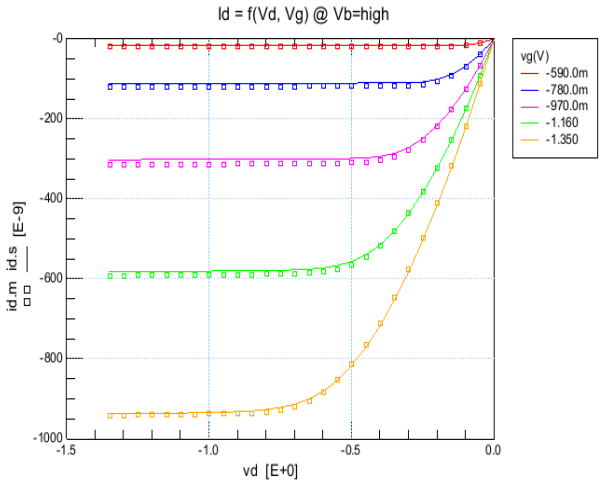
idvd, Vb = 0V, T = 27°C

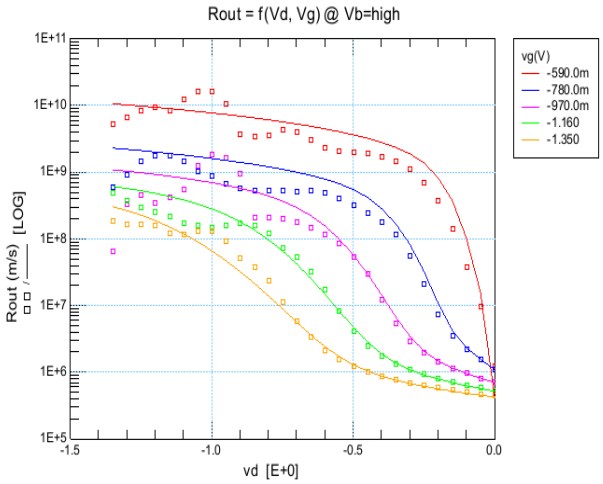


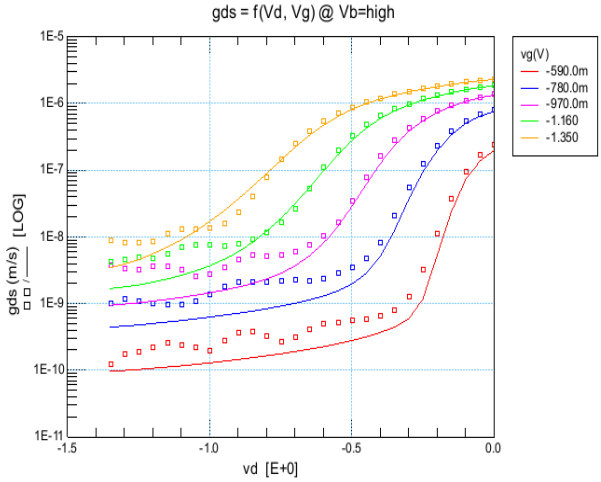




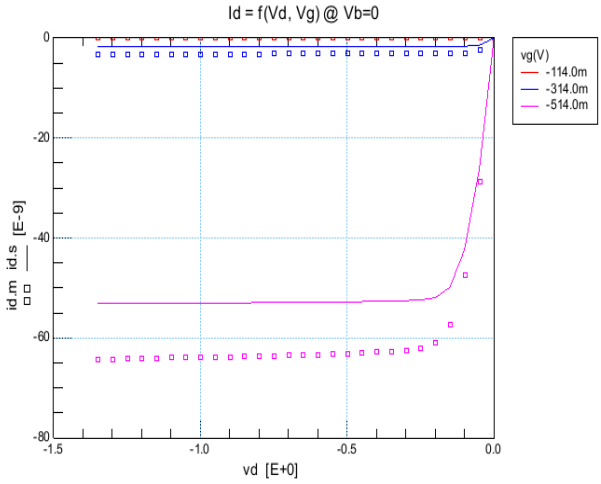
idvd_vbmin, Vb = 1.2V, T = 27°C

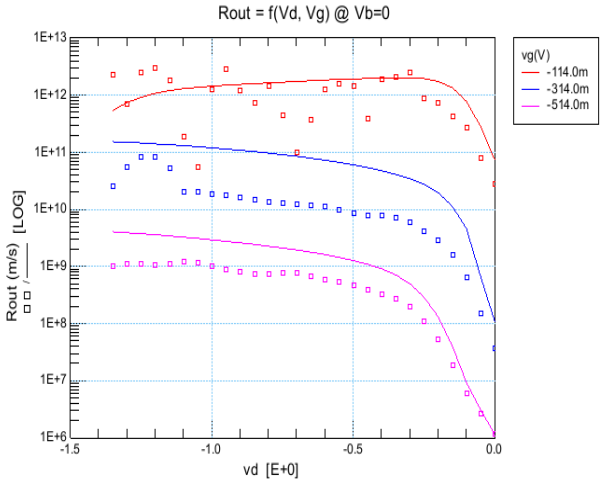


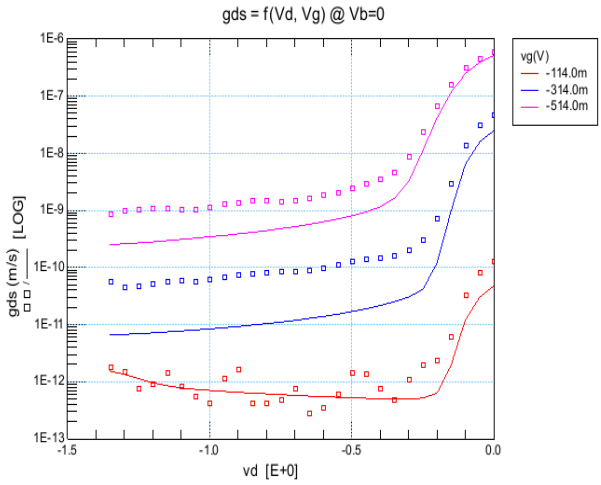




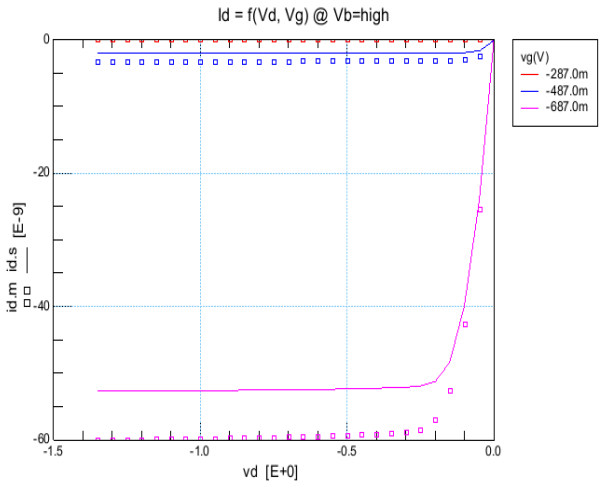
idvd_vth, Vb = 0V, T = 27°C

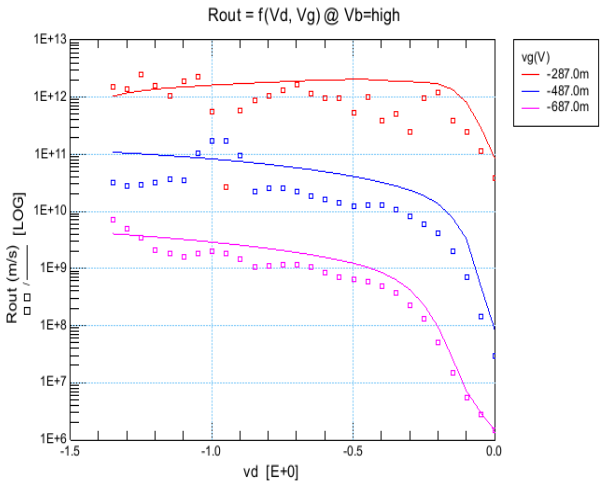


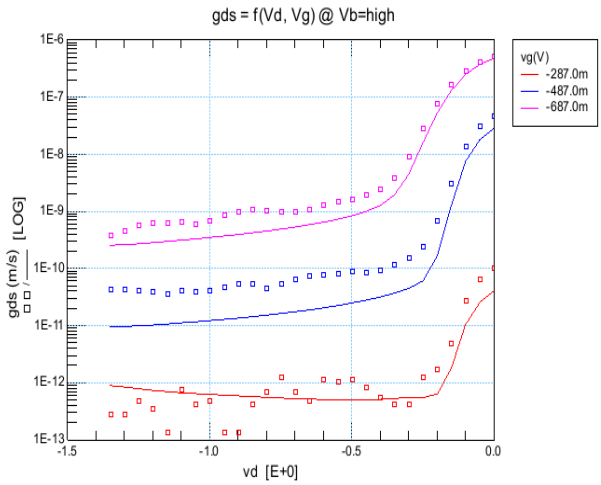




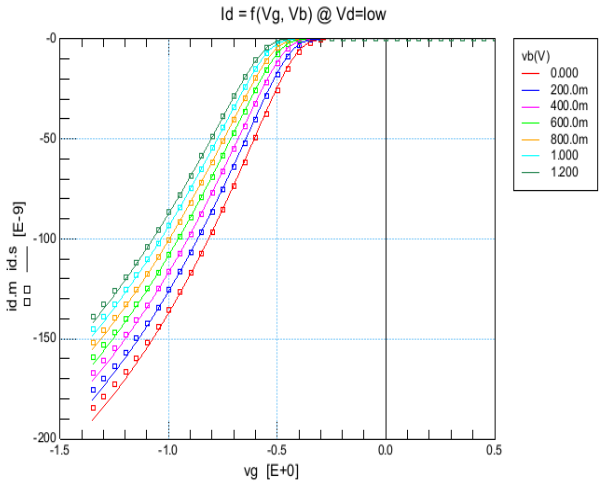
idvd_vth_vbmin, Vb = 1.2V, T = 27°C

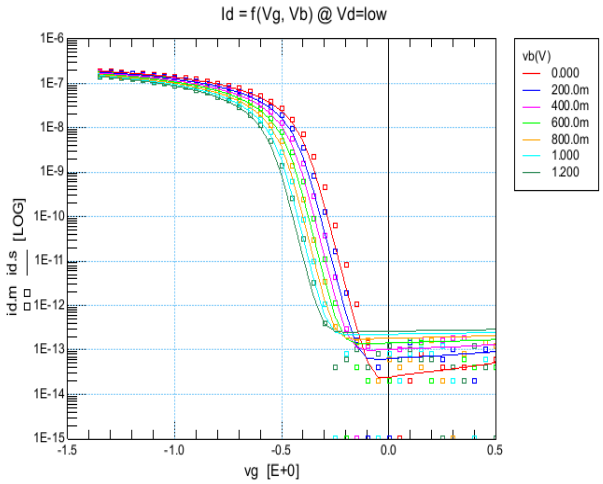


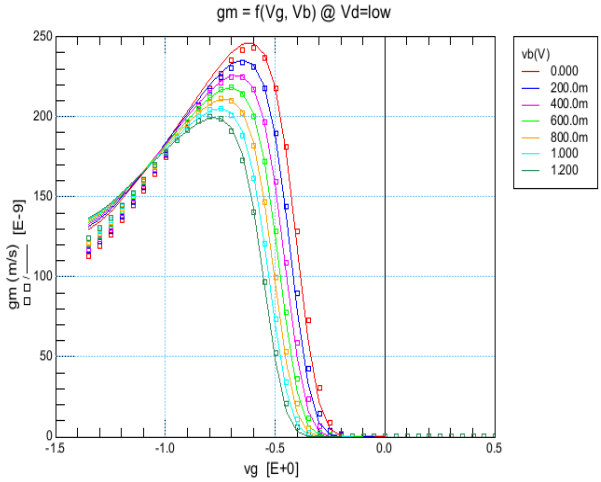


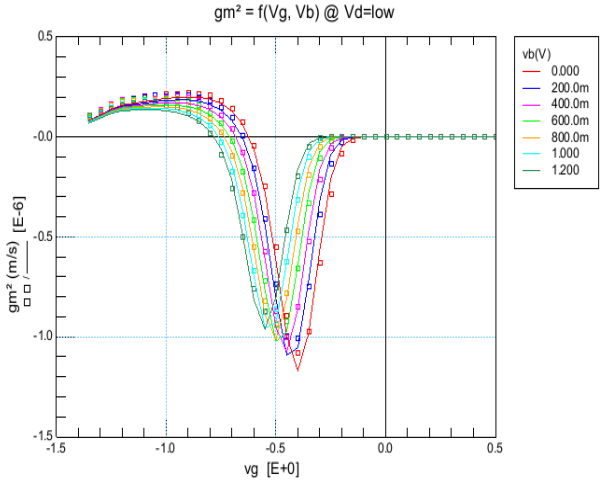


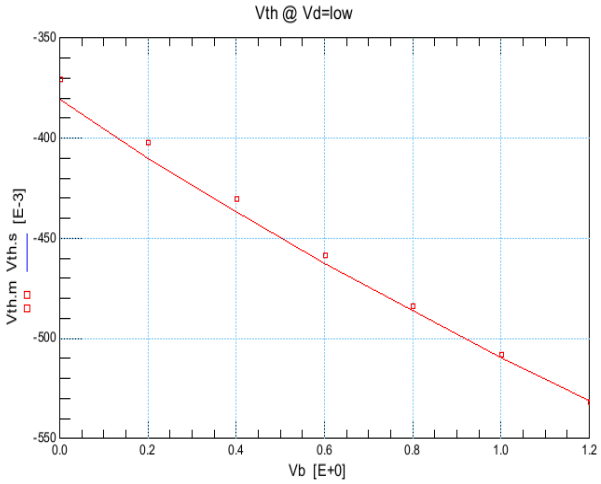
idvg, Vd = -0.05V, T = -40°C

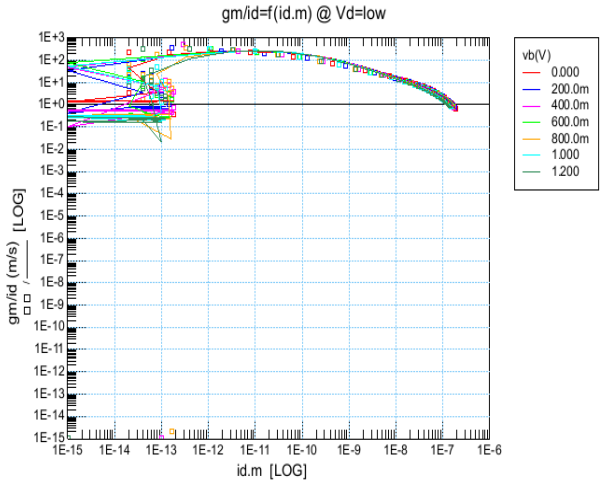


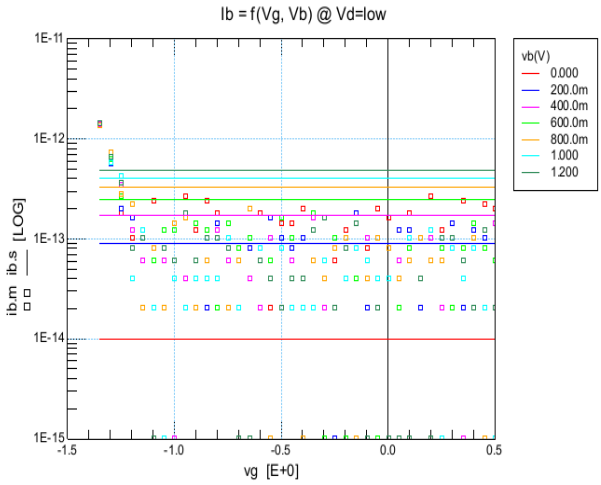


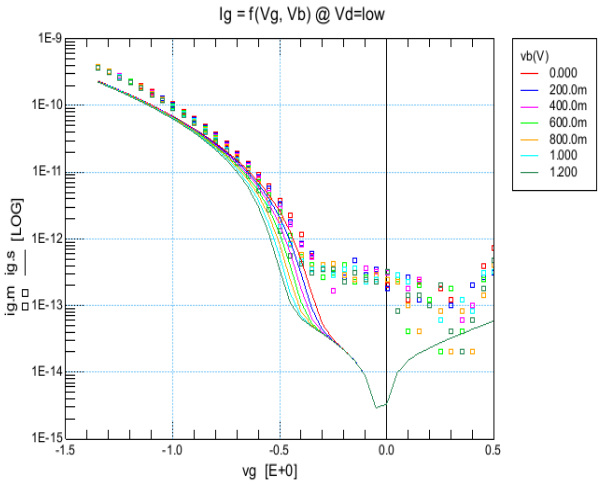




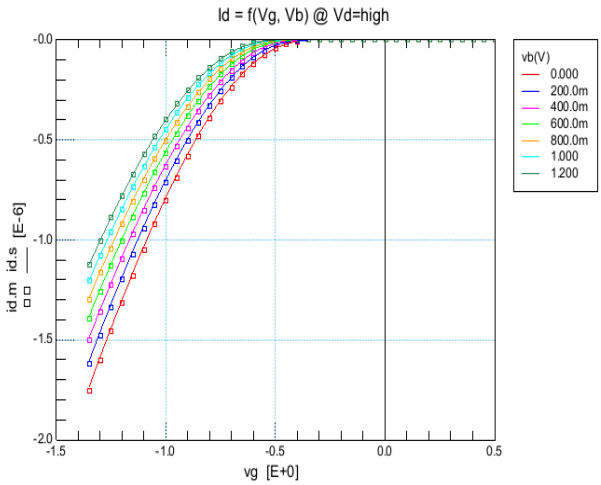


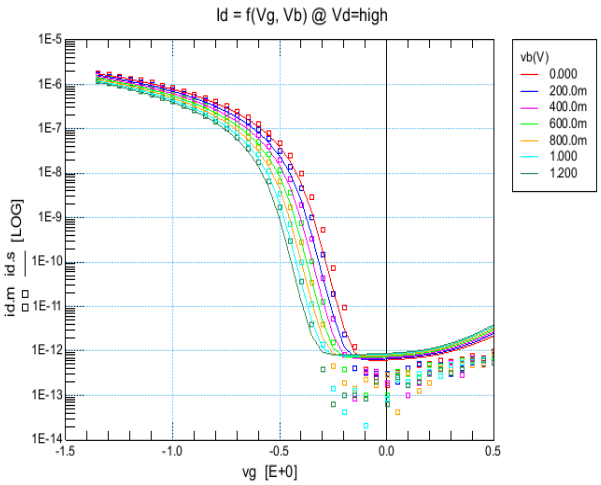


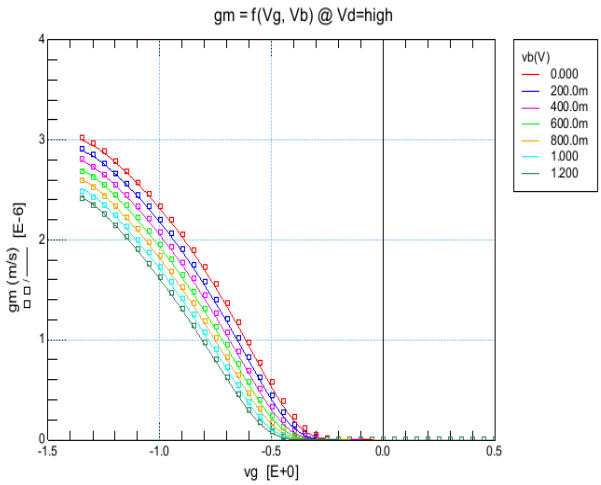


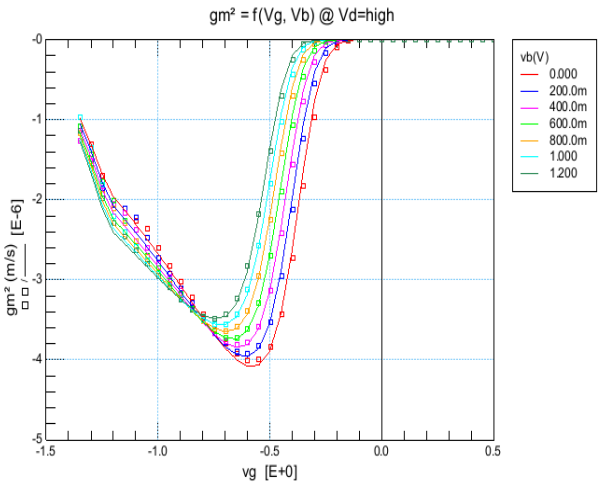


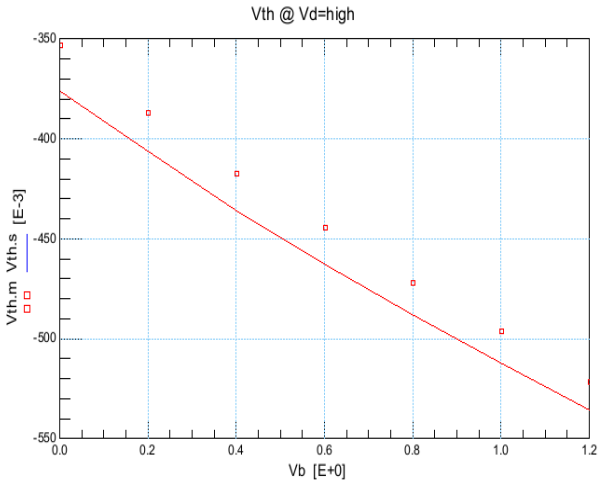
idvg, Vd = -1.2V, T = -40°C

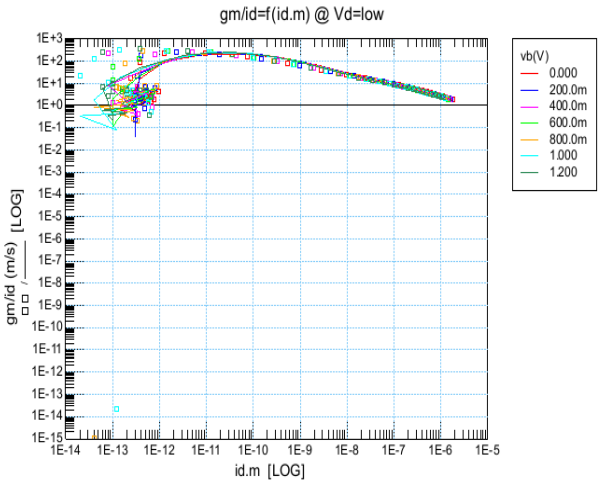


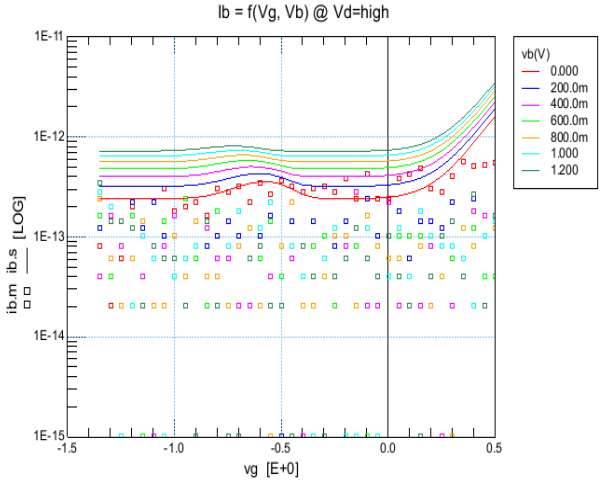


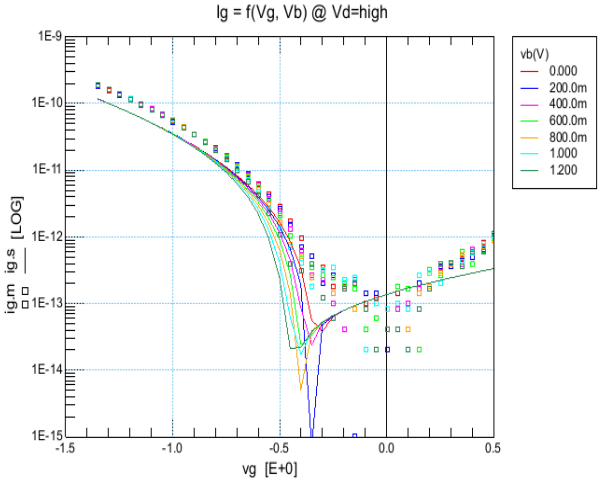




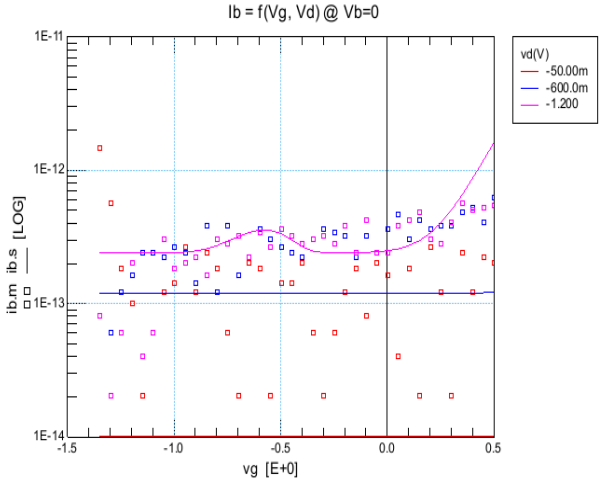


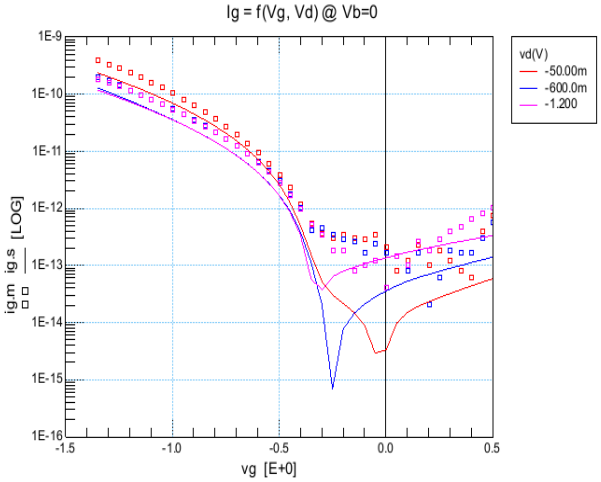




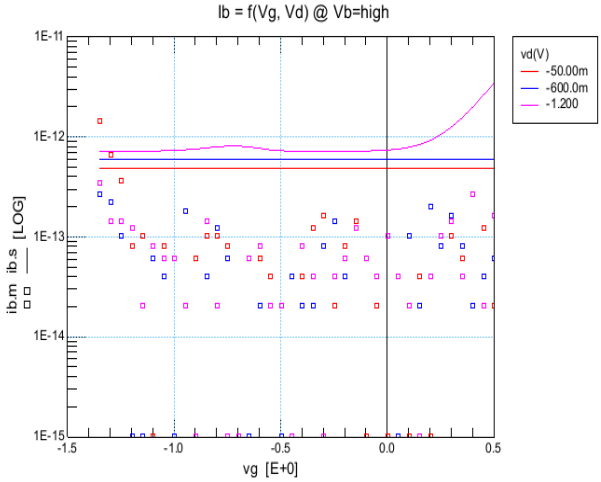


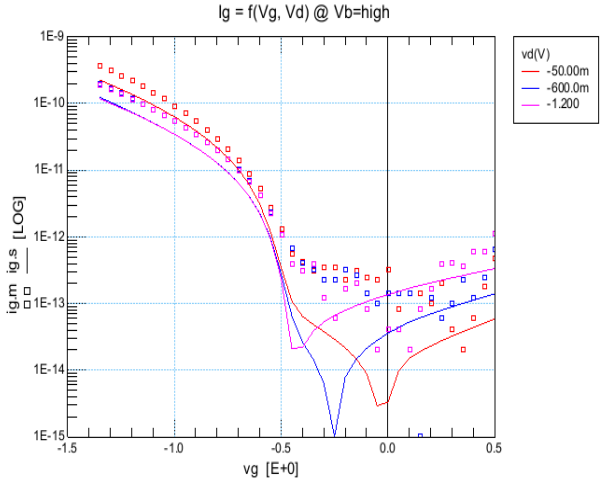
idvg, Ib,Ig, Vb = 0V, T = -40°C



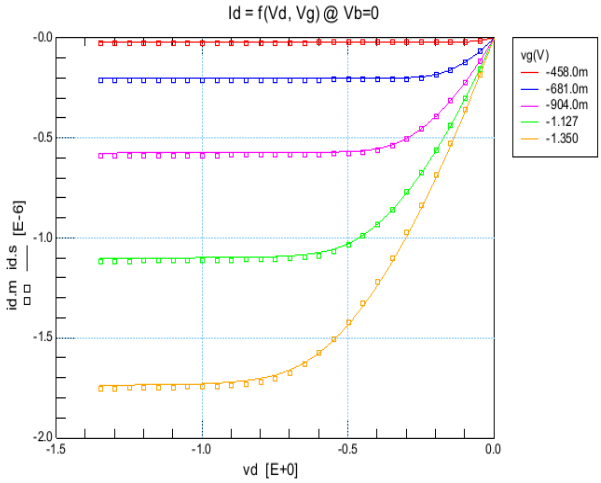


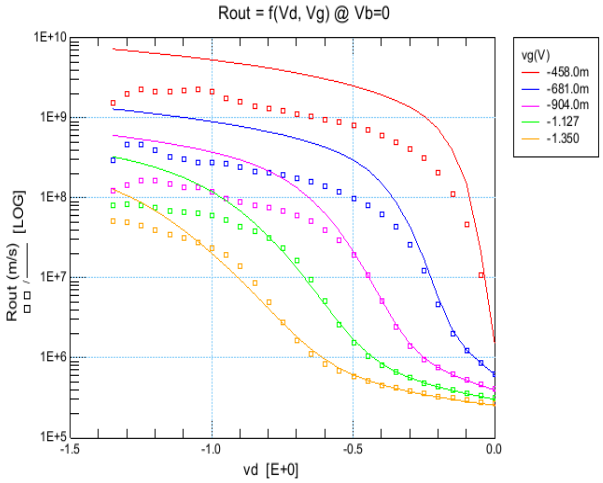
idvg, Ib,Ig, Vb = 1.2V, T = -40°C

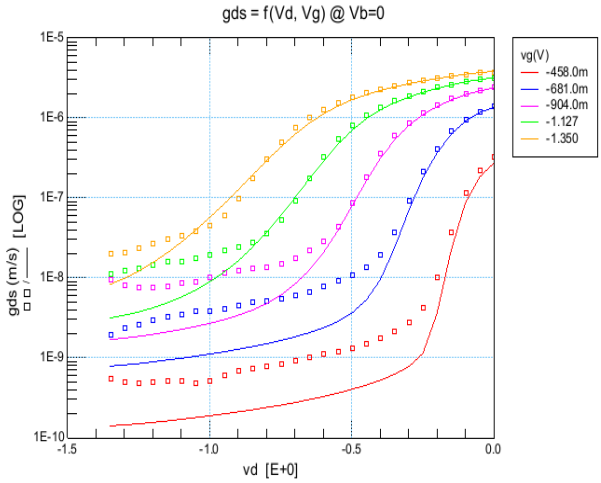




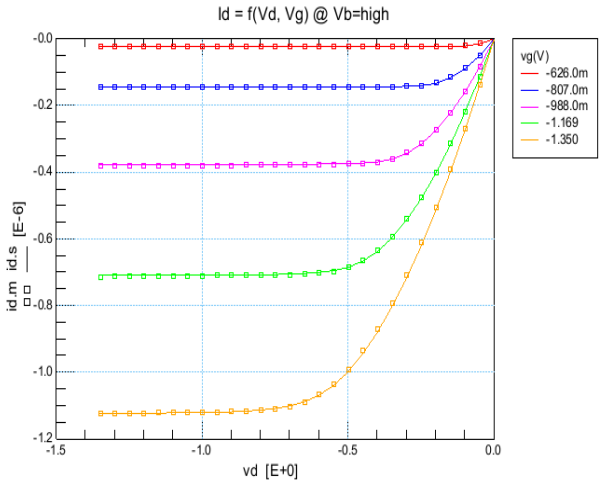
idvd, Vb = 0V, T = -40°C

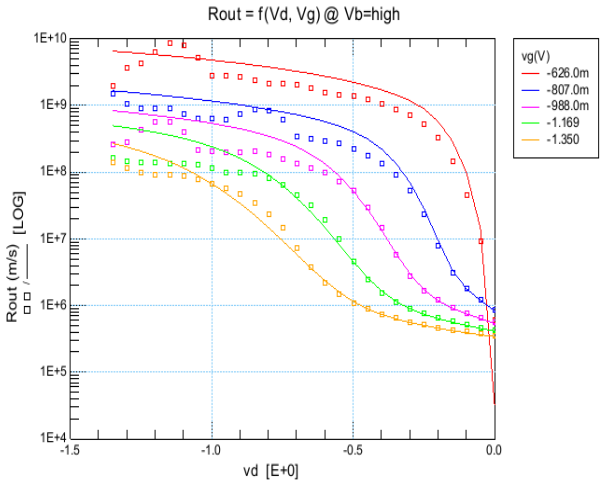


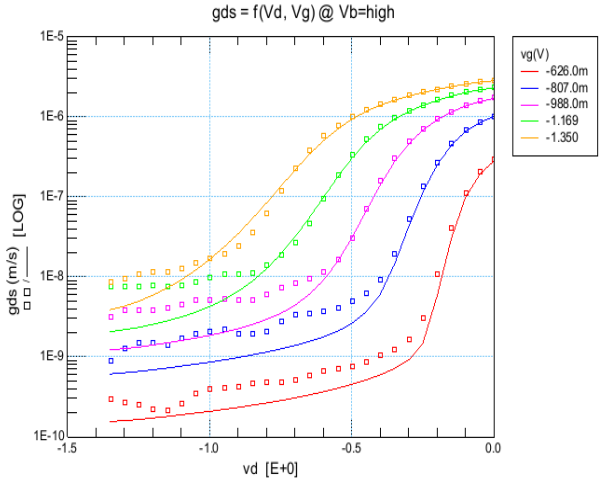




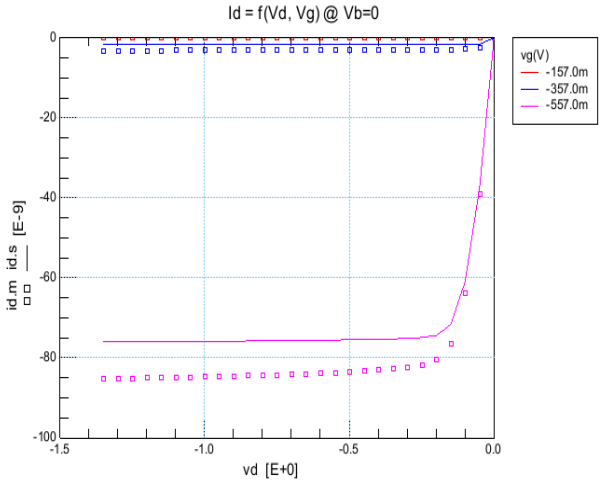
idvd_vbmin, Vb = 1.2V, T = -40°C

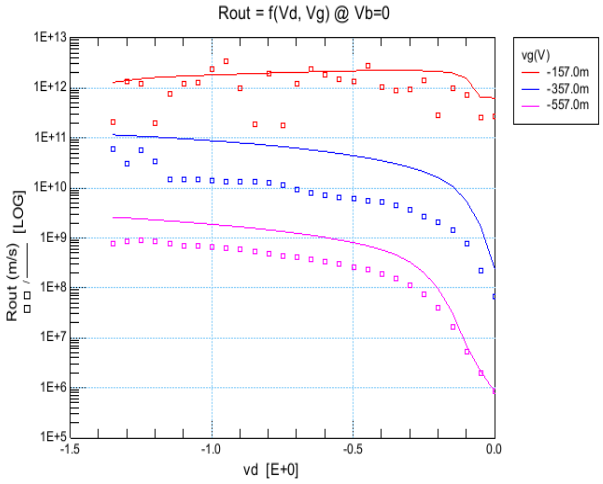


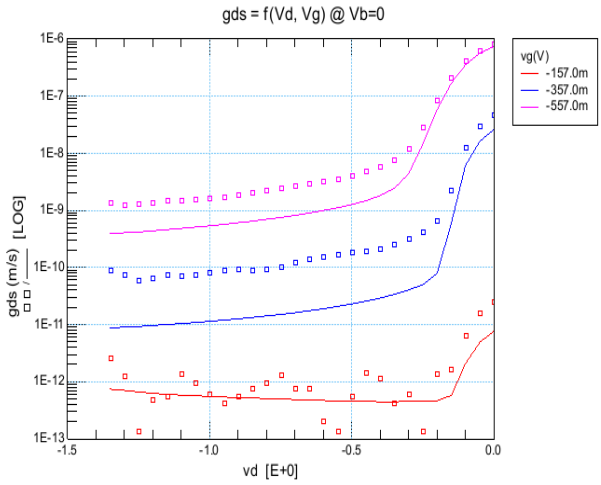




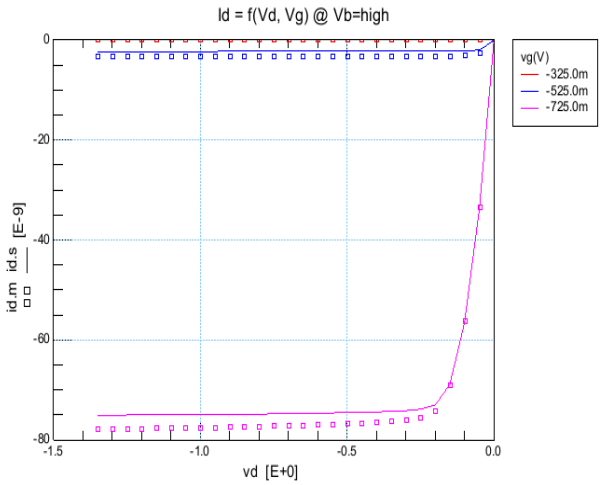
idvd_vth, Vb = 0V, T = -40°C

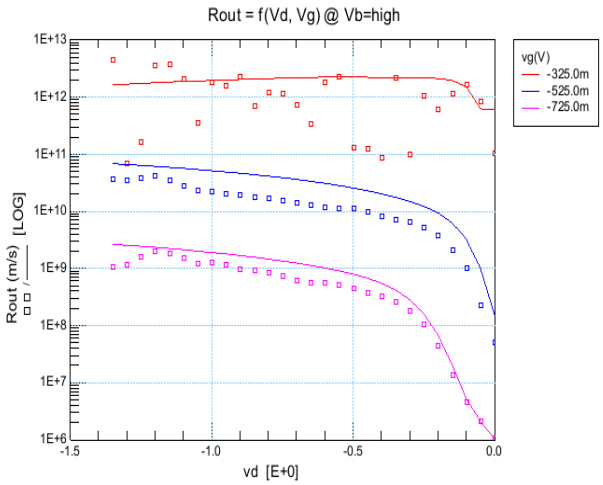


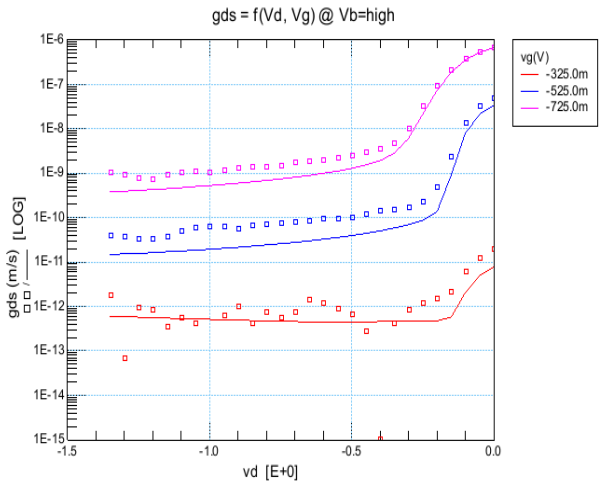




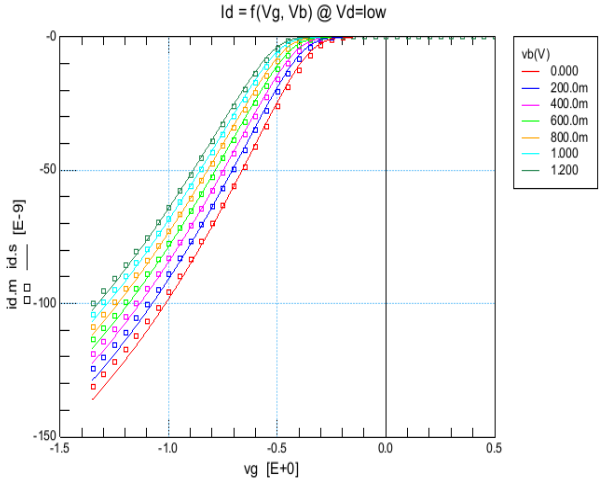
idvd_vth_vbmin, Vb = 1.2V, T = -40°C

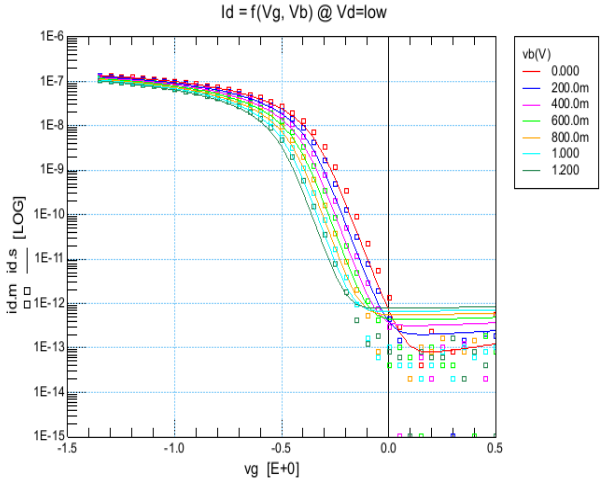


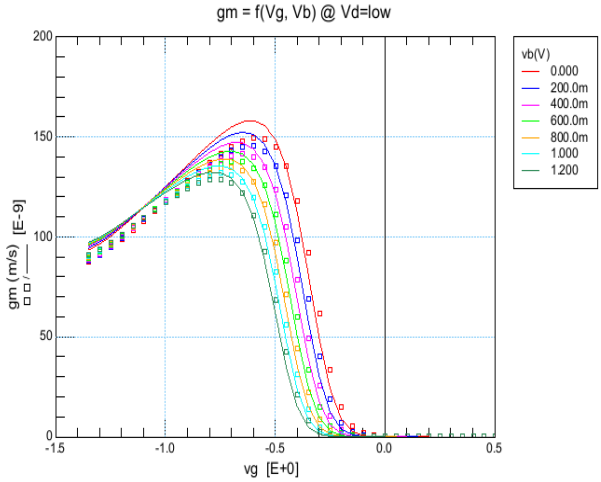


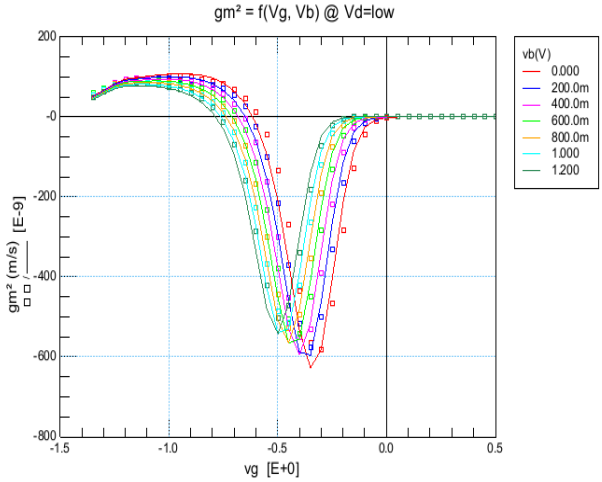


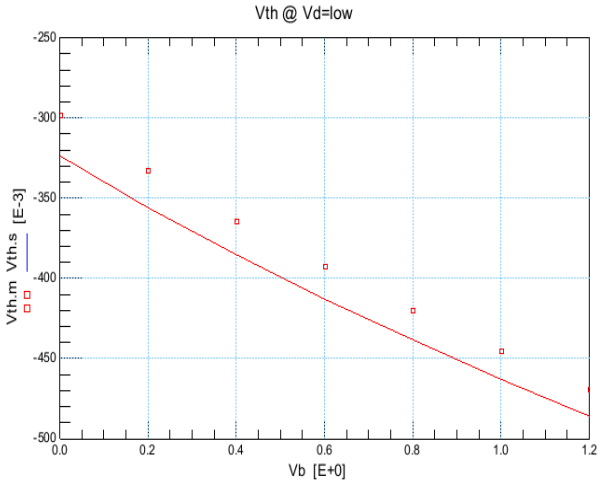
idvg, Vd = -0.05V, T = 70°C

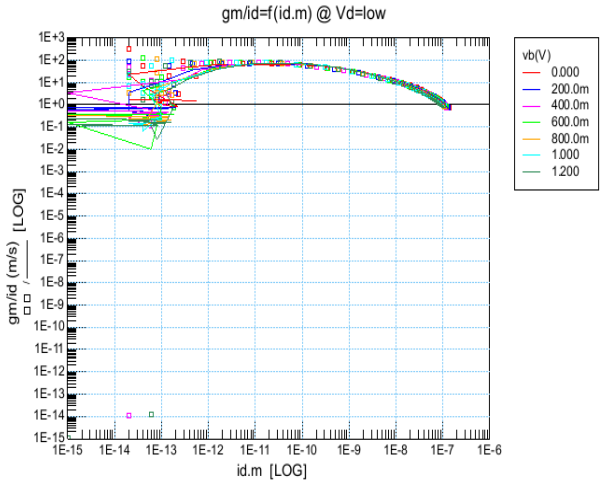


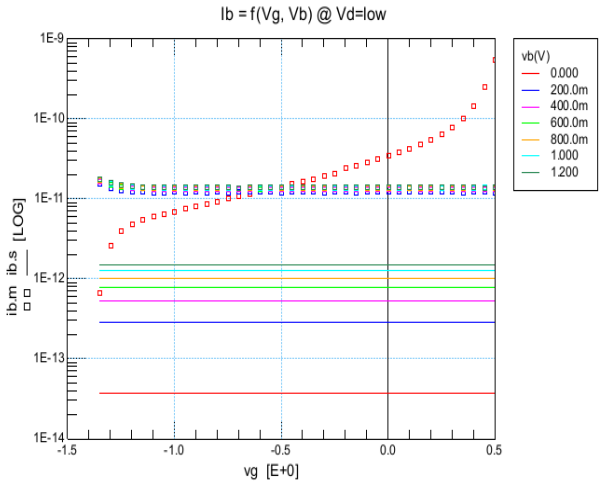


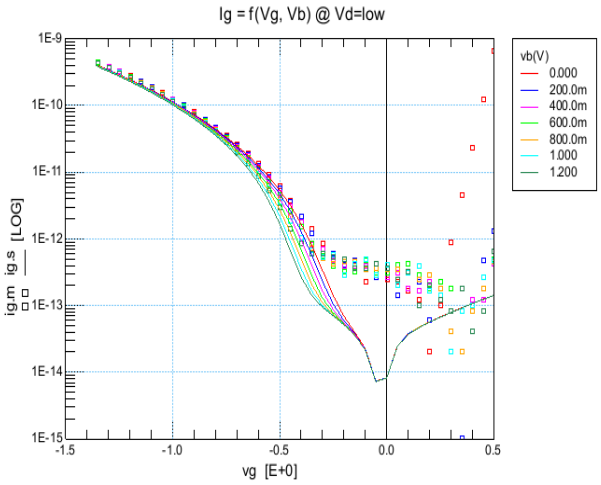




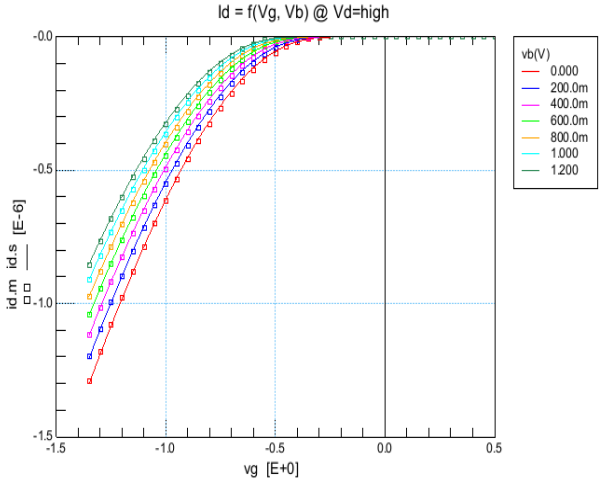


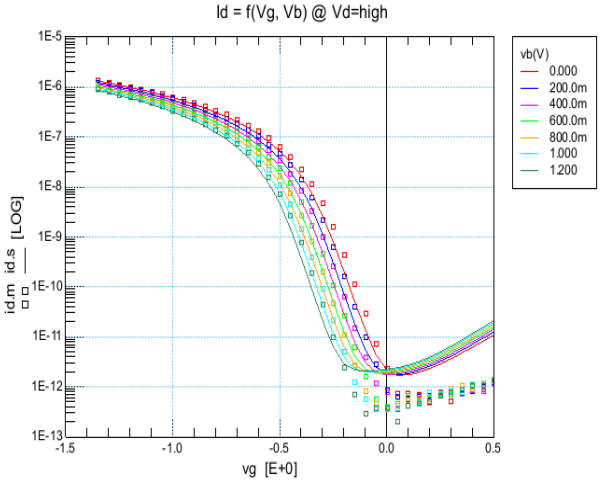


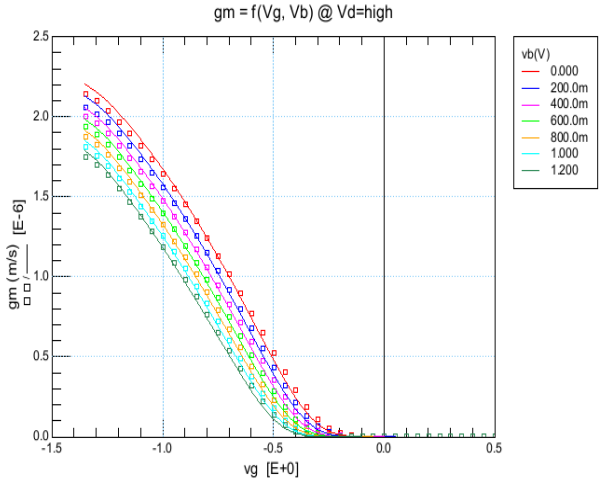


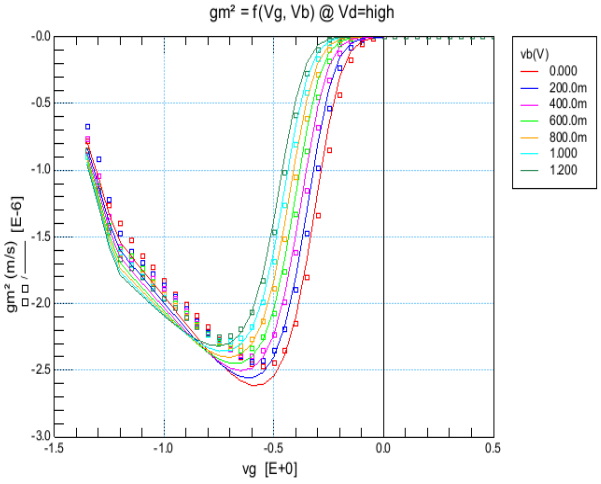


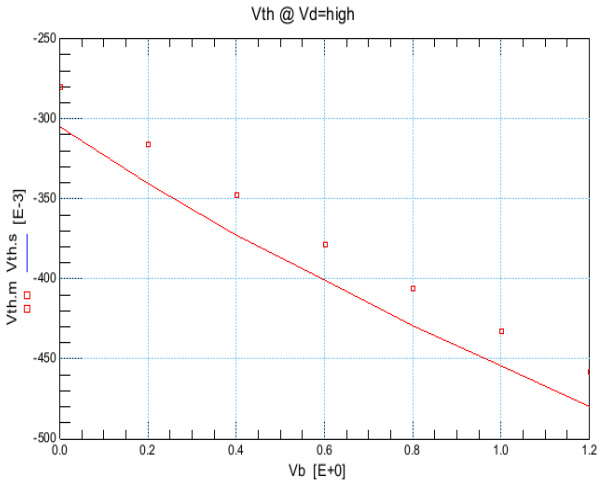
idvg, Vd = -1.2V, T = 70°C

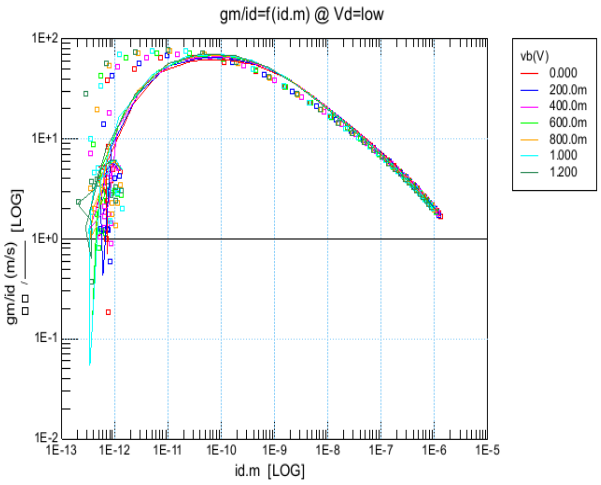


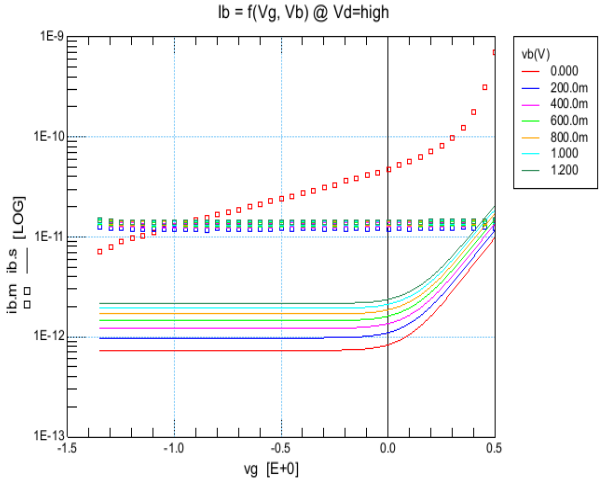


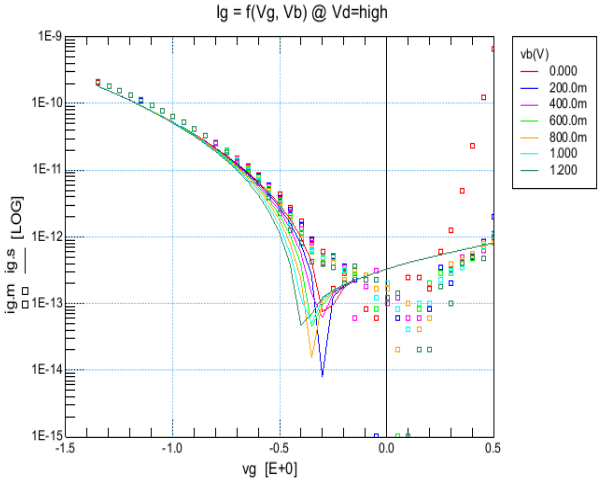




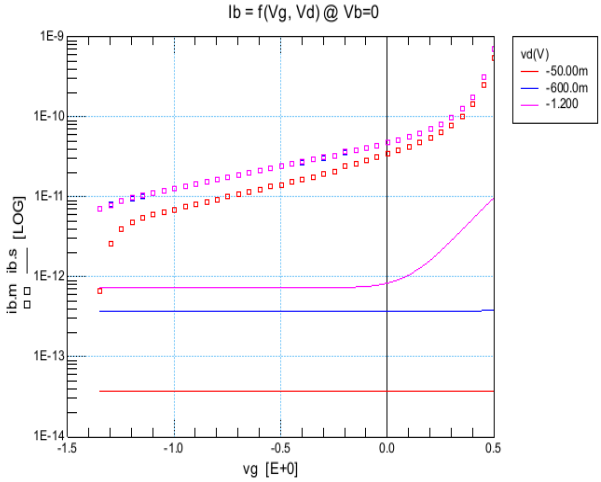


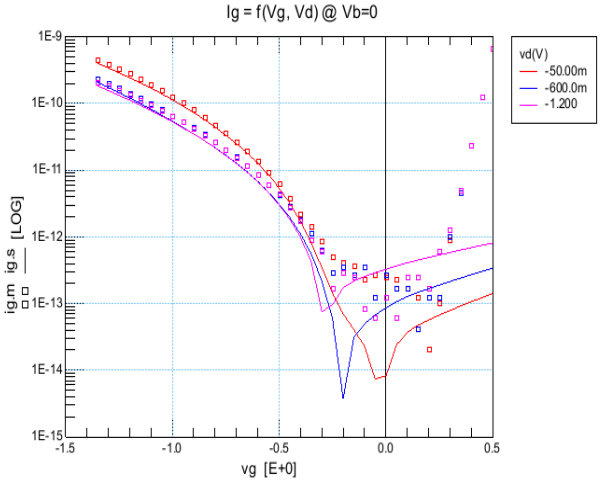




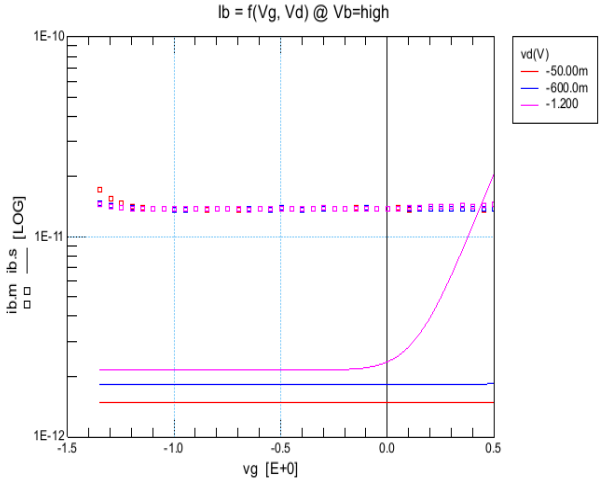


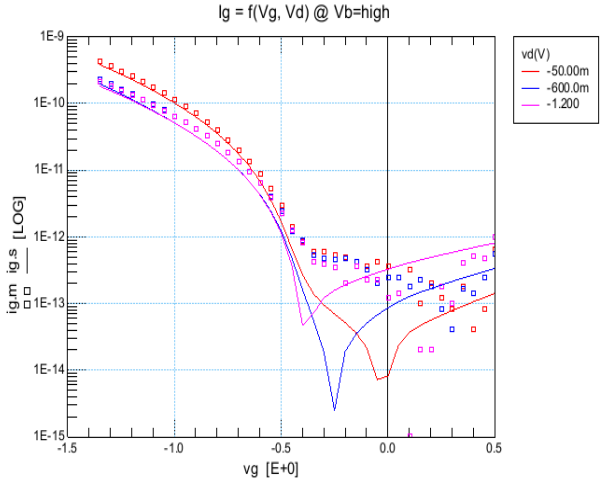
idvg, Ib,Ig, Vb = 0V, T = 70°C



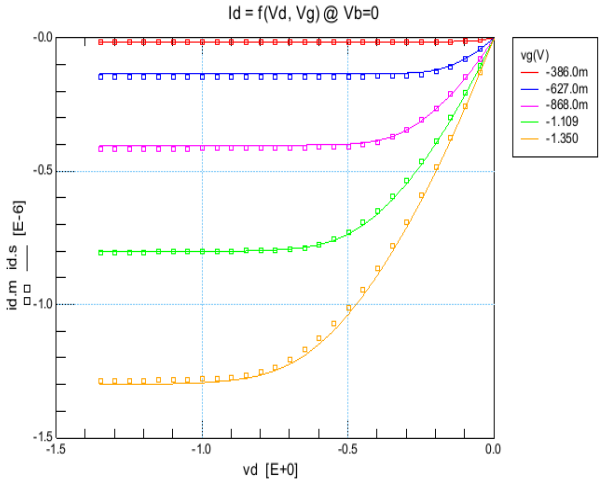


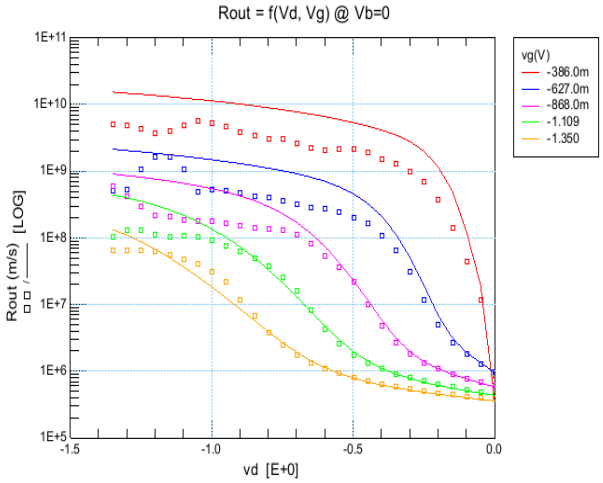
idvg, Ib,Ig, Vb = 1.2V, T = 70°C

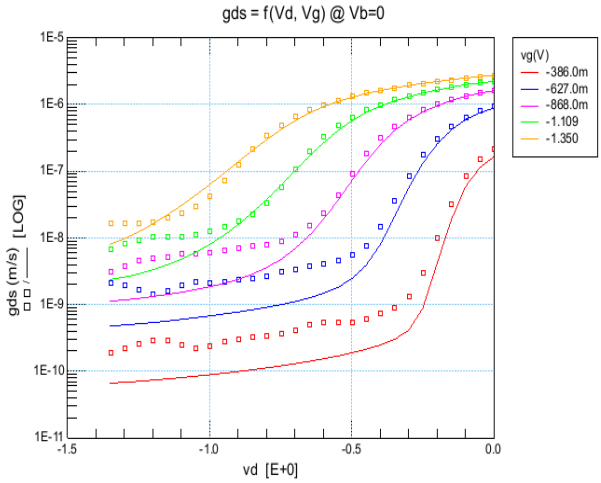




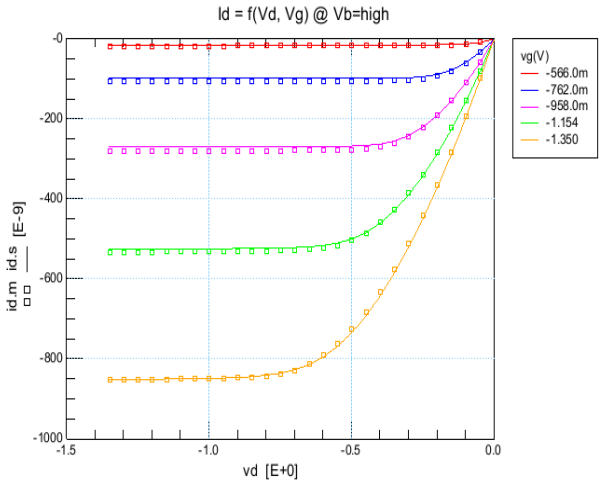
idvd, Vb = 0V, T = 70°C

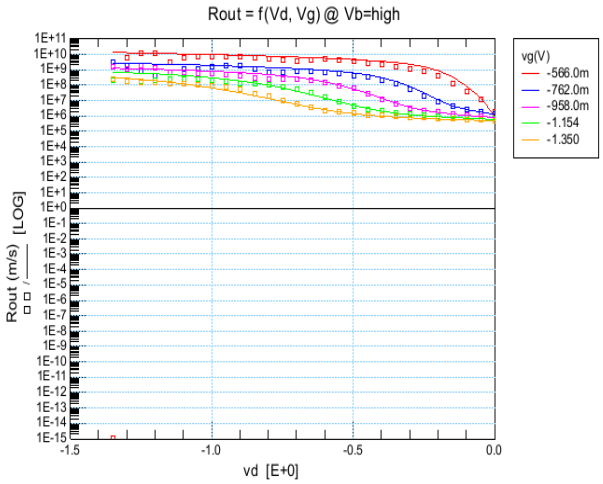


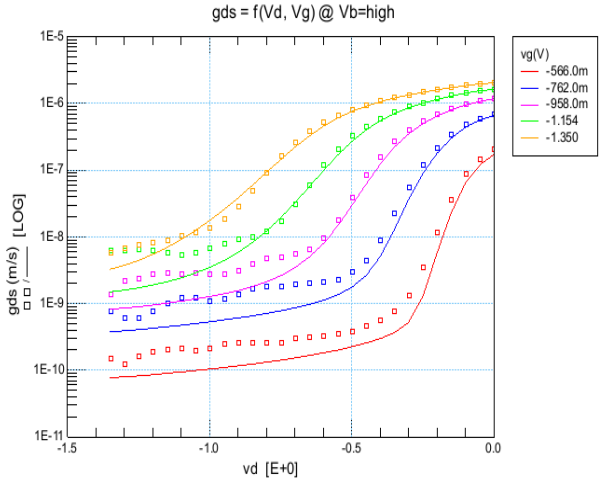




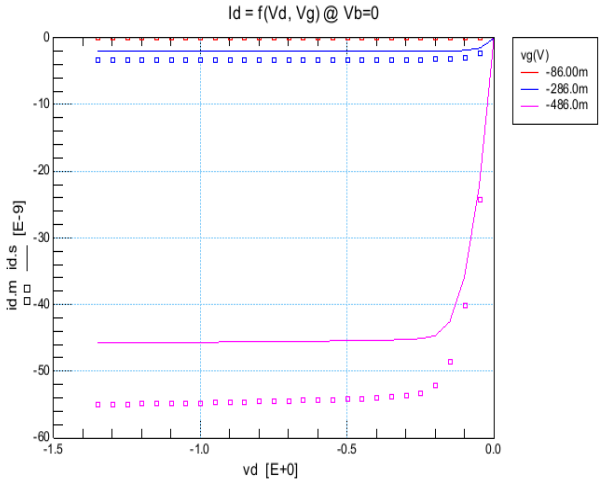
idvd_vbmin, Vb = 1.2V, T = 70°C

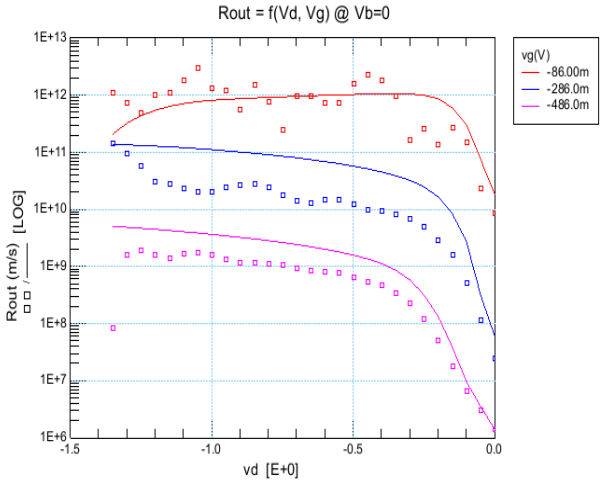


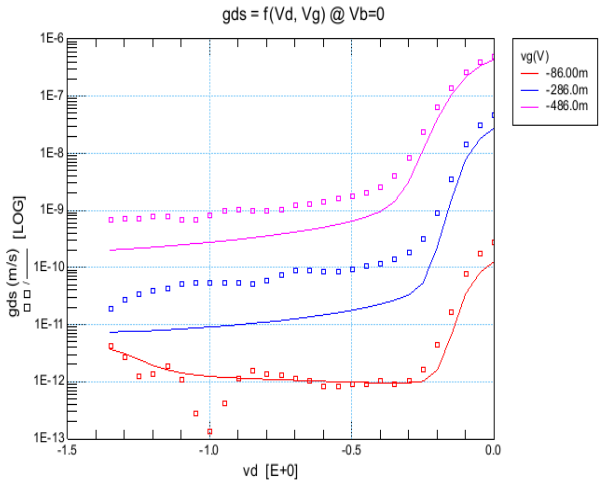




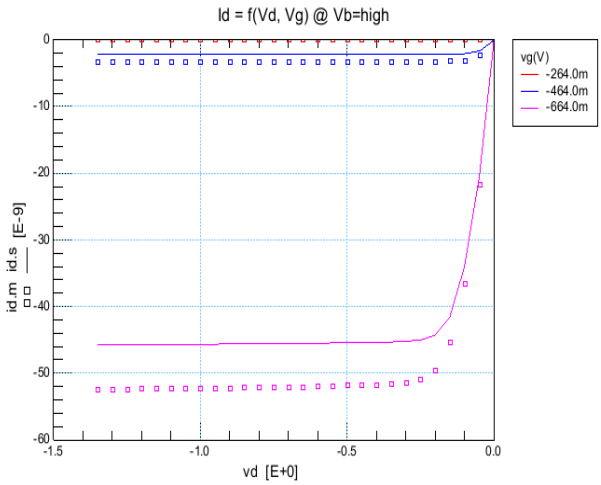
idvd_vth, Vb = 0V, T = 70°C

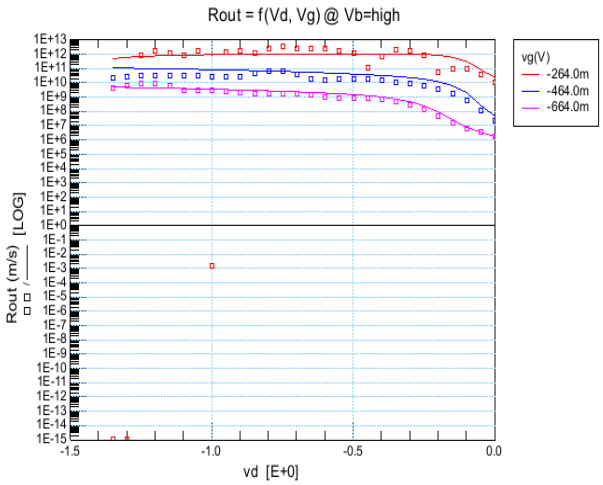


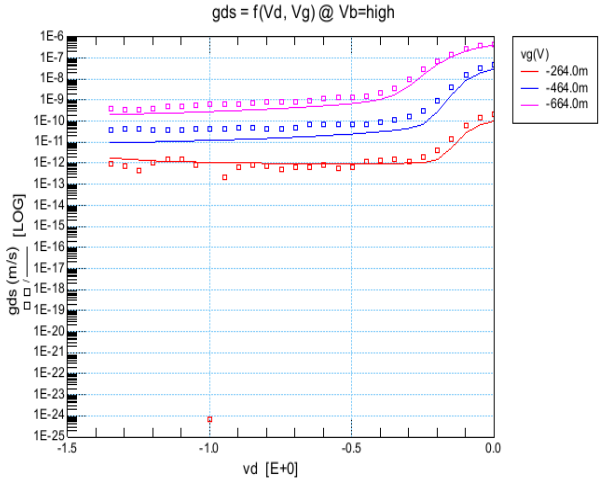




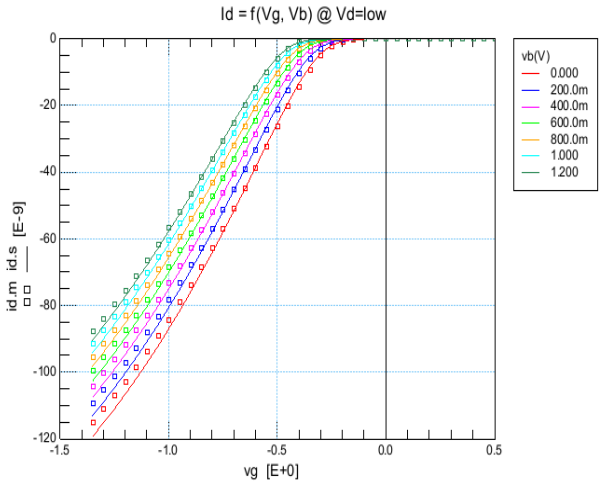
idvd_vth_vbmin, Vb = 1.2V, T = 70°C

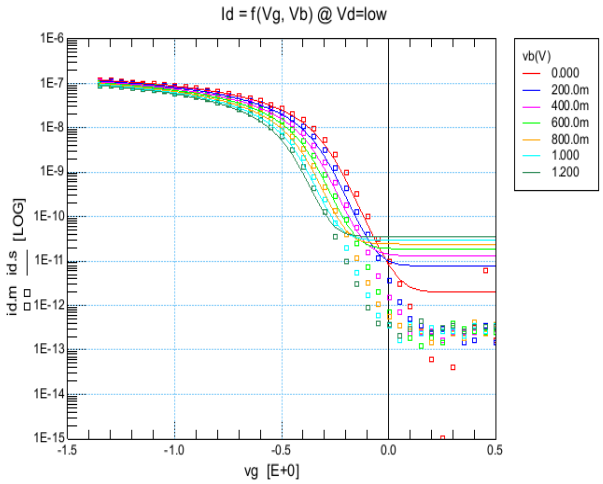


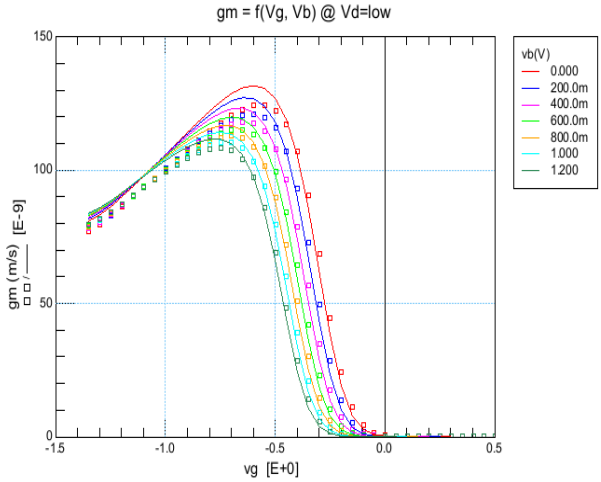


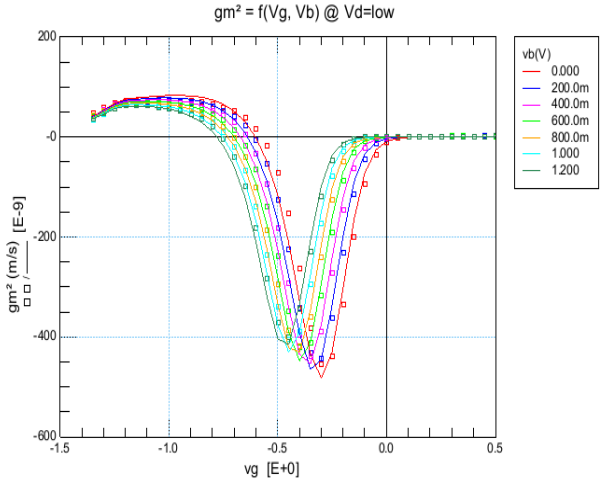


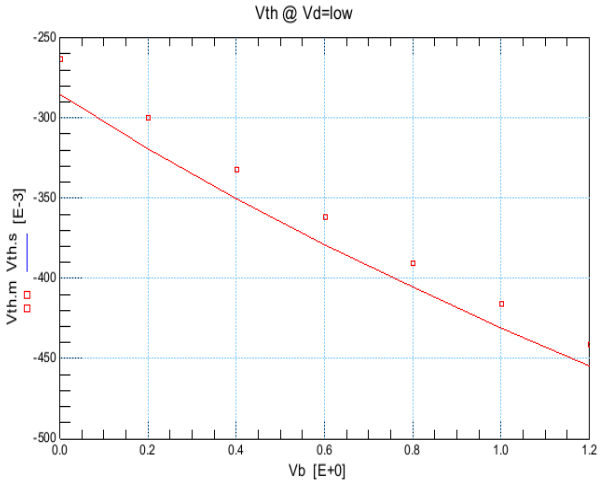
idvg, Vd = -0.05V, T = 125°C

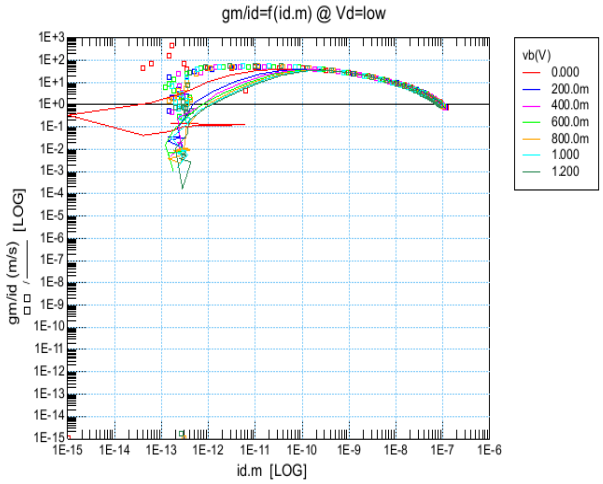


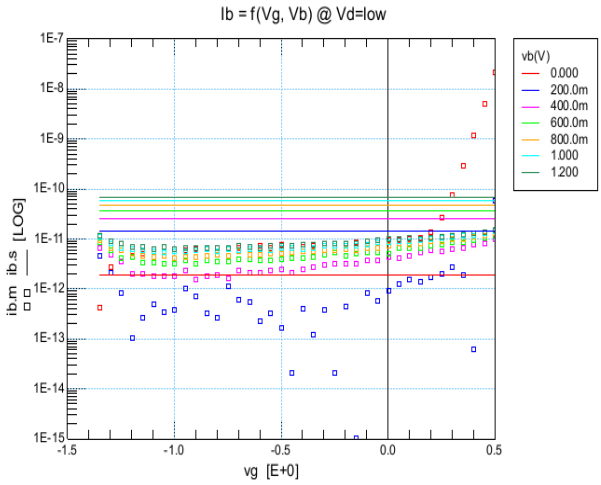


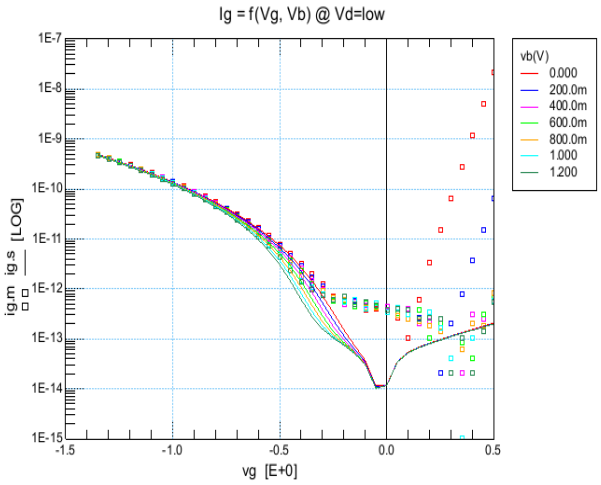




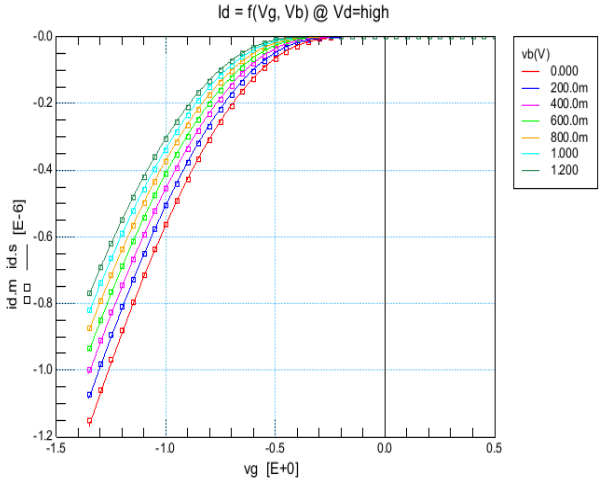


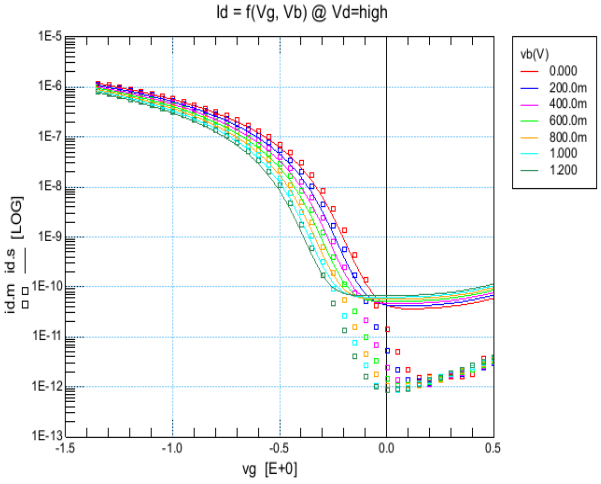


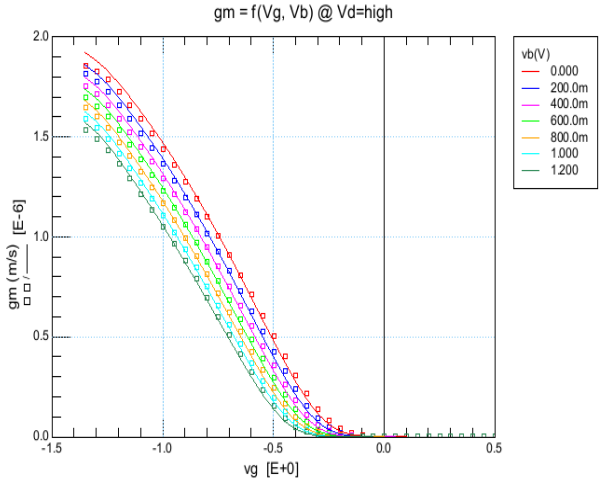


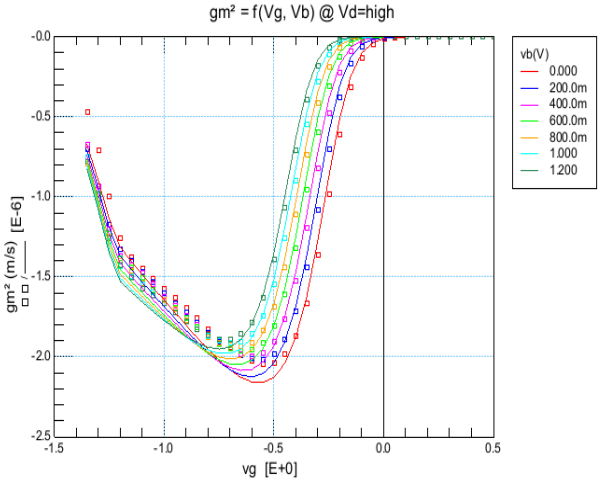


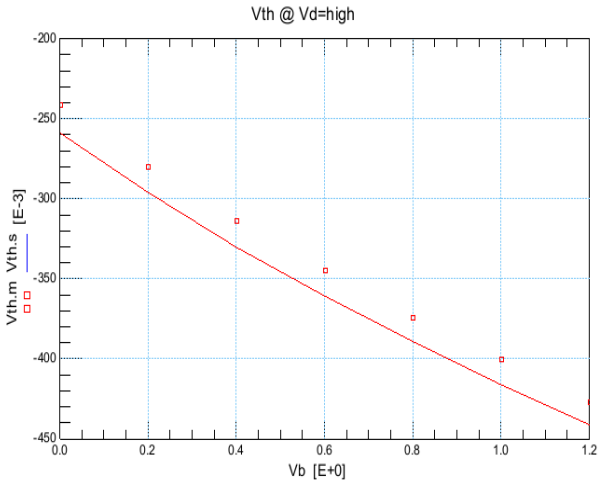
idvg, Vd = -1.2V, T = 125°C

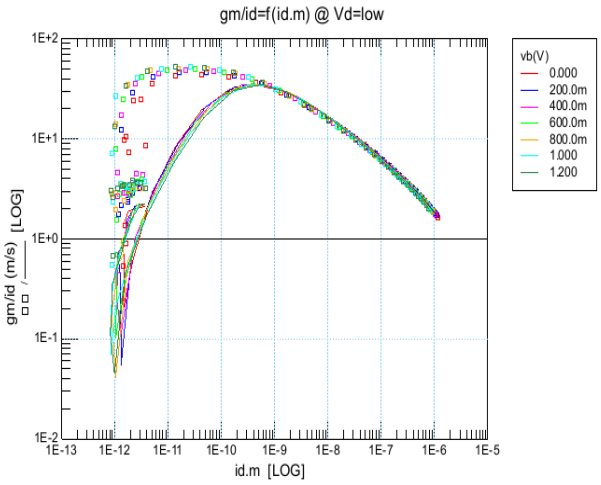


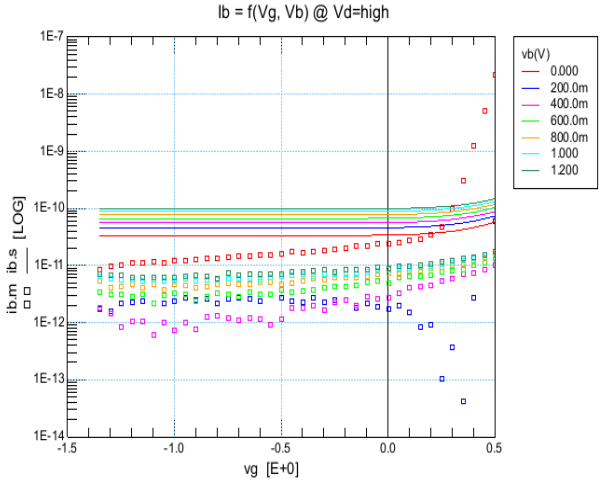


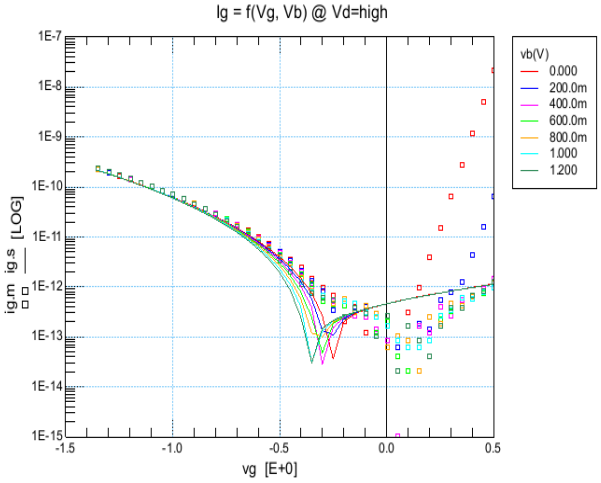




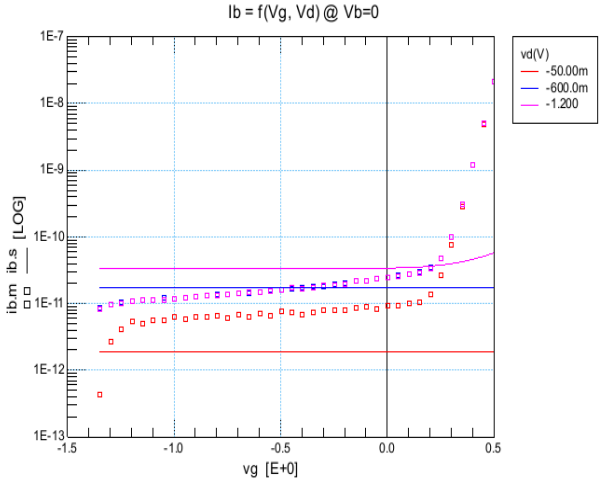


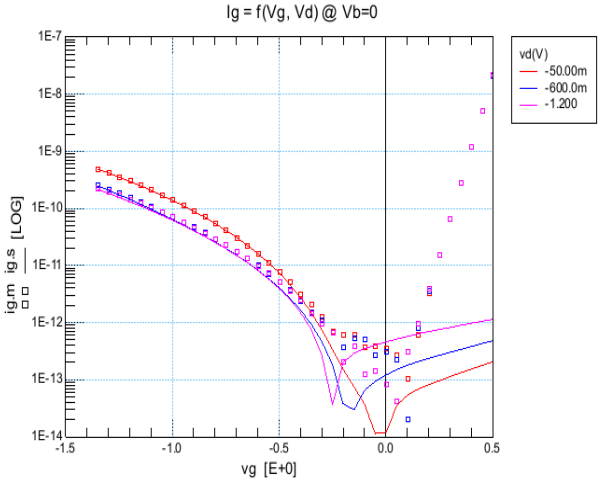




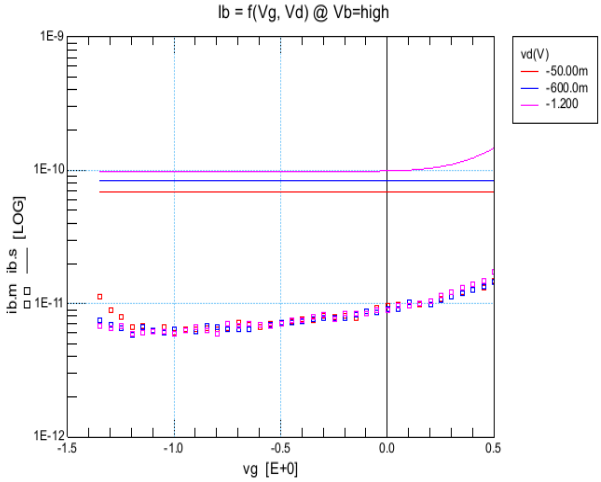


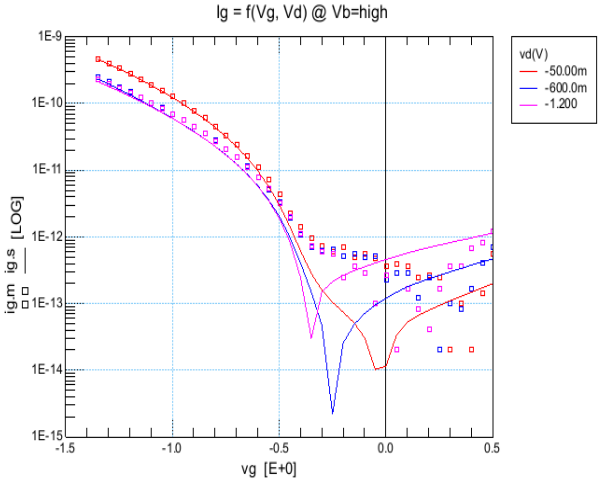
idvg, Ib,Ig, Vb = 0V, T = 125°C



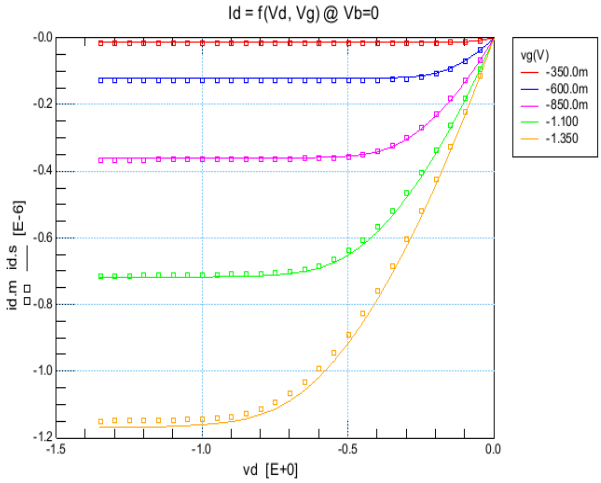


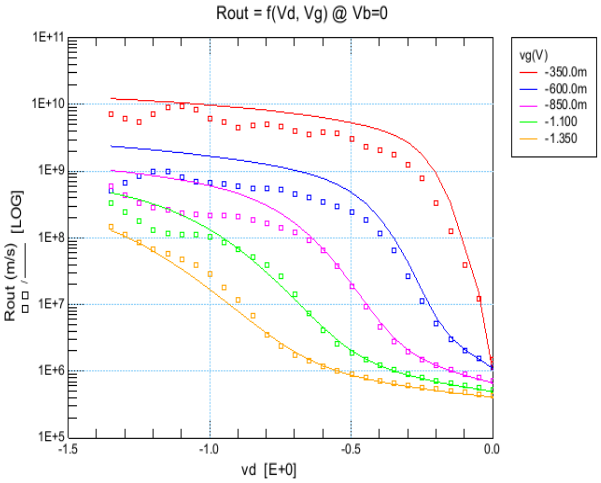
idvg, Ib,Ig, Vb = 1.2V, T = 125°C

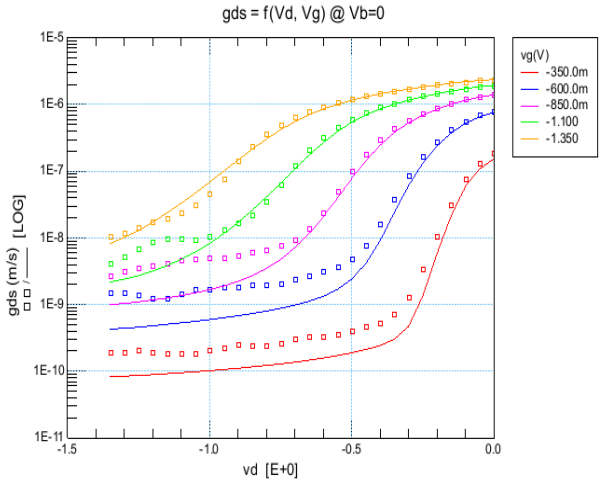




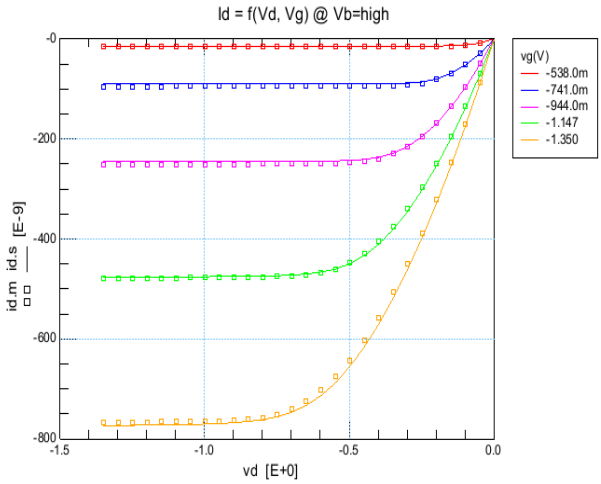
idvd, Vb = 0V, T = 125°C

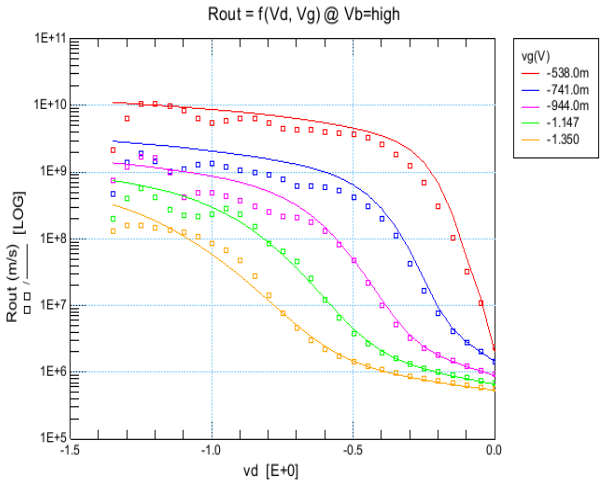


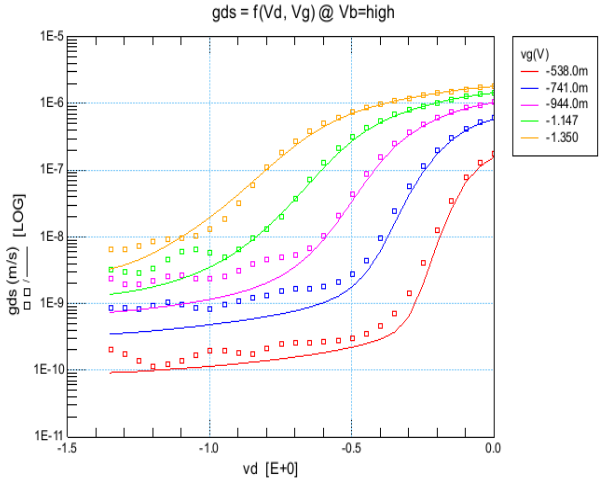




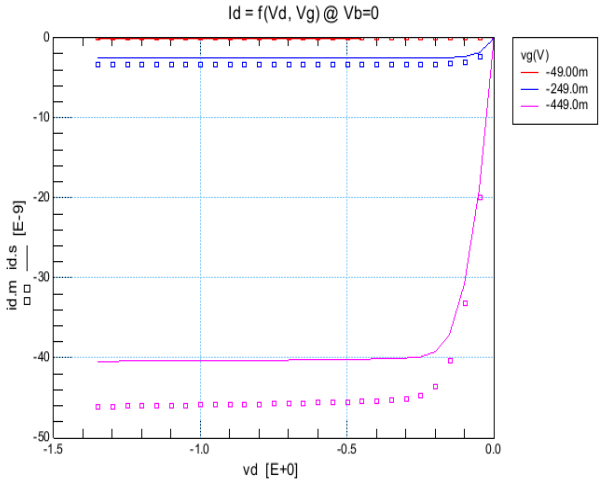
idvd_vbmin, Vb = 1.2V, T = 125°C

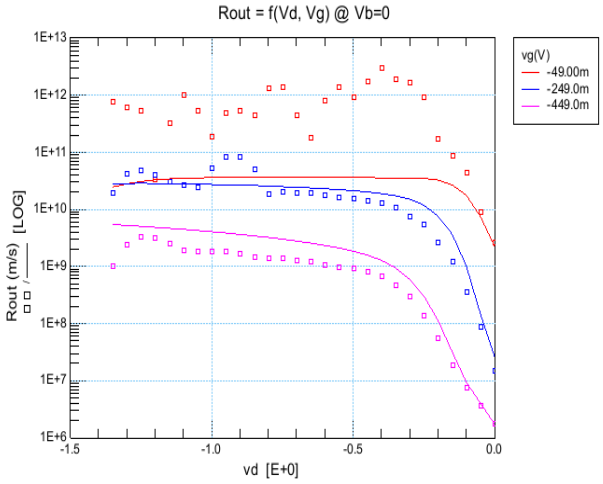


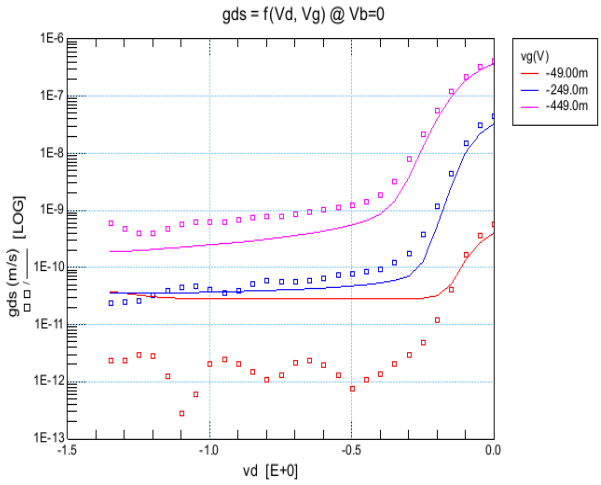




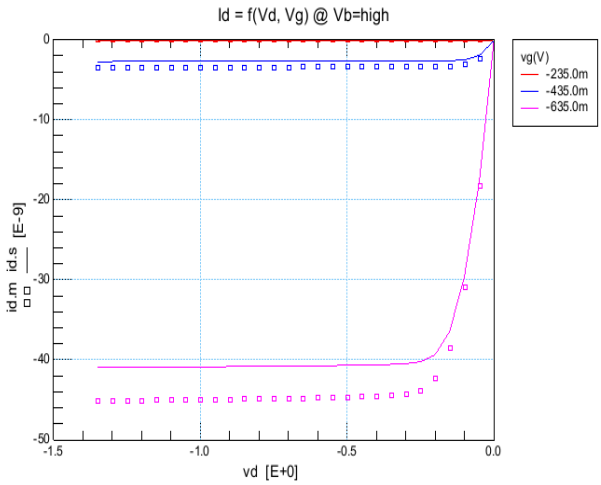
idvd_vth, Vb = 0V, T = 125°C

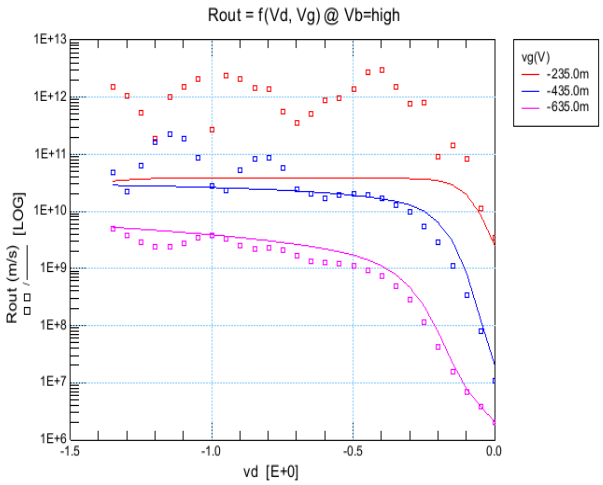


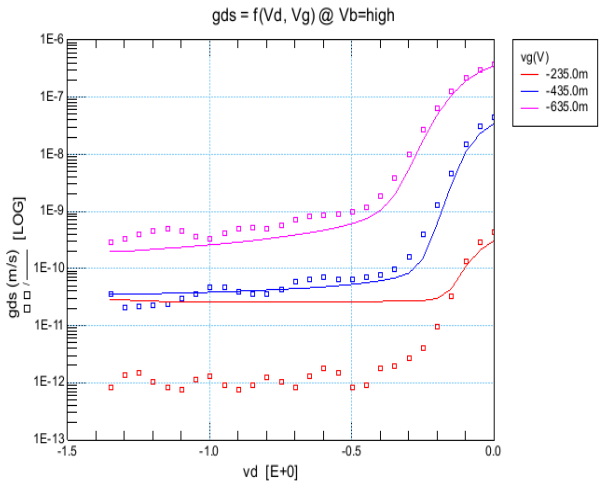




idvd_vth_vbmin, Vb = 1.2V, T = 125°C

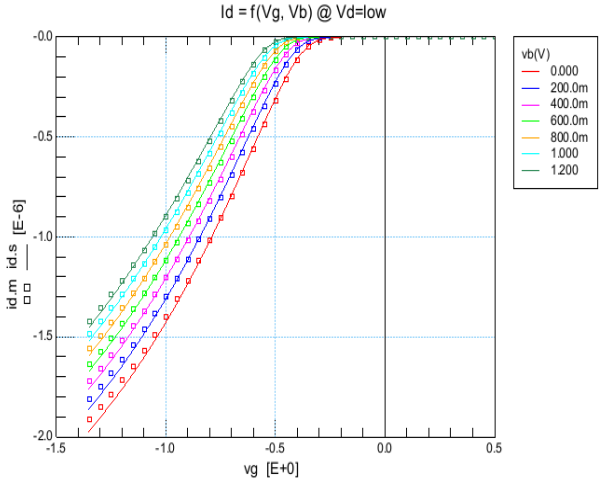


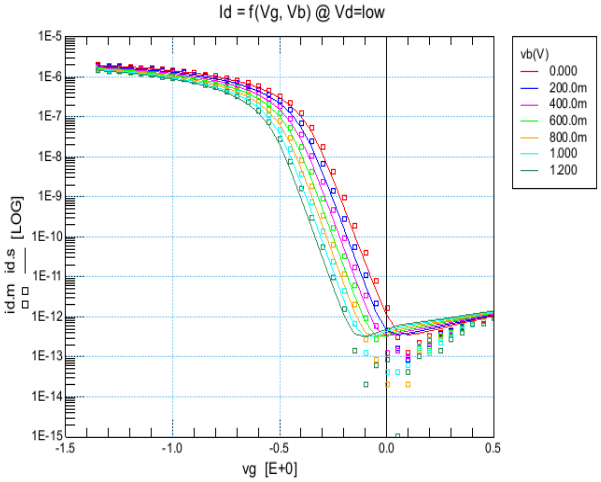


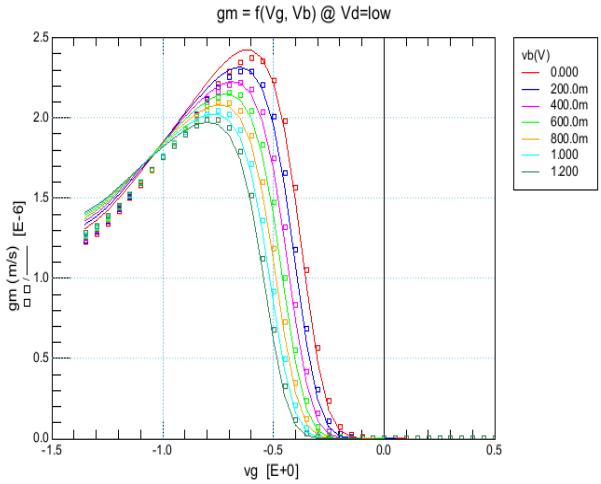


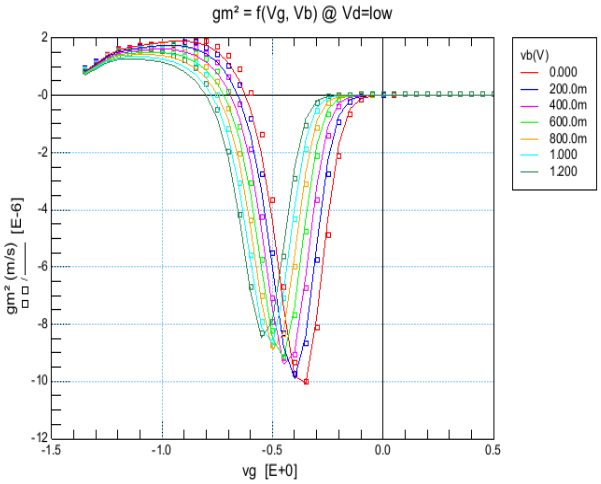
4.16W05u0_L10u0_S550_3

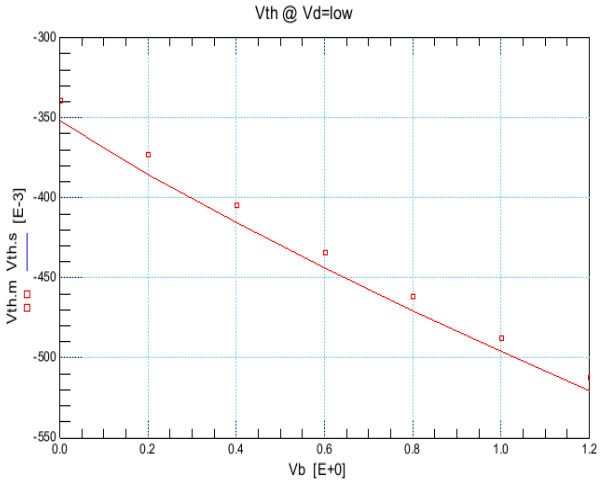
idvg, Vd = -0.05V, T = 27°C

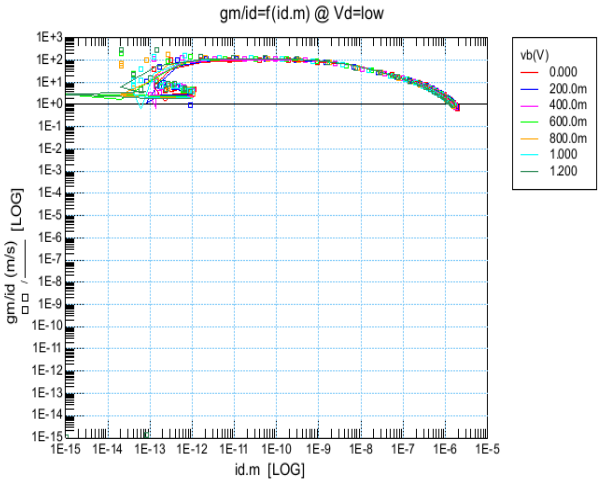


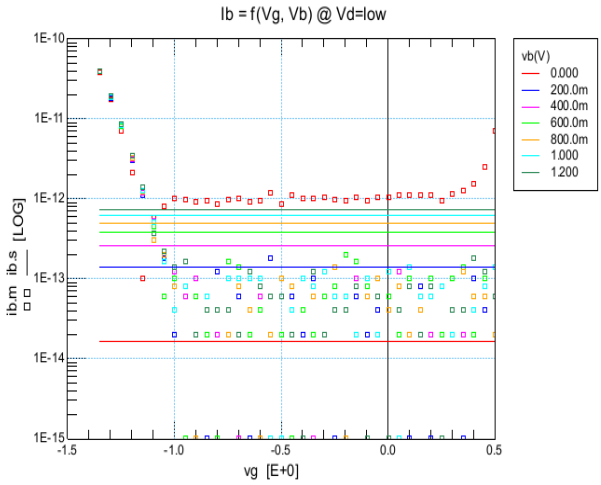


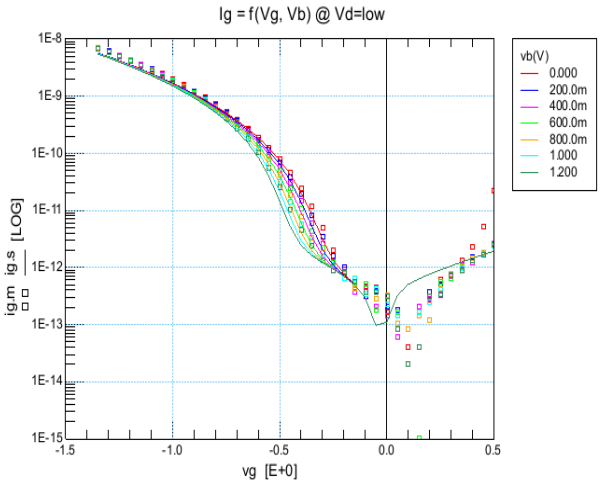




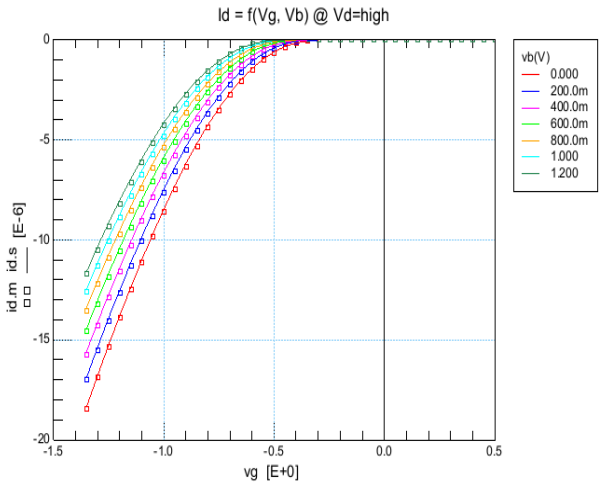


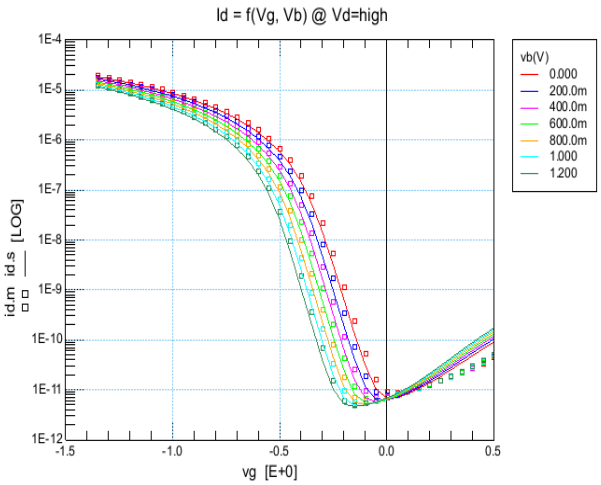


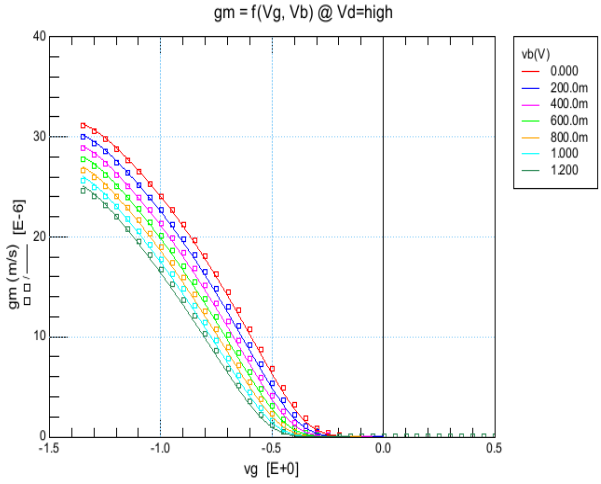


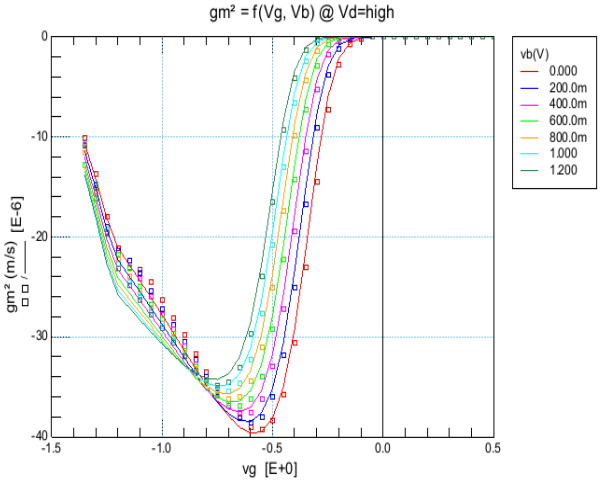


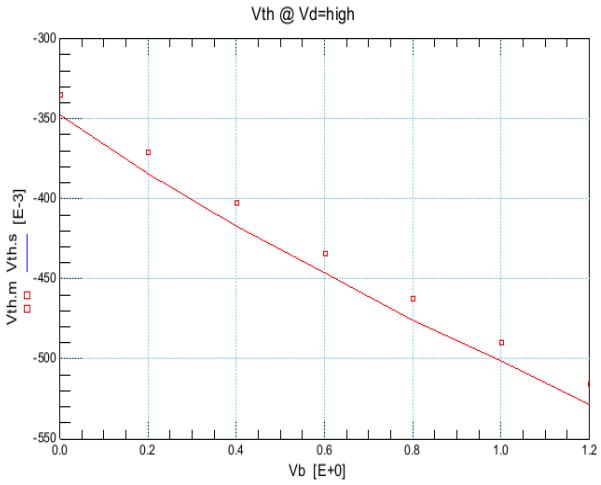
idvg, Vd = -1.2V, T = 27°C

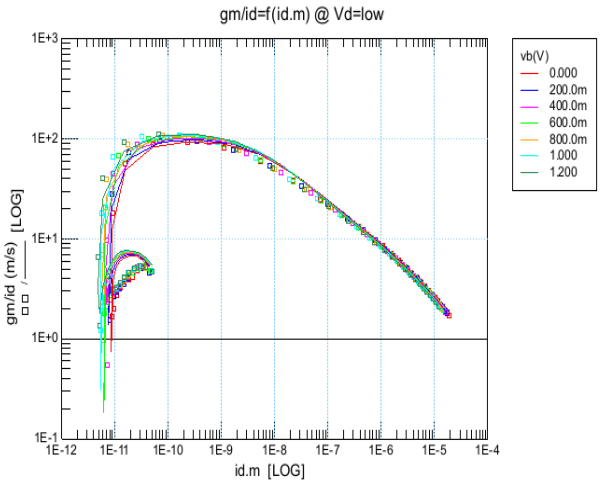


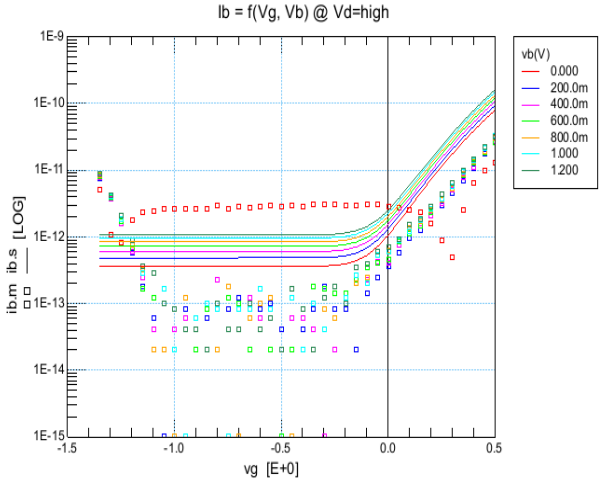


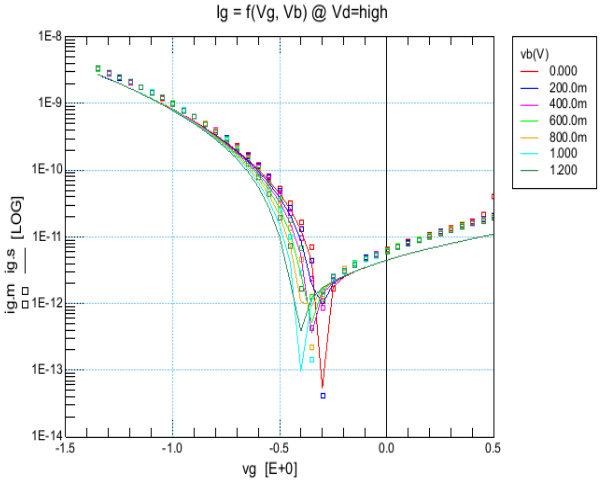




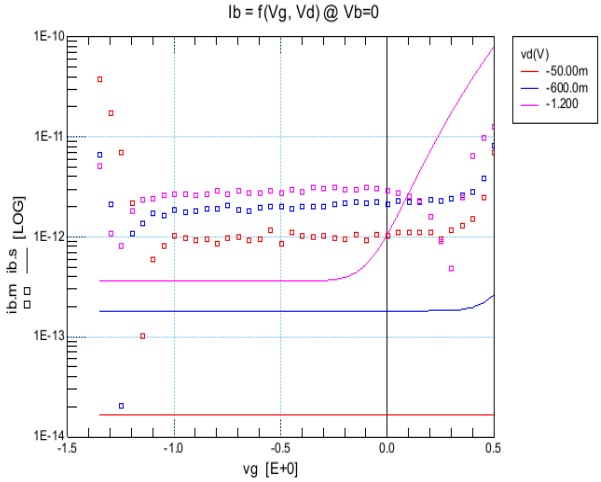


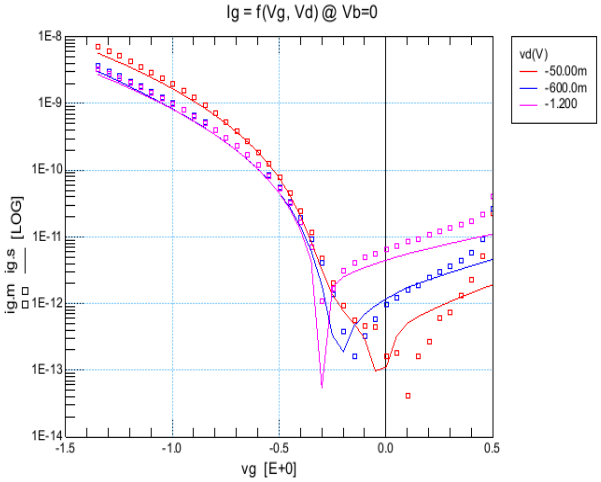




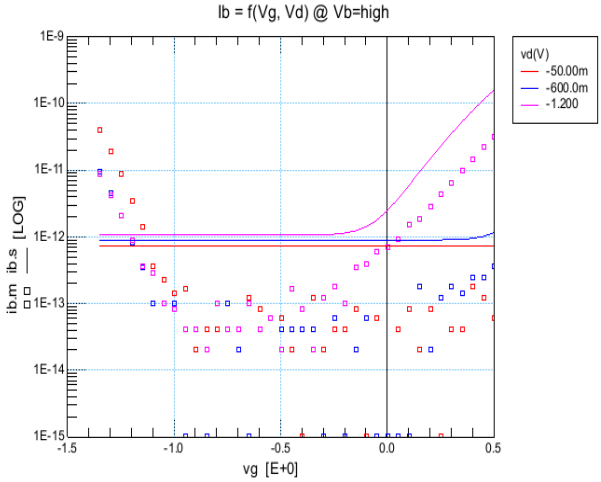


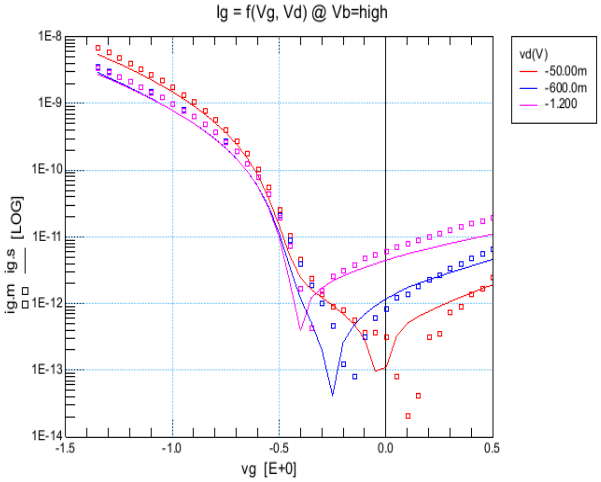
idvg, Ib,Ig, Vb = 0V, T = 27°C



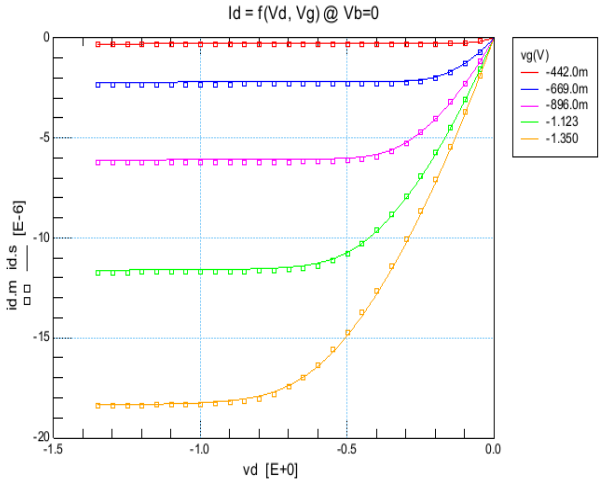


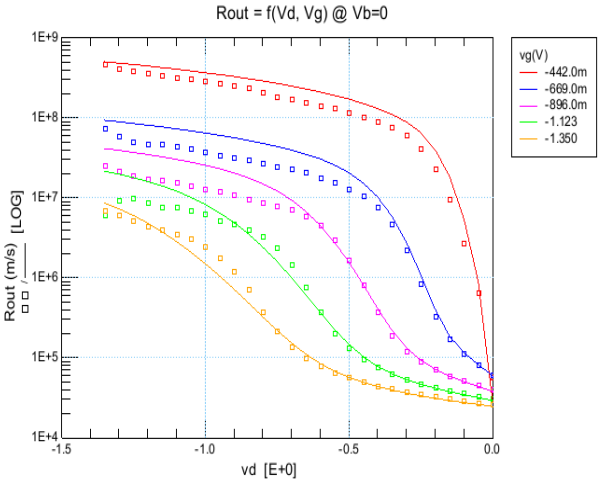
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

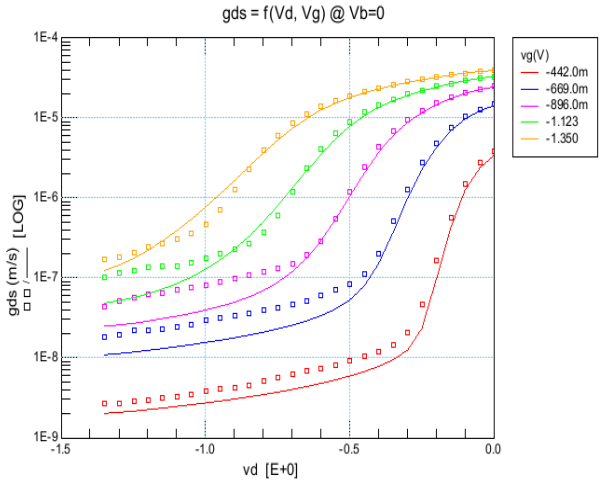




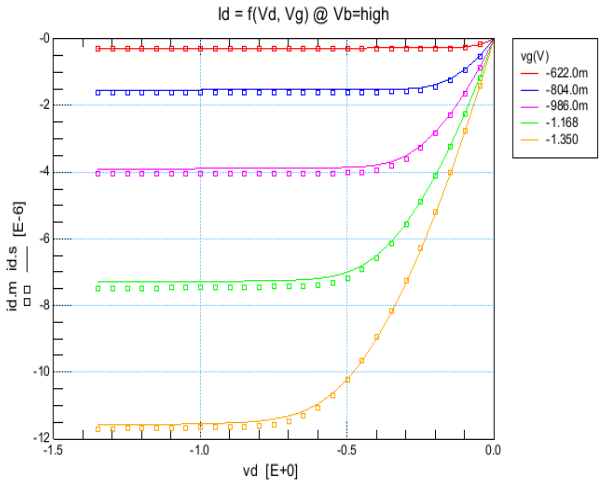
idvd, Vb = 0V, T = 27°C

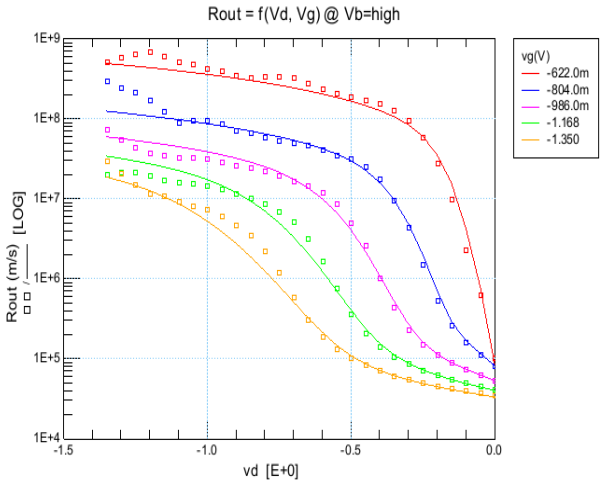


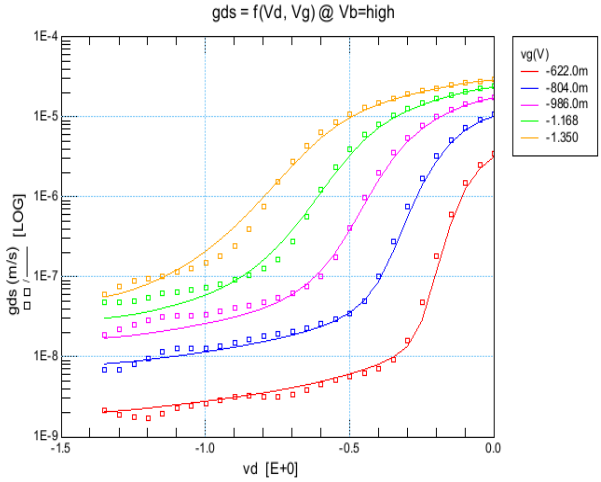




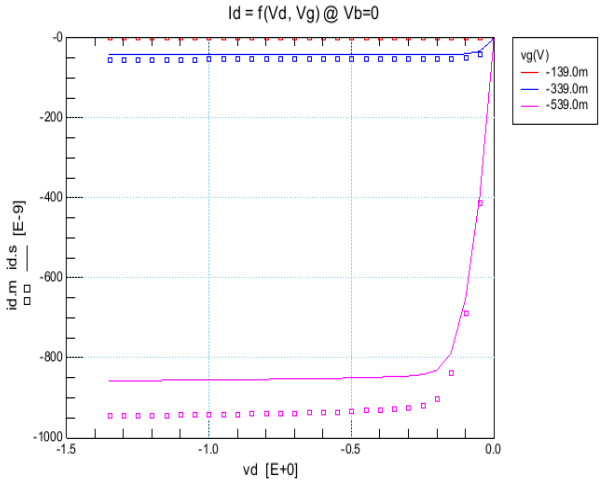
idvd_vbmin, Vb = 1.2V, T = 27°C

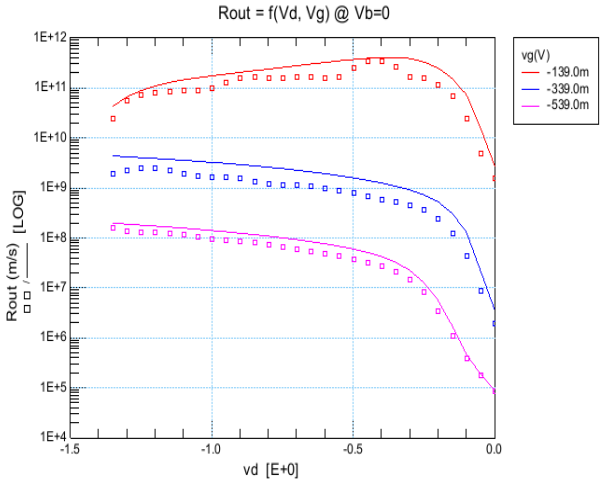


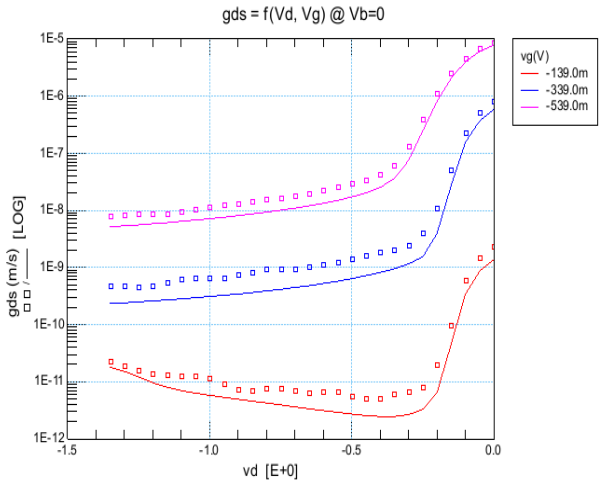




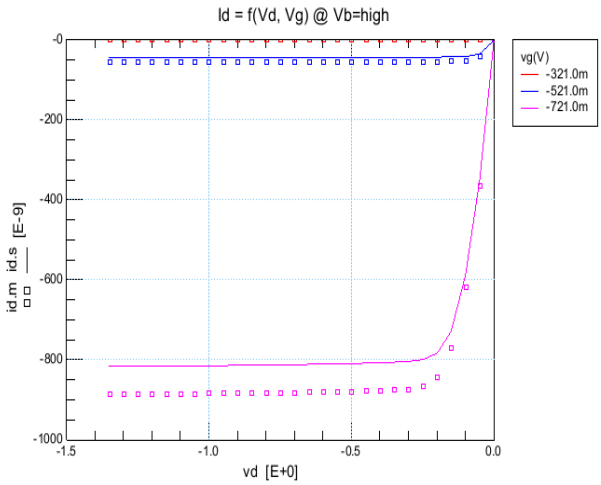
idvd_vth, Vb = 0V, T = 27°C

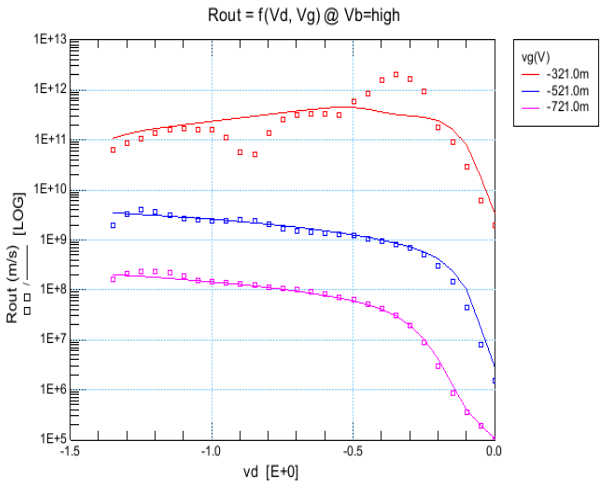


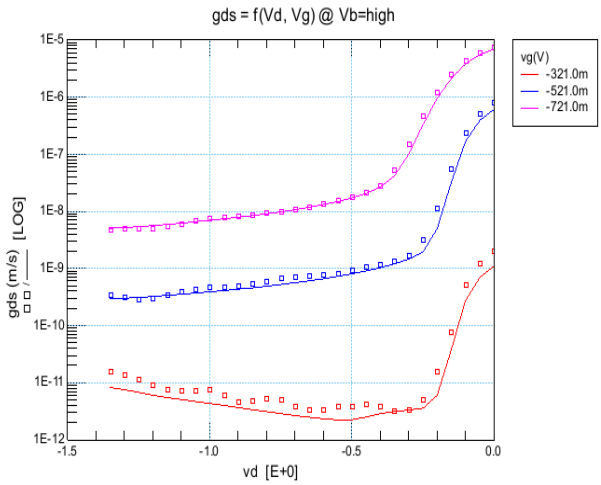




idvd_vth_vbmin, Vb = 1.2V, T = 27°C



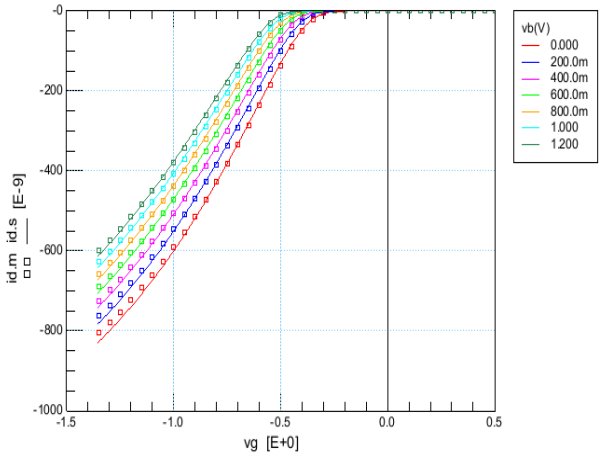




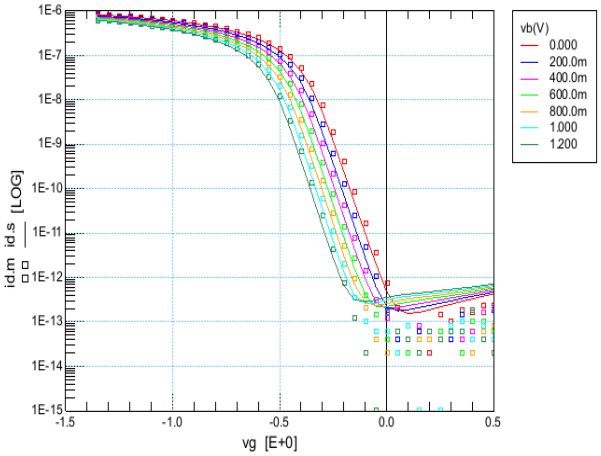
4.17W02u0_L10u0_S551_1

idvg, Vd = -0.05V, T = 27°C

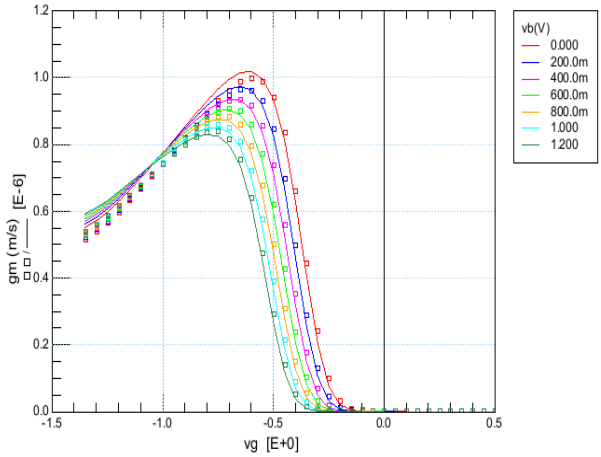
Id = f(Vg, Vb) @ Vd=low



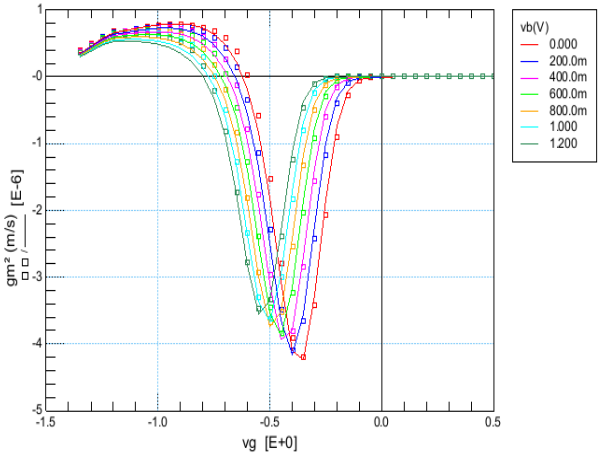
Id = f(Vg, Vb) @ Vd=low



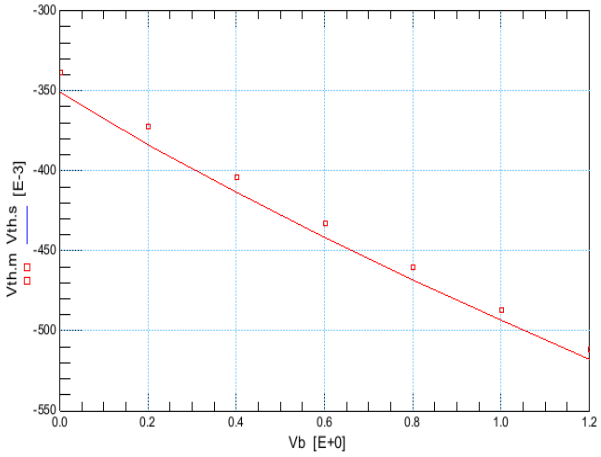
gm = f(Vg, Vb) @ Vd=low



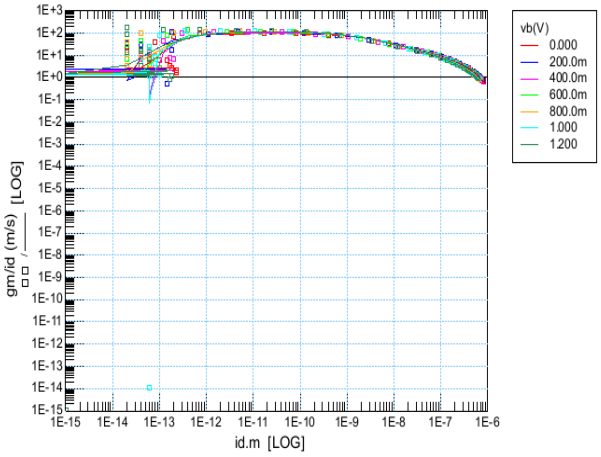
gm² = f(Vg, Vb) @ Vd=low

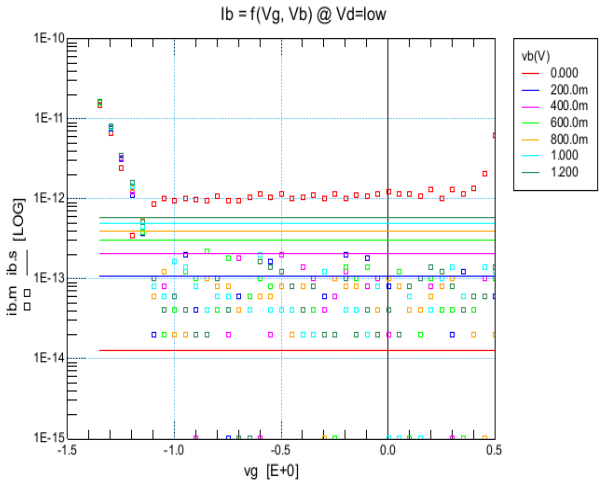


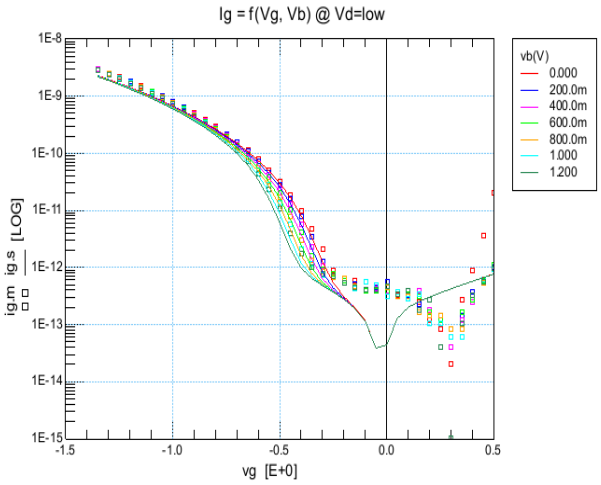
Vth @ Vd=low



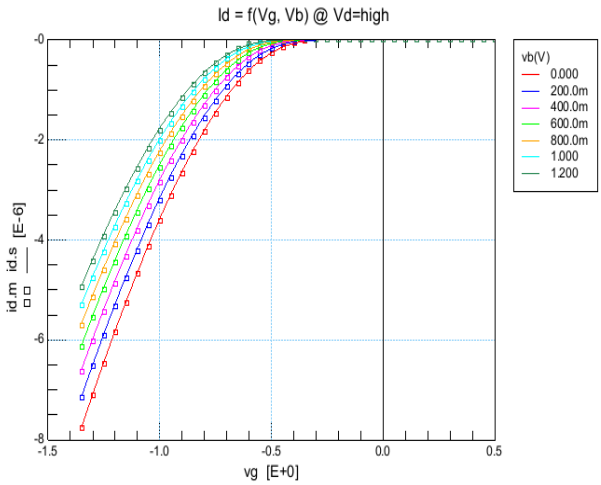
gm/id=f(id.m) @ Vd=low

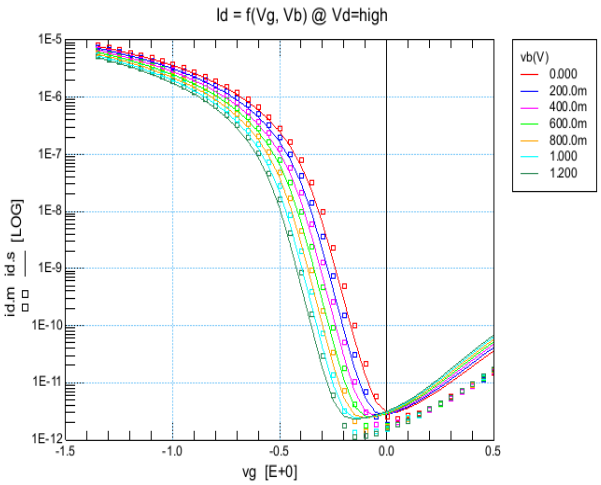


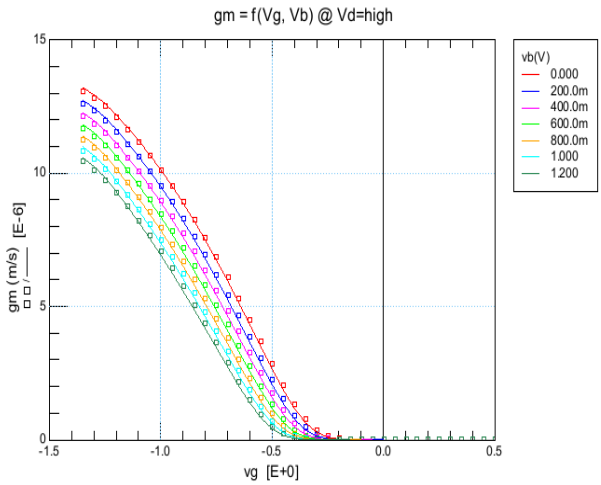


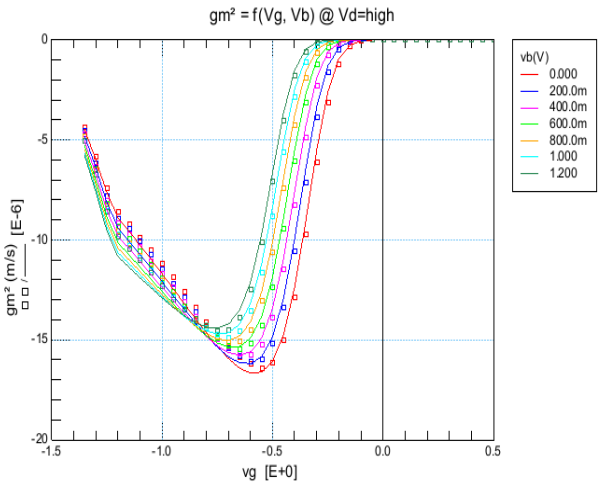


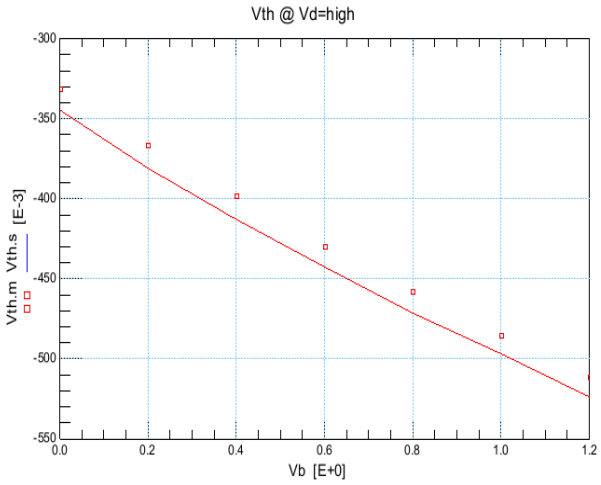
idvg, Vd = -1.2V, T = 27°C

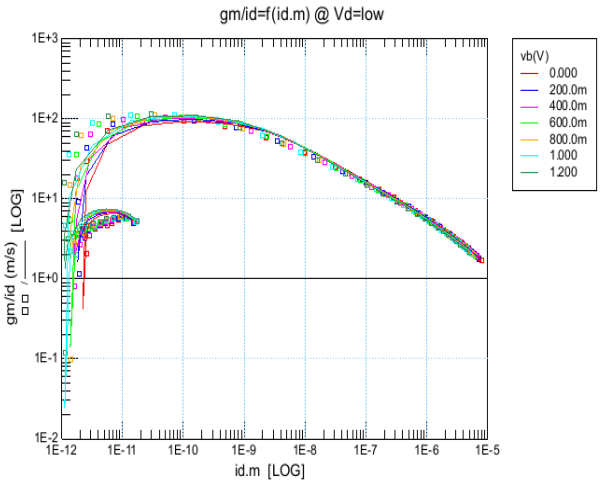


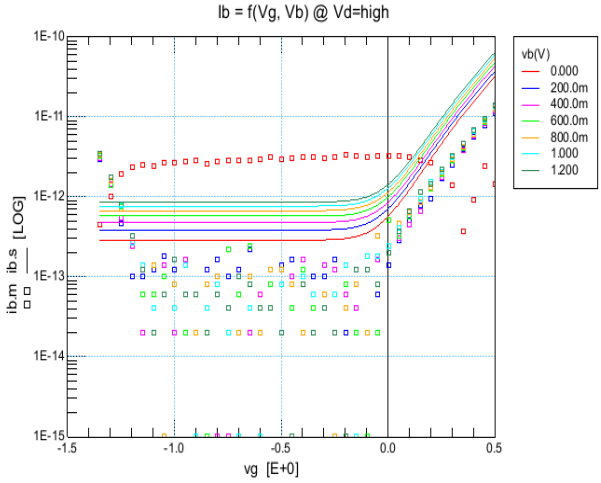


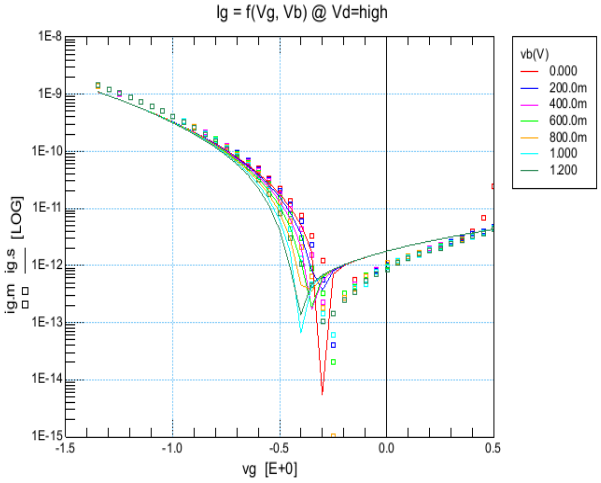




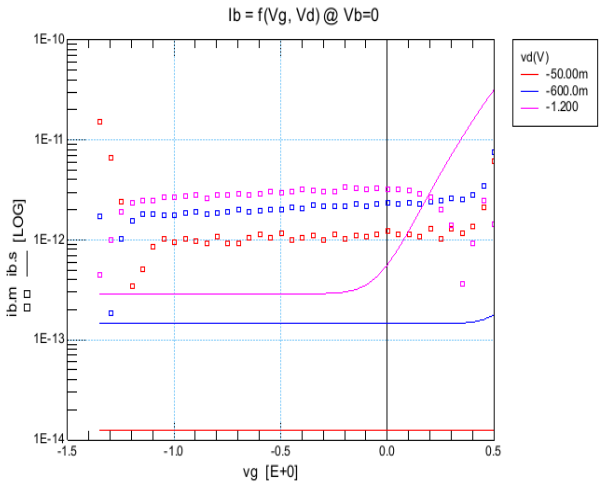


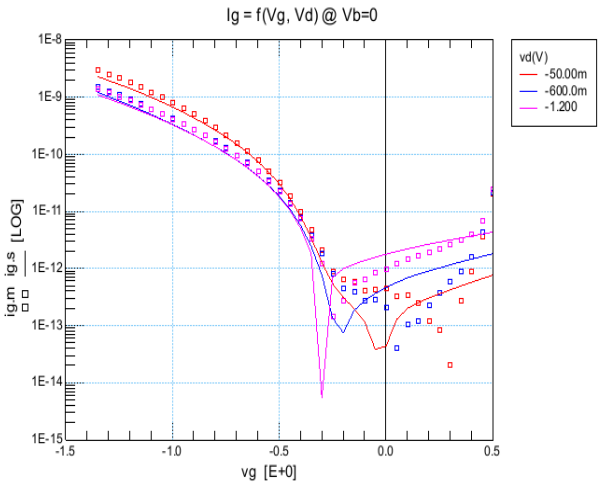




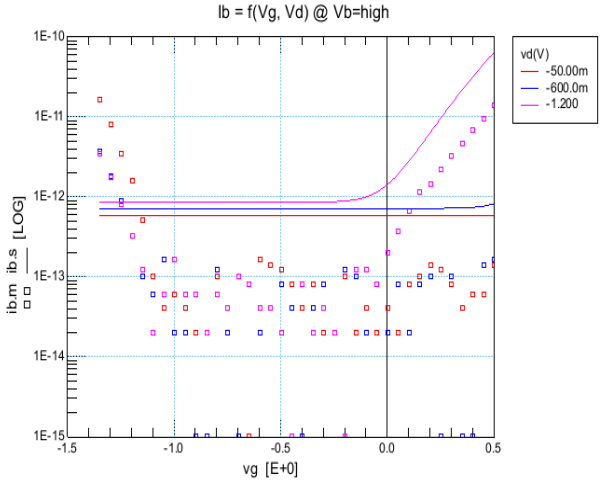


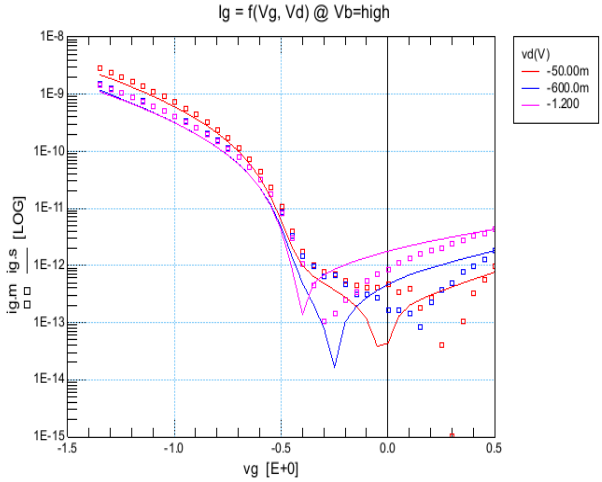
idvg, Ib,Ig, Vb = 0V, T = 27°C



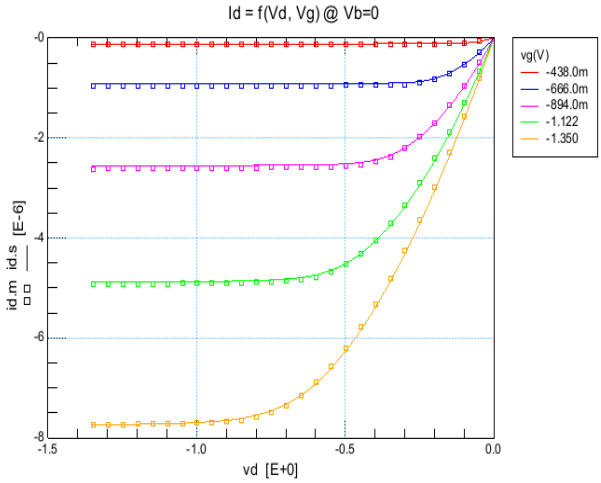


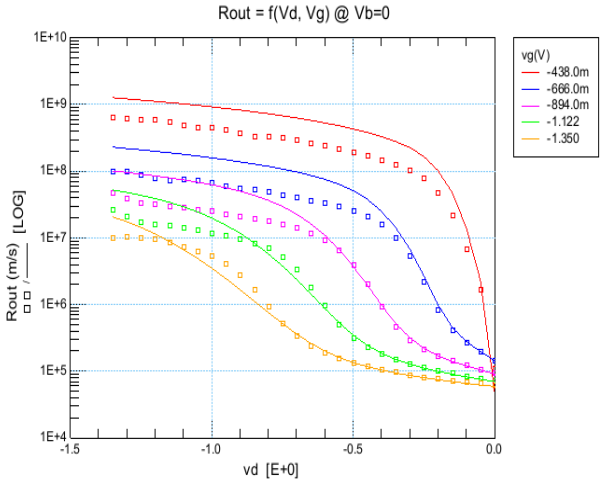
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

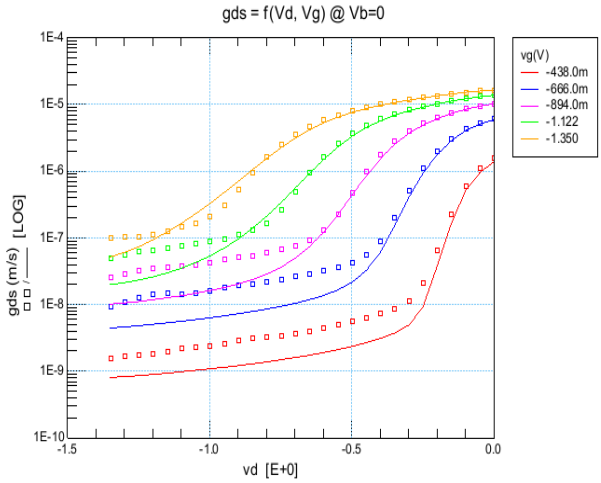




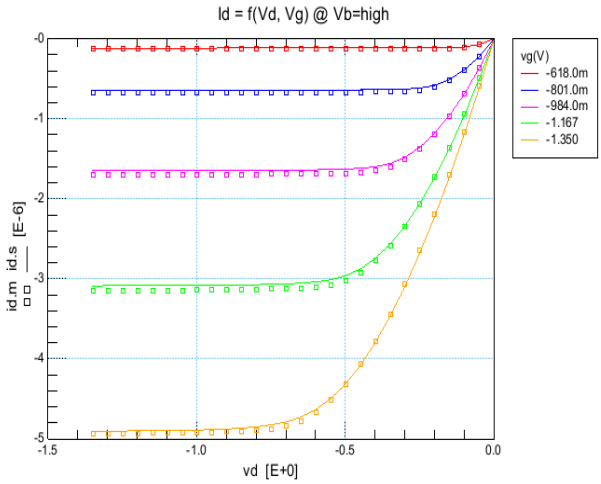
idvd, Vb = 0V, T = 27°C

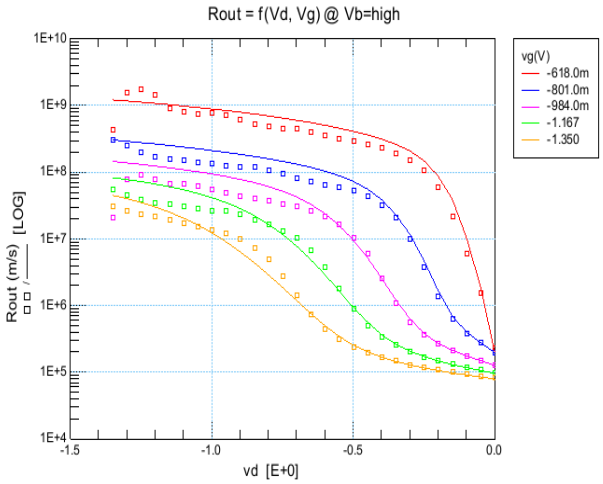


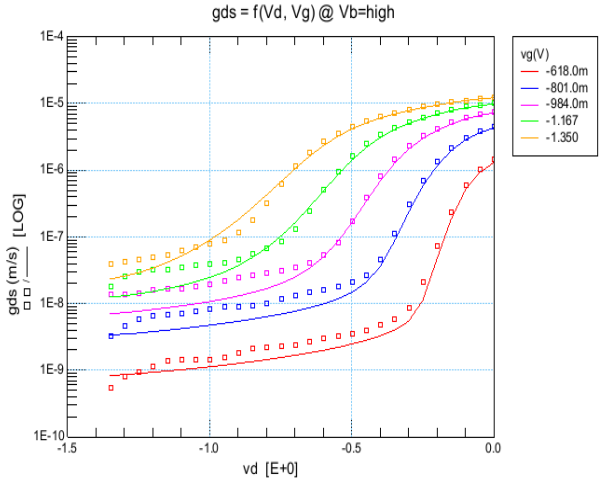




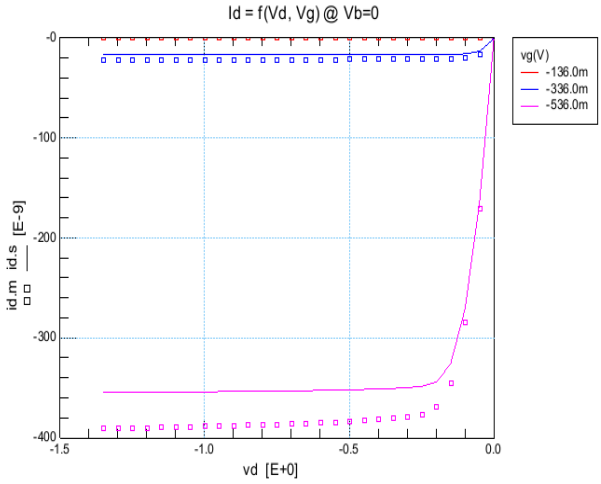
idvd_vbmin, Vb = 1.2V, T = 27°C

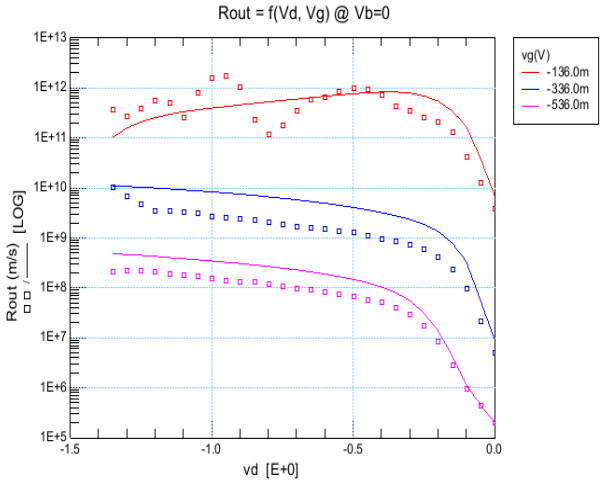


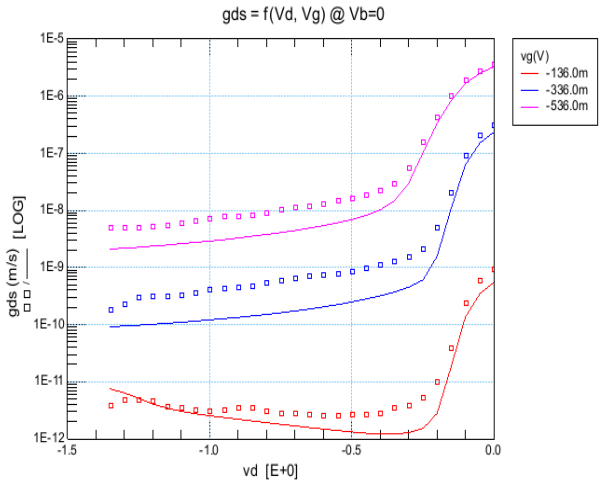




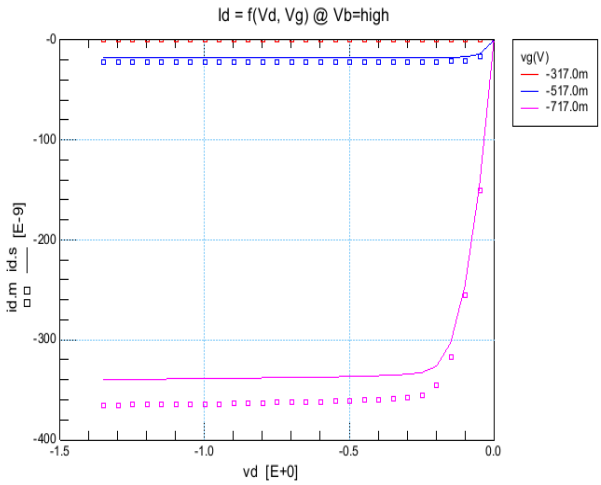
idvd_vth, Vb = 0V, T = 27°C

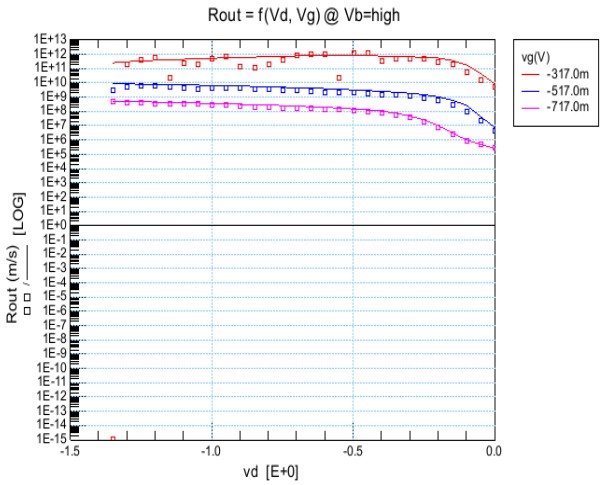


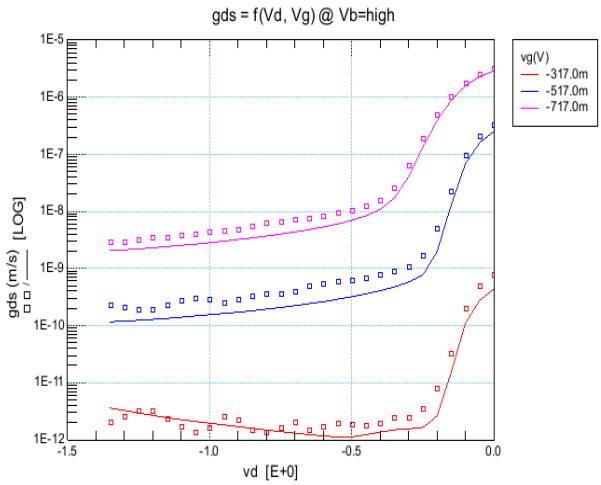




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

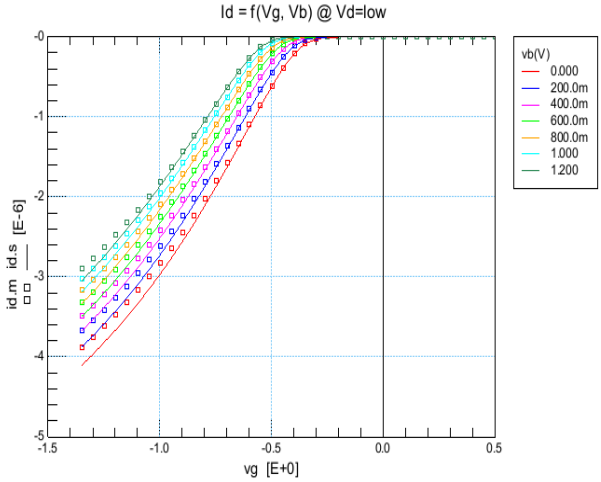


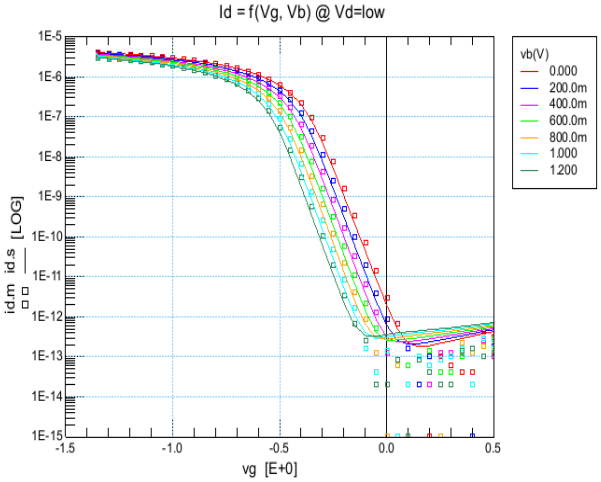


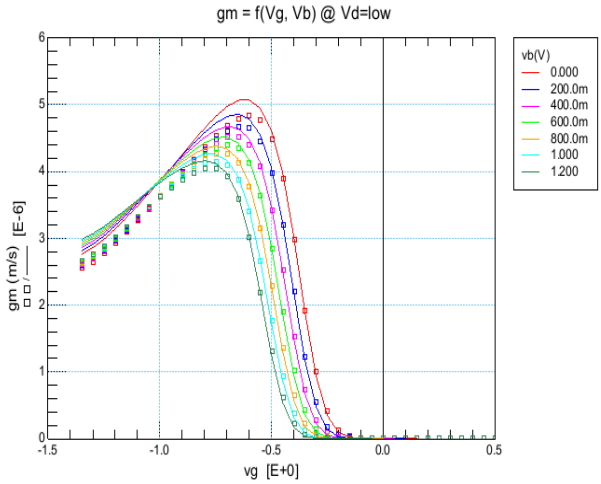


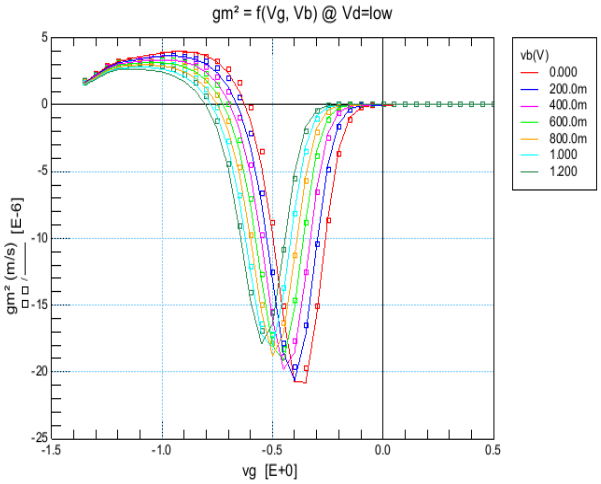
4.18 W02u0_L2u0_S550_5

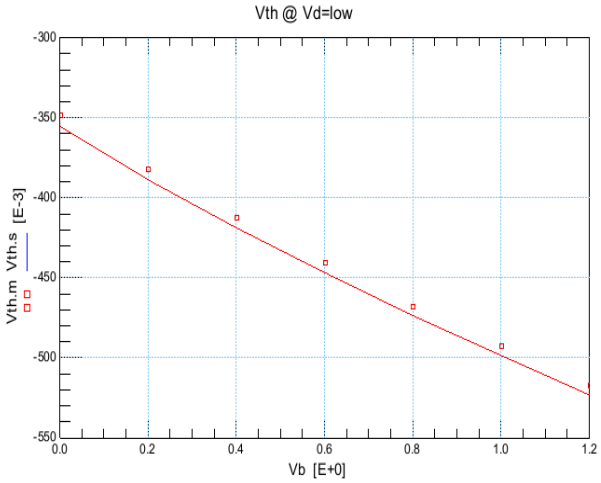
idvg, Vd = -0.05V, T = 27°C

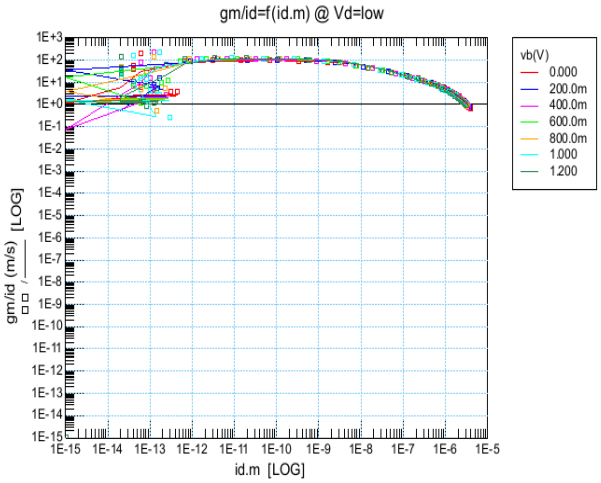


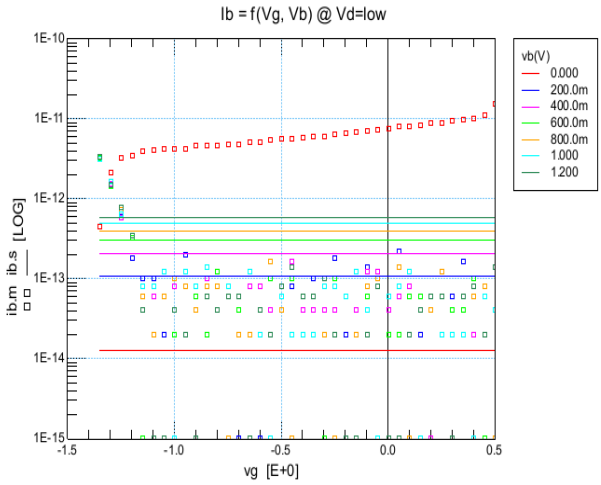


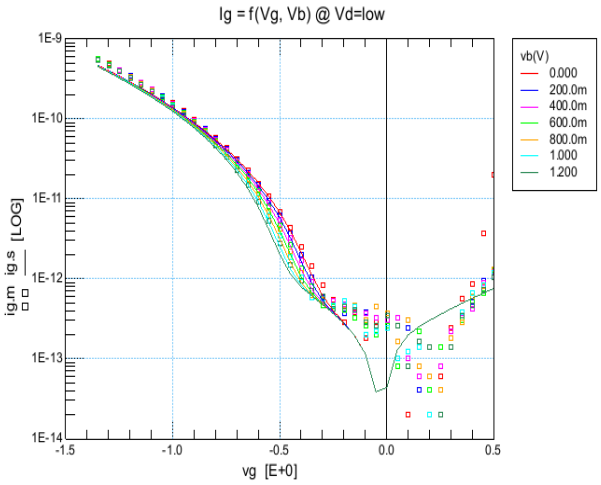




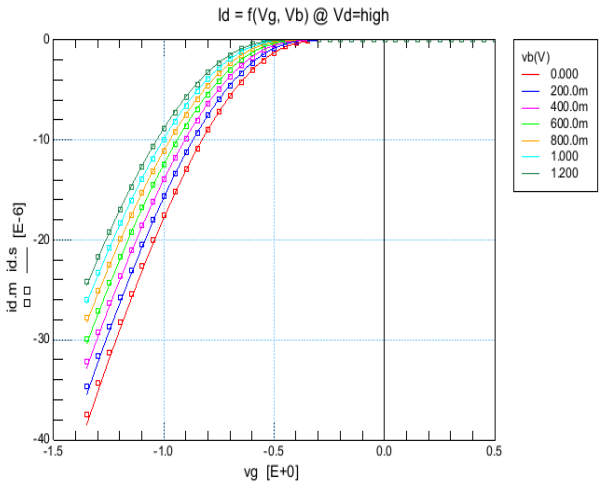


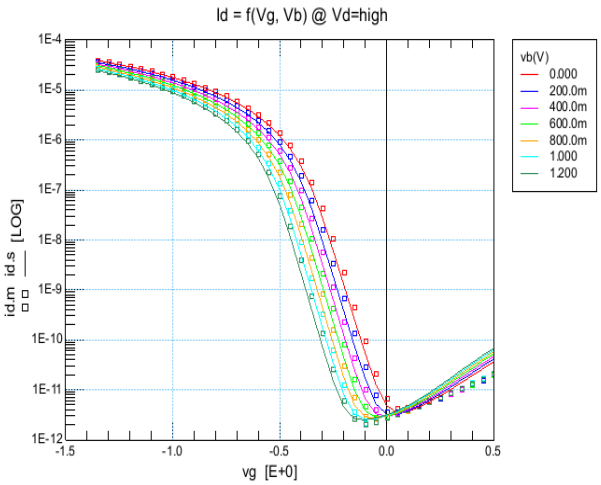


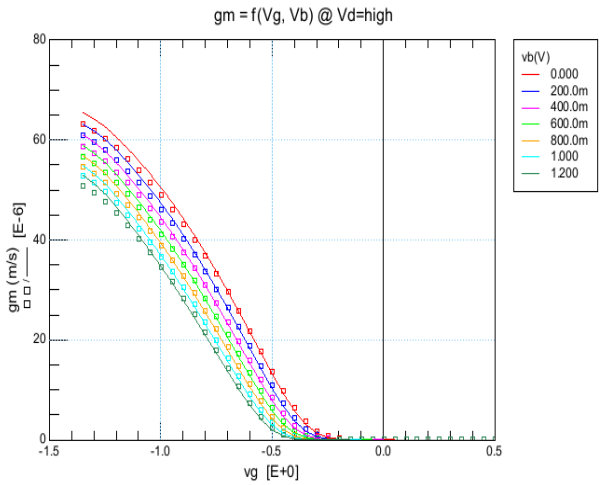


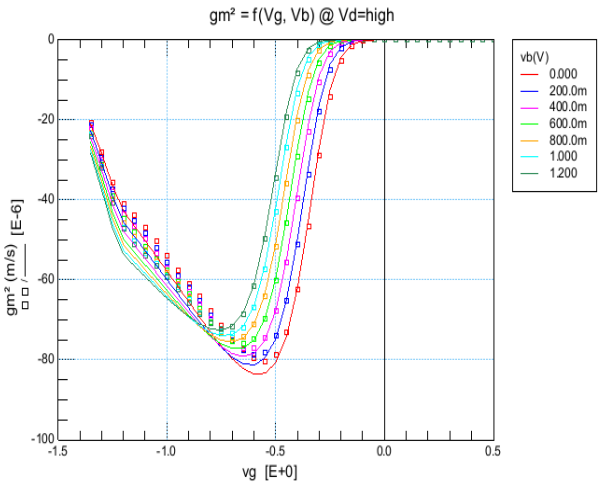


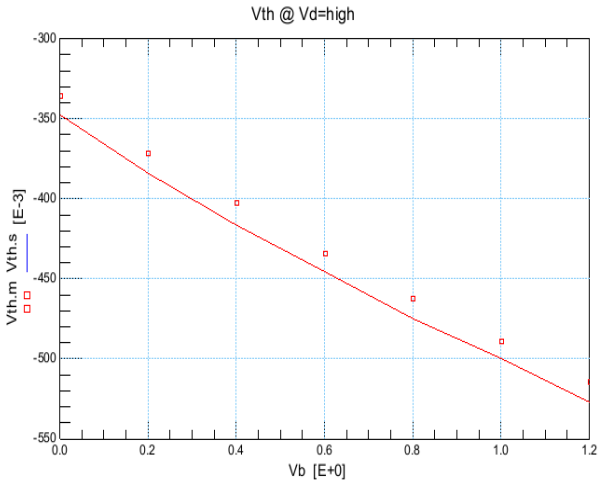
idvg, Vd = -1.2V, T = 27°C

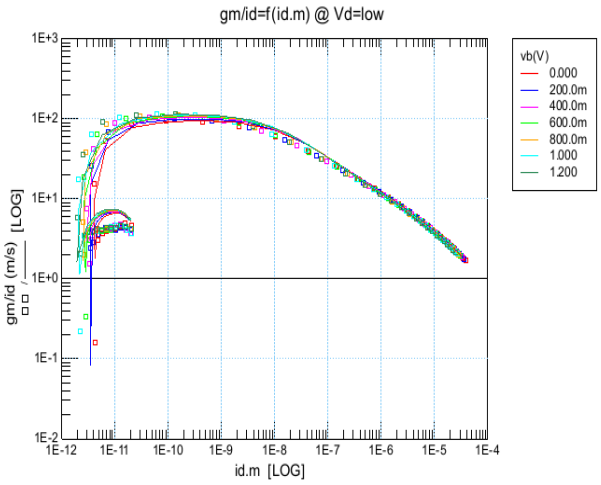


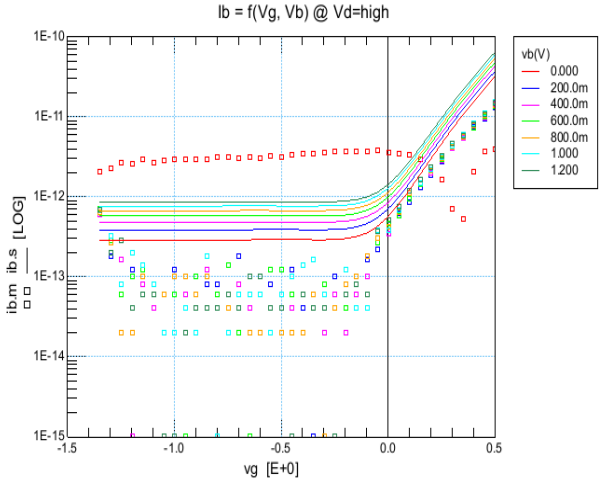


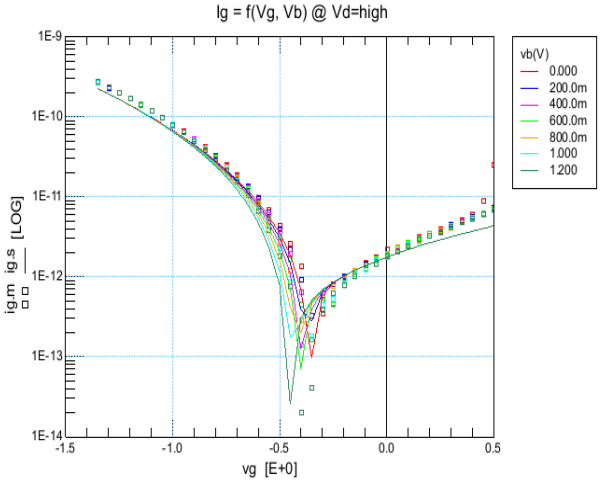




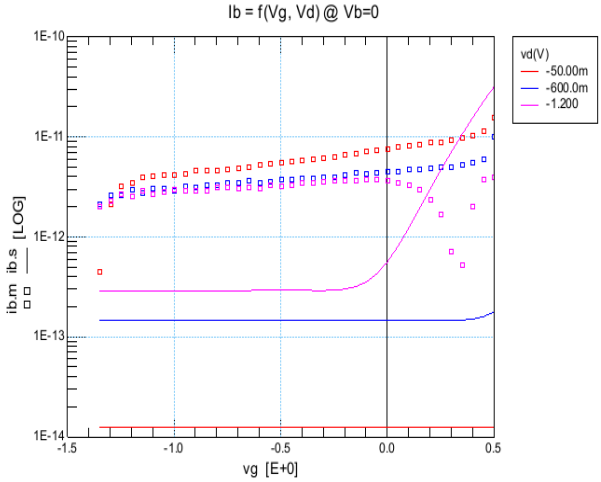


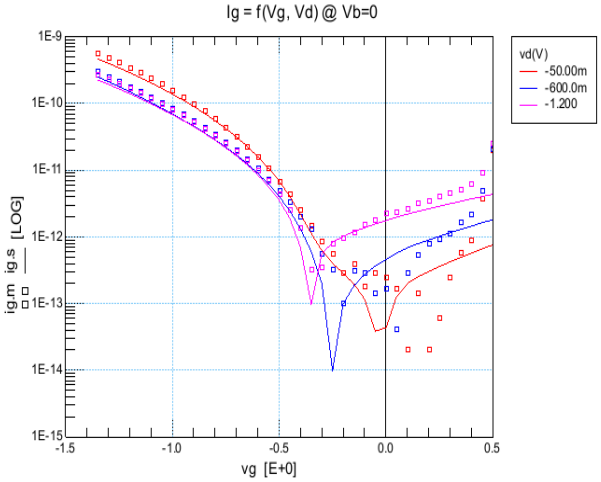




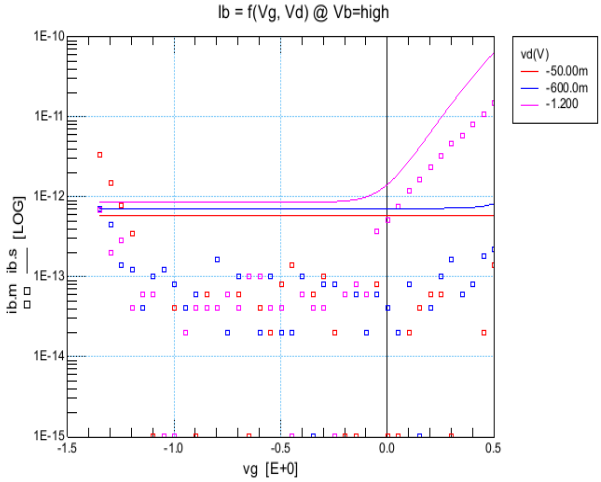


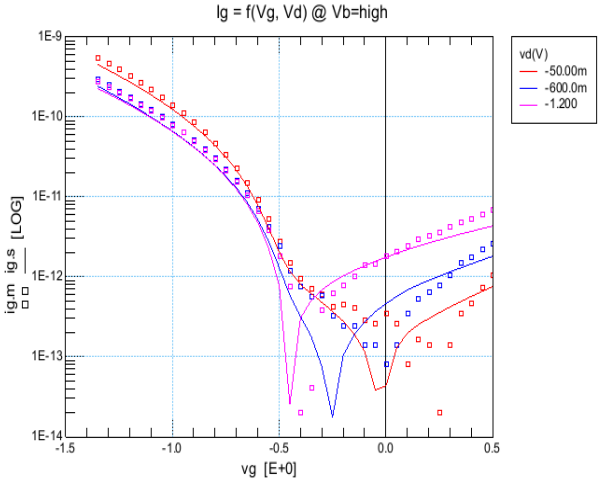
idvg, Ib,Ig, Vb = 0V, T = 27°C



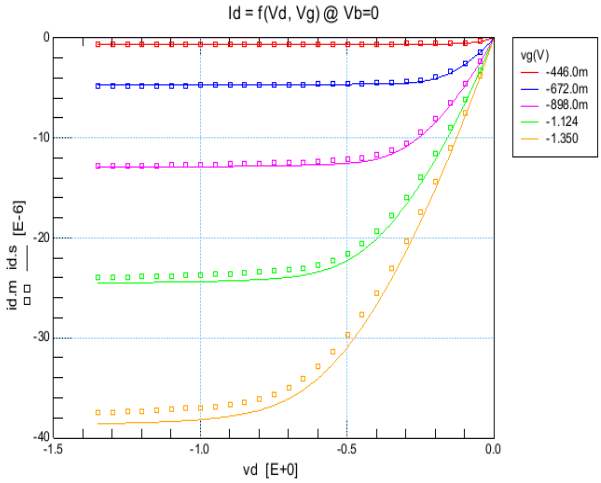


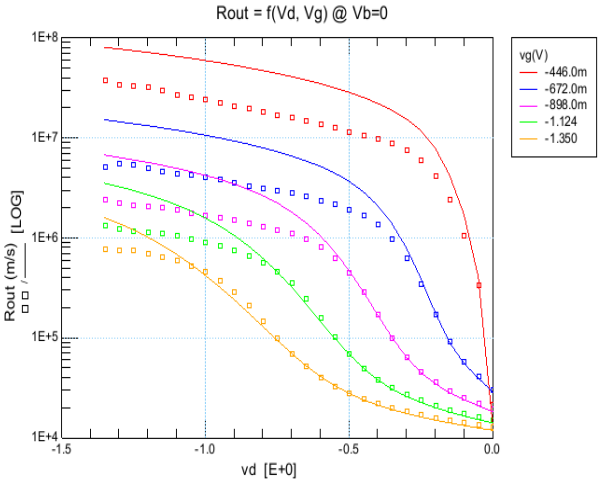
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

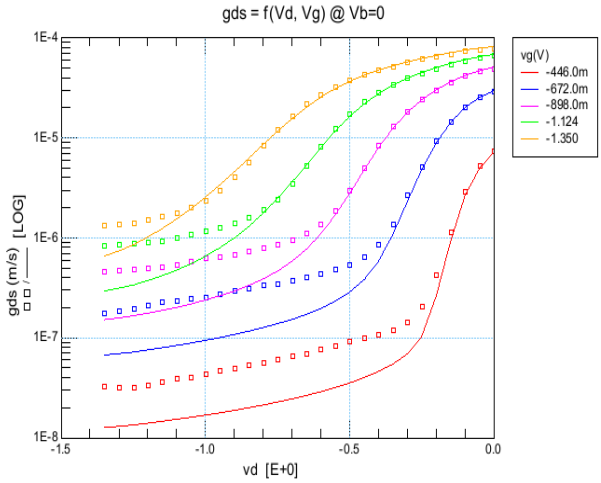




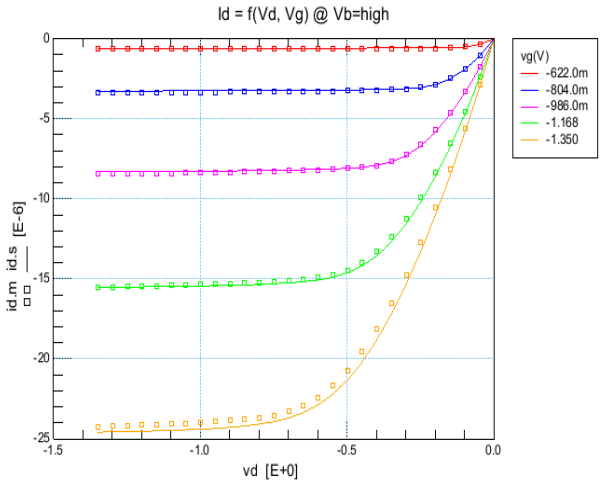
idvd, Vb = 0V, T = 27°C

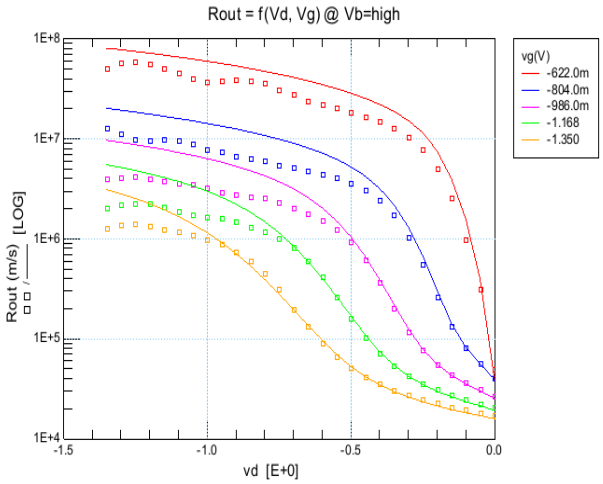


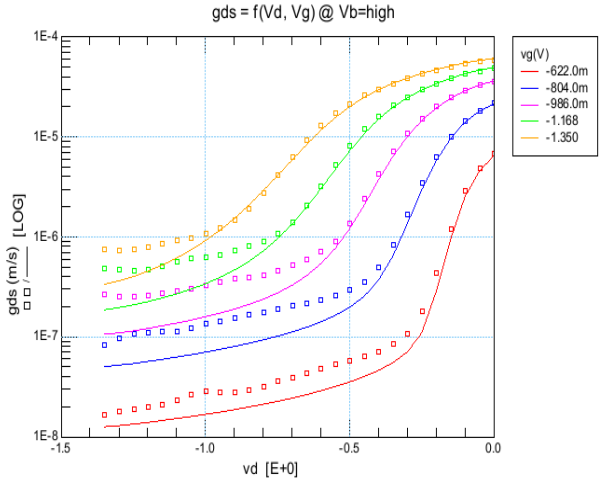




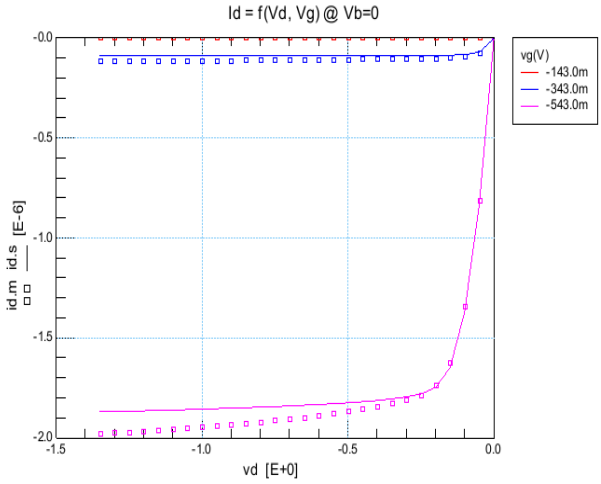
idvd_vbmin, Vb = 1.2V, T = 27°C

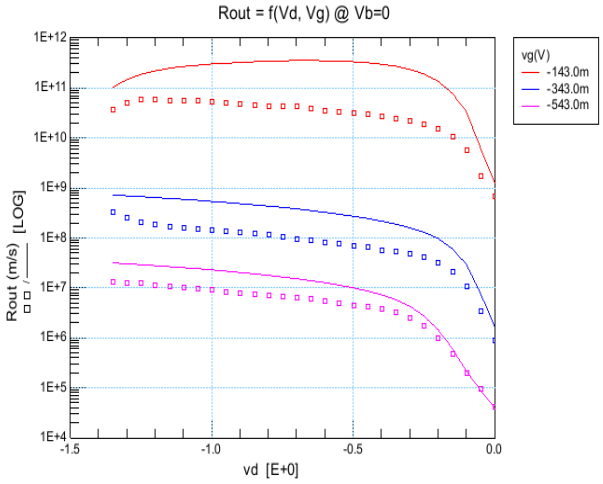


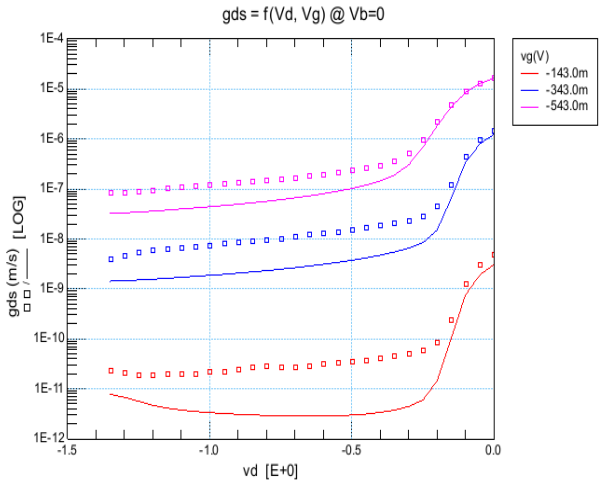




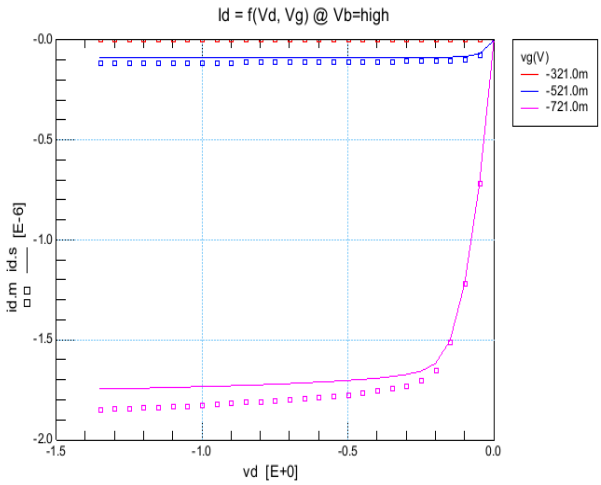
idvd_vth, Vb = 0V, T = 27°C

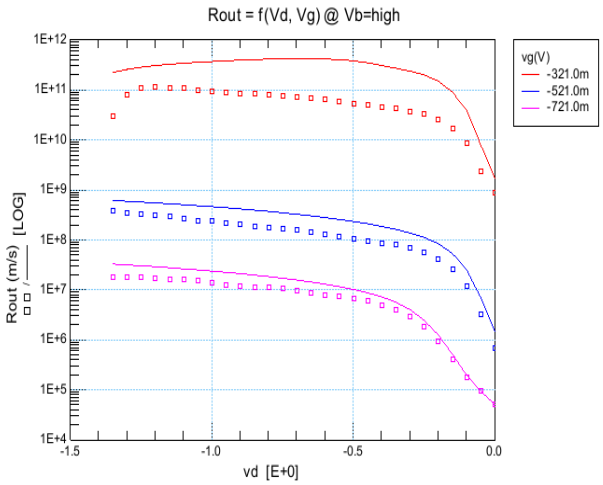


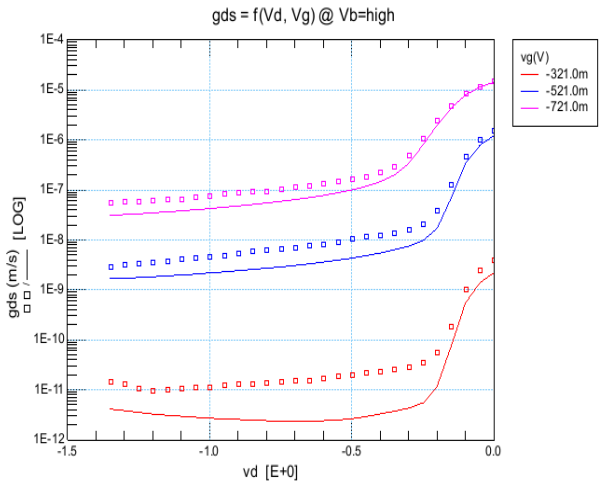




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

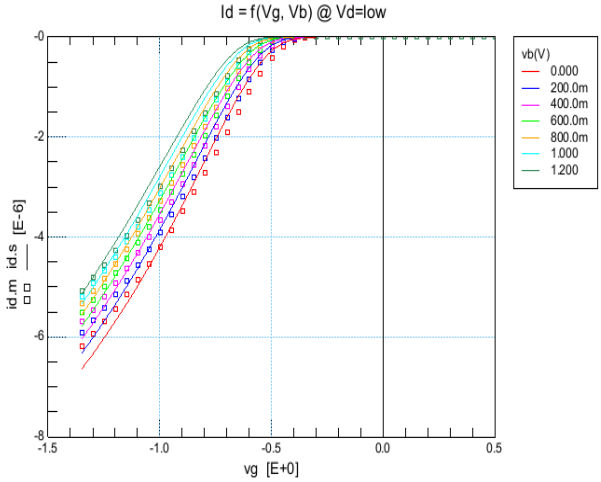


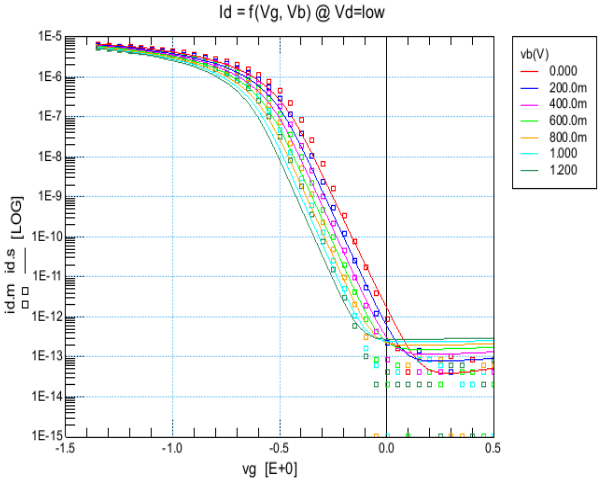


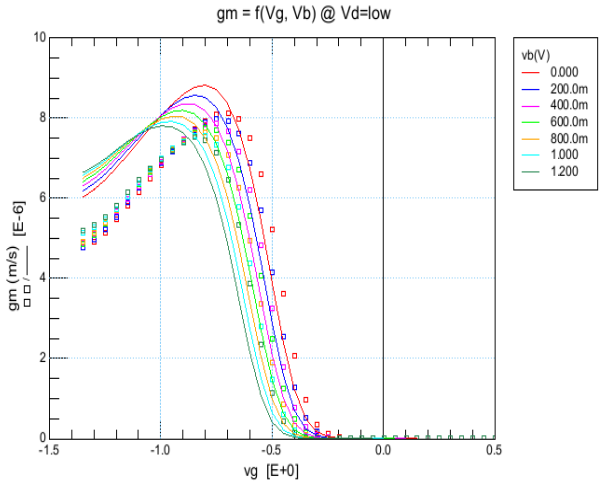


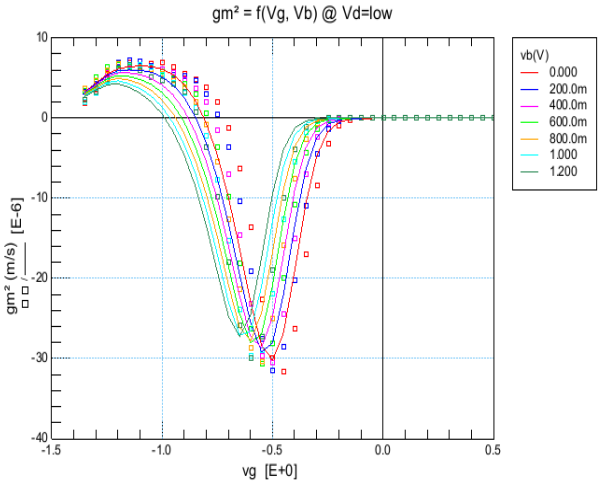
4.19W0u18_L0u14_S553_4

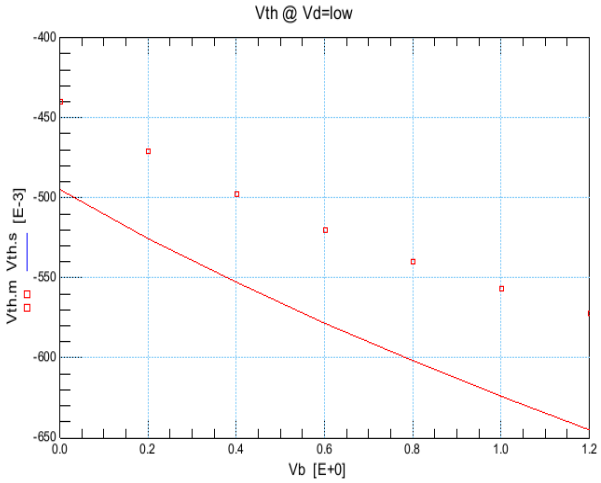
idvg, Vd = -0.05V, T = 27°C

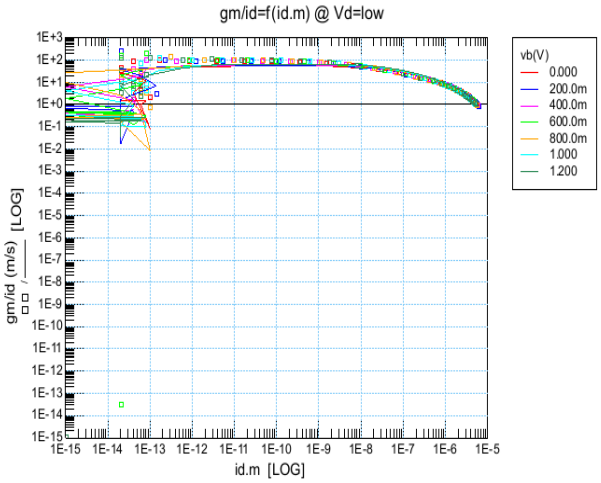


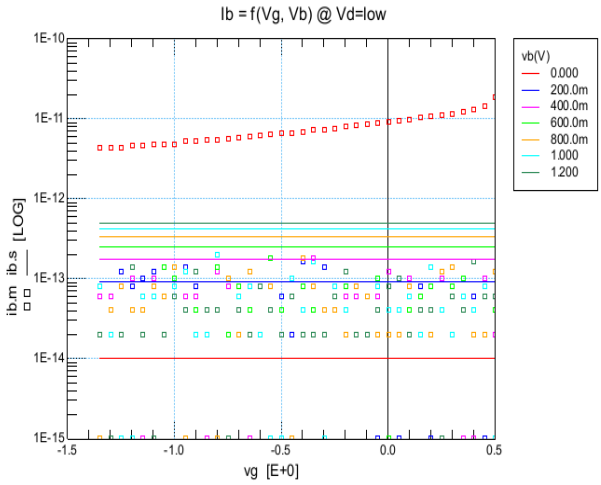


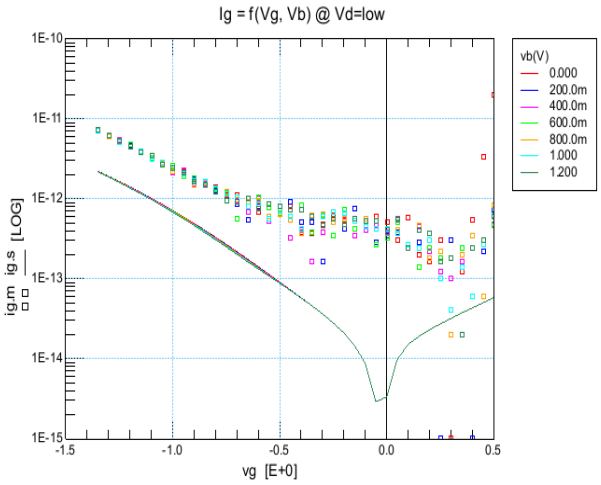




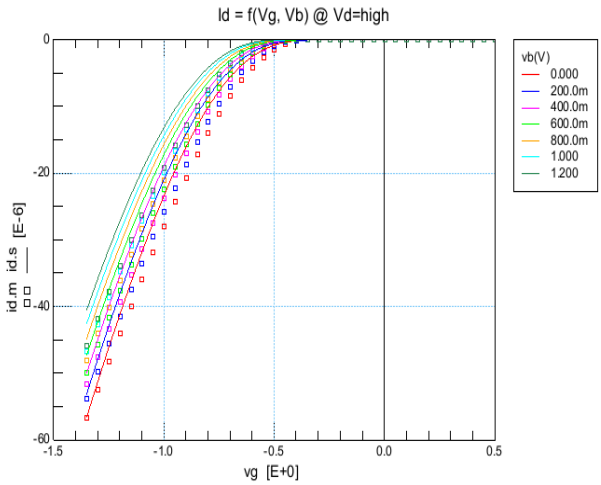


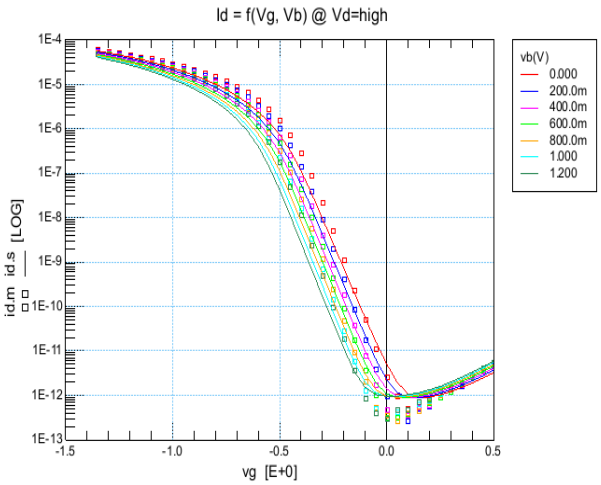


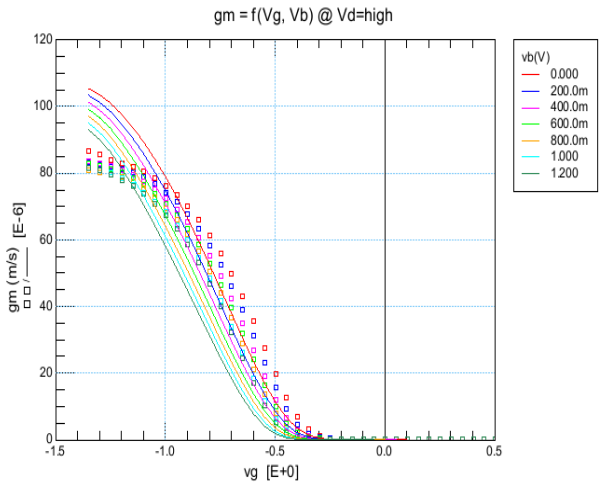


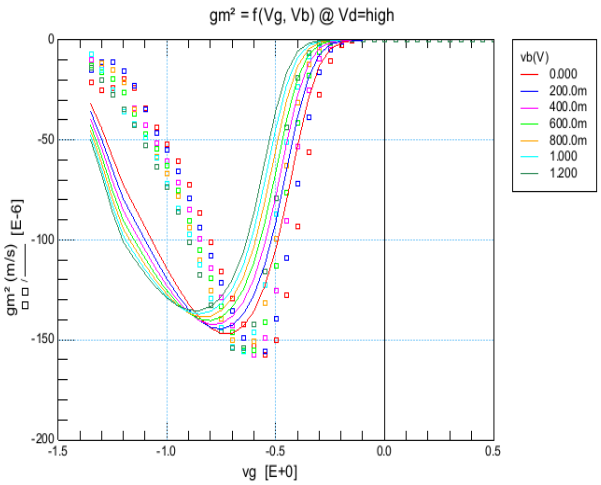


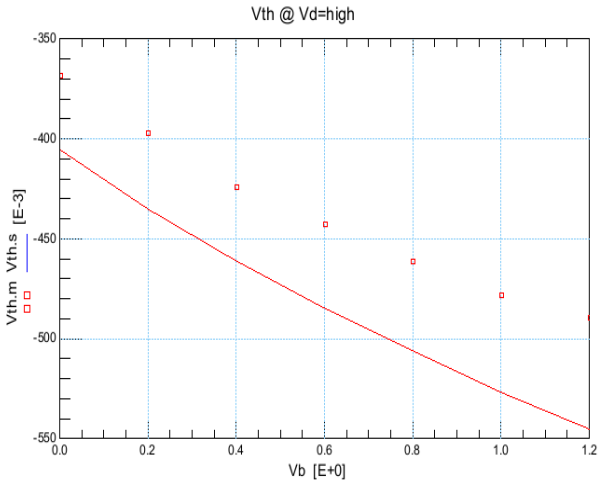
idvg, Vd = -1.2V, T = 27°C

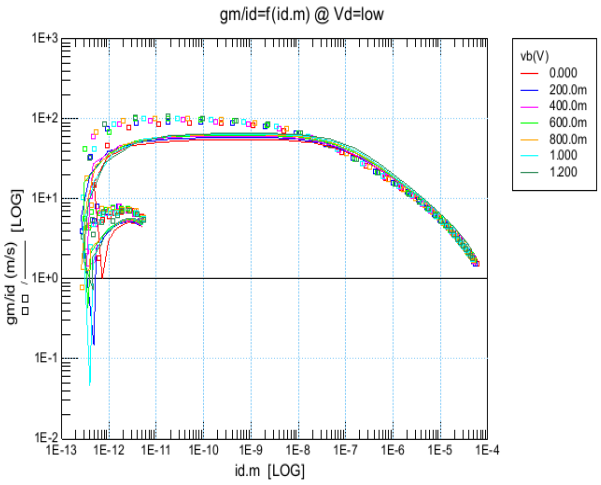


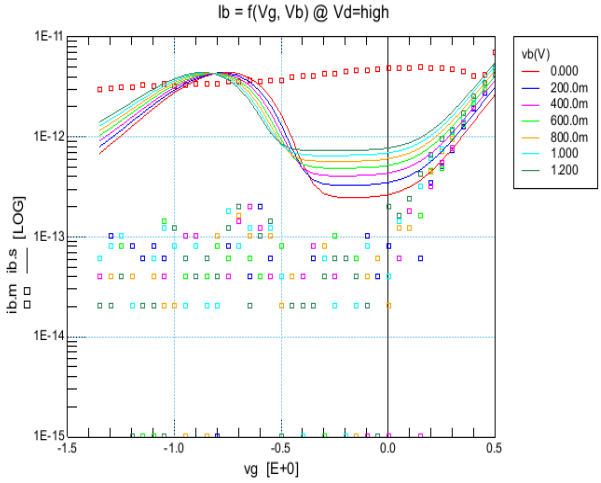


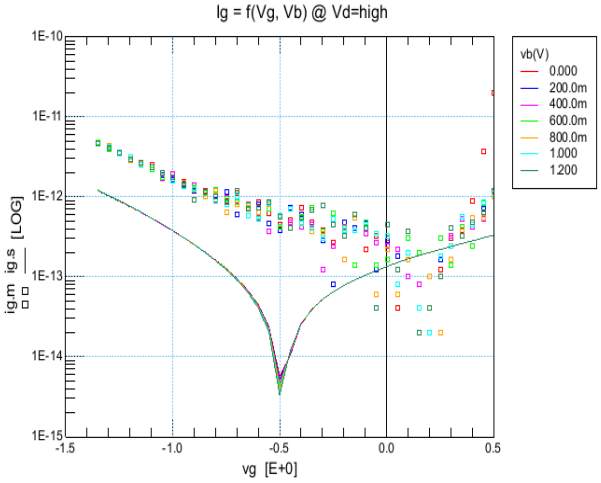




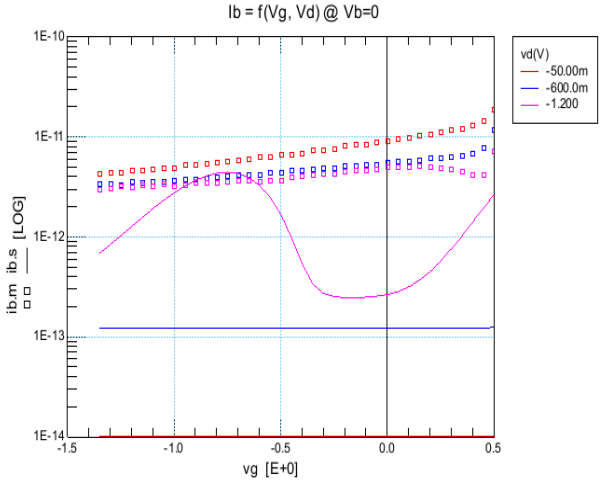


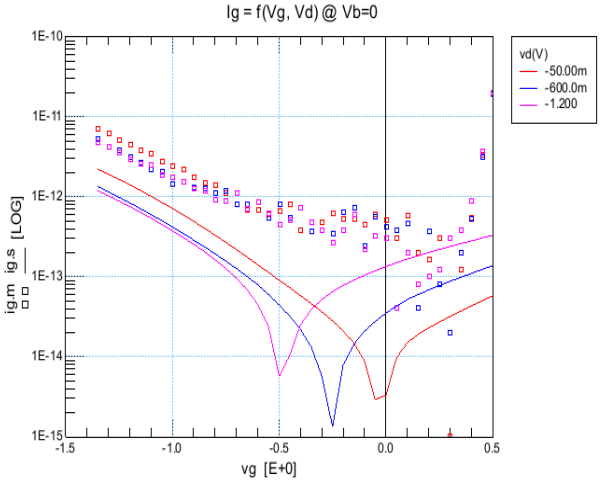




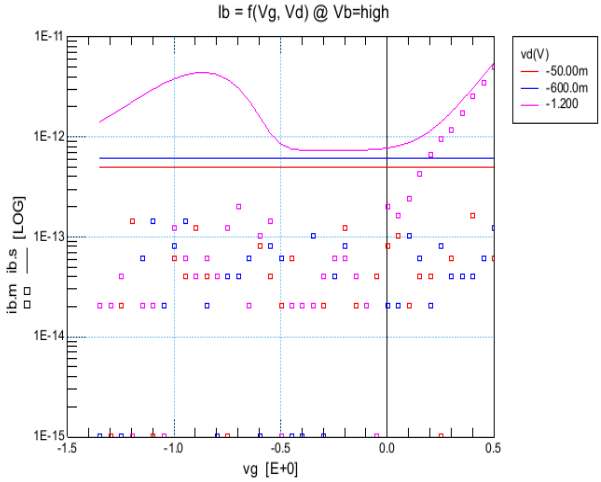


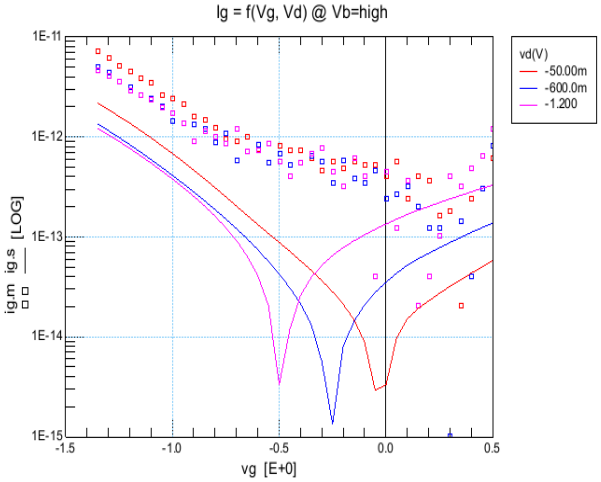
idvg, Ib,Ig, Vb = 0V, T = 27°C



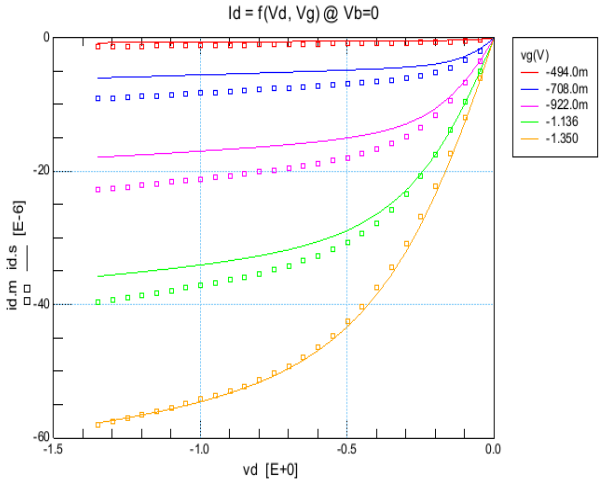


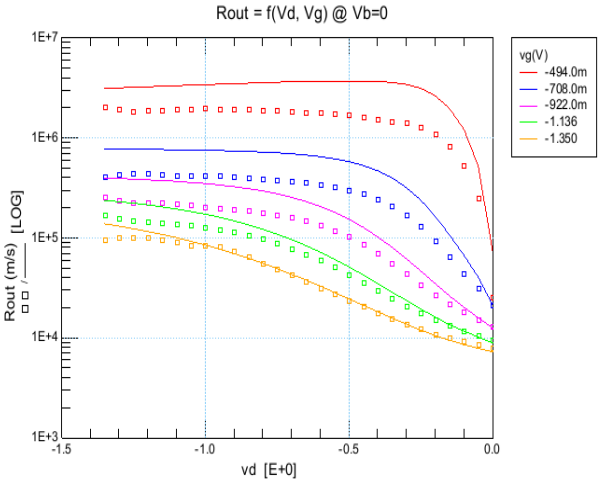
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

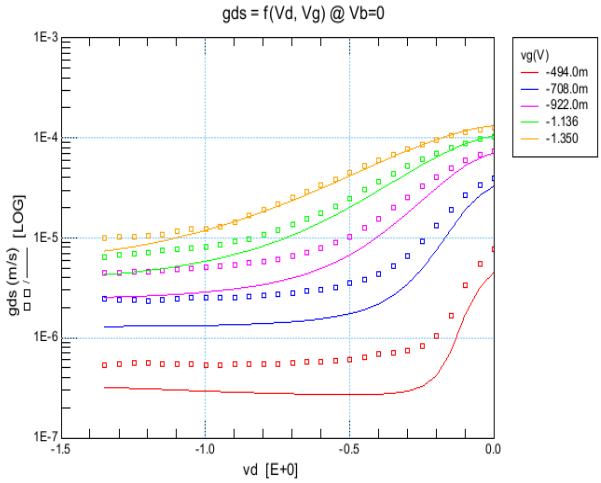




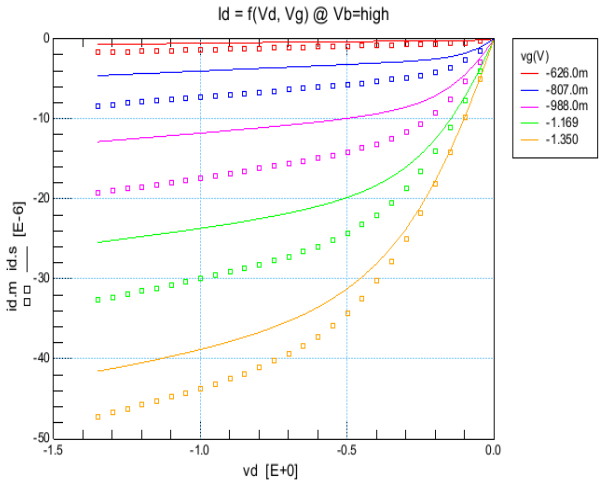
idvd, Vb = 0V, T = 27°C

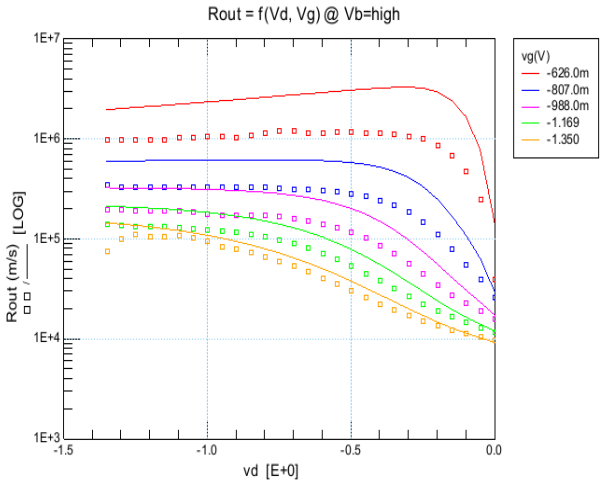


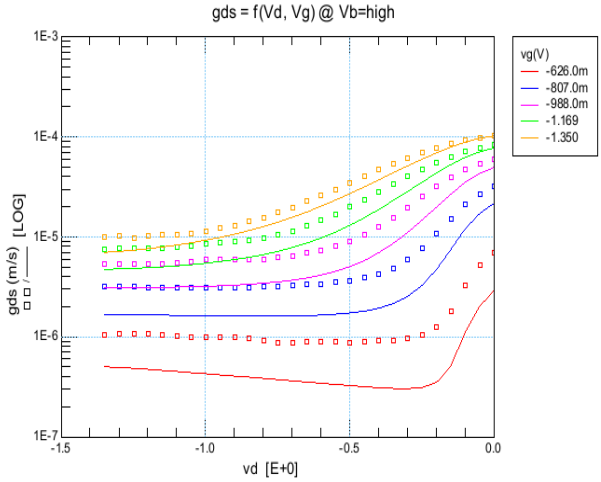




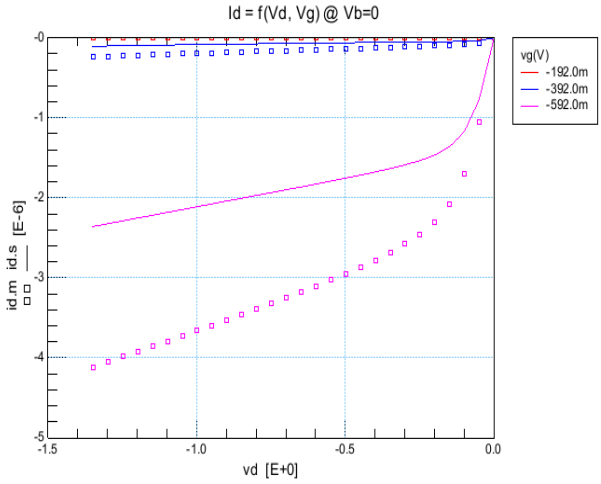
idvd_vbmin, Vb = 1.2V, T = 27°C

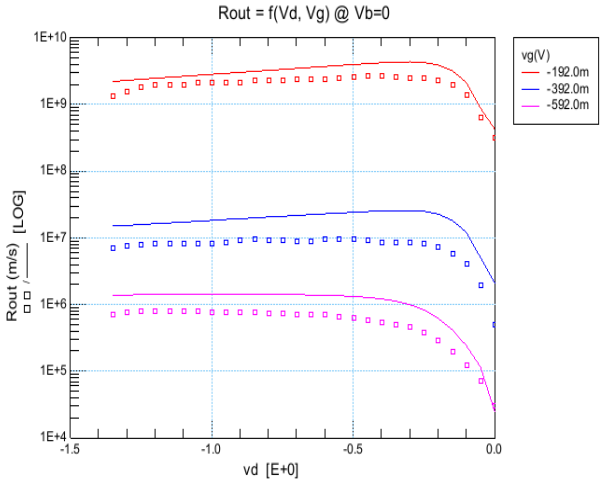


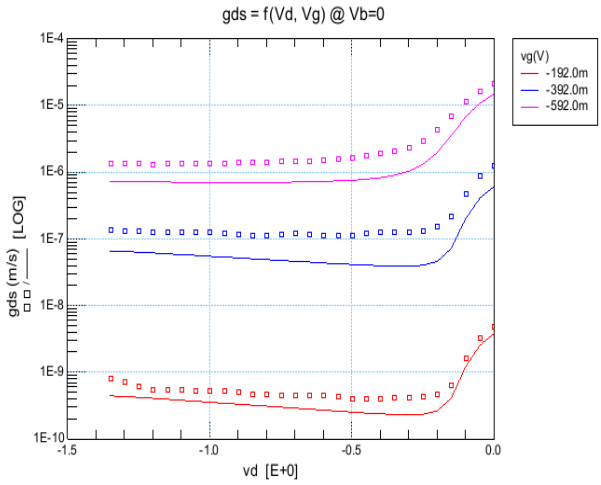




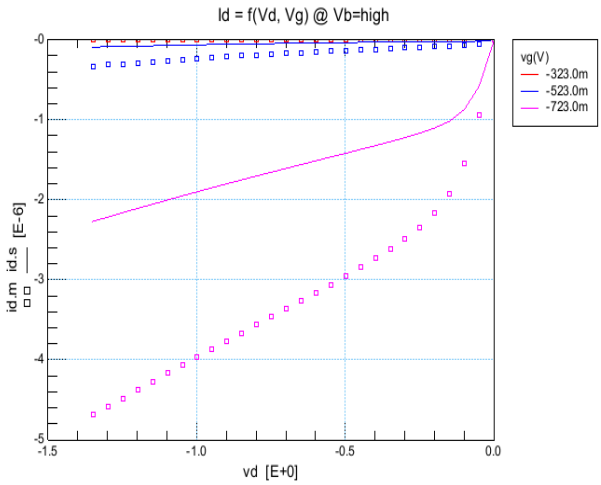
idvd_vth, Vb = 0V, T = 27°C

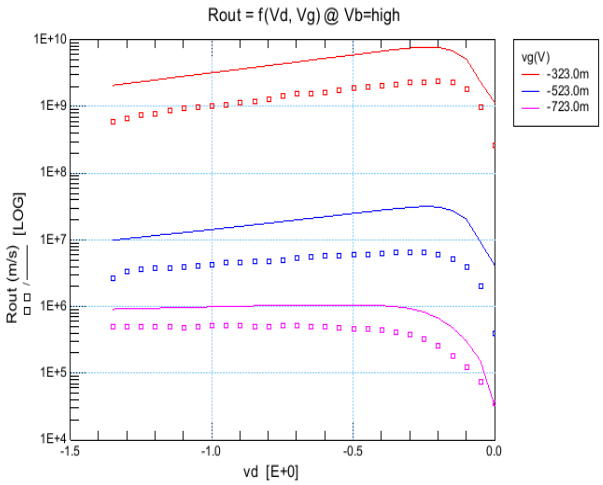


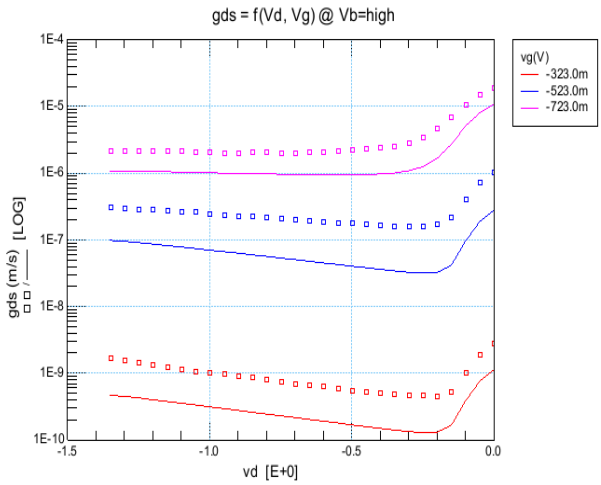




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

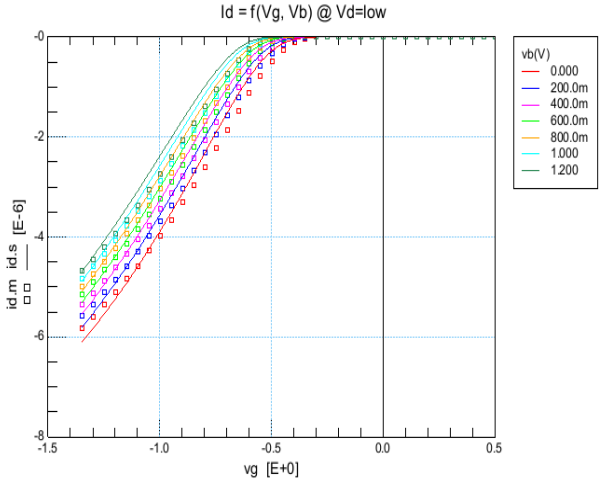


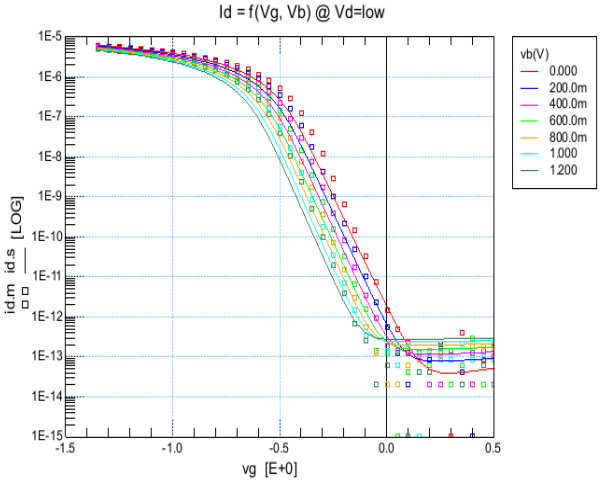


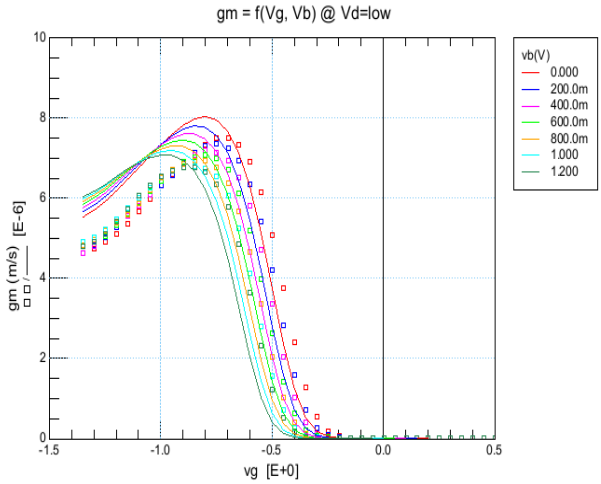


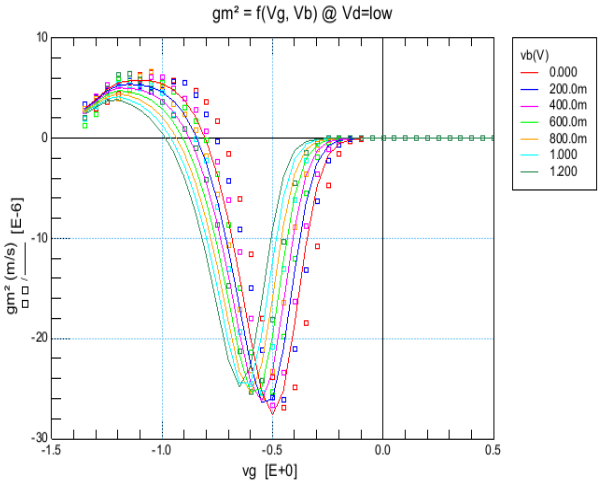
4.20W0u18_L0u15_S553_5

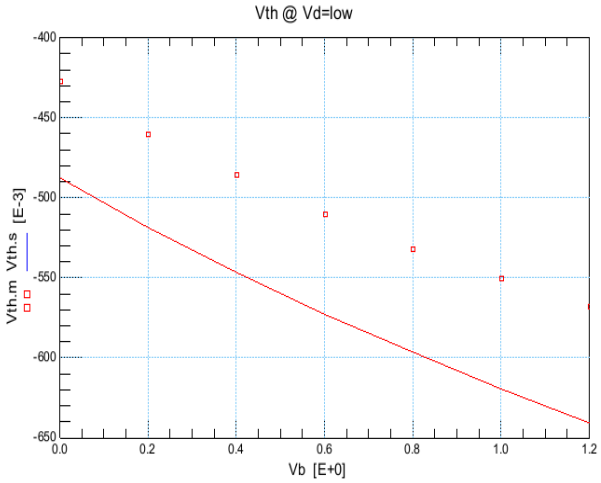
idvg, Vd = -0.05V, T = 27°C

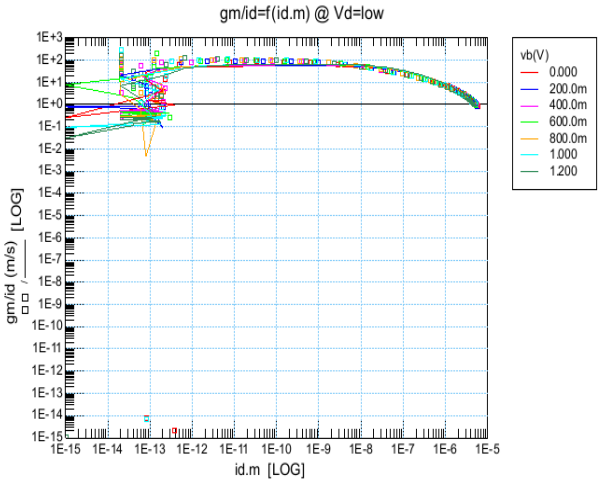


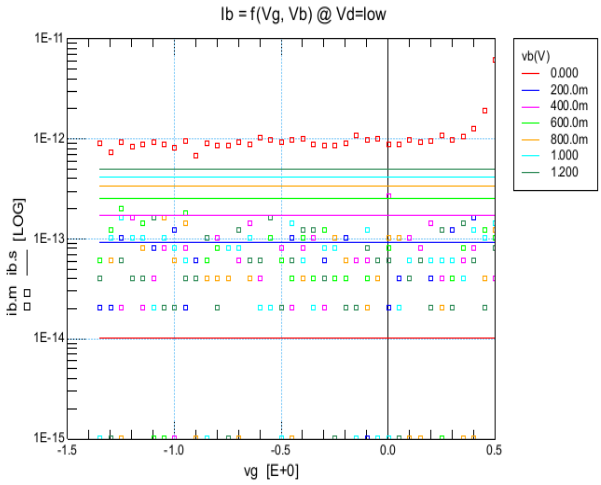


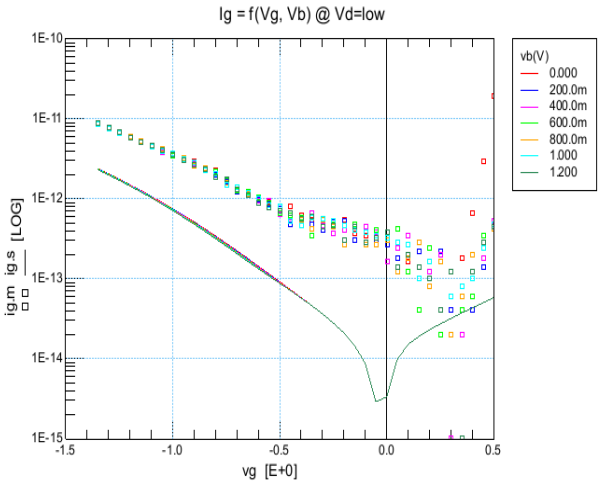




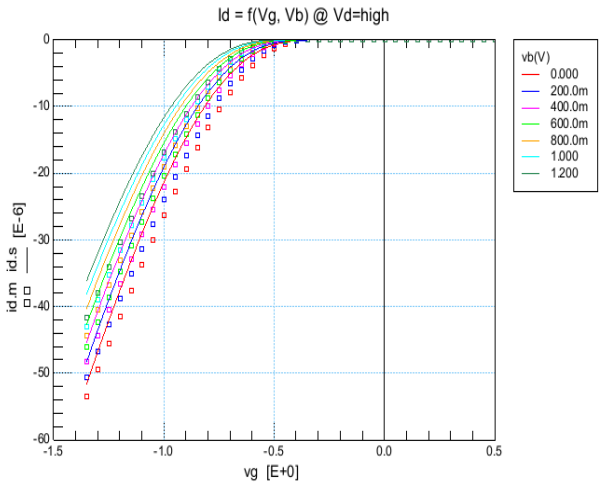


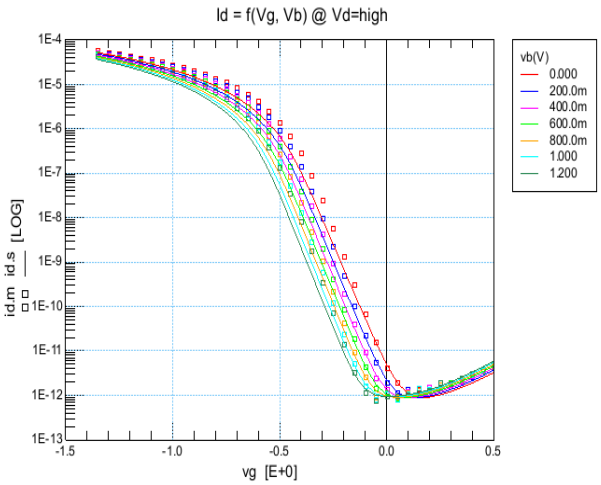


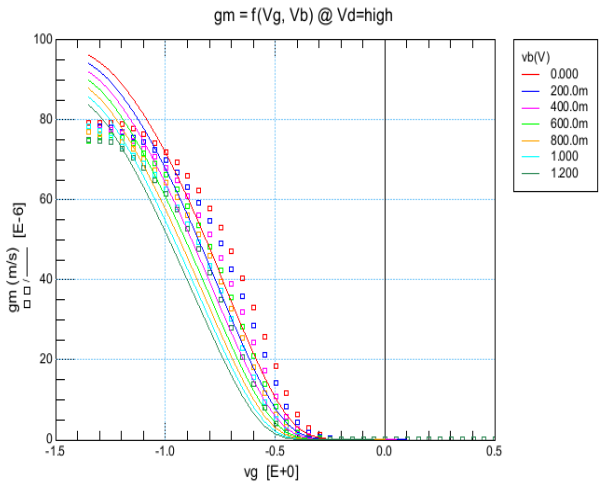


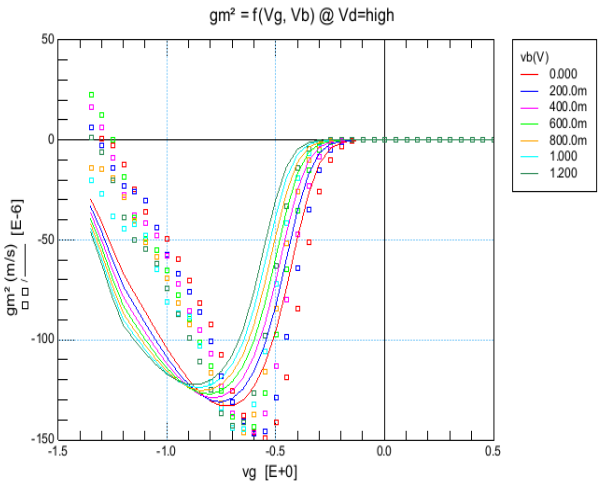


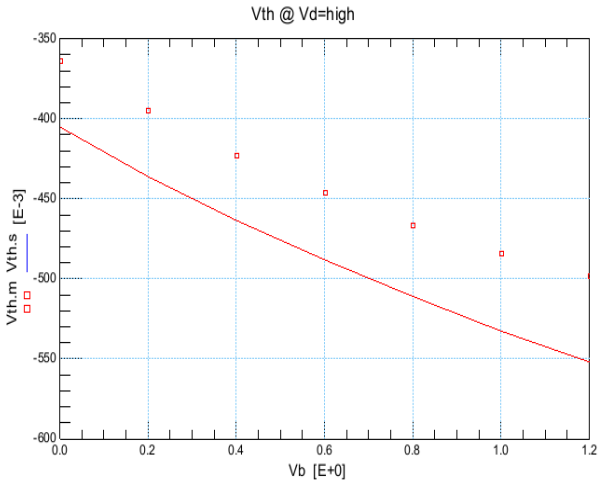
idvg, Vd = -1.2V, T = 27°C

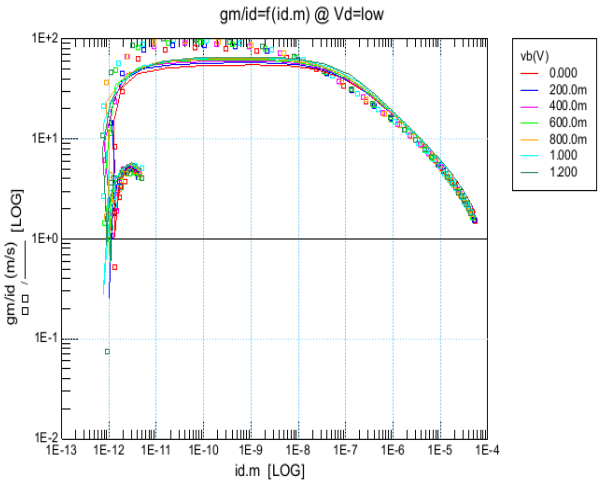


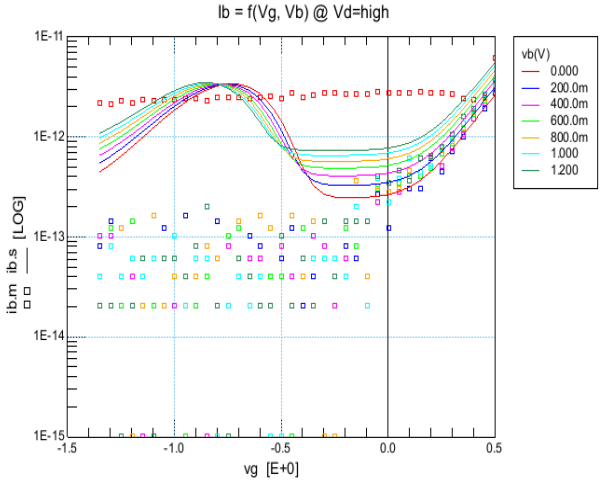


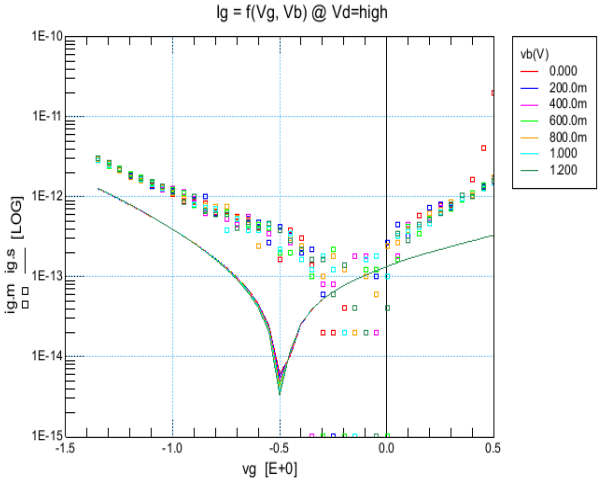




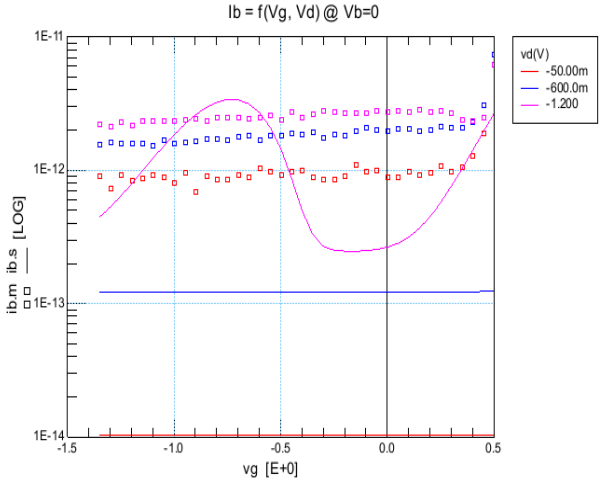


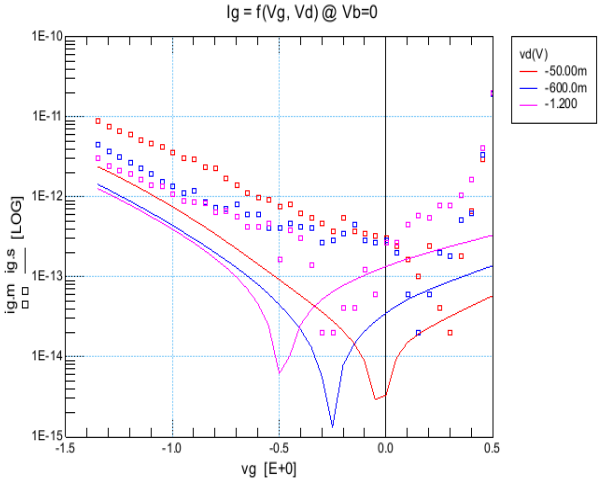




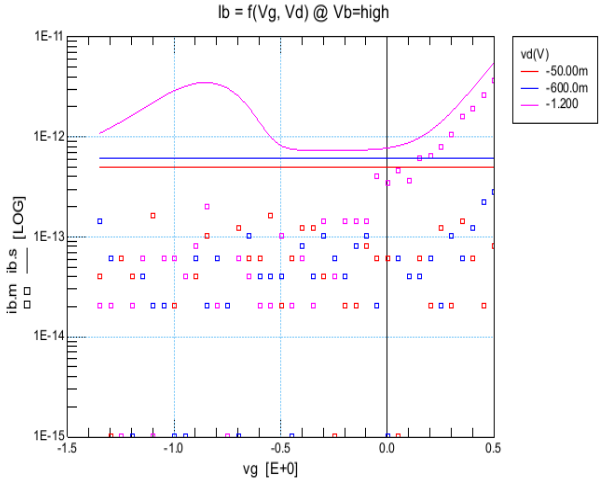


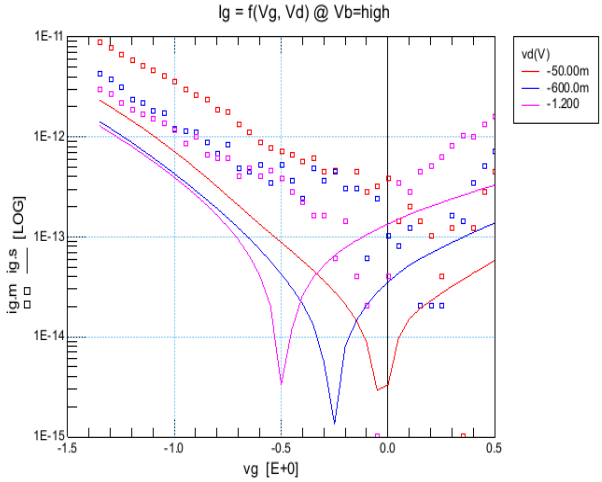
idvg, Ib,Ig, Vb = 0V, T = 27°C



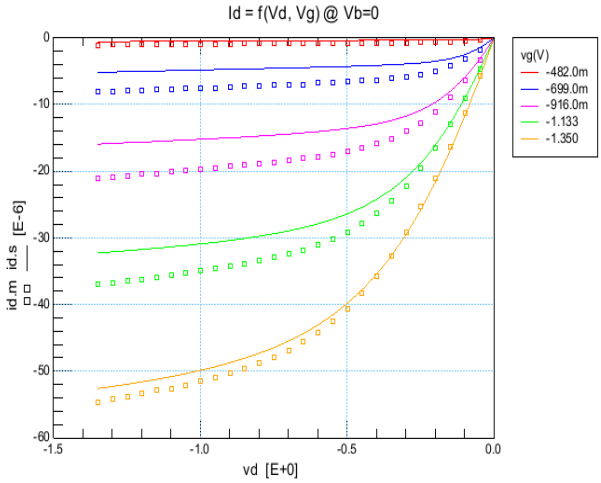


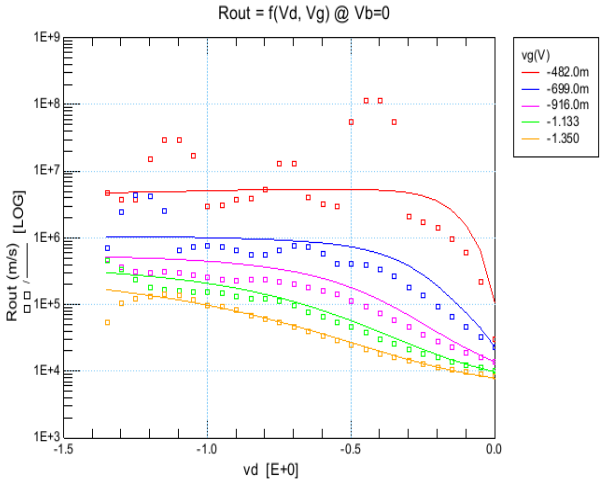
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

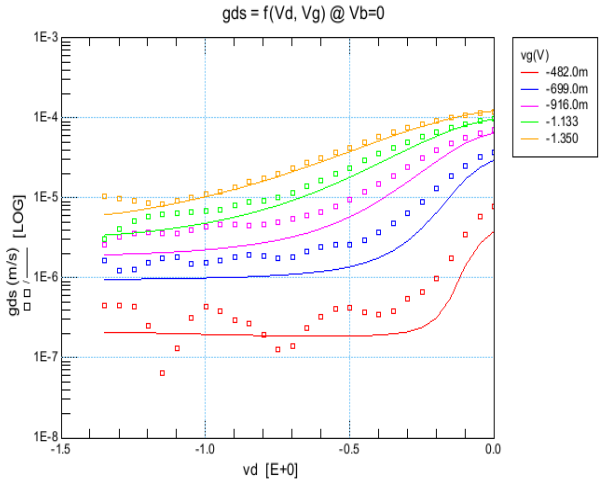




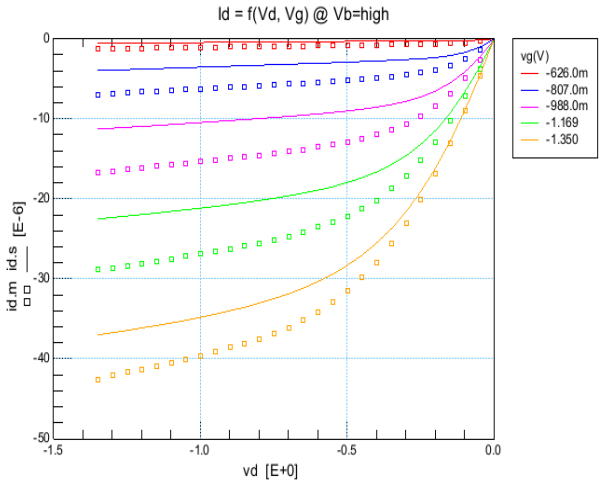
idvd, Vb = 0V, T = 27°C

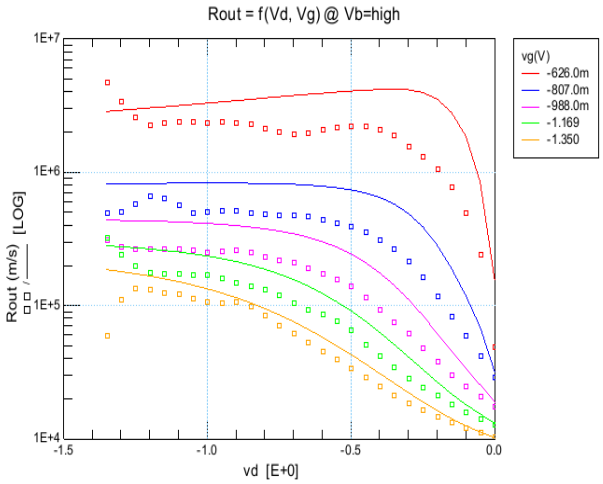


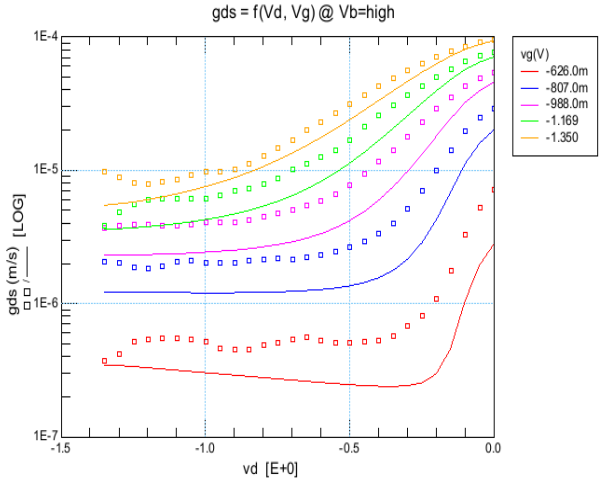




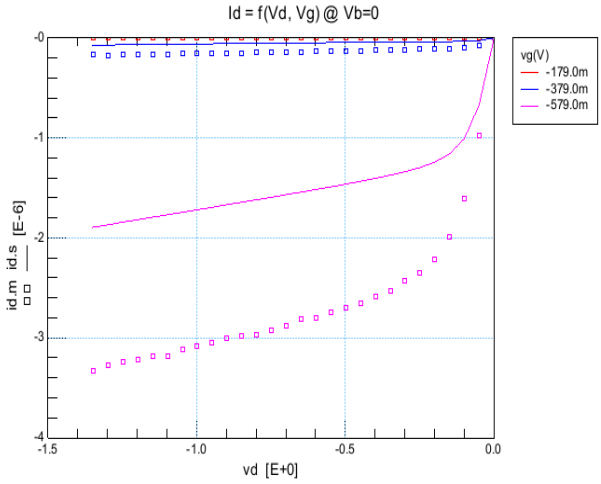
idvd_vbmin, Vb = 1.2V, T = 27°C

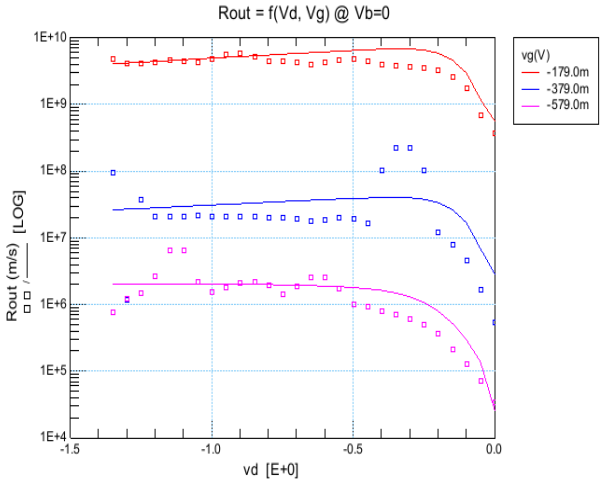


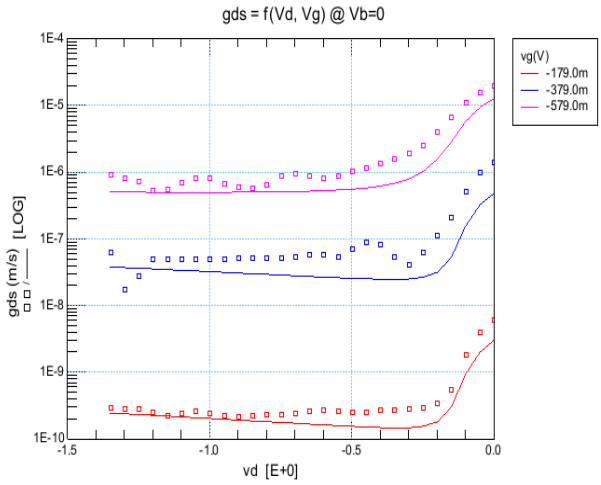




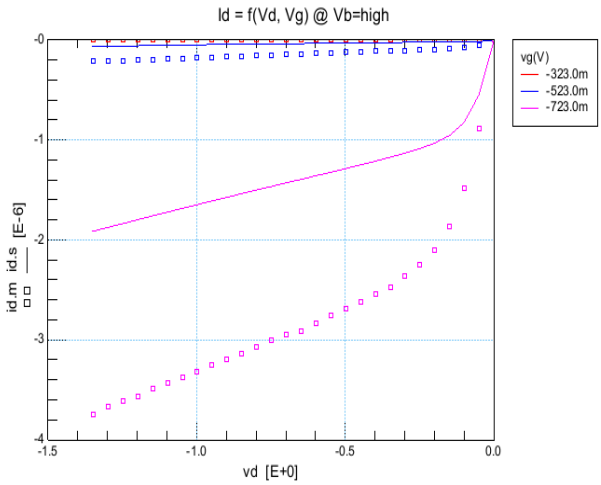
idvd_vth, Vb = 0V, T = 27°C

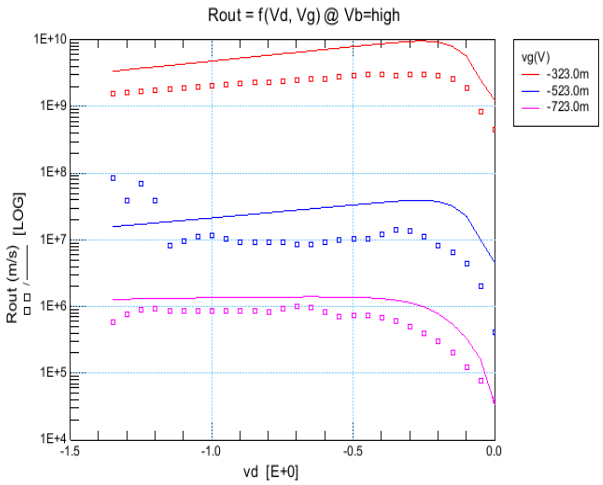


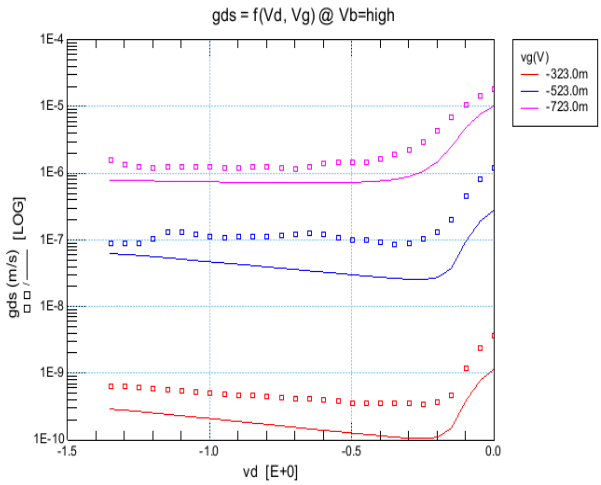




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

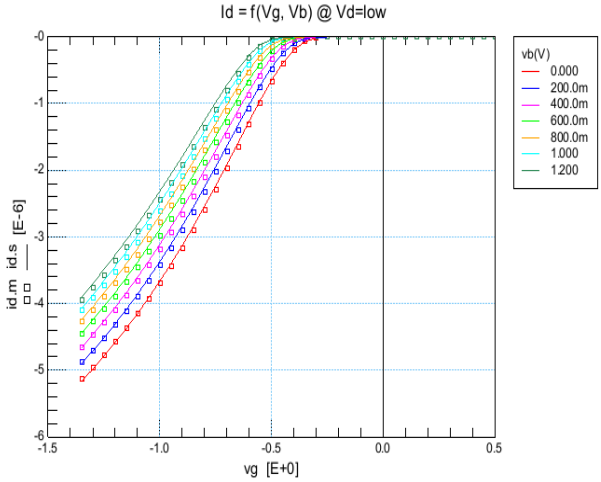


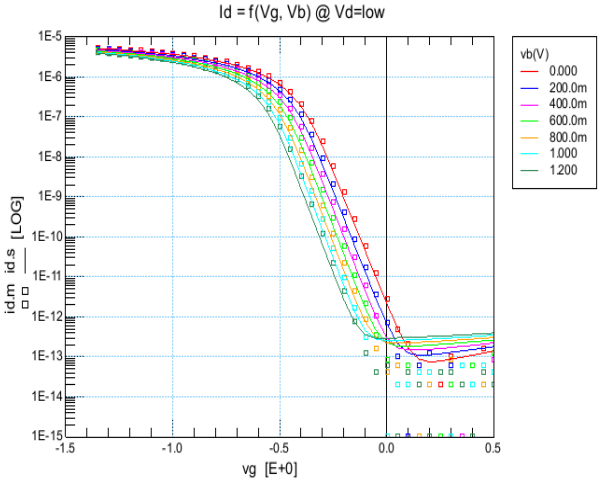


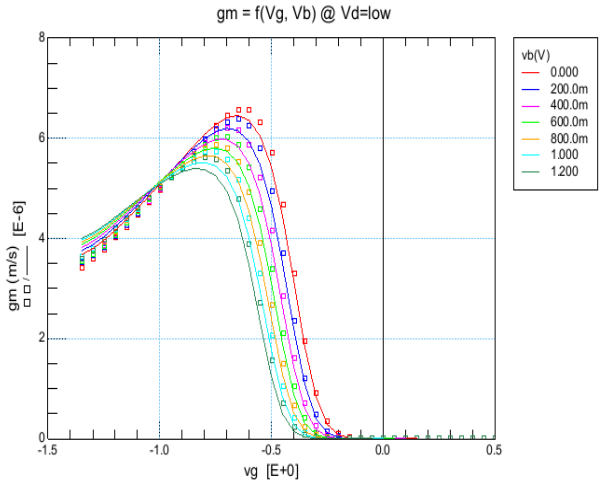


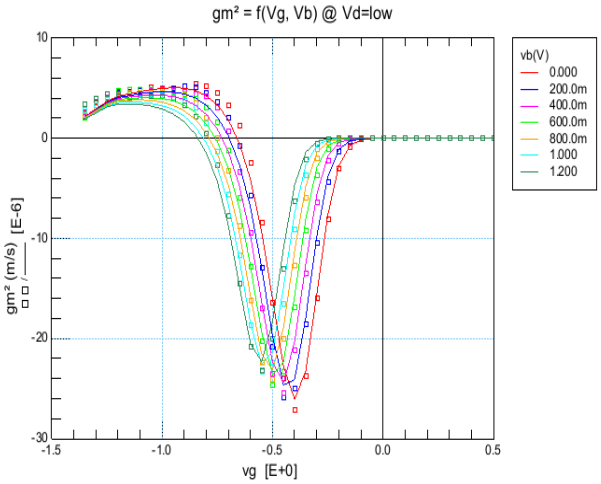
4.21 W0u6_L0u5_S551_4

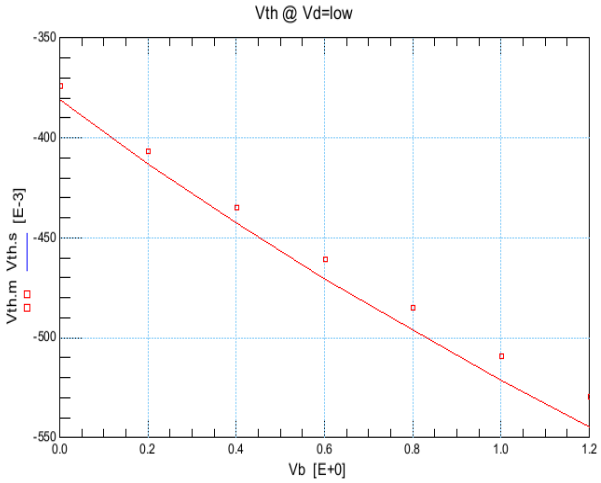
idvg, Vd = -0.05V, T = 27°C

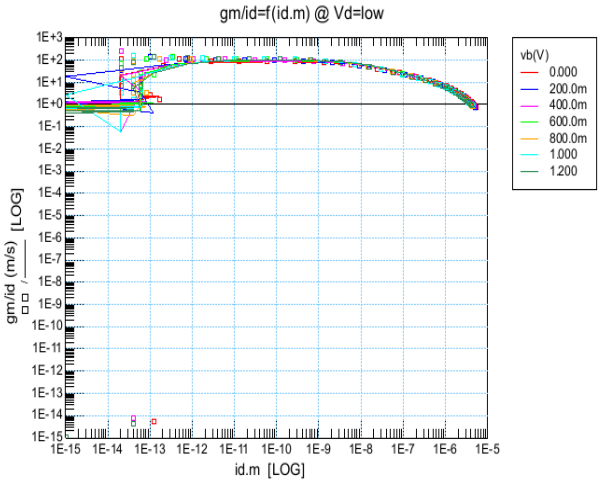


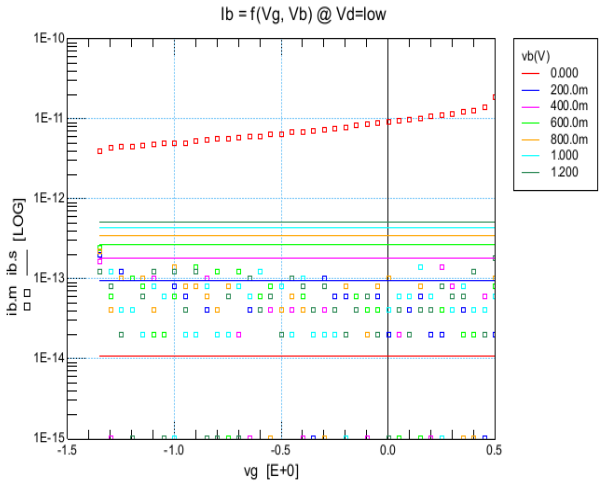


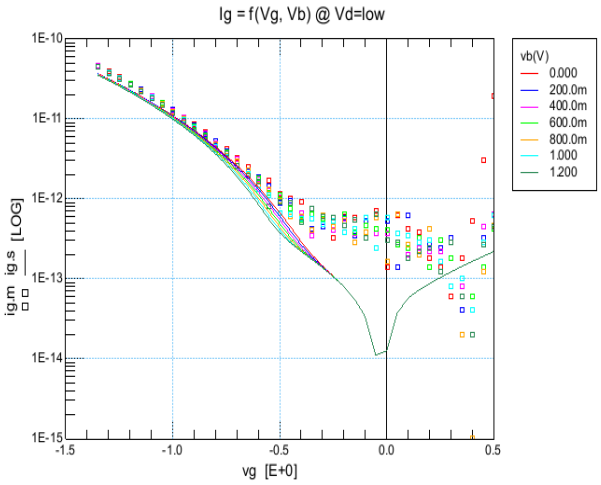




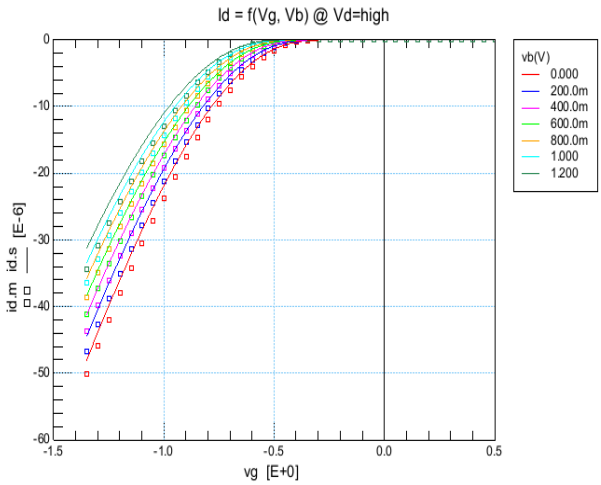


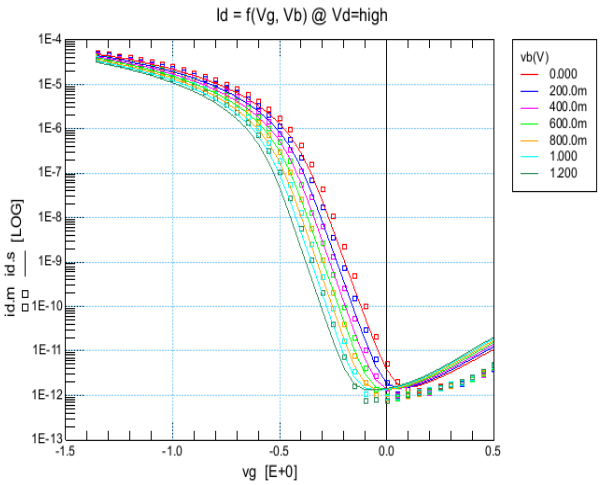


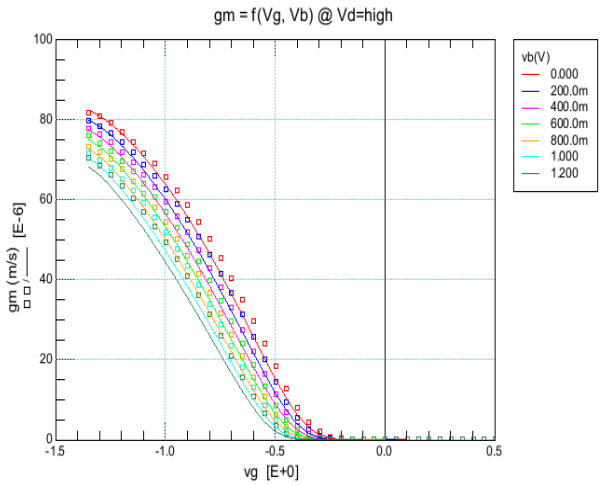


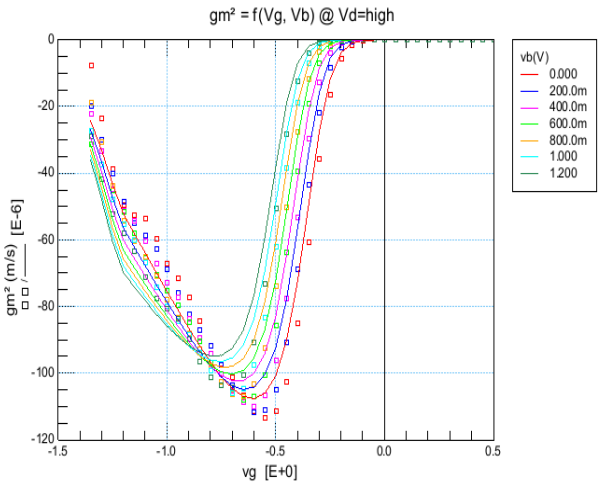


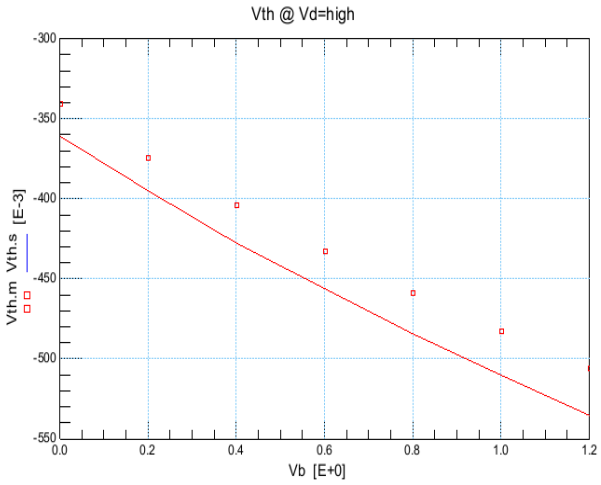
idvg, Vd = -1.2V, T = 27°C

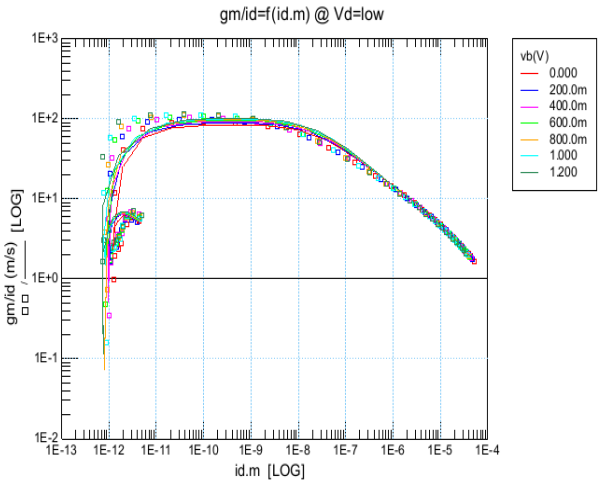


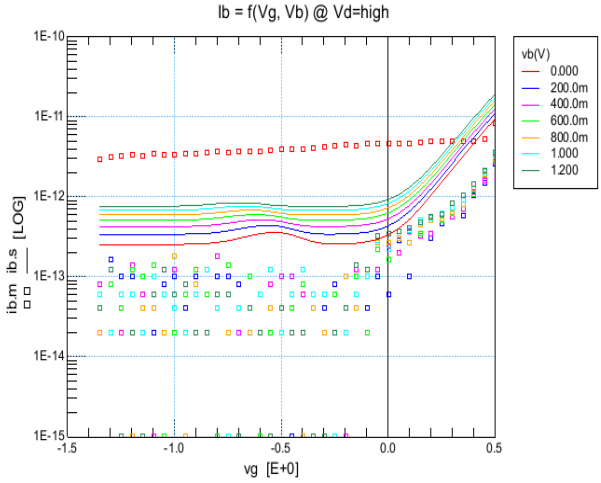


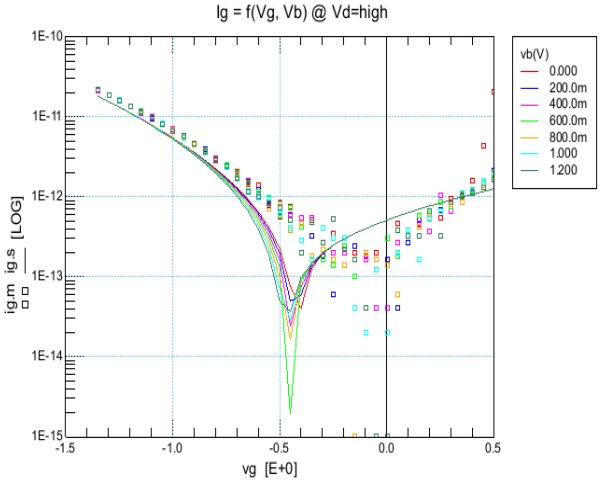




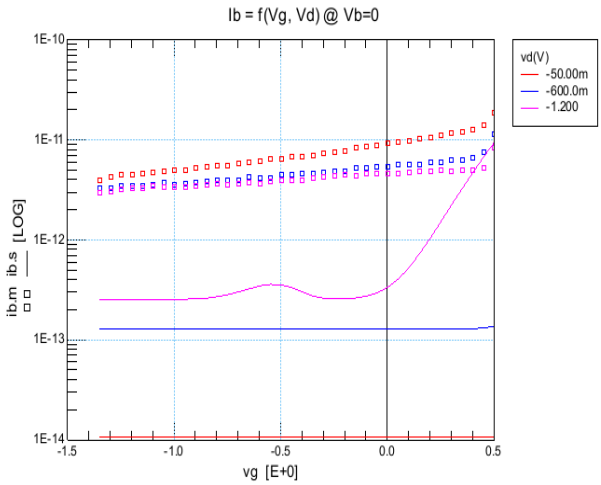


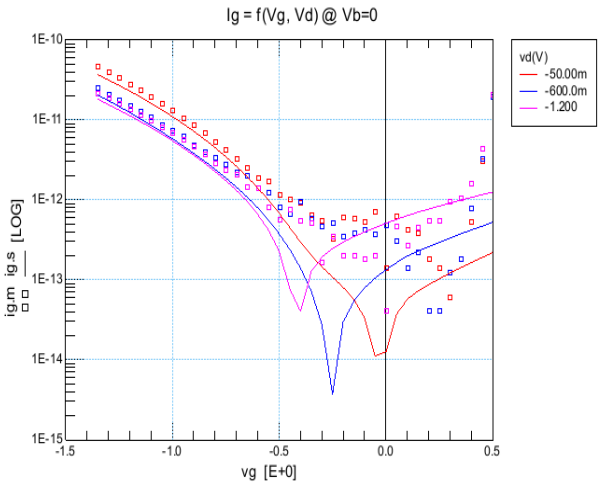




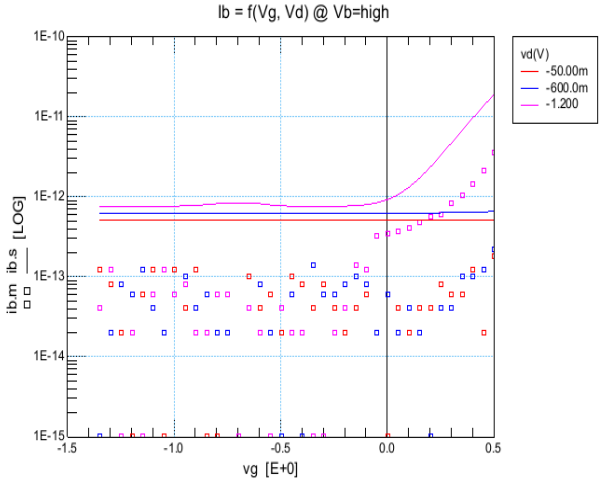


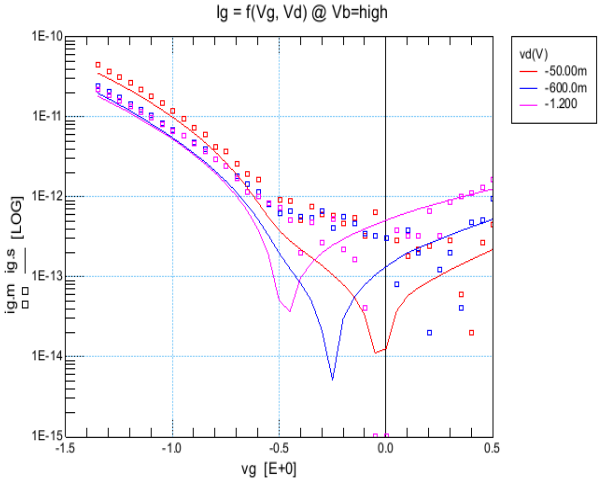
idvg, Ib,Ig, Vb = 0V, T = 27°C



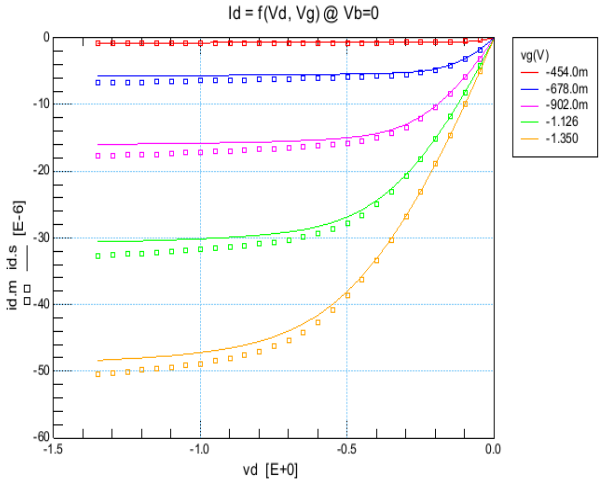


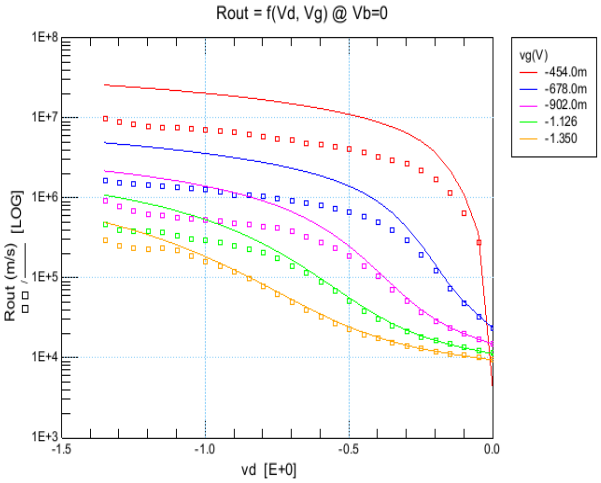
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

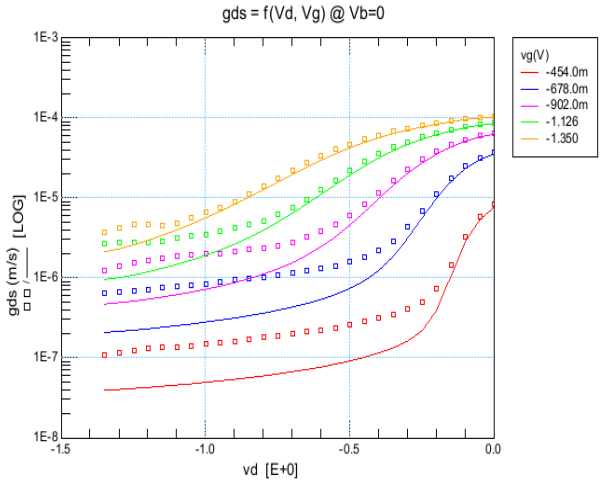




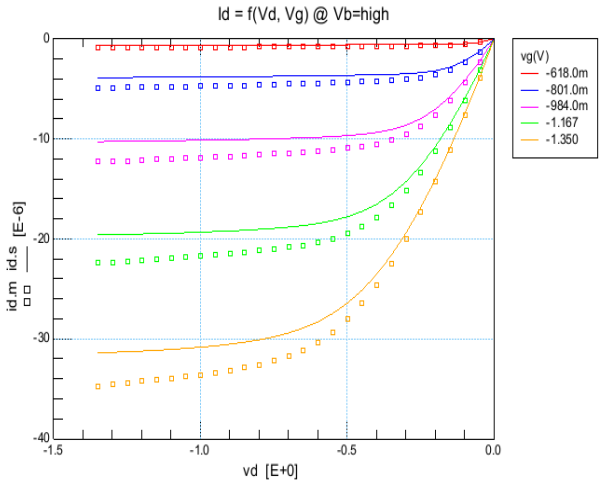
idvd, Vb = 0V, T = 27°C

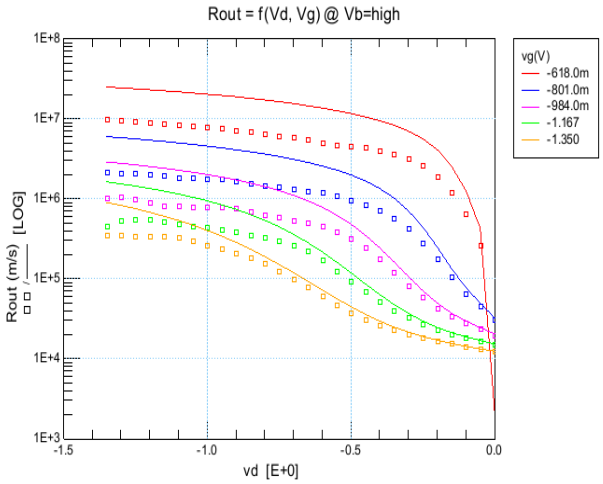


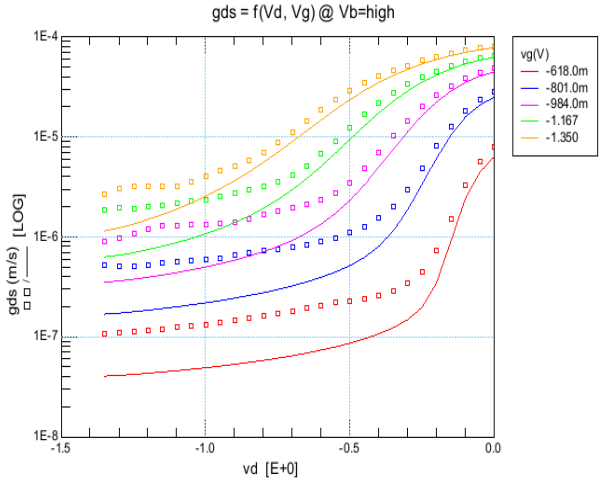




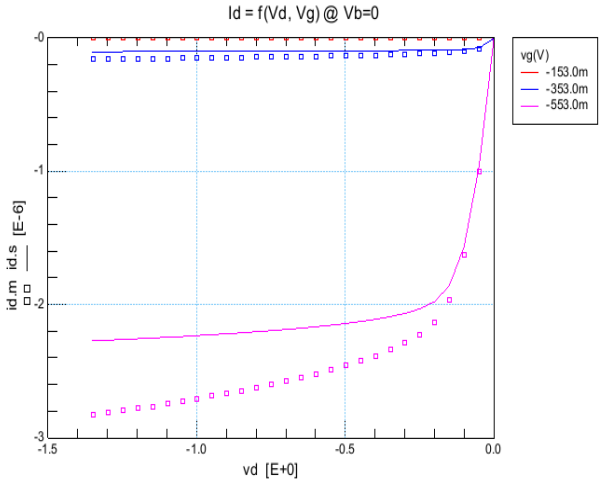
idvd_vbmin, Vb = 1.2V, T = 27°C

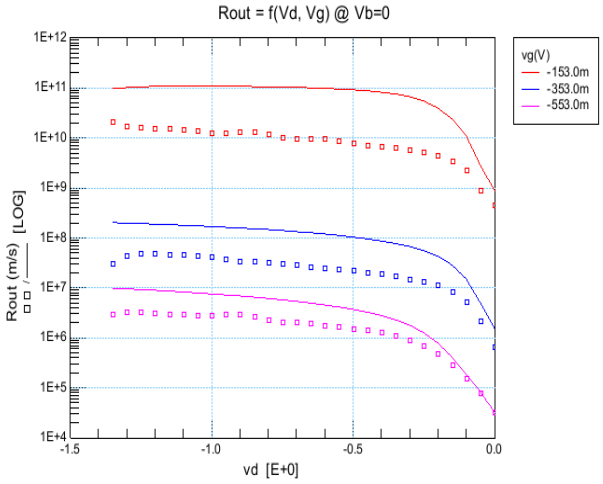


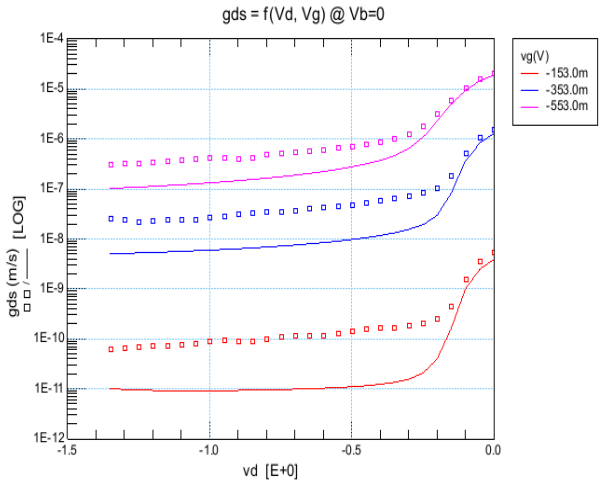




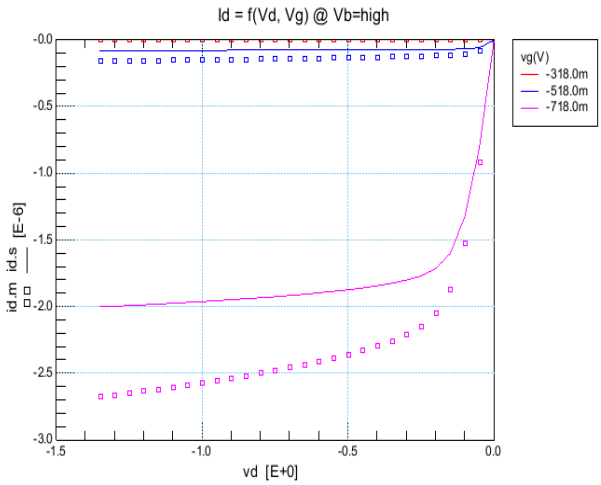
idvd_vth, Vb = 0V, T = 27°C

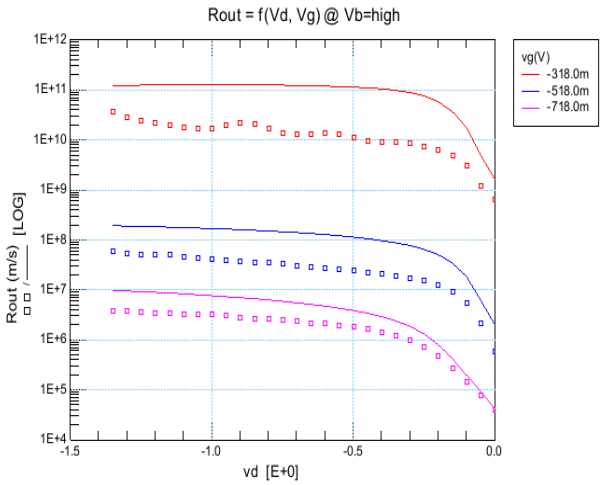


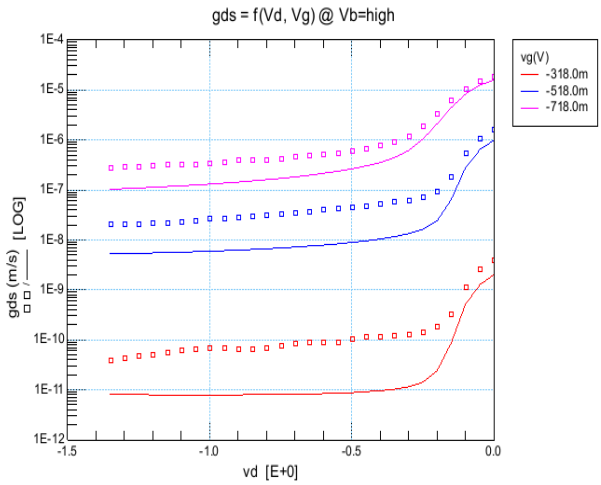




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

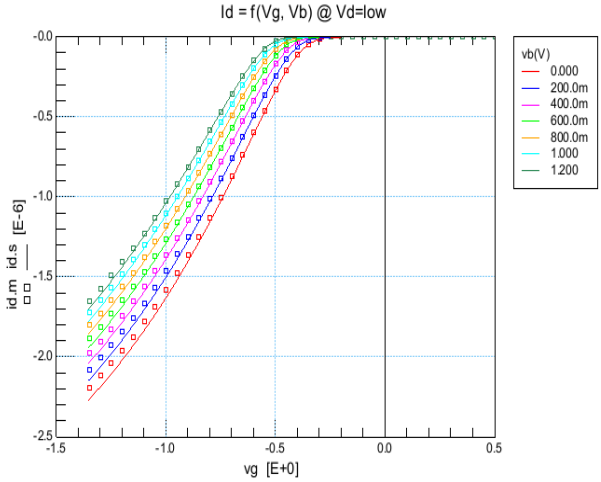


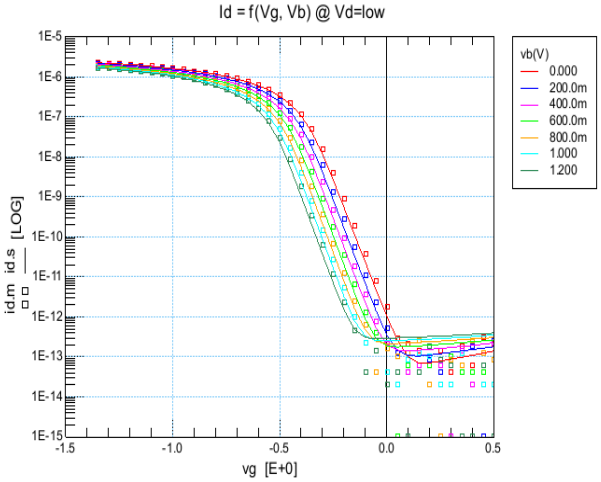


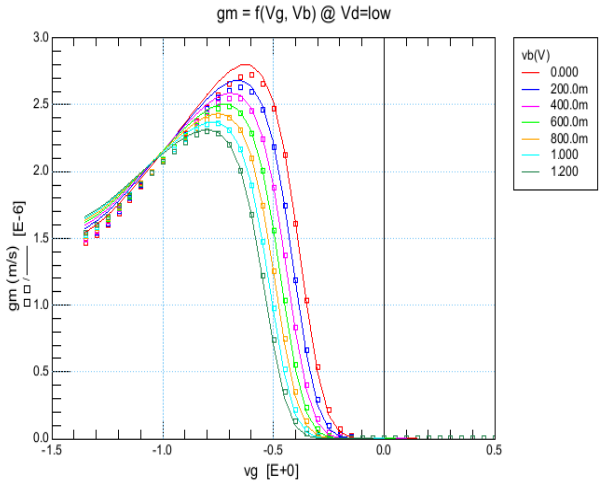


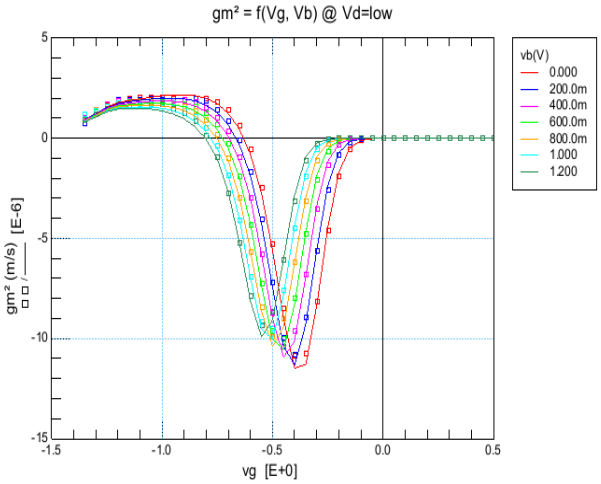
4.22 W0u6_L1u2_S551_5

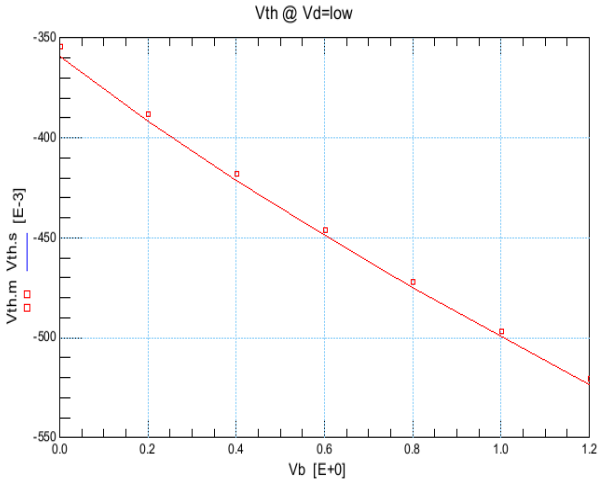
idvg, Vd = -0.05V, T = 27°C

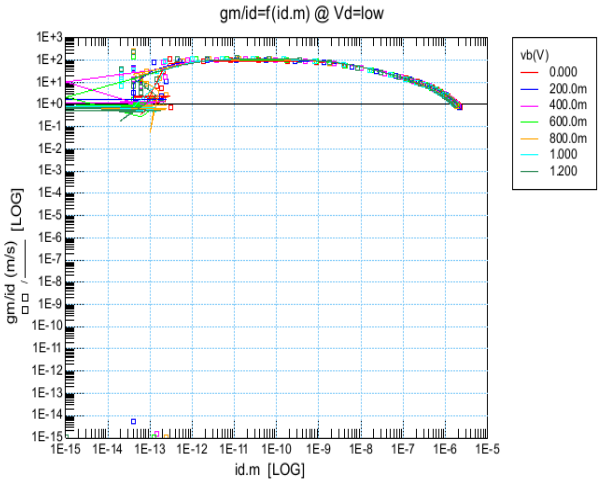


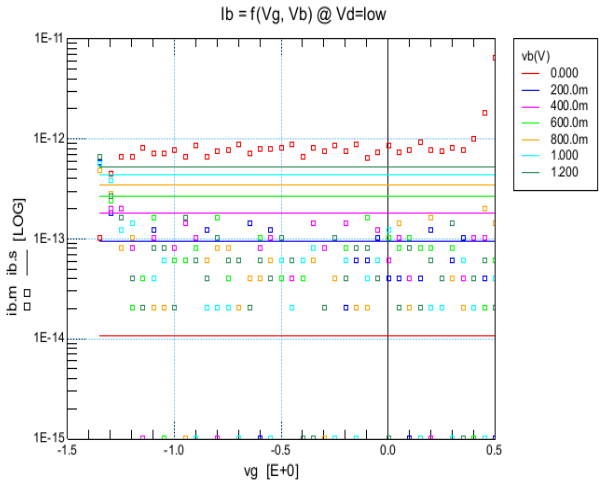


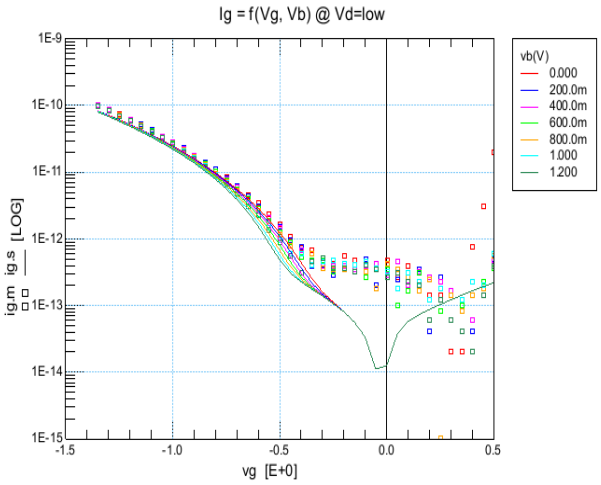




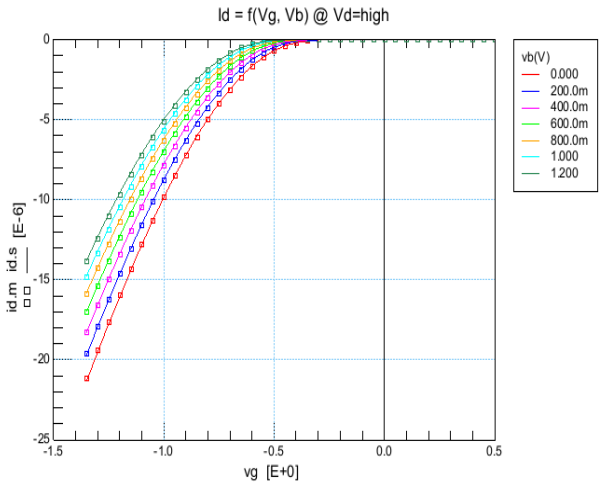


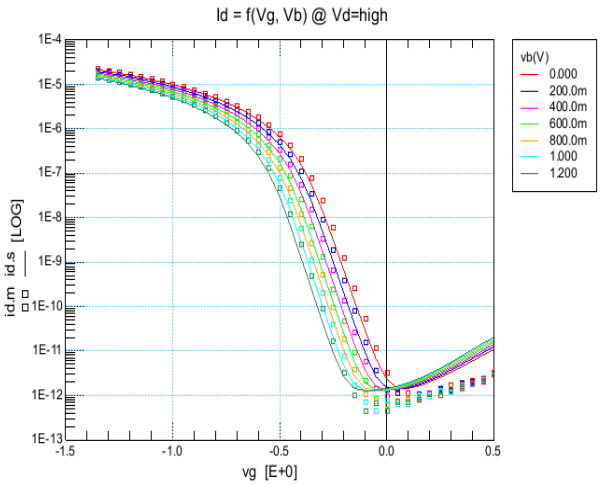


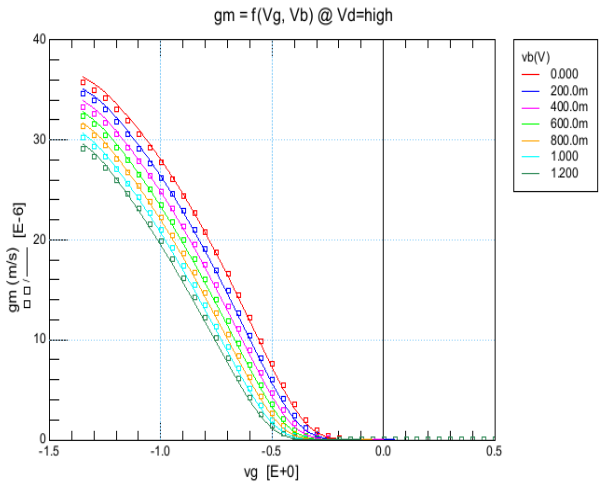


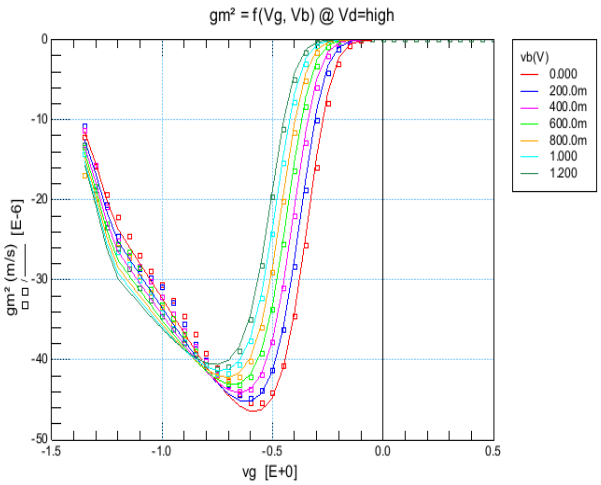


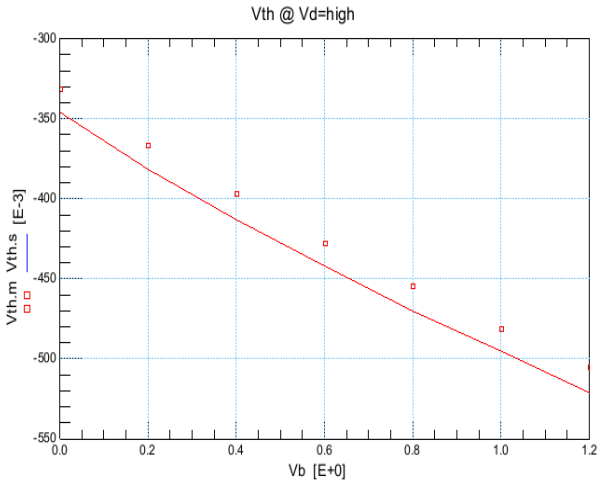
idvg, Vd = -1.2V, T = 27°C

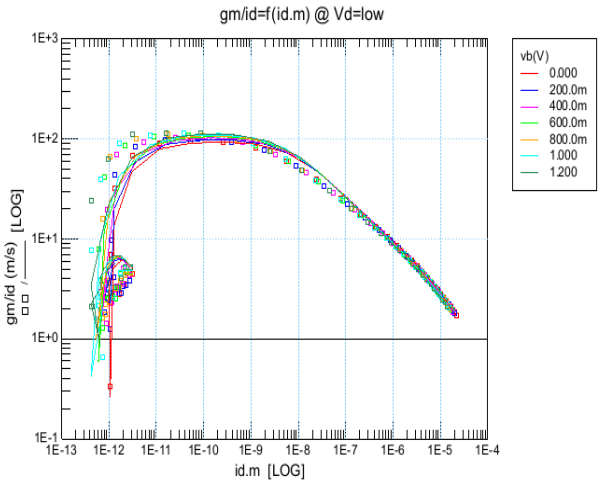


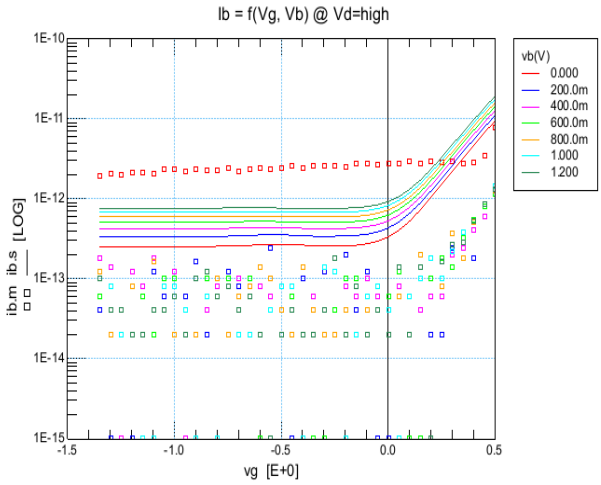


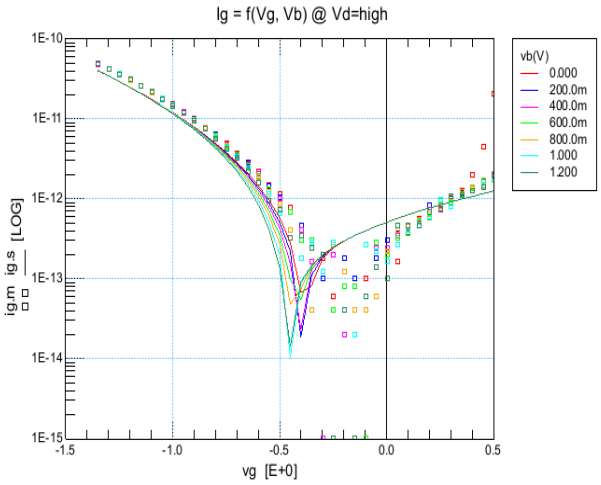




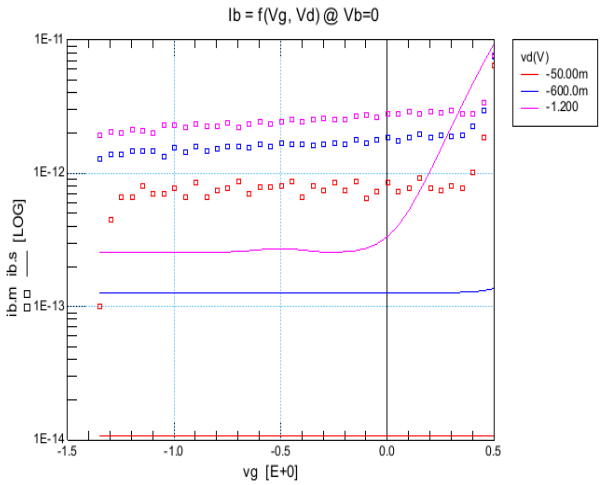


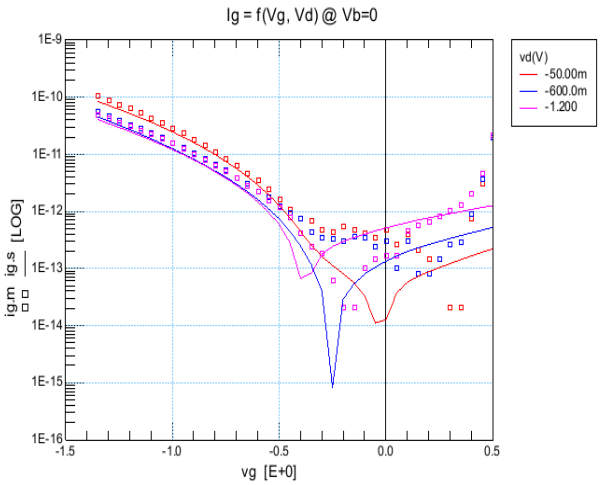




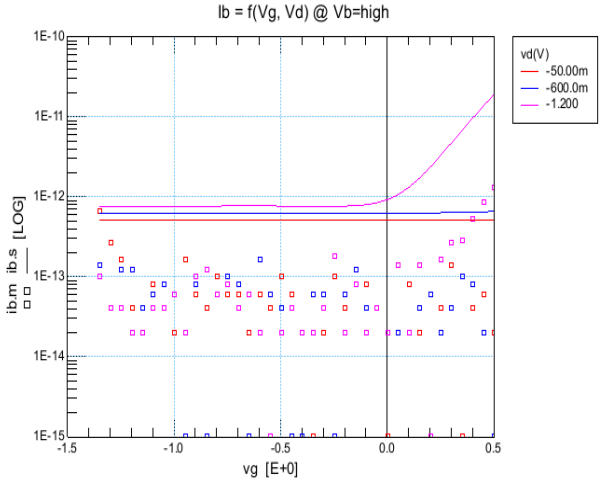


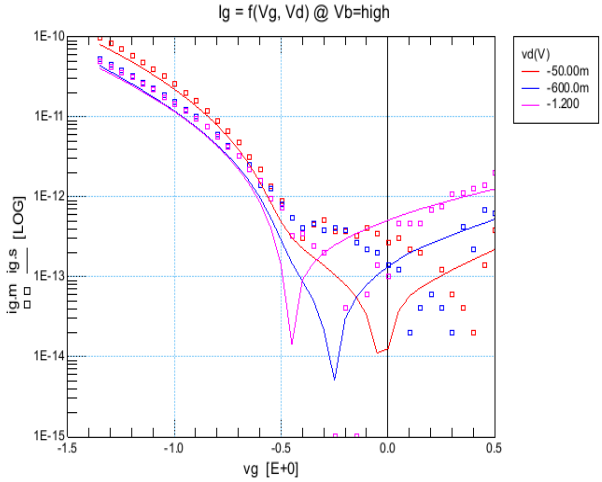
idvg, Ib,Ig, Vb = 0V, T = 27°C



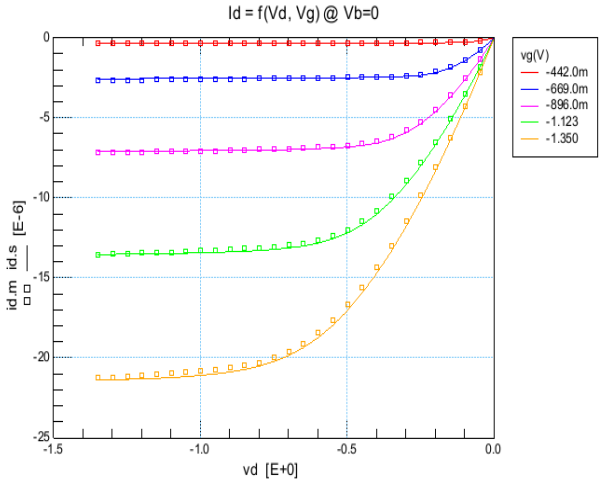


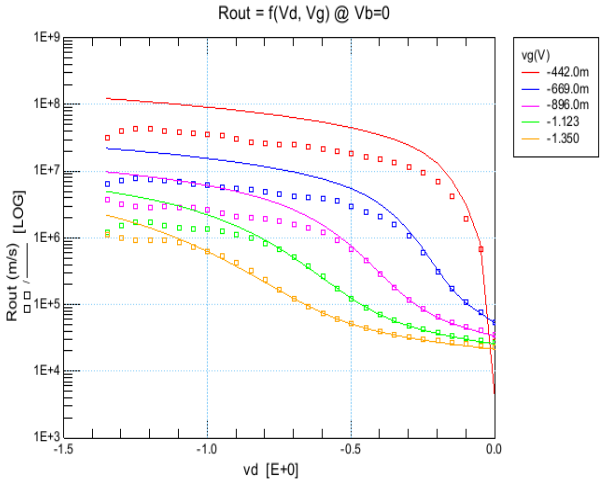
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

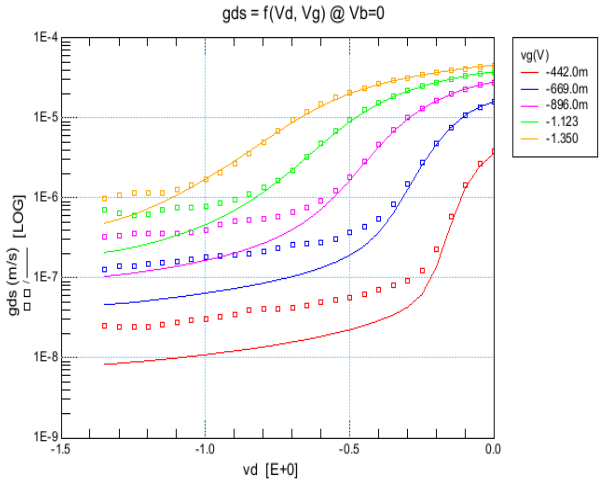




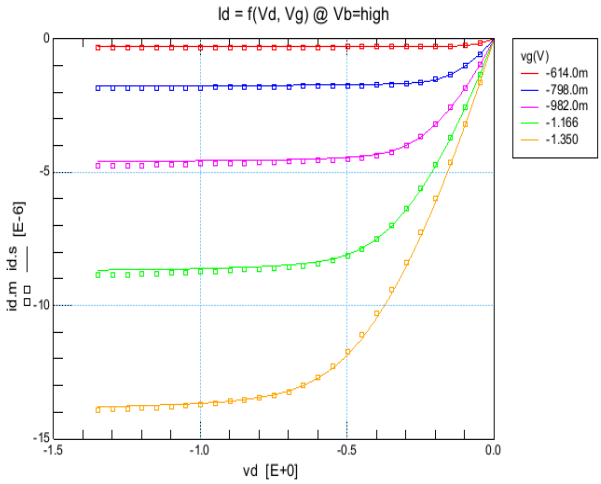
idvd, Vb = 0V, T = 27°C

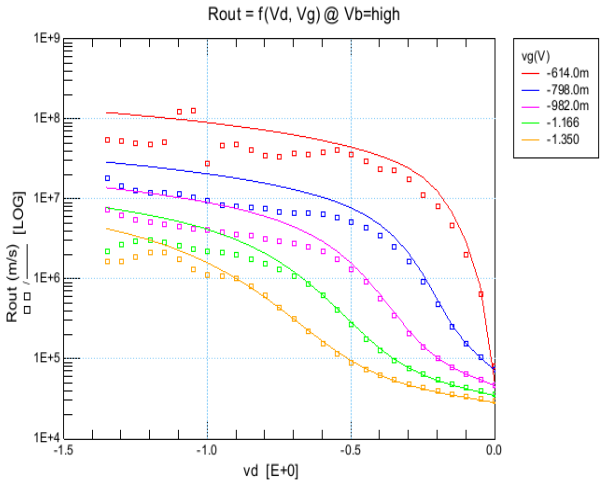


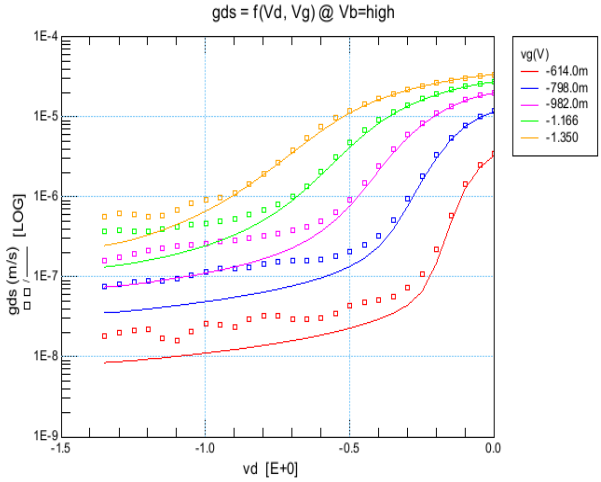




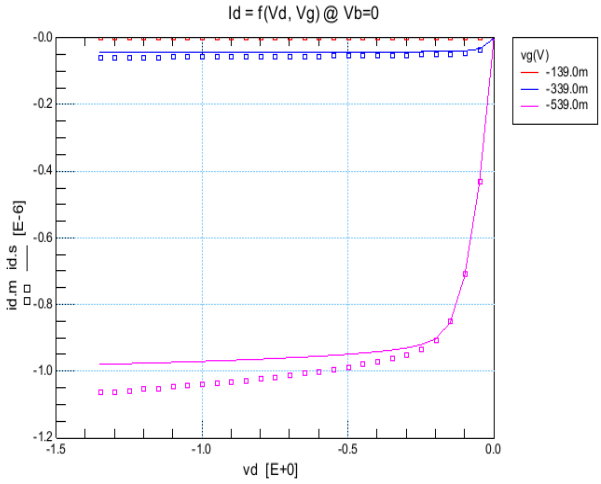
idvd_vbmin, Vb = 1.2V, T = 27°C

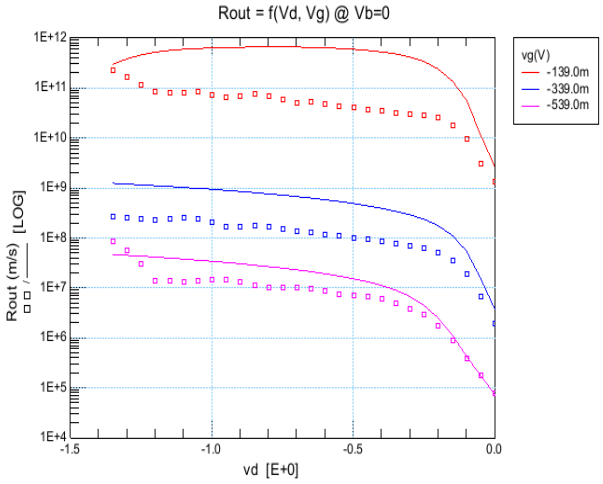


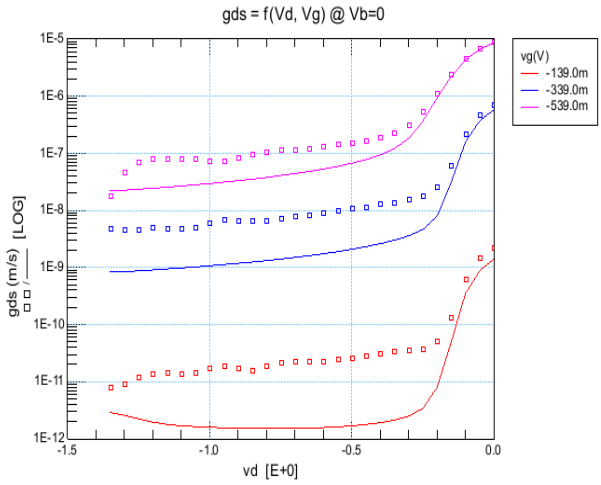




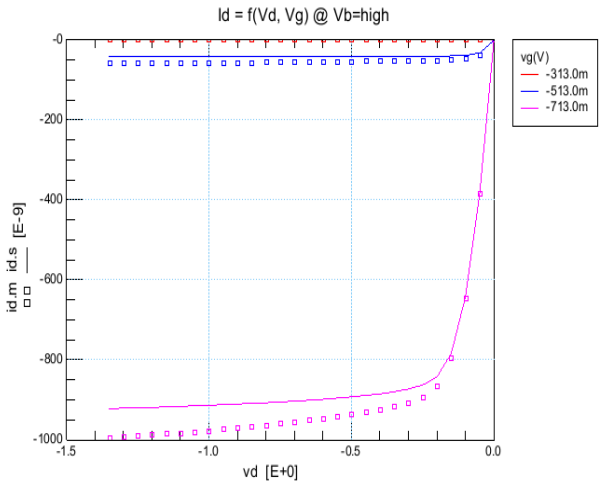
idvd_vth, Vb = 0V, T = 27°C

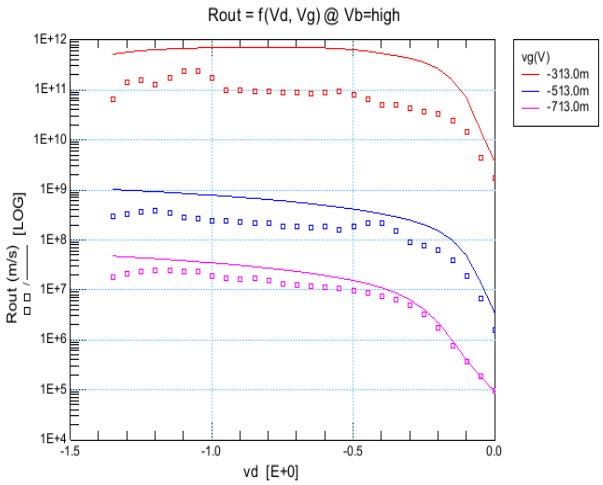


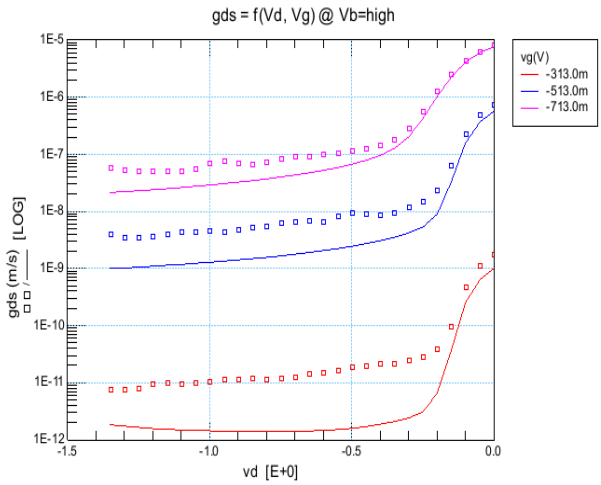




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

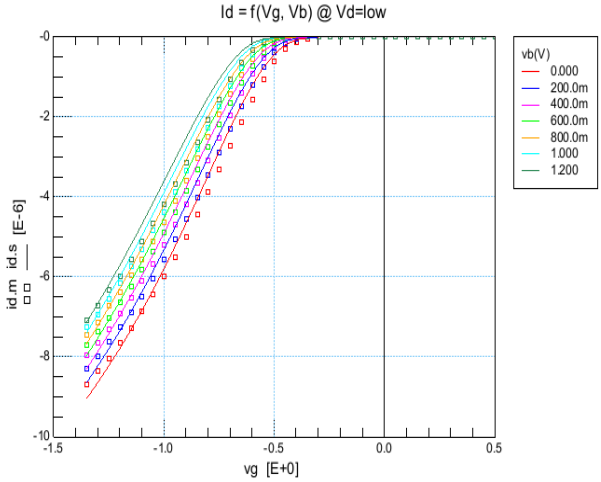


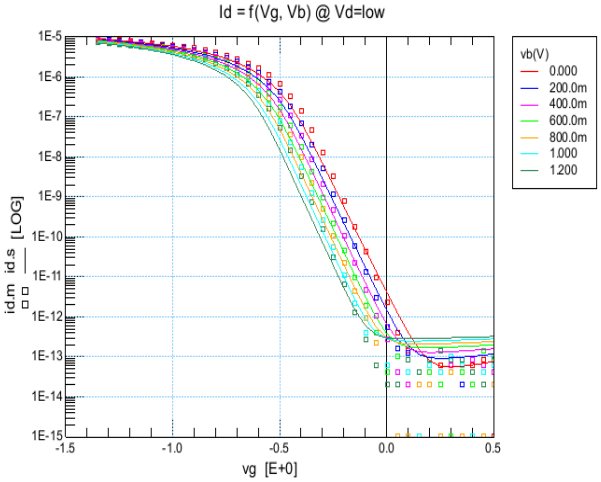


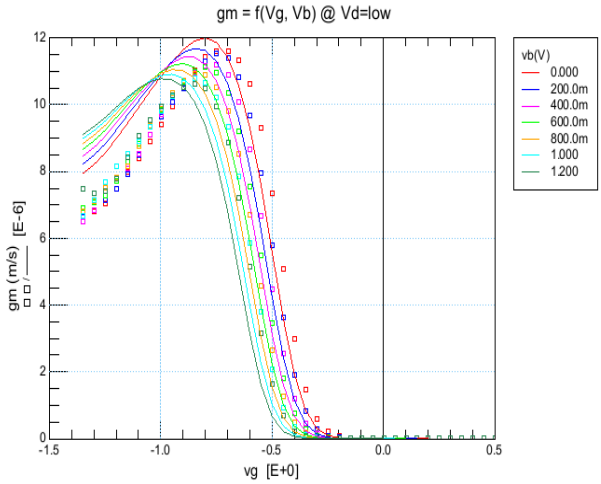


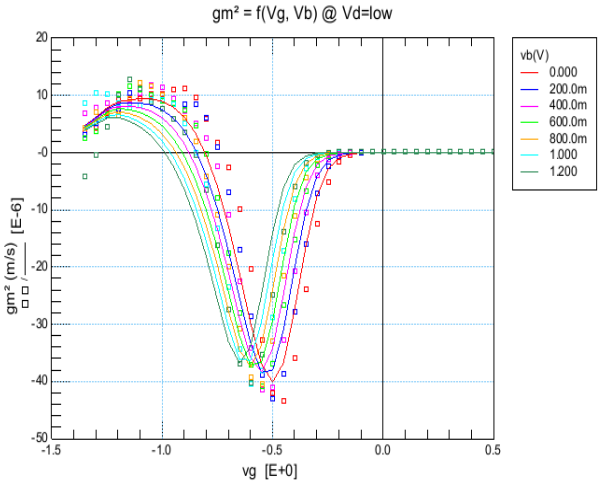
4.23 W0u3_L0u14_S552_3

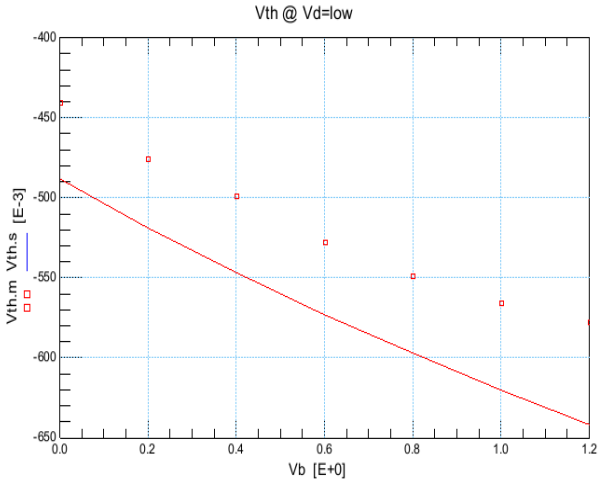
idvg, Vd = -0.05V, T = 27°C

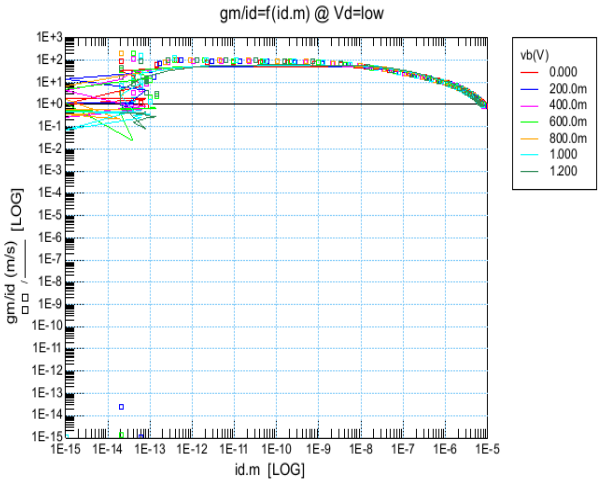


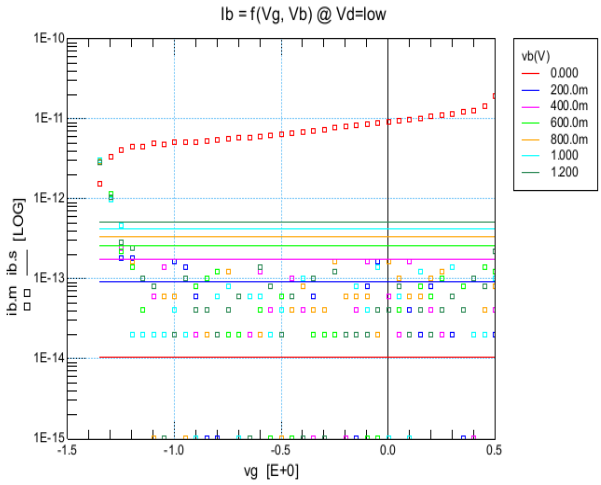


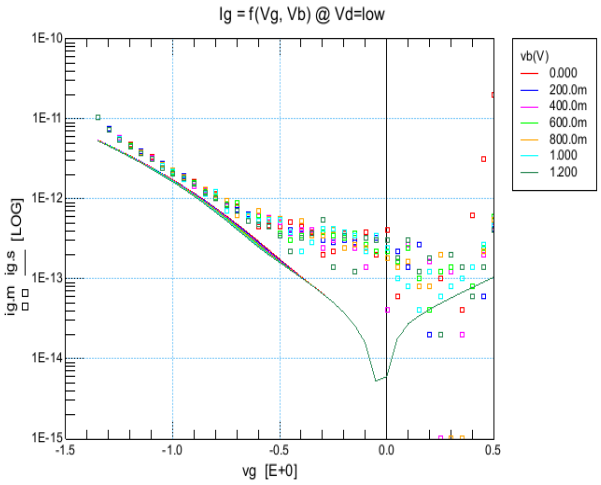




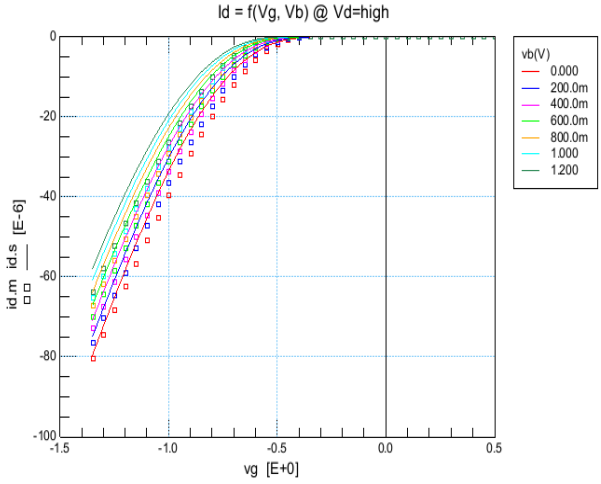


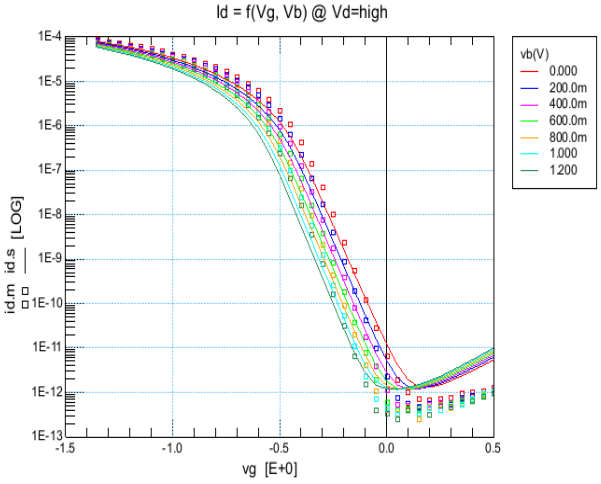


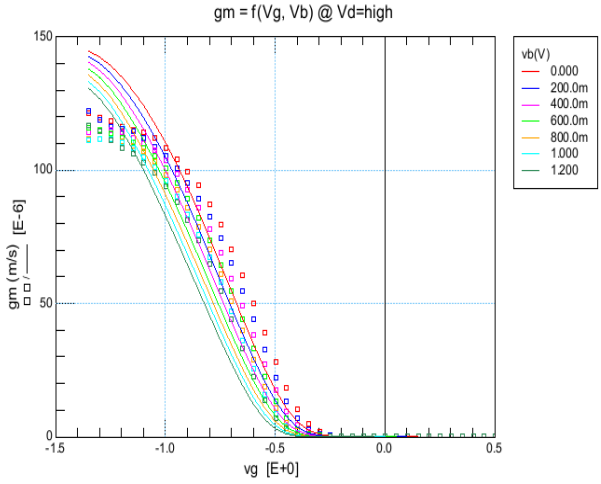


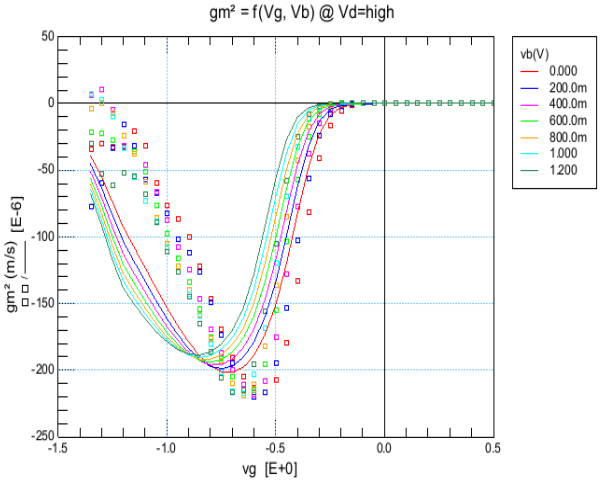


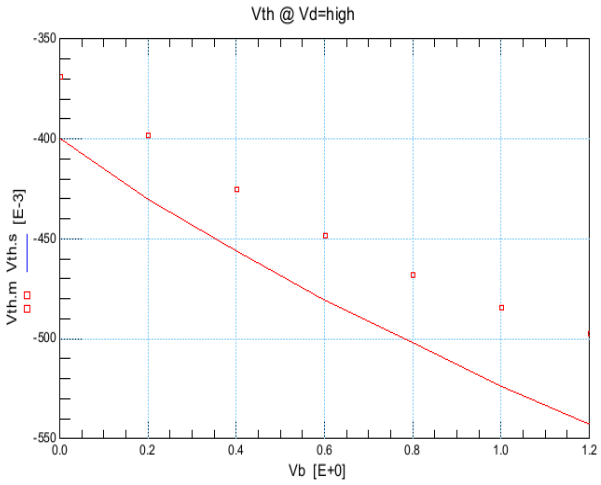
idvg, Vd = -1.2V, T = 27°C

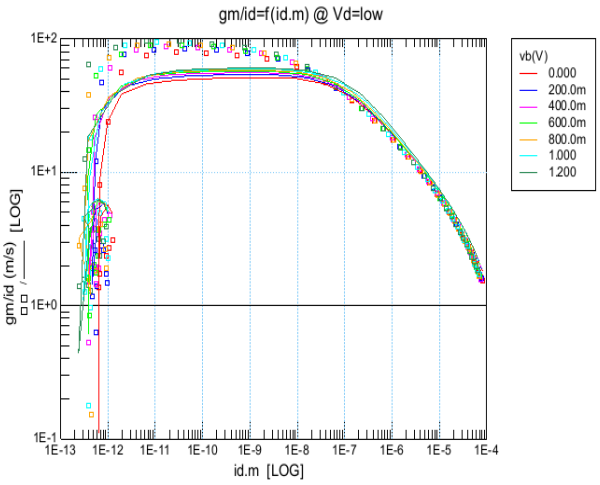


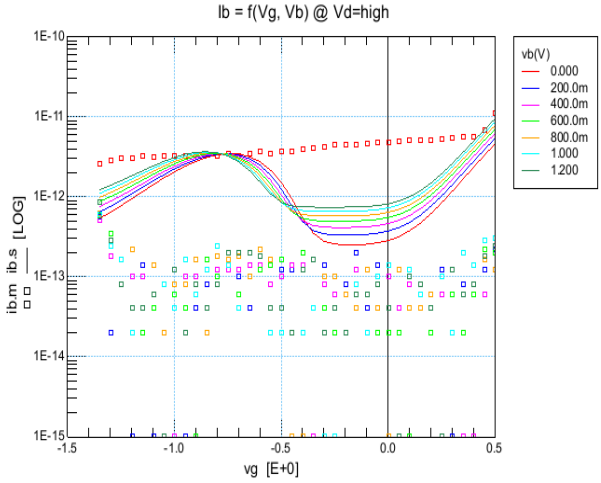


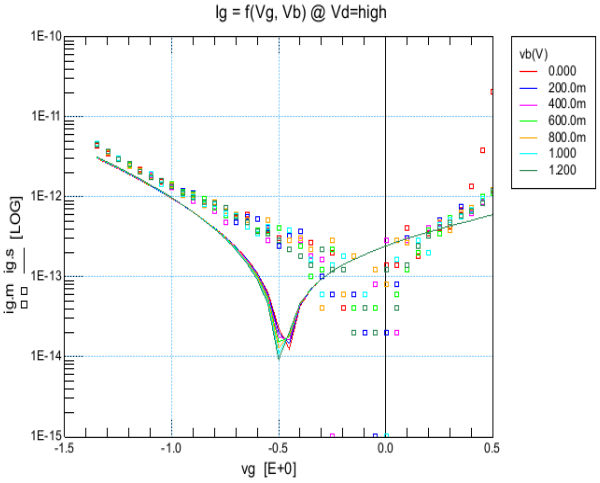




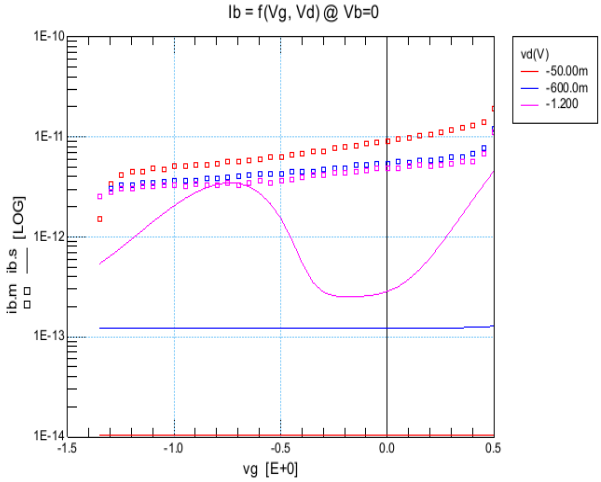


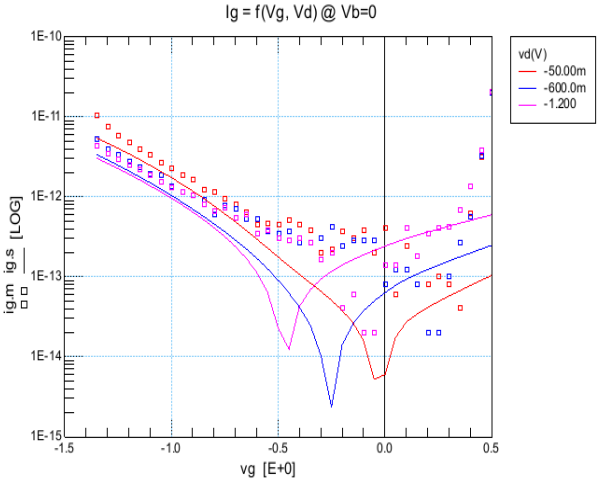




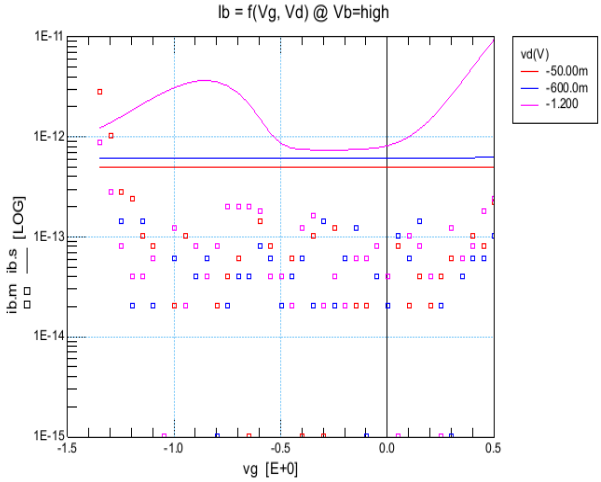


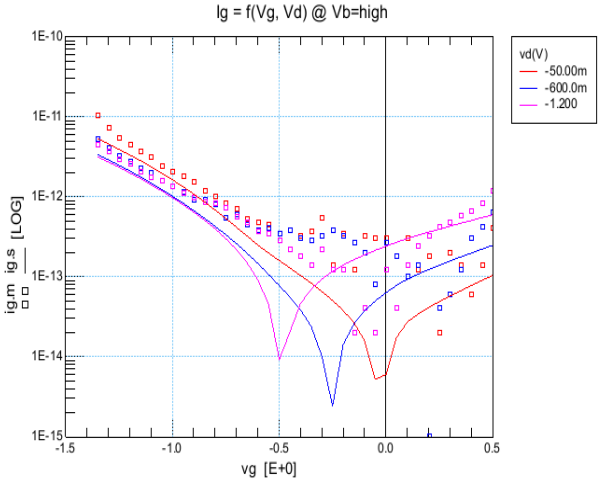
idvg, Ib,Ig, Vb = 0V, T = 27°C



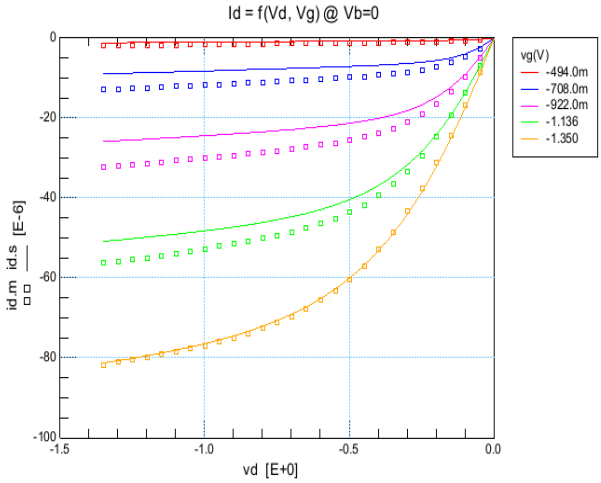


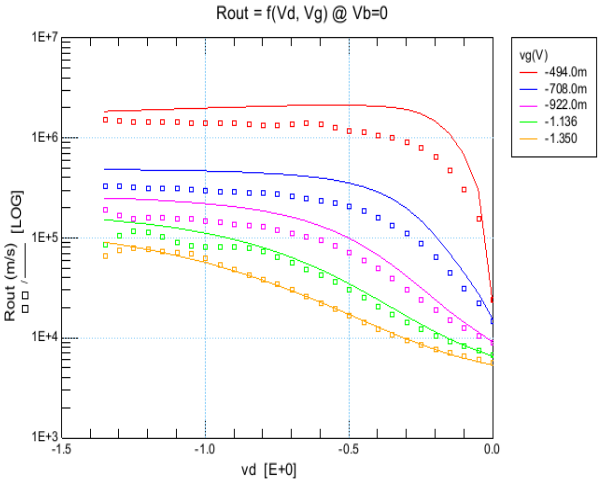
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

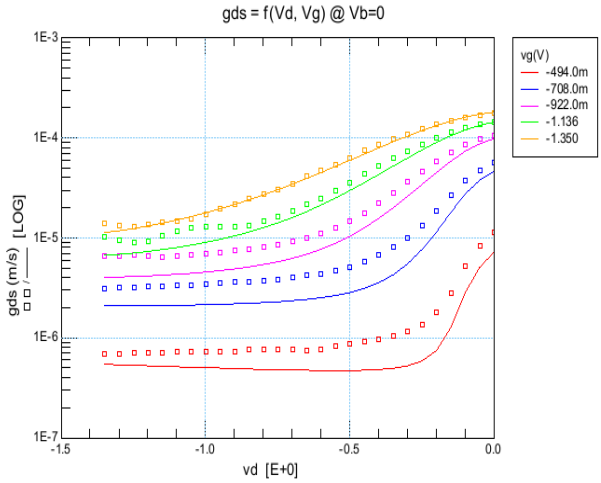




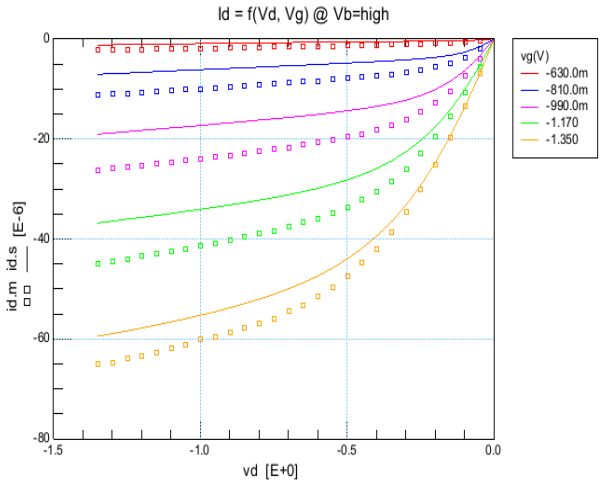
idvd, Vb = 0V, T = 27°C

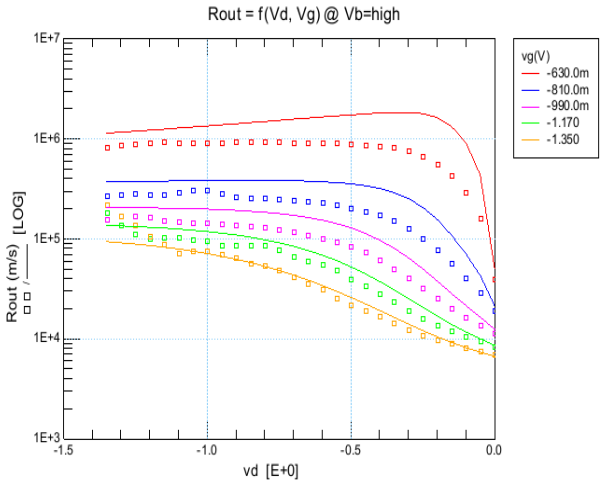


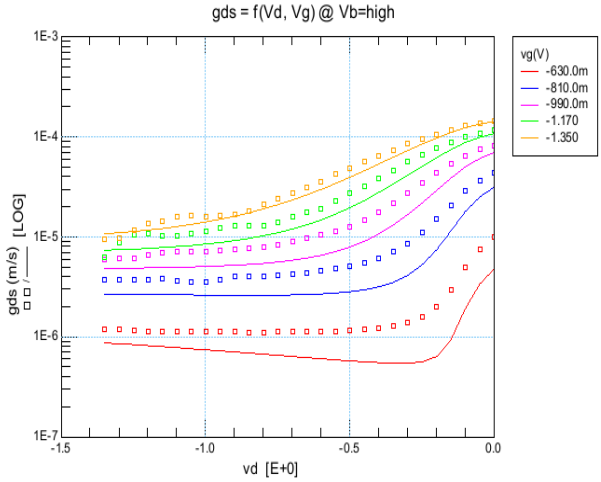




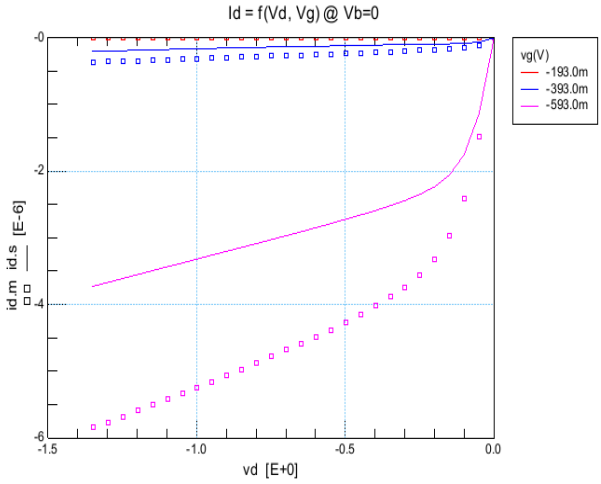
idvd_vbmin, Vb = 1.2V, T = 27°C

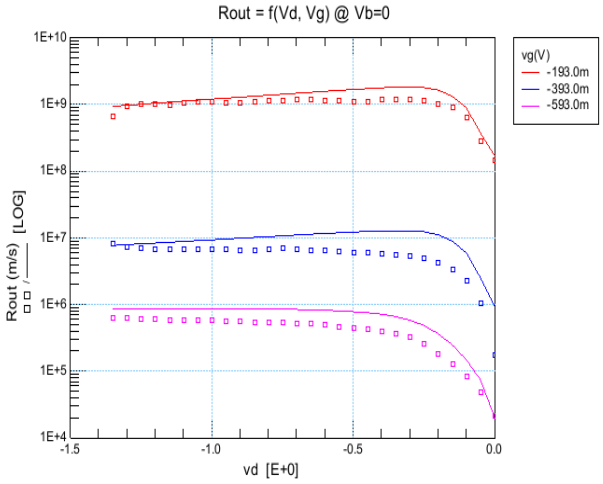


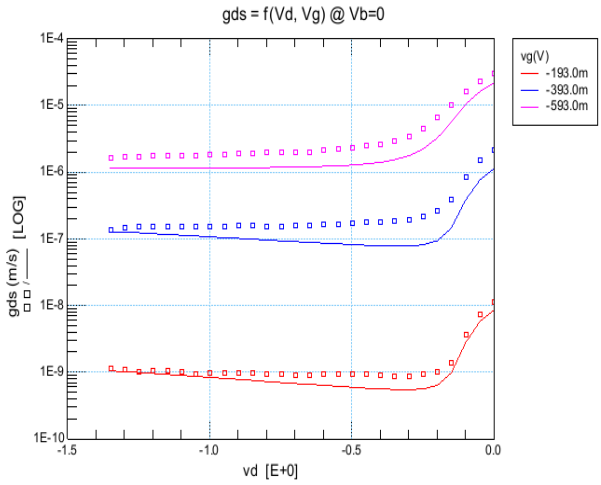




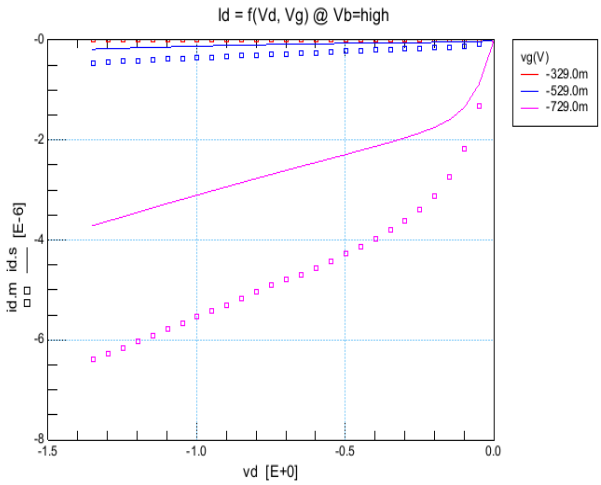
idvd_vth, Vb = 0V, T = 27°C

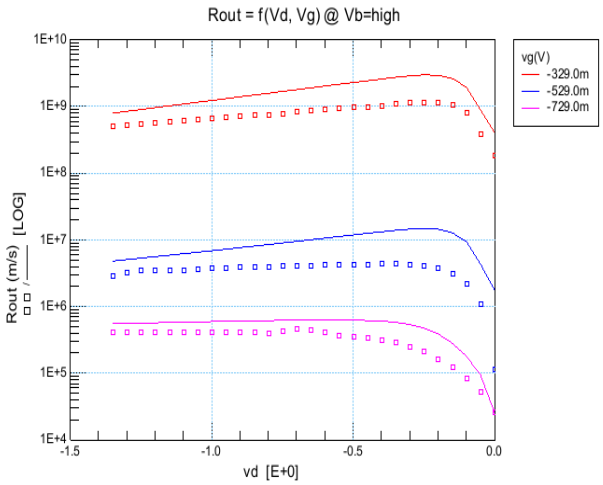


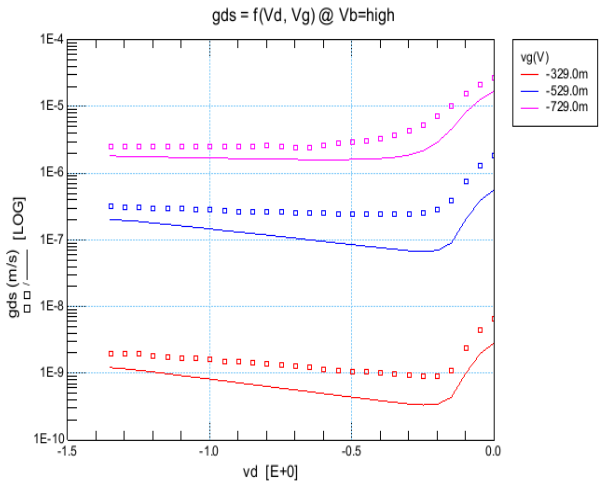




idvd_vth_vbmin, Vb = 1.2V, T = 27°C



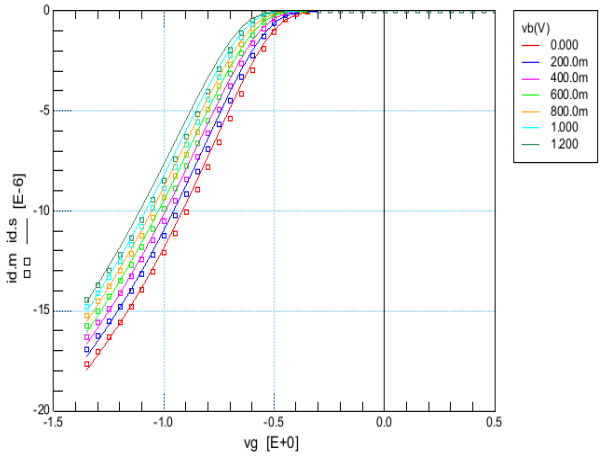




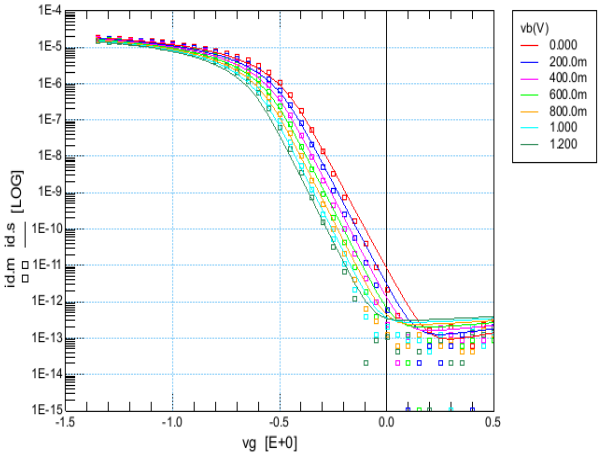
4.24 W0u6_L0u13_S551_2

idvg, Vd = -0.05V, T = 27°C

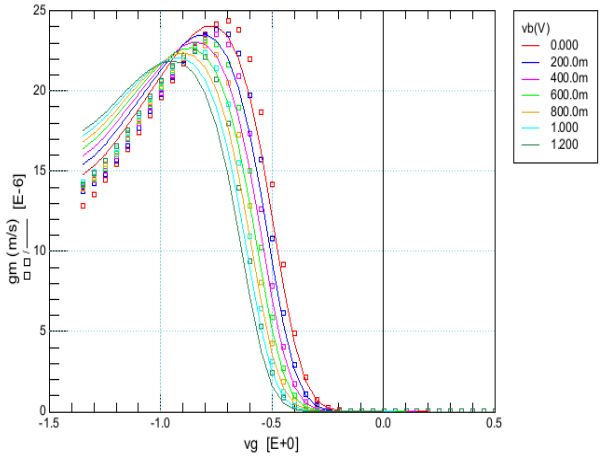
Id = f(Vg, Vb) @ Vd=low



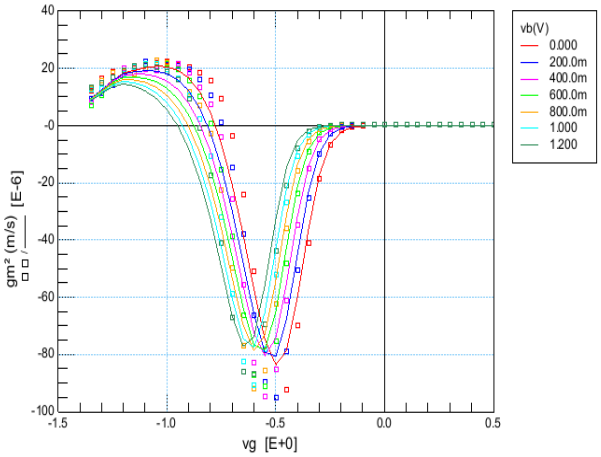
Id = f(Vg, Vb) @ Vd=low



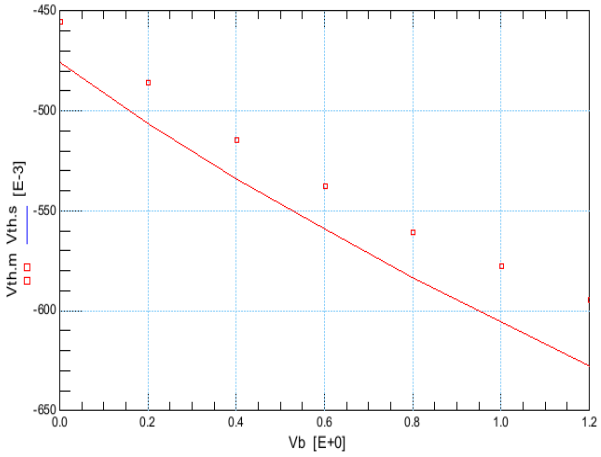
gm = f(Vg, Vb) @ Vd=low



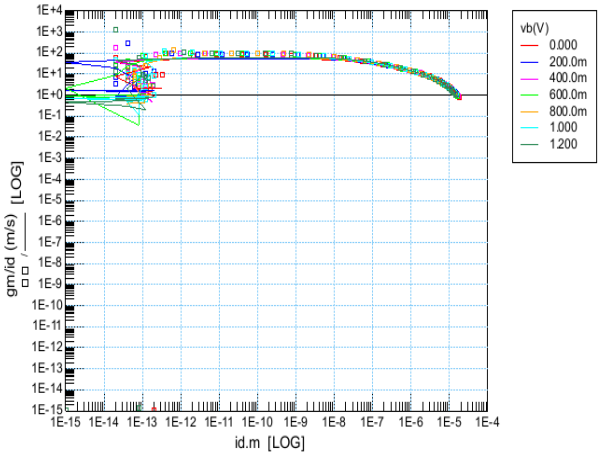
gm² = f(Vg, Vb) @ Vd=low

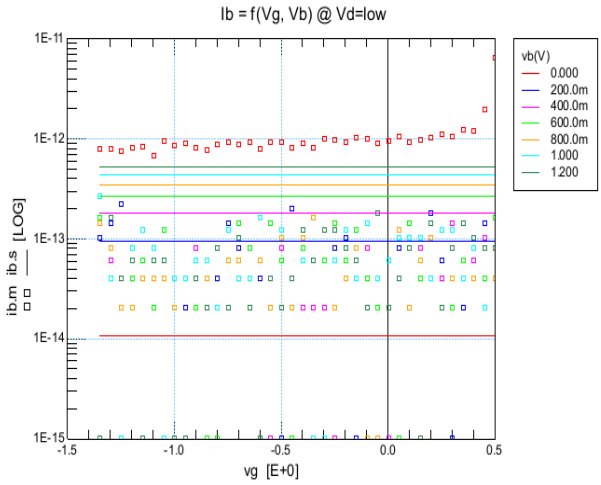


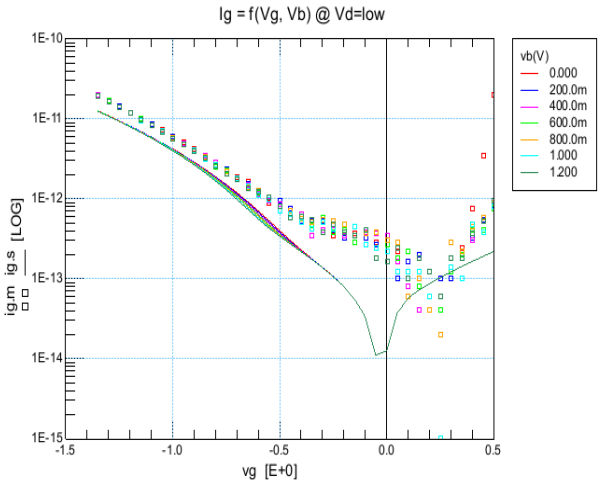
Vth @ Vd=low



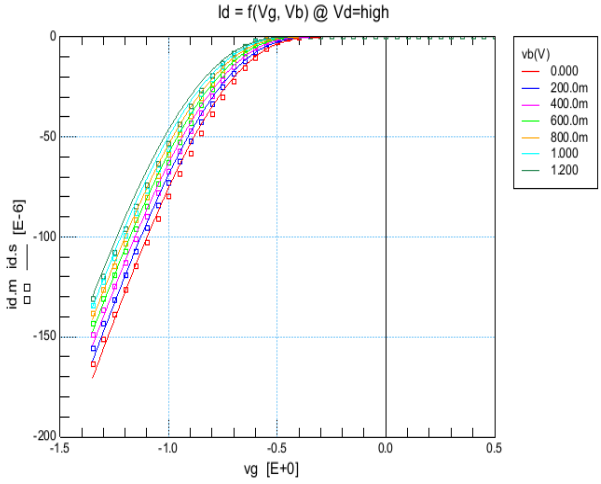
gm/id=f(id.m) @ Vd=low

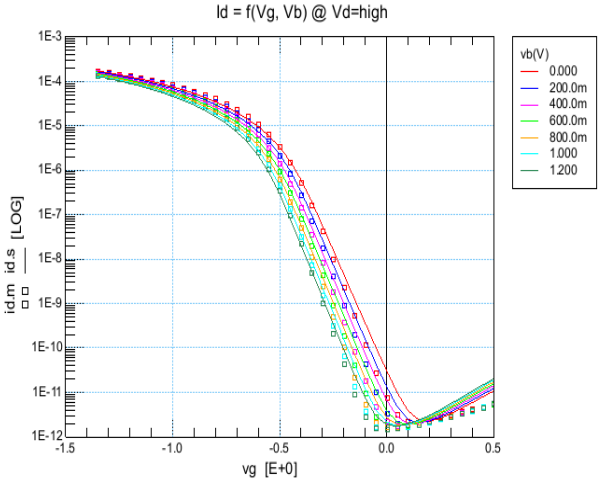


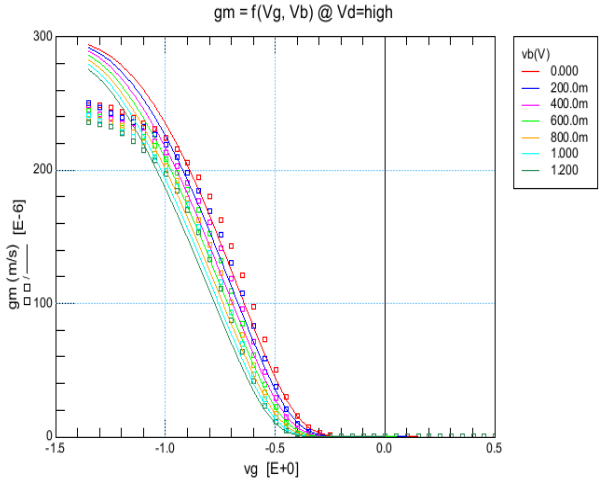


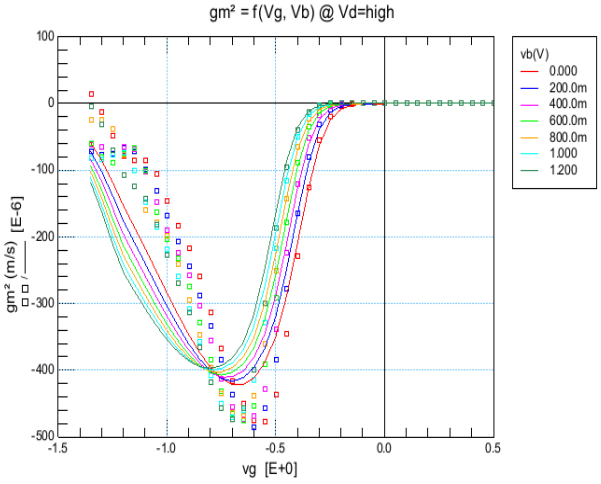


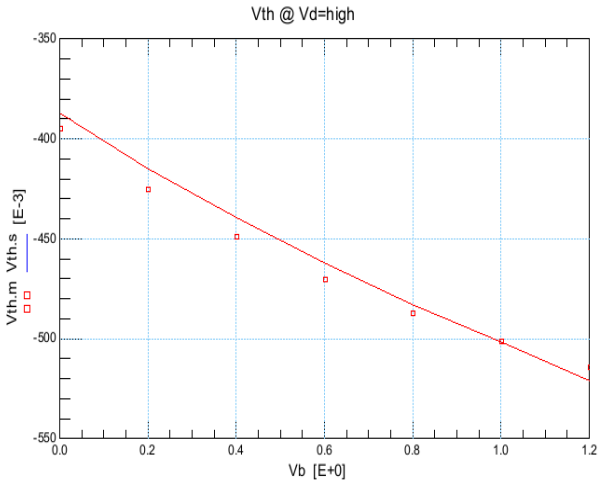
idvg, Vd = -1.2V, T = 27°C

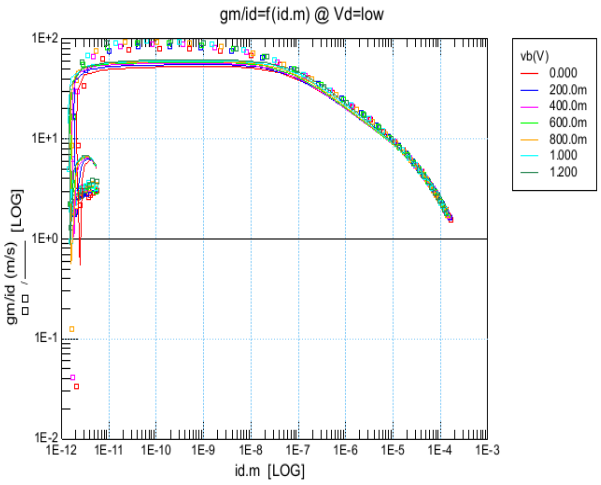


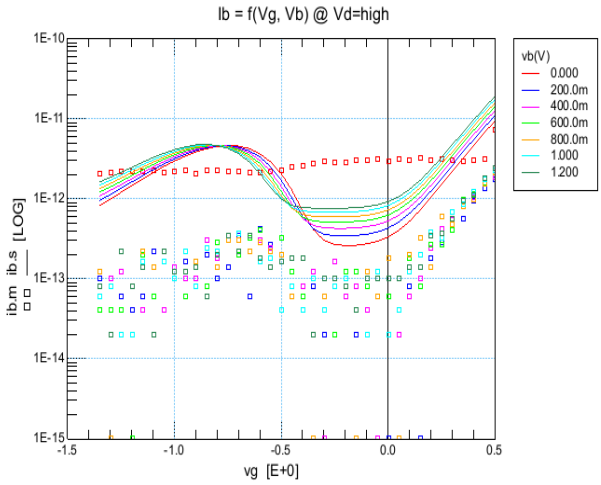


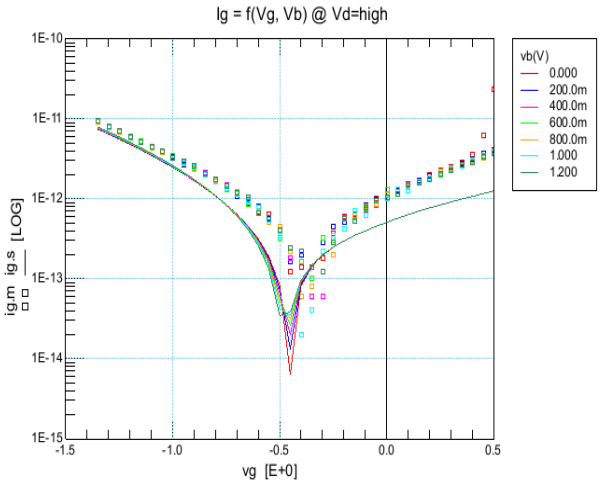




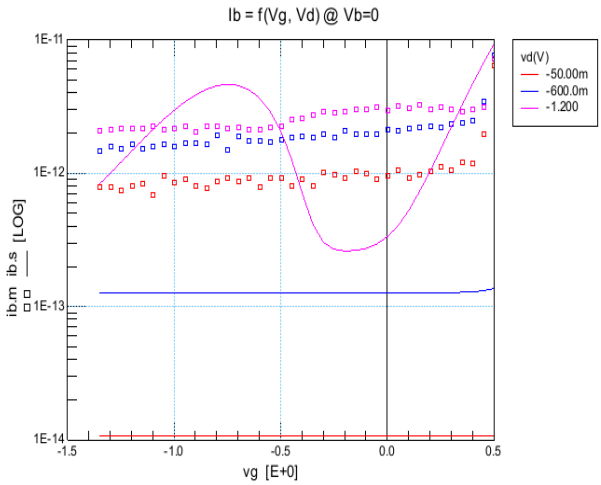


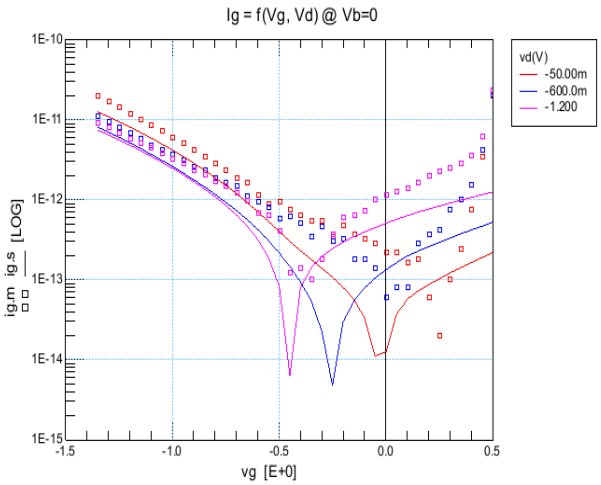




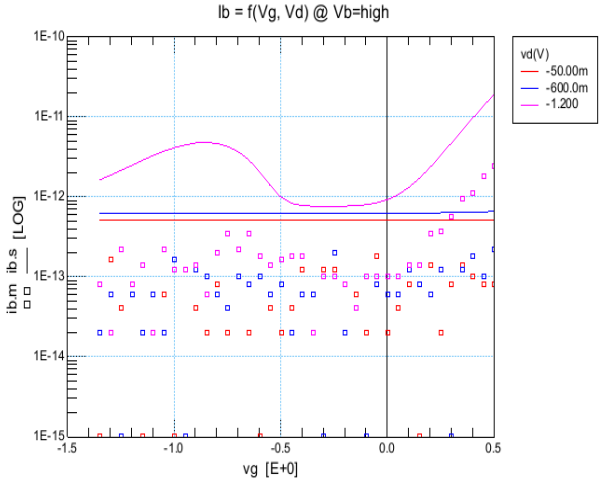


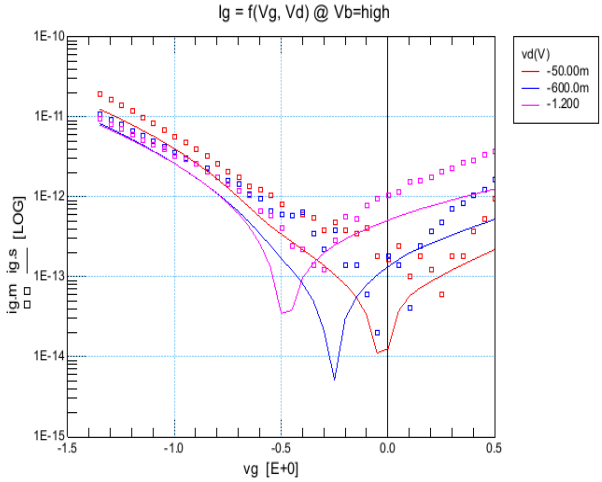
idvg, Ib,Ig, Vb = 0V, T = 27°C



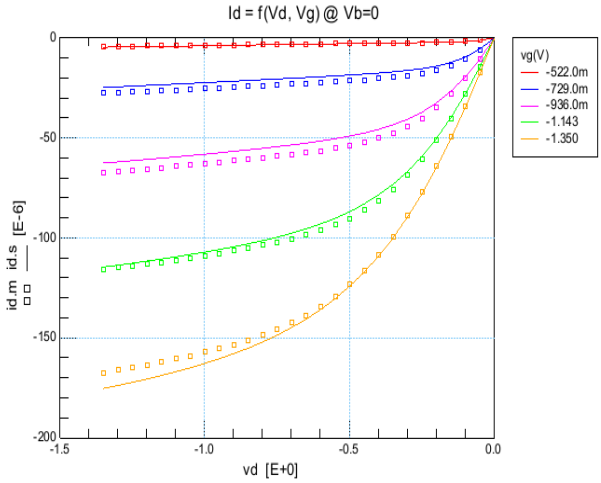


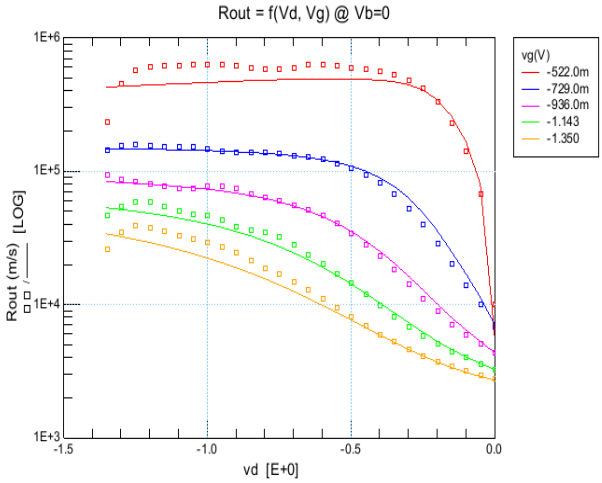
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

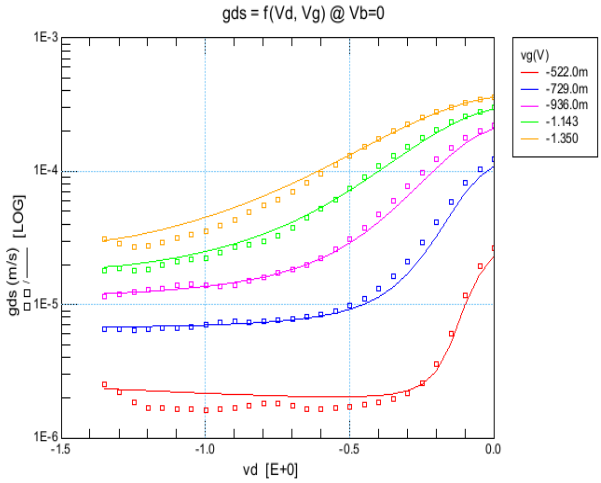




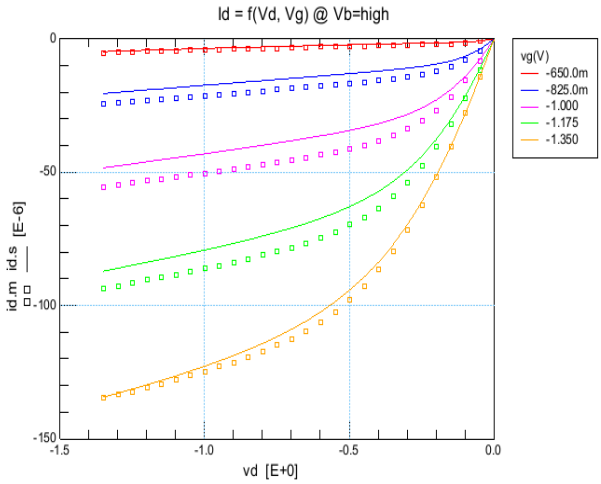
idvd, Vb = 0V, T = 27°C

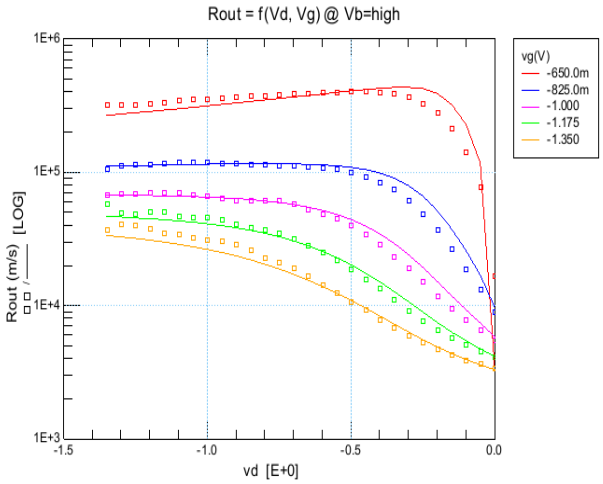


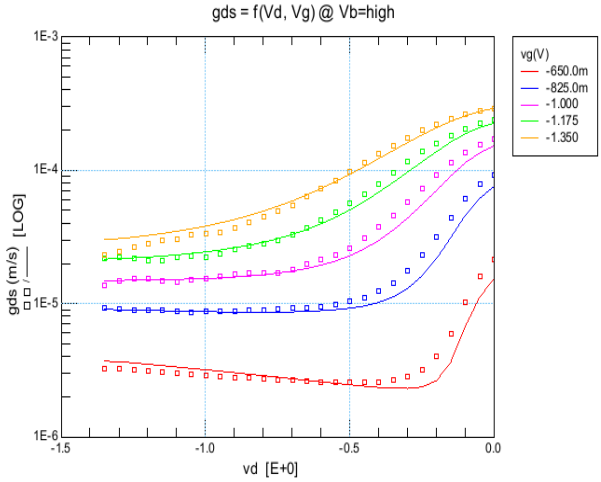




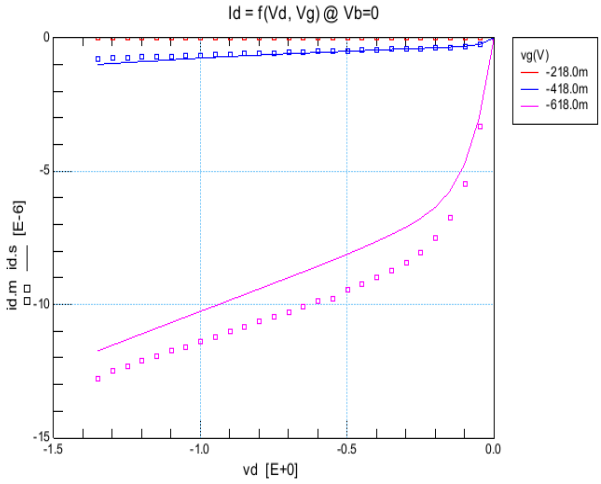
idvd_vbmin, Vb = 1.2V, T = 27°C

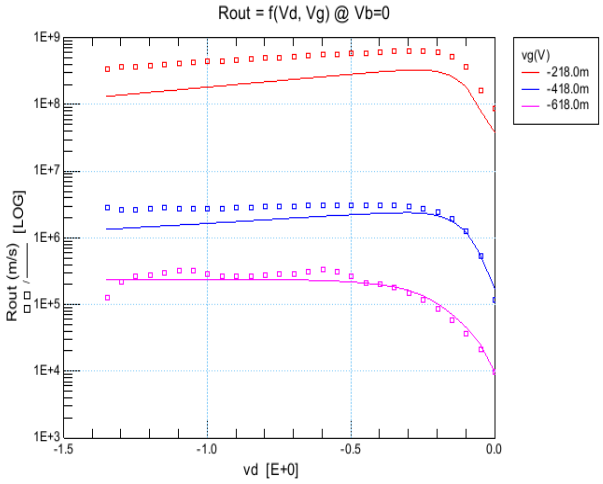


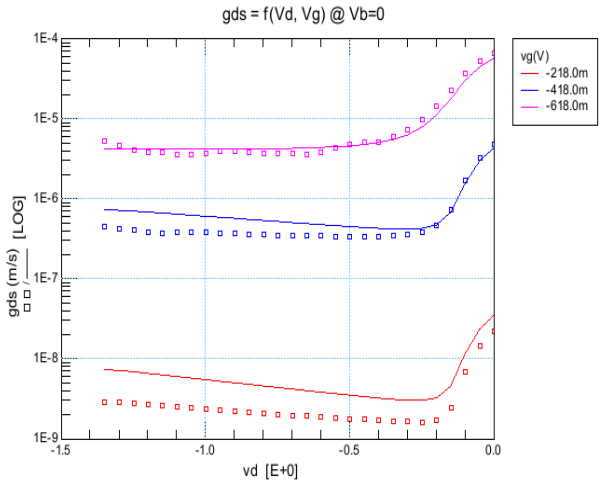




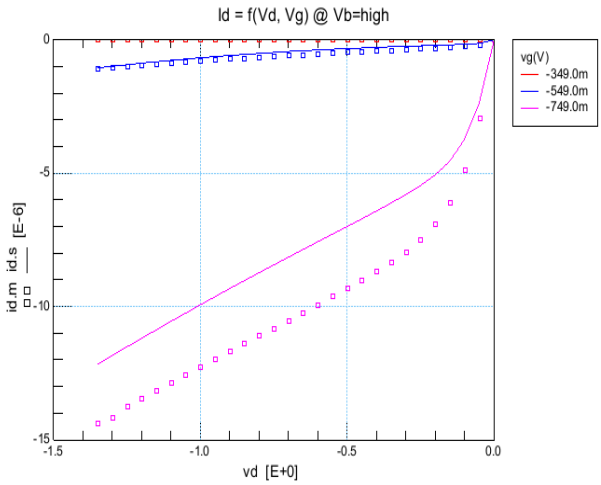
idvd_vth, Vb = 0V, T = 27°C

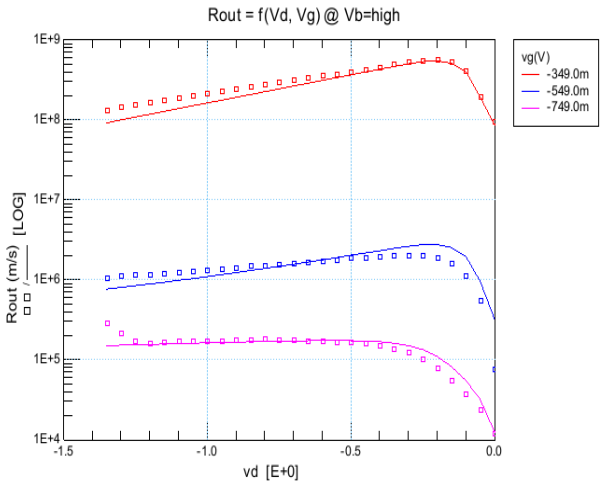


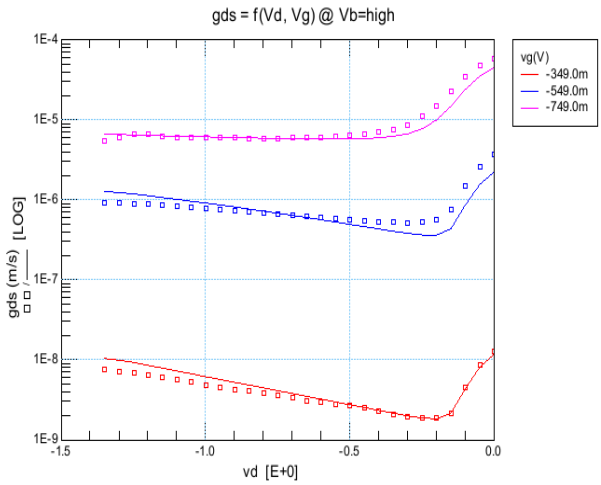




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

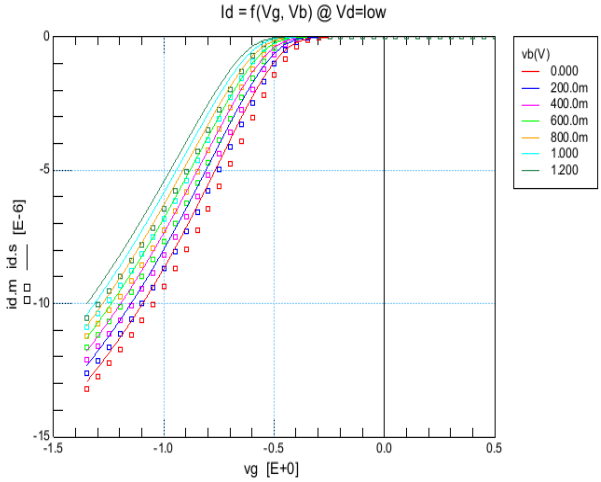


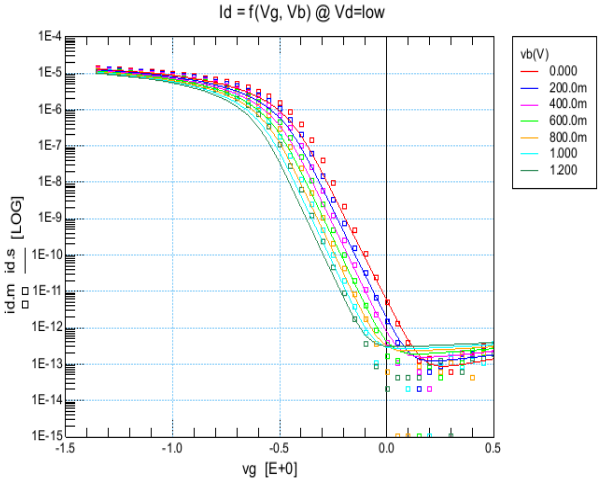


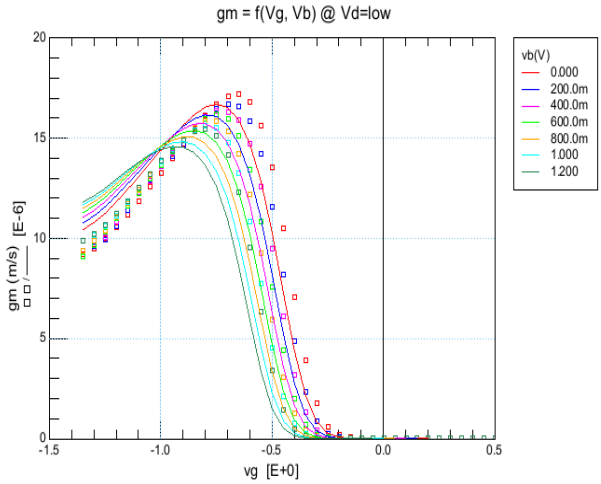


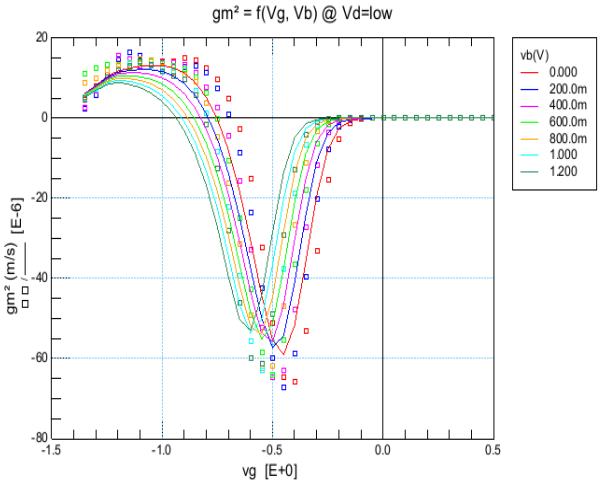
4.25W0u6_L0u18_S551_3

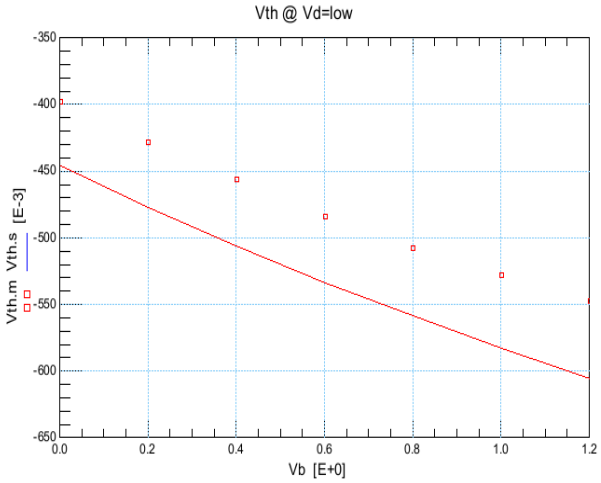
idvg, Vd = -0.05V, T = 27°C

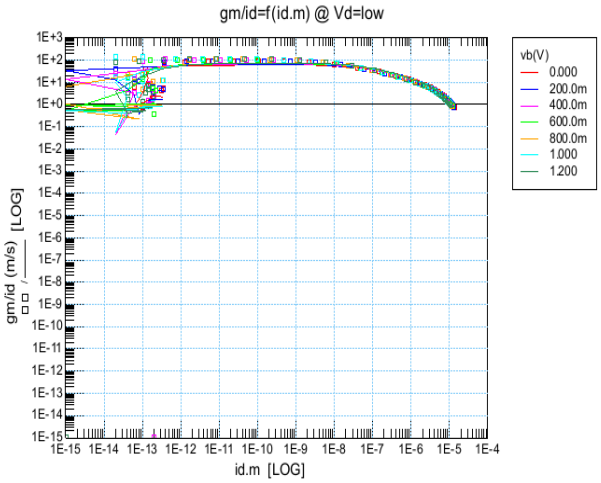


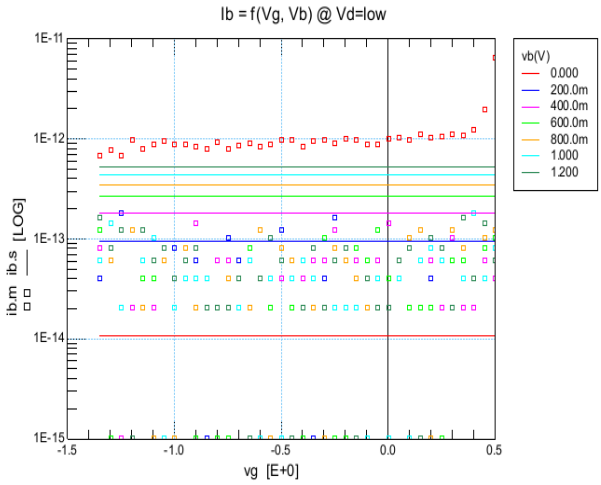


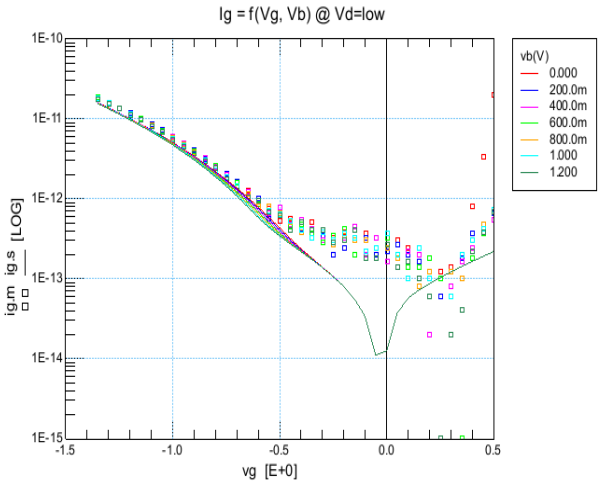




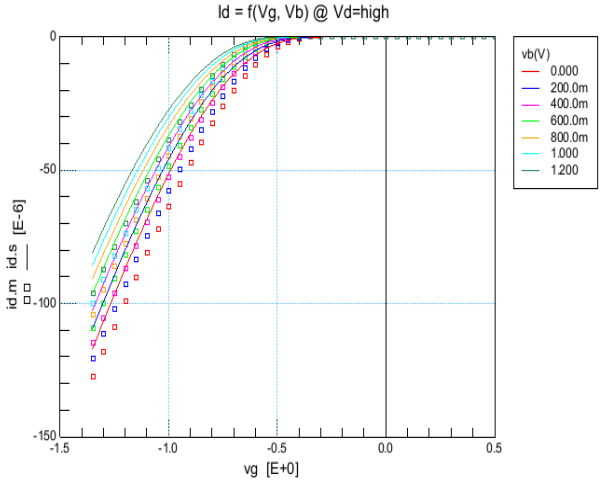


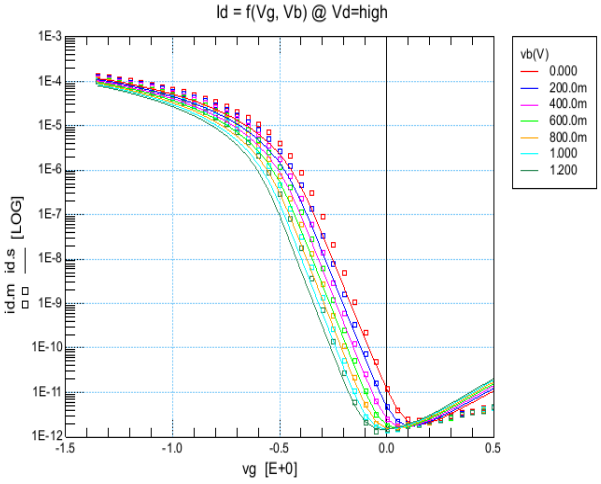


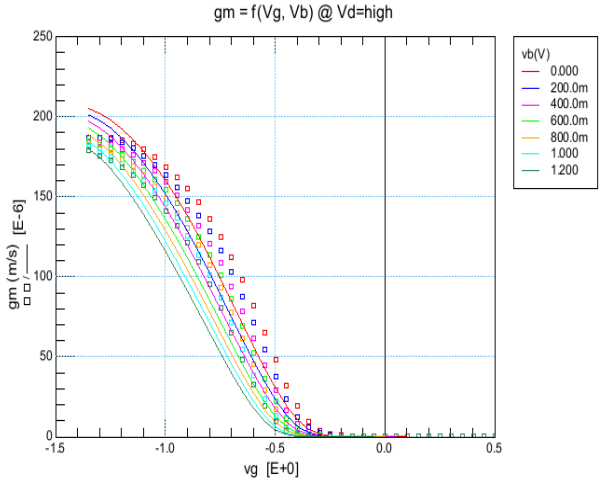


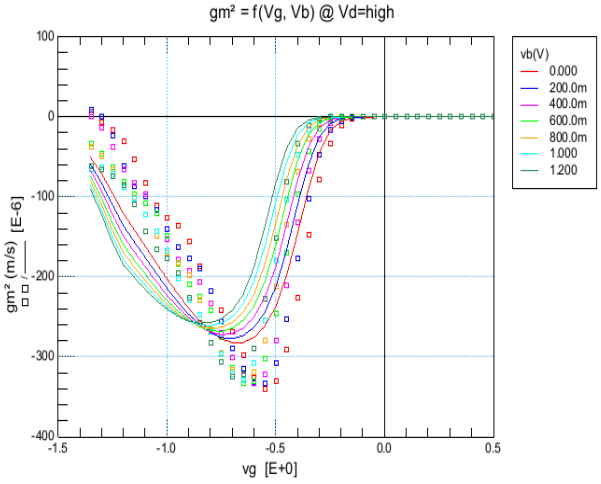


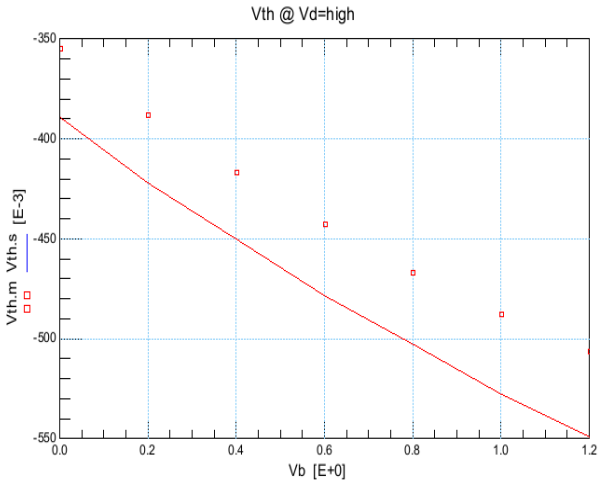
idvg, Vd = -1.2V, T = 27°C

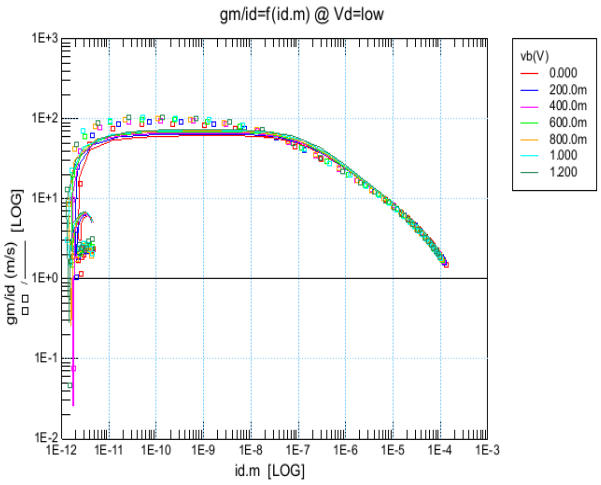


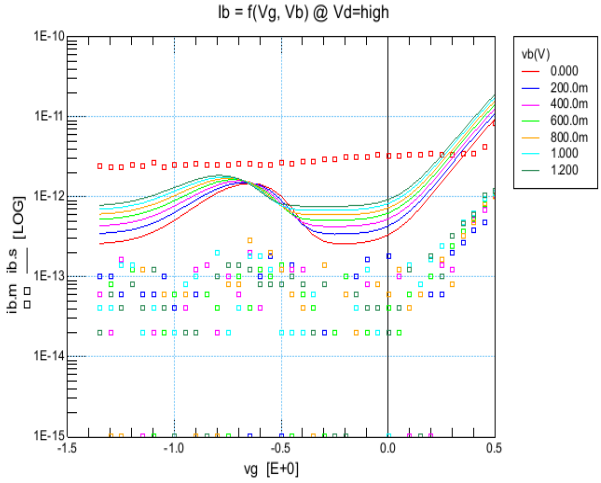


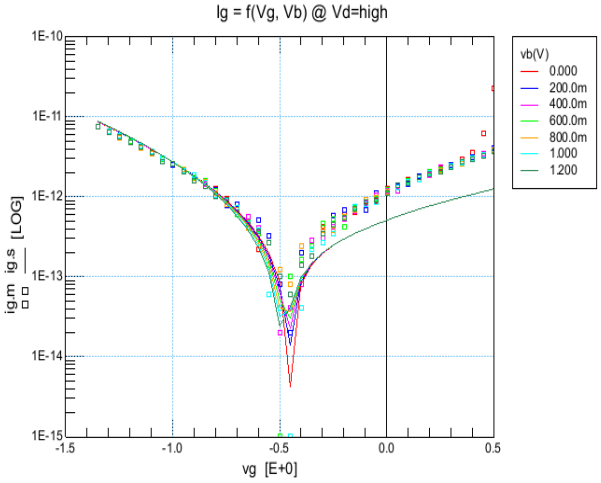




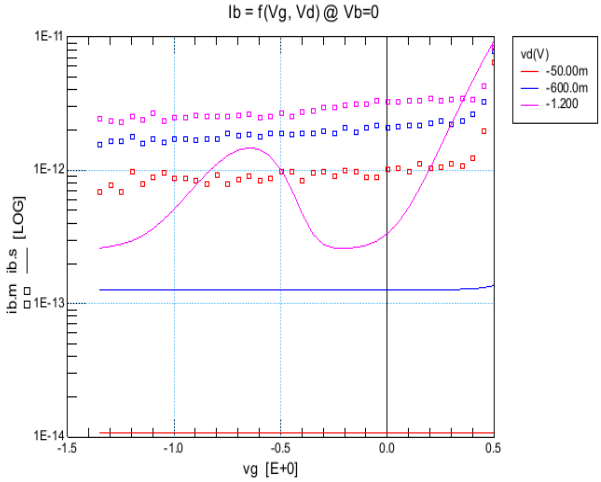


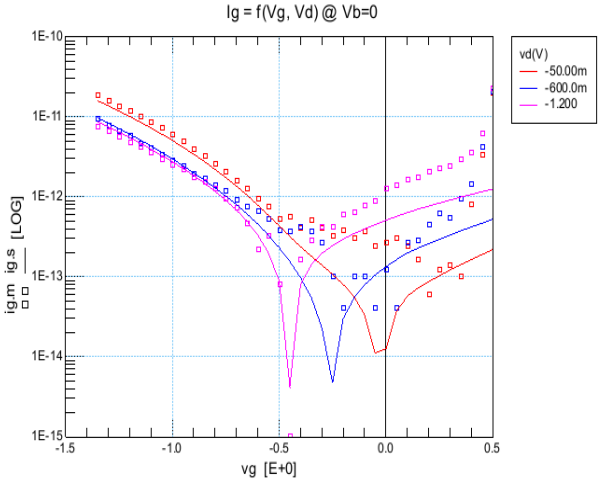




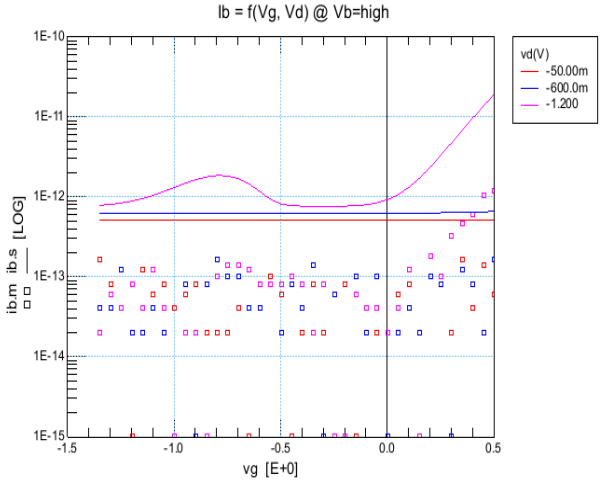


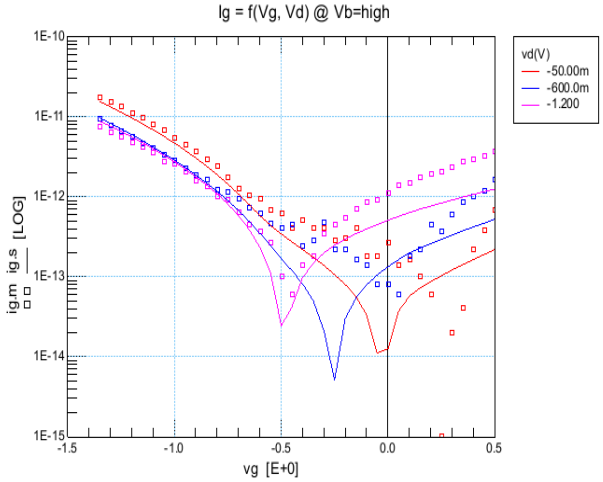
idvg, Ib,Ig, Vb = 0V, T = 27°C



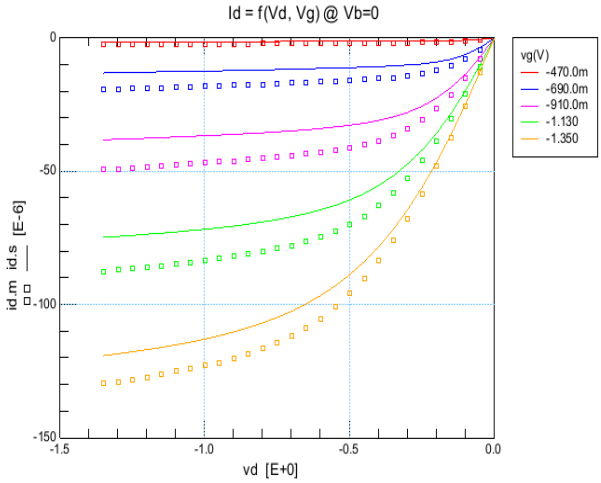


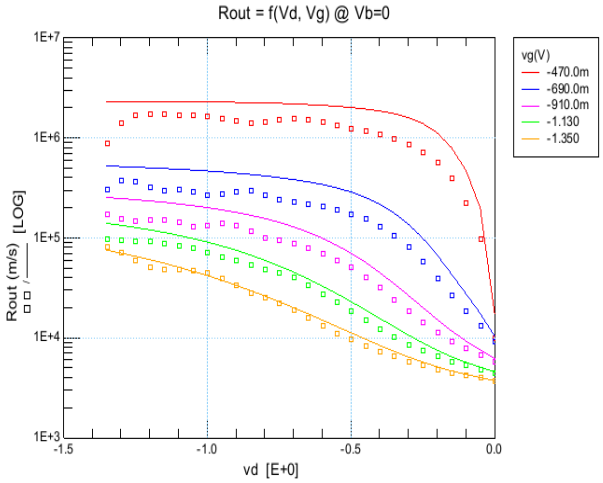
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

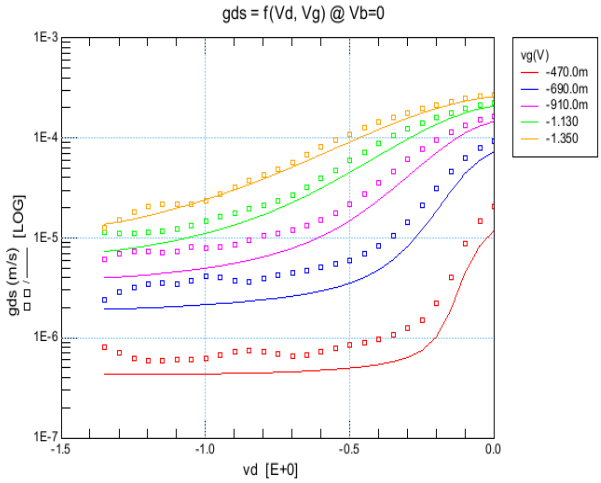




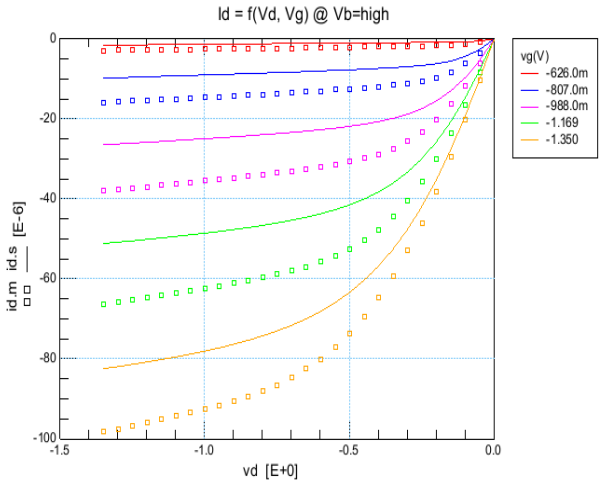
idvd, Vb = 0V, T = 27°C

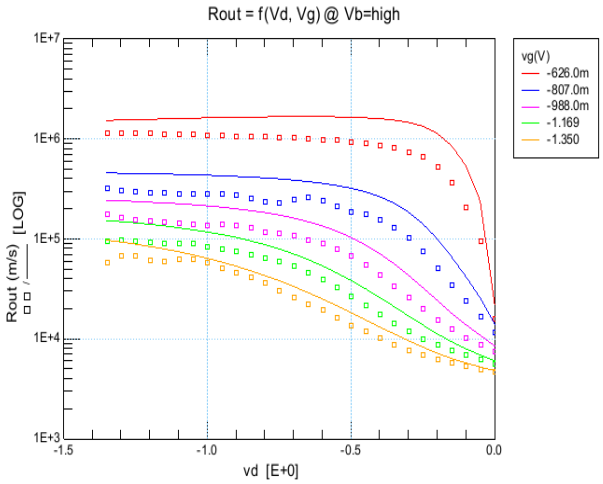


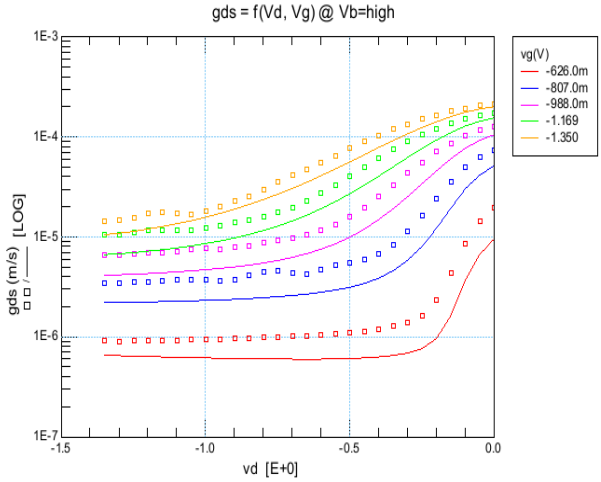




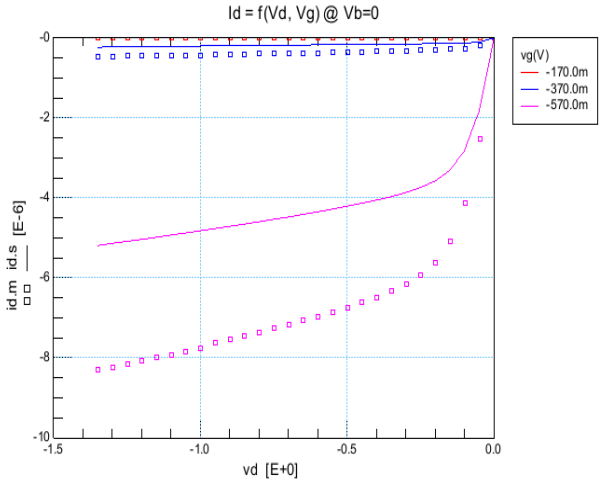
idvd_vbmin, Vb = 1.2V, T = 27°C

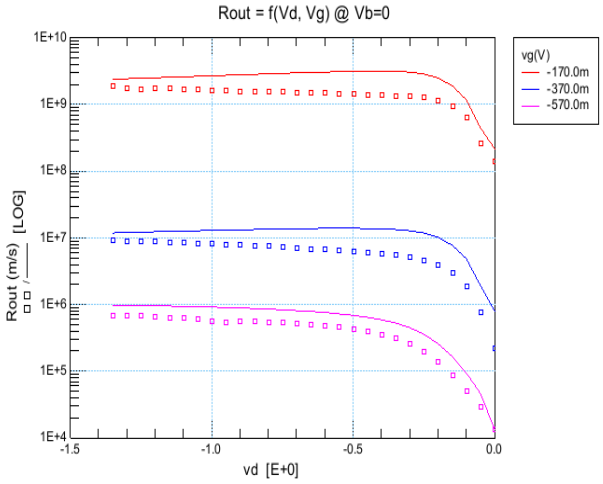


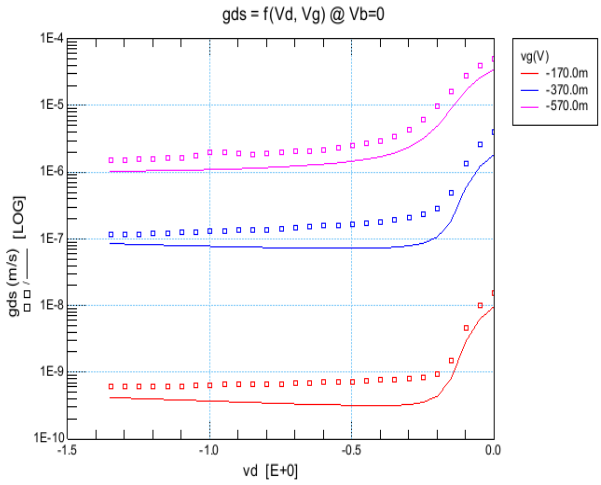




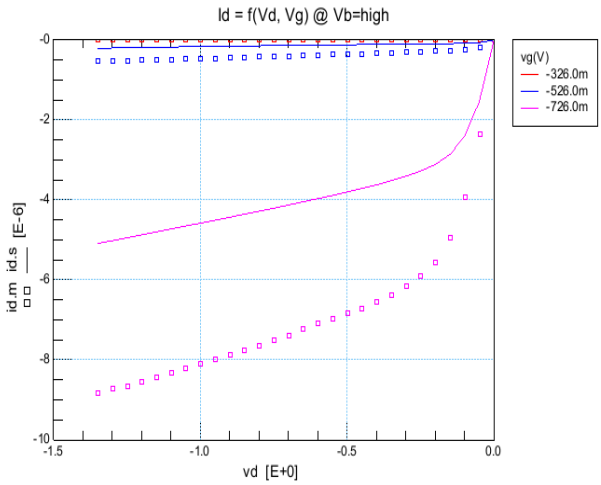
idvd_vth, Vb = 0V, T = 27°C

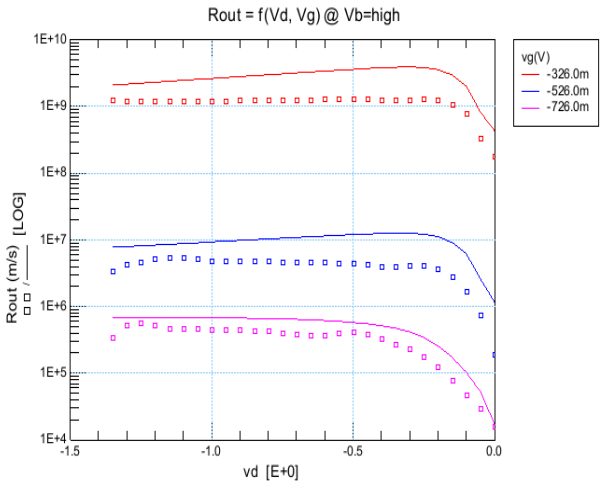


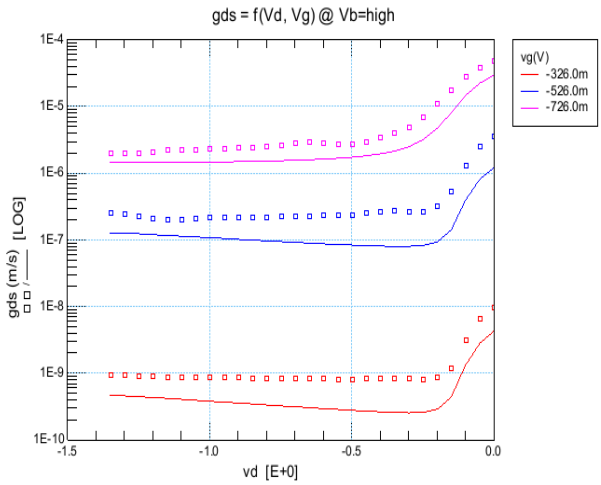




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

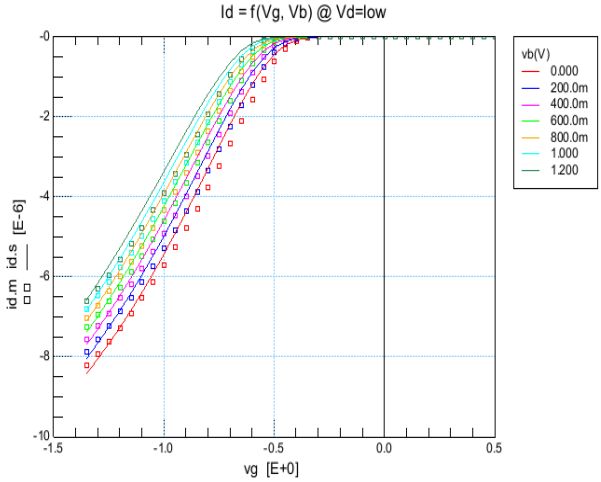


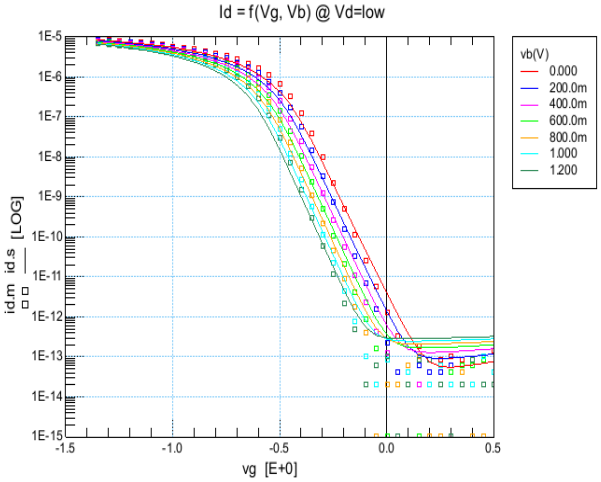


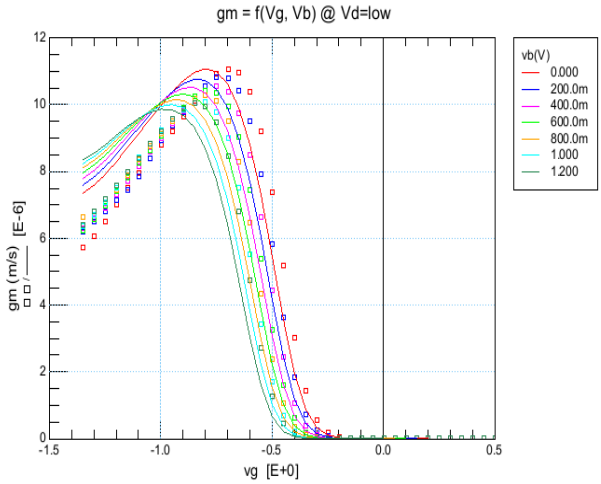


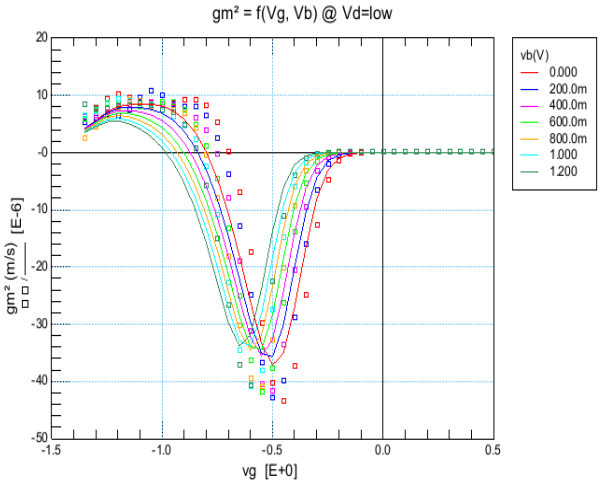
4.26W0u3_L0u15_S552_4

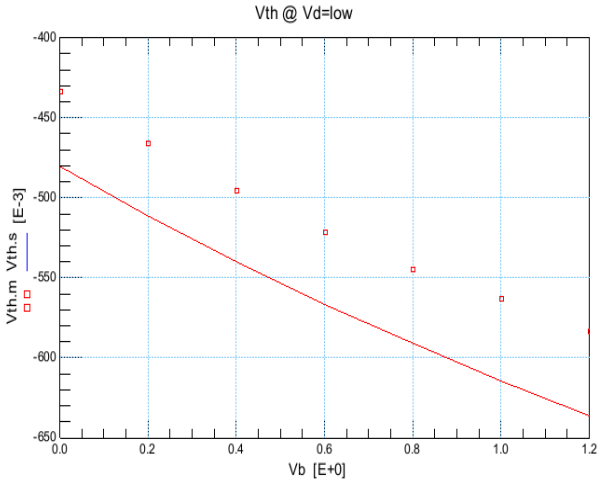
idvg, Vd = -0.05V, T = 27°C

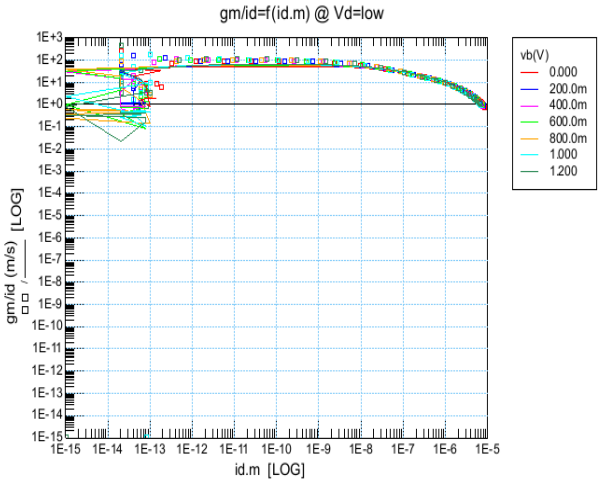


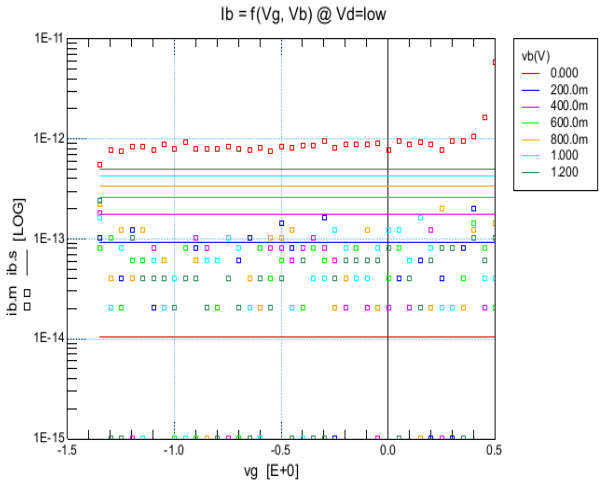


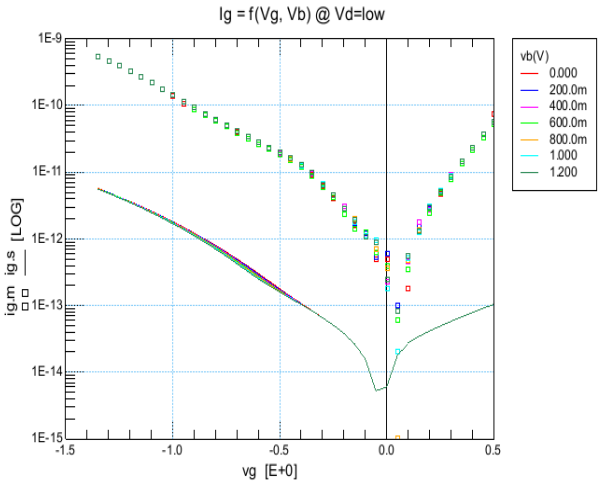




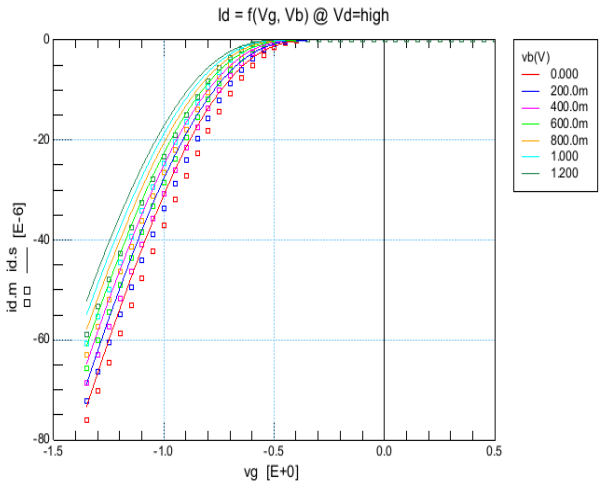


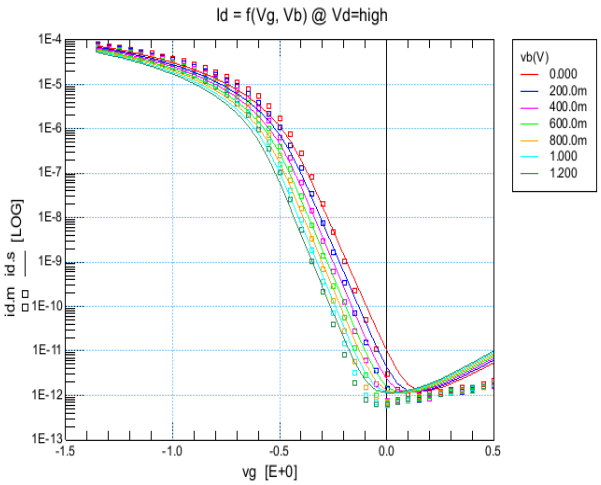


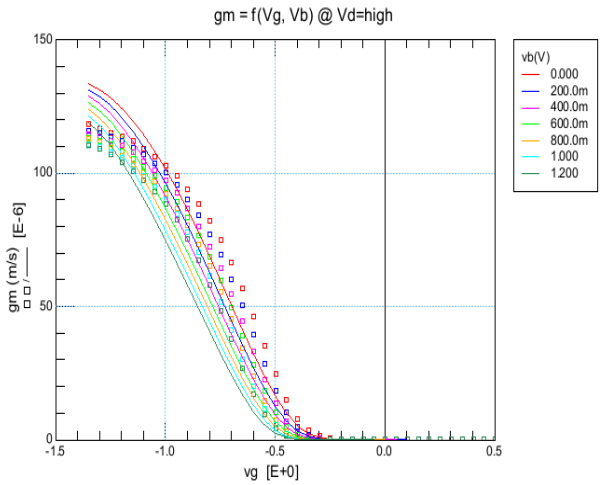


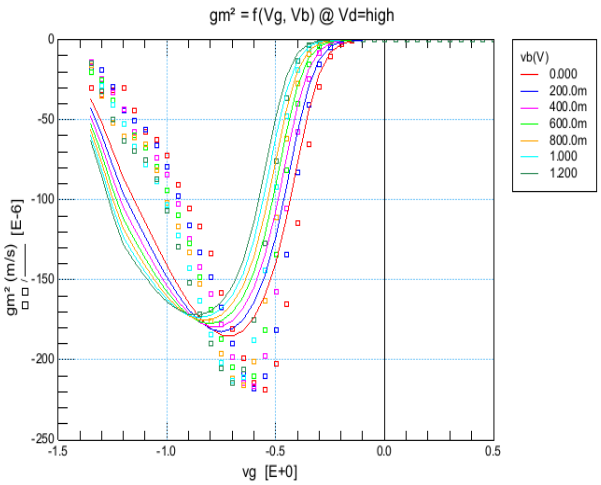


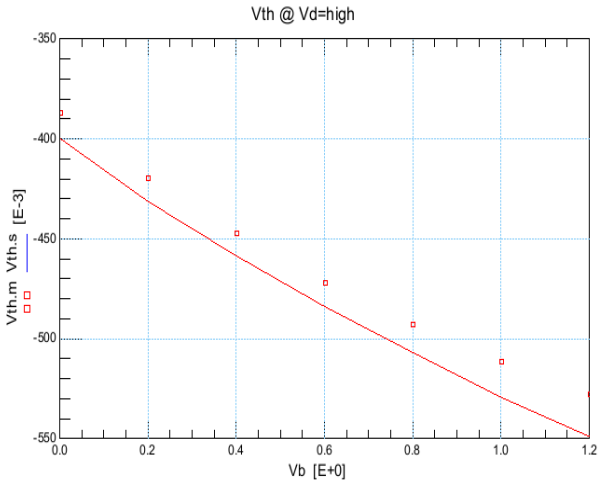
idvg, Vd = -1.2V, T = 27°C

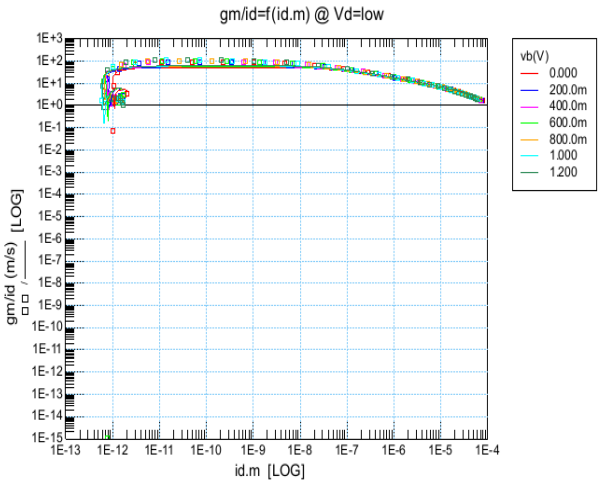


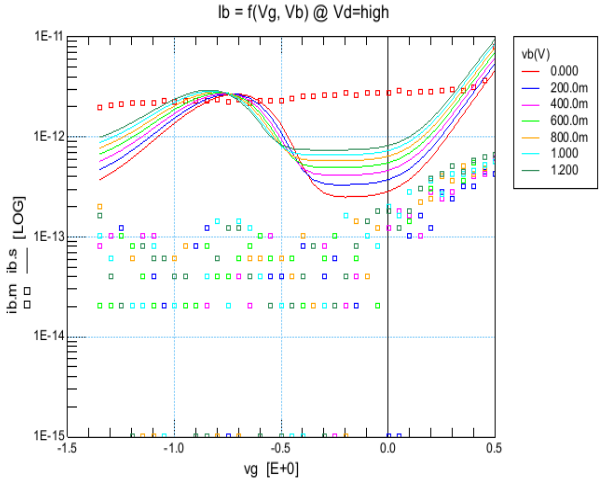


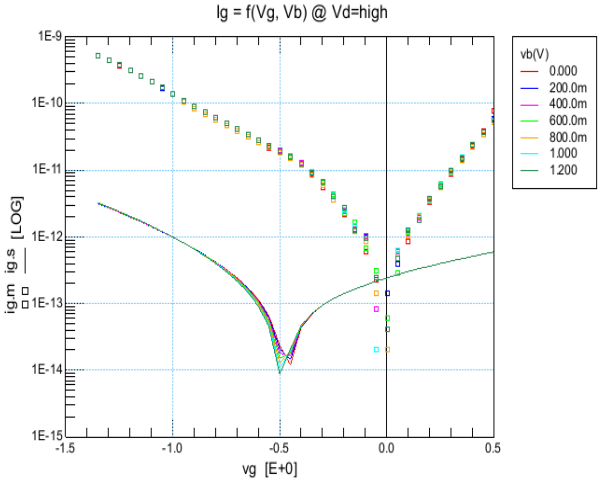




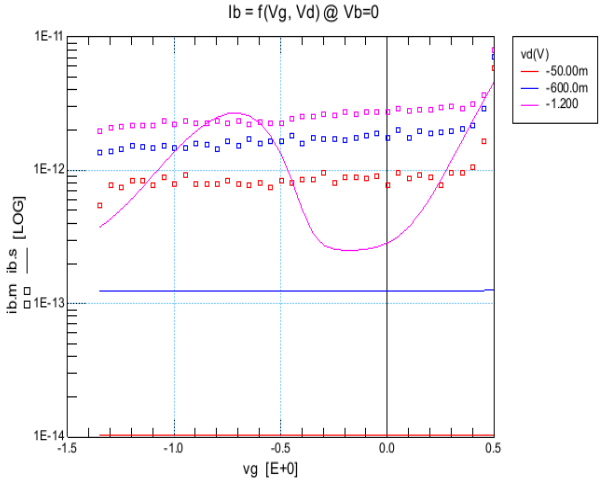


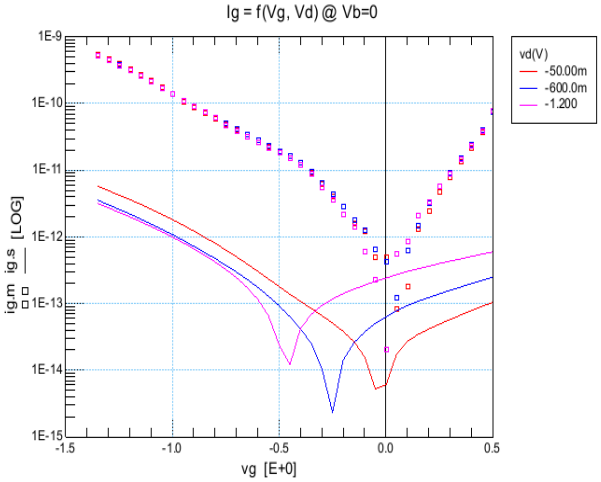




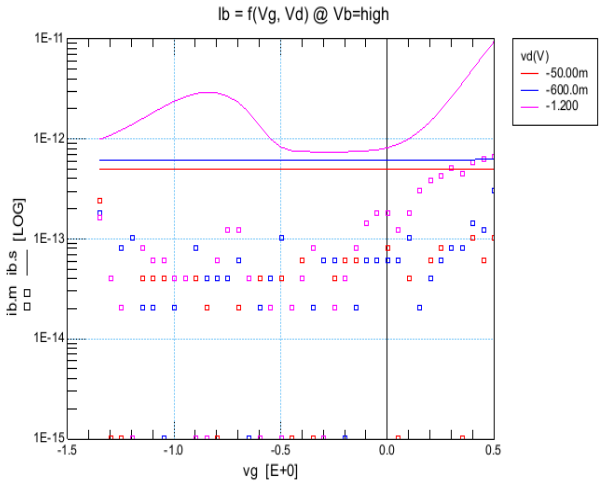


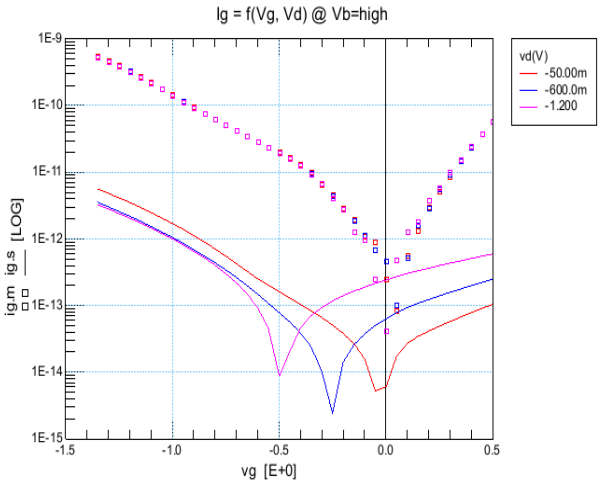
idvg, Ib,Ig, Vb = 0V, T = 27°C



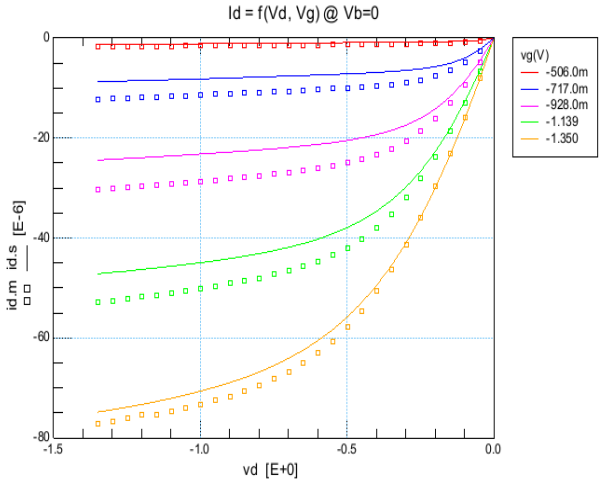


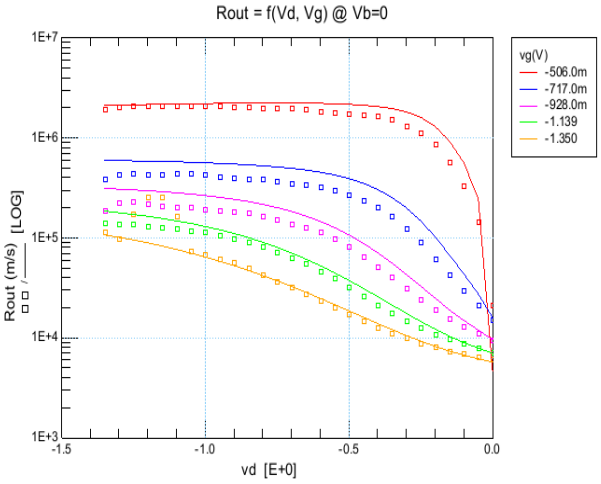
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

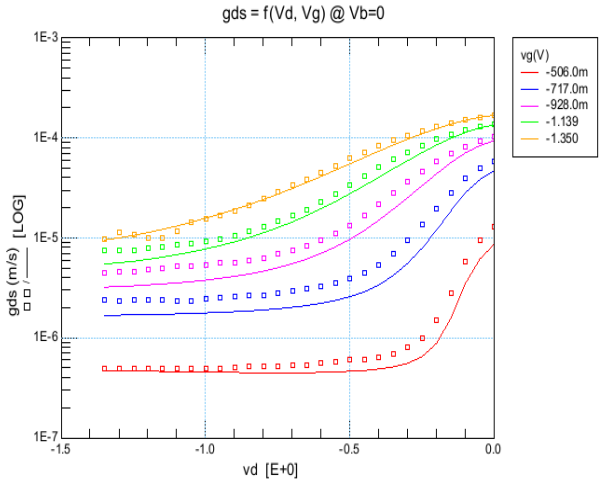




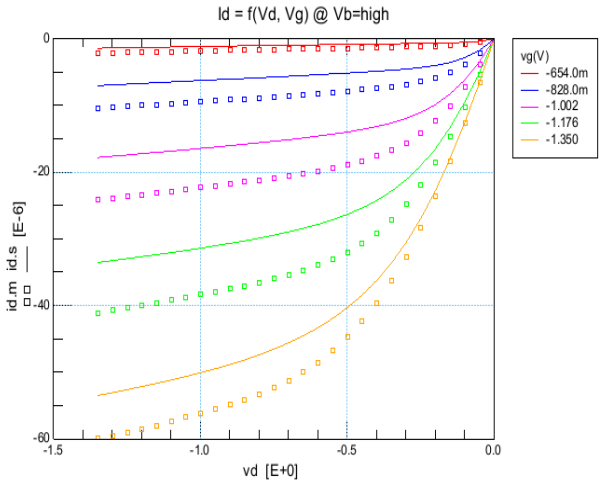
idvd, Vb = 0V, T = 27°C

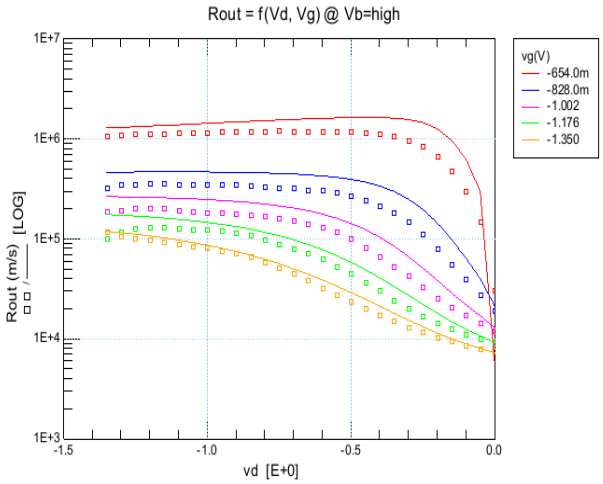


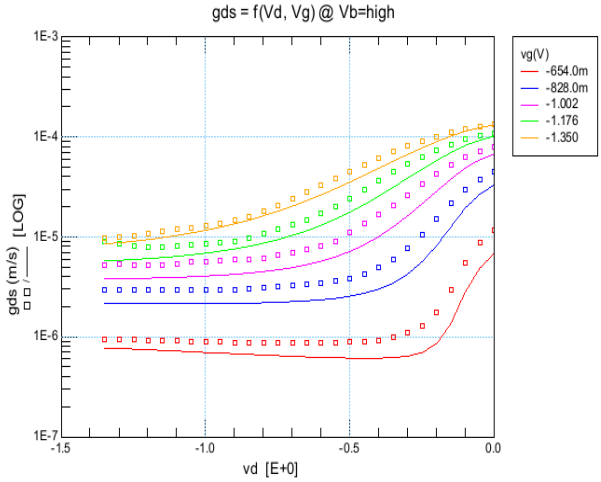




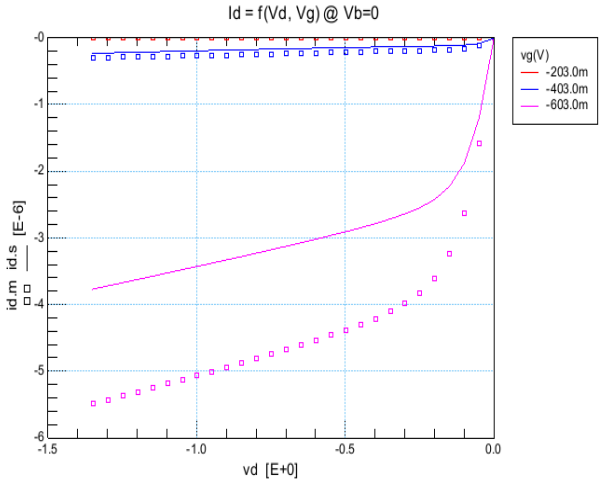
idvd_vbmin, Vb = 1.2V, T = 27°C

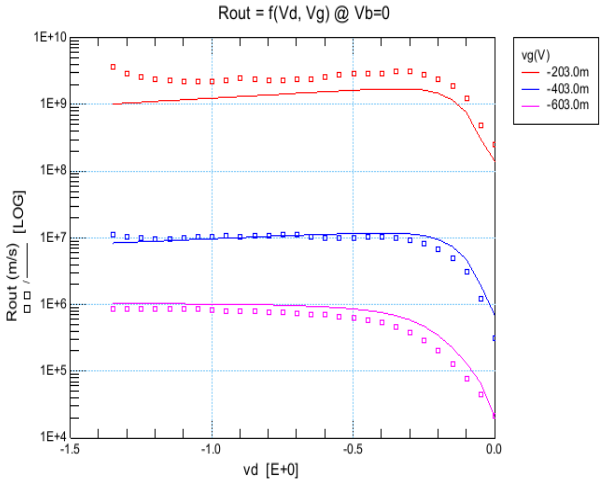


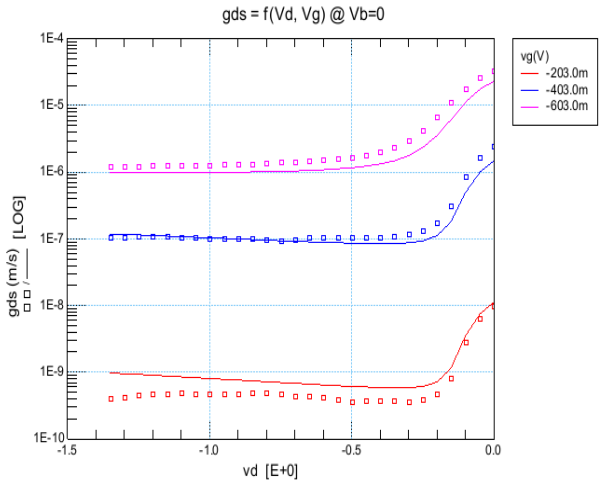




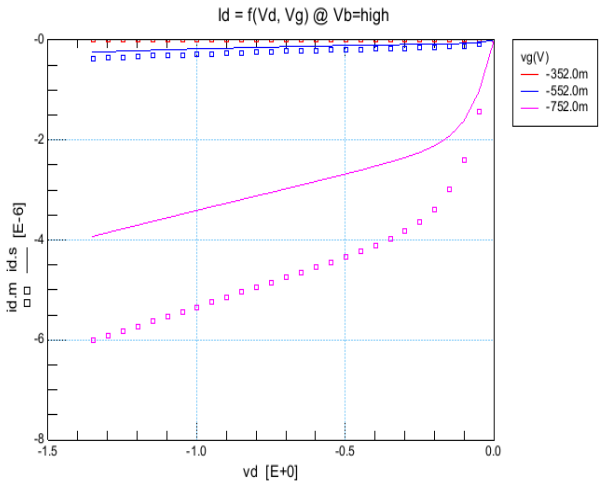
idvd_vth, Vb = 0V, T = 27°C

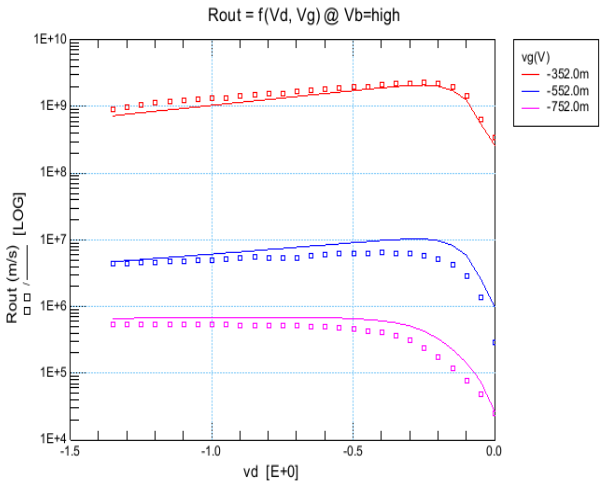


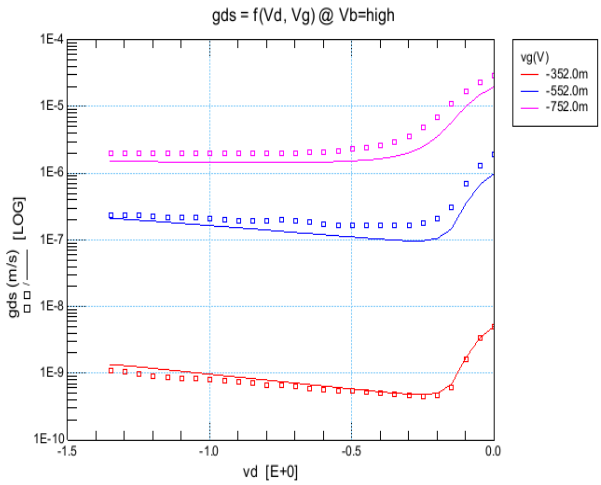




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

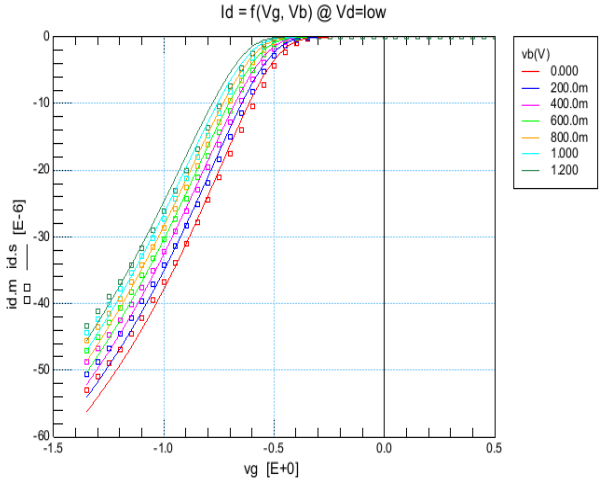


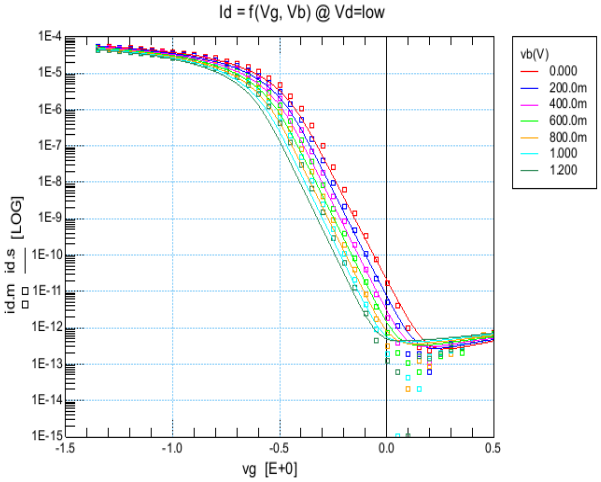


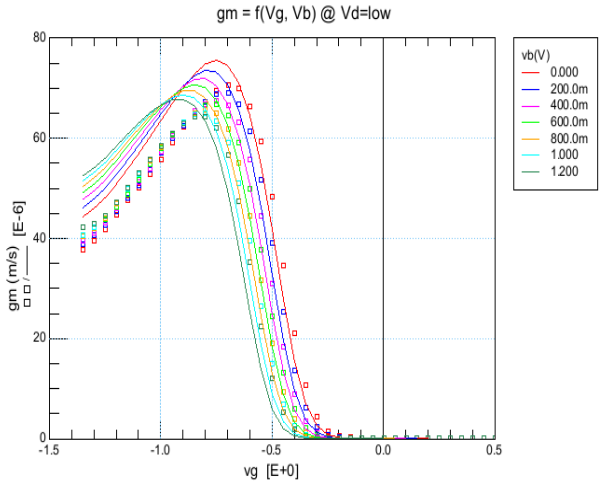


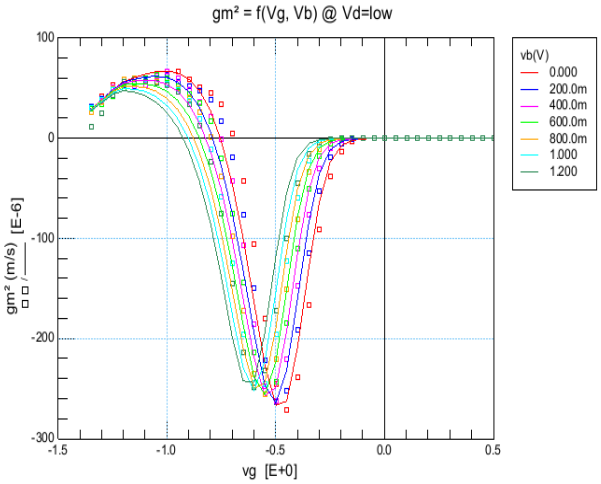
4.27W02u0_L0u13_S550_4

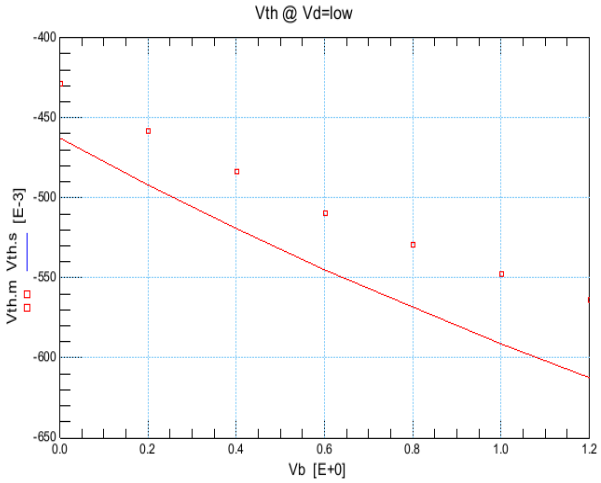
idvg, Vd = -0.05V, T = 27°C

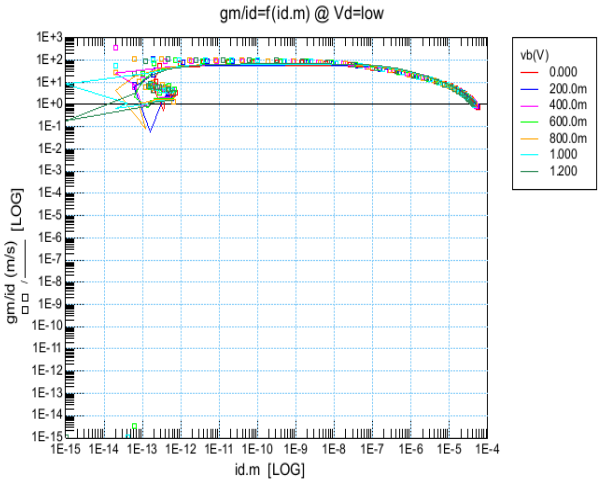


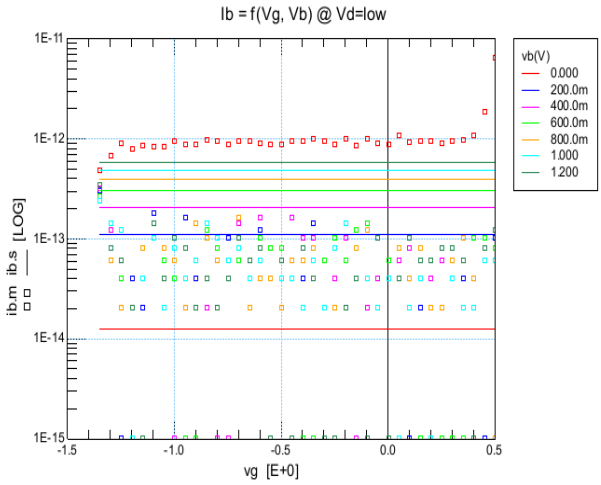


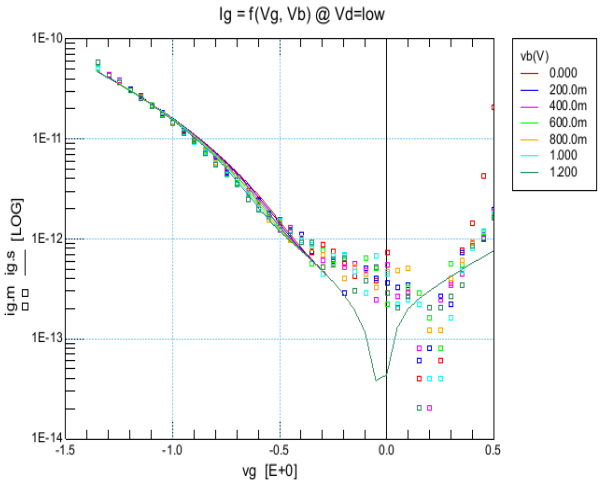




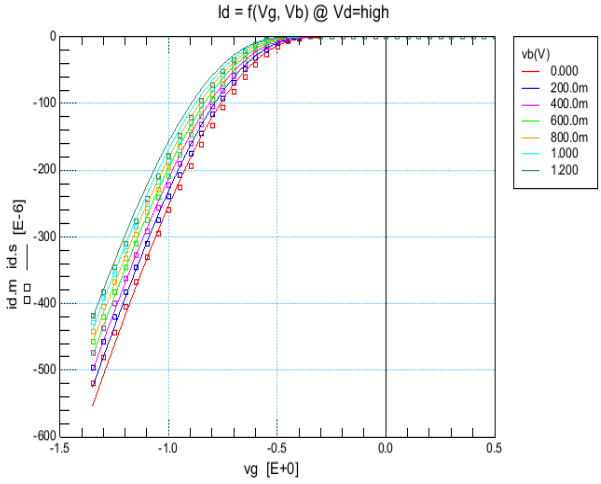


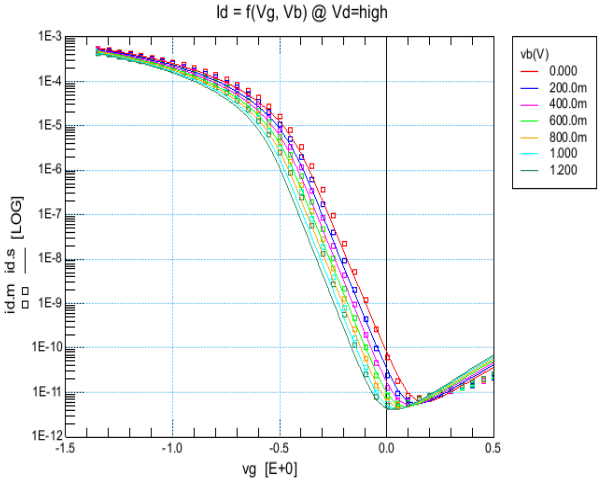


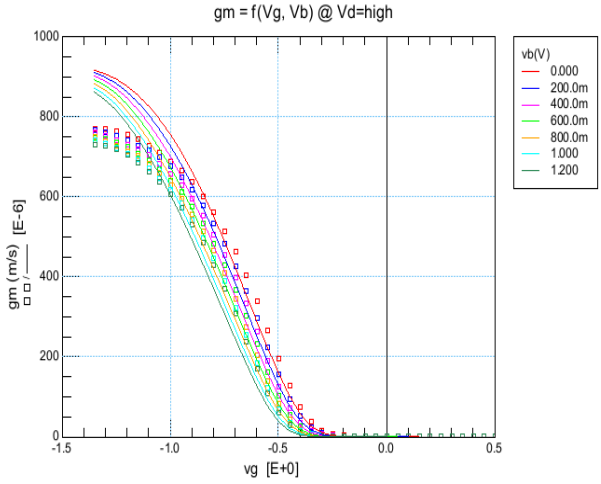


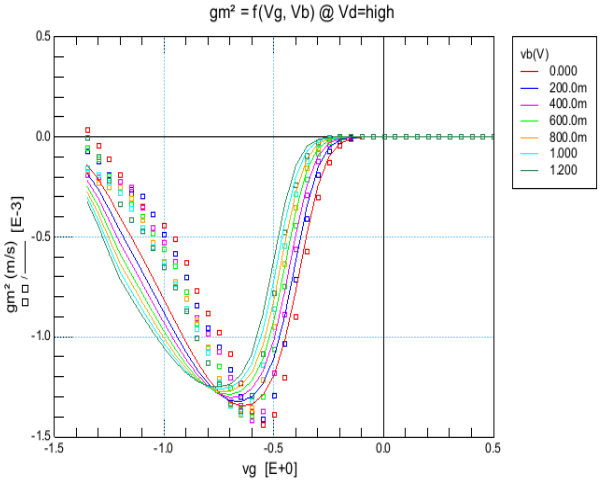


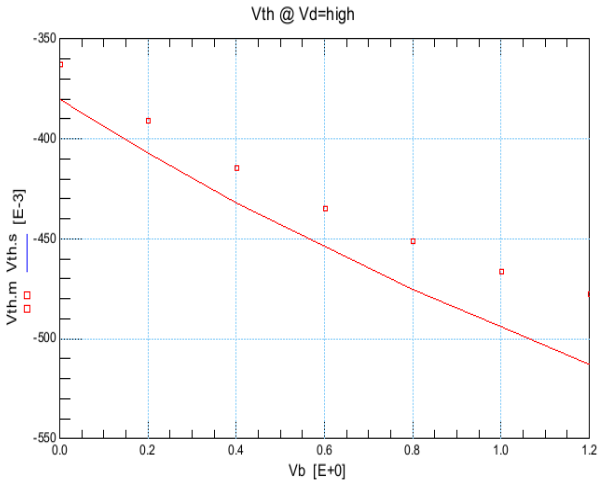
idvg, Vd = -1.2V, T = 27°C

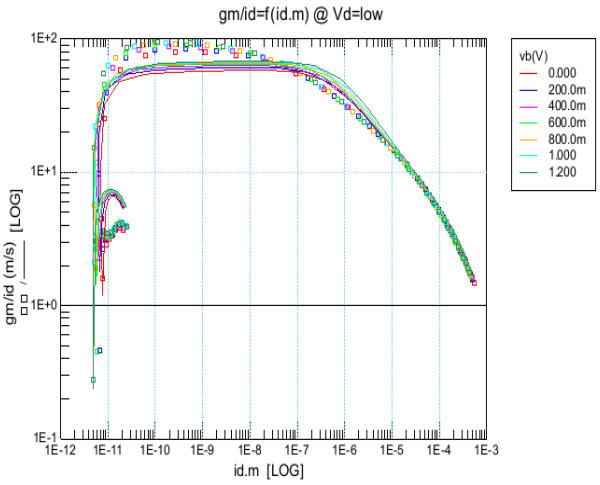


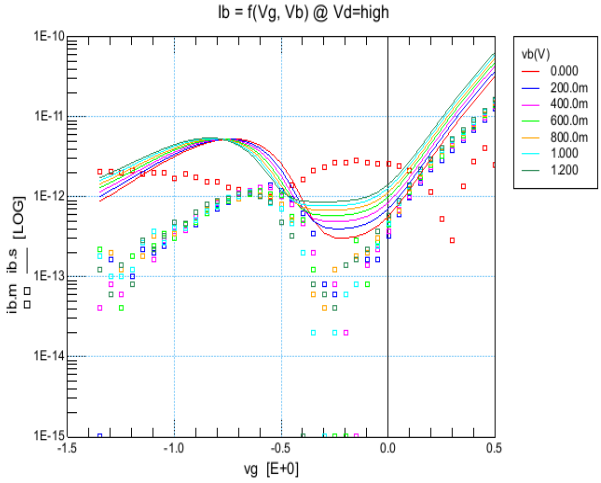


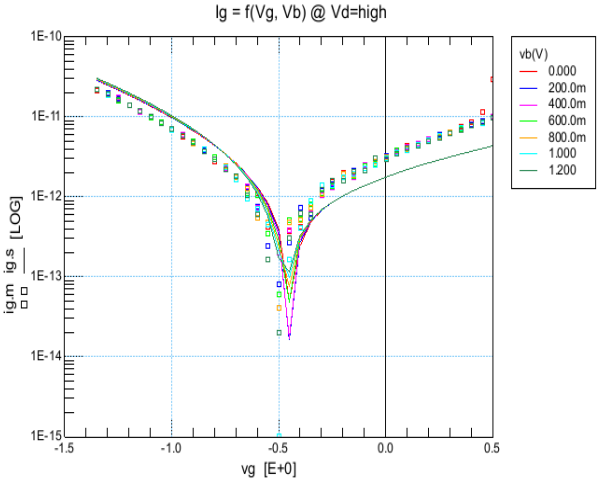




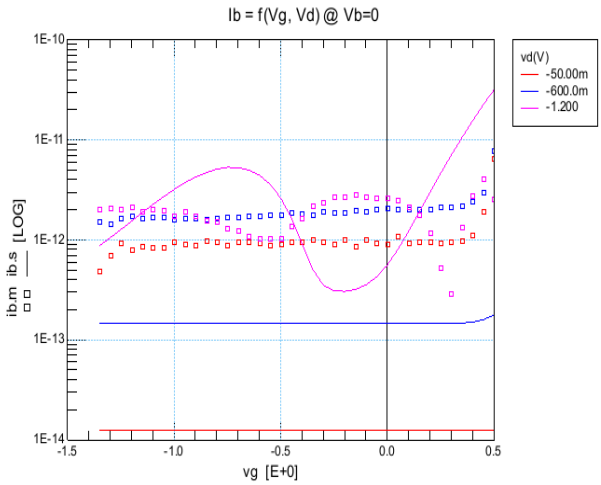


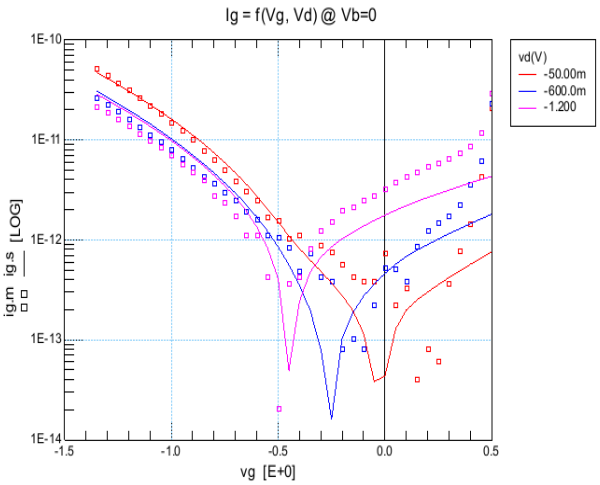




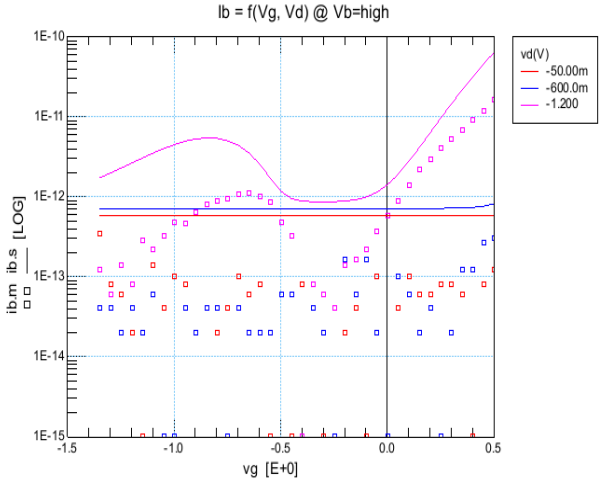


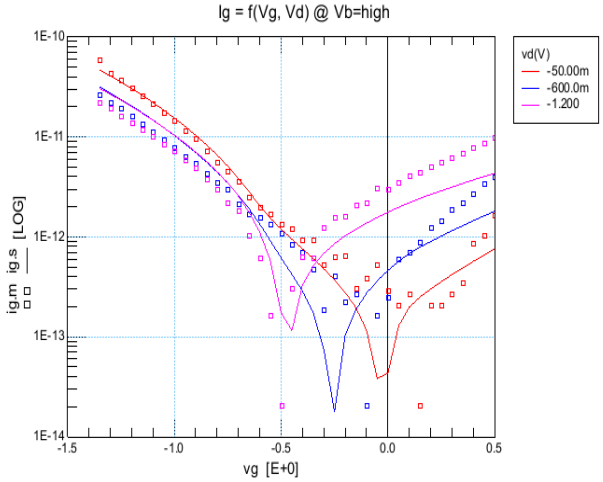
idvg, Ib,Ig, Vb = 0V, T = 27°C



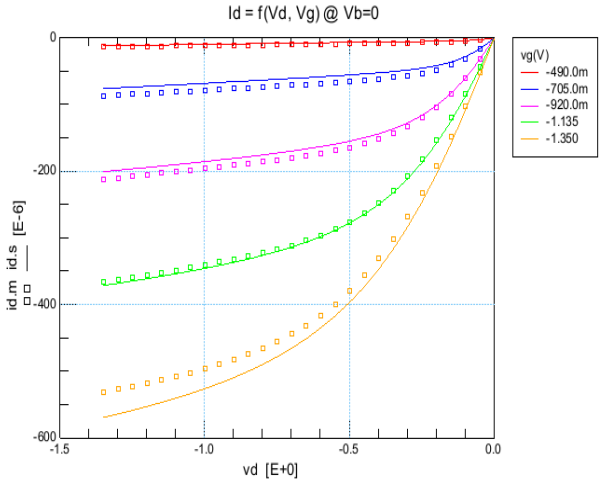


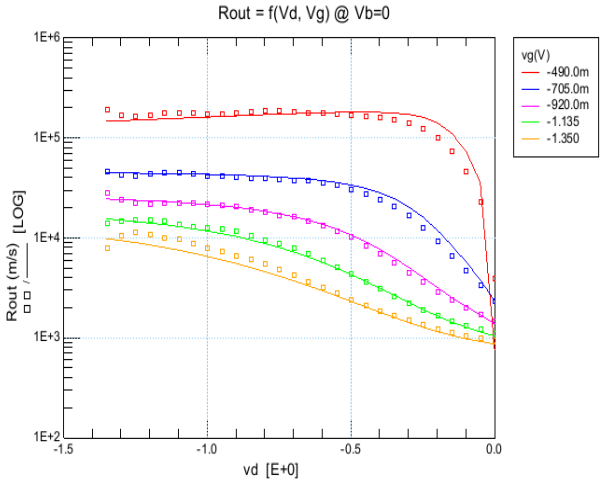
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

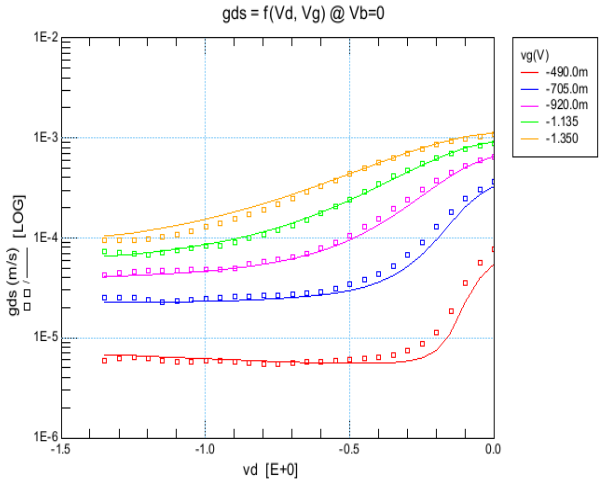




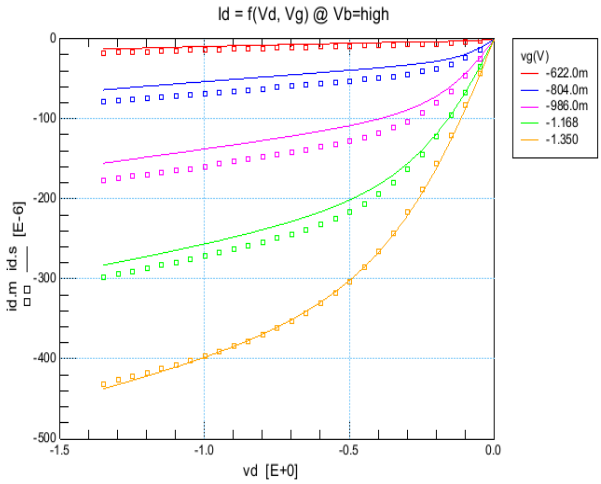
idvd, Vb = 0V, T = 27°C

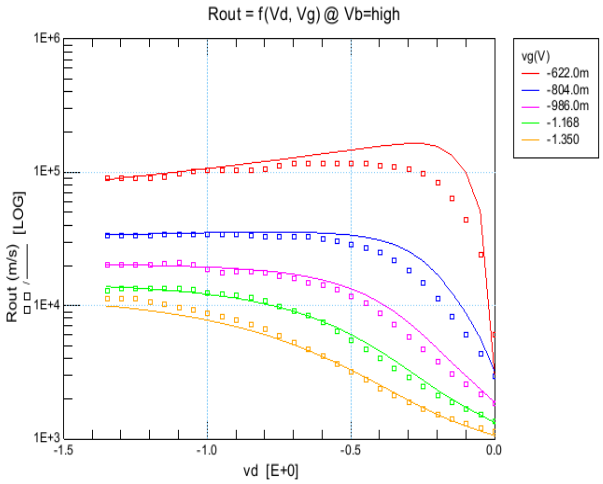


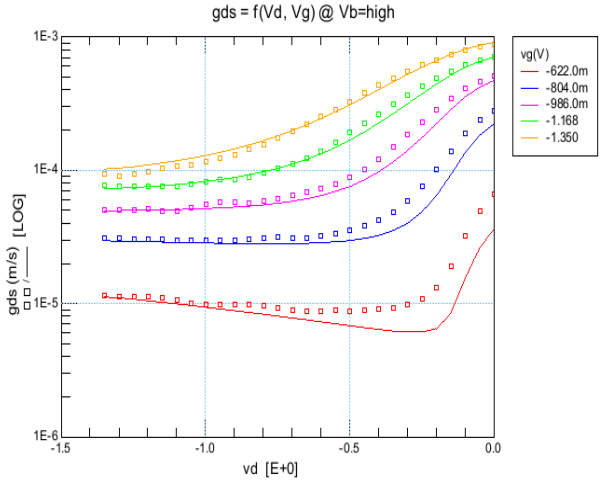




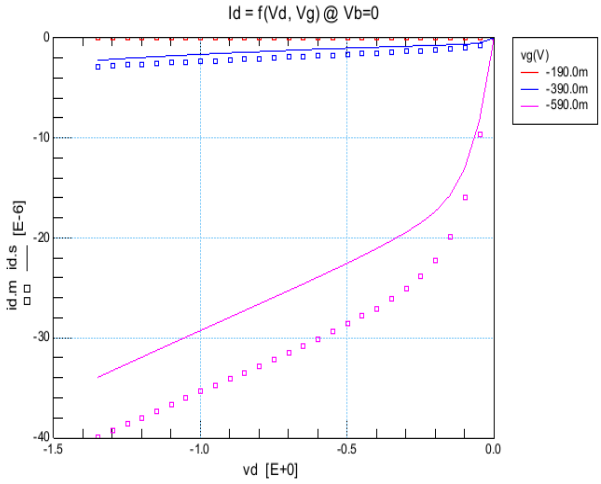
idvd_vbmin, Vb = 1.2V, T = 27°C

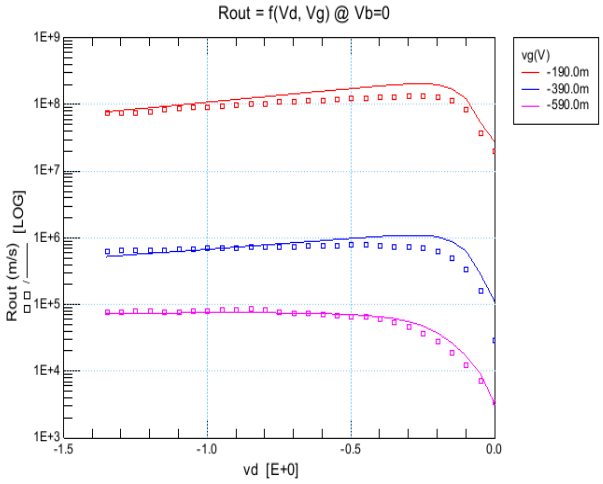


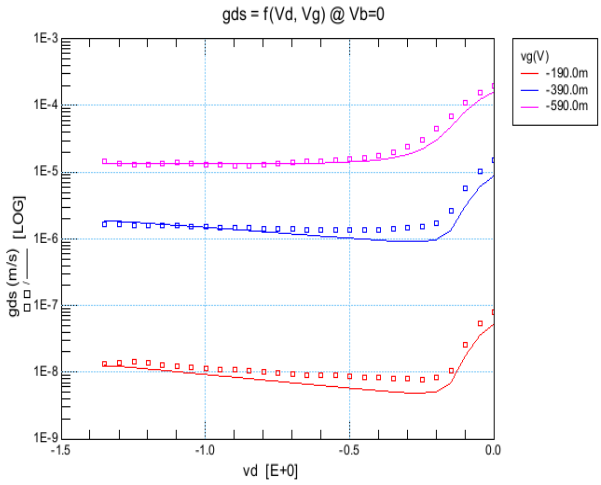




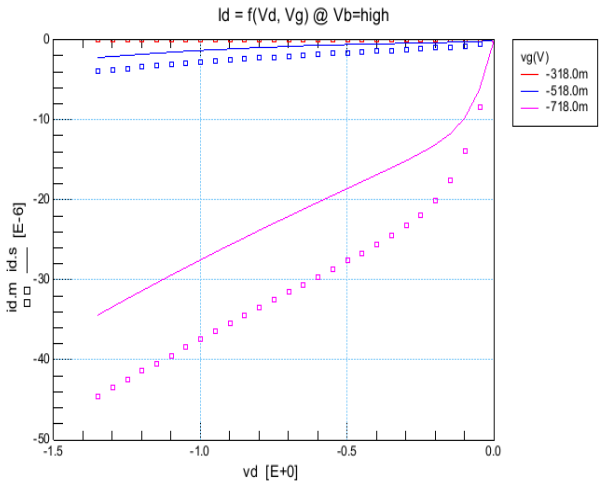
idvd_vth, Vb = 0V, T = 27°C

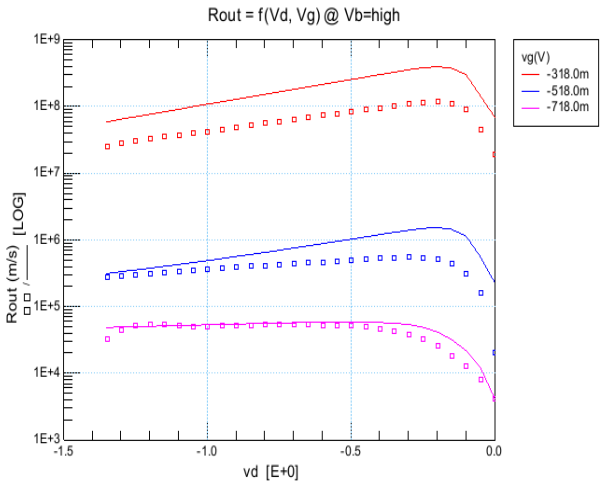


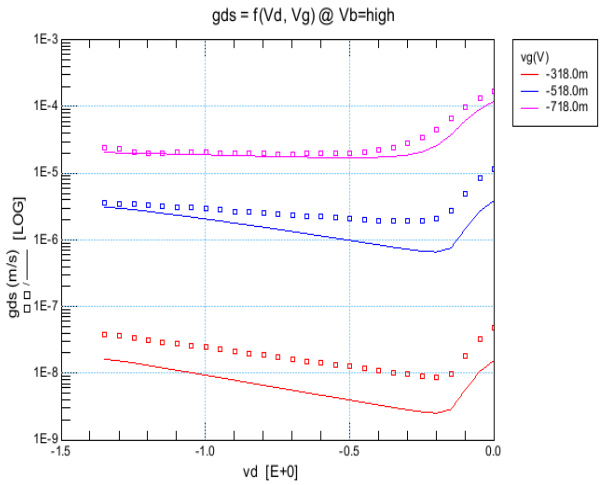




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

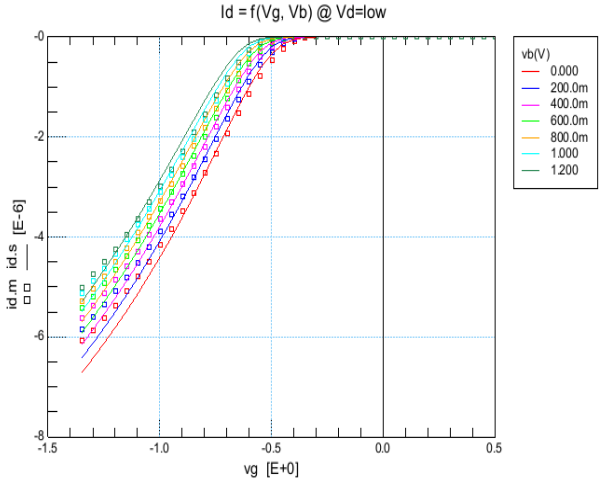


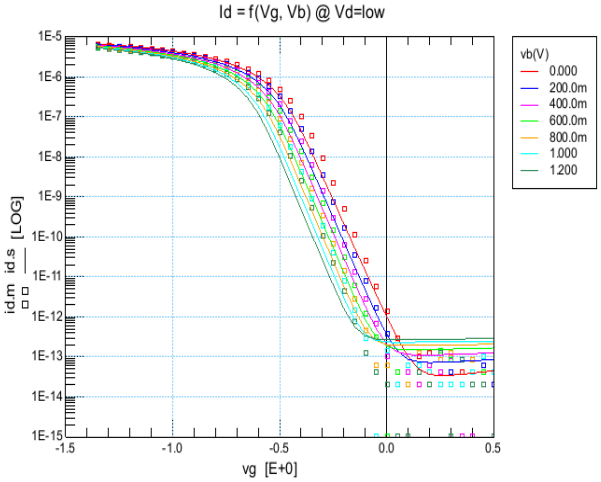


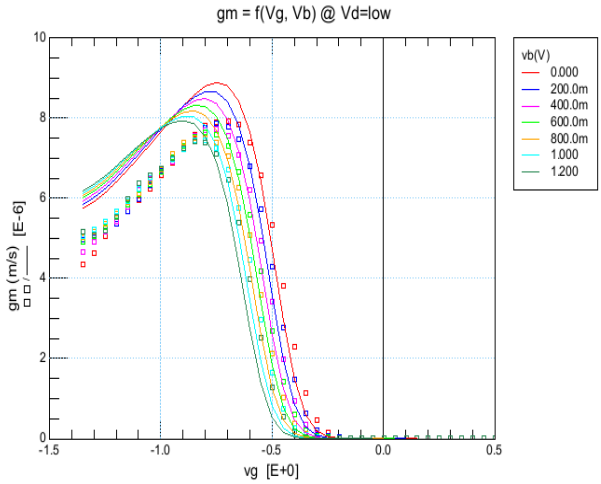


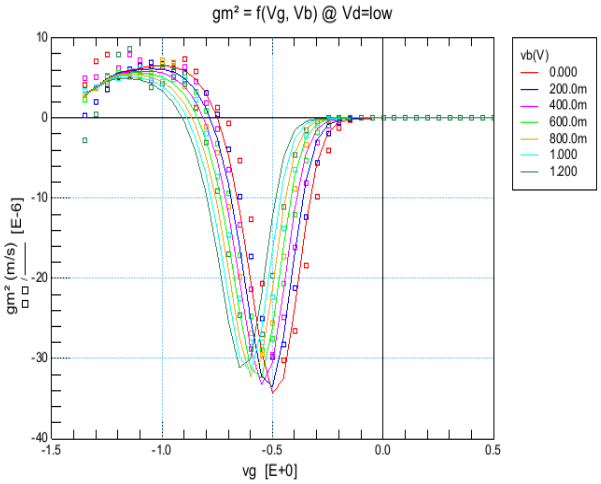
4.28W0u15_L0u14_S554_4

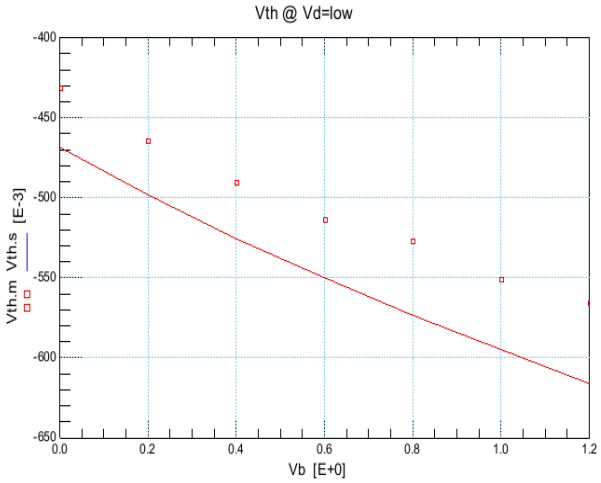
idvg, Vd = -0.05V, T = 27°C

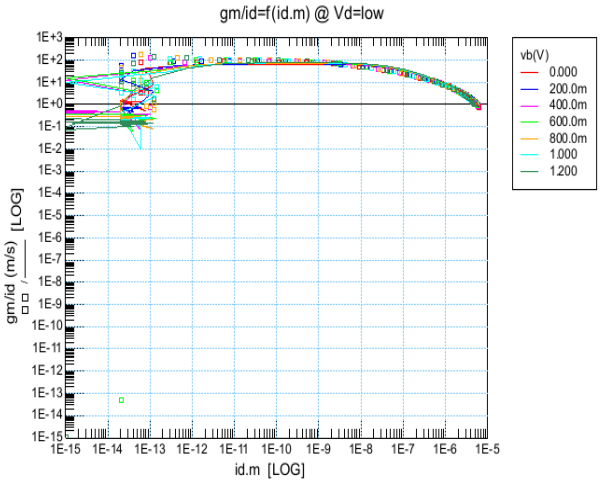


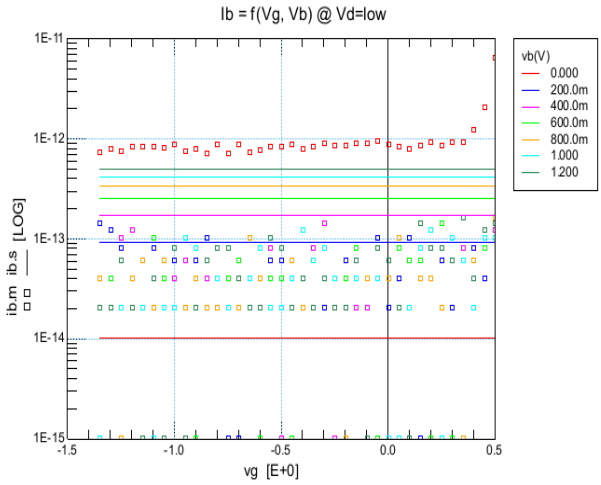


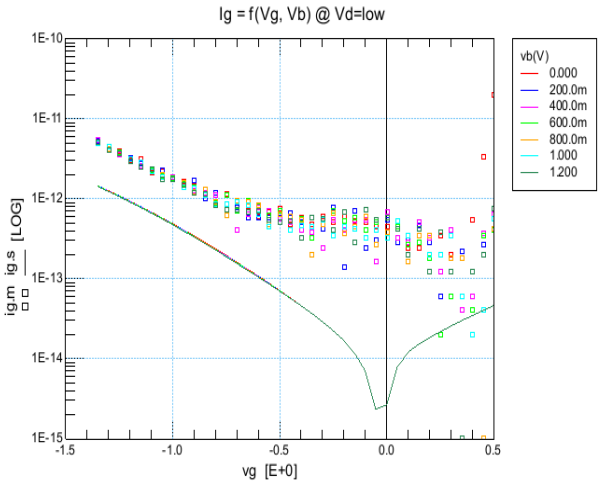




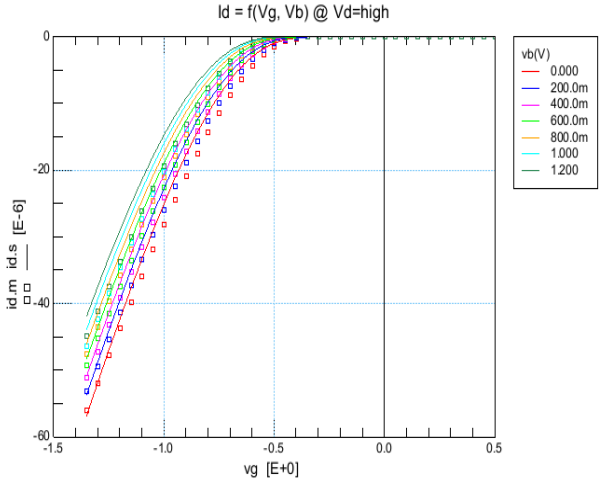


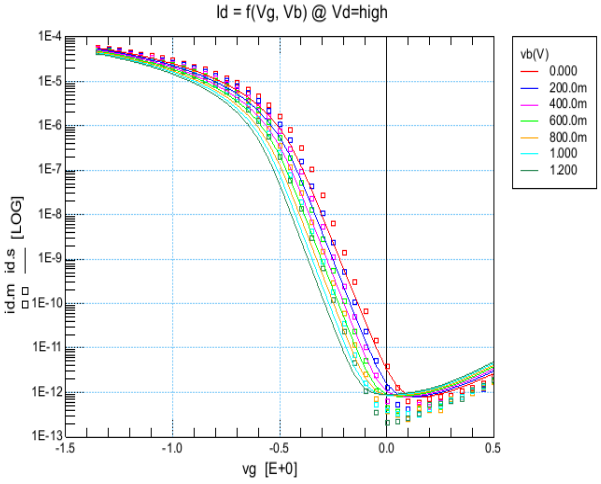


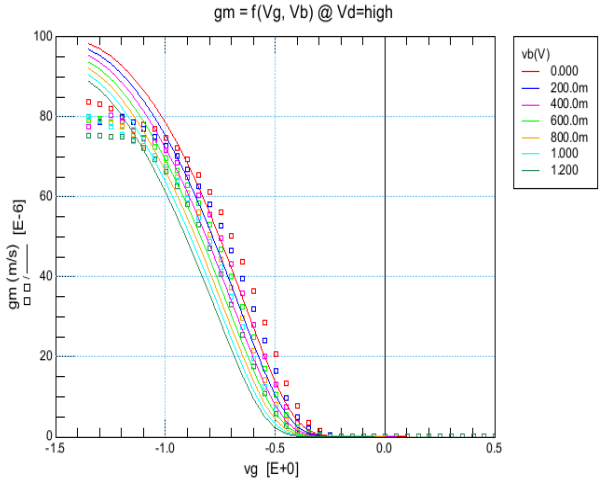


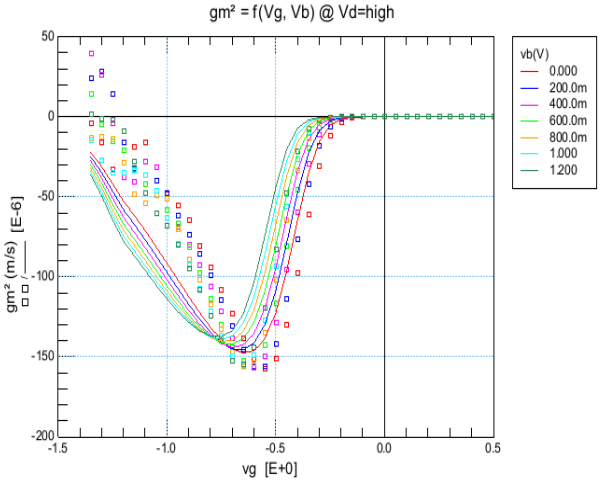


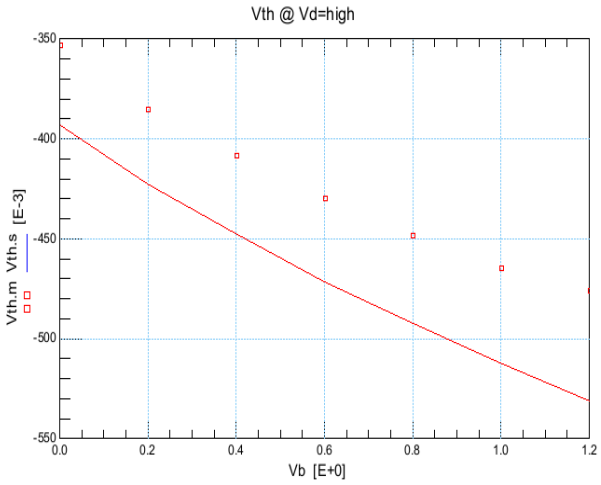
idvg, Vd = -1.2V, T = 27°C

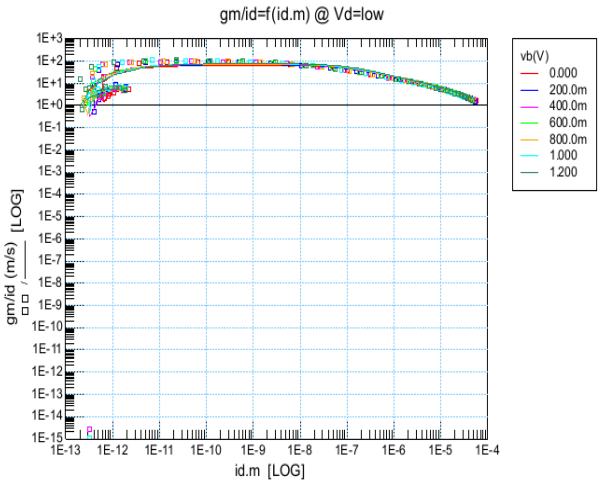


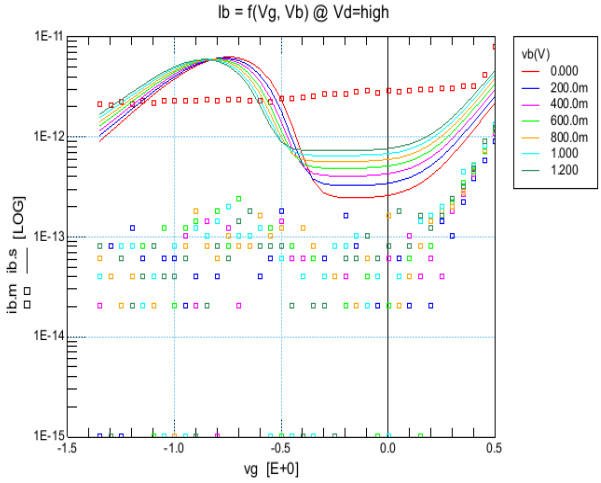


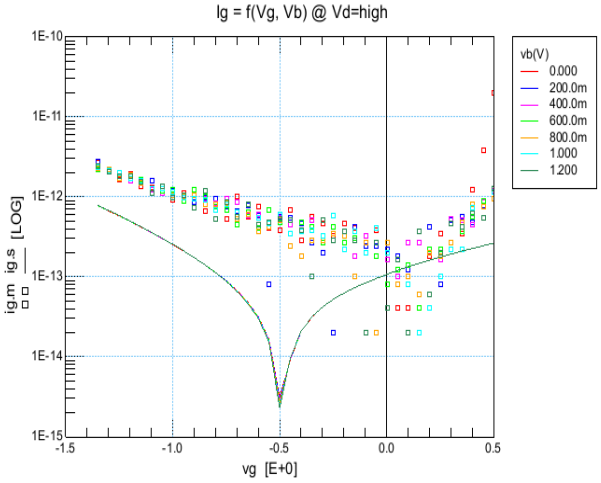




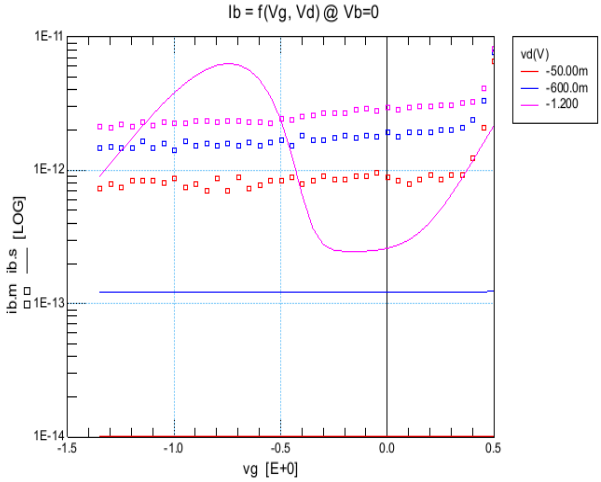


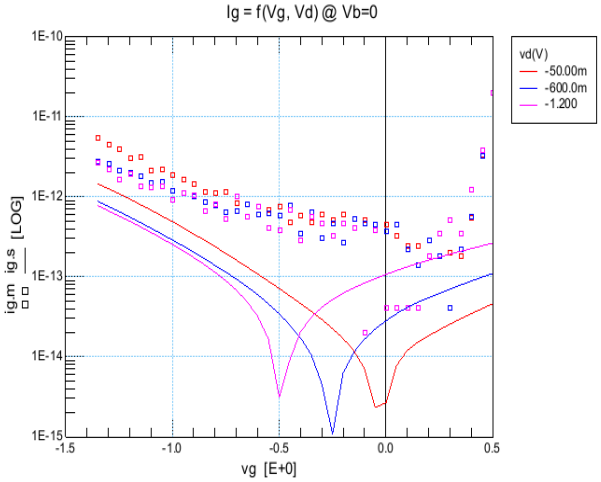




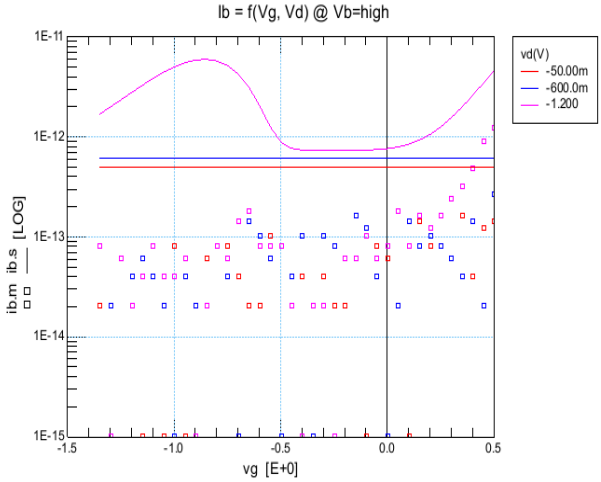


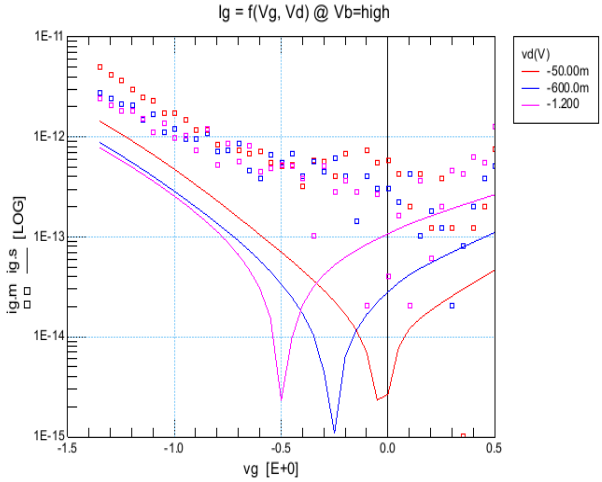
idvg, Ib,Ig, Vb = 0V, T = 27°C



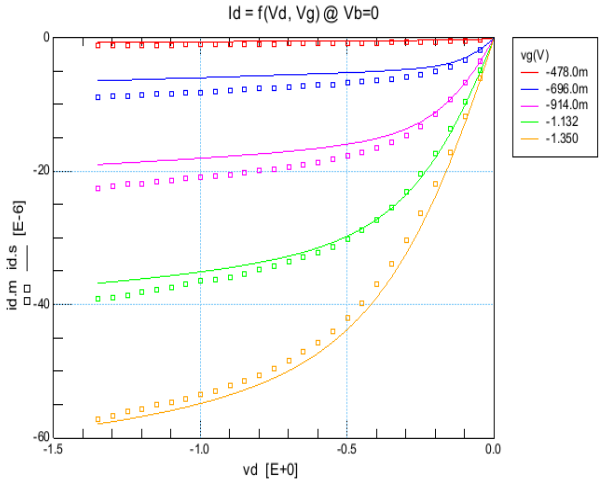


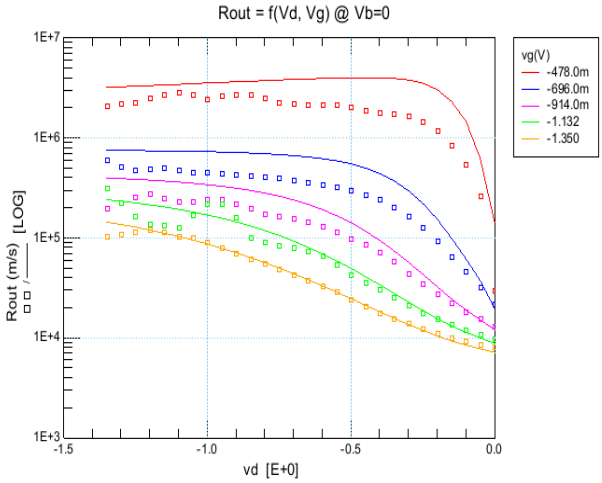
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

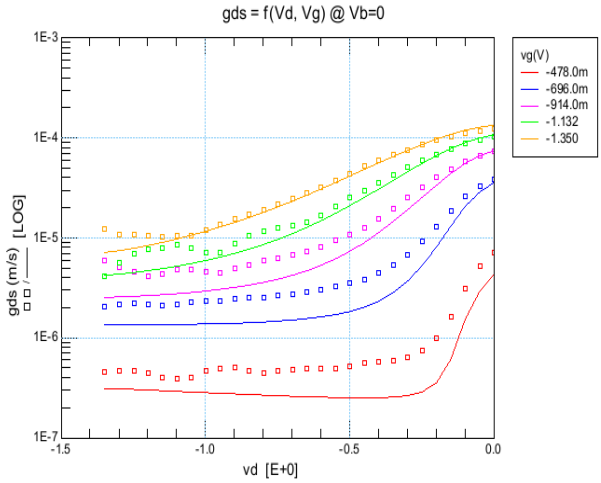




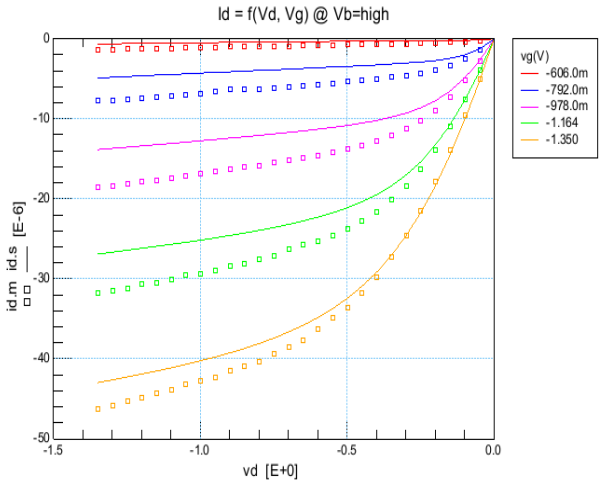
idvd, Vb = 0V, T = 27°C

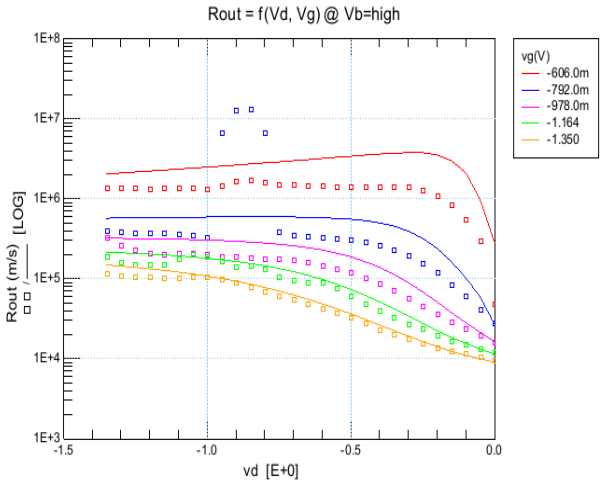


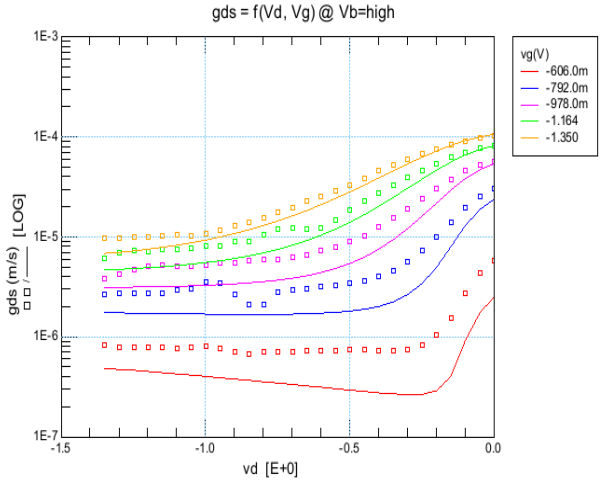




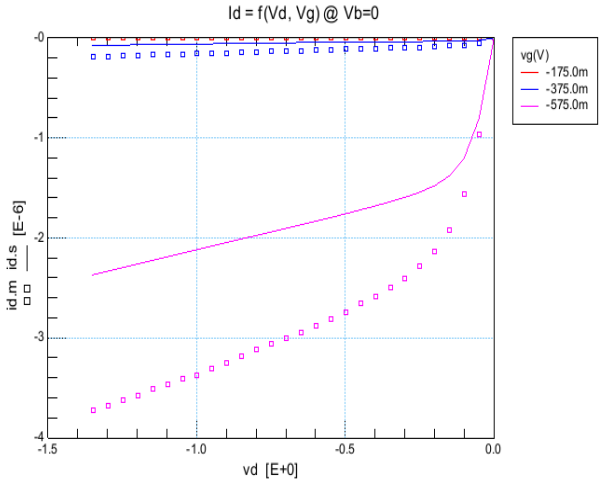
idvd_vbmin, Vb = 1.2V, T = 27°C

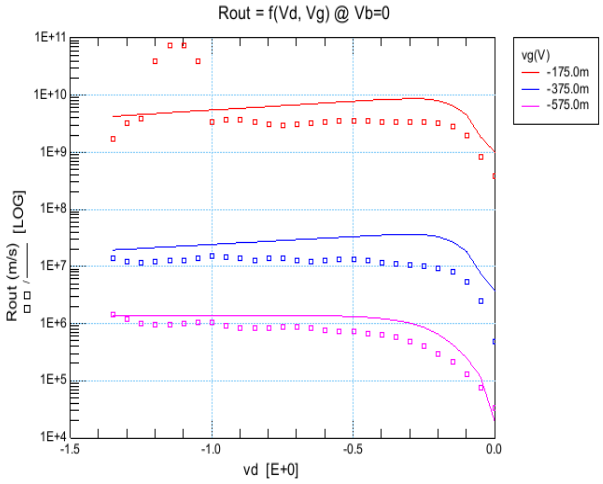


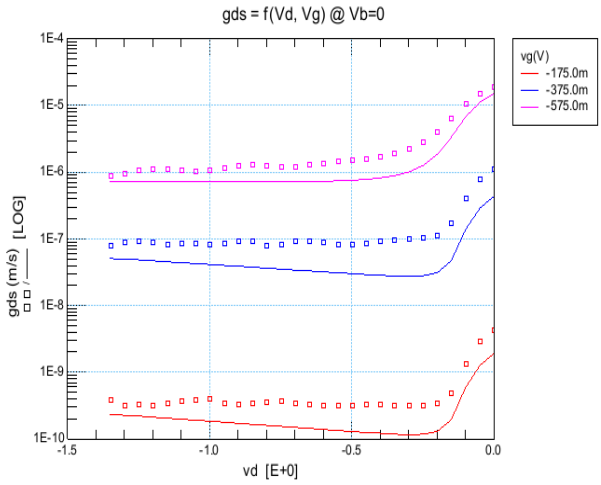




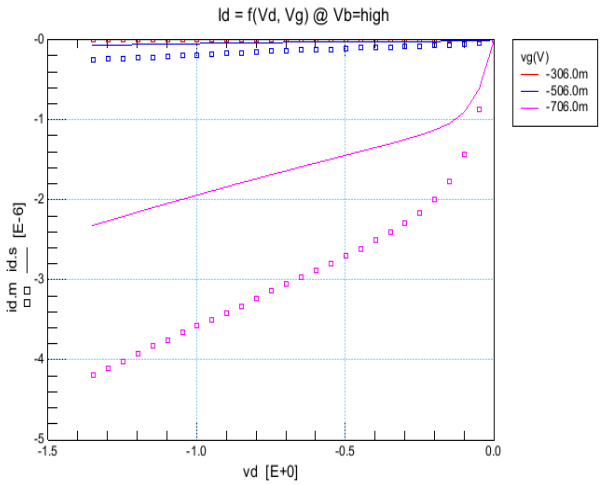
idvd_vth, Vb = 0V, T = 27°C

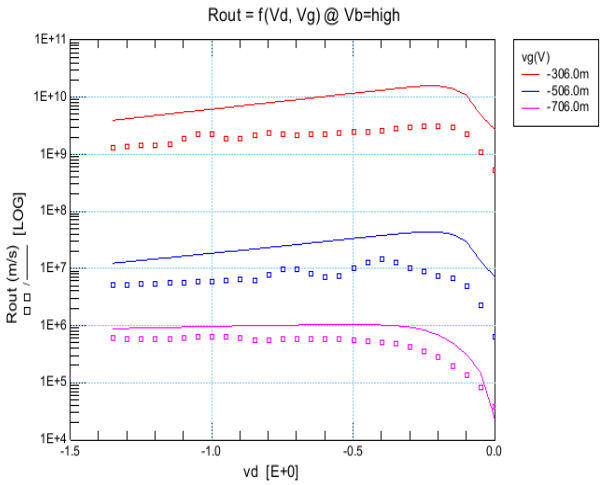


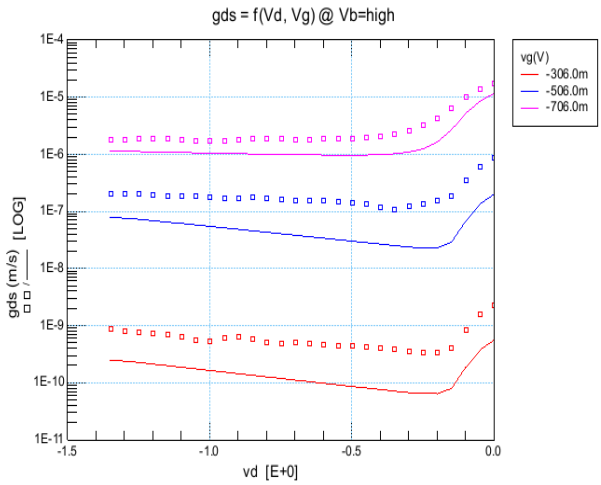




idvd_vth_vbmin, Vb = 1.2V, T = 27°C







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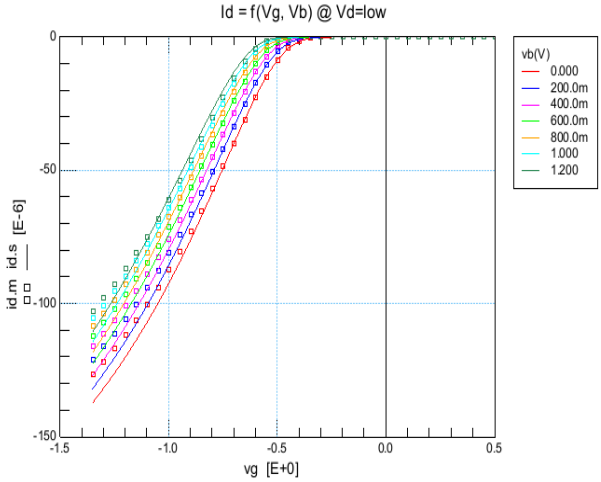
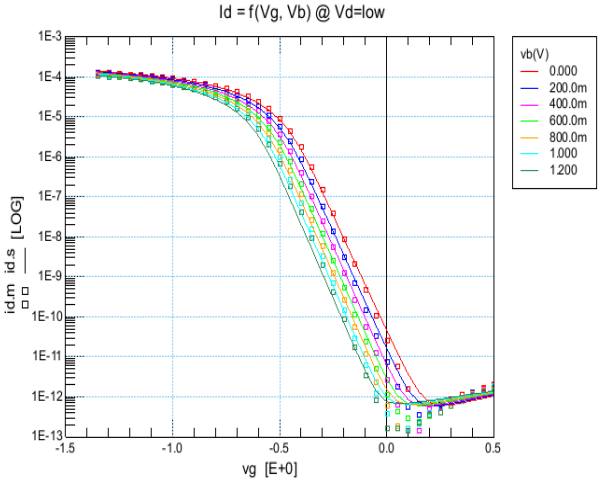
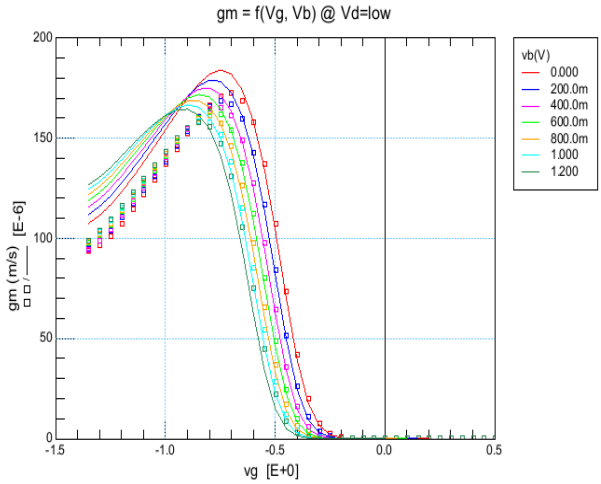
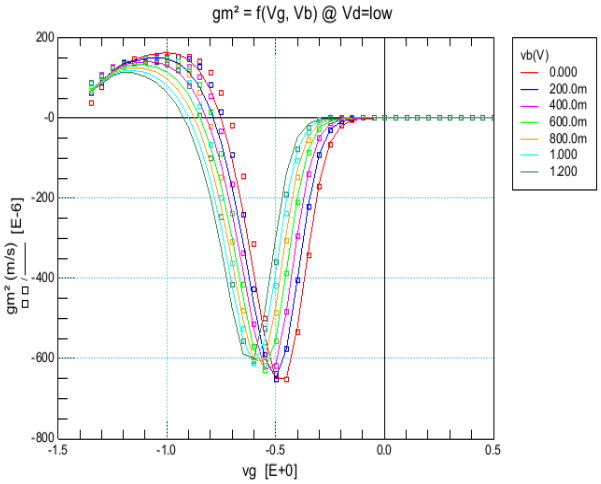
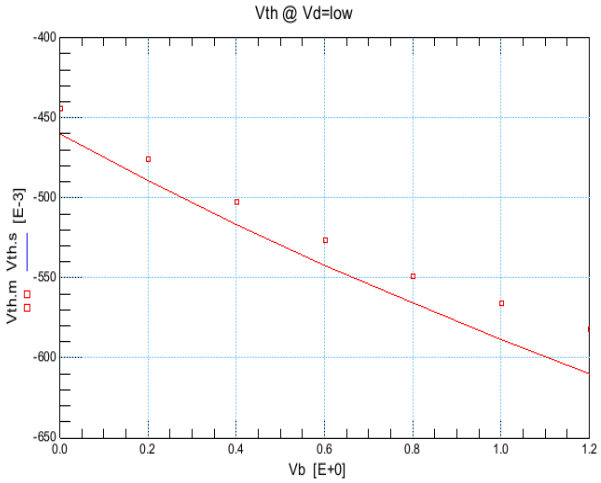
idvg, Vd = -0.05V, T = 27°C

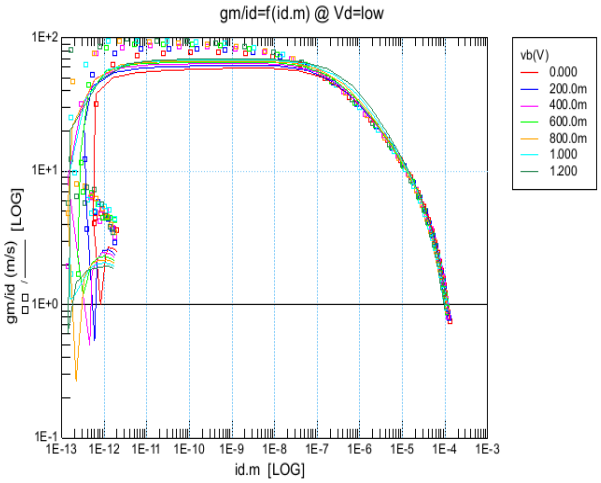


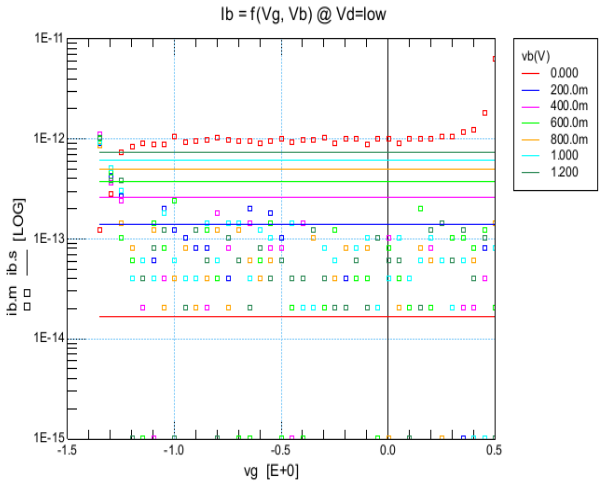


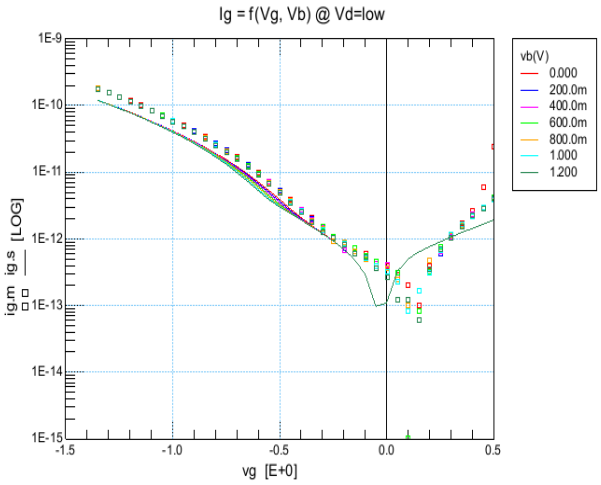




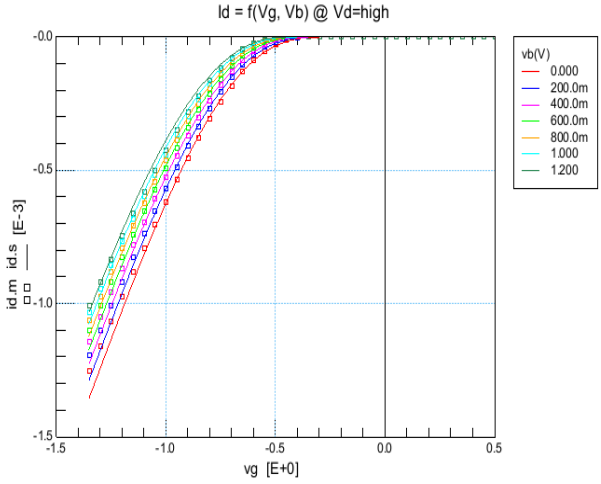


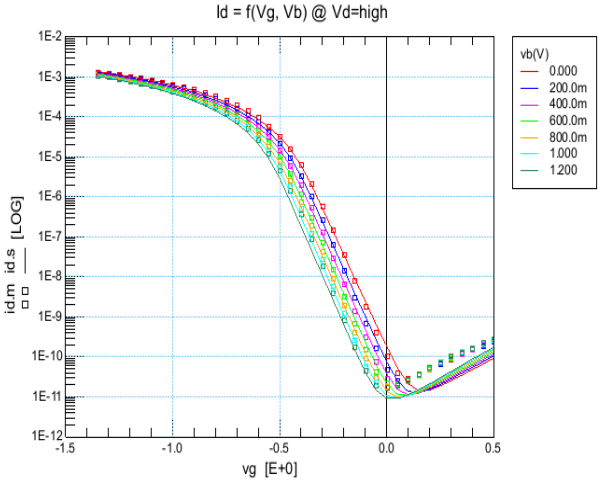


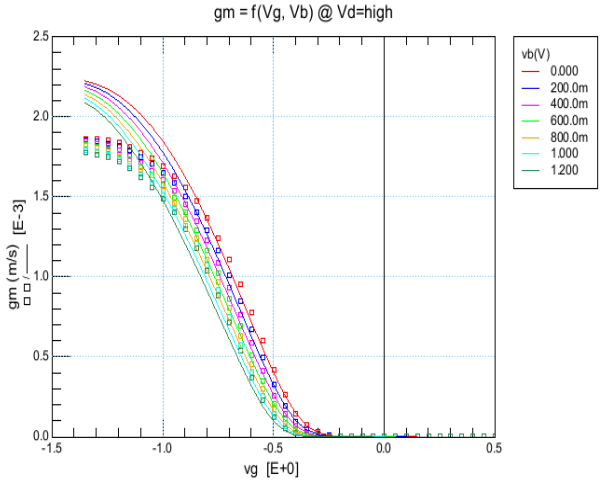


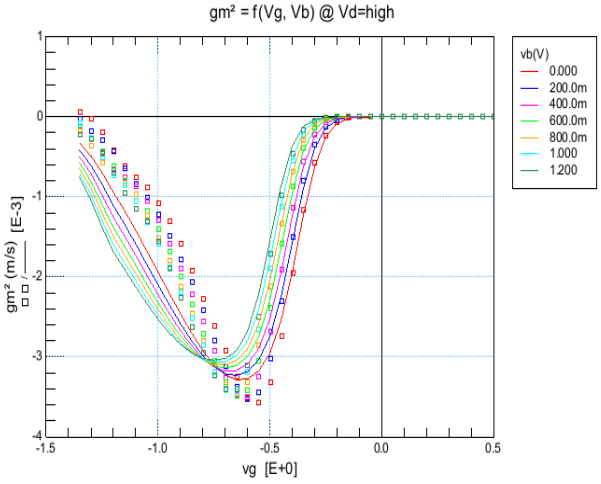


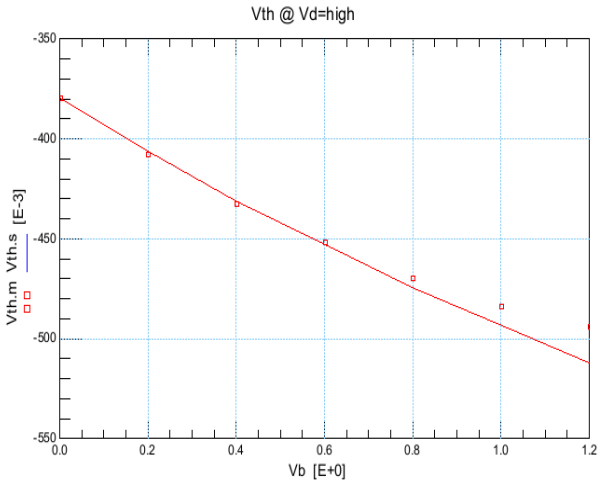
idvg, Vd = -1.2V, T = 27°C

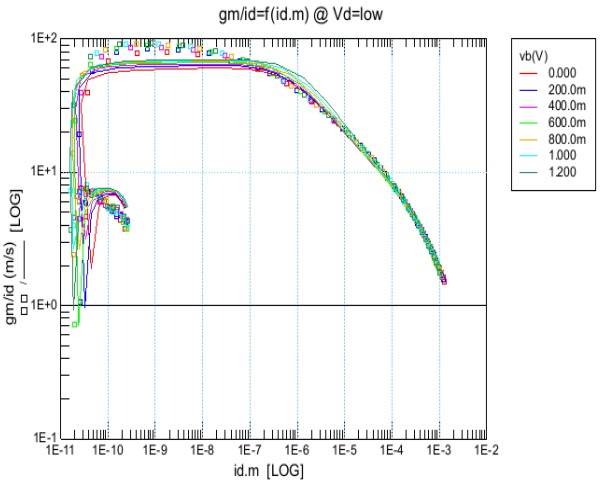


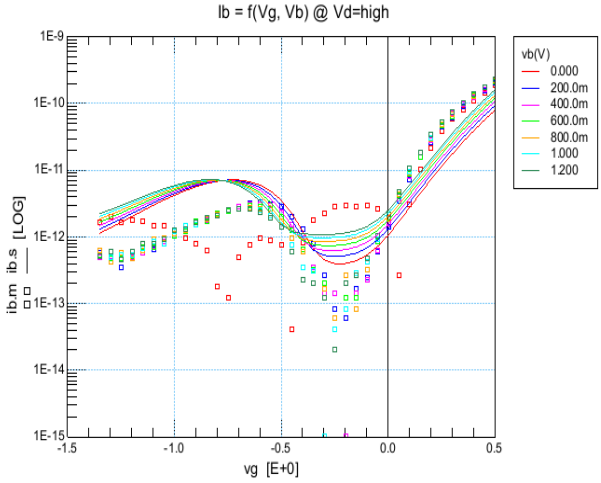


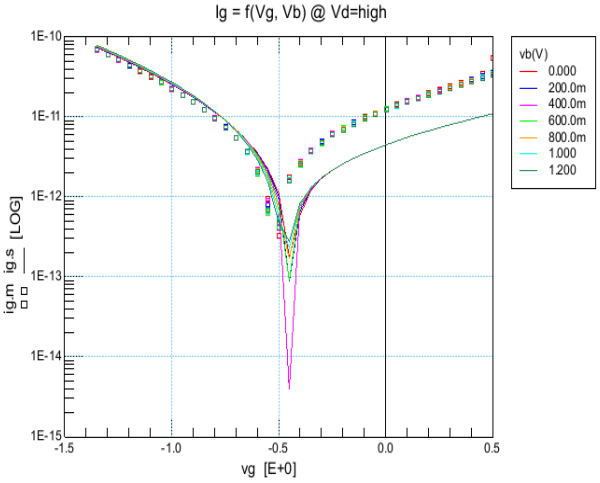




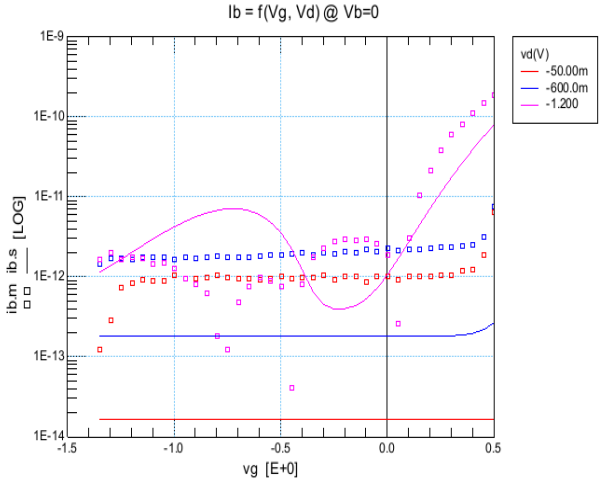


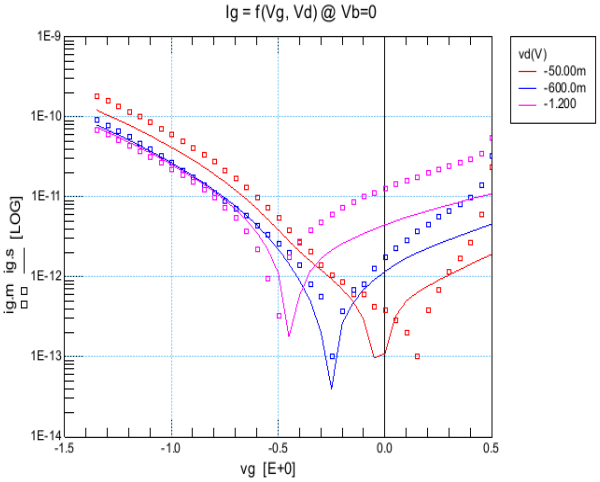




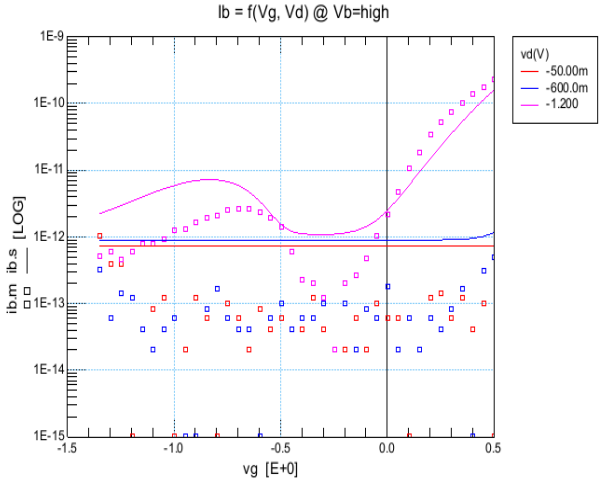


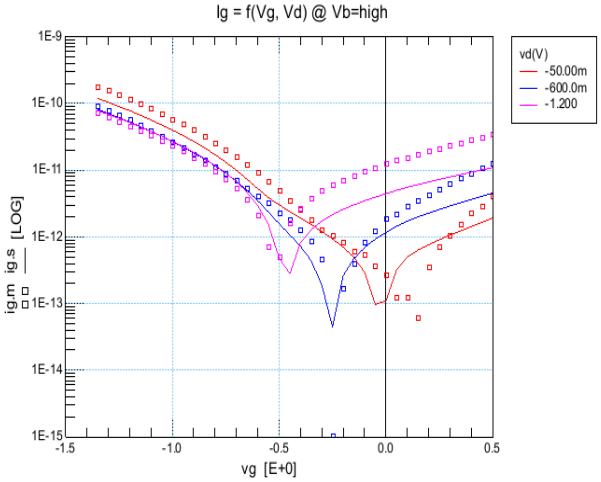
idvg, Ib,Ig, Vb = 0V, T = 27°C



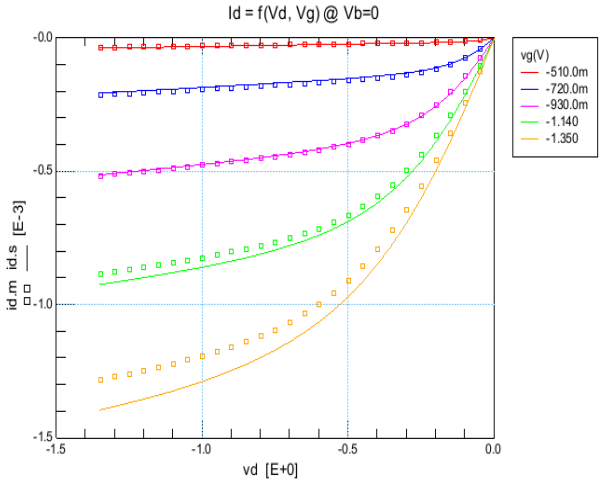


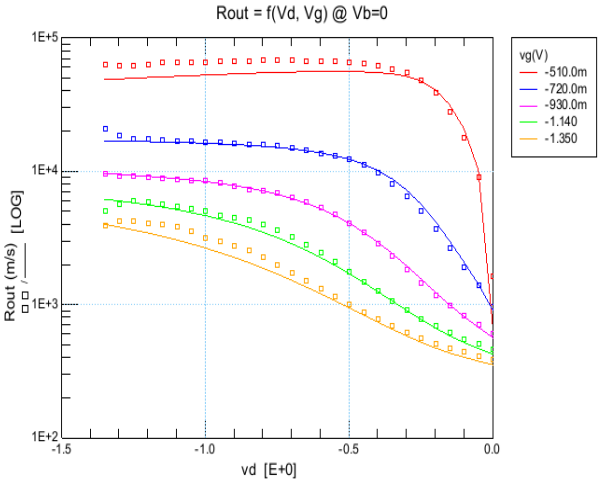
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

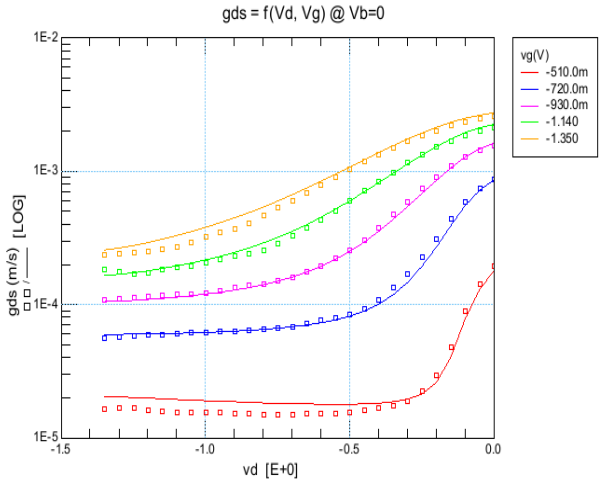




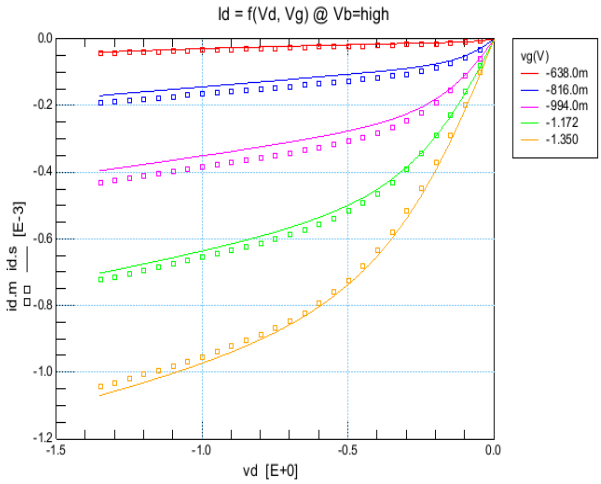
idvd, Vb = 0V, T = 27°C

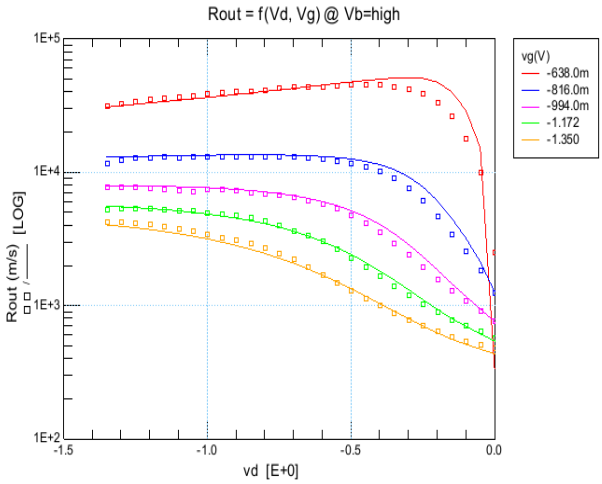


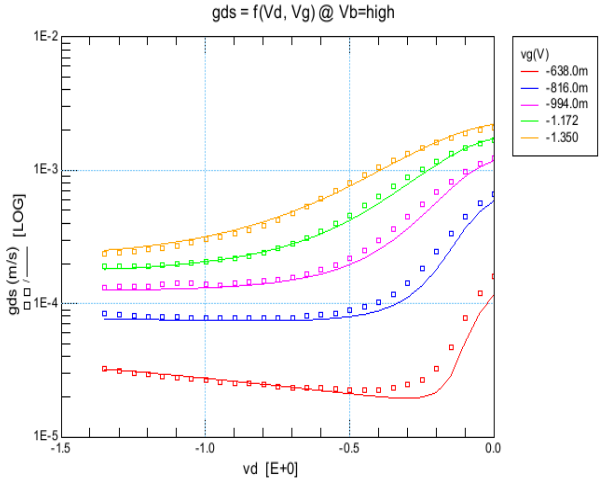




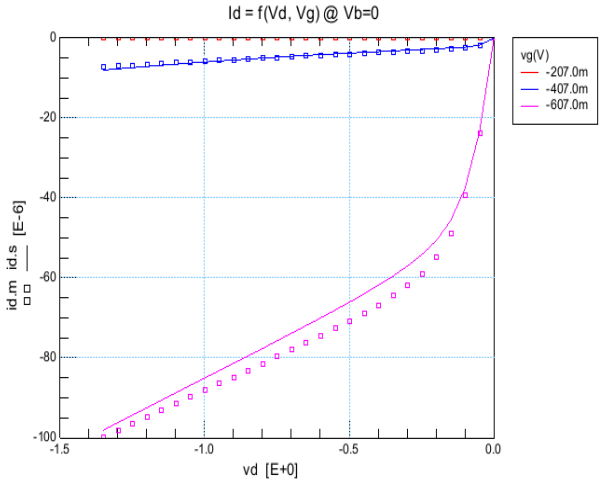
idvd_vbmin, Vb = 1.2V, T = 27°C

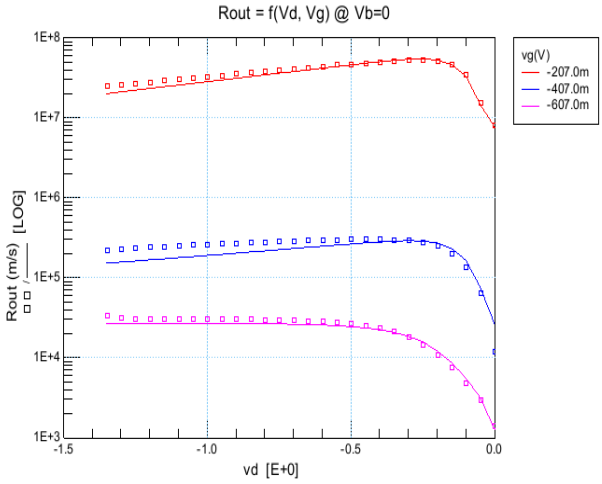


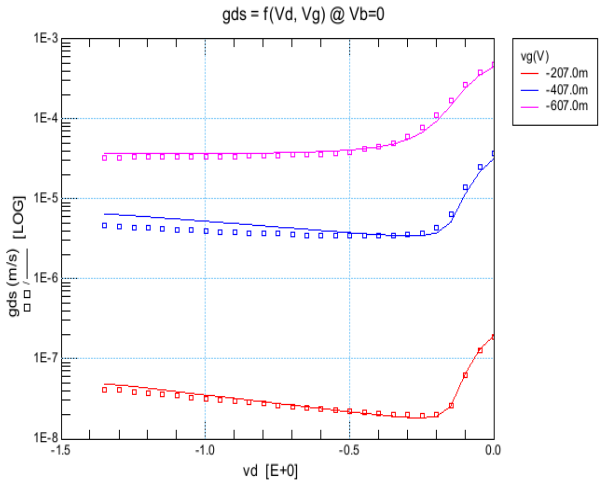




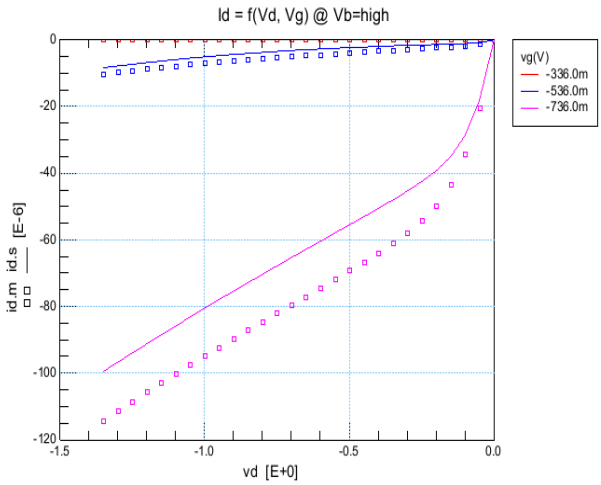
idvd_vth, Vb = 0V, T = 27°C

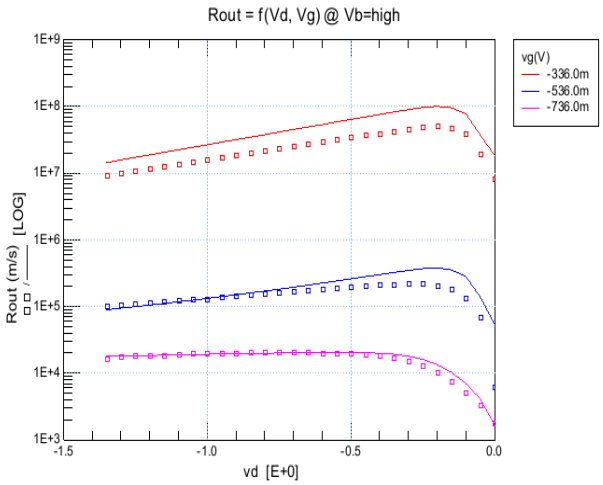


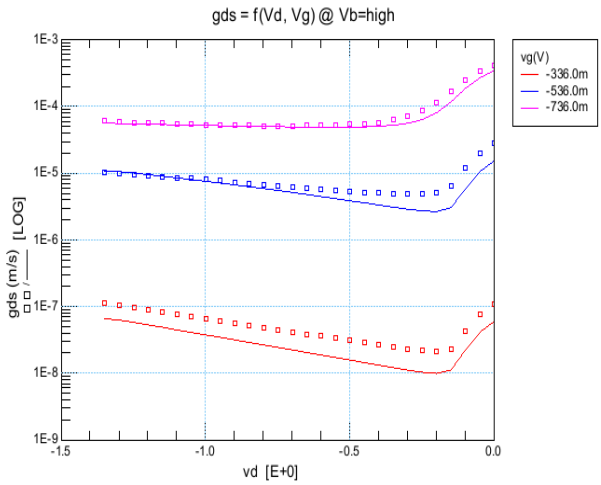




idvd_vth_vbmin, Vb = 1.2V, T = 27°C



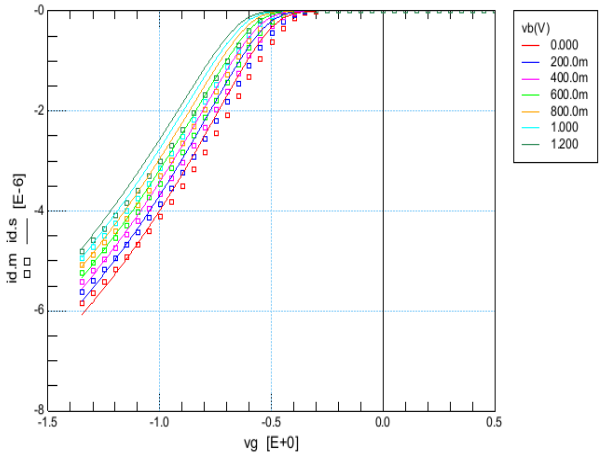




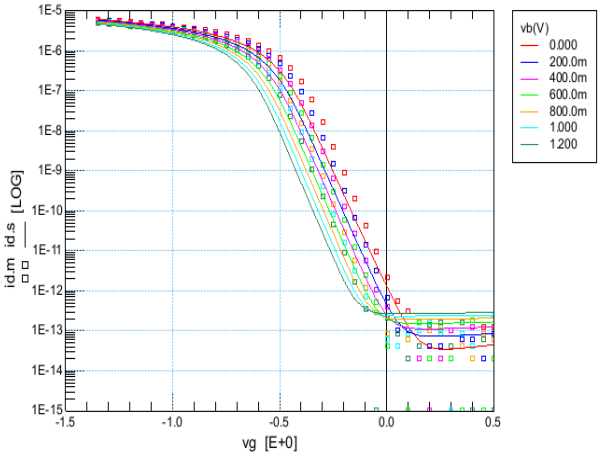
4.30 W0u15_L0u15_S554_5

idvg, Vd = -0.05V, T = 27°C

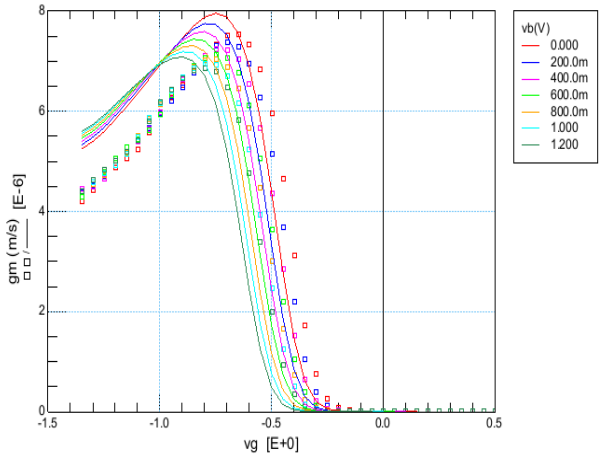
Id = f(Vg, Vb) @ Vd=low



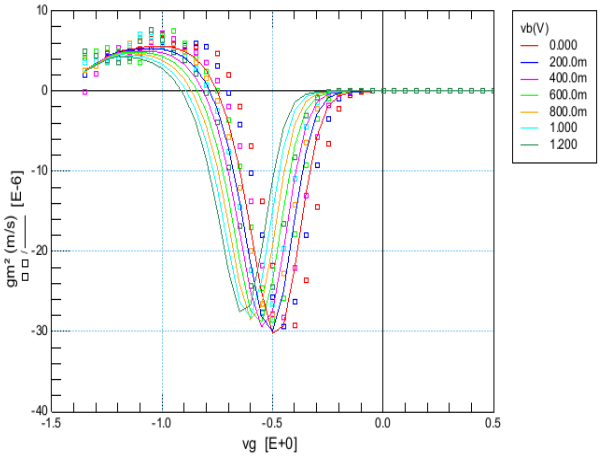
Id = f(Vg, Vb) @ Vd=low



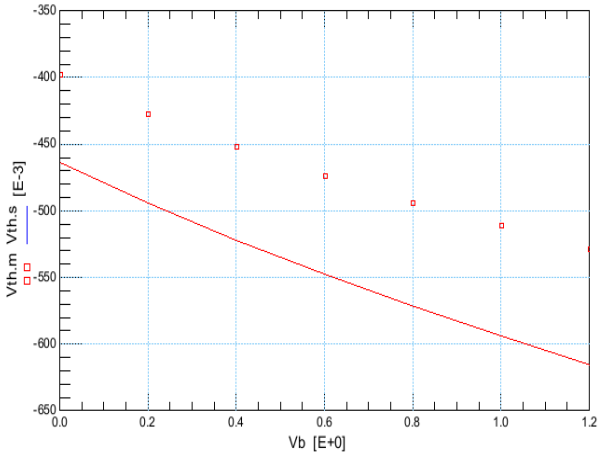
gm = f(Vg, Vb) @ Vd=low



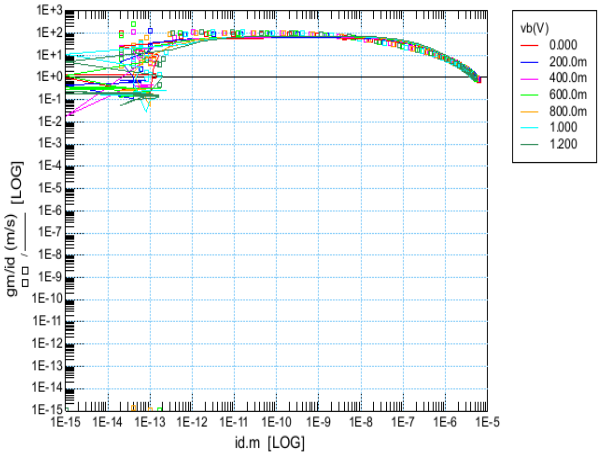
gm² = f(Vg, Vb) @ Vd=low

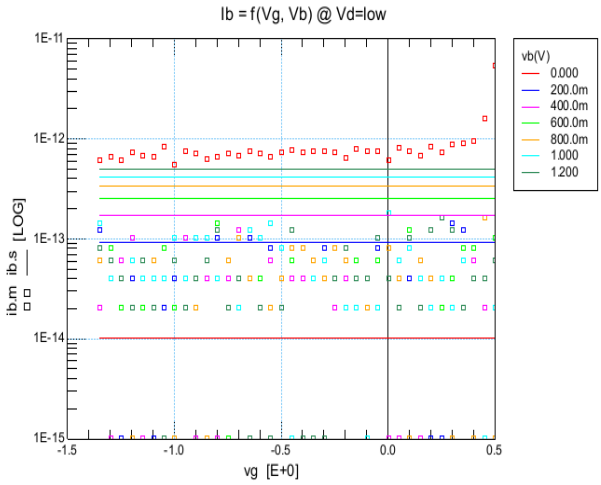


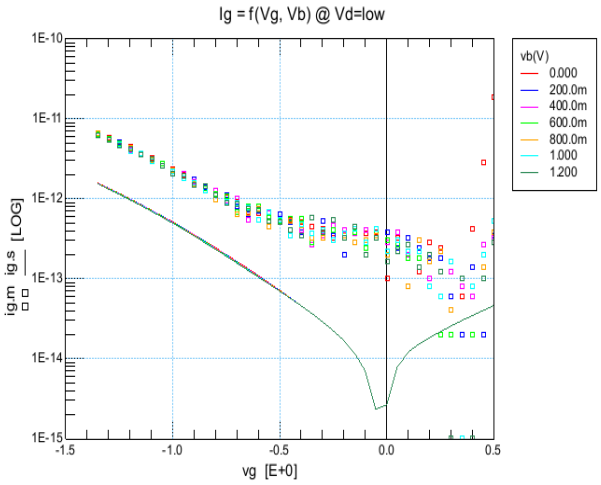
Vth @ Vd=low



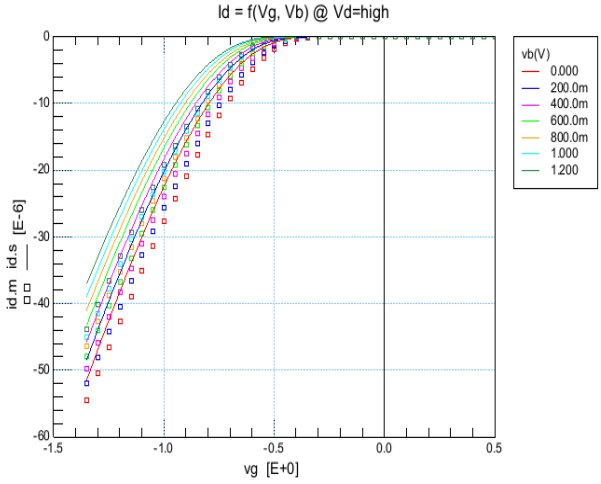
gm/id=f(id.m) @ Vd=low

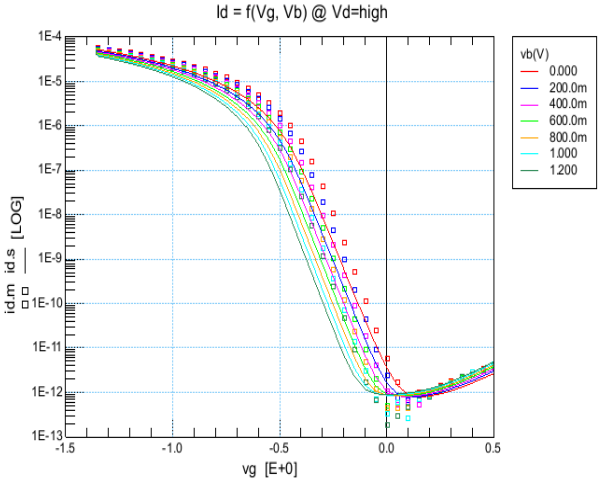


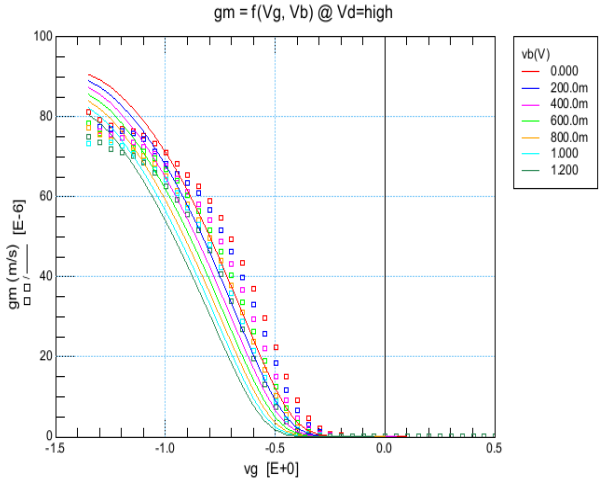


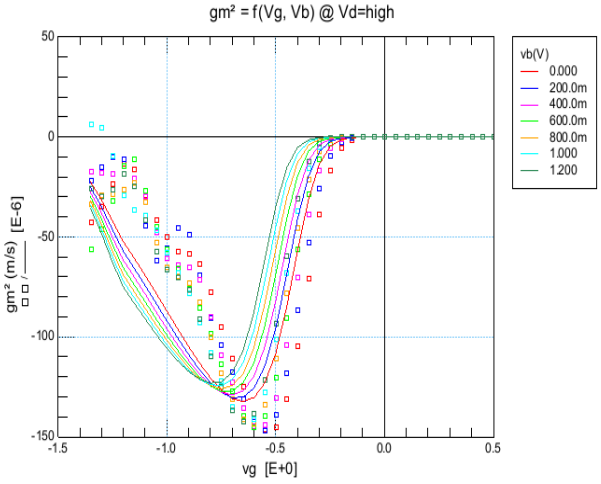


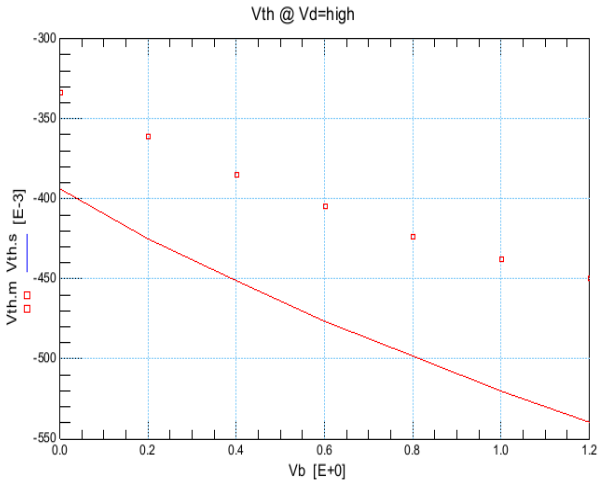
idvg, Vd = -1.2V, T = 27°C

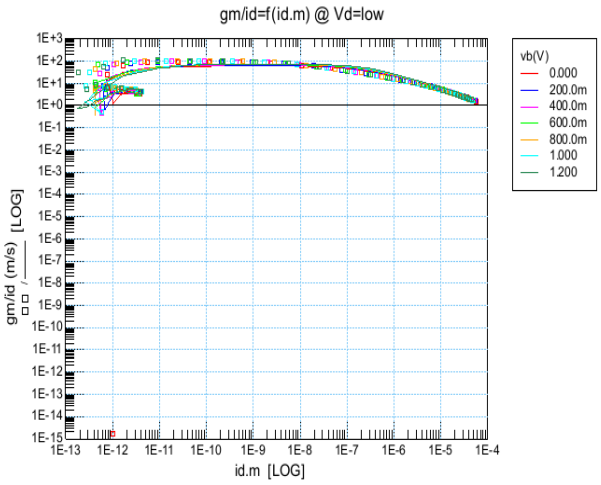


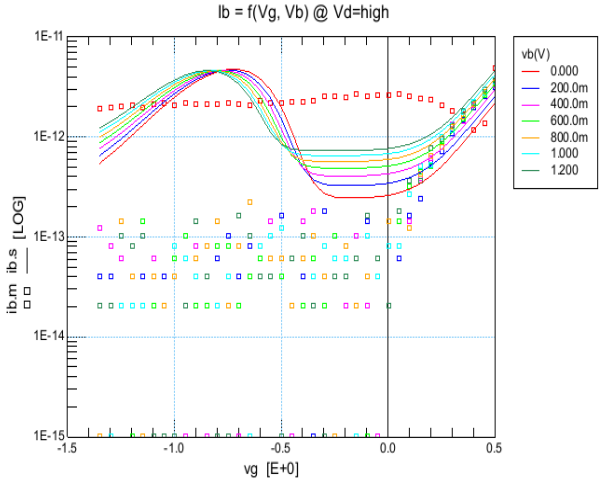


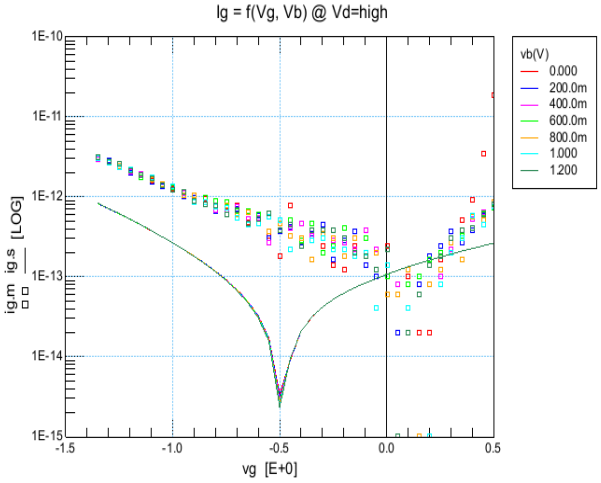




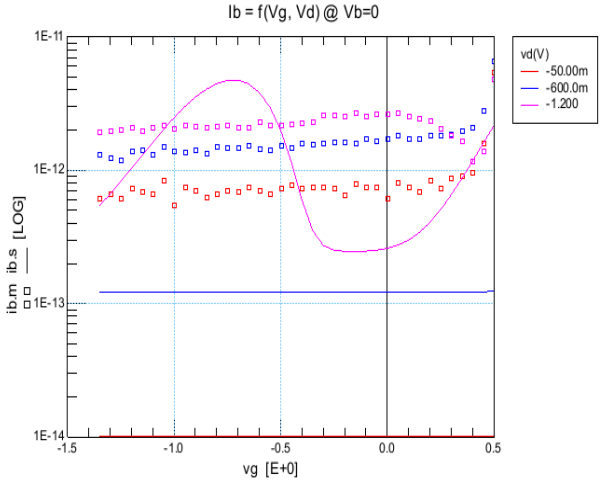


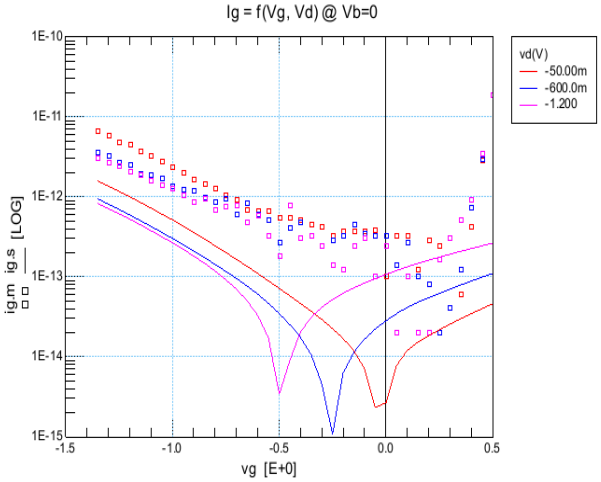




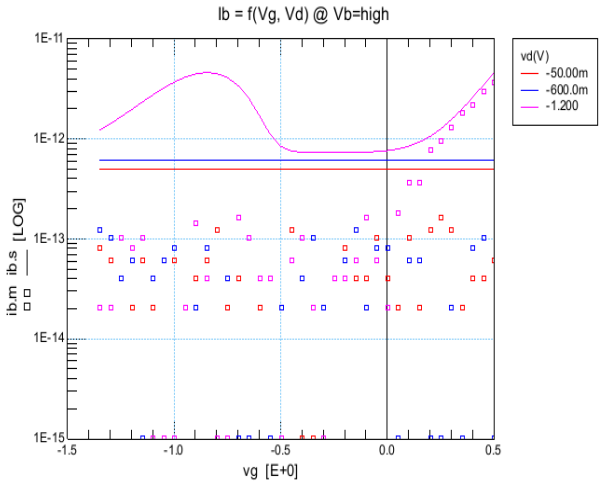


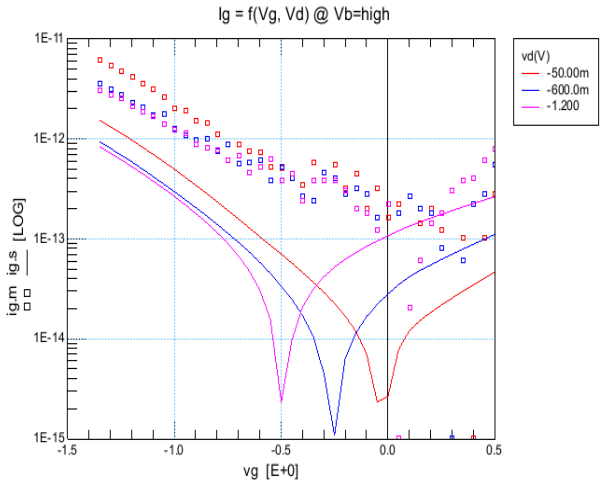
idvg, Ib,Ig, Vb = 0V, T = 27°C



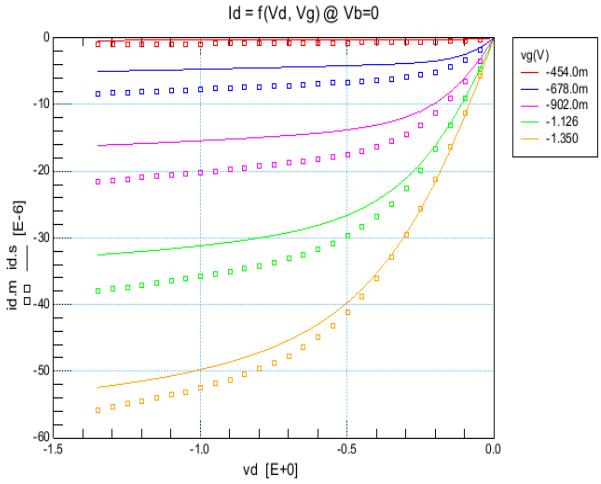


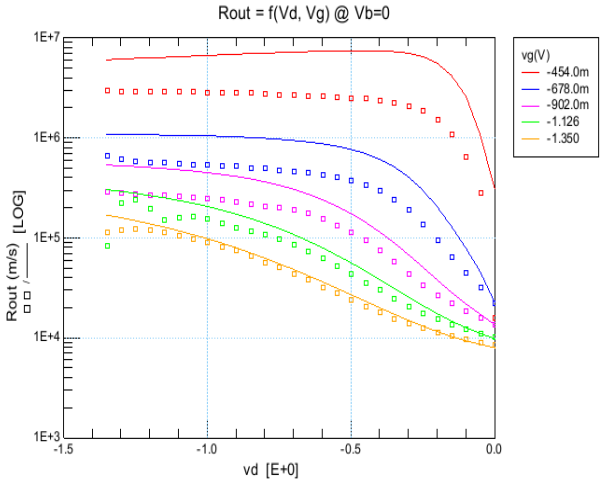
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

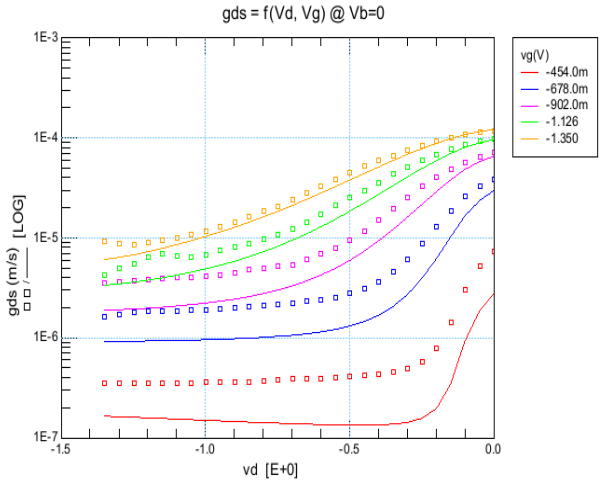




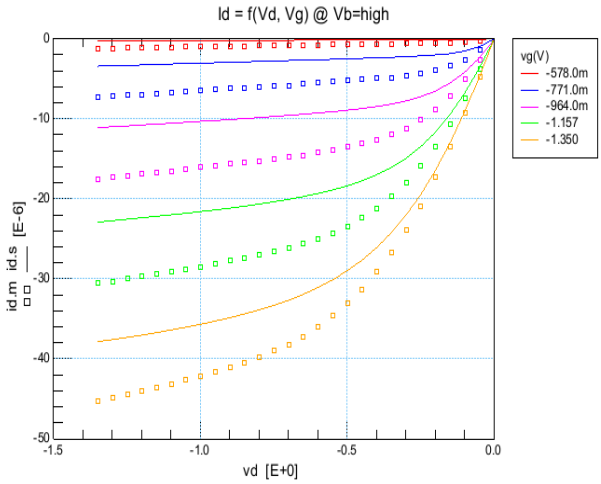
idvd, Vb = 0V, T = 27°C

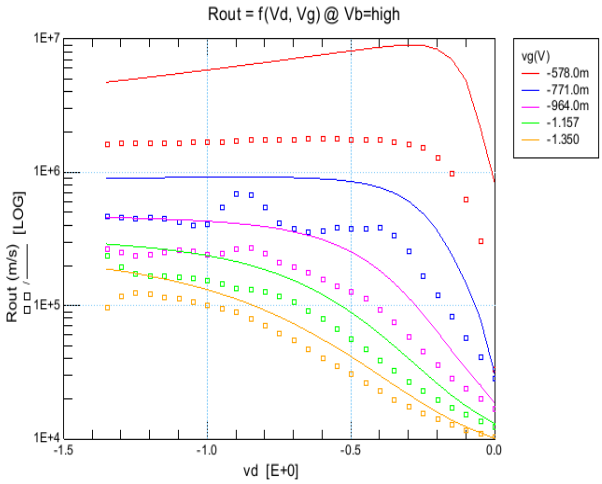


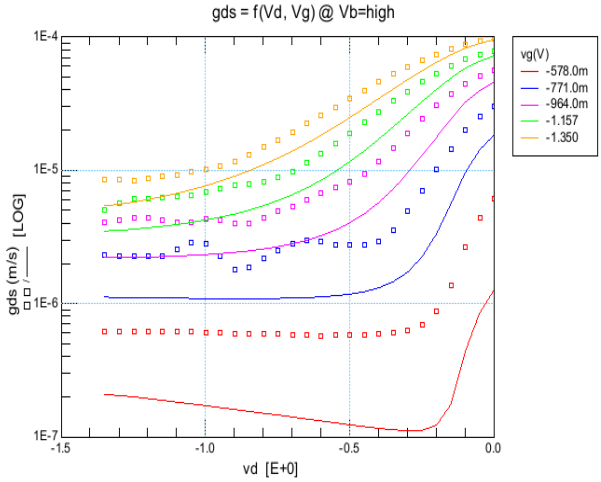




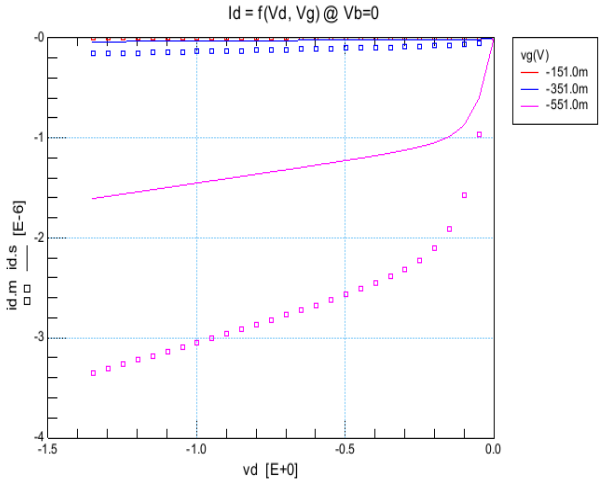
idvd_vbmin, Vb = 1.2V, T = 27°C

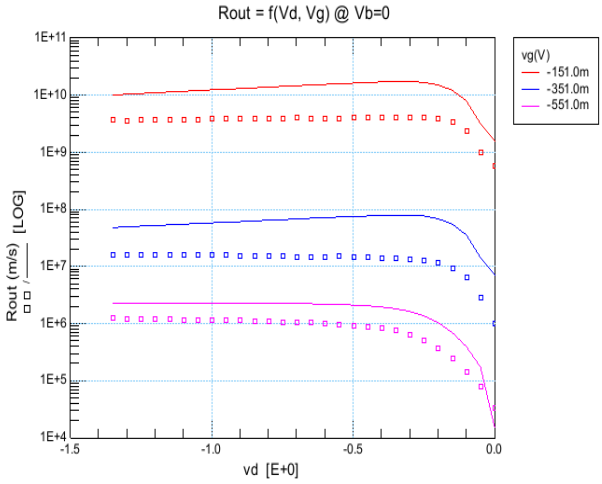


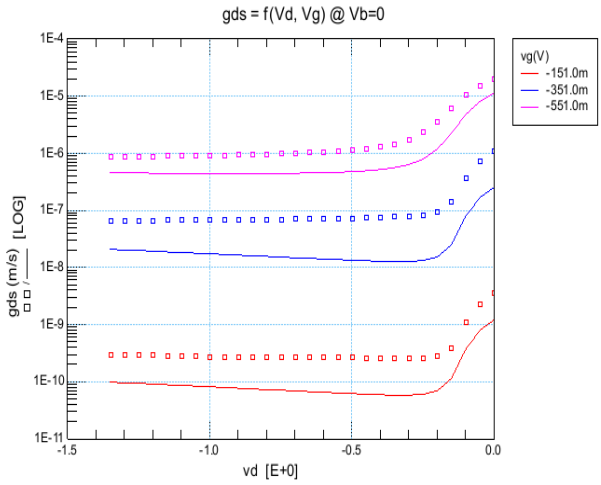




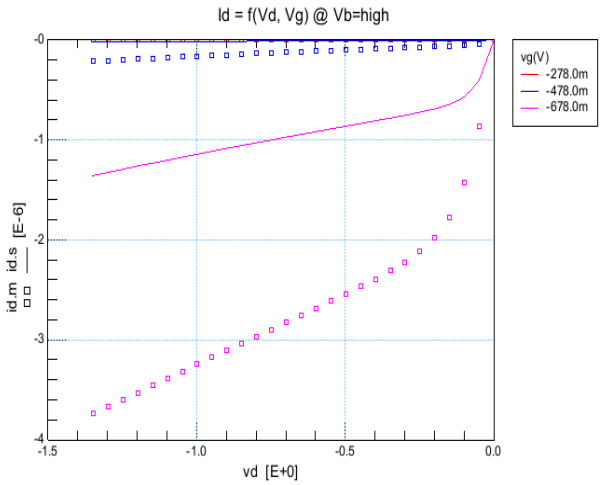
idvd_vth, Vb = 0V, T = 27°C

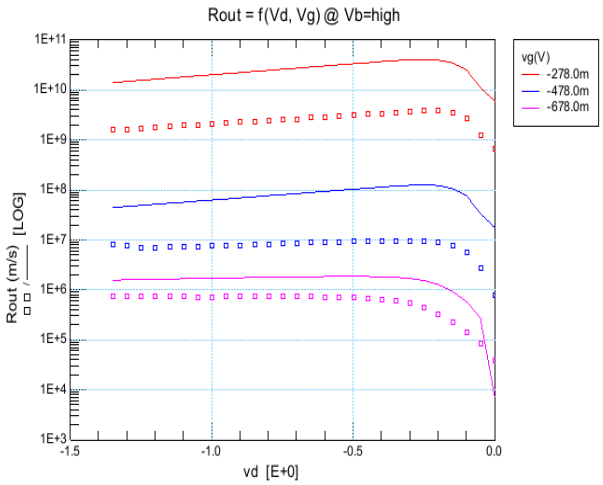


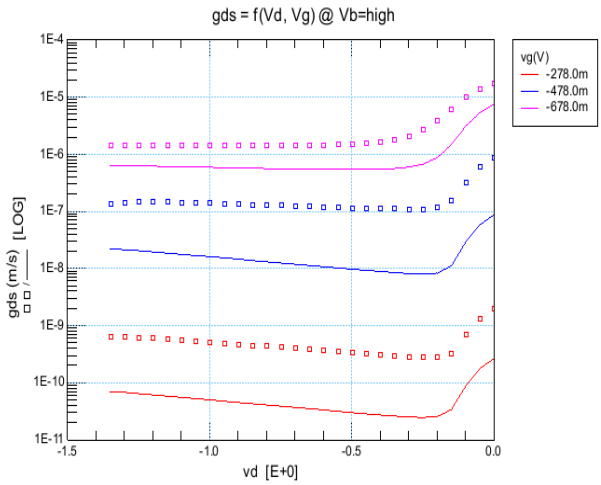




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

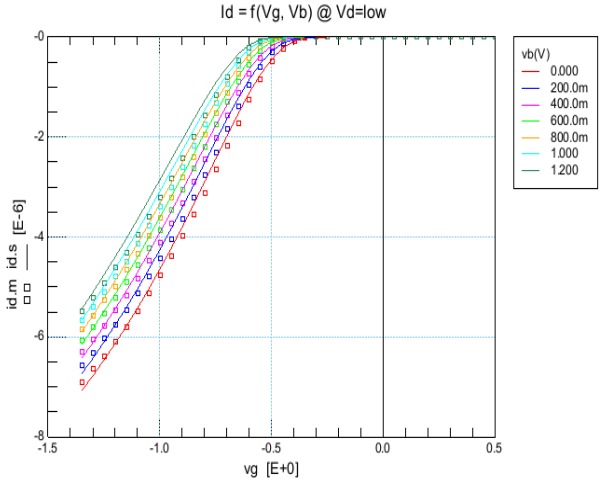


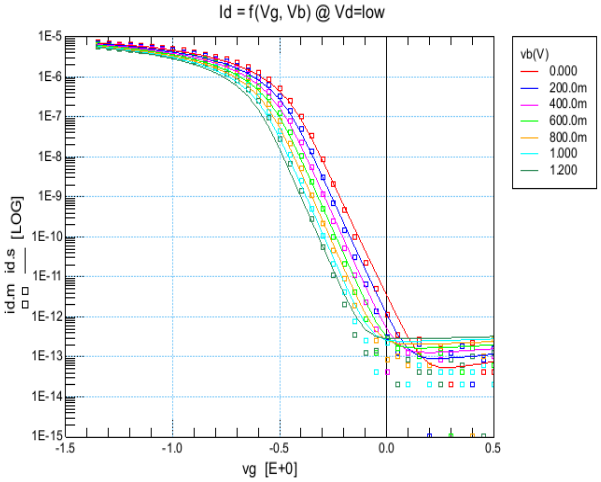


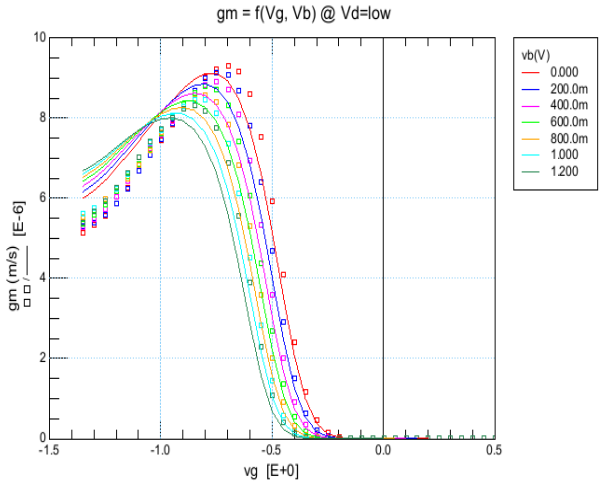


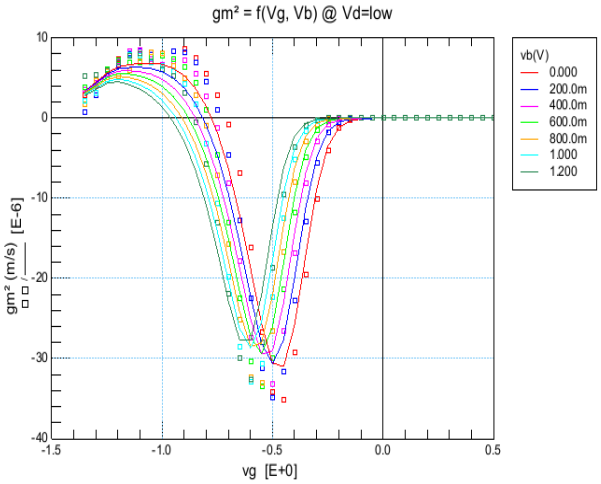
4.31 W0u3_L0u18_S552_5

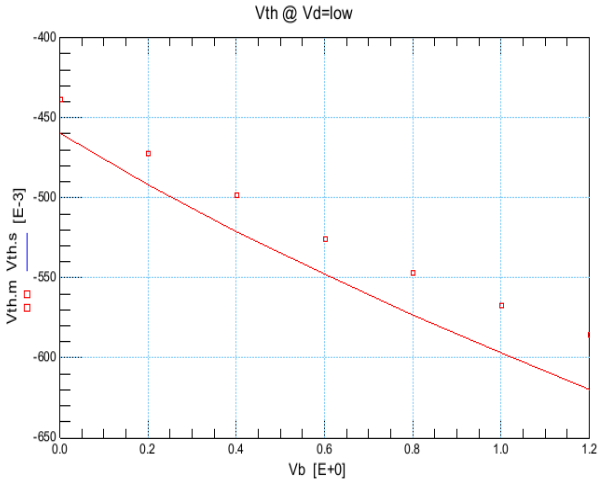
idvg, Vd = -0.05V, T = 27°C

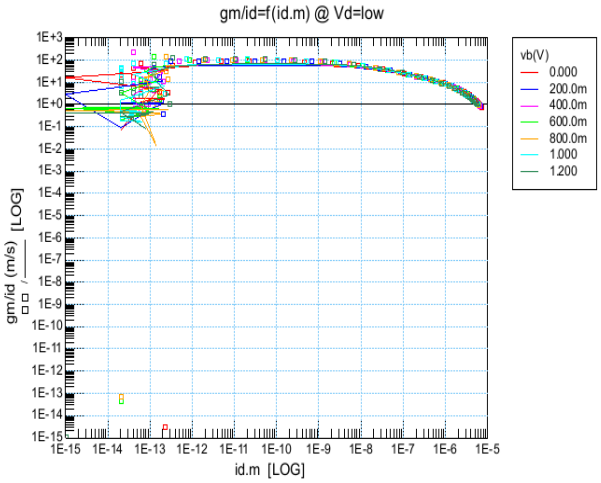


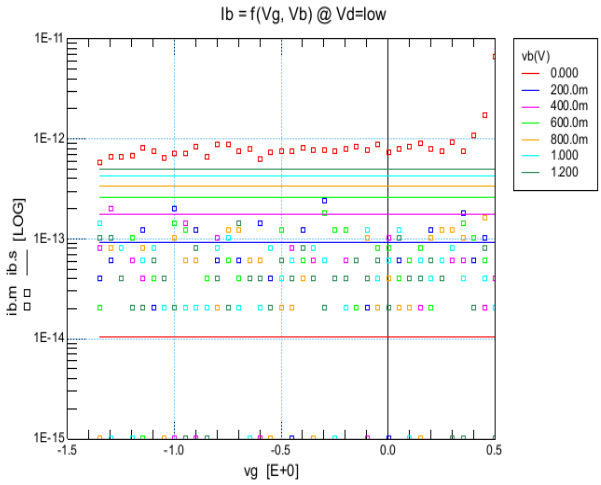


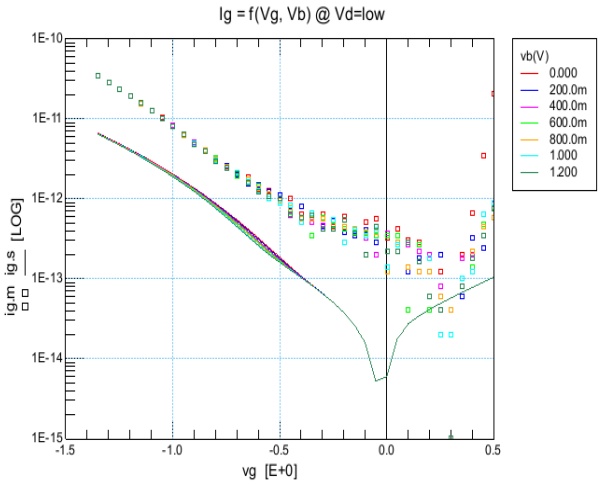




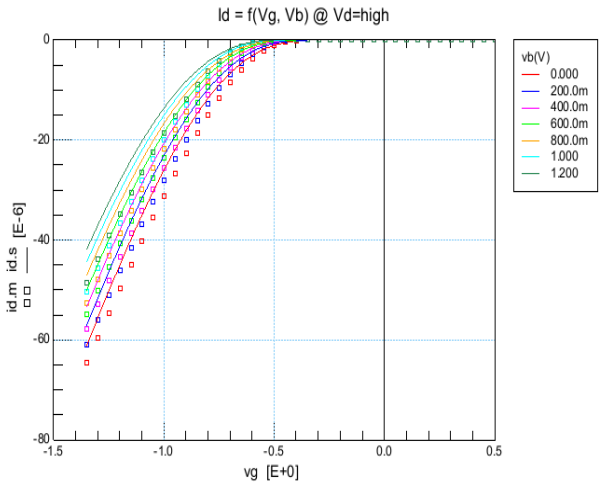


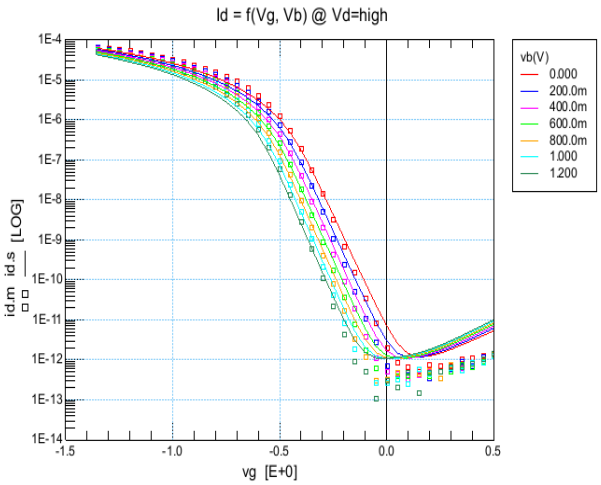


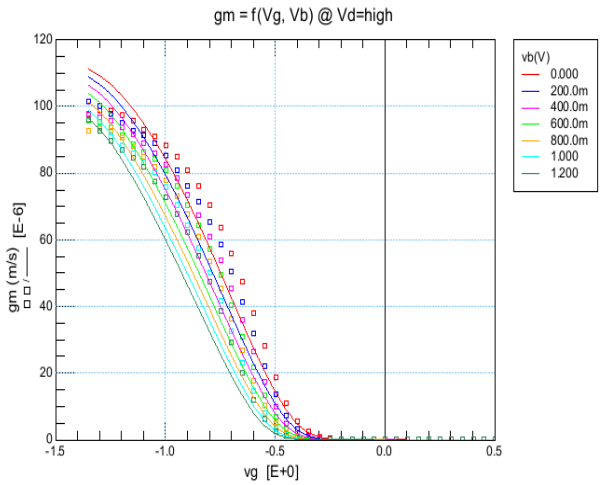


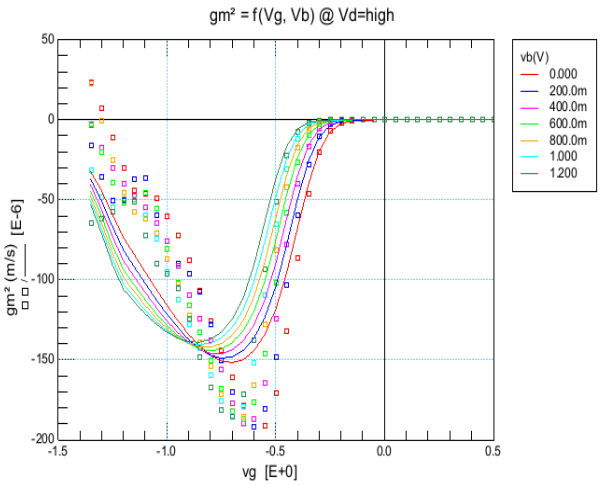


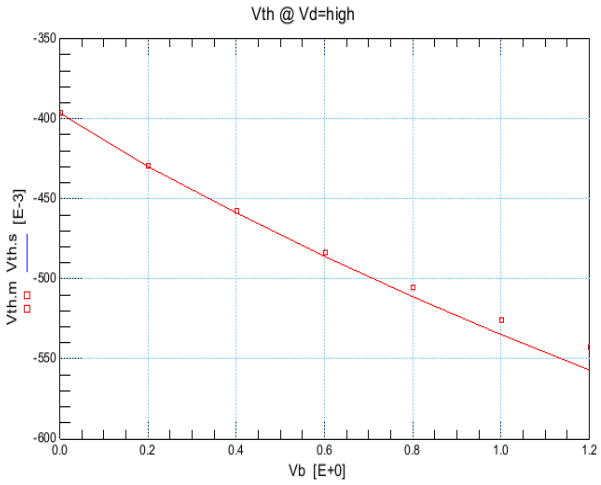
idvg, Vd = -1.2V, T = 27°C

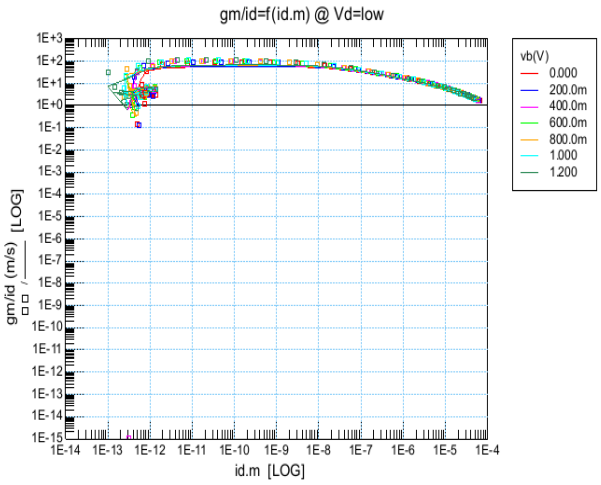


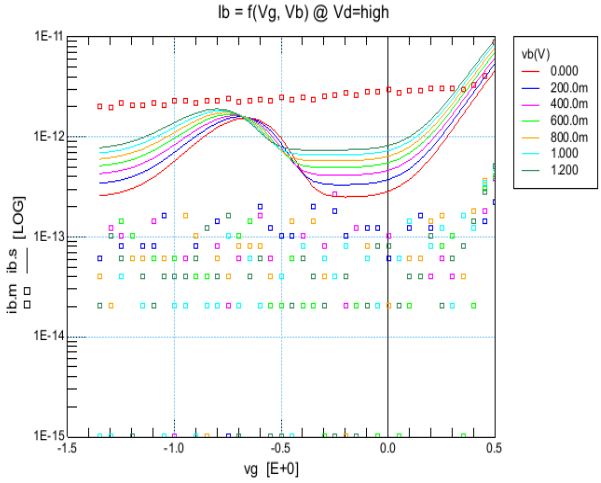


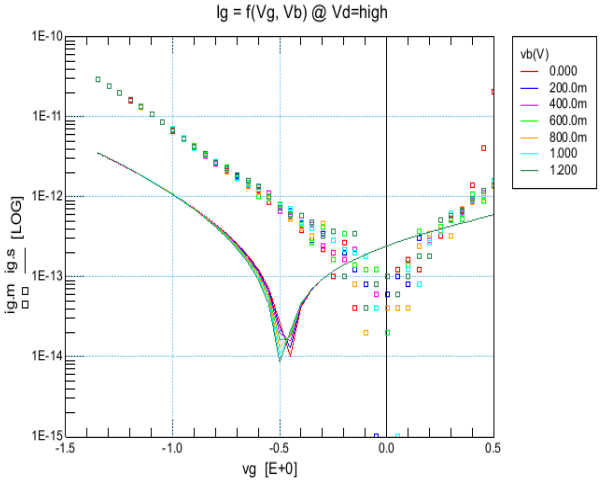




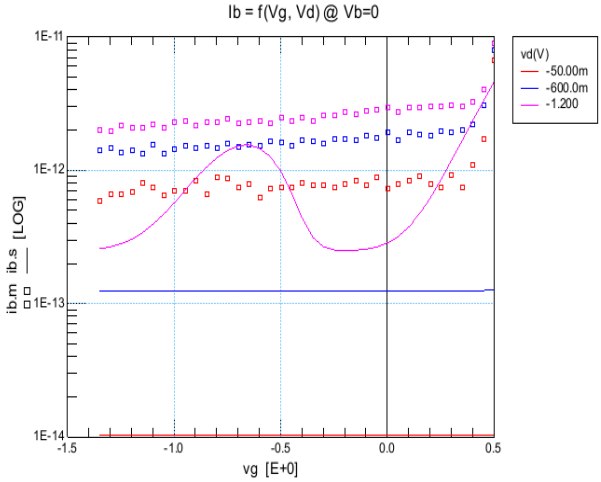


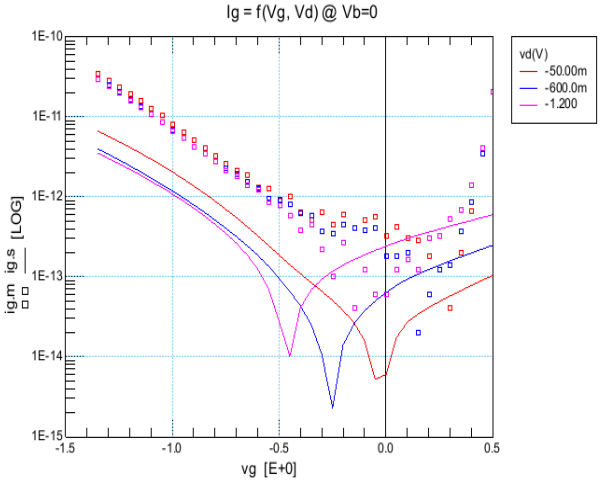




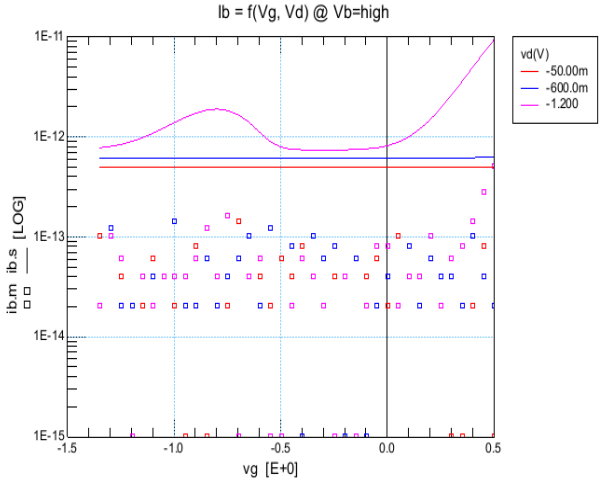


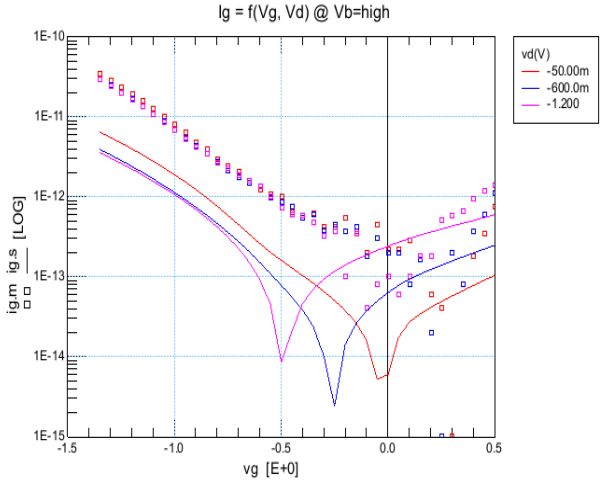
idvg, Ib,Ig, Vb = 0V, T = 27°C



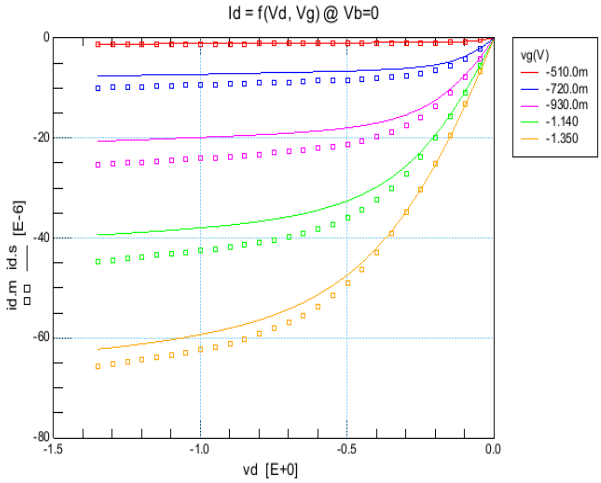


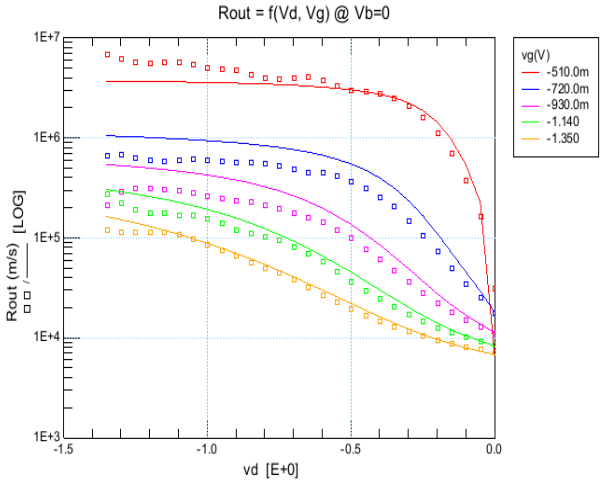
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

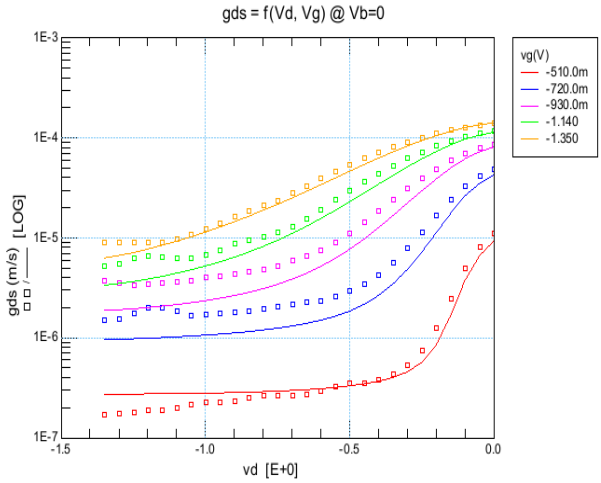




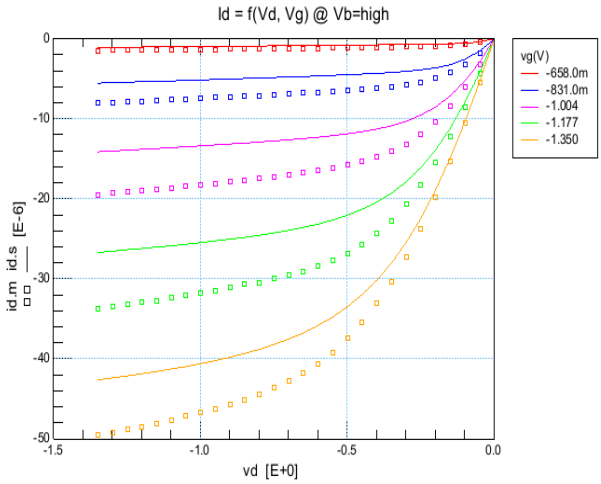
idvd, Vb = 0V, T = 27°C

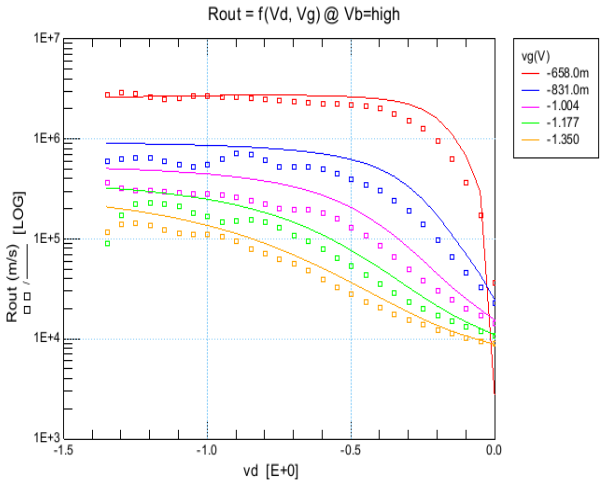


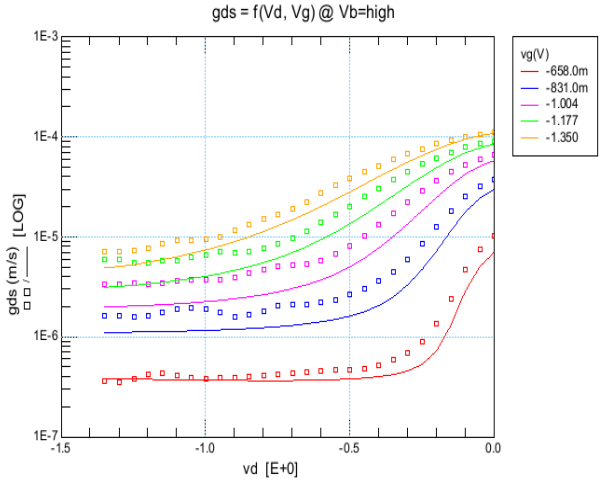




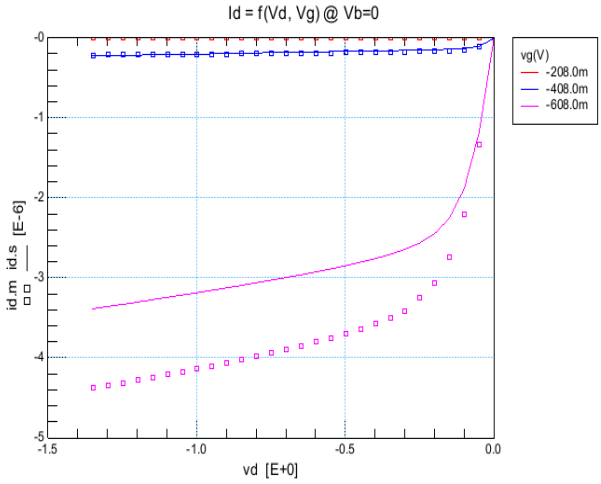
idvd_vbmin, Vb = 1.2V, T = 27°C

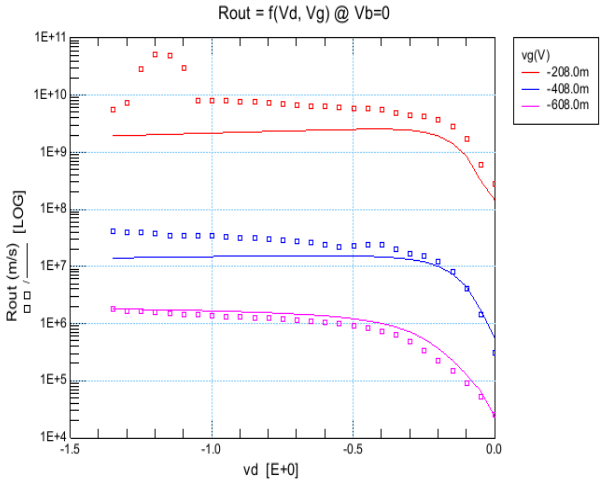


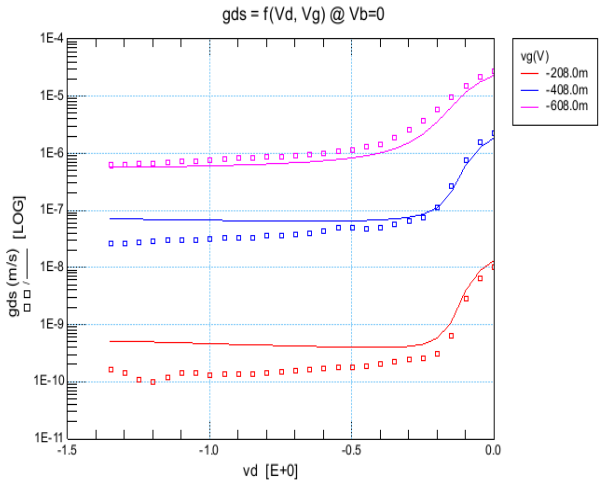




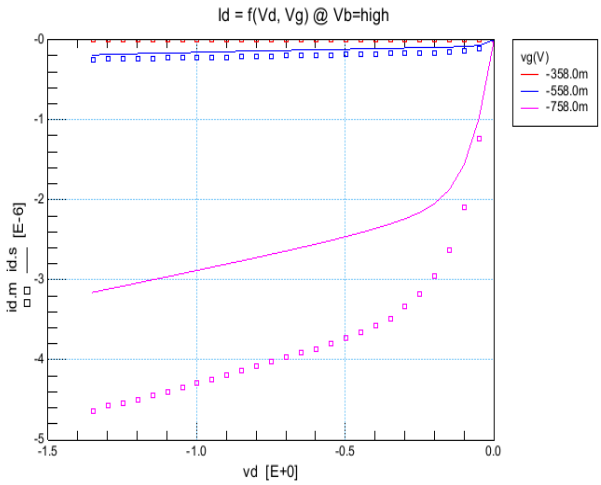
idvd_vth, Vb = 0V, T = 27°C

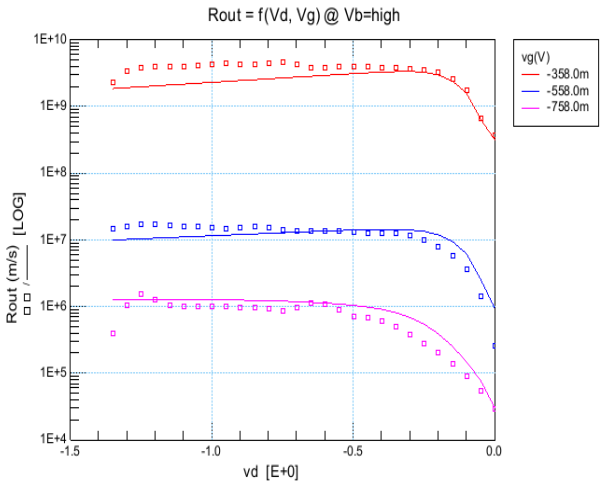


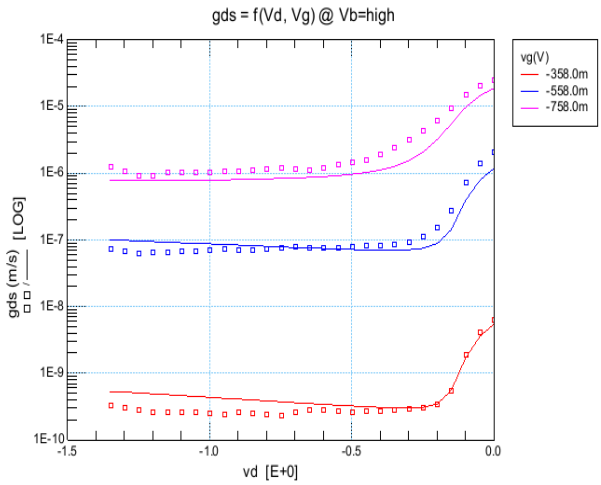




idvd_vth_vbmin, Vb = 1.2V, T = 27°C



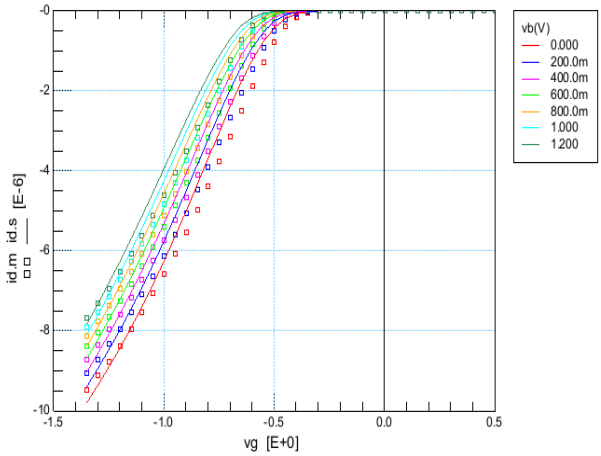




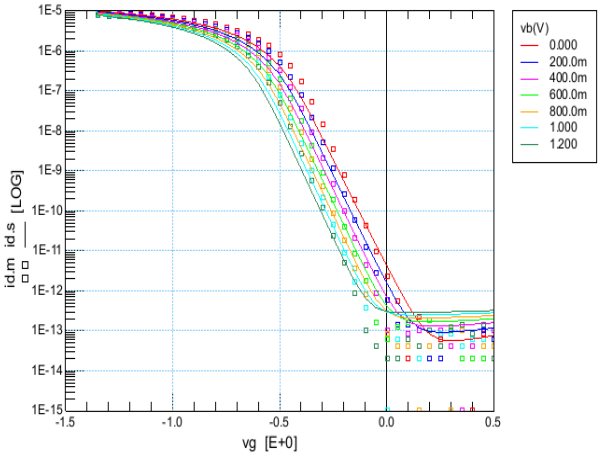
4.32 W0u3_L0u13_S552_2

idvg, Vd = -0.05V, T = 27°C

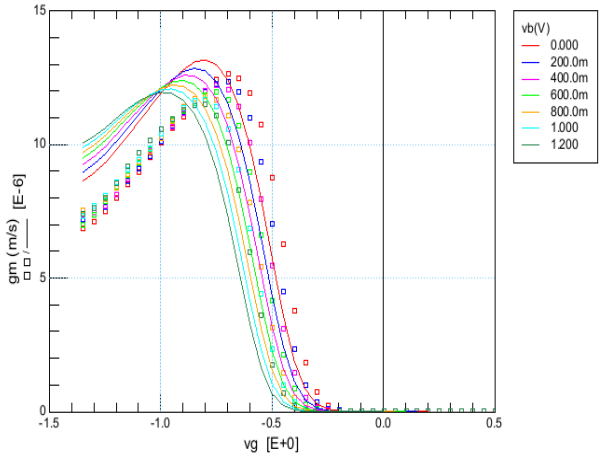
Id = f(Vg, Vb) @ Vd=low



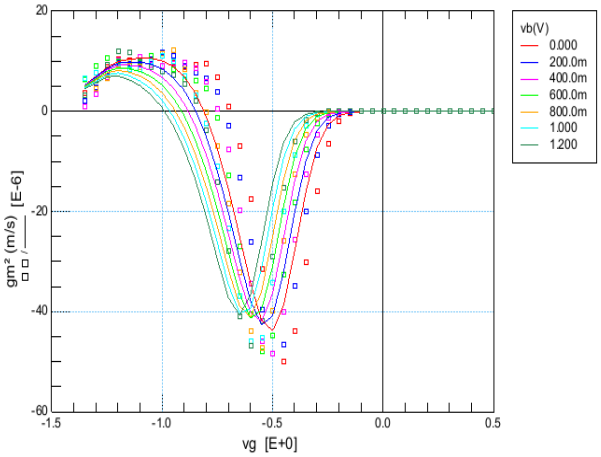
Id = f(Vg, Vb) @ Vd=low



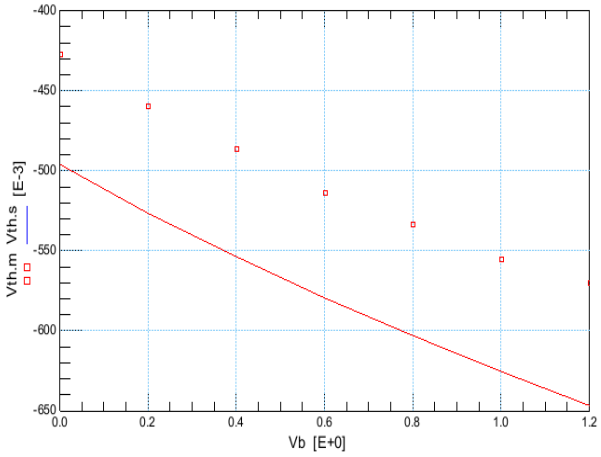
gm = f(Vg, Vb) @ Vd=low



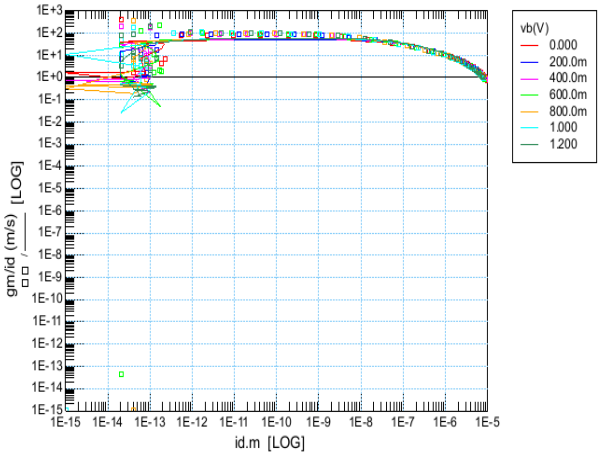
gm² = f(Vg, Vb) @ Vd=low

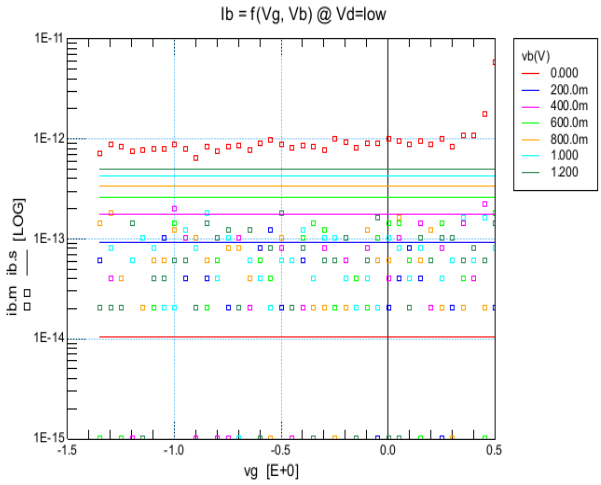


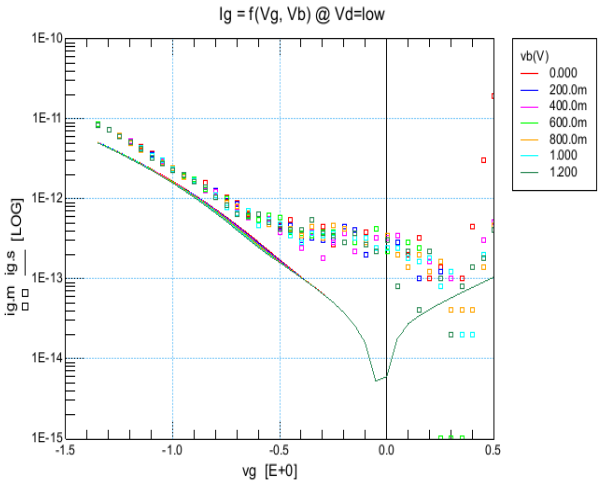
Vth @ Vd=low



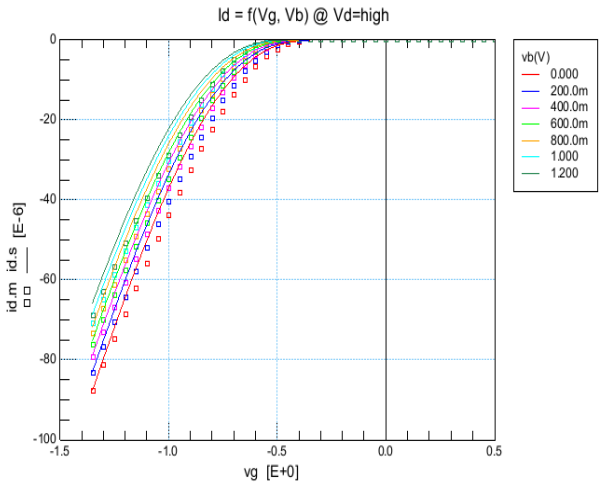
gm/id=f(id.m) @ Vd=low

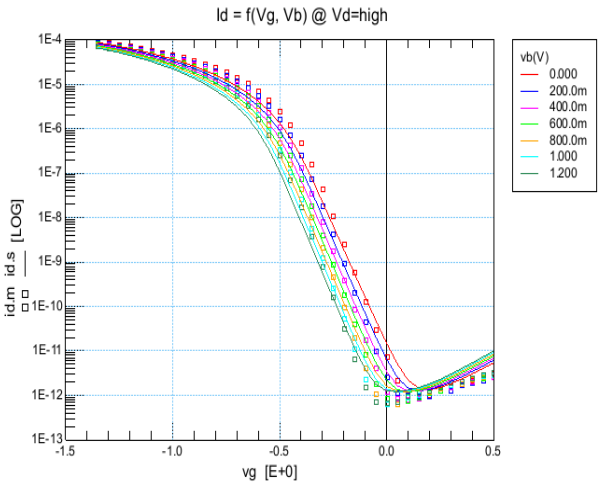


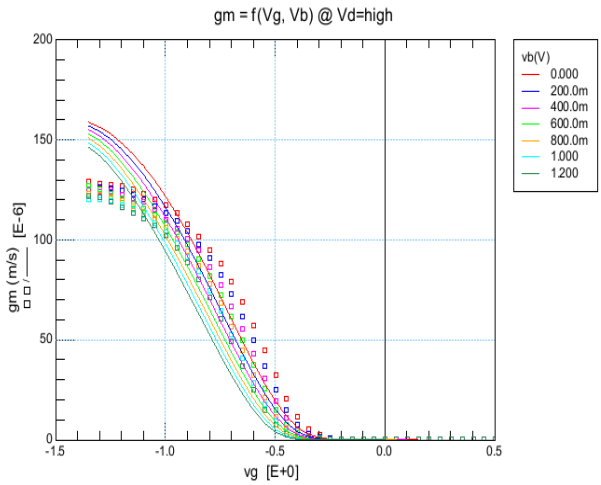


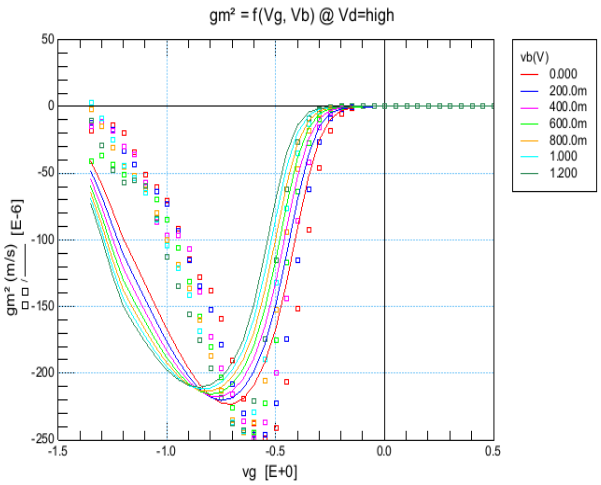


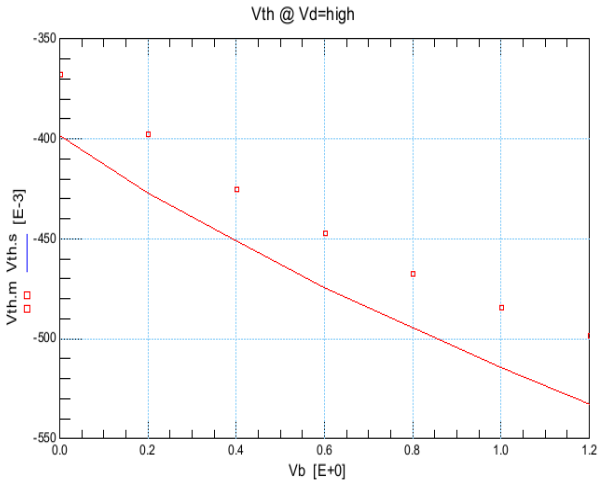
idvg, Vd = -1.2V, T = 27°C

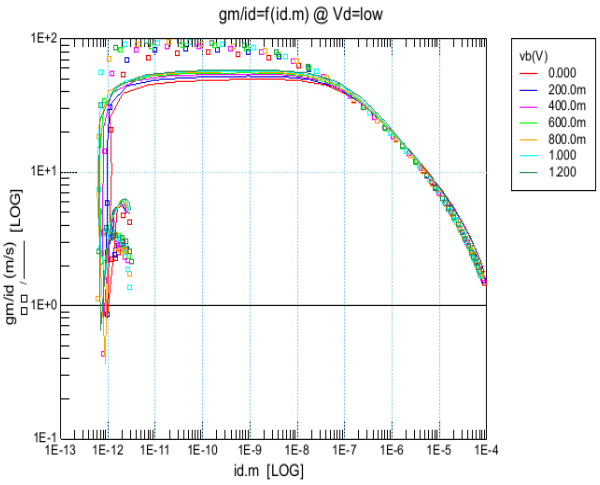


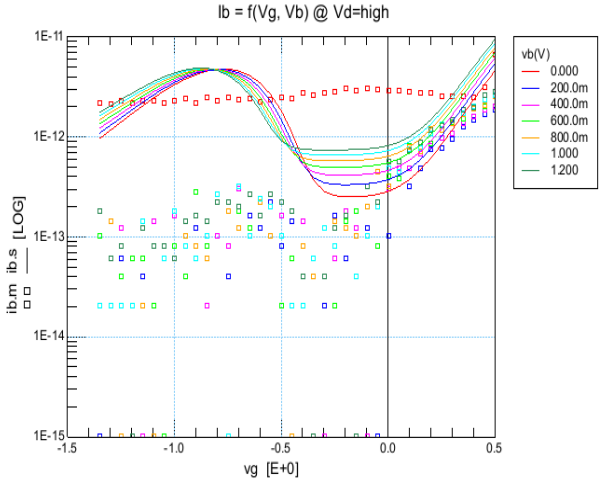


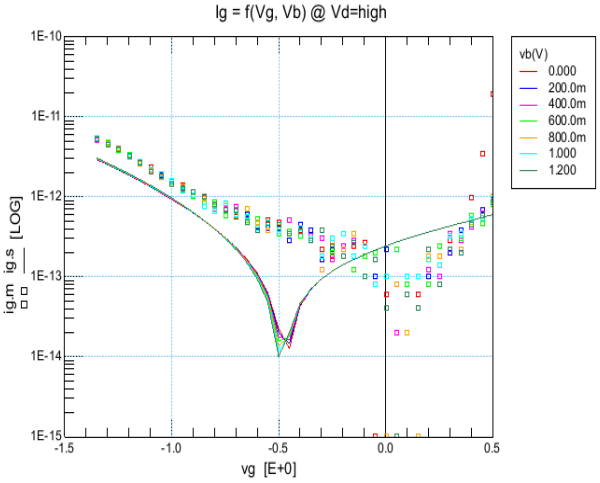




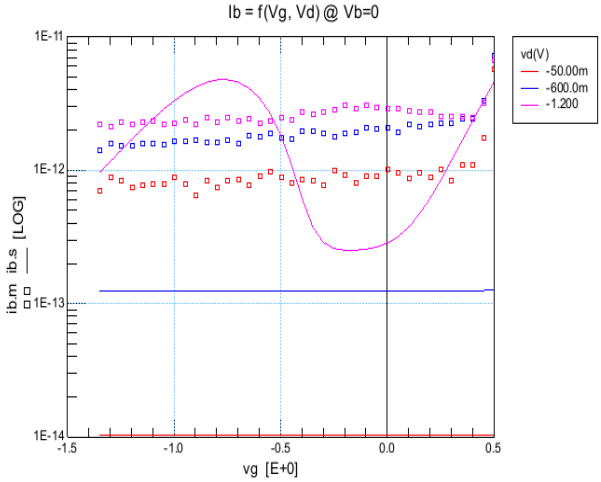


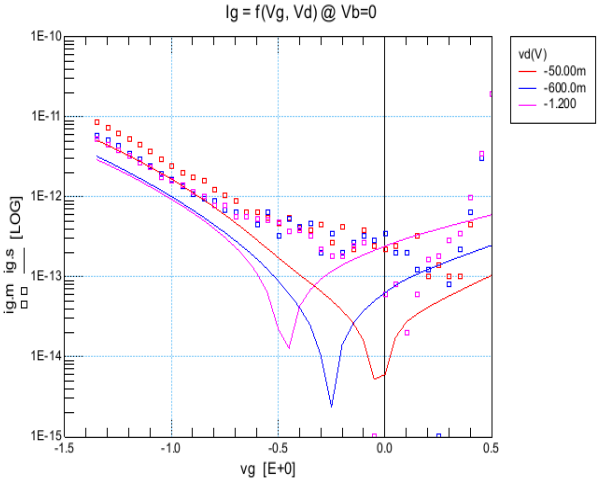




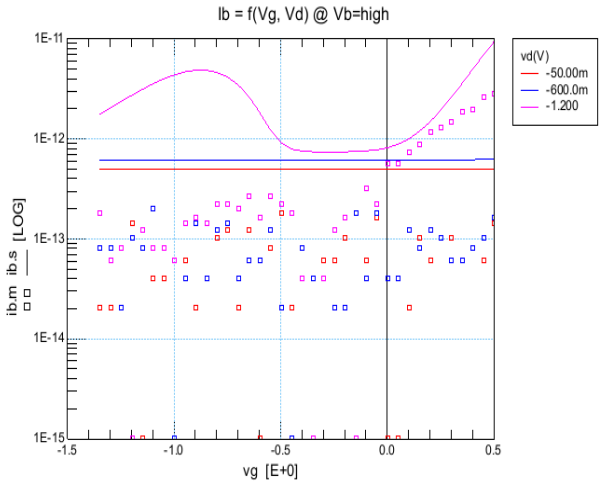


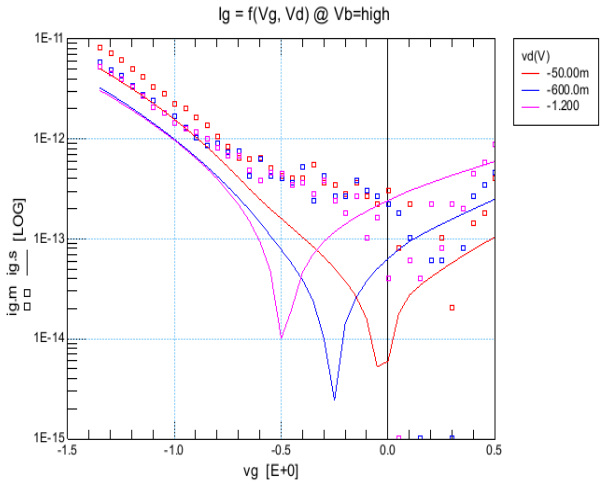
idvg, Ib,Ig, Vb = 0V, T = 27°C



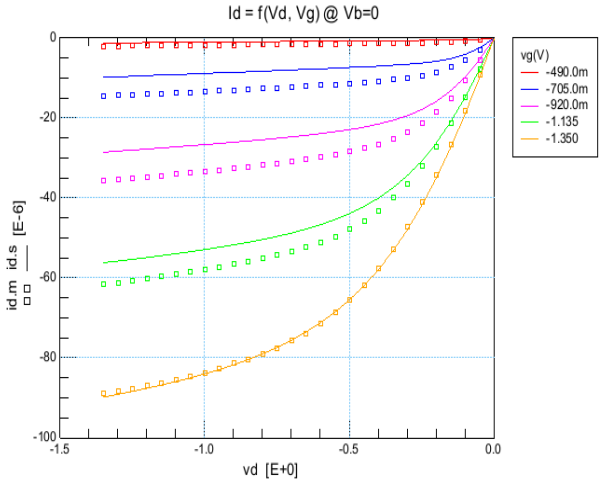


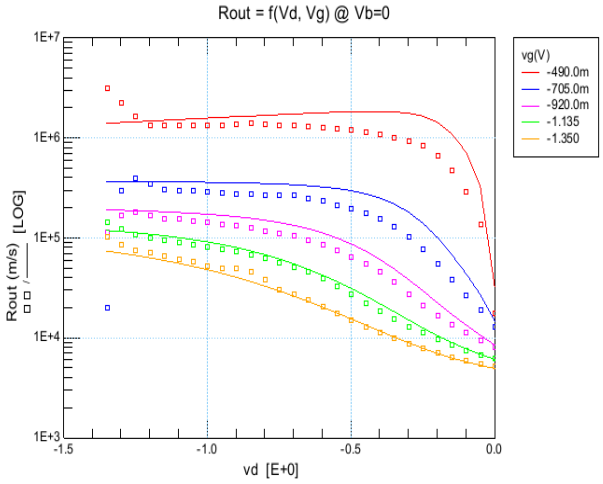
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

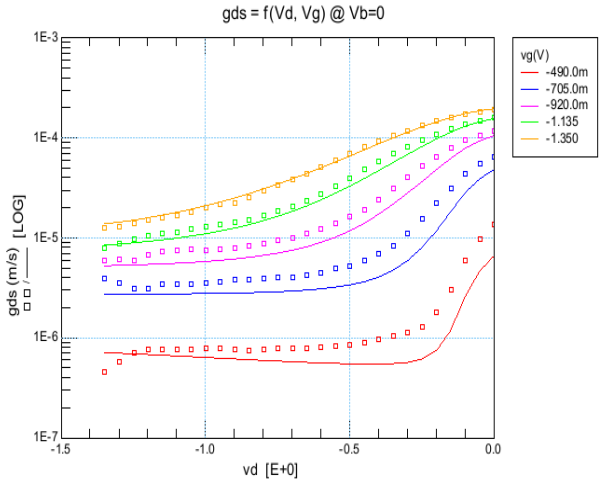




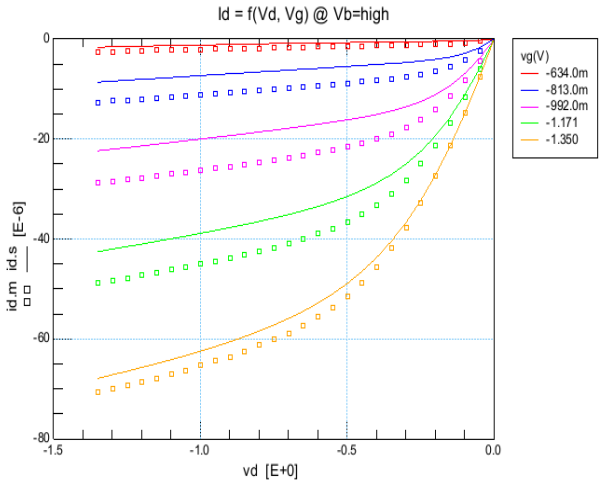
idvd, Vb = 0V, T = 27°C

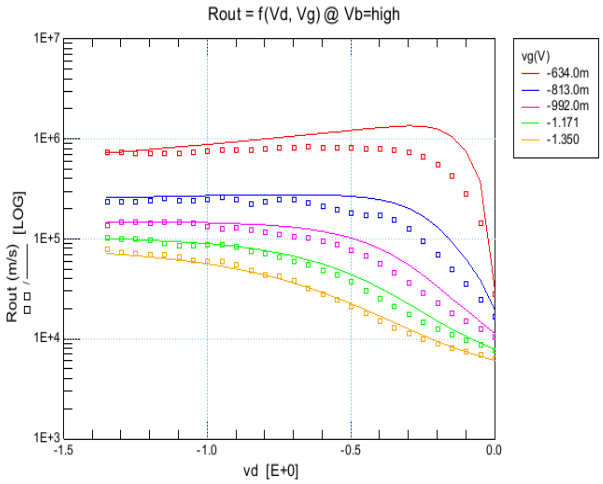


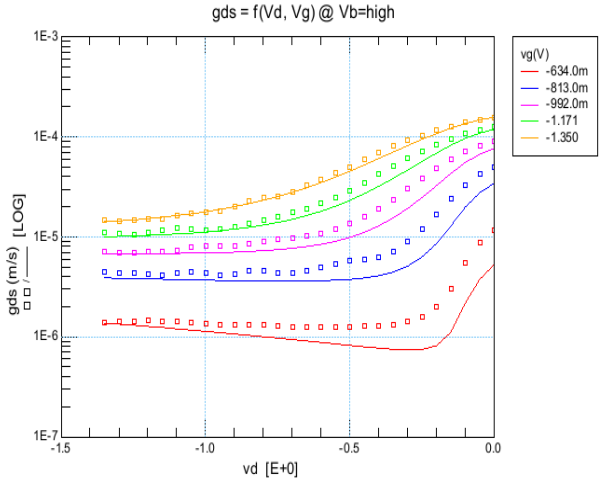




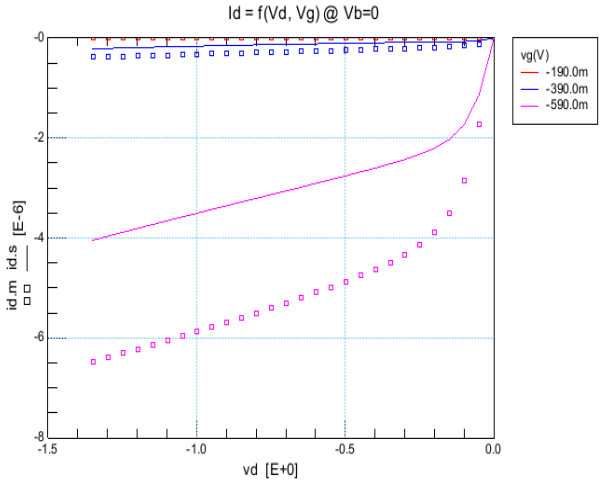
idvd_vbmin, Vb = 1.2V, T = 27°C

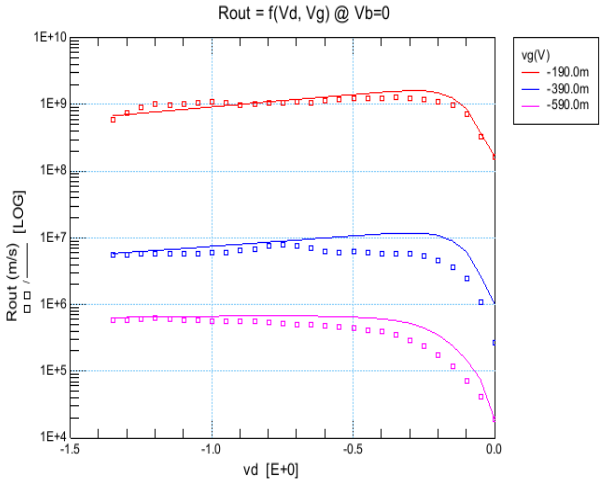


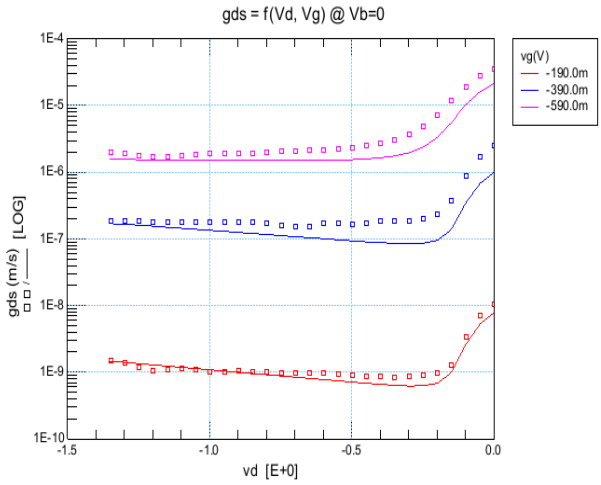




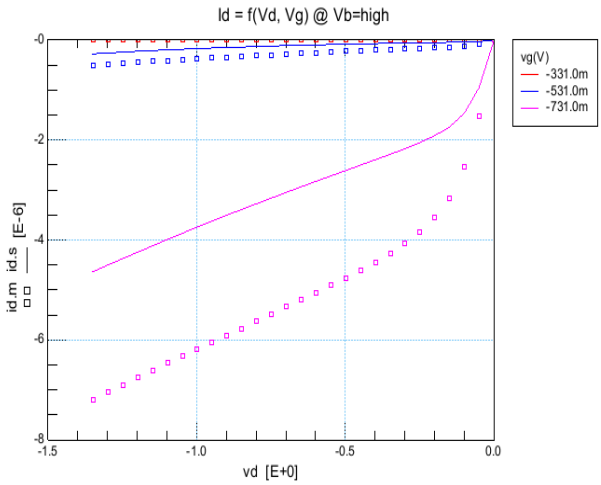
idvd_vth, Vb = 0V, T = 27°C

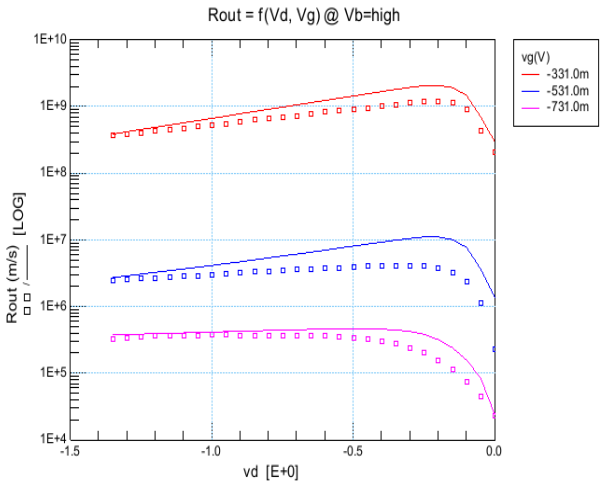


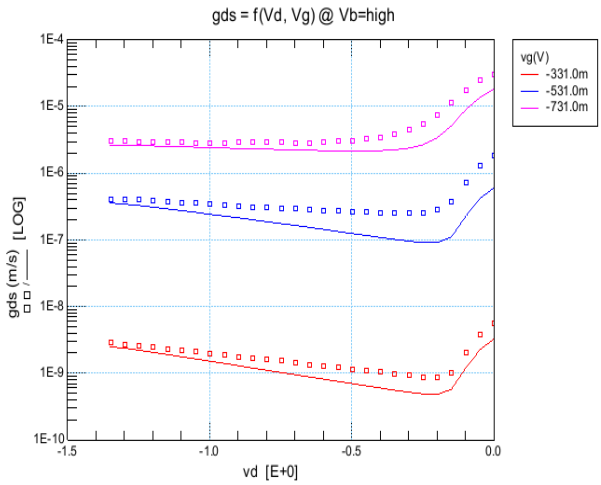




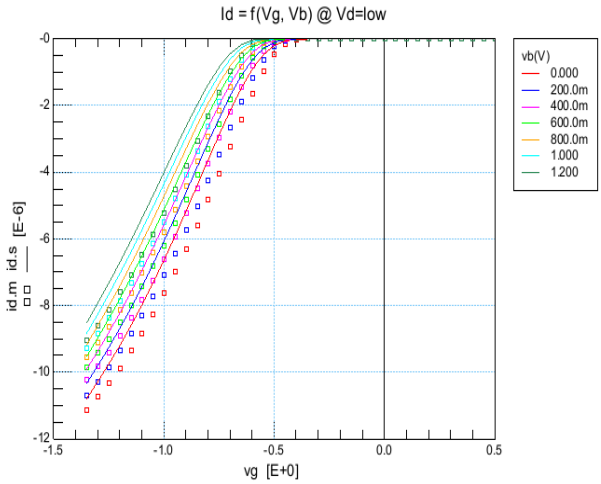
idvd_vth_vbmin, Vb = 1.2V, T = 27°C

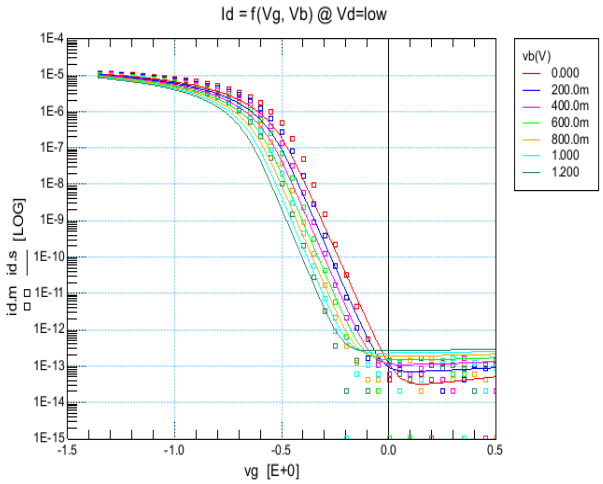


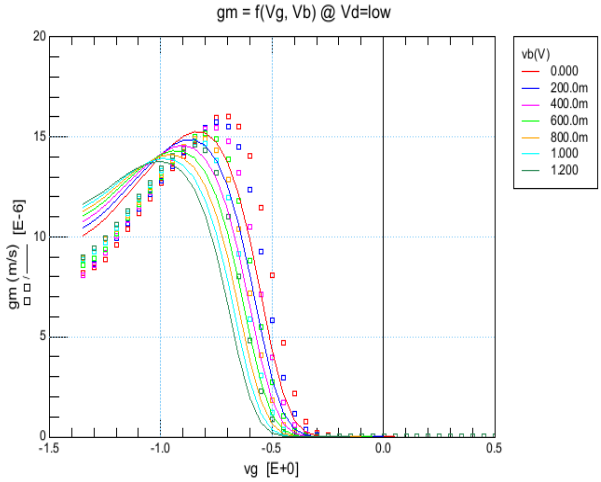


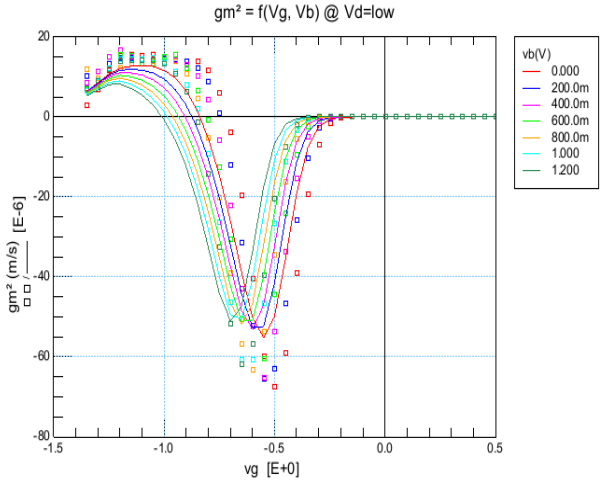


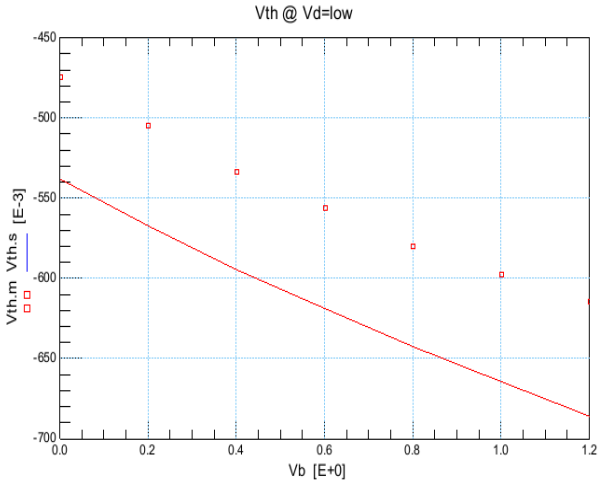
idvg, Vd = -0.05V, T = -40°C

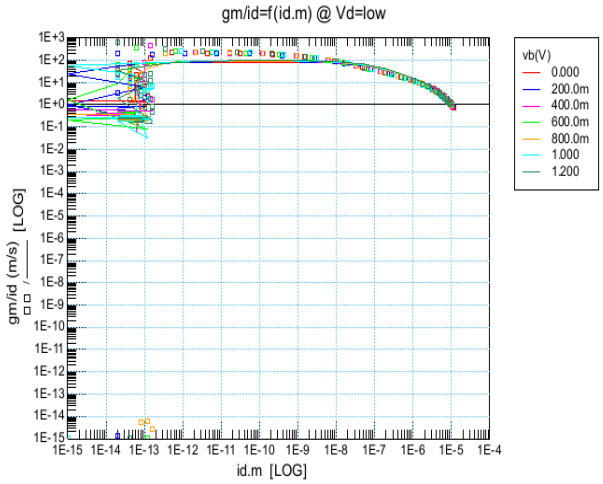


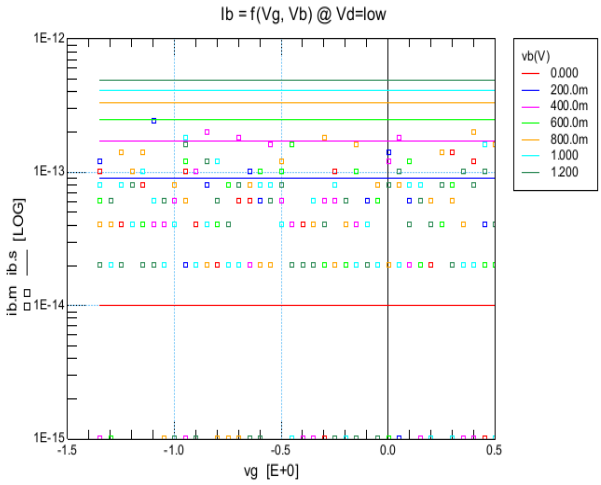


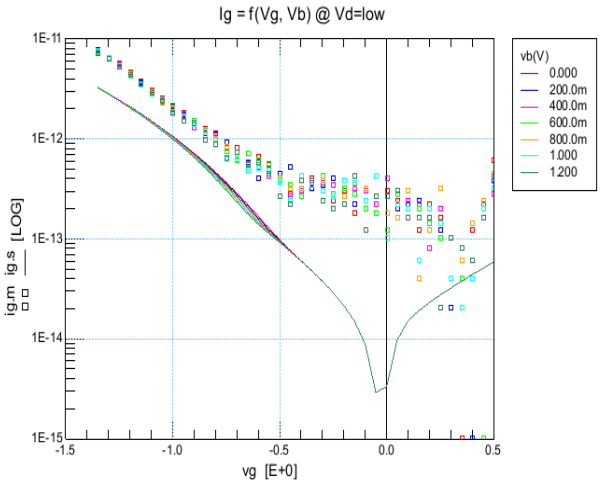




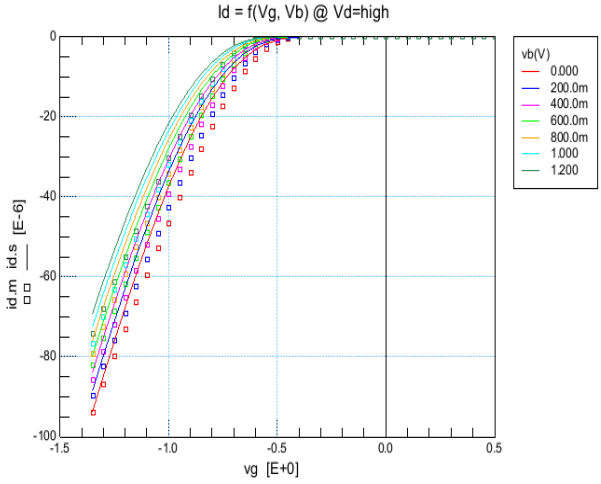


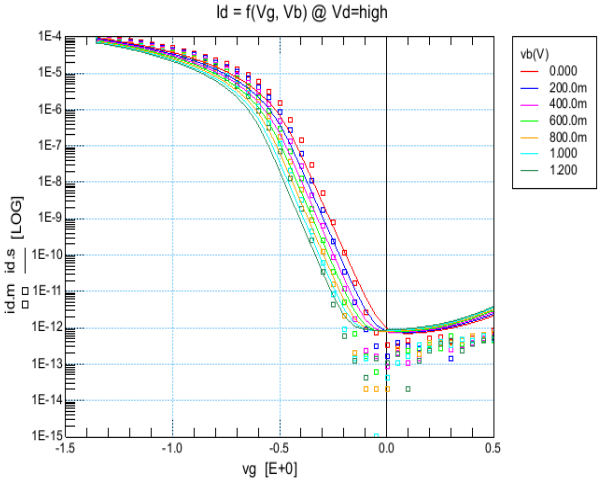


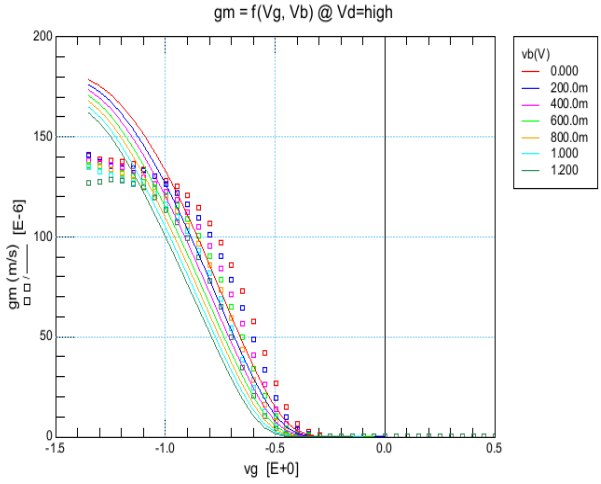


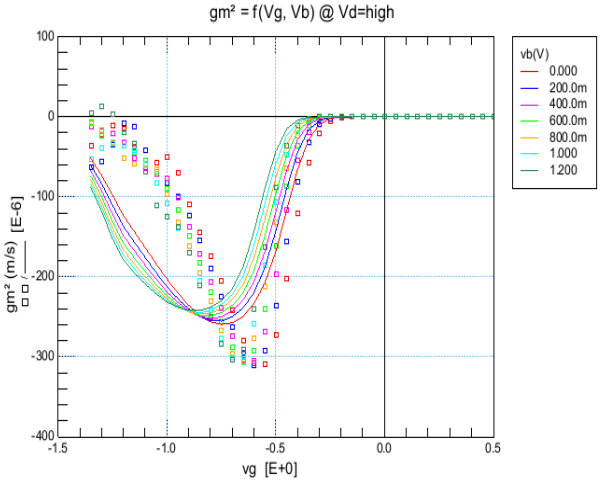


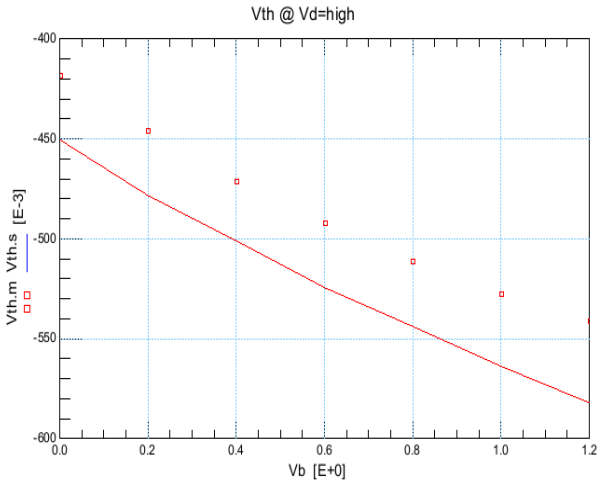
idvg, Vd = -1.2V, T = -40°C

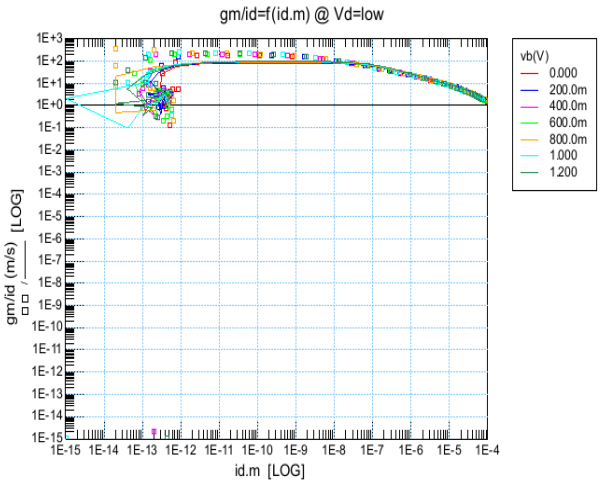


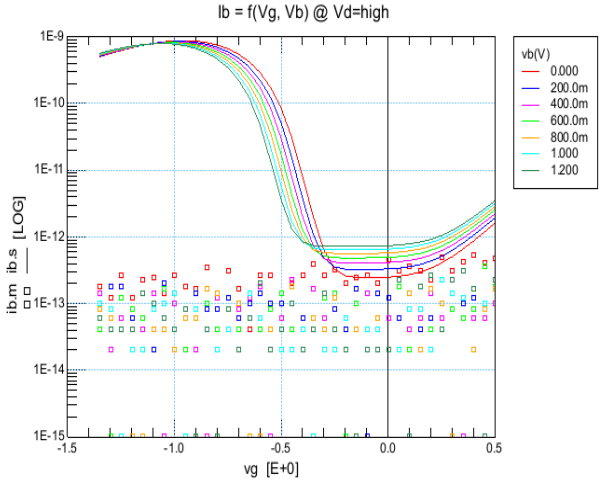


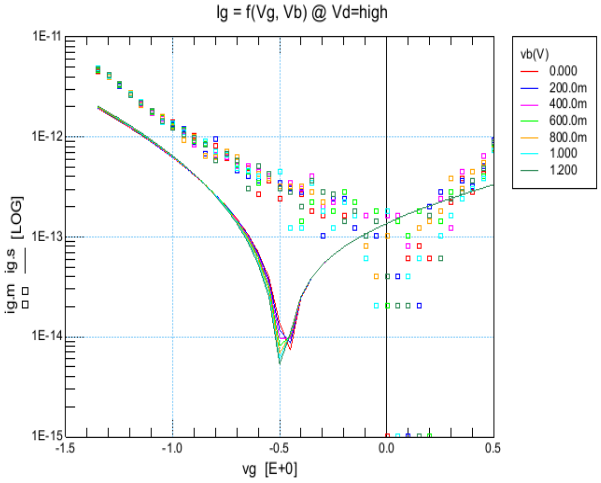




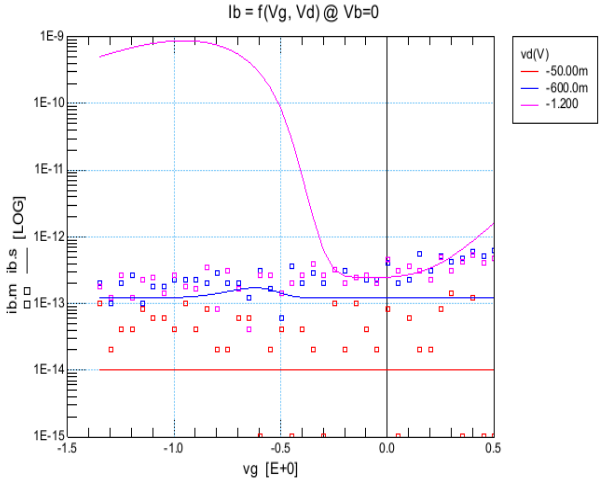


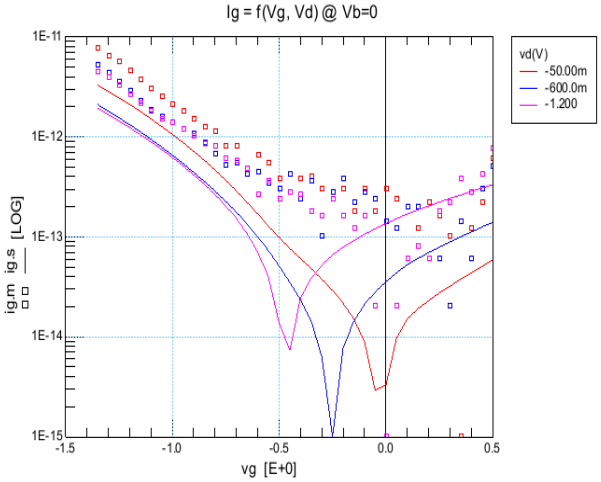




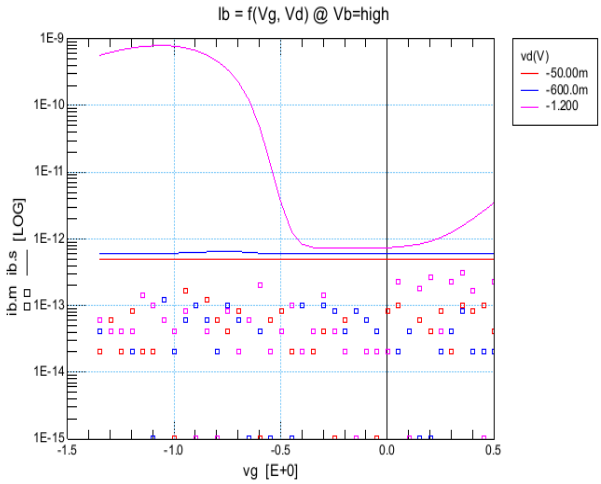


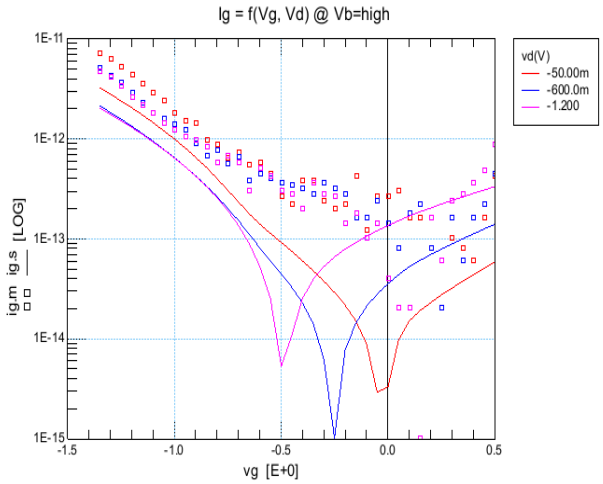
idvg, Ib,Ig, Vb = 0V, T = -40°C



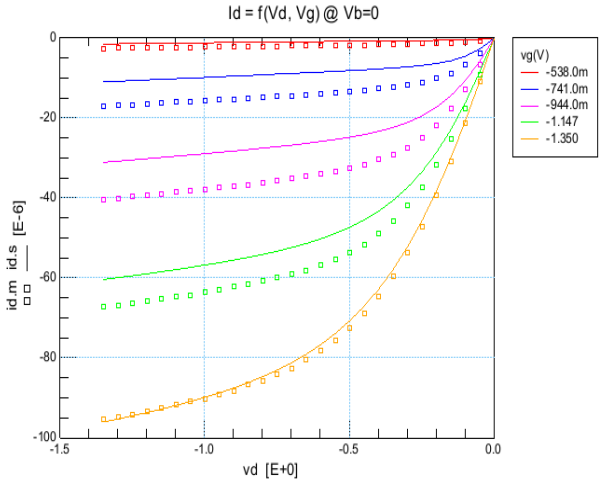


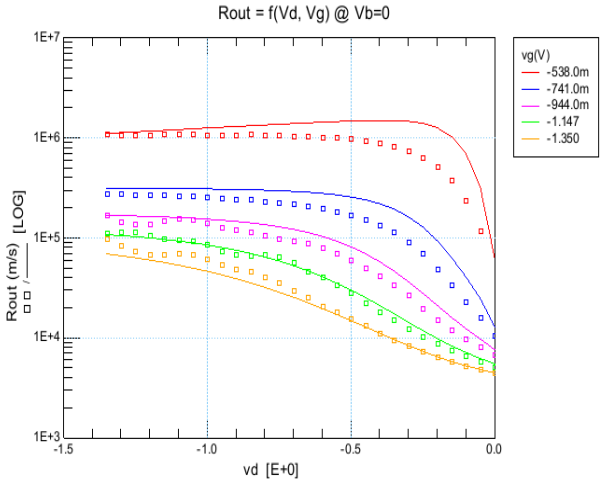
idvg, Ib,Ig, Vb = 1.2V, T = -40°C

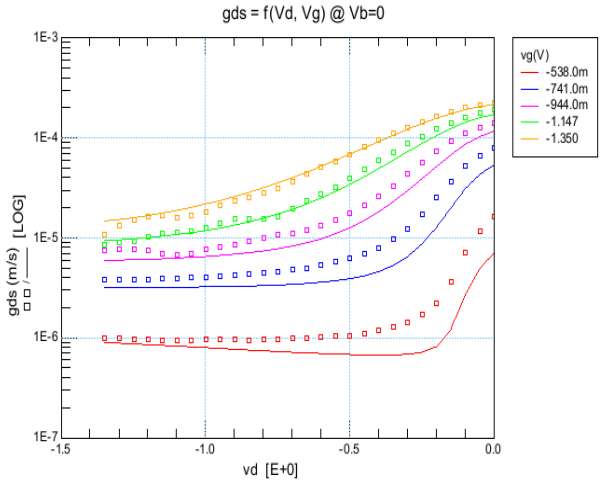




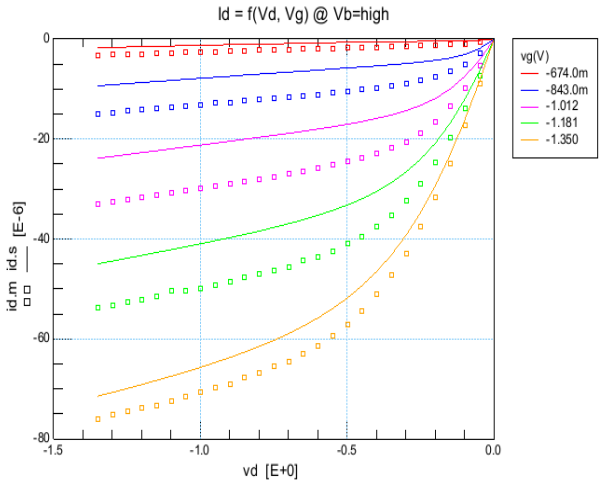
idvd, Vb = 0V, T = -40°C

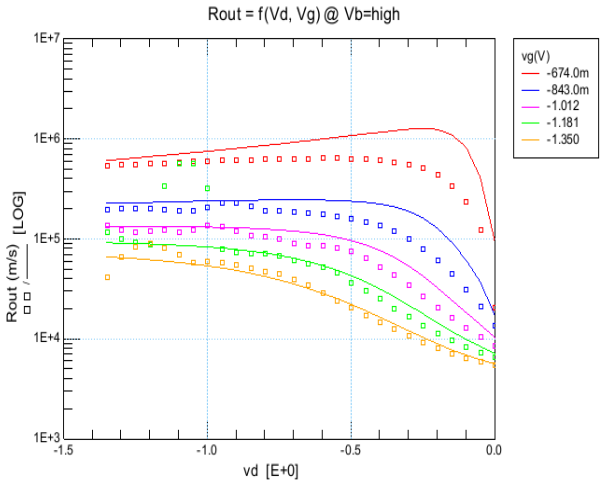


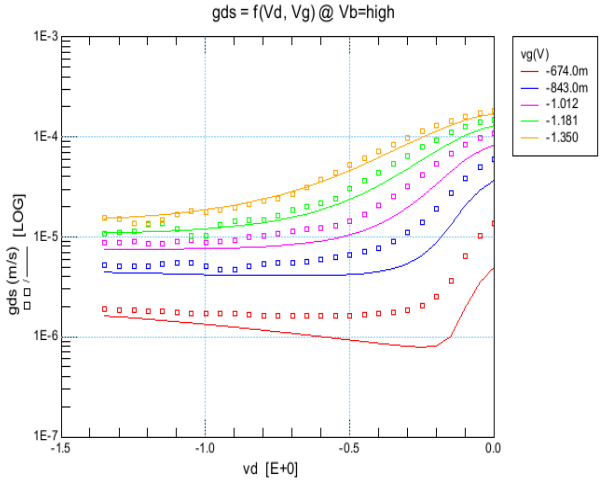




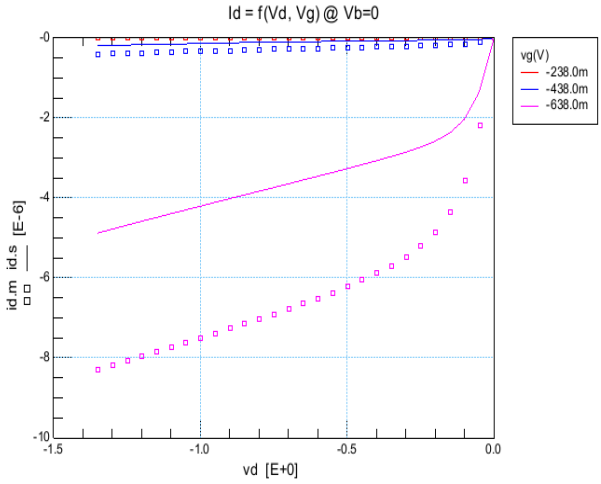
idvd_vbmin, Vb = 1.2V, T = -40°C

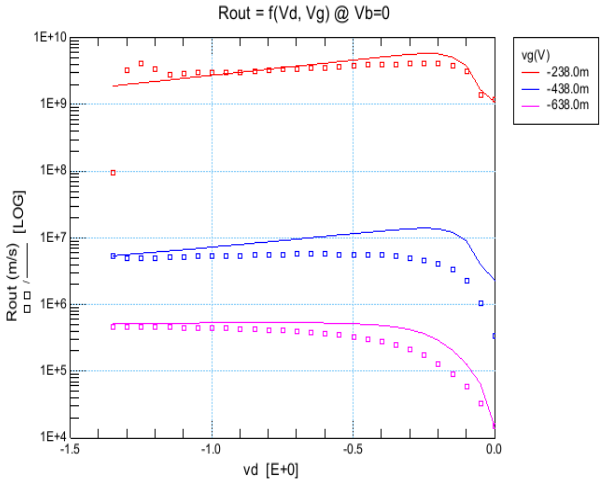


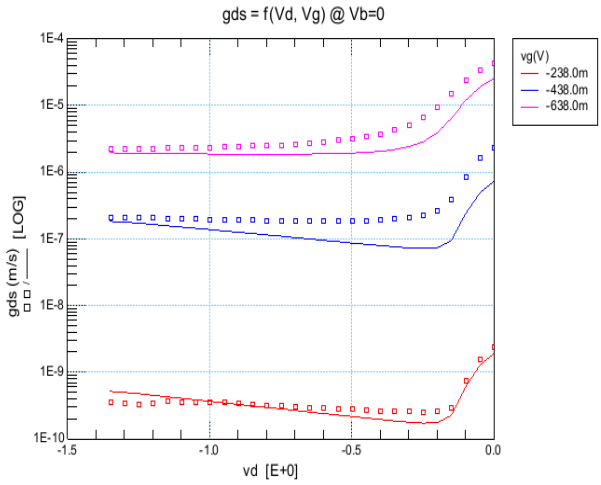




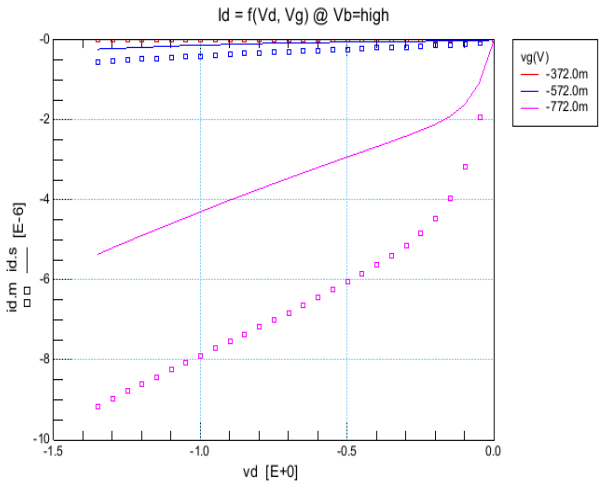
idvd_vth, Vb = 0V, T = -40°C

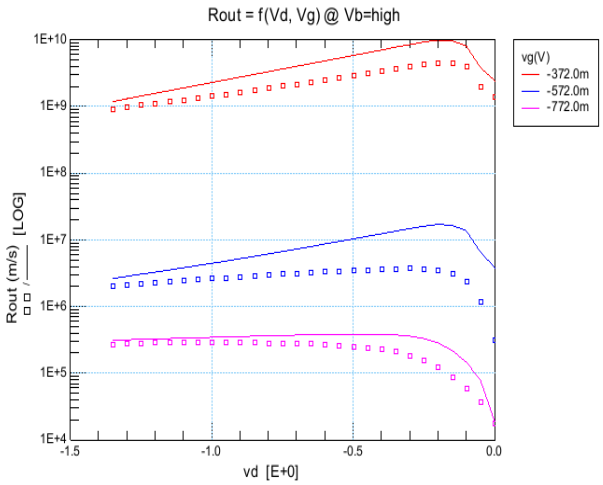


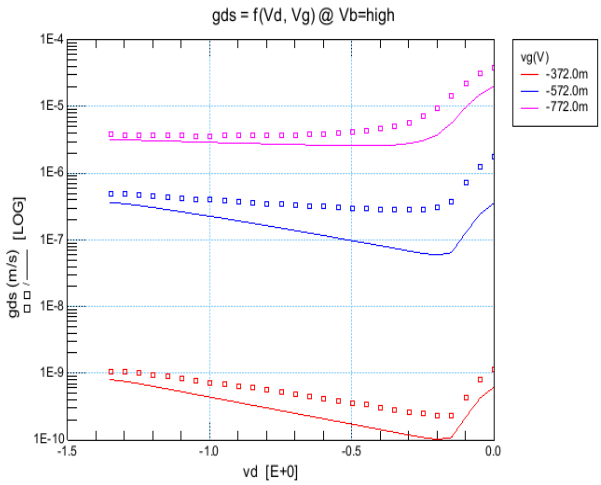




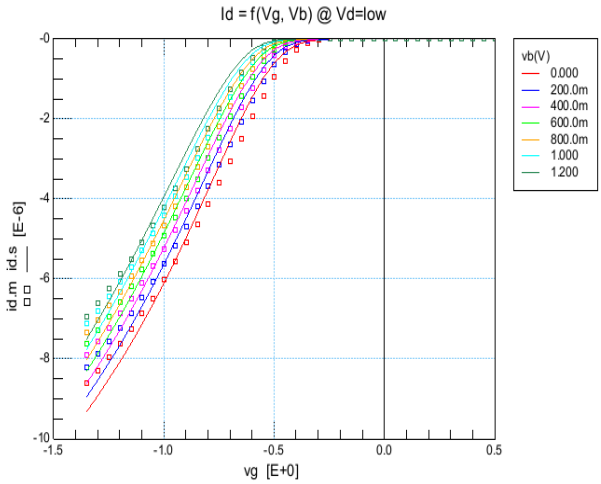
idvd_vth_vbmin, Vb = 1.2V, T = -40°C

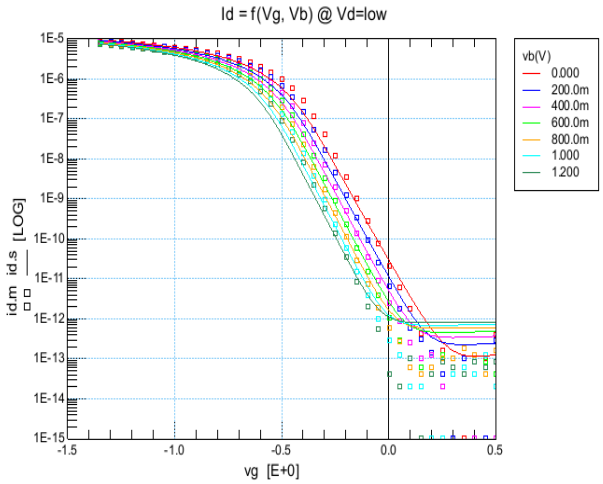


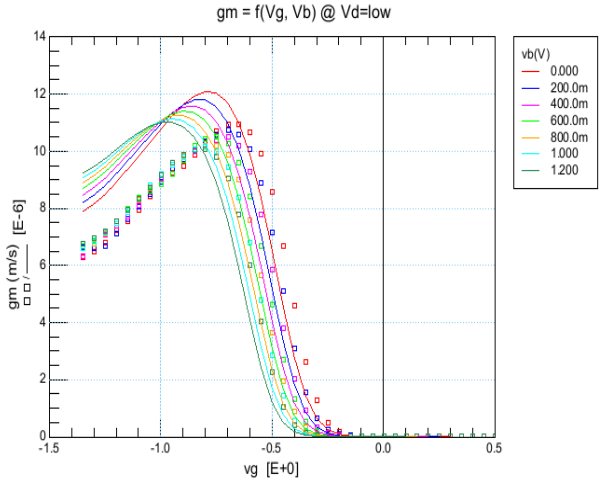


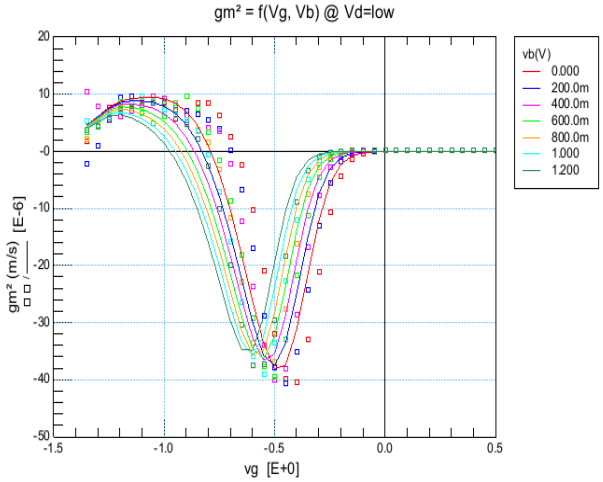


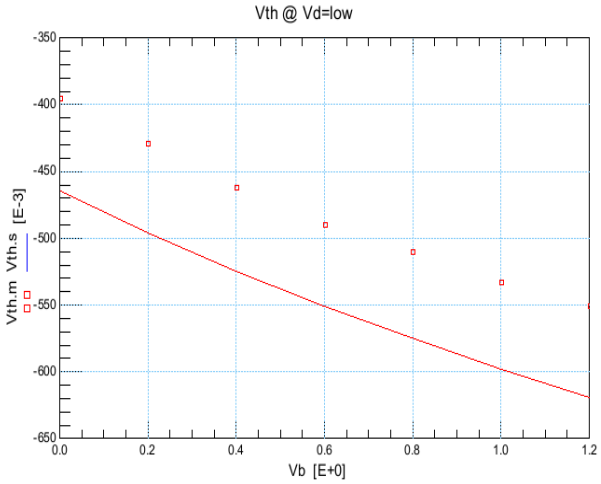
idvg, Vd = -0.05V, T = 70°C

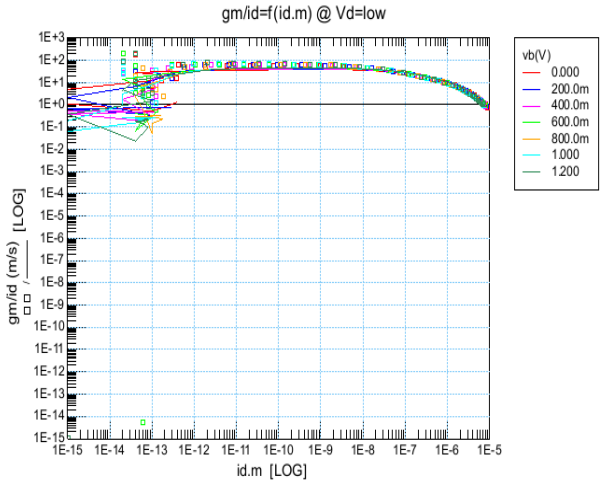


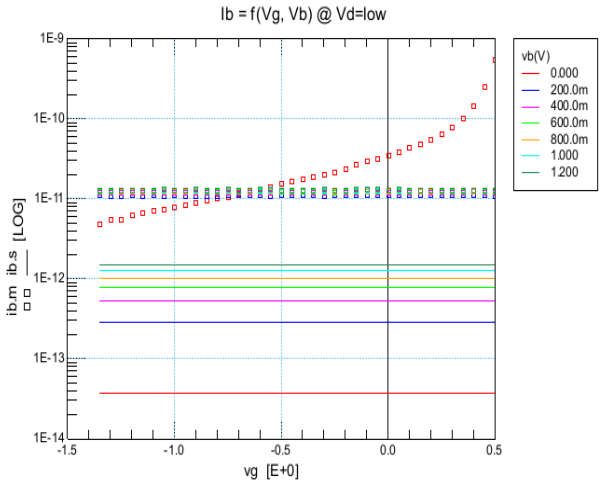


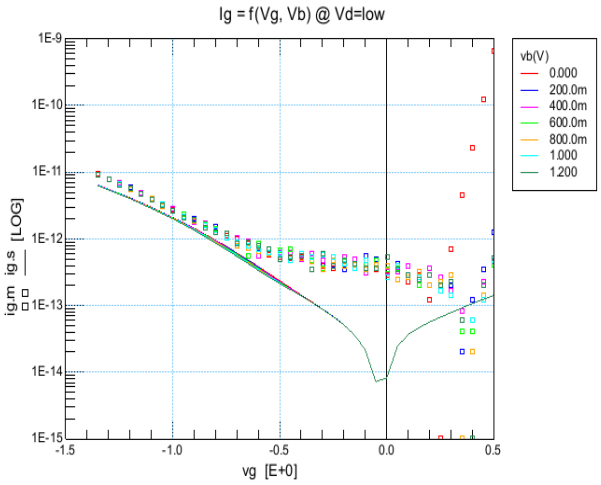




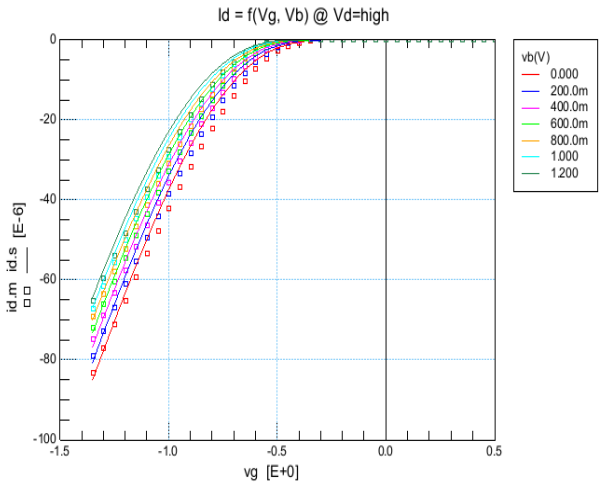


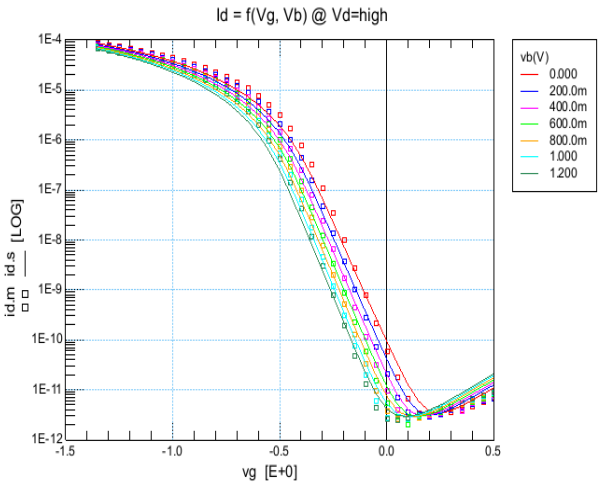


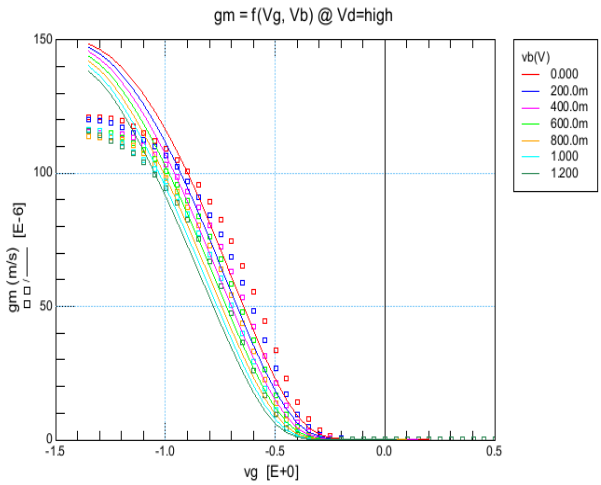


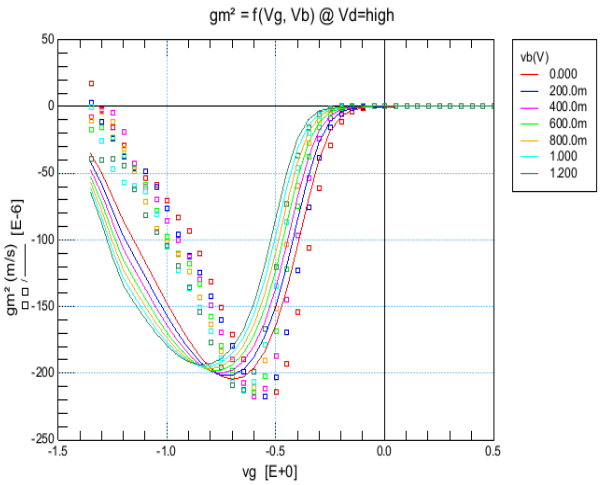


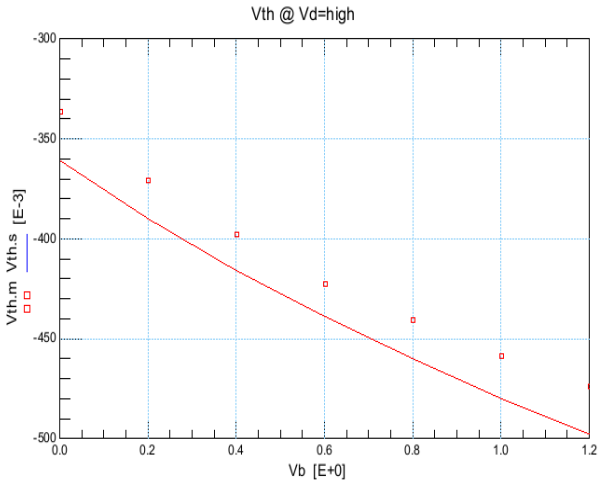
idvg, Vd = -1.2V, T = 70°C

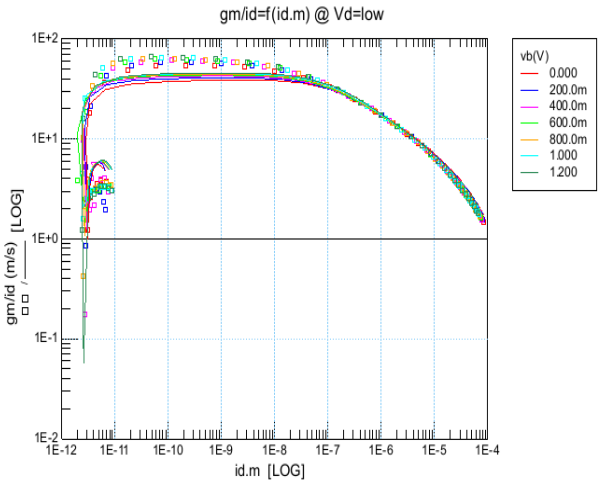


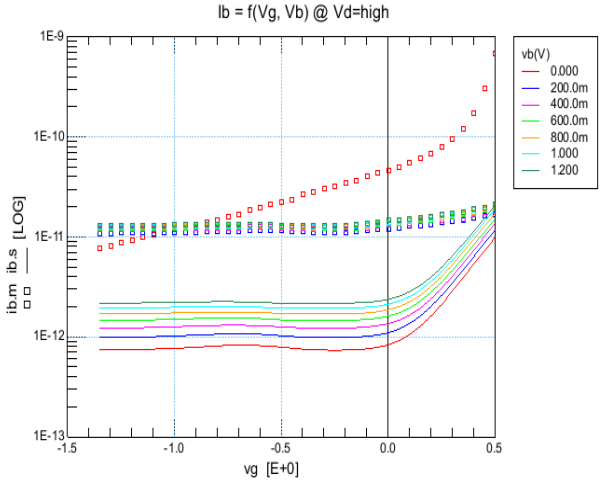


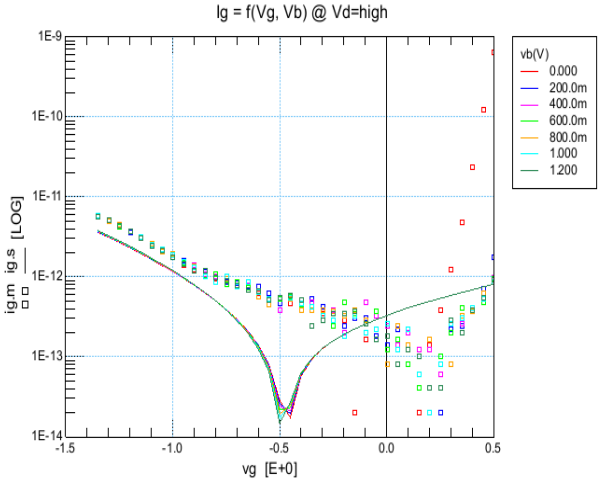




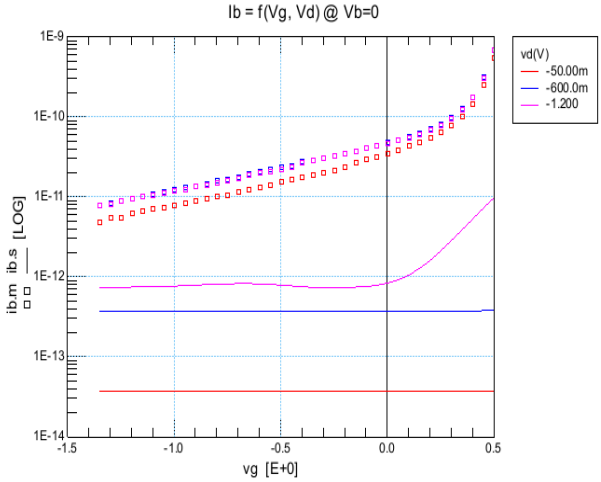


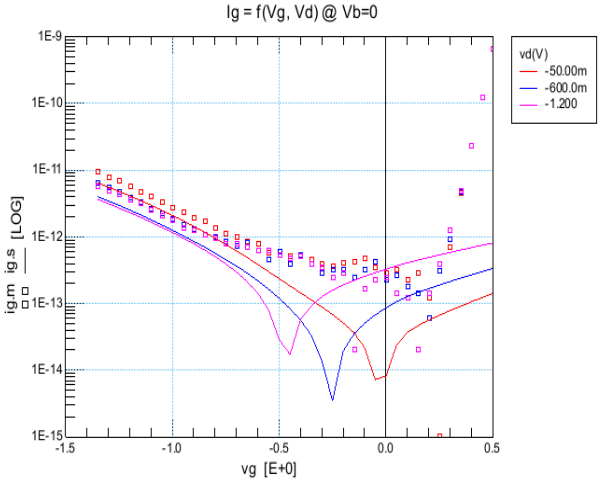




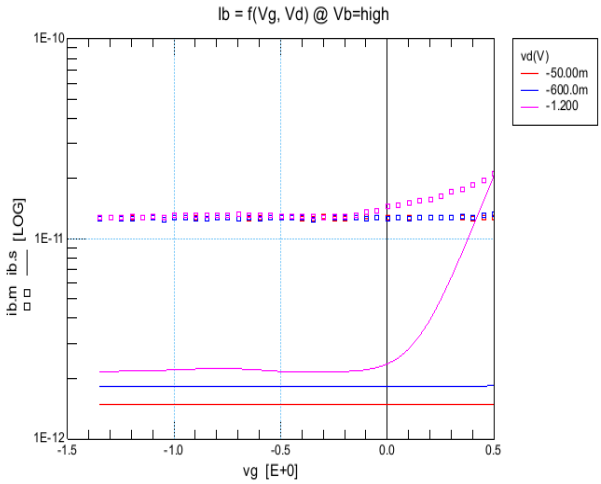


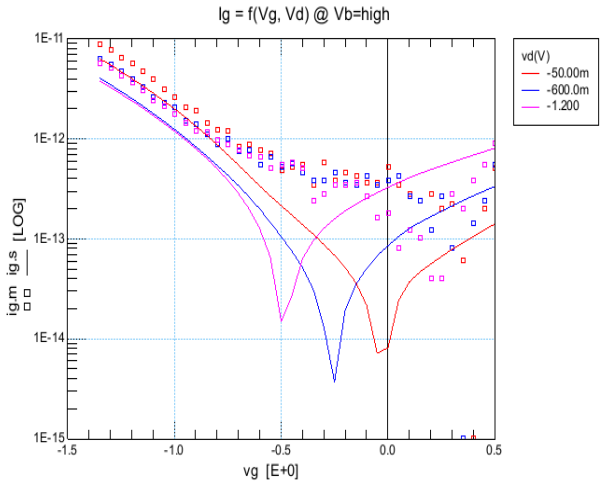
idvg, Ib,Ig, Vb = 0V, T = 70°C



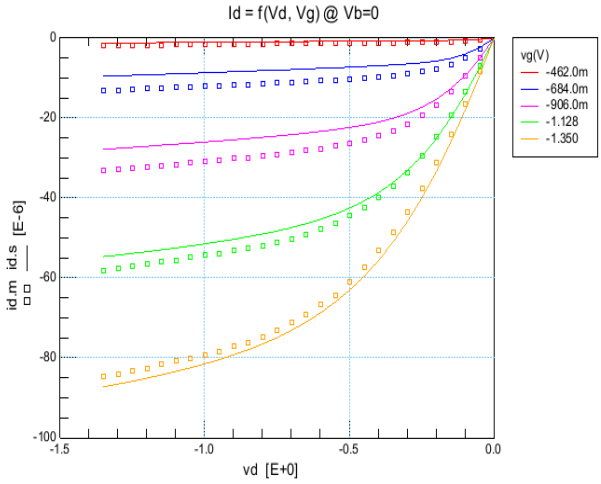


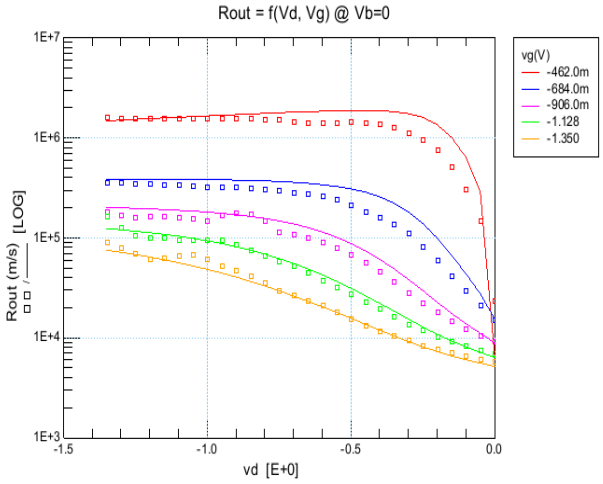
idvg, Ib,Ig, Vb = 1.2V, T = 70°C

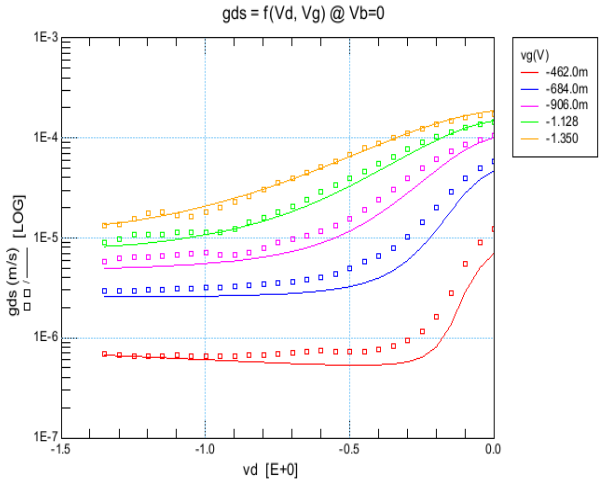




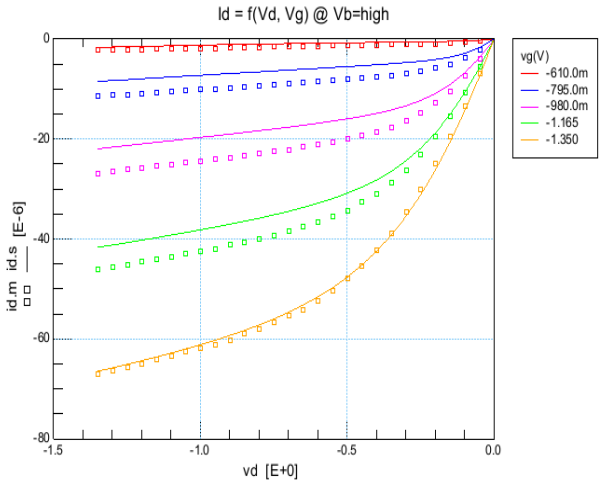
idvd, Vb = 0V, T = 70°C

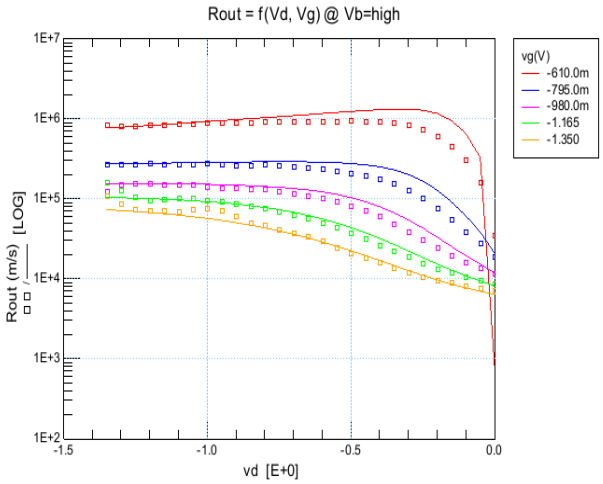


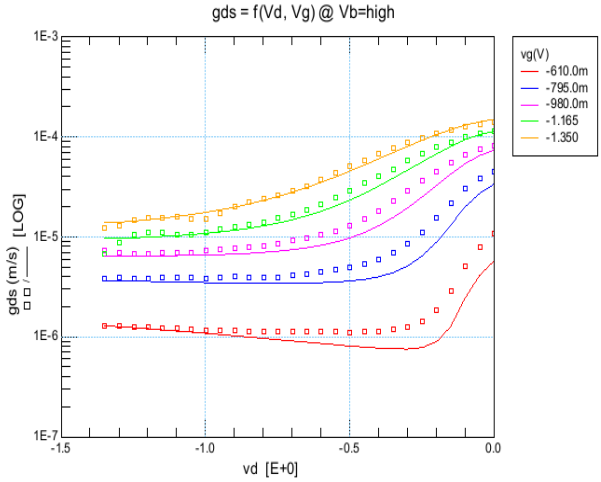




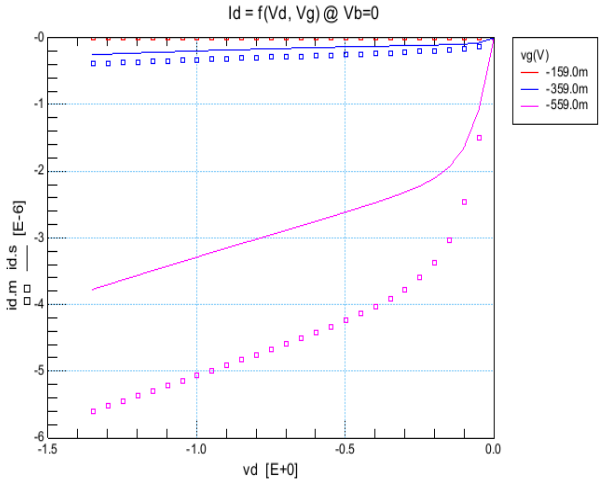
idvd_vbmin, Vb = 1.2V, T = 70°C

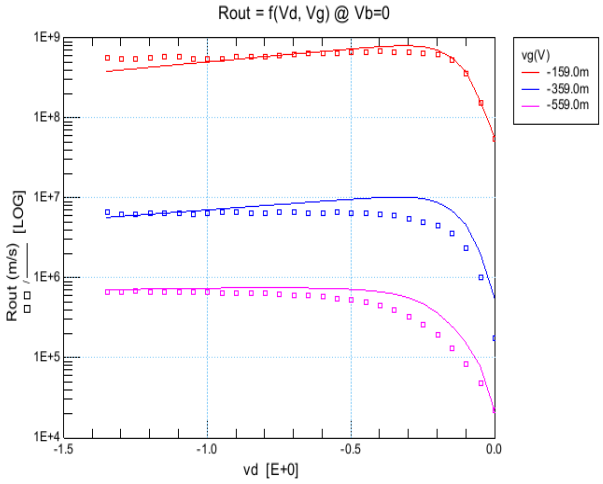


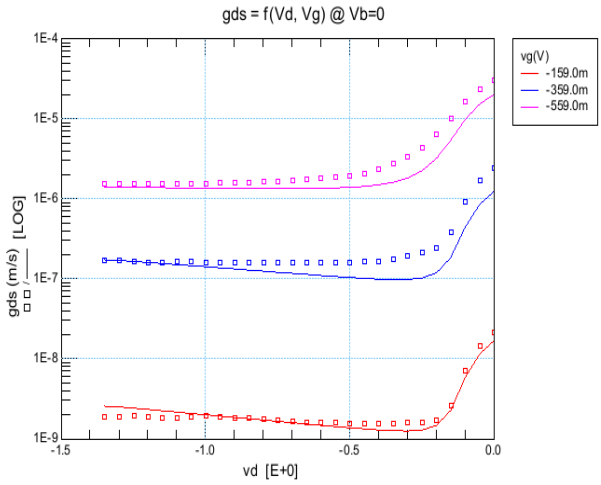




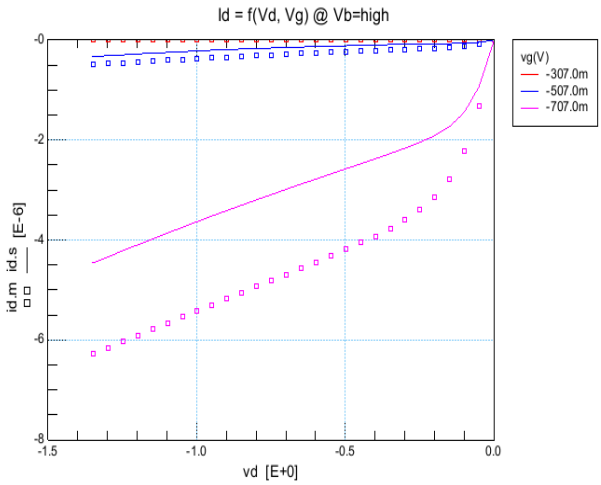
idvd_vth, Vb = 0V, T = 70°C

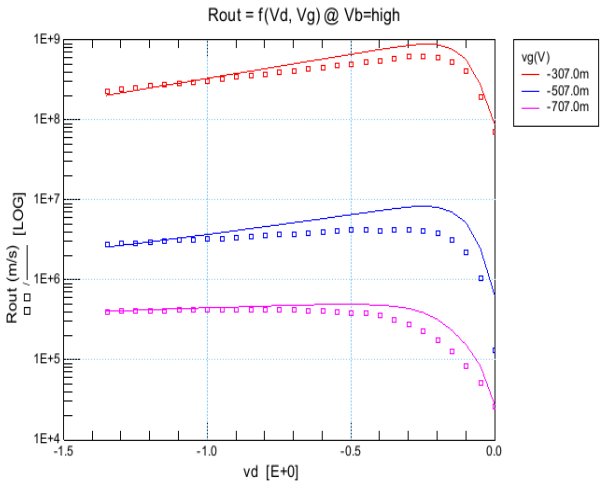


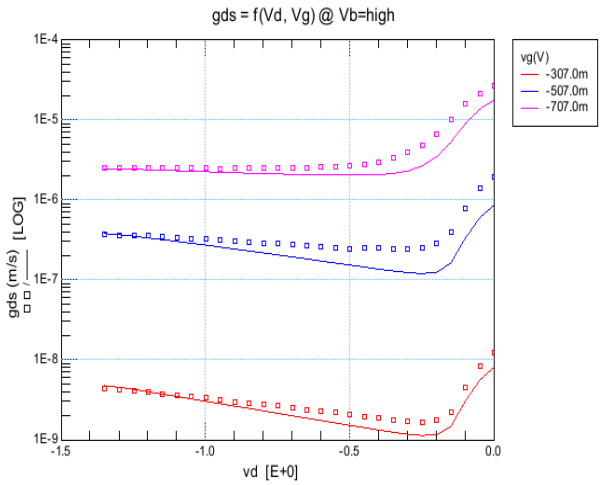




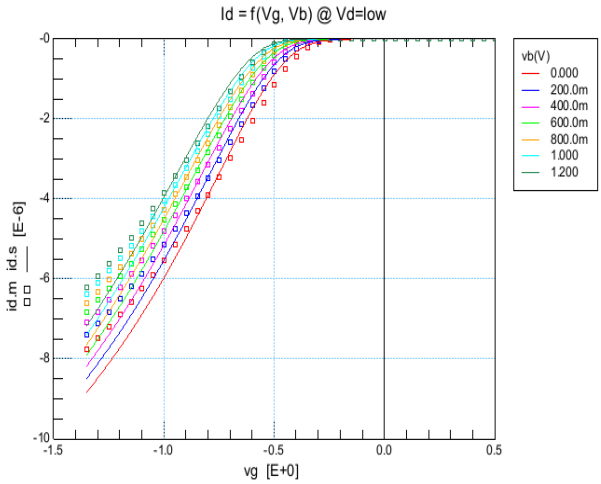
idvd_vth_vbmin, Vb = 1.2V, T = 70°C

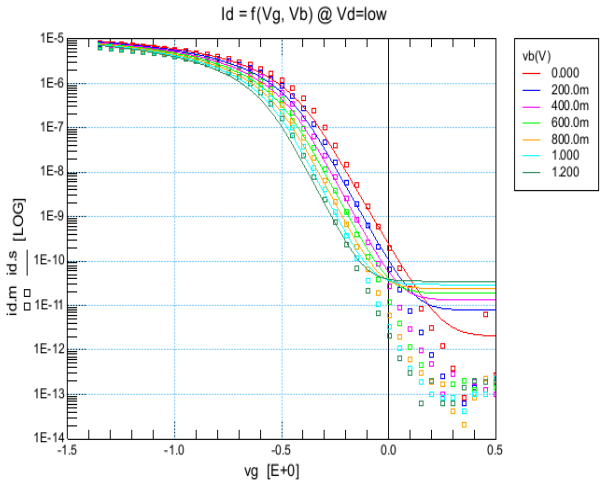


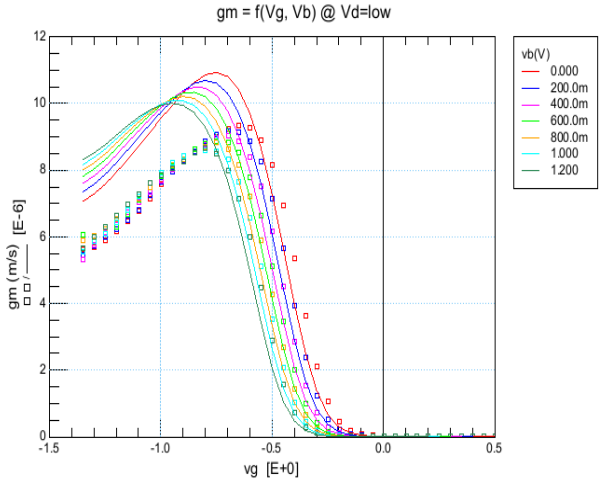


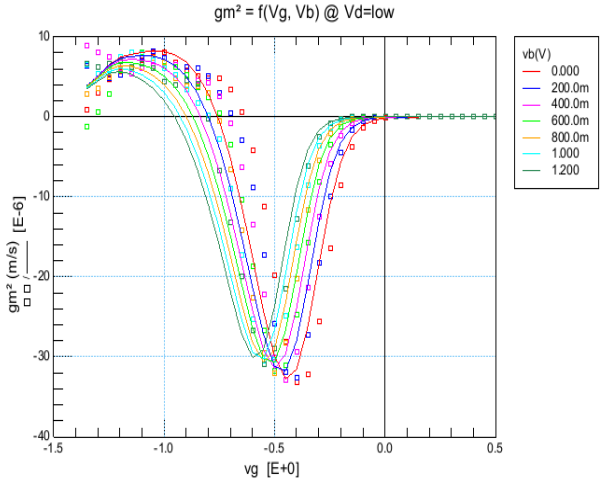


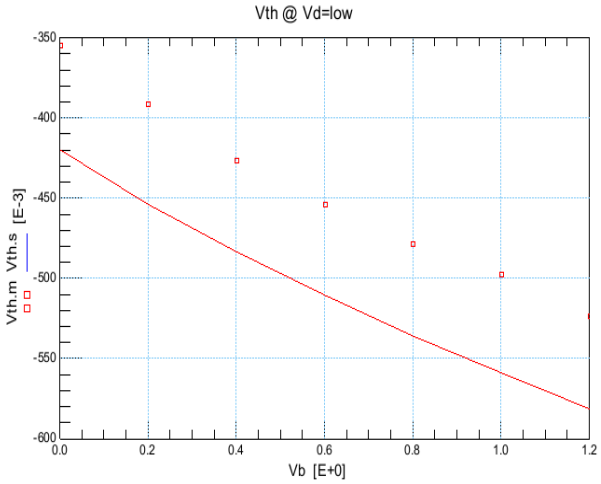
idvg, Vd = -0.05V, T = 125°C

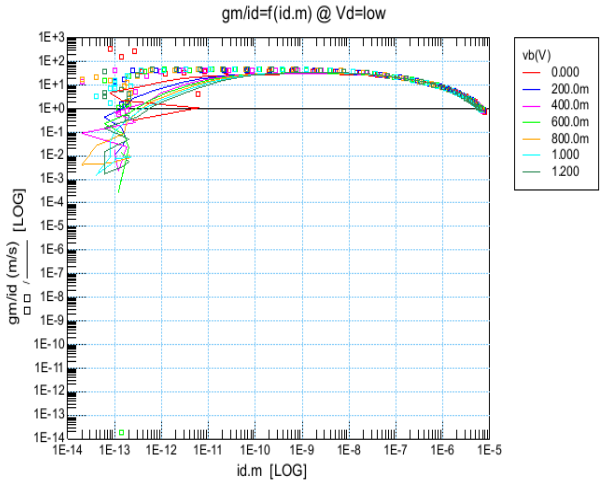


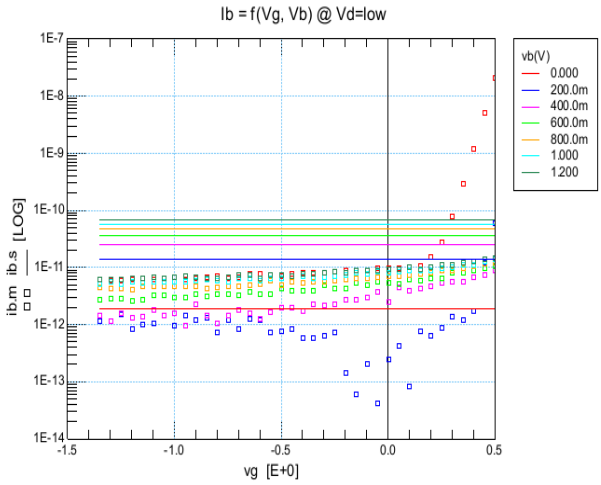


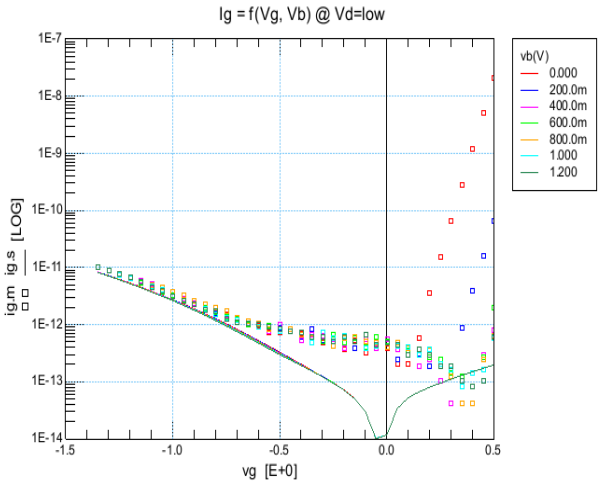




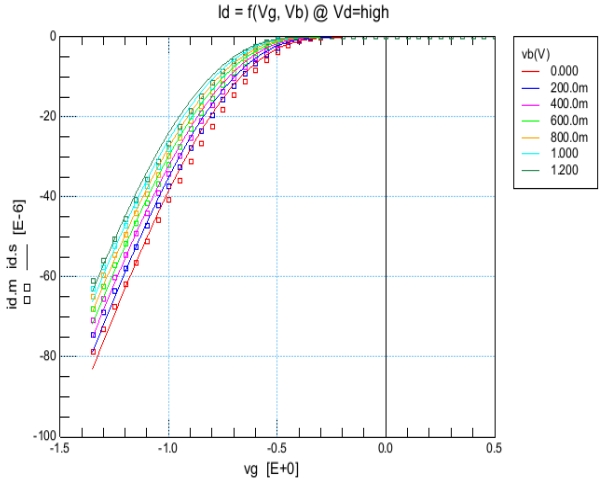


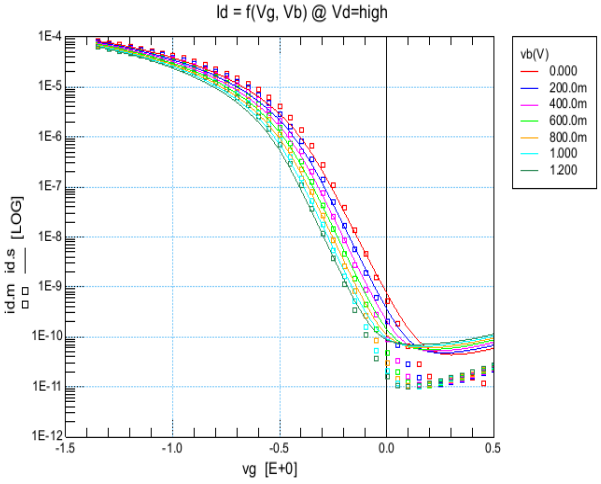


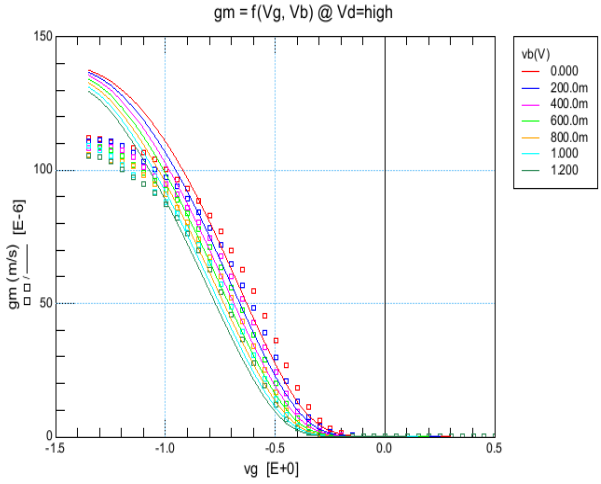


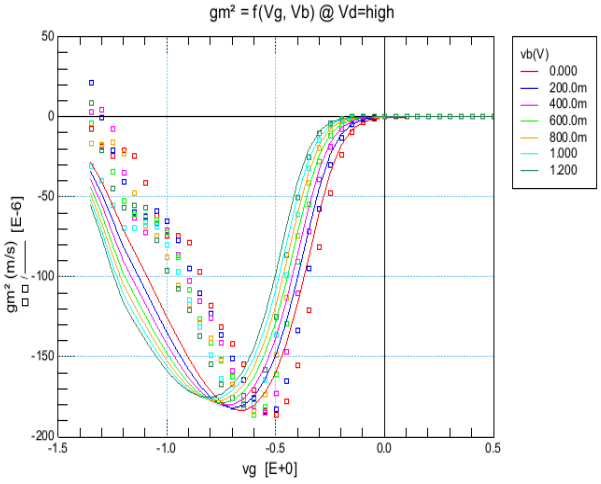


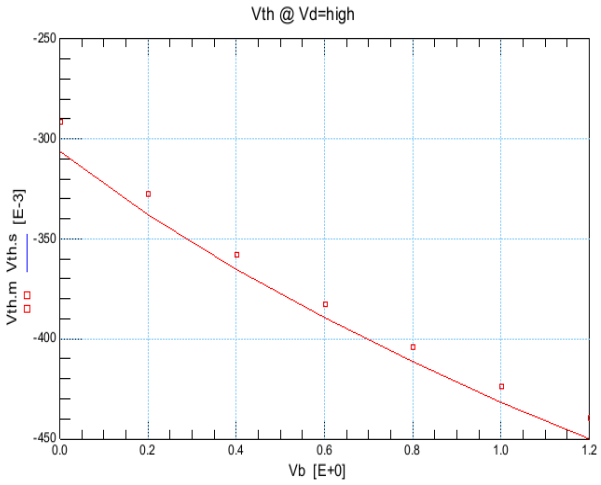
idvg, Vd = -1.2V, T = 125°C

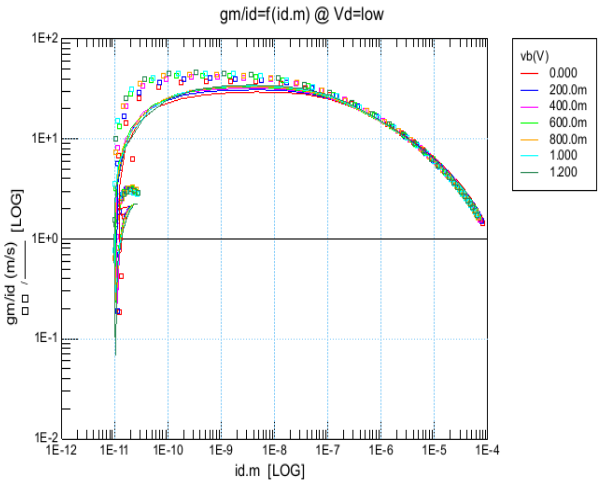


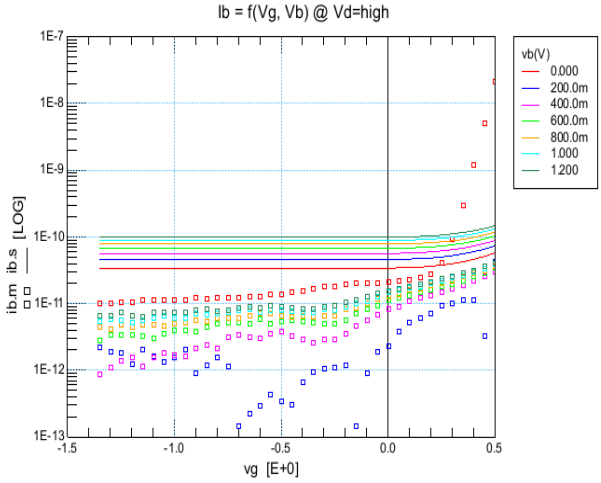


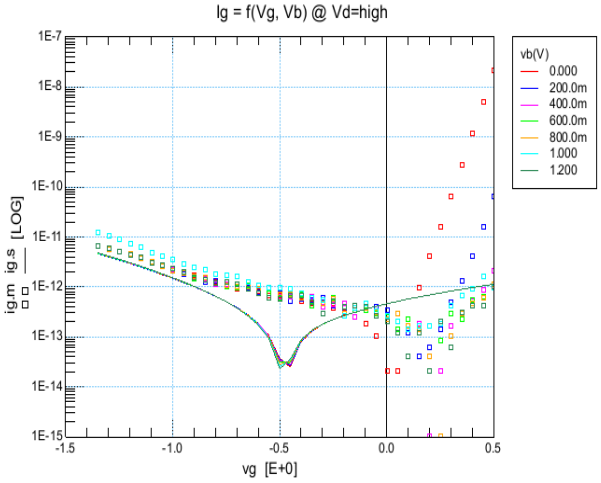




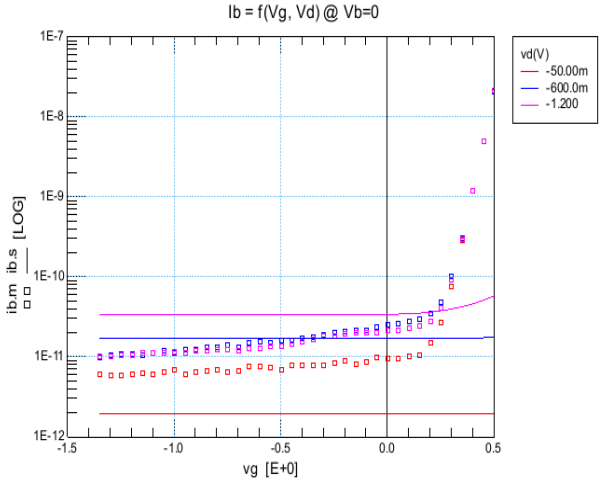


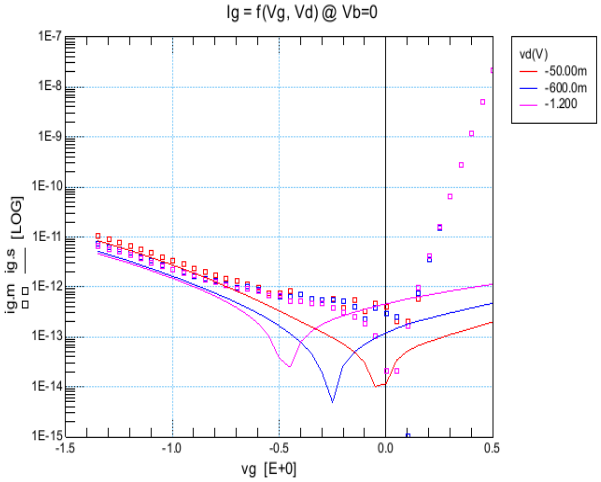




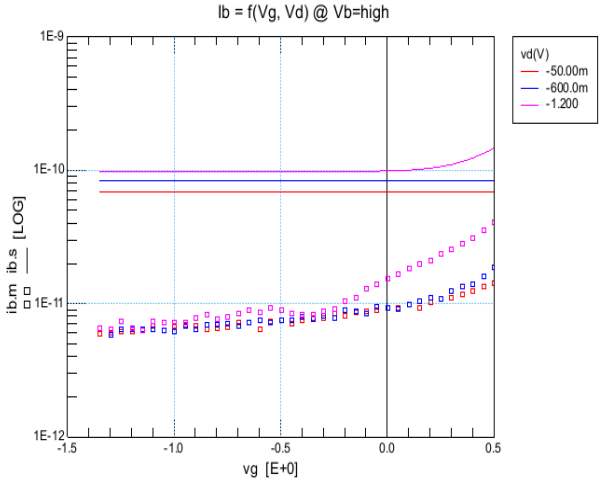


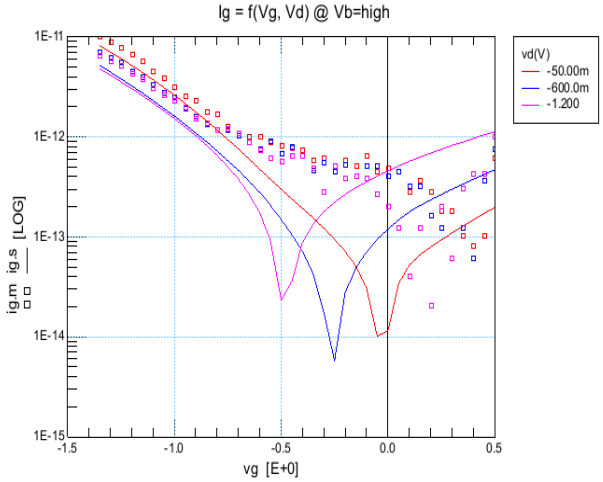
idvg, Ib,Ig, Vb = 0V, T = 125°C



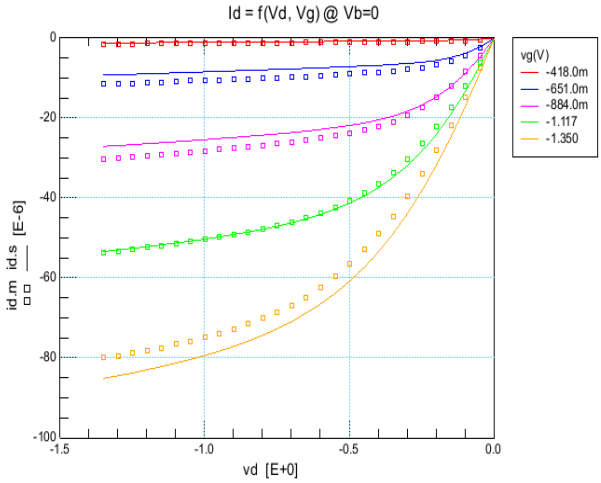


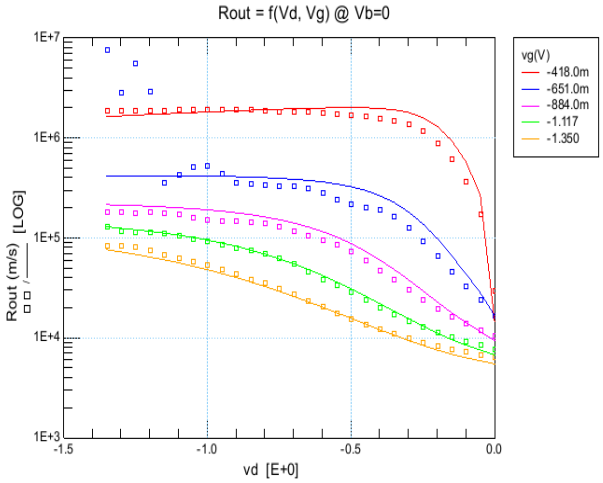
idvg, Ib,Ig, Vb = 1.2V, T = 125°C

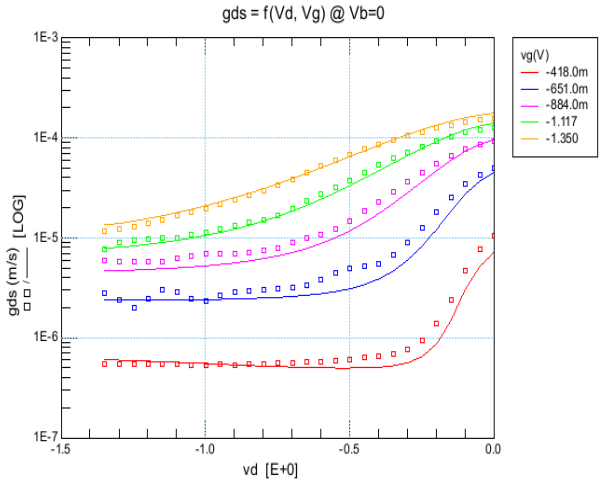




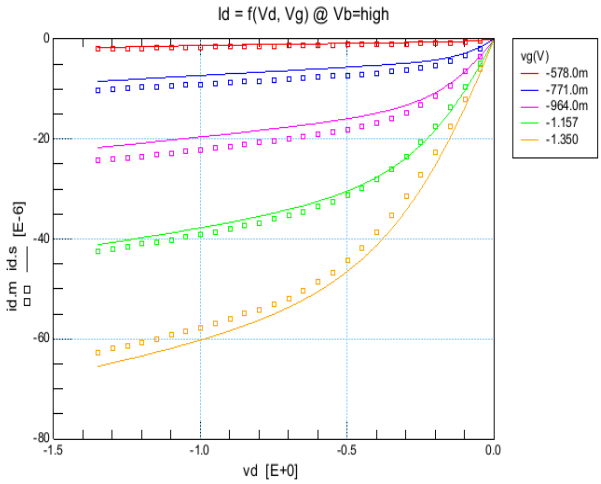
idvd, Vb = 0V, T = 125°C

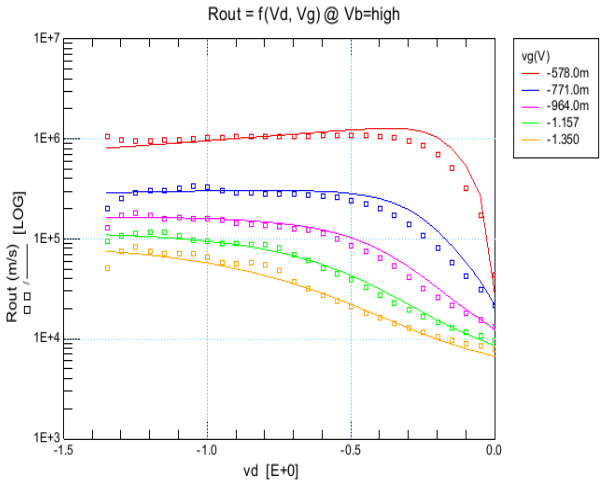


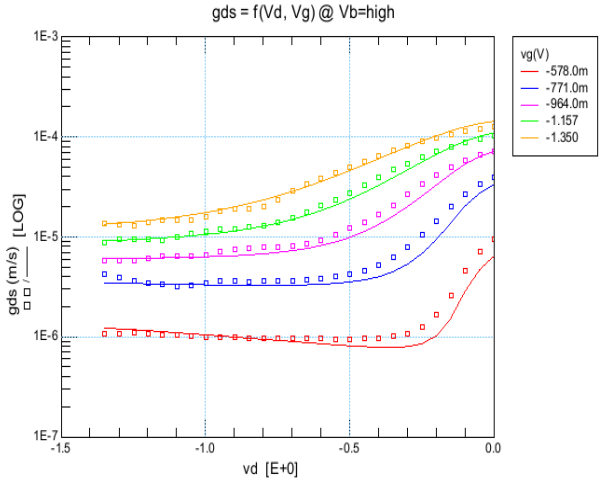




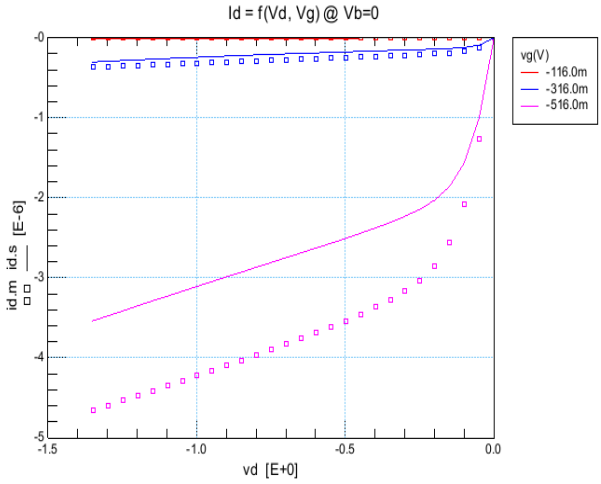
idvd_vbmin, Vb = 1.2V, T = 125°C

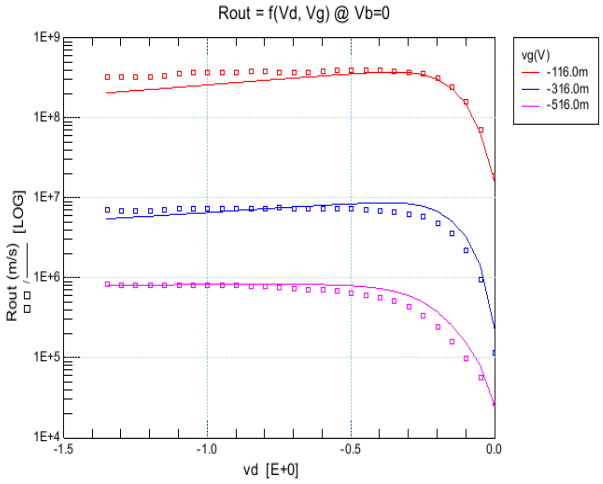


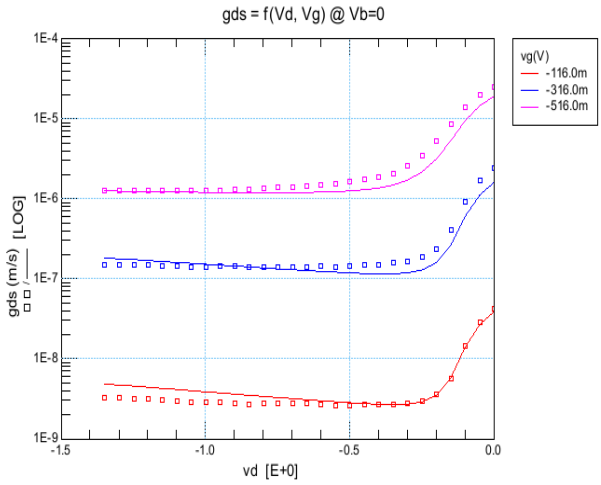




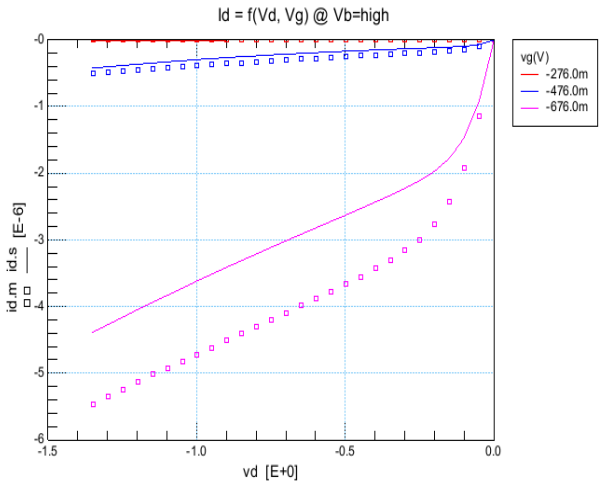
idvd_vth, Vb = 0V, T = 125°C

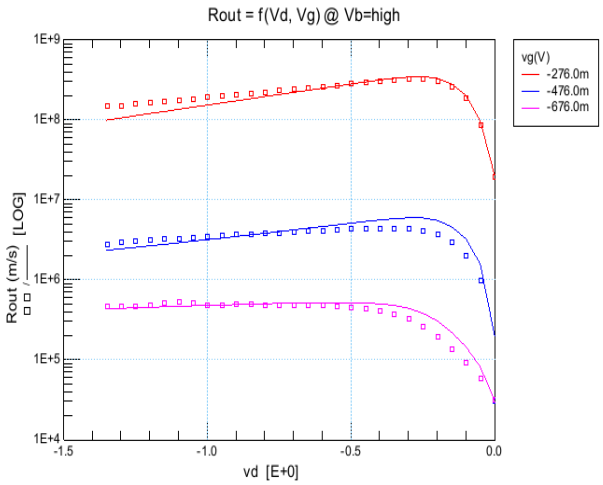


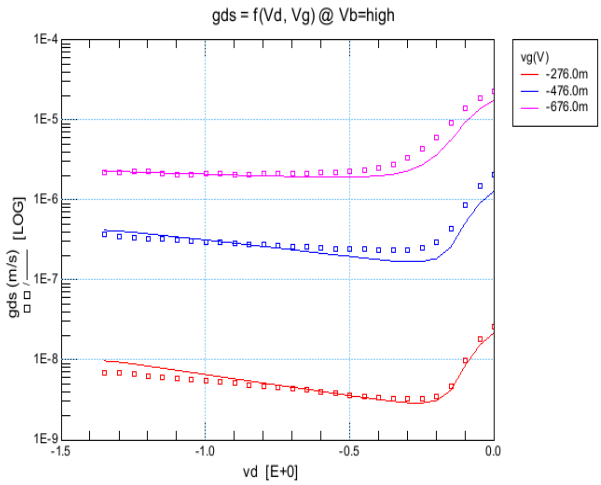




idvd_vth_vbmin, Vb = 1.2V, T = 125°C



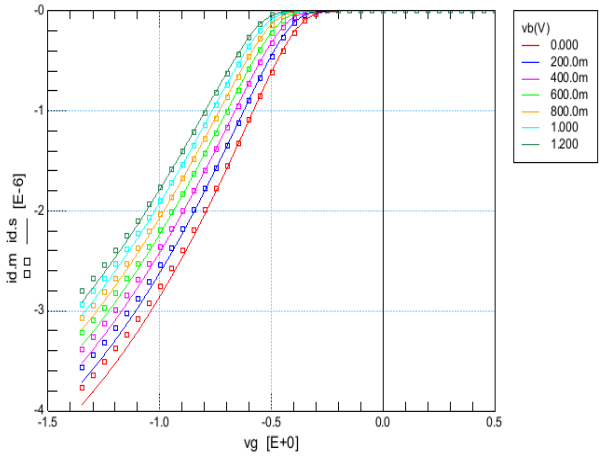




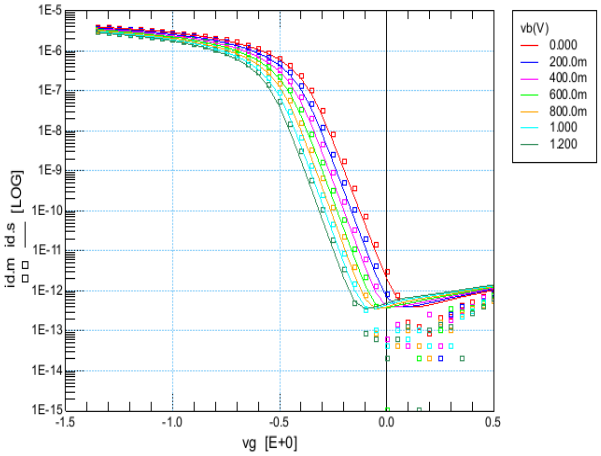
4.33 W05u0_L5u0_S550_2

idvg, Vd = -0.05V, T = 27°C

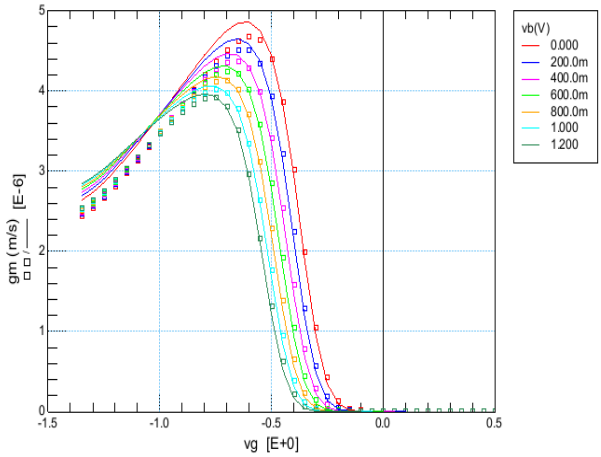
Id = f(Vg, Vb) @ Vd=low



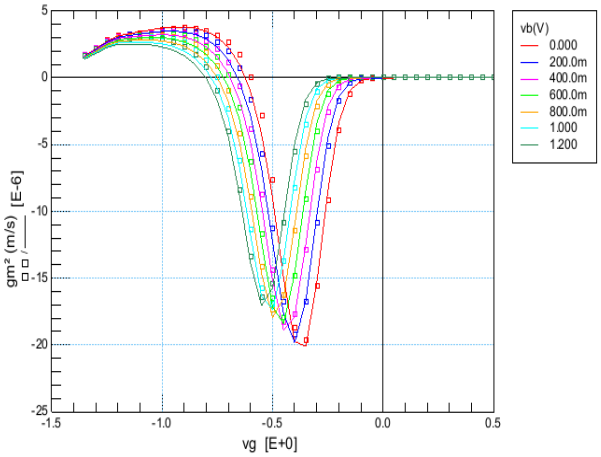
Id = f(Vg, Vb) @ Vd=low



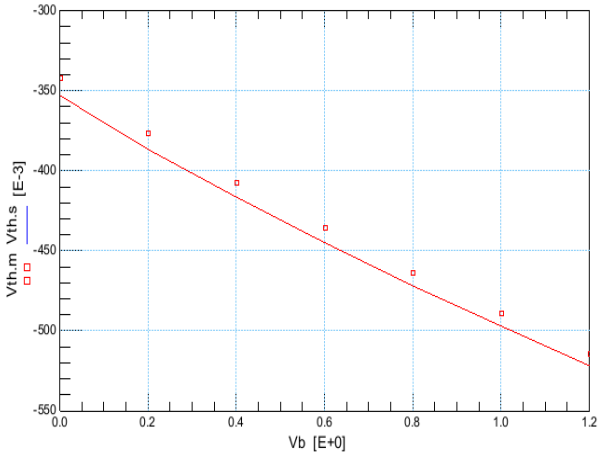
gm = f(Vg, Vb) @ Vd=low



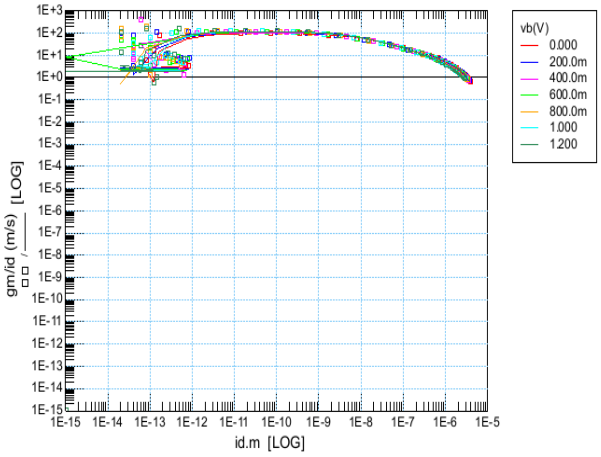
gm² = f(Vg, Vb) @ Vd=low

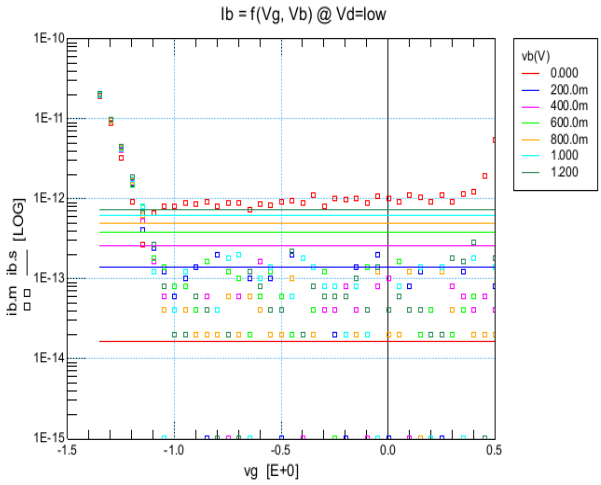


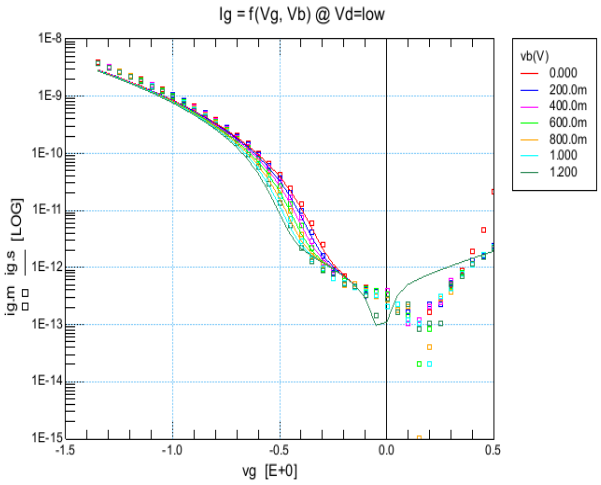
Vth @ Vd=low



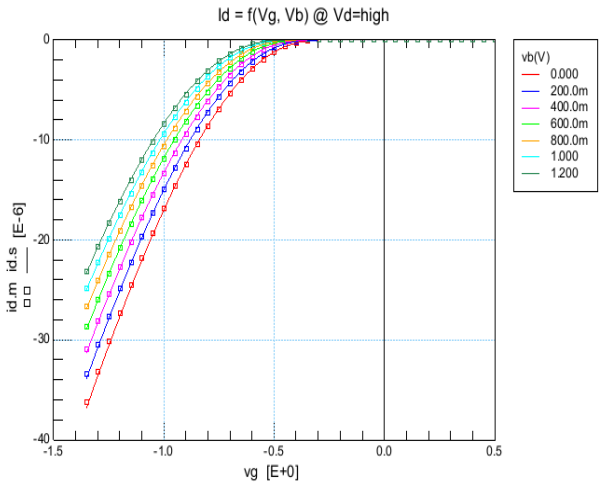
gm/id=f(id.m) @ Vd=low

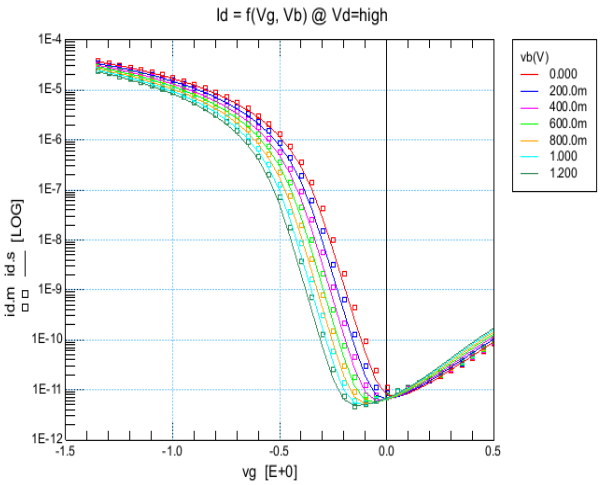


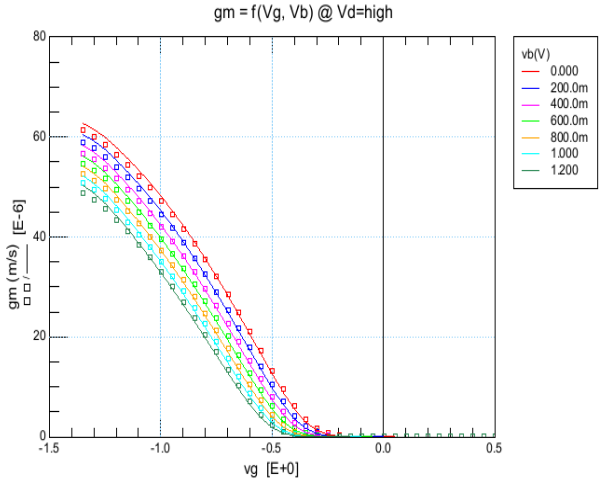


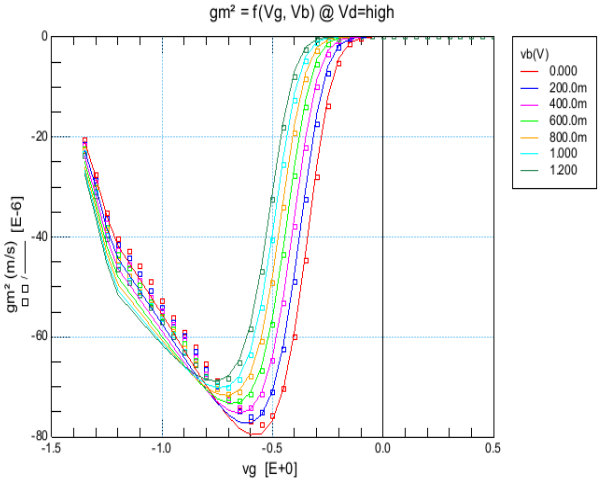


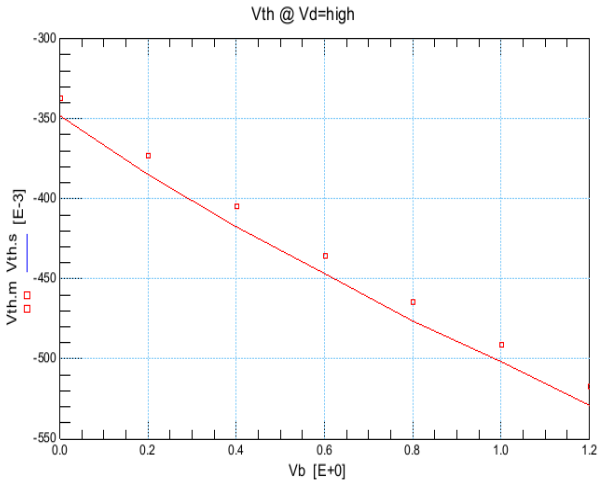
idvg, Vd = -1.2V, T = 27°C

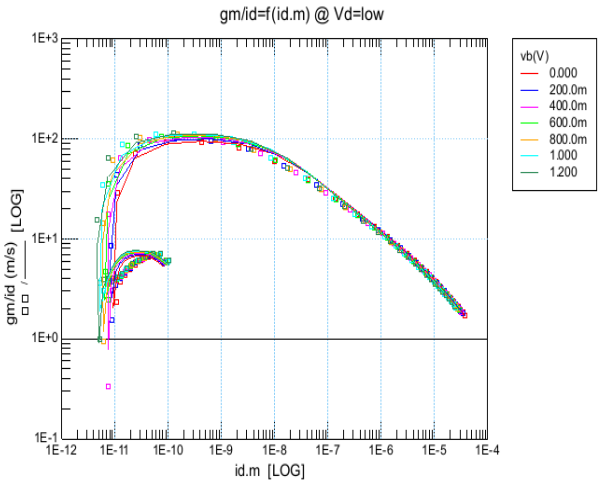


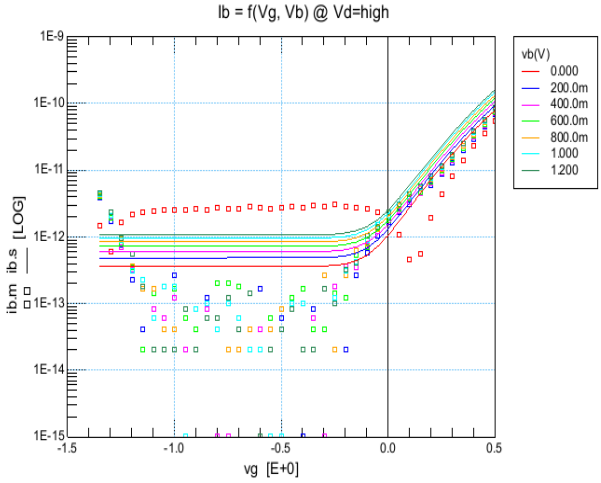


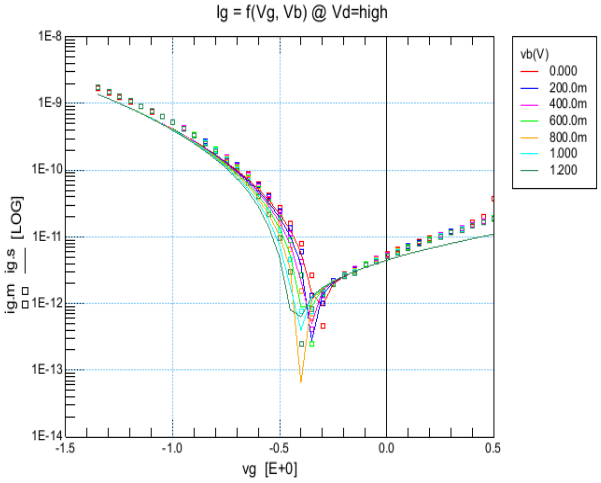




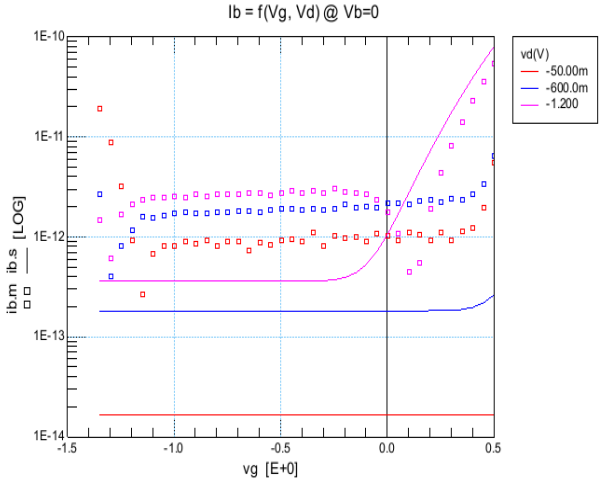


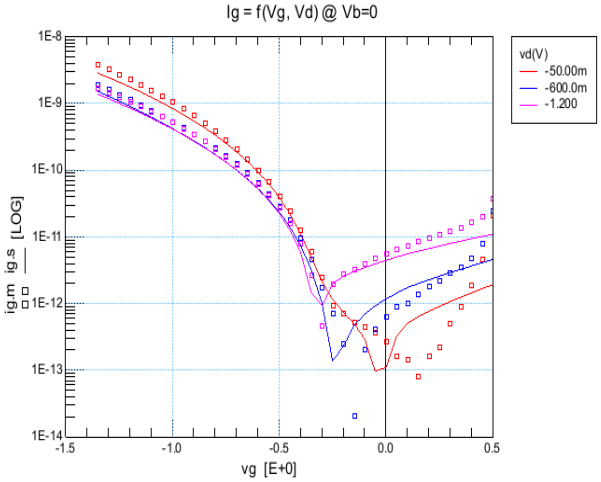




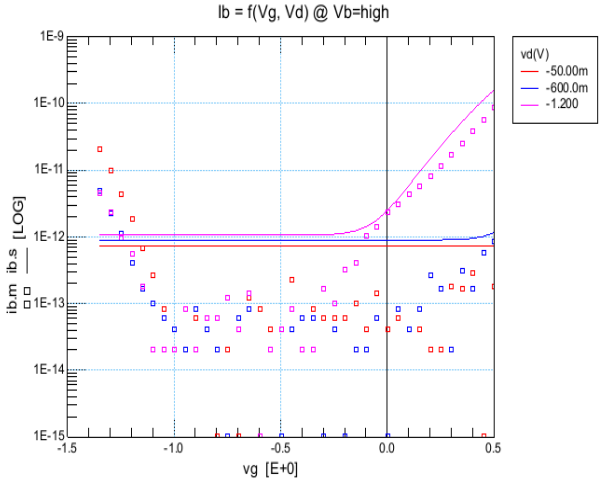


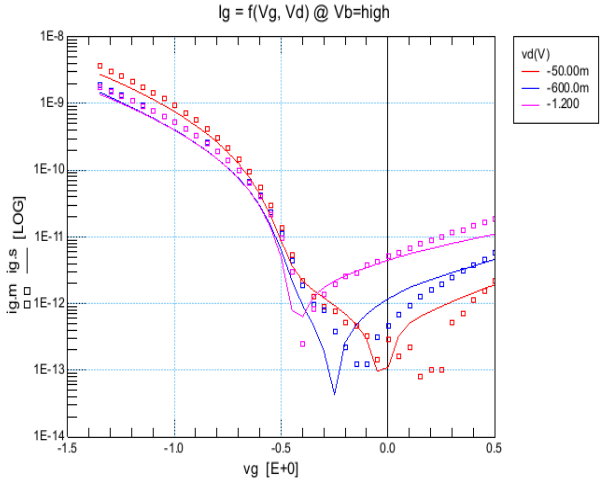
idvg, Ib,Ig, Vb = 0V, T = 27°C



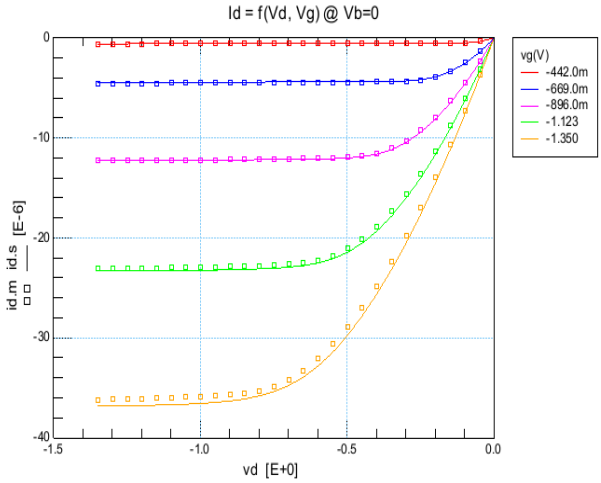


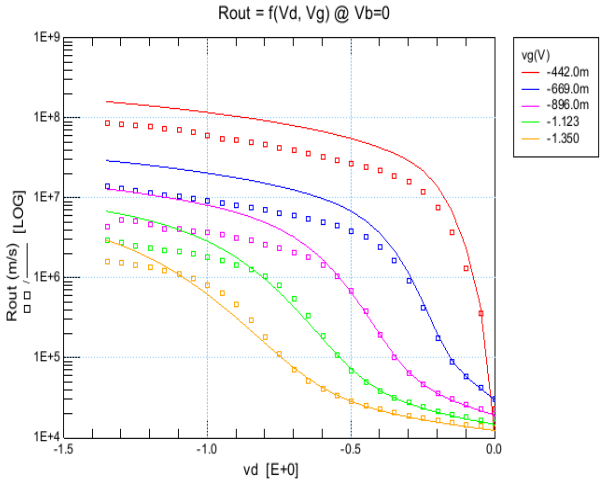
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

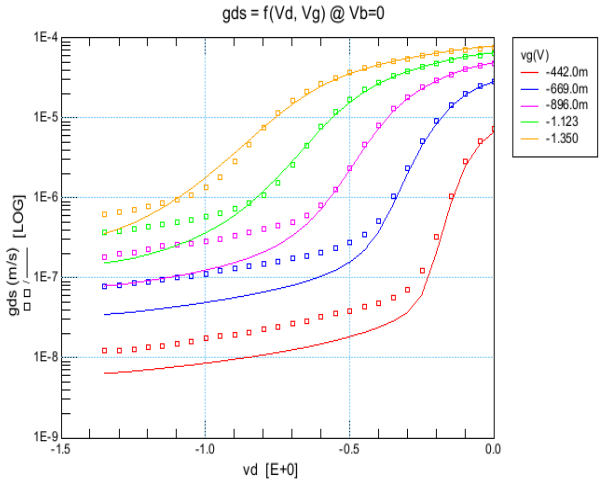




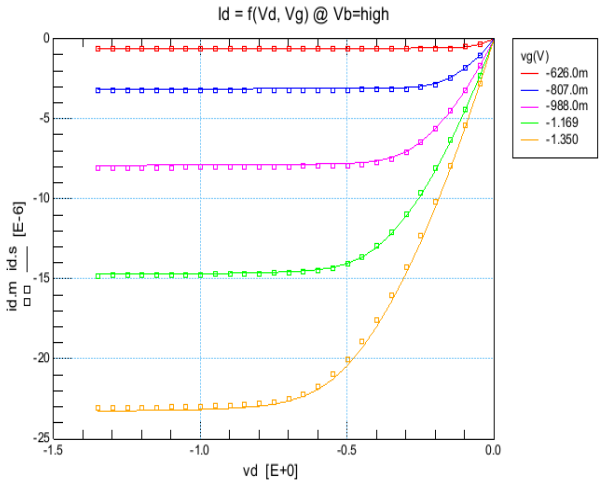
idvd, Vb = 0V, T = 27°C

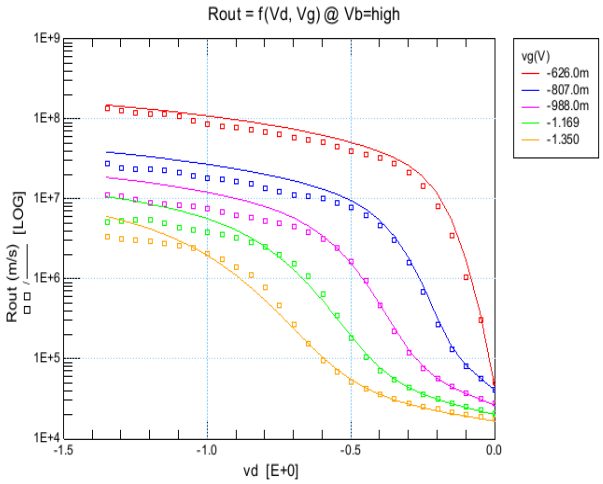


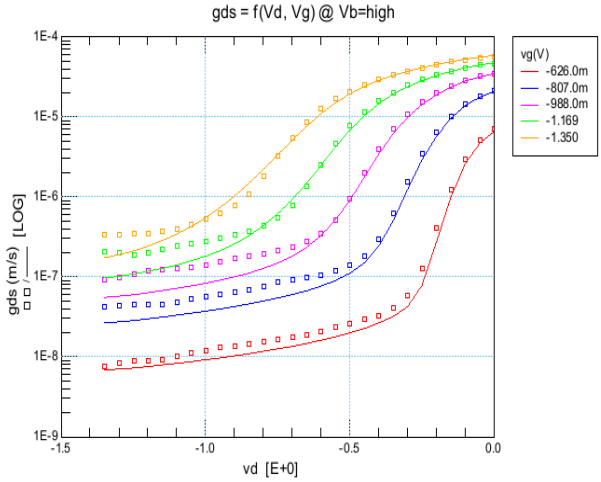




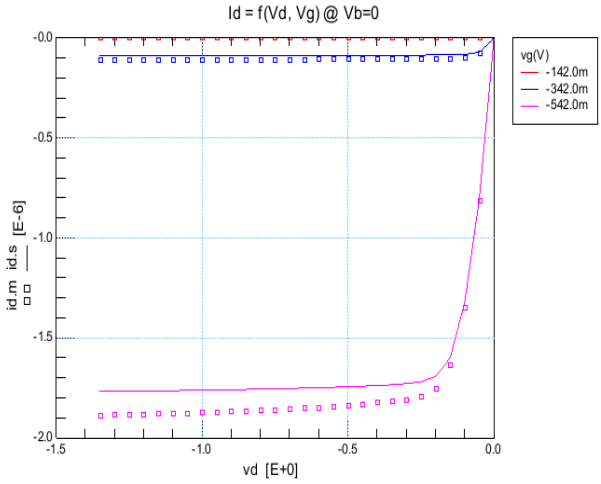
idvd_vbmin, Vb = 1.2V, T = 27°C

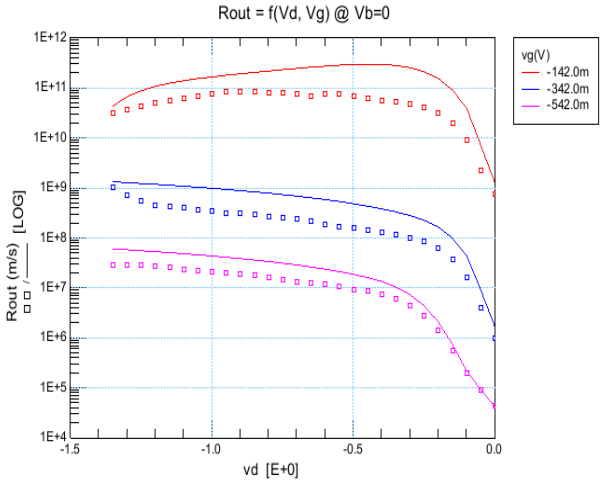


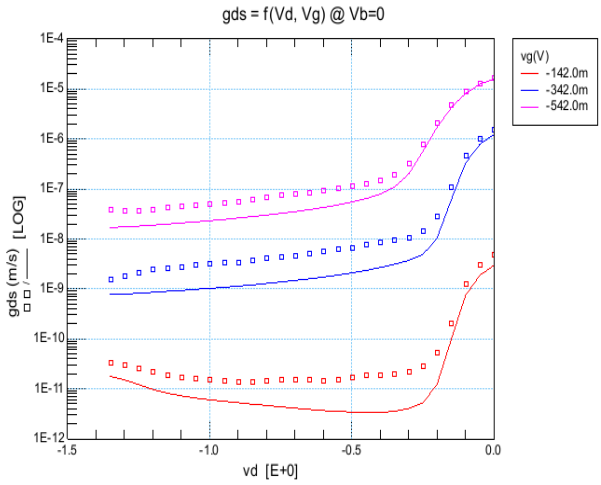




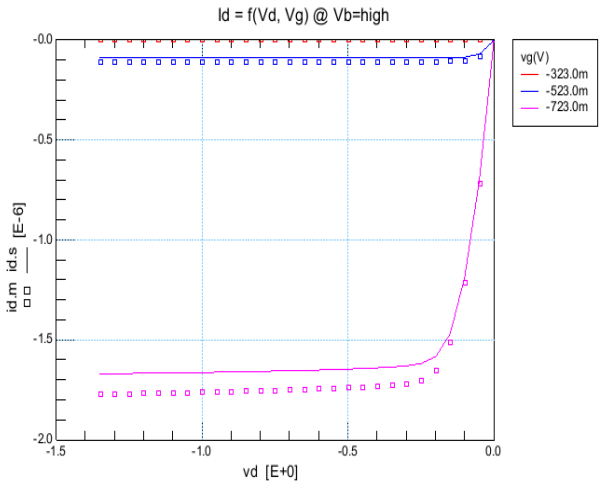
idvd_vth, Vb = 0V, T = 27°C

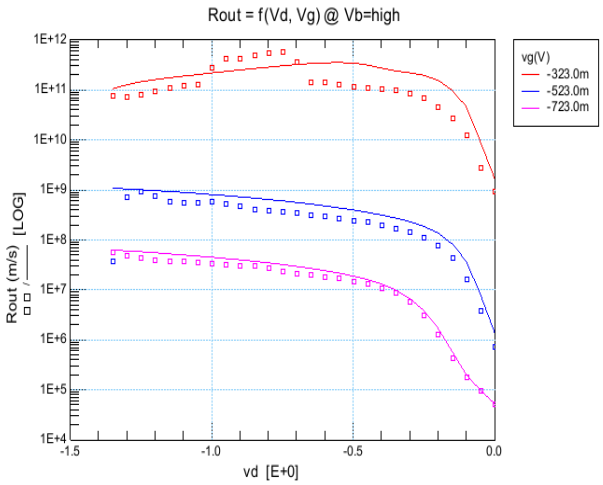


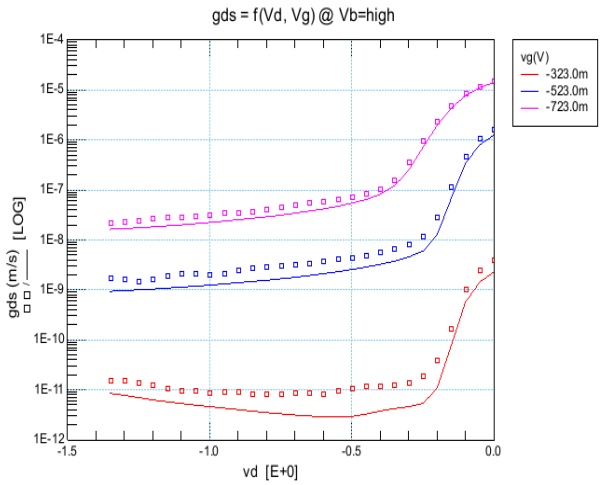




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

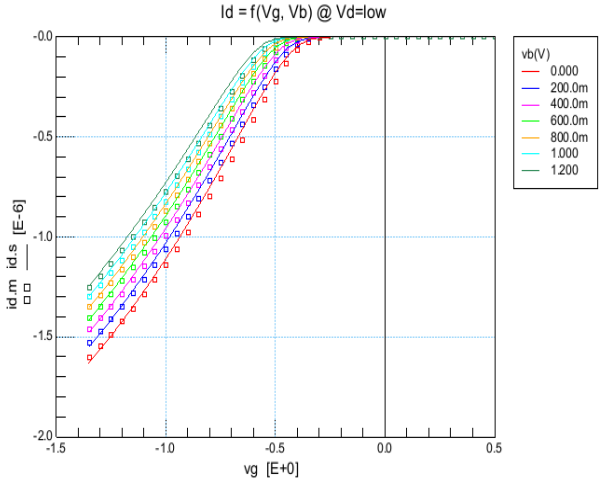


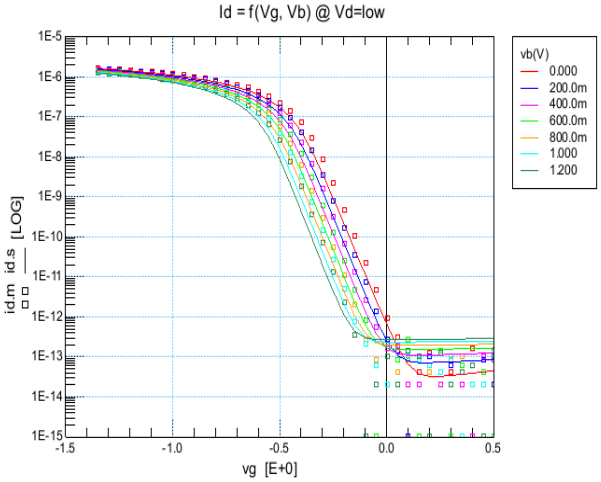


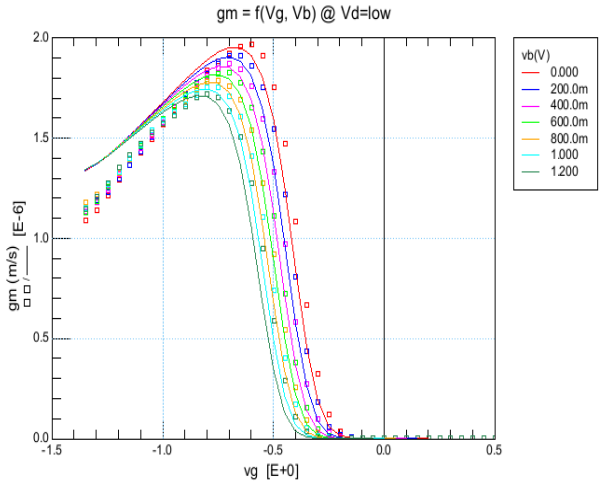


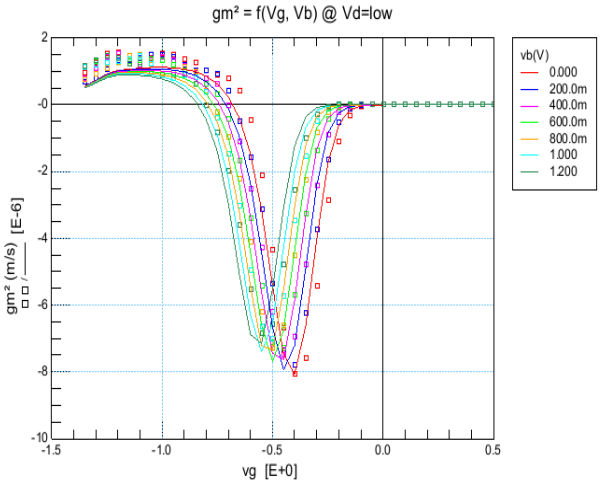
4.34 W0u15_L0u5_S555_2

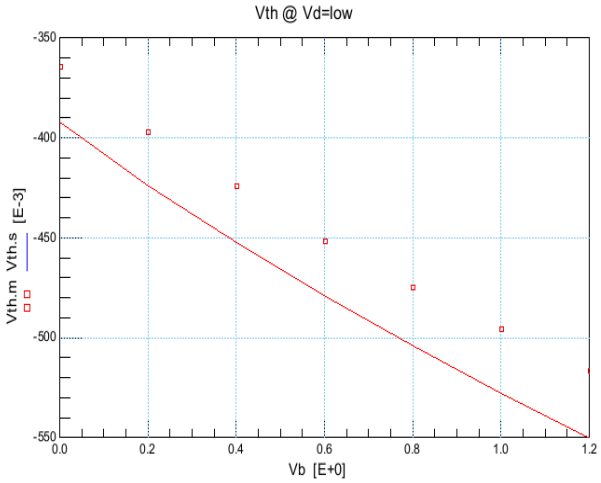
idvg, Vd = -0.05V, T = 27°C

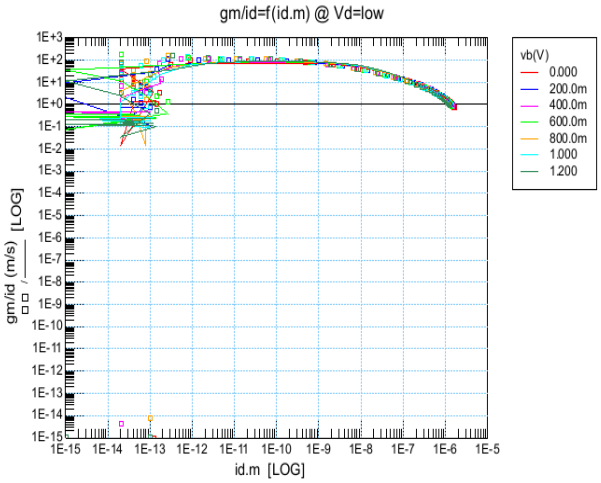


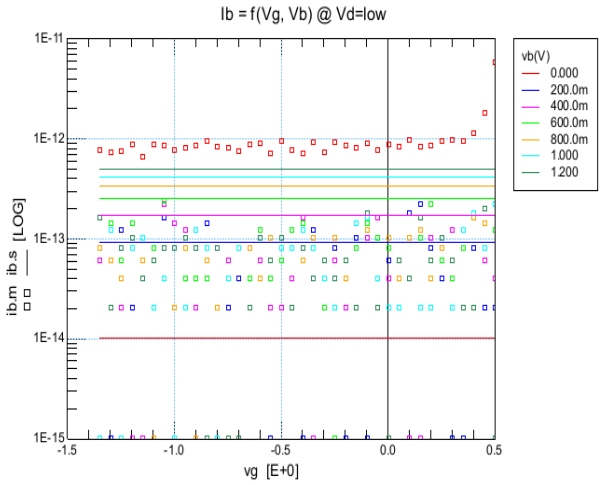


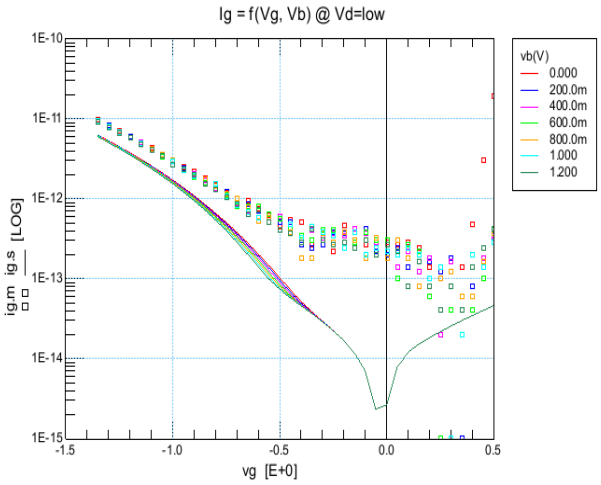




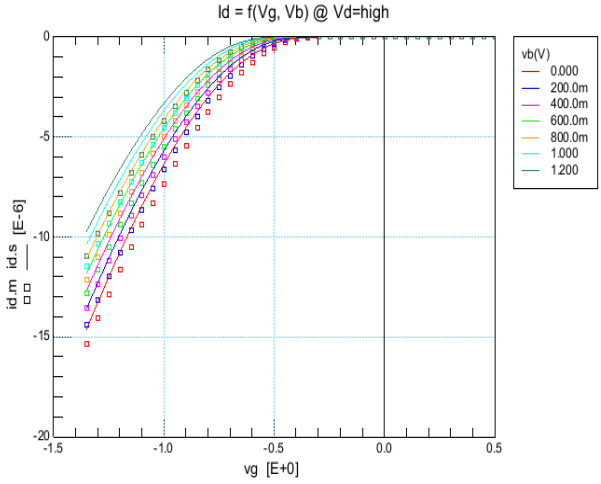


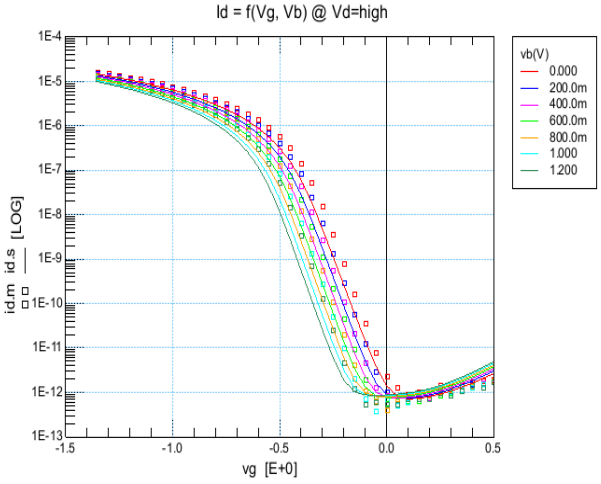


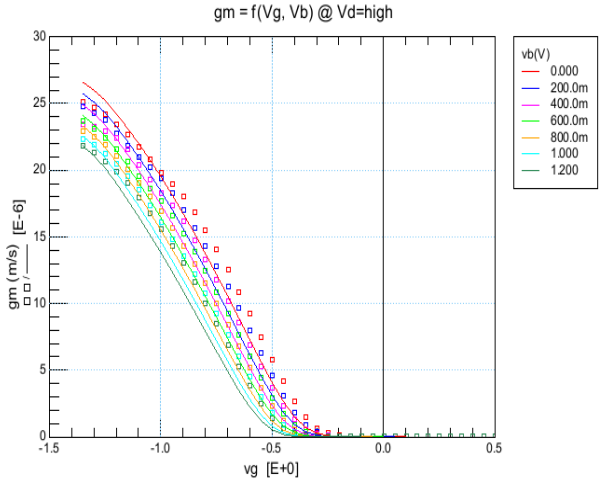


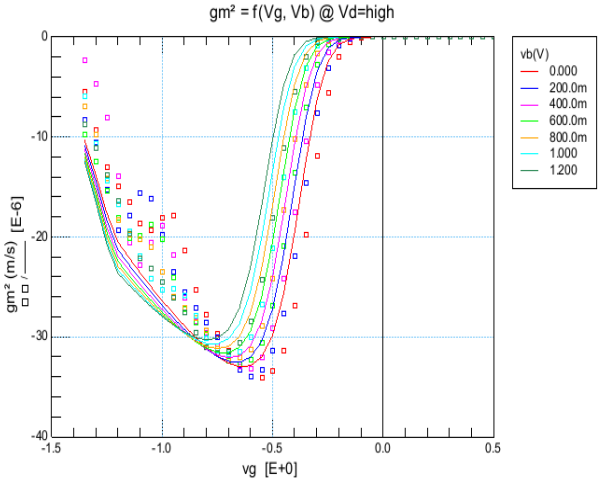


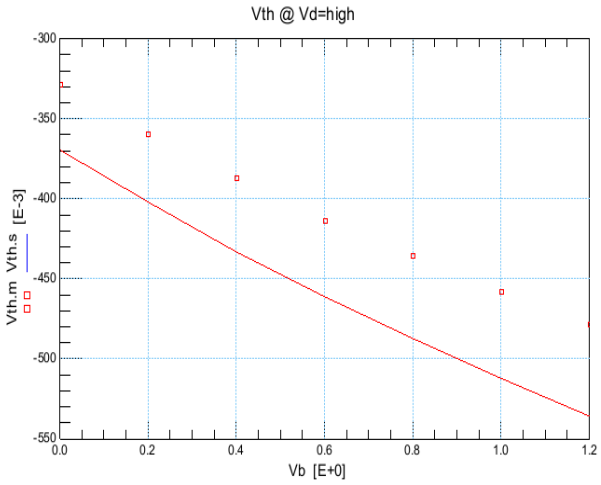
idvg, Vd = -1.2V, T = 27°C

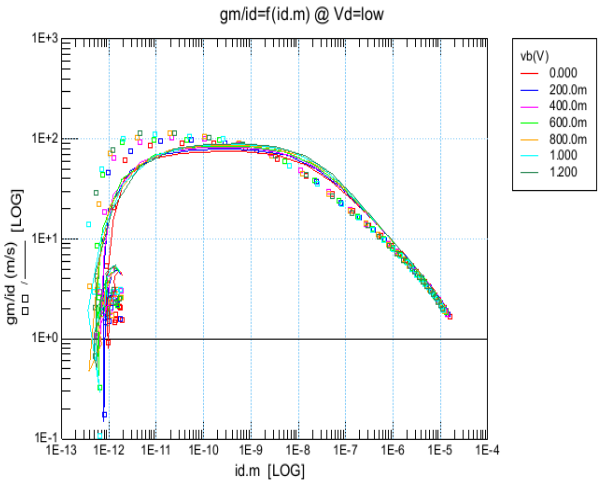


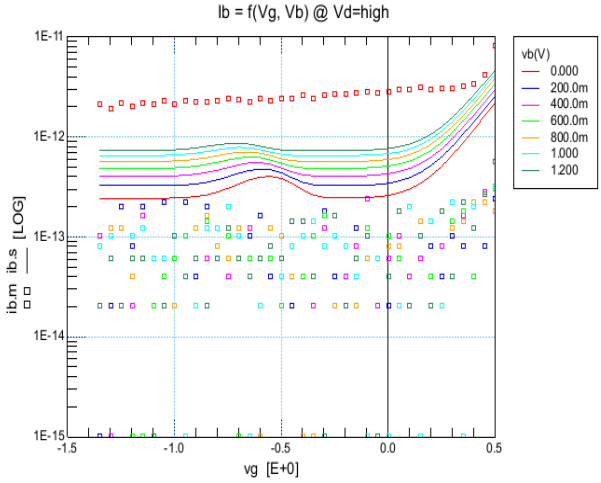


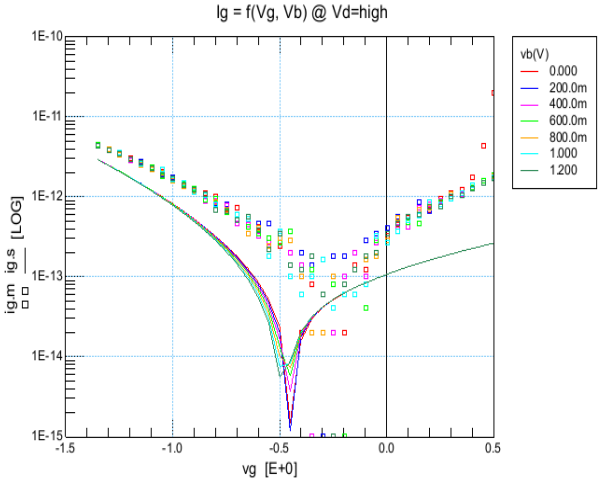




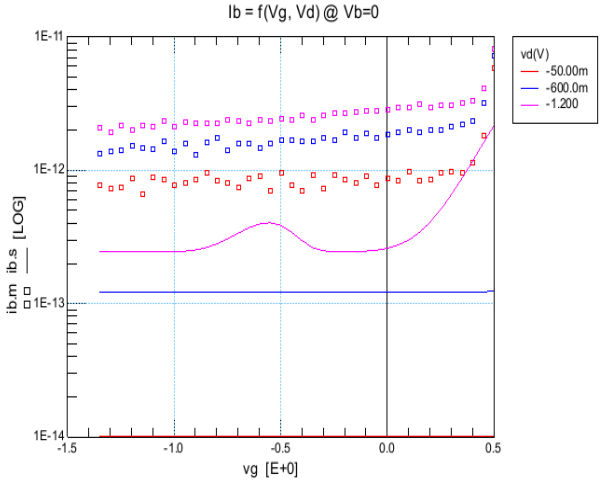


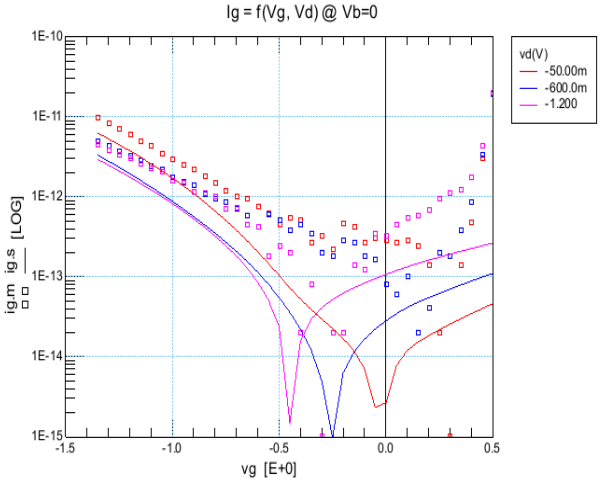




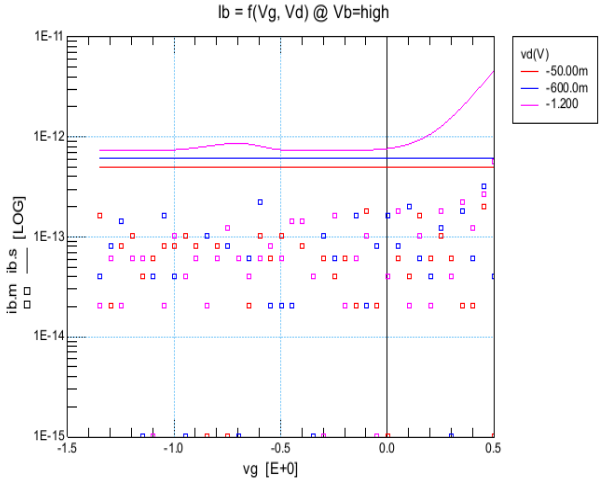


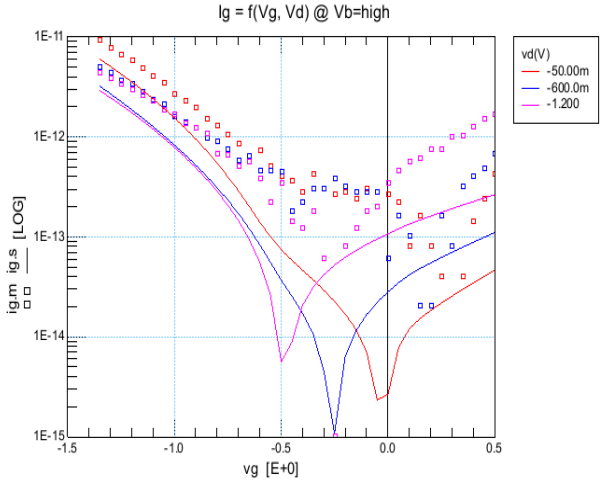
idvg, Ib,Ig, Vb = 0V, T = 27°C



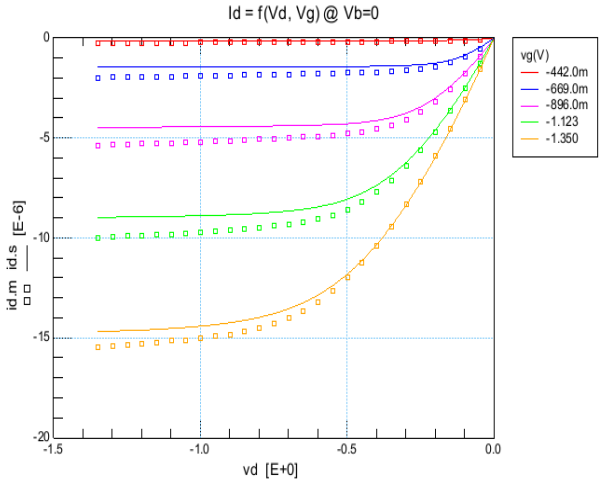


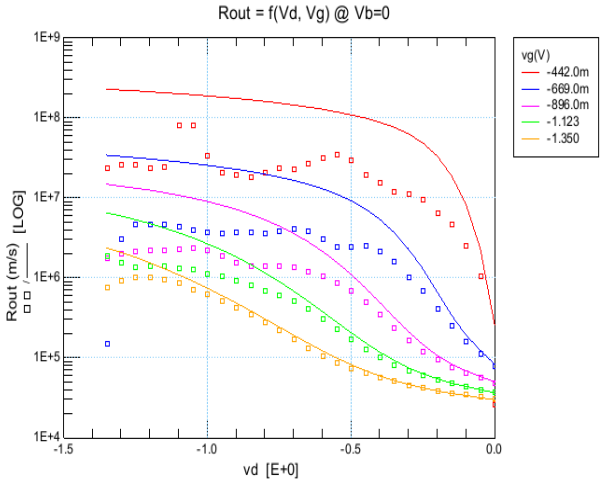
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

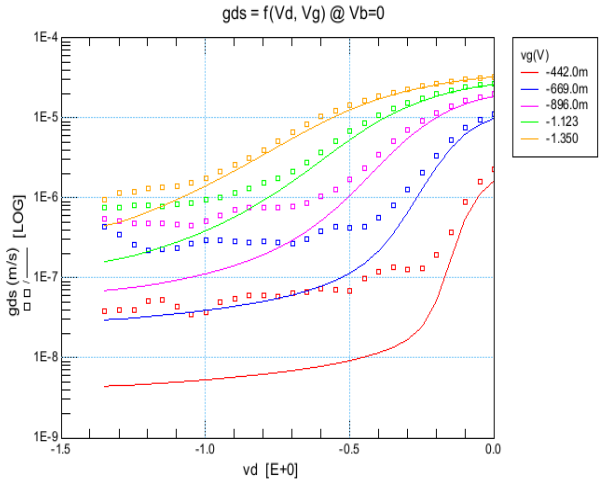




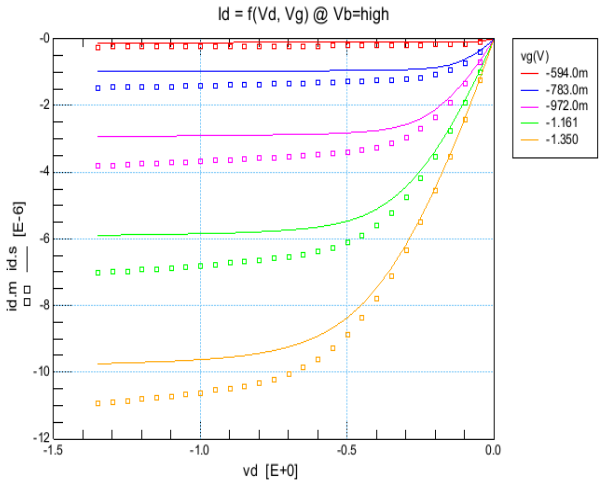
idvd, Vb = 0V, T = 27°C

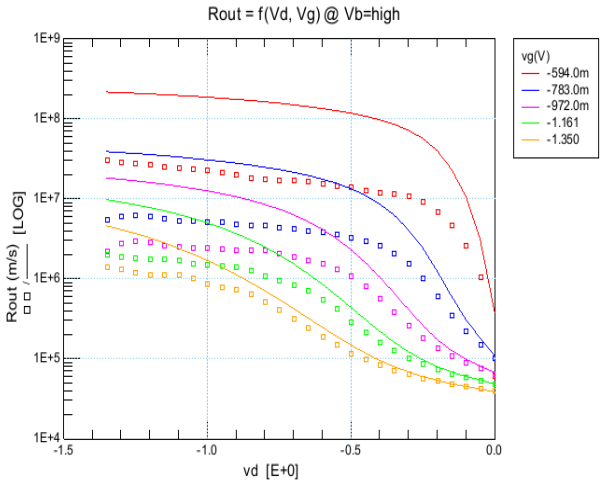


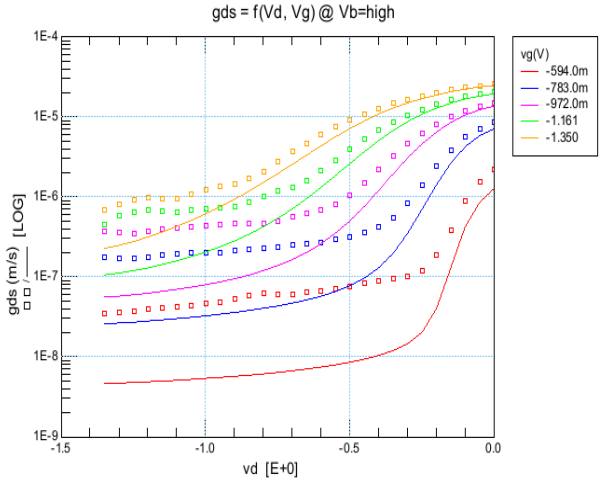




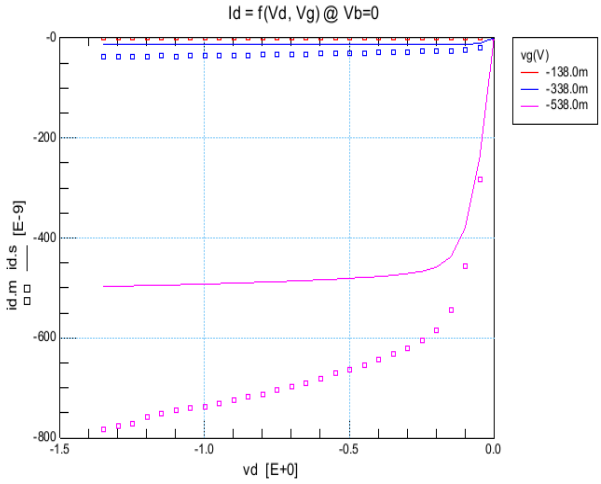
idvd_vbmin, Vb = 1.2V, T = 27°C

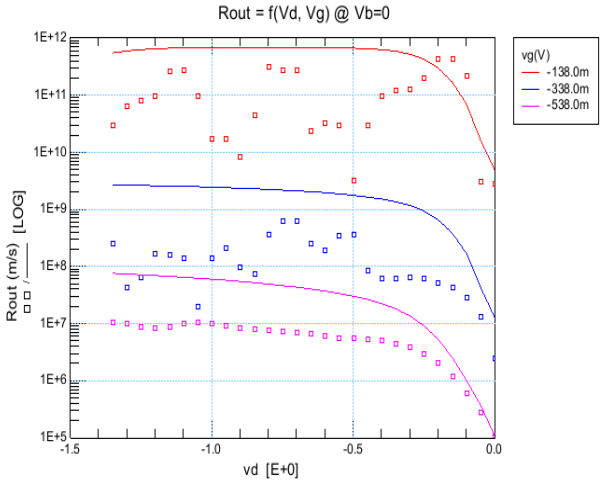


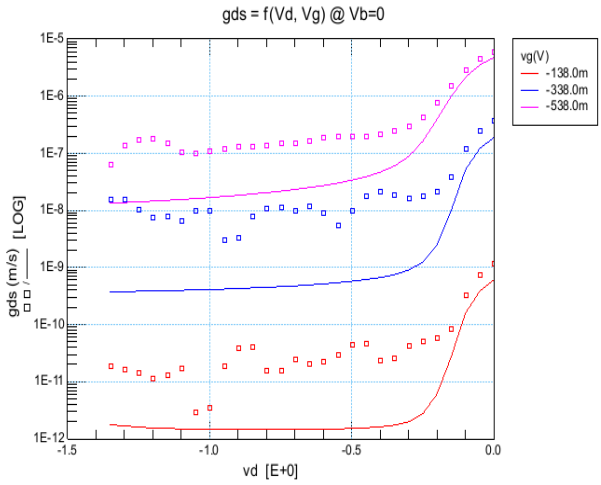




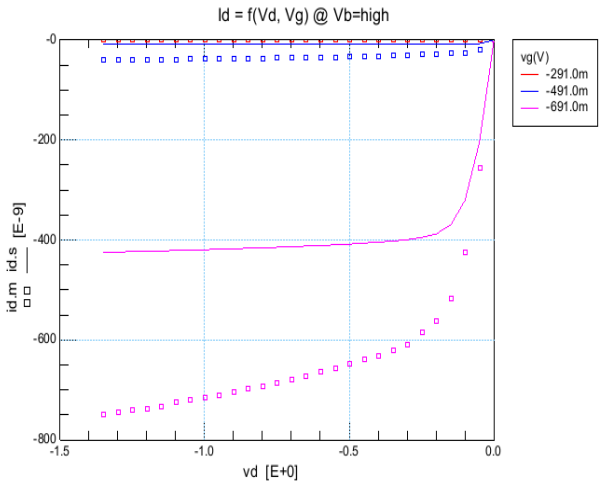
idvd_vth, Vb = 0V, T = 27°C

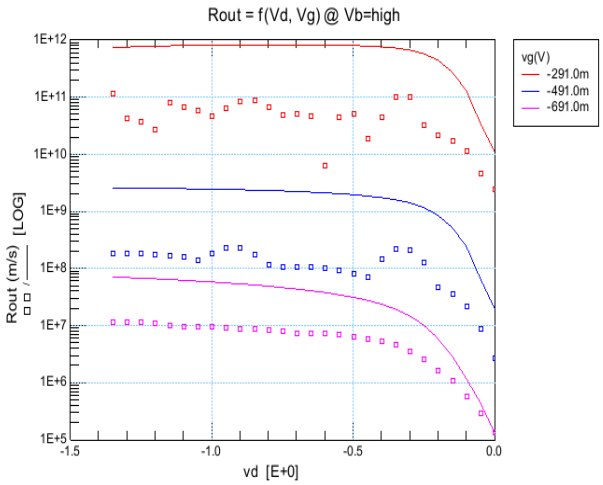


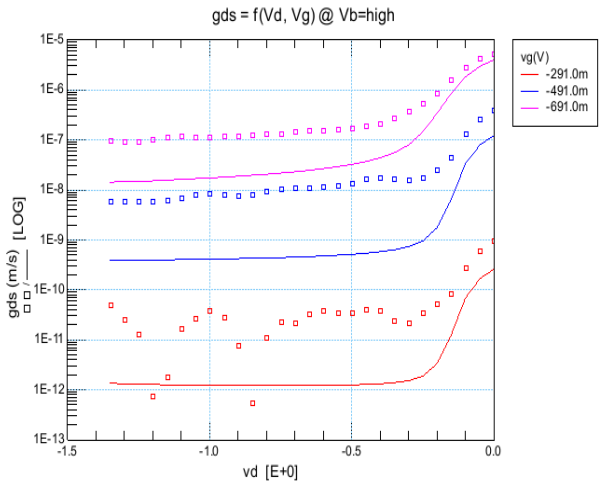




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

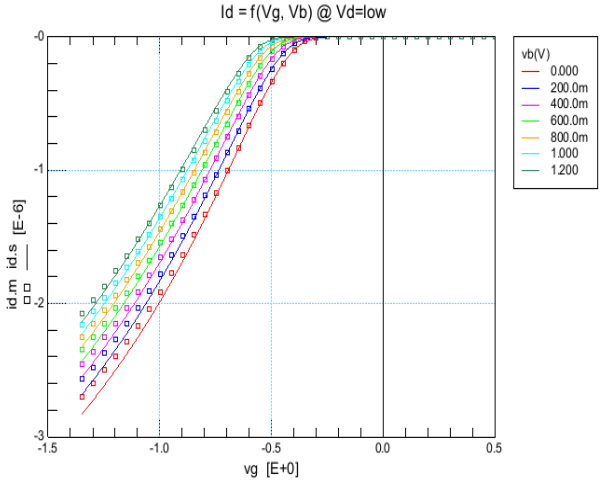


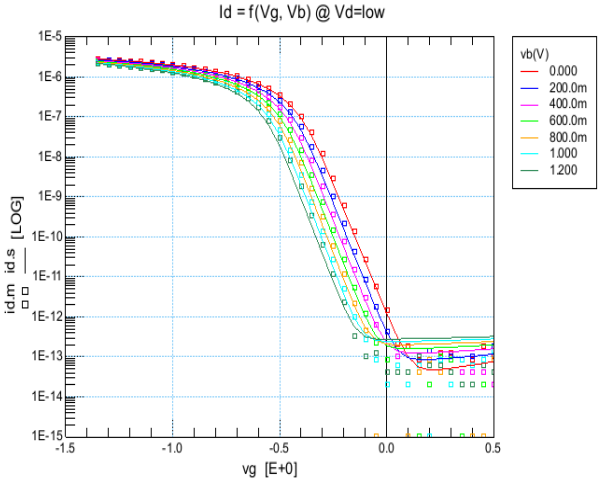


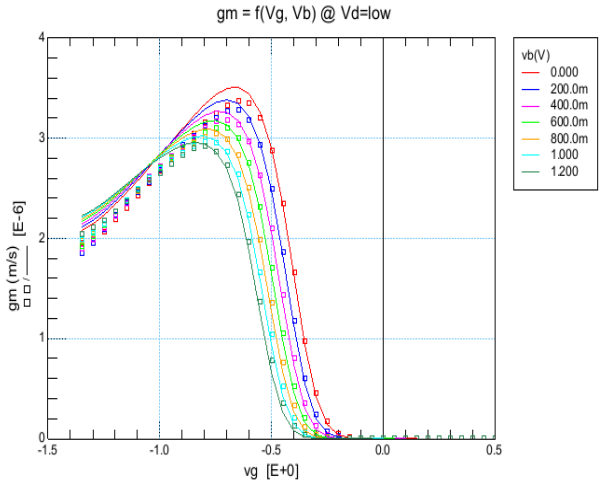


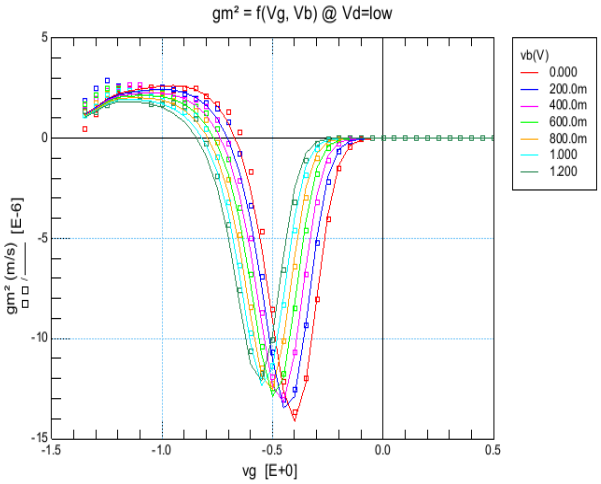
4.35W0u3_L0u5_S553_1

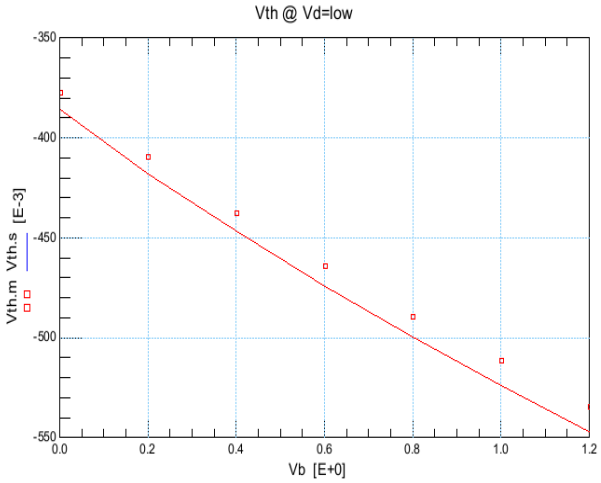
idvg, Vd = -0.05V, T = 27°C

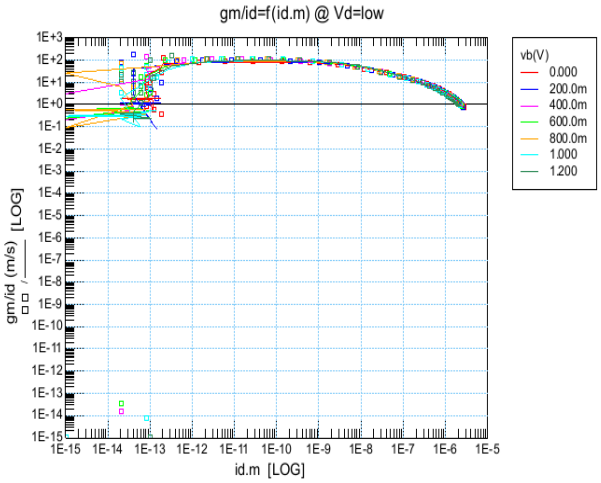


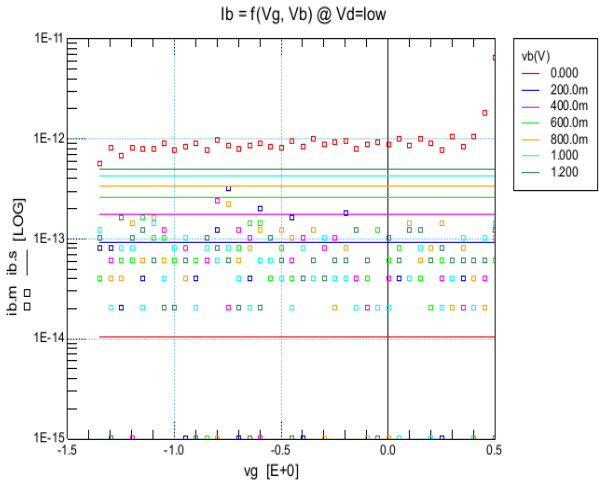


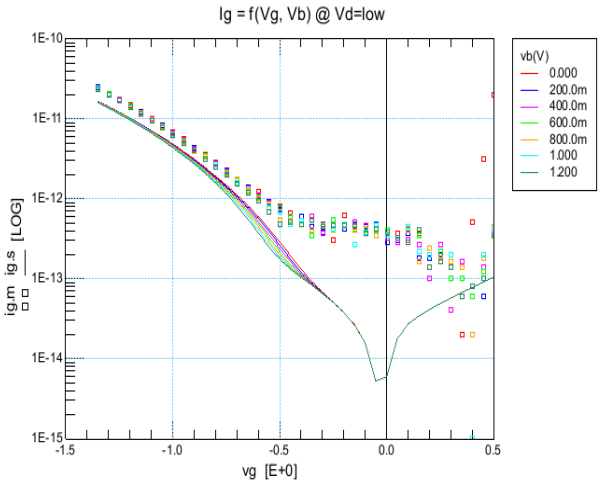




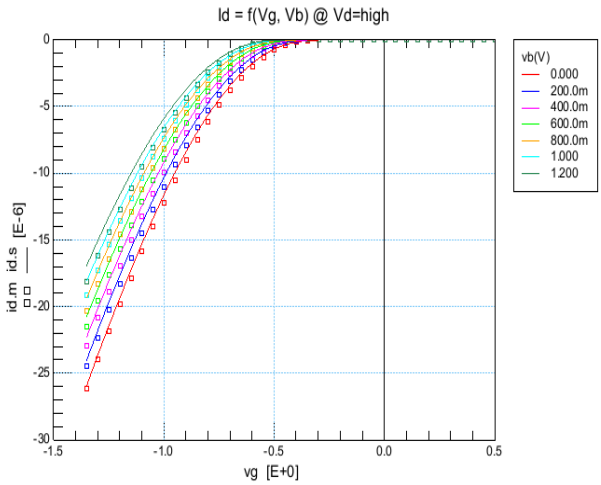


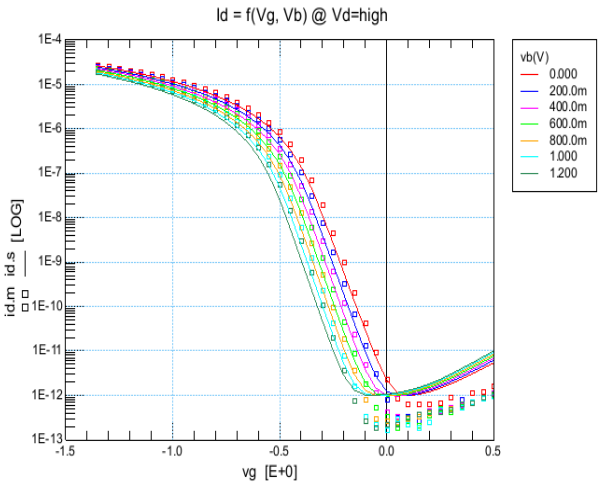


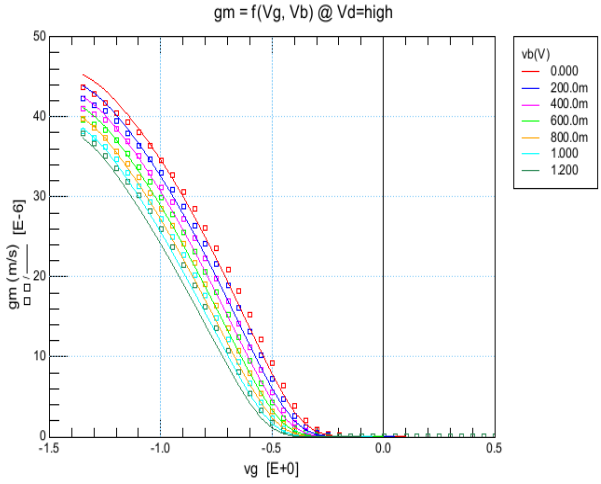


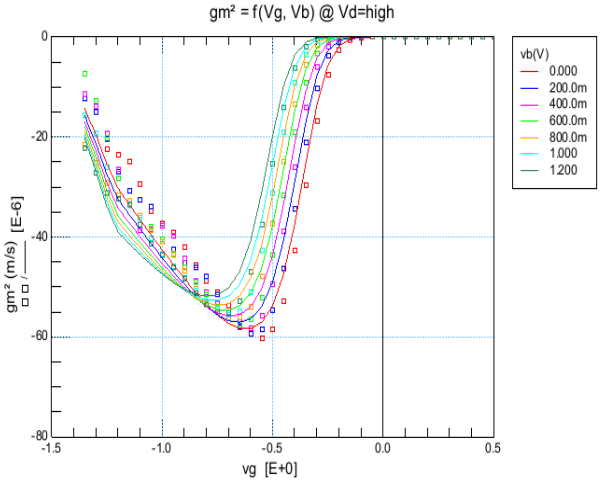


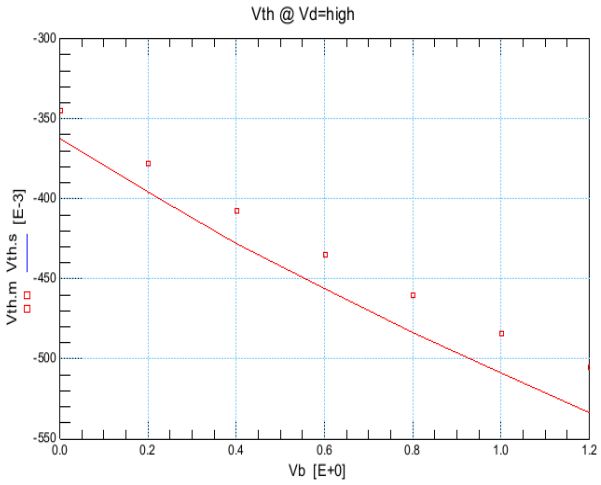
idvg, Vd = -1.2V, T = 27°C

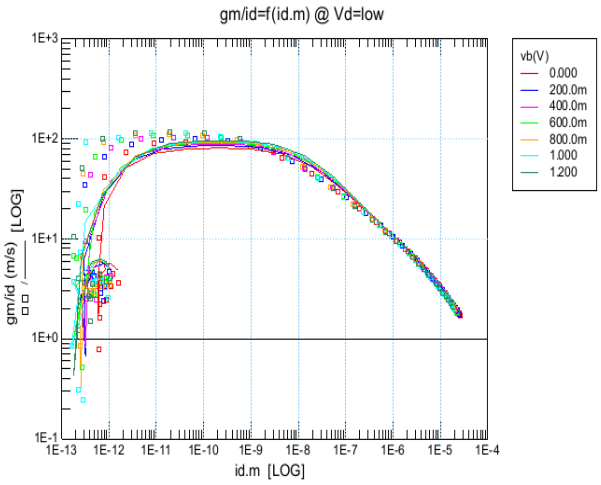


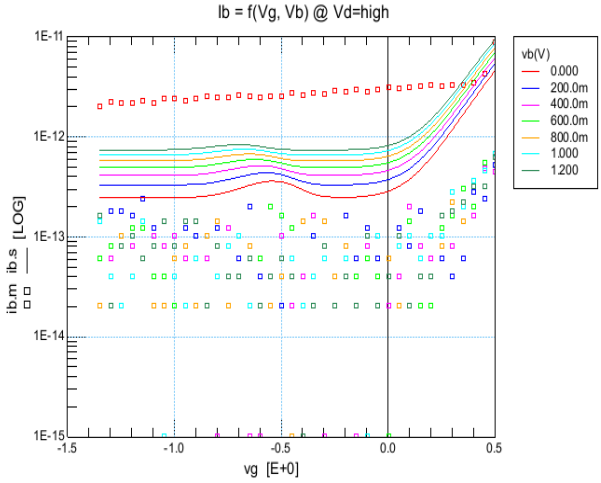


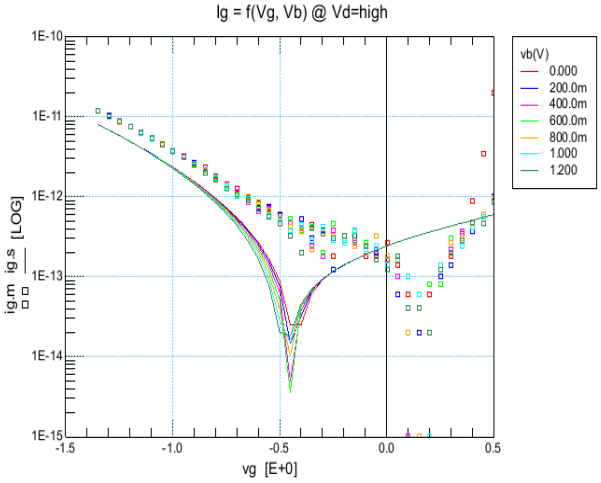




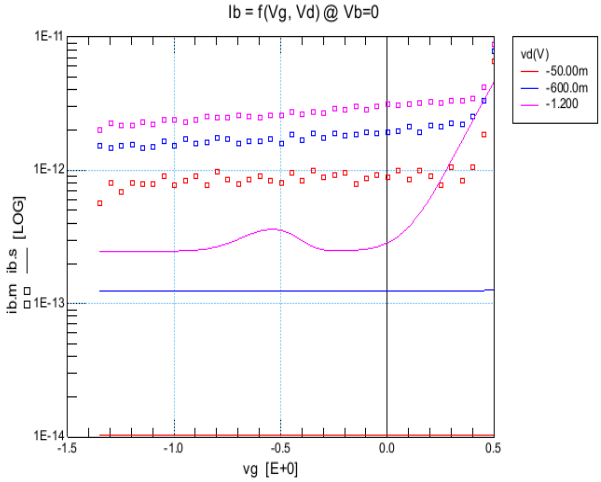


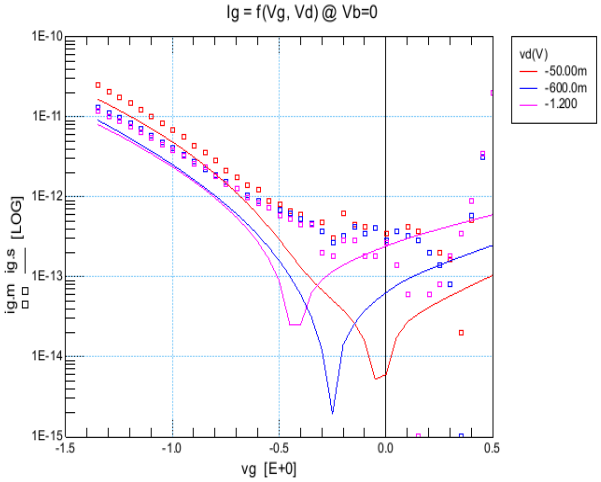




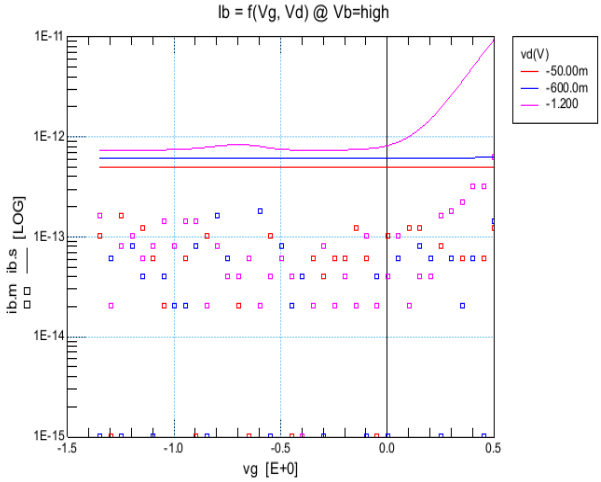


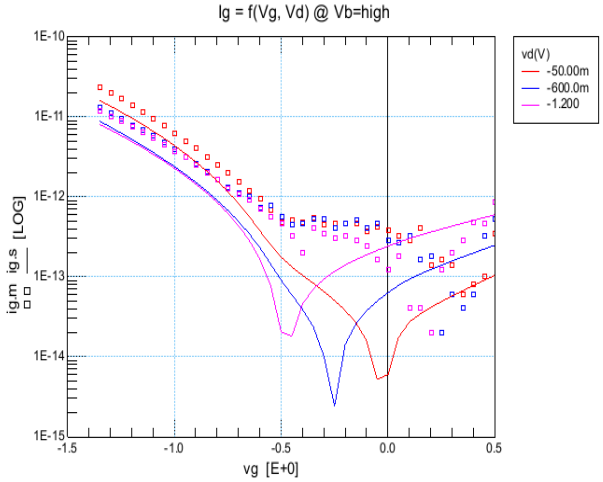
idvg, Ib,Ig, Vb = 0V, T = 27°C



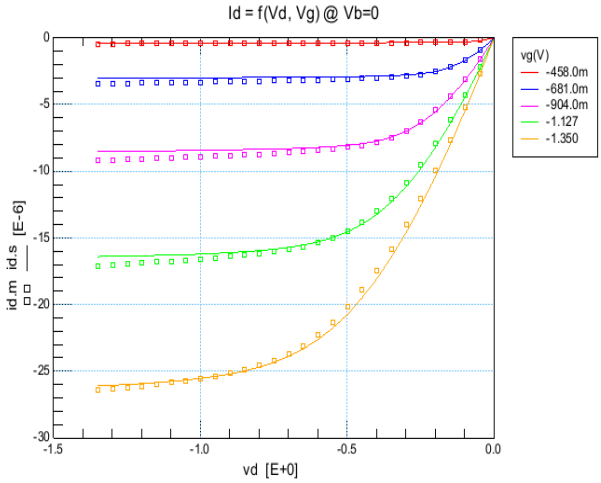


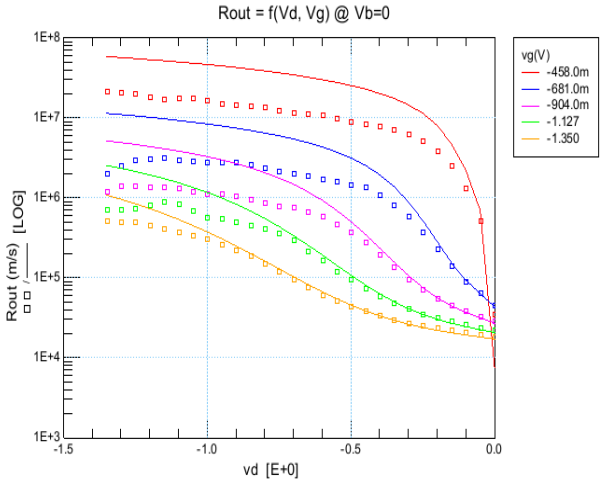
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

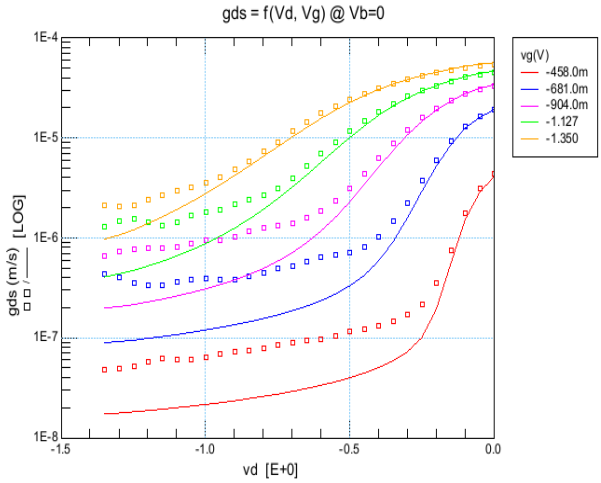




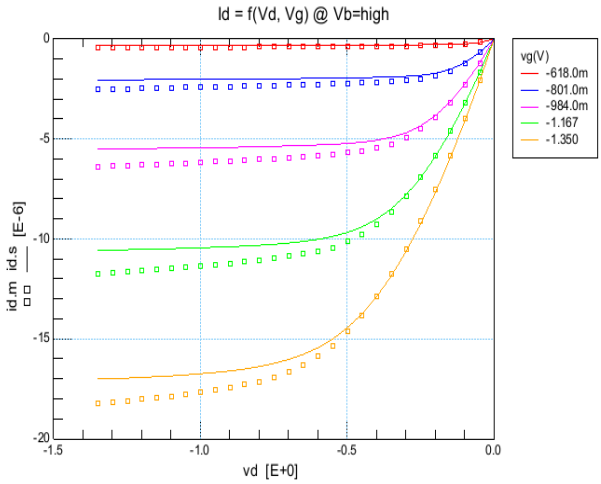
idvd, Vb = 0V, T = 27°C

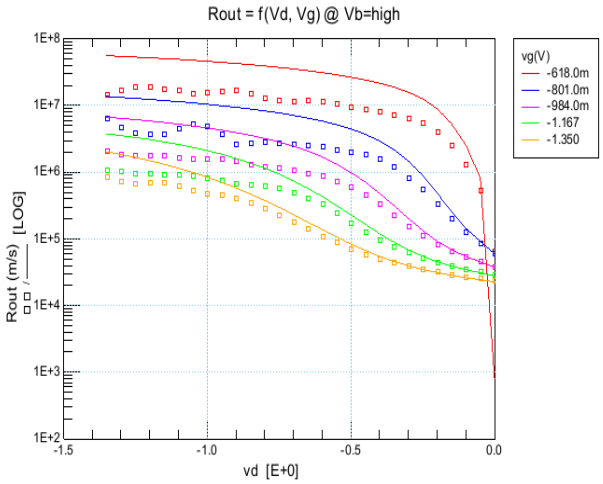


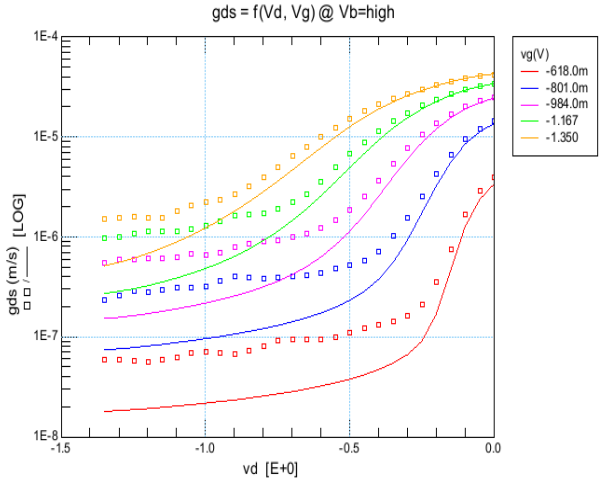




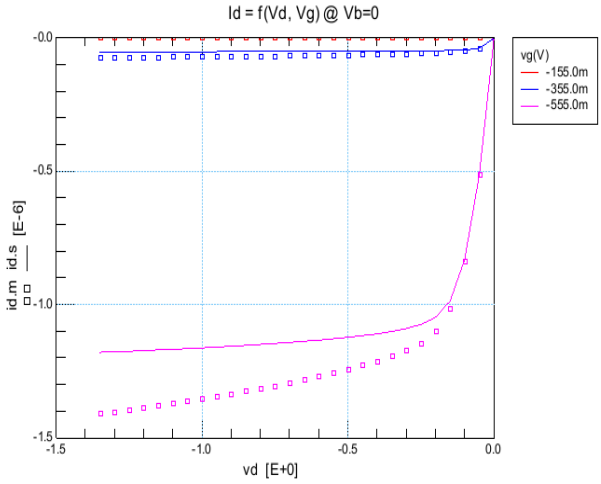
idvd_vbmin, Vb = 1.2V, T = 27°C

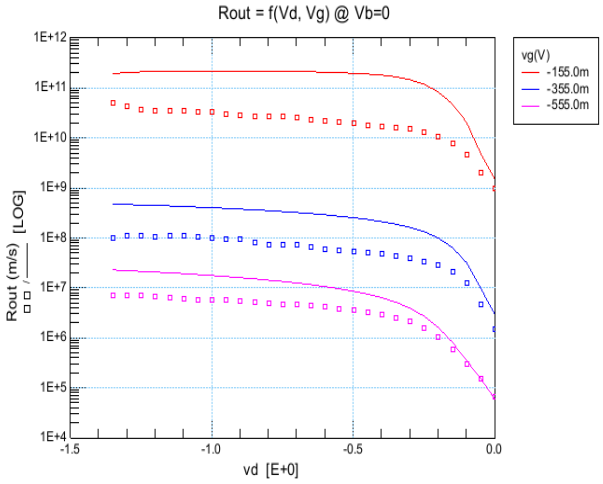


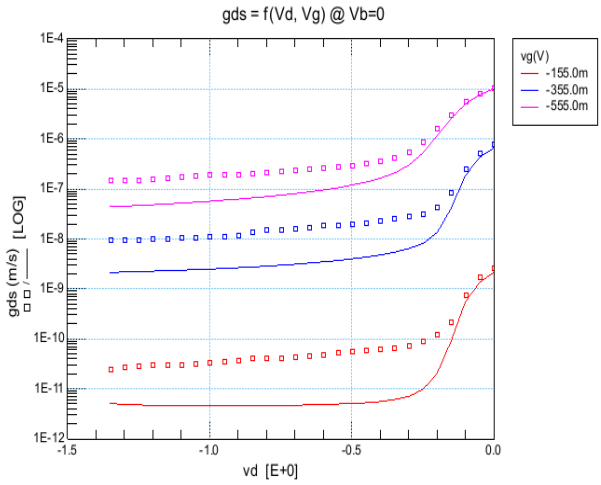




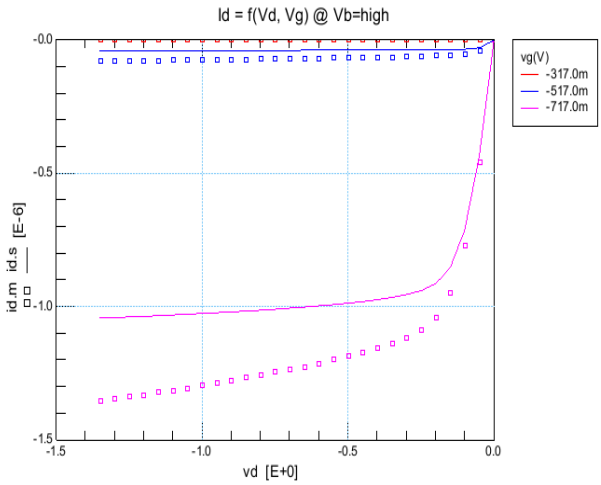
idvd_vth, Vb = 0V, T = 27°C

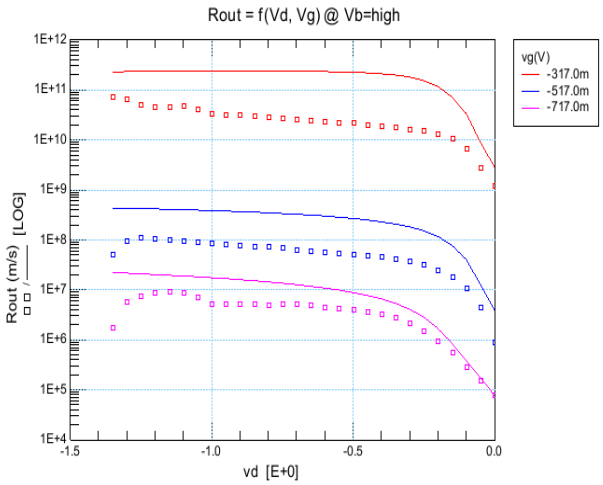


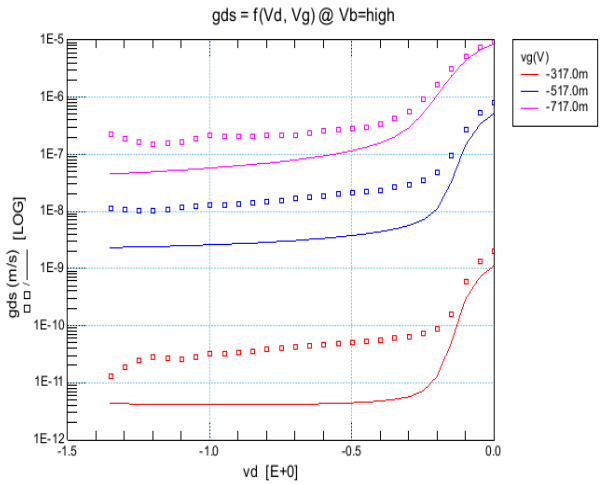




idvd_vth_vbmin, Vb = 1.2V, T = 27°C







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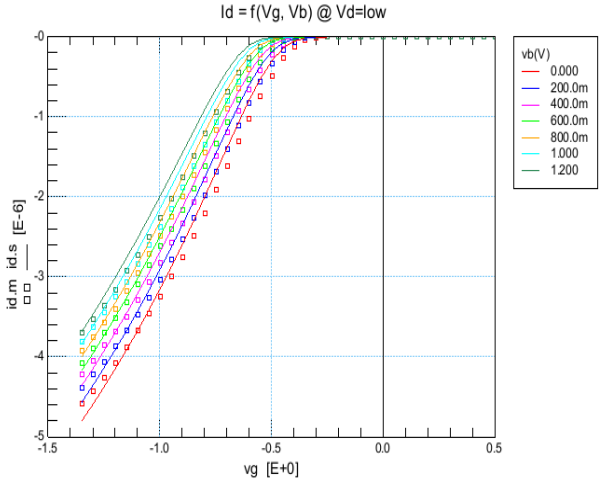
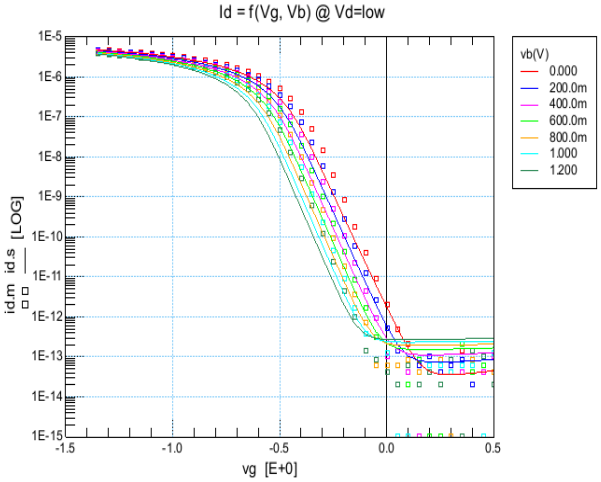
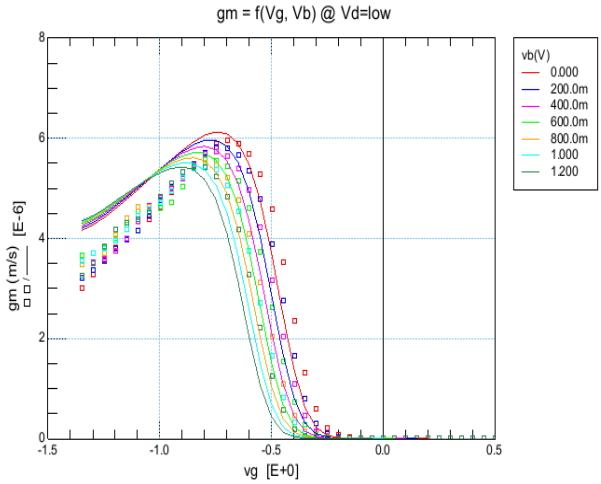
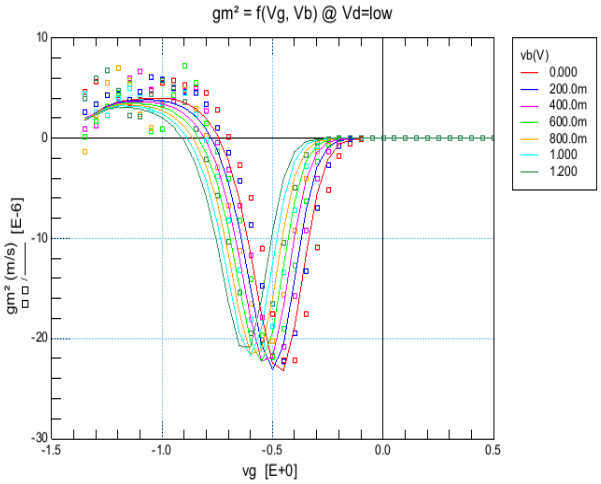
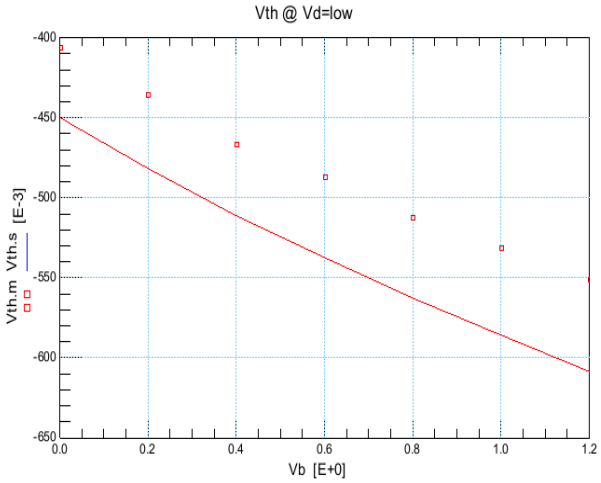
idvg, Vd = -0.05V, T = 27°C

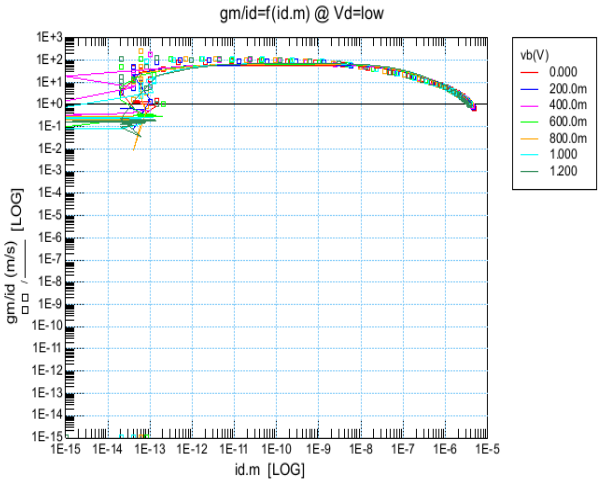


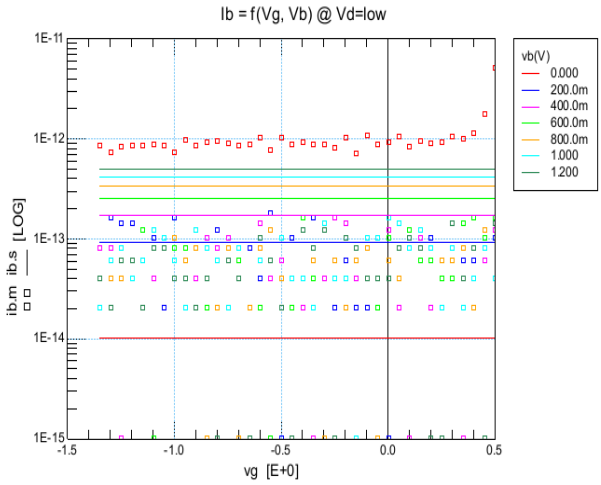


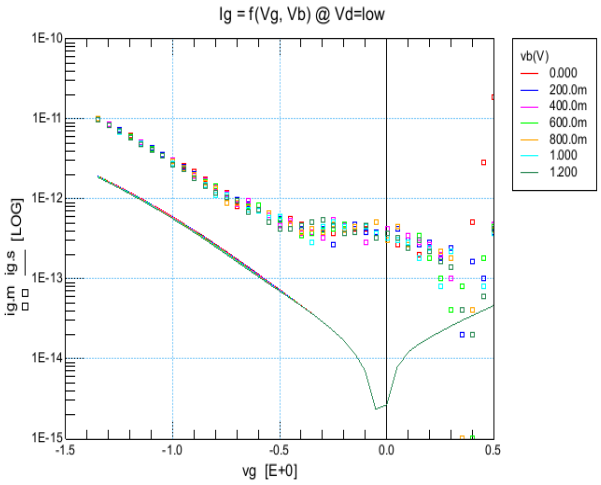




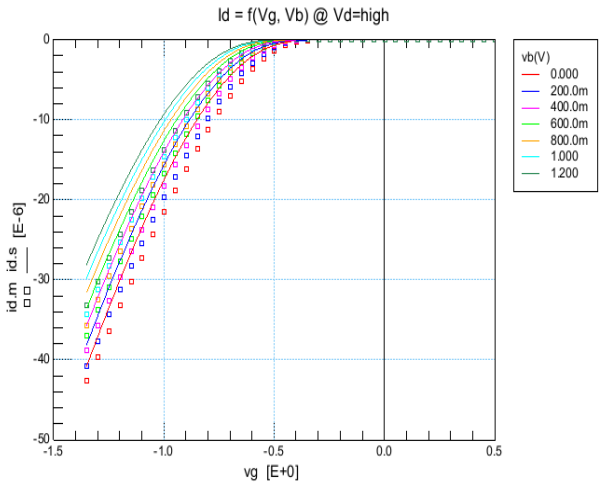


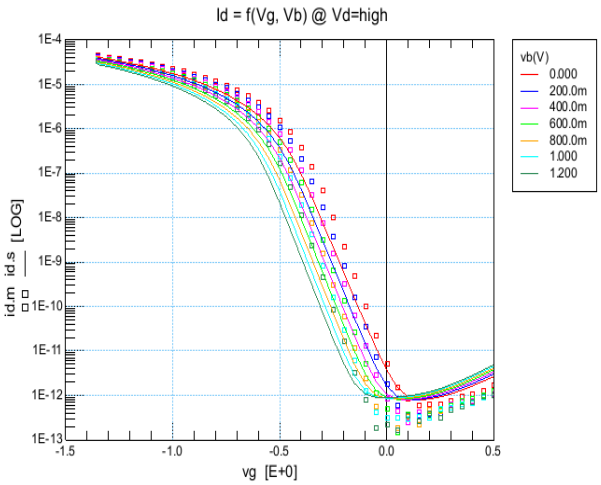


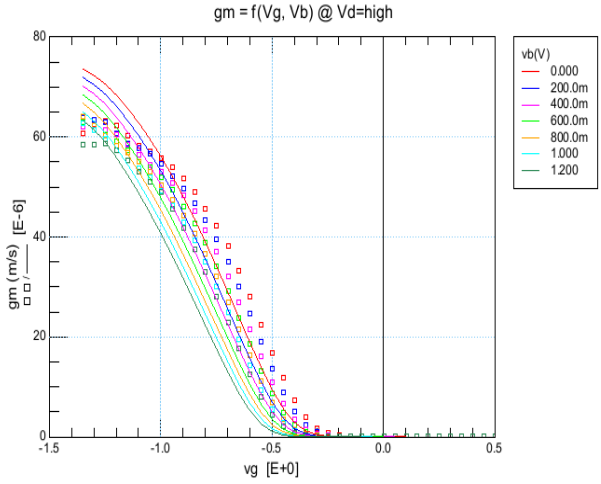


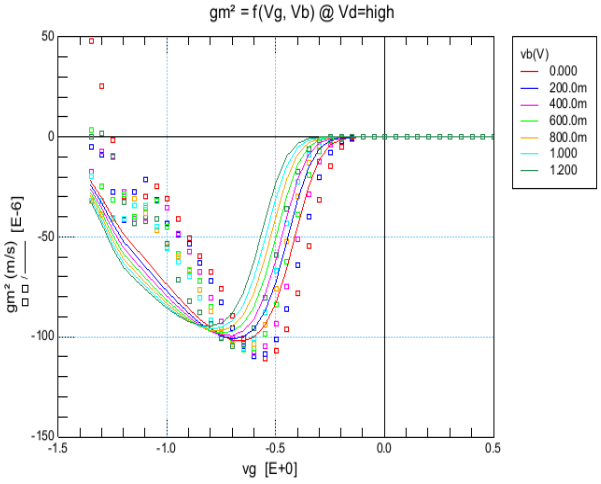


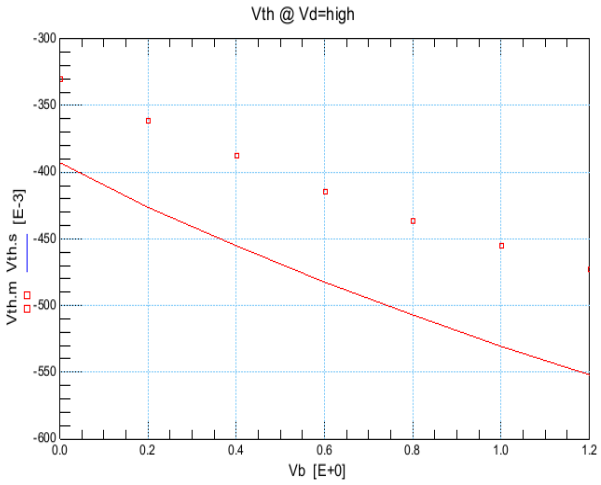
idvg, Vd = -1.2V, T = 27°C

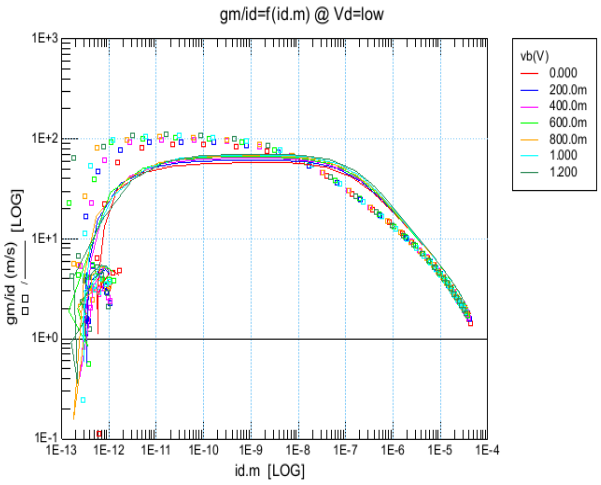


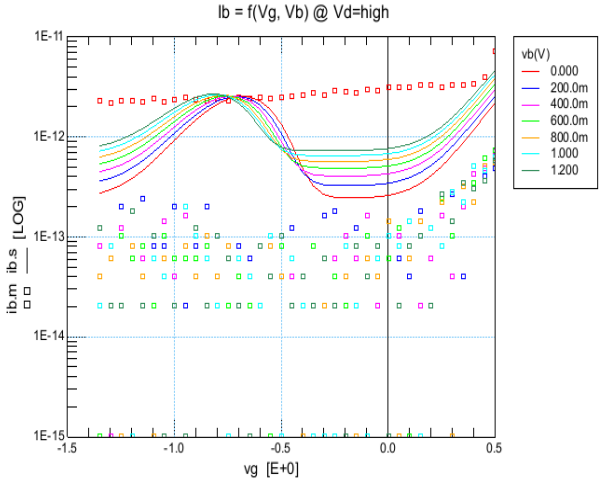


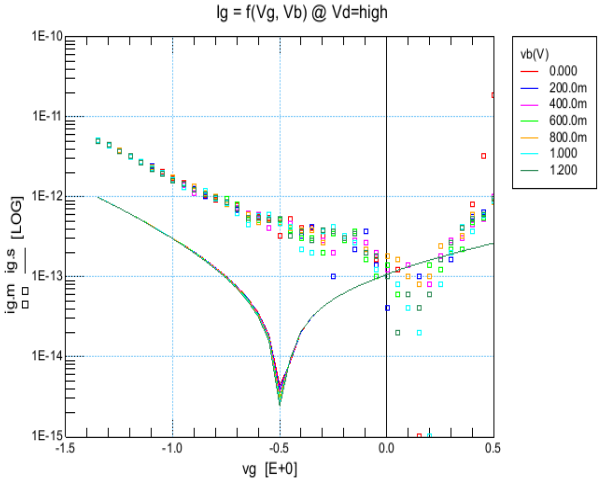




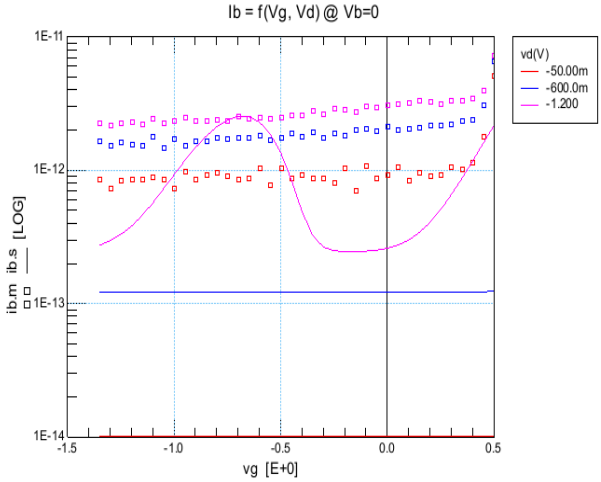


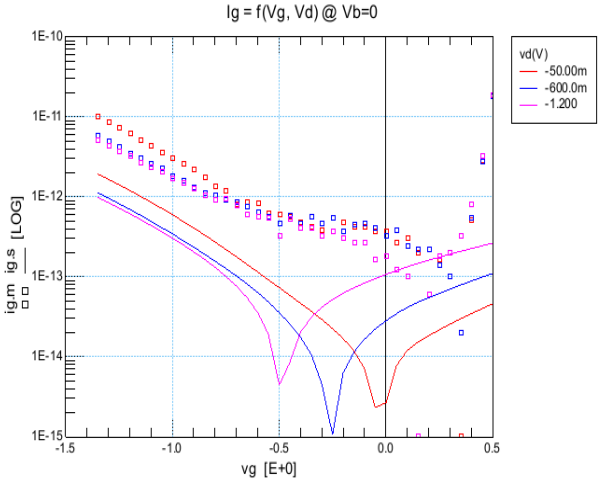




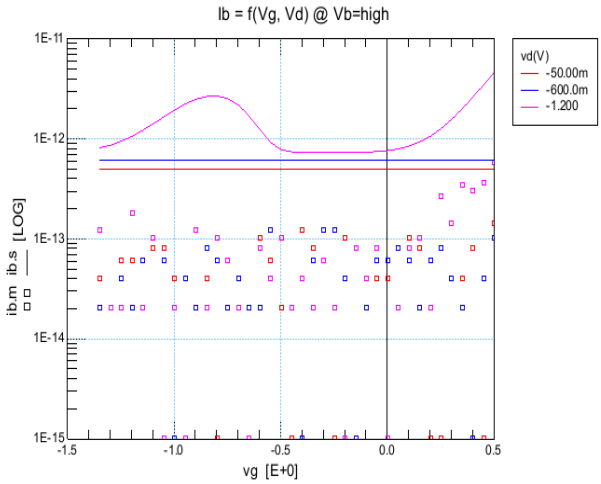


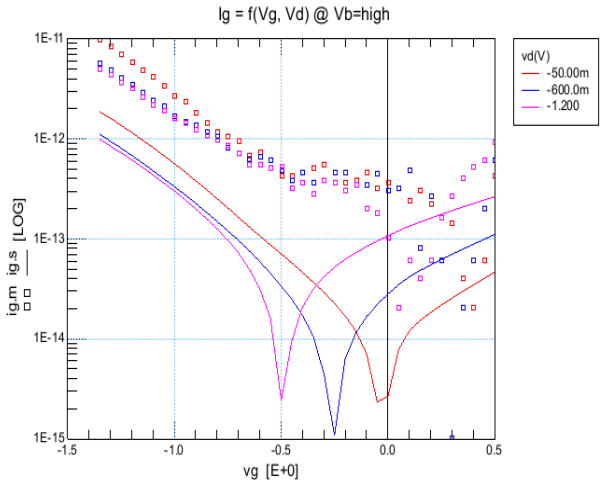
idvg, Ib,Ig, Vb = 0V, T = 27°C



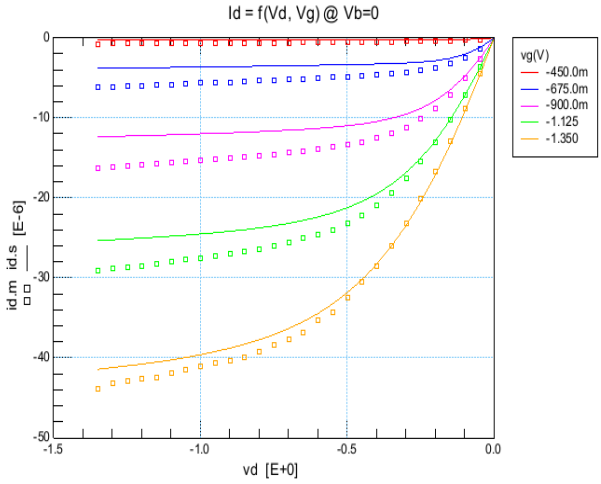


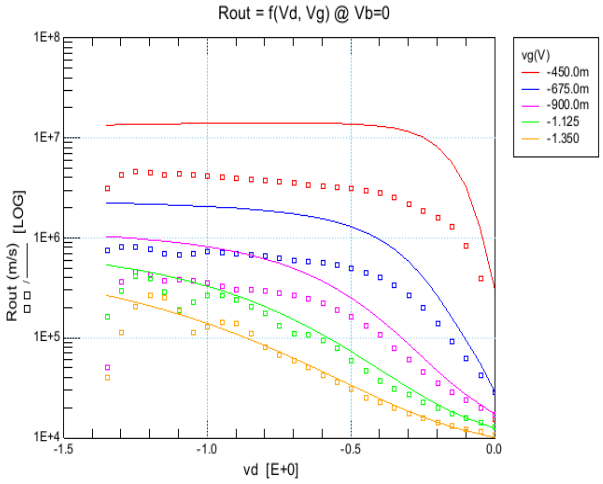
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

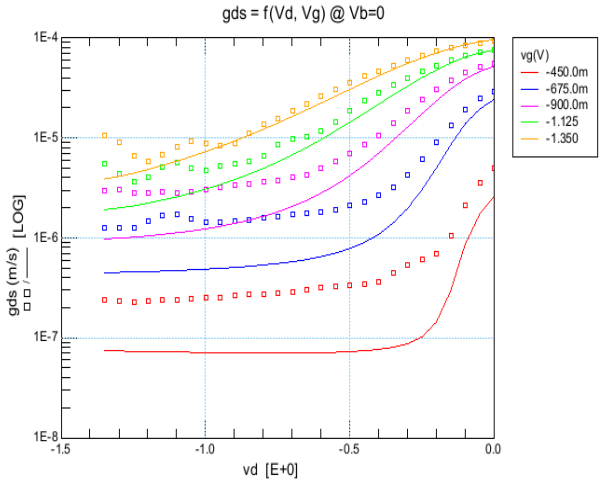




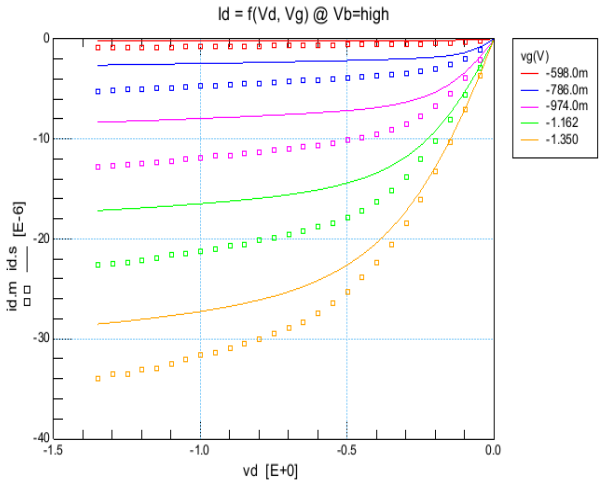
idvd, Vb = 0V, T = 27°C

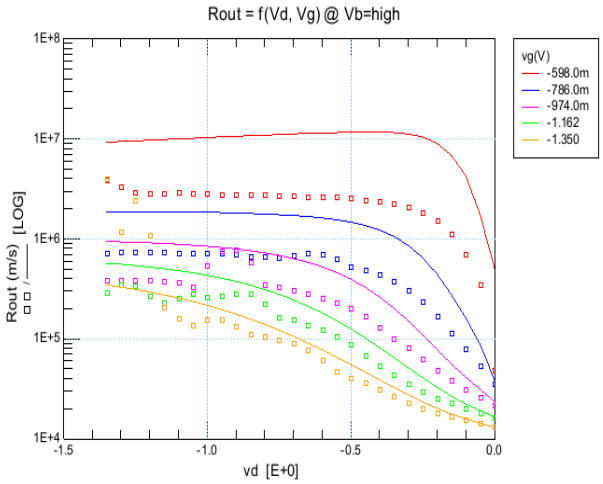


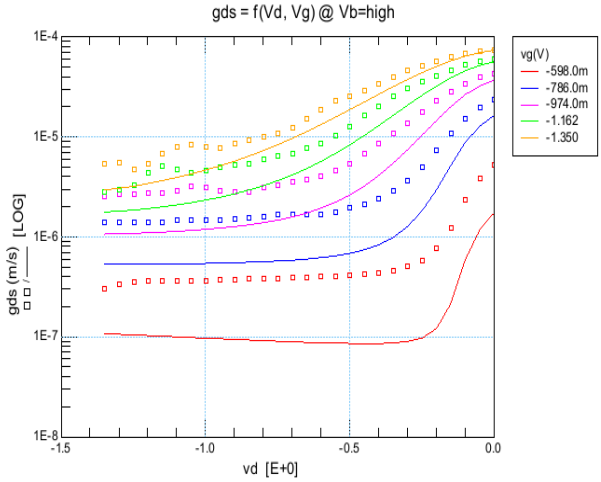




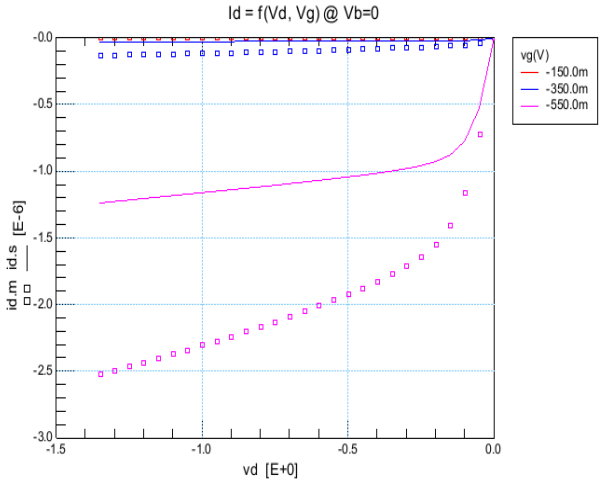
idvd_vbmin, Vb = 1.2V, T = 27°C

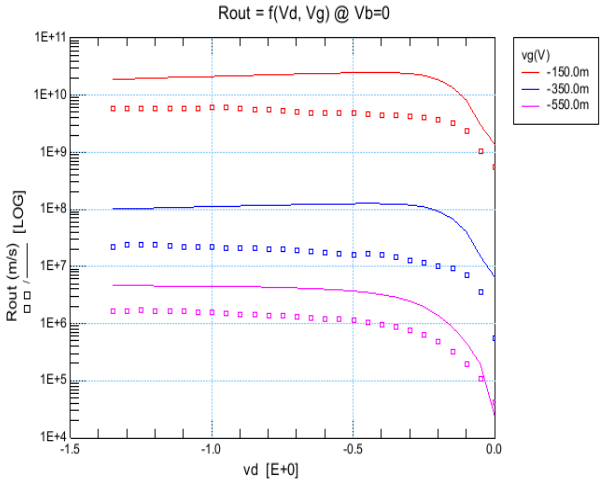


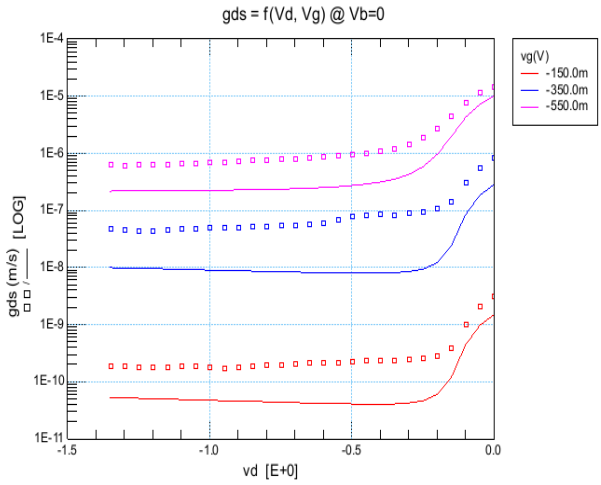




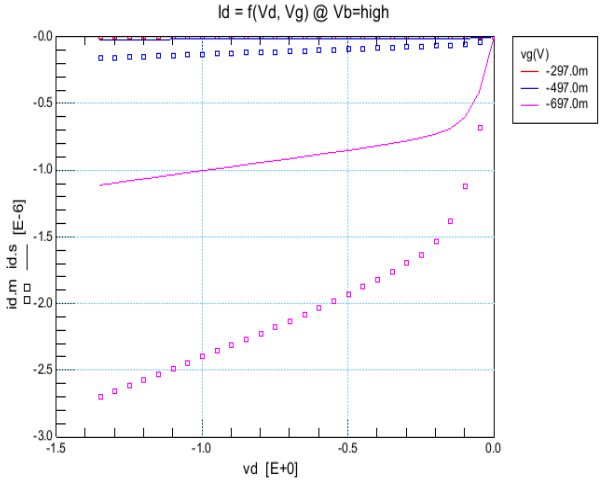
idvd_vth, Vb = 0V, T = 27°C

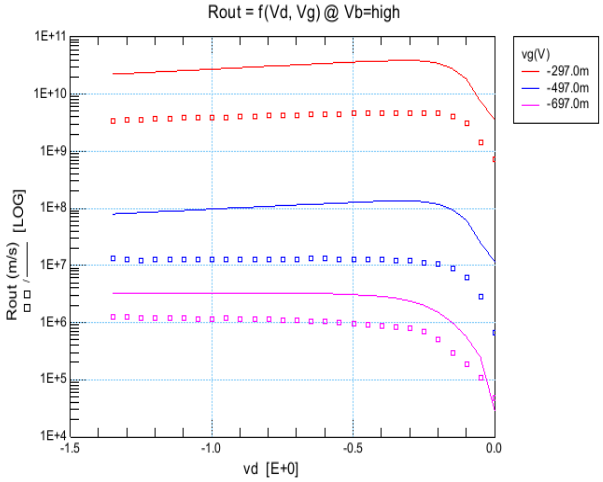


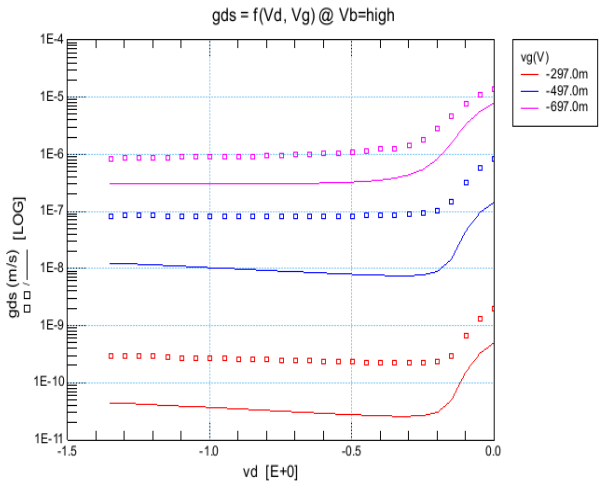




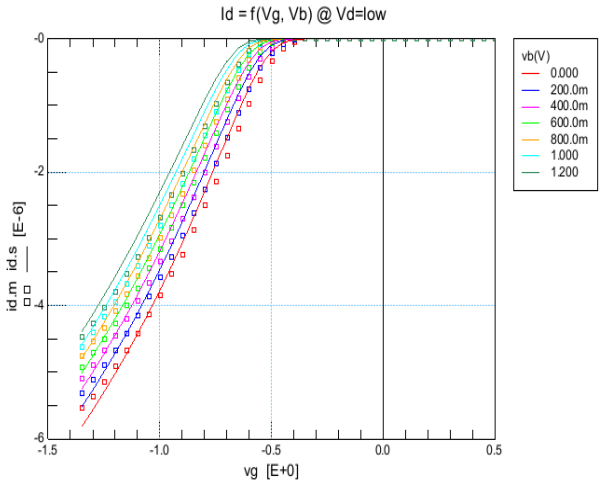
idvd_vth_vbmin, Vb = 1.2V, T = 27°C

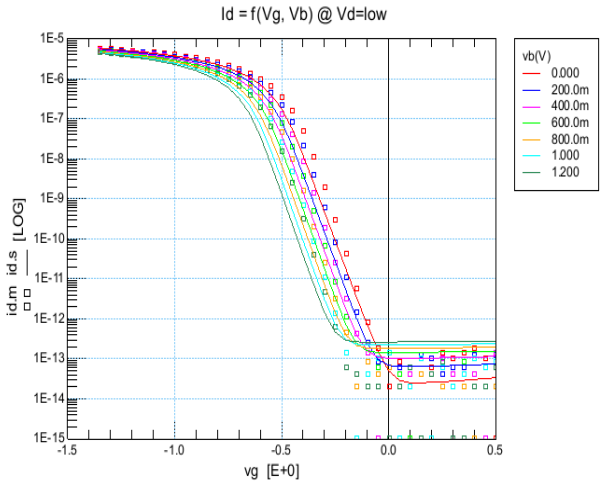


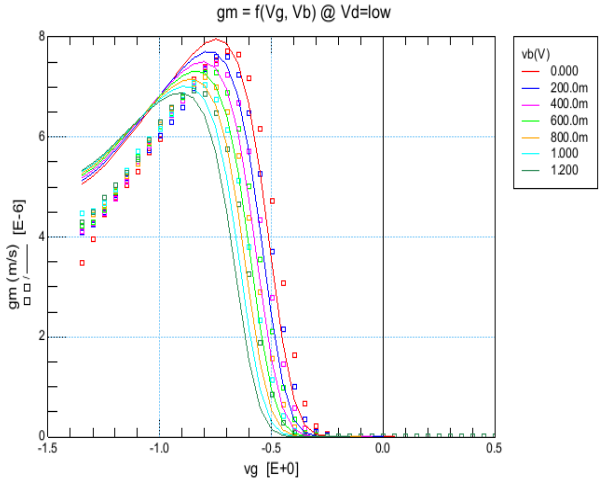


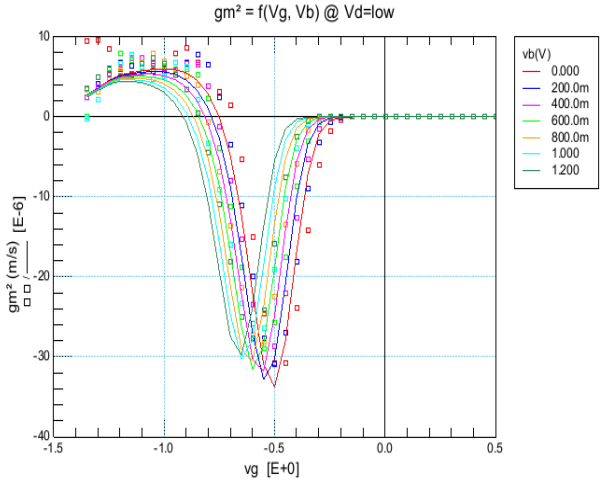


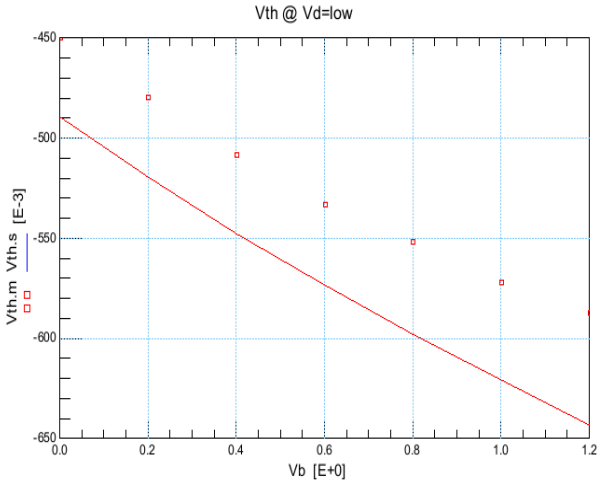
idvg, Vd = -0.05V, T = -40°C

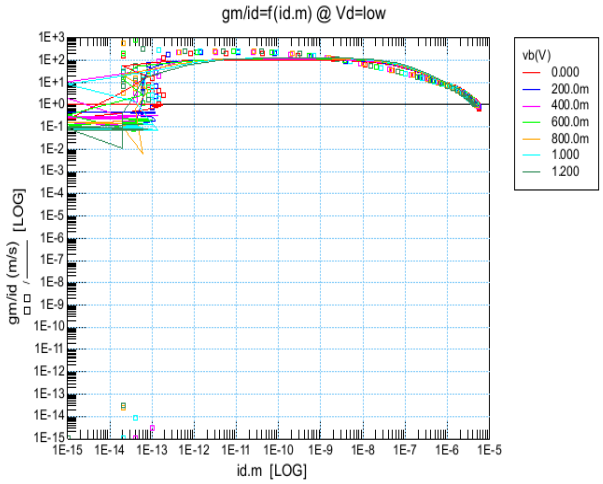


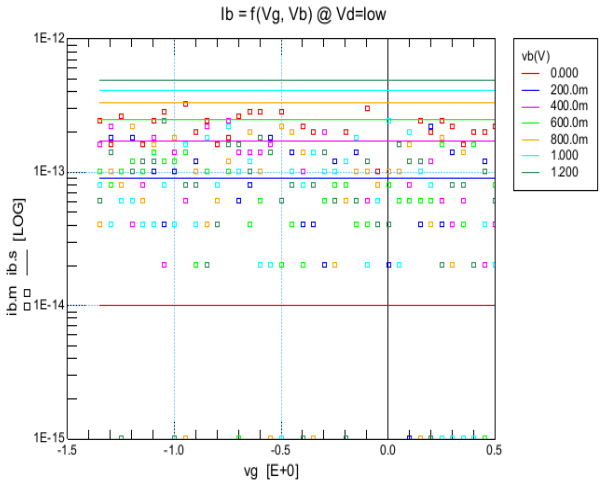


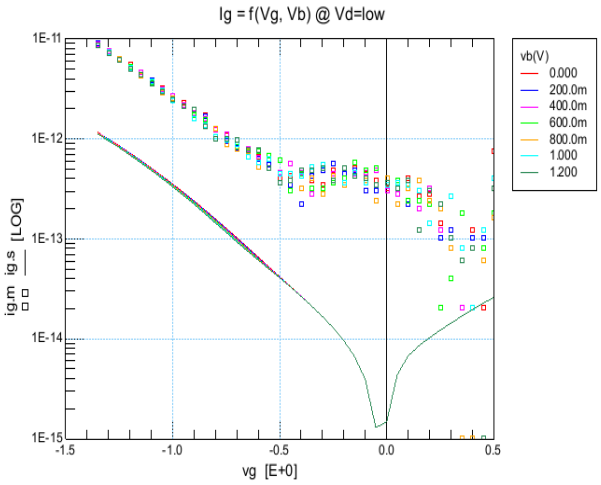




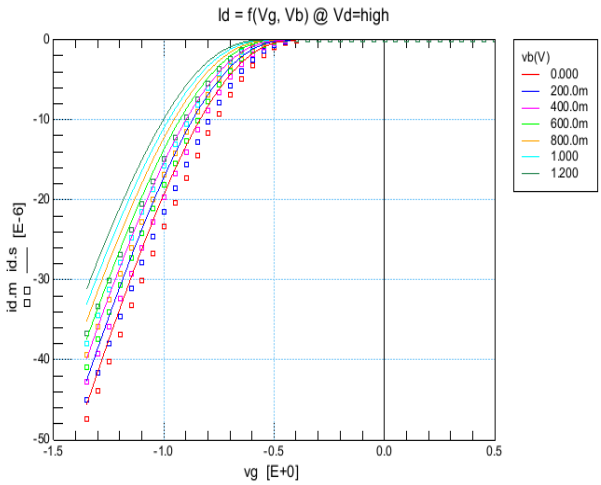


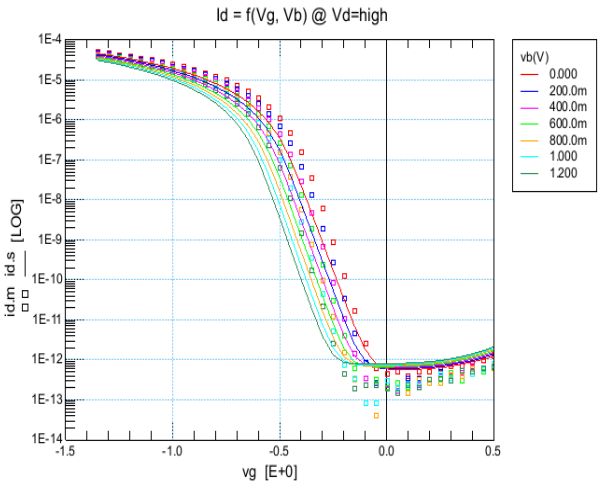


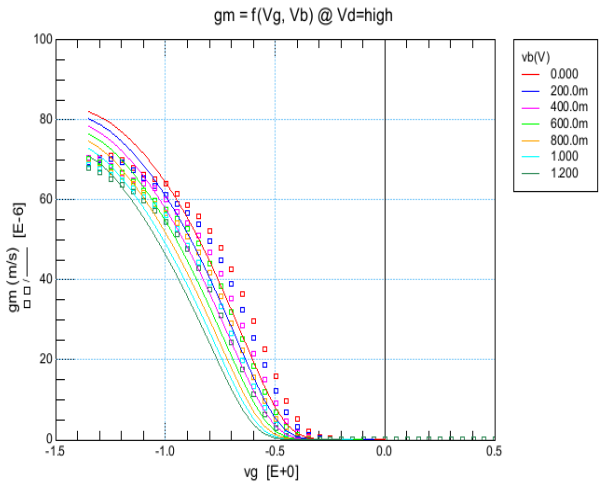


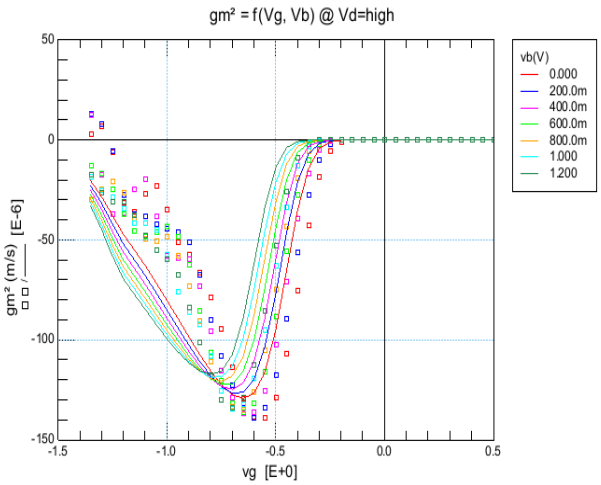


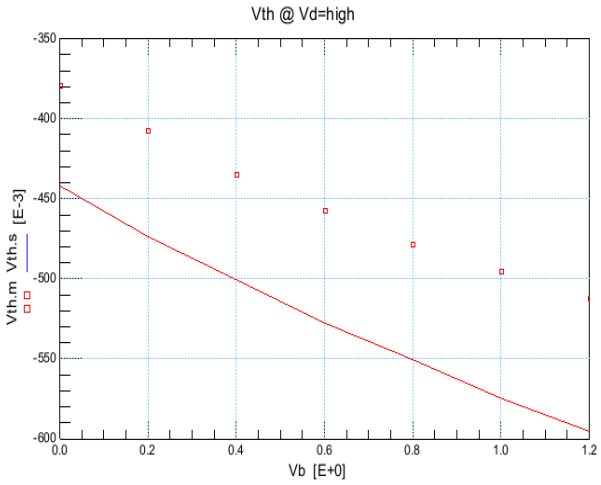
idvg, Vd = -1.2V, T = -40°C

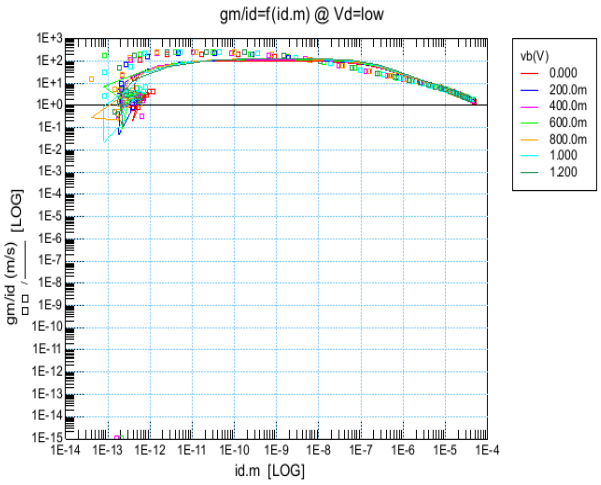


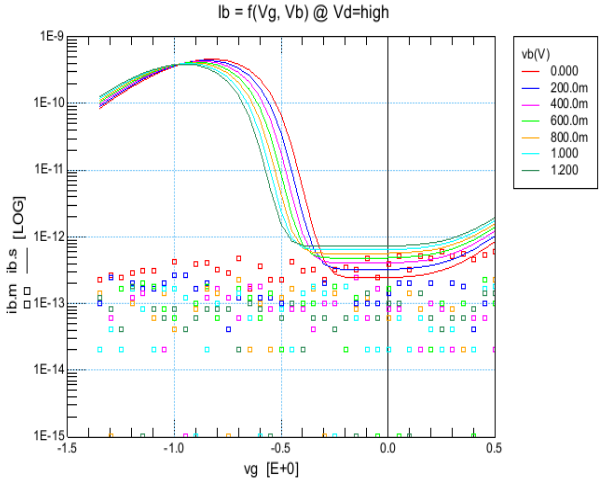


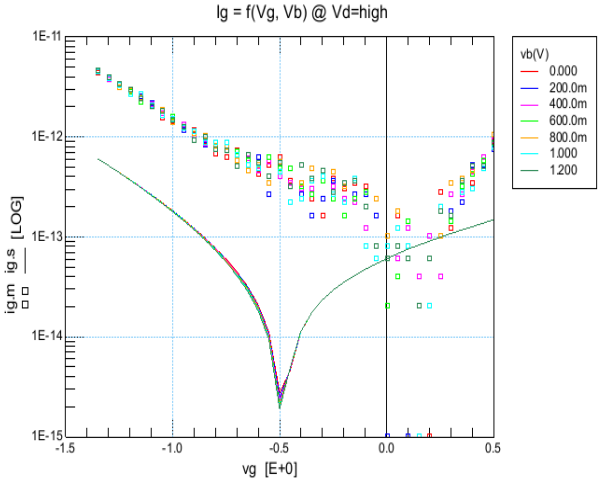




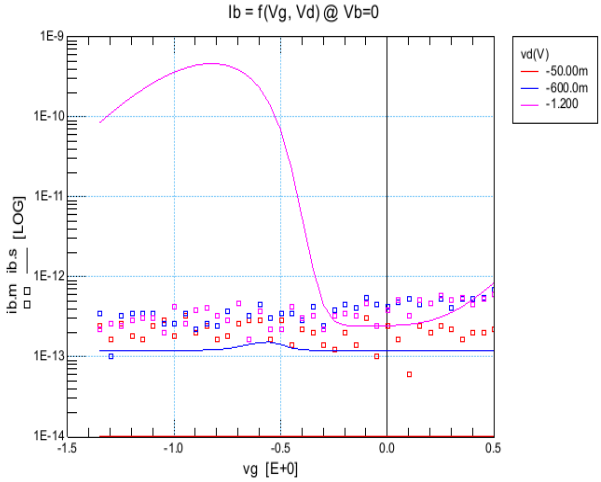


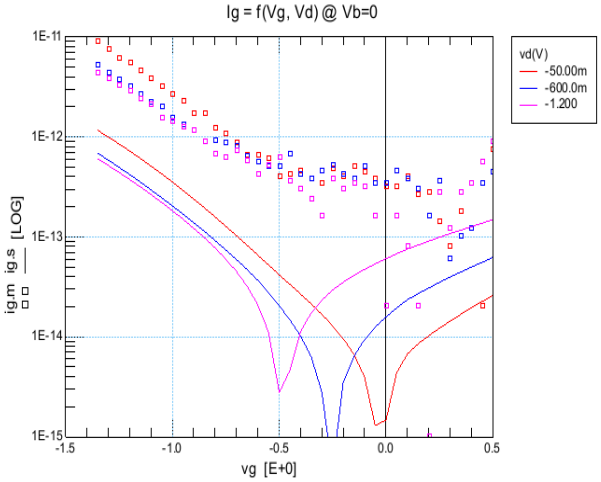




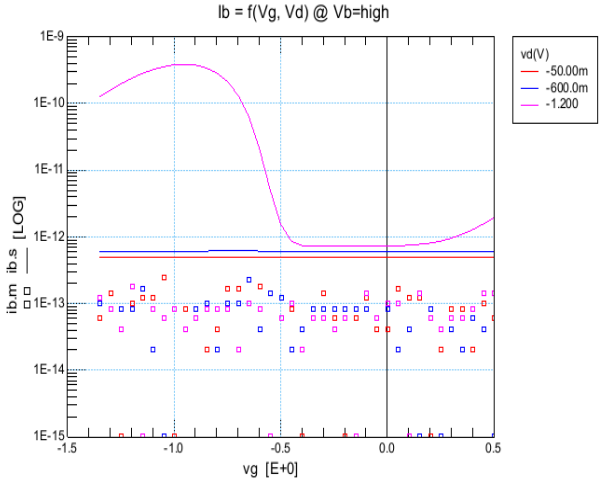


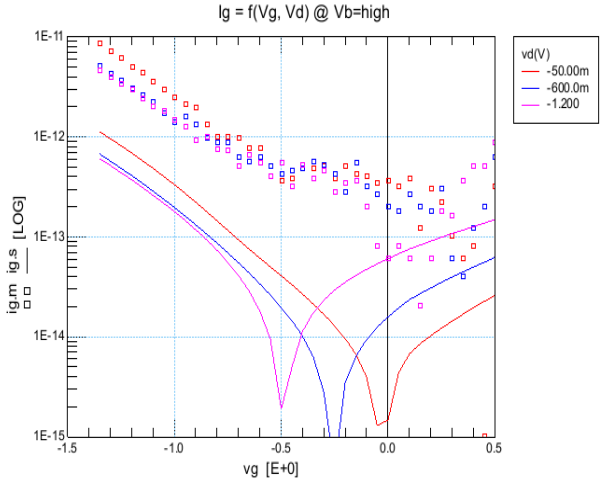
idvg, Ib,Ig, Vb = 0V, T = -40°C



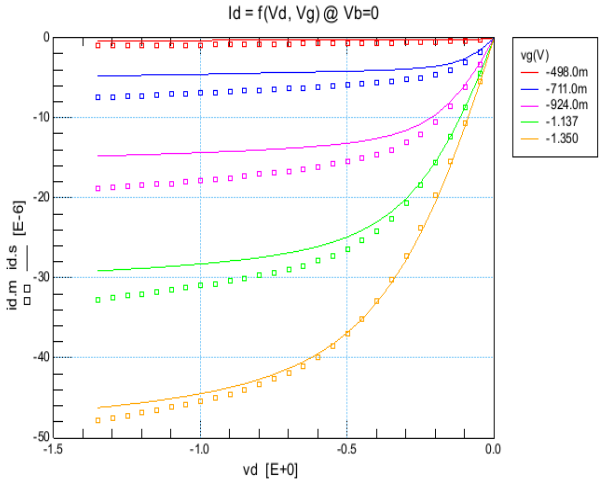


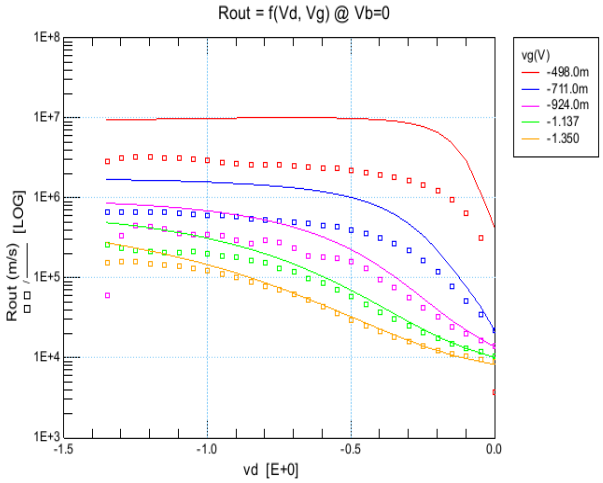
idvg, Ib,Ig, Vb = 1.2V, T = -40°C

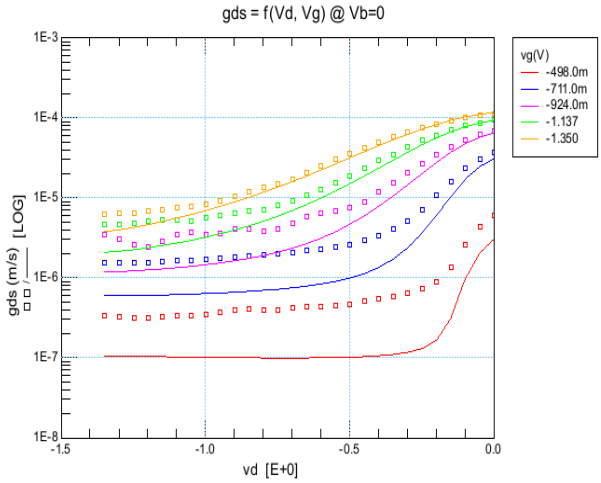




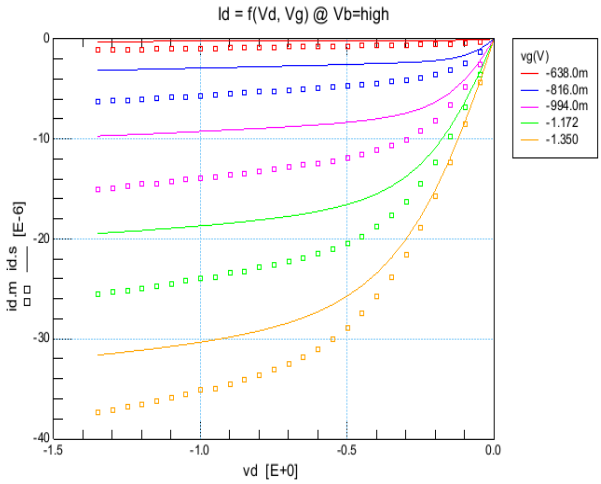
idvd, Vb = 0V, T = -40°C

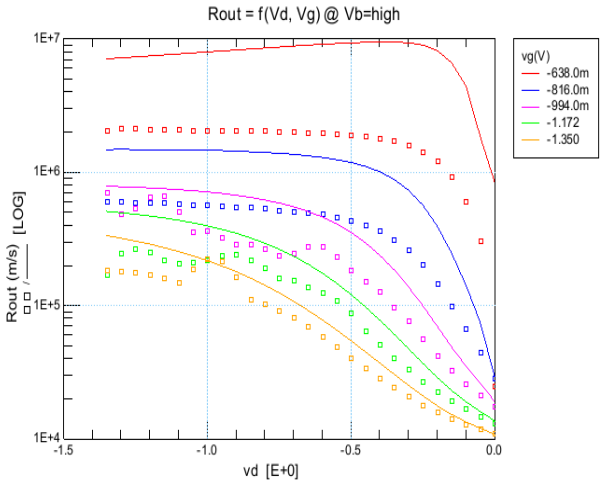


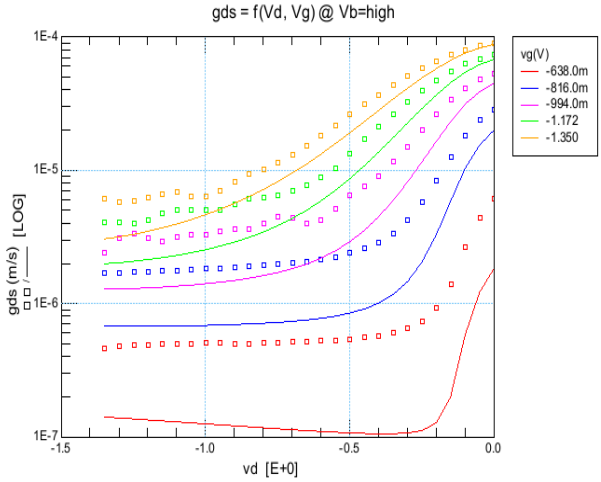




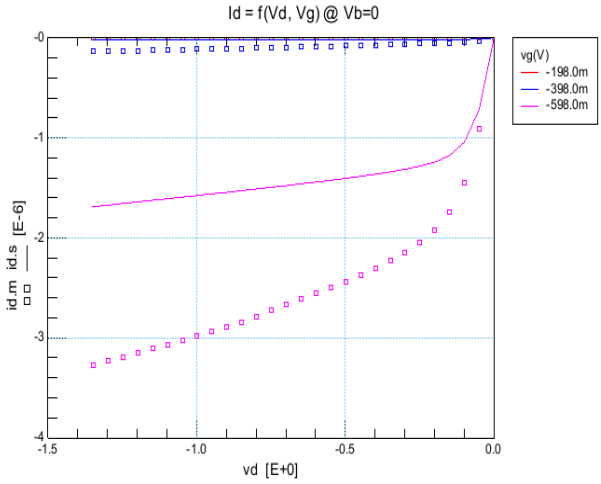
idvd_vbmin, Vb = 1.2V, T = -40°C

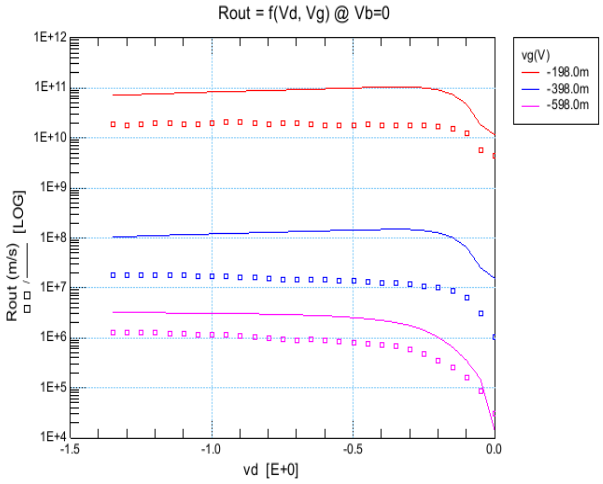


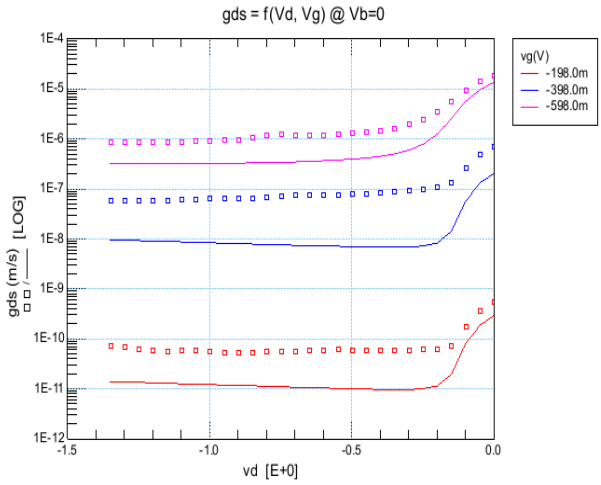




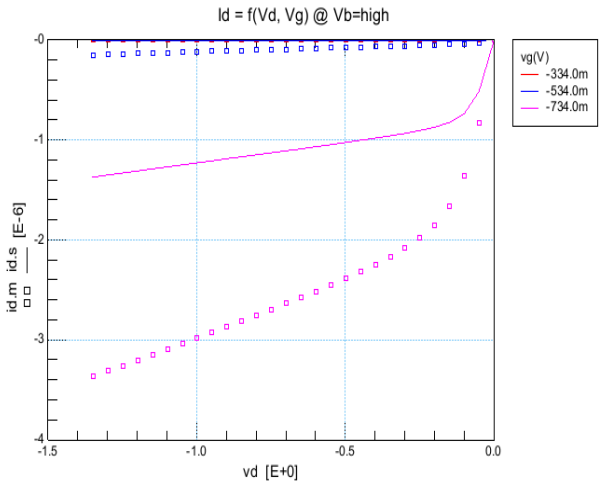
idvd_vth, Vb = 0V, T = -40°C

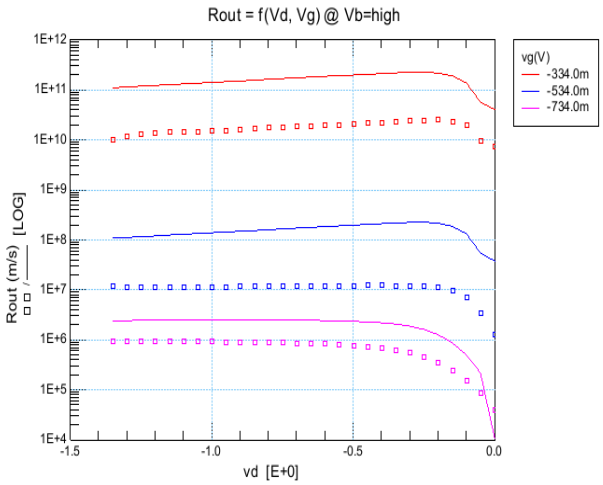


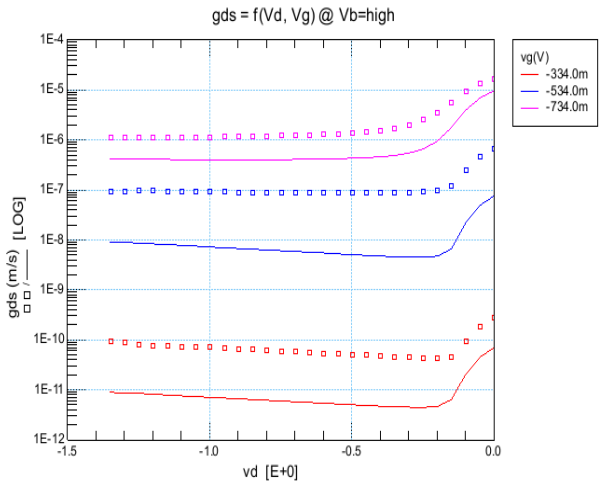




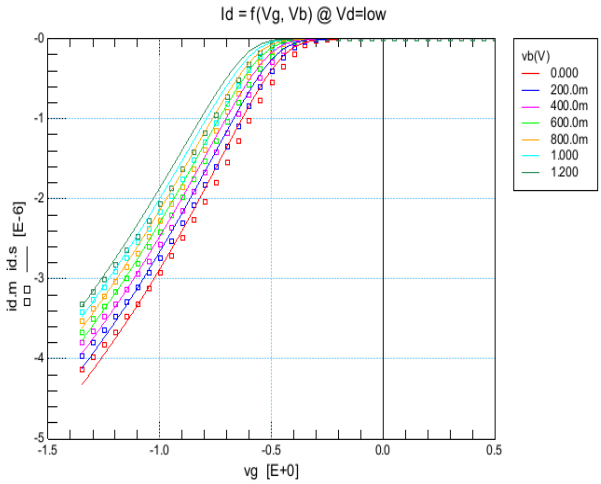
idvd_vth_vbmin, Vb = 1.2V, T = -40°C

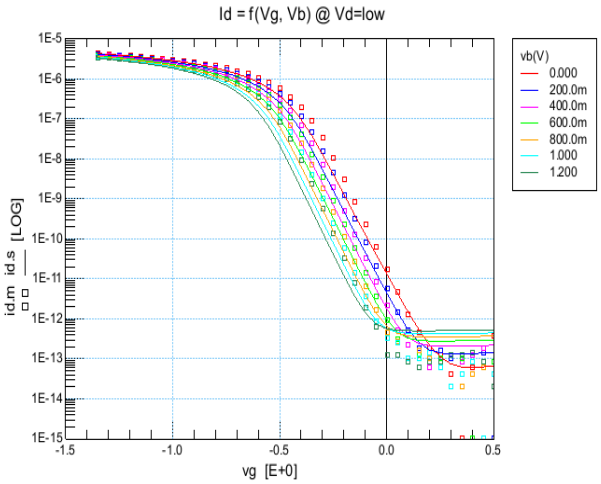


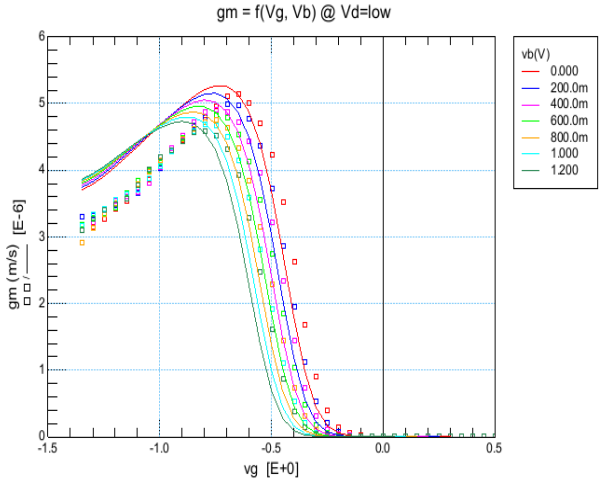


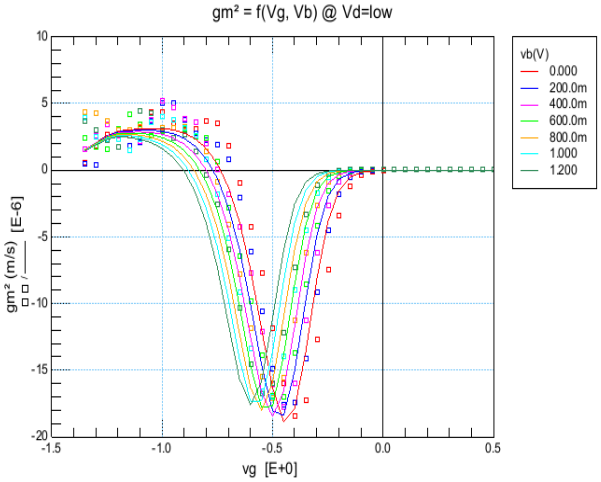


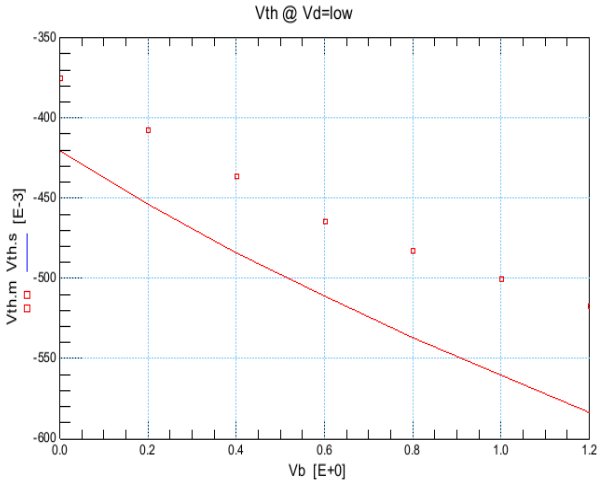
idvg, Vd = -0.05V, T = 70°C

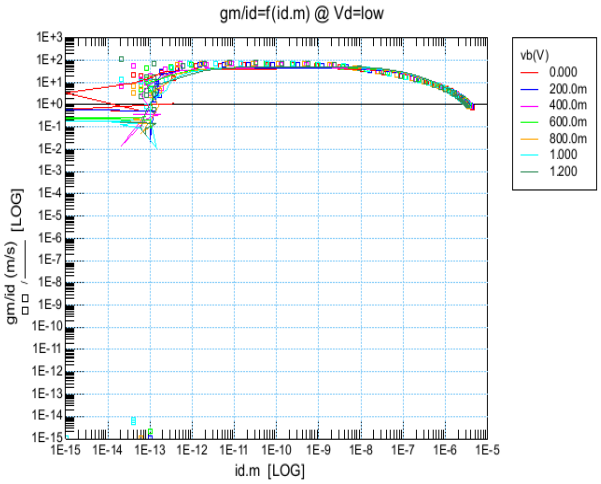


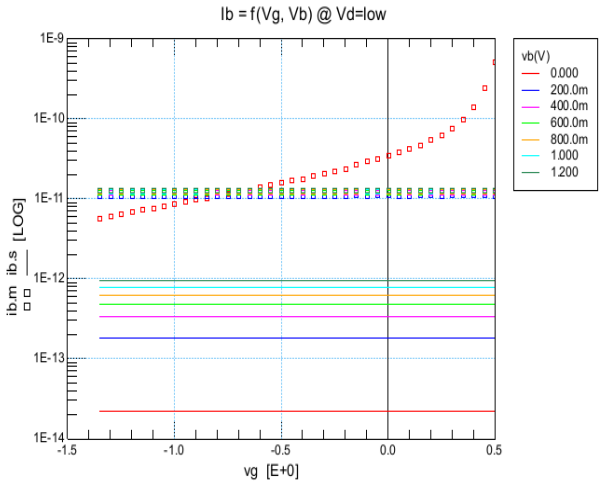


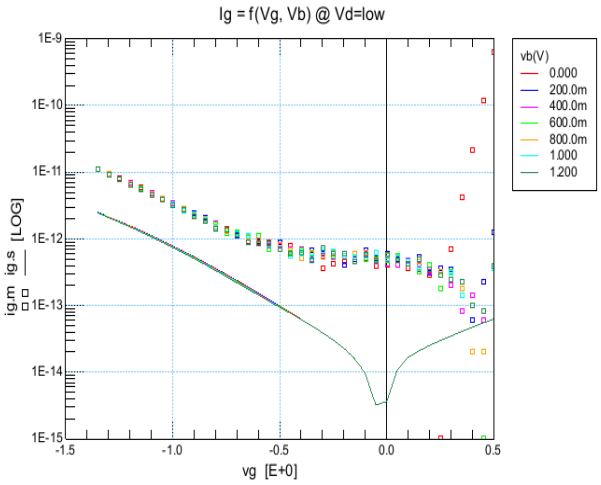




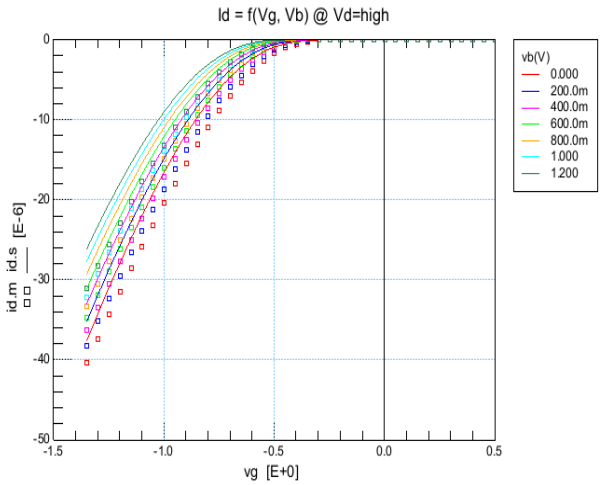


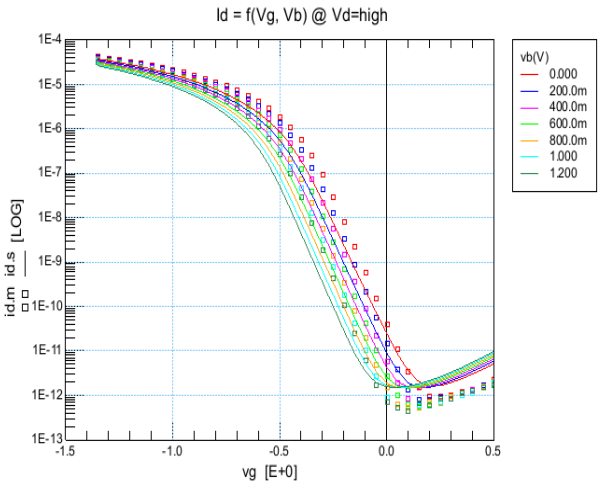


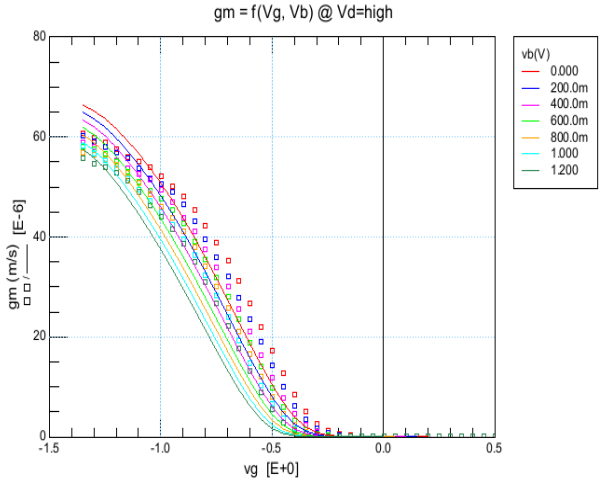


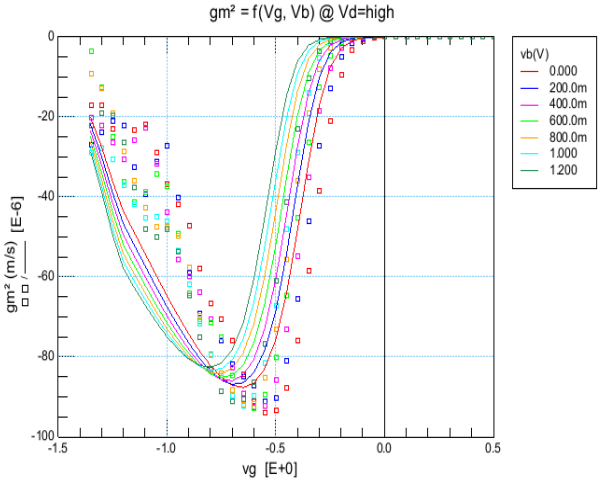


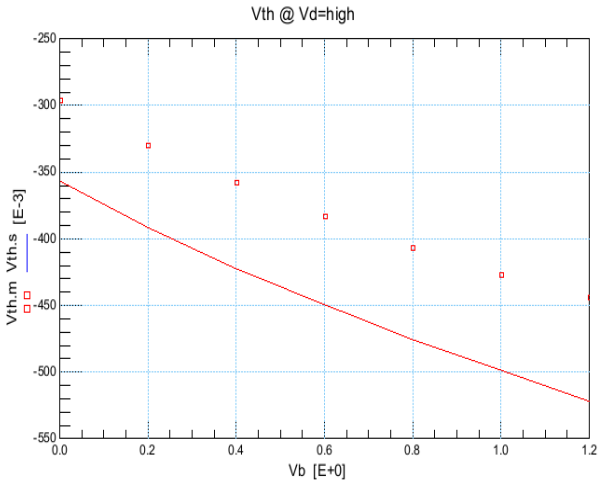
idvg, Vd = -1.2V, T = 70°C

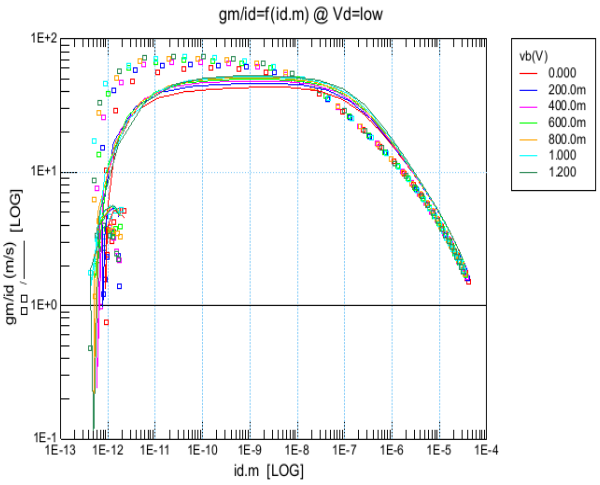


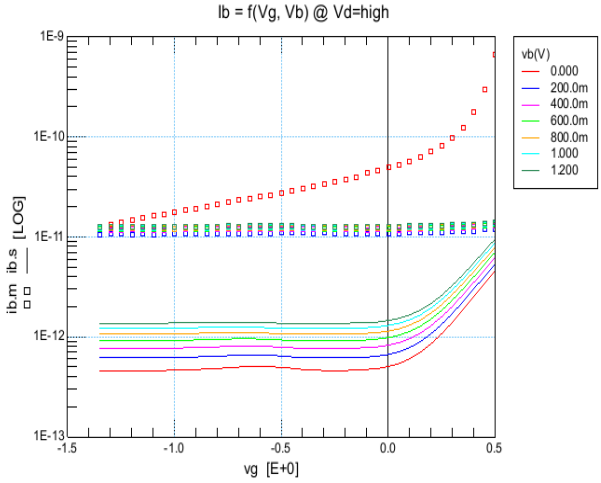


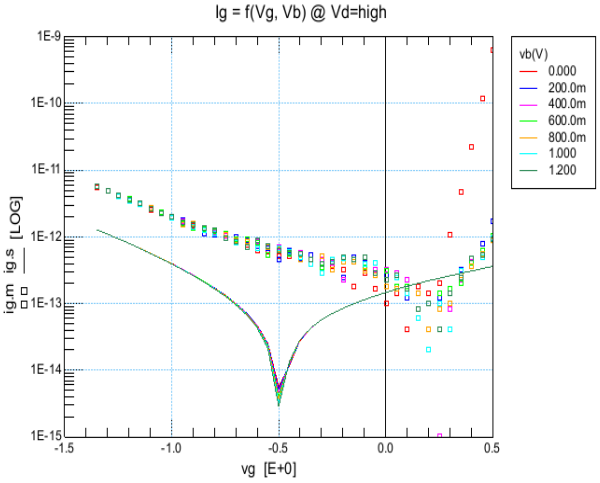




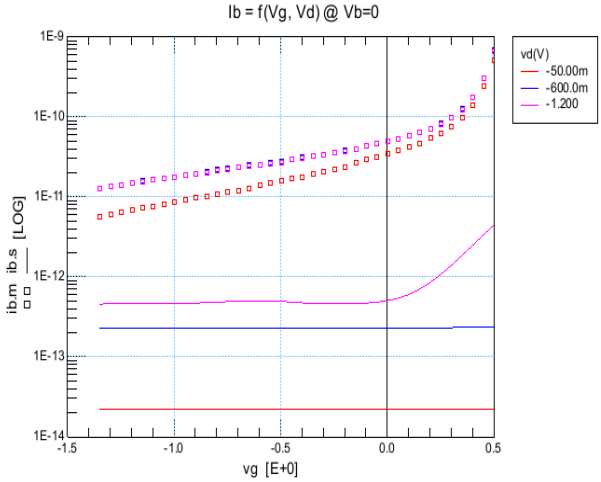


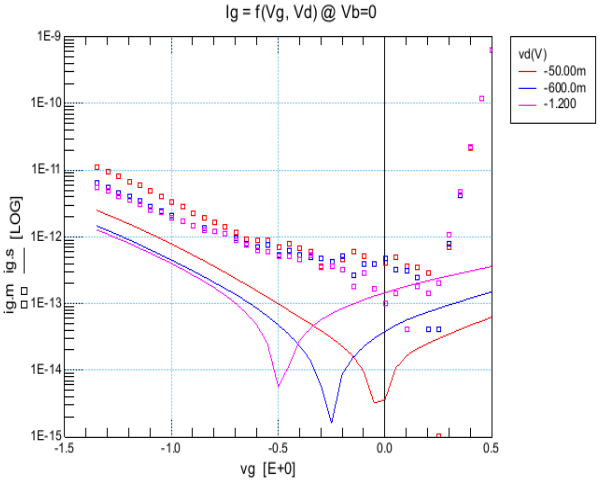




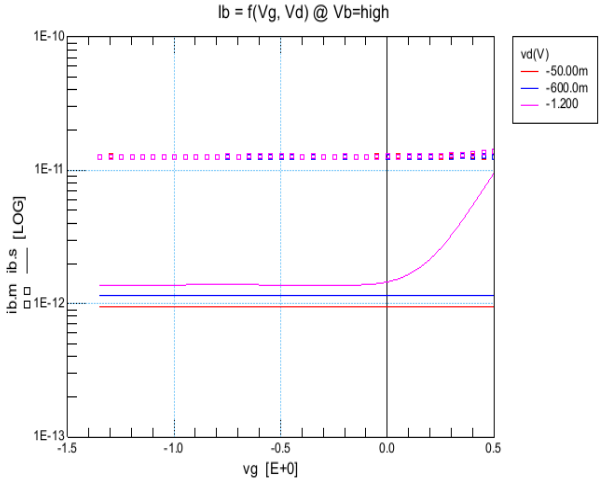


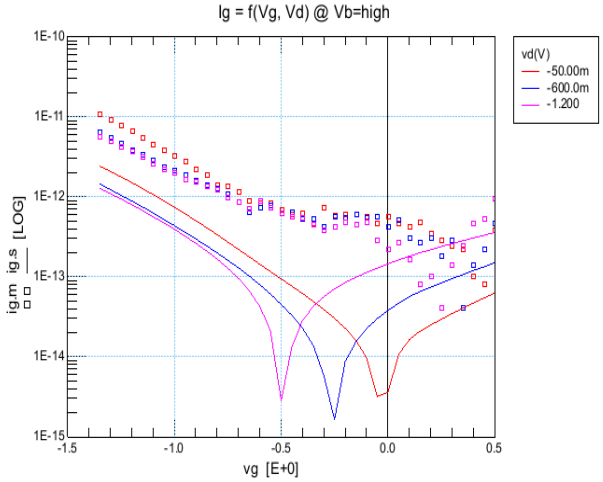
idvg, Ib,Ig, Vb = 0V, T = 70°C



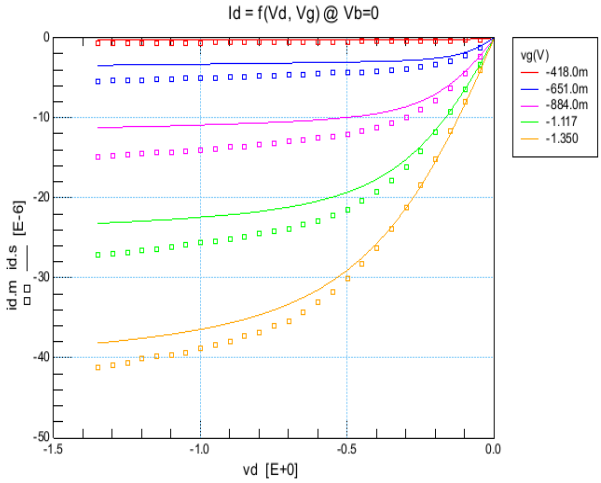


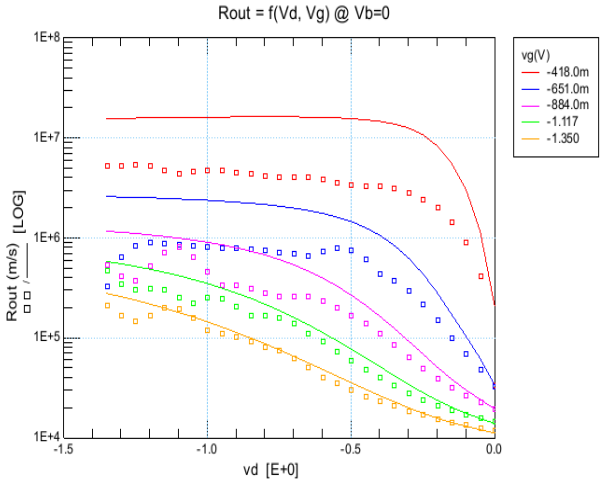
idvg, Ib,Ig, Vb = 1.2V, T = 70°C

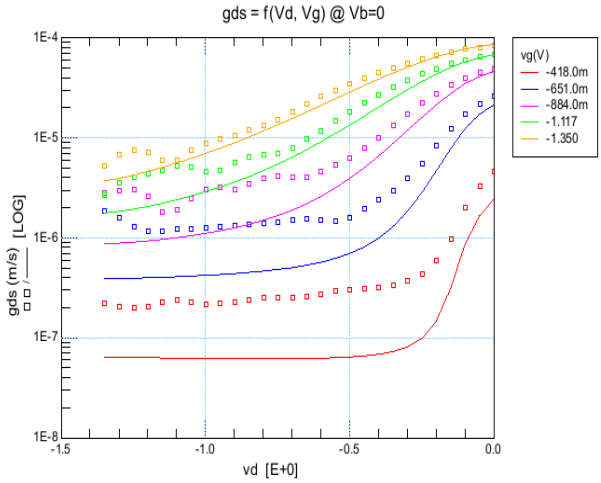




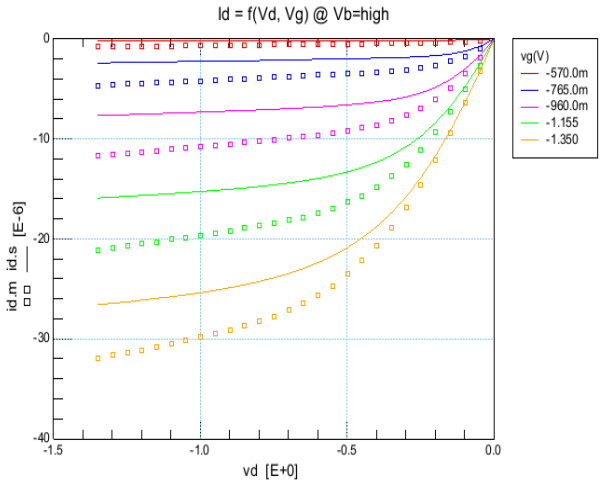
idvd, Vb = 0V, T = 70°C

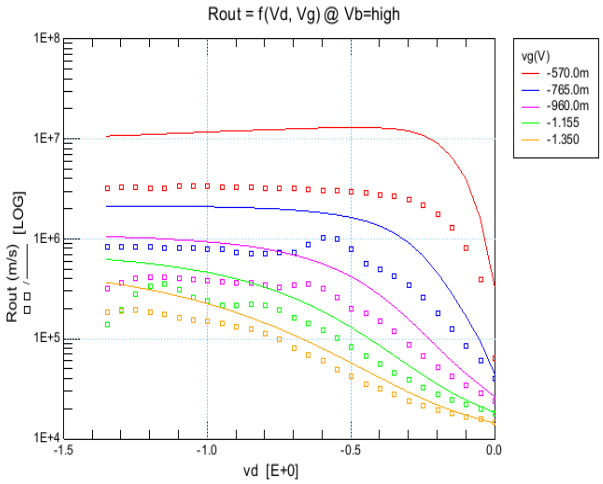


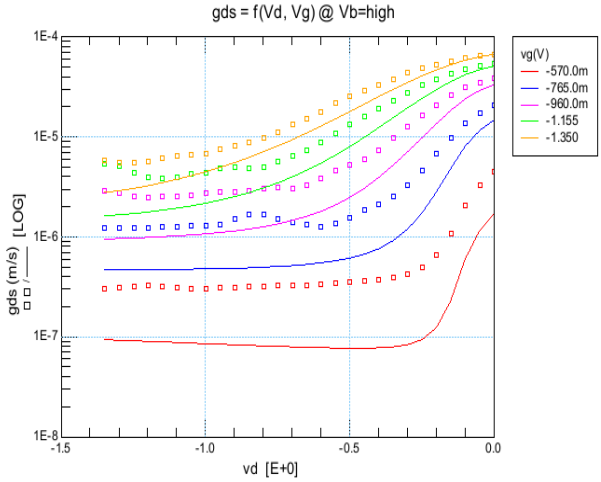




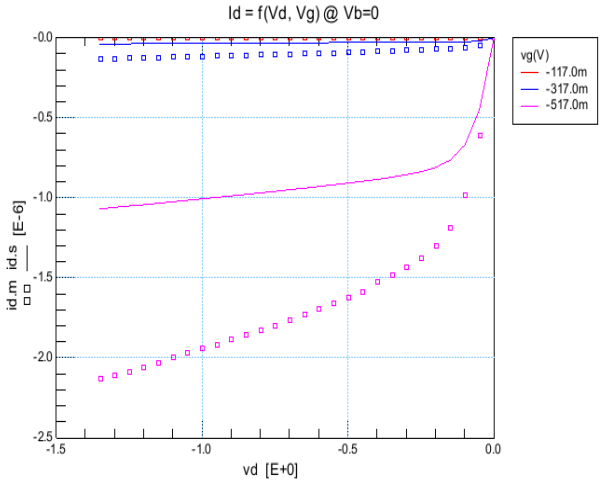
idvd_vbmin, Vb = 1.2V, T = 70°C

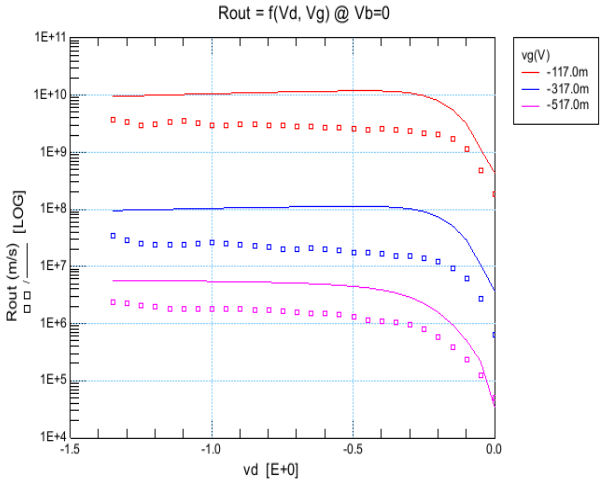


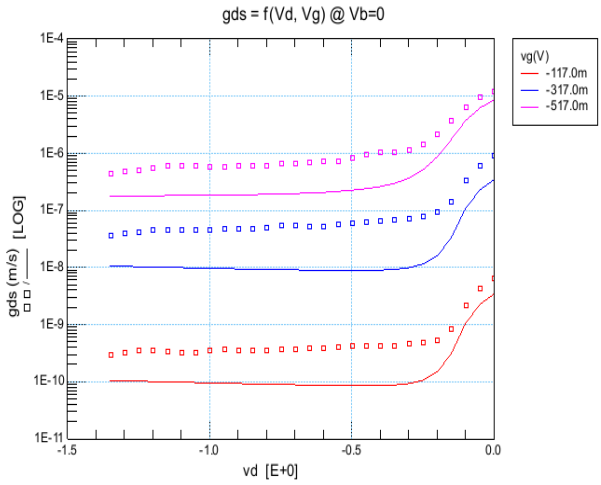




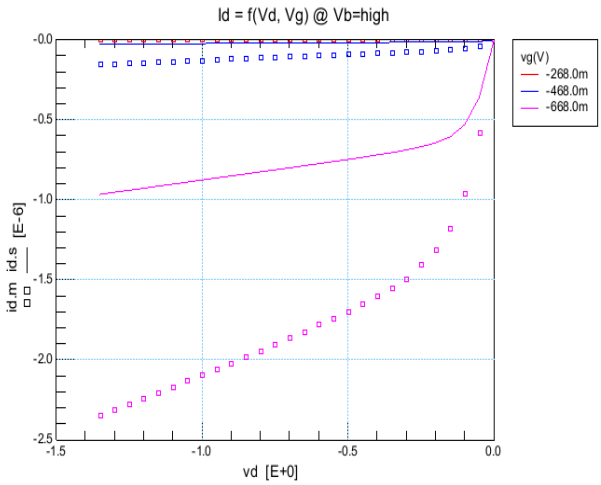
idvd_vth, Vb = 0V, T = 70°C

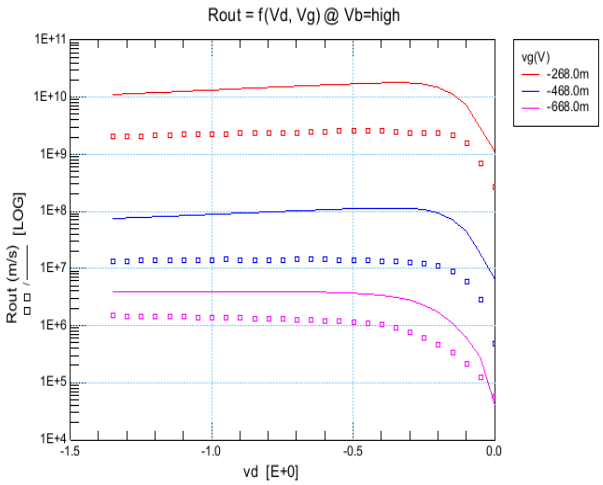


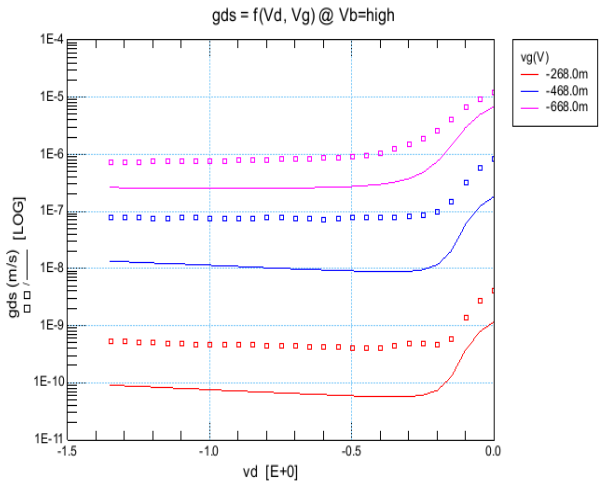




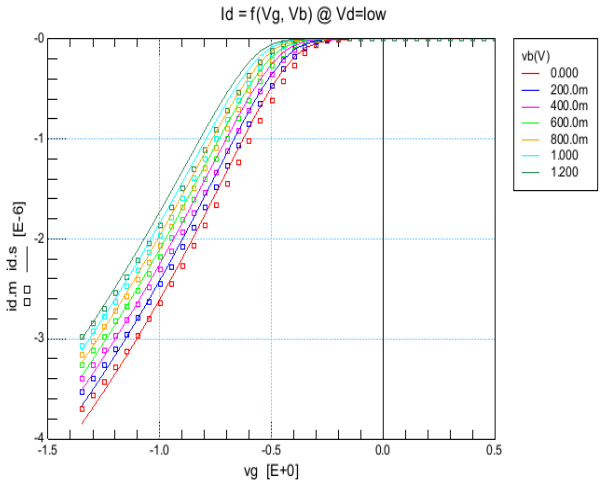
idvd_vth_vbmin, Vb = 1.2V, T = 70°C

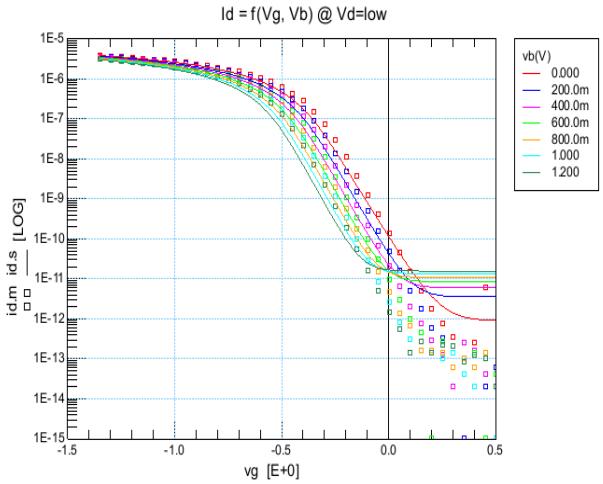


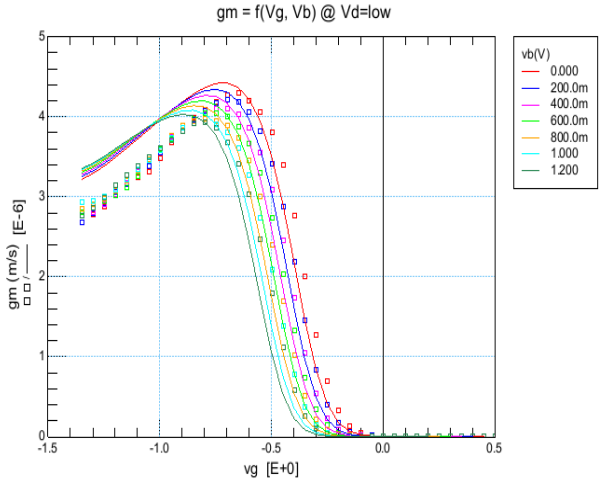


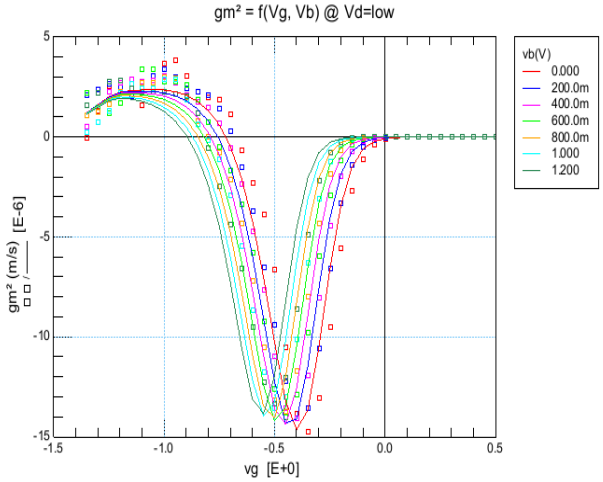


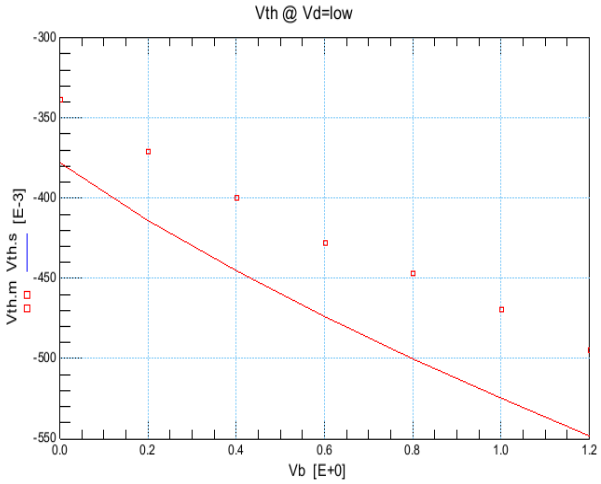
idvg, Vd = -0.05V, T = 125°C

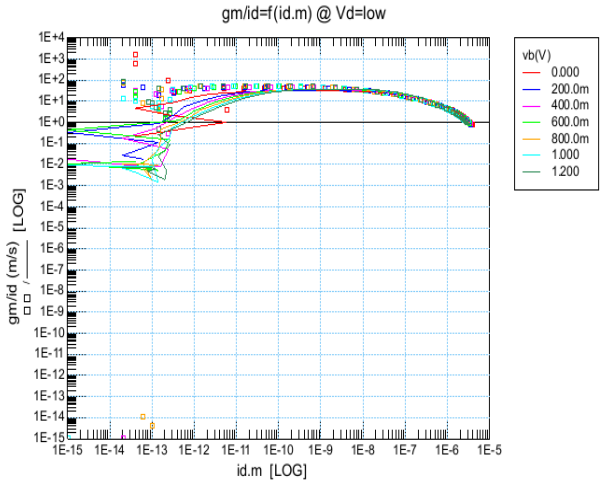


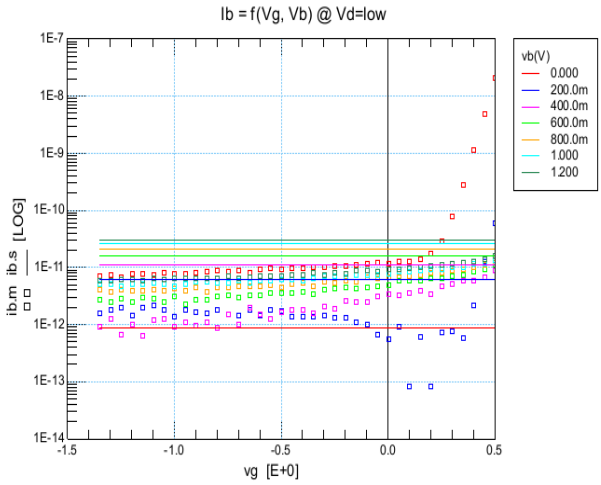


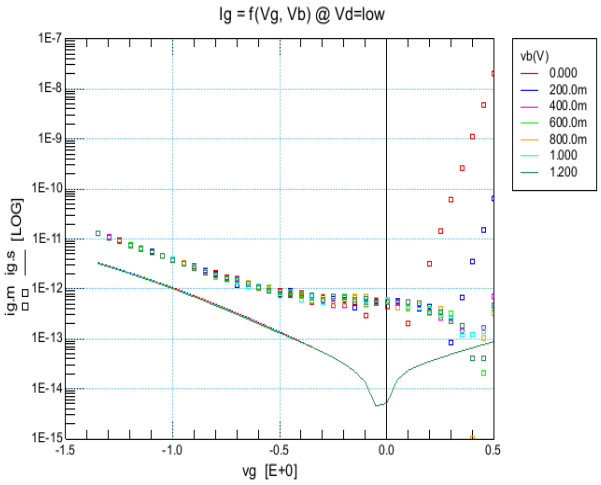




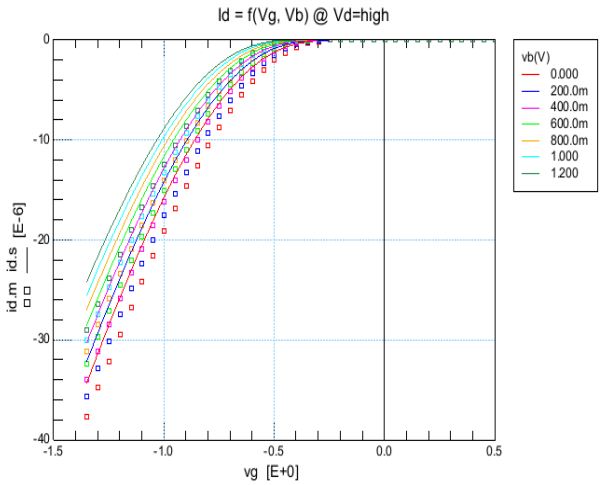


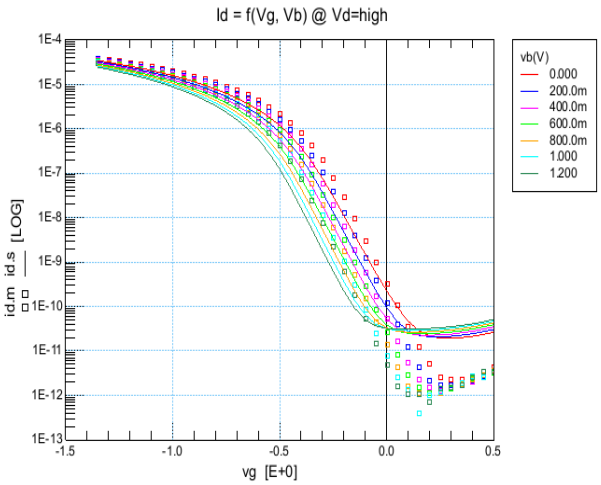


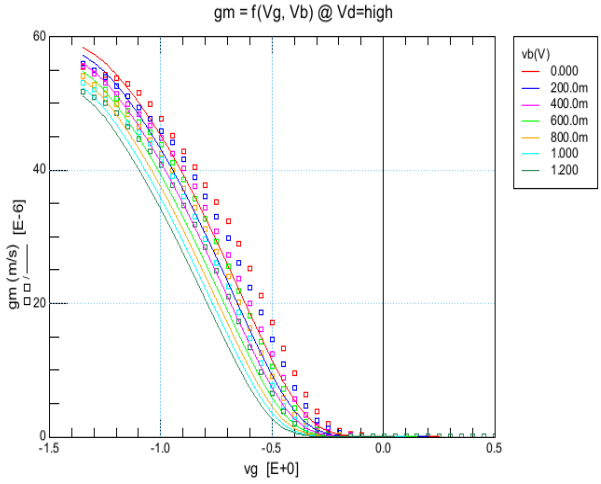


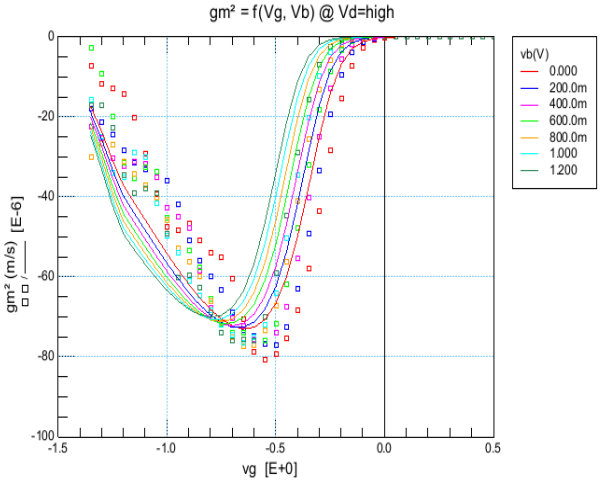


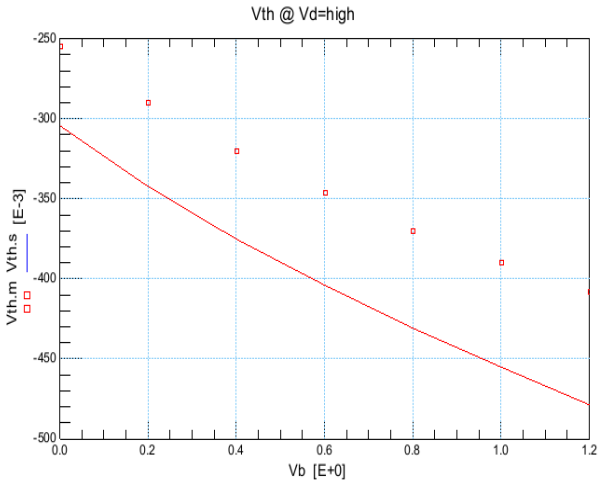
idvg, Vd = -1.2V, T = 125°C

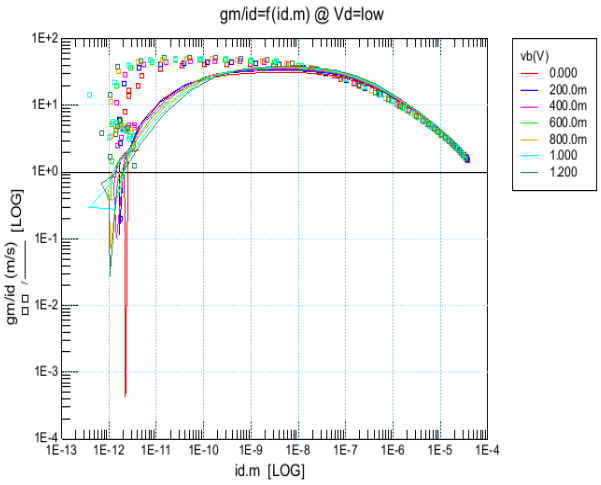


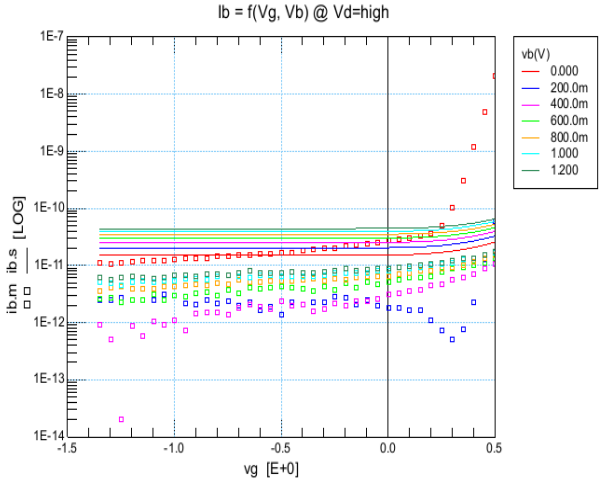


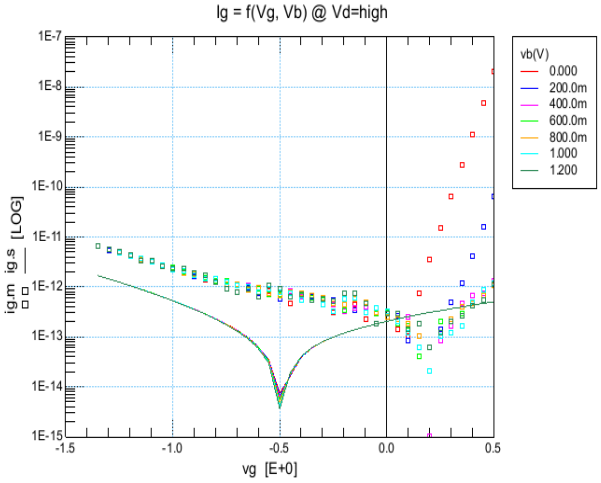




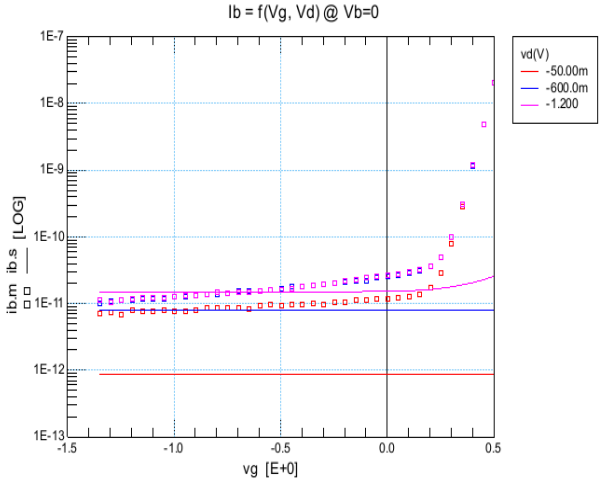


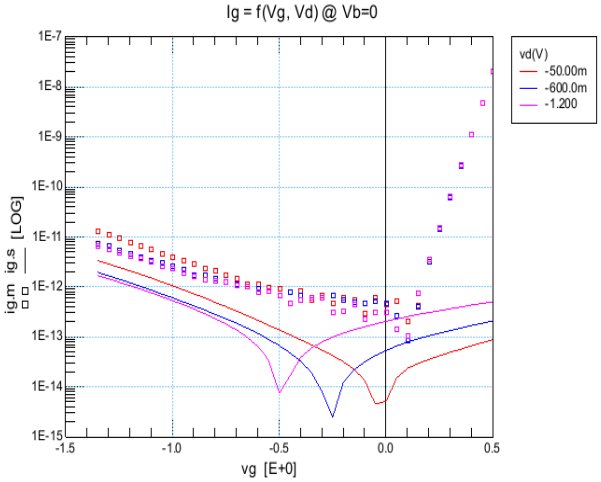




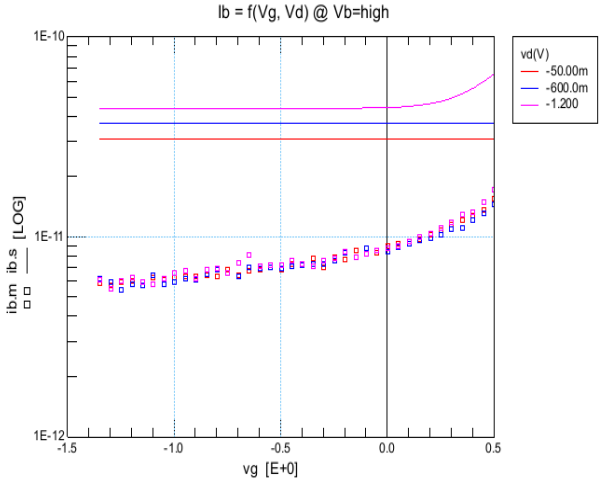


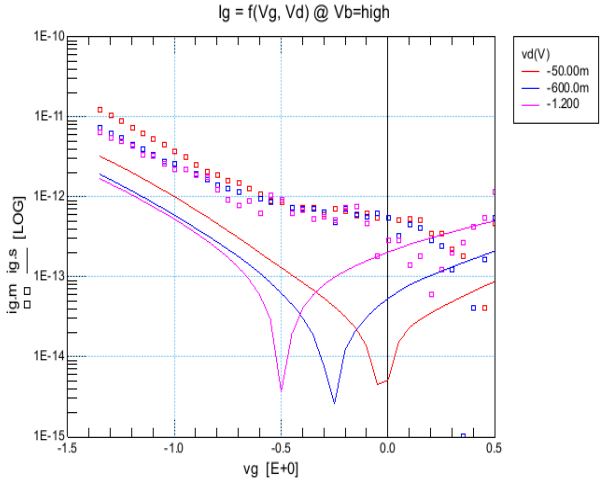
idvg, Ib,Ig, Vb = 0V, T = 125°C



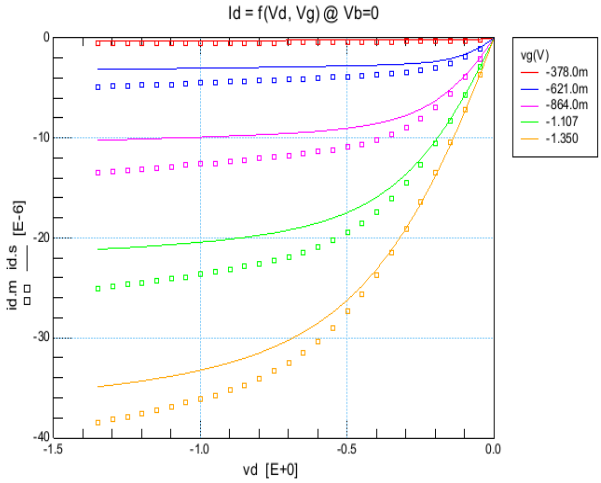


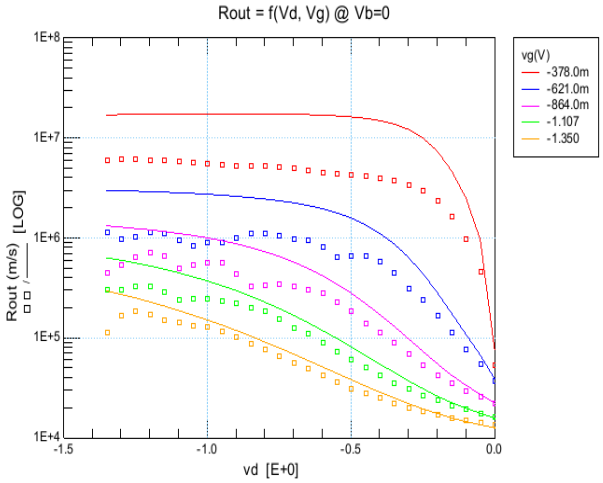
idvg, Ib,Ig, Vb = 1.2V, T = 125°C

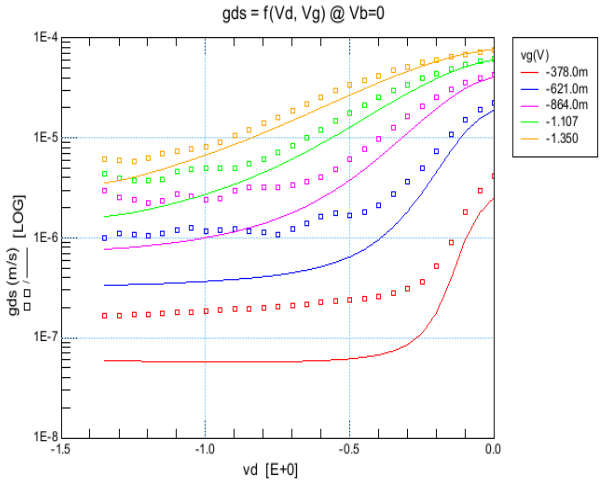




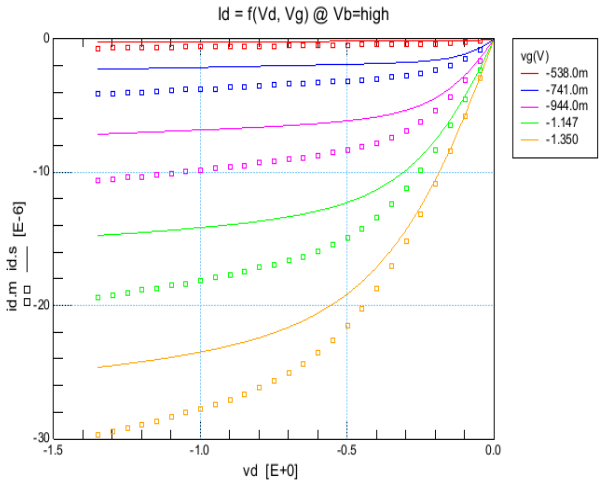
idvd, Vb = 0V, T = 125°C

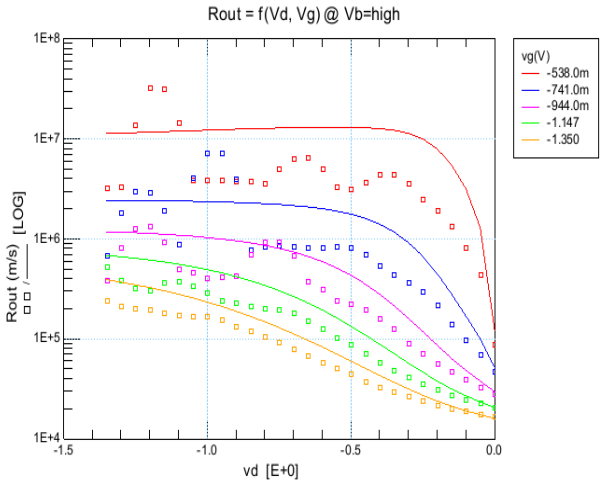


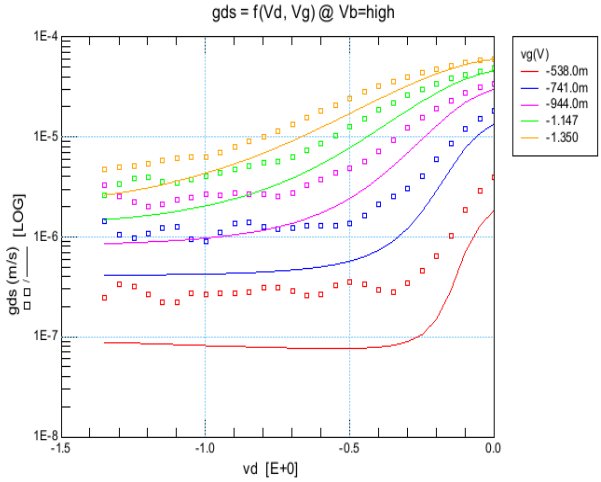




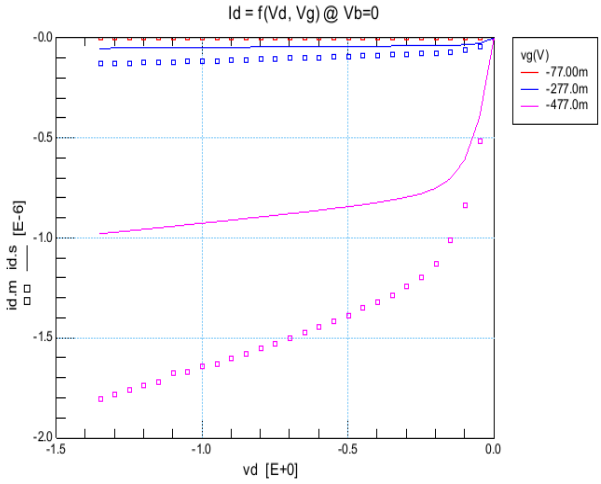
idvd_vbmin, Vb = 1.2V, T = 125°C

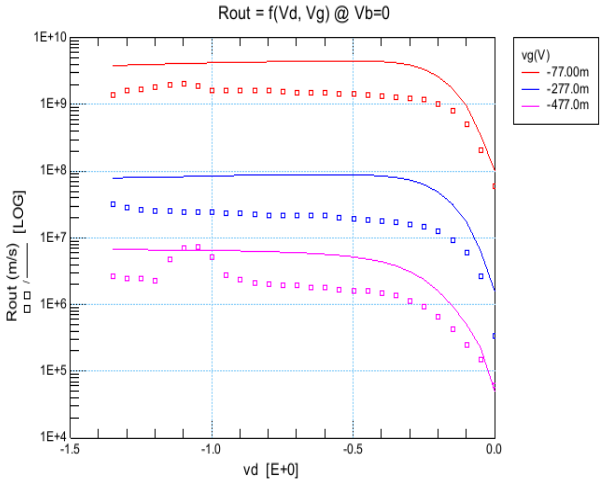


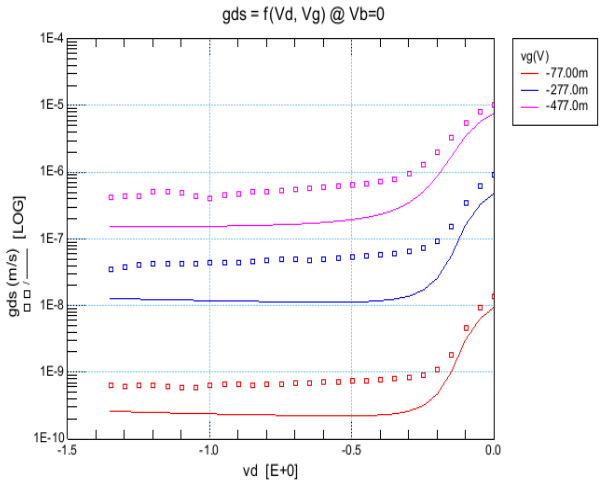




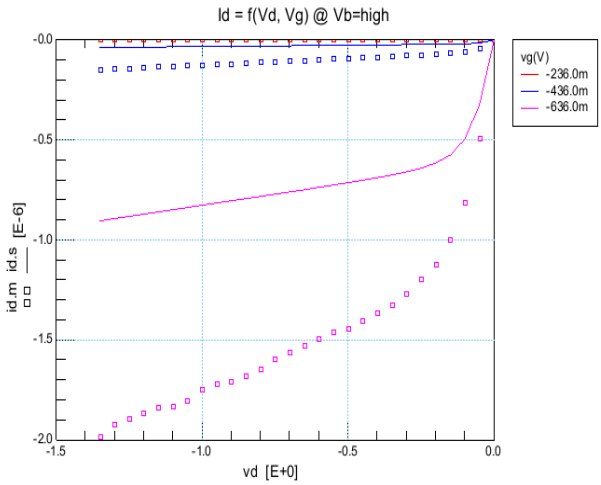
idvd_vth, Vb = 0V, T = 125°C

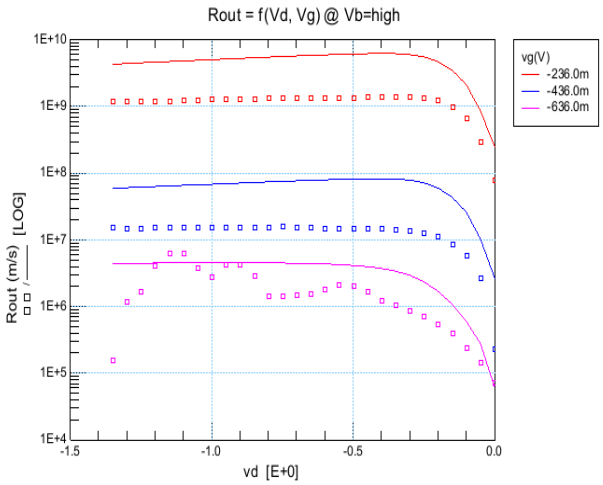


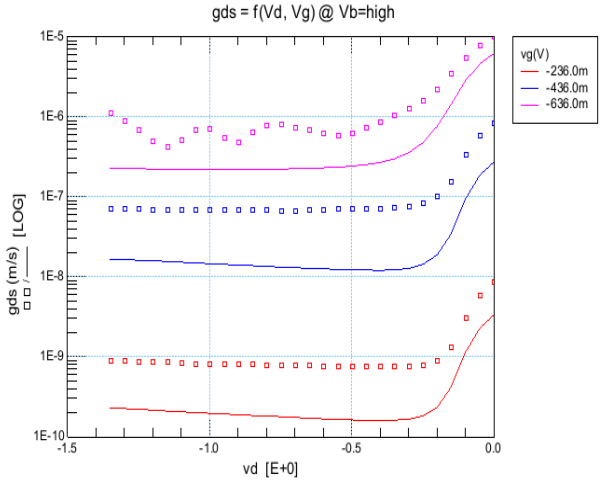




idvd_vth_vbmin, Vb = 1.2V, T = 125°C

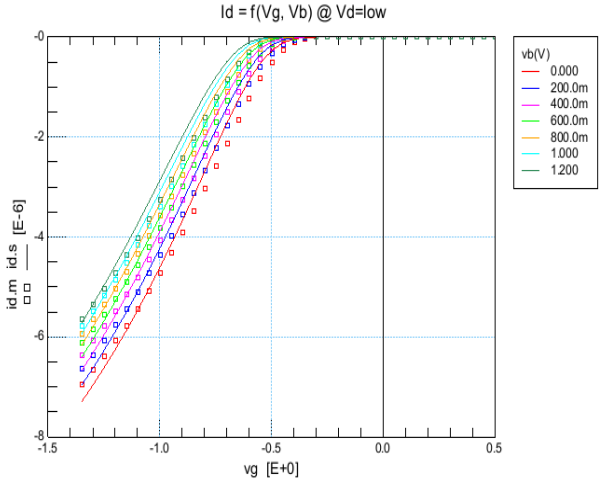


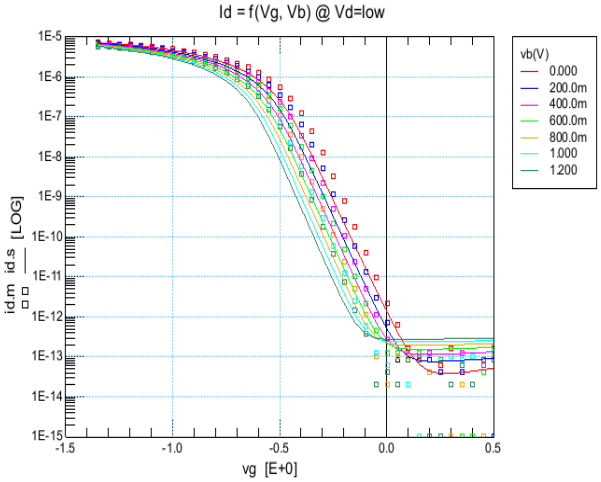


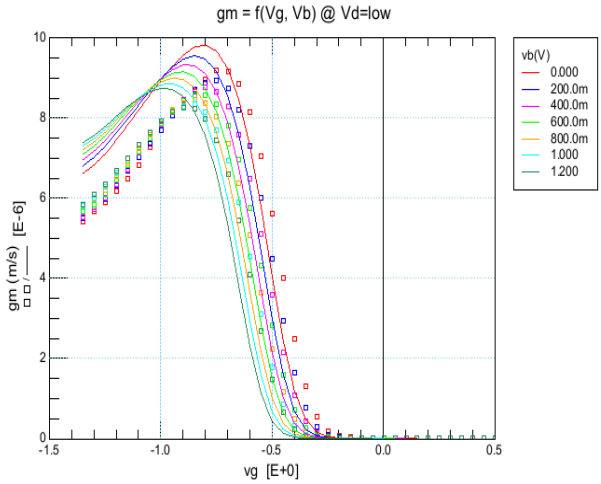


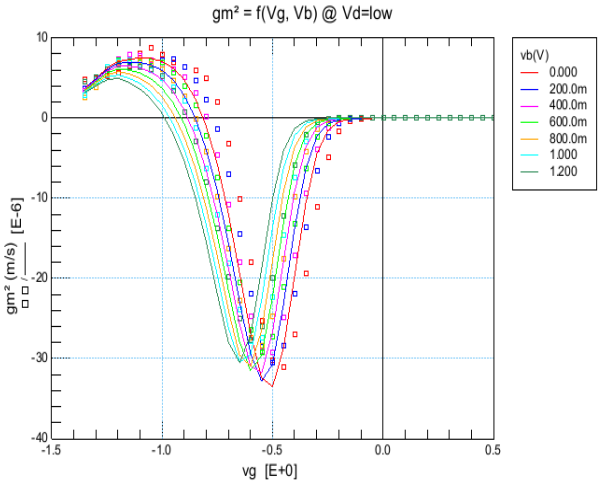
4.37W0u18_L0u13_S553_3

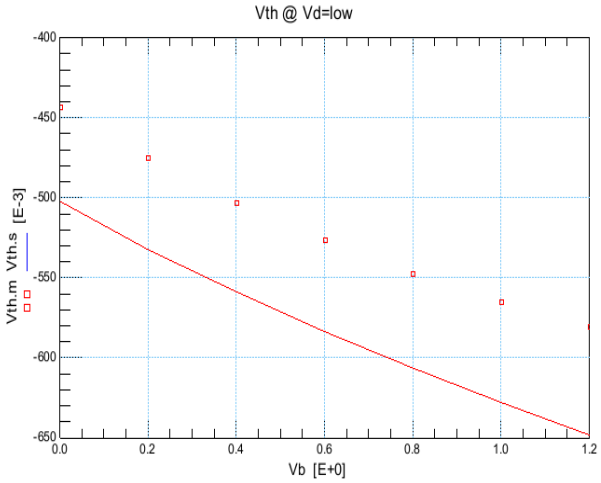
idvg, Vd = -0.05V, T = 27°C

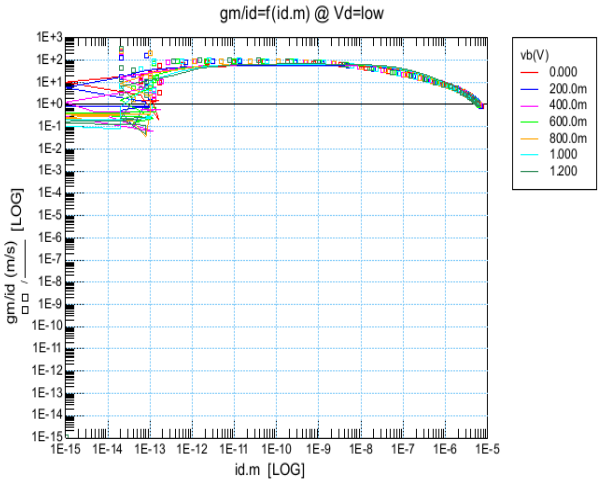


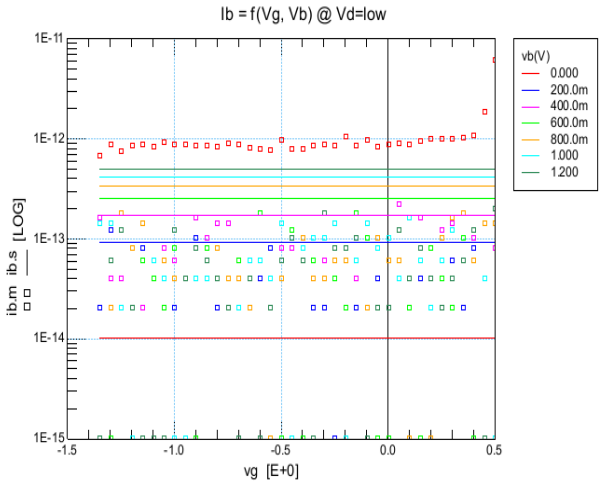


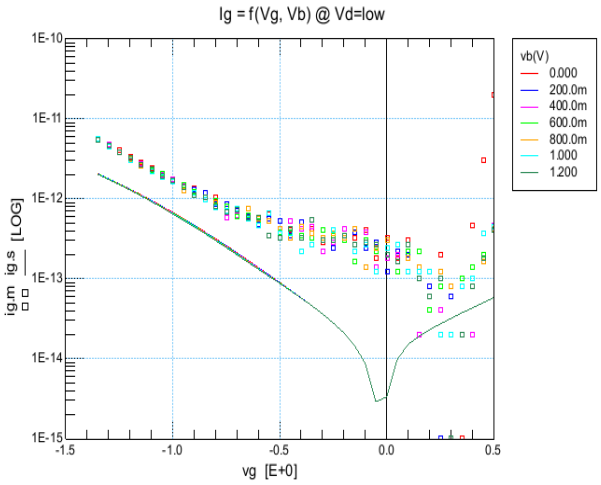




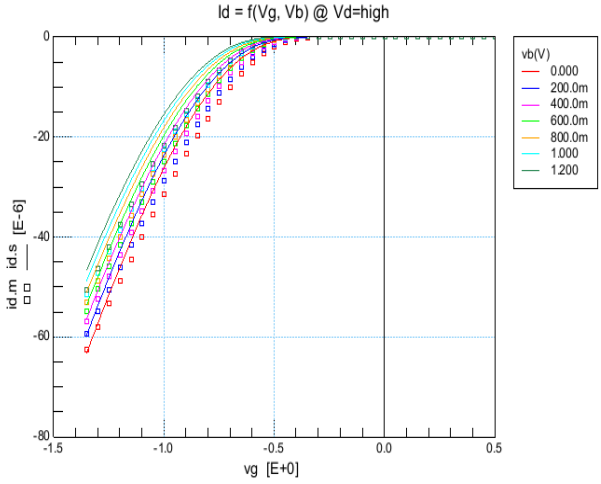


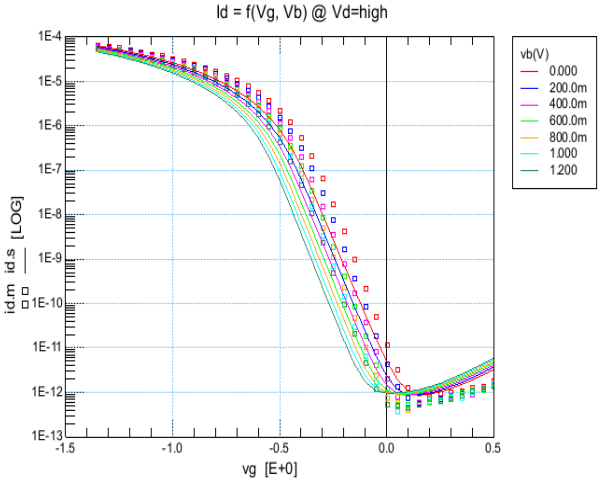


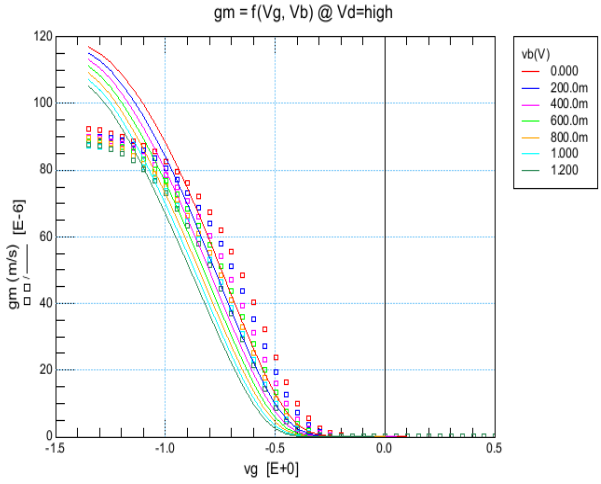


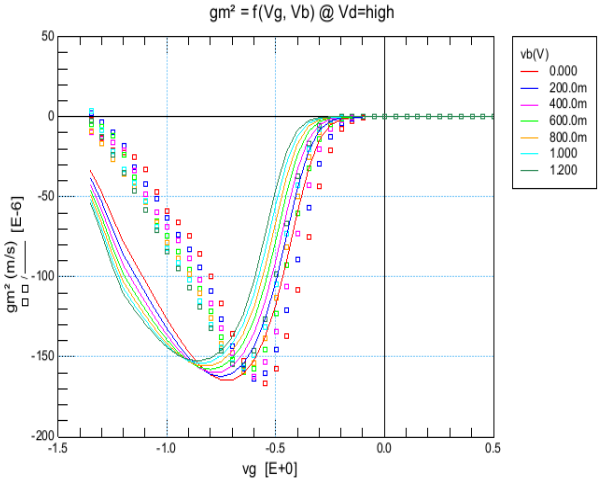


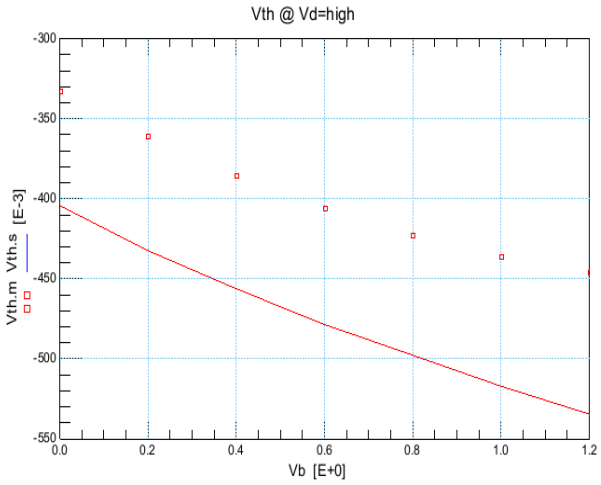
idvg, Vd = -1.2V, T = 27°C

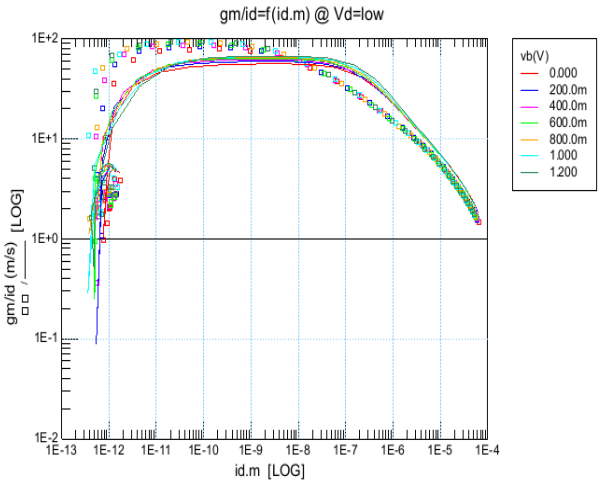


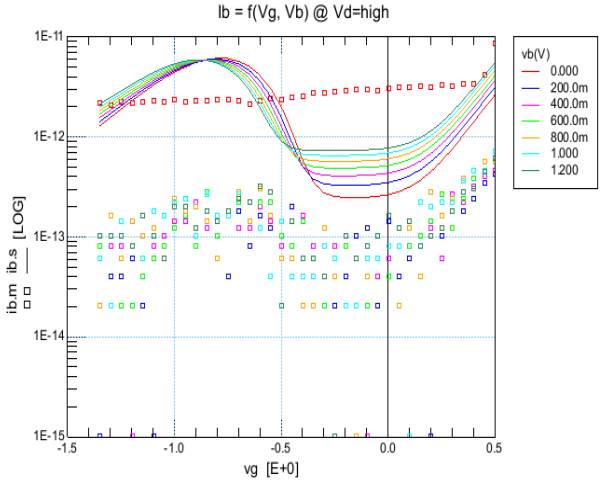


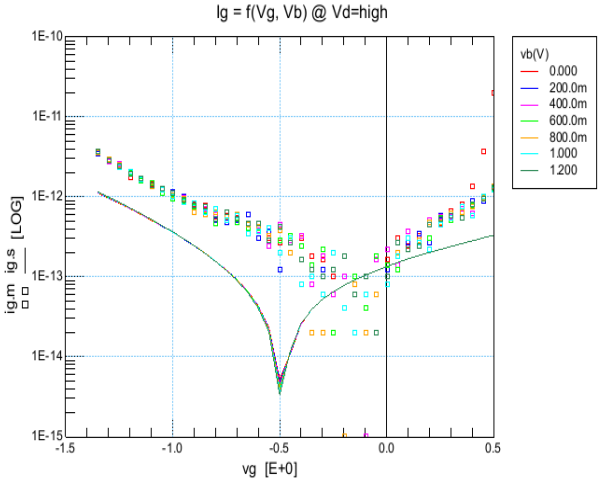




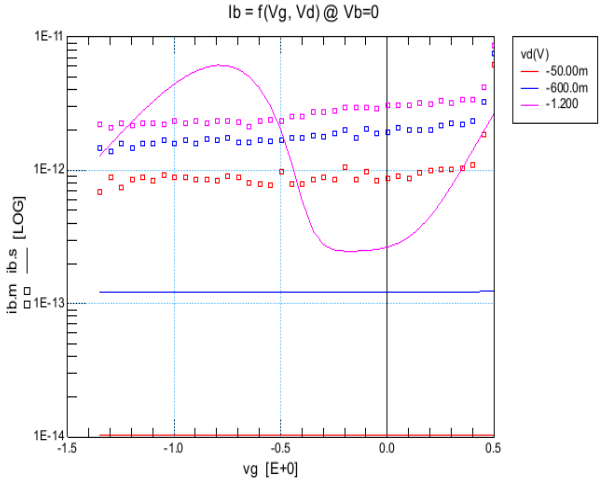


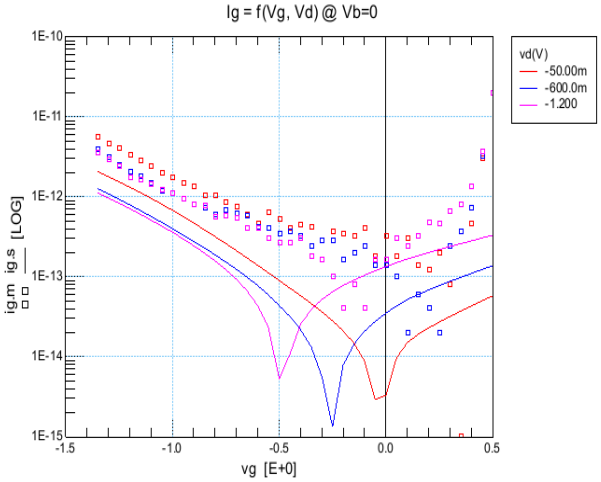




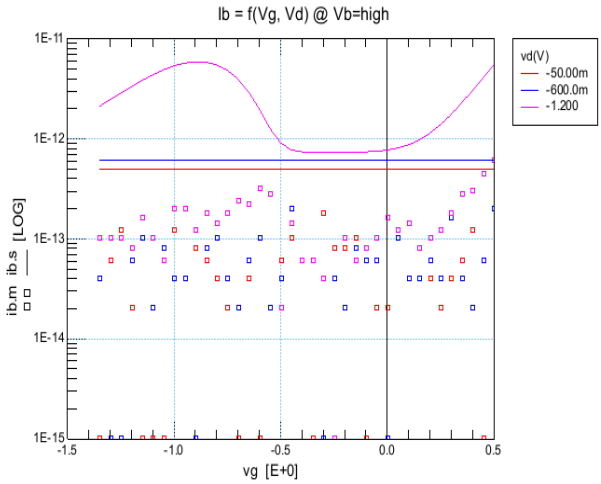


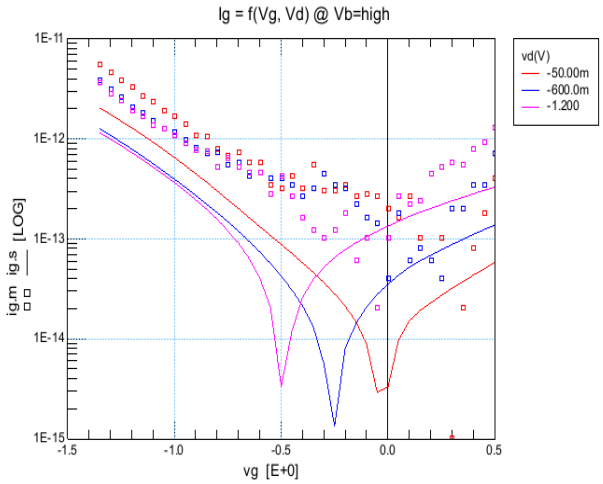
idvg, Ib,Ig, Vb = 0V, T = 27°C



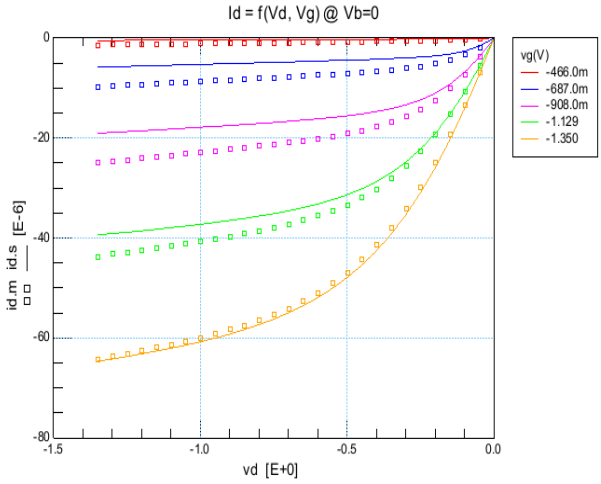


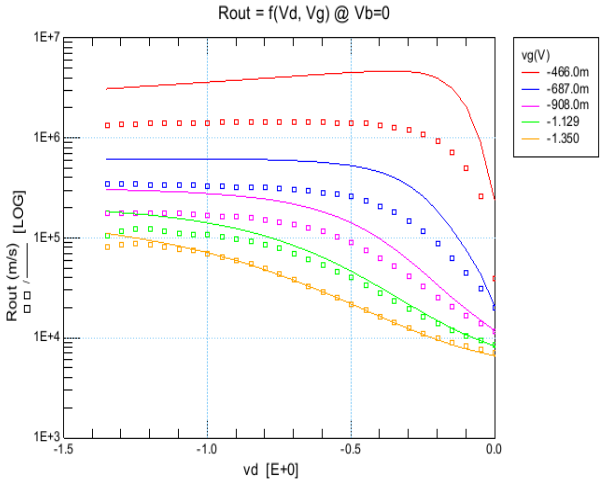
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

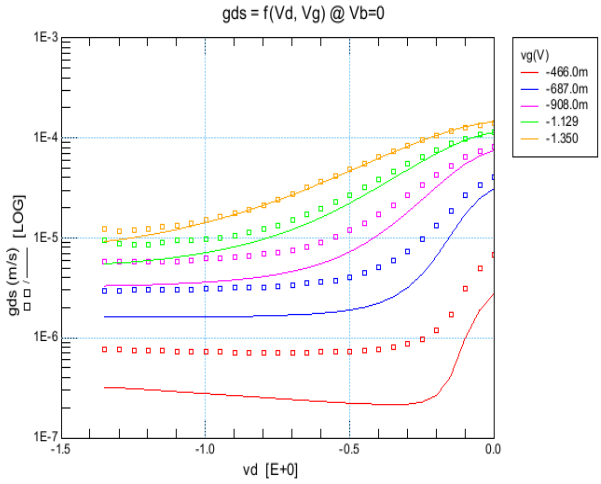




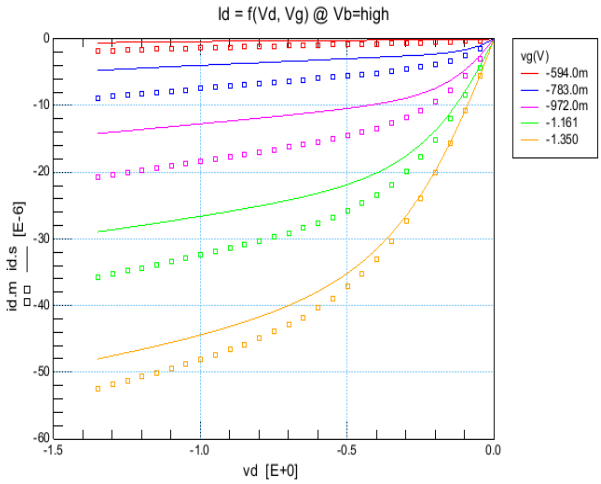
idvd, Vb = 0V, T = 27°C

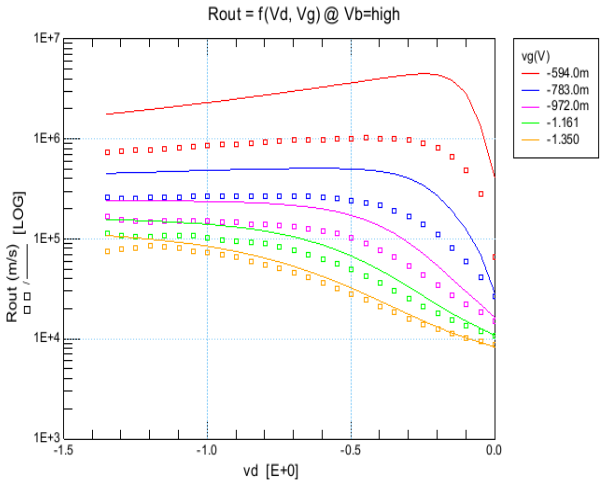


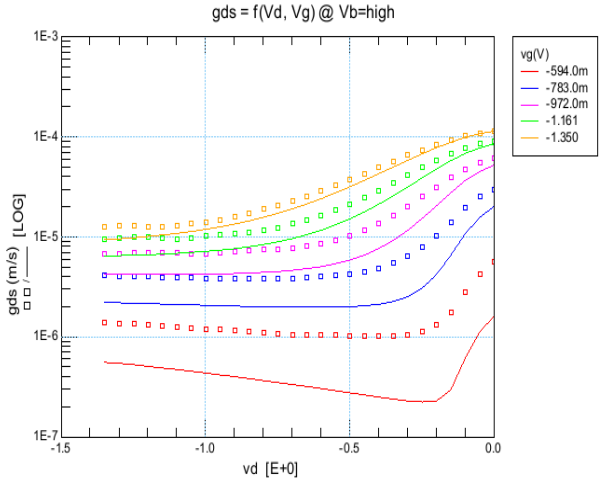




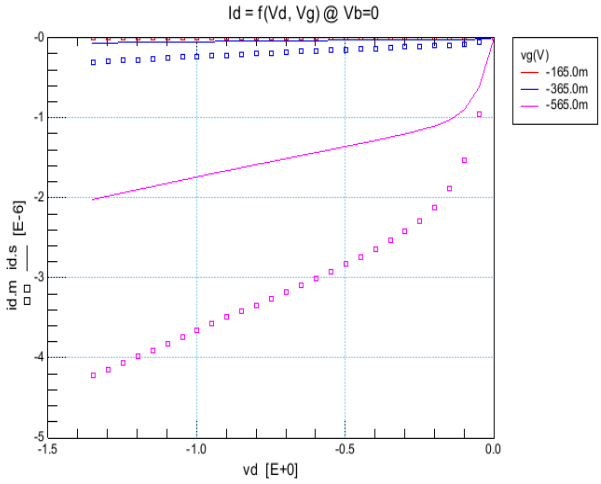
idvd_vbmin, Vb = 1.2V, T = 27°C

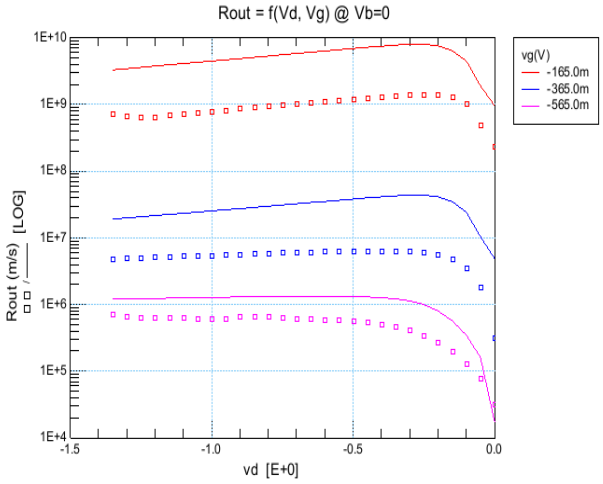


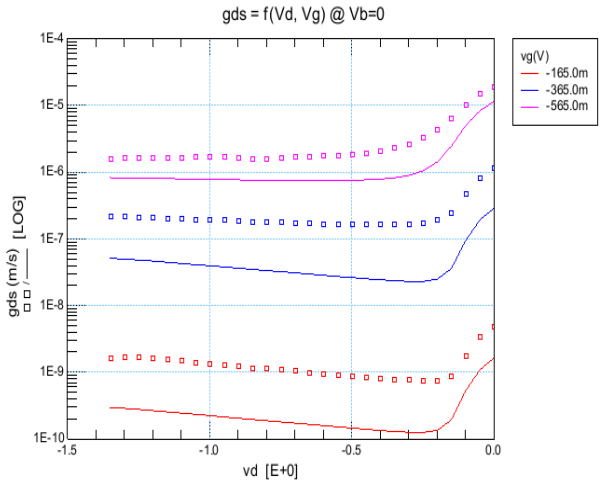




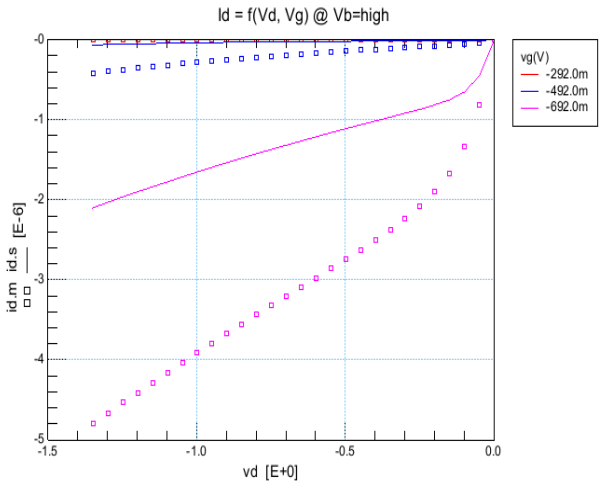
idvd_vth, Vb = 0V, T = 27°C

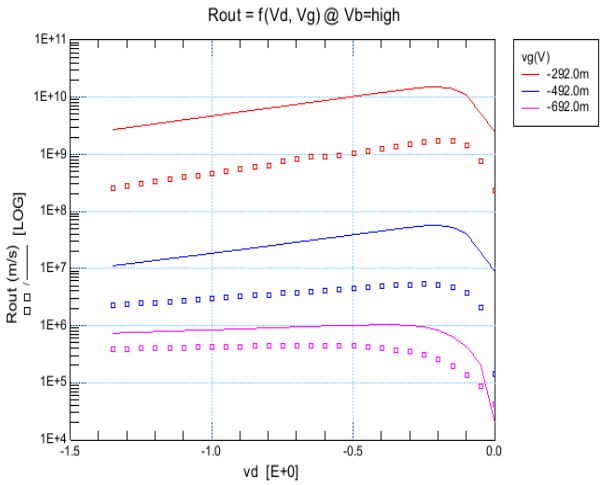


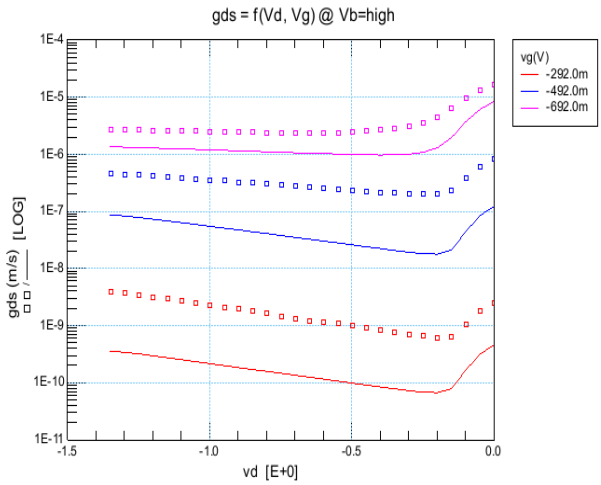




idvd_vth_vbmin, Vb = 1.2V, T = 27°C



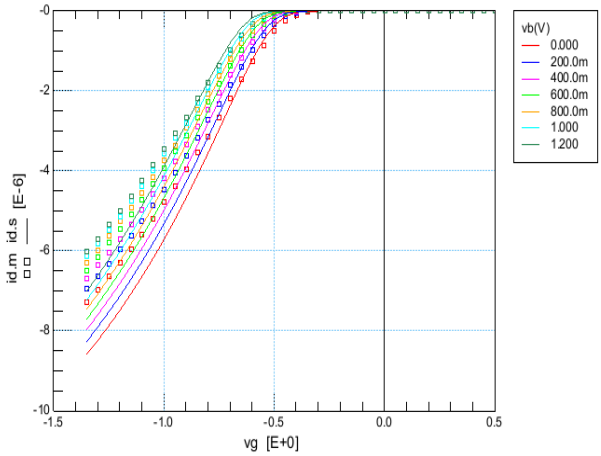




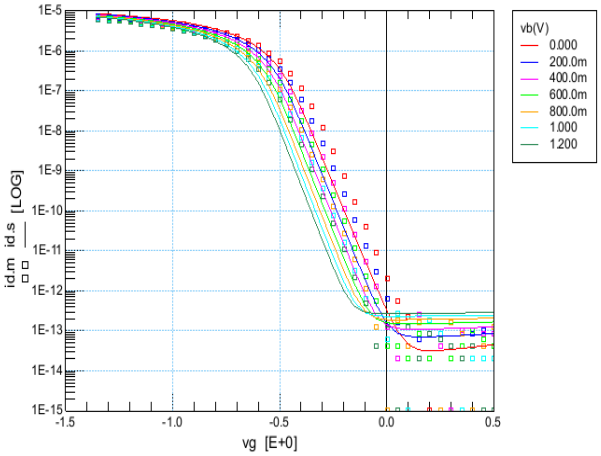
4.38 W0u15_L0u12_S554_2

idvg, Vd = -0.05V, T = 27°C

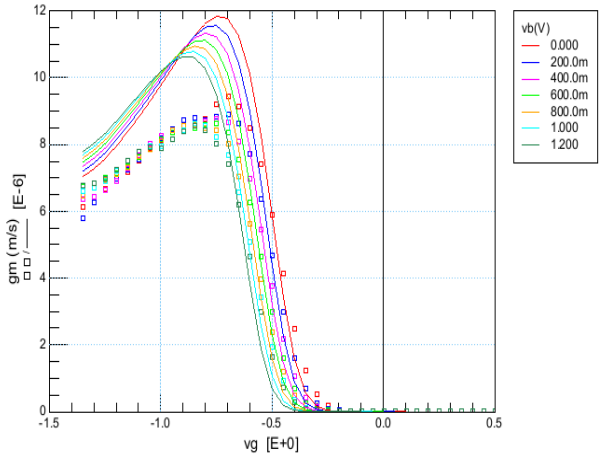
Id = f(Vg, Vb) @ Vd=low



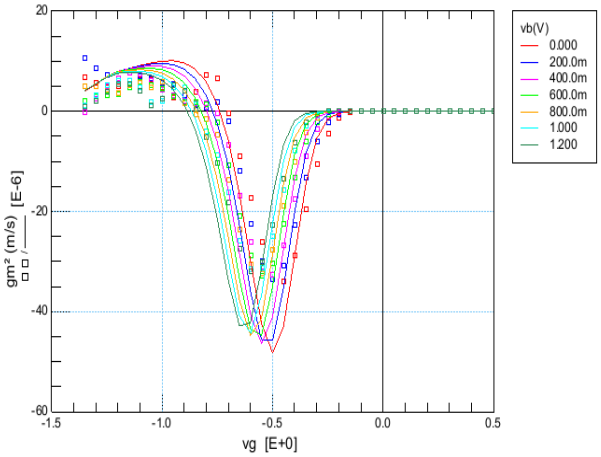
Id = f(Vg, Vb) @ Vd=low



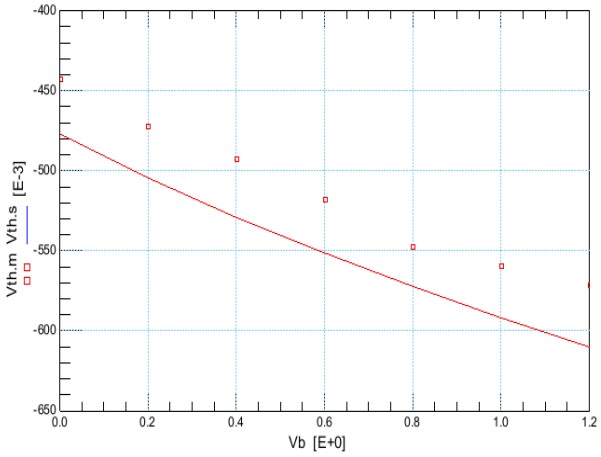
gm = f(Vg, Vb) @ Vd=low



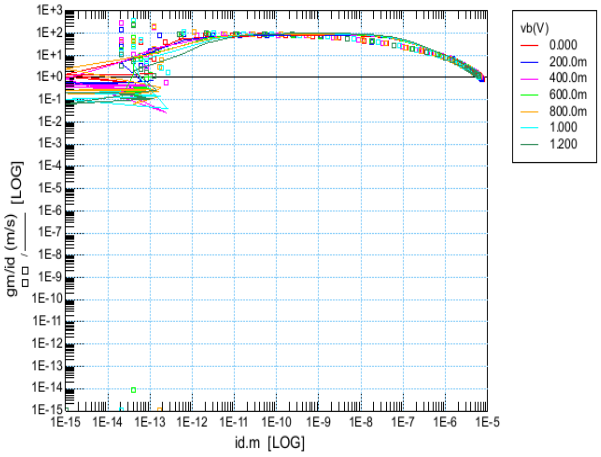
gm² = f(Vg, Vb) @ Vd=low

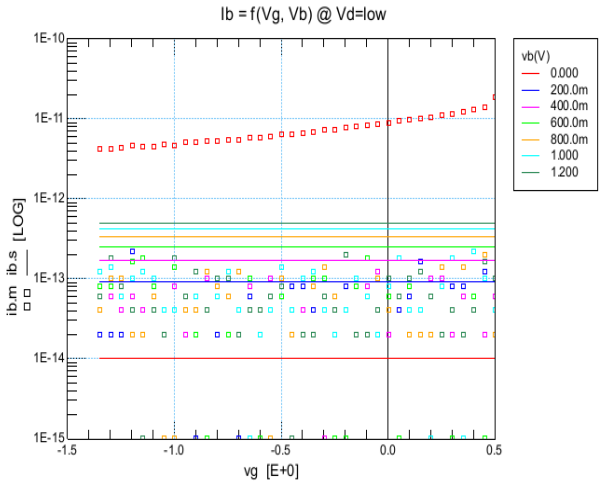


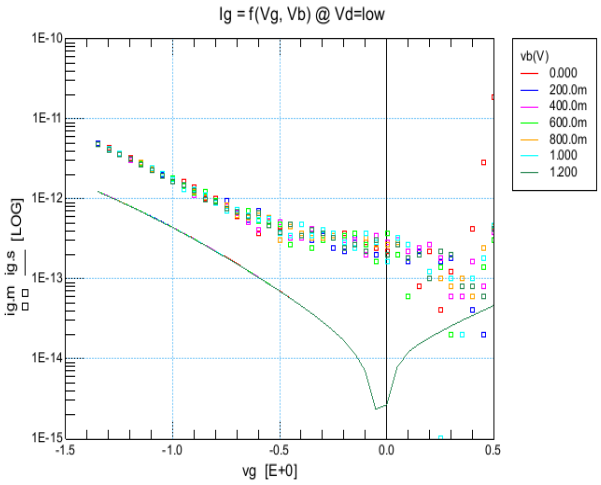
Vth @ Vd=low



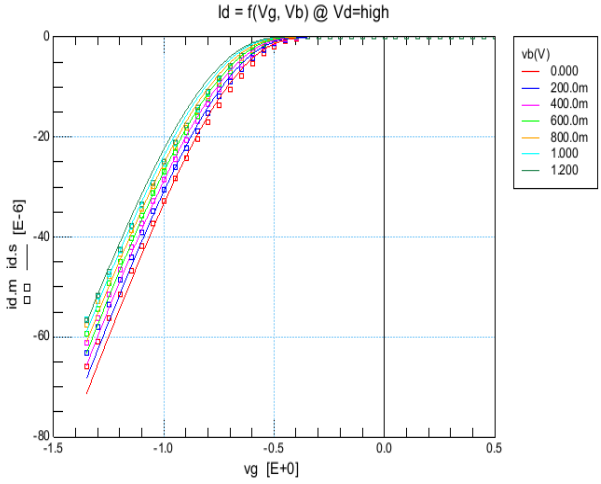
gm/id=f(id.m) @ Vd=low

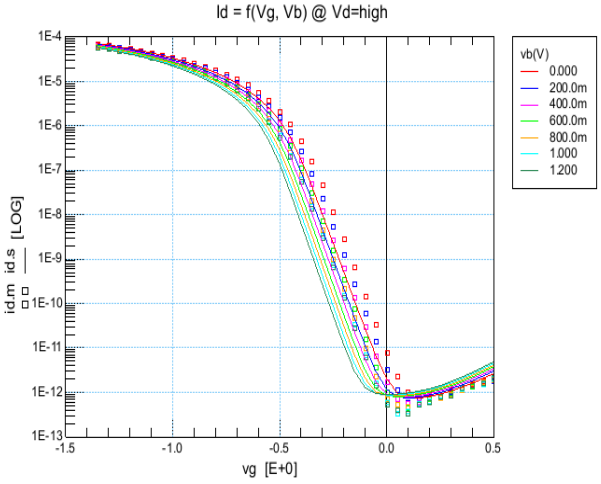


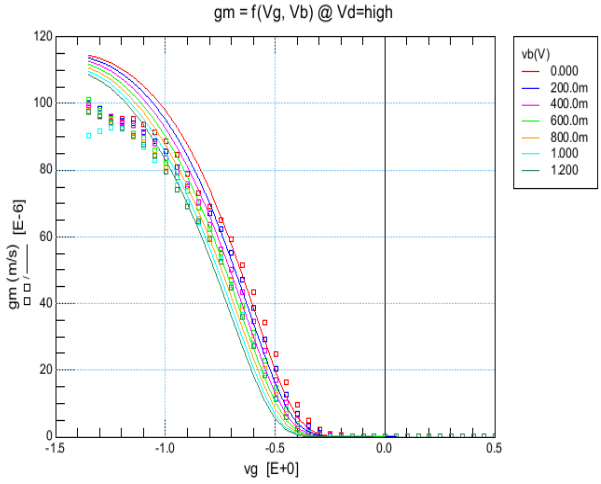


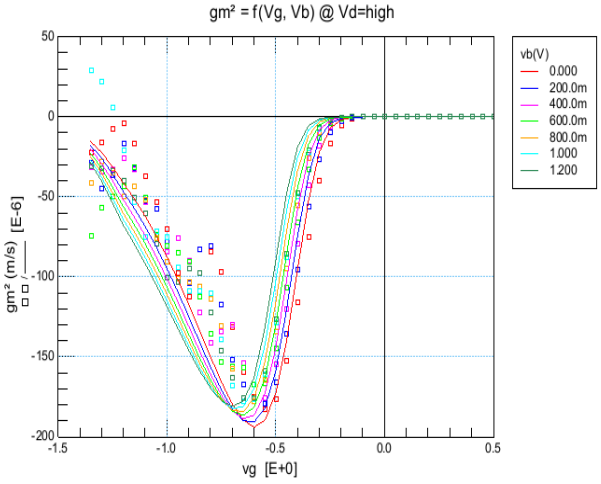


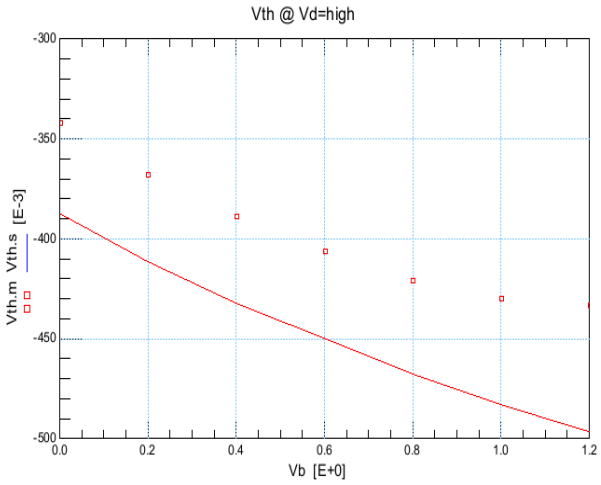
idvg, Vd = -1.2V, T = 27°C

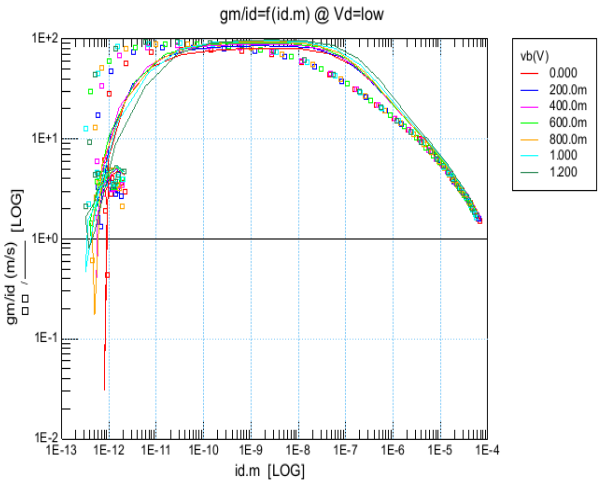


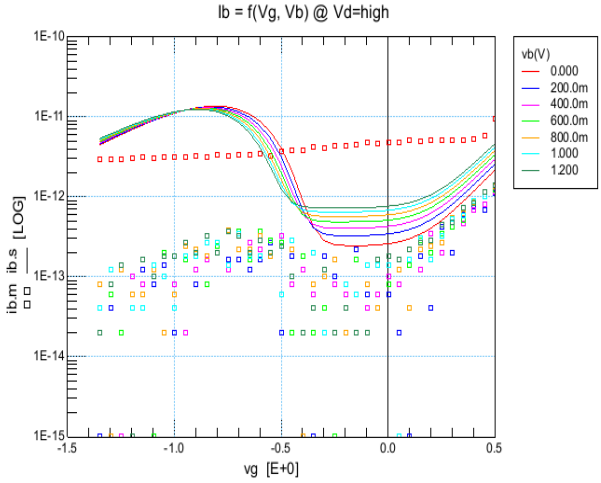


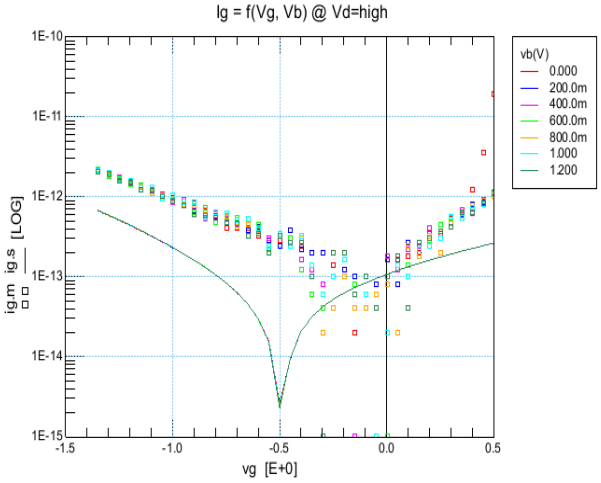




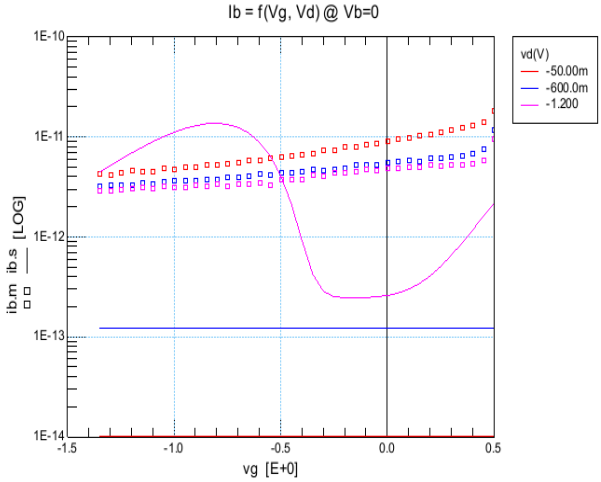


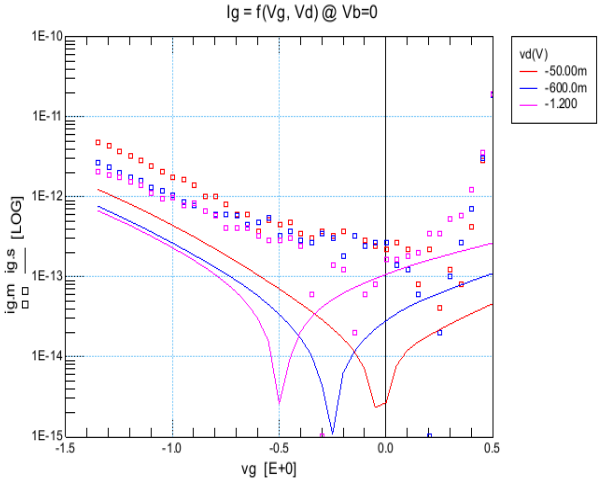




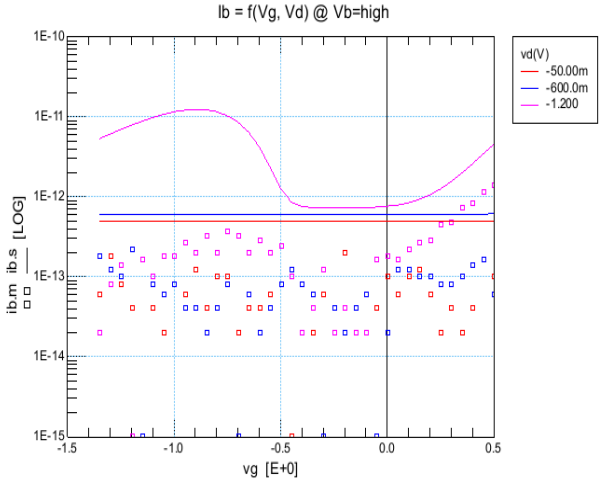


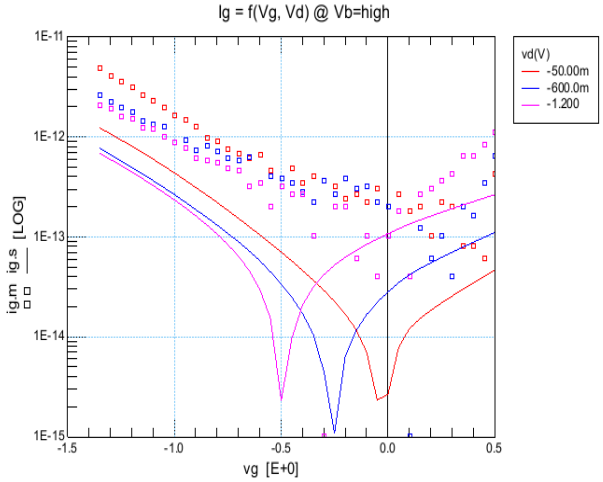
idvg, Ib,Ig, Vb = 0V, T = 27°C



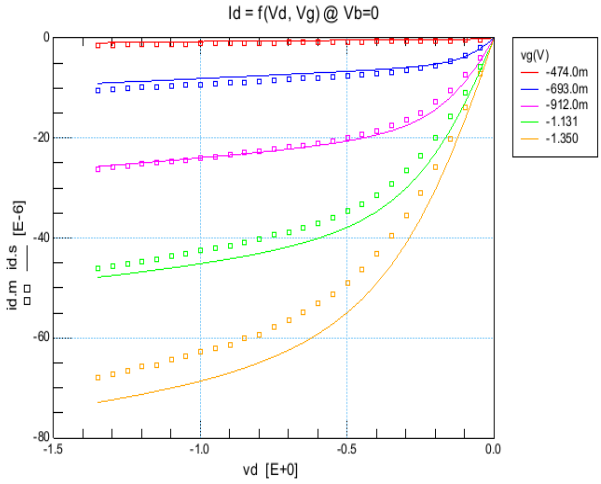


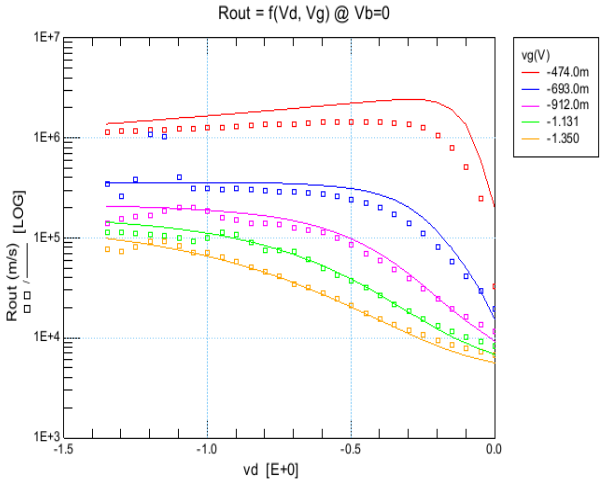
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

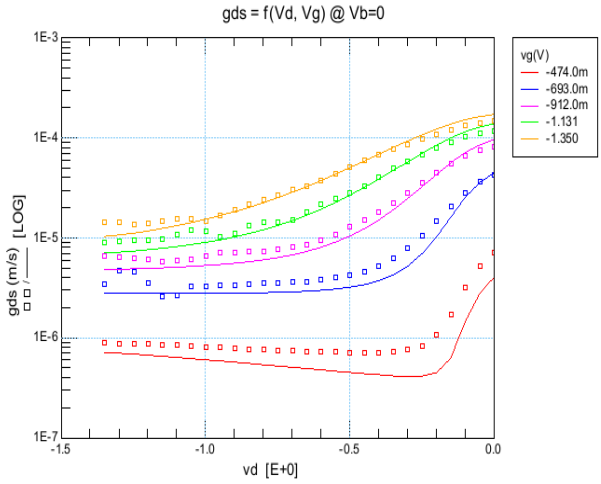




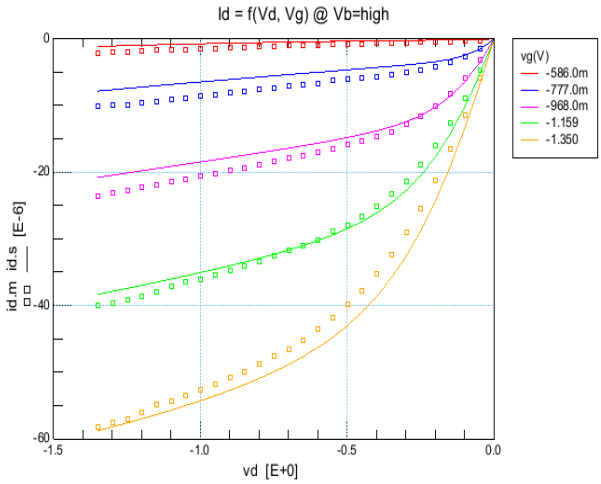
idvd, Vb = 0V, T = 27°C

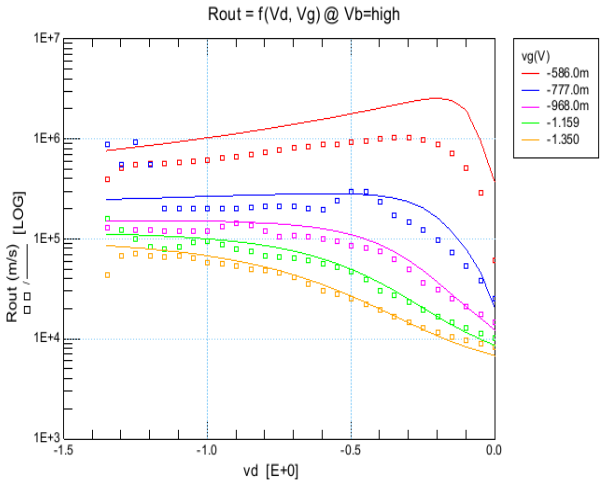


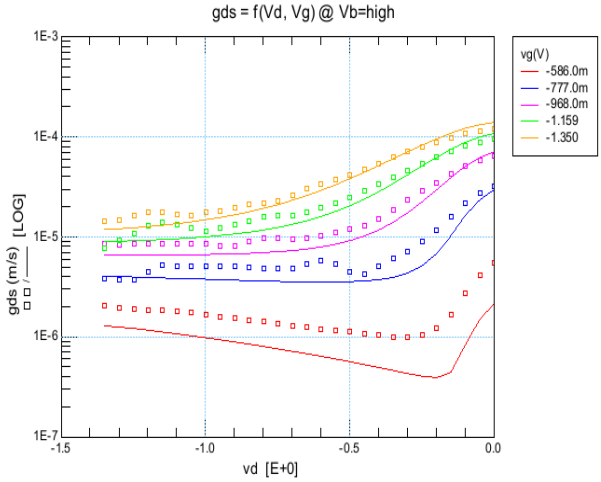




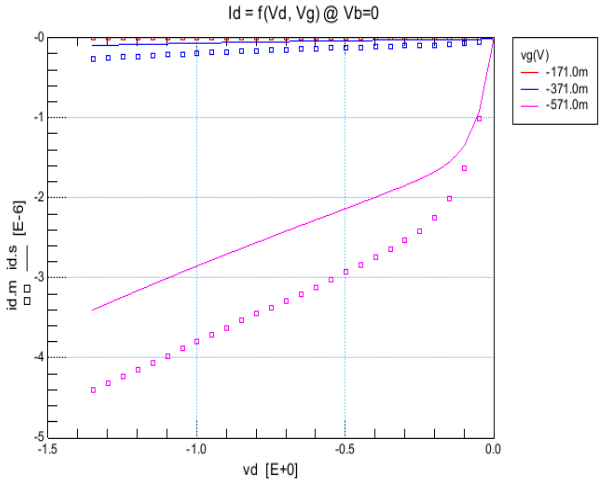
idvd_vbmin, Vb = 1.2V, T = 27°C

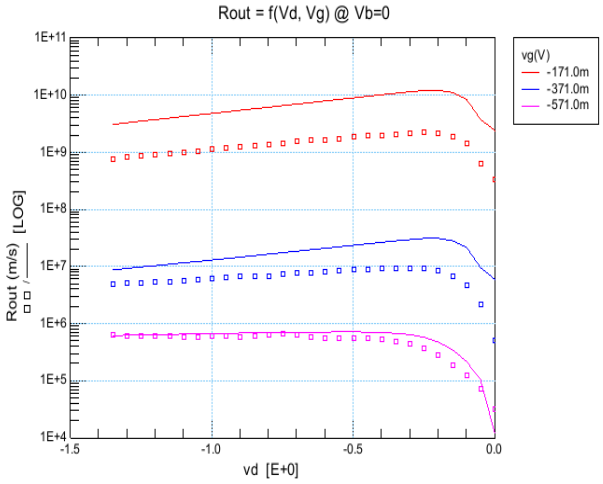


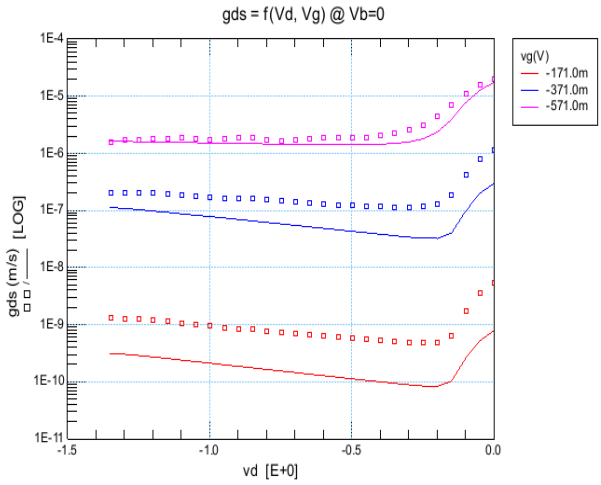




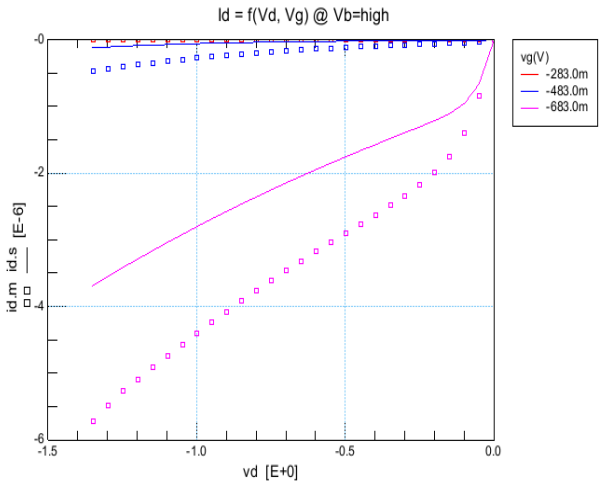
idvd_vth, Vb = 0V, T = 27°C

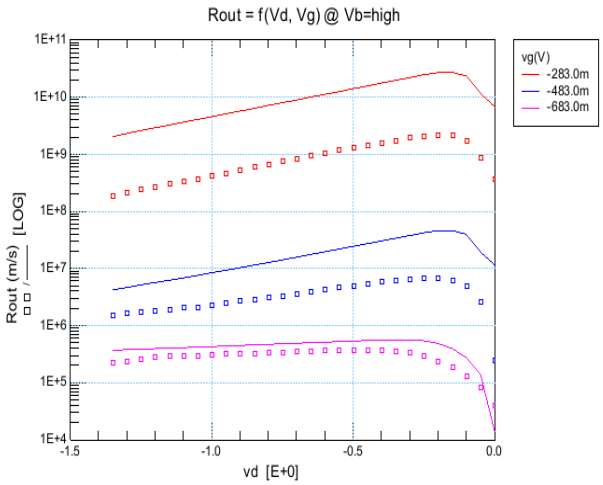


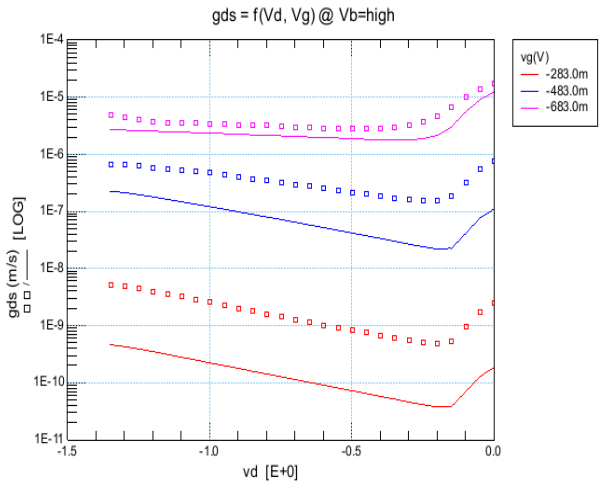




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

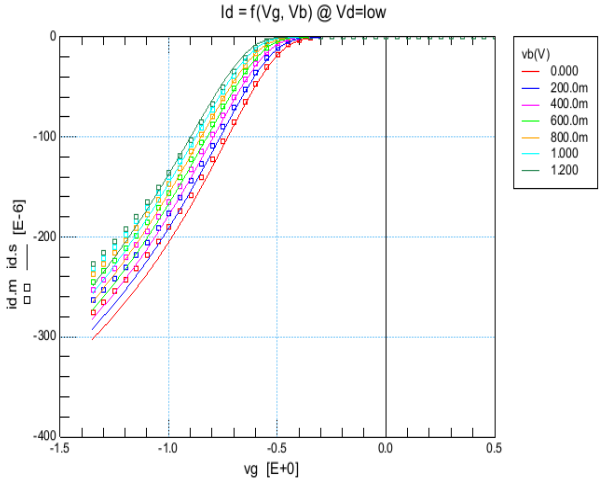


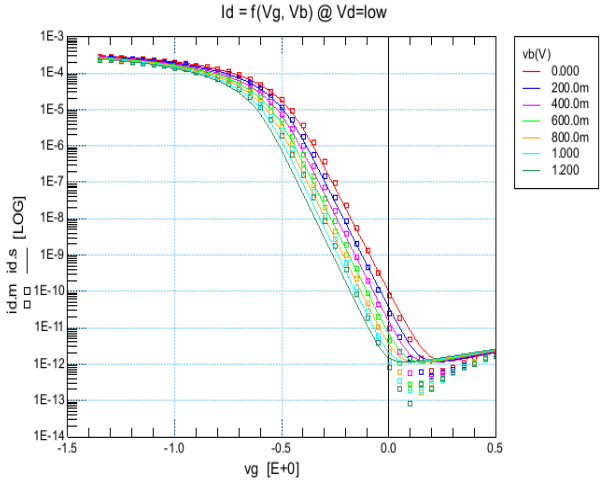


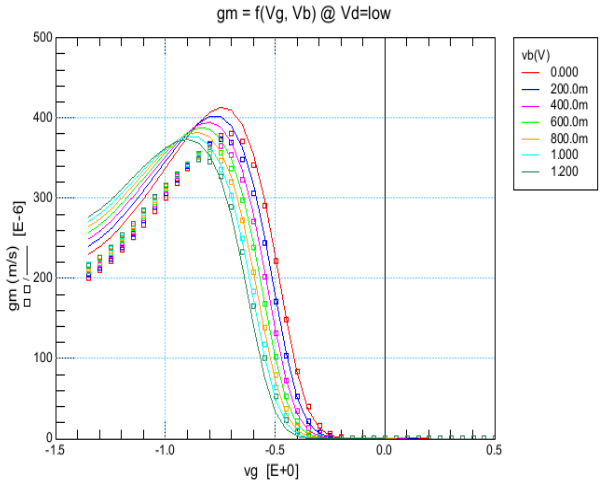


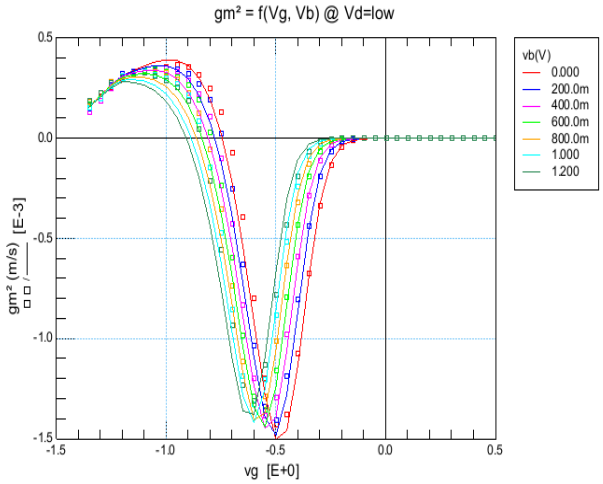
4.39W10u0_L0u12_S548_1

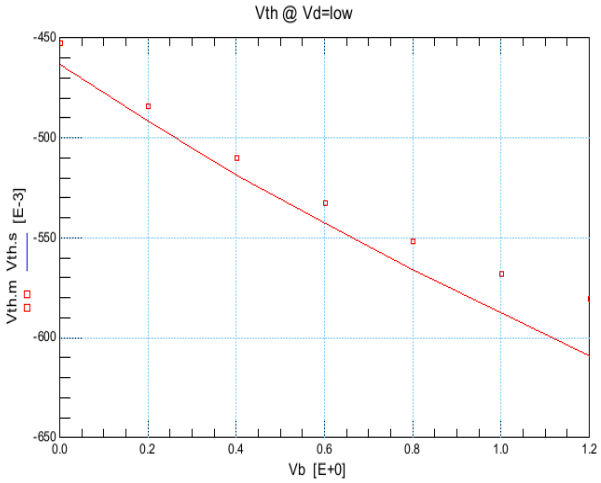
idvg, Vd = -0.05V, T = 27°C

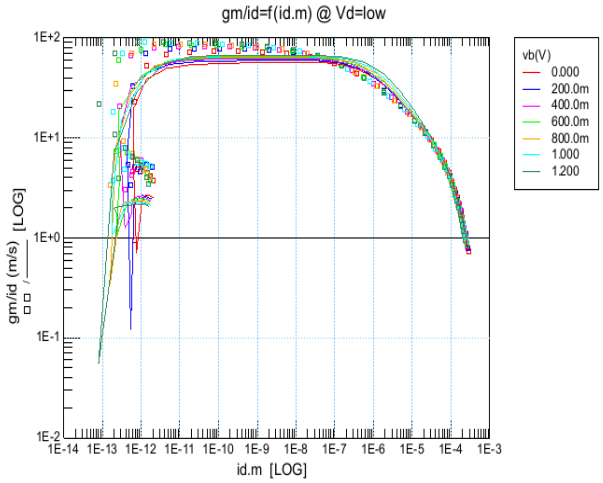


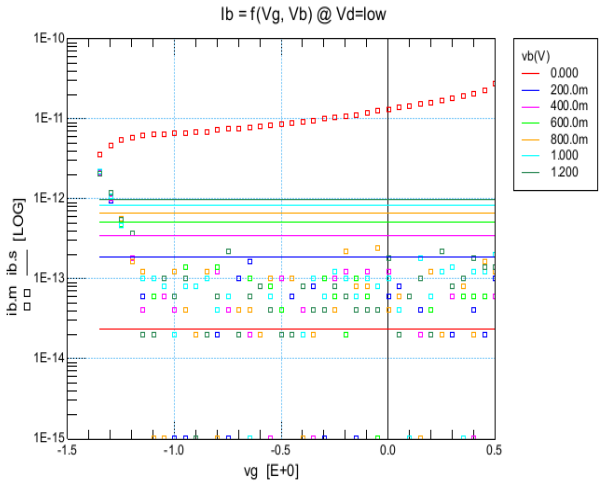


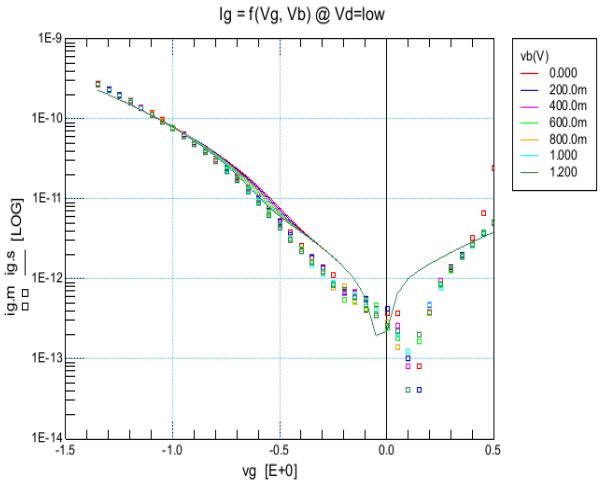




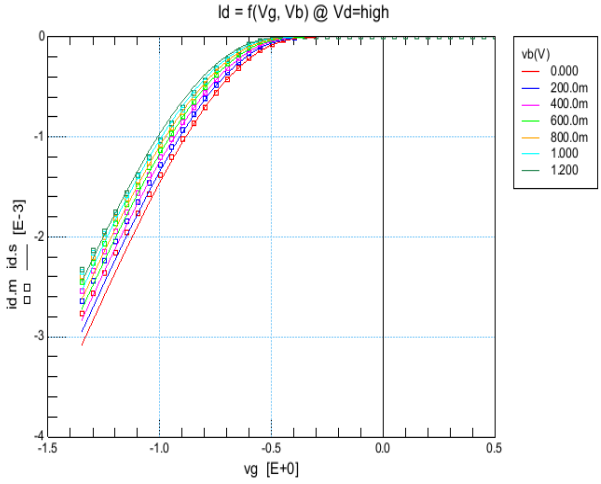


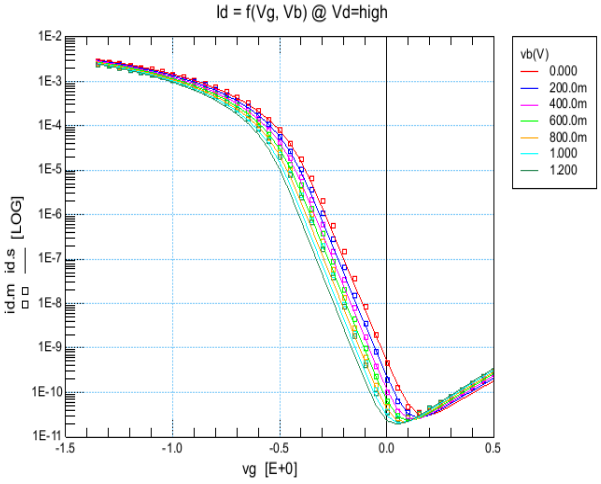


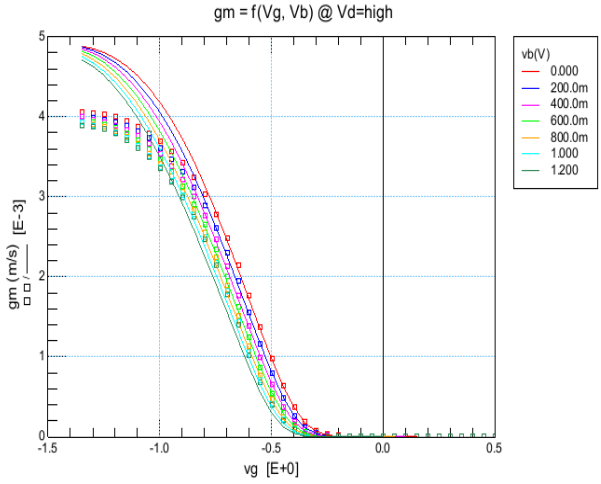


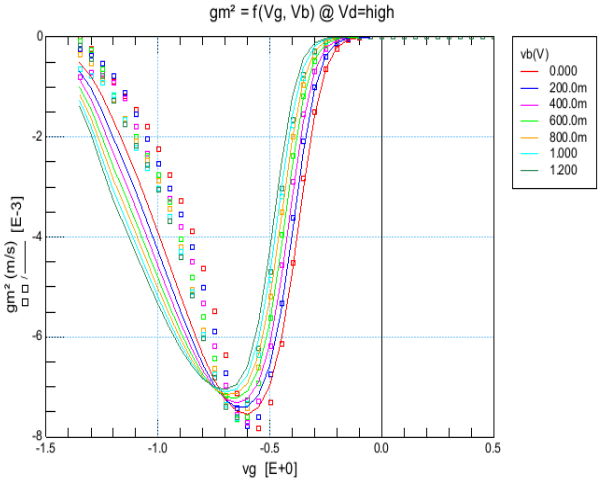


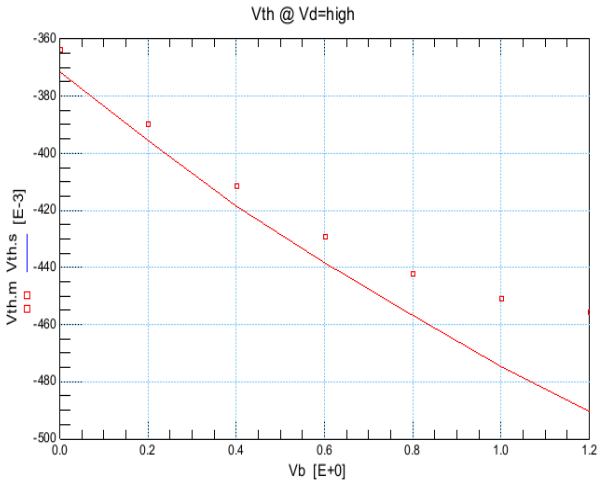
idvg, Vd = -1.2V, T = 27°C

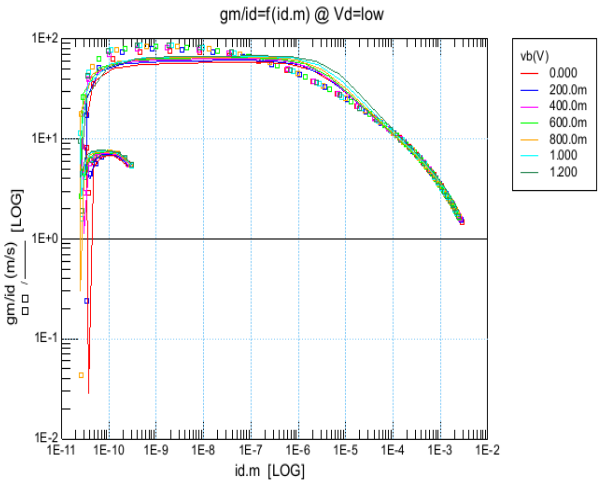


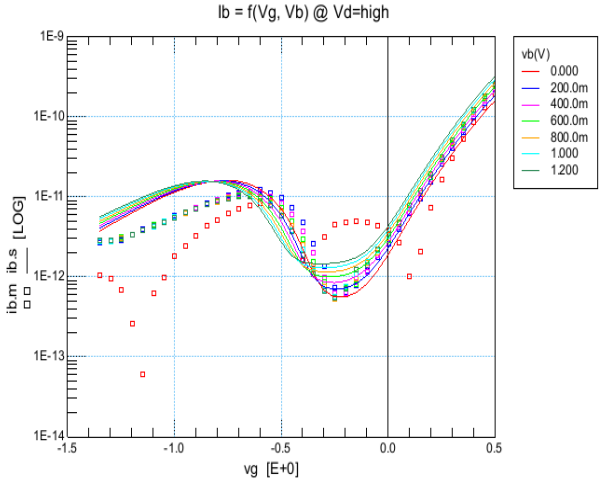


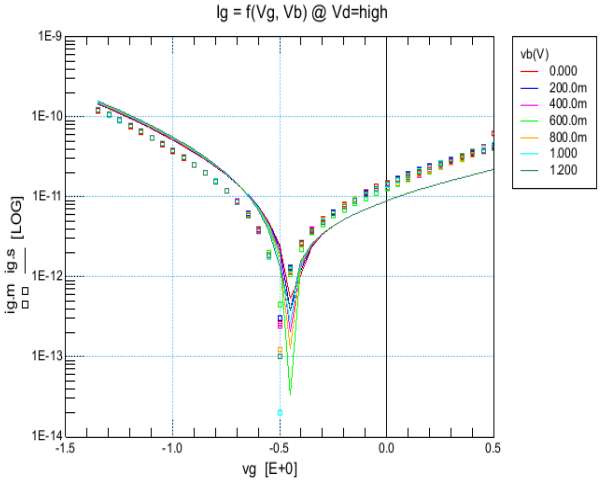




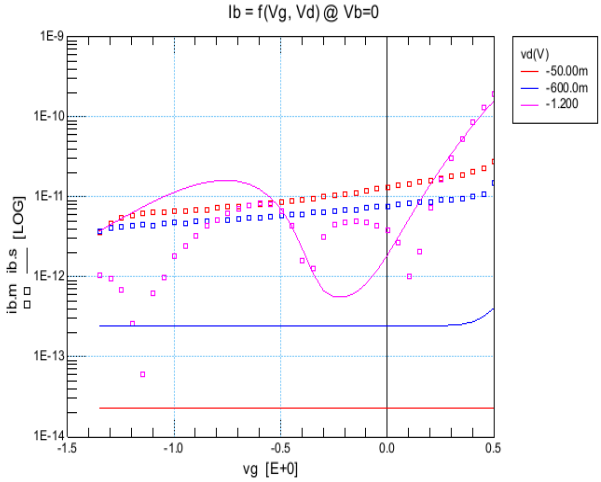


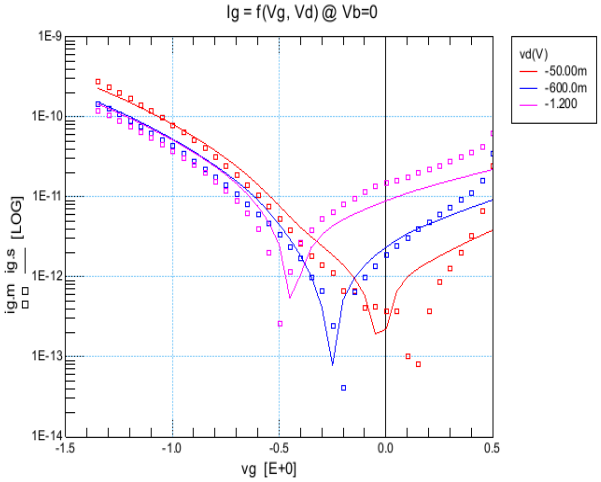




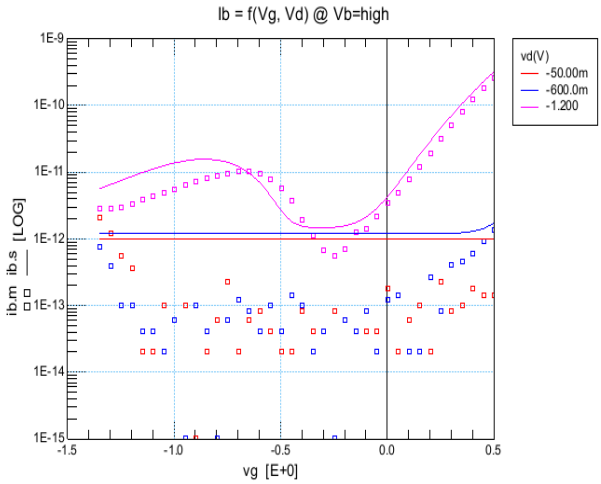


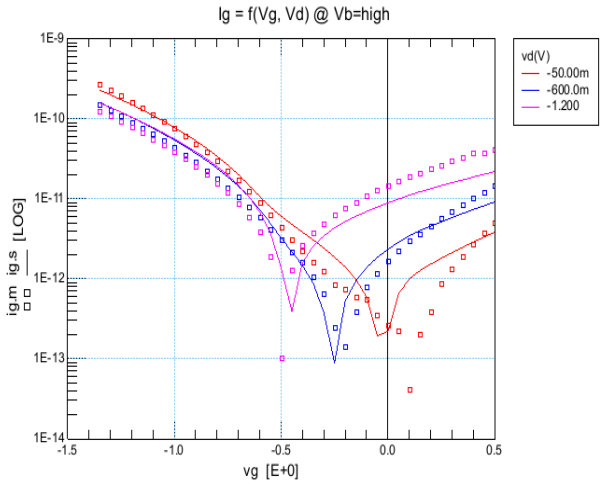
idvg, Ib,Ig, Vb = 0V, T = 27°C



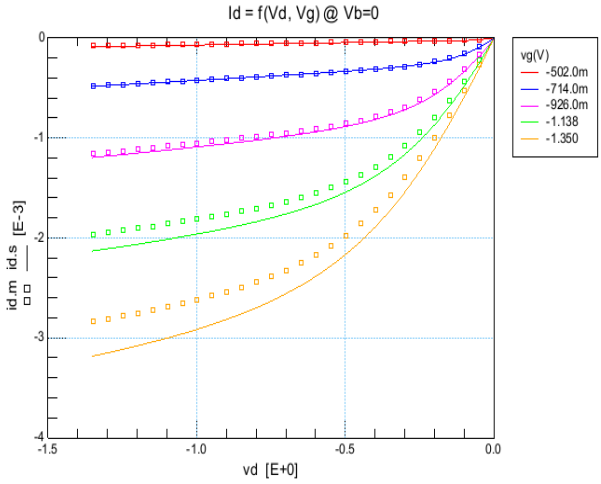


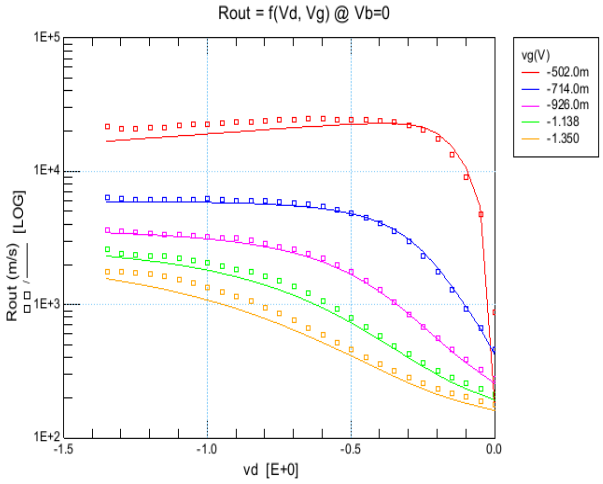
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

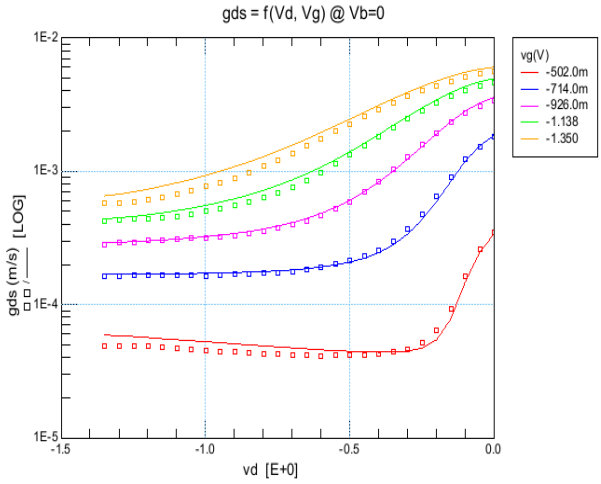




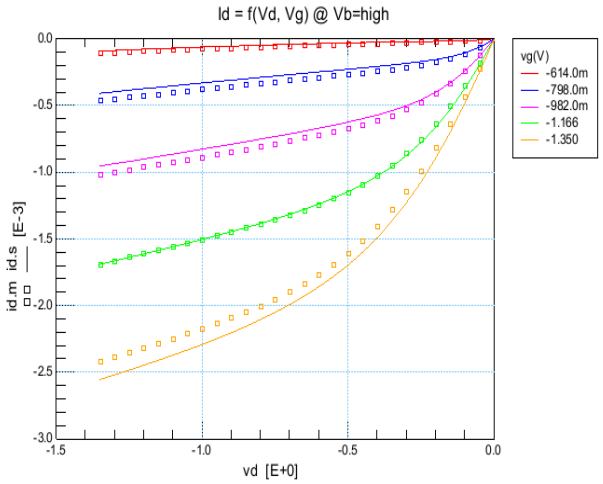
idvd, Vb = 0V, T = 27°C

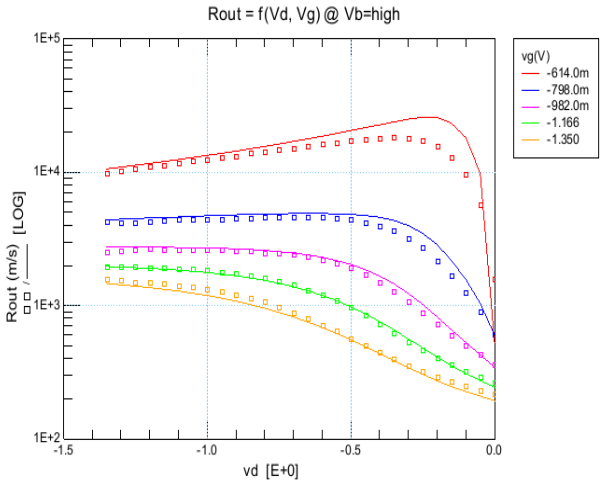


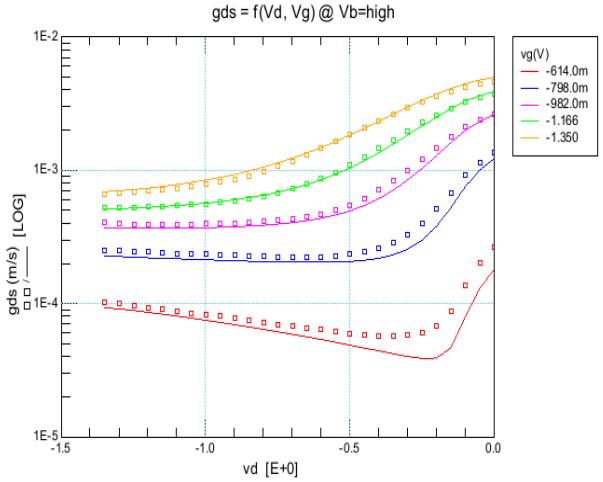




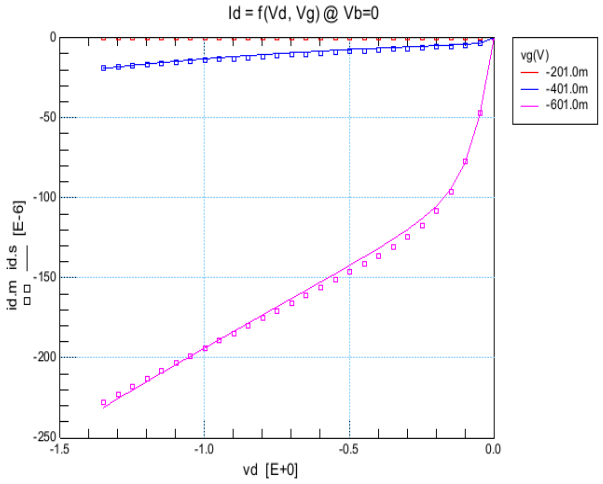
idvd_vbmin, Vb = 1.2V, T = 27°C

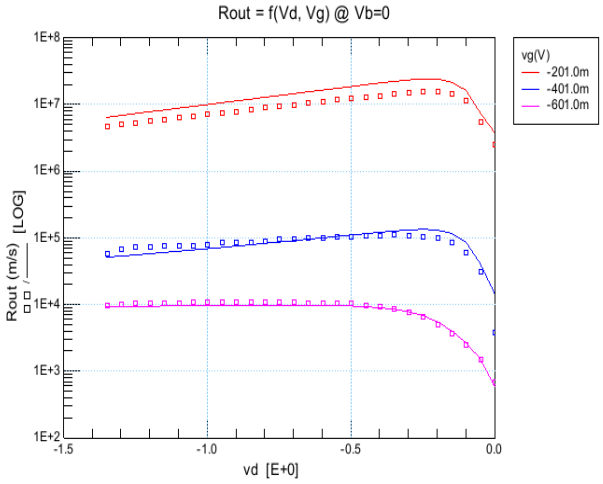


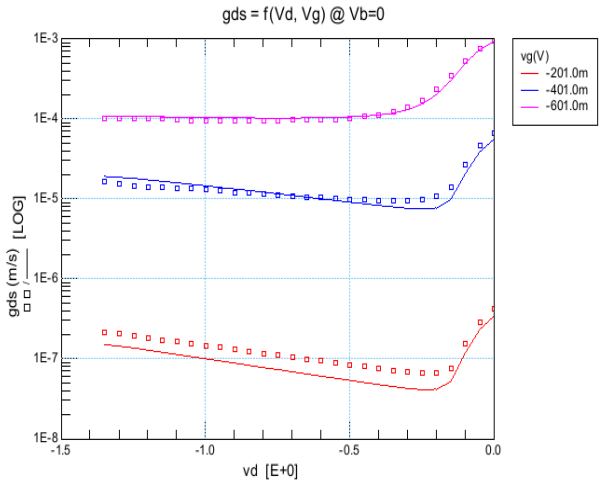




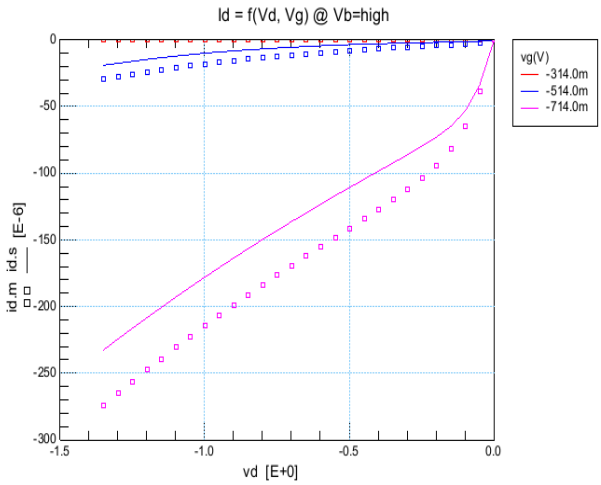
idvd_vth, Vb = 0V, T = 27°C

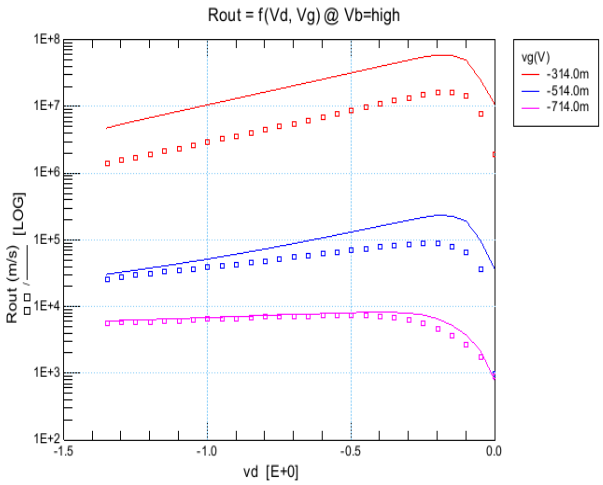


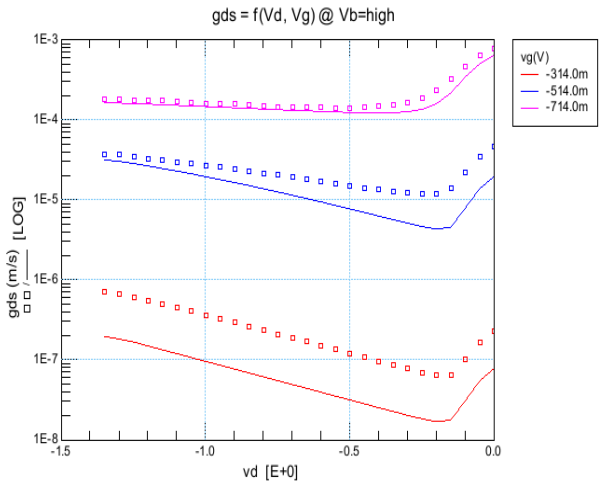




idvd_vth_vbmin, Vb = 1.2V, T = 27°C



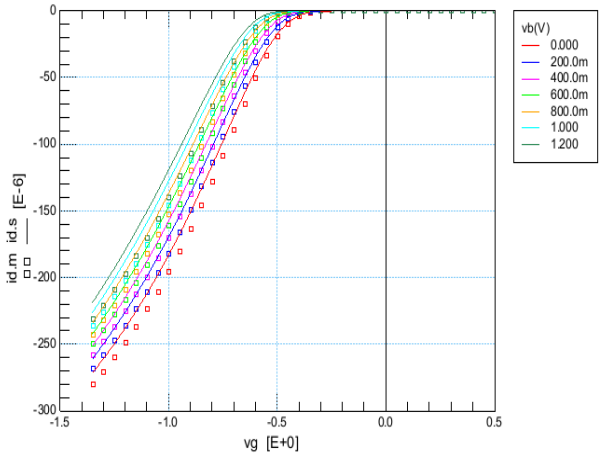




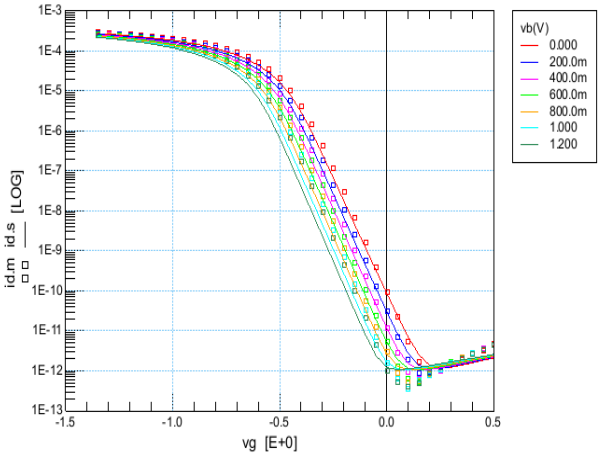
4.40W10u0_L0u13_S382_2

idvg, Vd = -0.05V, T = 27°C

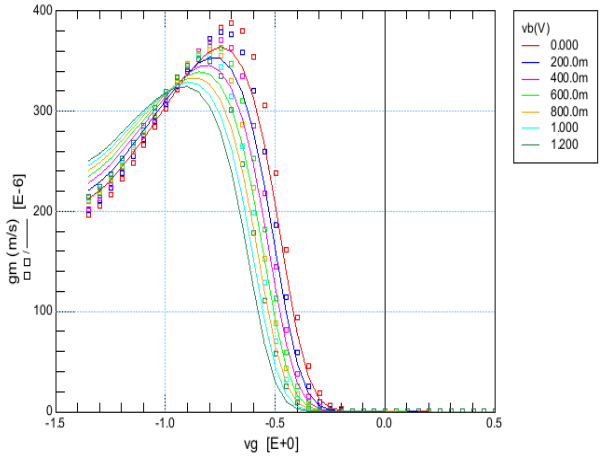
Id = f(Vg, Vb) @ Vd=low



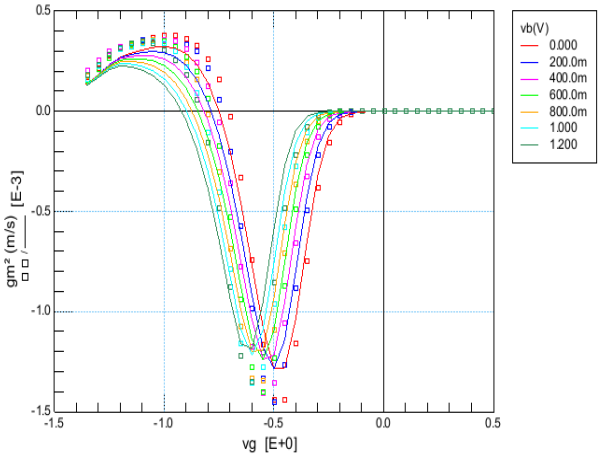
Id = f(Vg, Vb) @ Vd=low



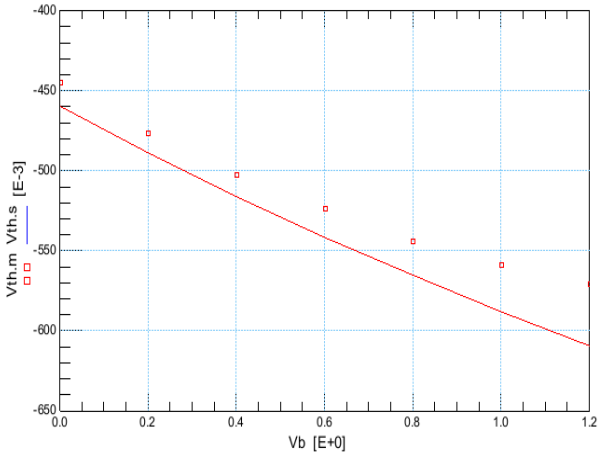
gm = f(Vg, Vb) @ Vd=low



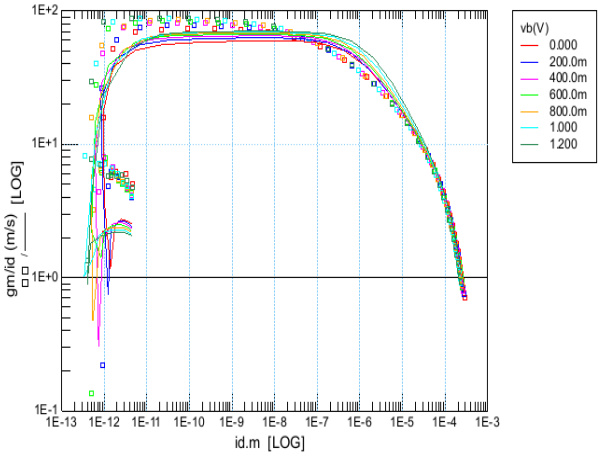
gm² = f(Vg, Vb) @ Vd=low

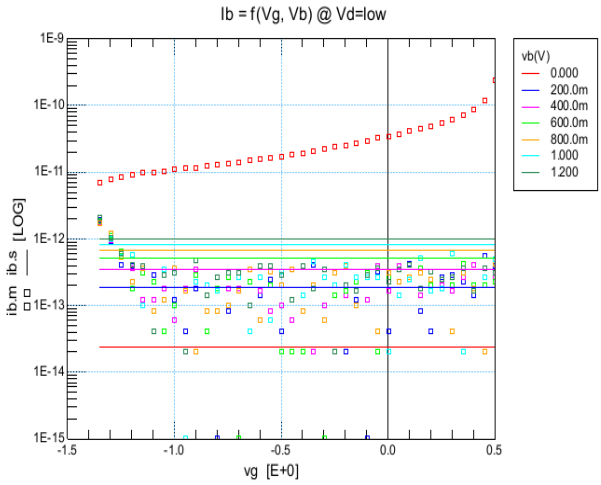


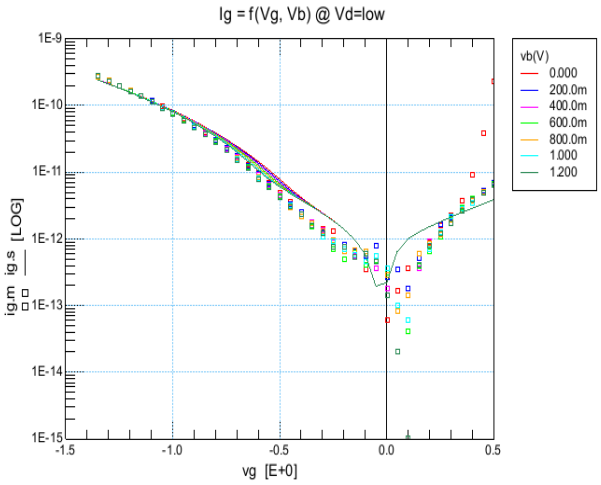
Vth @ Vd=low



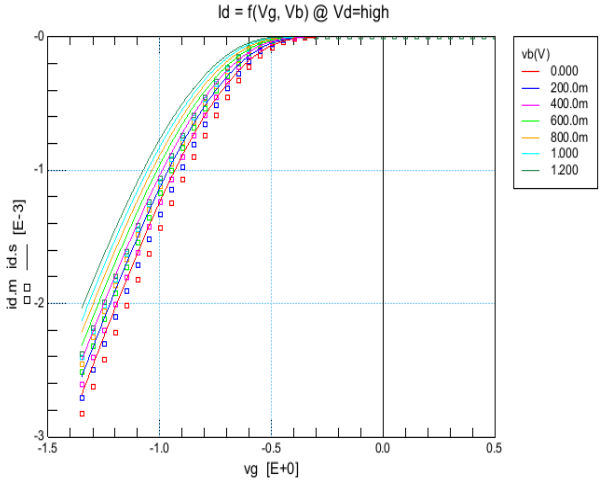
gm/id=f(id.m) @ Vd=low

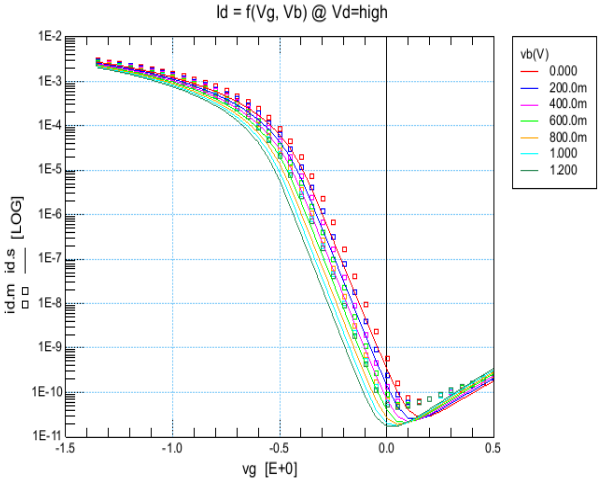


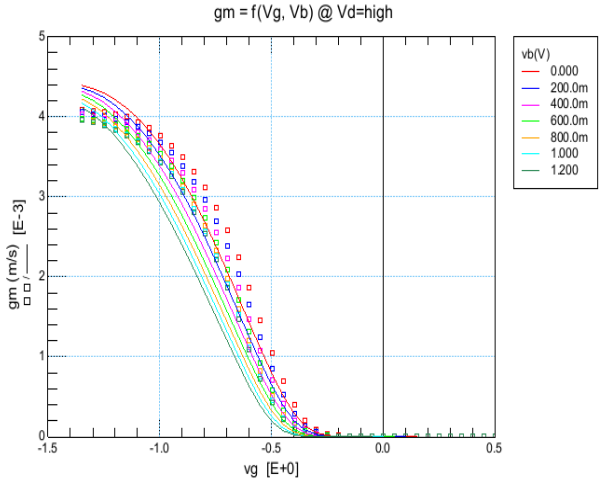


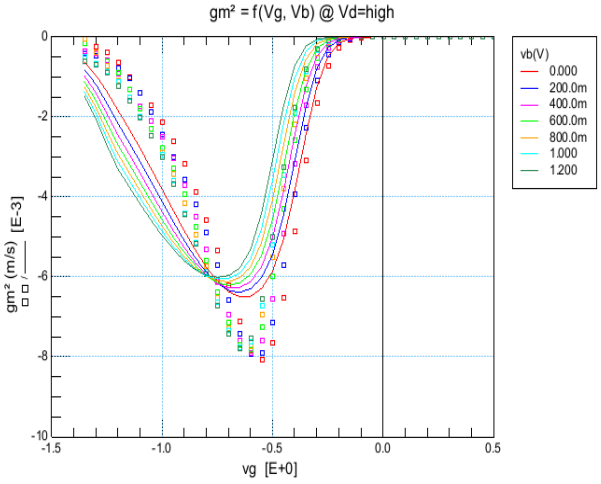


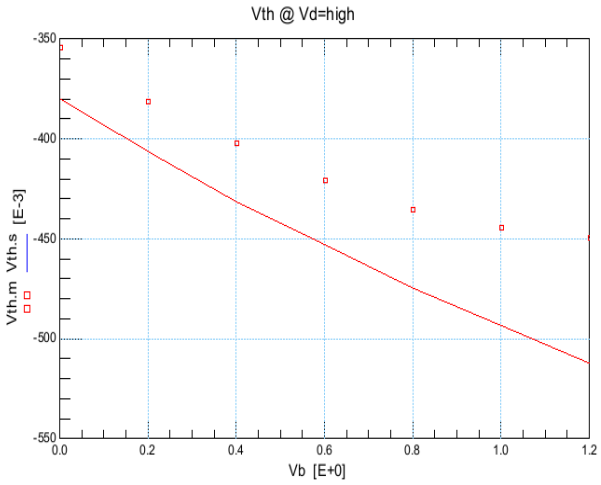
idvg, Vd = -1.2V, T = 27°C

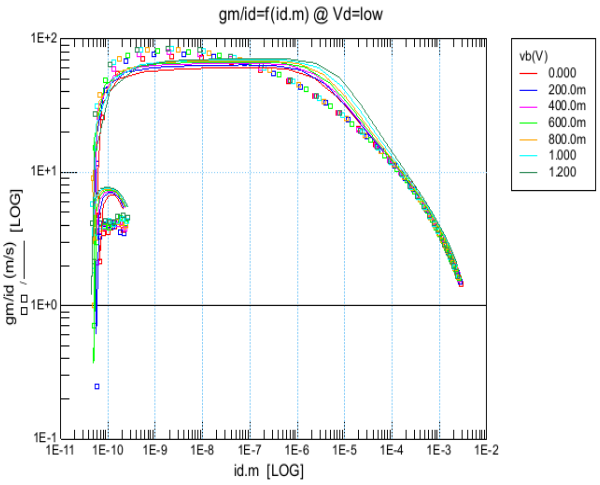


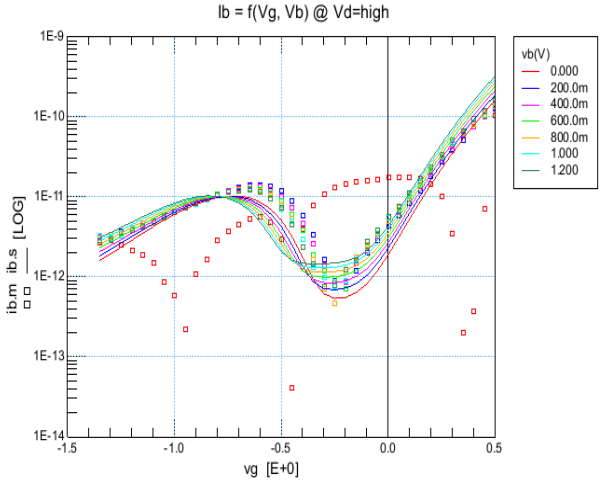


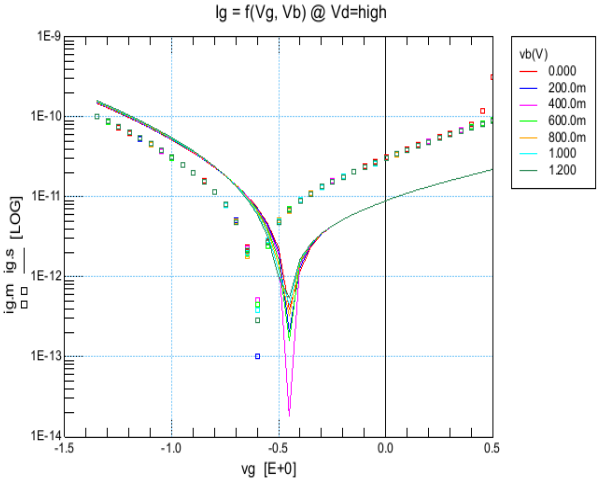




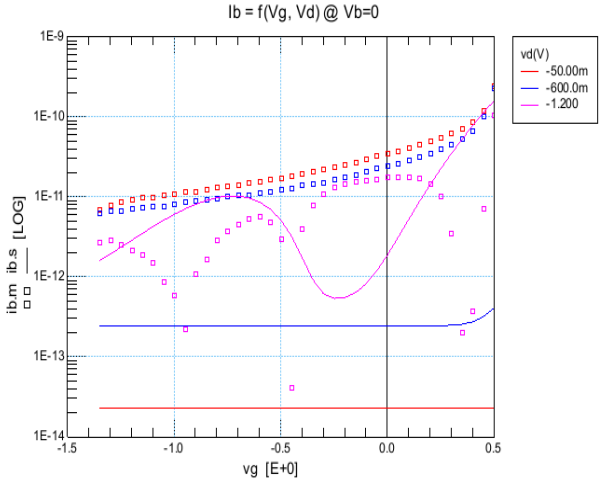


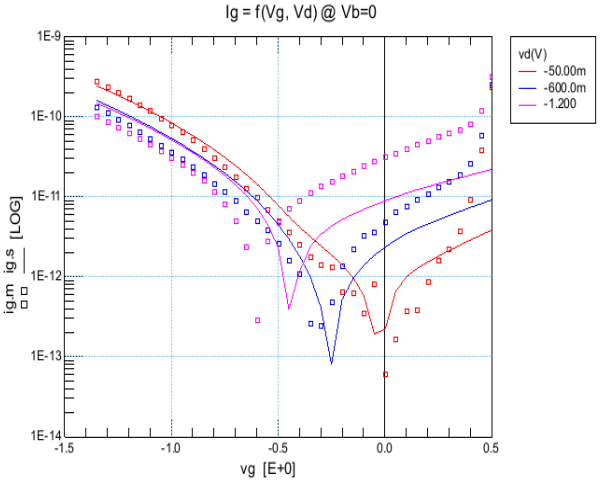




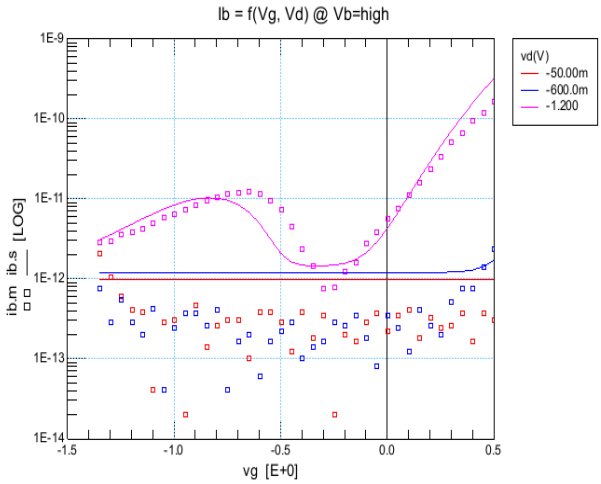


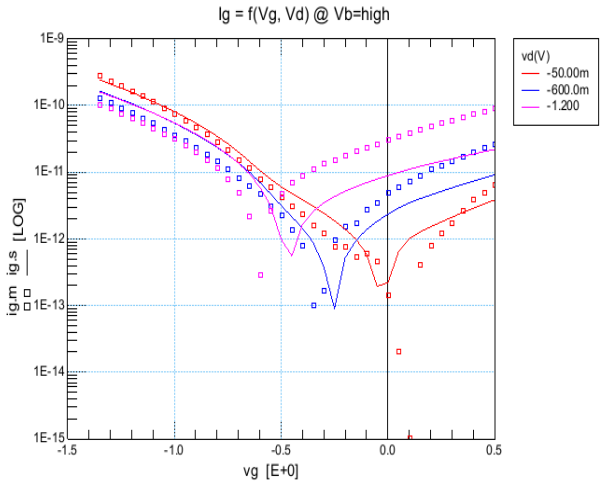
idvg, Ib,Ig, Vb = 0V, T = 27°C



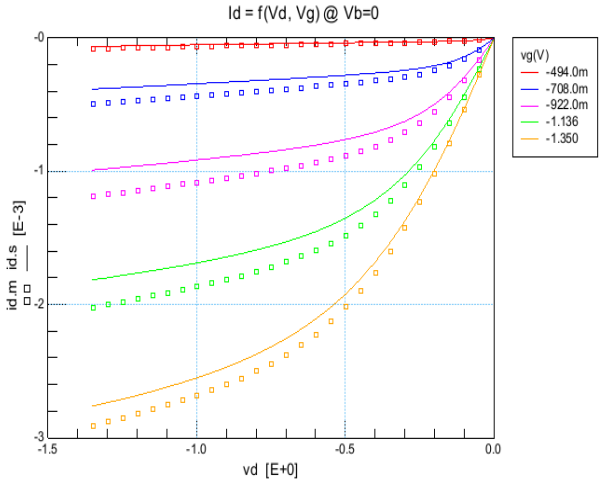


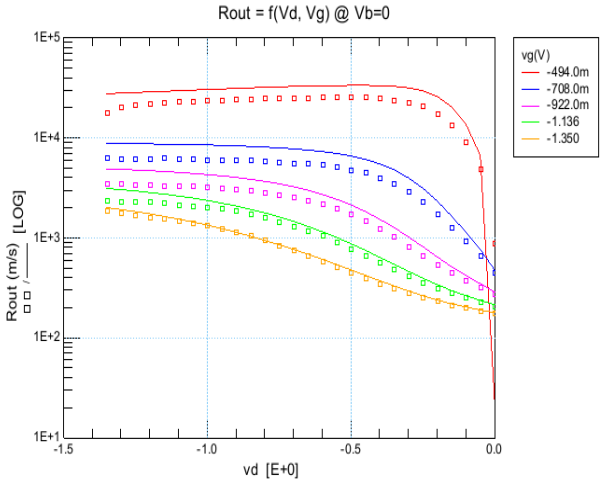
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

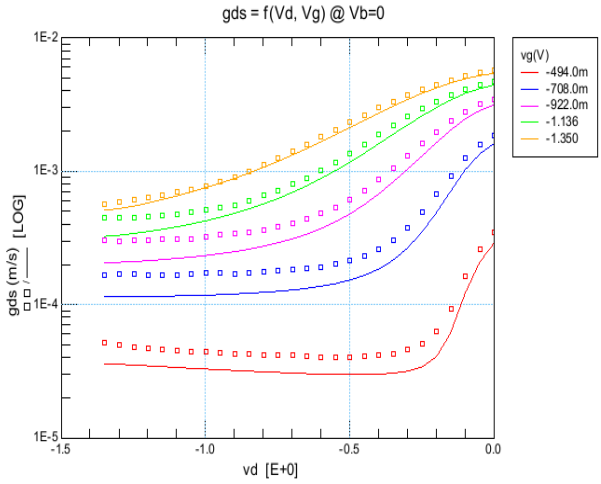




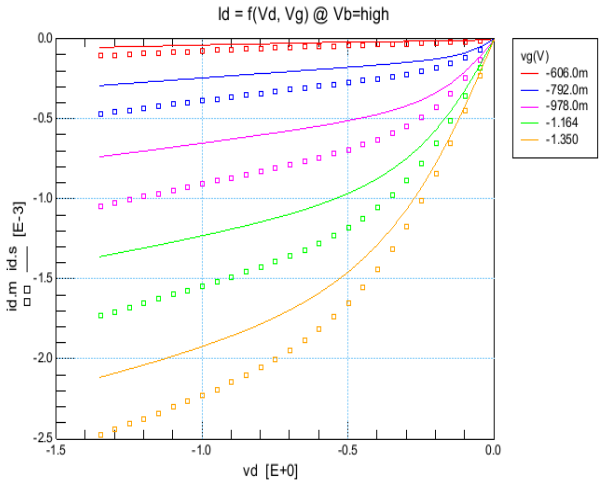
idvd, Vb = 0V, T = 27°C

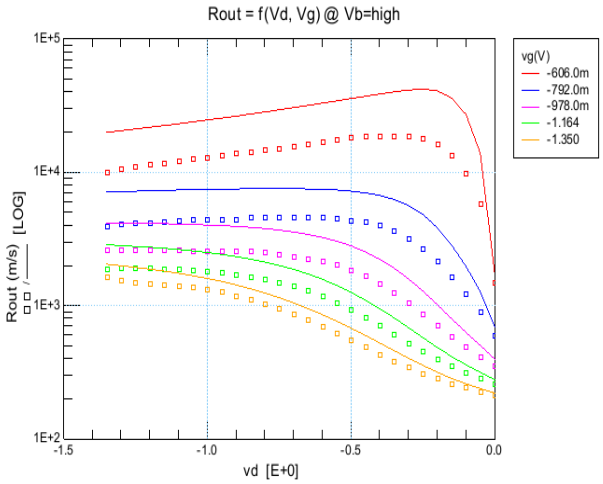


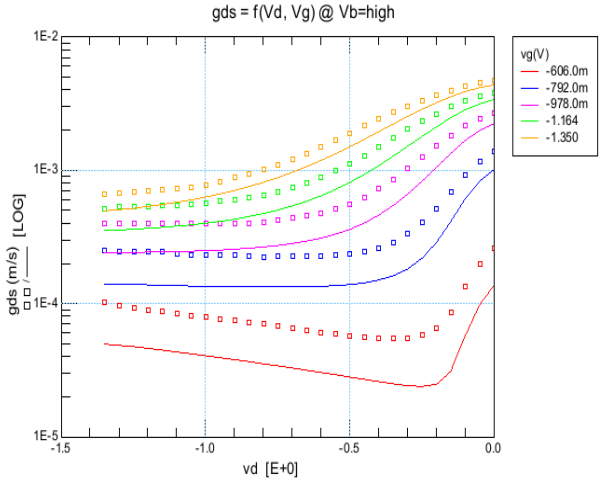




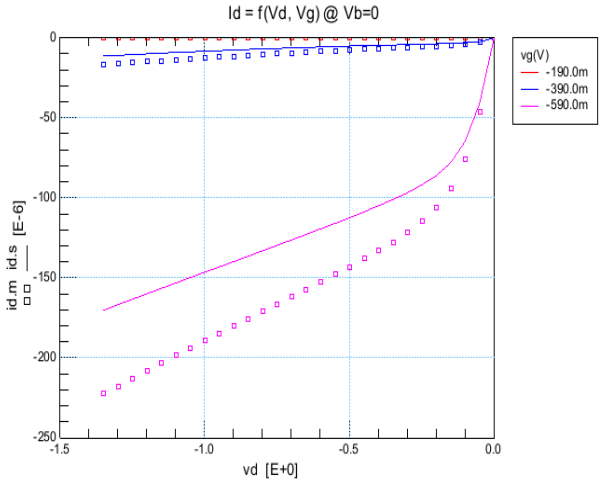
idvd_vbmin, Vb = 1.2V, T = 27°C

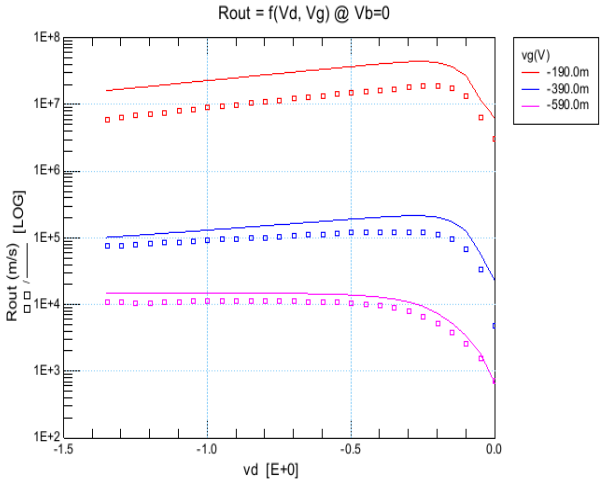


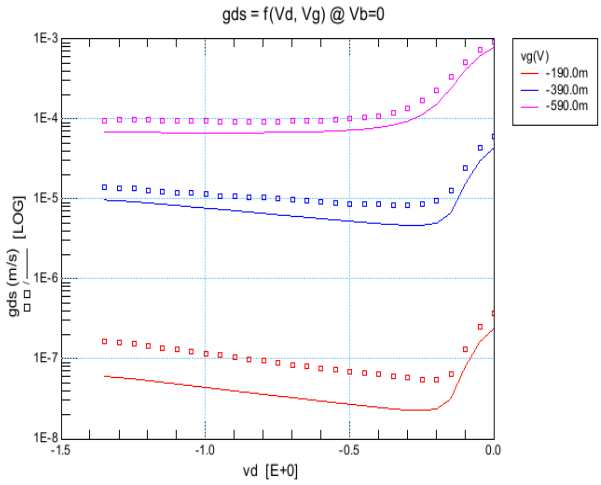




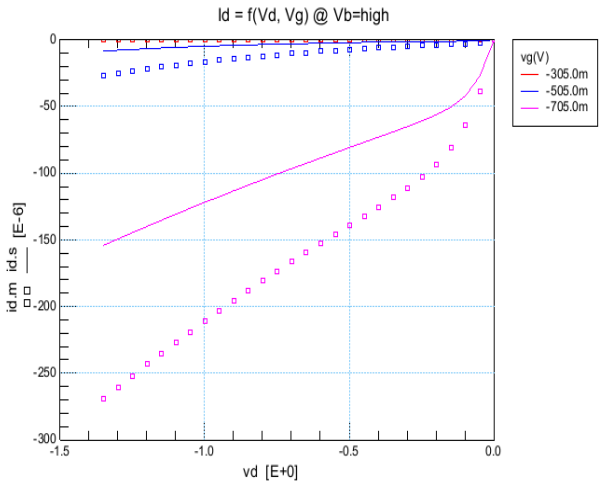
idvd_vth, Vb = 0V, T = 27°C

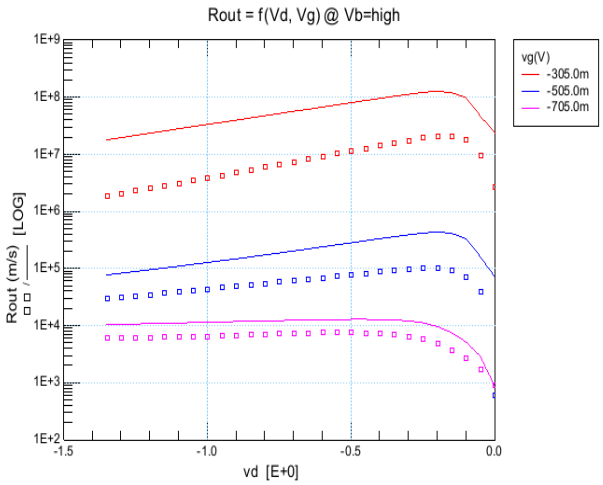


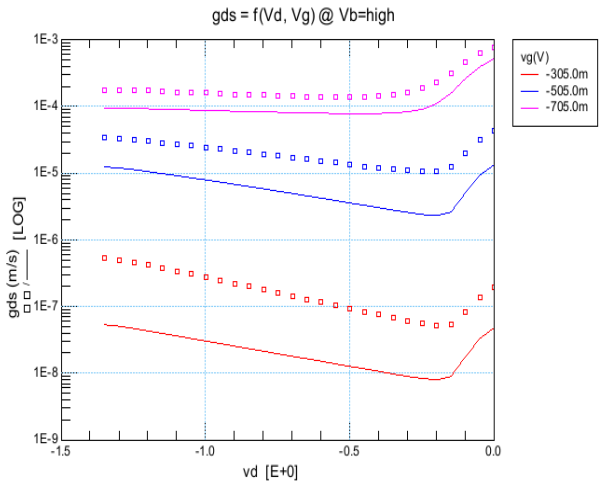




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

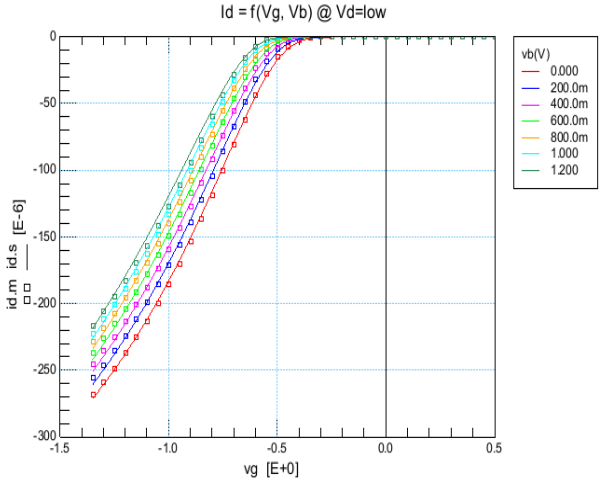


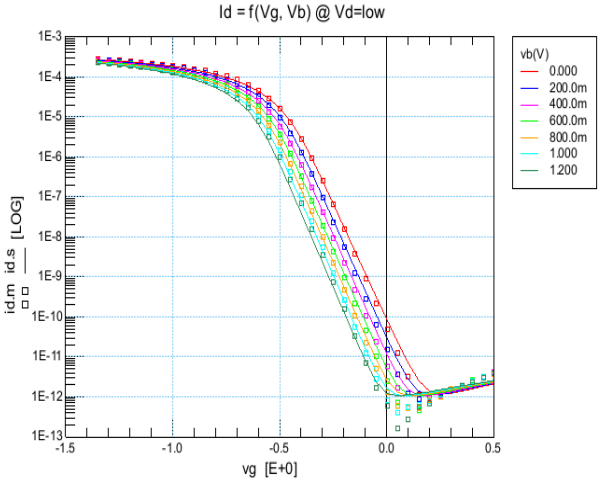


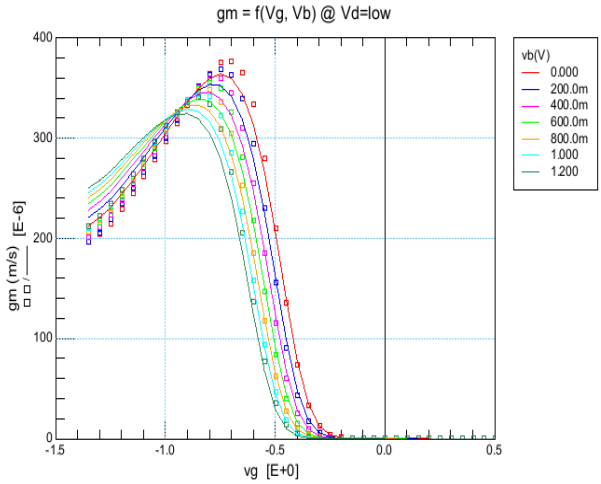


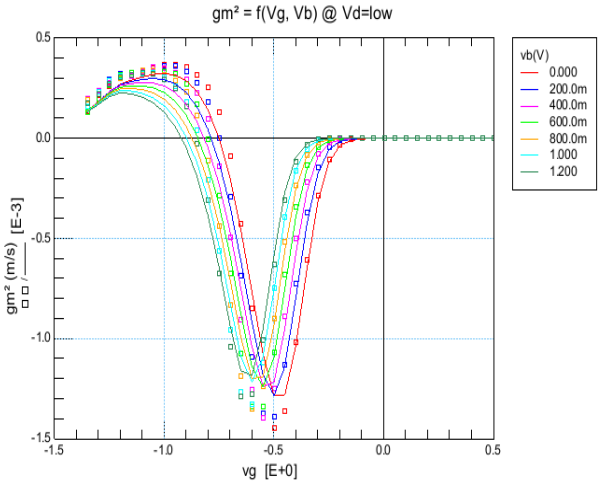
4.41 W10u0_L0u13_T356_S382_2

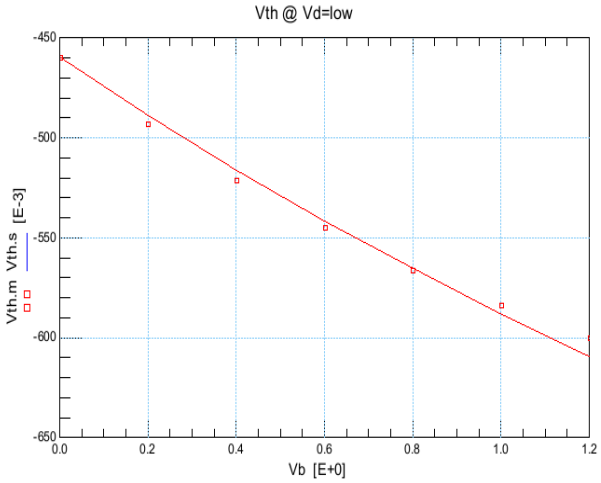
idvg, Vd = -0.05V, T = 27°C

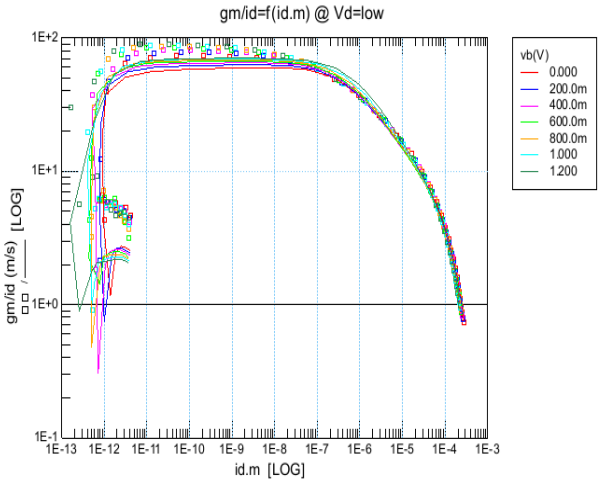


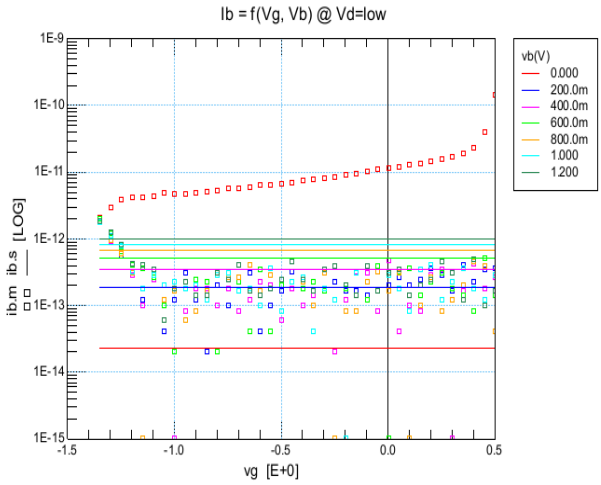


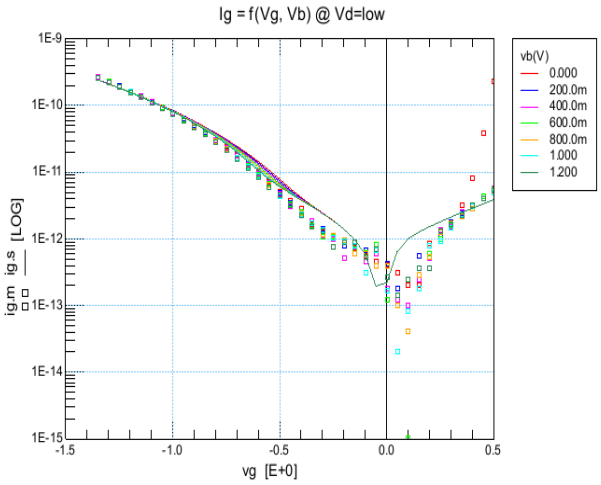




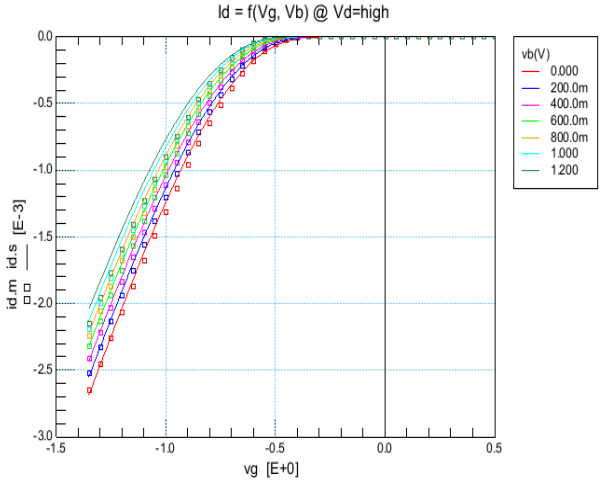


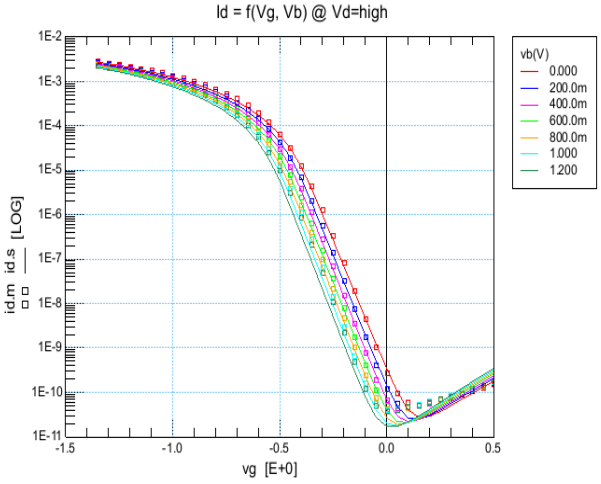


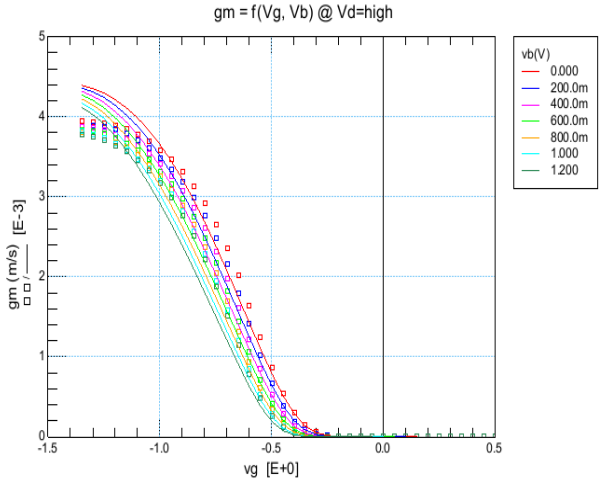


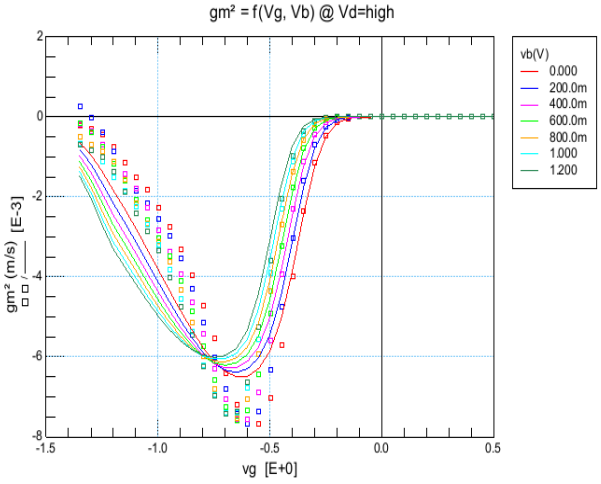


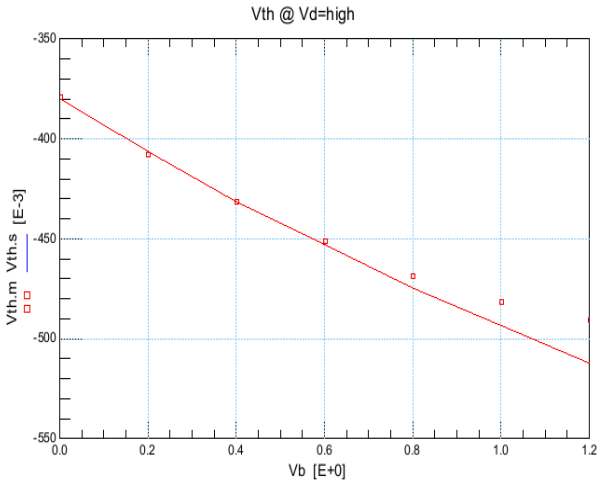
idvg, Vd = -1.2V, T = 27°C

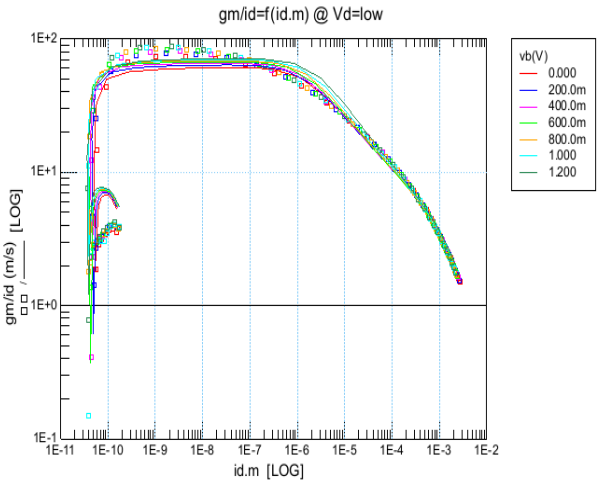


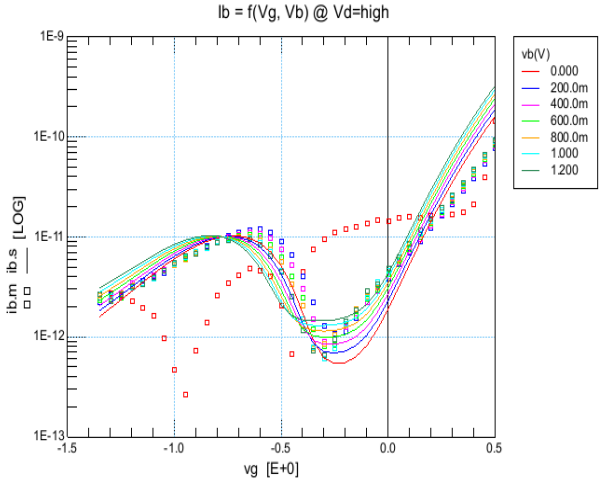


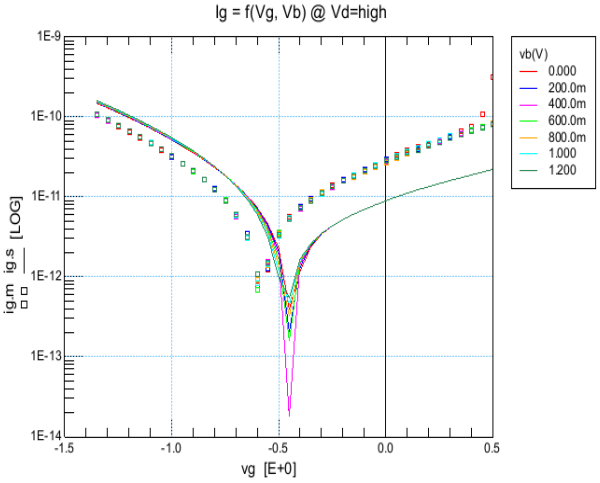




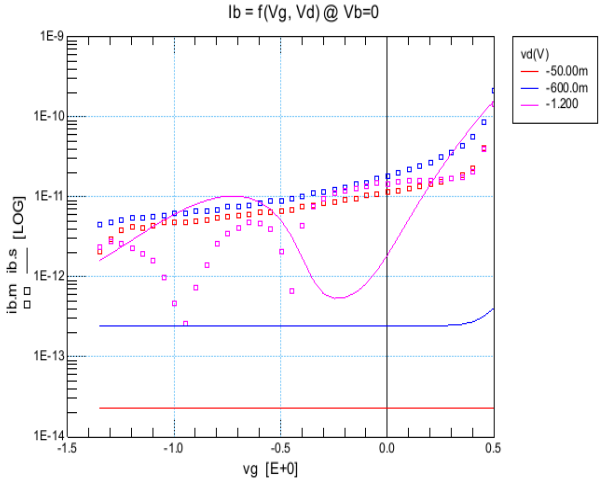


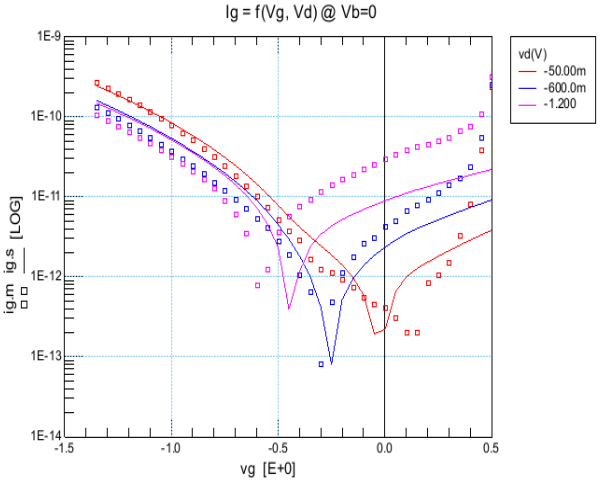




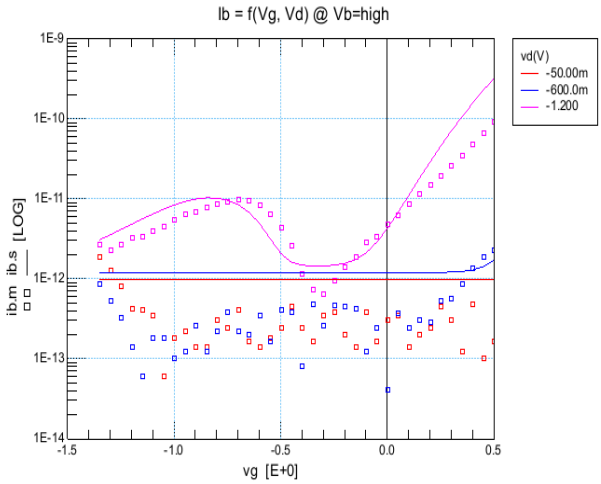


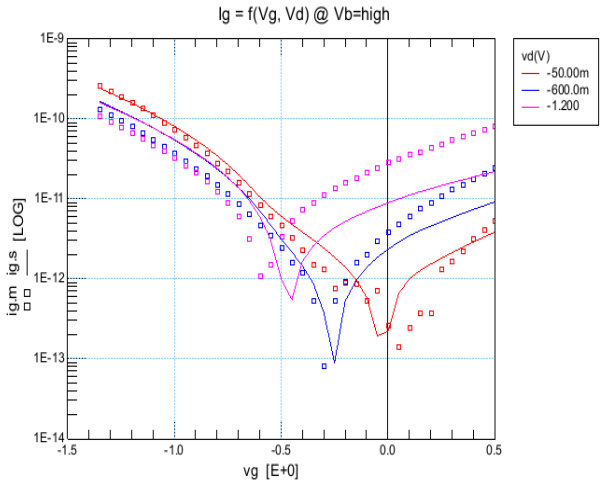
idvg, Ib,Ig, Vb = 0V, T = 27°C



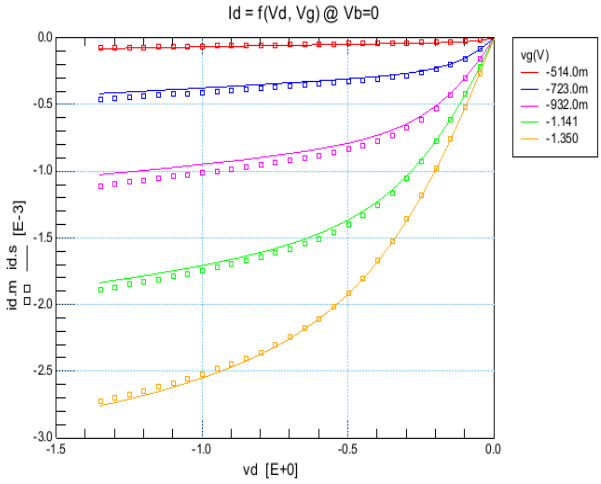


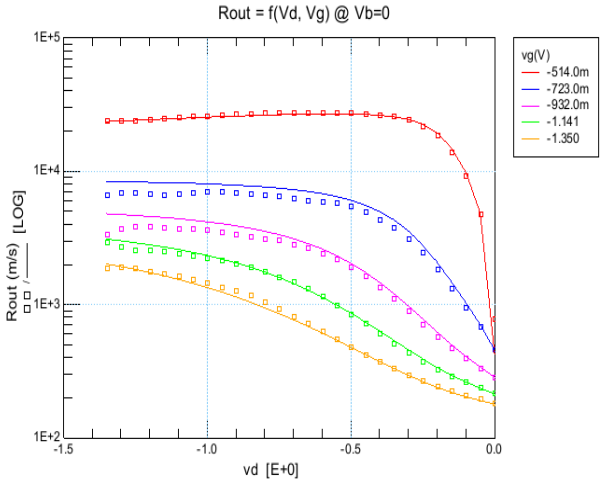
idvg, Ib,Ig, Vb = 1.2V, T = 27°C

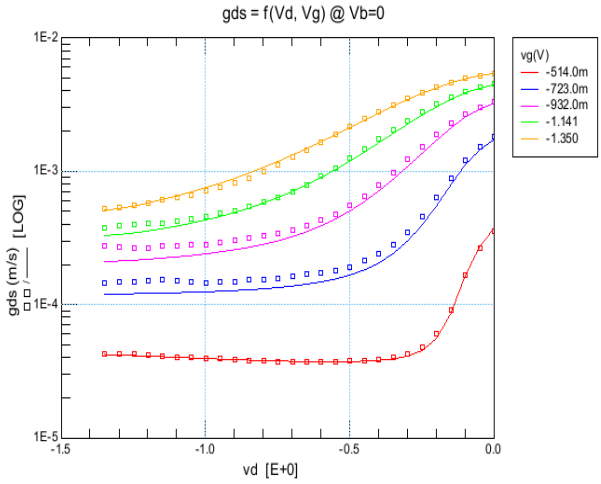




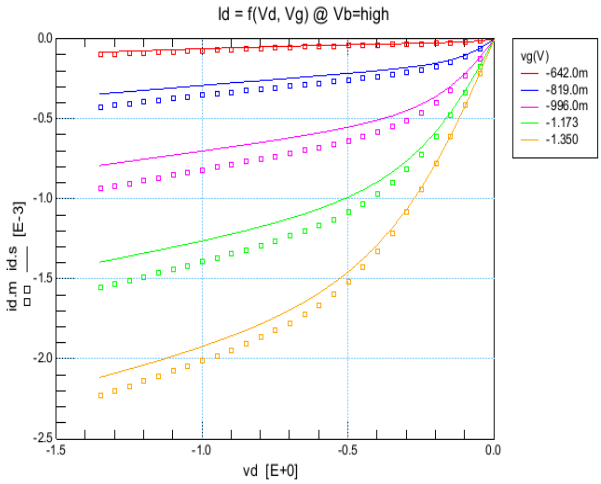
idvd, Vb = 0V, T = 27°C

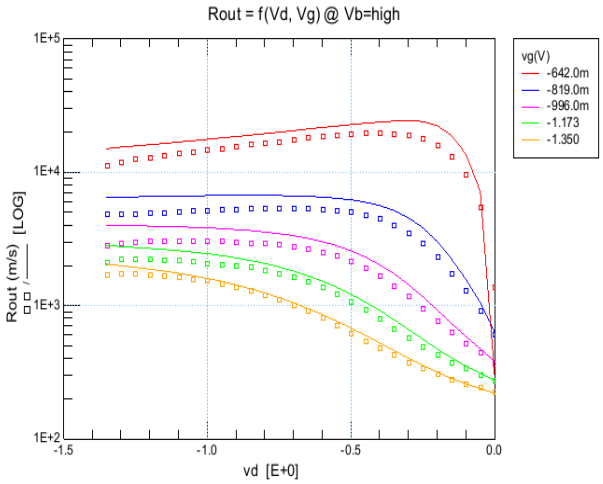


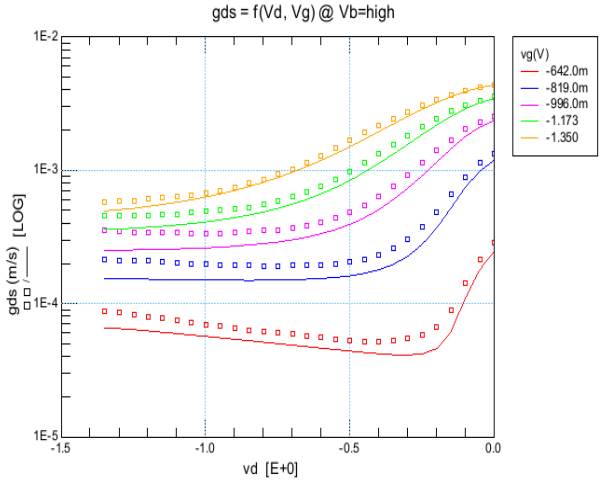




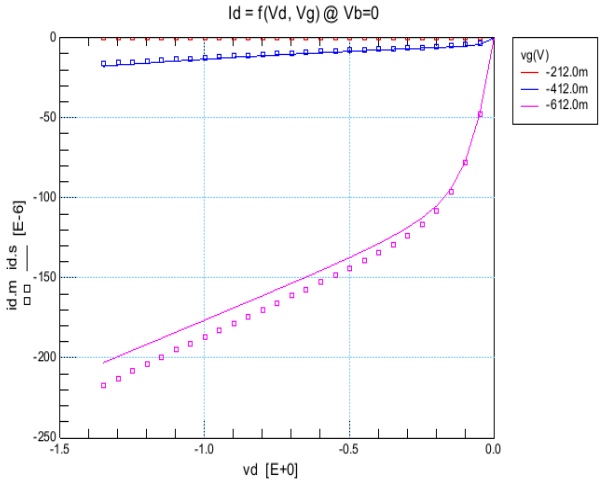
idvd_vbmin, Vb = 1.2V, T = 27°C

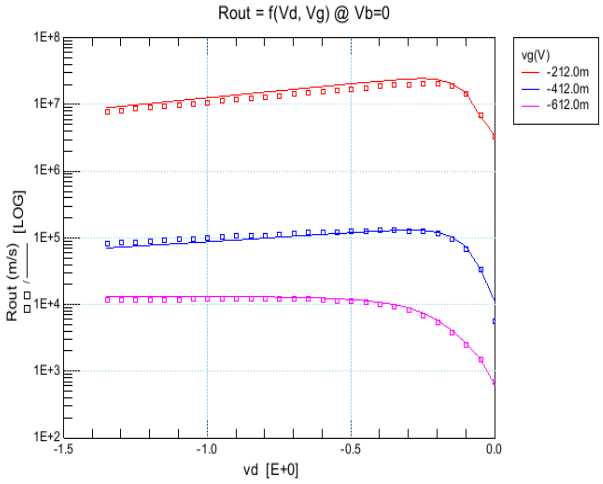


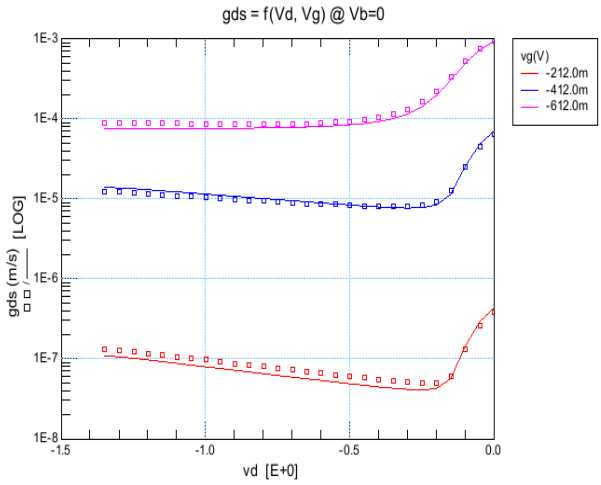




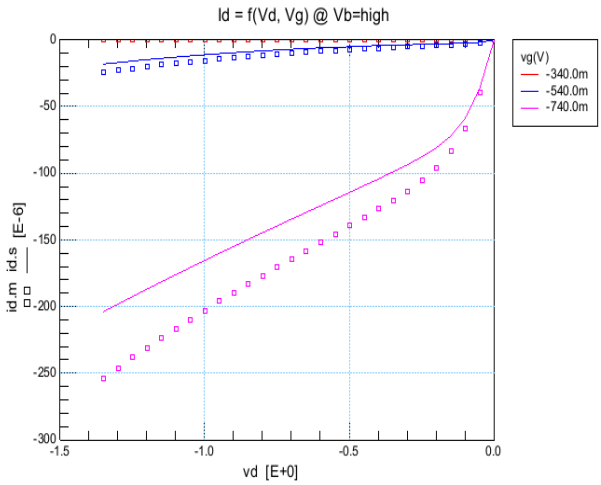
idvd_vth, Vb = 0V, T = 27°C

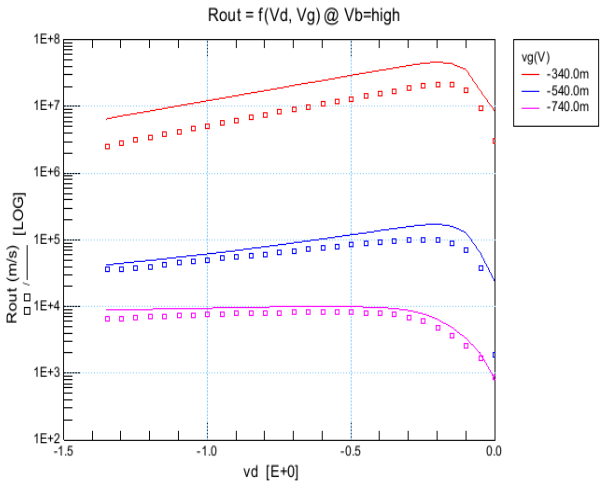


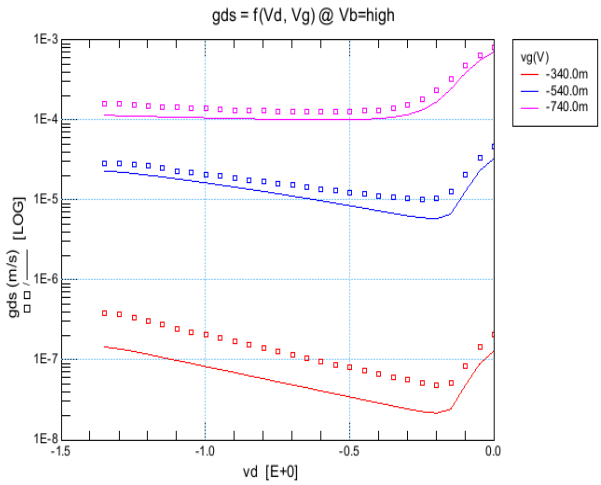




idvd_vth_vbmin, Vb = 1.2V, T = 27°C

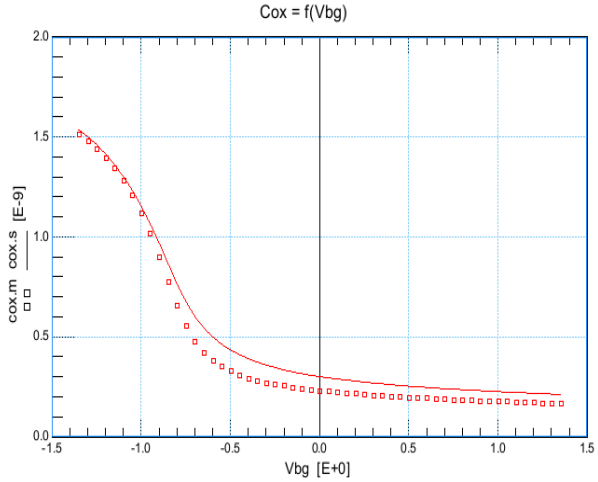




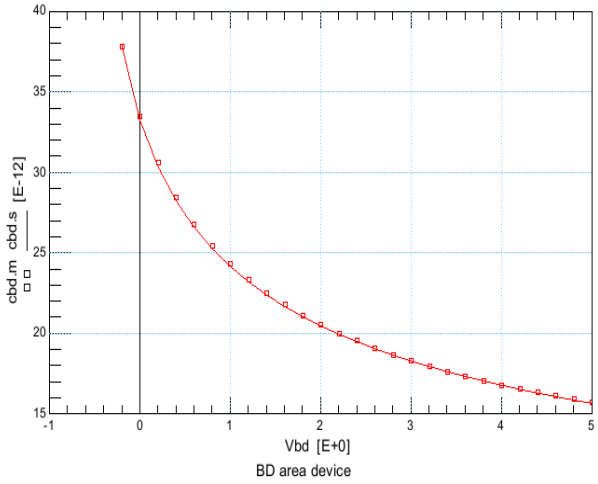


5 Capacitances

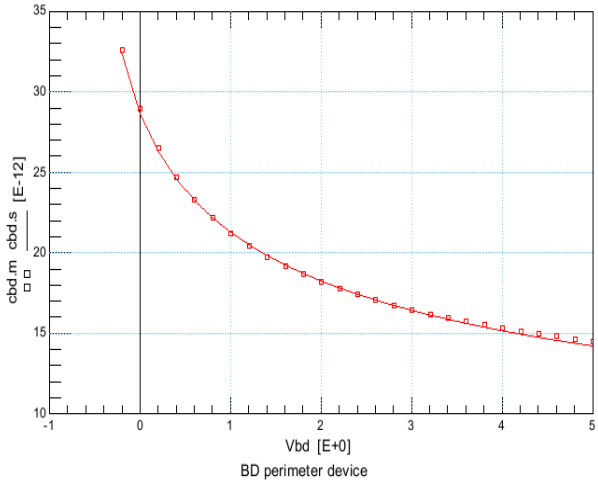
5.1 Gate Oxide Capacitance



5.2 Junction Capacitance Bulk - Drain

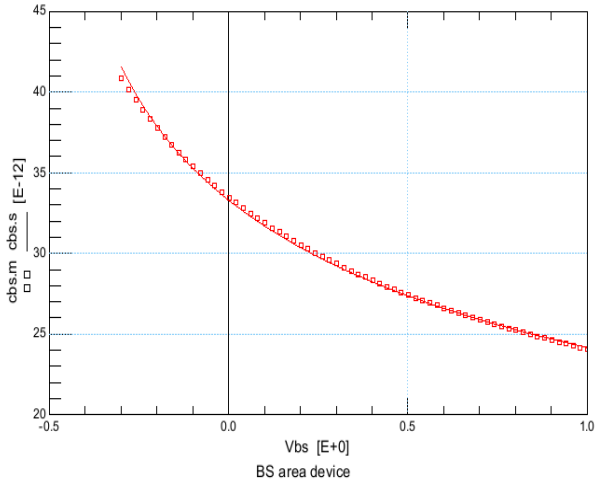


Area

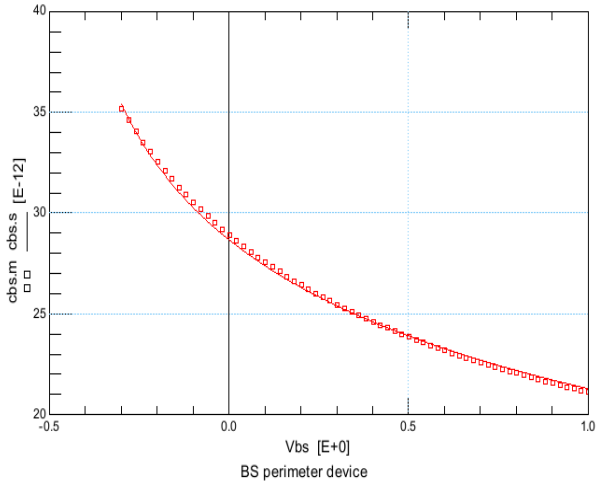


Perimeter

5.3 Junction Capacitance Bulk - Source

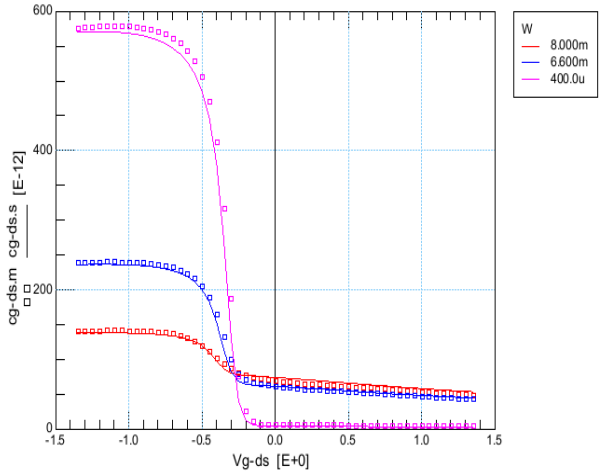


Area

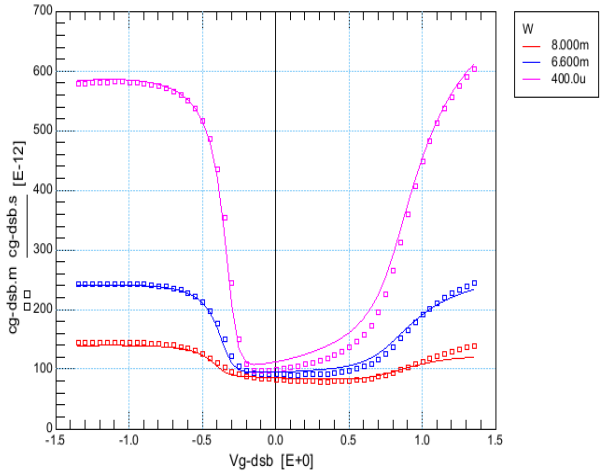


Perimeter

5.4 Overlap Capacitance



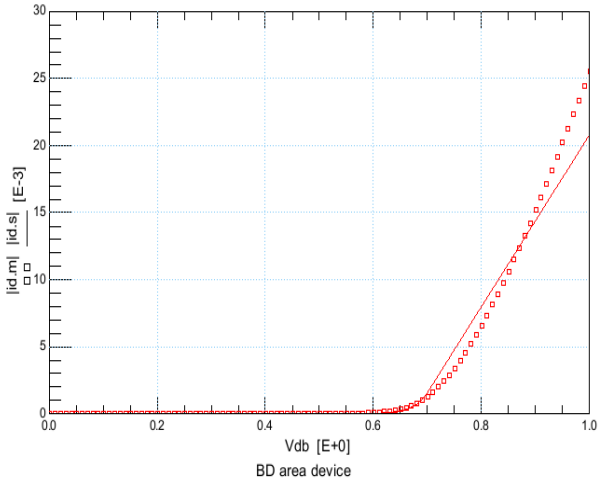
Gate - Drain,Source



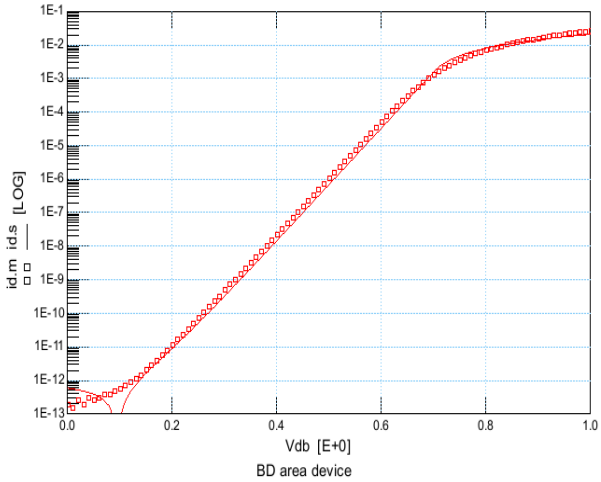
Gate - Drain,Source,Bulk

6 Diodes

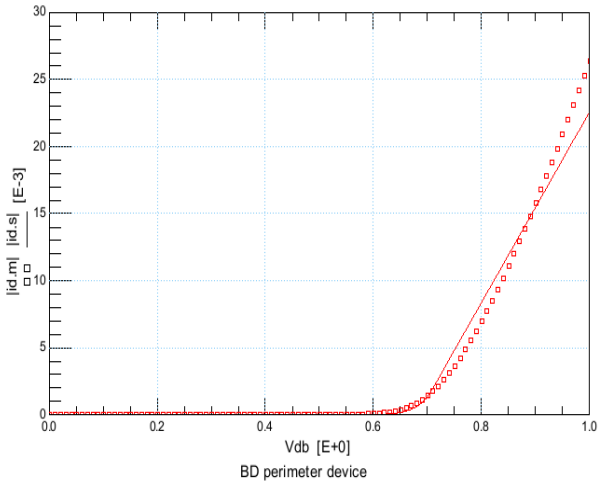
6.1 Diode Bulk - Drain



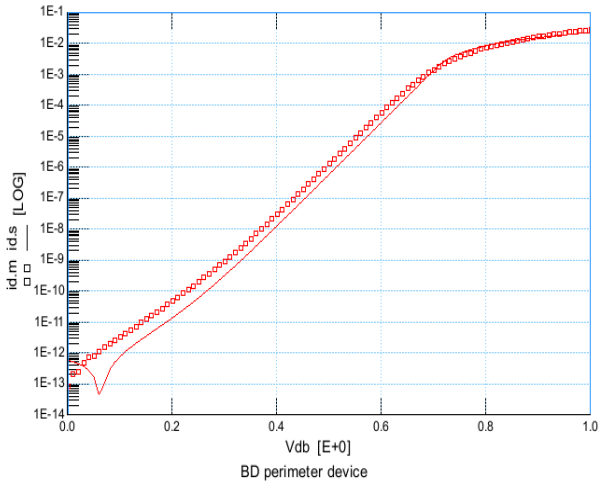
Area



Area

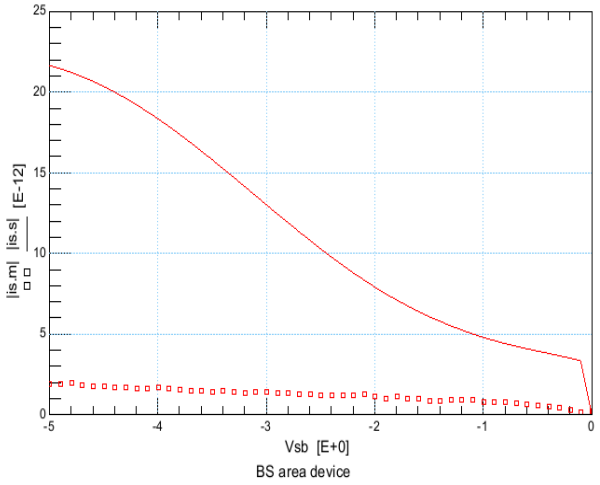


Perimeter

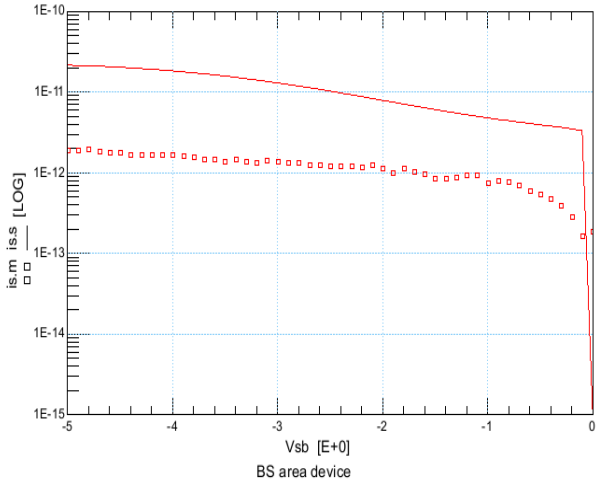


Perimeter

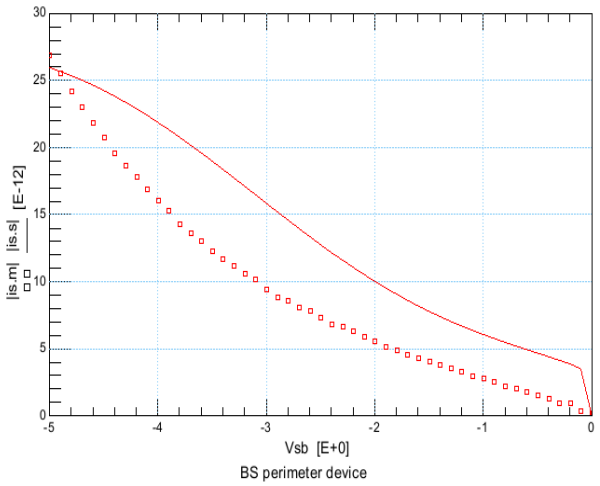
6.2 Diode Bulk - Source



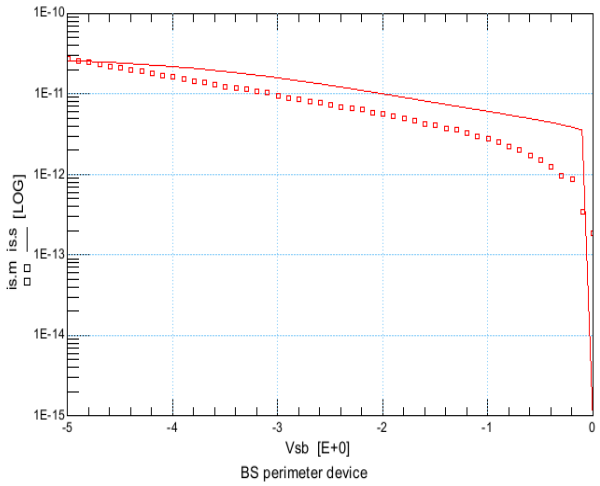
Area



Area



Perimeter

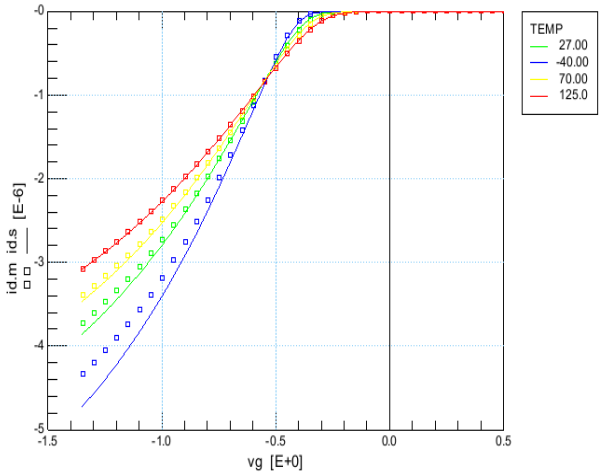


Perimeter

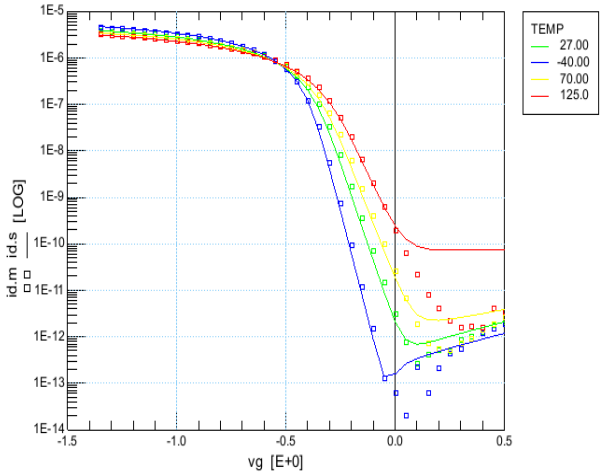
7 Temperature Measurement

7.1 Transistors

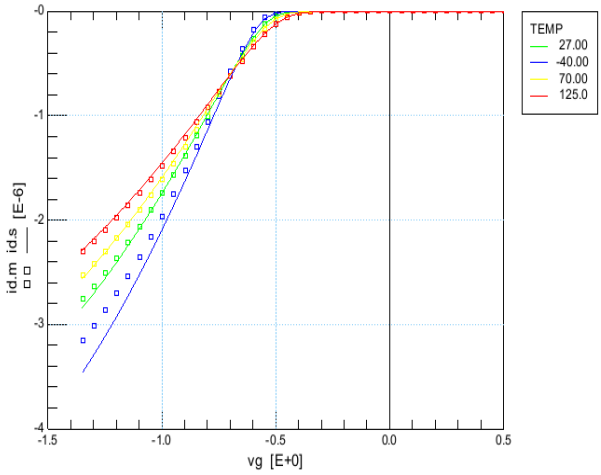
7.1.1 W10u0_L10u0_S549_5



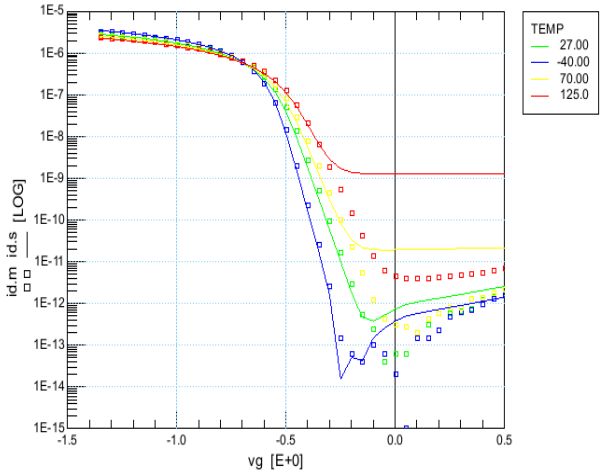
Id = f (vg, T) @ low Vds, Vbs = 0



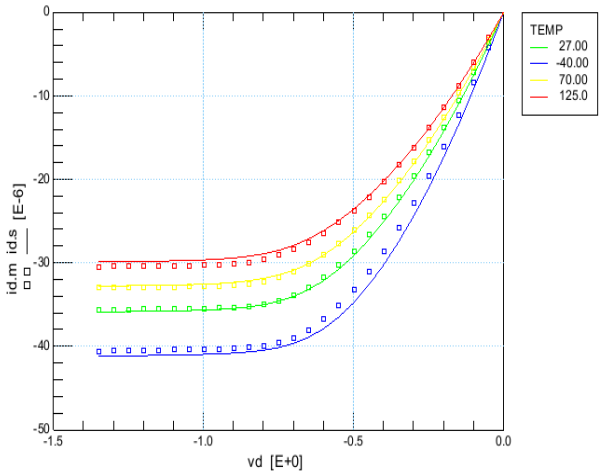
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

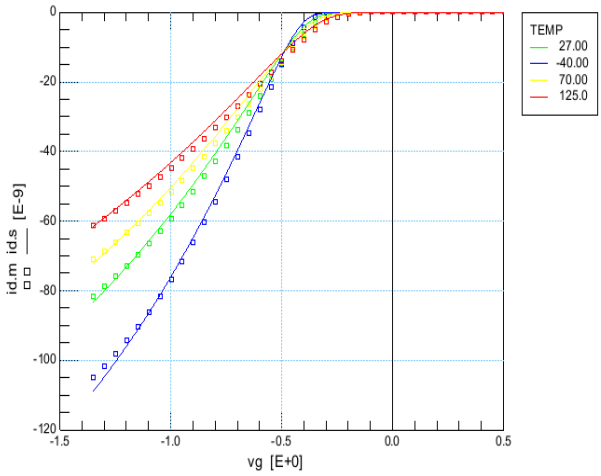


log(Id) = f (vg, T) @ low Vds, Vbs = max

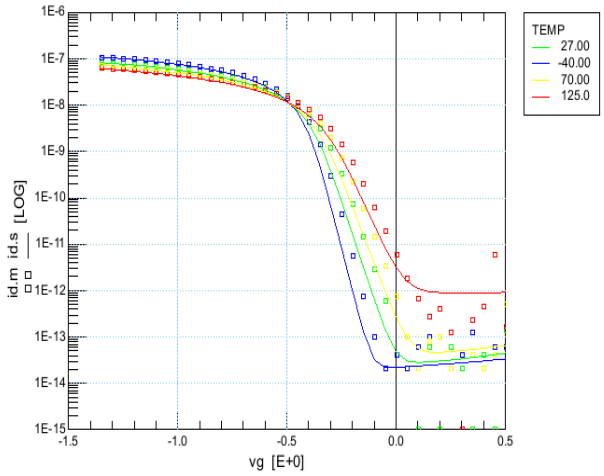


Id = f (Vd, Vg, T) @ Vbs = 0

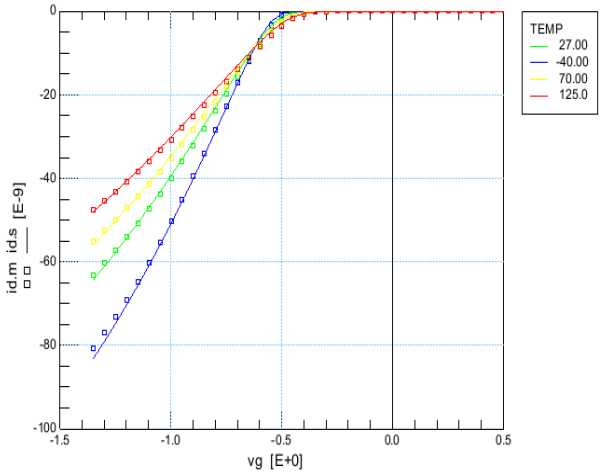
7.1.2 W0u15_L10u0_S555_3



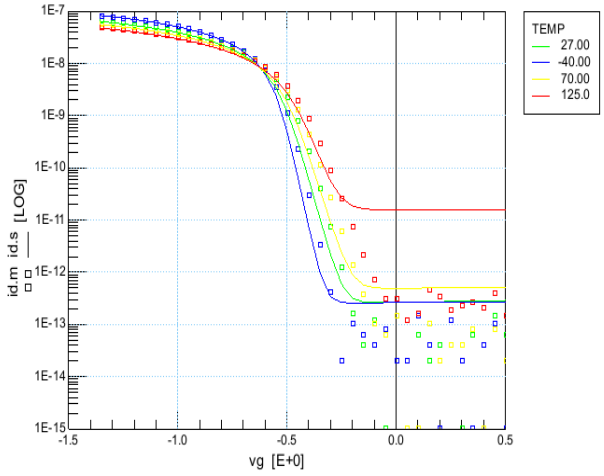
Id = f (vg, T) @ low Vds, Vbs = 0



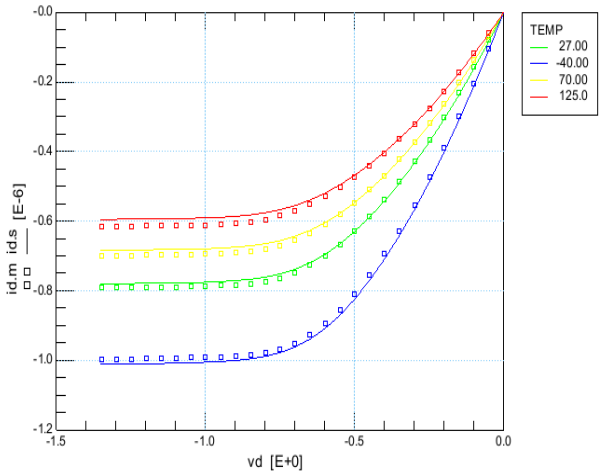
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

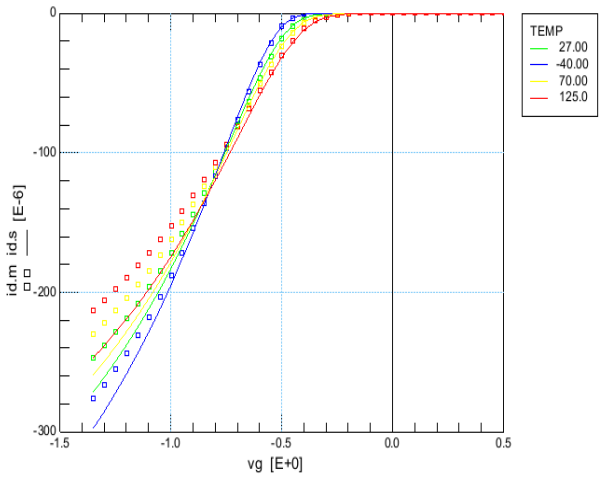


log(Id) = f (vg, T) @ low Vds, Vbs = max

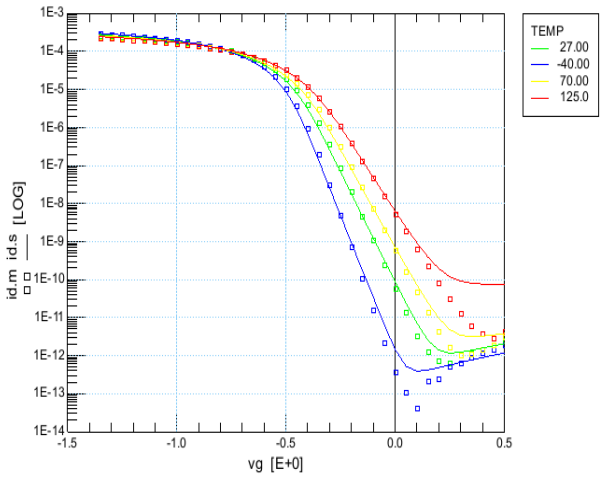


Id = f (Vd, Vg, T) @ Vbs = 0

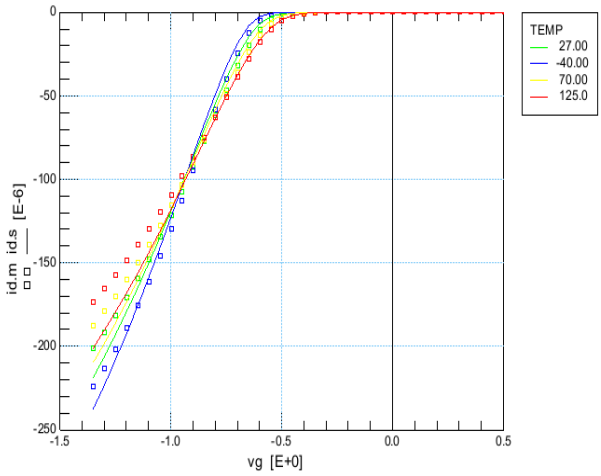
7.1.3 W10u0_L0u13_S548_2



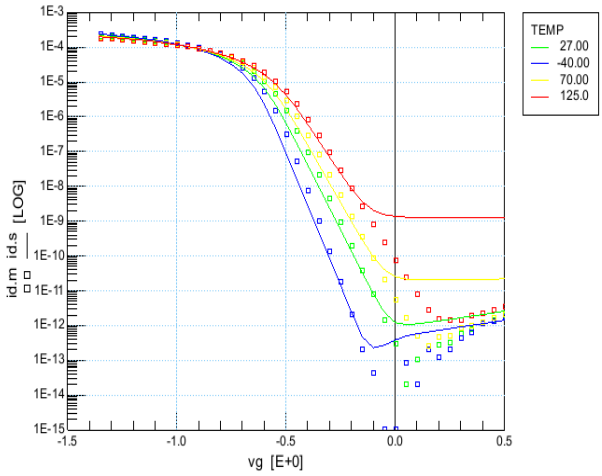
Id = f (vg, T) @ low Vds, Vbs = 0



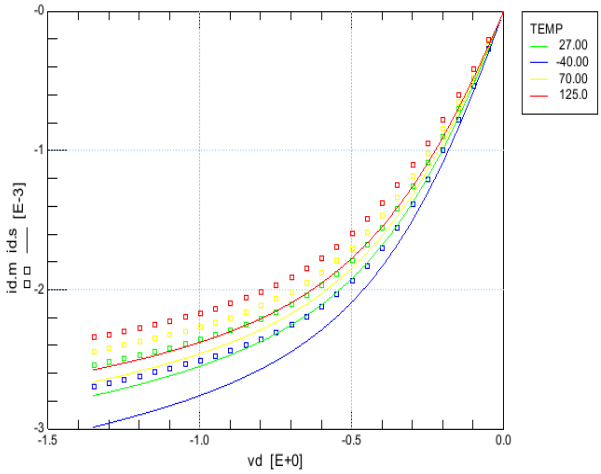
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

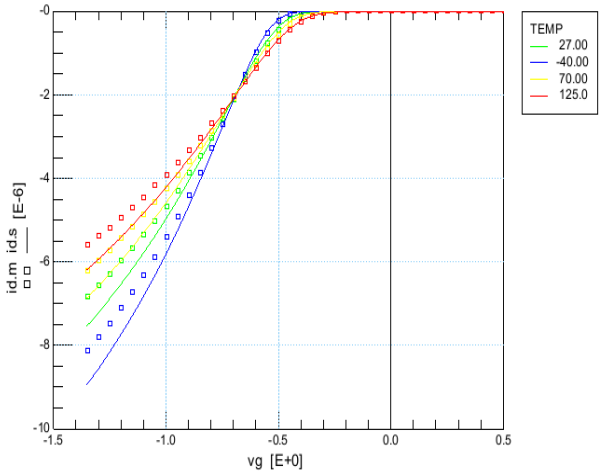


log(Id) = f (vg, T) @ low Vds, Vbs = max

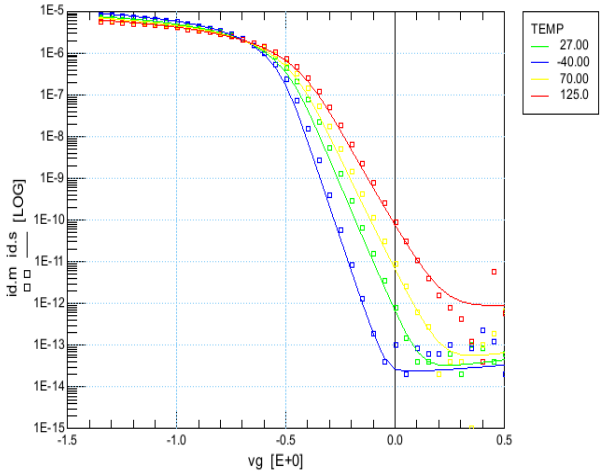


Id = f (Vd, Vg, T) @ Vbs = 0

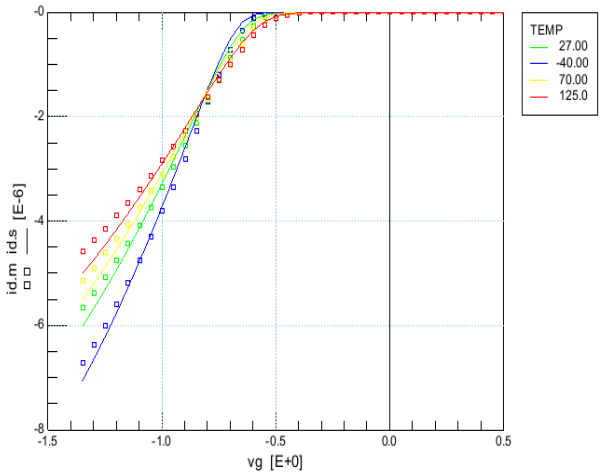
7.1.4 W0u15_L0u13_S554_3



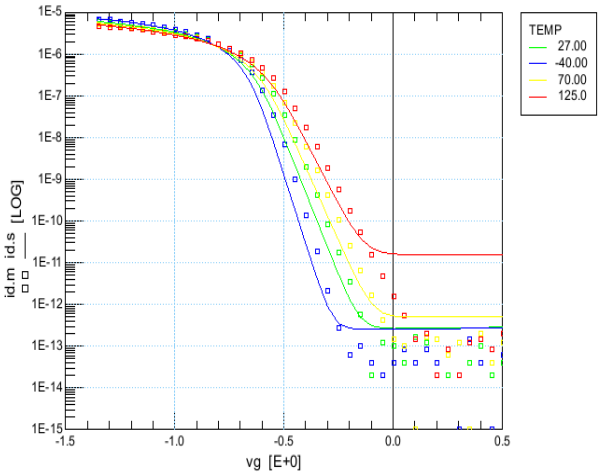
Id = f (vg, T) @ low Vds, Vbs = 0



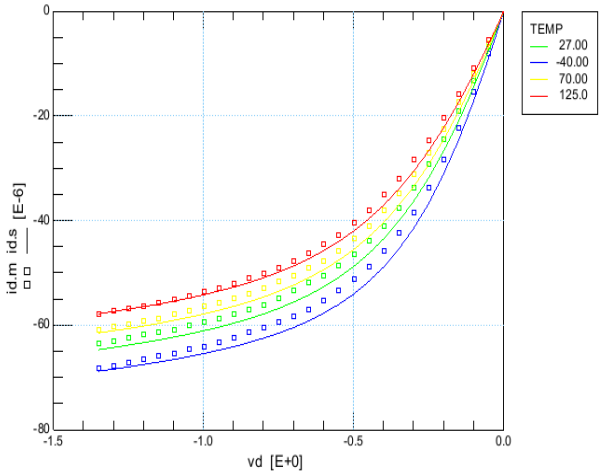
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

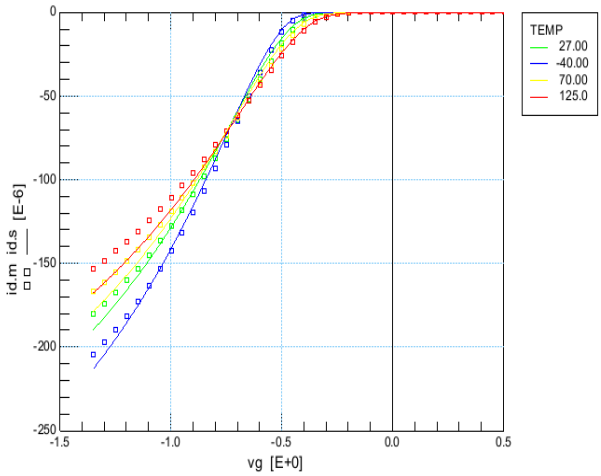


log(Id) = f (vg, T) @ low Vds, Vbs = max

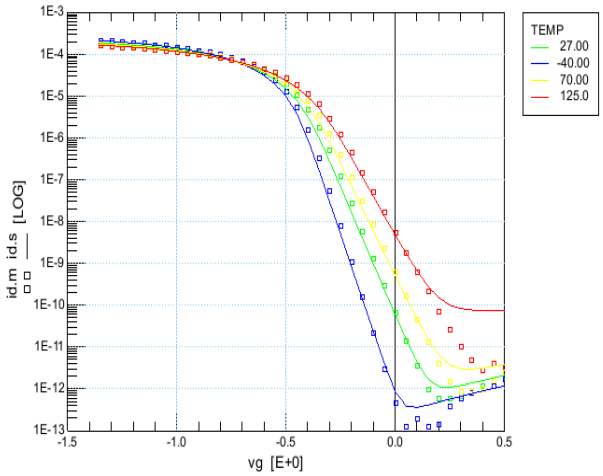


Id = f (Vd, Vg, T) @ Vbs = 0

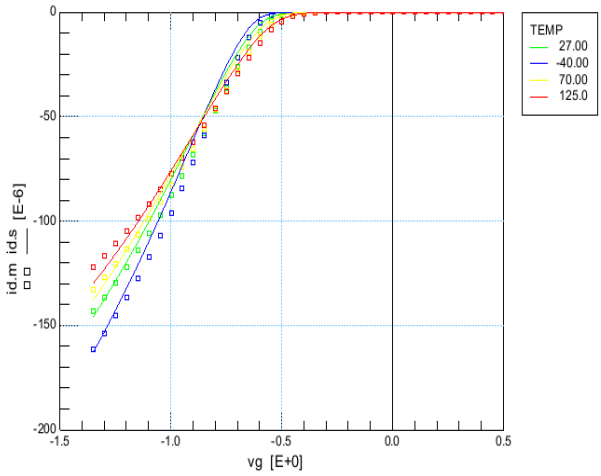
7.1.5 W10u0_L0u18_S548_5



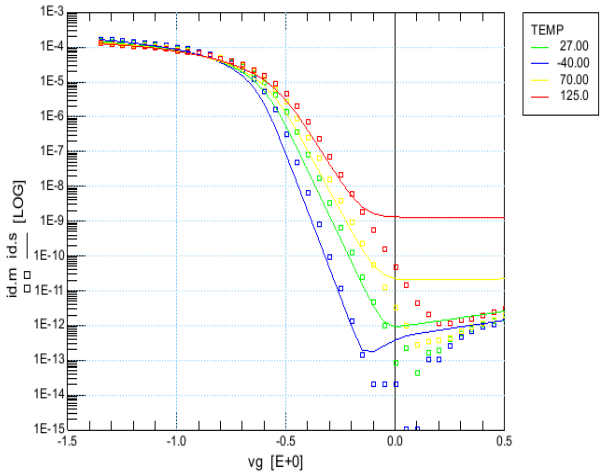
Id = f (vg, T) @ low Vds, Vbs = 0



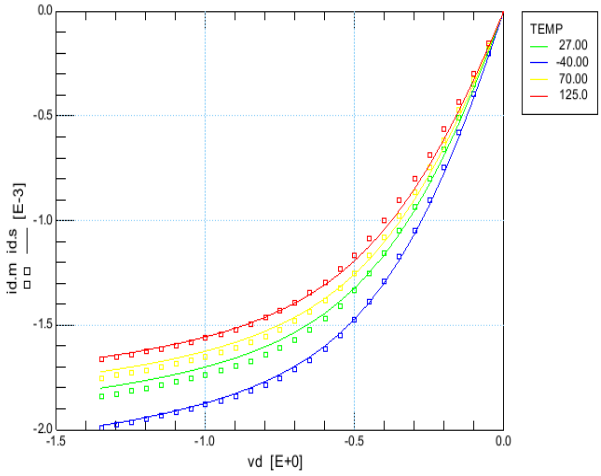
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

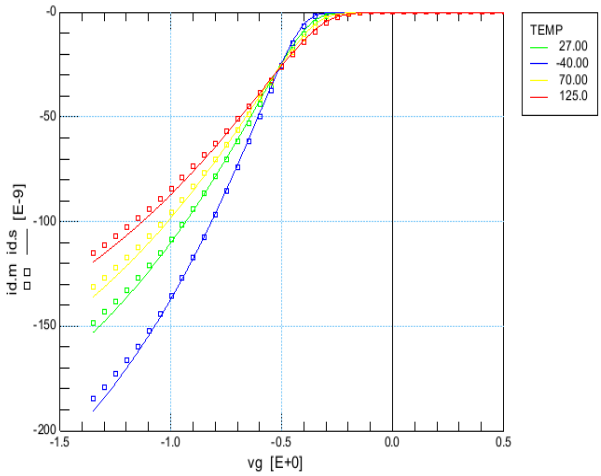


log(Id) = f (vg, T) @ low Vds, Vbs = max

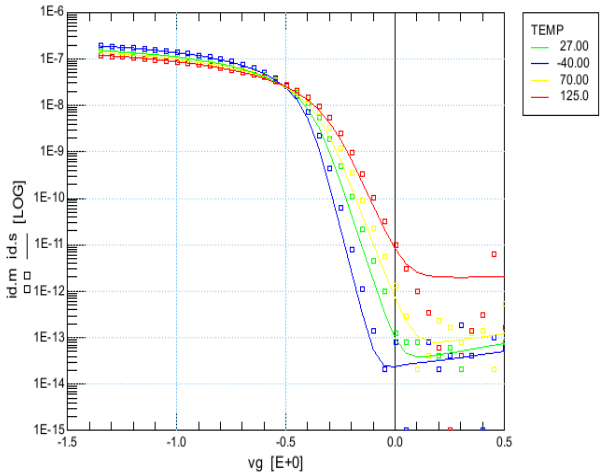


Id = f (Vd, Vg, T) @ Vbs = 0

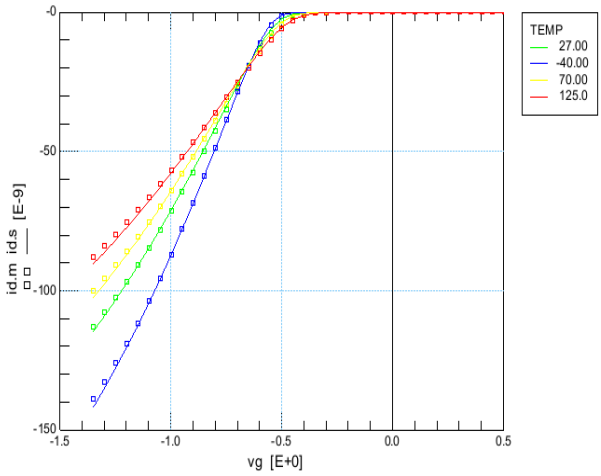
7.1.6 W0u3_L10u0_S553_2



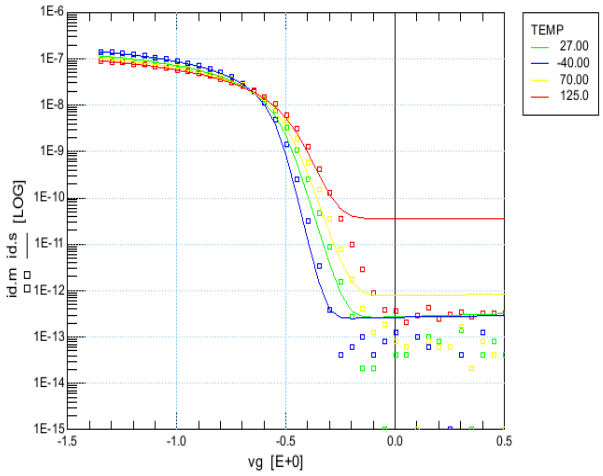
Id = f (vg, T) @ low Vds, Vbs = 0



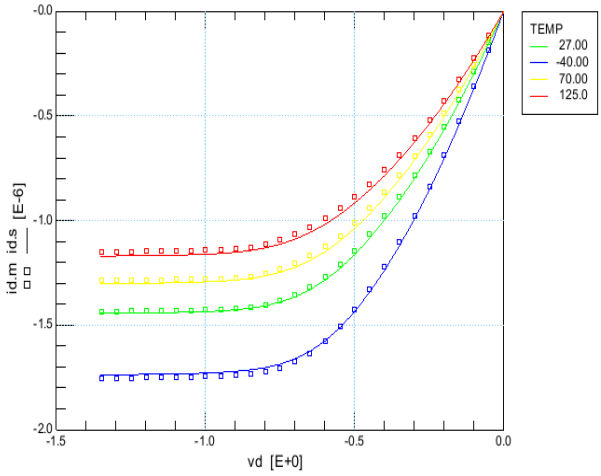
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

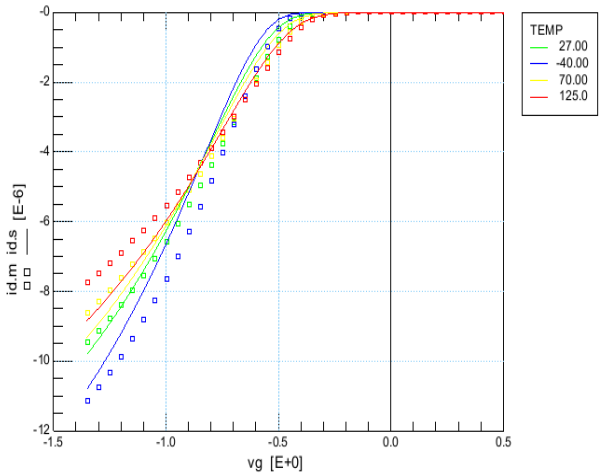


log(Id) = f (vg, T) @ low Vds, Vbs = max

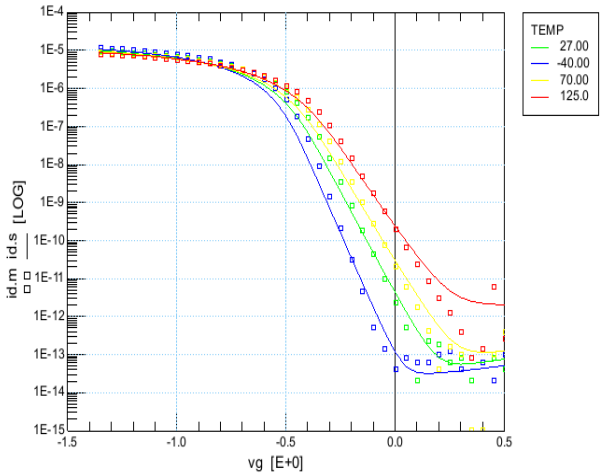


Id = f (Vd, Vg, T) @ Vbs = 0

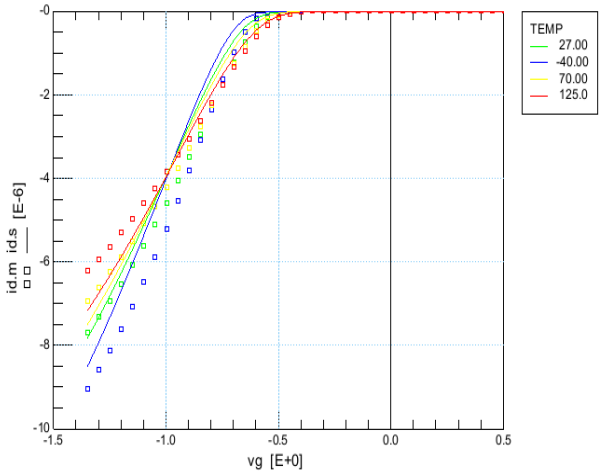
7.1.7 W0u3_L0u13_S552_2



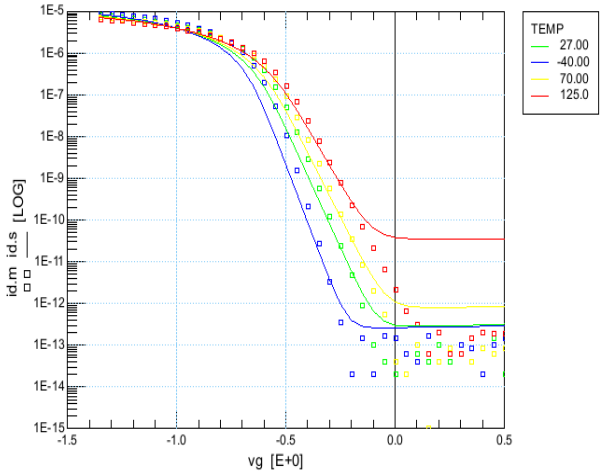
Id = f (vg, T) @ low Vds, Vbs = 0



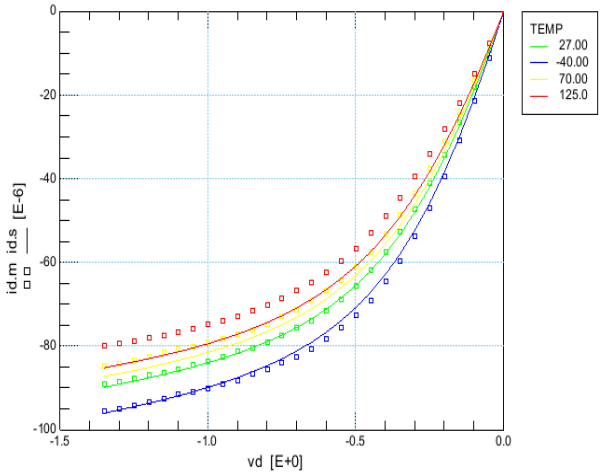
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max

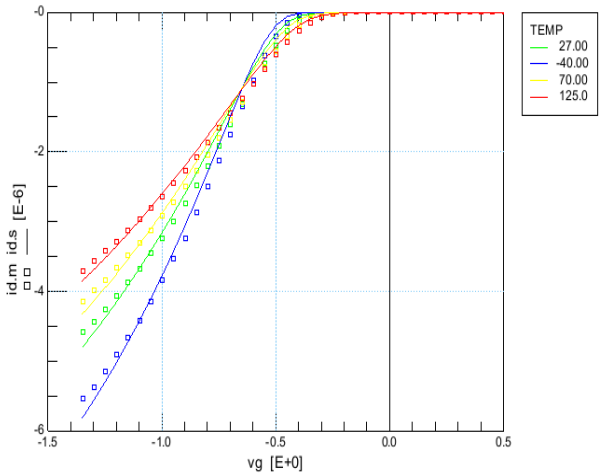


log(Id) = f (vg, T) @ low Vds, Vbs = max

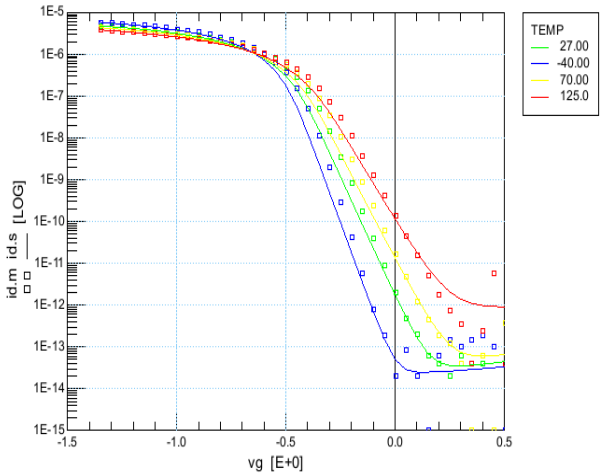


Id = f (Vd, Vg, T) @ Vbs = 0

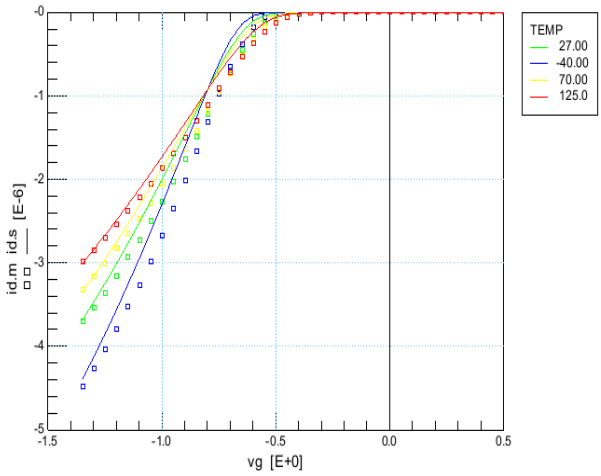
7.1.8 W0u15_L0u18_S555_1



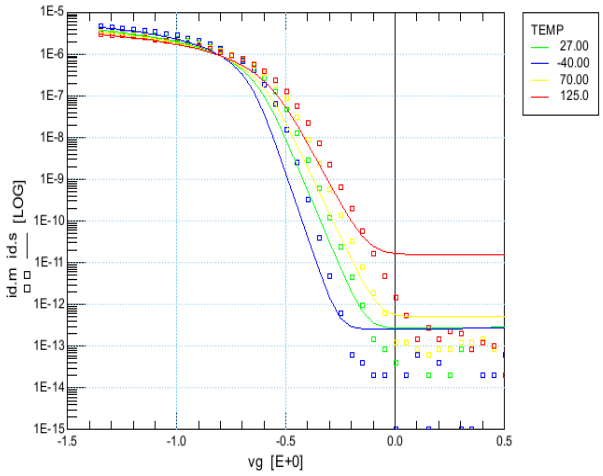
Id = f (vg, T) @ low Vds, Vbs = 0



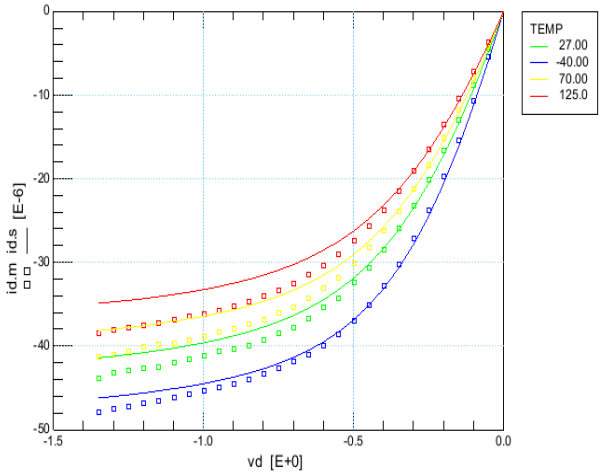
log(Id) = f (vg, T) @ low Vds, Vbs = 0



Id = f (vg, T) @ low Vds, Vbs = max



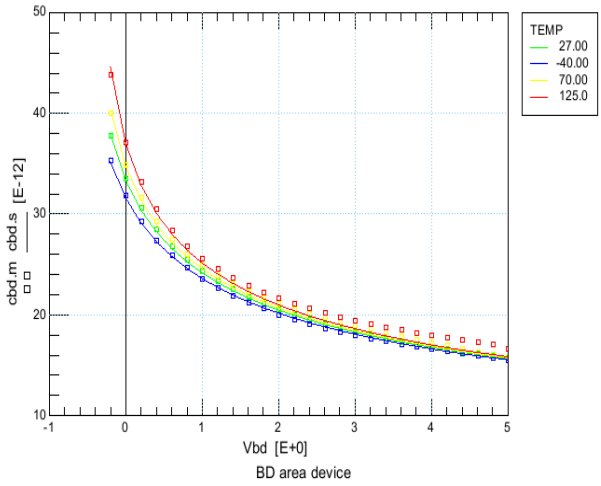
log(Id) = f (vg, T) @ low Vds, Vbs = max



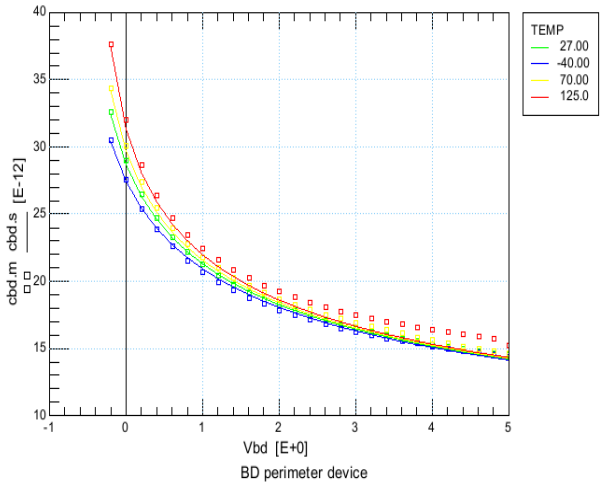
Id = f (Vd, Vg, T) @ Vbs = 0

7.2 Junction Capacitance

7.2.1 Bulk - Drain

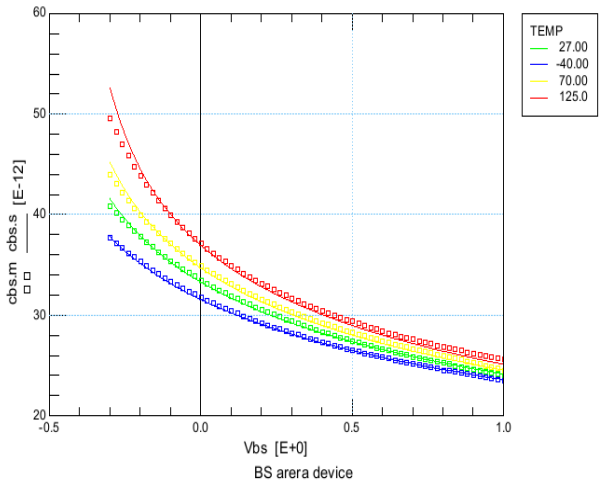


Area

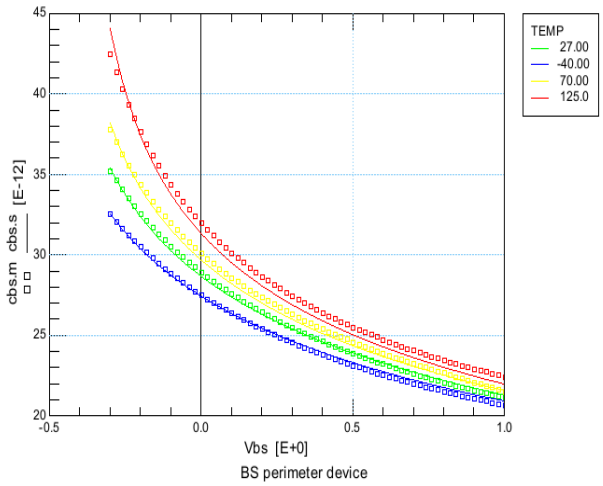


Perimeter

7.2.2 Bulk - Source



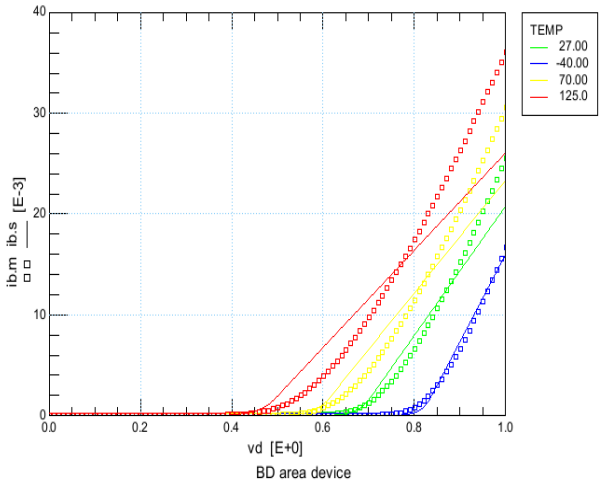
Area



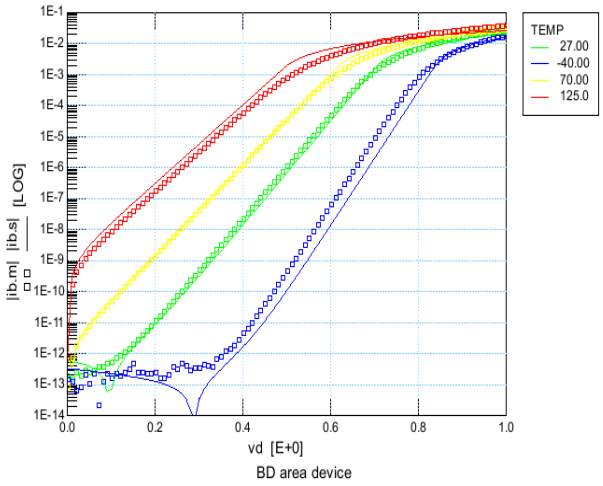
Perimeter

7.3 Parasitic Diodes

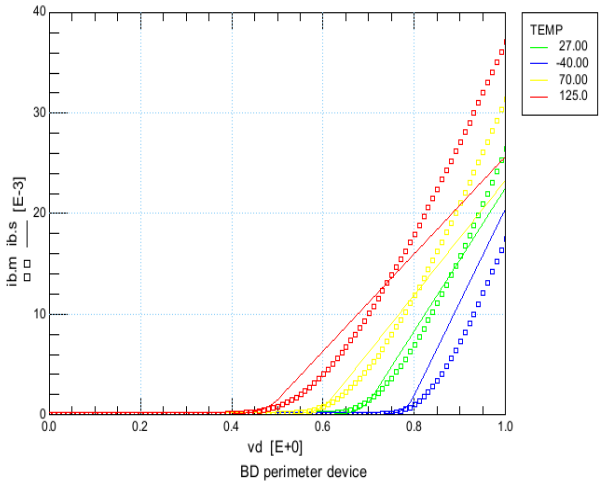
7.3.1 Bulk - Drain



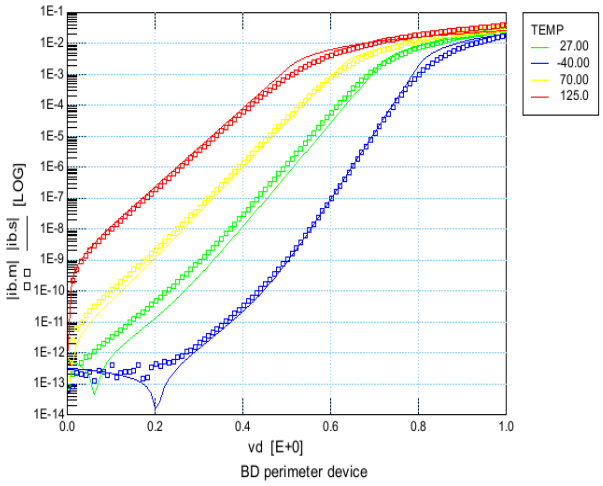
Area



Area

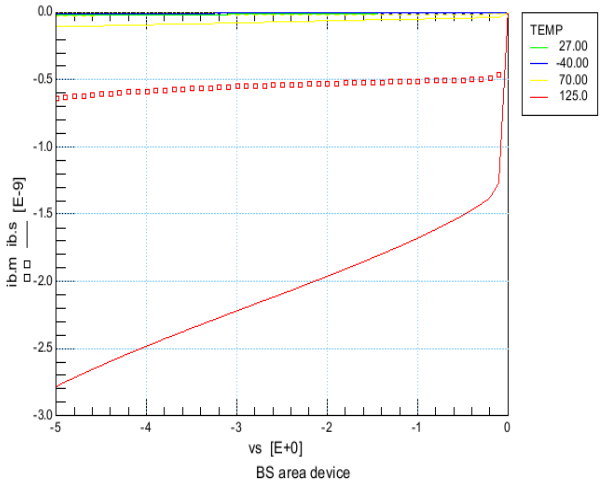


Perimeter

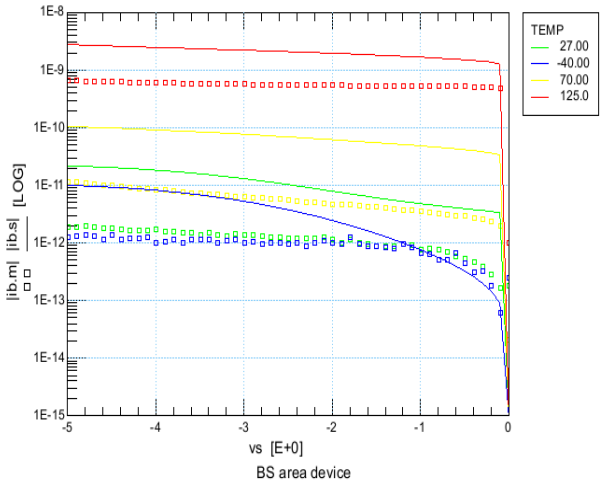


Perimeter

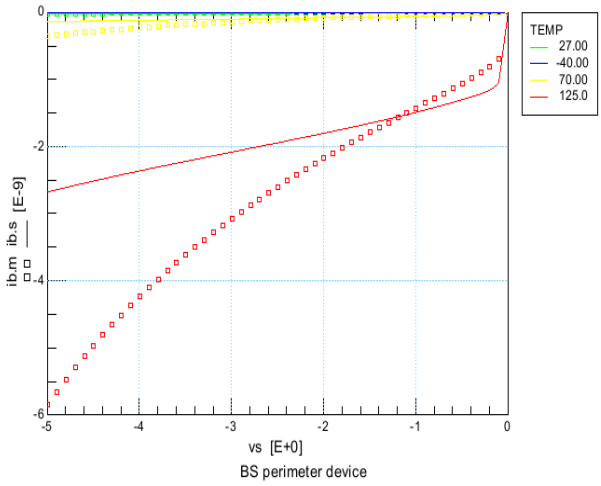
7.3.2 Bulk - Source



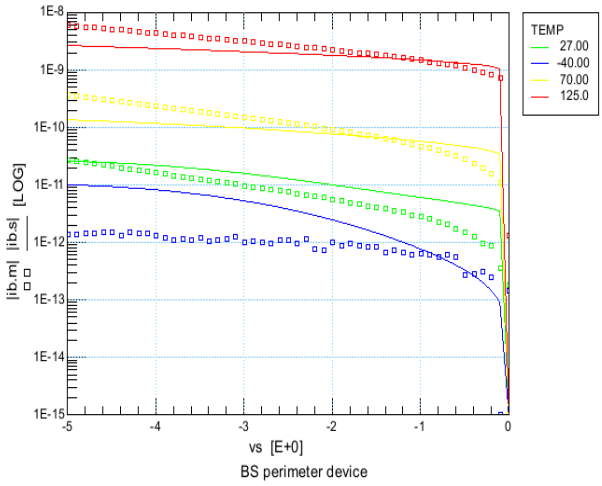
Area



Area



Perimeter



Perimeter

8 Flicker Noise

Flicker noise model parameters are taken from the SG13 simulation models.

8.1 Measurement Conditions

Not defined as no flicker noise measurements are yet done on SG13G2 devices.

8.2 Devices

Not defined as no flicker noise measurements are yet done on SG13G2 devices.

9 Change History

Revision	Device	Changes
190606	All	Initial version
190730	All	Adjust temperature and RF behavior
190924	Summarized Results	Legend added